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Katsuro Sakai

Geometric Aspects of General Topology

Second Edition

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Geometric Aspects of General Topology

Second Edition

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We can imagine and consider many mathematical concepts, such as numbers, spaces, maps, dimensions, etc., that can be indefinitely extended beyond infinity in our minds. Contemplating our mathematical ability in such a manner, I can recall this phrase from the Scriptures:

He has made everything beautiful in its time.
He has even put eternity in their heart;
yet mankind will never find out the work that
the true God has made from start to finish.

—Ecclesiastes 3:11

May our Maker be glorified! Our brain is the work of his hands, as in Psalm 100:3, "Know that Jehovah is God. He is the one who made us, and we belong to him." There are many reasons to give thanks to God and to praise his name. Our mathematical ability is one of them.

Note: Scripture quotations are from
New World Translation of the Holy Scriptures—Revised 2013
(Watchtower Bible and Tract Society of New York, Inc.)

Preface

This book is designed to provide graduate students with standard knowledge of Dimension Theory, ANR Theory (Theory of Retracts), and related topics.[1] As is widely known, these two theories are connected with various fields in Geometric Topology and Algebraic Topology as well as General Topology. Therefore, understanding them will benefit students who wish to study these fields. The following are excellent texts on these theories:

- W. Hurewicz and H. Wallman, *Dimension Theory* (Princeton University Press, Princeton, 1941)
- K. Borsuk, *Theory of Retracts,* MM **44** (Polish Scientific Publishers, Warsaw, 1967)
- S.-T. Hu, *Theory of Retracts* (Wayne State University Press, Detroit, 1965).

However, these classical texts must be updated in the light of new developments, which is our main purpose.

For a comprehensive study of Dimension Theory, we can refer the following:

- R. Engelking, *Theory of Dimensions, Finite and Infinite,* SSPM **10** (Heldermann Verlag, Lembo, 1995)
- M. G. Charalambous, *Dimension Theory, A Selection of Theorems and Counter-examples*, Atlantis Studies in Mathematics **7** (Springer Nature Switzerland AG, 2019).

Our main interest is in dimensions of metrizable spaces because Dimension Theory has been satisfactorily developed for metrizable spaces but there are many negative results for non-metrizable spaces. The above book by Charalambous provides such counter-examples in Dimension Theory of general spaces. In the above books or most other textbooks, no proof is given for the important theorem that $\dim X \times \mathbf{I} =$

[1] ANR Theory is known to serve as a link between Combinatorial Topology and Point-Set Topology. However, ANRs and spaces having the homotopy type of ANRs, like polyhedra or CW complexes, form a natural environment for Homotopy Theory.

$\dim X + 1$ for every metrizable space X.[2] So, we provide its proof. We also give a metrizable example illustrating the difference between the small and large inductive dimensions, and an infinite-dimensional compact metrizable space having no finite-dimensional subspaces except for zero-dimensional ones (i.e., being *hereditarily infinite-dimensional*). These are also contained in Charalambous' book.

In the 1980s and 1990s, famous longstanding problems in Dimension Theory and ANR Theory were finally resolved. In the process, it became clear that these theories are linked with each other through Shape Theory. In Dimension Theory, the Alexandroff Problem had long remained unsolved. This problem queried the existence of an infinite-dimensional space of finite cohomological dimension, which was posed in 1936. Besides, the CE Problem arose as a fascinating question in Shape Theory that asked whether there exists a cell-like map of a finite-dimensional space onto an infinite-dimensional space. In 1978, R. D. Edwards showed that these two problems are equivalent. Finally, in 1988, by constructing an infinite-dimensional compact metrizable space of finite cohomological dimension, A. N. Dranishnikov solved the Alexandroff Problem.

On the other hand, in ANR Theory, for many years, it was unknown whether a metrizable topological linear space is an AR (or more generally, whether a locally equi-connected metrizable space is an ANR). In 1994, using a cell-like map of a finite-dimensional compact manifold onto an infinite-dimensional space, R. Cauty constructed a separable metrizable topological linear space that is not an AR. These great results are discussed in the latter half of the final chapter of this book and provide an understanding of how deeply these theories are related to each other. This is the goal of this book.

A simplicial complex, the object of study in Combinatorial Topology, is a very useful tool in various fields of Topology, and indispensable for studying Dimension Theory and ANR Theory as well as (Co-)Homology Theory. In particular, using inverse systems, we can approximate spaces by simplicial complexes. Those simplicial complexes are not necessary locally finite. However, Piecewise Linear (PL) Topology treats only locally finite simplicial complexes. While there are many textbooks providing some knowledge on simplicial complexes, none discusses non-locally finite simplicial complexes in detail. Therefore, one should refer to the original papers when encountering them. For example, J. H. C. Whitehead's theorem on small subdivisions is very important, but its proof cannot be found in any other textbook.

This book properly treats simplicial complexes that are not locally finite nor countable. It also provides basic results on regular (or derived) neighborhoods and PL manifolds which are fundamental in PL Topology. The homotopy type of simplicial complexes is usually discussed in textbooks on Algebraic Topology using CW complexes, but our geometrical argument using simplicial complexes is simpler.

[2] This proof can be found in Kodama's appendix to the following book:

- K. Nagami, *Dimension Theory* (Academic Press, Inc., New York, 1970).

Thus, this book provides a *course on simplicial complexes and polyhedra* that can be used as a reference in various fields of Topology.

As prerequisites for studying infinite-dimensional manifolds, Jan van Mill provides three chapters (Chaps. 3–5) on simplicial complexes, dimensions, and ANRs in the book:

- J. van Mill, *Infinite-Dimensional Topology, Prerequisites and Introduction*, North-Holland Math. Library **43** (Elsevier Scientific Publishers B.V., Amsterdam, 1989)

Those chapters are similar to Chaps. 4–6 of the present book in content, but those are elementary (or introductory) courses and restricted to separable metrizable spaces. The important results mentioned above are not treated except for an example of a hereditarily infinite-dimensional space. Moreover, one can find an explanation of the Alexandroff Problem and the CE Problem in Chap. 3 of the following book:

- A. Chigogidze, *Inverse Spectra*, North-Holland Math. Library **53** (Elsevier Science B.V., Amsterdam, 1996)

Unfortunately, that book is harder for graduate students to read and understand.

The present book was written based on my lecture notes for graduate classes at the University of Tsukuba.[3] Every year, I had lectured on some topic corresponding to each chapter of this book except the final chapter, and also used the same material for my undergraduate seminar. Readers are required to have finished the initial courses of Set Theory and General Topology. Basic knowledge of Linear Algebra is also a prerequisite. Except for the latter half of the final chapter, this book is self-contained. Since drawing figures is a great help in thinking how to prove something and understanding a given proof, we encourage readers to draw figures when reading proofs, and we ourselves give many figures in this book. However, a figure only represents one aspect of the situation in which the matter to be proved is placed. So, we have tried to avoid saying "obvious from the figure." Instead, we provide detailed proofs.

I hope that this book will give a good insight on how General Topology interacts with other fields of mathematics. The following article provides us the same insight:

- T. Koetsier, J. van Mill, By their fruits ye shall know them: Some remarks on the interaction of General Topology with other areas of Mathematics, Chap. 8 in *History of Topology*, ed. by I. M. James (Elsevier Science B.V., Amsterdam, 1999), pp. 199–239.

Section 3 of that article gives us a brief description about the so-called golden age from the 1920s until the 1960s. The book by Hurewicz and Wallman is mentioned there as a beautiful survey of Dimension Theory until 1941. The other discussed there is what I wanted to deal with in Chap. 2 and Sect. 3.8 of this book. The results presented in its Subsects. 4.9 and 4.10 are dealt with later in our Chap. 7. The rest of the same Sect. 4 is covered in my second book: *Topology of Infinite-Dimensional Manifolds*, published in *Springer Monographs in Mathematics*, the same series as the first edition of this book.

* * *

Below is an overview of the book's contents. In Chap. 1, we fix the terminology and notation commonly used in this book. As an aside, we prove the well-known fact that $\mathbb{N}^{\mathbb{N}}$ is homeomorphic to $\mathbb{R}\backslash\mathbb{Q}$. We describe the Banach spaces $\ell_p(\Gamma)$, $p \in \mathbb{N} \cup \{\infty\}$,

[3] The lecture notes contained only about half of the book's content and did not include Chap. 7.

defined as linear subspaces of the infinite product $\mathbb{R}^\Gamma$ and prove Mazur's theorem that $\ell_p(\Gamma)$ is homeomorphic to $\ell_1(\Gamma)$ for every $1 < p < \infty$.[4]

Chapter 2 develops some general material concerning topological spaces appropriate to graduate students. It can be considered a *supplementary course in Topology* for students who finished a usual undergraduate course. We discuss compactly generated spaces (also called k-spaces),[5] paracompact spaces, and some metrization theorems for non-separable spaces, which are not treated in a typical undergraduate course.[6] This chapter also contains the Baire category theorem and Michael's theorem on local properties, which are very useful and applicable in various situations, even in fields outside Topology. We further discuss the direct limits of towers (increasing sequences) of spaces, which are often used in Algebraic Topology.[7] A non-Hausdorff direct limit of a closed tower of Hausdorff spaces is given together with various counter-examples showing differences in several kinds of normality. Topologies for function spaces and hyperspaces are also discussed. As for function spaces, we deal with the limitation topology as well as the compact-open topology, and prove the Ascoli–Arzela Theorem. As for hyperspaces, the Vietoris topology and the Hausdorff metric are described.

Chapter 3 is a *short basic course on topological linear spaces and convex sets*. There are many good textbooks on these subjects but this chapter is *a little different* from them. We focus on matters related to General Topology and Geometric Topology. Convex sets are then discussed in detail. Although not covered in many popular textbooks, it is shown that every metrizable topological linear space has an F-norm which is a generalization of a usual norm. This chapter also contains Michael's selection theorem, which is applied to show that every Banach space is a (topological) factor of $\ell_1(\Gamma)$. Moreover, for every Tychonoff space X, the existence of the free topological linear space over X and its basic properties are shown.

Chapter 4 is a *course on simplicial complexes and polyhedra.* As noted above, simplicial complexes are treated without assuming local finiteness. A simplicial complex introduces two standard topologies on its polyhedron, called the Whitehead (weak) topology and the metric topology, and we compare the advantages and

[4] These two facts on $\mathbb{N}^\mathbb{N}$ and $\ell_p(\Gamma)$ are not necessary to read this book.

[5] In Algebraic Topology, k-spaces commonly appear, whose class contains locally compact spaces, metrizable spaces, and CW complexes.

[6] These subjects are discussed in the following book, now a very popular textbook at the *senior undergraduate* or the *first-year graduate* level:

- J. R. Munkres, *Topology, 2nd ed.* (Prentice Hall, Inc., Upper Saddle River, 2000).

[7] The direct limits are discussed in the Appendix of Dugundji's book below but not in Engelking's book below, a comprehensive reference on General Topology:

- J. Dugundji, *Topology* (Allyn and Bacon, Inc., Boston, 1966).
- R. Engelking, *General Topology, Revised and completed edition*, SSPM **6** (Heldermann Verlag, Berlin, 1989).

disadvantages of each. As mentioned above, a proof of J. H. C. Whitehead's theorem on small subdivisions is provided. In general, the metric topology on a polyhedron depends on its triangulation, that is, many subdivisions change the metric topology. We characterize a simplicial subdivision that does not change the metric topology and show D. W. Henderson's metric topology version of Whitehead's theorem. A map between polyhedra is PL if it is simplicial with respect to some triangulations, but the converse is not necessarily true. A characterization of such a PL map is given. The topological uniqueness of regular neighborhoods is proved without local finiteness condition. We also prove the fact that a simplicial complex K is a combinatorial manifold if and only if $|K|$ is a PL manifold.

Using inverse systems, we can approximate spaces by simplicial complexes. In fact, we show that every paracompact space is homeomorphic to the limit of an inverse system of polyhedra with PL bonding maps, and that every completely metrizable space is homeomorphic to the inverse limit of an inverse *sequence* of locally finite-dimensional simplicial complexes with the metric topology with PL bonding maps. On the other hand, we also show Mardešić's result that a compactum X cannot be written as the limit of any inverse sequence of finite simplicial complexes with *simplicial bonding maps* if $\dim X > 0$ and X contains no arcs. The simplicial mapping cylinder is introduced and applied to prove the Whitehead–Milnor theorem on the homotopy type of simplicial complexes. It is also applied to prove that every weak homotopy equivalence between simplicial complexes is a homotopy equivalence. Those proofs cannot be found in any other book dealing with simplicial complexes. Knowledge of homotopy groups is not required, even when weak homotopy equivalences are discussed.

Chapters 5 and 6 are devoted to *Dimension Theory* and *ANR Theory*, respectively. We prove basic results and fundamental theorems on these theories. The contents are very similar to Chaps. 4 and 5 of van Mill's book: *Infinite-Dimensional Topology*. However, as mentioned previously, we do not restrict our arguments to separable metrizable spaces but our main concern is metrizable spaces. We discuss more results than are covered in van Mill's book. Below, we review some of them.

In Chap. 5, we exhibit a non-separable metrizable space such that the large inductive dimension does not coincide with the small inductive dimension. We describe Kulesza's example (with Levin's proof), which is also found in the book by Charalambous. The transfinite inductive dimension is also discussed, which is not treated in van Mill's book. We discuss dimension of subsets of Euclidean space $\mathbb{R}^n$, which is contained in the book by Hurewicz and Wallman but not in van Mill's book, that is, the invariance of domain, the countable dense homogeneity of $\mathbb{R}^n$, and the fact that for $X \subset \mathbb{R}^n$, $\dim X = n$ if and only if X has an interior point. It is also proved that every completely metrizable space with dimension $\leqslant n$ is homeomorphic to the inverse limit of an inverse sequence of metric simplicial complexes with dimension $\leqslant n$. Further, hereditarily infinite-dimensional spaces are discussed based on van Mill's book. As an addendum, we give a proof of the Hahn–Mazurkiewicz Theorem that a space X is the continuous image of the interval if and only if X is a locally connected non-empty continuum.

In Chap. 6, we discuss topics that are not treated in van Mill's book or in the two classical books by Hu and Borsuk. The following are examples of such topics: uniform ANRs in the sense of Michael and its completion; Kozlowski's theorem that the metrizable range of a fine homotopy equivalence of an ANR is also an ANR; Cauty's characterization that a metrizable space is an ANR if and only if every open set has the homotopy type of an ANR; Haver's theorem that every countable-dimensional locally contractible metrizable space is an ANR; and Bothe's theorem, with Kodama's proof, that every n-dimensional metrizable space can be embedded in an $(n+1)$-dimensional AR as a closed set. We also give a proof of Wojdysławski's Theorem on hyperspaces of Peano continua. Our proof follows Kelley's one applying a Whitney map, which is different from the proof given in Chap. 5 of van Mill's book: *Infinite-Dimensional Topology*. In that book, it is only proved that the hyperspace of compacta in a Peano continuum X is an AR. But, we also prove that the hyperspace of continua in X is an AR.

In Chap. 7, *cell-like maps and related topics* are discussed. The first half is self-contained, but the second half is not because some algebraic results are necessary. In the first half, we examine the existing relations between cell-like maps, soft maps, fine homotopy equivalences, etc. The second half is devoted to related topics. In particular, the CE Problem is explained and Cauty's example is presented. Note that Chigogidze's "*Inverse Spectra*" is the only book dealing with soft maps and provides an explanation of the Alexandroff Problem and the CE Problem.

In the second half of Chap. 7, using the K-theory result of Adams, we present the Taylor example. Eilenberg–MacLane spaces $K(\mathbb{Z}, n)$, $n \in \mathbb{N}$, are usually constructed as CW complexes, but here they are constructed as countable simplicial complexes. To avoid using cohomology, we define the cohomological dimension geometrically. By applying the cohomological dimension, we can prove the equality $\dim X \times \mathbf{I} = \dim X + 1$ for every metrizable space X. We also discuss the Alexandroff Problem and the CE Problem as mentioned above. The equivalence of these problems is proved. Next, we describe the Dydak–Walsh example (improving Dranishnikov's example) that gives an affirmative answer to the Alexandroff Problem. However, this part is not self-contained. As a corollary, we answer the CE Problem, i.e., we obtain a cell-like mapping of a finite-dimensional compact manifold onto an infinite-dimensional compactum. We also present Cauty's example, i.e., a metrizable topological linear space that is not an absolute extensor. In the proof, we need to make the above cell-like mapping an open map by applying Walsh's open mapping approximation theorem. But, a proof of Walsh's theorem is beyond the scope of this book.

In the Appendix, we give a short course on Homotopy Groups, which is helpful in reading the last section of Chap. 4 and the second half of Chap. 7. Using homotopy groups, we characterize n-equivalences and weak homotopy equivalences. We also discuss axioms of Čech cohomology theory and, using Eilenberg–MacLane spaces, we construct a cohomology theory which is equivalent to Čech theory.

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I would like to express my sincere appreciation to my teacher, Prof. Yukihiro Kodama, who introduced me to Shape Theory and Infinite-Dimensional Topology and warmly encouraged me to persevere in study. I warmly thank my graduate students, Yutaka Iwamoto, Yuji Akaike, Shigenori Uehara, Masayuki Kurihara, Masato Yaguchi, Kotaro Mine, Atsushi Yamashita, Minoru Nakamura, Atsushi Kogasaka, Katsuhisa Koshino, and Hanbiao Yang for their careful reading and helpful comments on lecture notes on which this book is based. I owe my gratitude to Ross Geoghegan for improving the written English text of the first edition. I am also grateful to Haruto Ohta, Taras Banakh, Zhongqiang Yang, Jan van Mill, and Leonard Rubin for their valuable comments and suggestions. Van Mill informed me that Charalambous' book was published after the first edition of this book was published. Rubin gave me information on Mardešić's result (Theorem 4.17.10). Among those people, I would like to express my special thanks to Atsushi Yamashita, who contributed immensely to the completion of this second edition by carefully reading and checking the whole manuscript in details, pointing out typos and logical gaps, and providing many helpful comments and suggestions to make it more readable. He spent over a couple of years doing so. Without his help, the book would not have been completed! Finally, I am indebted to Mr. Masayuki Nakamura, the editor at Springer Japan, for his work in the publication of the first and second editions of this book.

Tsukuba, Japan Katsuro Sakai
January 2025

Notes on the Second Edition

—To readers of the first edition of this book and the readers of the second book: *Topology of Infinite-Dimensional Manifolds*

This book is my first published book, which provides preliminaries and background for its companion, my second book, which is cited as [TIDM] below:

- *Topology of Infinite-Dimensional Manifolds*, SMM (Springer Nature, Singapore, 2020).

However, the first edition of this book was not sufficient to cover all preliminaries of [TIDM] and contained many typos, incorrect symbols/characters, and small gaps in some proofs. This second edition reflects all such corrections and contains supplementary results added to [TIDM]. To add more contents, new sections were inserted into Chaps. 2–6. Thus, it has about 40% more pages than the first edition.

In Chap. 2, compactly generated spaces (called k-spaces for short) are newly included (Sect. 2.1), which commonly appear in Algebraic Topology. There are more counter-examples than in the first edition. The compact-open topology is treated in the new Sect. 2.10 (it was included in Sect. 1.1 of the previous edition). Basic properties of free topological linear spaces over compacta were proved in Sect. 7.11 of the previous edition, some of which are now generalized to Tychonoff spaces in the current Sect. 3.9.

Chapter 4 was re-edited, that is, sections were reorganized and some new sections were inserted. Now, it contains nearly 60% more pages than before. The General Position Lemma was moved from the previous Chap. 5 (Lemma 5.8.4) to the present Chap. 4 (Lemma 4.14.1). In Sect. 1.9 of [TIDM], derived neighborhoods and regular neighborhoods are discussed; this is now completely rewritten in the present Sect. 4.12. PL manifolds and combinatorial manifolds are discussed in Appendix of [TIDM], which is now treated in new Sect. 4.13 except for Sect. A.4. Theorem 1.8.11 and Theorem A.4.1 in [TIDM] can be seen in Sect. 4.14.[8] Thus, this chapter is more useful and helpful for studying PL Topology (Combinatorial Topology). In Sect. 4.16, not only inverse sequences but also inverse systems are discussed. We give, in Sect. 4.17, Mardešić's result on an inverse limit of simplicial complexes mentioned in the Preface.

In Chap. 5, Sects. 5.2–5.4 are also reorganized and some contents are added. In Sect. 5.5, one more example is added to demonstrate pathological phenomena of the large (or small) inductive dimension. In Chap. 6, we add some results on a strong deformation retract in an ANR and locally equi-connected (LEC) spaces, which were supplemented in Sects. 1.13 and 1.14 of [TIDM]. A new Sect. 6.14 is established to contain a proof of Wojdysławski's Theorem on hyperspaces of Peano continua. In addition, supplementary results in Sect. 1.15 of [TIDM] are added to Sect. 7.4. The Appendix was added at the end of the book, where the section on homotopy groups (the previous Sect. 4.14) was moved and a homotopical definition of cohomology groups was added.

Thus, some contents in the previous edition were moved into different sections and the section numbers are different from the previous edition. The information below is helpful for readers of [TIDM] to confirm results cited from this book.

> The current Sects. 2.10, 2.12, 4.12, 4.13, 4.14, 4.16, and 6.14 are new additions, where Sect. 2.10 is an extension of the last part of the previous Sect. 1.1. The previous Sects. 2.10 and 2.9 were reversed in order, becoming Sects. 2.9 and 2.11. The previous Sects. 4.2 and 4.6 were reorganized into three sections: 4.2–4.4, where some contents were added. The previous Sects. 4.4 and 4.8 were split into two separate sections, the current Sects. 4.10, 4.11, and 4.7, 4.9, respectively, whose contents are increased. The previous Sects. 4.3, 4.5, 4.9, 4.10–4.13 are changed to 4.5, 4.6, 4.15, 4.17–4.20. As mentioned above, Sect. 4.14 was divided into two sections 8.1 and 8.2 in the final chapter, Appendix. The previous Sects. 5.2–5.4 were also reorganized and the current version of these sections deal with the subject matter a little differently. The previous Sects. 5.12 and 5.13 were combined into one section, Sect. 5.12. Thus, the previous Sect. 5.14 is the current 5.13.

[8] These are Theorem 1.8.11 and Theorem A.4.1 in [TIDM]. The former proof was wrong and is corrected here.

Contents

1 Preliminaries 1
1.1 Terminology and Notation 1
1.2 Banach Spaces in the Product of Real Lines 16
Notes for Chap. 1 28
References 28

2 Metrizability, Paracompactness, and Related Properties 29
2.1 Some Concepts Related to Compactness 29
2.2 The Tietze–Urysohn Extension Theorem and Normalities 40
2.3 A.H. Stone's Theorem and Metrization 44
2.4 Sequences of Open Covers and Metrization 49
2.5 Complete Metrizability 55
2.6 Paracompactness and Local Properties 61
2.7 Partitions of Unity 69
2.8 The Direct Limits of Towers of Spaces 75
2.9 Counter-Examples 81
2.10 The Compact-Open Topology for Function Spaces 94
2.11 The Limitation Topology for Function Spaces 103
2.12 The Vietoris Topology and the Hausdorff Metric for Hyperspaces 110
Notes for Chap. 2 123
References 125

3 Topology of Linear Spaces and Convex Sets 127
3.1 Flats and Affine Functions 127
3.2 Convex Sets 131
3.3 The Hahn–Banach Extension Theorem 142
3.4 Topological Linear Spaces 151
3.5 Finite Dimensionality 160
3.6 Metrizability and Normability 167
3.7 The Closed Graph and Open Mapping Theorems 177

3.8 Continuous Selections . . . 180
3.9 Free Topological Linear Spaces . . . 186
Notes for Chap. 3 . . . 199
References . . . 200

4 Simplicial Complexes and Polyhedra . . . 203
4.1 Simplexes and Cells . . . 204
4.2 Cell Complexes and Simplicial Complexes . . . 213
4.3 Subdivisions of Cell Complexes . . . 224
4.4 Combinatorial Equivalence and Triangulation of Polyhedra . . . 231
4.5 Product Complexes and Homotopy Extension . . . 241
4.6 The Metric Topology of Polyhedra . . . 248
4.7 Admissible Subdivisions . . . 260
4.8 Small Subdivisions and Paracompactness of Polyhedra . . . 270
4.9 Small Admissible Subdivisions . . . 277
4.10 Simplicial Maps . . . 285
4.11 Piecewise-Linear Maps . . . 292
4.12 Derived Neighborhoods and Regular Neighborhoods . . . 303
4.13 PL n-Manifolds and Combinatorial n-Manifolds . . . 317
4.14 PL Embedding and Relative PL Approximation Theorems . . . 328
4.15 The Nerves of Open Covers . . . 338
4.16 Inverse Systems of Polyhedra . . . 348
4.17 The Inverse Limits of Metric Polyhedra . . . 358
4.18 The Mapping Cylinders . . . 365
4.19 The Homotopy Type of Simplicial Complexes . . . 375
4.20 Weak Homotopy Equivalences . . . 380
Notes for Chap. 4 . . . 386
References . . . 388

5 Dimensions of Spaces . . . 391
5.1 The Brouwer Fixed Point Theorem . . . 392
5.2 Characterizations of Dimension . . . 397
5.3 Partitions and Essential Maps . . . 406
5.4 Dimension of Metrizable Spaces . . . 410
5.5 Inductive Dimensions . . . 419
5.6 Infinite Dimensions . . . 428
5.7 Dimension-Preserving Compactification Theorems . . . 436
5.8 Embeddings into Euclidean Spaces . . . 440
5.9 Universal Spaces . . . 449
5.10 Nöbeling Spaces and Menger Compacta . . . 455
5.11 Total Disconnectedness and the Cantor Set . . . 466
5.12 Totally Disconnected Examples and Infinite-Dimensional Examples . . . 470
5.13 Addendum: The Hahn–Mazurkiewicz Theorem . . . 477
Notes for Chap. 5 . . . 485
References . . . 487

6 Retracts and Extensors 491
6.1 The Dugundji Extension Theorem and ANEs 492
6.2 Embeddings of Metric Spaces and ANRs 500
6.3 Small Homotopies and LEC Spaces 511
6.4 The Homotopy Extension Property 519
6.5 Complementary Pairs of ANRs 525
6.6 Realizations of Simplicial Complexes 533
6.7 Fine Homotopy Equivalences 543
6.8 Completions of ANRs and Uniform ANRs 549
6.9 Homotopy Types of Open Sets in ANRs 557
6.10 Countable-Dimensional ANRs 563
6.11 The Local n-Connectedness 566
6.12 Finite-Dimensional ANRs 573
6.13 Embeddings into Finite-Dimensional ARs 584
6.14 Hyperspaces and Whitney Maps 592
Notes for Chap. 6 604
References 605

7 Cell-Like Maps and Related Topics 609
7.1 Trivial Shape and Related Properties 610
7.2 Soft Maps and the 0-Dimensional Selection Theorem 615
7.3 Hereditary n-Equivalence and Local Connections 621
7.4 Fine Homotopy Equivalences Between ANRs 629
7.5 Hereditary Shape Equivalences and UV^n Maps 634
7.6 The Near-Selection Theorem 640
7.7 The Suspensions and the Taylor Example 643
7.8 Eilenberg–MacLane Complexes 654
7.9 Cohomological Dimension 665
7.10 Alexandroff's Problem and the CE Problem 676
7.11 Continuous Metrics on Free Topological Linear Spaces 686
7.12 A Non-AR Metric Linear Space 691
Notes for Chap. 7 703
References 704

8 Appendix: Homotopy Groups and Čech Cohomology Groups 707
8.1 Definition and Basic Properties of Homotopy Groups 707
8.2 Characterizing n-Equivalences and Weak Homotopy Equivalences 717
8.3 Axioms of Čech Cohomology Theory 724
8.4 Homotopical Definition of Čech Cohomology Groups 729
Notes for Chap. 8 741
References 742

Index 743

Chapter 1
Preliminaries



The reader should have finished a first course in Set Theory and General Topology; basic knowledge of Linear Algebra is also a prerequisite. In this chapter, we introduce some terminology and notation. We also describe the standard Banach spaces contained in the product of real lines. Because of their interest, some minor topics are also described even if they will not be used hereafter.

1.1 Terminology and Notation

The standard sets are denoted as follows:

- $\mathbb{N}$ — the set of natural numbers (i.e., positive integers);
- $\omega = \mathbb{N} \cup \{0\}$ — the set of non-negative integers;
- $\mathbb{Z}$ — the set of integers;
- $\mathbb{Q}$ — the set of rationals;
- $\mathbb{R} = (-\infty, \infty)$ — the real line with the usual topology;
- $\mathbb{C}$ — the complex plane;
- $\mathbb{R}_+ = [0, \infty)$ — the half (real) line;
- $\mathbf{I} = [0, 1]$ — the unit closed interval.

A (topological) **space** is assumed to be **Hausdorff** and a **map** is a **continuous** function. A **singleton** is a space consisting of one point. A space is said to be **non-degenerate** if it has at least two points. A compact metrizable space is called a **compactum** and a connected compactum is called a **continuum**.[1]

[1] Their plurals are **compacta** and **continua**, respectively.

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For a space X and $A \subset X$, we use the following notation:

- $\mathrm{cl}_X A$ (or $\mathrm{cl}\, A$) — the closure of A in X;
- $\mathrm{int}_X A$ (or $\mathrm{int}\, A$) — the interior of A in X;
- $\mathrm{bd}_X A$ (or $\mathrm{bd}\, A$) — the boundary of A in X;
- id_X (or id) — the identity map of X.

We call A a **clopen** set in X if it is both closed and open in X. A countable union of closed sets is called an $\boldsymbol{F_\sigma}$ **set**; a countable intersection of open sets is called a $\boldsymbol{G_\delta}$ **set**.[2] The following notation is also used:

- $\mathrm{Metr}(X)$ — the set of all admissible metrics of X;
- $\mathrm{Metr}^B(X)$ — the set of all admissible bounded metrics of X;
- $\mathrm{Metr}^c(X)$ — the set of all admissible bounded complete metrics of X.

Then, $\mathrm{Metr}(X) \neq \emptyset$ (or $\mathrm{Metr}^c(X) \neq \emptyset$) means that X is metrizable (or completely metrizable).

For spaces X and Y with subspaces $X_1, \ldots, X_n \subset X$ and $Y_1, \ldots, Y_n \subset Y$,

- $X \approx Y$ means that X and Y are homeomorphic;
- $(X, X_1, \ldots, X_n) \approx (Y, Y_1, \ldots, Y_n)$ means that there exists a homeomorphism $h : X \to Y$ such that $h(X_1) = Y_1, \ldots, h(X_n) = Y_n$;
- $(X, x_0) \approx (Y, y_0)$ means $(X, \{x_0\}) \approx (Y, \{y_0\})$,

where (X, x_0) is called a **pointed space** and x_0 its **base point**.

The cardinality of a set Γ is denoted by $\mathrm{card}\, \Gamma$. The **weight** $w(X)$, the **density** $\mathrm{dens}\, X$, and the **cellularity** $c(X)$ of a space X are defined as follows:

- $w(X) \underset{\text{def}}{=} \min\{\mathrm{card}\, \mathcal{B} \mid \mathcal{B} \text{ is an open basis for } X\}$;
- $\mathrm{dens}\, X \underset{\text{def}}{=} \min\{\mathrm{card}\, D \mid D \text{ is a dense set in } X\}$;
- $c(X) \underset{\text{def}}{=} \sup\{\mathrm{card}\, \mathcal{G} \mid \mathcal{G} \text{ is a pairwise disjoint open collection}\}$.

As is easily observed, $c(X) \leqslant \mathrm{dens}\, X \leqslant w(X)$ in general. If X is metrizable, all these cardinalities coincide.

Indeed, let D be a dense set in X with $\mathrm{card}\, D = \mathrm{dens}\, X$, and $\mathcal{G}$ be a pairwise disjoint collection of non-empty open sets in X. Since each $G \in \mathcal{G}$ meets D, we have an injection $g : \mathcal{G} \to D$, hence $\mathrm{card}\, \mathcal{G} \leqslant \mathrm{card}\, D = \mathrm{dens}\, X$. It follows that $c(X) \leqslant \mathrm{dens}\, X$. Now, let $\mathcal{B}$ be an open basis for X with $\mathrm{card}\, \mathcal{B} = w(X)$. By taking any point $x_B \in B$ from each $B \in \mathcal{B}$, we have a dense set $\{x_B \mid B \in \mathcal{B}\}$ in X, which implies $\mathrm{dens}\, X \leqslant w(X)$.

When X is metrizable, we show the converse inequality. The case $\mathrm{card}\, X < \aleph_0$ is trivial. We may assume that $X = (X, d)$ is a metric space with $\mathrm{card}\, X \geqslant \aleph_0$. Here, we use the notation defined on the next page. Let D be a dense set in X with $\mathrm{card}\, D = \mathrm{dens}\, X$. Then, $\{\mathrm{B}(x, 1/n) \mid x \in D,\ n \in \mathbb{N}\}$ is an open basis for X, which implies $w(X) \leqslant \mathrm{dens}\, X$. For each $n \in \mathbb{N}$, using Zorn's Lemma, we can find a maximal 2^{-n}-discrete subset $X_n \subset X$, i.e., $d(x, y) \geqslant 2^{-n}$ for every pair of distinct points $x, y \in X_n$. Then, $\mathcal{G}_n = \{\mathrm{B}(x, 2^{-n-1}) \mid x \in$

[2] The term F_σ comes from French *fermé* (closed) and *somme* (sum, union); the term G_δ comes from German *Gebiet* (domain, region; meaning open set in this case) and *Durchschnitt* (intersection).

$X_n\}$ is a pairwise disjoint open collection, and hence $\operatorname{card} X_n = \operatorname{card} \mathcal{G}_n \leqslant c(X)$. Observe that $X_* = \bigcup_{n\in\mathbb{N}} X_n$ is dense in X, which implies

$$\operatorname{dens} X \leqslant \operatorname{card} X_* \leqslant \sum_{n\in\mathbb{N}} \operatorname{card} X_n \leqslant \aleph_0 c(X) = c(X).$$

An indexed collection of sets with an index set Γ is denoted as $(X_\gamma)_{\gamma\in\Gamma}$, which should be distinguished from the collection $\mathfrak{X} = \{X_\gamma \mid \gamma \in \Gamma\}$ and can be regarded as a surjection $\Gamma \ni \gamma \mapsto X_\gamma \in \mathfrak{X}$. The product $\prod_{\gamma\in\Gamma} X_\gamma$ is a set of all functions $x : \Gamma \to \bigcup_{\gamma\in\Gamma} X_\gamma$ such that $x(\gamma) \in X_\gamma$ for each $\gamma \in \Gamma$, where $x \in \prod_{\gamma\in\Gamma} X_\gamma$ is denoted by $(x(\gamma))_{\gamma\in\Gamma}$ and each $x(\gamma)$ is called the γ-coordinate of x. For each $\gamma \in \Gamma$, the projection $\mathrm{pr}_\gamma : \prod_{\gamma\in\Gamma} X_\gamma \to X_\gamma$ is defined by $\mathrm{pr}_\gamma(x) = x(\gamma)$. For each $\Lambda \subset \Gamma$, the projection $\mathrm{pr}_\Lambda : \prod_{\gamma\in\Gamma} X_\gamma \to \prod_{\lambda\in\Lambda} X_\lambda$ is defined by $\mathrm{pr}_\Lambda(x) = x|\Lambda\ (= (x(\lambda))_{\lambda\in\Lambda})$. In the case that every X_γ is the same set X, $\prod_{\gamma\in\Gamma} X_\gamma$ coincides with the set X^Γ of all function from Γ to X. In particular, if $\Gamma = \mathbb{N}$, the product $X^\mathbb{N}$ of countable infinite copies of X is just the set of all sequences $(x_i)_{i\in\mathbb{N}}$ in X. When $\Gamma = \{1, \ldots, n\}$, $X^\Gamma = X^n$ is the product of n copies of X.

For an indexed collection $(X_\gamma)_{\gamma\in\Gamma}$ of (topological) spaces, the **product topology** of $\prod_{\gamma\in\Gamma} X_\gamma$ is defined as the coarsest topology such that all projections pr_γ, $\gamma \in \Gamma$, are continuous, which is generated by $\{\mathrm{pr}_\gamma^{-1}(U) \mid \gamma \in \Gamma, U \text{ is open in } X_\gamma\}$. The **product space** $\prod_{\gamma\in\Gamma} X_\gamma$ is the space with the product topology. For each $\Lambda \subset \Gamma$, the projection $\mathrm{pr}_\Lambda : \prod_{\gamma\in\Gamma} X_\gamma \to \prod_{\lambda\in\Lambda} X_\lambda$ is continuous. For a space X, we regard X^Γ, $X^\mathbb{N}$, and X^n as the product spaces. For the product space $X \times Y$, the projections are denoted by $\mathrm{pr}_X = \mathrm{pr}_1 : X \times Y \to X$ and $\mathrm{pr}_Y = \mathrm{pr}_2 : X \times Y \to Y$.

Now, let $X = (X, d)$ be a metric space, $x \in X$, $\varepsilon > 0$, and $A, B \subset X$. Moreover, let $\mathcal{A}$ be a collection of subsets of X. We define:

- $\mathrm{B}_d(x, \varepsilon) \underset{\text{def}}{=} \{y \in X \mid d(x, y) < \varepsilon\}$ — the ε-neighborhood of x in X (i.e., the open ball with center x and radius ε);
- $\overline{\mathrm{B}}_d(x, \varepsilon) \underset{\text{def}}{=} \{y \in X \mid d(x, y) \leqslant \varepsilon\}$ — the closed ε-neighborhood of x in X (i.e., the closed ball with center x and radius ε);
- $d(x, A) \underset{\text{def}}{=} \inf\{d(x, y) \mid y \in A\}$ — the distance of x from A;
- $\mathrm{N}_d(A, \varepsilon) \underset{\text{def}}{=} \{x \in X \mid d(x, A) < \varepsilon\}$ $= \bigcup_{x\in A} \mathrm{B}_d(x, \varepsilon)$ — the ε-neighborhood of A in X;
- $\overline{\mathrm{N}}_d(A, \varepsilon) \underset{\text{def}}{=} \{x \in X \mid d(x, A) \leqslant \varepsilon\}$ — the closed ε-neighborhood of A in X;
- $\operatorname{dist}_d(A, B) \underset{\text{def}}{=} \inf\{d(x, y) \mid x \in A,\ y \in B\}$ $= \inf\{d(x, A) \mid x \in A\}$ — the distance of A and B;
- $\operatorname{diam}_d A \underset{\text{def}}{=} \sup\{d(x, y) \mid x, y \in A\}$ — the diameter of A;
- $\operatorname{mesh}_d \mathcal{A} \underset{\text{def}}{=} \sup\{\operatorname{diam}_d A \mid A \in \mathcal{A}\}$ — the mesh of $\mathcal{A}$.

It should be noted that $\mathrm{N}_d(\{x\}, \varepsilon) = \mathrm{B}_d(x, \varepsilon)$, $d(x, \{y\}) = d(x, y)$, $\mathrm{dist}_d(\{x\}, A) = d(x, A)$, and $\mathrm{mesh}\{A\} = \operatorname{diam} A$. If there is no possibility of confusion, we can drop the subscript d and simply write $\mathrm{B}(x, \varepsilon)$, $\overline{\mathrm{B}}(x, \varepsilon)$, $\mathrm{N}(A, \varepsilon)$, $\operatorname{diam} A$, $\operatorname{dist}(A, B)$, and $\operatorname{mesh} \mathcal{A}$. As special cases, $\mathrm{N}(\emptyset, \varepsilon) = \emptyset$, $d(x, \emptyset) = \infty$, $\operatorname{dist}(\emptyset, B) = \operatorname{dist}(A, \emptyset) = \infty$. By convention, $\operatorname{diam} \emptyset = \operatorname{mesh} \emptyset = 0$.

The standard spaces are listed below:

- $\mathbb{R}^n$ — Euclidean n-space with the following Euclidean norm:

$$\|x\| = \sqrt{x(1)^2 + \cdots + x(n)^2},$$

  $\mathbf{0} = (0, \ldots, 0) \in \mathbb{R}^n$ — the origin, the zero vector or the zero element,
  $\mathbf{e}_i \in \mathbb{R}^n$ — the unit vector defined by $\mathbf{e}_i(i) = 1$ and $\mathbf{e}_i(j) = 0$ for $j \neq i$;
- $\mathbb{R}^n_+ = \{x \in \mathbb{R}^n \mid x(n) \geqslant 0\} = \mathbb{R}^{n-1} \times \mathbb{R}_+$ — Euclidean half n-space[3];
- $\mathbf{S}^{n-1} = \{x \in \mathbb{R}^n \mid \|x\| = 1\}$ — the unit $(n-1)$-sphere;
- $\mathbf{B}^n = \{x \in \mathbb{R}^n \mid \|x\| \leqslant 1\}$ — the unit closed n-ball;
- $\Delta^n = \left\{x \in (\mathbb{R}_+)^{n+1} \;\middle|\; \sum_{i=1}^{n+1} x(i) = 1\right\}$ — the standard n-simplex;
- $\boldsymbol{Q} = [-1, 1]^{\mathbb{N}}$ — the Hilbert cube;
- $\boldsymbol{s} = \mathbb{R}^{\mathbb{N}}$ — the space of sequences;
- $\mu^0 = \left\{ \sum_{i=1}^{\infty} 2x_i/3^i \;\middle|\; x_i \in \{0, 1\}\right\}$ — the Cantor (ternary) set;
- $\nu^0 = \mathbb{R} \setminus \mathbb{Q}$ — the space of irrationals;
- $\mathbf{2} = \{0, 1\}$ — the discrete space of two points.

Note that $\mathbf{S}^{n-1}$, $\mathbf{B}^n$, and Δ^n are not product spaces, although they have the same symbols as product spaces. The indexes $n-1$ and n represent their dimensions (the same for indexes of μ^0 and ν^0).

As is well-known, the countable product $\mathbf{2}^{\mathbb{N}}$ of the discrete space $\mathbf{2} = \{0, 1\}$ is homeomorphic to the Cantor set μ^0 by the correspondence:

$$x \mapsto \sum_{i \in \mathbb{N}} \frac{2x(i)}{3^i}.$$

On the other hand, the countable product $\mathbb{N}^{\mathbb{N}}$ of the discrete space $\mathbb{N}$ of natural numbers is homeomorphic to the space ν^0 of irrationals.[4] In fact, $\mathbb{N}^{\mathbb{N}} \approx (0, 1) \setminus \mathbb{Q} \approx (-1, 1) \setminus \mathbb{Q} \approx \nu^0$. These three homeomorphisms are given as follows:

[3] Do not confuse $\mathbb{R}^n_+$ with $(\mathbb{R}_+)^n$ the product of n copies of $\mathbb{R}_+$.

[4] This fact will not be used hereafter in this book.

$$x \mapsto \cfrac{1}{x(1)+\cfrac{1}{x(2)+\cfrac{1}{x(3)+\cfrac{1}{\ddots}}}}; \quad t \mapsto 2t-1; \quad s \mapsto \frac{s}{1-|s|}.$$

The above first correspondence[5] is a homeomorphism can be verified as follows: For each $x \in \mathbb{N}^{\mathbb{N}}$, we define $x^* \in \mathbb{N}^{\mathbb{N}}$ by $x^*(i) = x(i+1)$. Then, using induction on n, we define a map $a_n : \mathbb{N}^{\mathbb{N}} \to \mathbf{I}$ by

$$a_1(x) = \frac{1}{x(1)} \text{ and } a_n(x) = \frac{1}{x(1)+a_{n-1}(x^*)} \text{ for } n > 1. \qquad (\star)$$

This definition can also be expressed as follows:

$$a_n(x) = \cfrac{1}{x(1)+\cfrac{1}{x(2)+\cfrac{1}{\ddots+\cfrac{1}{x(n)}}}}.$$

Observe that $0 < a_2(x) < a_4(x) < \cdots < a_3(x) < a_1(x) \leqslant 1$. By the following fact, the first correspondence $\mathbb{N}^{\mathbb{N}} \ni x \mapsto \alpha(x) = \lim_{n\to\infty} a_n(x) \in \mathbf{I}$ is well-defined and continuous:

Fact *For every $m > n$, $|a_n(x) - a_m(x)| \leqslant (n+1)^{-1}$.*

This fact can be shown by induction on $n \in \mathbb{N}$. First, observe that, for each $m > 1$,

$$|a_1(x) - a_m(x)| = a_1(x) - a_m(x) \leqslant a_1(x) - a_2(x) = \frac{1}{x(1)(x(1)x(2)+1)} \leqslant 1/2,$$

which implies the case $n = 1$. When $n > 1$, by the inductive assumption,

$$|a_{n-1}(x^*) - a_{m-1}(x^*)| \leqslant n^{-1} \text{ for } m > n,$$

which gives us

$$\begin{aligned} |a_n(x) - a_m(x)| &= \left| \frac{1}{x(1)+a_{n-1}(x^*)} - \frac{1}{x(1)+a_{m-1}(x^*)} \right| \\ &= \frac{|a_{n-1}(x^*) - a_{m-1}(x^*)|}{(x(1)+a_{n-1}(x^*))(x(1)+a_{m-1}(x^*))} \\ &\leqslant \frac{|a_{n-1}(x^*) - a_{m-1}(x^*)|}{(1+a_{n-1}(x^*))(1+a_{m-1}(x^*))} \\ &< \frac{|a_{n-1}(x^*) - a_{m-1}(x^*)|}{1+|a_{n-1}(x^*) - a_{m-1}(x^*)|} \leqslant \frac{n^{-1}}{1+n^{-1}} = \frac{1}{n+1}. \end{aligned}$$

Now, we introduce some notation. For $t \in \mathbb{R}_+$, let $\lfloor t \rfloor$ denote the unique integer $n \in \omega$ such that $n \leqslant t < n+1$, and define $\langle t \rangle = t - \lfloor t \rfloor$ (i.e., $t = \lfloor t \rfloor + \langle t \rangle$, where $\lfloor t \rfloor$ is the integer

[5] The expression of the assignment of x is called a **continued fraction**.

part of t and $\langle t\rangle$ is the decimal part of t). Letting $n \to \infty$ in the inductive definition of $a_n(x)$ in $(\star)$, we have $\alpha(x) = 1/(x(1) + \alpha(x^*))$, which means

$$x(1) = \lfloor 1/\alpha(x)\rfloor \text{ and } \alpha(x^*) = \langle 1/\alpha(x)\rangle. \qquad (\star\star)$$

To see that $\alpha(\mathbb{N}^{\mathbb{N}}) \subset \mathbf{I} \setminus \mathbb{Q} = (0,1) \setminus \mathbb{Q}$, suppose that $\alpha(x) \in \mathbb{Q}$ for some $x \in \mathbb{N}^{\mathbb{N}}$. Then, $\alpha(x) = r_1/r_0$ for some $r_1 < r_0 \in \mathbb{N}$. Since $\alpha(x^*) = \langle 1/\alpha(x)\rangle = \langle r_0/r_1\rangle$ by $(\star\star)$, we have $\alpha(x^*) = r_2/r_1$ for some $r_2 < r_1 \in \mathbb{N}$. Again by $(\star\star)$, $\alpha(x^{**}) = \langle 1/\alpha(x^*)\rangle = \langle r_1/r_2\rangle$, which implies that $\alpha(x^{**}) = r_3/r_2$ for some $r_3 < r_2 \in \mathbb{N}$. By induction, we can take a sequence $r_0 > r_1 > r_2 > r_3 > \cdots \in \mathbb{N}$, which is a contradiction.

To show that α is injective, let $x, y \in \mathbb{N}^{\mathbb{N}}$ and $\alpha(x) = \alpha(y)$. By $(\star\star)$, it follows that

$$x(1) = \lfloor 1/\alpha(x)\rfloor = \lfloor 1/\alpha(y)\rfloor = y(1) \text{ and } \alpha(x^*) = \langle 1/\alpha(x)\rangle = \langle 1/\alpha(y)\rangle = \alpha(y^*).$$

Then, we can apply the same argument as x, y to x^*, y^*. As a consequence, we have $x^*(1) = y^*(1)$, i.e., $x(2) = y(2)$, and $\alpha(x^{**}) = \alpha(y^{**})$. By induction, it can be obtained that $x(n) = y(n)$ for all $n \in \mathbb{N}$, which means $x = y$.

Thus, we have a continuous injection $\alpha : \mathbb{N}^{\mathbb{N}} \to (0,1) \setminus \mathbb{Q}$. To see that α is a homeomorphism, we shall define a map $\beta : (0,1) \setminus \mathbb{Q} \to \mathbb{N}^{\mathbb{N}}$ so that $\alpha\beta = \mathrm{id}_{(0,1)\setminus\mathbb{Q}}$. To this end, define $\varphi : (0,1) \setminus \mathbb{Q} \to (0,1) \setminus \mathbb{Q}$ by $\varphi(t) = \langle t^{-1}\rangle$. Then, φ is continuous. Indeed, since $\varphi(t) - \varphi(s) = t^{-1} - s^{-1}$ for $s, t \in ((n+1)^{-1}, n^{-1}) \setminus \mathbb{Q}$, $n \in \mathbb{N}$, it follows that φ is continuous on each $((n+1)^{-1}, n^{-1}) \setminus \mathbb{Q}$, so φ is continuous. Let $\varphi^0 = \mathrm{id}_{(0,1)\setminus\mathbb{Q}}$ and $\varphi^n = \varphi^{n-1}\varphi$ for $n \in \mathbb{N}$. Then, define maps $\beta_n : (0,1) \setminus \mathbb{Q} \to \mathbb{N}$, $n \in \mathbb{N}$, by $\beta_n(t) = \lfloor \varphi^{n-1}(t)^{-1}\rfloor$ $(= \varphi^{n-1}(t)^{-1} - \langle \varphi^{n-1}(t)^{-1}\rangle = \varphi^{n-1}(t)^{-1} - \varphi^n(t))$. The map $\beta : (0,1) \setminus \mathbb{Q} \to \mathbb{N}^{\mathbb{N}}$ can be defined by $\beta(x) = (\beta_n(x))_{n\in\mathbb{N}}$. Since $\beta_n\varphi(t) = \beta_{n+1}(t)$, $n \in \mathbb{N}$, we have $\beta\varphi(t) = \beta(t)^*$.

To prove $\alpha\beta = \mathrm{id}_{(0,1)\setminus\mathbb{Q}}$, since $\alpha(x) = \lim_{n\to\infty} a_n(x)$, it suffices to show that

$$a_{2n}\beta(t) < t < a_{2n-1}\beta(t) \text{ for all } t \in (0,1) \setminus \mathbb{Q} \text{ and } n \in \mathbb{N}.$$

For each $n \in \mathbb{N}$, let $P(n)$ be the assertion that $t < a_{2n-1}\beta(t)$ for all $t \in (0,1) \setminus \mathbb{Q}$ and $Q(n)$ be the assertion that $a_{2n}\beta(t) < t$ for all $t \in (0,1) \setminus \mathbb{Q}$ Since $\beta_1(t) = \lfloor t^{-1}\rfloor < t^{-1}$, we have $a_1\beta(t) = \beta_1(t)^{-1} > t$, that is, $P(1)$ is true. To see $P(n) \Rightarrow Q(n)$, applying $P(n)$ to $\varphi(t) \in (0,1) \setminus \mathbb{Q}$, we have $\varphi(t) < a_{2n-1}\beta\varphi(t)$. Then, by $(\star)$, we have

$$\begin{aligned} a_{2n}\beta(t) &= \frac{1}{\beta_1(t) + a_{2n-1}(\beta(t)^*)} = \frac{1}{\beta_1(t) + a_{2n-1}\beta\varphi(t)} \\ &< (\beta_1(t) + \varphi(t))^{-1} = (\lfloor t^{-1}\rfloor + \langle t^{-1}\rangle)^{-1} = (t^{-1})^{-1} = t. \end{aligned}$$

Thus, $P(n) \Rightarrow Q(n+1)$ is true. Similarly, we can show $Q(n) \Rightarrow P(n)$. By induction, all $P(n)$ and $Q(n)$ are true. The proof is completed.

A separable metrizable space M is called an ***n*-manifold** (or an ***n*-dimensional manifold**) if each $x \in M$ has a neighborhood that is homeomorphic to (an open set in) $\mathbf{I}^n$, where $\mathbf{I}^n$ can be replaced with $\mathbb{R}^n_+$.[6] An **interior point** of M is a point $x \in M$ which has a neighborhood homeomorphic to $(0,1)^n$ $(\approx \mathbb{R}^n)$. The set of all interior points of M is denoted by $\mathrm{Int}\, M$ and called the **interior** of M. A non-interior point $x \in M$ is called a **boundary point** of M. Then, *$x \in M$ is a boundary point if and only if x has a neighborhood N with a homeomorphism $h : N \to \mathbf{I}^n$ such that*

[6] Some literature does not assume the separability in the definition of an n-manifold. Even so, every connected n-manifold is separable. Then, the separability of an n-manifold means that it has at most countable components.

$h(x) \in \partial \mathbf{I}^n$.[7] The set of all boundary points of M is denoted by ∂M and called the **boundary** of M, where it can be shown that ∂M is an $(n-1)$-manifold, $\partial(\partial M) = \emptyset$, and closed in M. When $\partial M = \emptyset$, we call M an ***n*-manifold without boundary**. A compact n-manifold without boundary is called a **closed *n*-manifold**. The closed n-ball $\mathbf{B}^n$ is an n-manifold with $\partial \mathbf{B}^n = \mathbf{S}^{n-1}$ and the n-sphere $\mathbf{S}^n$ is a closed n-manifold; $\mathbf{B}^1 = [-1, 1]$, $\mathbf{S}^0 = \{\pm 1\} = \partial[-1, 1]$, and $\mathbf{2} = \{0, 1\} = \partial \mathbf{I}$.

Let $(X_\gamma)_{\gamma \in \Gamma}$ be an indexed collection of (topological) spaces and $X = \bigcup_{\gamma \in \Gamma} X_\gamma$. The **weak topology** on X with respect to $(X_\gamma)_{\gamma \in \Gamma}$ is defined as follows:

$$U \subset X \text{ is open in } X \underset{\text{def}}{\Leftrightarrow} \forall \gamma \in \Gamma,\ U \cap X_\gamma \text{ is open in } X_\gamma$$

$$\Big(A \subset X \text{ is closed in } X \underset{\text{def}}{\Leftrightarrow} \forall \gamma \in \Gamma,\ A \cap X_\gamma \text{ is closed in } X_\gamma\Big).$$

Suppose that X has the weak topology with respect to $(X_\gamma)_{\gamma \in \Gamma}$, and that the topologies of X_γ and $X_{\gamma'}$ agree on $X_\gamma \cap X_{\gamma'}$ for any $\gamma, \gamma' \in \Gamma$. If $X_\gamma \cap X_{\gamma'}$ is closed (resp. open) in X_γ for any $\gamma, \gamma' \in \Gamma$, then each X_γ is closed (resp. open) in X and the original topology of each X_γ is a subspace topology inherited from X. In the case that $X_\gamma \cap X_{\gamma'} = \emptyset$ for $\gamma \neq \gamma'$, we write $X = \bigoplus_{\gamma \in \Gamma} X_\gamma$, which is called the **topological sum** of $(X_\gamma)_{\gamma \in \Gamma}$, where each X_γ is clopen in X. More generally, even if the condition that $X_\gamma \cap X_{\gamma'} = \emptyset$ for $\gamma \neq \gamma'$ is not satisfied, identifying each X_γ with $X_\gamma \times \{\gamma\}$, we can define the topological sum $\bigoplus_{\gamma \in \Gamma} X_\gamma$.[8] For example, when $X_\gamma = X$ for every $\gamma \in \Gamma$, $\bigoplus_{\gamma \in \Gamma} X_\gamma$ can be defined as the product space $X \times \Gamma$, where Γ has the discrete topology. It should be noted that the weak topology of $X = \bigcup_{\gamma \in \Gamma} X_\gamma$ with respect to $(X_\gamma)_{\gamma \in \Gamma}$ is equal to the quotient topology induced by the natural surjection $q : \bigoplus_{\gamma \in \Gamma} X_\gamma \to X$ defined by $p|X_\gamma = \mathrm{id}$ for each $\gamma \in \Gamma$.

Let $f : A \to Y$ be a map from a closed set A in a space X to another space Y. The **adjunction space** $Y \cup_f X$ is the quotient space $(X \oplus Y)/\sim$, where $X \oplus Y$ is the topological sum and $\sim$ is the equivalence relation corresponding to the decomposition of $X \oplus Y$ into singletons $\{x\}$, $x \in X \setminus A$, and sets $\{y\} \cup f^{-1}(y)$, $y \in Y$ (the latter is a singleton $\{y\}$ if $y \in Y \setminus f(A)$). In the case that Y is a singleton, $Y \cup_f X$ is homeomorphic to the quotient space X/A of X defined by identifying A with a single point. One should note that, in general, the adjunction spaces are *not Hausdorff*. For example, for a non-regular space X, take a closed set A in X and $x \in X \setminus A$ so that if $U, V \subset X$ are open, $x \in U$, and $A \subset V$, then $U \cap V \neq \emptyset$. Then, X/A is not Hausdorff. Thus, for the adjunction space to be Hausdorff, some condition is required.

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[7] The "only if" part is trivial, but the "if" part is non-trivial. Using the Invariance of Domain (Theorem 5.1.10), we can show the "if" part as follows: If $x \in \mathrm{Int}\, M$, then x has a neighborhood V with a homeomorphism $g : V \to (0, 1)^n$. Choose $\delta > 0$ so that $g^{-1}(\mathrm{B}(g(x), \delta)) \subset N$. By the Invariance of Domain, $hg^{-1}(\mathrm{B}(g(x), \delta))$ is open in $\mathbb{R}^n$, which is a neighborhood of $h(x)$ in $\mathbb{R}^n$. Since $h(x) \in \partial \mathbf{I}^n = \mathrm{bd}_{\mathbb{R}^n} \mathbf{I}^n$, $hg^{-1}(\mathrm{B}(g(x), \delta)) \not\subset \mathbf{I}^n$ but $hg^{-1}(\mathrm{B}(g(x), \delta)) \subset h(N) = \mathbf{I}^n$, which is a contradiction.

[8] A non-indexed family $\mathfrak{A}$ of spaces is considered to be indexed by itself (namely, self-indexed). Identifying each $A \in \mathfrak{A}$ with $A \times \{A\}$ ($\{A\}$ is a singleton = a one-point space), $\bigoplus \mathfrak{A} = \bigoplus_{A \in \mathfrak{A}} A$ can be defined.

Let $\mathcal{A}$ and $\mathcal{B}$ be collections of subsets of X, and let $Y \subset X$ and $x \in X$. We define:

- $\mathcal{A} \wedge \mathcal{B} \underset{\text{def}}{=} \{A \cap B \mid A \in \mathcal{A},\ B \in \mathcal{B}\} \setminus \{\emptyset\}$;
- $\mathcal{A}[Y] \underset{\text{def}}{=} \{A \in \mathcal{A} \mid A \cap Y \neq \emptyset\}$;
- $\mathcal{A}|Y \underset{\text{def}}{=} \mathcal{A} \wedge \{Y\} = \{A \cap Y \mid A \in \mathcal{A}[Y]\}$;
- $\mathcal{A}[x] \underset{\text{def}}{=} \{A \in \mathcal{A} \mid x \in A\} = \mathcal{A}[\{x\}]$;
- $\mathcal{A}^{\mathrm{cl}} \underset{\text{def}}{=} \{\mathrm{cl}\, A \mid A \in \mathcal{A}\}$.

When each $A \in \mathcal{A}$ is contained in some $B \in \mathcal{B}$, it is said that $\mathcal{A}$ **refines** $\mathcal{B}$, and this is denoted as follows:

$$\mathcal{A} \prec \mathcal{B} \ \text{ or } \ \mathcal{B} \succ \mathcal{A}.$$

It is said that $\mathcal{A}$ is **star-finite** (resp. **point-finite**) in X if $\mathrm{card}\, \mathcal{A}[A] < \aleph_0$ for each $A \in \mathcal{A}$ (resp. $\mathrm{card}\, \mathcal{A}[x] < \aleph_0$ for each $x \in X$), that is, each $A \in \mathcal{A}$ meets only finitely many members of $\mathcal{A}$ (resp. each $x \in X$ is contained in only finitely many members of $\mathcal{A}$). Note that $\mathcal{A}$ is pairwise disjoint if and only if $\mathrm{card}\, \mathcal{A}[A] = 1$ for each $A \in \mathcal{A} \setminus \{\emptyset\}$. Moreover, $\mathcal{A}$ is said to be **locally finite** (resp. **discrete**) if each $x \in X$ has a neighborhood U such that $\mathrm{card}\, \mathcal{A}[U] < \aleph_0$ (resp. $\leqslant 1$), that is, U meets only finitely many members of $\mathcal{A}$ (resp. at most one member of $\mathcal{A}$).[9] The following easy fact will be used frequently:

Fact *If $\mathcal{A}$ is locally finite (or discrete) in X, then so is $\mathcal{A}^{\mathrm{cl}}$ and also*

$$\mathrm{cl} \bigcup \mathcal{A} = \bigcup \mathcal{A}^{\mathrm{cl}} \quad \left(= \bigcup_{A \in \mathcal{A}} \mathrm{cl}\, A \right).$$

To see this, just note that for each open set U in X and $A \subset X$, $U \cap A \neq \emptyset \Leftrightarrow U \cap \mathrm{cl}\, A \neq \emptyset$.

For an indexed collection $(A_\gamma)_{\gamma \in \Gamma}$ of subsets of X, the above concepts are differently defined. We define that $(A_\gamma)_{\gamma \in \Gamma}$ is **star-finite** (resp. **point-finite**) if $\mathrm{card}\{\gamma' \in \Gamma \mid A_\gamma \cap A_{\gamma'} \neq \emptyset\} < \aleph_0$ for each $\gamma \in \Gamma$ (resp. $\mathrm{card}\{\gamma \in \Gamma \mid x \in A_\gamma\} < \aleph_0$ for each $x \in X$). Evidently, if $(A_\gamma)_{\gamma \in \Gamma}$ is star-finite or point-finite, then so is $\mathcal{A} = \{A_\gamma \mid \gamma \in \Gamma\}$, where the converse is not true. It is said that $(A_\gamma)_{\gamma \in \Gamma}$ is **locally finite** (resp. **discrete**) in X if each $x \in X$ has a neighborhood U such that $\mathrm{card}\{\gamma \in \Gamma \mid A_\gamma \cap U \neq \emptyset\} < \aleph_0$ (resp. $\leqslant 1$). If $(A_\gamma)_{\gamma \in \Gamma}$ is locally finite or discrete in X, then so is $\mathcal{A} = \{A_\gamma \mid \gamma \in \Gamma\}$. But the converse does not hold. Similar to the above fact, we have the following:

[9] It is said that $A \subset X$ is **discrete** in X if each $x \in X$ has a neighborhood U with $\mathrm{card}(U \cap A) \leqslant 1$, which means that the collection $\{\{a\} \mid a \in A\}$ is discrete in X. It should be noted that A is discrete in X if and only if A is a closed subspace of X with the discrete topology.

Fact *If $(A_\gamma)_{\gamma\in\Gamma}$ is locally finite (or discrete) in X, then so is* $(\mathrm{cl}\, A_\gamma)_{\gamma\in\Gamma}$.

When $(A_\gamma)_{\gamma\in\Gamma}$ is locally finite in X, $\mathrm{cl}\bigcup_{\lambda\in\Lambda} A_\lambda = \bigcup_{\lambda\in\Lambda} \mathrm{cl}\, A_\lambda$ but this follows from the local finiteness of $\{A_\gamma \mid \gamma\in\Gamma\}$.

It is said that $\mathcal{A}$ **covers** Y (or $\mathcal{A}$ is a **cover** of Y in X) if $Y\subset\bigcup\mathcal{A}\,(=\bigcup_{A\in\mathcal{A}} A)$. When $Y=X$, a cover of Y in X is simply called a cover of X. A cover of Y in X is said to be **open** or **closed** in X depending on whether its members are open or closed in X. If $\mathcal{A}$ is an open (resp. closed) cover of X, then $\mathcal{A}|Y$ is an open (resp. closed) cover of Y and $\mathcal{A}[Y]$ is an open (resp. closed) cover of Y in X, where $\mathcal{A}[Y]\subset\mathcal{A}$ but $\mathcal{A}|Y\not\subset\mathcal{A}$ in general. We call $\mathcal{A}|Y$ the **restriction** of $\mathcal{A}$ onto Y. When $\mathcal{A}$ and $\mathcal{B}$ are open (resp. closed) covers of X, $\mathcal{A}\wedge\mathcal{B}$ is also an open (resp. closed) cover of X. For covers $\mathcal{A}$ and $\mathcal{B}$ of X, we call $\mathcal{A}$ a **refinement** of $\mathcal{B}$ if $\mathcal{A}\prec\mathcal{B}$, where $\mathcal{A}$ is an **open** (or **closed**) **refinement** if $\mathcal{A}$ is an open (or closed) cover. When $\mathcal{A}\subset\mathcal{B}$, $\mathcal{A}$ is called a **subcover** of $\mathcal{B}$. Note that every subcover of $\mathcal{B}$ is a refinement of $\mathcal{B}$. For a space X,

- $\mathrm{cov}(X)$ denotes the collection of all open covers of X.

An indexed collection $(A_\gamma)_{\gamma\in\Gamma}$ of subsets of X is called an **indexed cover** of Y in X if $\{A_\gamma \mid \gamma\in\Gamma\}$ is a cover of Y in X. An indexed cover of Y in X is said to be **open** or **closed** in X depending on whether its members are open or closed in X. For indexed collections $(A_\gamma)_{\gamma\in\Gamma}$, $(B_\lambda)_{\lambda\in\Lambda}$ of subsets of X, we define

$$(A_\gamma)_{\gamma\in\Gamma}\prec(B_\lambda)_{\lambda\in\Lambda} \underset{\text{def}}{\Leftrightarrow} \{A_\gamma \mid \gamma\in\Gamma\}\prec\{B_\lambda \mid \lambda\in\Lambda\},$$

which is also written as $(B_\lambda)_{\lambda\in\Lambda}\succ(A_\gamma)_{\gamma\in\Gamma}$. A **refinement** of an indexed cover of X is similarly defined.

Let $f: X\to Y$ be a map. For $A\subset X$ and $B\subset Y$, we define

$$f(A)=\big\{f(x)\;\big|\;x\in A\big\}\ \text{ and }\ f^{-1}(B)=\big\{x\in X\;\big|\;f(x)\in B\big\}.$$

For simplicity, we write $f^{-1}(\{y\})=f^{-1}(y)$. For collections $\mathcal{A}$ and $\mathcal{B}$ of subsets of X and Y, respectively, we define

$$f(\mathcal{A})=\big\{f(A)\;\big|\;A\in\mathcal{A}\big\}\ \text{ and }\ f^{-1}(\mathcal{B})=\big\{f^{-1}(B)\;\big|\;B\in\mathcal{B}\big\}.$$

The restriction of f to $A\subset X$ is denoted by $f|A$. It is said that a map $g: A\to Y$ **extends over** X if there is a map $f: X\to Y$ such that $f|A=g$. Such a map f is called an **extension** of g. The **graph** of f is defined as the following subset of $X\times Y$:

$$\mathrm{Gr}(f)\underset{\text{def}}{=}\big\{(x,f(x))\;\big|\;x\in X\big\}\subset X\times Y.$$

Then, $\mathrm{Gr}(f)$ is closed in the product space $X \times Y$ and $\mathrm{pr}_X | \mathrm{Gr}(f) : \mathrm{Gr}(f) \to X$ is a homeomorphism whose inverse is the natural injection $i_f : X \to \mathrm{Gr}(f)$ defined by $i_f(x) = (x, f(x))$ for each $x \in X$.[10]

When $X = (X, d)$ and $Y = (Y, \rho)$ are metric spaces, $f : X \to Y$ is called a **Lipschitz map** if there is some $\lambda \in \mathbb{R}_+$ such that

$$\rho(f(x), f(x')) \leqslant \lambda d(x, x') \text{ for each } x, x' \in X,$$

where λ is called a **Lipschitz constant**. A Lipschitz map with a Lipschitz constant λ is called a **λ-Lipschitz map**. Obviously, every Lipschitz map is uniformly continuous.

Let $[a, b]$ be a closed interval, where $a < b\ (\in \mathbb{R})$. A map $f : [a, b] \to X$ is called a **path** (from $f(a)$ to $f(b)$) in X, where it is said that two points $f(a)$ and $f(b)$ are **connected by** a path f in X. An embedding (i.e., an injective path) $f : [a, b] \to X$ is called an **arc** (from $f(a)$ to $f(b)$) in X, and the image $f([a, b])$ is also called an **arc**. Namely, a space is called an **arc** if it is homeomorphic to $\mathbf{I}$. A space X is said to be **path-connected** (or **arcwise connected**) if each pair of *distinct* points $x, y \in X$ can be connected by a path (or an arc). It is said that X is **locally path-connected** (or **locally arcwise connected**) if any neighborhood U of each point $x \in X$ contains a neighborhood V of x such that each pair of distinct points in V is connected by a path (or an arc) in U, i.e., for each $y \neq z \in V$, there is a path (or an arc) $f : \mathbf{I} \to U$ such that $f(0) = y$ and $f(1) = z$, where as is easily observed, V can be taken so as to be a path-connected (or an arcwise connected). The (local) arcwise connectedness looks to be stronger than the (local) path-connectedness, but they are the same. In fact, the following well-known fact will be shown in Corollary 5.13.6:

Fact *Each pair of distinct points $x, y \in X$ is connected by an arc if and only if the points are connected by a path.*

For spaces X and Y,

- $\mathrm{C}(X, Y)$ denotes the set of (continuous) maps from X to Y.

For maps $f, g : X \to Y$ (i.e., $f, g \in \mathrm{C}(X, Y)$),

- $f \simeq g$ means that f and g are **homotopic** (or f is **homotopic** to g),

that is, there is a map $h : X \times \mathbf{I} \to Y$ such that $h_0 = f$ and $h_1 = g$, where $h_t : X \to Y$, $t \in \mathbf{I}$, are defined by $h_t(x) = h(x, t)$, and h is called a **homotopy** from f to g (between f and g). Then, $h(\{x\} \times \mathbf{I})$, $x \in X$, is called the (**homotopy**) **tracks** of h, where each $h(\{x\} \times \mathbf{I})$ is the (**homotopy**) **track** of $x \in X$ by h. When g is a constant map, it is said that f is **null-homotopic**, denoted by $f \simeq 0$. The relation $\simeq$ is an equivalence relation on $\mathrm{C}(X, Y)$. The equivalence class $[f] = \{g \in \mathrm{C}(X, Y) \mid g \simeq f\}$ is called the **homotopy class** of f. We define

[10] In order that $\mathrm{Gr}(f)$ is closed in $X \times Y$ and i_f is continuous, we need the continuity of f and the Hausdorffness of Y.

- $[X, Y] \underset{\text{def}}{=} \{[f] \mid f \in \mathrm{C}(X, Y)\} = \mathrm{C}(X, Y)/\simeq$
  — the set of the homotopy classes of maps from X to Y.

For each $f, f' \in \mathrm{C}(X, Y)$ and $g, g' \in \mathrm{C}(Y, Z)$, we have the following:

$$f \simeq f',\ g \simeq g' \ \Rightarrow\ gf \simeq g'f'.$$

Thus, we can define the composition as follows:

$$[X, Y] \times [Y, Z] \ni ([f], [g]) \mapsto [g][f] = [gf] \in [X, Z].$$

Then, as is easily observed, for any $[f] \in [X, Y]$, $[g] \in [Y, Z]$, and $[h] \in [Z, W]$,

$$[h]([g][f]) = [hgf] = ([h][g])[f].$$

Moreover,

- $X \simeq Y$ means that X and Y are **homotopy equivalent** (or X is **homotopy equivalent** to Y),[11] i.e., there are maps $f : X \to Y$ and $g : Y \to X$ such that $gf \simeq \mathrm{id}_X$ and $fg \simeq \mathrm{id}_Y$, where f is called a **homotopy equivalence** and g is a **homotopy inverse** of f.

Given subspaces $X_1, \ldots, X_n \subset X$ and $Y_1, \ldots, Y_n \subset Y$, a map $f : X \to Y$ is said to be a map from $(X, X_1, \ldots, X_n)$ to $(Y, Y_1, \ldots, Y_n)$, written

$$f : (X, X_1, \ldots, X_n) \to (Y, Y_1, \ldots, Y_n),$$

if $f(X_1) \subset Y_1$, ..., $f(X_n) \subset Y_n$. The set of all such maps is denoted by

$$\mathrm{C}((X, X_1, \ldots, X_n), (Y, Y_1, \ldots, Y_n)).$$

For maps $f, g \in \mathrm{C}((X, X_1, \ldots, X_n), (Y, Y_1, \ldots, Y_n))$, we define $f \simeq g$ by requiring a homotopy h from f to g the following condition:

$$h_t \in \mathrm{C}((X, X_1, \ldots, X_n), (Y, Y_1, \ldots, Y_n)) \ \text{ for every } t \in \mathbf{I},$$

i.e., h is regarded as the map

$$h : (X \times \mathbf{I}, X_1 \times \mathbf{I}, \ldots, X_n \times \mathbf{I}) \to (Y, Y_1, \ldots, Y_n).$$

Then, $\simeq$ is an equivalence relation on $\mathrm{C}((X, X_1, \ldots, X_n), (Y, Y_1, \ldots, Y_n))$. We define

$$[(X, X_1, \ldots, X_n), (Y, Y_1, \ldots, Y_n)] \underset{\text{def}}{=} \mathrm{C}((X, X_1, \ldots, X_n), (Y, Y_1, \ldots, Y_n))/\simeq .$$

[11] It is also said that X and Y have the **same homotopy type** or X has the **homotopy type** of Y.

It is said that $(X, X_1, \ldots, X_n)$ and $(Y, Y_1, \ldots, Y_n)$ are **homotopy equivalent** (or $(X, X_1, \ldots, X_n)$ is **homotopy equivalent** to $(Y, Y_1, \ldots, Y_n)$), written

$$(X, X_1, \ldots, X_n) \simeq (Y, Y_1, \ldots, Y_n)$$

if there exist maps (called a **homotopy equivalence** and its **homotopy inverse**)

$$f : (X, X_1, \ldots, X_n) \to (Y, Y_1, \ldots, Y_n),\ g : (Y, Y_1, \ldots, Y_n) \to (X, X_1, \ldots, X_n)$$

such that $gf \simeq \mathrm{id}_X$ and $fg \simeq \mathrm{id}_Y$.

For pointed spaces (X, x_0) and (Y, y_0), we define the following:

- $\mathrm{C}((X, x_0), (Y, y_0)) = \mathrm{C}((X, \{x_0\}), (Y, \{y_0\}))$;
  a map $f \in \mathrm{C}((X, x_0), (Y, y_0))$ is called a **pointed map**.
- For $f, g \in \mathrm{C}((X, x_0), (Y, y_0))$, $f \simeq g$ means that f is **pointed homotopic** to g, that is, there is a **pointed homotopy** h from f to g, where $h(\{x_0\} \times \mathbf{I}) = \{y_0\}$.
- $[(X, x_0), (Y, y_0)] = \mathrm{C}((X, x_0), (Y, y_0))/\simeq$.
- $(X, x_0) \simeq (Y, y_0)$ means $(X, \{x_0\}) \simeq (Y, \{y_0\})$,
  that is, there is a **pointed homotopy equivalence** $f : (X, x_0) \to (Y, y_0)$ and a **pointed homotopy inverse** $g : (Y, y_0) \to (X, x_0)$, ($gf \simeq \mathrm{id}$ and $fg \simeq \mathrm{id}$).
  It is said that (X, x_0) and (Y, y_0) are **pointed homotopy equivalent**.

For $A \subset X$, a homotopy $h : X \times \mathbf{I} \to Y$ is called a **homotopy relative to** A if the homotopy track $h(\{x\} \times \mathbf{I})$ of each $x \in A$ is a singleton. When a homotopy from f to g is a homotopy relative to A (where $f|A = g|A$), we say that f and g are **homotopic relative to** A (or f is **homotopic** to g **relative to** A) and write

$$f \simeq g \ \text{ rel. } A.$$

Let $f, g : X \to Y$ be maps and $\mathcal{U}$ a collection of subsets of Y (usually, $\mathcal{U} \in \mathrm{cov}(Y)$). It is said that f and g are **$\mathcal{U}$-close** (or f is **$\mathcal{U}$-close** to g) if

$$\big\{\{f(x), g(x)\} \,\big|\, x \in X\big\} \prec \mathcal{U} \cup \big\{\{y\} \,\big|\, y \in Y\big\},$$

which implies that $\mathcal{U}$ covers the set $\{f(x), g(x) \mid f(x) \neq g(x)\}$. A homotopy h is called a **$\mathcal{U}$-homotopy** if

$$\big\{h(\{x\} \times \mathbf{I}) \,\big|\, x \in X\big\} \prec \mathcal{U} \cup \big\{\{y\} \,\big|\, y \in Y\big\},$$

which implies that $\mathcal{U}$ covers the following set:

$$\bigcup \big\{h(\{x\} \times \mathbf{I}) \,\big|\, h(\{x\} \times \mathbf{I}) \text{ is non-degenerate}\big\}.$$

When there exists a $\mathcal{U}$-homotopy $h : X \times \mathbf{I} \to Y$ from f to g (relative to A), it is said that f and g are **$\mathcal{U}$-homotopic** (**relative to** A) (or f is **$\mathcal{U}$-homotopic** to g (**relative to** A)), denoted as follows:

$$f \simeq_{\mathcal{U}} g \quad \big(f \simeq_{\mathcal{U}} g \ \text{ rel. } A\big).$$

In the case when $Y = (Y, d)$ is a metric space, a homotopy $h : X \times \mathbf{I} \to Y$ is called an **ε-homotopy** if

$$\operatorname{mesh}\big\{h(\{x\} \times \mathbf{I}) \;\big|\; x \in X\big\} = \sup_{x \in X} \operatorname{diam} h(\{x\} \times \mathbf{I}) < \varepsilon.$$

Note that h is an ε-homotopy if and only if h is a $\mathcal{U}$-homotopy for some open cover $\mathcal{U}$ of $h(X \times \mathbf{I})$ in Y such that $\operatorname{mesh} \mathcal{U} < \varepsilon$.

Indeed, for the "only if" part, choose $0 < \delta < (\varepsilon - \sup_{x \in X} \operatorname{diam} h(\{x\} \times \mathbf{I}))/2$ and let $\mathcal{U} = \{\mathrm{N}(h(\{x\} \times \mathbf{I}), \delta) \mid x \in X\}$. Then, h is a $\mathcal{U}$-homotopy and $\operatorname{mesh} \mathcal{U} < \varepsilon$.

It is said that f and g are **ε-homotopic** (**relative to** A) (or f is **ε-homotopic** to g (**relative to** A)), denoted by

$$f \simeq_{\varepsilon} g \quad \big(f \simeq_{\varepsilon} g \ \text{ rel. } A\big),$$

if there exists an ε-homotopy h from f to g. It is also said that f and g are **ε-close** (or f is **ε-close** to g) if

$$d_\infty(f, g) \underset{\text{def}}{=} \sup_{x \in X} d(f(x), g(x)) < \varepsilon.$$

In the above, it is possible that $d_\infty(f, g) = \infty$ for some $f, g \in \mathrm{C}(X, Y)$. In such a case, d_∞ is not a metric on the set $\mathrm{C}(X, Y)$.

When Y is bounded or X is compact, the above d_∞ is a metric, which is called the **sup-metric**. Even if d_∞ is not a metric (i.e., $d(f, g) = \infty$ for some $f, g \in \mathrm{C}(X, Y)$), d_∞ induces a topology on $\mathrm{C}(X, Y)$ such that each f has a neighborhood basis consisting of

$$\mathrm{B}_{d_\infty}(f, \varepsilon) = \big\{g \in \mathrm{C}(X, Y) \;\big|\; d_\infty(f, g) < \varepsilon\big\}, \ \varepsilon > 0.$$

This topology is metrizable. Indeed, consider the bounded metric $\bar{d} \in \operatorname{Metr}(Y)$ defined as follows:

$$\bar{d}(x, y) = \min\big\{d(x, y), 1\big\} \ \text{ for each } x, y \in Y.$$

The sup-metric $\bar{d}_\infty$ induces the same topology as d_∞ because $\bar{d}_\infty(f, g) = d_\infty(f, g)$ if $d_\infty(f, g) < 1$ and $\bar{d}_\infty(f, g) = 1$ if $1 \leqslant d_\infty(f, g) \leqslant \infty$. This metrizable topology is called the **uniform convergence topology**. The space $\mathrm{C}(X, Y)$ with this topology is denoted by $\mathrm{C}_u(X, Y)$. It is said that a sequence $(f_n)_{n \in \mathbb{N}}$ **uniformly converges** (or

is **uniformly convergent**) to f, denoted by $\mathrm{ulim}_{n\to\infty} f_n = f$ (or $d\text{-}\lim_{n\to\infty} f_n = f$), if $(f_n)_{n\in\mathbb{N}}$ converges to f in $\mathrm{C}_u(X, Y)$. Then,

$$\operatorname*{ulim}_{n\to\infty} f_n = f \;\Leftrightarrow\; \forall\varepsilon > 0,\ \exists n_0 \in \mathbb{N} \text{ such that } \forall n \geqslant n_0,\ \forall x \in X,\ d(f_n(x), f(x)) < \varepsilon.$$

Hence, if $d' \in \mathrm{Metr}(Y)$ is uniformly equivalent to d, then d' induces the same uniform convergence topology as d. Thus, we have the following:

Fact *The uniform convergence topology is metrizable. If d' is uniformly equivalent to d, then d'_∞ is equivalent to d_∞, which means that d' induces the same uniform convergence topology of $\mathrm{C}(X, Y)$ as d.*

It should be remarked that some $d' \in \mathrm{Metr}(Y)$ might induce a uniform convergence topology different from the one induced by d. Namely, d'_∞ is not equivalent to d_∞ even if d' is equivalent to d.

For example, let d be the usual metric for $\mathbb{R}$, i.e., $d(x, y) = |x - y|$. Since $\mathbb{R} \approx (-1, 1)$ by the correspondence $\mathbb{R} \ni x \mapsto x/(|x| + 1) \in (-1, 1)$, we can define $d' \in \mathrm{Metr}(\mathbb{R})$ as follows:

$$d'(x, y) = \left| \frac{x}{|x| + 1} - \frac{y}{|y| + 1} \right| \quad \text{for } x, y \in \mathbb{R}.$$

Then, the topology induced by d'_∞ is different from the one induced by d_∞. To see this, let $f_0 : (0, 1] \to \mathbb{R}$ be the map defined by $f_0(x) = 1/x$ for each $x \in (0, 1]$. For each $n \in \mathbb{N}$, let $f_n : (0, 1] \to \mathbb{R}$ be the map defined as follows:

$$f_n(x) = \begin{cases} x^{-1} & \text{if } x \geqslant 1/n, \\ n & \text{if } x \leqslant 1/n. \end{cases}$$

Then, $(f_n)_{n\in\mathbb{N}}$ does not converge to f_0 with respect to the uniform convergence topology induced by d because $\sup_{x\in(0,1]} d(f_n(x), f_0(x)) = \infty$ for every $n \in \mathbb{N}$.[12] On the other hand, $(f_n)_{n\in\mathbb{N}}$ converges to f_0 with respect to the uniform convergence topology induced by d' because

$$\sup_{x\in(0,1]} d'(f_0(x), f_n(x)) = 1 - \frac{n}{n+1} = \frac{1}{n+1} \to 0 \quad (n \to \infty).$$

Moreover, we have the following:

Proposition 1.1.1 *Let $Y = (Y, d)$ be a complete metric space. Suppose that Y is bounded or X is compact. Then, the sup-metric d_∞ on $\mathrm{C}(X, Y)$ is complete.*

Sketch of Proof Let $(f_n)_{n\in\mathbb{N}}$ be a d_∞-Cauchy sequence in $\mathrm{C}(X, Y)$. Since Y is complete and $(f_n(x))_{n\in\mathbb{N}}$ is d-Cauchy for each $x \in X$, we can define $f : X \to Y$ by $f(x) = \lim_{n\to\infty} f_n(x)$. Then, $f \in \mathrm{C}(X, Y)$ and $\lim_{n\to\infty} f_n = f$. To prove this, let $x_0 \in X$ and $\varepsilon > 0$. First, choose $n \in \mathbb{N}$ so that $i, j \geqslant n \Rightarrow d_\infty(f_i, f_j) < \varepsilon/3$. Then, show (1) $\forall x \in X$, $d(f_i(x), f(x)) \leqslant \varepsilon/3$, which implies that $d_\infty(f_i, f) < \varepsilon$ if $f \in \mathrm{C}(X, Y)$. Next, by the continuity of f_n, x_0 has a neighborhood U in X such that $f_n(U) \subset \mathrm{B}(f_n(x_0), \varepsilon/3)$. Then, show (2) $\forall x \in U$, $d(f(x), f(x_0)) < \varepsilon$.

[12] In fact, $\{f_n \mid n \in \mathbb{N}\}$ is discrete in $\mathrm{C}_u((0, 1], \mathbb{R})$ because $\sup_{x\in(0,1]} d(f_n(x), f_m(x)) = |n - m|$ for any $n, m \in \mathbb{N}$.

Now, we regard $\mathrm{C}(X, Y)$ as a subspace of the product space Y^X. The topology on $\mathrm{C}(X, Y)$ inherited from Y^X is called the **pointwise convergence topology**, which can be defined for an arbitrary space Y. The space $\mathrm{C}(X, Y)$ with this topology is denoted by $\mathrm{C}_p(X, Y)$.

The topology of X does not take part in the definition of the topology of $\mathrm{C}_p(X, Y)$ (the product topology of Y^X), but it is involved in defining the set $\mathrm{C}_p(X, Y)$. Thus, the space $\mathrm{C}_p(X, Y)$ is determined by the topologies of both X and Y.

It is said that a sequence $(f_n)_{n\in\mathbb{N}}$ **pointwise converges** (or is **pointwise convergent**) to f, denoted by $\mathrm{plim}_{n\to\infty} f_n = f$, if $(f_n)_{n\in\mathbb{N}}$ converges to f in $\mathrm{C}_p(X, Y)$. Then,

$$\operatorname*{plim}_{n\to\infty} f_n = f \;\Leftrightarrow\; \lim_{n\to\infty} f_n(x) = f(x) \;\text{ for every } x \in X.$$

When $Y = (Y, d)$ is a metric space, we have the following:

$$\operatorname*{plim}_{n\to\infty} f_n = f \;\Leftrightarrow\; \forall x \in X,\ \forall \varepsilon > 0,\ \exists n_x \in \mathbb{N} \text{ such that } \forall n \geqslant n_x,\ d(f_n(x), f(x)) < \varepsilon.$$

Evidently, $\mathrm{id} : \mathrm{C}_u(X, Y) \to \mathrm{C}_p(X, Y)$ is continuous, that is:

$$\boxed{\text{the pointwise convergence topology}} \subset \boxed{\text{the uniform convergence topology}}$$

In the above, the equality does not hold in general. Namely, $\mathrm{C}_p(X, Y) \neq \mathrm{C}_u(X, Y)$ as spaces in general.

For example, let $f_0 : \mathbf{I} \to \mathbb{R}$ be the constant map with $f_0(\mathbf{I}) = \{0\}$. For each $n \in \mathbb{N}$, let $f_n : \mathbf{I} \to \mathbb{R}$ be a map defined as follows:

$$f_n(x) = \begin{cases} 0 & \text{if } x \geqslant 1/n, \\ 1 - 4n^2(x - 1/2n)^2 & \text{if } x \leqslant 1/n. \end{cases}$$

Then, $d_\infty(f_n, f_0) = 1$ for every $n \in \mathbb{N}$ but $\lim_{n\to\infty} f_n(x) = 0 = f_0(x)$ for each $x \in \mathbf{I}$.

The **evaluation** $\mathrm{ev} : \mathrm{C}(X, Y) \times X \to Y$ is defined by $\mathrm{ev}(f, x) = f(x)$ for $f \in \mathrm{C}(X, Y)$ and $x \in X$. It is easy to see that $\mathrm{ev} : \mathrm{C}_u(X, Y) \times X \to Y$ is continuous.

Indeed, let $f \in \mathrm{C}(X, Y)$, $x \in X$, and $\varepsilon > 0$. Then, x has a neighborhood V in X such that $d(f(x), f(x')) < \varepsilon/2$ for any $x' \in V$. For any $f' \in \mathrm{C}(X, Y)$ with $d_\infty(f, f') < \varepsilon/2$ and $x' \in V$,

$$d(\mathrm{ev}(f, x), \mathrm{ev}(f', x')) = d(f(x), f'(x')) < \varepsilon.$$

However, $\mathrm{ev} : \mathrm{C}_p(X, Y) \times X \to Y$ is non-continuous in general.

For example, let $X = \{0, 1/n \mid n \in \mathbb{N}\}$, $Y = \mathbb{R}$, and $f_0 : X \to \mathbb{R}$ be the constant map with $f_0(X) = \{0\}$. Then, $\mathrm{ev}(f_0, 0) = 0$ but $\mathrm{ev}(\mathcal{U} \times V) \not\subset (-1, 1)$ for any neighborhood $\mathcal{U}$ of f_0 in $\mathrm{C}_p(X, R)$ and any neighborhood V of 0 in X. Indeed, we have a finite set $F \subset X$ and $0 < \varepsilon < 1$ such that

$$\left\{f \in \mathrm{C}(X, \mathbb{R}) \;\middle|\; \forall x \in F,\ |f(x)| < \varepsilon\right\} \subset \mathcal{U}.$$

Take $1/n_0 \in V \setminus F$ and define a map $f_1 : X \to \mathbb{R}$ by $f_1(1/n_0) = 1$ and $f_1(x) = 0$ if $x \in X \setminus \{1/n_0\}$. Then, $f_1(1/n_0) \in \mathcal{U} \times V$ but $\mathrm{ev}(f_1, 1/n_0) = f_1(1/n_0) = 1 \notin (-1, 1)$.

For each $x \in X$, we define $\mathrm{ev}_x : \mathrm{C}(X, Y) \to Y$ by $\mathrm{ev}_x(f) = f(x)$, which is called the **evaluation at** x. It should be noted that ev_x is the restriction of the projection $\mathrm{pr}_x : Y^X \to Y$ defined by $\mathrm{pr}_x(z) = z(x)$. Hence, we have the following:

Fact *The pointwise convergence topology is the coarsest topology such that the evaluations* ev_x, $x \in X$, *are continuous.*

Let $\mathcal{F} \subset \mathrm{C}(X, Y)$. The subspace topology of $\mathcal{F}$ inherited from $\mathrm{C}_u(X, Y)$ (resp. $\mathrm{C}_p(X, Y)$) is also called the **uniform convergence** (resp. **pointwise convergence**) **topology**.

1.2 Banach Spaces in the Product of Real Lines

Throughout this section, let Γ be an infinite set. We define:

- $\mathfrak{P}(\Gamma)$ — the power set of Γ, i.e., the set of all subsets of Γ;
- $\mathfrak{P}_0(\Gamma) = \mathfrak{P}(\Gamma) \setminus \{\emptyset\}$ — the set of all non-empty subsets of Γ;
- $\mathrm{Fin}(\Gamma)$ — the set of all non-empty finite subsets of Γ;
- $\mathrm{Fin}^*(\Gamma) = \mathrm{Fin}(\Gamma) \cup \{\emptyset\}$ — the set of all finite subsets of Γ.

It should be noted that $\mathrm{card}\,\mathrm{Fin}(\Gamma) = \mathrm{card}\,\Gamma$.

In this section, we survey Banach spaces,[13] which are linear subspaces of the product space $\mathbb{R}^\Gamma$, where $\mathbb{R}^\Gamma$ is a linear space with the following scalar multiplication and addition:

$$\mathbb{R}^\Gamma \times \mathbb{R} \ni (x, t) \mapsto tx = (tx(\gamma))_{\gamma\in\Gamma} \in \mathbb{R}^\Gamma;$$
$$\mathbb{R}^\Gamma \times \mathbb{R}^\Gamma \ni (x, y) \mapsto x + y = (x(\gamma) + y(\gamma))_{\gamma\in\Gamma} \in \mathbb{R}^\Gamma.$$

Before considering norms, we first discuss the product topology of $\mathbb{R}^\Gamma$. The scalar multiplication and addition are continuous with respect to the product topology. Namely, $\mathbb{R}^\Gamma$ with the product topology is a topological linear space.[14] Note that $w(\mathbb{R}^\Gamma) = \mathrm{card}\,\Gamma$.

Indeed, let $\mathcal{B}_0$ be a countable open basis for $\mathbb{R}$. Then, $\mathbb{R}^\Gamma$ has the following open basis:

$$\left\{\bigcap_{\gamma\in F} \mathrm{pr}_\gamma^{-1}(B_\gamma) \;\middle|\; F \in \mathrm{Fin}(\Gamma),\ B_\gamma \in \mathcal{B}_0\ (\gamma \in F)\right\}.$$

Thus, we have $w(\mathbb{R}^\Gamma) \leqslant \aleph_0\,\mathrm{card}\,\mathrm{Fin}(\Gamma) = \mathrm{card}\,\Gamma$. Let $\mathcal{B}$ be an open basis for $\mathbb{R}^\Gamma$. For each $B \in \mathcal{B}$, we can find $F_B \in \mathrm{Fin}(\Gamma)$ such that $\mathrm{pr}_\gamma(B) = \mathbb{R}$ for every $\gamma \in \Gamma \setminus F_B$.

[13] A **Banach space** is a complete normed linear space.

[14] For a topological linear space, refer to Sect. 3.4.

Then, $\operatorname{card}\bigcup_{B\in\mathcal{B}} F_B \leqslant \aleph_0 \operatorname{card}\mathcal{B}$. If $\operatorname{card}\mathcal{B} < \operatorname{card}\Gamma$ then $\operatorname{card}\bigcup_{B\in\mathcal{B}} F_B < \operatorname{card}\Gamma$, so we have $\gamma_0 \in \Gamma \setminus \bigcup_{B\in\mathcal{B}} F_B$. The open set $\mathrm{pr}_{\gamma_0}^{-1}((0,\infty)) \subset \mathbb{R}^\Gamma$ contains some $B \in \mathcal{B}$. Then, $\mathrm{pr}_{\gamma_0}(B) \subset (0,\infty)$, which means that $\gamma_0 \in F_B$. This is a contradiction. Therefore, $\operatorname{card}\mathcal{B} \geqslant \operatorname{card}\Gamma$, and thus we have $w(\mathbb{R}^\Gamma) \geqslant \operatorname{card}\Gamma$.

For each $\gamma \in \Gamma$, we define the unit vector $\mathbf{e}_\gamma \in \mathbb{R}^\Gamma$ by $\mathbf{e}_\gamma(\gamma) = 1$ and $\mathbf{e}_\gamma(\gamma') = 0$ for $\gamma' \neq \gamma$. It should be noted that $\{\mathbf{e}_\gamma \mid \gamma \in \Gamma\}$ is not a Hamel basis for $\mathbb{R}^\Gamma$, and the linear span of $\{\mathbf{e}_\gamma \mid \gamma \in \Gamma\}$ is the following:[15]

$$\mathbb{R}^\Gamma_f \underset{\text{def}}{=} \left\{x \in \mathbb{R}^\Gamma \;\middle|\; x(\gamma) = 0 \ \text{ except for finitely many } \gamma \in \Gamma\right\},$$

which is a dense linear subspace of $\mathbb{R}^\Gamma$. The subspace $\mathbb{R}^\mathbb{N}_f$ of $\boldsymbol{s} = \mathbb{R}^\mathbb{N}$ is also denoted by $\boldsymbol{s}_f$, which is the space of finite sequences (with the product topology). When $\operatorname{card}\Gamma = \aleph_0$, the space $\mathbb{R}^\Gamma$ is linearly homeomorphic to the space of sequences $\boldsymbol{s} = \mathbb{R}^\mathbb{N}$, i.e., there exists a linear homeomorphism between $\mathbb{R}^\Gamma$ and $\boldsymbol{s}$, where the linear subspace $\mathbb{R}^\Gamma_f$ is linearly homeomorphic to $\boldsymbol{s}_f$ by the same homeomorphism.

The following fact can easily be observed:

Proposition 1.2.1 *The following are equivalent:*

(a) $\mathbb{R}^\Gamma$ *is metrizable;*
(b) $\mathbb{R}^\Gamma_f$ *is metrizable;*
(c) $\mathbb{R}^\Gamma_f$ *is first countable;*
(d) $\operatorname{card}\Gamma \leqslant \aleph_0$.

Sketch of Proof The implications (a) $\Rightarrow$ (b) $\Rightarrow$ (c) are trivial and (d) $\Rightarrow$ (a) is elementary. To see (c) $\Rightarrow$ (d), take a countable neighborhood basis $\{U_i \mid i \in \mathbb{N}\}$ of $\mathbf{0}$ in $\mathbb{R}^\Gamma_f$. Show that each $\Gamma_i = \{\gamma \in \Gamma \mid \mathbb{R}\mathbf{e}_\gamma \not\subset U_i\}$ is finite. If Γ is uncountable then $\Gamma \setminus \bigcup_{i\in\mathbb{N}} \Gamma_i \neq \emptyset$, i.e., $\mathbb{R}\mathbf{e}_\gamma \subset \bigcap_{i\in\mathbb{N}} U_i$ for some $\gamma \in \Gamma$. In this case, $U_i \not\subset \mathrm{pr}_\gamma^{-1}((-1,1))$ for every $i \in \mathbb{N}$, which is a contradiction.

Thus, every linear subspace L of $\mathbb{R}^\Gamma$ containing $\mathbb{R}^\Gamma_f$ is non-metrizable if Γ is uncountable, and it is metrizable if Γ is countable. Besides, due to the following proposition, every linear subspaces L of $\mathbb{R}^\Gamma$ containing $\mathbb{R}^\Gamma_f$ is non-normable if Γ is infinite.

Proposition 1.2.2 *Let Γ be an infinite set. Any norm on $\mathbb{R}^\Gamma_f$ does not induce the topology inherited from the product topology of $\mathbb{R}^\Gamma$.*

Proof Suppose that a norm $\|\cdot\|$ on $\mathbb{R}^\Gamma_f$ induces the topology inherited from the product topology of $\mathbb{R}^\Gamma$. Since $U = \{x \in \mathbb{R}^\Gamma_f \mid \|x\| < 1\}$ is an open neighborhood of $\mathbf{0}$ in $\mathbb{R}^\Gamma_f$, we have a finite set $F \subset \Gamma$ and neighborhoods V_γ of $0 \in \mathbb{R}$, $\gamma \in F$, such that $\mathbb{R}^\Gamma_f \cap \bigcap_{\gamma\in F} \mathrm{pr}_\gamma^{-1}(V_\gamma) \subset U$. Then, we can take $\gamma_0 \in \Gamma \setminus F$. As $\mathbb{R}\mathbf{e}_{\gamma_0} \subset U$, we have $\|\mathbf{e}_{\gamma_0}\|^{-1}\mathbf{e}_{\gamma_0} \in U$ but $\|\|\mathbf{e}_{\gamma_0}\|^{-1}\mathbf{e}_{\gamma_0}\| = \|\mathbf{e}_{\gamma_0}\|^{-1}\|\mathbf{e}_{\gamma_0}\| = 1$, which is a contradiction. □

[15] The **linear span** of B is the linear subspace generated by B, i.e., the minimal linear subspace containing B.

We can consider various norms defined on linear subspaces of $\mathbb{R}^\Gamma$, which are not compatible with the product topology as just seen in Proposition 1.2.2 above. In general, the unit closed ball and the unit sphere of a normed linear space $X = (X, \|\cdot\|)$ are denoted by $\mathbf{B}_X$ and $\mathbf{S}_X$, respectively. Namely, let

$$\mathbf{B}_X = \bigl\{x \in X \bigm| \|x\| \leqslant 1\bigr\} \text{ and } \mathbf{S}_X = \bigl\{x \in X \bigm| \|x\| = 1\bigr\}.$$

When $X = \mathbb{R}^n$ (Euclidean n-space), we have $\mathbf{B}_{\mathbb{R}^n} = \mathbf{B}^n$ and $\mathbf{S}_{\mathbb{R}^n} = \mathbf{S}^{n-1}$. The zero vector (the zero element) of X is denoted by $\mathbf{0}_X$, or simply $\mathbf{0}$ if there is no possibility of confusion.

The Banach space $\ell_\infty(\Gamma)$ and its closed linear subspaces $\boldsymbol{c}(\Gamma) \supset \boldsymbol{c}_0(\Gamma)$ are defined as follows:

- $\ell_\infty(\Gamma) \underset{\text{def}}{=} \bigl\{x \in \mathbb{R}^\Gamma \bigm| \sup_{\gamma\in\Gamma} |x(\gamma)| < \infty\bigr\}$ with the sup-norm[16]

$$\|x\|_\infty \underset{\text{def}}{=} \sup_{\gamma\in\Gamma} |x(\gamma)|;$$

- $\boldsymbol{c}(\Gamma) \underset{\text{def}}{=} \bigl\{x \in \mathbb{R}^\Gamma \bigm| \exists t \in \mathbb{R} \text{ such that } \forall \varepsilon > 0,\ |x(\gamma) - t| < \varepsilon$ except for finitely many $\gamma \in \Gamma\bigr\}$;
- $\boldsymbol{c}_0(\Gamma) \underset{\text{def}}{=} \bigl\{x \in \mathbb{R}^\Gamma \bigm| \forall \varepsilon > 0,\ |x(\gamma)| < \varepsilon$ except for finitely many $\gamma \in \Gamma\bigr\}$.

These are linear subspaces of $\mathbb{R}^\Gamma$, but are not topological subspaces according to Proposition 1.2.2. The space $\boldsymbol{c}(\Gamma)$ is linearly homeomorphic to the product space $\boldsymbol{c}_0(\Gamma) \times \mathbb{R}$ by the correspondence

$$\boldsymbol{c}_0(\Gamma) \times \mathbb{R} \ni (x, t) \mapsto (x(\gamma) + t)_{\gamma\in\Gamma} \in \boldsymbol{c}(\Gamma).$$

This correspondence and its inverse are Lipschitz with respect to the norm $\|(x, t)\| = \max\{\|x\|_\infty, |t|\}$. Indeed, let $y = (x(\gamma) + t)_{\gamma\in\Gamma}$. Then, $\|y\|_\infty \leqslant \|x\|_\infty + |t| \leqslant 2\|(x, t)\|$. Moreover, observe that

$$|t| \leqslant |y(\gamma)| + |x(\gamma)| \leqslant \|y\|_\infty + |x(\gamma)| \text{ for every } \gamma \in \Gamma \text{ and } x \in \boldsymbol{c}_0(\Gamma),$$

where $\inf_{\gamma\in\Gamma} |x(\gamma)| = 0$. Hence, $|t| \leqslant \|y\|_\infty$. Since $|x(\gamma)| \leqslant |y(\gamma)| + |t| \leqslant 2\|y\|_\infty$ for every $\gamma \in \Gamma$, it follows that $\|x\|_\infty \leqslant 2\|y\|_\infty$. Therefore, $\|(x, t)\| \leqslant 2\|y\|_\infty$.

Furthermore, we define $\ell_\infty^f(\Gamma)$ as the linear space $\mathbb{R}_f^\Gamma$ with the norm inherited from $\ell_\infty(\Gamma)$. We then have the inclusions:

$$\ell_\infty^f(\Gamma) \subset \boldsymbol{c}_0(\Gamma) \subset \boldsymbol{c}(\Gamma) \subset \ell_\infty(\Gamma).$$

The topology of $\ell_\infty^f(\Gamma)$ is different from the one inherited from the product topology, that is, $\ell_\infty^f(\Gamma) \neq \mathbb{R}_f^\Gamma$ as (topological) spaces (Proposition 1.2.2).

[16] The completeness of $\ell_\infty(\Gamma)$ can be verified in the same way as Proposition 1.1.1.

This can be shown easily and directly. Indeed, $\{\mathbf{e}_\gamma \mid \gamma \in \Gamma\}$ is discrete in $\ell_\infty^f(\Gamma)$, but $\mathbf{0}$ is a cluster point of this set with respect to the product topology.

As is easily observed, $\ell_\infty^f(\Gamma)$ is dense in $\boldsymbol{c}_0(\Gamma)$. However, $\ell_\infty^f(\Gamma)$ is not dense in $\boldsymbol{c}(\Gamma)$ because $\boldsymbol{c}(\Gamma)$ is naturally identified with $\boldsymbol{c}_0(\Gamma) \times \mathbb{R}$. We must pay attention to the following fact concerning weight:

Proposition 1.2.3 *For an arbitrary infinite set* Γ,

$$w(\ell_\infty(\Gamma)) = 2^{\operatorname{card}\Gamma} \quad \textit{but} \quad w(\boldsymbol{c}(\Gamma)) = w(\boldsymbol{c}_0(\Gamma)) = w(\ell_\infty^f(\Gamma)) = \operatorname{card}\Gamma.$$

Proof The characteristic map $\chi_\Lambda : \Gamma \to \{0, 1\} \subset \mathbb{R}$ of $\Lambda \in \mathfrak{P}(\Gamma)$ belongs to $\ell_\infty(\Gamma)$ ($\chi_\emptyset = \mathbf{0} \in \ell_\infty(\Gamma)$), where $\|\chi_\Lambda - \chi_{\Lambda'}\|_\infty = 1$ if $\Lambda \neq \Lambda' \in \mathfrak{P}(\Gamma)$. It follows that $w(\ell_\infty(\Gamma)) = c(\ell_\infty(\Gamma)) \geqslant 2^{\operatorname{card}\Gamma}$. Since $\mathbb{Q}^\Gamma \cap \ell_\infty(\Gamma)$ is dense in $\ell_\infty(\Gamma)$, we have

$$w(\ell_\infty(\Gamma)) = \operatorname{dens}\ell_\infty(\Gamma) \leqslant \operatorname{card}\mathbb{Q}^\Gamma = \aleph_0^{\operatorname{card}\Gamma} = 2^{\operatorname{card}\Gamma}.$$

Therefore, $w(\ell_\infty(\Gamma)) = 2^{\operatorname{card}\Gamma}$.

On the other hand, $\mathbf{e}_\gamma \in \ell_\infty^f(\Gamma)$ for each $\gamma \in \Gamma$ and $\|\mathbf{e}_\gamma - \mathbf{e}_{\gamma'}\|_\infty = 1$ if $\gamma \neq \gamma'$. Since $\ell_\infty^f(\Gamma) \subset \boldsymbol{c}_0(\Gamma)$, it follows that

$$w(\boldsymbol{c}_0(\Gamma)) \geqslant w(\ell_\infty^f(\Gamma)) = c(\ell_\infty^f(\Gamma)) \geqslant \operatorname{card}\Gamma.$$

Moreover, $\boldsymbol{c}_0(\Gamma)$ has the following dense subset:

$$\mathbb{Q}_f^\Gamma = \left\{x \in \mathbb{Q}^\Gamma \;\middle|\; x(\gamma) = 0 \ \text{ except for finitely many } \gamma \in \Gamma\right\}.$$

Thus, it follows that

$$w(\boldsymbol{c}_0(\Gamma)) = \operatorname{dens}\boldsymbol{c}_0(\Gamma) \leqslant \operatorname{card}\mathbb{Q}_f^\Gamma \leqslant \aleph_0 \operatorname{card}\operatorname{Fin}(\Gamma) = \operatorname{card}\Gamma.$$

Hence, $w(\boldsymbol{c}_0(\Gamma)) = w(\ell_\infty^f(\Gamma)) = \operatorname{card}\Gamma$. As already observed, $\boldsymbol{c}(\Gamma) \approx \boldsymbol{c}_0(\Gamma) \times \mathbb{R}$, so we have $w(\boldsymbol{c}(\Gamma)) = w(\boldsymbol{c}_0(\Gamma))$. □

When $\Gamma = \mathbb{N}$, we write as follows:

- $\ell_\infty(\mathbb{N}) = \ell_\infty$ — the space of bounded sequences;
- $\boldsymbol{c}(\mathbb{N}) = \boldsymbol{c}$ — the space of convergent sequences;
- $\boldsymbol{c}_0(\mathbb{N}) = \boldsymbol{c}_0$ — the space of sequences convergent to 0;
- $\ell_\infty^f(\mathbb{N}) = \ell_\infty^f$ — the space of finite sequences with the sup-norm,

where $\ell_\infty^f \neq \boldsymbol{s}_f$ as (topological) spaces. According to Proposition 1.2.3, $\boldsymbol{c}$ and $\boldsymbol{c}_0$ are separable, but ℓ_∞ is non-separable. When $\operatorname{card}\Gamma = \aleph_0$, the spaces $\ell_\infty(\Gamma)$, $\boldsymbol{c}(\Gamma)$, $\boldsymbol{c}_0(\Gamma)$, and $\ell_\infty^f(\Gamma)$ are linearly isometric to these spaces ℓ_∞, $\boldsymbol{c}$, $\boldsymbol{c}_0$, and ℓ_∞^f, respectively.

Here, we regard $\mathrm{Fin}(\Gamma)$ as a directed set by $\subset$.[17] For $x \in \mathbb{R}^\Gamma$, we say that $\sum_{\gamma\in\Gamma} x(\gamma)$ is **convergent** if the net $\left(\sum_{\gamma\in F} x(\gamma)\right)_{F\in\mathrm{Fin}(\Gamma)}$ is convergent,[18] and define

$$\sum_{\gamma\in\Gamma} x(\gamma) = \lim_{F\in\mathrm{Fin}(\Gamma)} \sum_{\gamma\in F} x(\gamma),$$

where it is said that $\left(\sum_{\gamma\in F} x(\gamma)\right)_{F\in\mathrm{Fin}(\Gamma)}$ **converges to** $\alpha \in \mathbb{R}$ and α is called the **limit** if the following holds:

$$\forall \varepsilon > 0,\ \exists F_0 \in \mathrm{Fin}(\Gamma)\ \text{ such that }\ F_0 \subset F \in \mathrm{Fin}(\Gamma)\ \Rightarrow\ \left|\sum_{\gamma\in\Gamma} x(\gamma) - \alpha\right| < \varepsilon.$$

The following can be easily derived from the definition:

Proposition 1.2.4 *Let Γ be an infinite set and $x \in (\mathbb{R}_+)^\Gamma$ (i.e., $x \in \mathbb{R}^\Gamma$ such that $x(\gamma) \geqslant 0$ for all $\gamma \in \Gamma$). Then, $\sum_{\gamma\in\Gamma} x(\gamma)$ is convergent if and only if $\left(\sum_{\gamma\in F} x(\gamma)\right)_{F\in\mathrm{Fin}(\Gamma)}$ is upper bounded, and then*

$$\sum_{\gamma\in\Gamma} x(\gamma) = \sup_{F\in\mathrm{Fin}(\Gamma)} \sum_{\gamma\in F} x(\gamma). \qquad \square$$

For this reason, the following notation is suitable for $x \in (\mathbb{R}_+)^\Gamma$:

- $\sum_{\gamma\in\Gamma} x(\gamma) < \infty$ means that $\sum_{\gamma\in\Gamma} x(\gamma)$ is convergent.

For $x \in \mathbb{R}^{\mathbb{N}}$, $\sum_{i\in\mathbb{N}} x(i)$ and $\sum_{i=1}^{\infty} x(i)$ should be distinguished because they have different meanings. When the sequence $\left(\sum_{i=1}^{n} x(i)\right)_{n\in\mathbb{N}}$ is convergent, we say that $\sum_{i=1}^{\infty} x(i)$ is **convergent**, and define

$$\sum_{i=1}^{\infty} x(i) = \lim_{n\to\infty} \sum_{i=1}^{n} x(i).$$

Evidently, if $\sum_{i\in\mathbb{N}} x(i)$ is convergent, then $\sum_{i=1}^{\infty} x(i)$ is also convergent and the equality $\sum_{i=1}^{\infty} x(i) = \sum_{i\in\mathbb{N}} x(i)$ holds. But, $\sum_{i\in\mathbb{N}} x(i)$ is not necessarily convergent even if $\sum_{i=1}^{\infty} x(i)$ is convergent. In fact, we have the following equivalence:

$$\sum_{i\in\mathbb{N}} x(i)\ \text{ is convergent }\ \Leftrightarrow\ \sum_{i=1}^{\infty} |x(i)|\ \text{ is convergent.}$$

---

[17] A **directed set** is a preordered set $\Lambda = (\Lambda, \leqslant)$ (a set Λ given a preorder $\leqslant$) satisfying the following condition: $\forall a, b \in \Lambda,\ \exists c \in \Lambda$ such that $a, b \leqslant c$, where a reflexive and transitive binary relation is called a preorder.

[18] A **net** $x = (x_\lambda)_{\lambda\in\Lambda}$ in X is a function $x : \Lambda \to X$ (i.e., $x \in X^\Lambda$) of a directed set $\Lambda = (\Lambda, \leqslant)$ to X, which is a generalization of a sequence in X.

This follows from the more general proposition:

Proposition 1.2.5 *For an infinite set* Γ *and* $x \in \mathbb{R}^\Gamma$,

$$\sum_{\gamma\in\Gamma} x(\gamma) \ \textit{is convergent} \ \Leftrightarrow \ \sum_{\gamma\in\Gamma} |x(\gamma)| = \sup_{F\in \mathrm{Fin}(\Gamma)} \sum_{\gamma\in F} |x(\gamma)| < \infty.$$

In this case, $\Gamma_x = \{\gamma \in \Gamma \mid x(\gamma) \neq 0\}$ *is countable, i.e.,* $\mathrm{card}\, \Gamma_x \leqslant \aleph_0$, *and for any sequence* $(\gamma_i)_{i\in\mathbb{N}}$ *in* Γ *such that* $\Gamma_x \subset \{\gamma_i \mid i \in \mathbb{N}\}$ *and* $\gamma_i \neq \gamma_j$ *if* $i \neq j$,

$$\sum_{\gamma\in\Gamma} x(\gamma) = \sum_{i=1}^{\infty} x(\gamma_i).$$

Proof We can write $\Gamma_x = \Gamma_+ \cup \Gamma_-$, where

$$\Gamma_+ = \big\{\gamma \in \Gamma \,\big|\, x(\gamma) > 0\big\} \ \text{ and } \ \Gamma_- = \big\{\gamma \in \Gamma \,\big|\, x(\gamma) < 0\big\}.$$

($\Rightarrow$): Assume that $\sum_{\gamma\in\Gamma} x(\gamma)$ is convergent. Then, we have $F_0 \in \mathrm{Fin}(\Gamma)$ such that

$$F_0 \subset F \in \mathrm{Fin}(\Gamma) \ \Rightarrow \ \left|\sum_{\gamma\in\Gamma} x(\gamma) - \sum_{\gamma\in F} x(\gamma)\right| < 1.$$

For each $E \in \mathrm{Fin}(\Gamma_+) \cup \mathrm{Fin}(\Gamma_-)$,

$$\sum_{\gamma\in E\setminus F_0} |x(\gamma)| = \left|\sum_{\gamma\in E\setminus F_0} x(\gamma)\right| = \left|\sum_{\gamma\in E\cup F_0} x(\gamma) - \sum_{\gamma\in F_0} x(\gamma)\right| < 2.$$

For each $F \in \mathrm{Fin}(\Gamma)$ with $F_0 \subset F$, applying the above inequality, we have

$$\begin{aligned}\sum_{\gamma\in F} |x(\gamma)| &= \sum_{\gamma\in F_0} |x(\gamma)| + \sum_{\gamma\in F\cap\Gamma_+\setminus F_0} |x(\gamma)| + \sum_{\gamma\in F\cap\Gamma_-\setminus F_0} |x(\gamma)| \\ &< \sum_{\gamma\in F_0} |x(\gamma)| + 4,\end{aligned}$$

which implies that $\big(\sum_{\gamma\in F} |x(\gamma)|\big)_{F\in\mathrm{Fin}(\Gamma)}$ is upper bounded. Consequently,

$$\sum_{\gamma\in\Gamma} |x(\gamma)| = \sup_{F\in\mathrm{Fin}(\Gamma)} \sum_{\gamma\in F} |x(\gamma)| < \infty.$$

($\Leftarrow$): Assume that $\sum_{\gamma\in\Gamma} |x(\gamma)| < \infty$. We can write $\Gamma_x = \bigcup_{n\in\mathbb{N}} \Gamma_n$, where $\Gamma_n = \{\gamma \in \Gamma \mid |x(\gamma)| > 1/n\}$. Hence, Γ_x is countable because each Γ_n is finite. Note that $\sum_{\gamma\in\Gamma_+} |x(\gamma)| < \infty$ and $\sum_{\gamma\in\Gamma_-} |x(\gamma)| < \infty$. Then, it suffices to show the following:

$$\sum_{\gamma\in\Gamma} x(\gamma) = \sum_{\gamma\in\Gamma_+} |x(\gamma)| - \sum_{\gamma\in\Gamma_-} |x(\gamma)|.$$

For each $\varepsilon > 0$, we can find $F_+ \in \mathrm{Fin}(\Gamma_+)$ and $F_- \in \mathrm{Fin}(\Gamma_-)$ such that

$$F_\pm \subset E \in \mathrm{Fin}(\Gamma_\pm) \;\Rightarrow\; \sum_{\gamma\in\Gamma_\pm} |x(\gamma)| - \varepsilon/2 < \sum_{\gamma\in E} |x(\gamma)| \leqslant \sum_{\gamma\in\Gamma_\pm} |x(\gamma)|.$$

Then, for each $F \in \mathrm{Fin}(\Gamma)$ with $F \supset F_+ \cup F_-$, since $F \cap \Gamma_\pm \supset F_\pm$, it follows that

$$\begin{aligned}
&\left|\sum_{\gamma\in F} x(\gamma) - \left(\sum_{\gamma\in\Gamma_+} |x(\gamma)| - \sum_{\gamma\in\Gamma_-} |x(\gamma)|\right)\right| \\
&\qquad \leqslant \left|\sum_{\gamma\in F\cap\Gamma_+} |x(\gamma)| - \sum_{\gamma\in\Gamma_+} |x(\gamma)|\right| + \left|\sum_{\gamma\in F\cap\Gamma_-} |x(\gamma)| - \sum_{\gamma\in\Gamma_-} |x(\gamma)|\right| \\
&\qquad < \varepsilon/2 + \varepsilon/2 = \varepsilon.
\end{aligned}$$

Thus, the desired equality holds.

To show the additional statement, take any sequence $(\gamma_i)_{i\in\mathbb{N}}$ in Γ such that $\Gamma_x \subset \{\gamma_i \mid i \in \mathbb{N}\}$ and $\gamma_i \neq \gamma_j$ if $i \neq j$. Let $n_0 = \max\{i \in \mathbb{N} \mid \gamma_i \in F_+ \cup F_-\}$, where $F_\pm$ are the same as above. For each $n \geqslant n_0$, since $F_+ \cup F_- \subset \{\gamma_1, \ldots, \gamma_n\}$, we have

$$\left|\sum_{i=1}^{n} x(\gamma_i) - \left(\sum_{\gamma\in\Gamma_+} |x(\gamma)| - \sum_{\gamma\in\Gamma_-} |x(\gamma)|\right)\right| < \varepsilon.$$

Thus, we have $\sum_{\gamma\in\Gamma} x(\gamma) = \sum_{i=1}^{\infty} x(\gamma_i)$. □

For each $1 \leqslant p < \infty$, the Banach space $\ell_p(\Gamma)$ is defined as follows[19]:

- $\ell_p(\Gamma) \underset{\text{def}}{=} \left\{x \in \mathbb{R}^\Gamma \;\middle|\; \sum_{\gamma\in\Gamma} |x(\gamma)|^p < \infty\right\}$ with the norm

$$\|x\|_p \underset{\text{def}}{=} \left(\sum_{\gamma\in\Gamma} |x(\gamma)|^p\right)^{1/p}.$$

When $1 < p < \infty$, the triangle inequality for the norm $\|x\|_p$ is known as the MINKOWSKI INEQUALITY, which is derived from the following HÖLDER INEQUALITY:

$$\sum_{\gamma\in\Gamma} a_\gamma b_\gamma \leqslant \left(\sum_{\gamma\in\Gamma} a_\gamma^p\right)^{1/p} \left(\sum_{\gamma\in\Gamma} b_\gamma^{\frac{1}{1-1/p}}\right)^{1-1/p} \quad \text{for every } a_\gamma, b_\gamma \geqslant 0.$$

We can prove these inequalities as follows:

[19] For $2 < p < \infty$, $\ell_p(\Gamma)$ will not appear except in this chapter.

Proof (*The Hölder Inequality*) Let $q = p/(p-1)$ (i.e., $1/q = 1 - 1/p$). Since $\log x$ is an increasing concave function, for each $x, y \in \mathbb{R}$,

$$\log x^{1/p} y^{1/q} = \frac{1}{p}\log x + \frac{1}{q}\log y \leqslant \log\left(\frac{x}{p} + \frac{y}{q}\right), \quad \therefore\ x^{1/p} y^{1/q} \leqslant \frac{x}{p} + \frac{y}{q}.$$

Let $\alpha = \sum_{\gamma\in\Gamma} a_\gamma^p$ and $\beta = \sum_{\gamma\in\Gamma} b_\gamma^q$. If $\alpha = 0$ or $\beta = 0$, then $\sum_{\gamma\in\Gamma} a_\gamma b_\gamma = 0 = \alpha^{1/p}\beta^{1/q}$. When $\alpha, \beta > 0$, we can apply the above inequality to obtain the following:

$$\sum_{\gamma\in\Gamma}\left(\frac{a_\gamma^p}{\alpha}\right)^{1/p}\left(\frac{b_\gamma^q}{\beta}\right)^{1/q} \leqslant \sum_{\gamma\in\Gamma}\left(\frac{a_\gamma^p}{p\alpha} + \frac{b_\gamma^q}{q\beta}\right)$$

$$\therefore\ \frac{\sum_{\gamma\in\Gamma} a_\gamma b_\gamma}{\alpha^{1/p}\beta^{1/q}} \leqslant \frac{\sum_{\gamma\in\Gamma} a_\gamma^p}{p\alpha} + \frac{\sum_{\gamma\in\Gamma} b_\gamma^q}{q\beta} = \frac{1}{p} + \frac{1}{q} = 1.$$

Thus, we have $\sum_{\gamma\in\Gamma} a_\gamma b_\gamma \leqslant \alpha^{1/p}\beta^{1/q}$. □

Proof (*The Minkowski Inequality*) For every $x, y \in \ell_p(\Gamma)$,

$$\begin{aligned}\sum_{\gamma\in\Gamma} |x(\gamma)+y(\gamma)|^p &\leqslant \sum_{\gamma\in\Gamma} |x(\gamma)|\cdot|x(\gamma)+y(\gamma)|^{p-1} + \sum_{\gamma\in\Gamma} |y(\gamma)|\cdot|x(\gamma)+y(\gamma)|^{p-1}\\ &\leqslant \left(\sum_{\gamma\in\Gamma}|x(\gamma)|^p\right)^{1/p}\left(\sum_{\gamma\in\Gamma}|x(\gamma)+y(\gamma)|^{(p-1)\frac{1}{1-1/p}}\right)^{1-1/p}\\ &\quad + \left(\sum_{\gamma\in\Gamma}|y(\gamma)|^p\right)^{1/p}\left(\sum_{\gamma\in\Gamma}|x(\gamma)+y(\gamma)|^{(p-1)\frac{1}{1-1/p}}\right)^{1-1/p},\end{aligned}$$

$$\begin{aligned}\therefore\ \big(\|x+y\|_p\big)^p &\leqslant \big(\|x\|_p + \|y\|_p\big)\big(\|x+y\|_p\big)^{p(1-1/p)}\\ &= \big(\|x\|_p + \|y\|_p\big)\big(\|x+y\|_p\big)^{p-1}.\end{aligned}$$

Consequently, $\|x+y\|_p \leqslant \|x\|_p + \|y\|_p$. □

The completeness of $\ell_p(\Gamma)$ can be verified as follows:

Proof (*Completeness of* $\ell_p(\Gamma)$) Let $(x_n)_{n\in\mathbb{N}}$ be a $\|\cdot\|_p$-Cauchy sequence in $\ell_p(\Gamma)$. Since $(x_n(\gamma))_{n\in\mathbb{N}}$ is Cauchy in $\mathbb{R}$ for each $\gamma \in \Gamma$, we can define $x \in \mathbb{R}^\Gamma$ by $x(\gamma) = \lim_{n\to\infty} x_n(\gamma)$. We show that $x \in \ell_p(\Gamma)$ and $\|x_n - x\|_p \to 0$ $(n \to \infty)$.

Choose $n \in \mathbb{N}$ so that $i, j \geqslant n \Rightarrow \|x_i - x_j\|_p < 1$. Then, we have

$$i \geqslant n \ \Rightarrow\ \|x_i\|_p \leqslant \|x_i - x_n\|_p + \|x_n\|_p < \|x_n\|_p + 1.$$

For each $F \in \mathrm{Fin}(\Gamma)$, let $k = \mathrm{card}\, F$ and choose $i \geqslant n$ so that

$$\big||x(\gamma)|^p - |x_i(\gamma)|^p\big| < 1/k \ \text{ for every } \gamma \in F.$$

Then, it follows that

$$\begin{aligned}\sum_{\gamma\in F}|x(\gamma)|^p &\leqslant \sum_{\gamma\in F}\big||x(\gamma)|^p - |x_i(\gamma)|^p\big| + \sum_{\gamma\in F}|x_i(\gamma)|^p\\ &\leqslant \big(\|x_i\|_p\big)^p + 1 < \big(\|x_n\|_p + 1\big)^p + 1.\end{aligned}$$

Hence, $\sum_{\gamma\in\Gamma}|x(\gamma)|^p<\infty$, which means that $x\in\ell_p(\Gamma)$.

For each $\varepsilon>0$, choose $n\in\mathbb{N}$ so that $i,\,j\geqslant n\Rightarrow\|x_i-x_j\|_p<\varepsilon/4$. Now, choose $j\geqslant n$ so that

$$|x(\gamma)-x_j(\gamma)|^p<(\varepsilon/4)^p(1/k)\ \text{ for every } \gamma\in F.$$

Then, for each $i\geqslant n$, using the Minkowski inequality, we have

$$\begin{aligned}\left(\sum_{\gamma\in F}|x_i(\gamma)-x(\gamma)|^p\right)^{1/p}&\leqslant\left(\sum_{\gamma\in F}|x_i(\gamma)-x_j(\gamma)|^p\right)^{1/p}+\left(\sum_{\gamma\in F}|x_j(\gamma)-x(\gamma)|^p\right)^{1/p}\\&\leqslant\|x_i-x_j\|_p+\varepsilon/4<\varepsilon/4+\varepsilon/4=\varepsilon/2.\end{aligned}$$

Hence, $\|x_i-x\|_p\leqslant\varepsilon/2<\varepsilon$. Thus, $\|x_n-x\|_p\to0$ $(n\to\infty)$. □

The space $\ell_2(\Gamma)$ is the Hilbert space[20] with the inner product

$$\langle x,\,y\rangle=\sum_{\gamma\in\Gamma}x(\gamma)y(\gamma).$$

The above inner product is well-defined because

$$\sum_{\gamma\in\Gamma}|x(\gamma)y(\gamma)|\leqslant\frac{\|x\|_2^2+\|y\|_2^2}{2}<\infty.$$

When $\Gamma=\mathbb{N}$, we write $\ell_p(\mathbb{N})=\ell_p$ and $\ell_p^f(\mathbb{N})=\ell_p^f$.

For $1\leqslant p<q<\infty$, we have $\ell_p(\Gamma)\subsetneqq\ell_q(\Gamma)\subsetneqq\boldsymbol{c}_0(\Gamma)$ as sets (or linear spaces). These inclusions are continuous because $\|x\|_q\leqslant\|x\|_p$ for every $x\in\ell_p(\Gamma)$ and $\|x\|_\infty\leqslant\|x\|_q$ for every $x\in\ell_q(\Gamma)$. When Γ is infinite, the topology of $\ell_p(\Gamma)$ is different from that induced by the norm $\|\cdot\|_q$ or $\|\cdot\|_\infty$ (i.e., the topology inherited from $\ell_q(\Gamma)$ or $\boldsymbol{c}_0(\Gamma)$). In fact, the unit sphere $\mathbf{S}_{\ell_p(\Gamma)}$ is closed in $\ell_p(\Gamma)$ but not closed in $\ell_q(\Gamma)$ for any $q>p$, nor in $\boldsymbol{c}_0(\Gamma)$.

To see this, take distinct $\gamma_i\in\Gamma$, $i\in\mathbb{N}$, and define a sequence $(x_n)_{n\in\mathbb{N}}$ in $\mathbf{S}_{\ell_p(\Gamma)}$ by $x_n(\gamma_i)=n^{-1/p}$ for $i\leqslant n$ and $x_n(\gamma)=0$ for $\gamma\neq\gamma_1,\dots,\gamma_n$. Then, $\|x_n\|_\infty=n^{-1/p}\to0$ $(n\to\infty)$. Since $(p-q)/pq<0$, it follows that

$$\|x_n\|_q=\left(n\cdot n^{-q/p}\right)^{1/q}=n^{(p-q)/pq}\to0\ \ (n\to\infty).$$

For $1\leqslant p<\infty$, $\mathbb{R}_f^\Gamma\subset\ell_p(\Gamma)$ as sets (or linear spaces). Like $\ell_\infty^f(\Gamma)$, the notation $\ell_p^f(\Gamma)$ denotes the linear space $\mathbb{R}_f^\Gamma$ with the norm $\|\cdot\|_p$ inherited from $\ell_p(\Gamma)$. Then, for $1\leqslant p<q\leqslant\infty$, the topology of $\ell_p^f(\Gamma)$ is finer than that of $\ell_q^f(\Gamma)$, i.e., the identity $\mathrm{id}:\ell_p^f(\Gamma)\to\ell_q^f(\Gamma)$, where the unit sphere $\mathbf{S}_{\ell_p^f(\Gamma)}$ is closed in $\ell_p^f(\Gamma)$ but not closed in $\ell_q^f(\Gamma)$. Moreover, it is also finer than the topology of $\mathbb{R}_f^\Gamma$ inherited from the product space $\mathbb{R}^\Gamma$ because so is the topology of $\ell_\infty^f(\Gamma)$. As in the case $\boldsymbol{c}_0(\Gamma)$ and $\ell_\infty^f(\Gamma)$, we can show

[20] A **Hilbert space** is a complete inner product space.

$$w(\ell_p(\Gamma)) = w(\ell_p^f(\Gamma)) = \operatorname{card}\Gamma \ \text{ for any } 1 \leqslant p < \infty.$$

In fact, as is easily observed, $\ell_p^f(\Gamma)$ is dense in $\ell_p(\Gamma)$. When $\operatorname{card}\Gamma = \aleph_0$, the Banach space $\ell_p(\Gamma)$ is linearly isometric to $\ell_p = \ell_p(\mathbb{N})$, where ℓ_p and ℓ_p^f are separable.

Concerning the convergence of sequences in $\ell_p(\Gamma)$, we have the following:

Proposition 1.2.6 *For $1 \leqslant p < \infty$ and $x \in \ell_p(\Gamma)$, a sequence $(x_n)_{n\in\mathbb{N}}$ converges to x in $\ell_p(\Gamma)$ if and only if*

$$\|x\|_p = \lim_{n\to\infty} \|x_n\|_p \ \text{ and } \ x(\gamma) = \lim_{n\to\infty} x_n(\gamma) \ \text{ for every } \gamma \in \Gamma.$$

Proof The "only if" part is trivial, so we concern ourselves with proving the "if" part. For each $\varepsilon > 0$, we have $\gamma_1, \ldots, \gamma_k \in \Gamma$ such that

$$\sum_{\gamma\neq\gamma_i} |x(\gamma)|^p = \|x\|_p^p - \sum_{i=1}^k |x(\gamma_i)|^p < 2^{-p}\varepsilon^p/4.$$

Choose $n_0 \in \mathbb{N}$ so that

$$\begin{aligned} n \geqslant n_0 \ \Rightarrow \ & \left|\|x_n\|_p^p - \|x\|_p^p\right| < 2^{-p}\varepsilon^p/8, \\ & \left||x_n(\gamma_i)|^p - |x(\gamma_i)|^p\right| < 2^{-p}\varepsilon^p/8k \ \text{ and} \\ & |x_n(\gamma_i) - x(\gamma_i)|^p < \varepsilon^p/4k \ \text{ for each } i = 1, \ldots, k. \end{aligned}$$

Then, it follows that

$$\begin{aligned} \sum_{\gamma\neq\gamma_i} |x_n(\gamma)|^p &= \|x_n\|_p^p - \sum_{i=1}^k |x_n(\gamma_i)|^p \\ &= \|x_n\|_p^p - \|x\|_p^p + \|x\|_p^p - \sum_{i=1}^k |x(\gamma_i)|^p \\ &\qquad + \sum_{i=1}^k \left(|x(\gamma_i)|^p - |x_n(\gamma_i)|^p\right) \\ &< 2^{-p}\varepsilon^p/8 + 2^{-p}\varepsilon^p/4 + 2^{-p}\varepsilon^p/8 = 2^{-p}\varepsilon^p/2. \end{aligned}$$

Hence, we can estimate as follows:

$$\begin{aligned} \|x_n - x\|_p^p &\leqslant \sum_{i=1}^k |x_n(\gamma_i) - x(\gamma_i)|^p + \sum_{\gamma\neq\gamma_i} 2^p \max\left\{|x_n(\gamma)|,\ |x(\gamma)|\right\}^p \\ &< \varepsilon^p/4 + \sum_{\gamma\neq\gamma_i} 2^p |x_n(\gamma)|^p + \sum_{\gamma\neq\gamma_i} 2^p |x(\gamma)|^p \end{aligned}$$

$$< \varepsilon^p/4 + \varepsilon^p/2 + \varepsilon^p/4 = \varepsilon^p,$$

that is, $\|x_n - x\|_p < \varepsilon$. □

Remark 1.2.1 It should be noted that Proposition 1.2.6 is valid not only for sequences, but also for nets, which means that the unit spheres $\mathbf{S}_{\ell_p(\Gamma)}$, $1 \leqslant p < \infty$, are subspaces of the product space $\mathbb{R}^\Gamma$, whereas $\mathbb{R}^\Gamma$ and $\mathbb{R}^\Gamma_f$ are not metrizable if Γ is uncountable. When $1 \leqslant p \leqslant q < \infty$, we have the following commutative diagram:

$$\begin{array}{ccccc} \ell_p(\Gamma) & \supset & \mathbf{S}_{\ell_p(\Gamma)} & \xrightarrow{\text{cont.}} & \ell_q(\Gamma) \\ & & \cap & \searrow^{\text{cont.}} & \\ & & \mathbb{R}^\Gamma & & \end{array}$$

Then, $\mathbf{S}_{\ell_p(\Gamma)}$ is also a subspace of $\ell_q(\Gamma)$, although, as we have seen, $\mathbf{S}_{\ell_p(\Gamma)}$ of $\ell_p(\Gamma)$ is not closed in the space $\ell_q(\Gamma)$. Even if $p > q$, we have $\mathbf{S}_{\ell^f_p(\Gamma)} \subset \ell_q(\Gamma)$ as sets but it should be noted that $\mathbf{S}_{\ell^f_p(\Gamma)}$ is not a subspace of $\ell_q(\Gamma)$.[21]

Indeed, since Γ contains a countable infinite set, it suffices to show the case $\Gamma = \mathbb{N}$, that is, $\mathbf{S}_{\ell^f_p}$ is not a subspace of ℓ_q for $p > q$. Take $k \in \mathbb{N}$ so that $k > q/(p-q)$ and define $x_n \in \mathbf{S}_{\ell^f_p}$, $n \in \mathbb{N}$, as follows:

$$x_n(i) = \begin{cases} (1 - n^{-1})^{1/p} & \text{if } i = 1, \\ n^{-(k+1)/p} & \text{if } 2 \leqslant i \leqslant n^k + 1, \\ 0 & \text{if } i \geqslant n^k + 2. \end{cases}$$

Then, $(x_n)_{n\in\mathbb{N}}$ converges to $\mathbf{e}_1$ in $\mathbf{S}_{\ell^f_p}$ because $\mathbf{S}_{\ell^f_p}$ is a subspace of $\mathbb{R}^\mathbb{N}_f$. However,

$$\sum_{i=1}^{\infty} |x_n(i) - \mathbf{e}_1(i)|^q \geqslant \sum_{i=2}^{n^k+1} |x_n(i) - \mathbf{e}_1(i)|^q = n^k n^{-(k+1)q/p} \to \infty \quad (n \to \infty)$$

because $k - (k+1)q/p > 0$. Therefore, $(x_n)_{n\in\mathbb{N}}$ does not converges to $\mathbf{e}_1$ in ℓ^f_q.

Remark 1.2.2 The "if" part of Proposition 1.2.6 does not hold for the space $\boldsymbol{c}_0(\Gamma)$ (although the "only if" part obviously does hold), where Γ is infinite. For instance, take distinct $\gamma_n \in \Gamma$, $n \in \omega$, and let $(x_n)_{n\in\mathbb{N}}$ be the sequence in $\boldsymbol{c}_0(\Gamma)$ defined by $x_n = \mathbf{e}_{\gamma_n} + \mathbf{e}_{\gamma_0}$. Then, $\|x_n\|_\infty = 1$ for each $n \in \mathbb{N}$,

$$\lim_{n\to\infty} x_n(\gamma_0) = 1 = \mathbf{e}_{\gamma_0}(\gamma_0) \quad \text{and} \quad \lim_{n\to\infty} x_n(\gamma) = 0 = \mathbf{e}_{\gamma_0}(\gamma) \quad \text{for } \gamma \neq \gamma_0,$$

but $\|x_n - \mathbf{e}_{\gamma_0}\|_\infty = 1$ for every $n \in \mathbb{N}$. In addition, the unit sphere $\mathbf{S}_{\boldsymbol{c}_0(\Gamma)}$ of $\boldsymbol{c}_0(\Gamma)$ is not a subspace of $\mathbb{R}^\Gamma$, because $\mathbf{e}_{\gamma_n} \in \mathbf{S}_{\boldsymbol{c}_0(\Gamma)}$ but $(\mathbf{e}_{\gamma_n})_{n\in\mathbb{N}}$ converges to $\mathbf{0}$ in $\mathbb{R}^\Gamma$.

[21] For $p > q$, $\mathbf{S}_{\ell_p(\Gamma)} \not\subset \ell_q(\Gamma)$ as sets.

Concerning the topological classification of $\ell_p(\Gamma)$, we have MAZUR'S THEOREM, as follows:[22]

Theorem 1.2.7 (MAZUR) *For each $1 < p < \infty$, $\ell_p(\Gamma)$ is homeomorphic to $\ell_1(\Gamma)$. By the same homeomorphism, $\ell_p^f(\Gamma)$ is also homeomorphic to $\ell_1^f(\Gamma)$. In other words,*

$$(\ell_p(\Gamma), \ell_p^f(\Gamma)) \approx (\ell_1(\Gamma), \ell_1^f(\Gamma)).$$

Proof We define $\varphi : \ell_1(\Gamma) \to \ell_p(\Gamma)$ and $\psi : \ell_p(\Gamma) \to \ell_1(\Gamma)$ as follows:

$$\varphi(x)(\gamma) = \operatorname{sign} x(\gamma) \cdot |x(\gamma)|^{1/p} \quad \text{for } x \in \ell_1(\Gamma),$$
$$\psi(x)(\gamma) = \operatorname{sign} x(\gamma) \cdot |x(\gamma)|^{p} \quad \text{for } x \in \ell_p(\Gamma),$$

where $\operatorname{sign} 0 = 0$ and $\operatorname{sign} a = a/|a|$ for $a \neq 0$. We can apply Proposition 1.2.6 to verify the continuity of φ and ψ. In fact, the following functions are continuous:

$$\ell_1(\Gamma) \ni x \mapsto \|\varphi(x)\|_p = \big(\|x\|_1\big)^{1/p} \in \mathbb{R}, \quad \ell_1(\Gamma) \ni x \mapsto \varphi(x)(\gamma) \in \mathbb{R}, \ \gamma \in \Gamma;$$
$$\ell_p(\Gamma) \ni x \mapsto \|\psi(x)\|_1 = \big(\|x\|_p\big)^{p} \in \mathbb{R}, \quad \ell_p(\Gamma) \ni x \mapsto \psi(x)(\gamma) \in \mathbb{R}, \ \gamma \in \Gamma.$$

Observe that $\psi\varphi = \mathrm{id}$ and $\varphi\psi = \mathrm{id}$. Thus, φ is a homeomorphism with $\varphi^{-1} = \psi$, where $\varphi(\ell_1^f(\Gamma)) \subset \ell_p^f(\Gamma)$ and $\psi(\ell_p^f(\Gamma)) \subset \ell_1^f(\Gamma)$. □

For each space X, we denote $\mathrm{C}(X) = \mathrm{C}(X, \mathbb{R})$. The Banach space $\mathrm{C}^B(X)$ is defined as follows:

- $\mathrm{C}^B(X) \underset{\text{def}}{=} \big\{ f \in \mathrm{C}(X) \,\big|\, \sup_{x \in X} |f(x)| < \infty \big\}$ with the sup-norm

$$\|f\| \underset{\text{def}}{=} \sup_{x \in X} |f(x)|.$$

We can regard $\mathrm{C}^B(X)$ is a subspace of $\ell_\infty(X)$. This sup-norm of $\mathrm{C}^B(X)$ induces the uniform convergence topology (defined shortly before Proposition 1.1.1), that is, the Banach space $\mathrm{C}^B(X)$ is a subspace of the space $\mathrm{C}_u(X)$ $(= \mathrm{C}_u(X, \mathbb{R})$ with the uniform convergence topology). Then, as is easily observed, $\mathrm{C}^B(X)$ is closed and open in $\mathrm{C}_u(X)$. Note that $\mathrm{C}^B(X)$ is a component of the space $\mathrm{C}_u(X)$ because $\mathrm{C}(X)$ is path-connected as a normed linear space. If X is discrete and infinite, then we have $\mathrm{C}^B(X) = \ell_\infty(X)$. In particular, $\mathrm{C}^B(\mathbb{N}) = \ell_\infty$. Moreover, if $X = \{0, 1/n \mid n \in \mathbb{N}\}$, then $\mathrm{C}^B(X) = \mathrm{C}_u(X)$ and $\mathrm{C}^B(X)$ is linearly isometric to the space of convergent sequences $\boldsymbol{c}$ by the correspondence

$$\mathrm{C}^B(X) \ni f \mapsto (f(1/n))_{n \in \mathbb{N}} \in \boldsymbol{c}.$$

Besides, we have the space $\mathrm{C}_p(X) = \mathrm{C}_p(X, \mathbb{R})$ with the pointwise convergence topology, which is a subspace of the product space $\mathbb{R}^X$. Then, $\mathrm{C}_p(X)$ is a topological

[22] This theorem will not be used in this book.

linear space. The space $C_p(\mathbb{N})$ is simply the space of sequences, $\boldsymbol{s} = \mathbb{R}^{\mathbb{N}}$, that is, $C_p(\mathbb{N}) = \boldsymbol{s}$.

Notes for Chap. 1

Theorem 1.2.7 is due to Mazur [4]. Zhongqiang Yang pointed out that Proposition 1.2.6 can be applied to show the continuity of φ and ψ in the proof of Theorem 1.2.7. Related to Mazur's result, R.D. Anderson [1] proved that $\boldsymbol{s} = \mathbb{R}^{\mathbb{N}}$ is homeomorphic to the Hilbert space ℓ_2, which had been the longstanding problem posed by Fréchet [3] in 1928. For an elementary proof and a brief history of this problem, refer to [2].

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Chapter 2
Metrizability, Paracompactness, and Related Properties



This chapter provides advanced subjects concerning topological spaces for the reader who has finished the first course of General Topology. These will be background knowledge for studying various branches of Topology. We derive some properties of the products of compact spaces and define Stone–Čech compactification. As generalization of locally compact spaces and metrizable spaces, compactly generated spaces (i.e., k-spaces) are introduced, which commonly appear in Algebraic Topology. We discuss perfect (or proper) maps. Various normalities strengthening normality are also introduced. But, we are mainly concerned with metrization and paracompact spaces. Several metrization theorems are proved and complete metrizability is characterized. It is shown that a completely metrizable space has the Baire property, which is effective in showing various existence theorems. Paracompactness is a concept generalizing compactness and metrizability. A paracompact space has several different characteristics, that indicates, in many situations, the advantages of this concept. In particular, there exists a useful theorem showing that, if a paracompact space has a certain property *locally*, then it has the same property *globally*. This will be applied in later chapters several times. Furthermore, a paracompact space has a partition of unity, which is also a very useful property and will be applied later. The direct limit of a closed tower of paracompact spaces is also paracompact. In addition, we introduce the limitation topology on function spaces, which has the Baire property in some situations. The compact-open topology is standard for function spaces. We prove its basic properties together with the Ascoli–Arzela Theorem. Finally, the Vietoris topology and the Hausdorff metric on hyperspaces are expounded.

2.1 Some Concepts Related to Compactness

In this section, we present a number of theorems concerning the products of compact spaces and compactifications. Compactly generated spaces (i.e., k-spaces) are intro-

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duced. In addition, perfect maps and proper maps are discussed and their properties are demonstrated. First, we present a proof of the TYCHONOFF THEOREM:

Theorem 2.1.1 (TYCHONOFF) *The product space* $\prod_{\lambda\in\Lambda} X_\lambda$ *of compact spaces* X_λ, $\lambda \in \Lambda$, *is compact.*

Proof Let $X = \prod_{\lambda\in\Lambda} X_\lambda$. We may assume that $\Lambda = (\Lambda, \leqslant)$ is a well-ordered set. For each $\mu \in \Lambda$, let $p_\mu : X \to \prod_{\lambda\leqslant\mu} X_\lambda$ and $q_\mu : X \to \prod_{\lambda<\mu} X_\lambda$ be the projections.

Let $\mathcal{A}$ be a collection of subsets of X with the finite intersection property (f.i.p.). Using transfinite induction, we can find $x_\lambda \in X_\lambda$ such that $\mathcal{A}|\, p_\lambda^{-1}(U)$ has the f.i.p. for every neighborhood U of $(x_\nu)_{\nu\leqslant\lambda}$ in $\prod_{\nu\leqslant\lambda} X_\nu$. Indeed, suppose that $x_\lambda \in X_\lambda$, $\lambda < \mu$, have been found, but there exists no $x_\mu \in X_\mu$ with the above property, i.e., any $y \in X_\mu$ has an open neighborhood V_y with an open neighborhood U_y of $(x_\lambda)_{\lambda<\mu}$ in $\prod_{\lambda<\mu} X_\lambda$ such that $\mathcal{A}|q_\mu^{-1}(U_y) \cap \mathrm{pr}_\mu^{-1}(V_y)$ does not have the f.i.p. Because X_μ is compact, we have $y_1, \ldots, y_n \in X_\mu$ such that $X_\mu = \bigcup_{i=1}^n V_{y_i}$. Since $\bigcap_{i=1}^n U_{y_i}$ is a neighborhood of $(x_\lambda)_{\lambda<\mu}$ in $\prod_{\lambda<\mu} X_\lambda$, we have $\nu_1, \ldots, \nu_m < \mu$ and neighborhoods W_i of x_{ν_i} in X_{ν_i} such that $\bigcap_{i=1}^m \mathrm{pr}_{\nu_i}^{-1}(W_i) \subset q_\mu^{-1}(\bigcap_{i=1}^n U_{y_i})$. Let $\nu = \max\{\nu_1, \ldots, \nu_m\} < \mu$. Then, we can write $\bigcap_{i=1}^m \mathrm{pr}_{\nu_i}^{-1}(W_i) = p_\nu^{-1}(W)$ for some neighborhood W of $(x_\lambda)_{\lambda\leqslant\nu}$ in $\prod_{\lambda\leqslant\nu} X_\lambda$. Because $p_\nu^{-1}(W) \subset \bigcap_{i=1}^n q_\mu^{-1}(U_{y_i})$, no $\mathcal{A}|p_\nu^{-1}(W) \cap \mathrm{pr}_\mu^{-1}(V_{y_i})$ has the f.i.p. Since $X = \bigcup_{i=1}^n \mathrm{pr}_\mu^{-1}(V_{y_i})$, it follows that $\mathcal{A}|p_\nu^{-1}(W)$ does not have the f.i.p., which contradicts the inductive assumption.

Now, we have obtained the point $x = (x_\lambda)_{\lambda\in\Lambda} \in X$. For each neighborhood U of x in X, we have $\lambda_1, \ldots, \lambda_n \in \Lambda$ and neighborhoods U_i of x_{λ_i} in X_{λ_i} such that $\bigcap_{i=1}^n \mathrm{pr}_{\lambda_i}^{-1}(U_i) \subset U$. Let $\lambda_0 = \max\{\lambda_1, \ldots, \lambda_n\} \in \Lambda$. Then, we can write $\bigcap_{i=1}^n \mathrm{pr}_{\lambda_i}^{-1}(U_i) = p_{\lambda_0}^{-1}(U_0)$ for some neighborhood U_0 of $(x_\nu)_{\nu\leqslant\lambda_0}$ in $\prod_{\nu\leqslant\lambda_0} X_\nu$. Since $p_{\lambda_0}^{-1}(U_0) \subset U$, $\mathcal{A}|U$ has the f.i.p. Consequently, every neighborhood U of x in X meets every member of $\mathcal{A}$. This means $x \in \bigcap_{A\in\mathcal{A}} \mathrm{cl}\, A$, hence $\bigcap_{A\in\mathcal{A}} \mathrm{cl}\, A \neq \emptyset$. □

Note There are various proofs of the Tychonoff Theorem. In one familiar proof, Zorn's Lemma is applied instead of the transfinite induction. Let $\mathcal{A}$ be a collection of subsets of X with the f.i.p. and Φ be all of collections $\mathcal{A}'$ of subsets of X such that $\mathcal{A}'$ has the f.i.p. and $\mathcal{A} \subset \mathcal{A}'$. Applying Zorn's Lemma to the ordered set $\Phi = (\Phi, \subset)$, we can obtain a maximal element $\mathcal{A}^* \in \Phi$. Because of the maximality, $\mathcal{A}^*$ has the following properties:

(1) The intersection of any finite members of $\mathcal{A}^*$ belongs to $\mathcal{A}^*$;

(2) If $B \subset X$ meets every member of $\mathcal{A}^*$, then $B \in \mathcal{A}^*$.

For each $\lambda \in \Lambda$, $\mathrm{pr}_\lambda(\mathcal{A}^*)$ has the f.i.p. Since X_λ is compact, we have $x_\lambda \in \bigcap_{A\in\mathcal{A}^*} \mathrm{cl}\, \mathrm{pr}_\lambda(A)$. It follows from (2) that $\mathrm{pr}_\lambda^{-1}(V) \in \mathcal{A}^*$ for every neighborhood V of x_λ in X_λ. Now, it is easy to see that

$$x = (x_\lambda)_{\lambda\in\Lambda} \in \bigcap_{A\in\mathcal{A}^*} \mathrm{cl}\, A \subset \bigcap_{A\in\mathcal{A}} \mathrm{cl}\, A.$$

Next, we prove WALLACE'S THEOREM:

Theorem 2.1.2 (WALLACE) *Let* $A = \prod_{\lambda\in\Lambda} A_\lambda \subset X = \prod_{\lambda\in\Lambda} X_\lambda$, *where each* A_λ *is compact. Then, for each open set* W *in* X *with* $A \subset W$, *there exists a finite subset* $\Lambda_0 \subset \Lambda$ *and open sets* V_λ *in* X_λ, $\lambda \in \Lambda_0$, *such that* $A \subset \bigcap_{\lambda\in\Lambda_0} \mathrm{pr}_\lambda^{-1}(V_\lambda) \subset W$.

Proof When Λ is finite, we may take $\Lambda_0 = \Lambda$. Then, $\bigcap_{\lambda\in\Lambda_0} \mathrm{pr}_\lambda^{-1}(V_\lambda)$ coincides with $\prod_{\lambda\in\Lambda} V_\lambda$. This case can be proved by induction on card Λ, which is reduced to the case card $\Lambda = 2$. Proving the case card $\Lambda = 2$ is an excellent exercise.[1]

We will show that the general case is derived from the finite case. For each $x \in A$, we have a finite subset $\Lambda(x) \subset \Lambda$ and an open set $U(x)$ in $\prod_{\lambda\in\Lambda(x)} X_\lambda$ such that $x \in \mathrm{pr}_{\Lambda(x)}^{-1}(U(x)) \subset W$. Because of the compactness of A, there exist finitely many $x_1, \ldots, x_n \in A$ such that $A \subset \bigcup_{i=1}^n \mathrm{pr}_{\Lambda(x_i)}^{-1}(U(x_i))$. Thus, we have a finite subset $\Lambda_0 = \bigcup_{i=1}^n \Lambda(x_i) \subset \Lambda$. For each $i = 1, \ldots, n$, let $p_i : \prod_{\lambda\in\Lambda_0} X_\lambda \to \prod_{\lambda\in\Lambda(x_i)} X_\lambda$ be the projection. Then, $W_0 = \bigcup_{i=1}^n p_i^{-1}(U(x_i))$ is an open set in $\prod_{\lambda\in\Lambda_0} X_\lambda$.

Note that $\bigcup_{i=1}^n \mathrm{pr}_{\Lambda(x_i)}^{-1}(U(x_i)) = \mathrm{pr}_{\Lambda_0}^{-1}(W_0)$. From the finite case, we obtain open sets V_λ, $\lambda \in \Lambda_0$, such that $\prod_{\lambda\in\Lambda_0} A_\lambda \subset \prod_{\lambda\in\Lambda_0} V_\lambda \subset W_0$. Hence,

$$A \subset \bigcap_{\lambda\in\Lambda_0} \mathrm{pr}_\lambda^{-1}(V_\lambda) \subset \mathrm{pr}_{\Lambda_0}^{-1}(W_0) \subset W. \qquad \square$$

For any space X, we define the evaluation map $e_X : X \to \mathbf{I}^{\mathrm{C}(X,\mathbf{I})}$ by $e_X(x) = (f(x))_{f\in\mathrm{C}(X,\mathbf{I})}$ for each $x \in X$. The continuity of e_X follows from the fact that $\mathrm{pr}_f \circ e_X = f$ is continuous for each $f \in \mathrm{C}(X, \mathbf{I})$, where $\mathrm{pr}_f : \mathbf{I}^{\mathrm{C}(X,\mathbf{I})} \to \mathbf{I}$ is the projection (i.e., $\mathrm{pr}_f(\xi) = \xi(f)$).

Theorem 2.1.3 *For every Tychonoff space X, the map $e_X : X \to \mathbf{I}^{\mathrm{C}(X,\mathbf{I})}$ is an embedding.*

Proof Let U be an open set in X and $x \in U$. Since X is a Tychonoff space, we have some $f \in \mathrm{C}(X, \mathbf{I})$ such that $f(x) = 0$ and $f(X \setminus U) = \{1\}$. Then, $V = \mathrm{pr}_f^{-1}([0, 1))$ is an open set in $\mathbf{I}^{\mathrm{C}(X,\mathbf{I})}$. Since $\mathrm{pr}_f(e_X(x)) = f(x) = 0$, it follows that $e_X(x) \in V$. Since $\mathrm{pr}_f \circ e_X(X \setminus U) = f(X \setminus U) = \{1\}$, we have $e_X(X \setminus U) \cap V = \emptyset$. Therefore, $e_X(x) \in V \cap e_X(X) \subset e_X(U)$. This implies that $e_X : X \to e_X(X)$ is an open map.

For $x \neq y \in X$, applying the above argument to $U = X \setminus \{y\}$, we can see that $e_X(x)(f) = 0 \neq 1 = e_X(y)(f)$. Thus, e_X is an embedding. $\square$

Let X be a Tychonoff space. From Tychonoff's Theorem, it follows that the product space $\mathbf{I}^{\mathrm{C}(X,\mathbf{I})}$ is compact. Then, identifying X with $e_X(X)$, we define a compactification βX of X as follows:

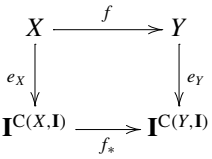
$$\beta X = \mathrm{cl}_{\mathbf{I}^{\mathrm{C}(X,\mathbf{I})}} e_X(X),$$

which is called the **Stone–Čech compactification**.

Now, let $f : X \to Y$ be a map between Tychonoff spaces. The map $f_* : \mathbf{I}^{\mathrm{C}(X,\mathbf{I})} \to \mathbf{I}^{\mathrm{C}(Y,\mathbf{I})}$ is defined as $f_*(\xi) = (\xi(kf))_{k\in\mathrm{C}(Y,\mathbf{I})}$ for each $\xi \in \mathbf{I}^{\mathrm{C}(X,\mathbf{I})}$. The continuity of f_* follows from the continuity of $\mathrm{pr}_k \circ f_* = \mathrm{pr}_{kf}$, $k \in \mathrm{C}(Y, \mathbf{I})$. Then, we have $f_* \circ e_X = e_Y \circ f$.

---

[1] Use the same strategy used in the proof of normality of a compact Hausdorff space.



Indeed, for each $x \in X$ and $k \in \mathrm{C}(Y, \mathbf{I})$,

$$f_*(e_X(x))(k) = e_X(x)(kf) = k(f(x)) = e_Y(f(x))(k).$$

Since f_* is continuous, it follows that $f_*(\beta X) \subset \beta Y$. Thus, f extends to the map $\beta f = f_*|\beta X : \beta X \to \beta Y$.

Further, let $g : Y \to Z$ be another map, where Z is Tychonoff. Then, for each $\xi \in \mathbf{I}^{\mathrm{C}(X,\mathbf{I})}$ and $k \in \mathrm{C}(Z, \mathbf{I})$,

$$g_*(f_*(\xi))(k) = f_*(\xi)(kg) = \xi(kgf) = (gf)_*(\xi)(k),$$

that is, $g_* f_* = (gf)_*$. Therefore, $\beta(gf) = \beta g \beta f$.

The Stone–Čech compactification βX can be characterized as follows:

Theorem 2.1.4 (STONE; CECH) *Let X be a Tychonoff space. For any compactification γX of X, there exists the (unique) map $f : \beta X \to \gamma X$ such that $f|X = \mathrm{id}_X$. If a compactification $\beta' X$ has the same property as above, then there exists a homeomorphism $h : \beta X \to \beta' X$ such that $h|X = \mathrm{id}_X$.*

Proof Note that $\beta(\gamma X) = \gamma X$ because γX is compact. Let $i : X \hookrightarrow \gamma X$ be the inclusion and let $f = \beta i : \beta X \to \beta(\gamma X) = \gamma X$. Then, $f|X = \mathrm{id}_X$ and f is unique because X is dense in βX.

If a compactification $\beta' X$ of X has the same property, then we have two maps $h : \beta X \to \beta' X$ and $h' : \beta' X \to \beta X$ such that $h|X = h'|X = \mathrm{id}_X$. It follows that $h'h = \mathrm{id}_{\beta X}$ and $hh' = \mathrm{id}_{\beta' X}$, hence h is a homeomorphism. □

Let $\mathrm{Comp}(X)$ be the collection of all non-empty compact sets in X. A space X is said to be **compactly generated** or simply called a ***k*-space** if X is equipped with the weak topology with respect to $\mathrm{Comp}(X)$, that is, U is open in X if $U \cap K$ is open in every compact set $K \subset X$ (equiv., A is closed in X if $A \cap K$ is closed in every compact set $K \subset X$). The class of k-spaces contains both metrizable spaces and locally compact spaces,[2] that is:

Theorem 2.1.5 *A space X is compactly generated (i.e., a k-space) if X is locally compact or first countable. In particular, every metrizable space is a k-space.*

[2] This class contains polyhedra (simplicial complexes) and also CW complexes.

Proof When X is locally compact, assume that $U \cap K$ is open in every compact set $K \subset X$. Each $x \in U$ has an open neighborhood V in X such that $\operatorname{cl} V$ is compact. Since $U \cap \operatorname{cl} V$ is open in $\operatorname{cl} V$, $U \cap V$ is open in V, so in X. Hence, U is a neighborhood of x in X. Therefore, U is open in X. Thus, X is a k-space.

When X is first countable, assume that $A \cap K$ is closed in every compact set $K \subset X$. Each $x \in \operatorname{cl} A$ is the limit point of some sequence $(x_n)_{n\in\mathbb{N}}$ in A. Then, $K = \{x_n \mid n \in \mathbb{N}\} \cup \{x\}$ is compact. Since $A \cap K$ is closed in K, it follows that $x \in A \cap K \subset A$. Hence, $\operatorname{cl} A = A$, so A is closed in X. Thus, X is a k-space. □

The following is easy to prove:

Proposition 2.1.6 *Let X be a k-space. Then, $f : X \to Y$ is continuous if and only if $f|K : K \to Y$ is continuous for any compact set K in X.*

Sketch of Proof To see the "if" part, observe that $f^{-1}(U) \cap K = (f|K)^{-1}(U)$ for each open set U in Y and each compact set K in X.

A k-space can be characterized as a quotient space of a locally compact space, that is:

Theorem 2.1.7 *A space X is compactly generated (i.e., a k-space) if and only if X is a quotient space of a locally compact space.*

Proof Let $\widetilde{X} = \bigoplus_{K\in\operatorname{Comp}(X)} K$ be the topological sum of all non-empty compact subspaces of X. Then, X is a k-space if and only if the natural map $q : \widetilde{X} \to X$ is a quotient map. Since $\widetilde{X}$ is locally compact, we have the "only if" part.

To see the "if" part, let $q : Y \to X$ be a quotient map of a locally compact space Y. For $A \subset X$, suppose that $A \cap K$ is closed in every $K \in \operatorname{Comp}(X)$. Then, it suffices to show that $q^{-1}(A)$ is closed in Y. Each $y \in \operatorname{cl} q^{-1}(A)$ has a compact neighborhood N in Y. Then, $y \in \operatorname{cl}(q^{-1}(A) \cap N)$ because $V \cap N \cap q^{-1}(A) \neq \emptyset$ for any neighborhood V of y. Since $A \cap q(N)$ is closed in $q(N)$, it follows that $q^{-1}(A) \cap q^{-1}(q(N)) = q^{-1}(A \cap q(N))$ is closed in $q^{-1}(q(N))$. Hence, we have

$$y \in \operatorname{cl}(q^{-1}(A) \cap N) \subset q^{-1}(A) \cap q^{-1}(q(N)) \subset q^{-1}(A).$$

Thus, $q^{-1}(A)$ is closed in Y. □

Since the composition of quotient maps is quotient, the following corollary can be obtained:

Corollary 2.1.8 *Every quotient space of a k-space is also a k-space.* □

Concerning quotient maps, we can prove the following:

Proposition 2.1.9 *Let $q : X \to Y$ be a quotient map and Z a locally compact space. Then, $q \times \mathrm{id}_Z : X \times Z \to Y \times Z$ is a quotient map.*

Proof For $U \subset Y \times Z$, we show that if $(q \times \mathrm{id}_Z)^{-1}(U)$ is open in $X \times Z$, then U is open in $Y \times Z$. For each $(y, z) \in U$, take $x \in X$ so that $y = q(x)$. Since $(x, z) \in (q \times \mathrm{id}_Z)^{-1}(U)$, z has a compact neighborhood N in Z such that $\{x\} \times N \subset (q \times \mathrm{id}_Z)^{-1}(U)$. Let

$$V = \left\{x' \in X \;\middle|\; \{x'\} \times N \subset (q \times \mathrm{id}_Z)^{-1}(U)\right\}.$$

Since $x \in V$, we have $y = q(x) \in q(V)$. Observe that $q^{-1}(q(V)) = V$. Because N is compact, it is easy to see that V is open in X. Hence, $q(V)$ is an open neighborhood of y in Y and $(y, z) \in q(V) \times N \subset U$. Therefore, U is open in $Y \times Z$. □

Combining Propositions 2.1.7 and 2.1.9, we can obtain the following:

Corollary 2.1.10 *The product $X \times Y$ of a k-space X and a locally compact space Y is also a k-space.* □

Remark 2.1.1 Proposition 2.1.9 does not hold even if $Z = \mathbb{Q}$, the space of rationals. For example, let $X = (\mathbb{N} \times \mathbf{I})/(\mathbb{N} \times \{0\})$ be the cone over $\mathbb{N}$ with the vertex v and the quotient map $q : \mathbb{N} \times \mathbf{I} \to X$, where $q(\mathbb{N} \times \{0\}) = \{v\}$. Then, the map $q \times \mathrm{id}_{\mathbb{Q}} : \mathbb{N} \times \mathbf{I} \times \mathbb{Q} \to X \times \mathbb{Q}$ is not quotient. Indeed, let

$$W = \left\{(n, t, s) \in \mathbb{N} \times \mathbf{I} \times \mathbb{Q} \;\middle|\; n \in \mathbb{N},\ s > \sqrt{2}/n + t \ \text{ or } \ s < \sqrt{2}/n - t\right\}.$$

Then, W is open in $\mathbb{N} \times \mathbf{I} \times \mathbb{Q}$ and

$$\mathbb{N} \times \{0\} \times \mathbb{Q} \subset W = (q \times \mathrm{id}_{\mathbb{Q}})^{-1}((q \times \mathrm{id}_{\mathbb{Q}})(W)).$$

To see that $q \times \mathrm{id}_{\mathbb{Q}}$ is not quotient, it suffices to show that $(q \times \mathrm{id}_{\mathbb{Q}})(W)$ is not open in $X \times \mathbb{Q}$. Assume that it is open in $X \times \mathbb{Q}$. Since $(v, 0) \in (q \times \mathrm{id}_{\mathbb{Q}})(W)$, we have a neighborhood V of v in X and $m \in \mathbb{N}$ such that

$$V \times \left(\left(-\sqrt{2}/m, \sqrt{2}/m\right) \cap \mathbb{Q}\right) \subset (q \times \mathrm{id}_{\mathbb{Q}})(W),$$
$$\text{i.e., } \ q^{-1}(V) \times \left(\left(-\sqrt{2}/m, \sqrt{2}/m\right) \cap \mathbb{Q}\right) \subset W.$$

Then, we can choose $0 < \delta < \sqrt{2}/m$ and $r \in \mathbb{Q}$ so that $\{m\} \times [0, \delta] \subset q^{-1}(V)$ and $\sqrt{2}/m - \delta < r < \sqrt{2}/m$. Then,

$$(m, \delta, r) \in q^{-1}(V) \times \left(\left(-\sqrt{2}/m, \sqrt{2}/m\right) \cap \mathbb{Q}\right) \ \text{ but } \ (m, \delta, r) \notin W,$$

which is a contradiction. See Fig. 2.1.

Moreover, this example shows that the product of a k-space and a separable metrizable space is not a k-space, that is, Corollary 2.1.10 does not hold even if $Y = \mathbb{Q}$. In fact, X is a k-space but $X \times \mathbb{Q}$ is not a k-space. To see that $X \times \mathbb{Q}$ is not a k-space, it suffices to show that if $K \subset X \times \mathbb{Q}$ is compact, then $(q \times \mathrm{id}_{\mathbb{Q}})(W) \cap K$ is open in K. We can find $n \in \mathbb{N}$ such that $\mathrm{pr}_X(K) \subset \bigcup_{i \leqslant n} q(\{i\} \times \mathbf{I})$.

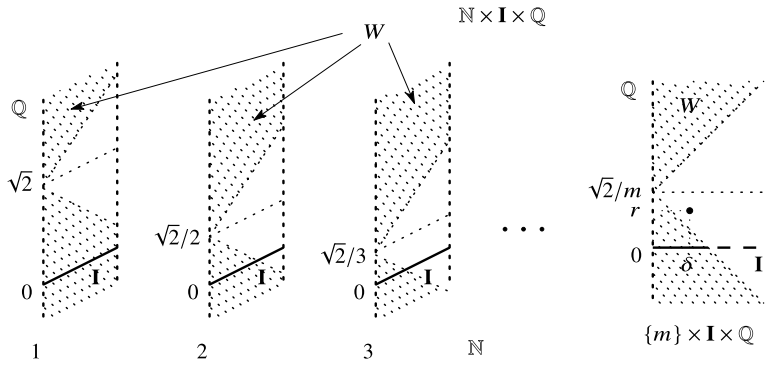


Fig. 2.1 $W \subset \mathbb{N} \times \mathbf{I} \times \mathbb{Q}$

In fact, every compact set in X meets only finitely many $\{n\} \times (0, 1]$ because for any $x \in (0, 1]^{\mathbb{N}}$, $\{(n, x(n)) \mid n \in \mathbb{N}\}$ is discrete in X.

Let $K_1 = \bigcup_{i \leqslant n} \{i\} \times \mathbf{I}$ and $K_2 = \mathrm{pr}_{\mathbb{Q}}(K)$. Then, K_1 and K_2 are compact and $K \subset q(K_1) \times K_2 = (q \times \mathrm{id}_{\mathbb{Q}})(K_1 \times K_2)$. Observe that

$$\begin{aligned} W \cap (K_1 \times K_2) &= (q \times \mathrm{id}_{\mathbb{Q}})^{-1}((q \times \mathrm{id}_{\mathbb{Q}})(W)) \cap (K_1 \times K_2) \\ &= (q \times \mathrm{id}_{\mathbb{Q}} | K_1 \times K_2)^{-1}(q \times \mathrm{id}_{\mathbb{Q}})(W \cap (K_1 \times K_2)). \end{aligned}$$

Since $q \times \mathrm{id}_{\mathbb{Q}} | K_1 \times K_2$ is quotient, it follows that $(q \times \mathrm{id}_{\mathbb{Q}})(W \cap (K_1 \times K_2))$ is open in $(q \times \mathrm{id}_{\mathbb{Q}})(K_1 \times K_2) = q(K_1) \times K_2$. which implies that $(q \times \mathrm{id}_{\mathbb{Q}})(W) \cap K$ is open in K.

Remark 2.1.2 Concerning Proposition 2.1.9 and Corollary 2.1.10, the following are known (cf. Notes for this chapter):

(1) *Z is locally compact if and only if $q \times \mathrm{id}_Z$ is quotient for every quotient map $q : X \to Y$.*
(2) *Y is locally compact if and only if $X \times Y$ is a k-space for every k-space X.*

A **perfect map** $f : X \to Y$ is a closed map such that $f^{-1}(y)$ is compact for each $y \in Y$. A map $f : X \to Y$ is said to be **proper** if $f^{-1}(K)$ is compact for every compact set $K \subset Y$. Obviously, *the composition of any two proper maps is proper*. To verify that *the composition of any two perfect maps is perfect*, it suffices to apply the following:

Proposition 2.1.11 *Every perfect map $f : X \to Y$ is proper.*

Proof Let $K \subset Y$ be compact and $\mathcal{U}$ an open cover of $f^{-1}(K)$ in X. For each $y \in K$, choose a finite subcollection $\mathcal{U}_y$ of $\mathcal{U}$ so that $f^{-1}(y) \subset \bigcup \mathcal{U}_y$. Since f is closed, each $V_y = Y \setminus f(X \setminus \bigcup \mathcal{U}_y)$ is an open neighborhood of y in Y, where

$f^{-1}(V_y) \subset \bigcup \mathcal{U}_y$. We can choose $y_1, \ldots, y_n \in K$ so that $K \subset \bigcup_{i=1}^n V_{y_i}$. Thus, we have a finite subcollection $\mathcal{U}_0 = \bigcup_{i=1}^n \mathcal{U}_{y_i} \subset \mathcal{U}$ such that $f^{-1}(K) \subset \bigcup \mathcal{U}_0$. Hence, $f^{-1}(K)$ is compact. □

In the above, if Y is a k-space, every proper map $f : X \to Y$ is perfect, that is:

Proposition 2.1.12 *Let $f : X \to Y$ be a proper map. If Y is a k-space, then $f : X \to Y$ is perfect and X is a k-space.*

Proof To show the perfectness of f, it suffices to prove that f is closed. Let $A \subset X$ be closed. For each compact set K in Y, $f^{-1}(K)$ is compact. Since $A \cap f^{-1}(K)$ is closed in $f^{-1}(K)$, $A \cap f^{-1}(K)$ is compact. Then, $f(A) \cap K = f(A \cap f^{-1}(K))$ is compact, hence it is closed in Y. Since Y is a k-space, $f(A)$ is closed in Y.

Assume that X is not a k-space, that is, X has a non-closed set A such that $A \cap K$ is closed in every compact set K in X. Then, we have $x \in \operatorname{cl} A \setminus A$. Since $f^{-1}(f(x))$ is compact, $A \cap f^{-1}(f(x))$ is a closed set in $f^{-1}(f(x))$. We can take a closed neighborhood N of x in X so that $(A \cap f^{-1}(f(x))) \cap N = \emptyset$. Then, observe that $x \in \operatorname{cl}(A \cap N)$. Replacing A with $A \cap N$, we can assume that $A \cap f^{-1}(f(x)) = \emptyset$. For each compact set K in Y, $A \cap f^{-1}(K)$ is closed in $f^{-1}(K)$ because $f^{-1}(K)$ is compact. Since $f|f^{-1}(K) : f^{-1}(K) \to K$ is a closed map, it follows that $f(A) \cap K = f(A \cap f^{-1}(K))$ is closed in K. Hence, $f(A)$ is closed in Y because Y is a k-space. Note that $f(x) \in f(\operatorname{cl} A) \subset \operatorname{cl} f(A) = f(A)$, which means $f^{-1}(f(x)) \cap A \neq \emptyset$. This is a contradiction. Consequently, X is a k-space. □

For metrizable spaces, we have the following characterization:

Proposition 2.1.13 *For a map $f : X \to Y$ between metrizable spaces, the following are equivalent:*

(a) *$f : X \to Y$ is perfect;*
(b) *$f : X \to Y$ is proper;*
(c) *Any sequence $(x_n)_{n\in\mathbb{N}}$ in X has a convergent subsequence if $(f(x_n))_{n\in\mathbb{N}}$ is convergent in Y.*

Proof The equivalence (a) ⇔ (b) was shown in Propositions 2.1.11 and 2.1.12.[3]

(b) ⇒ (c): Let $y = \lim_{n\to\infty} f(x_n) \in Y$ and $K = \{f(x_n) \mid n \in \mathbb{N}\} \cup \{y\}$. Since K is compact, (b) implies the compactness of $f^{-1}(K)$, whose sequence $(x_n)_{n\in\mathbb{N}}$ has a convergent subsequence.

(c) ⇒ (a): For each $y \in Y$, every sequence $(x_n)_{n\in\mathbb{N}}$ in $f^{-1}(y)$ has a convergent subsequence due to (c), which means that $f^{-1}(y)$ is compact because $f^{-1}(y)$ is metrizable.

To see that f is a closed map, let $A \subset X$ be a closed set and $y \in \operatorname{cl}_Y f(A)$. Then, we have a sequence $(x_n)_{n\in\mathbb{N}}$ in A such that $y = \lim_{n\to\infty} f(x_n)$. Due to (c), $(x_n)_{n\in\mathbb{N}}$ has a convergent subsequence $(x_{n_i})_{i\in\mathbb{N}}$, and since A is closed in X, we have $\lim_{i\to\infty} x_{n_i} = x \in A$. Then, $y = f(x) \in f(A)$, and therefore $f(A)$ is closed in Y. This completes the proof. □

[3] Here, we need only (a) ⇒ (b) (Proposition 2.1.11).

We show that the perfectness of maps implies the non-extendability:

Lemma 2.1.14 *Let D be a dense proper subset of X. Any perfect map $f : D \to Y$ cannot extend over X.*

Proof It suffices to show that f cannot extend over $D \cup \{x_0\}$ for any $x_0 \in X \setminus D$, that is, it suffices to show the case that $X = D \cup \{x_0\}$. Assume that f extends to a map $g : X \to Y$ and let $y_0 = g(x_0)$. Since $f^{-1}(y_0)$ is compact and $x_0 \notin f^{-1}(y_0)$, X has disjoint open sets U and V such that $x_0 \in U$ and $f^{-1}(y_0) \subset V$, where it is possible $f^{-1}(y_0) = \emptyset$. Since f is a closed map, $f(D \setminus V)$ is closed in Y, hence $g^{-1}(f(D \setminus V))$ is closed in $\widetilde{D}$. Because $g^{-1}(y) = f^{-1}(y)$ for any $y \in Y \setminus \{y_0\}$, we have

$$D \setminus V \subset g^{-1}(f(D \setminus V)) = f^{-1}(f(D \setminus V)) \subset D.$$

On the other hand, $x \notin \mathrm{cl}\, V$. Therefore, $D = \mathrm{cl}\, V \cup g^{-1}(f(D \setminus V))$ is closed in X, which contradicts the fact that D is dense in X. □

By using the Stone–Čech compactification, a perfect map can be characterized as follows:

Theorem 2.1.15 *For a map $f : X \to Y$ between Tychonoff spaces, the following are equivalent:*

(a) *f is perfect;*
(b) *For any compactification γY of Y, f extends to a map $\tilde{f} : \beta X \to \gamma Y$ so that $\tilde{f}(\beta X \setminus X) \subset \gamma Y \setminus Y$;*
(c) *$\beta f(\beta X \setminus X) \subset \beta Y \setminus Y$.*

Proof The implication (b) ⇒ (c) is obvious.

(a) ⇒ (b): Applying Theorem 2.1.4, we can obtain a map $g : \beta Y \to \gamma Y$ with $g|Y = \mathrm{id}$. Then, $\tilde{f} = g(\beta f)$ is an extension of f. Moreover, we can apply Lemma 2.1.14 to see that $\tilde{f}(\beta X \setminus X) \subset \gamma Y \setminus Y$.

(c) ⇒ (a): For each $y \in Y$, $f^{-1}(y) = (\beta f)^{-1}(y)$ is compact. For each closed set A in X,

$$(\beta f)(\mathrm{cl}_{\beta X} A) \cap Y = f(\mathrm{cl}_{\beta X} A \cap X) = f(A),$$

which implies that $f(A)$ is closed in Y. Therefore, f is perfect. □

Remark 2.1.3 In Theorem 2.1.4, the map $f : \beta X \to \gamma X$ with $f|X = \mathrm{id}_X$ satisfies the condition $f(\beta X \setminus X) \subset \gamma X \setminus X$ by Theorem 2.1.15.

We have also the following characterization of a perfect map:

Theorem 2.1.16 *A map $f : X \to Y$ between Tychonoff spaces is perfect if and only if f cannot extend over any space Z containing X as a dense proper subspace.*

Proof The "only if" part is nothing but Lemma 2.1.14. To see the "if" part, assume that f is not perfect. By Theorem 2.1.15, $\beta f(\beta X \setminus X) \cap Y \neq \emptyset$. Then, $Z = (\beta f)^{-1}(Y)$ $(\subset \beta X)$ is a space containing X as a dense proper subspace and f extends to a map $\beta f | Z : Z \to Y$. Thus, we have the "if" part. □

Using Tychonoff's Theorem 2.1.1 and Wallace's Theorem 2.1.2, we can prove the following:

Theorem 2.1.17 *For each $\lambda \in \Lambda$, let $f_\lambda : X_\lambda \to Y_\lambda$ be a perfect map. Then, the map $f = \prod_{\lambda\in\Lambda} f_\lambda : X = \prod_{\lambda\in\Lambda} X_\lambda \to Y = \prod_{\lambda\in\Lambda} Y_\lambda$ is also perfect.*

Proof Owing to Tychonoff's Theorem 2.1.1, $f^{-1}(y) = \prod_{\lambda\in\Lambda} f_\lambda^{-1}(y(\lambda))$ is compact for each $y \in Y$. To show that f is a closed map, let A be a closed set in X and $y \in Y \setminus f(A)$. If $f^{-1}(y) = \emptyset$, then $f_{\lambda_0}^{-1}(y(\lambda_0)) = \emptyset$ for some $\lambda_0 \in \Lambda$. In this case, $V = \mathrm{pr}_{\lambda_0}^{-1}(Y_{\lambda_0} \setminus f_{\lambda_0}(X_{\lambda_0}))$ is an open neighborhood of y in Y and $V \cap f(A) \subset V \cap f(X) = \emptyset$. When $f^{-1}(y) \neq \emptyset$, since $f^{-1}(y) \subset X \setminus A$, we can apply Wallace's Theorem 2.1.2 to obtain $\lambda_1, \ldots, \lambda_n \in \Lambda$ and open sets U_i in X_{λ_i}, $i = 1, \ldots, n$, such that

$$f^{-1}(y) = \prod_{\lambda\in\Lambda} f_\lambda^{-1}(y(\lambda)) \subset \bigcap_{i=1}^{n} \mathrm{pr}_{\lambda_i}^{-1}(U_i) \subset X \setminus A,$$

where $f_{\lambda_i}^{-1}(y(\lambda_i)) \subset \mathrm{pr}_{\lambda_i}^{-1}(U_i)$. Since f_{λ_i} is a closed map, $V_i = Y_{\lambda_i} \setminus f_{\lambda_i}(X_{\lambda_i} \setminus U_i)$ is an open neighborhood of $y(\lambda_i)$ in Y_{λ_i} and $f_{\lambda_i}^{-1}(V_i) \subset U_i$. Then, $V = \bigcap_{i=1}^{n} \mathrm{pr}_{\lambda_i}^{-1}(V_i)$ is a neighborhood of y in Y and $f^{-1}(V) \subset X \setminus A$, i.e., $V \cap f(A) = \emptyset$. Therefore, f is a closed map. □

Let X be a locally compact space that is not compact. The **one-point compactification** (or the **Alexandorff compactification**) αX of X is a space $X \cup \{\infty_X\}$ with a new added point $\infty_X \notin X$, called the **point at infinity**, and the topology defined as follows:

$$U \text{ is open in } \alpha X \underset{\text{def}}{\Leftrightarrow} U \text{ is an open set in } X \text{ or } \alpha X \setminus U \text{ is a compact set in } X,$$

where the complement of every compact set in X is a neighborhood of ∞_X. Then, αX is compact and X is a dense open subspace of αX. Thus, every locally compact (Hausdorff) space X is Tychonoff because so is αX.[4]

The proposition below follows from the definitions of a proper map and the one-point compactification:

Proposition 2.1.18 *Let X and Y be locally compact spaces that are not compact. Then, a map $f : X \to Y$ is proper if and only if f extends to a map $\alpha f : \alpha X \to \alpha Y$ defined by $\alpha f(\infty_X) = \infty_Y$, where ∞_X and ∞_Y are the points at infinity for X and Y, respectively.* □

[4] Although αX is normal, X is not normal in general (cf. Proposition 2.9.5).

Let $\mathrm{C}^P(X, Y)$ denote the set of proper maps from X to Y. It is said that $f, g \in \mathrm{C}^P(X, Y)$ are **proper homotopic** (or f is **proper homotopic** to g), written

$$f \simeq^p g,$$

if there is a proper homotopy $h : X \times \mathbf{I} \to Y$ from f to g. Evidently, $\simeq^p$ is an equivalence relation on $\mathrm{C}^P(X, Y)$. In the above, if h is a proper $\mathcal{U}$-homotopy (resp. a proper ε-homotopy when $Y = (Y, d)$ is a metric space), f and g are said to be **proper $\mathcal{U}$-homotopic** (resp. **proper ε-homotopic**), written

$$f \simeq^p_{\mathcal{U}} g \quad \big(\text{resp. } f \simeq^p_{\varepsilon} g\big).$$

When $f|A = g|A$ for $A \subset X$ and h is a proper homotopy relative to A, f and g are **proper homotopic relative to** A (or f is **proper homotopic** to g **relative to** A) and denoted by

$$f \simeq^p g \ \text{ rel. } A.$$

The following is easy:

- *For $f, f' \in \mathrm{C}^P(X, Y)$ and $g, g' \in \mathrm{C}^P(Y, Z)$,*

$$f \simeq^p f',\ g \simeq^p g' \ \Rightarrow\ gf \simeq^p g'f'.$$

It is said that X and Y are **proper homotopy equivalent** (or X is **proper homotopy equivalent** to Y), denoted by

$$X \simeq^p Y,$$

if there are proper maps $f : X \to Y$ and $g : Y \to X$ such that $gf \simeq^p \mathrm{id}_X$ and $fg \simeq^p \mathrm{id}_Y$, where f is called a **proper homotopy equivalence** and g is a **proper homotopy inverse** of f. Then, it is also said that X and Y have the **same proper homotopy type** (or X has the **proper homotopy type** of Y).

Roughly speaking, the following proposition states that a map between locally compact spaces is proper if it is close to a proper map in some sense:

Proposition 2.1.19 *Let $f : X \to Y$ be a proper map and $\mathcal{U}$ be a locally finite open cover of Y such that every $U \in \mathcal{U}$ has the compact closure in Y (so X and Y should be locally compact).*

(1) *If a map $g : X \to Y$ is $\mathcal{U}$-close to f, then g is also proper.*
(2) *If $h : X \times \mathbf{I} \to Y$ is a $\mathcal{U}$-homotopy with $h_0 = f$, then h is a proper homotopy.*

Proof (1): For each compact set A in Y, $g^{-1}(A) \subset f^{-1}(\mathrm{st}(A, \mathcal{U}^{\mathrm{cl}}))$. Since $\mathcal{U}^{\mathrm{cl}}$ is locally finite, it follows that $\mathcal{U}^{\mathrm{cl}}[A]$ is finite, and hence $\mathrm{st}(A, \mathcal{U}^{\mathrm{cl}})$ is compact. Then, $f^{-1}(\mathrm{st}(A, \mathcal{U}^{\mathrm{cl}}))$ is compact because f is proper. Thus, its closed subset $g^{-1}(A)$ is also compact.

(2): Applying (1) to a homotopy $h : X \times \mathbf{I} \to Y$ and a constant homotopy $f \mathrm{pr}_X : X \times \mathbf{I} \to Y$, we have (2). □

As a consequence of the above (2), we have:

Corollary 2.1.20 *Let X and Y be locally compact spaces and $\mathcal{U}$ be a locally finite open cover of Y such that every $U \in \mathcal{U}$ has the compact closure in Y. Then, for each $f, g \in \mathrm{C}^P(X, Y)$,*

$$f \simeq_{\mathcal{U}} g \;\Rightarrow\; f \simeq^p_{\mathcal{U}} g. \qquad \square$$

2.2 The Tietze–Urysohn Extension Theorem and Normalities

In this section, we prove the Tietze–Urysohn Extension Theorem and present some concepts that strengthen normality. For $A, B \subset X$, it is said that A and B are **separated** in X if $A \cap \mathrm{cl}\, B = \emptyset$ and $B \cap \mathrm{cl}\, A = \emptyset$.

Lemma 2.2.1 *Let A and B be separated F_σ sets in a normal space X. Then, X has disjoint open sets U and V such that $A \subset U$ and $B \subset V$.*

Proof Let $A = \bigcup_{n\in\mathbb{N}} A_n$ and $B = \bigcup_{n\in\mathbb{N}} B_n$, where $A_1 \subset A_2 \subset \cdots$ and $B_1 \subset B_2 \subset \cdots$ are closed in X. Set $U_0 = V_0 = \emptyset$. Using normality, we can inductively choose open sets $U_n, V_n \subset X$, $n \in \mathbb{N}$, so that

$$A_n \cup \mathrm{cl}\, U_{n-1} \subset U_n \subset \mathrm{cl}\, U_n \subset X \setminus (\mathrm{cl}\, B \cup \mathrm{cl}\, V_{n-1}) \quad \text{and}$$
$$B_n \cup \mathrm{cl}\, V_{n-1} \subset V_n \subset \mathrm{cl}\, V_n \subset X \setminus (\mathrm{cl}\, A \cup \mathrm{cl}\, U_n).$$

Then, $U = \bigcup_{n\in\mathbb{N}} U_n$ and $V = \bigcup_{n\in\mathbb{N}} V_n$ are disjoint open sets in X such that $A \subset U$ and $B \subset V$. See Fig. 2.2. □

We can now prove the following extension theorem:

Theorem 2.2.2 (TIETZE–URYSOHN EXTENSION) *Let A be a closed set in a normal space X. Then, every map $f : A \to \mathbf{I}$ extends over X.*

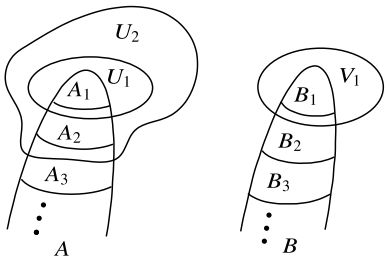


Fig. 2.2 Construction of U_n and V_n

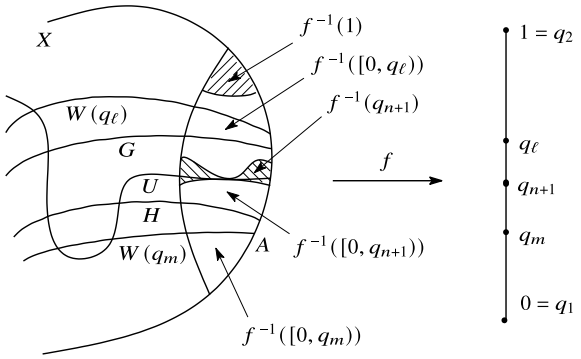


Fig. 2.3 Definition of $W(q_{n+1})$

Proof We first construct the open sets $W(q)$ in X, $q \in \mathbf{I} \cap \mathbb{Q}$, so that

(1) $q < q' \Rightarrow \operatorname{cl} W(q) \subset W(q')$,
(2) $A \cap W(q) = f^{-1}([0, q))$ and $A \cap \operatorname{cl} W(q) \subset f^{-1}([0, q])$.

To this end, let $\{q_n \mid n \in \mathbb{N}\} = \mathbf{I} \cap \mathbb{Q}$, where $q_1 = 0$, $q_2 = 1$ and $q_i \neq q_j$ if $i \neq j$. We define $W(q_1) = W(0) = \emptyset$ and $W(q_2) = W(1) = X \setminus f^{-1}(1)$. Assume that $W(q_1), W(q_2), \ldots, W(q_n)$ have been defined so as to satisfy (1) and (2). Let

$$q_l = \min\left\{q_i \;\middle|\; q_i > q_{n+1},\ i \leqslant n\right\} \text{ and } q_m = \max\left\{q_i \;\middle|\; q_i < q_{n+1},\ i \leqslant n\right\}.$$

Note that $f^{-1}([0, q_{n+1}))$ and $f^{-1}((q_{n+1}, 1])$ are separated F_σ sets in X. Applying Lemma 2.2.1, we can find an open set U in X such that

$$f^{-1}([0, q_{n+1})) \subset U \text{ and } f^{-1}((q_{n+1}, 1]) \cap \operatorname{cl} U = \emptyset.$$

Then, $V = U \setminus f^{-1}(q_{n+1})$ is open in X and $A \cap V = f^{-1}([0, q_{n+1}))$. Using normality, we can obtain an open set G in X such that

$$\operatorname{cl} W(q_m) \cup f^{-1}([0, q_{n+1}]) \subset G \subset \operatorname{cl} G \subset W(q_l).$$

Then, $A \cap (V \cap G) = f^{-1}([0, q_{n+1}))$ and $\operatorname{cl}(V \cap G) \subset W(q_l)$. Again, using normality, we can take an open set H in X such that

$$\operatorname{cl} W(q_m) \subset H \subset \operatorname{cl} H \subset G \setminus f^{-1}([q_{n+1}, 1]) \ (\subset W(q_l)).$$

Then, $W(q_{n+1}) = (V \cap G) \cup H$ is the desired open set in X. See Fig. 2.3.

Now, we define $\tilde{f} : X \to \mathbf{I}$ as follows:

$$\tilde{f}(x) = \begin{cases} 1 & \text{if } x \notin W(1), \\ \inf\left\{q \in \mathbf{I} \cap \mathbb{Q} \;\middle|\; x \in W(q)\right\} & \text{if } x \in W(1). \end{cases}$$

Then, $\tilde{f}|A = f$ because, for each $x \in A \cap W(1) = A \setminus f^{-1}(1)$,

$$\tilde{f}(x) = \inf\left\{q \in \mathbf{I} \cap \mathbb{Q} \;\middle|\; x \in f^{-1}([0, q))\right\} = f(x).$$

To see the continuity of $\tilde{f}$, let $0 < a \leqslant 1$ and $0 \leqslant b < 1$. Since $\tilde{f}(x) < a$ if and only if $x \in W(q)$ for some $q < a$, it follows that $\tilde{f}^{-1}([0, a)) = \bigcup_{q<a} W(q)$ is open in X. Moreover, from (1), it follows that $\tilde{f}(x) > b$ if and only if $x \notin \operatorname{cl} W(q)$ for some $q > b$. Then, $\tilde{f}^{-1}((b, 1]) = X \setminus \bigcap_{q>b} \operatorname{cl} W(q)$ is also open in X. Therefore, $\tilde{f}$ is continuous. □

As a corollary, we have Urysohn's Lemma:

Corollary 2.2.3 (Urysohn's Lemma) *For each disjoint pair of closed sets A and B in a normal space X, there exists a map $f : X \to \mathbf{I}$ such that $A \subset f^{-1}(0)$ and $B \subset f^{-1}(1)$.* □

Such a map f as in the above corollary is called a **Urysohn map**.

Note In the standard proof of the Tietze–Urysohn Extension Theorem 2.2.2, the desired extension is obtained as the uniform limit of a sequence of approximate extensions that are sums of Urysohn maps. On the other hand, Urysohn's Lemma is directly proved as follows:

Sketch of Proof Using the normality property yields the open sets $W(q)$ in X corresponding to all $q \in \mathbf{I} \cap \mathbb{Q}$ satisfying condition (1) in our proof of the Tietze–Urysohn Extension Theorem 2.2.2 and

$$A \subset W(0) \subset \operatorname{cl} W(0) \subset W(1) = X \setminus B.$$

A Urysohn map $f : X \to \mathbf{I}$ can be defined as follows:

$$f(x) = \begin{cases} 1 & \text{if } x \notin W(1), \\ \inf\{q \in \mathbf{I} \cap \mathbb{Q} \mid x \in W(q)\} & \text{if } x \in W(1). \end{cases}$$

Using a Urysohn map, we can prove the following $\mathbb{R}$-version of the Tietze–Urysohn Extension Theorem 2.2.2:

Theorem 2.2.4 (Tietze–Urysohn Extension for $\mathbb{R}$) *Let A be a closed set in a normal space X. Every map $f : A \to \mathbb{R}$ extends to a map $\tilde{f} : X \to \mathbb{R}$. When f is bounded, $\tilde{f}$ can be taken so that $\tilde{f}$ is bounded,* $\sup \tilde{f}(X) = \sup f(A)$, *and* $\inf \tilde{f}(X) = \inf f(A)$.

Proof The additional part is rather easy because $[\inf f(A), \sup f(A)] \approx \mathbf{I}$. For general case, $\mathbb{R}$ can be replaced by $(-1, 1)$. Then, applying the Tietze–Urysohn Extension Theorem 2.2.2, we can extend every map $f : A \to (-1, 1) \subset [-1, 1]$ to a map $\bar{f} : X \to [-1, 1]$. Since $\bar{f}^{-1}(\pm 1) = \bar{f}^{-1}(1) \cup \bar{f}^{-1}(-1)$ is closed in X and $A \cap \bar{f}^{-1}(\pm 1) = \emptyset$, there is a Urysohn map $k : X \to \mathbf{I}$ such that $A \subset k^{-1}(1)$ and $\bar{f}^{-1}(\pm 1) \subset k^{-1}(0)$. The desired extension $\tilde{f} : X \to (-1, 1)$ of f can be defined by $\tilde{f}(x) = k(x)\bar{f}(x)$ for each $x \in X$. □

As we will give an example in Sect. 2.9, in general, a subspace of a normal space is not normal. However, we have the following proposition:

Proposition 2.2.5 *Every F_σ set in a normal space is also normal.*

Proof Let Y be an F_σ set in a normal space X. All pairs of disjoint closed sets in Y are F_σ sets in X that are separated in X. Then, the normality of Y follows from Lemma 2.2.1. □

A space X is **hereditarily normal** if every subspace of X is normal. Obviously, *every metrizable space is hereditarily normal*. It is said that X is **completely normal** provided that, for each pair of separated subsets $A, B \subset X$, there exist disjoint open sets U and V in X such that $A \subset U$ and $B \subset V$. These concepts meet in the following theorem:

Theorem 2.2.6 *For a space X, the following are equivalent:*

(a) *X is hereditarily normal;*
(b) *Every open set in X is normal;*
(c) *X is completely normal.*

Proof The implication (a) $\Rightarrow$ (b) is obvious.

(c) $\Rightarrow$ (a): For an arbitrary subspace $Y \subset X$, each pair of disjoint closed sets A and B in Y is separated in X. Hence, we have (c) $\Rightarrow$ (a).

(b) $\Rightarrow$ (c): Let $A, B \subset X$ be separated, i.e., $A \cap \mathrm{cl}\, B = \emptyset$ and $B \cap \mathrm{cl}\, A = \emptyset$. Then, $W = X \setminus (\mathrm{cl}\, A \cap \mathrm{cl}\, B)$ is open in X and $A, B \subset W$. Moreover,

$$\mathrm{cl}_W A \cap \mathrm{cl}_W B = W \cap \mathrm{cl}\, A \cap \mathrm{cl}\, B = \emptyset.$$

From the normality of W, we have disjoint open sets U and V in W such that $A \subset U$ and $B \subset V$. Then, U and V are open in X, and hence we have (c). □

A normal space X is **perfectly normal** if every closed set in X is G_δ in X (equivalently, every open set in X is F_σ in X). Clearly, *every metrizable space is perfectly normal*. A closed set $A \subset X$ is called a **zero set** in X if $A = f^{-1}(0)$ for some map $f : X \to \mathbb{R}$, where $\mathbb{R}$ can be replaced by $\mathbf{I}$. The complement of a zero set in X is called a **cozero set**.

Theorem 2.2.7 *For a space X, the following conditions are equivalent:*

(a) *X is perfectly normal;*
(b) *Every closed set in X is a zero set (equiv., every open set in X is a cozero set);*
(c) *For every pair of disjoint closed sets A and B in X, there exists a map $f : X \to \mathbf{I}$ such that $A = f^{-1}(0)$ and $B = f^{-1}(1)$.*

Proof The implication (c) $\Rightarrow$ (a) is trivial.

(a) $\Rightarrow$ (b): Let A be a closed set in X. Then, we can write $A = \bigcap_{n\in\mathbb{N}} G_n$, where each G_n is open in X. Using Urysohn's Lemma, we take maps $f_n : X \to \mathbf{I}$, $n \in \mathbb{N}$, such that $f_n(A) = 0$ and $f_n(X \setminus G_n) = 1$. We can define a map $f : X \to \mathbf{I}$ as $f(x) = \sum_{n\in\mathbb{N}} 2^{-n} f_n(x)$. Then, it is easy to see that $A = f^{-1}(0)$.

(b) $\Rightarrow$ (c): Let A and B be disjoint closed sets in X. Condition (b) provides two maps $g, h : X \to \mathbf{I}$ such that $g^{-1}(0) = A$ and $h^{-1}(0) = B$. Then, the desired map $f : X \to \mathbf{I}$ can be defined as follows:

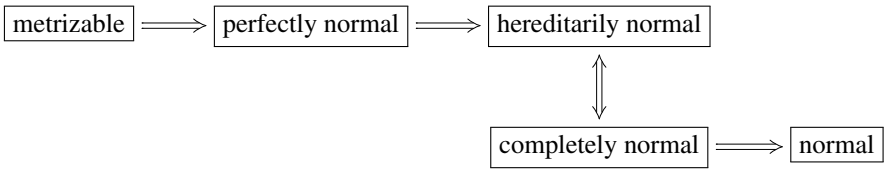
$$f(x) = \frac{g(x)}{g(x) + h(x)}.$$

□

Theorem 2.2.8 *Every perfectly normal space is hereditarily normal (= completely normal).*

Proof Let X be perfectly normal. Then, each open set in X is an F_σ set, which is normal as a consequence of Proposition 2.2.5. Hence, it follows from Theorem 2.2.6 that X is hereditarily normal. □

Consequently, we have the following implications:



Remark 2.2.1 Let $A_0, A_1, \ldots, A_n$ be pairwise disjoint closed sets in a normal space X. We can apply the Tietze–Urysohn Extension Theorem 2.2.2 to obtain a map $f : X \to \mathbf{I}$ such that $A_i \subset f^{-1}(i/n)$ (i.e., $f(A_i) = \{i/n\}$) for each $i = 0, 1, \ldots, n$. When X is perfectly normal and $n \geqslant 2$, the condition $A_i \subset f^{-1}(i/n)$ cannot be replaced by $A_i = f^{-1}(i/n)$. For example, let $X = \mathbf{S}^1$ be the unit circle (the unit 1-sphere of $\mathbb{R}^2$), $A_0 = \{\mathbf{e}_1\}$, $A_1 = \{\mathbf{e}_2\}$, and $A_2 = \{-\mathbf{e}_1\}$, where $\mathbf{e}_1 = (1, 0)$, $\mathbf{e}_2 = (0, 1) \in \mathbb{R}^2$. Since $X \setminus A_1$ is (path-)connected, there does not exist a map $f : X \to \mathbf{I}$ such that $A_0 = f^{-1}(0)$, $A_1 = f^{-1}(1/2)$ and $A_2 = f^{-1}(1)$.

2.3 A.H. Stone's Theorem and Metrization

In this section, we prove A.H. STONE'S THEOREM and characterize the metrizability using open bases. Let $\mathcal{A}$ be a collection of subsets of a space X. It is said that $\mathcal{A}$ is **σ-locally finite** (resp. **σ-discrete**) in X if $\mathcal{A} = \bigcup_{n\in\mathbb{N}} \mathcal{A}_n$ and each $\mathcal{A}_n$ is locally

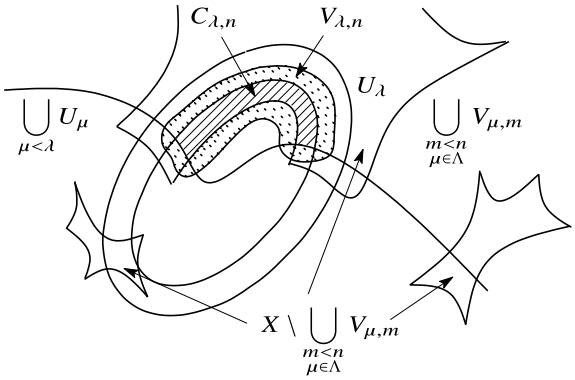


Fig. 2.4 Definition of $V_{\lambda,n}$

finite (resp. discrete) in X, that is, each $x \in X$ has a neighborhood U such that $\operatorname{card} \mathcal{A}_n[U] < \aleph_0$ (resp. $\operatorname{card} \mathcal{A}_n[U] \leqslant 1$).[5] First, we prove:

Theorem 2.3.1 (A.H. STONE) *Every open cover of a metrizable space has a locally finite and σ-discrete open refinement.*

Proof Let $X = (X, d)$ be a metric space and $\mathcal{U} \in \operatorname{cov}(X)$. Without loss of generality, we may assume that $X \notin \mathcal{U}$. We can index all members of $\mathcal{U}$ by a well-ordered set $\Lambda = (\Lambda, \leqslant)$, that is, $\mathcal{U} = \{U_\lambda \mid \lambda \in \Lambda\}$. By induction on $n \in \mathbb{N}$, we define open collections $\mathcal{V}_n = \{V_{\lambda,n} \mid \lambda \in \Lambda\}$ as follows:

$$V_{\lambda,n} = \mathrm{N}(C_{\lambda,n}, 2^{-n})\mathrm{N}(C_{\lambda,n}, 2^{-n}) = \bigcup_{x\in C_{\lambda,n}} \mathrm{B}(x, 2^{-n}),$$

where

$$C_{\lambda,n} = \left\{x \in X \;\middle|\; d(x, X \setminus U_\lambda) > 2^{-n}3\right\} \setminus \left(\bigcup_{\mu<\lambda} U_\mu \cup \bigcup_{\substack{m<n \\ \mu\in\Lambda}} V_{\mu,m}\right).$$

Refer to Fig. 2.4.

For each $x \in X$, let $\lambda(x) = \min\{\lambda \in \Lambda \mid x \in U_\lambda\}$ and choose $n \in \mathbb{N}$ so that $2^{-n}3 < d(x, X \setminus U_{\lambda(x)})$. Then, $x \in C_{\lambda(x),n} \subset V_{\lambda(x),n}$ or $x \in V_{\mu,m}$ for some $\mu \in \Lambda$ and $m < n$. Hence, we have $\mathcal{V} = \bigcup_{n\in\mathbb{N}} \mathcal{V}_n \setminus \{\emptyset\} \in \operatorname{cov}(X)$. Since each $V_{\lambda,n}$ is contained in U_λ, it follows that $\mathcal{V} \prec \mathcal{U}$.

The discreteness of each $\mathcal{V}_n$ is a consequence of the following:

Claim (1) If $\lambda \neq \mu$ then $\operatorname{dist}_d(V_{\lambda,n}, V_{\mu,n}) \geqslant 2^{-n}$.

[5] For notation $\mathcal{A}[B]$, refer to Sect. 1.1.

To prove this claim, we may assume $\mu < \lambda$. For each $x \in V_{\lambda,n}$ and $y \in V_{\mu,n}$, choose $x' \in C_{\lambda,n}$ and $y' \in C_{\mu,n}$ so that $d(x, x') < 2^{-n}$ and $d(y, y') < 2^{-n}$, respectively. Then, $x' \notin U_\mu$ and $d(y', X \setminus U_\mu) > 2^{-n}3$, hence $d(x', y') > 2^{-n}3$. Therefore,

$$d(x, y) \geqslant d(x', y') - d(x, x') - d(y, y') > 2^{-n}.$$

The local finiteness of $\mathcal{V}$ follows from the discreteness of each $\mathcal{V}_n$ and:

Claim (2) If $\mathrm{B}(x, 2^{-k}) \subset V_{\mu,m}$, then $\mathrm{B}(x, 2^{-k-1}) \cap V_{\lambda,n} = \emptyset$ for all $\lambda \in \Lambda$ and $n > \max\{k, m\}$.

For each $y \in V_{\lambda,n}$, choose $y' \in C_{\lambda,n}$ so that $d(y, y') < 2^{-n}$. Since $y' \notin V_{\mu,m}$, it follows that $d(x, y') \geqslant 2^{-k}$. Hence,

$$d(x, y) \geqslant d(x, y') - d(y, y') > 2^{-k} - 2^{-n} \geqslant 2^{-k-1}.$$

The proof is complete. □

Applying Theorem 2.3.1 to the open covers $\mathcal{B}_n = \{\mathrm{B}(x, 2^{-n}) \mid x \in X\}$, $n \in \mathbb{N}$, of a metric space $X = (X, d)$, we have the following corollary:

Corollary 2.3.2 *Every metrizable space has a σ-discrete open basis.* □

To demonstrate the metrization theorem, we need the following lemma:

Lemma 2.3.3 *A regular space X with a σ-locally finite open basis is perfectly normal.*

Proof Let $\mathcal{B} = \bigcup_{n\in\mathbb{N}} \mathcal{B}_n$ be an open basis for X where each $\mathcal{B}_n$ is locally finite in X. Instead of proving that every closed set in X is a G_δ set, we show that every open set $W \subset X$ is F_σ. For each $x \in W$, choose $k(x) \in \mathbb{N}$ and $B(x) \in \mathcal{B}_{k(x)}$ so that $x \in B(x) \subset \mathrm{cl}\, B(x) \subset W$. For each $n \in \mathbb{N}$, let

$$W_n = \bigcup \big\{ B(x) \;\big|\; x \in W,\ k(x) = n \big\}.$$

Because of the local finiteness of $\mathcal{B}_n$, we have

$$\mathrm{cl}\, W_n = \bigcup \big\{ \mathrm{cl}\, B(x) \;\big|\; x \in W,\ k(x) = n \big\} \subset W.$$

Since $W = \bigcup_{n\in\mathbb{N}} W_n$, it follows that $W = \bigcup_{n\in\mathbb{N}} \mathrm{cl}\, W_n$, which is F_σ in X.

To prove normality, let A and B be disjoint closed sets in X. As seen above, we have open sets $V_n, W_n \subset X$, $n \in \mathbb{N}$, such that $X \setminus A = \bigcup_{n\in\mathbb{N}} V_n = \bigcup_{n\in\mathbb{N}} \mathrm{cl}\, V_n$ and $X \setminus B = \bigcup_{n\in\mathbb{N}} W_n = \bigcup_{n\in\mathbb{N}} \mathrm{cl}\, W_n$. For each $n \in \mathbb{N}$, let

$$G_n = W_n \setminus \bigcup_{m\leqslant n} \mathrm{cl}\, V_m \quad \text{and} \quad H_n = V_n \setminus \bigcup_{m\leqslant n} \mathrm{cl}\, W_m.$$

Then, $G = \bigcup_{n\in\mathbb{N}} G_n$ and $H = \bigcup_{n\in\mathbb{N}} H_n$ are disjoint open sets in X such that $A \subset G$ and $B \subset H$. □

Theorem 2.3.4 (BING; NAGATA–SMIRNOV) *For a regular space X, the following conditions are equivalent:*

(a) *X is metrizable;*
(b) *X has a σ-discrete open basis;*
(c) *X has a σ-locally finite open basis.*

Proof The implication (a) $\Rightarrow$ (b) is Corollary 2.3.2 and (b) $\Rightarrow$ (c) is obvious. It remains to show the implication (c) $\Rightarrow$ (a).

(c) $\Rightarrow$ (a): Let $\mathcal{B} = \bigcup_{n\in\mathbb{N}} \mathcal{B}_n$ be an open basis for X where each $\mathcal{B}_n$ is locally finite in X. Since X is perfectly normal by Lemma 2.3.3, we have maps $f_B : X \to \mathbf{I}$, $B \in \mathcal{B}$, such that $f_B^{-1}(0) = X \setminus B$ (Theorem 2.2.7). For each $n \in \mathbb{N}$, since $\mathcal{B}_n$ is locally finite, we can define a map $f_n : X \to \ell_1(\mathcal{B}_n)$ by $f_n(x) = (f_B(x))_{B\in\mathcal{B}_n} \in \ell_1(\mathcal{B}_n)$. Let $f : X \to \prod_{n\in\mathbb{N}} \ell_1(\mathcal{B}_n)$ be the map defined by $f(x) = (f_n(x))_{n\in\mathbb{N}}$. Since $\prod_{n\in\mathbb{N}} \ell_1(\mathcal{B}_n)$ is metrizable, it suffices to show that f is an embedding.

For each $x \neq y \in X$, choose $B \in \mathcal{B}_n \subset \mathcal{B}$ so that $x \in B$ and $y \notin B$. Then, $f_B(x) > 0 = f_B(y)$, so $f_n(x) \neq f_n(y)$. Hence, f is an injection.

For each $n \in \mathbb{N}$ and $B \in \mathcal{B}_n$, $V_B = \{y \in \ell_1(\mathcal{B}_n) \mid y(B) > 0\}$ is open in $\ell_1(\mathcal{B}_n)$. Observe that for $x \in X$,

$$x \in B \;\Leftrightarrow\; f_n(x)(B) = f_B(x) > 0 \;\Leftrightarrow\; f_n(x) \in V_B.$$

Then, it follows that $f(B) = \mathrm{pr}_n^{-1}(V_B) \cap f(X)$ is open in $f(X)$, where $\mathrm{pr}_n : \prod_{n\in\mathbb{N}} \ell_1(\mathcal{B}_n) \to \ell_1(\mathcal{B}_n)$ is the projection. Thus, f is an embedding. □

The equivalence of (a) and (b) in Theorem 2.3.4 is called the BING METRIZATION THEOREM, and the equivalence of (a) and (c) is called the NAGATA–SMIRNOV METRIZATION THEOREM. As a corollary, we have the following URYSOHN METRIZATION THEOREM:

Corollary 2.3.5 (URYSOHN METRIZATION) *A space is separable and metrizable if and only if it is regular and second countable.* □

The following can also be obtained as an easy application of the above:

Corollary 2.3.6 *For every locally compact separable metrizable space X that is non-compact, the one-point compactification αX is metrizable.* □

Remark 2.3.1 In the definition of an n-manifold (Sect. 1.1), the separable metrizability is required, but it suffices to require the second countability by the Urysohn Metrization Theorem (Corollary 2.3.5).[6]

For a metrizable space X, let Γ be an infinite set with $w(X) \leqslant \mathrm{card}\,\Gamma$. In the proof of Theorem 2.3.4, $\mathrm{card}\,\mathcal{B}_n \leqslant \mathrm{card}\,\Gamma$ because of the local finiteness of $\mathcal{B}_n$ in X. Then, every $\ell_1(\mathcal{B}_n)$ can be embedded into $\ell_1(\Gamma)$, where $w(\ell_1(\Gamma)) = \mathrm{card}\,\Gamma$ (as is shown just before Proposition 1.2.6). Thus, the following corollary can be obtained:

[6] Every locally compact (Hausdorff) space is Tychonoff.

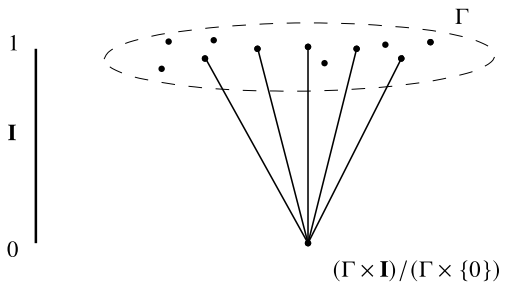


Fig. 2.5 The hedgehog $J(\Gamma)$

Corollary 2.3.7 *Let X be a metrizable space and Γ an infinite set such that $w(X) \leqslant \operatorname{card}\Gamma$. Then, X can be embedded in the completely metrizable topological linear space $\ell_1(\Gamma)^{\mathbb{N}}$, where $w(\ell_1(\Gamma)^{\mathbb{N}}) = \operatorname{card}\Gamma$.*[7] □

The **hedgehog** $J(\Gamma)$ is the closed subspace of $\ell_1(\Gamma)$ defined as follows:

$$J(\Gamma) = \bigcup_{\gamma\in\Gamma} \mathbf{I}\mathbf{e}_\gamma = \left\{x \in \ell_1(\Gamma) \cap \mathbf{I}^\Gamma \;\middle|\; x(\gamma) \neq 0 \text{ at most one } \gamma \in \Gamma\right\},$$

where $\mathbf{e}_\gamma \in \ell_1(\Gamma)$ is the unit vector defined by $\mathbf{e}_\gamma(\gamma) = 1$ and $\mathbf{e}_\gamma(\gamma') = 0$ for $\gamma' \neq \gamma$. The hedgehog $J(\Gamma)$ can also be defined as the space $(\Gamma \times \mathbf{I})/(\Gamma \times \{0\})$ with the metric induced from the pseudo-metric ρ on $\Gamma \times \mathbf{I}$ defined as follows:

$$\rho((\gamma, t), (\gamma', s)) = \begin{cases} |t - s| & \text{if } \gamma = \gamma', \\ t + s & \text{if } \gamma \neq \gamma'. \end{cases}$$

See Fig. 2.5. Then, we have $w(J(\Gamma)^{\mathbb{N}}) = \operatorname{card}\Gamma$.

In the proof of Theorem 2.3.4, if each $\mathcal{B}_n$ is discrete in X, then $f_n(X) \subset J(\mathcal{B}_n)$. Similar to Corollary 2.3.7, we have the following:

Corollary 2.3.8 *Let X be a metrizable space and Γ an infinite set such that $w(X) \leqslant \operatorname{card}\Gamma$. Then, X can be embedded in $J(\Gamma)^{\mathbb{N}}$.* □

In the second-countable case, X can be embedded in $\mathbf{I}^{\mathbb{N}}$, since we can take $\mathcal{B} = \bigcup_{n\in\mathbb{N}} \mathcal{B}_n$ in the proof of Theorem 2.3.4 so that each $\mathcal{B}_n$ contains only one open set. Thus, we have the following embedding theorem for separable metrizable spaces:

Corollary 2.3.9 *Every separable metrizable space can be embedded in the Hilbert cube $\mathbf{I}^{\mathbb{N}}$, and hence in $\mathbb{R}^{\mathbb{N}}$. Moreover, every locally compact separable metrizable space can be embedded in $\mathbf{I}^{\mathbb{N}} \times (0, 1]$ as a closed set.*[8]

---

[7] For topological linear spaces, refer to Sect. 3.4.

[8] In Theorem 2.5.4, it will be proved that every separable completely metrizable space can be embedded in $\mathbb{R}^{\mathbb{N}}$ as a closed set.

Sketch of Proof To construct an embedding of the additional statement, combine an embedding of αX into $\mathbf{I}^{\mathbb{N}}$ (Corollary 2.3.6) with a map $f : \alpha X \to \mathbf{I}$ with $f^{-1}(0) = \{\infty_X\}$.

In association with Corollary 2.3.7, we state the following theorem:

Theorem 2.3.10 *Every metric space $X = (X, d)$ can be isometrically embedded in the Banach space $C^B(X)$. Consequently, every completely metrizable space can be embedded in $C^B(X)$ as a closed set.*

Sketch of Proof Fix $x_0 \in X$ and define $\varphi : X \to C^B(X)$ as follows:

$$\varphi(x)(z) = d(x, z) - d(x_0, z),\ z \in X.$$

It is easy to see that $\|\varphi(x)\| = d(x, x_0)$ and $\|\varphi(x) - \varphi(y)\| = d(x, y)$.[9]

The (**metric**) **completion** of a metric space $X = (X, d)$ is a complete metric space $\widetilde{X} = (\widetilde{X}, \tilde{d})$ containing X as a dense set and as a metric subspace, that is, d is the restriction of $\tilde{d}$. Since a closed set in a complete metric space is also complete, Theorem 2.3.10 yields the following:

Corollary 2.3.11 *Every metric space has a completion.* □

2.4 Sequences of Open Covers and Metrization

In this section, we characterize metrizable spaces via sequences of open covers. Given a cover $\mathcal{V}$ of a space X and $A \subset X$, we define

$$\mathrm{st}(A, \mathcal{V}) = \bigcup \mathcal{V}[\mathcal{A}],$$

which is called the **star** of A with respect to $\mathcal{V}$. For each $n \in \mathbb{N}$, the ***n*-th star** $\mathrm{st}^n(A, \mathcal{V})$ of A with respect to $\mathcal{V}$ is inductively defined as follows:

$$\mathrm{st}^n(A, \mathcal{V}) = \mathrm{st}(\mathrm{st}^{n-1}(A, \mathcal{V}), \mathcal{V}),\ \mathrm{st}^0(A, \mathcal{V}) = A.$$

For simplicity, we write $\mathrm{st}(\{x\}, \mathcal{V}) = \mathrm{st}(x, \mathcal{V})$ and $\mathrm{st}^n(\{x\}, \mathcal{V}) = \mathrm{st}^n(x, \mathcal{V}), n \in \mathbb{N}$. For a collection $\mathcal{A}$ of subsets of X, the following is the **star** (***n*-th star**) of $\mathcal{A}$ with respect to $\mathcal{V}$:

$$\mathrm{st}(\mathcal{A}, \mathcal{V}) = \left\{ \mathrm{st}(A, \mathcal{V}) \;\middle|\; A \in \mathcal{A} \right\} \quad \left(\mathrm{st}^n(\mathcal{A}, \mathcal{V}) = \left\{ \mathrm{st}^n(A, \mathcal{V}) \;\middle|\; A \in \mathcal{A} \right\} \right).$$

The following is the main theorem of this section:

Theorem 2.4.1 (ALEXANDROFF–URYSOHN; FRINK) *For a space X, the following conditions are equivalent:*

[9] When d is bounded, the isometry φ can be simply defined as $\varphi(x)(z) = d(x, z)$. Refer to Notes for this chapter.

(a) *X is metrizable;*

(b) *X has a sequence of open covers $\mathcal{U}_1, \mathcal{U}_2, \ldots$ such that $\{\mathrm{st}(x, \mathcal{U}_n) \mid n \in \mathbb{N}\}$ is a neighborhood basis of each $x \in X$ and*

$$(*)\ \ U, U' \in \mathcal{U}_{n+1},\ U \cap U' \neq \emptyset \ \Rightarrow\ \exists U'' \in \mathcal{U}_n \ \text{ such that } \ U \cup U' \subset U'';$$

(c) *Each $x \in X$ has an open neighborhood basis $\{V_n(x) \mid n \in \mathbb{N}\}$ satisfying the condition that, for each $x \in X$ and $i \in \mathbb{N}$, there exists a $j(x, i) \geqslant i$ such that*

$$V_{j(x,i)}(x) \cap V_{j(x,i)}(y) \neq \emptyset \ \Rightarrow\ V_{j(x,i)}(y) \subset V_i(x).$$

Proof (a) $\Rightarrow$ (c): A metric space $X = (X, d)$ satisfies (c) because

$$\mathrm{B}(x, 3^{-n}) \cap \mathrm{B}(y, 3^{-n}) \neq \emptyset \ \Rightarrow\ \mathrm{B}(y, 3^{-n}) \subset \mathrm{B}(x, 3^{-n+1}).$$

(c) $\Rightarrow$ (b): For each $x \in X$, let $k(x, 1) = 1$ and inductively define

$$k(x, n) = \max\{n, j(x, i) \mid i = 1, \ldots, k(x, n-1)\} \geqslant n.$$

For each $n \in \mathbb{N}$, let $U_n(x) = \bigcap_{i=1}^{k(x,n)} V_i(x)$. Then, $\{U_n(x) \mid n \in \mathbb{N}\}$ is an open neighborhood basis of x and

$$U_n(x) \cap U_n(y) \neq \emptyset \ \Rightarrow U_n(x) \cup U_n(y) \subset U_{n-1}(x) \quad \text{or } \ U_n(x) \cup U_n(y) \subset U_{n-1}(y).$$

In fact, assume that $U_n(x) \cap U_n(y) \neq \emptyset$. In the case $k(x, n) \leqslant k(y, n)$, $V_{j(x,i)}(y) \subset V_i(x)$ for each $i = 1, \ldots, k(x, n-1)$ because $V_{j(x,i)}(x) \cap V_{j(x,i)}(y) \neq \emptyset$. Then, it follows that

$$U_n(y) \subset \bigcap_{i=1}^{k(x,n)} V_i(y) \subset \bigcap_{i=1}^{k(x,n-1)} V_{j(x,i)}(y) \subset \bigcap_{i=1}^{k(x,n-1)} V_i(x) = U_{n-1}(x).$$

Since $U_n(x) \subset U_{n-1}(x)$ by definition, we have $U_n(x) \cup U_n(y) \subset U_{n-1}(x)$. As above, $k(y, n) \leqslant k(x, n)$ implies $U_n(x) \cup U_n(y) \subset U_{n-1}(y)$.

For each $n \in \mathbb{N}$, we have $\mathcal{U}_n = \{U_n(x) \mid x \in X\} \in \mathrm{cov}(X)$. It remains to prove that $\{\mathrm{st}(x, \mathcal{U}_n) \mid n \in \mathbb{N}\}$ is a neighborhood basis of $x \in X$. Evidently, each $\mathrm{st}(x, \mathcal{U}_n)$ is a neighborhood of $x \in X$. Then, it suffices to show that $\mathrm{st}(x, \mathcal{U}_{j(x,n)}) \subset V_n(x)$. If $x \in U_{j(x,n)}(y)$, then

$$V_{j(x,n)}(x) \cap V_{j(x,n)}(y) \supset V_{j(x,n)}(x) \cap U_{j(x,n)}(y) \neq \emptyset,$$

and hence $U_{j(x,n)}(y) \subset V_{j(x,n)}(y) \subset V_n(x)$.

(b) $\Rightarrow$ (a): First, note that $\mathcal{U}_{i+1} \prec \mathcal{U}_i$ for each $i \in \mathbb{N}$. Let $\mathcal{U}_0 = \{X\} \in \mathrm{cov}(X)$. For each $x, y \in X$, define

$$\delta(x, y) = \inf \left\{ 2^{-i} \;\middle|\; \exists U \in \mathcal{U}_i \text{ such that } x, y \in U \right\}.$$

Note that if $\delta(x, y) > 0$, then $\delta(x, y) = 2^{-n}$ for some $n \geqslant 0$. As can easily be shown, the following hold for each $x, y, z \in X$:

(1) $\delta(x, y) = 0$ if and only if $x = y$;
(2) $\delta(x, y) = \delta(y, x)$;
(3) $\delta(x, y) \leqslant 2 \max\{\delta(x, z), \delta(z, y)\}$,

where (*) is used for (3). Furthermore, we claim that

(4) for every $n \geqslant 3$ and each $x_1, \ldots, x_n \in X$,

$$\delta(x_1, x_n) \leqslant 2(\delta(x_1, x_2) + \delta(x_{n-1}, x_n)) + 4 \sum_{i=2}^{n-2} \delta(x_i, x_{i+1}).$$

In fact, when $n = 3$, the inequality follows from (3). Assuming (4) holds for any $n < k$, we show (4) for $n = k$. Then, we may assume that $x_k \neq x_1$. For each $x_1, \ldots, x_k \in X$, let

$$m = \min \left\{ i \;\middle|\; \delta(x_1, x_k) \leqslant 2\delta(x_1, x_i) \right\} \geqslant 2.$$

Then, $\delta(x_1, x_k) \leqslant 2\delta(x_1, x_m)$. From (3) and the minimality of m, we have $\delta(x_1, x_k) \leqslant 2\delta(x_{m-1}, x_k)$. If $m = 2$ or $m = k$, then the inequality in (4) holds for $n = k$. In the case $2 < m < k$,

$$\delta(x_1, x_k) = 2^{-1}\delta(x_1, x_k) + 2^{-1}\delta(x_1, x_k) \leqslant \delta(x_1, x_m) + \delta(x_{m-1}, x_k).$$

By the inductive assumption, we have

$$\delta(x_1, x_m) \leqslant 2(\delta(x_1, x_2) + \delta(x_{m-1}, x_m)) + 4 \sum_{i=2}^{m-2} \delta(x_i, x_{i+1}) \quad \text{and}$$

$$\delta(x_{m-1}, x_k) \leqslant 2(\delta(x_{m-1}, x_m) + \delta(x_{k-1}, x_k)) + 4 \sum_{i=m}^{k-2} \delta(x_i, x_{i+1}),$$

so the desired inequality is obtained. By induction, (4) holds for all $n \geqslant 3$.

Now, we can define $d \in \mathrm{Metr}(X)$ as follows:

$$d(x, y) = \inf \left\{ \textstyle\sum_{i=1}^{n-1} \delta(x_i, x_{i+1}) \;\middle|\; n \in \mathbb{N},\ x_i \in X,\ x_1 = x,\ x_n = y \right\}.$$

In fact, $d(x, y) = d(y, x)$ by (2) and the above definition. The triangle inequality follows from the definition of d. Since $\delta(x, y) \leqslant 4d(x, y)$ by (4), it follows from (1)

that $d(x, y) = 0$ implies $x = y$. Obviously, $x = y$ implies $d(x, y) = 0$. Moreover, it follows that

$$d(x, y) \leqslant 2^{-n-2} \;\Rightarrow\; \exists U \in \mathcal{U}_n \text{ such that } x, y \in U,$$

which means that $\overline{\mathrm{B}}_d(x, 2^{-n-2}) \subset \mathrm{st}(x, \mathcal{U}_n)$ for each $x \in X$ and $n \in \mathbb{N}$. Since $d(x, y) \leqslant \delta(x, y)$, we have $\mathrm{mesh}_d\, \mathcal{U}_n \leqslant 2^{-n}$, so $\mathrm{st}(x, \mathcal{U}_n) \subset \overline{\mathrm{B}}_d(x, 2^{-n})$. Therefore, $\{\mathrm{B}_d(x, 2^{-n}) \mid n \in \mathbb{N}\}$ is a neighborhood basis of $x \in X$. □

Remark 2.4.1 In the above proof of (b) ⇒ (a), the obtained metric $d \in \mathrm{Metr}(X)$ has the following property:

$$\mathrm{st}(x, \mathcal{U}_{n+2}) \subset \overline{\mathrm{B}}_d(x, 2^{-n-2}) \subset \mathrm{st}(x, \mathcal{U}_n) \text{ for each } x \in X \text{ and } n \in \mathbb{N}.$$

Moreover, $d(x, y) \leqslant 1$ for every $x, y \in X$.

Remark 2.4.2 Let $\mathcal{U}_1, \mathcal{U}_2, \ldots$be a sequence of open covers of a space X which satisfies (*) in (b), where $\{\mathrm{st}(x, \mathcal{U}_n) \mid n \in \mathbb{N}\}$ is not necessary a neighborhood basis of $x \in X$. We define $\delta(x, y) \geqslant 0$, $x, y \in X$, as in the above proof of (b) ⇒ (a). Since δ satisfies the properties other than the "only if" part of (1), δ induces a *pseudo-metric* d on X by the same definition in the proof.[10] For this pseudo-metric d, the above Remark 2.4.1 is valid. Then, it is easy to see that d *is continuous*.

In Theorem 2.4.1, the equivalence of (a) and (b) is called the ALEXANDROFF–URYSOHN METRIZATION THEOREM and the equivalence of (a) and (c) is called the FRINK METRIZATION THEOREM.

Let $\mathcal{U}$ and $\mathcal{V}$ be covers of X. When $\{\mathrm{st}(x, \mathcal{V}) \mid x \in X\} \prec \mathcal{U}$, we call $\mathcal{V}$ a **Δ-refinement** (or **barycentric refinement**) of $\mathcal{U}$ and write

$$\mathcal{V} \overset{\Delta}{\prec} \mathcal{U} \quad (\text{or } \mathcal{U} \overset{\Delta}{\succ} \mathcal{V}).$$

We call $\mathrm{st}\,\mathcal{V} = \mathrm{st}(\mathcal{V}, \mathcal{V})$ the **star** of $\mathcal{V}$. When $\mathrm{st}\,\mathcal{V} \prec \mathcal{U}$, we call $\mathcal{V}$ a **star-refinement** of $\mathcal{U}$ and write

$$\mathcal{V} \overset{*}{\prec} \mathcal{U} \quad (\text{or } \mathcal{U} \overset{*}{\succ} \mathcal{V}).$$

For each $n \in \mathbb{N}$, $\mathrm{st}^n\,\mathcal{V} = \mathrm{st}^n(\mathcal{V}, \mathcal{V})$ $(= \mathrm{st}(\mathrm{st}^{n-1}\,\mathcal{V}, \mathcal{V}))$ is called the ***n*-th star** of $\mathcal{V}$. Observe that $\mathrm{st}\,\mathcal{V} = \mathrm{st}^1\,\mathcal{V}$, $\mathrm{st}(\mathcal{V}, \mathrm{st}\,\mathcal{V}) = \mathrm{st}^3\,\mathcal{V}$, and $\mathrm{st}(\mathrm{st}\,\mathcal{V}) = \mathrm{st}^4\,\mathcal{V}$. There is the following relation between Δ-refinements and star-refinements:

Proposition 2.4.2 *For every three open covers $\mathcal{U}, \mathcal{V}, \mathcal{W}$ of a space X,*

$$\mathcal{W} \overset{\Delta}{\prec} \mathcal{V} \overset{\Delta}{\prec} \mathcal{U} \;\Rightarrow\; \mathcal{W} \overset{*}{\prec} \mathcal{U}.$$

[10] A **pseudo-metric** is a function $d : X^2 \to \mathbb{R}_+$ such that $d(x, x) = 0$, $d(x, y) = d(y, x)$, and $d(x, z) \leqslant d(x, y) + d(y, x)$ for every $x, y, z \in X$.

Sketch of Proof For each $W \in \mathcal{W}$, take any $x \in W$ and choose $U \in \mathcal{U}$ so that $\mathrm{st}(x, \mathcal{V}) \subset U$. Then, we see that $\mathrm{st}(W, \mathcal{W}) \subset U$.

The following corollary follows from the Alexandroff–Urysohn Metrization Theorem 2.4.1 and Proposition 2.4.2:

Corollary 2.4.3 *For a space X, the following is equivalent:*

(a) *X is metrizable;*

(b) *There is a sequence of open covers $\mathcal{U}_1 \overset{\Delta}{\succ} \mathcal{U}_2 \overset{\Delta}{\succ} \mathcal{U}_3 \overset{\Delta}{\succ} \cdots$ of X such that $\{\mathrm{st}(x, \mathcal{U}_n) \mid n \in \mathbb{N}\}$ is a neighborhood basis of each $x \in X$;*

(c) *There is a sequence of open covers $\mathcal{U}_1 \overset{*}{\succ} \mathcal{U}_2 \overset{*}{\succ} \mathcal{U}_3 \overset{*}{\succ} \cdots$ of X such that $\{\mathrm{st}(x, \mathcal{U}_n) \mid n \in \mathbb{N}\}$ is a neighborhood basis of each $x \in X$.* □

Remark 2.4.3 By tracing the proof of Theorem 2.4.1, we can directly prove the implication (c) ⇒ (a) in Corollary 2.4.3. This direct proof is simpler than that of (b) ⇒ (a) in Theorem 2.4.1, and the obtained metric $d \in \mathrm{Metr}(X)$ has the following property, which is more acceptable than that in Remark 2.4.1:

$$\mathrm{st}(x, \mathcal{U}_n) \subset \overline{\mathrm{B}}_d(x, 2^{-n}) \subset \mathrm{st}(x, \mathcal{U}_{n-1}) \ \text{ for each } x \in X \text{ and } n \in \mathbb{N},$$

where $\mathcal{U}_0 = \{X\}$. Similar to Remark 2.4.1, $d(x, y) \leqslant 1$ for every $x, y \in X$.

Sketch of the direct proof of (c) ⇒ (a) in Corollary 2.4.3. Replicate the proof of (b) ⇒ (a) in Theorem 2.4.1 to construct $d \in \mathrm{Metr}(X)$. Let $\mathcal{U}_0 = \{X\}$. For each $x, y \in X$, we define

$$\delta(x, y) = \inf\left\{2^{-i} \;\middle|\; \exists U \in \mathcal{U}_i \text{ such that } x, y \in U\right\} \text{ and}$$
$$d(x, y) = \inf\left\{\textstyle\sum_{i=1}^{n} \delta(x_{i-1}, x_i) \;\middle|\; n \in \mathbb{N},\ x_0 = x,\ x_n = y\right\}.$$

The admissibility and additional property of d are derived from the inequality $d(x, y) \leqslant \delta(x, y) \leqslant 2d(x, y)$. To prove the right-hand inequality, it suffices to show the following:

$$\delta(x_0, x_n) \leqslant 2\sum_{i=1}^{n} \delta(x_{i-1}, x_i) \ \text{ for each } \ x_0, x_1, \ldots, x_n \in X.$$

This is proved by induction on $n \in \mathbb{N}$. Set $\sum_{i=1}^{n} \delta(x_{i-1}, x_i) = \alpha$ and let k be the largest number such that $\sum_{i=1}^{k} \delta(x_{i-1}, x_i) \leqslant \alpha/2$. Then, $\sum_{i=k+2}^{n} \delta(x_{i-1}, x_i) < \alpha/2$. By the inductive assumption, $\delta(x_0, x_k) \leqslant \alpha$ and $\delta(x_{k+1}, x_n) < \alpha$. Note that $\delta(x_k, x_{k+1}) \leqslant \alpha$. Let $m = \min\{i \in \mathbb{N} \mid 2^{-i} \leqslant \alpha\}$. Since $\mathrm{st}\,\mathcal{U}_m \prec \mathcal{U}_{m-1}$, we can find $U \in \mathcal{U}_{m-1}$ such that $x_0, x_n \in U$, and hence $\delta(x_0, x_n) \leqslant 2^{-m+1} \leqslant 2\alpha$.

Proposition 2.4.4 (Additional Results on Metrizability)

(1) *The perfect image of a metrizable space is metrizable, that is, if $f : X \to Y$ is a surjective perfect map of a metrizable space X, then Y is also metrizable.*

Sketch of Proof For each $y \in Y$ and $n \in \mathbb{N}$, let

$$U_n(y) = \mathrm{N}_d(f^{-1}(y), 2^{-n}) \ \text{ and } \ V_n(y) = Y \setminus f(X \setminus U_n(y)),$$

where d is an admissible metric for X. Since $f(X \setminus U_n(y))$ is closed in Y, $V_n(y)$ is open in Y. Show that $\{V_n(y) \mid n \in \mathbb{N}\}$ is a neighborhood basis of $y \in Y$ that satisfies condition 2.4.1(c). For each $y \in Y$ and $i \in \mathbb{N}$, since $f^{-1}(y)$ is compact, we can choose $j \geqslant i$ so that $U_j(y) \subset f^{-1}(V_{i+1}(y))$. Then, the following holds:

$$V_{j+1}(y) \cap V_{j+1}(z) \neq \emptyset \;\Rightarrow\; V_{j+1}(z) \subset V_i(y).$$

This can be shown as follows:

$$\begin{aligned} V_{j+1}(y) \cap V_{j+1}(z) \neq \emptyset \;&\Rightarrow\; U_{j+1}(y) \cap U_{j+1}(z) \neq \emptyset \;\Rightarrow\; U_j(y) \cap f^{-1}(z) \neq \emptyset \\ &\Rightarrow\; z \in f(U_j(y)) \subset V_{i+1}(y) \;\Rightarrow\; f^{-1}(z) \subset f^{-1}(V_{i+1}(y)) \subset U_{i+1}(y) \\ &\Rightarrow U_{j+1}(z) \subset U_i(y) \;\Rightarrow\; V_{j+1}(z) \subset V_i(y). \end{aligned}$$

(2) *A space X is metrizable if it is a locally finite union of metrizable closed subspaces.*

Sketch of Proof To apply (1) above, construct a surjective perfect map $f : \bigoplus_{\lambda\in\Lambda} X_\lambda \to X$ such that each X_λ is metrizable and $f|X_\lambda$ is a closed embedding. The metrizability of $\bigoplus_{\lambda\in\Lambda} X_\lambda$ easily follows from Theorem 2.3.4. (The metrizability of $\bigoplus_{\lambda\in\Lambda} X_\lambda$ can also be seen by embedding $\bigoplus_{\lambda\in\Lambda} X_\lambda$ into the product space $\Lambda \times \ell_1(\Gamma)^{\mathbb{N}}$ for some Γ, where we give Λ the discrete topology.)

A open cover $\mathcal{U}$ of a space X is called a **normal cover** if it has the following sequence of open refinements:

$$\mathcal{U} \overset{*}{\succ} \mathcal{U}_1 \overset{*}{\succ} \mathcal{U}_2 \overset{*}{\succ} \cdots \quad \big(\text{equiv., } \mathcal{U} \overset{\Delta}{\succ} \mathcal{U}_1 \overset{\Delta}{\succ} \mathcal{U}_2 \overset{\Delta}{\succ} \cdots\big).$$

By Remark 2.4.2 admitting the same adjustment as Remark 2.4.3, for the above sequence of open covers, we have a continuous pseudo-metric ρ on X such that

$$\mathrm{st}(x, \mathcal{U}_n) \subset \overline{\mathrm{B}}_\rho(x, 2^{-n}) \subset \mathrm{st}(x, \mathcal{U}_{n-1}) \;\text{ for each } x \in X \text{ and } n \in \mathbb{N},$$

where $\mathcal{U}_0 = \mathcal{U}$. Then, it is easy to prove the following:

Proposition 2.4.5 *For an open cover $\mathcal{U}$ of a space X, the following are equivalent:*

(a) *$\mathcal{U}$ is normal;*
(b) *There is a continuous pseudo-metric ρ on X such that $\{\overline{\mathrm{B}}_\rho(x, 1) \mid x \in X\} \prec \mathcal{U}$;*
(c) *There is a map $f : X \to Y$ onto a metric space $Y = (Y, d)$ such that*

$$\big\{ f^{-1}(\overline{\mathrm{B}}_d(y, 1)) \;\big|\; y \in Y \big\} \prec \mathcal{U}.$$

Sketch of Proof The implication (a) $\Rightarrow$ (b) has just shown in the above.

(b) $\Rightarrow$ (c): Consider the quotient $Y = X/\sim$ by the following equivalence relation:

$$x \sim x' \underset{\text{def}}{\Leftrightarrow} \rho(x, x') = 0,$$

and define a metric d on Y by $d([x],[y]) = \rho(x,y)$, where $[x] = \{x' \in X \mid x \sim x'\}$. Then, the canonical surjection $f : X \to Y = (Y,d)$ is the desired one.

(c) $\Rightarrow$ (b): Define $\rho(x,y) = d(f(x), f(y))$ for each $x, y \in X$.

(b) $\Rightarrow$ (a): For each $n \in \omega$, let $\mathcal{U}_n = \{\mathrm{B}_\rho(x, 3^{-n})) \mid x \in X\} \in \mathrm{cov}(X)$. Then, $\mathcal{U}_n \overset{*}{\prec} \mathcal{U}_{n-1}$ and $\mathcal{U}_0 \prec \mathcal{U}$.

2.5 Complete Metrizability

In this section, we consider complete metrizability. A space X is a **Baire space** if X has one of the following equivalent properties:

(a) The countable intersection of open dense sets in X is also dense;
(b) Every countable intersection of dense G_δ sets in X is also dense;
(c) The countable union of closed sets in X without interior points has no interior point;
(d) If a countable union of closed sets in X has an interior point, then at least one of them has an interior point.

Property (a) (or any one of above properties) is called the **Baire property**. This property is very valuable. In particular, it can be used to prove various existence theorems. The following statement is easily proved:

- *Every open subspace and every dense G_δ subspace of a Baire space is also Baire.*

Complete metrizability is preferable because of the BAIRE CATEGORY THEOREM:

Theorem 2.5.1 (BAIRE CATEGORY) *Every completely metrizable space X is a Baire space. Consequently, X cannot be written as a union of countably many closed sets without interior points.*

Proof For each $i \in \mathbb{N}$, let G_i be a dense open set in X and $d \in \mathrm{Metr}(X)$ be a complete metric. For each $x \in X$ and $\varepsilon > 0$, we inductively choose $y_i \in X$ and $\varepsilon_i > 0$, $i \in \mathbb{N}$, so that

$$y_i \in \mathrm{B}(y_{i-1}, \varepsilon_{i-1}/2) \cap G_i,\ \mathrm{B}(y_i, \varepsilon_i) \subset G_i \text{ and } \varepsilon_i \leqslant \varepsilon_{i-1}/2,$$

where $y_0 = x$ and $\varepsilon_0 = \varepsilon$ (Fig. 2.6). Then, $(y_i)_{i\in\mathbb{N}}$ is d-Cauchy, hence it converges to some $y \in X$. For each $n \in \omega$,

$$d(y_n, y) \leqslant \sum_{i=n}^{\infty} d(y_i, y_{i+1}) < \sum_{i=n}^{\infty} \tfrac{1}{2}\varepsilon_i \leqslant \sum_{i=1}^{\infty} 2^{-i}\varepsilon_n = \varepsilon_n.$$

Thus, $y \in \mathrm{B}(x,\varepsilon)$ and $y \in \mathrm{B}(y_i, \varepsilon_i) \subset G_i$ for each $i \in \mathbb{N}$, i.e., $y \in \mathrm{B}(x,\varepsilon) \cap \bigcap_{i\in\mathbb{N}} G_i$. Therefore, $\bigcap_{i\in\mathbb{N}} G_i$ is dense in X. □

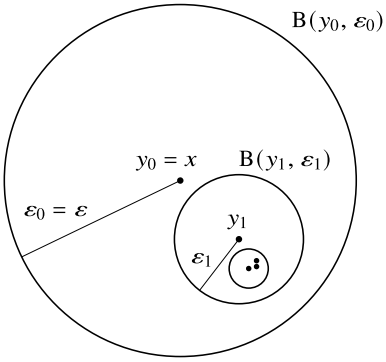


Fig. 2.6 Definition of $y_n \in X$ and $\varepsilon_n > 0$

A metrizable space X is said to be **absolutely G_δ** if X is G_δ in an arbitrary metrizable space that contains X as a subspace. This concept characterizes complete metrizability, which leads us to the following:

Theorem 2.5.2 *A metrizable space is completely metrizable if and only if it is absolutely G_δ.*

This follows from Corollary 2.3.7 (or 2.3.11) and the following theorem:

Theorem 2.5.3 *Let $X = (X, d)$ be a metric space and $A \subset X$.*

(1) *If A is completely metrizable, then A is G_δ in X.*
(2) *If X is complete and A is G_δ in X, then A is completely metrizable.*

Proof (1): Since $\operatorname{cl} A$ is G_δ in X, it suffices to show that A is G_δ in $\operatorname{cl} A$. Let $\rho \in \operatorname{Metr}(A)$ be a complete metric. For each $n \in \mathbb{N}$, let

$$G_n = \Big\{ x \in \operatorname{cl} A \;\Big|\; x \text{ has a neighborhood } U \text{ in } X \text{ with } \operatorname{diam}_d U < 2^{-n} \text{ and } \operatorname{diam}_\rho U \cap A < 2^{-n} \Big\}.$$

Then, each G_n is clearly open in $\operatorname{cl} A$ and $A \subset \bigcap_{n\in\mathbb{N}} G_n$. Each $x \in \bigcap_{n\in\mathbb{N}} G_n$ has neighborhoods $U_1 \supset U_2 \supset \cdots$ in X such that $\operatorname{diam}_d U_n < 2^{-n}$ and $\operatorname{diam}_\rho U_n \cap A < 2^{-n}$. Since $x \in \operatorname{cl} A$, we have points $x_n \in U_n \cap A$, $n \in \mathbb{N}$. Then, $(x_n)_{n\in\mathbb{N}}$ converges to x. Since $(x_n)_{n\in\mathbb{N}}$ is ρ-Cauchy, it is convergent in A. Thus, we can conclude that $x \in A$. Therefore, $A = \bigcap_{n\in\mathbb{N}} G_n$, which is G_δ in $\operatorname{cl} A$.

(2): First, we show that any open set U in X is completely metrizable. We can define an admissible metric ρ for U as follows:

$$\rho(x, y) = d(x, y) + \big| d(x, X \setminus U)^{-1} - d(y, X \setminus U)^{-1} \big|.$$

Every ρ-Cauchy sequence $(x_n)_{n\in\mathbb{N}}$ in U is d-Cauchy, so it converges to some $x \in X$. Since $(d(x_n, X \setminus U)^{-1})_{n\in\mathbb{N}}$ is a Cauchy sequence in $\mathbb{R}$, it is bounded. Then,

$$d(x, X \setminus U) = \lim_{n\to\infty} d(x_n, X \setminus U) > 0.$$

This means that $x \in U$, and hence $(x_n)_{n\in\mathbb{N}}$ is convergent in U. Thus, ρ is complete.

Next, we show that an arbitrary G_δ set A in X is completely metrizable. Write $A = \bigcap_{n\in\mathbb{N}} U_n$, where $U_1, U_2, \ldots$ are open in X. As we saw above, each U_n admits a complete metric $d_n \in \mathrm{Metr}(U_n)$. Now, we can define a metric $\rho \in \mathrm{Metr}(A)$ as follows:

$$\rho(x, y) = \sum_{n\in\mathbb{N}} \min\left\{2^{-n},\ d_n(x, y)\right\}.$$

Every ρ-Cauchy sequence in A is d_n-Cauchy, which is convergent in U_n. Hence, it is convergent in $A = \bigcap_{n\in\mathbb{N}} U_n$. Therefore, ρ is complete. □

Applying Theorem 2.5.2, we can show the following:

Theorem 2.5.4 *Every separable completely metrizable space X can be embedded into $\mathbb{R}^{\mathbb{N}}$ as a closed set.*[11]

Proof Since X is separable metrizable, we may assume that $X \subset \mathbb{R}^{\mathbb{N}}$ (Corollary 2.3.9). Then, X is not necessarily closed in $\mathbb{R}^{\mathbb{N}}$, but X is G_δ in $\mathbb{R}^{\mathbb{N}}$ by Theorem 2.5.2, that is, there are open sets $U_i \subset \mathbb{R}^{\mathbb{N}}$, $i \in \mathbb{N}$, such that $X = \bigcap_{i\in\mathbb{N}} U_i$. Since $\mathbb{R}^{\mathbb{N}} \approx \mathbb{R}^{\mathbb{N}} \times \mathbb{R}^{\mathbb{N}}$, we may construct a closed embedding $h : X \to \mathbb{R}^{\mathbb{N}} \times \mathbb{R}^{\mathbb{N}}$. Let $d \in \mathrm{Metr}(\mathbb{R}^{\mathbb{N}})$ and define maps $f_i : U_i \to \mathbb{R}$, $i \in \mathbb{N}$, by $f_i(x) = d(x, \mathbb{R}^{\mathbb{N}} \setminus U_i)^{-1}$. Then, we have an embedding $h : X \to \mathbb{R}^{\mathbb{N}} \times \mathbb{R}^{\mathbb{N}}$ defined by $h(x) = (x, (f_i(x))_{i\in\mathbb{N}})$.

All that remains is to show that $h(X)$ is closed in $\mathbb{R}^{\mathbb{N}} \times \mathbb{R}^{\mathbb{N}}$. To see this, let $(x_n)_{n\in\mathbb{N}}$ be a sequence in X such that $\lim_{n\to\infty} h(x_n) = (x, (t_i)_{i\in\mathbb{N}}) \in \mathbb{R}^{\mathbb{N}} \times \mathbb{R}^{\mathbb{N}}$. Then, $\lim_{n\to\infty} x_n = x$ and $\lim_{n\to\infty} f_i(x_n) = t_i$ for each $i \in \mathbb{N}$. Now, suppose that $x \notin U_i$ for some $i \in \mathbb{N}$. Since $\lim_{n\to\infty} d(x_n, \mathbb{R}^{\mathbb{N}} \setminus U_i) = d(x, \mathbb{R}^{\mathbb{N}} \setminus U_i) = 0$, it follows that

$$\lim_{n\to\infty} f_i(x_n) = \lim_{n\to\infty} d(x_n, \mathbb{R}^{\mathbb{N}} \setminus U_i)^{-1} = \infty,$$

which contradicts $\lim_{n\to\infty} f_i(x_n) = t_i$. Thus, we have $x \in \bigcap_{i\in\mathbb{N}} U_i = X$. Hence, $t_i = \lim_{n\to\infty} f_i(x_n) = f_i(x)$ for each $i \in \mathbb{N}$, which means that $(x, (t_i)_{i\in\mathbb{N}}) = (x, (f_i(x))_{i\in\mathbb{N}}) = h(x) \in h(X)$. Therefore, $h(X)$ is closed in $\mathbb{R}^{\mathbb{N}} \times \mathbb{R}^{\mathbb{N}}$. □

Analogous to compactness, the completeness of metric spaces can be characterized by the finite intersection property (f.i.p.), as follows:

Theorem 2.5.5 *In order for a metric space $X = (X, d)$ to be complete, it is necessary and sufficient that, if a family $\mathcal{F}$ of subsets of X has the finite intersection property and contains sets with arbitrarily small diameter, then $\mathcal{F}^{\mathrm{cl}}$ has a non-empty intersection, which is a singleton.*

[11] Although $\mathbb{R}^{\mathbb{N}} \approx \ell_2$ as mentioned in Notes for Chap. 1, we will directly prove that every separable completely metrizable space can be embedded in ℓ_2 a closed set (Theorem 6.2.4).

Proof (*Necessity*) Let $\mathcal{F}$ be a family of subsets of X with the f.i.p. such that $\mathcal{F}$ contains sets with arbitrarily small diameter. For each $n \in \mathbb{N}$, choose $F_n \in \mathcal{F}$ so that $\operatorname{diam} F_n < 2^{-n}$, and take $x_n \in F_n$. For any $n < m$, $F_n \cap F_m \neq \emptyset$, hence

$$d(x_n, x_m) \leqslant \operatorname{diam} F_n + \operatorname{diam} F_m < 2^{-n} + 2^{-m} < 2^{-n+1}.$$

Thus, $(x_n)_{n\in\mathbb{N}}$ is a Cauchy sequence, therefore it converges to a point $x \in X$. Then, $x \in \bigcap \mathcal{F}^{\mathrm{cl}}$. Otherwise, $x \notin \operatorname{cl} F$ for some $F \in \mathcal{F}$. Choose $n \in \mathbb{N}$ so that $d(x, x_n),\ 2^{-n} < \frac{1}{2} d(x, F)$. Since $F \cap F_n \neq \emptyset$, it follows that

$$d(x, F) \leqslant d(x, x_n) + \operatorname{diam} F_n < d(x, x_n) + 2^{-n} < d(x, F),$$

which is a contradiction.

(*Sufficiency*) Let $(x_n)_{n\in\mathbb{N}}$ be a Cauchy sequence in X. For each $n \in \mathbb{N}$, let $F_n = \{x_i \mid i \geqslant n\}$. Then, $F_1 \supset F_2 \supset \cdots$ and $\operatorname{diam} F_n \to 0\ (n \to \infty)$. From this condition, we have $x \in \bigcap_{n\in\mathbb{N}} \operatorname{cl} F_n$. For each $\varepsilon > 0$, choose $n \in \mathbb{N}$ so that $\operatorname{diam} \operatorname{cl} F_n = \operatorname{diam} F_n < \varepsilon$. Then, $d(x_i, x) < \varepsilon$ for $i \geqslant n$. Therefore, $\lim_{n\to\infty} x_n = x$. Thus, X is complete. □

Using compactifications, we can characterize complete metrizability as follows:

Theorem 2.5.6 *For a metrizable space X, the following are equivalent:*

(a) *X is completely metrizable;*
(b) *X is G_δ in an arbitrary compactification of X;*
(c) *X is G_δ in the Stone–Čech compactification βX;*
(d) *X has a compactification in which X is G_δ.*

Proof The implications (b) ⇒ (c) ⇒ (d) are obvious. We show the converse (d) ⇒ (c) ⇒ (b) and the equivalence (a) ⇔ (b).

(d) ⇒ (c): Let γX be a compactification of X and $X = \bigcap_{n\in\mathbb{N}} G_n$, where each G_n is open in γX. Then, by Theorem 2.1.4, we have a map $f : \beta X \to \gamma X$ such that $f|X = \mathrm{id}$, where $X = f^{-1}(X)$ by Theorem 2.1.15. Consequently, $X = \bigcap_{n\in\mathbb{N}} f^{-1}(G_n)$ is G_δ in βX.

(c) ⇒ (b): By condition (c), we can write $\beta X \setminus X = \bigcup_{n\in\mathbb{N}} F_n$, where each F_n is closed in βX. For any compactification γX of X, we have a map $f : \beta X \to \gamma X$ such that $f|X = \mathrm{id}$ (Theorem 2.1.4). From Theorem 2.1.15, $\gamma X \setminus X = f(\beta X \setminus X) = \bigcup_{n\in\mathbb{N}} f(F_n)$ is F_σ in γX, hence X is G_δ in γX.

(b) ⇒ (a): To prove the complete metrizability of X, we show that X is absolutely G_δ (Theorem 2.5.2). Let X be contained in a metrizable space Y. Since $\operatorname{cl}_{\beta Y} X$ is a compactification of X, it follows from (b) that X is G_δ in $\operatorname{cl}_{\beta Y} X$, and hence it is G_δ in $Y \cap \operatorname{cl}_{\beta Y} X = \operatorname{cl}_Y X$, where $\operatorname{cl}_Y X$ is also G_δ in Y. Therefore, X is G_δ in Y.

(a) ⇒ (b): Let γX be a compactification of X and d an admissible complete metric for X. For each $n \in \mathbb{N}$ and $x \in X$, let $G_n(x)$ be an open set in γX such that $G_n(x) \cap X = \mathrm{B}_d(x, 2^{-n})$. Then, $G_n = \bigcup_{x\in X} G_n(x)$ is open in γX and $X \subset G_n$. We

will show that each $y \in \bigcap_{n\in\mathbb{N}} G_n$ is contained in X. This implies that $X = \bigcap_{n\in\mathbb{N}} G_n$ is G_δ in γX.

For each $n \in \mathbb{N}$, choose $x_n \in X$ so that $y \in G_n(x_n)$. Since $y \in \mathrm{cl}_{\gamma X} X$ and $G_n(x_n) \cap X = \mathrm{B}_d(x_n, 2^{-n})$, it follows that $\{\mathrm{B}_d(x_n, 2^{-n}) \mid n \in \mathbb{N}\}$ has the f.i.p. By Theorem 2.5.5, we have $x \in \bigcap_{n\in\mathbb{N}} \mathrm{cl}_X \mathrm{B}_d(x_n, 2^{-n})$, where $\lim_{n\to\infty} x_n = x$ because $d(x_n, x) \leqslant 2^{-n}$. Thus, we have $y = x \in X$. Otherwise, there would be disjoint open sets U and V in γX such that $x \in U$ and $y \in V$. Since $y \in \bigcap_{n\in\mathbb{N}} G_n \cap V$, $\{\mathrm{B}_d(x_n, 2^{-n}) \cap V \mid n \in \mathbb{N}\}$ has the f.i.p. Again, by Theorem 2.5.5, we have

$$x' \in \bigcap_{n\in\mathbb{N}} \mathrm{cl}_X(\mathrm{B}_d(x_n, 2^{-n}) \cap V) \subset \mathrm{cl}_X V.$$

Since $\lim_{n\to\infty} x_n = x'$ is the same as x, it follows that $x' = x \in U$, which is a contradiction. □

Note that conditions (b), (c), and (d) in Theorem 2.5.6 are equivalent without the metrizability of X, but X should be assumed to be Tychonoff in order that X has a compactification. A Tychonoff space X is said to be **Čech-complete** if X satisfies one of these conditions.

Evidently, every compact metric space is complete. A non-compact locally compact metrizable space X is open in the one-point compactification $\alpha X = X \cup \{\infty\}$. Then, X is completely metrizable by Theorem 2.5.6. Thus, we have the following corollary:

Corollary 2.5.7 *Every locally compact metrizable space is completely metrizable.* □

Remark 2.5.1 When X is a locally compact separable metrizable space, the one-point compactification αX is metrizable by Corollary 2.3.6. Then, applying Theorem 2.5.3 (2), we can see the complete metrizability of X. Furthermore, the above corollary can also obtained by the following fact:

- *Every locally compact metrizable space X can be written as a topological sum of locally compact separable metrizable spaces.*

  Indeed, by Stone's Theorem 2.3.1, X has a locally finite open cover $\mathcal{U}$ such that each $U \in \mathcal{U}$ has the compact closure. Then, for each $U \in \mathcal{U}$, $\mathcal{U}[\mathrm{cl}U]$ is finite because $\mathrm{cl}\, U$ is compact. Thus, $\mathcal{U}$ is star-finite. We define an equivalence relation $\sim$ on X as follows:

$$x \sim y \underset{\text{def}}{\Leftrightarrow} \exists U_1, \ldots, U_n \in \mathcal{U} \text{ such that } x \in U_1, y \in U_n, \\ U_i \cap U_{i+1} \neq \emptyset \text{ for each } i = 1, \ldots, n-1.$$

  Then, each equivalence class is a clopen set in X, which is a countable union of compact sets, so it is a locally compact separable metrizable.

We now state and prove the LAVRENTIEFF G_δ- EXTENSION THEOREM:

Theorem 2.5.8 (LAVRENTIEFF G_δ- EXTENSION) *Let $f : A \to Y$ be a map from a subset A of a space X to a completely metrizable space Y. Then, f extends over a G_δ set G in* cl A *with $A \subset G$.*

Proof We may assume that $Y = (Y, d)$ is a complete metric space. The **oscillation** of f at $x \in \operatorname{cl} A$ is defined as follows:

$$\operatorname{osc}_f(x) = \inf \left\{ \operatorname{diam} f(A \cap U) \;\middle|\; U \text{ is an open neighborhood of } x \right\}.$$

Let $G = \{x \in \operatorname{cl} A \mid \operatorname{osc}_f(x) = 0\}$. Then, $A \subset G$ because f is continuous. Since each $\{x \in \operatorname{cl} A \mid \operatorname{osc}_f(x) < 1/n\}$ is open in $\operatorname{cl} A$, it follows that G is G_δ in $\operatorname{cl} A$. For each $x \in G$, consider the following:

$$\mathcal{F}_x = \left\{ f(A \cap U) \;\middle|\; U \text{ is an open neighborhood of } x \right\},$$

which has the f.i.p. and contains sets with arbitrarily small diameter. By Theorem 2.5.5, we have $\bigcap \mathcal{F}_x^{\mathrm{cl}} \neq \emptyset$, which is a singleton because $\operatorname{diam} \bigcap \mathcal{F}_x^{\mathrm{cl}} = 0$. The desired extension $\tilde{f} : G \to Y$ of f can be defined by $\tilde{f}(x) \in \bigcap \mathcal{F}_x^{\mathrm{cl}}$. □

If A is a subspace of a metric space X and Y is a complete metric space, then every uniformly continuous map $f : A \to Y$ extends over $\operatorname{cl} A$. This result can be obtained by showing that $G = \operatorname{cl} A$ in the above proof. But, a direct proof is easier.

We will modify Theorem 2.5.8 into the following, known as the LAVRENTIEFF HOMEOMORPHISM EXTENSION THEOREM:

Theorem 2.5.9 (LAVRENTIEFF HOMEOMORPHISM EXTENSION) *Let X and Y be completely metrizable spaces and let $f : A \to B$ be a homeomorphism between $A \subset X$ and $B \subset Y$. Then, f extends to a homeomorphism $\tilde{f} : G \to H$ between G_δ sets in X and Y such that $A \subset G \subset \operatorname{cl} A$ and $B \subset H \subset \operatorname{cl} B$.*

Proof By Theorem 2.5.8, f and f^{-1} extend to maps $g : G' \to Y$ and $h : H' \to X$, where $A \subset G' \subset \operatorname{cl} A$, $B \subset H' \subset \operatorname{cl} B$ and G', H' are G_δ in X and Y, respectively. Then, we have G_δ sets $G = g^{-1}(H')$ and $H = h^{-1}(G')$ that contain A and B as dense subsets, respectively. Consider the maps $h(g|G) : G \to X$ and $g(h|H) : H \to Y$. Since $h(g|G)|A = \mathrm{id}_A$ and $g(h|H)|B = \mathrm{id}_B$, it follows that $h(g|G) = \mathrm{id}_G$ and $g(h|H) = \mathrm{id}_H$. Then, as is easily observed, we have $g(G) \subset H$ and $h(H) \subset G$. Hence, $\tilde{f} = g|G : G \to H$ is a homeomorphism extending f. □

In the above, when $X = Y$ and $A = B$, we can take $G = H$, that is, we can show the following:

Corollary 2.5.10 *Let X be a completely metrizable space and $A \subset X$. Then, every homeomorphism $f : A \to A$ extends to a homeomorphism $\tilde{f} : G \to G$ over a G_δ set G in X with $A \subset G \subset \operatorname{cl} A$.*

Proof Using Theorem 2.5.9, we extend f to a homeomorphism $g : G' \to G''$ between G_δ sets G', $G'' \subset X$ with $A \subset G' \cap G''$ and G', $G'' \subset \operatorname{cl} A$. We inductively define a sequence of G_δ sets $G' \cap G'' = G_1 \supset G_2 \supset \cdots$ in X as follows:

$$G_{n+1} = G_n \cap g(G_n) \cap g^{-1}(G_n).$$

Then, $G = \bigcap_{n\in\mathbb{N}} G_n$ is G_δ in X and $g(x), g^{-1}(x) \in G$ for each $x \in G$. Indeed, for each $n \in \mathbb{N}$, since $x \in G_{n+1}$, it follows that $g(x) \in G_n$ and $g^{-1}(x) \in G_n$. Thus, $\tilde{f} = g|G : G \to G$ is the desired extension of f. □

Proposition 2.5.11 (Additional Results on Complete Metrizability)

(1) *Let $f : X \to Y$ be a surjective perfect map between Tychonoff spaces. Then, X is Čech-complete if and only if Y is Čech-complete. When X is metrizable, X is completely metrizable if and only if Y is completely metrizable.*

Sketch of Proof See Theorem 2.1.15.

(2) *A space X is completely metrizable if it is a locally finite union of completely metrizable closed subspaces.*

Sketch of Proof Imitate the proof of Proposition 2.4.4(2). To prove the complete metrizability of the topological sum $\bigoplus_{\lambda\in\Lambda} X_\lambda$ of completely metrizable spaces, embed $\bigoplus_{\lambda\in\Lambda} X_\lambda$ into the product space $\Lambda \times \ell_1(\Gamma)^{\mathbb{N}}$ for some Γ.

2.6 Paracompactness and Local Properties

A space X is **paracompact** if each open cover of X has a locally finite open refinement.[12] According to Stone's Theorem 2.3.1, we have the following fact:

- *Every metrizable space is paracompact.*

A space X is **collectionwise normal** if, for each discrete collection $\mathcal{F}$ of closed sets in X, there is a pairwise disjoint collection $\{U_F \mid F \in \mathcal{F}\}$ of open sets in X such that $F \subset U_F$ for each $F \in \mathcal{F}$. Obviously, every collectionwise normal space is normal. In the definition of collectionwise normality, $\{U_F \mid F \in \mathcal{F}\}$ can be discrete in X.

Indeed, choose an open set V in X so that $\bigcup\mathcal{F} \subset V \subset \operatorname{cl} V \subset \bigcup_{F\in\mathcal{F}} U_F$. Then, $F \subset V \cap U_F$ for each $F \in \mathcal{F}$, and $\{V \cap U_F \mid F \in \mathcal{F}\}$ is discrete in X.

Theorem 2.6.1 *Every paracompact space X is collectionwise normal.*

Proof To see the regularity of X, let A be a closed set in X and $x \in X \setminus A$. Each $a \in A$ has an open neighborhood U_a in X so that $x \notin \operatorname{cl} U_a$. Let $\mathcal{U}$ be a locally finite open refinement of

$$\{U_a \mid a \in A\} \cup \{X \setminus A\} \in \operatorname{cov}(X).$$

Then, $V = \operatorname{st}(A, \mathcal{U}) = \bigcup \mathcal{U}[A]$ is an open neighborhood of A. Since $\mathcal{U}$ is locally finite, it follows that $\operatorname{cl} V = \bigcup \mathcal{U}[A]^{\mathrm{cl}}$. Since each $U \in \mathcal{U}[A]$ is contained in some U_a, it follows that $x \notin \operatorname{cl} U$, and hence $x \notin \operatorname{cl} V$.

[12] Recall that spaces are assumed to be **Hausdorff**.

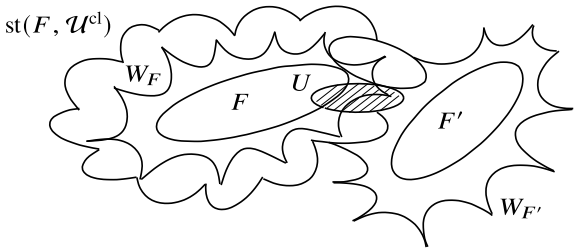


Fig. 2.7 The pairwise disjoint collection $\{W_F \mid F \in \mathcal{F}\}$

We now show that X is collectionwise normal. Let $\mathcal{F}$ be a discrete collection of closed sets in X. Since X is regular, each $x \in X$ has an open neighborhood V_x in X such that $\operatorname{card} \mathcal{F}[\operatorname{cl} V_x] \leqslant 1$. Let $\mathcal{U}$ be a locally finite open refinement of $\{V_x \mid x \in X\} \in \operatorname{cov}(X)$. For each $F \in \mathcal{F}$, we define

$$W_F = X \setminus \bigcup \big\{ \operatorname{cl} U \;\big|\; U \in \mathcal{U},\ F \cap \operatorname{cl} U = \emptyset \big\}.$$

Then, W_F is open in X and $F \subset W_F \subset \operatorname{st}(F, \mathcal{U}^{\mathrm{cl}})$ (cf. Fig. 2.7). Since $\operatorname{card} \mathcal{F}[\operatorname{cl} U] \leqslant 1$ for each $U \in \mathcal{U}$, it follows that $\operatorname{st}(F, \mathcal{U}^{\mathrm{cl}}) \cap W_{F'} = \emptyset$ if $F' \neq F \in \mathcal{F}$. Therefore, $\{W_F \mid F \in \mathcal{F}\}$ is pairwise disjoint. □

A **shrinking** of an open cover $\mathcal{U}$ (or $\{U_\lambda \mid \lambda \in \Lambda\}$) of X is a cover $\mathcal{A} = \{A_U \mid U \in \mathcal{U}\}$ (or $\mathcal{A} = \{A_\lambda \mid \lambda \in \Lambda\}$) of X such that $A_U \subset U$ for each $U \in \mathcal{U}$ (or $A_\lambda \subset U_\lambda$ for each $\lambda \in \Lambda$). A shrinking $\mathcal{A}$ is said to be **open** or **closed** depending on whether all members of $\mathcal{A}$ are open or closed in X. An open shrinking $\mathcal{A}$ is said to be **strong** if $\operatorname{cl} A_U \subset U$ for each $U \in \mathcal{U}$ (or $\operatorname{cl} A_\lambda \subset U_\lambda$ for each $\lambda \in \Lambda$). Evidently, every shrinking $\mathcal{A}$ of $\mathcal{U}$ is a refinement of $\mathcal{U}$, which is locally finite or point-finite in X if so is $\mathcal{U}$.

Lemma 2.6.2 *Let X be regular. Suppose that each open cover of X has a locally finite refinement (consisting of arbitrary sets). Then, every open cover $\mathcal{U}$ of X has a locally finite closed shrinking $\{F_U \mid U \in \mathcal{U}\}$.*

Proof Since X is regular, we have $\mathcal{V} \in \operatorname{cov}(X)$ such that $\mathcal{V}^{\mathrm{cl}} \prec \mathcal{U}$. Let $\mathcal{A}$ be a locally finite refinement of $\mathcal{V}$. There exists a function $\varphi : \mathcal{A} \to \mathcal{U}$ such that $\operatorname{cl} A \subset \varphi(A)$ for each $A \in \mathcal{A}$. For each $U \in \mathcal{U}$, define

$$F_U = \bigcup \big\{ \operatorname{cl} A \;\big|\; A \in \varphi^{-1}(U) \big\} \subset U.$$

Since each $x \in X$ is contained in some $A \in [A]$ and $A \subset F_{\varphi(A)}$, $\{F_U \mid U \in \mathcal{U}\}$ is a cover of X. Since $\mathcal{A}$ is locally finite, each F_U is closed in X and $\{F_U \mid U \in \mathcal{U}\}$ is locally finite. □

Now, we can characterize paracompactness as follows:

Theorem 2.6.3 *For a space X, the following conditions are equivalent:*

(a) *X is paracompact;*
(b) *Every open cover of X has an open Δ-refinement;*
(c) *Every open cover of X has an open star-refinement;*
(d) *Every open cover of X is a normal cover, that is, there exists a sequence of open star-refinements*

$$\mathcal{U} \overset{*}{\succ} \mathcal{U}_1 \overset{*}{\succ} \mathcal{U}_2 \overset{*}{\succ} \cdots;$$

(e) *X is regular and each open cover of X has a σ-discrete open refinement;*
(f) *X is regular and each open cover of X has a locally finite refinement.*

Proof The equivalences (b) $\Leftrightarrow$ (c) $\Leftrightarrow$ (d) are trivial. Indeed, due to Proposition 2.4.2, for $\mathcal{U}, \mathcal{V}, \mathcal{W} \in \mathrm{cov}(X)$,

$$\mathcal{W} \overset{\Delta}{\prec} \mathcal{V} \overset{\Delta}{\prec} \mathcal{U} \;\Rightarrow\; \mathcal{W} \overset{*}{\prec} \mathcal{U}.$$

(a) $\Rightarrow$ (b): Let $\mathcal{U} \in \mathrm{cov}(X)$. By Lemma 2.6.2, $\mathcal{U}$ has a locally finite closed shrinking $\{F_U \mid U \in \mathcal{U}\}$. For each $x \in X$, define

$$W_x = \bigcap \{U \in \mathcal{U} \mid x \in F_U\} \setminus \bigcup \{F_U \mid U \in \mathcal{U},\ x \notin F_U\}.$$

Then, W_x is an open neighborhood of x in X, hence $\mathcal{W} = \{W_x \mid x \in X\} \in \mathrm{cov}(X)$. For each $x \in X$, choose $U \in \mathcal{U}$ so that $x \in F_U$. If $x \in W_y$ then $y \in F_U$, which implies that $W_y \subset U$. Therefore, $\mathrm{st}(x, \mathcal{W}) \subset U$ for each $x \in X$, which means that $\mathcal{W}$ is a Δ-refinement of $\mathcal{U}$.

(c), (d) $\Rightarrow$ (e): To prove the regularity of X, let $\mathrm{A} \subset X$ be closed and $x \in X \setminus A$. Then, $\{X \setminus A, X \setminus \{x\}\} \in \mathrm{cov}(X)$ has an open star-refinement $\mathcal{W}$. Choose $W \in \mathcal{W}$ so that $x \in W$. Then, $\mathrm{st}(W, \mathcal{W}) \subset X \setminus A$, i.e., $W \cap \mathrm{st}(A, \mathcal{W}) = \emptyset$. Hence, X is regular.

Next, we show that each $\mathcal{U} \in \mathrm{cov}(X)$ has a σ-discrete open refinement. We may assume that $\mathcal{U} = \{U_\lambda \mid \lambda \in \Lambda\}$, where $\Lambda = (\Lambda, \leqslant)$ is a well-ordered set. Using the sequence of open star-refinements in (d), for each $(\lambda, n) \in \Lambda \times \mathbb{N}$, we define

$$U_{\lambda,n} = \bigcup \{U \in \mathcal{U}_n \mid \mathrm{st}(U, \mathcal{U}_n) \subset U_\lambda\} \subset U_\lambda.$$

Then, we have

$(*)$ $\mathrm{st}(U_{\lambda,n}, \mathcal{U}_{n+1}) \subset U_{\lambda,n+1}$ for each $(\lambda, n) \in \Lambda \times \mathbb{N}$.

Indeed, each $U \in \mathcal{U}_{n+1}[U_{\lambda,n}]$ meets some $U' \in \mathcal{U}_n$ such that $\mathrm{st}(U', \mathcal{U}_n) \subset U_\lambda$. Since $U \subset \mathrm{st}(U', \mathcal{U}_{n+1})$, it follows that

$$\mathrm{st}(U, \mathcal{U}_{n+1}) \subset \mathrm{st}^2(U', \mathcal{U}_{n+1}) \subset \mathrm{st}(U', \mathrm{st}\,\mathcal{U}_{n+1}) \subset \mathrm{st}(U', \mathcal{U}_n) \subset U_\lambda,$$

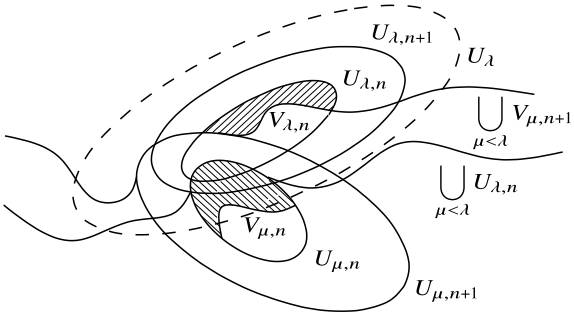


Fig. 2.8 Definition of $V_{\lambda,n}$

which implies that $U \subset U_{\lambda,n+1}$. Thus, we have $(*)$.

Now, for each $(\lambda, n) \in \Lambda \times \mathbb{N}$, let

$$V_{\lambda,n} = U_{\lambda,n} \setminus \operatorname{cl} \textstyle\bigcup_{\mu<\lambda} U_{\mu,n+1} \subset U_\lambda.$$

Then, each $\mathcal{V}_n = \{V_{\lambda,n} \mid \lambda \in \Lambda\}$ is discrete in X. Indeed, each $x \in X$ is contained in some $U \in \mathcal{U}_{n+1}$. If $U \cap V_{\mu,n} \neq \emptyset$, then $U \subset \operatorname{st}(U_{\mu,n}, \mathcal{U}_{n+1}) \subset U_{\mu,n+1}$ by $(*)$. Hence, $U \cap V_{\lambda,n} = \emptyset$ for all $\lambda > \mu$. This implies that U meets at most one member of $\mathcal{V}_n$. See Fig. 2.8.

It remains to be proved that $\mathcal{V} = \bigcup_{n\in\mathbb{N}} \mathcal{V}_n \in \operatorname{cov}(X)$. Each $x \in X$ is contained in some $U \in \mathcal{U}_1$. Since $\operatorname{st}(U, \mathcal{U}_1) \subset U_\lambda$ for some $\lambda \in \Lambda$, it follows that $x \in U_{\lambda,1}$. Thus, we can define

$$\lambda(x) = \min \left\{ \lambda \in \Lambda \;\middle|\; x \in U_{\lambda,n} \text{ for some } n \in \mathbb{N} \right\}.$$

Then, $x \in U_{\lambda(x),n}$ for some $n \in \mathbb{N}$. It follows from $(*)$ that

$$\begin{aligned} \operatorname{cl} \textstyle\bigcup_{\mu<\lambda(x)} U_{\mu,n+1} &\subset \operatorname{st}\left(\textstyle\bigcup_{\mu<\lambda(x)} U_{\mu,n+1}, \mathcal{U}_{n+2} \right) \\ &= \bigcup_{\mu<\lambda(x)} \operatorname{st}(U_{\mu,n+1}, \mathcal{U}_{n+2}) \subset \bigcup_{\mu<\lambda(x)} U_{\mu,n+2}, \end{aligned}$$

hence $x \notin \operatorname{cl} \bigcup_{\mu<\lambda(x)} U_{\mu,n+1}$. Therefore, $x \in V_{\lambda(x),n}$, and hence $\mathcal{V} \in \operatorname{cov}(X)$. Consequently, $\mathcal{V}$ is a σ-discrete open refinement of $\mathcal{U}$.

(e) $\Rightarrow$ (f): It suffices to show that every σ-discrete open cover $\mathcal{U}$ of X has a locally finite refinement. Let $\mathcal{U} = \bigcup_{n\in\mathbb{N}} \mathcal{U}_n$, where each $\mathcal{U}_n$ is discrete in X and $\mathcal{U}_n \cap \mathcal{U}_m = \emptyset$ if $n \neq m$. For each $U \in \mathcal{U}_n$, let $A_U = U \setminus \bigcup_{m<n} (\bigcup \mathcal{U}_m)$. Then, $\mathcal{A} = \{A_U \mid U \in \mathcal{U}\}$ is a cover of X that refines $\mathcal{U}$. For each $x \in X$, choose the smallest $n \in \mathbb{N}$ such that $x \in \bigcup \mathcal{U}_n$ and let $x \in U_0 \in \mathcal{U}_n$. Then, U_0 misses A_U for all $U \in \bigcup_{m>n} \mathcal{U}_m$. For each $m \leqslant n$, since $\mathcal{U}_m$ is discrete, x has a neighborhood V_m in X

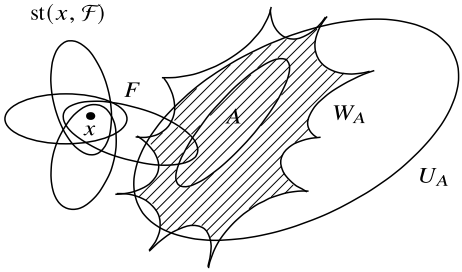


Fig. 2.9 Definition of W_A

such that $\operatorname{card} \mathcal{U}_m[V_m] \leqslant 1$. Then, $V = U_0 \cap V_1 \cap \cdots \cap V_n$ is a neighborhood of x in X such that $\operatorname{card} \mathcal{A}[V] \leqslant n$. Hence, $\mathcal{A}$ is locally finite in X.

(e) $\Rightarrow$ (a): Let $\mathcal{U} \in \operatorname{cov}(X)$. Then $\mathcal{U}$ has a locally finite refinement $\mathcal{A}$. For each $x \in X$, choose an open neighborhood V_x of x in X so that $\operatorname{card} \mathcal{A}[V_x] < \aleph_0$. According to Lemma 2.6.2, $\{V_x \mid x \in X\} \in \operatorname{cov}(X)$ has a locally finite closed refinement $\mathcal{F}$. Then, $\operatorname{card} \mathcal{A}[F] < \aleph_0$ for each $F \in \mathcal{F}$. For each $A \in \mathcal{A}$, choose $U_A \in \mathcal{U}$ so that $A \subset U_A$ and define

$$W_A = U_A \setminus \bigcup \{F \in \mathcal{F} \mid A \cap F = \emptyset\}.$$

Then, $A \subset W_A \subset U_A$ and W_A is open in X, hence $\mathcal{W} = \{W_A \mid A \in \mathcal{A}\}$ is an open refinement of $\mathcal{U}$. Since $\mathcal{F}$ is a locally finite closed cover of X, $\operatorname{st}(x, \mathcal{F})$ is a neighborhood of $x \in X$. For each $F \in \mathcal{F}$ and $A \in \mathcal{A}$, $F \cap W_A \neq \emptyset$ implies $F \cap A \neq \emptyset$. Then, $\operatorname{card} \mathcal{W}[F] \leqslant \operatorname{card} \mathcal{A}[F] < \aleph_0$ for each $F \in \mathcal{F}$. Since $\operatorname{card} \mathcal{F}[x] < \aleph_0$, $\operatorname{st}(x, \mathcal{F})$ meets only finitely many members of $\mathcal{W}$. Therefore, $\mathcal{W}$ is locally finite in X. See Fig. 2.9. □

In the above, the implication (d) $\Rightarrow$ (a) can be also obtained as a direct consequence of the following proposition:

Proposition 2.6.4 *Every normal cover $\mathcal{U}$ of a space X has a locally finite refinement consisting of cozero sets in X.*

Proof By Proposition 2.4.5, we have a map $f : X \to Y$ onto a metric space $Y = (Y, d)$ such that $\{f^{-1}(\overline{\mathrm{B}}_d(y, 1)) \mid y \in Y\} \prec \mathcal{U}$. Since Y is paracompact by Stone's Theorem 2.3.1, $\{\mathrm{B}_d(y, 1) \mid y \in Y\}$ has a locally finite open refinement $\mathcal{V}$. Then, $f^{-1}(\mathcal{V})$ is a locally finite open refinement of $\mathcal{U}$. Since every open set in a metric space is a cozero set, $f^{-1}(V)$ is a cozero set in X for each $V \in \mathcal{V}$. □

A space X is **Lindelöf** if every open cover of X has a countable subcover. By verifying condition (e) above, we have the following:

Corollary 2.6.5 *Every regular Lindelöf space is paracompact.* □

A condition weaker than (c) in Theorem 2.6.3 characterizes normality, that is:

Proposition 2.6.6 *A space X is normal if and only if each finite open cover $\mathcal{U} = \{U_1, \ldots, U_n\}$ of X has a (finite) open star-refinement.*

Sketch of Proof (*The "if" Part*) Let A and B be disjoint closed sets in X. The open cover $\{X \setminus A, X \setminus B\}$ of X has an open star-refinement $\mathcal{V}$. Then, $\mathrm{st}(A, \mathcal{V})$ and $\mathrm{st}(B, \mathcal{V})$ are disjoint open neighborhoods of A and B in X, respectively.

(*The "only if" Part*) We first show the case $n = 2$. For each $\mathcal{U} = \{U_1, U_2\} \in \mathrm{cov}(X)$, by the normality of X, we can take disjoint open sets V_1 and V_2 in X so that $X \setminus U_i \subset V_i$. Moreover, take open sets W_1 and W_2 in X so that $X \setminus U_i \subset W_i \subset \mathrm{cl}\, W_i \subset V_i$. Then, the following is the desired (finite) refinement:

$$\mathcal{V} = \left\{ W_i, V_i \cap U_i, X \setminus (\mathrm{cl}\, W_1 \cup \mathrm{cl}\, W_2) \;\middle|\; i = 1, 2 \right\}.$$

In the case $n > 2$, inductively choose open sets W_k so that

$$X \setminus \left(\bigcup_{i<k} W_i \cup \bigcup_{j>k} U_j \right) \subset W_k \subset \mathrm{cl}\, W_k \subset U_k.$$

Then, $X = W_1 \cup \cdots \cup W_n$. Applying the case $n = 2$, we can obtain a finite open star-refinement $\mathcal{V}_i$ of $\{U_i, X \setminus \mathrm{cl}\, W_i\}$. Then, $\mathcal{V}_1 \wedge \cdots \wedge \mathcal{V}_n$ is a finite open star-refinement of $\mathcal{U}$. Indeed,

$$\mathrm{st}(V_1 \cap \cdots \cap V_n, \mathcal{V}_1 \wedge \cdots \wedge \mathcal{V}_n) \subset \bigcap_{i=1}^{n} \mathrm{st}(V_i, \mathcal{V}_i).$$

When some $\mathrm{st}(V_k, \mathcal{V}_k)$ is contained in U_k, $\bigcap_{i=1}^{n} \mathrm{st}(V_i, \mathcal{V}_i) \subset U_k$. Otherwise, every $\mathrm{st}(V_i, \mathcal{V}_i)$ is contained in $X \setminus \mathrm{cl}\, W_i$. Then, $\bigcap_{i=1}^{n} \mathrm{st}(V_i, \mathcal{V}_i) \subset X \setminus \bigcup_{i=1}^{n} \mathrm{cl}\, W_i = \emptyset$.

According to Proposition 2.6.6 above, condition (c) in Theorem 2.6.3 can be regarded as a condition for a certain kind of normality. So, it is said that a space X is **fully normal** if X satisfies (c) (equiv. (d)) in Theorem 2.6.3, where the term "*fully*" is suitable because a normal space can be characterized as a space such that every *finite* open cover is normal.

Let $\mathcal{P}$ be a property of subsets of a space X. It is said that X has property $\mathcal{P}$ **locally** if each $x \in X$ has a neighborhood U in X that has property $\mathcal{P}$. Occasionally, we need to determine whether X has some property $\mathcal{P}$ if X has property $\mathcal{P}$ locally. Let us consider this problem now. A property $\mathcal{P}$ of *open* sets in X is said to be ***G*-hereditary** if the following conditions are satisfied:

(G-1) If U has property $\mathcal{P}$, then every open subset of U has $\mathcal{P}$;
(G-2) If U and V have property $\mathcal{P}$, then $U \cup V$ has property $\mathcal{P}$;
(G-3) If $\{U_\lambda \mid \lambda \in \Lambda\}$ is discrete in X and each U_λ has property $\mathcal{P}$, then $\bigcup_{\lambda\in\Lambda} U_\lambda$ has property $\mathcal{P}$.

The following theorem, called MICHAEL'S THEOREM on local property, is very useful to show that a space has a certain property:

Theorem 2.6.7 (E. MICHAEL) *Let $\mathcal{P}$ be a G-hereditary property of open sets in a paracompact space X. If X has property $\mathcal{P}$ locally, then X itself has property $\mathcal{P}$.*

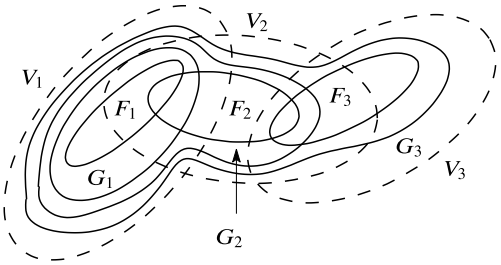


Fig. 2.10 Construction of G_n

Proof Since X has property $\mathcal{P}$ locally, there exists $\mathcal{U} \in \mathrm{cov}(X)$ such that each $U \in \mathcal{U}$ has property $\mathcal{P}$. According to Theorem 2.6.3, $\mathcal{U}$ has an open refinement $\mathcal{V} = \bigcup_{n\in\mathbb{N}} \mathcal{V}_n$ such that each $\mathcal{V}_n$ is discrete in X. Each $V \in \mathcal{V}$ has property $\mathcal{P}$ by (G-1). For each $n \in \mathbb{N}$, let $V_n = \bigcup \mathcal{V}_n$. Then, each V_n has property $\mathcal{P}$ by (G-3), hence $V_1 \cup \cdots \cup V_n$ has property $\mathcal{P}$ by (G-2). Due to Lemma 2.6.2, the open cover $\{V_n \mid n \in \mathbb{N}\}$ has a closed shrinking $\{F_n \mid n \in \mathbb{N}\}$.[13] Inductively choose open sets G_n $(n \in \mathbb{N})$ so that

$$F_n \cup \mathrm{cl}\, G_{n-1} \subset G_n \subset \mathrm{cl}\, G_n \subset V_1 \cup \cdots \cup V_n,$$

where $G_0 = \emptyset$ (Fig. 2.10). For each $n \in \mathbb{N}$, let $W_n = G_n \setminus \mathrm{cl}\, G_{n-2}$, where $G_{-1} = \emptyset$. Then, each W_n also has property $\mathcal{P}$ by (G-1). Let $X_i = \bigcup_{n\in\omega} W_{3n+i}$, where $i = 1, 2, 3$. Since $\{W_{3n+i} \mid n \in \omega\}$ is discrete in X, each X_i has property $\mathcal{P}$ by (G-3). Hence, $X = X_1 \cup X_2 \cup X_3$ also has property $\mathcal{P}$ by (G-2). □

There are many cases where we consider properties of closed sets rather than open sets. In such cases, Theorem 2.6.7 can also be applied. In fact, let $\mathcal{P}$ be a property of *closed* sets of X. We define the property $\mathcal{P}^\circ$ of *open* sets in X as follows:

$$U \text{ has property } \mathcal{P}^\circ \underset{\text{def}}{\Leftrightarrow} \mathrm{cl}\, U \text{ has property } \mathcal{P}.$$

It is said that $\mathcal{P}$ is ***F*-hereditary** if it satisfies the following conditions:

(F-1) If A has property $\mathcal{P}$, then every closed subset of A has property $\mathcal{P}$;
(F-2) If A and B have property $\mathcal{P}$, then $A \cup B$ has property $\mathcal{P}$;
(F-3) If $\{A_\lambda \mid \lambda \in \Lambda\}$ is discrete in X and each A_λ has property $\mathcal{P}$, then $\bigcup_{\lambda\in\Lambda} A_\lambda$ has property $\mathcal{P}$.

Evidently, if property $\mathcal{P}$ is F-hereditary, then $\mathcal{P}^\circ$ is G-hereditary. Thus, Theorem 2.6.7 yields the following corollary:

Corollary 2.6.8 (E. MICHAEL) *Let $\mathcal{P}$ be an F-hereditary property of closed sets in a paracompact space X. If X has property $\mathcal{P}$ locally, then X itself has property $\mathcal{P}$.* □

[13] In this case, without applying Lemma 2.6.2, we can inductively choose closed sets $F_n \subset X$, $n \in \mathbb{N}$, so that $X \setminus \left(\bigcup_{i<n} \mathrm{int}\, F_i \cup \bigcup_{j>n} V_j\right) \subset \mathrm{int}\, F_n \subset F_n \subset V_n$.

Proposition 2.6.9 (Additional Results on Paracompact Spaces)

(1) *A space X is paracompact if it is a locally finite union of paracompact closed subspaces.*

Sketch of Proof Let $\mathcal{F}$ be a locally finite closed cover of X such that each $F \in \mathcal{F}$ is paracompact. To prove regularity, let $x \in X$ and U an open neighborhood of x in X. Since each $F \in \mathcal{F}[x]$ is regular, we have an open neighborhood U_F of x in X such that $\mathrm{cl}(F \cap U_F) \subset U$. The following U_0 is an open neighborhood of x in X:

$$U_0 = \bigcap_{F \in \mathcal{F}[x]} U_F \setminus \bigcup (\mathcal{F} \setminus \mathcal{F}[x]) \left(\subset \bigcup \mathcal{F}[x] = \mathrm{st}(x, \mathcal{F}) \right).$$

Observe that $\mathrm{cl}_X U_0 = \mathrm{cl} \bigcup_{F \in \mathcal{F}[x]} (U_0 \cap F) = \bigcup_{F \in \mathcal{F}[x]} \mathrm{cl}(U_0 \cap F) \subset U$. Thus, it suffices to show that X satisfies condition 2.6.3 (f).

(2) *Every F_σ subspace A of a paracompact space X is paracompact.*

Sketch of Proof It suffices to show that A satisfies condition 2.6.3 (e). Let $A = \bigcup_{n \in \mathbb{N}} A_n$, where each A_n is closed in X. For each $\mathcal{V} \in \mathrm{cov}(A)$ and $n \in \mathbb{N}$, let

$$\mathcal{U}_n = \left\{ X \setminus A_n \right\} \cup \left\{ \widetilde{V} \;\middle|\; V \in \mathcal{V} \right\} \in \mathrm{cov}(X),$$

where each $\widetilde{V}$ is open in X with $\widetilde{V} \cap A = V$. Note that $\mathcal{V}_n \prec \mathcal{U}_n$ implies that $\mathcal{V}_n[A_n] | A \prec \mathcal{V}$.

(3) *If every open subspace of X is paracompact, then every subspace A of X is also paracompact.*

Sketch of Proof To find a locally finite open refinement of $\mathcal{U} \in \mathrm{cov}(A)$, take an open collection $\widetilde{\mathcal{U}}$ in X such that $\widetilde{\mathcal{U}} | A = \mathcal{U}$ and use the paracompactness of $\bigcup \widetilde{\mathcal{U}}$.

(4) *The product $X \times Y$ is paracompact for any paracompact space X and any compact space Y.*

Sketch of Proof Let $\mathcal{U} \in \mathrm{cov}(X \times Y)$. Since Y is compact, it is not hard to show that each $x \in X$ has an open neighborhood $O(x)$ in X and there are finitely many open sets $W_1^x, \ldots, W_{n(x)}^x$ in Y such that each $O(x) \times W_i^x$ is contained in some member of $\mathcal{U}$ and $Y = \bigcup_{i=1}^{n(x)} W_i^x$. Since X is paracompact, $\{O(x) \mid x \in X\} \in \mathrm{cov}(X)$ has a locally finite open refinement $\mathcal{V} \in \mathrm{cov}(X)$. For each $V \in \mathcal{V}$, choose $x_V \in X$ so that $V \subset O(x_V)$. Then, the following is a locally finite open refinement of $\mathcal{U}$:

$$\left\{ V \times W_i^{x_V} \;\middle|\; V \in \mathcal{V}, i = 1, \ldots, n(x_V) \right\} \in \mathrm{cov}(X \times Y).$$

(5) *A paracompact space X is (completely) metrizable if it is locally (completely) metrizable.*

Sketch of Proof To apply Proposition 2.4.4 (2) (2.5.11 (2)), construct a locally finite cover of X consisting of (completely) metrizable closed sets.

Note When defining n-manifolds in Sect. 1.1, we assumed metrizability, but many textbooks instead assume paracompactness. See Footnote 6 in Sect. 1.1.

A space X is **hereditarily paracompact** if every subspace of X is paracompact. Clearly, *every metrizable space is hereditarily paracompact*. By (2) and (3) above, we have the following:

Theorem 2.6.10 *Every perfectly normal paracompact space is hereditarily paracompact.* □

2.7 Partitions of Unity

The **support** of a map $f : X \to \mathbb{R}$ is the following closed set in X:

$$\operatorname{supp} f \underset{\text{def}}{=} \operatorname{cl}\left\{x \in X \;\middle|\; f(x) \neq 0\right\} \subset X.$$

For a map $f : X \to \mathbb{R}_+$, $\operatorname{supp} f = \operatorname{cl} f^{-1}((0, \infty))$. A **partition of unity** on X is an indexed family $(f_\lambda)_{\lambda\in\Lambda}$ of maps $f_\lambda : X \to \mathbf{I}$ such that $\sum_{\lambda\in\Lambda} f_\lambda(x) = 1$ for each $x \in X$,[14] where, by Proposition 1.2.5, we have

$$\operatorname{card}\left\{\lambda \in \Lambda \;\middle|\; f_\lambda(x) \neq 0\right\} \leqslant \aleph_0.$$

It is said that $(f_\lambda)_{\lambda\in\Lambda}$ is **locally finite** if $(\operatorname{supp} f_\lambda)_{\lambda\in\Lambda}$ is locally finite in X, (equiv., $(f_\lambda^{-1}((0, 1]))_{\lambda\in\Lambda}$ is locally finite in X).[15]

There exists a partition of unity which is not locally finite. For example, the hedgehog $J(\mathbb{N})$ has a non-locally finite partition of unity $(f_n)_{n\in\omega}$ defined as follows: $f_0(x) = 1 - \|x\|_1$ and $f_n(x) = x(n)$ for each $n \in \mathbb{N}$, where

$$J(\mathbb{N}) = \bigcup_{n\in\mathbb{N}} \mathbf{I}\mathbf{e}_n = \left\{x \in \ell_1 \;\middle|\; x(n) \in \mathbf{I} \text{ for all } n \in \mathbb{N} \text{ and } x(n) \neq 0 \text{ at most one } n \in \mathbb{N}\right\} \subset \ell_1.$$

However, as we can see from the following proposition, the existence of a partition of unity implies the existence of a locally finite one:

Proposition 2.7.1 *If X has a partition of unity $(f_\lambda)_{\lambda\in\Lambda}$, then X has a locally finite partition of unity $(g_\lambda)_{\lambda\in\Lambda}$ such that $\operatorname{supp} g_\lambda \subset f_\lambda^{-1}((0, 1])$ for each $\lambda \in \Lambda$.*

Proof We define $h : X \to \mathbf{I}$ by $h(x) = \sup_{\lambda\in\Lambda} f_\lambda(x) > 0$. To see the continuity of h, for each $x \in X$, choose $\Lambda(x) \in \operatorname{Fin}(\Lambda)$ so that $\sum_{\lambda\in\Lambda(x)} f_\lambda(x) > 1 - h(x)/2$. Then, $f_\lambda(x) < h(x)/2$ for every $\lambda \in \Lambda \setminus \Lambda(x)$, so $h(x) = f_{\lambda(x)}(x)$ for some $\lambda(x) \in \Lambda(x)$. Since $\sum_{\lambda\in\Lambda(x)} f_\lambda$ and $f_{\lambda(x)}$ are continuous, x has a neighborhood U_x in X such that

[14] For infinite sum, refer to Proposition 1.2.4.

[15] This does not mean that $\{\operatorname{supp} f_\lambda \mid \lambda \in \Lambda\}$ is locally finite in X. Refer to Sect. 1.1.

$$\sum_{\lambda\in\Lambda(x)} f_\lambda(y) > 1 - h(x)/2 \text{ and } f_{\lambda(x)}(y) > h(x)/2 \text{ for all } y \in U_x.$$

Thus, $f_\lambda(y) < h(x)/2 < f_{\lambda(x)}(y)$ for $\lambda \in \Lambda \setminus \Lambda(x)$ and $y \in U_x$. Therefore,

$$h(y) = \max\left\{ f_\lambda(y) \;\middle|\; \lambda \in \Lambda(x)\right\} \text{ for each } y \in U_x.$$

Hence, h is continuous.

For each $\lambda \in \Lambda$, let $k_\lambda : X \to \mathbf{I}$ be a map defined by

$$k_\lambda(x) = \max\left\{0, f_\lambda(x) - 2h(x)/3\right\}.$$

Then, $\operatorname{supp} k_\lambda \subset f_\lambda^{-1}((0, 1])$. Indeed, if $f_\lambda(x) = 0$, then x has a neighborhood U such that $f_\lambda(y) < 2h(y)/3$ for every $y \in U$, which implies $x \notin \operatorname{supp} k_\lambda$. For each $x \in X$, take U_x and $\Lambda(x)$ as in the proof of the continuity of h. Choose an open neighborhood V_x of x in X so that $V_x \subset U_x$ and $h(y) > 3h(x)/4$ for all $y \in V_x$. If $\lambda \in \Lambda \setminus \Lambda(x)$ and $y \in V_x$, then

$$f_\lambda(y) - 2h(y)/3 < f_\lambda(y) - h(x)/2 < 0,$$

which implies that $V_x \cap \operatorname{supp} k_\lambda = \emptyset$ for any $\lambda \in \Lambda \setminus \Lambda(x)$. Thus, $(\operatorname{supp} k_\lambda)_{\lambda\in\Lambda}$ is locally finite. Let $\varphi : X \to \mathbb{R}_+$ be the map defined by $\varphi(x) = \sum_{\lambda\in\Lambda} k_\lambda(x)$. The desired partition of unity $(g_\lambda)_{\lambda\in\Lambda}$ on X can be defined by $g_\lambda(x) = k_\lambda(x)/\varphi(x)$ for each $\lambda \in \Lambda$. □

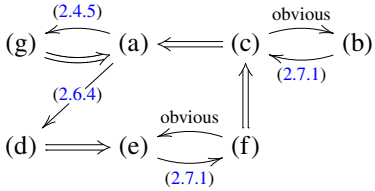
Let $\mathcal{U}$ be an open cover of a space X. A partition of unity $(f_\lambda)_{\lambda\in\Lambda}$ on X is said to be **subordinated** (resp. **weakly subordinated**) to $\mathcal{U}$ if $\{\operatorname{supp} f_\lambda \mid \lambda \in \Lambda\} \prec \mathcal{U}$ (resp. $\{f_\lambda^{-1}((0, 1]) \mid \lambda \in \Lambda\} \prec \mathcal{U}$). We call $\mathcal{U}$ a **numerable cover** of X if there is a partition of unity $(f_U)_{U\in\mathcal{U}}$ such that $f_U^{-1}((0, 1]) \subset U$ for each $U \in \mathcal{U}$. We have the following characterization:

Theorem 2.7.2 *For an open cover $\mathcal{U}$ of a space X, the following are equivalent:*

(a) *$\mathcal{U}$ is a normal cover;*
(b) *$\mathcal{U}$ is a numerable cover;*
(c) *There exists a locally finite partition of unity $(f_U)_{U\in\mathcal{U}}$ such that* $\operatorname{supp} f_U \subset U$ *for each $U \in \mathcal{U}$;*
(d) *$\mathcal{U}$ has a locally finite refinement consisting of cozero sets in X;*
(e) *There exists a partition of unity weakly subordinated to $\mathcal{U}$;*
(f) *There exists a locally finite partition of unity subordinated to $\mathcal{U}$;*
(g) *There exist a paracompact space Y with a map $f : X \to Y$ and an open cover $\mathcal{V}$ such that $f^{-1}(\mathcal{V}) \prec \mathcal{U}$.*

Proof The implications (c) ⇒ (b) and (f) ⇒ (e) are obvious and their converses (b) ⇒ (c) and (e) ⇒ (f) are direct consequences of Proposition 2.7.1. The implication (a) ⇒ (d) is none other than Proposition 2.6.4 and (a) ⇒ (g) is a direct consequence

of Proposition 2.4.5. We will show (g) ⇒ (a), (d) ⇒ (e), and (f) ⇒ (c) ⇒ (a). See the following diagram:



(g) ⇒ (a): In (g), $\mathcal{V}$ is a normal cover by Theorem 2.6.3, that is, there is a sequence of open star-refinements $\mathcal{V} \overset{*}{\succ} \mathcal{V}_1 \overset{*}{\succ} \mathcal{V}_2 \overset{*}{\succ} \cdots$. Then, $\mathcal{U}$ is a normal cover because

$$\mathcal{U} \succ f^{-1}(\mathcal{V}) \overset{*}{\succ} f^{-1}(\mathcal{V}_1) \overset{*}{\succ} f^{-1}(\mathcal{V}_2) \overset{*}{\succ} \cdots .$$

(d) ⇒ (e): By (d), we can take maps $f_\lambda : X \to \mathbf{I}$, $\lambda \in \Lambda$, such that $\{f_\lambda^{-1}((0,1]) \mid \lambda \in \Lambda\} \in \mathrm{cov}(X)$ is a locally finite refinement of $\mathcal{U}$. Then, we can define a map $\varphi : X \to (0,\infty)$ by $\varphi(x) = \sum_{\lambda\in\Lambda} f_\lambda(x)$. For each $\lambda \in \Lambda$, let $g_\lambda : X \to \mathbf{I}$ be a map defined by $g_\lambda(x) = f_\lambda(x)/\varphi(x)$. Then, $(g_\lambda)_{\lambda\in\Lambda}$ is a locally finite partition of unity on X weakly subordinated to $\mathcal{U}$ because $g_\lambda^{-1}((0,1]) = f_\lambda^{-1}((0,1])$ for each $\lambda \in \Lambda$.

(f) ⇒ (c): Let $(f_\lambda)_{\lambda\in\Lambda}$ be a locally finite partition of unity subordinated to $\mathcal{U}$. Then, we have a function $\Psi : \Lambda \to \mathcal{U}$ such that $\mathrm{supp}\, f_\lambda \subset \Psi(\lambda) \in \mathcal{U}$ for each $\lambda \in \Lambda$. For each $U \in \mathcal{U}$, we define $g_U = \sum_{\lambda\in\Psi^{-1}(U)} f_\lambda : X \to \mathbf{I}$, where $g_U = 0$ if $\Psi^{-1}(U) = \emptyset$. Then, $(g_U)_{U\in\mathcal{U}}$ is a partition of unity such that $\mathrm{supp}\, g_U \subset U$ for each $U \in \mathcal{U}$.

(c) ⇒ (a): There is a locally finite partition of unity $(f_U)_{U\in\mathcal{U}}$ such that $\mathrm{supp}\, f_U \subset U$ for every $U \in \mathcal{U}$. Let $Y = \mathbf{S}_{\ell_1(\mathcal{U})} \cap (\mathbb{R}_+)^{\mathcal{U}}$, where $\mathbf{S}_{\ell_1(\mathcal{U})}$ is the unit sphere of $\ell_1(\mathcal{U})$. Then, $\mathcal{V} = \{\mathrm{pr}_U^{-1}((0,\infty)) \cap Y \mid U \in \mathcal{U}\}$ is an open cover of Y, where $\mathrm{pr}_U : \mathbb{R}^{\mathcal{U}} \to \mathbb{R}$ is the projection, i.e., $\mathrm{pr}_U(x) = x(U)$ for each $x = (x(U))_{U\in\mathcal{U}} \in \mathbb{R}^{\mathcal{U}}$. We define map $\varphi : X \to Y$ by $\varphi(x) = (f_U(x))_{U\in\mathcal{U}}$ for each $x \in X$. Observe that

$$\varphi^{-1}(\mathrm{pr}_U^{-1}((0,\infty)) \cap Y) = f_U^{-1}((0,1]) \subset U \quad \text{for each } U \in \mathcal{U},$$

which implies that $\varphi^{-1}(\mathcal{V}) \prec \mathcal{U}$. Since Y is paracompact (Stone's Theorem 2.3.1), we can inductively apply Theorem 2.6.3 to obtain a sequence of open covers $\mathcal{V} \overset{*}{\succ} \mathcal{V}_1 \overset{*}{\succ} \mathcal{V}_2 \overset{*}{\succ} \cdots$. Then, we have

$$\mathcal{U} \succ \varphi^{-1}(\mathcal{V}) \overset{*}{\succ} \varphi^{-1}(\mathcal{V}_1) \overset{*}{\succ} \varphi^{-1}(\mathcal{V}_2) \overset{*}{\succ} \cdots ,$$

which means that $\mathcal{U}$ is a normal cover of X. □

By using the above characterization, it is easy to show the following:

Corollary 2.7.3 *For each pair of numerable (= normal) covers $\mathcal{U}$, $\mathcal{V} \in \mathrm{cov}(X)$, $\mathcal{U} \wedge \mathcal{V}$ is numerable (= normal).* □

Corollary 2.7.4 *Let $f : X \to Y$ be a map. For each numerable (= normal) cover $\mathcal{V}$ of Y, $f^{-1}(\mathcal{V})$ is a numerable (= normal) cover of X.* □

By Theorem 2.7.2, we can add some more conditions to the list of conditions characterizing paracompactness in Theorem 2.6.3, that is:

Theorem 2.7.5 *For a space X, the following are equivalent:*

(a) *X is paracompact;*
(g) *Every open cover of X is numerable;*
(h) *Every open cover of X has a locally finite refinement consisting of cozero sets;*
(i) *For each $\mathcal{U} \in \mathrm{cov}(X)$, there is a partition of unity weakly subordinated to $\mathcal{U}$;*
(j) *For each $\mathcal{U} \in \mathrm{cov}(X)$, there is a locally finite partition of unity subordinated to $\mathcal{U}$.* □

Recall that a collection $\mathcal{A}$ of subsets of a space X is **point-finite** if $\mathrm{card}\, \mathcal{A}[x] < \aleph_0$ for each $x \in X$ (Sect. 1.1). Obviously, every locally finite collection is point-finite. We prove below the OPEN COVER SHRINKING LEMMA, which will be also used in Chaps. 4, 5, and 6:

Lemma 2.7.6 (OPEN COVER SHRINKING) *Each point-finite open cover $\mathcal{U}$ of a normal space X has a strong open shrinking $\{V_U \mid U \in \mathcal{U}\}$.*

Proof Let $\mathcal{T}$ be the topology of X (i.e., the collection of all open sets in X) and define an ordered set $\Phi = (\Phi, \leqslant)$ as follows:

$$\Phi = \Big\{\varphi : \mathcal{U} \to \mathcal{T} \;\Big|\; \textstyle\bigcup_{U\in\mathcal{U}} \varphi(U) = X;\ \mathrm{cl}\,\varphi(U) \subset U \ \text{ if } \ \varphi(U) \neq U\Big\},$$
$$\varphi_1 \leqslant \varphi_2 \underset{\text{def}}{\Leftrightarrow} \varphi_1(U) \neq U \ \text{ implies } \ \varphi_1(U) = \varphi_2(U).$$

If Φ has a maximal element φ_0, then the desired open refinement $\{V_U \mid U \in \mathcal{U}\}$ can be defined by $V_U = \varphi_0(U)$. Indeed, if $\mathrm{cl}\,\varphi_0(U_0) \not\subset U_0$ for some $U_0 \in \mathcal{U}$, then $\varphi(U_0) = U_0$. Hence, because of the normality of X, we have $V_0 \in \mathcal{T}$ such that

$$X \setminus \bigcup_{U\in\mathcal{U}\setminus\{U_0\}} \varphi(U) \subset V_0 \subset \mathrm{cl}\, V_0 \subset U_0.$$

We define $\varphi : \mathcal{U} \to \mathcal{T}$ so that $\varphi(U) = \varphi_0(U)$ for each $U \in \mathcal{U} \setminus \{U_0\}$ and $\varphi(U_0) = V_0$. Then, $\varphi \in \Phi$ and $\varphi_0 < \varphi$, which contradicts the maximality of φ_0.

It remains to show that Φ has a maximal element. Applying Zorn's Lemma, we show this. It suffices to show that every totally ordered subset $\Psi \subset \Phi$ is upper bounded in Φ. For each $U \in \mathcal{U}$, let $\varphi(U) = \bigcap_{\psi\in\Psi} \psi(U)$. Then, $\varphi(U) \neq U$ implies $\psi_U(U) \neq U$ for some $\psi_U \in \Psi$, which means that $\varphi(U) = \psi_U(U)$ because $\psi(U) = \psi_U(U)$ or $\psi(U) = U$ for every $\psi \in \Psi$. Thus, we have $\varphi : \mathcal{U} \to \mathcal{T}$ such that $\mathrm{cl}\,\varphi(U) \subset U$ if $\varphi(U) \neq U$. To verify $X = \bigcup_{U\in\mathcal{U}} \varphi(U)$, let $x \in X$. If $\varphi(U) = U$ for some $U \in \mathcal{U}[x]$ then $x \in U = \varphi(U)$. When $\varphi(U) \neq U$ for every $U \in \mathcal{U}[x]$, by the same argument as above, we can see that $\varphi(U) = \psi_U(U)$ for each $U \in \mathcal{U}[x]$.

Since $\mathcal{U}[x]$ is finite, we have $\psi_0 = \max\{\psi_U \mid U \in \mathcal{U}[x]\} \in \Psi$. Then, $\varphi(U) = \psi_U(U) = \psi_0(U)$ for each $U \in \mathcal{U}[x]$. Since $X = \bigcup_{U\in\mathcal{U}} \psi_0(U)$, it follows that $x \in \psi_0(U)$ $(\subset U)$ for some $U \in \mathcal{U}$, which implies $x \in \varphi(U)$ because $U \in \mathcal{U}[x]$. Consequently, $\varphi \in \Phi$. It follows from the definition that $\psi \leqslant \varphi$ for any $\psi \in \Psi$. □

Remark 2.7.1 The above lemma can be proved using the transfinite induction instead of Zorn's Lemma.

Theorem 2.7.7 *Every locally finite open cover $\mathcal{U}$ of a normal space X is numerable, that is, there exists a (locally finite) partition of unity $(f_U)_{U\in\mathcal{U}}$ on X such that $f_U^{-1}((0, 1]) \subset U$ (or* supp $f_U \subset U$*) for each $U \in \mathcal{U}$.*

Proof By the Open Cover Shrinking Lemma 2.7.6, we can take $\{V_U \mid U \in \mathcal{U}\}$, $\{W_U \mid U \in \mathcal{U}\} \in \mathrm{cov}(X)$ such that $\mathrm{cl}\, W_U \subset V_U \subset \mathrm{cl}\, V_U \subset U$ for each $U \in \mathcal{U}$. For each $U \in \mathcal{U}$, let $f_U : X \to \mathbf{I}$ be a Urysohn map with $f_U(\mathrm{cl}\, W_U) = 1$ and $f_U(X \setminus V_U) = 0$. Since $\mathcal{U}$ is locally finite and $f_U^{-1}((0, 1]) \subset \mathrm{cl}\, V_U \subset U$ for each $U \in \mathcal{U}$, $\{f_U^{-1}((0, 1]) \mid U \in \mathcal{U}\}$ is a locally finite refinement of $\mathcal{U}$ consisting of cozero sets. The result follows from Theorem 2.7.2. □

It is said that a real-valued function $f : X \to \mathbb{R}$ is **lower semi-continuous**, abbreviated as **l.s.c.** (or **upper semi-continuous**, **u.s.c.**) if $f^{-1}((t, \infty))$ (or $f^{-1}((-\infty, t))$) is open in X for each $t \in \mathbb{R}$. Then, $f : X \to \mathbb{R}$ is continuous if and only if f is l.s.c. and u.s.c.

Theorem 2.7.8 *Let $g, h : X \to \mathbb{R}$ be real-valued functions on a paracompact space X such that g is u.s.c., h is l.s.c. and $g(x) < h(x)$ for each $x \in X$. Then, there exists a map $f : X \to \mathbb{R}$ such that $g(x) < f(x) < h(x)$ for each $x \in X$. Moreover, given a map $f_0 : A \to \mathbb{R}$ of a closed set A in X such that $g(x) < f_0(x) < h(x)$ for each $x \in A$, the map f can be an extension of f_0.*

Proof For each $q \in \mathbb{Q}$, let

$$U_q = g^{-1}((-\infty, q)) \cap h^{-1}((q, \infty)).$$

For each $x \in X$, we have $q \in \mathbb{Q}$ such that $g(x) < q < h(x)$, which means that $\mathcal{U} = \{U_q \mid q \in \mathbb{Q}\} \in \mathrm{cov}(X)$. By Theorem 2.7.5, X has a locally finite partition of unity $(f_\lambda)_{\lambda\in\Lambda}$ subordinated to $\mathcal{U}$. For each $\lambda \in \Lambda$, choose $q(\lambda) \in \mathbb{Q}$ so that $\mathrm{supp}\, f_\lambda \subset U_{q(\lambda)}$. Then, we define a map $f : X \to \mathbb{R}$ as follows:

$$f(x) = \sum_{\lambda\in\Lambda} q(\lambda) f_\lambda(x).$$

For each $x \in X$, let $\{\lambda \in \Lambda \mid x \in \mathrm{supp}\, f_\lambda\} = \{\lambda_1, \cdots, \lambda_n\}$. Since $x \in \bigcap_{i=1}^n U_{q(\lambda_i)}$, we have $g(x) < q(\lambda_i) < h(x)$ for each $i = 1, \cdots, n$, hence it follows that

$$g(x) = \sum_{i=1}^{n} g(x) f_{\lambda_i}(x) < f(x) = \sum_{i=1}^{n} q(\lambda_i) f_{\lambda_i}(x)$$
$$< h(x) = \sum_{i=1}^{n} h(x) f_{\lambda_i}(x).$$

To prove the additional statement, apply the $\mathbb{R}$-version of the Tietze–Urysohn Extension Theorem 2.2.4 to extend f_0 to a map $f' : X \to \mathbb{R}$. Then, we have an open neighborhood U of A in X such that $g(x) < f'(x) < h(x)$ for each $x \in U$. Let $k : X \to \mathbf{I}$ be a Urysohn map with $k(A) = 1$ and $k(X \setminus U) = 0$. We can define $\tilde{f} : X \to \mathbb{R}$ as follows:

$$\tilde{f}(x) = (1 - k(x)) f(x) + k(x) f'(x).$$

Then, $\tilde{f}|A = f_0$ and $g(x) < \tilde{f}(x) < h(x)$ for each $x \in X$. □

Proposition 2.7.9 (Refinements by Open Balls)

(1) *Let X be a metrizable space and $\mathcal{U}$ an open cover of X. Then, X has an admissible metric ρ such that*

$$\left\{ \overline{\mathrm{B}}_\rho(x, 1) \;\middle|\; x \in X \right\} \prec \mathcal{U}.$$

Moreover, for a given $d \in \mathrm{Metr}(X)$, ρ can be chosen so that $\rho \geqslant d$ (hence, if d is complete then ρ is) and if d is bounded then ρ is also bounded.

Sketch of Proof Take an open Δ-refinement $\mathcal{V}$ of $\mathcal{U}$ and a locally finite partition of unity $(f_\lambda)_{\lambda \in \Lambda}$ on X subordinated to $\mathcal{V}$. For a given $d \in \mathrm{Metr}(X)$, the desired metric $\rho \in \mathrm{Metr}(X)$ can be defined as follows:

$$\rho(x, y) = d(x, y) + \sum_{\lambda \in \Lambda} |f_\lambda(x) - f_\lambda(y)| \geqslant d(x, y).$$

If $\rho(x, y) \leqslant 1$ then $x, y \in f_\lambda^{-1}((0, 1]) \subset \mathrm{supp}\, f_\lambda$ for some $\lambda \in \Lambda$, otherwise we have

$$\sum_{\lambda \in \Lambda} |f_\lambda(x) - f_\lambda(y)| = \sum_{\lambda \in \Lambda} f_\lambda(x) + \sum_{\lambda \in \Lambda} f_\lambda(y) = 2 > 1.$$

Then, it follows that $\overline{\mathrm{B}}_\rho(x, 1) \subset \mathrm{st}(x, \mathcal{V})$.

Sketch of another Proof The above can be obtained as a corollary of Theorem 2.6.3 and Corollary 2.4.3 as follows: By Corollary 2.4.3, X has a sequence of open covers

$$\mathcal{U}_1 \overset{\Delta}{\succ} \mathcal{U}_2 \overset{\Delta}{\succ} \mathcal{U}_3 \overset{\Delta}{\succ} \cdots \quad \left(\text{or } \mathcal{U}_1 \overset{*}{\succ} \mathcal{U}_2 \overset{*}{\succ} \mathcal{U}_3 \overset{*}{\succ} \cdots \right)$$

such that $\{\mathrm{st}(x, \mathcal{U}_n) \mid n \in \mathbb{N}\}$ is a neighborhood basis of each $x \in X$. By Theorem 2.6.3, we can inductively define $\mathcal{V}_n \in \mathrm{cov}(X)$, $n \in \mathbb{N}$, such that

$$\mathcal{V}_n \prec \mathcal{U}_n \text{ and } \mathcal{V}_n \overset{\Delta}{\prec} \mathcal{V}_{n-1} \quad \left(\mathcal{V}_n \overset{*}{\prec} \mathcal{V}_{n-1} \right),$$

where $\mathcal{V}_0 = \mathcal{U}$. Let $d' \in \mathrm{Metr}(X)$ be the bounded metric obtained by applying Corollary 2.4.3 with Remark 2.4.1 (or 2.4.3). For a given $d \in \mathrm{Metr}(X)$, the desired $\rho \in \mathrm{Metr}(X)$ can be defined by $\rho = 8d' + d$ (or $\rho = 4d' + d$).

(2) *Let $X = (X, d)$ be a metric space. For each open cover $\mathcal{U}$ of X, there is a map $\gamma : X \to (0, 1)$ such that*

$$\left\{\overline{\mathrm{B}}(x, \gamma(x)) \;\middle|\; x \in X\right\} \prec \mathcal{U}.$$

Sketch of Proof For each $x \in X$, let

$$r(x) = \sup_{U \in \mathcal{U}} \min\{1,\ d(x, X \setminus U)\} = \sup_{U \in \mathcal{U}} \bar{d}(x, X \setminus U),$$

where $\bar{d} = \min\{1, d\}$. Show that $r : X \to (0, \infty)$ is l.s.c. Then, we can apply Theorem 2.7.8 to obtain a map $\gamma : X \to (0, 1)$ such that $\gamma(x) < r(x)$ for each $x \in X$.

Note Actually, it can be proved that r is 1-Lipschitz (hence continuous), so we can define $\gamma = r/2$.

2.8 The Direct Limits of Towers of Spaces

In this section, we consider the direct limit of a tower $X_1 \subset X_2 \subset \cdots$ of spaces, where each X_n is a subspace of X_{n+1}. The **direct limit** $\varinjlim X_n$ is the space $\bigcup_{n\in\mathbb{N}} X_n$ endowed with the weak topology with respect to the tower $(X_n)_{n\in\mathbb{N}}$, that is,

$$U \subset \varinjlim X_n \text{ is open in } \varinjlim X_n \;\Leftrightarrow\; \forall n \in \mathbb{N},\ U \cap X_n \text{ is open in } X_n$$

$$\Big(\text{equiv. } A \subset \varinjlim X_n \text{ is closed in } \varinjlim X_n \;\Leftrightarrow\; \forall n \in \mathbb{N},\ A \cap X_n \text{ is closed in } X_n\Big).$$

In other words, the topology of $\varinjlim X_n$ is the finest topology such that every inclusion $X_n \subset \varinjlim X_n$ is continuous. The direct limit $\varinjlim X_n$ can be also defined as a quotient space of the topological sum $\bigoplus_{n\in\mathbb{N}} X_n$ with the quotient map $q : \bigoplus_{n\in\mathbb{N}} X_n \to \varinjlim X_n$ defined by $q|X_n = \mathrm{id}$ for each $n \in \mathbb{N}$. It should be noted that every X_n is a subspace of $\varinjlim X_n$.

Indeed, for each open set V in X_n, we can inductively define open sets U_{n+i} in X_{n+i} so that $U_{n+i} \cap X_{n+i-1} = U_{n+i-1}$, where $U_n = V$. Then, $U = \bigcup_{i\in\mathbb{N}} U_{n+i}$ is an open set in $\varinjlim X_n$ and $U \cap X_n = U_n = V$.

For an arbitrary space Y,

$$f : \varinjlim X_n \to Y \text{ is continuous} \;\Leftrightarrow\; \forall n \in \mathbb{N},\ f|X_n \text{ is continuous.}$$

Remark 2.8.1 Each point $x \in \varinjlim X_n$ belongs to some $X_{n(x)}$. If V is a neighborhood of x in $\varinjlim X_n$, then $V \cap X_n$ is a neighborhood x in X_n for every $n \geqslant n(x)$. However, it should be noted that the converse does not hold. For example, consider the direct limit $\mathbb{R}^\infty = \varinjlim \mathbb{R}^n$ of the tower $\mathbb{R} \subset \mathbb{R}^2 \subset \mathbb{R}^3 \subset \cdots$, where each $\mathbb{R}^n$ is identified with $\mathbb{R}^n \times \{(0, 0, \ldots)\} \subset \mathbb{R}^\mathbb{N}_f$. Let $W = \bigcup_{n\in\mathbb{N}}(-2^{-n}, 2^{-n})^n \subset \mathbb{R}^\infty$. Then, every $W \cap \mathbb{R}^n$ is a neighborhood of $0 \in \mathbb{R}^n$ because it contains $(-2^{-n}, 2^{-n})^n$. Nevertheless, W is not a neighborhood of 0 in $\mathbb{R}^\infty$. Indeed,

$$(\mathrm{int}_{\mathbb{R}^\infty} W) \cap \mathbb{R}^n \subset \mathrm{int}_{\mathbb{R}^n}(W \cap \mathbb{R}^n) = (-2^{-n}, 2^{-n})^n \quad \text{for each } n \in \mathbb{N}.$$

Then, it follows that $(\mathrm{int}_{\mathbb{R}^\infty} W) \cap \mathbb{R} \subset \bigcap_{n\in\mathbb{N}}(-2^{-n}, 2^{-n}) = \{0\}$, which means that $(\mathrm{int}_{\mathbb{R}^\infty} W) \cap \mathbb{R} = \emptyset$, and hence $0 \notin \mathrm{int}_{\mathbb{R}^\infty} W$.

It should also be noted that the direct limit $\varinjlim X_n$ is T_1 but, in general, non-Hausdorff. Such an example will be shown in Proposition 2.9.11.

As is easily observed, $\varinjlim X_{n(i)} = \varinjlim X_n$ for any $n(1) < n(2) < \cdots \in \mathbb{N}$. It is also easy to prove the following proposition:

Proposition 2.8.1 *Let $X_1 \subset X_2 \subset \cdots$ and $Y_1 \subset Y_2 \subset \cdots$ be towers of spaces. Suppose that there exist $n(1) < n(2) < \cdots$, $m(1) < m(2) < \cdots \in \mathbb{N}$ and maps $f_i : X_{n(i)} \to Y_{m(i)}$ and $g_i : Y_{m(i)} \to X_{n(i+1)}$ such that $g_i f_i = \mathrm{id}_{X_{n(i)}}$ and $f_{i+1} g_i = \mathrm{id}_{Y_{m(i)}}$, that is, the following diagram is commutative:*

$$\begin{array}{ccccccccc}
X_{n(1)} & \subset & X_{n(2)} & \subset & X_{n(3)} & \subset & \cdots \\
{\scriptstyle f_1}\downarrow & \overset{g_1}{\nearrow} & {\scriptstyle f_2}\downarrow & \overset{g_2}{\nearrow} & {\scriptstyle f_3}\downarrow & & \\
Y_{m(1)} & \subset & Y_{m(2)} & \subset & Y_{m(3)} & \subset & \cdots
\end{array}$$

Then, $\varinjlim X_n$ is homeomorphic to $\varinjlim Y_n$. □

Remark 2.8.2 It should be noted that $\varinjlim X_n$ is not a subspace of $\varinjlim Y_n$ even if each X_n is a closed subspace of Y_n. For example, let $Y_n = \mathbb{R}$ be the real line and

$$X_n = \{0\} \cup [n^{-1}, 1] \subset Y_n = \mathbb{R}.$$

Then, $\mathbf{I} = \bigcup_{n\in\mathbb{N}} X_n$, $\mathbb{R} = \varinjlim Y_n$, and 0 is an isolated point of $\varinjlim X_n$ but is not in the subspace $\mathbf{I} \subset \mathbb{R}$.

On the other hand, as is easily observed, if each X_n is an open subspace of Y_n then $\varinjlim X_n$ is an open subspace of $\varinjlim Y_n$.

The following proposition is also rather obvious:

Proposition 2.8.2 *Let $Y_1 \subset Y_2 \subset \cdots$ be a tower of spaces. If X is a closed (resp. open) subspace of $Y = \varinjlim Y_n$, then $X = \varinjlim(X \cap Y_n)$. Equivalently, if each $X \cap Y_n$ is closed (resp. open) in Y_n, then $\varinjlim(X \cap Y_n)$ is a closed (resp. open) subspace of Y.* □

Remark 2.8.3 In general, $X \neq \varinjlim(X \cap Y_n)$ for a subspace $X \subset \varinjlim Y_n$. For example, let Y_n be a subspace of Euclidean plane $\mathbb{R}^2$ defined by

$$Y_n = \left\{(0,0),\ (i^{-1},0),\ (j^{-1},k^{-1}) \;\middle|\; i,k \in \mathbb{N},\ j = 1,\ldots,n\right\}.$$

Observe that $A = \{(j^{-1},k^{-1}) \mid j,k \in \mathbb{N}\}$ is dense in $\varinjlim Y_n$, hence it is not closed in the following subspace X of $\varinjlim Y_n$:

$$X = \{(0,0)\} \cup \{(j^{-1},k^{-1}) \mid j,k \in \mathbb{N}\},$$

whereas A is closed in $\varinjlim(X \cap Y_n)$.

By the definition, the direct limit of k-spaces is also a k-space, that is:

Proposition 2.8.3 *Let $X_1 \subset X_2 \subset \cdots$ be a tower of k-spaces. Then, $\varinjlim X_n$ is also a k-space.* □

With regard to products of direct limits, we have:

Proposition 2.8.4 *Let $X_1 \subset X_2 \subset \cdots$ be a tower of spaces. If Y is locally compact then $(\varinjlim X_n) \times Y = \varinjlim(X_n \times Y)$ as spaces.*

Proof First of all, note that

$$\left(\varinjlim X_n\right) \times Y = \varinjlim(X_n \times Y) = \bigcup_{n\in\mathbb{N}}(X_n \times Y) \text{ as sets.}$$

Recall that $\varinjlim X_n$ is a quotient space of $\bigoplus_{n\in\mathbb{N}} X_n$ with the quotient map $q : \bigoplus_{n\in\mathbb{N}} X_n \to \varinjlim X_n$ defined $q|X_n = \mathrm{id}$, $n \in \mathbb{N}$. Due to Proposition 2.1.9, the following is a quotient map:

$$q \times \mathrm{id}_Y : \bigoplus_{n\in\mathbb{N}}(X_n \times Y) = \left(\bigoplus_{n\in\mathbb{N}} X_n\right) \times Y \to \left(\varinjlim X_n\right) \times Y,$$

which means $(\varinjlim X_n) \times Y = \varinjlim(X_n \times Y)$ as spaces. □

Proposition 2.8.5 *Let $X_1 \subset X_2 \subset \cdots$ and $Y_1 \subset Y_2 \subset \cdots$ be towers of spaces. If each X_n and Y_n are locally compact, then*

$$\varinjlim X_n \times \varinjlim Y_n = \varinjlim(X_n \times Y_n) \text{ as spaces.}$$

Proof First of all, note that

$$\varinjlim X_n \times \varinjlim Y_n = \varinjlim(X_n \times Y_n) = \bigcup_{n\in\mathbb{N}}(X_n \times Y_n) \text{ as sets.}$$

It is easy to see that $\mathrm{id} : \varinjlim(X_n \times Y_n) \to \varinjlim X_n \times \varinjlim Y_n$ is continuous. To see that this is open, let W be an open set in $\varinjlim(X_n \times Y_n)$. For each $(x, y) \in W$, choose $m \in \mathbb{N}$ so that $(x, y) \in X_m \times Y_m$. Since X_m and Y_m are locally compact, we have open sets $U_m \subset X_m$ and $V_m \subset Y_m$ such that

$$x \in U_m,\ \ y \in V_m,\ \ \mathrm{cl}_{X_m} U_m \times \mathrm{cl}_{Y_m} V_m \subset W$$

and both $\mathrm{cl}_{X_m} U_m$ and $\mathrm{cl}_{Y_m} V_m$ are compact. Then, by the compactness of $\mathrm{cl}_{X_m} U_m$ and $\mathrm{cl}_{Y_m} V_m$, we can easily find open sets $U_{m+1} \subset X_{m+1}$ and $V_{m+1} \subset Y_{m+1}$ such that

$$\mathrm{cl}_{X_m} U_m \subset U_{m+1},\ \ \mathrm{cl}_{Y_m} V_m \subset V_{m+1},\ \ \mathrm{cl}_{X_{m+1}} U_{m+1} \times \mathrm{cl}_{Y_{m+1}} V_{m+1} \subset W$$

and both $\mathrm{cl}_{X_{m+1}} U_{m+1}$ and $\mathrm{cl}_{Y_{m+1}} V_{m+1}$ are compact. Inductively, we can obtain $U_m \subset U_{m+1} \subset U_{m+2} \subset \cdots$ and $V_m \subset V_{m+1} \subset V_{m+2} \subset \cdots$ such that U_n and V_n are open in X_n and Y_n, respectively, $\mathrm{cl}_{X_n} U_n$ and $\mathrm{cl}_{Y_n} V_n$ are compact, and $\mathrm{cl}_{X_n} U_n \times \mathrm{cl}_{Y_n} V_n \subset W$. Then, $U = \bigcup_{n \geqslant m} U_n$ and $V = \bigcup_{n \geqslant m} V_n$ are open in $\varinjlim X_n$ and $\varinjlim Y_n$, respectively, and $(x, y) \in U \times V \subset W$. Therefore, W is open in $\varinjlim X_n \times \varinjlim Y_n$. □

A tower $X_1 \subset X_2 \subset \cdots$ of spaces is said to be **closed** if each X_n is closed in X_{n+1}; equivalently, each X_n is closed in the direct limit $\varinjlim X_n$. For a pointed space $X = (X, *)$, let

$$X_f^{\mathbb{N}} = \left\{ x \in X^{\mathbb{N}} \;\middle|\; x(n) = * \ \text{ except for finitely many } n \in \mathbb{N} \right\} \subset X^{\mathbb{N}}.$$

Identifying each X^n with $X^n \times \{(*, *, \ldots)\} \subset X_f^{\mathbb{N}}$, we have a closed tower $X \subset X^2 \subset X^3 \subset \cdots$ with $X_f^{\mathbb{N}} = \bigcup_{n \in \mathbb{N}} X^n$. We write $X^\infty = \varinjlim X^n$, which is the space $X_f^{\mathbb{N}}$ with the weak topology with respect to the tower $(X^n)_{n \in \mathbb{N}}$. A typical example is $\mathbb{R}^\infty$, which appeared in Remark 2.8.1. Applying by Proposition 2.8.2, we can easily obtain the following:

Proposition 2.8.6 *Let X be a closed (resp. open) subspace of a pointed space $Y = (Y, *)$. Assume that X has the same base point $*$ as Y. Then, the topology of $X^\infty = \varinjlim X^n$ is equal to the subspace topology induced by $Y^\infty = \varinjlim Y^n$ and X^∞ is a closed (resp. open) subspace of Y^∞.* □

When X is locally compact, the topology of $X^\infty = \varinjlim X^n$ is simple as follows:

Proposition 2.8.7 *Let $X = (X, *)$ be a pointed locally compact space. Then, each $x \in X^\infty = \varinjlim X^n$ has a neighborhood basis consisting of $X^\infty \cap \prod_{n \in \mathbb{N}} V_n$, where each V_n is a neighborhood of $x(n)$ in X^n.*[16]

Sketch of Proof Let U be an open neighborhood of x in X^∞. Choose $n_0 \in \mathbb{N}$ so that $x \in X^{n_0}$. For each $i = 1, \ldots, n_0$, each $x(i)$ has a neighborhood V_i in X such that $\mathrm{cl}\, V_i$ is compact and $\prod_{i=1}^{n_0} \mathrm{cl}\, V_i \subset U \cap X^{n_0}$. Recall that we identify $X^{n-1} = X^{n-1} \times \{*\} \subset X^n$. For $n > n_0$,

[16] I.e., the topology of X^∞ is a subspace topology inherited from the box topology of $X^{\mathbb{N}}$.

we can inductively choose a neighborhood V_n of $x(n) = *$ in X so that $\operatorname{cl} V_n$ is compact and $\prod_{i=1}^{n} \operatorname{cl} V_i \subset U \cap X^n$, where we use the compactness of $\prod_{i=1}^{n-1} \operatorname{cl} V_i \ \big(= \prod_{i=1}^{n-1} \operatorname{cl} V_i \times \{*\}\big)$.[17]

Remark 2.8.4 Proposition 2.8.4 does not hold without the local compactness of Y even if each X_n is locally compact. For example, $(\varinjlim \mathbb{R}^n) \times \ell_2 \neq \varinjlim(\mathbb{R}^n \times \ell_2)$. Indeed, each $\mathbb{R}^n$ is identified with $\mathbb{R}^n \times \{\mathbf{0}\} \subset \mathbb{R}^{\mathbb{N}}_f \subset \ell_2$. Then, we regard

$$(\varinjlim \mathbb{R}^n) \times \ell_2 = \varinjlim(\mathbb{R}^n \times \ell_2) = \mathbb{R}^{\mathbb{N}}_f \times \ell_2 \ \text{ as sets.}$$

Consider the following set:

$$D = \big\{(k^{-1}\mathbf{e}_n, n^{-1}\mathbf{e}_k) \in \mathbb{R}^{\mathbb{N}}_f \times \ell_2 \ \big|\ k, n \in \mathbb{N}\big\},$$

where each $\mathbf{e}_i \in \mathbb{R}^{\mathbb{N}}_f \subset \ell_2$ is the unit vector defined by $\mathbf{e}_i(i) = 1$ and $\mathbf{e}_i(j) = 0$ for $j \neq i$. For each $n \in \mathbb{N}$, let

$$D_n = \big\{(k^{-1}\mathbf{e}_n, n^{-1}\mathbf{e}_k) \ \big|\ k \in \mathbb{N}\big\}.$$

Since $\{n^{-1}\mathbf{e}_k \mid k \in \mathbb{N}\}$ is discrete in ℓ_2, it follows that D_n is discrete (so closed) in $\mathbb{R}^n \times \ell_2$, hence it is also closed in $\mathbb{R}^m \times \ell_2$ for every $m \geqslant n$.[18] Observe that $D \cap (\mathbb{R}^n \times \ell_2) = \bigcup_{i=1}^{n} D_i$. Then, D is closed in $\varinjlim(\mathbb{R}^n \times \ell_2)$. On the other hand, for each neighborhood U of $(0, 0)$ in $(\varinjlim \mathbb{R}^n) \times \ell_2$, we can apply Proposition 2.8.7 to take $\delta_i > 0$ $(i \in \mathbb{N})$ and $n \in \mathbb{N}$ so that

$$\left(\mathbb{R}^{\mathbb{N}}_f \cap \prod_{i \in \mathbb{N}} [-\delta_i, \delta_i]\right) \times n^{-1}\mathbf{B}_{\ell_2} \subset U,$$

where $\mathbf{B}_{\ell_2}$ is the unit closed ball of ℓ_2. Choose $k \in \mathbb{N}$ so that $k^{-1} < \delta_n$. Then, $(k^{-1}\mathbf{e}_n, n^{-1}\mathbf{e}_k) \in U$, which implies $U \cap D \neq \emptyset$. Consequently, D is not closed in $(\varinjlim \mathbb{R}^n) \times \ell_2$.

Remark 2.8.5 In Proposition 2.8.5, it is necessary to assume that both X_n and Y_n are locally compact. Indeed, let $X_n = \mathbb{R}^n$ and $Y_n = \ell_2$ for every $n \in \mathbb{N}$. Then, as seen above in Remark 2.8.4, it follows that $\varinjlim X_n \times \varinjlim Y_n \neq \varinjlim(X_n \times Y_n)$. Using this example, we can show that $\varinjlim(\ell_2)^n \times \varinjlim(\ell_2)^n \neq \varinjlim((\ell_2)^n \times (\ell_2)^n)$, so Proposition 2.8.5 does not hold in general even if $X_n = Y_n$ for every $n \in \mathbb{N}$. Identifying $\mathbb{R}^n = (\mathbb{R}\mathbf{e}_1)^n \subset (\ell_2)^n$ and $\ell_2 = \ell_2 \times \{\mathbf{0}\} \subset (\ell_2)^n$, we can regard $\mathbb{R}^n$ and ℓ_2 are closed subspaces of $(\ell_2)^n$. By Proposition 2.8.2, $(\varinjlim \mathbb{R}^n) \times \ell_2$ is a closed subspace of $\varinjlim(\ell_2)^n \times \varinjlim(\ell_2)^n$. On the other hand, $\mathbb{R}^n \times \ell_2$ is a closed subspace of $(\ell_2)^n \times (\ell_2)^n$. Again by Proposition 2.8.2, $\varinjlim(\mathbb{R}^n \times \ell_2)$ is a closed subspace

[17] This is an excellent exercise (cf. the first part of the proof of Wallace's Theorem 2.1.2).

[18] For a discrete subset, refer to Footnote 9 in Sect. 1.1.

of $\varinjlim((\ell_2)^n \times (\ell_2)^n)$. Since $(\varinjlim \mathbb{R}^n) \times \ell_2 \neq \varinjlim(\mathbb{R}^n \times \ell_2)$, we have $\varinjlim(\ell_2)^n \times \varinjlim(\ell_2)^n \neq \varinjlim((\ell_2)^n \times (\ell_2)^n)$.

Concerning compact subsets and metrizable subsets of the direct limit of a tower of spaces, we have the following:

Theorem 2.8.8 *For the direct limit $X = \varinjlim X_n$ of a tower $X_1 \subset X_2 \subset \cdots$ of spaces, the following hold:*

(1) *Every compact set $A \subset X$ is contained in some X_n.*
(2) *For each map $f : Y \to X$ from a first-countable space Y to X, each point of Y has a neighborhood V in Y such that the image $f(V)$ is contained in some X_n. In particular, if $A \subset X$ is a metrizable subspace, then each point of A has a neighborhood in A that is contained in some X_n.*

Proof (1): Assume that A is not contained in any X_n. For each $n \in \mathbb{N}$, take $x_n \in A \setminus X_n$ and let $D = \{x_n \mid n \in \mathbb{N}\} \subset A$. Then, D is infinite and discrete in $\varinjlim X_n$. Indeed, every $C \subset D$ is closed in $\varinjlim X_n$ because $C \cap X_n$ is finite for each $n \in \mathbb{N}$. This contradicts the compactness of A.

(2): Let $\{V_n \mid n \in \mathbb{N}\}$ be a neighborhood basis of y_0 in Y such that $V_n \subset V_{n-1}$. Assume that $f(V_n) \not\subset X_n$ for every $n \in \mathbb{N}$. Then, taking $y_n \in V_n \setminus f^{-1}(X_n)$, we have a compact set $A = \{y_n \mid n \in \omega\}$ in Y. Due to (1), $f(A)$ is contained in some X_m, and hence $f(y_m) \in X_m$. Thich is a contradiction. Therefore, $f(V_n) \subset X_n$ for some $n \in \mathbb{N}$. □

Due to Theorem 2.8.8 (2), the direct limit of metrizable spaces is non-metrizable in general (e.g., $\varinjlim \mathbb{R}^n$ is non-metrizable). However, it has some favorable properties, which we now discuss.

Theorem 2.8.9 *For the direct limit $X = \varinjlim X_n$ of a closed tower $X_1 \subset X_2 \subset \cdots$ of spaces, the following properties hold:*

(1) *If each X_n is normal, then X is also normal;*
(2) *If each X_n is perfectly normal, then X is also perfectly normal;*
(3) *If each X_n is collectionwise normal, then X is also collectionwise normal;*
(4) *If each X_n is paracompact, then X is also paracompact.*

Proof (1): Obviously, every singleton of X is closed, so X is T_1. Let A and B be disjoint closed sets in X. Then, we have a map $f_1 : X_1 \to \mathbf{I}$ such that $f_1(A \cap X_1) = 0$ and $f_1(B \cap X_1) = 1$. Using the Tietze–Urysohn Extension Theorem 2.2.2, we can extend f_1 to a map $f_2 : X_2 \to \mathbf{I}$ such that $f_2(A \cap X_2) = 0$ and $f_2(B \cap X_2) = 1$. Thus, we inductively obtain maps $f_n : X_n \to \mathbf{I}$, $n \in \mathbb{N}$, such that

$$f_n | X_{n-1} = f_{n-1}, \ \ f_n(A \cap X_n) = 0 \ \text{ and } \ f_n(B \cap X_n) = 1.$$

Let $f : X \to \mathbf{I}$ be the map defined by $f | X_n = f_n$ for $n \in \mathbb{N}$. Evidently, $f(A) = 0$ and $f(B) = 1$. Therefore, X is normal.

(2): From (1), it suffices to show that every closed set A in X is a G_δ set. Each X_n has open sets $G_{n,m}$, $m \in \mathbb{N}$, such that $A \cap X_n = \bigcap_{m\in\mathbb{N}} G_{n,m}$. For each $n, m \in \mathbb{N}$, let $G^*_{n,m} = G_{n,m} \cup (X \setminus X_n)$. Since X_n is closed in X, each $G^*_{n,m}$ is open in X. Observe that $A = \bigcap_{n,m\in\mathbb{N}} G^*_{n,m}$. Hence, A is G_δ in X.

(3): Let $\mathcal{F}$ be a discrete collection of closed sets in X. By induction on $n \in \mathbb{N}$, we have discrete collections $\{U^F_n \mid F \in \mathcal{F}\}$ of open sets in X_n such that $(F \cap X_n) \cup \operatorname{cl} U^F_{n-1} \subset U^F_n$ for each $F \in \mathcal{F}$, where $U^F_0 = \emptyset$. For each $F \in \mathcal{F}$, let $U_F = \bigcup_{n\in\mathbb{N}} U^F_n$. Then, $F \subset U_F$ and U_F is open in X because $U_F \cap X_n = \bigcup_{i\geqslant n} U^F_i \cap X_n$ is open in X_n for each $n \in \mathbb{N}$. If $F \neq F'$, then $U_F \cap U_{F'} = \emptyset$ because

$$U^F_i \cap U^{F'}_j \subset U^F_{\max\{i,j\}} \cap U^{F'}_{\max\{i,j\}} = \emptyset \quad \text{for each } i, j \in \mathbb{N}.$$

Therefore, X is collectionwise normal.

(4): Since every paracompact space is collectionwise normal (Theorem 2.6.1), X is also collectionwise normal by (3), so it is regular. Then, due to Theorem 2.6.3, it suffices to show that each $\mathcal{U} \in \operatorname{cov}(X)$ has a σ-discrete open refinement. By Theorem 2.6.3, we have $\bigcup_{m\in\mathbb{N}} \mathcal{V}_{n,m} \in \operatorname{cov}(X_n)$, $n \in \mathbb{N}$, such that each $\mathcal{V}_{n,m}$ is discrete in X_n and $\mathcal{V}_{n,m} \prec \mathcal{U}$. For each $V \in \mathcal{V}_{n,m}$, choose $U_V \in \mathcal{U}$ so that $V \subset U_V$. Note that each $\mathcal{V}^{\mathrm{cl}}_{n,m}$ is discrete in X, and recall that X is collectionwise normal. So, X has a discrete open collection $\{W_V \mid V \in \mathcal{V}_{n,m}\}$ such that $\operatorname{cl} V \subset W_V$. Let $\mathcal{W}_{n,m} = \{W_V \cap U_V \mid V \in \mathcal{V}_{n,m}\}$. Then, $\mathcal{W} = \bigcup_{n,m\in\mathbb{N}} \mathcal{W}_{n,m} \in \operatorname{cov}(X)$ is a σ-discrete open cover refinement of $\mathcal{U}$. □

From Theorems 2.8.9 and 2.6.10 and Proposition 2.8.3, we have the following:

Corollary 2.8.10 *The direct limit of a closed tower of metrizable spaces is a perfectly normal and paracompact k-space, and so it is hereditarily paracompact.* □

2.9 Counter-Examples

In this section, we show that the converses of Theorems 2.2.8 and 2.6.1 do not hold, and that perfect normality does not imply collectionwise normality. It is also shown that paracompactness does not imply hereditary normality (= complete normality). Consequently, normality does not imply complete normality. Thus, the converses of horizontal implications in the diagram just before Remark 2.1.1 do not hold. We also show that normality, collectionwise normality, and paracompactness are neither hereditary nor productive, and that perfect normality and hereditary normality are not productive either, where a topological property $\mathcal{P}$ is said to be **hereditary** if every subspace of a space X with $\mathcal{P}$ has also $\mathcal{P}$, and to be **productive** if a product space $X \times Y$ of two spaces X and Y with $\mathcal{P}$ has also $\mathcal{P}$. Additionally, we show that the direct limit of a closed tower of Hausdorff spaces need not be Hausdorff. Finally, we give an example showing the importance of paracompactness (or metrizability) in the definition of a manifold.

The Spaces $[0, \omega_1]$ and $[0, \omega_1)$

We first give examples showing that the converses of Theorems 2.2.8 and 2.6.1 do not hold, that is:

- *Complete normality (= hereditary normality) does not imply perfect normality.*
- *Collectionwise normality does not imply paracompactness.*

Let ω_1 be the first uncountable ordinal and $[0, \omega_1]$ the space of all ordinals $a \leqslant \omega_1$ with the order topology, while $[0, \omega_1) = [0, \omega_1] \setminus \{\omega_1\}$ is the space of all countable ordinals.

Proposition 2.9.1 *The space $[0, \omega_1]$ is compact, completely normal but not perfectly normal.*

Proof (*Non-perfect Normality*) The singleton $\{\omega_1\}$ is closed but not G_δ in $[0, \omega_1]$, which means that $[0, \omega_1]$ is not perfectly normal.

Indeed, if there are open sets U_n, $n \in \mathbb{N}$, such that $\{\omega_1\} = \bigcap_{n\in\mathbb{N}} U_n$, then each U_n contains $(\alpha_n, \omega_1]$, where $\alpha_n < \omega_1$. Let $\alpha = \sup_{n\in\mathbb{N}} \alpha_n \leqslant \omega_1$. Then, α is countable because $[0, \alpha) \subset \bigcup_{n\in\mathbb{N}}[0, \alpha_n]$ is countable. Since $\alpha < \alpha + 1 < \omega_1$, it follows that $\alpha + 1 \in \bigcap_{n\in\mathbb{N}}(\alpha_n, \omega_1] \subset \bigcap_{n\in\mathbb{N}} U_n = \{\omega_1\}$, which is a contradiction.

(*Complete Normality*) For each pair of non-empty separated sets A, B in $[0, \omega_1]$, let

$$A_0 = \{\alpha \in A \mid B \cap [0, \alpha] \neq \emptyset\} \text{ and } B_0 = \{\beta \in B \mid A \cap [0, \beta] \neq \emptyset\}.$$

For each $\alpha \in A_0$, define $\beta(\alpha) = \sup B \cap [0, \alpha]$. Since $\beta(\alpha) \in \operatorname{cl} B$, we have $\beta(\alpha) < \alpha$ and $(\beta(\alpha), \alpha] \cap B = \emptyset$. Similarly, for each $\beta \in B_0$, we have $\alpha(\beta) = \sup A \cap [0, \beta] < \beta$ such that $(\alpha(\beta), \beta] \cap A = \emptyset$. Then, A and B have the following open neighborhoods in $[0, \omega_1]$:

$$U = \bigcup_{\alpha\in A\setminus A_0} [0, \alpha] \cup \bigcup_{\alpha\in A_0} (\beta(\alpha), \alpha] \text{ and } V = \bigcup_{\beta\in B\setminus B_0} [0, \beta] \cup \bigcup_{\beta\in B_0} (\alpha(\beta), \beta].$$

For each $\alpha \in A$ and $\beta \in B$, $\alpha < \beta$ or $\beta < \alpha$. When $\alpha < \beta$, since $\beta \in B_0$ and $\alpha \leqslant \alpha(\beta)$, it follows that $[0, \alpha] \cap (\alpha(\beta), \beta] = \emptyset$, where $(\beta(\alpha), \alpha] \cap (\alpha(\beta), \beta] = \emptyset$ if $\alpha \in A_0$. When $\beta < \alpha$, it similarly follows that $[0, \beta] \cap (\beta(\alpha), \alpha] = \emptyset$, where $(\alpha(\beta), \beta] \cap (\beta(\alpha), \alpha] = \emptyset$ if $\beta \in B_0$. Therefore, we have $U \cap V = \emptyset$.

(*Compactness*) Let $\mathcal{U}$ be an open cover of $[0, \omega_1]$. Then, we have a function $\gamma : [1, \omega_1] \to [0, \omega_1)$ such that $(\gamma(\alpha), \alpha]$ is contained in some $U \in \mathcal{U}$. Let $\alpha_0 = \omega_1$ and inductively define $\alpha_i = \gamma(\alpha_{i-1})$ if $\alpha_{i-1} > 0$. Then, $\alpha_n = 0$ for some $n > 0$. Otherwise, we have an infinite decreasing sequence $\alpha_0 > \alpha_1 > \alpha_2 >$, which is a contradiction because $[0, \omega_1]$ is well-ordered. Observe that $\bigcup_{i=1}^{n}(\alpha_i, \alpha_{i-1}] \cup \{0\} = [0, \omega_1]$. Thus, $\mathcal{U}$ has a finite open refinement $\{\{0\},\ (\alpha_i, \alpha_{i-1}] \mid i = 1, \ldots, n\}$. □

Remark 2.9.1 By the same arguments as in the above proof, we can show that $(\alpha, \beta] = [\alpha + 1, \beta]$ is compact for every ordinal $\alpha < \beta$. Hence, $[0, \omega_1)$ is locally compact. Thus, $[0, \omega_1]$ is the one-point compactification of $[0, \omega_1)$.

Proposition 2.9.2 *The space $[0, \omega_1)$ is countably compact, completely normal and collectionwise normal but not paracompact.*[19]

Proof (*Countable Compactness*) Let $\mathcal{U} = \{U_n \mid n \in \mathbb{N}\}$ be a countable open cover of $[0, w_1)$. If $\mathcal{U}$ has no finite subcover, then we can take $\alpha_n \in [0, w_1) \setminus \bigcup_{i \leqslant n} U_n$, $n \in \mathbb{N}$. Let $\alpha = \sup_{n\in\mathbb{N}} \alpha_n < \omega_1$. Since $[0, \alpha]$ is compact (Remark 2.9.1), $[0, \alpha] \subset \bigcup_{i=1}^{n} U_n$ for some $n \in \mathbb{N}$, which is a contradiction because $\alpha_n \in [0, \alpha] \subset \bigcup_{i=1}^{n} U_n$. Hence, $\mathcal{U}$ has a finite subcover.

(*Complete Normality*) Since $[0, \omega_1]$ is completely normal (= hereditarily normal) by Proposition 2.9.1, it follows that $[0, \omega_1)$ is completely normal.

(*Collectionwise Normality*) Since $[0, \omega_1)$ is normal, it suffices to show that $[0, \omega_1)$ has no infinite discrete collection of closed sets. On the contrary, assume that $\mathcal{F}$ is an infinite discrete collection of closed sets in $[0, \omega_1)$. For each $F \in \mathcal{F}$, choose $\alpha_F \in F$. Then, $\{\alpha_F \mid F \in \mathcal{F}\}$ is discrete in $[0, \omega_1)$. For each $F \in \mathcal{F}$, $\{F' \in \mathcal{F} \mid \alpha_{F'} < \alpha_F\}$ is finite because $[0, \alpha_F]$ is compact. Hence, we can take $F_n \in \mathcal{F}$, $n \in \mathbb{N}$, so that $\alpha_{F_1} < \alpha_{F_2} < \cdots$. Let $\sup_{n\in\mathbb{N}} \alpha_{F_n} = \alpha_0$. Since each $[0, \alpha_{F_n})$ is countable, $[0, \alpha_0) = \bigcup_{n\in\mathbb{N}} [0, \alpha_{F_n})$ is also countable, which means $\alpha_0 < \omega_1$. For any $\beta < \alpha_0$, there is some $n \in \mathbb{N}$ such that $\beta < \alpha_{F_i} \leqslant \alpha_0$ (i.e., $\alpha_{F_i} \in (b, \alpha_0]$) for all $i \geqslant n$. Hence, $(\beta, \alpha_0]$ meets infinitely many $F \in \mathcal{F}$, which contradicts the discreteness of $\mathcal{F}$.

(*Non-paracompactness*) Consider the open cover $\mathcal{U} = \{[0, \alpha) \mid 0 < \alpha < \omega_1\}$ of $[0, \omega_1)$. We show that any refinement $\mathcal{V}$ of $\mathcal{U}$ is not locally finite in $[0, \omega_1)$. First, take any non-empty $V_1 \in \mathcal{V}$ and let $\alpha_1 = \sup V_1 < \omega_1$. Choose $V_2 \in \mathcal{V}$ so that $\alpha_1 + 1 \in V_2$, and let $\alpha_2 = \sup V_2 < \omega_1$. Inductively we can take $V_i \in \mathcal{V}$, $i \in \mathbb{N}$, so that $\alpha_i = \sup V_i$ and $\alpha_i + 1 \in V_{i+1}$. Then, $\alpha_1 < \alpha_2 < \alpha_3 < \cdots < \omega_1$. Let $\alpha = \sup_{i\in\mathbb{N}} \alpha_i < \omega_1$. For any ordinal $\beta < \alpha$, we have $n \in \mathbb{N}$ such that $\beta < \alpha_n < \alpha$. Then, $\alpha_{i-1} + 1 \in V_i \cap (\beta, \alpha]$ for every $i > n$, that is, $(\beta, \alpha]$ meets V_i, $i > n$. Consequently, $\mathcal{V}$ is not locally finite in $[0, \omega_1)$. □

By Proposition 2.9.2 above, we know that:

- *Compactness implies countable compactness but the converse is not true.*

Note that every compact space is paracompact. Due to Propositions 2.9.1 and 2.9.2, the following has also been shown:

- *Paracompactness is not hereditary.*

Here, we show the additional property of the space $[0, \omega_1)$:

Proposition 2.9.3 *For any map $f : [0, \omega_1) \to \mathbb{R}$, there exists some $\alpha < \omega_1$ such that $f|[\alpha, \omega_1)$ is constant.*

---

[19] X is **countably compact** if every countable open cover of X has a finite subcover. It is known that X is countably compact if and only if every infinite set in X has an accumulation point in X.

Proof First, we show that for each $\varepsilon > 0$, there is $\alpha_0 < \omega_1$ such that $|f(\alpha_0) - f(\alpha)| < \varepsilon$ for every $\alpha \in (\alpha_0, \omega_1)$. Assume that this does not hold. Then, there exists $\delta > 0$ with $\beta_1 < \beta_2 < \cdots < \omega_1$ such that $|f(\beta_n) - f(\beta_{n+1})| \geqslant \delta$ for each $n \in \mathbb{N}$. Let $\beta = \sup_{n\in\mathbb{N}} \beta_n < \omega_1$. Then, $\beta = \lim_{n\to\infty} \beta_n$ but $(f(\beta_n))_{n\in\mathbb{N}}$ is not convergent, which is a contradiction.

Now, for each $n \in \mathbb{N}$, we have $\alpha_n < \omega_1$ such that $|f(\alpha_n) - f(\alpha)| < 1/n$ for every $\alpha \in (\alpha_n, \omega_1)$. Let $\alpha = \sup_{n\in\mathbb{N}} \alpha_n < \omega_1$. For each $\beta \in (\alpha, \omega_1)$,

$$|f(\alpha) - f(\beta)| \leqslant |f(\alpha_n) - f(\alpha)| + |f(\alpha_n) - f(\beta)| < 2/n \quad \text{for any } n \in \mathbb{N},$$

which means that $f(\alpha) = f(\beta)$. Thus, $f|[\alpha, \omega_1)$ is constant. □

Using the above proposition, we can show the following:

Proposition 2.9.4 *The space* $[0, \omega_1)$ *is not perfectly normal.*

Proof Let S be the set of all limit ordinals $< \omega_1$. Then, S is closed in $[0, \omega_1)$ but not G_δ in $[0, \omega_1)$. Indeed, let U_n, $n \in \mathbb{N}$, open sets in $[0, \omega_1)$ containing S. For each $n \in \mathbb{N}$, take an Urysohn map $f_n : [0, \omega_1) \to \mathbf{I}$ such that $S \subset f_n^{-1}(0)$ and $[0, \omega_1) \setminus U_n \subset f_n^{-1}(1)$. Due to Proposition 2.9.3, $f_n|[\alpha_n, \omega_1)$ is constant for some $\alpha_n < \omega_1$. Since S is unbounded, we have $\alpha \in S$ such that $\alpha > \sup_{n\in\mathbb{N}} \alpha_n$. Then, $f_n(\alpha + 1) = f_n(\alpha) = 0$ for every $n \in \mathbb{N}$, which implies that $\alpha + 1 \in \bigcap_{n\in\mathbb{N}} U_n$. Hence, $S \subsetneqq \bigcap_{n\in\mathbb{N}} U_n$. □

The (Deleted) Tychonoff Plank

We give examples below showing:

- *Tychonoff-ness does not imply normality.*
- Paracompactness does not imply hereditary normality (= complete normality).

Let $[0, \omega_1]$ be the space of Proposition 2.9.1 and $[0, \omega]$ be the space of all ordinals $a \leqslant \omega$ with the order topology, where $[0, \omega]$ is the one-point compactification of the space $\omega = [0, \omega)$ of non-negative integers. The product space $T = [0, \omega_1] \times [0, \omega]$ is called the **Tychonoff plank**, which is a compact Hausdorff space, hence it is paracompact. The following dense subspace of the Tychonoff plank T is called the **deleted Tychonoff plank**:

$$T_\infty = [0, \omega_1] \times [0, \omega] \setminus \{(\omega_1, \omega)\},$$

where T_∞ is Tychonoff because every subspace of a Tychonoff space is Tychonoff. We now prove the following:

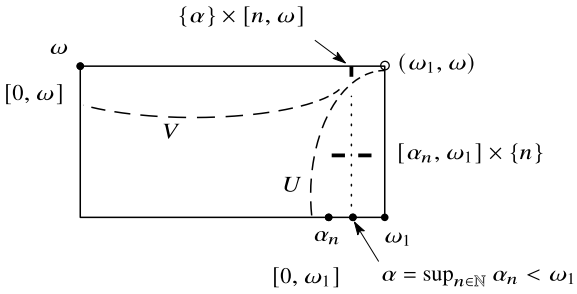


Fig. 2.11 The deleted Tychonoff plank T_∞

Proposition 2.9.5 *The deleted Tychonoff plank T_∞ is not normal.*

Proof We have disjoint closed sets $\{\omega_1\} \times [0, \omega)$ and $[0, \omega_1) \times \{\omega\}$ in T_∞. Assume that T_∞ has disjoint open sets U, V such that $\{\omega_1\} \times [0, \omega) \subset U$ and $[0, \omega_1) \times \{\omega\} \subset V$. For each $n \in \omega$, choose $\alpha_n < \omega_1$ so that $[\alpha_n, \omega_1] \times \{n\} \subset U$. Let $\alpha = \sup_{n\in\mathbb{N}} \alpha_n < \omega_1$. Then, $[\alpha, \omega_1] \times [0, \omega) \subset U$. On the other hand, we can choose $n \in \omega$ so that $\{\alpha\} \times [n, \omega] \subset V$. Thus, $U \cap V \neq \emptyset$, which is a contradiction. See Fig. 2.11. □

As a direct consequence of Proposition 2.9.5 above, we have the following:

Corollary 2.9.6 *The Tychonoff plank T is compact but not hereditarily normal (equiv., not completely normal).* □

Note that every compact space is collectionwise normal. By Proposition 2.9.5 and Corollary 2.9.6, we also know the following:

- *Normality and collectionwise normality are not hereditary.*

The Uncountable Products $\mathbb{N}^{\mathbf{I}}$ and $\mathbf{I}^{\mathbf{I}}$

Here is given other examples with similar properties as $T_\infty \subset T$. Such examples can be obtained as uncountable products of very standard spaces which are the spaces of all functions (not necessarily continuous) between very standard spaces with pointwise convergence topology.

Proposition 2.9.7 *The uncountable product $\mathbb{N}^I$ is Tychonoff but not normal.*

Proof It is easy to see that an arbitrary product of Tychonoff spaces is Tychonoff. In particular, $\mathbb{N}^{\mathbf{I}}$ is Tychonoff.

Indeed, let $X = \prod_{\lambda\in\Lambda} X_\lambda$ be a product space, where each X_λ is Tychonoff. For each closed set $F \subset X$ and each $y \in X \setminus F$, we can take $\lambda_1, \ldots, \lambda_n \in \Lambda$ and open neighborhoods U_i of $y(\lambda_i) \in X_{\lambda_i}$ so that $\bigcap_{i=1}^n \mathrm{pr}_{\lambda_i}^{-1}(U_i) = X \setminus F$. Then, we have maps $f_i : X_{\lambda_i} \to \mathbf{I}$ such that $X_{\lambda_i} \setminus U_i \subset f_i^{-1}(0)$ and $f_i(y(\lambda_i)) = 1$. Let $f : X \to \mathbf{I}$ be a map defined by $f(x) = \prod_{i=1}^n f_i \mathrm{pr}_{\lambda_i}$. It follows that

$$F \subset X \setminus \bigcap_{i=1}^n \mathrm{pr}_{\lambda_i}^{-1}(U_i) = \bigcup_{i=1}^n \mathrm{pr}_{\lambda_i}^{-1}(X_{\lambda_i} \setminus U_i) \subset \bigcup_{i=1}^n \mathrm{pr}_{\lambda_i}^{-1}(f_i^{-1}(0)) = f^{-1}(0)$$

and $f(y) = 1$. Therefore, X is also Tychonoff.

To show that $\mathbb{N}^{\mathbf{I}}$ is not normal, for $i = 1, 2$, let $P_i \subset \mathbb{N}^{\mathbf{I}}$ be the subset consisting of all points with the property that each integer except i appears at most once as a coordinate. Then, $P_1 \cap P_2 = \emptyset$ because every point of $\mathbb{N}^{\mathbf{I}}$ has uncountably many coordinates. Observe that

$$\mathbb{N}^{\mathbf{I}} \setminus P_i = \bigcup_{\substack{0\leqslant s<t\leqslant 1\\ n\in\mathbb{N}\setminus\{i\}}} \big(\mathrm{pr}_s^{-1}(n) \cap \mathrm{pr}_t^{-1}(n)\big).$$

Hence, P_1 and P_2 are closed in $\mathbb{N}^{\mathbf{I}}$. Then, it suffices to show that $U \cap V \neq \emptyset$ for any open neighborhoods U and V of P_1 and P_2, respectively.

For each $A \in \mathrm{Fin}(\mathbf{I})$ and $x \in \mathbb{N}^{\mathbf{I}}$, let $A(x) = \bigcap_{t\in A} \mathrm{pr}_t^{-1}(x(t))$. Then, note that $\{A(x) \mid A \in \mathrm{Fin}(\mathbf{I})\}$ is a neighborhood basis of x in $\mathbb{N}^{\mathbf{I}}$. By induction, we define $A_n = \{t_1, \ldots, t_{k(n)}\} \in \mathrm{Fin}(\mathbf{I})$, $x_n \in P_1$, $n \in \mathbb{N}$, so that $A_n(x_{n-1}) \subset U$ and $k(n) = \mathrm{card}\, A_n$, where $x_0 \in P_1$ is defined by $x_0(t) = 1$ for all $t \in \mathbf{I}$. Suppose that $x_{n-1} \in P_1$ and $A_{n-1} = \{t_1, \ldots, t_{k(n-1)}\} \in \mathrm{Fin}(\mathbf{I})$ have defined, where $A_0 = \emptyset$. Choose $A_n = \{t_1, \ldots, t_{k(n)}\} \in \mathrm{Fin}(\mathbf{I})$ so that $k(n-1) < k(n)$ (so $A_{n-1} \subsetneqq A_n$) and $A_n(x_{n-1}) \subset U$, and define $x_n \in P_1$ as follows:

$$x_n(t) = \begin{cases} j & \text{if } t = t_j,\ j = 1, \ldots, k(n),\\ 1 & \text{if } t \notin A_n. \end{cases}$$

Note that $\bigcup_{n\in\mathbb{N}} A_n = \{t_j \mid j \in \mathbb{N}\}$. We define $y \in P_2$ as follows:

$$y(t) = \begin{cases} j & \text{if } t = t_j,\ j \in \mathbb{N},\\ 2 & \text{if } t \notin \bigcup_{n\in\mathbb{N}} A_n. \end{cases}$$

Choose $B \in \mathrm{Fin}(\mathbf{I})$ so that $B(y) = \bigcap_{t\in B} \mathrm{pr}_t^{-1}(y(t)) \subset V$. Since B is finite and $A_n \subset A_{n+1}$, it follows that $B \cap \bigcup_{n\in\mathbb{N}} A_n = B \cap A_m$ for some $m \in \mathbb{N}$. We define $z \in \mathbb{N}^{\mathbf{I}}$ as follows:

$$z(t) = \begin{cases} j & \text{if } t = t_j,\ j = 1, \ldots, k(m),\\ 1 & \text{if } t \in A_{m+1} \setminus A_m,\\ 2 & \text{if } t \notin A_{m+1}. \end{cases}$$

Then, $z(t) = y(t)$ for $t \in B \cap A_m$ and $z(t) = 2 = y(t)$ for $t \in B \setminus A_m = B \setminus A_{m+1}$, which implies that $z \in B(z) = B(y) \subset V$. Moreover, $z(t) = x_m(t)$ for $t \in A_m$ and $z(t) = 1 = x_m(t)$ for $t \in A_{m+1} \setminus A_m$, which implies that $z \in A_{m+1}(z) = A_{m+1}(x_m) \subset U$. Thus, we have $z \in U \cap V$, so $U \cap V \neq \emptyset$. □

Corollary 2.9.8 *The uncountable product* $\mathbf{I}^{\mathbf{I}}$ *is compact but not hereditarily normal (equiv., not completely normal).*

Proof By the Tychonoff Theorem 2.1.1, $\mathbf{I}^{\mathbf{I}}$ is compact. Let $A = \{1/n \mid n \in \mathbb{N}\}$. Then, $A^{\mathbf{I}}$ is a subspace of $\mathbf{I}^{\mathbf{I}}$. Observe that $A^{\mathbf{I}} \approx \mathbb{N}^{\mathbf{I}}$, which is not normal by Proposition 2.9.7 above. Thus, $\mathbf{I}^{\mathbf{I}}$ contains a non-normal subspace, namely it is not hereditarily normal. □

The Space $F = \omega^{\mathfrak{P}(\mathbf{I})}$

Here, we will construct an example showing the following:

- *Perfect normality does not imply collectionwise normality.*

Let $F = \omega^{\mathfrak{P}(\mathbf{I})}$ be the space with the topology defined in the following. We define $f_t \in F$ (i.e., $f_t : \mathfrak{P}(\mathbf{I}) \to \omega$) as follows:

$$f_t(A) = \begin{cases} 1 & \text{if } t \in A, \\ 0 & \text{if } t \notin A, \end{cases}$$

that is, $\mathrm{pr}_A(f_t) = 1$ for $A \in \mathfrak{P}(\mathbf{I})[t]$ and $\mathrm{pr}_A(f_t) = 0$ for $A \notin \mathfrak{P}(\mathbf{I})[t]$. For each non-empty finite set $\mathcal{A} \in \mathrm{Fin}(\mathfrak{P}(\mathbf{I}))$ and $n \in \mathbb{N}$, let

$$\mathcal{A}(f_t, n) = \{f_t\} \cup \left\{ f \in F \;\middle|\; \forall A \in \mathcal{A},\ f(A) > n,\ f(A) \equiv f_t(A) \bmod 2 \right\}.$$

Then, for each $\mathcal{A}, \mathcal{A}' \in \mathrm{Fin}(\mathfrak{P}(\mathbf{I}))$ and $n, n' \in \mathbb{N}$,

$$\mathcal{A}(f_t, n) \cap \mathcal{A}'(f_t, n') \supset (\mathcal{A} \cup \mathcal{A}')(f_t, \max\{n, n'\}).$$

Hence, we can topologize F so that the following are neighborhood bases:

$$\left\{ \mathcal{A}(f_t, n) \;\middle|\; \mathcal{A} \in \mathrm{Fin}(\mathfrak{P}(\mathbf{I})),\ n \in \mathbb{N} \right\} \text{ for } f_t,\ t \in \mathbf{I}, \text{ and } \{\{f\}\} \text{ for } f \in F \setminus F_{\mathbf{I}},$$

where $F_{\mathbf{I}} = \{f_t \mid t \in \mathbf{I}\}$.

Proposition 2.9.9 *The space* F *is perfectly normal but not collectionwise normal.*

Proof (*Hausdorffness*) For each $f \neq f' \in F \setminus F_{\mathbf{I}}$, $\{f\} \cap \{f'\} = \emptyset$. For each $t \neq t' \in \mathbf{I}$, let $\mathcal{A} = \{\{t\}\} \in \mathrm{Fin}(\mathfrak{P}(\mathbf{I}))$. Then, $\mathcal{A}(f_t, 1) \cap \mathcal{A}(f_{t'}, 1) = \emptyset$ because $f_t(\{t\}) = 1 \not\equiv 0 = f_{t'}(\{t\})$. For each $t \in \mathbf{I}$ and $f \in F \setminus F_{\mathbf{I}}$, let $\mathcal{A} = \{\emptyset\} \in \mathrm{Fin}(\mathfrak{P}(\mathbf{I}))$ and $n = \max\{1, f(\emptyset)\} \in \mathbb{N}$. Then, $f \notin \mathcal{A}(f_t, n)$, which means $\mathcal{A}(f_t, n) \cap \{f\} = \emptyset$.

(*Normality*) For each disjoint pair of closed sets H_1 and H_2 in F, let

$$A_i = \left\{ t \in \mathbf{I} \;\middle|\; f_t \in H_i \right\} \subset \mathbf{I},\ i = 1, 2,$$

that is, $H_i \cap F_{\mathbf{I}} = \{f_t \mid t \in A_i\}$. Then, $A_1 \cap A_2 = \emptyset$. Let $\mathcal{A} = \{A_1\} \in \mathrm{Fin}(\mathfrak{P}(\mathbf{I}))$. For each $t \in A_1$ and $t' \in A_2$, $f_t(A_1) = 1 \not\equiv 0 = f_{t'}(A_1)$, which implies that $\mathcal{A}(f_t, 1) \cap \mathcal{A}(f_{t'}, 1) = \emptyset$. Moreover, for each $t \in A_i$, we can choose $\mathcal{A}_t \in \mathrm{Fin}(\mathfrak{P}(\mathbf{I}))$ and $n_t \in \mathbb{N}$ so that $\mathcal{A} \subset \mathcal{A}_t$ and $\mathcal{A}_t(f_t, n_t) \cap H_{3-i} = \emptyset$. Then,

$$\mathcal{A}_t(f_t, n_t) \subset \mathcal{A}(f_t, 1) \cap \left(F \setminus H_{3-i}\right) \text{ for each } t \in A_i.$$

Thus, we can define disjoint open neighborhoods of H_1 and H_2 in F as follows:

$$W_i = (H_i \setminus F_{\mathbf{I}}) \cup \bigcup_{t \in A_i} \mathcal{A}_t(f_t, n_t),\ i = 1, 2.$$

(*Perfect Normality*) It suffices to show that every closed set H in F is G_δ. Let $A = \{t \in \mathbf{I} \mid f_t \in H\}$, i.e., $H \cap F_{\mathbf{I}} = \{f_t \mid t \in A\}$, and $\mathcal{A} = \{\emptyset\} \in \mathrm{Fin}(\mathfrak{P}(\mathbf{I})$. The following are open neighborhoods of H in F:

$$W_n = (H \setminus F_{\mathbf{I}}) \cup \bigcup_{t \in A} \mathcal{A}(f_t, n),\ n \in \mathbb{N}.$$

Then, $\bigcap_{n \in \mathbb{N}} W_n = H$. Indeed, for each $f \in \bigcap_{n \in \mathbb{N}} W_n$, let $n_0 = \max\{1, f(\emptyset)\} \in \mathbb{N}$. Since $f \in W_{n_0}$, it follows that $f \in H \setminus F_{\mathbf{I}}$ or $f \in \mathcal{A}(f_t, n_0)$ for some $t \in A$. In the latter case, $f = f_t \in H \cap F_{\mathbf{I}}$ because $f(\emptyset) \leqslant n_0$. Hence, $f \in H$.

(*Non-collectionwise Normality*) Since $F_{\mathbf{I}}$ is closed in F and $\mathcal{A}(f_t, n) \cap F_{\mathbf{I}} = \{f_t\}$ for every $t \in \mathbf{I}$ and $n \in \mathbb{N}$, it follows that $\{\{f_t\} \mid t \in \mathbf{I}\}$ is a discrete collection of closed sets in F (cf. Footnote 9 in Sect. 1.1). Suppose that F is collectionwise normal. Then, there exist $\mathcal{A}_t \in \mathrm{Fin}(\mathfrak{P}(\mathbf{I}))$ and $n_t \in \mathbb{N}$ for each $t \in \mathbf{I}$ so that

$$\mathcal{A}_t(f_t, n_t) \cap \mathcal{A}_{t'}(f_{t'}, n_{t'}) = \emptyset \text{ if } t \neq t' \in \mathbf{I}.$$

Because $\mathbf{I}$ is uncountable, there exists an uncountable subset $J_0 \subset \mathbf{I}$ with $k, n \in \mathbb{N}$ such that $\mathrm{card}\, \mathcal{A}_t = k$ and $n_t = n$ for every $t \in J_0$. Then,

$$\mathcal{A}_t \cap \mathcal{A}_{t'} \neq \emptyset \text{ for each } t \neq t' \in J_0. \tag{$\sharp_1$}$$

Otherwise, we can define $f \in F$ (i.e., $f : \mathfrak{P}(\mathbf{I}) \to \omega$) as follows:

$$f(A) = \begin{cases} f_t(A) + 2n & \text{if } A \in \mathcal{A}_t, \\ f_{t'}(A) + 2n & \text{if } A \in \mathcal{A}_{t'}, \\ 0 & \text{if } A \notin \mathcal{A}_t \cup \mathcal{A}_{t'}. \end{cases} \tag{$*$}$$

Then, $f \in \mathcal{A}_t(f_t, n) \cap \mathcal{A}_{t'}(f_{t'}, n)$, which is a contradiction.

Now, take one $t_1 \in J_0$. Then, $\operatorname{card} \mathcal{A}_{t_1} = k$ and $f_t(\mathcal{A}_{t_1}) \subset \{0, 1\}$ for every $t \in J_0$. Because of $(\sharp_1)$ and uncountability of J_0, we have an uncountable set $J_1 \subset J_0$ and $A_1 \in \mathcal{A}_{t_1}$ such that $A_1 \in \mathcal{A}_t$ for every $t \in J_1$ and $f_t(A_1) = f_{t'}(A_1)$ for each $t \neq t' \in J_1$. Then,

$$\big(\mathcal{A}_t \setminus \{A_1\}\big) \cap \big(\mathcal{A}_{t'} \setminus \{A_1\}\big) \neq \emptyset \text{ for each } t \neq t' \in J_1. \tag{$\sharp_2$}$$

Otherwise, $\mathcal{A}_t \cap \mathcal{A}_{t'} = \{A_1\}$ for some $t \neq t' \in J_1$. Since $f_t(A_1) = f_{t'}(A_1)$, we can define $f \in F$ (i.e., $f : \mathfrak{P}(\mathbf{I}) \to \omega$) similarly to the above $(*)$. Then, $f \in \mathcal{A}_t(f_t, n) \cap \mathcal{A}_{t'}(f_{t'}, n)$, which is a contradiction.

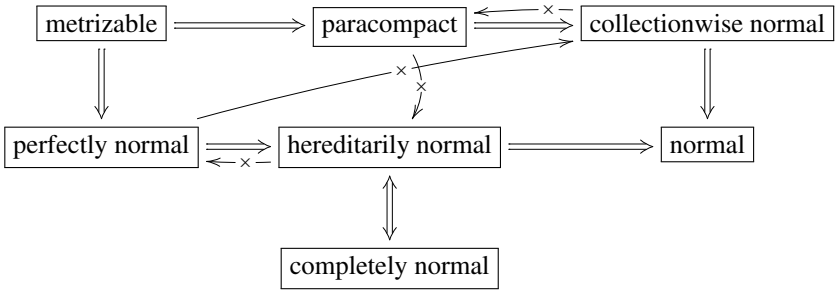
In the same way as above, taking $t_2 \in J_1$, by $(\sharp_2)$ and uncountability of J_1, we can find an uncountable set $J_2 \subset J_1$ and $A_2 \in \mathcal{A}_{t_2} \setminus \{A_1\}$ such that $A_1, A_2 \in \mathcal{A}_t$ for every $t \in J_2$ and $f_t(A_i) = f_{t'}(A_i)$, $i = 1, 2$, for each $t \neq t' \in J_2$.

By iteration, we can obtain uncountable sets $J_0 \supset J_1 \supset \cdots \supset J_k$ and distinct $A_1, \ldots, A_k \in \mathfrak{P}(\mathbf{I})$ such that $A_1, \ldots, A_k \in \mathcal{A}_t$ for every $t \in J_k$ and $f_t(A_i) = f_{t'}(A_i)$, $i = 1, \ldots, k$ for each $t \neq t' \in J_k$. Then, $\mathcal{A}_t = \{A_1, \ldots, A_k\}$ for every $t \in J_k$ because $\operatorname{card} \mathcal{A}_t = k$. By the definition, for each $t \neq t' \in J_k$,

$$\mathcal{A}_t(f_t, n) \cap \mathcal{A}_{t'}(f_{t'}, n) = \mathcal{A}_t(f_t, n) \setminus \{f_t\} = \mathcal{A}_{t'}(f_{t'}, n) \setminus \{f_{t'}\},$$

which contains $f \in F$ defined by $f(A_i) = f_t(A_i) + 2n, i = 1, \ldots, k$, and $f(A) = 0$ if $A \notin \mathcal{A}_t$. This is a contradiction. Thus, F is not collectionwise normal. □

We have seen that the complete normality (= hereditary normality) does not imply the perfect normality (Proposition 2.9.1), the collectionwise normality does not imply the paracompactness (Proposition 2.9.2), the paracompactness does not imply the hereditary normality (Corollary 2.9.6 or 2.9.8), and the perfect normality does not imply the collectionwise normality (Proposition 2.9.9), which are indicated by the arrows with cross $-\!\times\!\rightarrow$ in the following diagram. Consequently, in the diagram, the logical implication is not valid except for those indicated by the thick arrows $\Longrightarrow$ and their combinations:



The Sorgenfrey Line and Plane

Now, let us give an example showing:

- *The concepts of normality, perfect normality, hereditary normality, collectionwise normality, and paracompactness are not productive.*

The **Sorgenfrey line** S is the space $\mathbb{R}$ with the topology generated by $[a, b), a < b$. The product S^2 is called the **Sorgenfrey plane**. We show the following proposition concerning these spaces:

Proposition 2.9.10 *The Sorgenfrey line S has the following properties:*

(i) *S is a separable regular Lindelöf space, hence it is paracompact, and so is collectionwise normal;*
(ii) *S is perfectly normal, and so is hereditarily normal;*
(iii) S^2 *is not normal.*

Proof (i): The Hausdorffness of S is obvious. Since each basic open set $[a, b)$ is also closed in S, it is easy to see that S is regular. Clearly, $\mathbb{Q}$ is dense in S, so S is separable. To see that S is Lindelöf, let $\mathcal{U} \in \mathrm{cov}(S)$. Take a function $\gamma : S \to \mathbb{Q}$ so that $\gamma(x) > x$ and $[x, \gamma(x)) \subset U$ for some $U \in \mathcal{U}$. Then, $\{[x, \gamma(x)) \mid x \in S\} \in \mathrm{cov}(S)$ is an open refinement of $\mathcal{U}$. For each $q \in \gamma(S)$, if there exists $\min \gamma^{-1}(q)$, let $R(q) = \{\min \gamma^{-1}(q)\}$. Otherwise, choose a countable subset $R(q) \subset \gamma^{-1}(q)$ so that $\inf R(q) = \inf \gamma^{-1}(q)$, where we mean $\inf \gamma^{-1}(q) = -\infty$ if $\gamma^{-1}(q)$ is unbounded below. Then, the following is a subcover of $\{[x, \gamma(x)) \mid x \in S\} \in \mathrm{cov}(S)$:

$$\left\{[z, q) \;\middle|\; q \in \gamma(S),\ z \in R(q)\right\} \in \mathrm{cov}(S),$$

which is a countable open refinement of $\mathcal{U}$.

(ii): Let U be an open set in S. We have a function $\gamma : U \to \mathbb{Q}$ so that $\gamma(x) > x$ and $[x, \gamma(x)) \subset U$. Then, $U = \bigcup_{x \in U} [x, \gamma(x))$. By the same argument as the proof of (i), we can find a countable subcollection

$$\big\{[a_i, b_i) \;\big|\; i \in \mathbb{N}\big\} \subset \big\{[x, \gamma(x)) \;\big|\; x \in U\big\}$$

such that $U = \bigcup_{i\in\mathbb{N}}[a_i, b_i)$, hence U is F_σ in S. Thus, S is perfectly normal.

(iii): As we saw in the proof of (i), $\mathbb{Q}$ is dense in S, hence $\mathbb{Q}^2$ is dense in S^2. It follows that the restriction $\mathrm{C}(S^2, \mathbb{R}) \ni f \mapsto f|\mathbb{Q}^2 \in \mathbb{R}^{\mathbb{Q}^2}$ is injective. Therefore,

$$\operatorname{card} \mathrm{C}(S^2, \mathbb{R}) \leqslant \operatorname{card} \mathbb{R}^{\mathbb{Q}^2} = 2^{\aleph_0} = \mathfrak{c}.$$

On the other hand, $D = \{(x, y) \in S^2 \mid x + y = 0\}$ is a discrete set in S^2. Then, we have

$$\operatorname{card} \mathrm{C}(D, \mathbb{R}) = \operatorname{card} \mathbb{R}^D = 2^{\mathfrak{c}} > \mathfrak{c} \geqslant \operatorname{card} \mathrm{C}(S^2, \mathbb{R}).$$

If S^2 is normal, it would follow from the Tietze–Urysohn Extension Theorem 2.2.2 that the restriction $\mathrm{C}(S^2, \mathbb{R}) \ni f \mapsto f|D \in \mathrm{C}(D, \mathbb{R})$ is surjective, which is a contradiction. Consequently, S^2 is not normal. □

Remark 2.9.2 By Proposition 2.9.10, the product space $X \times Y$ is not necessarily normal (resp. paracompact) even if X and Y are normal (resp. paracompact). As is shown in Proposition 2.6.9 (4), $X \times Y$ is paracompact if X is paracompact and Y is compact. However, $X \times \mathbf{I}$ *is not necessarily normal for a normal space* X. Such a space X is called a **Dowker space**, which is characterized as a countably paracompact normal space.[20] It is difficult to construct such a space and prove that it would be a Dowker space, so unfortunately we will not do so here. Refer to Notes of this chapter.

A Non-Hausdorff Direct Limit

Additionally, we will construct a closed tower showing the following:

Proposition 2.9.11 *There exists a closed tower of spaces such that the direct limit is not Hausdorff.*

Proof Let Y be a space which is Hausdorff but non-normal, such as the deleted Tychonoff plank or the uncountable product $\mathbb{N}^{\mathbf{I}}$. Let A_0, A_1 be disjoint closed sets in Y that have no disjoint neighborhoods. We define $X = (Y \times \mathbb{N}) \cup \{0, 1\}$ with the topology generated by open sets in the product space $Y \times \mathbb{N}$ and sets of the form

$$\bigcup_{k>n}(U_k \times \{k\}) \cup \{i\},$$

[20] A space X is **countably paracompact** if every countable open cover of X has a locally finite open refinement.

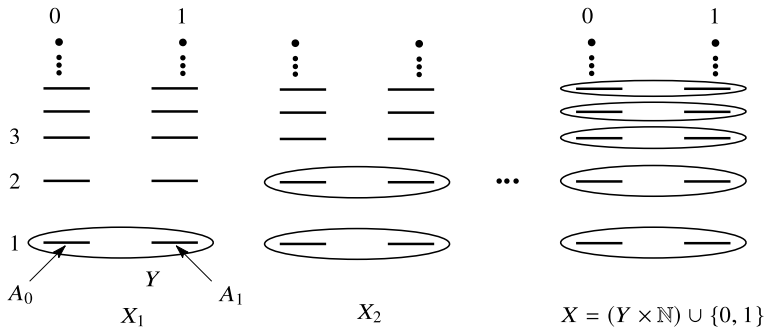


Fig. 2.12 Non-Hausdorff direct limit

where $i = 0, 1$ and each U_k is an open neighborhood of A_i (cf. Fig. 2.12). Then, X is not Hausdorff because 0 and 1 have no disjoint neighborhoods in X. For each $n \in \mathbb{N}$, let

$$X_n = Y \times \{1, \ldots, n\} \cup (A_0 \cup A_1) \times \{k \mid k > n\} \cup \{0, 1\}.$$

Then, $X_1 \subset X_2 \subset \cdots$ are closed in X and $X = \bigcup_{n\in\mathbb{N}} X_n$ (Fig. 2.12). As is easily observed, every X_n is Hausdorff.

All that remains is to prove that $X = \varinjlim X_n$, that is:

Claim X has the weak topology with respect to the tower $(X_n)_{n\in\mathbb{N}}$.

Since $\mathrm{id} : \varinjlim X_n \to X$ is obviously continuous, it suffices to show that every open set V in $\varinjlim X_n$ is open in X. To this end, assume that $V \cap X_n$ is open in X_n for each $n \in \mathbb{N}$. Each $x \in V \setminus \{0, 1\}$ is contained in some $Y \times \{n\} \subset X_n$. Then, $V \cap (Y \times \{n\})$ is an open neighborhood of x in $Y \times \{n\}$, and so is an open neighborhood in X. When $0 \in V$, $A_0 \times \{k \mid k > n\} \subset V$ for some $n \in \mathbb{N}$ because $V \cap X_1$ is open in X_1. For each $k > n$, since $V \cap (Y \times \{k\})$ is open in $Y \times \{k\}$, there is an open set U_k in Y such that $V \cap (Y \times \{k\}) = U_k \times \{k\}$. Note that $A_0 \subset U_k$. Then, $\bigcup_{k>n}(U_k \times \{k\}) \cup \{0\} \subset V$, hence V is a neighborhood of 0 in X. Similarly, V is a neighborhood of 1 in X if $1 \in V$. Thus, V is open in X. □

The Long Line and the Long Segment

Finally, we give an example showing the importance of assuming paracompactness (or metrizability) in the definition of a manifold. The **long line** $\mathbb{L}$ is constructed from the ordinal space $[0, \omega_1)$ by placing between each ordinal α and its successor $\alpha + 1$ a copy of the unit open interval $(0, 1)$. Precisely, $\mathbb{L}$ is defined as a space $[0, \omega_1) \times [0, 1)$ with the order topology defined by the lexicographical order, i.e.,

$$(\alpha, t) \leqslant (\alpha', t') \underset{\text{def}}{\Leftrightarrow} \alpha < \alpha' \text{ or } \alpha = \alpha', t \leqslant t'.$$

Identifying each $\alpha \in [0, \omega_1)$ with $(\alpha, 0) \in \mathbb{L}$, we regard $[0, \omega_1)$ as a closed subspace of $\mathbb{L}$. Writing $(\alpha, t) = \alpha + t$ makes it easy to consider the order relation on $\mathbb{L}$ and to regard $[0, \omega_1) \subset \mathbb{L}$. For $a < b \in \mathbb{L}$, the following sets are written $[a, b]_{\mathbb{L}}$, $(a, b]_{\mathbb{L}}$, $[a, b)_{\mathbb{L}}$, $(a, b)_{\mathbb{L}}$, respectively:

$$\{x \in \mathbb{L} \mid a \leqslant x \leqslant b\}, \{x \in \mathbb{L} \mid a < x \leqslant b\},$$
$$\{x \in \mathbb{L} \mid a \leqslant x < b\}, \{x \in \mathbb{L} \mid a < x < b\},$$

where $\mathbb{L} \setminus [0, a)_{\mathbb{L}}$ and $\mathbb{L} \setminus [0, a]_{\mathbb{L}}$ are written as $[a, \omega_1)_{\mathbb{L}}$ and $(a, \omega_1)_{\mathbb{L}}$, respectively. Note that $[\alpha, \alpha + 1) = \{\alpha\}$ but $[\alpha, \alpha + 1)_{\mathbb{L}} \cong [0, 1)$ (order isomorphic). When an ordinal $\alpha < \omega_1$ has a predecessor α', i.e., $\alpha = \alpha' + 1$, α has a neighborhood basis in $\mathbb{L}$ consisting of the following sets:

$$(\alpha' + t', \alpha + t)_{\mathbb{L}},\ 0 < t, t' < 1,$$

which are homeomorphic to the unit open interval $(0, 1)$. When $\alpha < \omega_1$ is a limit ordinal, α has a neighborhood basis in $\mathbb{L}$ consisting of the following sets:

$$(\alpha', \alpha + t)_{\mathbb{L}},\ \alpha' < \alpha,\ 0 < t < 1,$$

which are also homeomorphic to the unit open interval $(0, 1)$ by the following proposition:

Proposition 2.9.12 *For each $a < b \in \mathbb{L}$, $[a, b]_{\mathbb{L}} \approx \mathbf{I}$ and $(a, b)_{\mathbb{L}} \approx (0, 1)$. As a consequence, $\mathbb{L}$ is arcwise connected and each $x \in \mathbb{L}$ has a neighborhood that is homeomorphic to $\mathbf{I}$.*

Proof Because of similarity, we show only that $[a, b]_{\mathbb{L}} \approx \mathbf{I}$. Let $a = \alpha + s$ and $b = \beta + t$, where α and β are ordinals and $0 \leqslant s, t < 1$. When $\alpha = \beta$, since $s < t \in [0, 1)$, we have $[a, b]_{\mathbb{L}} \approx [s, t] \approx \mathbf{I}$. When $\alpha + 1 = \beta$, we have $[a, b]_{\mathbb{L}} = [a, \beta]_{\mathbb{L}} \cup [\beta, b]_{\mathbb{L}} \approx [s, 1] \cup [1, 1 + t] \approx \mathbf{I}$. When $\alpha + 1 < \beta$, we have $a < \beta$ and $[\beta, b]_{\mathbb{L}} = \{\beta\}$ if $t = 0$ or $[\beta, b]_{\mathbb{L}} \approx [0, t] \approx \mathbf{I}$ if $t > 0$. Then, it suffices to show the following:

$(*)_\beta$ $[a, \beta]_{\mathbb{L}} \approx \mathbf{I}$ for each $a < \beta$.

By transfinite induction on $\beta < \omega_1$, we prove this $(*)_\beta$. If $\beta = 1$, then $\alpha = 0$, and hence $[a, \beta]_{\mathbb{L}} = [s, 1] \approx \mathbf{I}$. Thus, $(*)_1$ holds. Assume that $(*)_{\beta'}$ holds for any $\beta' < \beta$. When β has a predecessor of β', i.e., $\beta = \beta' + 1$, if $\alpha = \beta'$, this case was already shown because $\alpha + 1 = \beta$. We may assume that $\alpha < \beta'$. Then, since $a < \beta'$, we have $[a, \beta']_{\mathbb{L}} \approx \mathbf{I}$ by $(*)_{\beta'}$ and $[\beta', \beta]_{\mathbb{L}} \approx \mathbf{I}$ by the case $\alpha + 1 = \beta$. Hence, $(*)_\beta$ can be easily obtained. When β is a limit ordinal ($< \omega_1$), we can take ordinals $a < \beta_1 < \beta_2 < \cdots < \beta$ such that $\lim_{n\to\infty} \beta_n = \sup_{n\in\mathbb{N}} \beta_n = \beta$. By $(*)_{\beta_n}$, we have

$$[a, \beta_1]_{\mathbb{L}} \approx [0, 1/2] \text{ and } [\beta_{n-1}, \beta_n]_{\mathbb{L}} \approx \left[\frac{n-1}{n}, \frac{n}{n+1}\right],\ n > 1.$$

Then, it follows that $[\alpha, \beta)_{\mathbb{L}} \approx [0, 1)$. Taking the one-point compactifications of both sides, we have $(*)_\beta$. □

Now, Proposition 2.9.13 below shows that, in the definition of a manifold in Sect. 1.1, assuming metrizability (or paracompactness) is important to avoid pathological phenomena.

Proposition 2.9.13 *The long line $\mathbb{L}$ is countably compact, but it is non-Lindelöf, non-separable, non-paracompact, and non-perfectly normal.*

Proof (*Countable Compactness*) We can prove countable compactness of $\mathbb{L}$ by small modification of the proof for $[0, \omega_1)$ (Proposition 2.9.2).

(*Non-Lindelöfness and Non-separability*) The open cover $\{[0, \alpha)_{\mathbb{L}} \mid \alpha < \omega_1\} \in \mathrm{cov}(\mathbb{L})$ has no countable subcover because $\sup_{n\in\mathbb{N}} \alpha_n + 1 < \omega_1$ for any α_n, $n \in \mathbb{N}$. Hence, $\mathbb{L}$ is non-Lindelöf. By the same reasoning, $\mathbb{L}$ is non-separable.

(*Non-paracompactness and Non-perfect Nomality*) Since $\mathbb{L}$ contains a closed set $[0, \omega_1)$ which is not paracompact nor perfectly normal (Propositions 2.9.2, 2.9.4), $\mathbb{L}$ is not paracompact nor perfectly normal, either. □

Let $\overline{\mathbb{L}} = \mathbb{L} \cup \{\omega_1\}$ and extend the order on $\mathbb{L}$ onto $\overline{\mathbb{L}}$ so that $\omega_1 = \max \overline{\mathbb{L}}$. The space $\overline{\mathbb{L}}$ with the order topology is called the **long segment** or the **extended long line**. Then, $\mathbb{L}$ is an open subspace of $\overline{\mathbb{L}}$ and $[0, \omega_1]$ is regarded as a closed subspace of $\overline{\mathbb{L}}$. In fact, $\overline{\mathbb{L}}$ is the one-point compactification of $\mathbb{L}$.

Indeed, ω_1 has a neighborhood basis in $\overline{\mathbb{L}}$ consisting of $(\alpha, \omega_1]_{\overline{\mathbb{L}}} = \{x \in \overline{\mathbb{L}} \mid \alpha < x \leqslant \omega_1\}$, $\alpha < \omega_1$, where $\mathbb{L} \setminus (\alpha, \omega_1]_{\overline{\mathbb{L}}} = [0, \alpha]_{\mathbb{L}}$ is compact. Conversely, for each compact set A in $\mathbb{L}$, $A \subset [0, \alpha]_{\mathbb{L}}$ for some $\alpha < \omega_1$ because the open cover $\{[0, \alpha)_{\mathbb{L}} \mid \alpha < \omega_1\}$ of A in $\mathbb{L}$ has a finite subcover. Then, $\overline{\mathbb{L}} \setminus A$ is a neighborhood of ω_1 in $\overline{\mathbb{L}}$ since $(\alpha, \omega_1]_{\overline{\mathbb{L}}} \subset \overline{\mathbb{L}} \setminus A$.

Proposition 2.9.14 *The long segment $\overline{\mathbb{L}}$ is compact but not perfectly normal. Moreover, $\overline{\mathbb{L}}$ is connected but not path-connected.*

Proof Since $\overline{\mathbb{L}}$ contains a closed set $[0, \omega_1]$ which is not perfectly normal (Proposition 2.9.1), $\overline{\mathbb{L}}$ is not perfectly normal, either. Since $\mathbb{L}$ is connected and dense in $\overline{\mathbb{L}}$, it follows that $\overline{\mathbb{L}}$ is connected.

(*Non-path-Connectedness*) Assume that $\overline{\mathbb{L}}$ is path-connected. Then, we have a path $f : \mathbf{I} \to \overline{\mathbb{L}}$ from 0 to ω_1, i.e., $f(0) = 0$ and $f(1) = \omega_1$. If $f(\mathbf{I}) = \overline{\mathbb{L}}$, then $f(\mathbf{I} \cap \mathbb{Q})$ is dense in $\overline{\mathbb{L}}$, which contradicts the non-separability of $\overline{\mathbb{L}}$. If $f(\mathbf{I}) \neq \overline{\mathbb{L}}$, then there exists a point $a \in \overline{\mathbb{L}} \setminus f(\mathbf{I})$. Since $0 < a < \omega_1$, $[0, a)_{\overline{\mathbb{L}}}$ and $(a, \omega_1]_{\overline{\mathbb{L}}}$ are disjoint open sets in $\overline{\mathbb{L}}$, and $f(\mathbf{I}) \subset [0, a)_{\overline{\mathbb{L}}} \cup (a, \omega_1]_{\overline{\mathbb{L}}}$, which contradicts the connectedness of $f(\mathbf{I})$. Consequently, $\overline{\mathbb{L}}$ is not path-connected. □

2.10 The Compact-Open Topology for Function Spaces

Let X and Y be spaces. Recall that $\mathrm{C}(X, Y)$ denotes the set of all maps from X to Y. On this set $\mathrm{C}(X, Y)$, various topologies can be introduced. In Chapter 1, we described two topologies among them, the uniform convergence topology and pointwise convergence topology. In this section, we discuss the compact-open topology,

which is the most standard topology. Here, we prove fundamental properties of this topology and also the Ascoli–Arzelà Theorem 2.10.11.

The **compact-open topology** on $\mathrm{C}(X, Y)$ is generated by the sets

$$\langle K; U\rangle = \big\{ f \in \mathrm{C}(X, Y) \;\big|\; f(K) \subset U \big\},$$

where K is any compact set in X and U is any open set in Y. For simplicity, we write $\langle \{x\}; U\rangle = \langle x; U\rangle$. The pointwise convergence topology is generated by the sets:

$$\langle x; U\rangle = \big\{ f \in \mathrm{C}(X, Y) \;\big|\; f(x) \in U \big\},$$

where $x \in X$ and U is open in Y. Thus, when $\mathrm{C}(X, Y)$ is considered as a space with the compact-open topology, $\mathrm{id} : \mathrm{C}(X, Y) \to \mathrm{C}_p(X, Y)$ is continuous, that is:

$$\boxed{\text{the pointwise convergence topology}} \subset \boxed{\text{the compact-open topology}}$$

Moreover, for every metric space $Y = (Y, d)$, $\mathrm{id} : \mathrm{C}_u(X, Y) \to \mathrm{C}(X, Y)$ is continuous, that is:

$$\boxed{\text{the compact-open topology}} \subset \boxed{\text{the uniform convergence topology}}$$

This is a direct consequence of the following:

Proposition 2.10.1 *When $Y = (Y, d)$ is a metric space, the following sets form a neighborhood basis of $f \in \mathrm{C}(X, Y)$ in the compact-open topology:*

$$\mathrm{B}_K(f, \varepsilon) = \big\{ g \in \mathrm{C}(X, Y) \;\big|\; d_\infty(f|K, g|K) < \varepsilon \big\},$$

where K is any compact set in X and $\varepsilon > 0$.

Proof Let K be a compact set in X and U an open set in Y. For each $f \in \langle K; U\rangle$, choose $\varepsilon > 0$ so that $\varepsilon < \mathrm{dist}(f(K), Y \setminus U)$. Then, $\mathrm{B}_K(f, \varepsilon) \subset \langle K; U\rangle$. Indeed,

$$g \in \mathrm{C}(X, Y),\ d_\infty(g|K, f|K) < \varepsilon \ \Rightarrow\ g(K) \subset U.$$

Thus, it suffices that for each compact set $K \subset X$ and $\varepsilon > 0$, $\mathrm{B}_K(f, \varepsilon)$ is a neighborhood of $f \in \mathrm{C}(X, Y)$ in the compact-open topology.

Each $x \in K$ has an open neighborhood $U(x)$ in K such that $f(\mathrm{cl}_K\, U(x)) \subset \mathrm{B}(f(x), \varepsilon/3)$. Since K is compact, there are finitely many $x_1, \ldots, x_n \in K$ such that $K = \bigcup_{i=1}^n U(x_i)$. Then, each $\mathrm{cl}_K\, U(x_i)$ is compact and f belongs to the following set:

$$\mathcal{U} = \bigcap_{i=1}^n \langle \mathrm{cl}_K\, U(x_i); \mathrm{B}(f(x_i), \varepsilon/3)\rangle.$$

It remains to show $\mathcal{U} \subset \mathrm{B}_K(f, \varepsilon)$. Take $g \in \mathcal{U}$. Each $x \in K$ is contained in some $\mathrm{cl}_K\, U(x_i)$. Since $f(x), g(x) \in \mathrm{B}(f(x_i), \varepsilon/3)$, it follows that

$$d(f(x), g(x)) \leqslant d(f(x), f(x_i)) + d(f(x_i), g(x)) < 2\varepsilon/3.$$

Hence, $d_\infty(f|K, g|K) \leqslant 2\varepsilon/3 < \varepsilon$, that is, $g \in \mathrm{B}_K(f, \varepsilon)$. □

Furthermore, we have the following:

Proposition 2.10.2 *Let X be a Tychonoff space and Y have a non-trivial path.*

(1) *The compact-open topology of $\mathrm{C}(X, Y)$ coincides with the pointwise convergence topology (i.e., $\mathrm{C}(X, Y) = \mathrm{C}_p(X, Y)$ as spaces) if and only if every compact set in X is finite.*
(2) *When $Y = (Y, d)$ is a metric space, the compact-open topology of $\mathrm{C}(X, Y)$ coincides with the uniform convergence topology (i.e., $\mathrm{C}(X, Y) = \mathrm{C}_u(X, Y)$ as spaces) if and only if X is compact, where the sup-metric d_∞ is admissible for the compact-open topology.*

Proof In both (1) and (2), the "if" part can be easily shown (for (2) use Proposition 2.10.1). Thus, we prove the "only if" part. To this end, take a path $\alpha : \mathbf{I} \to Y$ with $\alpha(0) \neq \alpha(1)$ and let $f_0 : X \to Y$ be a constant map with $f_0(X) = \{\alpha(0)\}$.

(1): We have an open set U in Y such that $\alpha(0) \in U$ and $\alpha(1) \notin U$. Assume that X has a compact set K containing infinitely many points. Then, $\langle K; U\rangle$ is an open neighborhood of f_0 in the compact-open topology but it does not contain any neighborhood $\mathcal{V}$ of f_0 in the pointwise convergence topology. Indeed, since $f_0(X) = \{\alpha(0)\}$, we can take a finite set $F \subset X$ and an open neighborhood V of $\alpha(0)$ in Y so that $\langle F; V\rangle \subset \mathcal{V}$. Then, we have $x_0 \in K \setminus F$. Since X is Tychonoff, there is a map $k : X \to \mathbf{I}$ such that $k(x_0) = 1$ and $k(F) = \{0\}$. Then, $\alpha k \in \langle F; V\rangle \subset \mathcal{V}$ but $\alpha k \notin \langle K; U\rangle$.

(2): Let $\delta = d(\alpha(0), \alpha(1)) > 0$. Assume that X is not compact. Then, $\mathrm{B}_{d_\infty}(f_0, \delta)$ is an open neighborhood of f_0 in the uniform convergence topology but it does not contain any neighborhood $\mathcal{V}$ of f_0 in the compact-open topology. Indeed, by Proposition 2.10.1, we have a compact set $K \subset X$ and $\varepsilon > 0$ such that $\mathrm{B}_K(f_0, \varepsilon) \subset \mathcal{V}$. Then, we can take $x_0 \in X \setminus K$. Since X is Tychonoff, we have a map $k : X \to \mathbf{I}$ such that $k(x_0) = 1$ and $k(K) = \{0\}$. Then,

$$\alpha k|K = f_0|K \ \text{ but } \ d(f(x_0), \alpha k(x_0)) = d(\alpha(0), \alpha(1)) = \delta,$$

which means that $\alpha k \in \mathrm{B}_K(f_0, \varepsilon) \subset \mathcal{V}$ but $\alpha k \notin \mathrm{B}_{d_\infty}(f_0, \delta)$. □

Let $\mathcal{F} \subset \mathrm{C}(X, Y)$. The subspace topology of $\mathcal{F}$ inherited from the compact-open topology of $\mathrm{C}(X, Y)$ is also the **compact-open topology**. Henceforth, in this section, we assume the following:

- *The space $\mathrm{C}(X, Y)$ and its subspaces have the compact-open topology.*

Proposition 2.10.3 *Every map $f : Z \times X \to Y$ (or $f : X \times Z \to Y$) induces the map $\bar{f} : Z \to \mathrm{C}(X, Y)$ defined by $\bar{f}(z)(x) = f(z, x)$ (or $\bar{f}(z)(x) = f(x, z)$). Moreover, the function $\varphi : \mathrm{C}(Z \times X, Y) \to \mathrm{C}(Z, \mathrm{C}(X, Y))$ defined by $\varphi(f) = \bar{f}$ is continuous and preserves the relation $\simeq$, i.e., for $f, g \in \mathrm{C}(Z \times X, Y)$, if $f \simeq g$, then $\varphi(f) \simeq \varphi(g)$.*

Proof For each $z \in Z$, it is easy to see that $\bar{f}(z) : X \to Y$ is continuous, i.e., $\bar{f}(z) \in \mathrm{C}(X, Y)$. Thus, $\bar{f}$ is well-defined.

To verify the continuity of $\bar{f} : Z \to \mathrm{C}(X, Y)$, it suffices to show that $\bar{f}^{-1}(\langle K; U\rangle)$ is open in Z for each compact set K in X and each open set U in Y. Let $z \in \bar{f}^{-1}(\langle K; U\rangle)$, i.e., $f(\{z\} \times K) \subset U$. Using the compactness of K, we can easily find an open neighborhood V of z in Z such that $f(V \times K) \subset U$, which means that $V \subset \bar{f}^{-1}(\langle K; U\rangle)$.

For the continuity of φ, observe that for a compact set K in X, an open set U in Y, and a compact set K' in Z, $\varphi(\langle K' \times K; U\rangle) \subset \langle K'; \langle K; U\rangle\rangle$.

For $f, g \in \mathrm{C}(Z \times X, Y)$, let $h : Z \times X \times \mathbf{I} \to Y$ be a homotopy from f to g. In the above, replacing Z with $Z \times \mathbf{I}$, we have a map $\bar{h} : Z \times \mathbf{I} \to \mathrm{C}(X, Y)$, which is a homotopy from $\bar{f}$ to $\bar{g}$. □

The following is the proper version of Proposition 2.10.3 above:

Proposition 2.10.4 *Every proper map $f : Z \times X \to Y$ (or $f : X \times Z \to Y$) induces the map $\bar{f} : Z \to \mathrm{C}^P(X, Y)$ defined by $\bar{f}(z)(x) = f(z, x)$ (or $\bar{f}(z)(x) = f(x, z)$).*

Sketch of Proof It suffices to show that $\bar{f}(z) : X \to Y$ is proper ($\bar{f}(z) \in \mathrm{C}^P(X, Y)$) for each $z \in Z$. For each compact set $K \subset Y$, $\bar{f}(z)^{-1}(K)$ is compact because

$$\{z\} \times \bar{f}(z)^{-1}(K) = f^{-1}(K) \cap (\{z\} \times X) \ \big(\text{or } \bar{f}(z)^{-1}(K) \times \{z\} = f^{-1}(K) \cap (X \times \{z\})\big).$$

Concerning the homotopy relation $\simeq$ on $\mathrm{C}(X, Y)$, we have the following:

Proposition 2.10.5 *For each $f, g \in \mathrm{C}(X, Y)$, if $f \simeq g$, then f and g are connected by a path in $\mathrm{C}(X, Y)$. For a k-space X, the converse is also true, that is, $f \simeq g$ if and only if f and g are connected by a path in $\mathrm{C}(X, Y)$.*

Proof By Proposition 2.10.3, a homotopy $h : X \times \mathbf{I} \to Y$ from f to g induces the path $\bar{h} : \mathbf{I} \to \mathrm{C}(X, Y)$ defined as $\bar{h}(t)(x) = h(x, t)$ for each $t \in \mathbf{I}$ and $x \in X$, where $\bar{h}(0) = h_0 = f$ and $\bar{h}(1) = h_1 = g$.

Now, we assume that X is a k-space. For a path $\varphi : \mathbf{I} \to \mathrm{C}(X, Y)$ from f to g, we define a homotopy $\tilde{\varphi} : X \times \mathbf{I} \to Y$ by $\tilde{\varphi}(x, t) = \varphi(t)(x)$ for each $(x, t) \in X \times \mathbf{I}$. Then, $\tilde{\varphi}_0 = \varphi(0) = f$ and $\tilde{\varphi}_1 = \varphi(1) = g$. To see the continuity of $\tilde{\varphi}$, it suffices to verify the continuity of $\tilde{\varphi}|K$ for each compact set K in $X \times \mathbf{I}$ because $X \times \mathbf{I}$ is a k-space by Corollary 2.1.10. For each $(x, t) \in K$ and each open neighborhood U of $\tilde{\varphi}(x, t) = \varphi(t)(x)$ in Y, since $\varphi(t)|\mathrm{pr}_X(K)$ is continuous, x has a closed neighborhood N in $\mathrm{pr}_X(K)$ such that $\varphi(t)(N) \subset U$, where N is compact and $\varphi(t) \in \langle N; U\rangle$. Since φ is continuous, t has a neighborhood V in $\mathbf{I}$ such that $\varphi(V) \subset \langle N; U\rangle$. Then, $(N \times V) \cap K$ is a neighborhood of (x, t) in K and $\tilde{\varphi}((N \times V) \cap K) \subset U$. This means that $\tilde{\varphi}|K$ is continuous. □

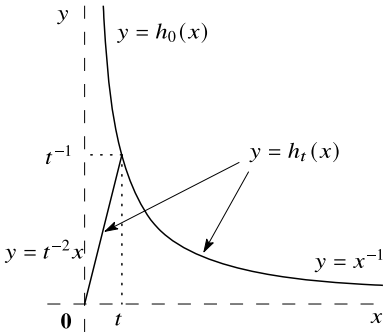


Fig. 2.13 The homotopy $h : (0, \infty) \times \mathbf{I} \to (0, \infty)$

Remark 2.10.1 It is easily observed that Proposition 2.10.5 is also valid for

$$\mathrm{C}((X, X_1, \ldots, X_n), (Y, Y_1, \ldots, Y_n)).$$

As a direct consequence of Proposition 2.10.4, we have the following proper version of the first half of Proposition 2.10.5:

- *Each* $f, g \in \mathrm{C}^P(X, Y)$ *are connected by a path in* $\mathrm{C}^P(X, Y)$ *if* $f \simeq^p g$.

The converse of the above is false even if $X = (0, \infty)$. Namely, the second half of Proposition 2.10.5 does not hold for $\mathrm{C}^P(X, Y)$. For example, let $h : (0, \infty) \times \mathbf{I} \to (0, \infty)$ be a homotopy defined as follows:

$$h(x, t) = \begin{cases} t^{-2}x & \text{if } x \leqslant t, \\ x^{-1} & \text{if } x \geqslant t, \end{cases}$$

where $h_0(x) = x^{-1}$ (see Fig. 2.13). Since $h_t|(0, t]$ and $h_t|[t, \infty)$ are closed embeddings for $t > 0$ and h_0 is homeomorphism, it follows that each h_t is a proper (perfect) map. Hence, h induces a path $\bar{h} : \mathbf{I} \to \mathrm{C}^P((0, \infty), (0, \infty))$ from h_0 to h_1. However, h is not proper because

$$\mathrm{pr}_1 h^{-1}(1) = \bigcup_{t \in \mathbf{I}} h_t^{-1}(1) = \left\{1,\ t^2 \;\middle|\; 0 < t \leqslant 1\right\} = (0, 1].$$

Furthermore, any other homotopy $h' : (0, \infty) \times \mathbf{I} \to (0, \infty)$ from h_0 to h_1 is not proper. Indeed, suppose that h' is proper. Since $h'^{-1}(1)$ is a compact set in the product space $(0, \infty) \times \mathbf{I}$, we have $a = \min \mathrm{pr}_1(h'^{-1}(1))/2 > 0$. Then,

$$h'(a, 0) = h_0(a) = a^{-1} > 1 \ \text{ and } \ h'(a, 1) = h_1(a) = a < 1.$$

By the Intermediate Value Theorem, we have $0 < t < 1$ such that $h'(a, t) = 1$, that is, $(a, t) \in h'^{-1}(1)$. It follows that $a \geqslant \min \mathrm{pr}_1(h'^{-1}(1))$, which is a contradiction. Consequently, h_0 and h_1 are connected by a path in $\mathrm{C}^P((0, \infty), (0, \infty))$ but $h_0 \not\simeq^p h_1$.

In the proof of Proposition 2.10.5, $\mathbf{I}$ can be replaced with any locally compact space. If X is locally compact, then $\mathbf{I}$ can be replaced with an arbitrary space Z. Namely, we can prove the following:

Proposition 2.10.6 *For a map $f : Z \to \mathrm{C}(X, Y)$, let $\tilde{f} : X \times Z \to Y$ be a function defined by $\tilde{f}(x, z) = f(z)(x)$. Then, $\tilde{f}$ is continuous in the following cases:*

(i) *X is a k-space and Z is locally compact;* (ii) *X is locally compact.*

In both cases, the function $\psi : \mathrm{C}(Z, \mathrm{C}(X, Y)) \to \mathrm{C}(X \times Z, Y)$ defined by $\psi(f) = \tilde{f}$ is continuous and preserves the relation $\simeq$, i.e., for $f, g \in \mathrm{C}(Z, \mathrm{C}(X, Y))$, if $f \simeq g$, then $\psi(f) \simeq \psi(g)$.

Proof (i): Since $X \times Z$ is a k-space by Corollary 2.1.10, the continuity of $\tilde{f}$ can be proved by the same argument as in the proof Proposition 2.10.5.

(ii): For each open set U in Y let $(x, z) \in \tilde{f}^{-1}(U)$. Since $f(z) \in \mathrm{C}(X, Y)$ and $f(z)(x) = \tilde{f}(x, z) \in U$, we have a neighborhood V of x in X such that $\mathrm{cl}\, V$ is compact and $f(z)(\mathrm{cl}\, V) \subset U$. Then, $f(z) \in \langle \mathrm{cl}\, V; U \rangle$. Hence, we have a neighborhood W of z in Z such that $f(W) \subset \langle \mathrm{cl}\, V; U \rangle$.

(*The Continuity of ψ*) Let K be a compact set in $X \times Z$ and U an open set in Y. It suffices to show that $\psi^{-1}(\langle K; U \rangle)$ is open in $\mathrm{C}(Z, \mathrm{C}(X, Y))$. Note that $K_1 = \mathrm{pr}_X(K)$ and $K_2 = \mathrm{pr}_Z(K)$ are compact and $K \subset K_1 \times K_2$. For each $f \in \psi^{-1}(\langle K; U \rangle)$, we can find finite open sets $V_1, \ldots, V_n$ in K_1 and $W_1, \ldots, W_n$ in K_2 such that $\mathrm{cl}\, V_i$ and $\mathrm{cl}\, W_i$ are compact and $\tilde{f}(\mathrm{cl}\, V_i \times \mathrm{cl}\, W_i) \subset U$, where $\tilde{f} = \psi(f)$. Then, $\bigcap_{i=1}^n \langle \mathrm{cl}\, W_i; \langle \mathrm{cl}\, V_i; U \rangle \rangle$ is a neighborhood of f in $\mathrm{C}(Z, \mathrm{C}(X, Y))$ which is contained in $\psi^{-1}(\langle K; U \rangle)$. Indeed,

$$\begin{aligned} \psi\Big(\textstyle\bigcap_{i=1}^n \langle \mathrm{cl}\, W_i; \langle \mathrm{cl}\, V_i; U \rangle \rangle\Big) &\subset \bigcap_{i=1}^n \psi(\langle \mathrm{cl}\, W_i; \langle \mathrm{cl}\, V_i; U \rangle \rangle) \\ &\subset \bigcap_{i=1}^n \langle \mathrm{cl}\, V_i \times \mathrm{cl}\, W_i; U \rangle \subset \langle K; U \rangle. \end{aligned}$$

Hence, $\psi^{-1}(\langle K; U \rangle)$ is open in $\mathrm{C}(Z, \mathrm{C}(X, Y))$.

For the additional statement, let $h : Z \times \mathbf{I} \to \mathrm{C}(X, Y)$ be a homotopy from f to g. In the above, replacing Z with $Z \times \mathbf{I}$, we have a map $\tilde{h} : X \times Z \times \mathbf{I} \to Y$ which is a homotopy from $\tilde{f}$ to $\tilde{g}$. □

In Proposition 2.10.6, the map $\psi : \mathrm{C}(Z, \mathrm{C}(X, Y)) \to \mathrm{C}(X \times Z, Y)$ is the inverse of the map $\varphi : \mathrm{C}(X \times Z, Y) \to \mathrm{C}(Z, \mathrm{C}(X, Y))$ in Proposition 2.10.3 because $\psi\varphi = \mathrm{id}$ and $\varphi\psi = \mathrm{id}$. Hence, we have the following:

Corollary 2.10.7 *In cases (i) and (ii) of Proposition 2.10.6, the map φ (resp. ψ) obtained in Proposition 2.10.3 (resp. 2.10.6) is a homeomorphism. Namely,*

$$\mathrm{C}(X \times Z, Y) \approx \mathrm{C}(Z, \mathrm{C}(X, Y)).$$

Moreover, for $f, g \in \mathrm{C}(X \times Z, Y)$, $f \simeq g$ if and only if $\varphi(f) \simeq \varphi(g)$ (resp. for $f, g \in \mathrm{C}(Z, \mathrm{C}(X, Y))$, $f \simeq g$ if and only if $\varphi(f) \simeq \varphi(g)$). □

We list some more properties of the compact-open topology:

Proposition 2.10.8 (Some Properties of the Compact-Open Topology) *The following hold with respect to the compact-open topology:*

(1) *For $f \in \mathrm{C}(Z, X)$ and $g \in \mathrm{C}(Y, Z)$, the following are continuous:*

$$f^* : \mathrm{C}(X, Y) \to \mathrm{C}(Z, Y),\ f^*(h) = h \circ f;$$
$$g_* : \mathrm{C}(X, Y) \to \mathrm{C}(X, Z),\ g_*(h) = g \circ h.$$

(2) *In the case that Y is locally compact, the following (composition) is continuous:*

$$\mathrm{C}(X, Y) \times \mathrm{C}(Y, Z) \ni (f, g) \mapsto g \circ f \in \mathrm{C}(X, Z).$$

Sketch of Proof Let K be a compact set in X and U an open set in Z with $f \in \mathrm{C}(X, Y)$ and $g \in \mathrm{C}(Y, Z)$ such that $g \circ f(K) \subset U$. Since Y is locally compact, we have an open set V in Y such that $\mathrm{cl}\, V$ is compact, $f(K) \subset V$ and $g(\mathrm{cl}\, V) \subset U$. Then, $f'(K) \subset V$ and $g'(\mathrm{cl}\, V) \subset U$ imply $g' \circ f'(K) \subset U$.

(3) *For each $x \in X$, the following (evaluation) is continuous:*

$$\mathrm{ev}_x : \mathrm{C}(X, Y) \to Y,\ \mathrm{ev}_x(f) = f(x)$$

(4) *When X is locally compact, the following (evaluation) is continuous:*

$$\mathrm{ev} : \mathrm{C}(X, Y) \times X \to Y,\ \mathrm{ev}(f, x) = f(x).$$

In this case, for every map $f : Z \to \mathrm{C}(X, Y)$, the following is continuous:

$$Z \times X \ni (z, x) \mapsto f(z)(x) \in Y.$$

(5) *In the case that X is locally compact,*

$$w(Y) \leqslant w(\mathrm{C}(X, Y)) \leqslant \aleph_0 w(X) w(Y).$$

Sketch of Proof By embedding Y into $\mathrm{C}(X, Y)$, we obtain the first inequality. For the second, we take open bases $\mathcal{B}_X$ and $\mathcal{B}_Y$ for X and Y, respectively, such that $\operatorname{card}\mathcal{B}_X = w(X)$, $\operatorname{card}\mathcal{B}_Y = w(Y)$, and $\operatorname{cl} A$ is compact for every $A \in \mathcal{B}_X$. The following is an open sub-basis for $\mathrm{C}(X, Y)$:

$$\mathcal{B} = \left\{ \langle \operatorname{cl} A, B\rangle \;\middle|\; (A, B) \in \mathcal{B}_X \times \mathcal{B}_Y \right\}.$$

Indeed, let K be a compact set in X, U be an open set in Y, and $f \in \mathrm{C}(X, Y)$ with $f(K) \subset U$, i.e., $f \in \langle K, U\rangle$. First, find $B_1, \ldots, B_n \in \mathcal{B}_Y$ so that $f(K) \subset B_1 \cup \cdots \cup B_n \subset U$. Next, find $A_1, \ldots, A_m \in \mathcal{B}_X$ so that $K \subset A_1 \cup \cdots \cup A_m$ and each $\operatorname{cl} A_i$ is contained in some $f^{-1}(B_{j(i)})$. Then, $f \in \bigcap_{i=1}^{m} \langle \operatorname{cl} A_i, B_{j(i)}\rangle \subset \langle K, U\rangle$.

(6) *Let $X = \bigcup_{n\in\mathbb{N}} X_n$, where X_n is compact and $X_n \subset \operatorname{int} X_{n+1}$. If $Y = (Y, d)$ is a metric space, then $\mathrm{C}(X, Y)$ with the compact-open topology is metrizable.*

Sketch of Proof We define a metric ρ on $\mathrm{C}(X, Y)$ as follows:

$$\rho(f, g) = \sup_{n\in\mathbb{N}} \min\left\{ n^{-1}, \sup_{x\in X_n} d(f(x), g(x)) \right\}.$$

Then, ρ is admissible for the compact-open topology on $\mathrm{C}(X, Y)$. To see this, use Proposition 2.10.1 and the fact that each compact set in X is contained in some X_n.

(7) *Let $X = \bigoplus_{\lambda\in\Lambda} X_\lambda$. Then, $\mathrm{C}(X, Y)$ is homeomorphic to $\prod_{\lambda\in\Lambda} \mathrm{C}(X_\lambda, Y)$ by the correspondence $f \mapsto (f|X_\lambda)_{\lambda\in\Lambda}$.*

Sketch of Proof The correspondence is obviously bijective. For each $\lambda \in \Lambda$, let $i_\lambda : X_\lambda \subset X$ be the inclusion. Since each $f \mapsto f|X_\lambda = i_\lambda^*(f)$ is continuous by the above (1), the correspondence is continuous. To see an open map, let K be a compact set in X and U an open set in Y. Choose $\lambda_1, \ldots, \lambda_n \in \Lambda$ so that $K \subset \bigcup_{i=1}^{n} X_{\lambda_i}$. Then, by the correspondence, $\langle K, U\rangle$ maps onto $\bigcap_{i=1}^{n} \mathrm{pr}_{\lambda_i}^{-1}\langle K \cap X_{\lambda_i}, U\rangle$, where $\mathrm{pr}_\lambda : \prod_{\lambda\in\Lambda} \mathrm{C}(X_\lambda, Y) \to \mathrm{C}(X_\lambda, Y)$ is the projection.

Let $\mathcal{F}$ be a set of functions from X to Y, that is, $\mathcal{F} \subset Y^X$. When $Y = (Y, d)$ is a metric space, $\mathcal{F}$ is said to be **equi-continuous** at $x_0 \in X$ provided that for each $\varepsilon > 0$, there is a neighborhood U of x_0 in X such that $f(U) \subset \mathrm{B}_d(f(x_0), \varepsilon)$ for all $f \in \mathcal{F}$. It is said that $\mathcal{F}$ is **equi-continuous** on X if it is equi-continuous at each point of X. Then, $\mathcal{F} \subset \mathrm{C}(X, Y)$.

Proposition 2.10.9 *Let X be a locally compact space and $Y = (Y, d)$ a metric space. Every compact subset $\mathcal{F} \subset \mathrm{C}(X, Y)$ is equi-continuous on X.*

Proof To show that $\mathcal{F}$ is equi-continuous at $x_0 \in X$, let $\varepsilon > 0$. For each $f \in \mathcal{F}$, since X is locally compact and f is continuous, x_0 has a compact neighborhood N_f in X such that $f(N_f) \subset \mathrm{B}(f(x_0), \varepsilon/2)$. Note that $\langle N_f; \mathrm{B}(f(x_0), \varepsilon/2)\rangle$ is an open neighborhood of f in $\mathrm{C}(X, Y)$. Since $\mathcal{F}$ is compact, we have $f_1, \ldots, f_n \in \mathcal{F}$ such that

$$\mathcal{F} \subset \bigcup_{i=1}^{n} \langle N_{f_i}; \mathrm{B}(f_i(x_0), \varepsilon/2)\rangle.$$

Then, $N = \bigcap_{i=1}^{n} N_{f_i}$ is a neighborhood of x_0 in X. Each $f \in \mathcal{F}$ is contained in some $\langle N_{f_i}; \mathrm{B}(f_i(x_0), \varepsilon/2)\rangle$, that is, $f(N_{f_i}) \subset \mathrm{B}(f_i(x_0), \varepsilon/2)$. For each $x \in N$, since $f(x), f(x_0) \in \mathrm{B}(f_i(x_0), \varepsilon/2)$, it follows that

$$\begin{aligned} d(f(x), f(x_0)) &\leqslant d(f(x), f_i(x_0)) + d(f_i(x_0), f(x_0)) \\ &< \varepsilon/2 + \varepsilon/2 = \varepsilon. \end{aligned}$$

Hence, $f(N) \subset \mathrm{B}(f(x_0), \varepsilon)$. This means that $\mathcal{F}$ is equi-continuous at x_0. □

For $\mathcal{F} \subset \mathrm{C}(X, Y)$ $(\subset Y^X)$, let $\mathrm{cl}_p \mathcal{F}$ denote the closure of $\mathcal{F}$ in the product space Y^X. Then, it does not necessarily follow that $\mathrm{cl}_p \mathcal{F} \subset \mathrm{C}(X, Y)$. Since $\mathrm{id} : \mathrm{C}(X, Y) \to \mathrm{C}_p(X, Y) \subset Y^X$ is continuous, it follows that $\mathrm{cl}\,\mathcal{F} \subset \mathrm{cl}_p \mathcal{F}$. When $\mathcal{F}$ is equi-continuous, we have the following:

Proposition 2.10.10 *Let X be an arbitrary space and $Y = (Y, d)$ a metric space. If $\mathcal{F} \subset \mathrm{C}(X, Y)$ is equi-continuous on X, then $\mathrm{cl}_p \mathcal{F}$ is also equi-continuous on X. Consequently, $\mathrm{cl}_p \mathcal{F} \subset \mathrm{C}(X, Y)$, which is equal to the closure of $\mathcal{F}$ in $\mathrm{C}_p(X, Y)$.*

Proof We show the equi-continuity of $\mathrm{cl}_p \mathcal{F}$ at each $x_0 \in X$. For each $\varepsilon > 0$, since $\mathcal{F}$ is equi-continuous at x_0, there is a neighborhood U of x_0 in X such that $f(U) \subset \mathrm{B}_d(f(x_0), \varepsilon/3)$ for all $f \in \mathcal{F}$. Then, it suffices to show that $g(U) \subset \mathrm{B}_d(g(x_0), \varepsilon)$ for all $g \in \mathrm{cl}_p \mathcal{F}$. On the contrary, assume that $g(U) \not\subset \mathrm{B}_d(g(x_0), \varepsilon)$ for some $g \in \mathrm{cl}_p \mathcal{F}$. Then, we have $x_1 \in U$ such that $d(g(x_0), g(x_1)) \geqslant \varepsilon$. Because $g \in \mathrm{cl}_p \mathcal{F}$,

$$\langle x_0, \mathrm{B}_d(g(x_0), \varepsilon/3)\rangle \cap \langle x_1, \mathrm{B}_d(g(x_1), \varepsilon/3)\rangle \cap \mathcal{F} \neq \emptyset,$$

that is, there is some $f \in \mathcal{F}$ such that

$$d(f(x_0), g(x_0)),\ d(f(x_1), g(x_1)) < \varepsilon/3.$$

Since $x_1 \in U$, it follows that $d(f(x_0), f(x_1)) < \varepsilon/3$. Hence, we have

$$\begin{aligned} d(g(x_0), g(x_1)) &\leqslant d(g(x_0), f(x_0)) + d(f(x_0), f(x_1)) + d(f(x_1, g(x_1)) \\ &< \varepsilon/3 + \varepsilon/3 + \varepsilon/3 = \varepsilon, \end{aligned}$$

which is a contradiction because $d(g(x_0), g(x_1)) \geqslant \varepsilon$. Thus, $g(U) \subset \mathrm{B}_d(g(x_0), \varepsilon)$ for all $g \in \mathrm{cl}_p \mathcal{F}$. □

Using the above proposition, we can show the following well-known theorem called the ASCOLI–ARZELÀ THEOREM:

Theorem 2.10.11 (ASCOLI–ARZELÀ) *Let X be an arbitrary space and $Y = (Y, d)$ a metric space. Suppose that $\mathcal{F} \subset \mathrm{C}(X, Y)$ satisfies the following two conditions:*

(i) *$\mathcal{F}$ is equi-continuous on X;*
(ii) *$\mathrm{cl}_Y \mathcal{F}(x) = \mathrm{cl}_Y\{f(x) \mid f \in \mathcal{F}\}$ is compact for each $x \in X$.*

Then, $\mathrm{cl}\,\mathcal{F}$ is compact, that is, $\mathcal{F}$ has the compact closure with respect to the compact-open topology.

Proof The product space $\prod_{x\in X} \mathrm{cl}_Y \mathcal{F}(x)$ is compact by the Tychonoff Theorem 2.1.1. Since $\mathrm{ev}_x : \mathrm{C}_p(X, Y) \to Y$ is continuous, we have

$$(\mathrm{cl}_p \mathcal{F})(x) = \mathrm{ev}_x(\mathrm{cl}_p \mathcal{F}) \subset \mathrm{cl}_Y \mathrm{ev}_x(\mathcal{F}) = \mathrm{cl}_Y \mathcal{F}(x),$$

which implies $\mathrm{cl}_p \mathcal{F} \subset \prod_{x\in X} \mathrm{cl}_Y \mathcal{F}(x)$. Since $\mathrm{cl}_p \mathcal{F}$ is closed in $\prod_{x\in X} \mathrm{cl}_Y \mathcal{F}(x)$, it is compact. Thus, it suffices to show that $\mathrm{cl}\,\mathcal{F} = \mathrm{cl}_p \mathcal{F}$ as spaces. Since $\mathrm{cl}\,\mathcal{F} \subset \mathrm{cl}_p \mathcal{F} \subset \mathrm{C}(X, Y)$ as observed before, it is enough to verify the continuity of the inclusion of $\mathrm{cl}_p \mathcal{F}$ with the pointwise convergent topology into $\mathrm{C}(X, Y)$ with the compact-open topology.

By Proposition 2.10.10, condition (i) implies that $\mathrm{cl}_p \mathcal{F}$ is equi-continuous on X. The evaluation $\mathrm{ev} : \mathrm{C}_p(X, Y) \times X \to Y$ is not continuous in general as saw in Sect. 1.2. However, the restriction $\mathrm{ev}|\,\mathrm{cl}_p \mathcal{F} \times X$ is continuous, where $\mathrm{cl}_p \mathcal{F}$ has the pointwise convergent topology. Indeed, let $(f, x) \in \mathrm{cl}_p \mathcal{F} \times X$. For each $\varepsilon > 0$, $\langle x; \mathrm{B}_d(f(x), \varepsilon/2)\rangle$ is a neighborhood of f in $\mathrm{C}_p(X, Y)$. By the equi-continuity of $\mathrm{cl}_p \mathcal{F}$, there is a neighborhood U of x in X such that $f'(U) \subset \mathrm{B}_d(f'(x), \varepsilon/2)$ for all $f' \in \mathrm{cl}_p \mathcal{F}$. Then, for each $(f', x') \in \langle x; \mathrm{B}_d(f(x), \varepsilon/2)\rangle \times U$,

$$d(f'(x'), f(x)) \leqslant d(f'(x'), f'(x)) + d(f'(x), f(x)) < \varepsilon/2 + \varepsilon/2 = \varepsilon,$$
$$\text{i.e., } \mathrm{ev}(f', x') = f'(x') \in \mathrm{B}_d(f(x), \varepsilon) = \mathrm{B}_d(\mathrm{ev}(f, x), \varepsilon).$$

By Proposition 2.10.3, $\mathrm{ev}|\,\mathrm{cl}_p \mathcal{F} \times X$ induces the map of $\mathrm{cl}_p \mathcal{F}$ to $\mathrm{C}(X, Y)$, which is the inclusion. Hence, $\mathrm{cl}_p \mathcal{F} \subset \mathrm{cl}\,\mathcal{F}$ and this inclusion is continuous. Consequently, $\mathrm{cl}_p \mathcal{F} = \mathrm{cl}\,\mathcal{F}$ as spaces. □

2.11 The Limitation Topology for Function Spaces

For the set $\mathrm{C}(X, Y)$ of all maps from X to Y, three topologies have been considered, the pointwise convergence topology and the uniform convergence topology in Chap. 1, and the compact-open topology in the previous section. Here, we introduce a new topology called the limitation topology, which is finer than the uniform convergence topology. Moreover, when Y is completely metrizable, it will be proved that the space $\mathrm{C}(X, Y)$ with this topology is a Baire space and so is the subspace $\mathrm{Homeo}(Y)$ of $\mathrm{C}(Y, Y)$ consisting of all homeomorphisms of Y onto itself. This fact is very useful to show the existence of some types of maps or homeomorphisms.

For each $f \in \mathrm{C}(X, Y)$ and $\mathcal{U} \in \mathrm{cov}(Y)$, we define

$$\mathcal{U}(f) = \left\{ g \in \mathrm{C}(X, Y) \;\middle|\; g \text{ is } \mathcal{U}\text{-close to } f \right\}.$$

As is easily observed, if $\mathcal{V} \in \mathrm{cov}(Y)$ is a Δ-refinement (or a star-refinement) of $\mathcal{U}$, then $\mathcal{V}(g) \subset \mathcal{U}(f)$ for each $g \in \mathcal{V}(f)$. Thus, by Theorem 2.6.3, we have the following:

- *In the case when Y is paracompact, $\mathrm{C}(X, Y)$ has a topology such that $\{\mathcal{U}(f) \mid \mathcal{U} \in \mathrm{cov}(Y)\}$ is a neighborhood basis of each $f \in \mathrm{C}(X, Y)$.*

Such a topology is called the **limitation topology**.

The limitation topology is Hausdorff. Indeed, let $f \neq g \in \mathrm{C}(X, Y)$. Then $f(x_0) \neq g(x_0)$ for some $x_0 \in X$. Take disjoint open sets $U, V \subset Y$ with $f(x_0) \in U$ and $g(x_0) \in V$, and define

$$\mathcal{U} = \{U,\ Y \setminus \{f(x_0)\}\},\ \mathcal{V} = \{V,\ Y \setminus \{g(x_0)\}\} \in \mathrm{cov}(Y).$$

Then, $\mathcal{U}(f) \cap \mathcal{V}(g) = \emptyset$.

Remark 2.11.1 In the above, $\mathcal{U}(f)$ is not open in general. For example, consider the hedgehog $J(\mathbb{N}) = \bigcup_{n\in\mathbb{N}} \mathbf{I}\mathbf{e}_n \subset \ell_1$ (see Sect. 2.3) and the map $f : \mathbb{N} \to J(\mathbb{N})$ defined by $f(n) = \mathbf{e}_n$ for each $n \in \mathbb{N}$, where $\mathbf{e}_n$ is the unit vector of ℓ_1 defined by $\mathbf{e}_n(n) = 1$ and $\mathbf{e}_n(i) = 0$ if $i \neq n$. For each $n \in \mathbb{N}$, let

$$U_n = \mathbf{I}\mathbf{e}_n \cup \mathrm{B}(\mathbf{0}, n^{-1}) \subset J(\mathbb{N}).$$

Then, $\mathcal{U} = \{U_n \mid n \in \mathbb{N}\} \in \mathrm{cov}(J(\mathbb{N}))$. We show that $\mathcal{U}(f)$ is not open in $\mathrm{C}(\mathbb{N}, J(\mathbb{N}))$ with respect to the limitation topology. Indeed, $\mathcal{U}(f)$ contains the constant map f_0 with $f_0(\mathbb{N}) = \{\mathbf{0}\}$. For each $\mathcal{V} \in \mathrm{cov}(J(\mathbb{N}))$, choose $k \in \mathbb{N}$ so that $\mathrm{B}(\mathbf{0}, k^{-1}) \subset V_0$ for some $V_0 \in \mathcal{V}$. Then, $\mathcal{V}(f_0)$ contains the map $g : \mathbb{N} \to J(\mathbb{N})$ defined by $g(n) = (k+1)^{-1}\mathbf{e}_{n+1}$ for each $n \in \mathbb{N}$. Observe that

$$g(k+1) = (k+1)^{-1}\mathbf{e}_{k+2} \notin U_{k+1} \ \text{ but } \ f(k+1) = \mathbf{e}_{k+1} \in U_{k+1} \setminus \bigcup_{n\neq k+1} U_n,$$

which means that $g \notin \mathcal{U}(f)$. Thus, $\mathcal{V}(f_0) \not\subset \mathcal{U}(f)$. Hence, $\mathcal{U}(f)$ is not open.

Recall that $\mathrm{Metr}^B(Y)$ denotes the set of all admissible bounded metrics of a metrizable space Y. When Y is completely metrizable, $\mathrm{Metr}^c(Y)$ denotes the set of all admissible bounded complete metrics of Y. The sup-metric on $\mathrm{C}(X, Y)$ defined by $d \in \mathrm{Metr}^B(Y)$ is denoted by d_∞. For each $f \in \mathrm{C}(X, Y)$ and $d \in \mathrm{Metr}^B(Y)$, let

$$U_d(f) = \mathrm{B}_{d_\infty}(f, 1) = \left\{ g \in \mathrm{C}(X, Y) \;\middle|\; d_\infty(f, g) < 1 \right\}.$$

Then, $U_{n\cdot d}(f) = \mathrm{B}_{d_\infty}(f, n^{-1})$ for each $n \in \mathbb{N}$.

Proposition 2.11.1 *When Y is metrizable, $\{U_d(f) \mid d \in \mathrm{Metr}^B(Y)\}$ is a neighborhood basis of $f \in \mathrm{C}(X, Y)$ in the space $\mathrm{C}(X, Y)$ with the limitation topology. If Y is completely metrizable, then $\{U_d(f) \mid d \in \mathrm{Metr}^c(Y)\}$ is also a neighborhood basis of $f \in \mathrm{C}(X, Y)$.*

Proof For each $d \in \mathrm{Metr}^B(Y)$, let

$$\mathcal{U} = \left\{ \mathrm{B}_d(y, 1/3) \;\middle|\; y \in Y \right\} \in \mathrm{cov}(Y).$$

Then, clearly $\mathcal{U}(f) \subset U_d(f)$ for each $f \in \mathrm{C}(X, Y)$. Conversely, for each $\mathcal{U} \in \mathrm{cov}(Y)$, choose $d \in \mathrm{Metr}^B(Y)$ (or $d \in \mathrm{Metr}^c(Y)$) so that $\{\mathrm{B}_d(y, 1) \mid y \in Y\} \prec \mathcal{U}$ (cf. Proposition 2.7.9 (1)). Thus, $U_d(f) \subset \mathcal{U}(f)$ for each $f \in \mathrm{C}(X, Y)$. □

For a *paracompact* space X, let $\mathrm{Homeo}(X)$ be the set of all homeomorphisms of X onto itself. The **limitation topology** on $\mathrm{Homeo}(X)$ is the subspace topology inherited from the space $\mathrm{C}(X, X)$ with the limitation topology. When X is metrizable, for each $f \in \mathrm{Homeo}(X)$ and $d \in \mathrm{Metr}^B(X)$, we define

$$U_{d^*}(f) \underset{\text{def}}{=} \mathrm{B}_{d^*}(f, 1) = \left\{ g \in \mathrm{Homeo}(X) \;\middle|\; d^*(f, g) < 1 \right\},$$

where d^* is the metric on $\mathrm{Homeo}(X)$ defined as follows:

$$d^*(f, g) \underset{\text{def}}{=} d_\infty(f, g) + d_\infty(f^{-1}, g^{-1}).$$

When d is complete, the sup-metric d_∞ defined by d is a complete metric on $\mathrm{C}(X, X)$ (Proposition 1.1.1), but the restriction of d_∞ on $\mathrm{Homeo}(X)$ is not complete.

For example, let d be the usual metric for $\mathbf{I} = [0, 1]$. We define $f_n \in \mathrm{Homeo}(\mathbf{I})$, $n \in \mathbb{N}$, and $f \in \mathrm{C}(\mathbf{I}, \mathbf{I})$ as follows:

$$f_n(x) = \begin{cases} x/n & \text{if } x \leqslant 1/2, \\ ((2n-1)x - n + 1)/n & \text{if } x \geqslant 1/2, \end{cases} \quad \text{and} \quad f(x) = \begin{cases} 0 & \text{if } x \leqslant 1/2, \\ 2x - 1 & \text{if } x \geqslant 1/2. \end{cases}$$

Then, $(f_n)_{n\in\mathbb{N}}$ is a d_∞-Cauchy sequence in $\mathrm{Homeo}(\mathbf{I})$ but it does not converge in $\mathrm{Homeo}(\mathbf{I})$. Indeed, $d_\infty(f_n, f) = 1/2n \to 0$ as $n \to \infty$, i.e., $\mathrm{ulim}_{n\to\infty} f_n = f$, but $f \notin \mathrm{Homeo}(\mathbf{I})$.

It should be remarked that d^* is a complete metric on $\mathrm{Homeo}(X)$ if d is complete.

Indeed, let $(f_n)_{n\in\mathbb{N}}$ be a d^*-Cauchy sequence in $\mathrm{Homeo}(X)$. Since $(f_n)_{n\in\mathbb{N}}$ and $(f_n^{-1})_{n\in\mathbb{N}}$ are d_∞-Cauchy, they converge to $f, g \in \mathrm{C}(X, X)$. Then, it follows that

$$\begin{aligned} d_\infty(fg, \mathrm{id}) &\leqslant d_\infty(fg, ff_n^{-1}) + d_\infty(ff_n^{-1}, \mathrm{id}) \\ &= d_\infty(fg, ff_n^{-1}) + d_\infty(f, f_n) \to 0 \quad (n \to \infty), \\ d_\infty(gf, \mathrm{id}) &\leqslant d_\infty(gf, gf_n) + d_\infty(gf_n, \mathrm{id}) \\ &= d_\infty(gf, gf_n) + d_\infty(g, f_n^{-1}) \to 0 \quad (n \to \infty). \end{aligned}$$

Hence, $fg = \mathrm{id}$ and $gf = \mathrm{id}$. Thus, we have $f \in \mathrm{Homeo}(X)$ and $f^{-1} = g$. Moreover,

$$d^*(f_n, f) = d_\infty(f_n, f) + d_\infty(f_n^{-1}, f^{-1}) \to 0 \quad (n \to \infty).$$

The following is the homeomorphism space version of Proposition 2.11.1:

Proposition 2.11.2 *When X is metrizable, $\{U_{d^*}(f) \mid d \in \mathrm{Metr}^B(X)\}$ is a neighborhood basis of $f \in \mathrm{Homeo}(X)$ in the space $\mathrm{Homeo}(X)$ with the limitation topology. If X is completely metrizable, then $\{U_{d^*}(f) \mid d \in \mathrm{Metr}^c(X)\}$ is also a neighborhood basis of $f \in \mathrm{Homeo}(X)$.*

Proof For each $f \in \mathrm{Homeo}(X)$ and $d \in \mathrm{Metr}^B(X)$, let

$$\mathcal{U} = \left\{ \mathrm{B}_d(x, 1/5) \cap f\left(\mathrm{B}_d(f^{-1}(x), 1/5)\right) \;\middle|\; x \in X \right\} \in \mathrm{cov}(X).$$

Then, $\mathcal{U}(f) \cap \mathrm{Homeo}(X) \subset U_{d^*}(f)$. Indeed, for each $g \in \mathcal{U}(f) \cap \mathrm{Homeo}(X)$ and $x \in X$, we can find $y \in X$ such that

$$f(g^{-1}(x)), x = g(g^{-1}(x)) \in f\left(\mathrm{B}_d(f^{-1}(y), 1/5)\right),$$

which means that $d(g^{-1}(x), f^{-1}(y)) < 1/5$ and $d(f^{-1}(x), f^{-1}(y)) < 1/5$, hence $d(f^{-1}(x), g^{-1}(x)) < 2/5$. Therefore, $d_\infty(f^{-1}, g^{-1}) \leqslant 2/5$. On the other hand, it is easy to see that $d_\infty(f, g) \leqslant 2/5$. Thus, we have $d^*(f, g) < 1$, that is, $g \in U_{d^*}(f)$.

Conversely, for each $f \in \mathrm{Homeo}(X)$ and $\mathcal{U} \in \mathrm{cov}(X)$, choose $d \in \mathrm{Metr}^B(X)$ (or $d \in \mathrm{Metr}^c(X)$) so that $\{\mathrm{B}_d(x, 1) \mid x \in X\} \prec \mathcal{U}$ (cf. Proposition 2.7.9 (1)). Then, $U_{d^*}(f) \subset \mathcal{U}(f)$. Indeed, for each $g \in U_{d^*}(f)$ and $x \in X$, $d(f(x), g(x)) < 1$ and $\mathrm{B}_d(f(x), 1)$ is contained in some $U \in \mathcal{U}$, hence $f(x), g(x) \in U$. Therefore, $g \in \mathcal{U}(f)$. □

When $Y = (Y, d)$ is a metric space, for each $f \in \mathrm{C}(X, Y)$ and $\alpha \in \mathrm{C}(Y, (0, \infty))$, we define

$$N_\alpha(f) \underset{\text{def}}{=} \left\{ g \in \mathrm{C}(X, Y) \;\middle|\; \forall x \in X,\ d(f(x), g(x)) < \alpha(f(x)) \right\}.$$

Let $\mathcal{U} = \{\alpha^{-1}((t, 2t)) \mid t > 0\} \in \mathrm{cov}(Y)$. Applying Proposition 2.7.9 (2), we can take $\beta \in \mathrm{C}(Y, (0, \infty))$ so that $\beta(y) \leqslant \alpha(y)/3$ and $\{\mathrm{B}(y, \beta(y) \mid y \in Y\} \prec \mathcal{U}$. Then, it is easy to see the following:

$$d(y, y') < \beta(y) \;\Rightarrow\; \alpha(y') < 2\alpha(y).$$

We show that $N_\beta(g) \subset N_\alpha(f)$ for each $g \in N_\beta(f)$. Let $h \in N_\beta(g)$. For each $x \in X$, since $d(f(x), g(x)) < \beta(f(x))$, we have $\beta(g(x)) < \alpha(g(x))/3 < 2\alpha(f(x))/3$. Then, it follows that

$$\begin{aligned} d(f(x), h(x)) &\leqslant d(f(x), g(x)) + d(g(x), h(x)) < \beta(f(x)) + \beta(g(x)) \\ &< \alpha(f(x))/3 + 2\alpha(f(x))/3 = \alpha(f(x)), \end{aligned}$$

which means that $h \in N_\alpha(f)$. Therefore, $N_\beta(g) \subset N_\alpha(f)$. This shows that $\mathrm{C}(X, Y)$ has a topology such that $\{N_\alpha(f) \mid \alpha \in \mathrm{C}(Y, (0, \infty))\}$ is a neighborhood basis of $f \in \mathrm{C}(X, Y)$, which is called the **majorant topology**. By the definition, we have the following:

the uniform convergence topology	$\subset$	the majorant topology

As is easily observed, if $f \in \mathrm{C}(X, Y)$ has the compact image (i.e., $f(X)$ is compact), then the neighborhoods of f in the majorant topology coincides with the ones in the uniform convergence topology. In particular, when X is compact, the majorant topology is equal to the uniform convergence topology, and so it is metrizable. However, they are not equal for $\mathrm{C}(X, \mathbb{R})$ when X is a non-compact metrizable space.

In fact, X contains an infinite set $D = \{x_i \mid i \in \mathbb{N}\}$ that is discrete in X, where $x_i \neq x_j$ if $i \neq j$. Using the $\mathbb{R}$-version of the Tietze–Urysohn Theorem 2.2.4, we can obtain a map $f_0 : X \to \mathbb{R}$ such that $f_0(x_i) = i$ for every $i \in \mathbb{N}$. Let $\alpha \in \mathrm{C}(Y, (0, \infty))$ be a map defined by $\alpha(t) = (|t| + 1)^{-1}$. If the limitation topology is equal to the uniform convergence topology, then we have $\varepsilon > 0$ such that $\mathrm{B}_{d_\infty}(f_0, \varepsilon) \subset N_\alpha$ where d_∞ is the sup-metric induced by the usual metric d for $\mathbb{R}$. Choose $n \in \mathbb{N}$ so that $\alpha(n) = (n + 1)^{-1} < \varepsilon$ and define a map $f : X \to \mathbb{R}$ by $f(x) = f_0(x) + (n + 1)^{-1}$. Then, $f \in \mathrm{B}_{d_\infty}(f_0, \varepsilon)$ but $f \neq N_\alpha(f_0)$. This is a contradiction.

In the following proposition, we will show the equality of the limitation topology and the majorant topology:

$$\boxed{\text{the limitation topology}} \;=\; \boxed{\text{the majorant topology}}$$

Proposition 2.11.3 *When $Y = (Y, d)$ is a metric space, the limitation topology is equal to the majorant topology, that is, $\{N_\alpha(f) \mid \alpha \in \mathrm{C}(Y, (0, \infty))\}$ is a neighborhood basis of $f \in \mathrm{C}(X, Y)$ in the space $\mathrm{C}(X, Y)$ with the limitation topology.*

Proof Let $\alpha \in \mathrm{C}(Y, (0, \infty))$. For each $y \in Y$, choose an open neighborhood U_y so that $\operatorname{diam} U_y \leqslant \alpha(y)/2$ and $\alpha(y') > \alpha(y)/2$ for all $y' \in U_y$. Thus, we have $\mathcal{U} = \{U_y \mid y \in Y\} \in \operatorname{cov}(Y)$. Let $f \in \mathrm{C}(X, Y)$ and $g \in \mathcal{U}(f)$. Then, for each $x \in X$, we have some $y \in Y$ such that $f(x), g(x) \in U_y$, which implies $d(f(x), g(x)) \leqslant \alpha(y)/2 < \alpha(f(x))$. Therefore, $\mathcal{U}(f) \subset N_\alpha(f)$.

Conversely, let $f \in \mathrm{C}(X, Y)$ and $\mathcal{U} \in \operatorname{cov}(Y)$. By Proposition 2.7.9 (2), we have a map $\alpha : Y \to (0, 1)$ such that $\{\mathrm{B}(y, \alpha(y)) \mid y \in Y\} \prec \mathcal{U}$. For each $g \in N_\alpha(f)$ and each $x \in X$, $d(f(x), g(x)) < \alpha(f(x))$ and $\mathrm{B}(f(x), \alpha(f(x)))$ is contained in some $U \in \mathcal{U}$. Then, $g(x) \in \mathrm{B}(f(x), \alpha(f(x))) \subset U$. Thus, we have $N_\alpha(f) \subset \mathcal{U}(f)$. □

The following two theorems are very useful to show the existence of some types of maps or homeomorphisms:

Theorem 2.11.4 *For a completely metrizable space Y, the space $\mathrm{C}(X, Y)$ with the limitation topology is a Baire space.*

Proof Let G_n, $n \in \mathbb{N}$, be dense open sets in $\mathrm{C}(X, Y)$. To see that $\bigcap_{n\in\mathbb{N}} G_n$ is dense in $\mathrm{C}(X, Y)$, let $f \in \mathrm{C}(X, Y)$ and $d \in \mathrm{Metr}^c(Y)$. We can inductively choose $g_n \in \mathrm{C}(X, Y)$ and $d_n \in \mathrm{Metr}(Y)$, $n \in \mathbb{N}$, so that

$$g_n \in U_{2d_{n-1}}(g_{n-1}) \cap G_n, \;\; U_{d_n}(g_n) \subset G_n \text{ and } d_n \geqslant 2d_{n-1},$$

where $g_0 = f$ and $d_0 = d$. Observe that $d_m \leqslant 2^{-n} d_{m+n}$ for each $m, n \in \omega$. For simplicity, we omit the subscript ∞ and use the same d and d_n to describe the sup-metric induced by d and d_n. Observe that

$$d(g_{n-1}, g_n) \leqslant 2^{-n+1} d_{n-1}(g_{n-1}, g_n) < 2^{-n} \quad \text{for each } n \in \mathbb{N}.$$

Hence, $(g_n)_{n\in\mathbb{N}}$ is d-Cauchy. From the completeness of d, $(g_n)_{n\in\mathbb{N}}$ converges uniformly to $g \in \mathrm{C}(X, Y)$ with respect to d. Then,

$$d(f, g) \leqslant \sum_{n\in\mathbb{N}} d(g_{n-1}, g_n) < \sum_{n\in\mathbb{N}} 2^{-n} = 1,$$

which implies that $g \in U_d(f)$ and, for each $n \in \mathbb{N}$,

$$\begin{aligned} d_n(g_n, g) &\leqslant \sum_{i\in\mathbb{N}} d_n(g_{n+i-1}, g_{n+i}) \\ &\leqslant \sum_{i\in\mathbb{N}} 2^{-i+1} d_{n+i-1}(g_{n+i-1}, g_{n+i}) < \sum_{i\in\mathbb{N}} 2^{-i} = 1, \end{aligned}$$

hence $g \in U_{d_n}(g_n) \subset G_n$. Thus, $U_d(f) \cap \bigcap_{n\in\mathbb{N}} G_n \neq \emptyset$, hence $\bigcap_{n\in\mathbb{N}} G_n$ is dense in $\mathrm{C}(X, Y)$. □

In the above proof, replace $\mathrm{C}(X, Y)$ and U_{d_n} with $\mathrm{Homeo}(X)$ and $U_{d_n^*}$, respectively. Then, we can see that $(g_n)_{n\in\mathbb{N}}$ is d^*-Cauchy. From the completeness of d^*, we have $g \in \mathrm{Homeo}(X)$ with $\lim_{n\to\infty} d^*(g_n, g) = 0$. By the same calculation, we can see $d_n^*(g_n, g) < 1$, that is, $g \in U_{d_n^*}(g) \subset G_n$ for every $n \in \mathbb{N}$. Then, $U_{d^*}(f) \cap \bigcap_{n\in\mathbb{N}} G_n \neq \emptyset$. Therefore, we have:

Theorem 2.11.5 *For a completely metrizable space X, the space* $\mathrm{Homeo}(X)$ *with the limitation topology is a Baire space.* □

Now, we consider the subspace $\mathrm{C}^P(X, Y)$ of $\mathrm{C}(X, Y)$ consisting of all proper maps.[21] The following is a direct consequence of Proposition 2.1.19 (1):

Proposition 2.11.6 *If Y is locally compact and paracompact, then $\mathrm{C}^P(X, Y)$ is clopen (i.e., closed and open) in the space $\mathrm{C}(X, Y)$ with the limitation topology, where X is also locally compact if $\mathrm{C}^P(X, Y) \neq \emptyset$.* □

From Theorem 2.11.4 and Proposition 2.11.6, we have:

Theorem 2.11.7 *For every pair of locally compact metrizable spaces X and Y, the space $\mathrm{C}^P(X, Y)$ with the limitation topology is a Baire space.* □

We list some properties of the limitation topology, which can be easily proved:

Proposition 2.11.8 (Some Properties of the Limitation Topology) *The following hold with respect to the limitation topology:*

[21] If Y is locally compact, $\mathrm{C}^P(X, Y)$ is the subspace of $\mathrm{C}(X, Y)$ consisting of all perfect maps (Propositions 2.1.11 and 2.1.12).

(1) *For each paracompact space Y, the following evaluation is continuous:*[22]

$$\mathrm{ev} : X \times \mathrm{C}(X, Y) \ni (x, f) \mapsto f(x) \in Y.$$

Sketch of Proof For each $(x, f) \in X \times \mathrm{C}(X, Y)$ and each open neighborhood V of $f(x)$ in Y, take an open neighborhood W of $f(x)$ in Y so that $\mathrm{cl}\, W \subset V$ and let $\mathcal{V} = \{V, Y \setminus \mathrm{cl}\, W\} \in \mathrm{cov}(Y)$. Show the following:

$$(x', f') \in f^{-1}(W) \times \mathcal{V}(f) \;\Rightarrow\; f'(x') \in V.$$

(2) *If both Y and Z are paracompact, the following composition is continuous:*

$$\mathrm{C}(X, Y) \times \mathrm{C}(Y, Z) \ni (f, g) \mapsto g \circ f \in \mathrm{C}(X, Z).$$

Sketch of Proof For each $(f, g) \in \mathrm{C}(X, Y) \times \mathrm{C}(Y, Z)$ and $\mathcal{U} \in \mathrm{cov}(Z)$, let $\mathcal{V} \in \mathrm{cov}(Z)$ be a star-refinement of $\mathcal{U}$. Show the following:

$$f' \in g^{-1}(\mathcal{V})(f),\ g' \in \mathcal{V}(g) \;\Rightarrow\; g' \circ f' \in \mathcal{U}(g \circ f).$$

(3) *For every paracompact space X, the following inverse operation is continuous:*

$$\mathrm{Homeo}(X) \ni h \mapsto h^{-1} \in \mathrm{Homeo}(X).$$

Sketch of Proof *Let $h \in \mathrm{Homeo}(X)$ and $\mathcal{U} \in \mathrm{cov}(X)$. Show the following:*

$$g \in h(\mathcal{U})(h) \cap \mathrm{Homeo}(X) \;\Rightarrow\; g^{-1} \in \mathcal{U}(h^{-1}).$$

(4) *For every paracompact space X, the group $\mathrm{Homeo}(X)$ is a topological group.*

This is just combination of (2) and (3) in the above.

Remark 2.11.2 When $Y = (Y, d)$ is a metric space, for each $f \in \mathrm{C}(X, Y)$ and $\gamma \in \mathrm{C}(X, (0, \infty))$, let

$$V_\gamma(f) = \left\{ g \in \mathrm{C}(X, Y) \;\middle|\; \forall x \in X,\ d(f(x), g(x)) < \gamma(x) \right\}.$$

We have the topology of $\mathrm{C}(X, Y)$ such that $\{V_\gamma(f) \mid \gamma \in \mathrm{C}(X, (0, \infty))\}$ is a neighborhood basis of f, which is called the **source majorant topology**. This is finer than the majorant topology (= the limitation topology). To distinguish the source majorant topology, the previous one is also called the **target majorant topology**. In general, these majorant topologies are not equal. For example, let $\gamma \in \mathrm{C}(\mathbb{N}, (0, \infty))$ be the map defined by $\gamma(n) = 2^{-n}$ for $n \in \mathbb{N}$. Then, $V_\gamma(\mathbf{0})$ is not a neighborhood of $\mathbf{0} \in \mathrm{C}(\mathbb{N}, \mathbb{R})$ in the limitation topology. Indeed, for any $\alpha \in \mathrm{C}(\mathbb{R}, (0, \infty))$, we define

[22] Refer to the last part of Sect. 1.1

$g \in \mathrm{C}(\mathbb{N}, \mathbb{R})$ by $g(n) = \alpha(0)/2$ for every $n \in \mathbb{N}$. Then, $g \in N_\alpha(\mathbf{0})$ but $g \notin V_\gamma(\mathbf{0})$. Thus, $N_\alpha(\mathbf{0}) \not\subset V_\gamma(\mathbf{0})$. Moreover, the composition

$$\mathrm{C}(\mathbb{N}, \mathbb{R}) \times \mathrm{C}(\mathbb{R}, \mathbb{R}) \ni (f, g) \mapsto g \circ f \in \mathrm{C}(\mathbb{N}, \mathbb{R})$$

is not continuous with respect to the source majorant topology (See (2) in the above).

Indeed, let γ be the above map. For any $\alpha \in \mathrm{C}(\mathbb{R}, (0, \infty))$, we have $n \in \mathbb{N}$ such that $2^{-n} < \alpha(0)/2$. Let $h = \mathrm{id} + \alpha/2 \in \mathrm{C}(\mathbb{R}, \mathbb{R})$. Then, $h \in V_\alpha(\mathrm{id})$ but $h \circ \mathbf{0} \notin V_\gamma(\mathrm{id} \circ \mathbf{0})$ because $h \circ \mathbf{0}(n) = h(0) = \alpha(0)/2 > 2^{-n} = \gamma(n)$. (Here, id can be replaced by any $g \in \mathrm{C}(\mathbb{R}, \mathbb{R})$.)

2.12 The Vietoris Topology and the Hausdorff Metric for Hyperspaces

A **hyperspace** is a space consisting of some kind of non-empty subsets of a space X, that is, a space obtained by topologizing a subset of $\mathfrak{P}_0(X) = \mathfrak{P}(X) \setminus \{\emptyset\}$, where $\mathfrak{P}(X)$ is the power set of X. There are various ways to topologize such a set. In this section, we introduce the Vietoris topology and the Hausdorff metric. The former is defined for the set $\mathfrak{P}_0(X)$ of an arbitrary space X and the latter is defined for the set $\mathrm{Cld}^B(X)$ of all bounded non-empty closed sets of a metric space $X = (X, d)$. The **Vietoris topology**[23] is the most popular topology, which is defined on $\mathfrak{P}_0(X)$ as the coarsest topology satisfying the following two conditions:

(V-1) $\{A \in \mathfrak{P}_0(X) \mid A \subset U\}$ is open for each open set U in X;
(V-2) $\{A \in \mathfrak{P}_0(X) \mid A \subset F\}$ is closed for each closed set F in X.

The condition (V-2) is equivalent to the following:

(V-2') $\{A \in \mathfrak{P}_0(X) \mid A \cap U \neq \emptyset\}$ is open for each open set U in X.

Indeed, this comes from the following equality:

$$\mathfrak{P}_0(X) \setminus \{A \in \mathfrak{P}_0(X) \mid A \subset F\} = \{A \in \mathfrak{P}_0(X) \mid A \cap (X \setminus F) \neq \emptyset\}.$$

Thus, the Vietoris topology is generated by the sets

$$U^+ = \{A \in \mathfrak{P}_0(X) \mid A \subset U\} \text{ and } U^- = \{A \in \mathfrak{P}_0(X) \mid A \cap U \neq \emptyset\},$$

where U is open in X. Note that $\emptyset^+ = \emptyset^- = \emptyset$, $U^+ \subset U^-$ for every open U in X, and $X^+ = X^- = \mathfrak{P}_0(X)$. Moreover, for open sets $U_1, \ldots, U_n$ in X,

$$U_1^+ \cap \cdots \cap U_n^+ = (U_1 \cap \cdots \cap U_n)^+.$$

[23] It is also called the **finite topology**.

Thus, as is easily observed, the Vietoris topology has an open basis consisting of the following sets:

$$\begin{aligned}\mathrm{V}(U_1,\dots,U_n) &= \big\{A \in \mathfrak{P}_0(X) \;\big|\; A \subset U_1 \cup \cdots \cup U_n \text{ and} \\ &\qquad\qquad A \cap U_i \neq \emptyset \text{ for each } i = 1,\dots,n\big\} \\ &= (U_1 \cup \cdots \cup U_n)^+ \cap U_1^- \cap \cdots \cap U_n^-,\end{aligned}$$

where $U_1, \dots, U_n$ are open sets in X. Using this notation, we have

$$\mathrm{V}(U) = U^+ \text{ and } \mathrm{V}(U, X) = U^-.$$

Note The Vietoris topology can be defined on the power set $\mathfrak{P}(X)$. In this case, U^+ contains $\emptyset$ as a point. In addition, $\emptyset^+ = \{\emptyset\}$, $\emptyset^- = \emptyset$, $X^+ = \mathfrak{P}(X)$, and $X^- = \mathfrak{P}_0(X)$. Then, $\emptyset$ is an isolated point of the space $\mathfrak{P}(X)$ with the Vietoris topology. In other words, the space $\mathfrak{P}(X)$ is the topological sum of the space $\mathfrak{P}_0(X)$ and the singleton $\{\emptyset\}$.

It should be noted that $\mathfrak{P}_0(X)$ is not Hausdorff for almost all X. In fact, $\mathfrak{P}_0(X)$ is not T_1 if X is non-discrete.

When X has a non-closed set A (i.e., $\mathrm{cl}\, A \neq A$), every neighborhood $\mathcal{U}$ of $\mathrm{cl}\, A$ in $\mathfrak{P}_0(X)$ contains A. Indeed, we have open sets $U_1, \dots, U_n$ in X so that $\mathrm{cl}\, A \in \mathrm{V}(U_1, \dots, U_n) \subset \mathcal{U}$. Then, $A \subset \mathrm{cl}\, A \subset \bigcup_{i=1}^n U_i$. For each $i = 1, \dots, n$, since $U_i \cap \mathrm{cl}\, A \neq \emptyset$, it follows that $U_i \cap A \neq \emptyset$. Hence, $A \in \mathrm{V}(U_1, \dots, U_n) \subset \mathcal{U}$. Thus, $\{A\}$ is not closed in $\mathfrak{P}_0(X)$.

We consider the following hyperspaces which are subspaces of $\mathfrak{P}_0(X)$:

- $\mathrm{Cld}(X)$ — the space of all non-empty closed sets in X;
- $\mathrm{F}_n(X)$ — the space of non-empty sets consisting of at most n many points of X, $\mathrm{F}_1(X) = \{\{x\} \mid x \in X\}$ is the space of all singletons in X;
- $\mathrm{Fin}(X) = \bigcup_{n\in\mathbb{N}} \mathrm{F}_n(X)$ — the space of all non-empty finite sets in X;
- $\mathrm{Comp}(X)$ — the space of all non-empty compact sets in X, $\mathrm{Comp}(X)$ consists of all non-empty compacta in X if X is metrizable;
- $\mathrm{Conn}(X)$ — the space of all non-empty connected closed sets in X;
- $\mathrm{Cont}(X)$ — the space of all non-empty continua in X, $\mathrm{Cont}(X) = \mathrm{Comp}(X) \cap \mathrm{Conn}(X)$ if X is metrizable.[24]

Recall that X is assumed to be Hausdorff. Then, the following inclusions hold:

$$\mathrm{Fin}(X) \subset \mathrm{Comp}(X) \subset \mathrm{Cld}(X).$$

For a subspace $\mathcal{H}$ of $\mathfrak{P}_0(X)$, the topology of $\mathcal{H}$ inherited from the Vietoris topology of $\mathfrak{P}_0(X)$ is generated by the sets

$$U_{\mathcal{H}}^+ = U^+ \cap \mathcal{H} \text{ and } U_{\mathcal{H}}^- = U^- \cap \mathcal{H},$$

where U is open in X. This topology has an open basis consisting of the sets

[24] $\mathrm{Cld}(X) \cup \{\emptyset\}$, $\mathrm{Fin}(X) \cup \{\emptyset\}$, …, are denoted by $\mathrm{Cld}^*(X)$, $\mathrm{Fin}^*(X)$, …, respectively.

$$\mathrm{V}_{\mathcal{H}}(U_1, \ldots, U_n) = \mathrm{V}(U_1, \ldots, U_n) \cap \mathcal{H}.$$

where $U_1, \ldots, U_n$ are open in X. In the above, if there is no confusion, we simply write U^+, U^-, and $\mathrm{V}(U_1, \ldots, U_n)$ without the subscript $\mathcal{H}$ instead of $U^+_{\mathcal{H}}$, $U^-_{\mathcal{H}}$, and $\mathrm{V}_{\mathcal{H}}(U_1, \ldots, U_n)$.

Proposition 2.12.1 (Basic Properties of the Vietoris Topology) *For every space X and its subspace $Y \subset X$, the following hold:*[25]

(1) *If Y is closed in X, then* $\mathrm{Cld}(Y)$ *is a closed subspace of* $\mathrm{Cld}(X)$.
(2) $\mathrm{Comp}(Y)$ *is a subspace of* $\mathrm{Cld}(X)$.[26]
(3) $\mathrm{Comp}(X)$ *is Hausdorff.*
(4) *If Y is dense in X and* $\mathrm{Fin}(Y) \subset \mathcal{H} \subset \mathrm{Cld}(X)$, *then $\mathcal{H}$ is dense in* $\mathrm{Cld}(X)$.
(5) *For each* $n \in \mathbb{N}$, $\mathrm{F}_n(X)$ *is closed in* $\mathrm{Cld}(X)$.
(6) *The natural correspondence* $X \ni x \mapsto \{x\} \in \mathrm{F}_1(X)$ *is a homeomorphism, so* $X \approx \mathrm{F}_1(X)$. *Consequently, X can be embedded into* $\mathrm{Cld}(X)$ *as a closed set.*
(7) *If X is normal, then* $\mathrm{Conn}(X)$ *is closed in* $\mathrm{Cld}(X)$.

Sketch of Proof (1): The inclusion $\mathrm{Cld}(Y) \subset \mathrm{Cld}(X)$ is obvious. Since $\mathrm{Cld}(X) \setminus \mathrm{Cld}(Y) = (X \setminus Y)^-_{\mathrm{Cld}(X)}$, $\mathrm{Cld}(Y)$ is closed in $\mathrm{Cld}(X)$. For each open set U in X, $(U \cap Y)^+_{\mathrm{Cld}(Y)} = U^+_{\mathrm{Cld}(Y)}$ and $(U \cap Y)^-_{\mathrm{Cld}(Y)} = U^-_{\mathrm{Cld}(Y)}$. Then, it is easy to see that the Vietoris topology of $\mathrm{Cld}(Y)$ is a subspace topology inherited from that of $\mathrm{Cld}(X)$.

(2): Since every compact set in X is closed in X, we have $\mathrm{Comp}(Y) \subset \mathrm{Cld}(X)$. In the same way as (1), it can be shown that the Vietoris topology of $\mathrm{Comp}(Y)$ is a subspace topology inherited from that of $\mathrm{Cld}(X)$.

(3): Let $A \neq B \in \mathrm{Comp}(X)$. We may assume that $B \setminus A \neq \emptyset$. Take $x \in B \setminus A$ and choose disjoint open sets U and V in X so that $A \subset U$ and $x \in V$.[27] Then, $A \in U^+_{\mathrm{Comp}(X)}$, $B \in V^-_{\mathrm{Comp}(X)}$, and $U^+_{\mathrm{Comp}(X)} \cap V^-_{\mathrm{Comp}(X)} = \emptyset$.

(4): For each finitely many non-empty open sets $U_1, \ldots, U_n$ in X, we have $y_i \in U_i \cap Y$, $i = 1, \ldots, n$. Then, $\{y_1, \ldots, y_n\} \in \mathrm{V}(U_1, \ldots, U_n)$. Hence, every non-empty open set in $\mathrm{Cld}(X)$ meets $\mathrm{Fin}(Y)$.

(5): Each $A \in \mathrm{Cld}(X) \setminus \mathrm{F}_n(X)$ has at least $n+1$ points, $x_1, \ldots, x_{n+1}$. Choose open sets U_i, $i = 1, \ldots, n+1$, in X so that $x_i \in U_i$ and $U_i \cap U_j = \emptyset$ if $i \neq j$. Then, $\bigcap_{i=1}^{n+1} U_i^-$ is an open neighborhood of A in $\mathrm{Cld}(X)$. Since every $B \in \bigcap_{i=1}^{n+1} U_i^-$ has at least $n+1$ points, we have $\mathrm{F}_n(X) \cap \bigcap_{i=1}^{n+1} U_i^- = \emptyset$.

(6): For each open set U in X, $U^+_{\mathrm{F}_1(X)} = U^-_{\mathrm{F}_1(X)} = \big\{\{x\} \bigm| x \in U\big\}$.

(7): Each $A \in \mathrm{Cld}(X) \setminus \mathrm{Conn}(X)$ is written as $A = A_1 \cup A_2$, where $A_1, A_2 \in \mathrm{Cld}(X)$ are disjoint. By the normality of X, there are disjoint open sets U_1, U_2 in X such that $A_i \subset U_i$, $i = 1, 2$. Then, $A \in \mathrm{V}(U_1, U_2)$. Observe that every $B \in \mathrm{V}(U_1, U_2)$ is disconnected. Hence, $\mathrm{V}(U_1, U_2) \subset \mathrm{Cld}(X) \setminus \mathrm{Conn}(X)$. Thus, $\mathrm{Cld}(X) \setminus \mathrm{Conn}(X)$ is open in $\mathrm{Cld}(X)$.

Remark 2.12.1 Due to (5) and (6) above, each space X can be regarded a closed subspace of $\mathrm{Cld}(X)$ identifying each $x \in X$ with $\{x\} \in \mathrm{Cld}(X)$, we can regard $X =$

[25] Recall that spaces are assumed to be Hausdorff.

[26] It should be noted that $\mathrm{Cld}(Y) \not\subset \mathrm{Cld}(X)$ unless Y is a closed set in X.

[27] Here the compactness of A is used.

$\mathrm{F}_1(X) \subset \mathrm{Cld}(X)$. However, it should be remarked so as not to confuse the subset $X = \mathrm{F}_1(X) \subset \mathrm{Cld}(X)$ and the point $X \in \mathrm{Cld}(X)$, which are different objects.

Note Even if X is not Hausdorff, the above proof of (6) is valid (except for the closedness of X in $\mathrm{Cld}(X)$), that is, $X \approx \mathrm{F}_1(X) \subset \mathrm{Comp}(X)$. Hence, in the case when X is not assumed to be Hausdorff, (3) above can be stated as follows:

- $\mathrm{Comp}(X)$ *is Hausdorff if and only if* X *is Hausdorff.*

Theorem 2.12.2 *For a space* X, $\mathrm{Cld}(X)$ *is Hausdorff if and only if* X *is regular.*

Proof To prove the "if" part, we can apply the same proof as Proposition 2.12.1 (3).

In the proof as Proposition 2.12.1 (3), the Hausdorffness of X was enough to separate a compact set and its outer point, but now the regularity of X is required to separate a closed set and its outer point.

To prove the "only if" part, when X is not regular, we may show that $\mathrm{Cld}(X)$ is not Hausdorff. Then, there exist $x \in X$ and $A \in \mathrm{Cld}(X)$ such that $x \notin A$ but they do not have disjoint open neighborhoods in X. Let $\mathcal{U}$ and $\mathcal{V}$ be open neighborhoods of A and $B = A \cup \{x\}$ in $\mathrm{Cld}(X)$, respectively. Since $X \setminus A$ and $X \setminus \{x\}$ are open neighborhoods of x and A in X, respectively, we can easily find open sets $V_0, V_1, \ldots, V_n$ in X such that

$$B \in \mathrm{V}(V_0, V_1, \ldots, V_n) \subset \mathcal{V},\ x \in V_0 \subset X \setminus A,$$
$$x \notin V_i,\ \text{and}\ A \cap V_i \neq \emptyset\ \text{for each}\ i = 1, \ldots, n,$$

where it should be noted that $A \notin \mathrm{V}(V_0, V_1, \ldots, V_n)$ but

$$A \cup \{y\} \in \mathrm{V}(V_0, V_1, \ldots, V_n)\ \text{ for any } y \in V_0.$$

We can take open sets $U_1, \ldots, U_m$ in X so that $A \in \mathrm{V}(U_1 \ldots, U_m) \subset \mathcal{U}$. From our choice of x and A, it follows that

$$V_0 \cap (U_1 \cup \cdots \cup U_m) \neq \emptyset,$$

that is, $V_0 \cap U_i \neq \emptyset$ for some $i = 1, \ldots, m$. Taking $y \in V_0 \cap U_i$, we have

$$A \cup \{y\} \in \mathrm{V}(U_1, \ldots, U_m) \cap \mathrm{V}(V_0, V_1, \ldots, V_n) \subset \mathcal{U} \cap \mathcal{V}.$$

Therefore, $\mathrm{Cld}(X)$ is not Hausdorff. □

Note For $\mathrm{Comp}(X)$, it can be stated that $\mathrm{Comp}(X)$ *is regular if and only if* X *is regular*. The "if" part can be proved as follows: Let $A \in \mathrm{Comp}(X)$ and $U_1, \ldots, U_n$ open sets in X such that $A \in \mathrm{V}(U_1, \ldots, U_n)$. For each $i = 1, \ldots, n$, take $x_i \in A \cap U_i$ and choose an open neighborhood V_i of x_i in X so that $\mathrm{cl}\, V_i \subset U_i$. Since A is compact, we can find an open neighborhood V_0 of A in X such that $\mathrm{cl}\, V_0 \subset \bigcup_{i=1}^{n} U_i$. Then,

$$A \in V_0^+ \cap \bigcap_{i=1}^n V_i^- \subset (\mathrm{cl}\, V_0)^+ \cap \bigcap_{i=1}^n (\mathrm{cl}\, V_i)^- \subset \mathrm{V}(U_1, \ldots, U_n),$$

where it should be noted that for any closed set $A \subset X$, A^+ and A^- are closed as the complements of $(X \setminus A)^-$ and $(X \setminus A)^+$, respectively.[28]

Concerning the density and the weight of hyperspaces, we have:

Proposition 2.12.3 *For a space X with infinitely many points, the following hold:*

(i) $\mathrm{dens}\, \mathrm{Cld}(X) = \mathrm{dens}\, X$;
(ii) $w(\mathrm{Comp}(X)) = w(X)$.

Proof (i): Let D be a dense set in X with $\mathrm{card}\, D = \mathrm{dens}\, X$ ($\geqslant \aleph_0$). Since $\mathrm{Fin}(D)$ is dense in $\mathrm{Cld}(X)$ by Proposition 2.12.1(4), it follows that

$$\mathrm{dens}\, \mathrm{Cld}(X) \leqslant \mathrm{card}\, \mathrm{Fin}(D) = \mathrm{card}\, D = \mathrm{dens}\, X.$$

Now, let $\mathcal{D}$ be a dense set in $\mathrm{Cld}(X)$ with $\mathrm{card}\, \mathcal{D} = \mathrm{dens}\, \mathrm{Cld}(X)$. For each $A \in \mathcal{D}$, take $x_A \in A$ and let $D = \{x_A \mid A \in \mathcal{D}\} \subset X$. Then, D is dense in X. Indeed, for each non-empty open set U in X, $U^+_{\mathrm{Cld}(X)} \cap \mathcal{D} \neq \emptyset$, that is, $A \in U^+_{\mathrm{Cld}(X)}$ for some $A \in \mathcal{D}$. Since $x_A \in A \subset U$, we have $U \cap D \neq \emptyset$. Thus, it follows that

$$\mathrm{dens}\, X \leqslant \mathrm{card}\, D \leqslant \mathrm{card}\, \mathcal{D} = \mathrm{dens}\, \mathrm{Cld}(X).$$

(ii): Since X can be regarded as a subspace of $\mathrm{Comp}(X)$, we have $w(X) \leqslant w(\mathrm{Comp}(X))$. To see $w(\mathrm{Comp}(X)) \leqslant w(X)$, let $\mathcal{B}$ be an open basis for X with $\mathrm{card}\, \mathcal{B} = w(X)$ ($\geqslant \aleph_0$), and define $\mathfrak{B} = \{\mathrm{V}(B_1, \ldots, B_n) \mid B_1, \ldots, B_n \in \mathcal{B}\}$. Then,

$$\mathrm{card}\, \mathfrak{B} \leqslant \mathrm{card}\, \mathrm{Fin}(\mathcal{B}) = \mathrm{card}\, \mathcal{B} = w(X).$$

So, all that remains is to show that $\mathfrak{B}$ is an open basis for $\mathrm{Comp}(X)$. For each open set $\mathcal{U}$ in $\mathrm{Comp}(X)$ and $A \in \mathcal{U}$, there are open sets $U_1, \ldots, U_n$ in X such that $A \in \mathrm{V}(U_1, \ldots, U_n) \subset \mathcal{U}$. Since $\{U_1, \ldots, U_n\}$ is an open cover of A and A is compact, we can find $B_1, \ldots, B_m \in \mathcal{B}$ such that $A \subset B_1 \cup \cdots \cup B_m \subset U_1 \cup \cdots \cup U_n$, each B_i meets A, and each U_j contains some B_i, which means that

$$A \in \mathrm{V}(B_1, \ldots, B_m) \subset \mathrm{V}(U_1, \ldots, U_n) \subset \mathcal{U}.$$

Hence, $\mathfrak{B}$ is an open basis for $\mathrm{Comp}(X)$. □

In (ii) above, $\mathrm{Comp}(X)$ cannot be replaced with $\mathrm{Cld}(X)$. In fact, $w(\mathrm{Cld}(\mathbb{R})) \geqslant 2^{\aleph_0} = 2^{w(\mathbb{R})}$ by the following proposition:

Proposition 2.12.4 *For a space X, $w(\mathrm{Cld}(X)) \geqslant 2^{\mathrm{card}\, D}$ if D is an infinite discrete set in X. In particular, $w(\mathrm{Cld}(X)) \geqslant 2^{\mathrm{card}\, X}$ if X is infinite and discrete.*

[28] Here, $A^+ = \{C \in \mathrm{Comp}(X) \mid C \subset A\}$ and $A^- = \{C \in \mathrm{Comp}(X) \mid C \cap A \neq \emptyset\}$.

Proof Let $\mathfrak{B}$ be an open basis for $\mathrm{Cld}(X)$ with $\operatorname{card}\mathfrak{B} = w(\mathrm{Cld}(X))$. Each $x \in D$ has an open neighborhood U_x in X such that $U_x \cap D = \{x\}$. Note that every subset of D is closed in X. For each non-empty $A \subset D$ (i.e., $A \in \mathfrak{P}_0(D)$), $U_A = \bigcup_{x\in A} U_x$ is open in X and $U_A \cap D = A$. Take $\mathcal{B}_A \in \mathfrak{B}$ so that $A \in \mathcal{B}_A \subset U_A^+$. Then, $D \cap \bigcup \mathcal{B}_A = A$ because

$$A \subset D \cap \bigcup \mathcal{B}_A \subset D \cap \bigcup U_A^+ = D \cap U_A = A.$$

Thus, $A \neq A'$ implies $\mathcal{B}_A \neq \mathcal{B}_{A'}$. Therefore, we have

$$2^{\operatorname{card} D} = \operatorname{card}\mathfrak{P}(D) = \operatorname{card}\mathfrak{P}_0(D) \leqslant \operatorname{card}\mathfrak{B} = w(\mathrm{Cld}(X)). \qquad \square$$

Now, we will discuss the metrizability of hyperspaces. For a metric space $X = (X, d)$, let $\mathrm{Cld}^B(X)$ be the set of all non-empty bounded closed sets in X. In general, $\mathrm{Comp}(X) \subset \mathrm{Cld}^B(X) \subset \mathrm{Cld}(X)$, but $\mathrm{Comp}(X) = \mathrm{Cld}^B(X) = \mathrm{Cld}(X)$ if X is compact. When X is bounded and non-compact, $\mathrm{Comp}(X) \subsetneqq \mathrm{Cld}^B(X) = \mathrm{Cld}(X)$. For Euclidean n-space $\mathbb{R}^n$, we have $\mathrm{Comp}(\mathbb{R}^n) = \mathrm{Cld}^B(\mathbb{R}^n) \subsetneqq \mathrm{Cld}(\mathbb{R}^n)$. For each $A, B \in \mathrm{Cld}^B(X)$, define

$$\begin{aligned} d_H(A, B) &\underset{\text{def}}{=} \max\left\{ \sup_{x\in A} d(x, B),\ \sup_{y\in B} d(y, A) \right\} \\ &= \inf\left\{ r > 0 \;\middle|\; A \subset \mathrm{N}_d(B, r),\ B \subset \mathrm{N}_d(A, r) \right\}. \end{aligned}$$

To verify the second equality, let γ be the value of the upper side and R the set of the lower side. For each $r \in R$, since $A \subset \mathrm{N}_d(B, r)$, it follows that $d(x, B) < r$ for every $x \in A$, which implies $\sup_{x\in A} d(x, B) \leqslant r$. Similarly, $\sup_{y\in B} d(y, A) \leqslant r$. Hence, $\gamma \leqslant \inf R$. Conversely, for any $r > \gamma$, since $\sup_{x\in A} d(x, B) < r$, it follows that $A \subset \mathrm{N}_d(B, r)$. Similarly, $B \subset \mathrm{N}_d(A, r)$. Thus, we have $r \in R$. Hence, $\gamma \geqslant \inf R$.

In the following proposition, we will prove that d_H is a metric on $\mathrm{Cld}^B(X)$, which is called the **Hausdorff metric** (or **Hausdorff distance**) induced by d. When X is unbounded, since $d_H(A, B) = \infty$ for some $A, B \in \mathrm{Cld}(X)$ (e.g., $d_H(\{x\}, X) = \infty$ for any $x \in X$), d_H is not a metric on $\mathrm{Cld}(X)$ but it defines a topology on $\mathrm{Cld}(X)$. In this topology, $\mathrm{Cld}^B(X)$ is closed and open in $\mathrm{Cld}(X)$.

Proposition 2.12.5 *For a metric space $X = (X, d)$, d_H is a metric on $\mathrm{Cld}^B(X)$ and the following hold:*

(i) *d_H is complete if d is complete;*
(ii) *d_H is totally bounded if d is totally bounded.*

Consequently, $(\mathrm{Cld}(X), d_H) = (\mathrm{Comp}(X), d_H)$ is compact if X is compact.

Proof First, we verify d_H is a metric on $\mathrm{Cld}^B(X)$. For each $A, B \in \mathrm{Cld}^B(X)$, $d_H(A, A) = 0$ and $d_H(A, B) = d_H(B, A)$ are obvious from the definition. When $A \neq B$, we may assume that B has a point x_0 that does not belong to A. Then, $\sup_{x\in B} d(x, A) \geqslant d(x_0, A) > 0$, which implies $d_H(A, B) > 0$. Thus, $d_H(A, B) = 0$ implies $A = B$.

The triangle inequality can be shown as follows: For each $A, B, C \in \mathrm{Cld}^B(X)$, let $r > d_H(A, B) + d_H(B, C)$. We can choose $r_1 > d_H(A, B)$, $r_2 > d_H(B, C)$ so that $r_1 + r_2 < r$. Then, it follows that

$$C \subset \mathrm{N}_d(B, r_2) \subset \mathrm{N}_d(\mathrm{N}_d(A, r_1), r_2) \subset \mathrm{N}_d(A, r).$$

Similarly, we have $A \subset \mathrm{N}_d(C, r)$. Therefore, $d_H(A, C) \leqslant d_H(A, B) + d_H(B, C)$.

Next, we will prove the additional statements (i) and (ii).

(i): Let $(A_i)_{i\in\mathbb{N}}$ be a d_H-Cauchy sequence in $\mathrm{Cld}^B(X)$. For each $n \in \mathbb{N}$, let $B_n = \mathrm{cl}\bigcup_{i\geqslant n} A_i$. Since $X = (X, d)$ is complete, each B_n is also complete. We can inductively choose $n_1 < n_2 < \cdots \in \mathbb{N}$ so that $d_H(A_j, A_k) < 2^{-i}$ for any $j, k \geqslant n_i$. Take any point $x_1 \in A_{n_1}$ and inductively choose $x_i \in A_{n_i}$, $i > 1$, so that $d(x_i, x_{i+1}) < 2^{-i}$. Then, $(x_i)_{i\in\mathbb{N}}$ is d-Cauchy, which converges to some $x_0 \in X$. Since $x_j \in A_{n_j} \subset B_{n_i}$ for any $j \geqslant i$, it follows that $x_0 \in B_{n_i}$ for any $i \in \mathbb{N}$, that is, $x_0 \in \bigcap_{i\in\mathbb{N}} B_{n_i} = \bigcap_{n\in\mathbb{N}} B_n$. Thus, we have $A_0 = \bigcap_{n\in\mathbb{N}} B_n \in \mathrm{Cld}^B(X)$. To see $\lim_{i\to\infty} A_i = A_0$, for each $\varepsilon > 0$, choose $n \in \mathbb{N}$ so that $d_H(A_i, A_j) < \varepsilon/3$ for any $i, j \geqslant n$. Then, we show that $d_H(A_m, A_0) < \varepsilon$ for any $m \geqslant n$.

To verify $\sup_{x\in A_0} d(x, A_m) < \varepsilon$, let $x \in A_0$. Because $x \in B_m = \mathrm{cl}\bigcup_{i\geqslant m} A_i$, $d(x, y) < \varepsilon/3$ for some $y \in A_i$, $i \geqslant m$. Since $d_H(A_m, A_i) < \varepsilon/3$, we have $z \in A_m$ such that $d(y, z) < \varepsilon/3$. Then, it follows that

$$d(x, A_m) \leqslant d(x, z) \leqslant d(x, y) + d(y, z) < 2\varepsilon/3.$$

Thus, we have $\sup_{x\in A_0} d(x, A_m) < \varepsilon$.

It remains for us to show that $\sup_{x\in A_m} d(x, A_0) < \varepsilon$. Since $(A_i)_{i\in\mathbb{N}}$ is d_H-Cauchy, we can inductively choose $m < n_1 < n_2 < \cdots \in \mathbb{N}$ so that $d_H(A_j, A_k) < 3^{-i}\varepsilon$ for any $j, k \geqslant n_i$. For each $x \in A_m$, we can inductively choose $x_i \in A_{n_i}$, $i \in \mathbb{N}$, so that $d(x, x_1) < \varepsilon/3$ and $d(x_i, x_{i+1}) < 3^{-i}\varepsilon$. Since $(x_i)_{i\in\mathbb{N}}$ is d-Cauchy, we have $x_0 = \lim_{i\to\infty} x_i \in \bigcap_{i\in\mathbb{N}} B_{n_i} = A_0$. Then, it follows that

$$d(x, x_0) \leqslant d(x, x_1) + \sum_{i=1}^{\infty} d(x_i, x_{i+1}) < \varepsilon/3 + \varepsilon/2 = 5\varepsilon/6.$$

Hence, $\sup_{x\in A_m} d(x, A_0) < \varepsilon$. Consequently, $d_H(A_m, A_0) < \varepsilon$ for any $m \geqslant n$.

(ii): For each $\varepsilon > 0$, there is a finite set $F \subset X$ such that $d(x, F) < \varepsilon$ for every $x \in X$, that is, $X = \bigcup_{x\in F} \mathrm{B}(x, \varepsilon)$. Then, $\mathrm{Fin}(F)$ $(= \mathfrak{P}_0(F))$ is a finite set in $\mathrm{Comp}(X)$. For each $A \in \mathrm{Cld}(X)$, let

$$F_A = \left\{ x \in F \;\middle|\; \mathrm{B}_d(x, \varepsilon) \cap A \neq \emptyset \right\} \in \mathrm{Fin}(F).$$

It follows that $A \subset \mathrm{N}_d(F_A, \varepsilon)$ and $F_A \subset \mathrm{N}_d(A, \varepsilon)$, which means $d_H(A, F_A) < \varepsilon$. Hence, $d_H(A, \mathrm{Fin}(F)) < \varepsilon$. Thus, d_H is totally bounded. □

In the proof of (i) above, if each A_n is compact, then $A_0 = \lim_{n\to\infty} A_n$ is compact, which means d_H is complete on $\mathrm{Comp}(X)$. To see this, because $A_0 = \bigcap_{n\in\mathbb{N}} B_n$, it

suffices to show that B_1 is compact. Since every totally bounded complete metric space is compact, we may show that B_1 is totally bounded. For each $\varepsilon > 0$, we have $n \in \mathbb{N}$ such that $d_H(A_i, A_j) < \varepsilon/3$ for any $i, j \geqslant n$. Since $A_1, \ldots, A_n$ are compact, it is totally bounded, that is, $\bigcup_{i=1}^{n} A_i \subset \bigcup_{j=1}^{m} \mathrm{B}_d(x_j, \varepsilon/3)$ for some $x_1, \ldots, x_m \in \bigcup_{i=1}^{n} A_i$. For any $x \in \mathrm{cl} \bigcup_{i>n} A_i$, we have $y \in A_k$ for some $k > n$ such that $d(x, y) < \varepsilon/3$. Since $d_H(A_k, A_n) < \varepsilon/3$, we can find $z \in A_n$ such that $d(y, z) < \varepsilon/3$. Then, $d(z, x_j) < \varepsilon/3$ for some $j = 1, \ldots, m$, which implies that

$$d(x, x_j) \leqslant d(x, y) + d(y, z) + d(z, x_j) < \varepsilon,$$

that is, $x \in \mathrm{B}_d(x_j, \varepsilon)$. Hence,

$$B_1 = \mathrm{cl} \bigcup_{i \geqslant 1} A_i = \bigcup_{i=1}^{n} A_i \cup \mathrm{cl} \bigcup_{k>n} A_k \subset \bigcup_{j=1}^{m} \mathrm{B}_d(x_j, \varepsilon).$$

Thus, we have the following:

Proposition 2.12.6 *For a complete metric space $X = (X, d)$, the Hausdorff metric d_H on* $\mathrm{Comp}(X)$ *is complete.* □

Due to Proposition 2.12.6 above, $\mathrm{Comp}(X)$ is closed in the metric space $(\mathrm{Cld}^B(X), d_H)$ if (X, d) is complete. Hence, $\mathrm{Comp}(X)$ is not dense in $(\mathrm{Cld}^B(X), d_H)$ if (X, d) is complete but non-compact. Then, d_H is not admissible for the space $\mathrm{Cld}^B(X)$ with the Vietoris topology because $\mathrm{Comp}(X)$ is dense $\mathrm{Cld}(X)$ by Proposition 2.12.1 (4). However, we can prove the following:

Proposition 2.12.7 *For a metric space $X = (X, d)$, the Hausdorff metric d_H induces the Vietoris topology of* $\mathrm{Comp}(X)$*, that is, d_H is admissible for the space* $\mathrm{Comp}(X)$ *with the Vietoris topology.*

Proof First, we show that $\mathrm{B}_{d_H}(A, r) = \{B \in \mathrm{Comp}(X) \mid d_H(A, B) < r\}$ is open for each $A \in \mathrm{Comp}(X)$ and $r > 0$. For each $B \in \mathrm{B}_{d_H}(A, r)$, let $0 < \delta < r - d_H(A, B)$. Since $B \in \mathrm{Comp}(X)$, there are finitely many open sets $U_1, \ldots, U_n$ in X such that each U_i meets B, $\mathrm{diam}\, U_i < \delta$, and $B \subset \bigcup_{i=1}^{n} U_i$, which implies $\bigcup_{i=1}^{n} U_i \subset \mathrm{N}_d(B, \delta)$ and $B \in \mathrm{V}(U_1, \ldots, U_n)$. For every $C \in \mathrm{V}(U_1, \ldots, U_n)$, we have $C \subset \bigcup_{i=1}^{n} U_i \subset \mathrm{N}_d(B, \delta)$. Since each U_i meets C, it follows that $B \subset \bigcup_{i=1}^{n} U_i \subset \mathrm{N}_d(C, \delta)$. Hence, $d_H(B, C) \leqslant \delta < r - d_H(A, B)$, which implies

$$d_H(A, C) \leqslant d_H(A, B) + d_H(B, C) < r,$$

that is, $C \in \mathrm{B}_{d_H}(A, r)$. Consequently, we have $\mathrm{V}(U_1, \ldots, U_n) \subset \mathrm{B}_{d_H}(A, r)$.

To see that $\{\mathrm{B}_{d_H}(A, r) \mid r > 0\}$ is a neighborhood basis of A in $\mathrm{Comp}(X)$, let $U_1, \ldots, U_n$ be open sets in X such that $A \in \mathrm{V}(U_1, \ldots, U_n)$. Since A is compact and $A \subset \bigcup_{i=1}^{n} U_i$, we have $r_0 = \mathrm{dist}(A, X \setminus \bigcup_{i=1}^{n} U_i) > 0$. For each $i = 1, \ldots, n$, since $A \cap U_i \neq \emptyset$, we can take $a_i \in A \cap U_i$. Then, $\mathrm{B}_d(a_i, r_i) \subset U_i$ for some $r_i > 0$. Let $\delta = \min\{r_i \mid i = 0, 1, \ldots, n\} > 0$. For each $B \in \mathrm{B}_{d_H}(A, \delta)$, we have

$$B \subset \mathrm{N}_d(A, \delta) \subset \mathrm{N}_d(A, r_0) \subset \bigcup_{i=1}^{n} U_i.$$

For each $i = 1, \ldots, n$,

$$d(a_i, B) \leqslant \sup_{x \in A} d(x, B) \leqslant d_H(A, B) < \delta \leqslant r_i,$$

which implies $B \cap U_i \supset B \cap \mathrm{B}_d(a_i, r_i) \neq \emptyset$. Hence, $B \in \mathrm{V}(U_1, \ldots, U_n)$. Thus, we have $\mathrm{B}_{d_H}(A, \delta) \subset \mathrm{V}(U_1, \ldots, U_n)$. □

Since X can be regarded as a closed subspace of $\mathrm{Comp}(X)$, the (complete or compact) metrizability of $\mathrm{Comp}(X)$ implies the (complete or compact) metrizability of X. Thus, we have the following:

Theorem 2.12.8 *For a space X, $\mathrm{Comp}(X)$ is (completely or compact) metrizable if and only if X is (completely or compact) metrizable.* □

For the metrizability of $\mathrm{Cld}(X)$, we have the following:

Theorem 2.12.9 *For a space X, the following are equivalent:*

(a) $\mathrm{Cld}(X)$ *is metrizable;*
(b) $\mathrm{Cld}(X)$ *is a compactum;*
(c) X *is a compactum.*

Proof The implication (b) $\Rightarrow$ (a) is obvious and (c) $\Rightarrow$ (b) follows from Theorem 2.12.8 because $\mathrm{Cld}(X) = \mathrm{Comp}(X)$ for any compactum X.

(a) $\Rightarrow$ (c): Since $X \approx \mathrm{F}_1(X) \subset \mathrm{Cld}(X)$, the metrizability of $\mathrm{Cld}(X)$ implies the metrizability of X. To see the compactness of X, suppose that X is not compact. Then, X contains a discrete set Y with $\mathrm{card}\, Y = \aleph_0$. Since Y is closed in X, $\mathrm{Cld}(Y)$ is also metrizable as a subspace of $\mathrm{Cld}(X)$ (Proposition 2.12.1 (1)). Hence, we have $w(\mathrm{Cld}(Y)) = \mathrm{dens}\, \mathrm{Cld}(Y)$ as saw in Sect. 1.1. However, applying Proposition 2.12.4, we can obtain $w(\mathrm{Cld}(Y)) \geqslant 2^{\aleph_0}$. On the other hand, $\mathrm{dens}\, \mathrm{Cld}(Y) = \mathrm{dens}(Y) = \aleph_0$ by Proposition 2.12.3. This is a contradiction. Thus, X is compact. □

Remark 2.12.2 For a totally bounded metric space $X = (X, d)$, the metric completion $\widetilde{X} = (\widetilde{X}, \tilde{d})$ of X is a compactification of X. Then, $\mathrm{Comp}(\widetilde{X}) = \mathrm{Cld}(\widetilde{X})$ and $\tilde{d}_H$ is admissible for the Vietoris topology of $\mathrm{Cld}(\widetilde{X})$ by Proposition 2.12.7 but d_H is not admissible for the Vietoris topology of $\mathrm{Cld}(X)$ unless X is compact (equiv. complete) by Theorem 2.12.9. As is easily observed, the correspondence $\mathrm{Cld}(X) \ni A \mapsto \mathrm{cl}_{\widetilde{X}} A \in \mathrm{Comp}(\widetilde{X})$ is an isometry (isometric embedding) with respect to Hausdorff metrics, but not an embedding with respect to the Vietoris topologies.

Remark 2.12.3 Even if two metrics d and ρ on X induce the same topology, the Hausdorff metrics d_H and ρ_H do not necessarily induce the same topology. For example, the real line $\mathbb{R}$ has the admissible bounded metrics d, ρ defined as follows:

$$d(x, y) = \min\{|x - y|,\ 1\};\quad \rho(x, y) = \left|\frac{x}{1 + |x|} - \frac{y}{1 + |y|}\right|.$$

Then, d_H and ρ_H induce the same topology on $\mathrm{Comp}(\mathbb{R})$ by Proposition 2.12.7 but different topologies on $\mathrm{Cld}(\mathbb{R})$. In fact, $\{[-i, i] \mid i \in \mathbb{N}\}$ is discrete in $(\mathrm{Cld}(\mathbb{R}), d_H)$ because $d_H([-i, i], [-j, j]) = 1$ if $i \neq j$. However, $([-i, i])_{i\in\mathbb{N}}$ converges to $\mathbb{R}$ in $(\mathrm{Cld}(\mathbb{R}), \rho_H)$ because $\rho_H([-i, i], \mathbb{R}) = (i + 1)^{-1}$ for each $i \in \mathbb{N}$. By the way, the topology induced by d_H on $\mathrm{Cld}(\mathbb{R})$ is not comparable with the Vietoris topology, that is, both $\mathrm{id} : (\mathrm{Cld}(\mathbb{R}), d_H) \to \mathrm{Cld}(\mathbb{R})$ and $\mathrm{id} : \mathrm{Cld}(\mathbb{R}) \to (\mathrm{Cld}(\mathbb{R}), d_H)$ are not continuous. Indeed, let

$$U = \bigcup_{n\in\mathbb{N}} \left(n - n^{-1}, n + n^{-1}\right).$$

Then, U^+ is an open neighborhood of $\mathbb{N}$ in $\mathrm{Cld}(\mathbb{R})$. For any $\varepsilon > 0$, choose $k \in \mathbb{N}$ so that $k^{-1} < \varepsilon$ and let

$$A = \mathbb{N} + k^{-1} = \left\{n + k^{-1} \;\middle|\; n \in \mathbb{N}\right\} \in \mathrm{Cld}(\mathbb{R}).$$

Then, $d_H(\mathbb{N}, A) = k^{-1} < \varepsilon$ but $A \not\subset U$, that is, $A \in \mathrm{B}_{d_H}(\mathbb{N}, \varepsilon)$ but $A \notin U^+$, which implies that $\mathrm{B}_{d_H}(\mathbb{N}, \varepsilon) \not\subset U^+$. Hence, $\mathrm{id} : (\mathrm{Cld}(\mathbb{R}), d_H) \to \mathrm{Cld}(\mathbb{R})$ is not continuous.

Let $U_1,\ldots,U_n$ be open sets in $\mathbb{R}$ such that $\mathbb{R} \in \mathrm{V}(U_1, \ldots, U_n)$. Since $U_1 \cup \cdots \cup U_n = \mathbb{R}$, $[-i, i] \in (U_1 \cup \cdots \cup U_n)^+$ for every $i \in \mathbb{N}$. For sufficiently large $k \in \mathbb{N}$, $[-k, k]$ meets every U_j, that is, $[-k, k] \in U_j^-$. Hence, $[-k, k] \in \mathrm{V}(U_1, \ldots, U_n)$. However, $d_H([-k, k], \mathbb{R}) = 1$, so $[-k, k] \notin \mathrm{B}_{d_H}(\mathbb{R}, 1)$. Thus, $\mathrm{V}(U_1, \ldots, U_n) \not\subset \mathrm{B}_{d_H}(\mathbb{R}, 1)$. Therefore, $\mathrm{id} : \mathrm{Cld}(\mathbb{R}) \to (\mathrm{Cld}(\mathbb{R}), d_H)$ is not continuous.

Since every connected closed subset of $\mathbf{I}$ is a closed subinterval, we can identify the space $\mathrm{Conn}(\mathbf{I})$ $(= \mathrm{Cont}(\mathbf{I}))$ as follows:

Theorem 2.12.10 $\mathrm{Cont}(\mathbf{I}) = \mathrm{Conn}(\mathbf{I}) \approx \mathbf{B}^2$.

Proof As noted in the above, $\mathrm{Conn}(\mathbf{I}) = \{[a, b] \mid 0 \leqslant a \leqslant b \leqslant 1\}$. For every two intervals $[a, b], [a', b'] \in \mathrm{Conn}(\mathbf{I})$,

$$d_H([a, b], [a', b']) = \max\{|a - a'|,\ |b - b'|\}.$$

Hence, $\mathrm{Conn}(\mathbf{I})$ is homeomorphic to $\{(a, b) \in \mathbb{R}^2 \mid 0 \leqslant a \leqslant b \leqslant 1\}$, which is an isosceles right triangle in Euclidean plane $\mathbb{R}^2$. □

Hyperspaces have union operators, whose continuity can be easily proved:

Proposition 2.12.11 *For a space X, the following union operators are continuous:*

(1) $u_2 : \mathfrak{P}_0(X)^2 \to \mathfrak{P}_0(X)$ *defined by* $u_2(A, B) = A \cup B$;
(2) $u_{\mathfrak{P}_0} : \mathfrak{P}_0(\mathfrak{P}_0(X)) \to \mathfrak{P}_0(X)$ *defined by* $u_{\mathfrak{P}_0}(\mathcal{A}) = \bigcup \mathcal{A}$.

Proof (1): For each open set U in X, the following are open in $\mathfrak{P}_0(X)^2$:

$$u_2^{-1}(U^+) = U^+ \times U^+ \text{ and } u_2^{-1}(U^-) = \big(U^- \times \mathfrak{P}_0(X)\big) \cup \big(\mathfrak{P}_0(X) \times U^-\big).$$

(2): For each open set U in X, $u_{\mathfrak{P}_0}^{-1}(U^+) = (U^+)^+$ and $u_{\mathfrak{P}_0}^{-1}(U^-) = (U^-)^-$. Indeed,

$$\begin{gathered}\mathcal{A} \in u_{\mathfrak{P}_0}^{-1}(U^+) \Leftrightarrow \bigcup \mathcal{A} \subset U \Leftrightarrow \forall A \in \mathcal{A},\ A \subset U \\ \Leftrightarrow \forall A \in \mathcal{A},\ A \in U^+ \Leftrightarrow \mathcal{A} \subset U^+ \Leftrightarrow \mathcal{A} \in (U^+)^+ \text{ and} \\ \mathcal{A} \in u_{\mathfrak{P}_0}^{-1}(U^-) \Leftrightarrow \bigcup \mathcal{A} \cap U \neq \emptyset \Leftrightarrow \exists A \in \mathcal{A} \text{ such that } A \cap U \neq \emptyset \\ \Leftrightarrow \exists A \in \mathcal{A} \text{ such that } A \in U^- \Leftrightarrow \mathcal{A} \cap U^- \neq \emptyset \Leftrightarrow \mathcal{A} \in (U^-)^-.\end{gathered}$$

Since U^+ and U^- are open in $\mathfrak{P}_0(X)$, it follows that $u_{\mathfrak{P}_0}^{-1}(U^+) = (U^+)^+$ and $u_{\mathfrak{P}_0}^{-1}(U^-) = (U^-)^-$ are open in $\mathfrak{P}_0(\mathfrak{P}_0(X))$. □

In Proposition 2.12.11(1), $\mathfrak{P}_0$ can be replaced by Cld, Comp, and Fin. Namely, the restriction of u_2 to each of the subspaces $\mathrm{Cld}(X)^2$, $\mathrm{Comp}(X)^2$, and $\mathrm{Fin}(X)^2$ is a continuous operator of each subspace. Concerning (2), $\mathfrak{P}_0$ can be replaced by Fin because $u_{\mathfrak{P}_0}(\mathrm{Fin}(\mathrm{Fin}(X))) = \mathrm{Fin}(X)$. However, $u_{\mathfrak{P}_0}(\mathrm{Cld}(\mathrm{Cld}(X))) \not\subset \mathrm{Cld}(X)$ in general. For example, let $X = \mathbf{I}^2 \setminus (\{0\} \times (0, 1])$ ($\subset \mathbb{R}^2$ a subspace). For each $n \in \mathbb{N}$, let $A_n = \{n^{-1}\} \times \mathbf{I}$. Then, $\mathcal{A} = \{A_n \mid n \in \mathbb{N}\}$ is a closed set in $\mathrm{Cld}(X)$ but $\bigcup \mathcal{A} = \bigcup_{n\in\mathbb{N}} A_n$ is not closed in X.

The closedness of $\mathcal{A}$ in $\mathrm{Cld}(X)$ can be shown as follows: For each $B \in \mathrm{Cld}(X) \setminus \mathcal{A}$, one of the following four cases occurs:

$$\begin{gathered}(1)\ \mathrm{pr}_1(B) \text{ has at least two points } a,\ b; \quad (2)\ \mathrm{pr}_1(B) = \{n^{-1}\},\ n \in \mathbb{N}; \\ (3)\ \mathrm{pr}_1(B) = \{a\},\ a \notin \{n^{-1} \mid n \in \mathbb{N}\} \cup \{0\}; \quad (4)\ B = \{\mathbf{0}\}.\end{gathered}$$

Case (1): Let $c = (a + b)/2$ and $U = ([0, c) \times \mathbf{I}) \cap X$ and $V = (c, 1] \times \mathbf{I}$. Then, $U^-_{\mathrm{Cld}(X)} \cap V^-_{\mathrm{Cld}(X)}$ is an open neighborhood of B in $\mathrm{Cld}(X)$, which does not meet $\mathcal{A}$.

Case (2): Since $B \notin \mathcal{A}$, B is a proper subset of A_n. Take $x = (n^{-1}, x_2) \in A_n \setminus B$ and let

$$U = \big((n + 1)^{-1}, (n - 1)^{-1}\big) \times (\mathbf{I} \setminus \{x_2\}).$$

Then, $U^+_{\mathrm{Cld}(X)}$ is an open neighborhood of B in $\mathrm{Cld}(X)$, which does not meet $\mathcal{A}$.

Case (3): Choose $n \in \mathbb{N}$ so that $(n + 1)^{-1} < a < n^{-1}$, and let $U = \big((n + 1)^{-1}, n^{-1}\big) \times \mathbf{I}$. Then, $U^-_{\mathrm{Cld}(X)}$ is an open neighborhood of B in $\mathrm{Cld}(X)$, which does not meet $\mathcal{A}$.

Case (4): Let $U = [0, 1/2)^2 \cap X = \{\mathbf{0}\} \cup ((0, 1/2) \times [0, 1/2))$. Then, $U^+_{\mathrm{Cld}(X)}$ is an open neighborhood of $B = \{\mathbf{0}\}$ in $\mathrm{Cld}(X)$ which does not meet $\mathcal{A}$.

In the above example, we can define a map $f : (0, 1] \to \mathrm{Cont}(X) \subset \mathrm{Conn}(X)$ by $f(t) = \{t\} \times \mathbf{I}$. Then, $u_{\mathfrak{P}_0}(f((0, 1])) = \bigcup_{t\in(0,1]} f(t) = (0, 1] \times \mathbf{I}$ is not closed in X. However, in the same way as $\mathcal{A}$ in the above, we can see that $f((0, 1]) = \{f(t) \mid t \in (0, 1]\}$ is closed in $\mathrm{Cld}(X)$, where only three cases can be consid-

ered. Since $f((0,1])$ is connected, we have $f((0,1]) \in \mathrm{Conn}(\mathrm{Conn}(X))$. Thus, $u_{\mathfrak{P}_0}(\mathrm{Conn}(\mathrm{Conn}(X))) \not\subset \mathrm{Conn}(X)$.

The continuity of f can be verified as follows: Since $\mathrm{Cont}(X) \subset \mathrm{Comp}(X)$, the topology of $\mathrm{Cont}(X)$ induced by the Hausdorff metric d_H (Theorem 2.12.7). For each $s, t \in (0,1]$,

$$d_H(f(s), f(t)) = d_H(\{s\} \times \mathbf{I}, \{t\} \times \mathbf{I}) = |s - t|,$$

which means f is an isometry, so it is continuous.

When X is metrizable, $\mathfrak{P}_0$ in Proposition 2.12.11 (2) can be replaced by Comp and Cont. To show this, we need the following lemmas:

Lemma 2.12.12 *For a metrizable space X, if $\mathcal{A} \subset \mathrm{Comp}(X)$ is compact, then $\bigcup \mathcal{A}$ is compact. Namely, $u_{\mathfrak{P}_0}(\mathcal{A}) \in \mathrm{Comp}(X)$ for each $\mathcal{A} \in \mathrm{Comp}(\mathrm{Comp}(X))$.*

Proof Let $(x_n)_{n\in\mathbb{N}}$ be a sequence in $\bigcup \mathcal{A}$. For each $n \in \mathbb{N}$, choose $A_n \in \mathcal{A}$ so that $x_n \in A_n$. Due to Theorem 2.12.7, $\mathrm{Comp}(X)$ is metrizable because so is X. Since $\mathcal{A}$ is a compactum (i.e., a compact metrizable space), we have $n(1) < n(2) < \cdots \in \mathbb{N}$ such that $(A_{n(i)})_{i\in\mathbb{N}}$ converges to $A \in \mathcal{A}$. For each $i \in \mathbb{N}$, choose $y_i \in A$ so that

$$d(x_{n(i)}, y_i) \leqslant 2d(x_{n(i)}, A) \leqslant 2d_H(A_{n(i)}, A).$$

Since A is compact, we have $i(1) < i(2) < \cdots \in \mathbb{N}$ such that $(y_{i(j)})_{j\in\mathbb{N}}$ converges to x in A. Then, $(x_{n(i(j))})_{j\in\mathbb{N}}$ converges to x because

$$\begin{aligned} d(x_{n(i(j))}, x) &\leqslant d(x_{n(i(j))}, y_{i(j)}) + d(y_{i(j)}, x) \\ &\leqslant 2d_H(A_{n(i(j))}, A) + d(y_{i(j)}, x). \end{aligned}$$

Therefore, $\bigcup \mathcal{A}$ is compact. □

Lemma 2.12.13 *If $\mathcal{A} \subset \mathrm{Cld}(X)$ is connected and $\mathcal{A} \cap \mathrm{Conn}(X) \neq \emptyset$, then $\bigcup \mathcal{A}$ is connected. In particular, if $\emptyset \neq \mathcal{A} \subset \mathrm{Conn}(X)$ is connected, then $\bigcup \mathcal{A}$ is connected, which implies* $\mathrm{cl} \bigcup \mathcal{A} \in \mathrm{Conn}(X)$.

Proof Let U_1 and U_2 be open sets in X such that

$$\bigcup \mathcal{A} \subset U_1 \cup U_2 \quad \text{and} \quad U_1 \cap U_2 \cap \bigcup \mathcal{A} = \emptyset.$$

Since $\mathcal{A} \cap \mathrm{Conn}(X) \neq \emptyset$, we have $A_0 \in \mathcal{A} \cap \mathrm{Conn}(X)$. Then, it can be assumed that $A_0 \subset U_1$, i.e., $A_0 \in U_1^+$. For every $A \in \mathcal{A}$, since $A \subset U_1 \cup U_2$, it follows that $A \in U_1^+$ or $A \in U_2^-$. Thus, $\mathcal{A} \subset U_1^+ \cup U_2^-$ and $U_1^+ \cap U_2^- \cap \mathcal{A} = \emptyset$. Since $\mathcal{A}$ is connected and $U_1^+ \cap \mathcal{A} \neq \emptyset$, we have $U_2^- \cap \mathcal{A} = \emptyset$, which means that $U_2 \cap A = \emptyset$ for any $A \in \mathcal{A}$. Hence, $U_2 \cap \bigcup \mathcal{A} = \emptyset$. Consequently, $\bigcup \mathcal{A}$ is connected. □

Combining the above two lemmas, we have the following:

Lemma 2.12.14 *For a metrizable space X, if $\mathcal{A} \subset \mathrm{Comp}(X)$ is a continuum and $\mathcal{A} \cap \mathrm{Cont}(X) \neq \emptyset$, then $\bigcup \mathcal{A} \in \mathrm{Cont}(X)$. In particular, $u_{\mathfrak{P}_0}(\mathcal{A}) \in \mathrm{Cont}(X)$ for each $\mathcal{A} \in \mathrm{Cont}(\mathrm{Cont}(X))$.* □

By the above lemmas, restricting $u_{\mathfrak{P}_0}$ to the subspaces $\mathrm{Comp}(\mathrm{Comp}(X))$ and $\mathrm{Cont}(\mathrm{Cont}(X))$, we have the following version of Proposition 2.12.11(2):

Proposition 2.12.15 *For each metrizable space X, the following union operators are continuous:*

$$u_{\mathrm{Comp}} : \mathrm{Comp}(\mathrm{Comp}(X)) \to \mathrm{Comp}(X),\ u_{\mathrm{Cont}} : \mathrm{Cont}(\mathrm{Cont}(X)) \to \mathrm{Cont}(X),$$

which are restrictions of $u_{\mathfrak{P}_0}$. □

Concerning the Hausdorff metric, we have the following:

Proposition 2.12.16 *For a metric space $X = (X, d)$, $u_{\mathrm{Comp}} : \mathrm{Comp}(\mathrm{Comp}(X)) \to \mathrm{Comp}(X)$ is 1-Lipschitz with respect to the Hausdorff metric d_H, that is,*

$$d_H(u_{\mathrm{Comp}}(\mathcal{A}), u_{\mathrm{Comp}}(\mathcal{B})) \leqslant (d_H)_H(\mathcal{A}, \mathcal{B}) \quad \text{for each } \mathcal{A}, \mathcal{B} \in \mathrm{Comp}(\mathrm{Comp}(X)).$$

Proof It suffices to show the following:

$$\sup_{x \in \bigcup \mathcal{A}} d(x, \bigcup \mathcal{B}),\ \sup_{y \in \bigcup \mathcal{B}} d(y, \bigcup \mathcal{A}) \leqslant (d_H)_H(\mathcal{A}, \mathcal{B}).$$

Each $x \in \bigcup \mathcal{A}$ is contained in some $A \in \mathcal{A}$. Then, it follows that

$$d(x, \bigcup \mathcal{B}) = \inf_{B \in \mathcal{B}} d(x, B) \leqslant \inf_{B \in \mathcal{B}} d_H(A, B) = d_H(A, \mathcal{B}) \leqslant (d_H)_H(\mathcal{A}, \mathcal{B}).$$

Hence, $\sup_{x \in \bigcup \mathcal{A}} d_H(x, \bigcup \mathcal{B}) \leqslant (d_H)_H(\mathcal{A}, \mathcal{B})$. The other one is the same. □

The following is easy to show:

Proposition 2.12.17 *For a metric space $X = (X, d)$, $\mathrm{diam}_d : \mathrm{Comp}(X) \to \mathbb{R}_+$ is 2-Lipschitz with respect to the Hausdorff metric d_H.*

Sketch of Proof For each $A, B \in \mathrm{Comp}(X)$, show the following:

$$\mathrm{diam}_d A \leqslant \mathrm{diam}_d B + 2d_H(A, B) \ \text{ and } \ \mathrm{diam}_d B \leqslant \mathrm{diam}_d A + 2d_H(A, B),$$

which implies that $|\mathrm{diam}_d A - \mathrm{diam}_d B| \leqslant 2d_H(A, B)$.

Each map $f : X \to Y$ induces a map $\hat{f} : \mathfrak{P}_0(X) \to \mathfrak{P}_0(Y)$ defined by $\hat{f}(A) = f(A) = \{f(a) \mid a \in A\}$ for each $A \in \mathfrak{P}_0(X)$. The continuity of $\hat{f}$ comes from the following:

$$\hat{f}^{-1}(U^+) = f^{-1}(U)^+ \ \text{ and } \ \hat{f}^{-1}(U^-) = f^{-1}(U)^-.$$

where U is open in Y. Then, $\hat{f}(\mathrm{Cld}(X)) \not\subset \mathrm{Cld}(Y)$ unless f is closed. However, the following inclusions always hold:

$$\hat{f}(\mathrm{Comp}(X)) \subset \mathrm{Comp}(Y) \ \text{ and } \ \hat{f}(\mathrm{Cont}(X)) \subset \mathrm{Cont}(Y).$$

To verify the second inclusion, it is necessary to show the metrizability of $f(A)$ for each $A \in \mathrm{Cont}(X)$. This can be shown as follows: for each $A \in \mathrm{Cont}(X)$, $f|A : A \to f(A)$ is a perfect map because A is compact. Since A is metrizable, $f(A)$ is also metrizable by Proposition 2.4.4 (1).

The following maps are restrictions of $\hat{f}$ but written using the same notation:

$$\hat{f} : \mathrm{Comp}(X) \to \mathrm{Comp}(Y), \quad \hat{f} : \mathrm{Cont}(X) \to \mathrm{Cont}(Y).$$

Notes for Chap. 2

There are many textbooks for a first course of General Topology. Among those recently published, the following appears to be popular and readable:

- J.R. Munkres, *Topology, 2nd edition* (Prentice Hall, Inc., Upper Saddle River, 2000)

About half of what is covered in this chapter is contained in Chaps. 4–8 of the above text, although it does not discuss the Frink Metrization Theorem (cf. 2.4.1) and Michael's Theorem 2.6.7 on local properties. Munkres does not treat the direct limit of a tower of spaces (Sect. 2.8) nor the limitation topology of function spaces (Sect. 2.11) nor the hyperspaces (Sect. 2.12). Moreover, perfect maps are only dealt with in Exercises. However, almost all elementary textbooks for General Topology contain a few of the contents in this chapter. Thus, this chapter might be considered as supplements of such textbooks.

For more comprehensive studies on General Topology, see Engelking's book, which contains excellent historical and bibliographic notes at the end of each section.

- R. Engelking, *General Topology, Revised and complete edition*, Sigma Ser. in Pure Math. **6** (Heldermann Verlag, Berlin, 1989)

The following classical books are still good sources:

- J. Dugundji, *Topology* (Allyn and Bacon, Inc., Boston, 1966)
- J.L. Kelley, *General Topology*, GTM **27** (Springer-Verlag, Berlin, 1975), Reprint of the original edition (Van Nostrand, Princeton, 1955)

For counter-examples, the following is a good reference:

- L.A. Steen and J.A. Seebach, Jr., *Counterexamples in Topology, 2nd edition* (Springer-Verlag, New York, 1978) — cited below as [SS]

The following notes are mainly based on historical and bibliographic notes in Engelking's book mentioned above.

According to Engelking (his book, p. 145), the finite case of Wallace's Theorem 2.1.2 appears in the above Kelley's book,[29] where it is attributed to Wallace; the full general case was proved by Frolík [18] and by Lin [29]. The Tychonoff Theorem 2.1.1 appeared in [48], but Engelking stated that its validity was seen in the proof of the compactness of $\mathbf{I}^{\Gamma}$ in [47]. Among various proofs of the Tychonoff Theorem 2.1.1, our proof is a modification of the proof due to D.G. Wright [54]. The Stone–Čech compactification appeared in works of Stone [44] and Čech [10]. The one-point compactification was established by Alexandorff [1]. A k-space first appeared in Gale's paper [19] but this notion was originally introduced by Hurewicz as stated there. Proposition 2.1.9 and Corollary 2.1.10 were proved by Whitehead [53] and Cohen [11], respectively. Statements (1) and (2) in Remark 2.1.2 were proved by Michael [34]. A perfect map between metric spaces was first

[29] Engelking mentions that it also appears in the book by Gottschalk and Hedlund [20] but I cannot find it.

introduced by Vaĭnšteĭn [51] and a proper map between locally compact spaces was introduced by Leray [28] and Bourbaki [9], independently. For various properties of perfect maps, refer to Sect. 3.7 of Engelking's book.

The Tietze–Urysohn Extension Theorem 2.2.2 is simply called the Tietze Extension Theorem very often; it was proved by Tietze [45] for metric spaces and generalized by Urysohn [50] for normal spaces. In 1907, it had been proved by Lebesgue [26] for the plane. Our proof of the Tietze–Urysohn Extension Theorem 2.2.2 is due to Scott [40]. Theorem 2.3.1 was established by Stone [43], but the proof presented here is due to Rudin [38].

The Nagata–Smirnov Metrization Theorem (cf. 2.3.4) was independently proved by Nagata [36] and Smirnov [41]. The Bing Metrization Theorem (cf. 2.3.4) was proved in [8]. The Urysohn Metrization Theorem 2.3.5 and the Alexandroff–Urysohn Metrization Theorem (cf. 2.4.1) were established in [49] and [2], respectively. The Frink Metrization Theorem (cf. 2.4.1) was proved by Frink [17]. The isometry φ in Footnote 9 (for Sketch of Proof of Theorem 2.3.10) was first introduced by Kuratowski [24]. A normal cover was introduced by Tukey [46].

The term "Baire space" was introduced by Bourbaki and the Baire Category Theorem 2.5.1, versions of which were first proved independently in 1897 and 1899 by Osgood [37] and Baire [6] (cf. Introduction of [22], which is different from Engelking's information). The equivalence of (a) and (b) in Theorem 2.5.6 was shown by Čech [10]. Theorems 2.5.8 and 2.5.9 were established by Lavrentieff [25].

The concept of paracompactness was introduced by J. Dieudonné [12]. In [8], R.H. Bing introduced the concept of collectionwise normality and showed that a paracompact space has this property (Theorem 2.6.1). The equivalence (b) $\Leftrightarrow$ (c) in Theorem 2.6.3 was proved by Tukey [46], who called condition (c) "fully normal" (similarly to "normal cover" in (d)). The equivalence (a) $\Leftrightarrow$ (c) and the equivalence of (a), (e), and (f) were, respectively, proved by Stone [43] and Michael [32]. Theorem 2.6.7 on local properties was established by Michael [33].

Proposition 2.7.1 was shown by E. Michael [32]. The simple proof of Proposition 2.7.1 presented here is due to Mather [30]. The term "numerable cover" was given by Dord [13]. Theorem 2.7.2 was formulated by Morita [35] (cf. Theorem 1.2 of [35]), but as he mentioned, this is combined work as Theorem 2.7.5 by Stone [43], Michael [32], and Morita [35]. The Open Cover Shrinking Lemma 2.7.6 appeared in [27]. Theorem 2.7.8 was proved by Dieudonné [12].

In Sect. 2.8, we discussed the direct limits of towers of spaces. However, the direct limits can be defined in more general setting, that is, they can be defined for directed systems of spaces and maps. For such direct limits one can refer to Appendix two of Dugungji's book. In some literature, it is mentioned that the direct limit of a closed tower of Hausdorff spaces need not be Hausdorff. Such an example was given by Herrlich [23]. The author was informed by H. Ohta about the construction which is given in Sect. 2.9 (Proposition 2.9.11).

In Sect. 2.9, other than the example of a non-Hausdorff direct limit mentioned above, we gave examples showing differences among concepts of various normalities and paracompactness. Propositions 2.9.2 and 2.9.9 were given by Bing [8]. For the spaces $[0, \omega_1)$ and $[0, \omega_1]$, refer to Examples 42 and 43 in the book [SS]. For the (deleted) Tychonoff plank, refer to Examples 86 and 87 in [SS]. Proposition 2.9.10 was proved by Sorgenfrey [42] (cf. Example 84 in [SS] or Examples in Sections 30 and Sect. 31 in Munkres' textbook). In Remark 2.9.2, we mentioned a Dowker space, whose characterization was given by Dowker [14]. The first example of Dowker spaces was constructed by Rudin [39]. For more information concerning Dowker spaces, refer to the article [7].

The compact-open topology was defined by Fox [15]. The notion of equi-continuity was introduced by Arzelà [3] and Ascoli [5]. In the Ascoli–Arzelà Theorem 2.10.11, the case that X is a bounded closed interval and $Y = \mathbb{R}$ was established by Ascoli [5] and Arzelà [4], where the former showed the first half and the latter the additional part (the converse of the first half). Fréchet [16] generalized the theorem to the case X is a compactum. For a more general form of the theorem, refer to Kelley's book (p. 234).

The Vietoris topology was introduced by Vietoris [52] and its fundamental properties were studied by Michael [31]. The Hausdorff metric was introduced by Hausdorff in his book [21]. To study hyperspaces in earnest, the following textbooks are recommended:

- A. Illanes and S.B. Nadler, Jr., *Hyperspaces, Fundamentals and Recent Advances*, PAM **216** (Marcel Dekker, Inc., New York, 1999)
- S.B. Nadler, Jr., *Hyperspaces of Sets, A Text with Research Questions*, PAM **49** (Marcel Dekker, Inc., New York, 1978)

The following book emphasizes the interplay between Theory of Hyperspaces and Geometric Functional Analysis in a different way from the above books and also discusses other various topologies:

- G. Beer, *Topologies on Closed and Closed Convex Sets*, MIA **268** (Kluwer Academic Publ., Dordrecht, 1993)

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Chapter 3
Topology of Linear Spaces and Convex Sets



In this chapter, several basic results on topological linear spaces and convex sets are presented. In order to provide background knowledge for Chap. 4, we will also detail matters that are omitted in an ordinary course on topological linear spaces. Here, we deal only with linear spaces over the real field $\mathbb{R}$. We will characterize finite dimensionality, metrizability, and normability of topological linear spaces. Among important results, we prove the Hahn–Banach Extension Theorem, the Separation Theorem, the Closed Graph Theorem, and the Open Mapping Theorem. The Michael Selection Theorem is also proved and applied to prove the Bartle–Graves Theorem. Finally, although a peculiar topic, we will discuss free topological spaces and prove some basic properties, which will be applied in Sect. 7.12.

3.1 Flats and Affine Functions

In this section, we present the basic properties of flats and affine functions. Let E be a linear space (over $\mathbb{R}$). We call a non-empty subset $F \subset E$ a **flat**[1] if

$$(1-t)x + ty \in F \quad \text{for each } x, y \in F \text{ and } t \in \mathbb{R}.$$

Evidently, a non-empty intersection and any product of flats are also flats. We have the following characterization of flats:

Proposition 3.1.1 *Let E be a linear space. For each non-empty subset $F \subset E$, the following are equivalent:*

(a) *F is a flat.*
(b) *For each $n \in \mathbb{N}$, if $v_1, \ldots, v_n \in F$ and $\sum_{i=1}^n t_i = 1$, then $\sum_{i=1}^n t_i v_i \in F$.*

[1] A flat is also called an **affine set**, a **linear manifold**, or a **linear variety**.

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K. Sakai, *Geometric Aspects of General Topology*, Springer Asia Pacific Mathematics Series 2, https://doi.org/10.1007/978-981-96-8570-7_3

(c) $F - x$ *is a linear subspace of* E *for any* $x \in F$.
(d) $F - x_0$ *is a linear subspace of* E *for some* $x_0 \in E$.

Proof By induction on $n \in \mathbb{N}$, we can obtain (a) $\Rightarrow$ (b). Condition (c) follows from the case $n = 3$ of (b) because, for each $x, y, z \in F$ and $a, b \in \mathbb{R}$,

$$a(y - x) + b(z - x) + x = (1 - a - b)x + ay + bz.$$

To see (c) $\Rightarrow$ (a), let $x, y \in F$ and $t \in \mathbb{R}$. Since $F - x$ is a linear subspace of E by (c), we have $t(y - x) \in F - x$, which means $(1 - t)x + ty \in F$. The implication (c) $\Rightarrow$ (d) is obvious.

(d) $\Rightarrow$ (c): It suffices to show that if $F - x_0$ is a linear subspace of E, then $F - x = F - x_0$ for any $x \in F$. For every $z \in F$, we have

$$z - x = (z - x_0) - (x - x_0) \in F - x_0.$$

Here, take $z' \in F$ so that $(z - x_0) + (x - x_0) = z' - x_0$. Then, we have

$$z - x_0 = (z' - x_0) - (x - x_0) = z' - x \in F - x.$$

Consequently, we have $F - x = F - x_0$. □

In the above proof of (d) $\Rightarrow$ (c), we have actually proved the following:

Corollary 3.1.2 *Let* F *be a flat in a linear space* E. *Then,* $F - x = F - y$ *for any* $x, y \in F$. □

A maximal proper flat $H \subsetneqq E$ is called a **hyperplane** of E. The following proposition shows the relationship between hyperplanes and linear functionals:[2]

Proposition 3.1.3 *Let* E *be a linear space.*

(1) *For each hyperplane* $H \subset E$, *there is a linear functional* $f : E \to \mathbb{R}$ *such that* $H = f^{-1}(s)$ *for some* $s \in \mathbb{R}$.
(2) *For each non-trivial linear functional* $f : E \to \mathbb{R}$ *and* $s \in \mathbb{R}$, $f^{-1}(s)$ *is a hyperplane of* E.
(3) *For any two linear functionals* $f_1, f_2 : E \to \mathbb{R}$, *if* $f_1^{-1}(s_1) = f_2^{-1}(s_2)$ *for some* $s_1, s_2 \in \mathbb{R}$, *then* $f_2 = rf_1$ *for some* $r \in \mathbb{R}$.

Proof (1): For a given $x_0 \in H$, $H_0 = H - x_0$ is a maximal proper linear subspace of E (Proposition 3.1.1). Let $x_1 \in E \setminus H_0$. For each $x \in E$, there exists a unique $t \in \mathbb{R}$ such that $x - tx_1 \in H_0$. Indeed, $E = H_0 + \mathbb{R}x_1$ because of the maximality of H_0. Hence, we can write $x = z + tx_1$ for some $z \in H_0$ and $t \in \mathbb{R}$. Then, $x - tx_1 \in H_0$. Moreover, if $x - t'x_1 \in H_0$ and $t' \in \mathbb{R}$, then $(t - t')x_1 \in H_0$. Since $x_1 \notin H_0$, it follows that $t = t'$. Therefore, we have a function $f : E \to \mathbb{R}$ such that $x - f(x)x_1 \in H_0$. For each $x, y \in E$ and $a, b \in \mathbb{R}$,

[2] A **functional** means a real-valued (or complex-valued) function.

$$(ax + by) - (af(x) + bf(y))x_1 = a(x - f(x)x_1) + b(y - f(y)x_1) \in H_0,$$

which means $f(ax + by) = af(x) + bf(y)$, i.e., f is linear. Observe that $f^{-1}(0) = H_0 = H - x_0$, hence it follows that $H = f^{-1}(f(x_0))$.

(2): From the non-triviality of f, it follows that $f(E) = \mathbb{R}$, and hence $\emptyset \subsetneqq f^{-1}(s) \subsetneqq E$. A simple calculation shows that $f^{-1}(s)$ is a flat. To prove the maximality, let $F \subset E$ be a flat with $f^{-1}(s) \subsetneqq F$. Take $x_0 \in f^{-1}(s)$ and $x_1 \in F \setminus f^{-1}(s)$. Since $f(x_1) \neq f(x_0)$ and F is a flat, it follows that $f(F) = \mathbb{R}$. For each $x \in E$, we can choose $y \in F \setminus f^{-1}(s)$ so that $f(y) \neq f(x)$. Note that $s = tf(x) + (1-t)f(y)$ for some $t \in \mathbb{R} \setminus \{0\}$. Let $z = tx + (1-t)y \in f^{-1}(s) \subset F$. Then, $x = (1 - t^{-1})y + t^{-1}z \in F$. Accordingly, we have $F = E$.

(3): If $f_1^{-1}(s_1) = f_2^{-1}(s_2) = \emptyset$ or $= E$, then both f_1 and f_2 are trivial (i.e., $f_1(E) = f_2(E) = \{0\}$), and hence $f_1 = f_2$. When $f_1^{-1}(s_1) = f_2^{-1}(s_2) \neq \emptyset$ and $\neq E$, take $x_0 \in f_1^{-1}(s_1) = f_2^{-1}(s_2)$. Then, it follows that

$$f_1^{-1}(0) = f_1^{-1}(s_1) - x_0 = f_2^{-1}(s_2) - x_0 = f_2^{-1}(0).$$

Let $H_0 = f_1^{-1}(0) = f_2^{-1}(0)$ and $x_1 \in E \setminus H_0$. Analogous to (1), each $x \in E$ can be uniquely written as $x = y + tx_1$, where $y \in H_0$ and $t \in \mathbb{R}$. Then, $f_1(x) = tf_1(x_1)$ and $f_2(x) = tf_2(x_1)$, hence $f_2(x) = f_1(x)f_1(x_1)^{-1}f_2(x_1)$. Let $r = f_1(x_1)^{-1}f_2(x_1)$. It follows that $f_2 = rf_1$. □

It is said that finitely many distinct points $v_1, \ldots, v_n \in E$ are **affinely** (or **geometrically**) **independent** provided that, for $t_1, \ldots, t_n \in \mathbb{R}$,

$$\sum_{i=1}^{n} t_i v_i = \mathbf{0},\ \sum_{i=1}^{n} t_i = 0 \ \Rightarrow\ t_1 = \cdots = t_n = 0,$$

i.e., $v_1 - v_n, \ldots, v_{n-1} - v_n$ are linearly independent. In this case, the subset $\{v_1, \ldots, v_n\} \subset E$ is also said to be **affinely** (or **geometrically**) **independent**. For convenience, we also consider the empty set to be affinely independent. An (infinite) subset $A \subset E$ is said to be **affinely** (or **geometrically**) **independent** if every finite subset of A is affinely independent. This condition is equivalent to the condition that $(A - v) \setminus \{\mathbf{0}\}$ is linearly independent for some/any $v \in A$.[3]

For a non-empty subset $A \subset E$, the smallest flat containing A is called the **flat hull**[4] of A and is denoted by $\mathrm{fl}\, A$. Then, $\mathbb{R}^n = \mathrm{fl}\{\mathbf{0}, \mathbf{e}_1, \ldots, \mathbf{e}_n\}$, where $\{\mathbf{e}_1, \ldots, \mathbf{e}_n\}$ is the canonical orthonormal basis for $\mathbb{R}^n$ (i.e., $\mathbf{e}_i(i) = 1$ and $\mathbf{e}_i(j) = 0$ for $j \neq i$). Observe that

$$\mathrm{fl}\{v_1, \ldots, v_n\} = \Big\{ \sum_{i=1}^{n} t_i v_i \ \Big|\ \sum_{i=1}^{n} t_i = 1 \Big\} \ \text{and}$$

[3] The phase "for some/any" means that we can choose either "some" or "any" in the sentence. By this choice, we have two different conditions. The condition using "some" is weaker than the one using "any" in general. However, these two conditions are equivalent in the context of the sentence.

[4] The flat hull is also called the **affine hull**.

$$\mathrm{fl}\, A = \bigcup \big\{ \mathrm{fl}\{x_1, \dots, x_n\} \;\big|\; n \in \mathbb{N},\ x_1, \dots, x_n \in A \big\}.$$

By Zorn's Lemma, every non-empty subset $A \subset E$ contains a maximal affinely independent subset $A_0 \subset A$. Then, $\mathrm{fl}\, A_0 = \mathrm{fl}\, A$ and each $x \in \mathrm{fl}\, A$ can be uniquely written as $x = \sum_{i=1}^{n} t_i v_i$, where $v_1, \dots, v_n \in A_0$ and $t_1, \dots, t_n \in \mathbb{R} \setminus \{0\}$ such that $\sum_{i=1}^{n} t_i = 1$. In fact, for any $v \in A_0$, $(A_0 - v) \setminus \{\mathbf{0}\}$ $(= (A_0 \setminus \{v\}) - v)$ is a Hamel basis for the linear subspace $\mathrm{fl}\, A - v$ $(= \mathrm{fl}\, A_0 - v)$ of E.

The **dimension** of a flat $F \subset E$ is denoted by $\dim F$, and is defined by the dimension of the linear space $F - x$ for some/any $x \in F$, i.e., $\dim F = \dim(F - x)$. When $\dim F = n$ (resp. $\dim F < \infty$ or $\dim F = \infty$), it is said that F is ***n*-dimensional** (resp. **finite-dimensional** (abbrev. **f.d.**) or **infinite-dimensional** (abbrev. **i.d.**)). Every n-dimensional flat $F \subset E$ contains $n+1$ points $v_1, \dots, v_{n+1}$ such that $F = \mathrm{fl}\{v_1, \dots, v_{n+1}\}$, where $v_1, \dots, v_{n+1}$ are affinely independent. Conversely, if $F = \mathrm{fl}\{v_1, \dots, v_{n+1}\}$ for some $n+1$ affinely independent points $v_1, \dots, v_{n+1} \in F$, then $\dim F = n$.

Let F and F' be flats in linear spaces E and E', respectively. A function $f : F \to F'$ is said to be **affine** if it satisfies the following condition:

$$f((1-t)x + ty) = (1-t)f(x) + tf(y) \quad \text{for each } x, y \in F \text{ and } t \in \mathbb{R},$$

which is equivalent to the following:

$$f\Big(\sum_{i=1}^{n} t_i v_i\Big) = \sum_{i=1}^{n} t_i f(v_i) \quad \text{for each } n \in \mathbb{N},\ v_i \in F,\ t_i \in \mathbb{R} \text{ with } \sum_{i=1}^{n} t_i = 1.$$

Recall that a non-empty subset $F \subset E$ is a flat if and only if $F - x_0$ is a linear subspace of E for some/any $x_0 \in F$ (Proposition 3.1.1).

Proposition 3.1.4 *Let $f : F \to F'$ be a function between flats F and F' in linear spaces E and E', respectively. In order that f is affine, it is necessary and sufficient that the following $f^{x_0} : F - x_0 \to F' - f(x_0)$ is linear for some/any $x_0 \in F$:*

$$f^{x_0}(x) = f(x + x_0) - f(x_0) \quad \textit{for each } x \in F - x_0.$$

Proof (*Necessity*) For each $x, y \in F - x_0$ and $a, b \in \mathbb{R}$,

$$\begin{aligned} f^{x_0}(ax + by) &= f(ax + by + x_0) - f(x_0) \\ &= f(a(x + x_0) + b(y + x_0) + (1 - a - b)x_0) - f(x_0) \\ &= af(x + x_0) + bf(y + x_0) + (1 - a - b)f(x_0) - f(x_0) \\ &= a(f(x + x_0) - f(x_0)) + b(f(y + x_0) - f(x_0)) \\ &= af^{x_0}(x) + bf^{x_0}(y). \end{aligned}$$

(*Sufficiency*) For each $x, y \in F$ and $t \in \mathbb{R}$,

$$\begin{aligned} f((1-t)x + ty) &= f^{x_0}((1-t)x + ty - x_0) + f(x_0) \\ &= f^{x_0}((1-t)(x - x_0) + t(y - x_0)) + f(x_0) \end{aligned}$$

$$= (1-t)f^{x_0}(x-x_0) + tf^{x_0}(y-x_0) + f(x_0)$$
$$= (1-t)(f^{x_0}(x-x_0) + f(x_0)) + t(f^{x_0}(y-x_0) + f(x_0))$$
$$= (1-t)f(x) + tf(y). \qquad \square$$

Proposition 3.1.5 *Let A be a non-empty affinely independent subset of a linear space E. Then, every function $g : A \to E'$ to another linear space E' uniquely extends to an affine function $\tilde{g} : \mathrm{fl}\, A \to E'$ such that $\tilde{g}(\mathrm{fl}\, A) = \mathrm{fl}\, g(A)$. Accordingly, every affine function f defined on $F = \mathrm{fl}\, A$ is uniquely determined by $f|A$ and the image $f(F)$ is a flat.*

Proof Let $F' = \mathrm{fl}\, g(A)$ and take $v_0 \in A$. Since $(A \setminus \{v_0\}) - v_0$ is a Hamel basis of the linear subspace $\mathrm{fl}\, A - v_0$ of E, we have the unique linear function $h : \mathrm{fl}\, A - v_0 \to F' - g(v_0)$ such that $h(v - v_0) = g(v) - g(v_0)$ for each $v \in A \setminus \{v_0\}$. Then, g uniquely extends to the affine function $\tilde{g} : \mathrm{fl}\, A \to F'$ defined by

$$\tilde{g}(x) = h(x - v_0) + g(v_0) \quad \text{for each } x \in \mathrm{fl}\, A.$$

It is easy to see that $\tilde{g}(\mathrm{fl}\, A) = \mathrm{fl}\, g(A)$. $\square$

The following easily follows from definition. The details are left to the reader.

Proposition 3.1.6 (Additional Properties of Flats and Affine Functions) *Let E and E' be linear spaces and $f : F \to E'$ be a function of a flat F in E. Then, the following hold:*

(1) *If f is affine, then $f(F)$ is a flat in E' and $f^{-1}(F')$ is a flat in E' for each flat F' in E' with $F' \cap f(F) \neq \emptyset$.*
(2) *A function f is affine if and only if the graph $\mathrm{Gr}(f) = \{(x, f(x)) \mid x \in F\}$ of f is a flat in $E \times E'$.*

Let F and F' be flats in a linear space E. When $F' \subset F$, F' is called a **subflat** of F. A maximal proper subflat $H \subsetneqq F$ is called a **hyperplane** of F. As is easily observed, Proposition 3.1.3 is valid for hyperplanes of a flat F, where linear functionals $f, f_1, f_2 : E \to \mathbb{R}$ are replaced by affine functionals $f, f_1, f_2 : F \to \mathbb{R}$.

3.2 Convex Sets

In this section, we introduce the basic concepts of convex sets. A subset $C \subset E$ is said to be **convex** provided

$$(1-t)x + ty \in C \quad \text{for each } x, y \in C \text{ and } t \in \mathbf{I}.$$

By induction on n, we can prove that every convex set $C \subset E$ satisfies the following condition:

$$\sum_{i=1}^{n} z(i)v_i \in C \quad \text{for each } n \in \mathbb{N},\ v_i \in C \text{ and } z \in \Delta^{n-1},$$

where $\Delta^{n-1} = \{z \in \mathbf{I}^n \mid \sum_{i=1}^n z(i) = 1\}$ is the standard $(n-1)$-simplex. The following is easy:

- *If $A, B \subset E$ are convex, then $aA + bB$ is also convex for each $a, b \in \mathbb{R}$.*

The **dimension** of a non-empty convex set $C \subset E$ is defined by the dimension of the flat hull $\mathrm{fl}\, C$, i.e., $\dim C = \dim \mathrm{fl}\, C$. As convention, we define $\dim \emptyset = -1$. Concerning the flat hull of a convex set, we have the following proposition:

Proposition 3.2.1 *For each non-empty convex set $C \subset E$,*

$$\mathrm{fl}\, C = \big\{(1-t)x + ty \;\big|\; x, y \in C,\ t \in \mathbb{R}\big\}.$$

Proof Each $z \in \mathrm{fl}\, C$ can be written as $z = \sum_{i=1}^n t_i x_i$, where $x_i \in C$ and $\sum_{i=1}^n t_i = 1$. We may assume that $t_1 \leqslant \cdots \leqslant t_n \in \mathbb{R} \setminus \{0\}$. If $t_1 \geqslant 0$ then $z \in C$. Otherwise, $t_k < 0$ and $t_{k+1} > 0$ for some $k = 1, \ldots, n-1$. Then, we have $t = \sum_{i=1}^{n-k} t_{k+i} > 0$, where $1 - t = \sum_{i=1}^k t_i < 0$. Let

$$x = \sum_{i=1}^{k} (1-t)^{-1} t_i x_i,\ \ y = \sum_{i=1}^{n-k} t^{-1} t_{k+i} x_{k+i} \in C.$$

Then, $z = (1-t)x + ty$. Accordingly, we have

$$\mathrm{fl}\, C \subset \big\{(1-t)x + ty \;\big|\; x, y \in C,\ t \in \mathbb{R}\big\}.$$

The converse inclusion is obvious. □

The smallest convex set containing $A \subset E$ is called the **convex hull** of A and is denoted by $\langle A \rangle$. We simply write $\langle v_1, \ldots, v_n \rangle = \langle \{v_1, \ldots, v_n\} \rangle$. Then, $\Delta^{n-1} = \langle \mathbf{e}_1, \ldots, \mathbf{e}_n \rangle$. Observe that

$$\langle v_1, \ldots, v_n \rangle = \Big\{ \sum_{i=1}^{n} z(i) v_i \;\Big|\; z \in \Delta^{n-1} \Big\} \quad \text{and}$$
$$\langle A \rangle = \bigcup \big\{ \langle x_1, \ldots, x_n \rangle \;\big|\; n \in \mathbb{N},\ x_1, \ldots, x_n \in A \big\}.$$

For every two non-empty subsets $A, B \subset E$,

$$\langle A \cup B \rangle = \big\{(1-t)x + ty \;\big|\; x \in \langle A \rangle,\ y \in \langle B \rangle,\ t \in \mathbf{I}\big\} \quad \text{and}$$
$$\langle aA + bB \rangle = a\langle A \rangle + b\langle B \rangle \quad \text{for } a, b \in \mathbb{R}.$$

The second equality can be proved as follows: Because $a\langle A\rangle + b\langle B\rangle$ is convex and $aA + bB \subset a\langle A\rangle + b\langle B\rangle$, we have $\langle aA + bB\rangle \subset a\langle A\rangle + b\langle B\rangle$. To show that $a\langle A\rangle + b\langle B\rangle \subset \langle aA + bB\rangle$, let $x \in \langle A\rangle$ and $y \in \langle B\rangle$. Then, $x = \sum_{i=1}^{n} t_i x_i$ and $y = \sum_{j=1}^{m} s_j y_j$ for some $x_i \in A$, $y_j \in B$, and $t_i, s_j > 0$ with $\sum_{i=1}^{n} t_i = \sum_{j=1}^{m} s_j = 1$. Since $ax_i + by_j \in aA + bB$ and $\sum_{i=1}^{n}\sum_{j=1}^{m} t_i s_j = 1$, it follows that

$$ax + by = \sum_{i=1}^{n} t_i(ax_i + by) = \sum_{i=1}^{n}\sum_{j=1}^{m} t_i s_j(ax_i + by_j) \in \langle aA + bB\rangle.$$

The following is easily verified:

Proposition 3.2.2 *For every (bounded) subset A of a normed linear space $E = (E, \|\cdot\|)$, the following holds:*

(i) $\|x - y\| \leqslant \sup_{z\in A} \|x - z\|$ *for each* $x \in E$ *and* $y \in \langle A\rangle$;
(ii) $\operatorname{diam}\langle A\rangle = \operatorname{diam} A$.

Sketch of Proof (i): Write $y = \sum_{i=1}^{n} z(i)x_i$ for some $x_1, \ldots, x_n \in A$ and $z \in \Delta^{n-1}$.
(ii): For each $x, y \in \langle A\rangle$,

$$\|x - y\| \leqslant \sup_{z\in A} \|x - z\| \leqslant \sup_{z\in A}\sup_{z'\in A} \|z - z'\| = \operatorname{diam} A.$$

Let C and C' be non-empty convex sets in the linear spaces E and E', respectively. A function $f : C \to C'$ is said to be **affine** (or **linear in the affine sense**) if

$$f((1-t)x + ty) = (1-t)f(x) + tf(y) \quad \text{for each } x, y \in C \text{ and } t \in \mathbf{I},$$

where $\mathbf{I}$ can be replaced by $\mathbb{R}$, i.e.:

$$x, y \in C,\ t \in \mathbb{R},\ (1-t)x + ty \in C \ \Rightarrow\ f((1-t)x + ty) = (1-t)f(x) + tf(y).$$

Indeed, let $z = (1-t)x + ty \in C$ in the above expression. When $t < 0$ (resp. $t > 1$), consider

$$x = \frac{1}{1-t}z + \frac{-t}{1-t}y,\ \frac{1}{1-t} \in \mathbf{I},\ \frac{-t}{1-t} = 1 - \frac{1}{1-t}$$
$$\left(\text{resp. } y = \frac{1}{t}z + \frac{t-1}{t}x,\ \frac{1}{t} \in \mathbf{I},\ \frac{t-1}{t} = 1 - \frac{1}{t}\right).$$

Then, $f(x) = (1-t)^{-1}f(z) - (1-t)^{-1}tf(y)$ (resp. $f(y) = t^{-1}f(z) - t^{-1}(1-t)f(x)$, which implies that $f(z) = (1-t)f(x) + tf(y)$.

As is easily seen, $f : C \to C'$ is affine if and only if

$$f\Big(\sum_{i=1}^{n} z(i)v_i\Big) = \sum_{i=1}^{n} z(i)f(v_i) \quad \text{for each } n \in \mathbb{N},\ v_i \in C \text{ and } z \in \Delta^{n-1},$$

which is equivalent to the following:

$$v_i \in C,\ t_i \in \mathbb{R},\ \sum_{i=1}^{n} t_i v_i \in C,\ \sum_{i=1}^{n} t_i = 1 \ \Rightarrow\ f\big(\textstyle\sum_{i=1}^{n} t_i v_i\big) = \displaystyle\sum_{i=1}^{n} t_i f(v_i).$$

Evidently, for every affine function $f : C \to E'$ of a convex set $C \subset E$ into another linear space E', the image $f(C)$ is also convex.

Proposition 3.2.3 *Let C and D be non-empty convex sets in the linear spaces E and E', respectively. Every affine function $f : C \to D$ uniquely extends to an affine function $\tilde{f} : \mathrm{fl}\, C \to \mathrm{fl}\, D$. Moreover, if f is injective (or surjective), then so is $\tilde{f}$.*

Proof Let C_0 be a maximal affinely independent subset of C. Then, $\mathrm{fl}\, C = \mathrm{fl}\, C_0$. Due to Proposition 3.1.5, $f|C_0$ uniquely extends to an affine function $\tilde{f} : \mathrm{fl}\, C \to \mathrm{fl}\, D$. From the above remark, we can see that $\tilde{f}|C = f$.

When f is injective, we show that $\tilde{f}$ is also injective. By the definition of $\tilde{f}$ in the proof of Proposition 3.1.5, it suffices to show that $f(C_0)$ is affinely independent. Assume that $f(C_0)$ is not affinely independent, i.e., there are distinct points $v_1, \ldots, v_n \in C_0$ and $t_1, \ldots, t_n \in \mathbb{R} \setminus \{0\}$ such that $\sum_{i=1}^{n} t_i f(v_i) = \mathbf{0}$ and $\sum_{i=1}^{n} t_i = 0$. Without loss of generality, it can be assumed that $t_1, \ldots, t_k > 0$ and $t_{k+1}, \ldots, t_n < 0$. Let $s = \sum_{i=1}^{k} t_i = -\sum_{j=k+1}^{n} t_j > 0$ and define

$$x = \sum_{i=1}^{k} \frac{t_i}{s} v_i,\ \ y = \sum_{j=k+1}^{n} \frac{-t_j}{s} v_j \in C.$$

Then, $f(x) = f(y)$ because

$$f(x) - f(y) = \frac{1}{s} \sum_{i=1}^{n} t_i f(v_i) = \mathbf{0}.$$

Since f is injective, we have $x = y$, which implies $\sum_{i=1}^{k} t_i v_i = -\sum_{j=k+1}^{n} t_j v_j$, i.e., $\sum_{i=1}^{n} t_i v_i = \mathbf{0}$. Since $v_1, \ldots, v_n$ are affinely independent, we have $t_1 = \cdots = t_n = 0$, which is a contradiction.

Finally, we show that if f is surjective then so is $\tilde{f}$. By Proposition 3.2.1, each $z \in \mathrm{fl}\, D$ can be written as follows:

$$z = (1-t)y + ty',\ \ y, y' \in D,\ t \in \mathbb{R}.$$

Since f is surjective, we have $x, x' \in C$ such that $f(x) = y$ and $f(x') = y'$. Then, $(1-t)x + tx' \in \mathrm{fl}\, C$ and

$$\tilde{f}((1-t)x + tx') = (1-t)y + ty' = z.$$

Therefore, $\tilde{f}$ is also surjective. □

Let C be a convex set in a linear space E. The following set is called the **radial interior** of C:[5]

$$\operatorname{rint} C \underset{\text{def}}{=} \big\{x \in C \;\big|\; \forall y \in C,\ \exists \delta > 0 \ \text{ such that } \ (1+\delta)x - \delta y \in C\big\}.$$

In the case $C = \langle v_1, \dots, v_n\rangle$, observe that

$$\operatorname{rint}\langle v_1, \dots, v_n\rangle = \Big\{ \sum_{i=1}^{n} z(i) v_i \;\Big|\; z \in \Delta^{n-1} \cap (0,\infty)^n \Big\}.$$

Indeed, let $x_0 = \sum_{i=1}^n n^{-1} v_i \in \langle v_1, \dots, v_n\rangle$. For each $x \in \operatorname{rint}\langle v_1, \dots, v_n\rangle$, we have $y \in \langle v_1, \dots, v_n\rangle$ such that $x = (1-t)x_0 + ty$ for some $t \in (0,1)$. Then, $y = \sum_{i=1}^n z(i) v_i$ for some $z \in \Delta^{n-1}$. It follows that $x = \sum_{i=1}^n ((1-t)n^{-1} + tz(i))v_i$, where $\sum_{i=1}^n ((1-t)n^{-1} + tz(i)) = 1$ and $(1-t)n^{-1} + tz(i) > 0$ for all $i = 1, \dots, n$. Thus, x is a point of the set on the right side. Conversely, it is straightforward to prove that each point of this set belongs to $\operatorname{rint}\langle v_1, \dots, v_n\rangle$.

In particular, $\operatorname{rint}\{v\} = \{v\}$ and, for each $v_1 \neq v_2$,

$$\operatorname{rint}\langle v_1, v_2\rangle = \{(1-t)v_1 + tv_2 \mid 0 < t < 1\} = \langle v_1, v_2\rangle \setminus \{v_1, v_2\}.$$

The radial interior of C can also be defined as

$$\operatorname{rint} C = \big\{x \in C \;\big|\; \forall y \in C,\ \exists z \in C \ \text{ such that } x \in \operatorname{rint}\langle y, z\rangle\big\}.$$

It is said that $x \in E$ is **linearly accessible** from C if there is some $y \in C$ such that

$$\operatorname{rint}\langle x, y\rangle \subset C \quad \text{i.e., } \langle x, y\rangle \setminus \{x\} \subset C \ \text{ if } x \neq y.$$

The **radial closure** $\operatorname{rcl} C$ of C is the set of all linearly accessible points from C,[6] i.e.:

$$\begin{aligned}\operatorname{rcl} C &\underset{\text{def}}{=} \big\{x \in E \;\big|\; \exists y \in C \ \text{ such that } \ \operatorname{rint}\langle x, y\rangle \subset C\big\} \\ &= \big\{x \in E \;\big|\; \exists y \in C \ \text{ such that } \ 0 < \forall t < 1,\ (1-t)x + ty \in C\big\}.\end{aligned}$$

When $C \neq \emptyset$, it should be noted that $\operatorname{rcl} C \subset \operatorname{fl} C$ by Proposition 3.2.1, hence $\operatorname{fl}\operatorname{rcl} C = \operatorname{fl} C$. Consequently, we have the following inclusions:

$$\operatorname{rint} C \subset C \subset \operatorname{rcl} C \subset \operatorname{fl} C.$$

The set $\partial C = \operatorname{rcl} C \setminus \operatorname{rint} C$ is called the **radial boundary** of C.

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[5] In Köthe's book (see Notes for this chapter), this is called the **algebraic kernel** of C and denoted by C^i.

[6] In Köthe's book, $\operatorname{rcl} C$ is denoted by C^a and called the **algebraic hull** of C.

Proposition 3.2.4 *Let $C \subset E$ be a convex set. For $x \in E$, the following holds:*

$$x \in \partial C \;\Leftrightarrow\; \exists y \in C \;\; \textit{such that} \;\; \text{(i)}\; 0 < \forall t < 1,\; (1-t)x + ty \in C \;\; \textit{and} \;\; \text{(ii)}\; \forall s > 0,\; (1+s)x - sy \notin C.$$

Proof The implication $\Leftarrow$ is obvious, but the converse implication $\Rightarrow$ needs to be verified. If $x \in \partial C$, then we have $y, z \in C$ satisfying (i) and (ii), respectively, that is,

$$\text{(i)}\; 0 < \forall t < 1,\; (1-t)x + ty \in C,\; \text{(ii)}\; \forall s > 0,\; (1+s)x - sz \notin C.$$

It suffices to show that one of y and z satisfies both of (i) and (ii). When y does not satisfy (ii), that is, there is some $r > 0$ such that $(1+r)x - ry \in C$, it suffices to show that z satisfies (i). Let $y' = (1+r)x - ry \in C$. Then,

$$x = \frac{r}{1+r}y + \frac{1}{1+r}y' \in \langle y, y' \rangle \subset C.$$

Since C is convex, z satisfies (i). □

As might expected, we have the following:

Proposition 3.2.5 *For each normed linear space $E = (E, \|\cdot\|)$,*

$$\operatorname{rcl} \mathbf{B}_E = \mathbf{B}_E,\; \operatorname{rint} \mathbf{B}_E = \mathbf{B}_E \setminus \mathbf{S}_E, \;\; \textit{and} \;\; \partial \mathbf{B}_E = \mathbf{S}_E.$$

Proof To see $\operatorname{rcl} \mathbf{B}_E = \mathbf{B}_E$, it suffices to show $\operatorname{rcl} \mathbf{B}_E \subset \mathbf{B}_E$. For each $x \in \operatorname{rcl} \mathbf{B}_E$, we have $y \in \mathbf{B}_E$ such that $\operatorname{rint}\langle x, y\rangle \subset \mathbf{B}_E$. Then, for $0 < t < 1$,

$$\|x\| = \|(1-t)x + ty + t(x-y)\| \leqslant \|(1-t)x + ty\| + t\|x-y\| \leqslant 1 + t\|x-y\|,$$

where $t\|x-y\| \to 0$ if $t \to 0$. Hence, $\|x\| \leqslant 1$, i.e., $x \in \mathbf{B}_E$.

To see $\mathbf{S}_E \subset \partial \mathbf{B}_E$, let $x \in \mathbf{S}_E$, i.e., $\|x\| = 1$. Then, $(1-t)x + t\mathbf{0} \in \mathbf{B}_E$ for each $0 < t < 1$, and $(1+s)x - s\mathbf{0} \notin \mathbf{B}_E$ for each $s > 0$. Hence, $x \in \partial \mathbf{B}_E$ (Proposition 3.2.4).

To see $\mathbf{B}_E \setminus \mathbf{S}_E \subset \operatorname{rint} \mathbf{B}_E$, let $x \in \mathbf{B}_E \setminus \mathbf{S}_E$, i.e., $\|x\| < 1$, and let $\delta = (1 - \|x\|)/(1 + \|x\|) > 0$. For each $y \in \mathbf{B}_E$,

$$\|(1+\delta)x - \delta y\| \leqslant (1+\delta)\|x\| + \delta\|y\| \leqslant (1+\delta)\|x\| + \delta = \|x\| + \delta(\|x\| + 1) = 1,$$

that is, $(1+\delta)x - \delta y \in \mathbf{B}_E$. Hence, $x \in \operatorname{rint} \mathbf{B}_E$.

Now, we have shown that $\operatorname{rint} \mathbf{B}_E = \operatorname{rcl} \mathbf{B}_E \setminus \partial \mathbf{B}_E \subset \mathbf{B}_E \setminus \mathbf{S}_E \subset \operatorname{rint} \mathbf{B}_E$. Then, it follows that $\operatorname{rint} \mathbf{B}_E = \mathbf{B}_E \setminus \mathbf{S}_E$ and $\partial \mathbf{B}_E = \mathbf{B}_E \setminus \operatorname{rint} \mathbf{B}_E = \mathbf{S}_E$. □

Remark 3.2.1 For convex sets C and D,

$$\text{(i)}\;\; C \subset D \;\Rightarrow\; \operatorname{rcl} C \subset \operatorname{rcl} D \;\; \text{but} \;\; \text{(ii)}\;\; C \subset D \;\not\Rightarrow\; \operatorname{rint} C \subset \operatorname{rint} D.$$

The implication (i) is trivial. For (ii), consider $C = \mathbf{I}^n \times \{0\} \subset D = \mathbf{I}^{n+1}$. Then, $C \cap \operatorname{rint} D = \emptyset$.

For the Hilbert cube $\boldsymbol{Q} = [-1, 1]^{\mathbb{N}}$, we have

$$\operatorname{rint} \boldsymbol{Q} = \left\{ x \in \boldsymbol{Q} \;\middle|\; \sup_{i\in\mathbb{N}} |x(i)| < 1 \right\} \subsetneqq (-1, 1)^{\mathbb{N}}.$$

Observe that $\operatorname{rint}[-1, 1]_f^{\mathbb{N}} = (-1, 1)_f^{\mathbb{N}}$ but $\operatorname{rint} \mathbf{I}_f^{\mathbb{N}} = \emptyset$,[7] where

$$[-1, 1]_f^{\mathbb{N}} = \mathbb{R}_f^{\mathbb{N}} \cap [-1, 1]^{\mathbb{N}},\ (-1, 1)_f^{\mathbb{N}} = \mathbb{R}_f^{\mathbb{N}} \cap (-1, 1)^{\mathbb{N}},\ \text{and}\ \mathbf{I}_f^{\mathbb{N}} = \mathbb{R}_f^{\mathbb{N}} \cap \mathbf{I}^{\mathbb{N}}.$$

As is easily observed, $\mathbf{I}_f^{\mathbb{N}} = \operatorname{rcl}(\mathbf{I}_f^{\mathbb{N}} \setminus \{\mathbf{0}\})$. It will be shown in Remark 3.2.3 that $\mathbf{I}_f^{\mathbb{N}} \setminus \{\mathbf{0}\} = \operatorname{rcl} C$ for some convex set $C \subset \mathbb{R}_f^{\mathbb{N}}$.

Proposition 3.2.6 *Let $C \subset E$ be a convex set. If $x \in \operatorname{rint} C$, $y \in \operatorname{rcl} C$, and $0 \leqslant t < 1$, then $(1 - t)x + ty \in \operatorname{rint} C$, i.e., $\langle x, y\rangle \setminus \{y\} \subset \operatorname{rint} C$.*

Proof For each $z \in C$, we have to find $v \in C$ and $0 < s < 1$ such that

$$(1 - t)x + ty = (1 - s)z + sv \in \operatorname{rint}\langle z, v\rangle.$$

Take $w \in C$ so that $\operatorname{rint}\langle w, y\rangle \subset C$, and choose $0 < r < 1$ so that

$$z' = (1 + r)x - rz,\ w' = (1 + r)x - rw \in C.$$

The desired v is written as

$$v = t_1 y + t_2 w + t_3 w' + t_4 z' = (t_1 + t_2)u + (t_3 + t_4)u' \in C,$$

where $t_1 + t_2 + t_3 + t_4 = 1, t_1, t_2, t_3, t_4 > 0$,

$$u = \frac{t_1}{t_1 + t_2} y + \frac{t_2}{t_1 + t_2} w,\ u' = \frac{t_3}{t_3 + t_4} w' + \frac{t_4}{t_3 + t_4} z' \in C.$$

Then, we have

$$\begin{aligned}(1 - s)z + sv &= (1 - s)z + s(t_1 y + t_2 w + t_3 w' + t_4 z') \\ &= st_1 y + s(t_2 - t_3 r)w + s(t_3 + t_4)(1 + r)x + (1 - s - st_4 r)z.\end{aligned}$$

To obtain $(1 - s)z + sv = (1 - t)x + ty$, it is enough to find $t_1, t_2, t_3, t_4 > 0$ and $0 < s < 1$ satisfying the equations: $st_1 = t, t_2 = t_3 r, s(t_3 + t_4)(1 + r) = 1 - t$, and $1 - s = st_4 r$, i.e.,

$$t_1 = \frac{t}{s},\ t_4 = \frac{1 - s}{rs},\ t_3 = \frac{1}{r} - \frac{1 + rt}{(1 + r)rs},\ t_2 = 1 - \frac{1 + rt}{(1 + r)s}. \qquad (*)$$

[7] It is noteworthy that $[-1, 1]_f^{\mathbb{N}} \approx \mathbf{I}_f^{\mathbb{N}}$ is known.

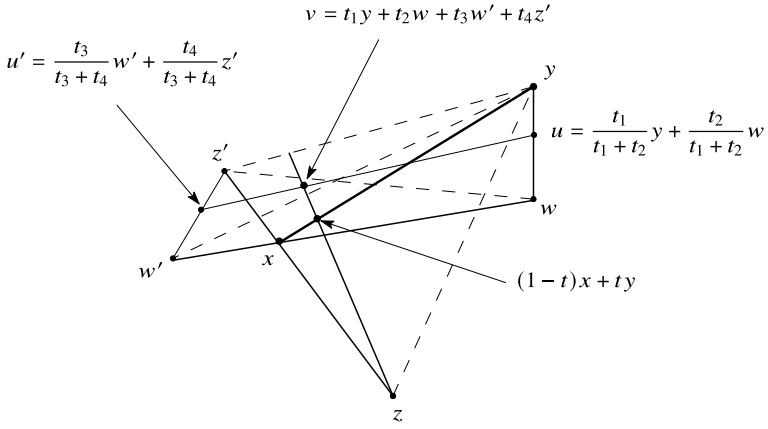


Fig. 3.1 $(1 - t)x + ty \in \operatorname{rint} C$

If $0 < s < 1$, then $t_1, t_4 > 0$. Since $0 < r < 1$, we have $t_3 = t_2/r > t_2$. Choose s so that

$$\frac{1 + rt}{1 + r} < s < 1.$$

We can take such an s because the left side of the above inequality is less than 1. Then, $t_1, t_2, t_3, t_4 > 0$ can be defined as in $(*)$, which satisfy $t_1 + t_2 + t_3 + t_4 = 1$. Thus, we have the desired $v = t_1 y + t_2 w + t_3 w' + t_4 z' \in C$. See Fig. 3.1. □

Contrary to Remark 3.2.1 (ii), we have the following:

Corollary 3.2.7 *Let C and D be non-empty convex sets in E. If $C \subset D$ and $C \cap \operatorname{rint} D \neq \emptyset$, then* $\operatorname{rint} C \subset \operatorname{rint} D$.

Proof Let $x \in C \cap \operatorname{rint} D$. For each $y \in \operatorname{rint} C$, we have $z \in C$ such that $y \in \operatorname{rint}\langle x, z\rangle$. Since $\operatorname{rint}\langle x, z\rangle \subset \operatorname{rint} D$ by Proposition 3.2.6, it follows that $y \in \operatorname{rint} D$. □

Let $C \subset E$ be a convex set. The **face** of C at $x \in C$ is defined as follows:[8]

$$\begin{aligned} C_x &\underset{\text{def}}{=} \left\{ y \in C \;\middle|\; \exists \delta > 0 \text{ such that } (1 + \delta)x - \delta y \in C \right\} \\ &= \left\{ y \in C \;\middle|\; \exists z \in C \text{ such that } x \in \operatorname{rint}\langle y, z\rangle \right\}. \end{aligned}$$

As is easily observed, $C \subset D$ implies $C_x \subset D_x$ for each $x \in C$. Moreover, we have

$$\operatorname{rint} C = \{x \in C \mid C_x = C\}, \quad \text{i.e., } x \in \operatorname{rint} C \Leftrightarrow C_x = C.$$

[8] The face C_x is a little differently defined than the **supporting facet** of C through x in Köthe's book.

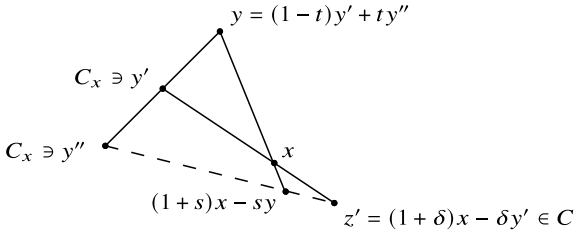


Fig. 3.2 $C \cap \mathrm{fl}\, C_x \subset C_x$

A point $x \in C$ is called an **extreme point** of C if $C_x = \{x\}$.

Remark 3.2.2 The unit closed ball $\mathbf{B}_{\boldsymbol{c}_0}$ of the Banach space $\boldsymbol{c}_0$ has no extreme points. In fact, every $x \in \mathbf{B}_{\boldsymbol{c}_0}$ is the mid-point of two distinct points $y, z \in \mathbf{B}_{\boldsymbol{c}_0}$, i.e., $x = (y+z)/2$. For example, choose $n \in \mathbb{N}$ so that $|x(n)| < 1/2$ and let $y, z \in \mathbf{B}_{\boldsymbol{c}_0}$ such that $y(i) = z(i) = x(i)$ for $i \neq n$, $y(n) = x(n) + 1/2$, and $z(n) = x(n) - 1/2$.

Proposition 3.2.8 *Let $C \subset E$ be a convex set. For each $x \in C$, the face C_x is convex and $C_x = C \cap \mathrm{fl}\, C_x$.*

Proof First, we show that C_x is convex. For each $y, z \in C_x$, we can choose $\delta > 0$ so that $(1+\delta)x - \delta y \in C$ and $(1+\delta)x - \delta z \in C$. Then, for each $t \in \mathbf{I}$,

$$(1+\delta)x - \delta\big((1-t)y + tz\big) = (1-t)\big((1+\delta)x - \delta y\big) + t\big((1+\delta)x - \delta z\big) \in C,$$

which means $(1-t)y + tz \in C_x$.

Since $C_x \subset C \cap \mathrm{fl}\, C_x$, it remains to show $C \cap \mathrm{fl}\, C_x \subset C_x$. By Proposition 3.2.1, each $y \in C \cap \mathrm{fl}\, C_x$ can be written as $y = (1-t)y' + ty''$ for some $y', y'' \in C_x$ and $t \in \mathbb{R}$. If $t \in \mathbf{I}$, then $y \in C_x$ because C_x is convex. Otherwise, we may assume that $t < 0$ (when $t > 1$, exchange y' with y''). Since $y' \in C_x$, we have $\delta > 0$ such that $z' = (1+\delta)x - \delta y' \in C$. See Fig. 3.2. Observe that

$$\begin{aligned}(1+s)x - sy &= (1+s)\left(\frac{\delta}{1+\delta}y' + \frac{1}{1+\delta}z'\right) - s\big((1-t)y' + ty''\big)\\ &= \left(\frac{(1+s)\delta}{1+\delta} - s(1-t)\right)y' + \frac{1+s}{1+\delta}z' - sty''.\end{aligned}$$

Let $s = \delta/(1-t-t\delta) > 0$. Then, $1+s = (1-t)(1+\delta)/(1-t-t\delta)$. It follows that

$$(1+s)x - sy = \frac{1-t}{1-t-t\delta}z' + \frac{-t\delta}{1-t-t\delta}y'' \in C,$$

which implies that $y \in C_x$. □

In the following, we will further list basic properties on the radial interior, closure, boundary, and faces of convex sets:

Proposition 3.2.9 (Basic Properties of Convex Sets) *Let $C, D \subset E$ be convex sets. Then, the following statements hold:*

(i) *Both* $\operatorname{rint} C$ *and* $\operatorname{rcl} C$ *are convex.*
(ii) $\operatorname{rint} \operatorname{rint} C = \operatorname{rint} C \subset \operatorname{rint} \operatorname{rcl} C$.
(iii) $\operatorname{rint} C \neq \emptyset \Rightarrow \operatorname{rint} \operatorname{rcl} C = \operatorname{rint} C$, $\operatorname{rcl} \operatorname{rint} C = \operatorname{rcl} \operatorname{rcl} C = \operatorname{rcl} C$, *in which case* $\partial \operatorname{rint} C = \partial \operatorname{rcl} C = \partial C$.
(iv) $\operatorname{rint} C \neq \emptyset \Rightarrow \operatorname{fl} C = \operatorname{fl} \operatorname{rint} C$.
(v) $\operatorname{rint} C \neq \emptyset$, $\operatorname{rcl} C = \operatorname{fl} C \Rightarrow \operatorname{rint} C = C = \operatorname{fl} C$.
(vi) $\partial C \neq \emptyset \Leftrightarrow \emptyset \neq C \subsetneqq \operatorname{fl} C$.
(vii) $\operatorname{rint} C \cap \operatorname{rint} D \neq \emptyset \Rightarrow \operatorname{rint}(C \cap D) = \operatorname{rint} C \cap \operatorname{rint} D$.
(viii) *Each* $x \in C$ *is contained in* $\operatorname{rint} C_x$ *and* $(C_x)_x = C_x$.
(xi) $(C_x)_y = C_y$ *for each* $x \in C$ *and* $y \in C_x$.
(x) $C_x = C_y$ *for each* $x \in C$ *and* $y \in \operatorname{rint} C_x$.
(xi) *For each* $x, y \in C$, $\operatorname{rint} C_x \cap \operatorname{rint} C_y \neq \emptyset \Rightarrow C_x = C_y$.
(xii) $(C \cap D)_x = C_x \cap D_x$ *for each* $x \in C \cap D$.

Sketch of Proof (i): To prove the convexity of $\operatorname{rint} C$, apply Proposition 3.2.6. Showing the convexity of $\operatorname{rcl} C$ is quite straightforward.

(ii): By definition, $\operatorname{rint} \operatorname{rint} C \subset \operatorname{rint} C$. Apply Proposition 3.2.6 to show $\operatorname{rint} C \subset \operatorname{rint} \operatorname{rint} C$. To see $\operatorname{rint} C \subset \operatorname{rint} \operatorname{rcl} C$, let $x \in \operatorname{rint} C$. For each $y \in \operatorname{rcl} C$, $\frac{1}{2}x + \frac{1}{2}y \in \operatorname{rint} C \subset C$ by Proposition 3.2.6. Then, we have $\delta > 0$ such that $(1+\delta)x - \delta(\frac{1}{2}x + \frac{1}{2}y) \in C$, equivalently, $(1 + \frac{1}{2}\delta)x - \frac{1}{2}\delta y \in C$.

(iii): Let $x_0 \in \operatorname{rint} C$. For each $x \in \operatorname{rint} \operatorname{rcl} C$, we have $y \in \operatorname{rcl} C$ such that $x \in \operatorname{rint}\langle x_0, y\rangle$, which implies that $x \in \operatorname{rint} C$ by Proposition 3.2.6. The converse is (ii).

Now, $x_0 \in \operatorname{rint} C = \operatorname{rint} \operatorname{rcl} C$. If $x \in \operatorname{rcl} \operatorname{rcl} C$, then $\operatorname{rint}\langle x_0, x\rangle \subset \operatorname{rint} \operatorname{rcl} C = \operatorname{rint} C$ by Proposition 3.2.6, which means that $x \in \operatorname{rcl} \operatorname{rint} C$. Since $\operatorname{rcl} \operatorname{rint} C \subset \operatorname{rcl} C \subset \operatorname{rcl} \operatorname{rcl} C$, we have $\operatorname{rcl} \operatorname{rint} C = \operatorname{rcl} C = \operatorname{rcl} \operatorname{rcl} C$.

(iv): Let $x_0 \in \operatorname{rint} C$. For each $x \in C$, combining Propositions 3.2.6 and 3.2.1, we have $x = 2(\frac{1}{2}x + \frac{1}{2}x_0) - x_0 \in \operatorname{fl} \operatorname{rint} C$. Hence, $C \subset \operatorname{fl} \operatorname{rint} C$, which implies $\operatorname{fl} C \subset \operatorname{fl} \operatorname{rint} C$. The converse inclusion is obvious.

(v): Let $x_0 \in \operatorname{rint} C$. For each $x \in \operatorname{fl} C$, since $2x - x_0 \in \operatorname{fl} C = \operatorname{rcl} C$, it follows that $x = \frac{1}{2}x_0 + \frac{1}{2}(2x - x_0) \in \operatorname{rint} C \subset C$ by Proposition 3.2.6.

(vi): If $C = \operatorname{fl} C$, i.e., C is a flat, then $\operatorname{rcl} C = \operatorname{rint} C = C$ by definition. Thus, $\Rightarrow$ holds. To see $\Leftarrow$, take $x \in \operatorname{fl} C \setminus C$ and write $x = (1+t)y - tz$, $y \neq z \in C$, $t > 0$ (Proposition 3.2.1). Then, we have $(1+s_0)y - s_0 z \in \operatorname{rcl} C \setminus \operatorname{rint} C = \partial C$, where

$$s_0 = \inf\left\{ s > 0 \;\middle|\; (1+s)y - sz \notin C \right\} \geqslant 0.$$

(vii): Since $C \cap D \cap \operatorname{rint} D \supset \operatorname{rint} C \cap \operatorname{rint} D \neq \emptyset$, we have $C \cap D \subset \operatorname{rint} D$ by Corollary 3.2.7. The same for $\operatorname{rint}(C \cap D) \subset \operatorname{rint} C$. Thus, we have $\operatorname{rint}(C \cap D) \subset \operatorname{rint} C \cap \operatorname{rint} D$. The converse inclusion is trivial.

(viii): This follows from the definition of $\operatorname{rint} C_x$.

(ix): Since $C_x \subset C$, we have $(C_x)_y \subset C_y$ by definition. We will show that $C_y \subset C_x$, which implies $C_y = (C_y)_y \subset (C_x)_y$ by (viii) and the definition. For each $z \in C_y$, choose $\delta_1 > 0$

so that $u = (1+\delta_1)y - \delta_1 z \in C$. Since $y \in C_x$, we have $\delta_2 > 0$ such that $v = (1+\delta_2)x - \delta_2 y \in C$. Then, observe that

$$\frac{(1+\delta_1)(1+\delta_2)}{1+\delta_1+\delta_2}x - \frac{\delta_1\delta_2}{1+\delta_1+\delta_2}z = \frac{1+\delta_1}{1+\delta_1+\delta_2}v + \frac{\delta_2}{1+\delta_1+\delta_2}u \in C,$$

which means $z \in C_x$.

(x): Since $y \in \operatorname{rint} C_x$, we have $(C_x)_y = C_x$. On the other hand, $(C_x)_y = C_y$ by (ix).

(xi): Take $z \in \operatorname{rint} C_x \cap \operatorname{rint} C_y$. Applying (x), we have $C_x = C_z = C_y$.

(xii): Evidently, $(C \cap D)_x \subset C_x \cap D_x$. For each $y \in C_x \cap D_x$, we can choose $\delta > 0$ so that $(1+\delta)x - \delta y \in C \cap D$, which means $y \in (C \cap D)_x$.

Remark 3.2.3 As in (ii) above, $\operatorname{rint}\operatorname{rint} C = \operatorname{rint} C$ but $\operatorname{rcl}\operatorname{rcl} C \neq \operatorname{rcl} C$ in general. For example, let C be the convex set in $\mathbb{R}^{\mathbb{N}}_f$ defined as follows:

$$C = \Big\{ x \in \mathbf{I}^{\mathbb{N}}_f \;\Big|\; \exists k \in \mathbb{N} \text{ such that } \sum_{i\in\mathbb{N}} x(i) \geqslant k^{-1}, \; x(i) \neq 0 \text{ for at least } k\, i \in \mathbb{N} \Big\}.$$

Then, $\operatorname{rcl} C = \mathbf{I}^{\mathbb{N}}_f \setminus \{\mathbf{0}\}$.

Indeed, as is easily seen, $\mathbf{0} \notin \operatorname{rcl} C$, i.e., $\operatorname{rcl} C \subset \mathbf{I}^{\mathbb{N}}_f \setminus \{\mathbf{0}\}$. For each $x \in \mathbf{I}^{\mathbb{N}}_f \setminus \{\mathbf{0}\}$, choose $k \in \mathbb{N}$ so that $k^{-1} \leqslant \sum_{i\in\mathbb{N}} x(i)$, and define $y \in C$ by $y(i) = k^{-2}$ for $i \leqslant k$ and $y(i) = 0$ for $i > k$. For $0 < t \leqslant 1$, $(1-t)x + ty \in C$ because $(1-t)x(i) + ty(i) \neq 0$ for at least k $i \in \mathbb{N}$ and

$$\sum_{i\in\mathbb{N}} \big((1-t)x(i) + ty(i)\big) = (1-t)\sum_{i\in\mathbb{N}} x(i) + t\sum_{i\in\mathbb{N}} y(i) \geqslant k^{-1}.$$

As observed in Remark 3.2.1, $\operatorname{rcl}\big(\mathbf{I}^{\mathbb{N}}_f \setminus \{\mathbf{0}\}\big) = \mathbf{I}^{\mathbb{N}}_f$. Thus, we have $\operatorname{rcl}\operatorname{rcl} C \neq \operatorname{rcl} C$. It should also be noted that $\operatorname{rint} C = \emptyset$ (cf. (iii) in the above).

Remark 3.2.4 In the above (vii), $\operatorname{rint}(C \cap D) \neq \operatorname{rint} C \cap \operatorname{rint} D$ in general. In fact, for the convex sets $C = \mathbf{I}^n \times \{0\} \subset D = \mathbf{I}^{n+1}$ in Remark 3.2.1, $\operatorname{rint}(C \cap D) = \operatorname{rint} C \neq \emptyset$ but $\operatorname{rint} C \cap \operatorname{rint} D = \emptyset$.

Proposition 3.2.10 *Every non-empty finite-dimensional convex set C has a non-empty radial interior, i.e.,* $\operatorname{rint} C \neq \emptyset$*, and therefore*

$$\operatorname{rcl}\operatorname{rint} C = \operatorname{rcl}\operatorname{rcl} C = \operatorname{rcl} C \;\; \textit{and} \;\; \partial \operatorname{rint} C = \partial \operatorname{rcl} C = \partial C.$$

Proof We have a maximal affinely independent finite subset $\{v_1, \cdots, v_n\} \subset C$. Then, $v_0 = \sum_{i=1}^n n^{-1} v_i \in \operatorname{rint} C$. Indeed, since $C \subset \operatorname{fl}\{v_1, \cdots, v_n\}$, each $x \in C$ can be uniquely written as $x = \sum_{i=1}^n t_i v_i$, where $t_i \in \mathbb{R}$ and $\sum_{i=1}^n t_i = 1$. Observe that

$$(1+\delta)v_0 - \delta x = (1+\delta)\sum_{i=1}^n n^{-1} v_i - \delta \sum_{i=1}^n t_i v_i = \sum_{i=1}^n (n^{-1} + \delta(n^{-1} - t_i))v_i.$$

If $v_0 \neq x$, then $s = \min\{n^{-1} - t_i \mid i = 1, \ldots, n\} < 0$. Let $\delta = (-sn)^{-1} > 0$. Then, $n^{-1} + \delta(n^{-1} - t_i) \geqslant n^{-1} + \delta s = 0$ for every $i = 1, \ldots, n$, which implies that $(1 + \delta)v_0 - \delta x \in C$. □

Concerning the product of convex sets and the affine (pre-)image of a convex set, the following can be easily proved:

Proposition 3.2.11 (Additional Results for Convex Sets) *Let C and D be convex sets in linear spaces E and E', respectively. Then, the following holds:*

(1) *$C \times D$ is also convex,* $\operatorname{rint}(C \times D) = \operatorname{rint} C \times \operatorname{rint} D$, $\operatorname{rcl}(C \times D) = \operatorname{rcl} C \times \operatorname{rcl} D$, *and* $(C \times D)_{(x,y)} = C_x \times D_y$ *for each* $(x, y) \in C \times D$.

Sketch of Proof It is trivial to verify except the inclusions $\operatorname{rint}(C \times D) \supset \operatorname{rint} C \times \operatorname{rint} D$ and $(C \times D)_{(x,y)} \supset C_x \times D_y$.

To see the second inclusion, let $(x', y') \in C_x \times D_y$. Then, we have $\delta_1, \delta_2 > 0$ so that $(1 + \delta_1)x - \delta_1 x' \in C$ and $(1 + \delta_2)y - \delta_2 y' \in D$. Then, $(1 + \delta)(x, y) - \delta(x', y') \in C \times D$, where $\delta = \min\{\delta_1, \delta_2\} > 0$.

The first inclusion can be verified in the same way. But, $\operatorname{rint}(C \times D) \supset \operatorname{rint} C \times \operatorname{rint} D$ can be obtained by the following equivalences:

$$\begin{aligned}(x, y) \in \operatorname{rint} C \times \operatorname{rint} D &\Leftrightarrow C_x = C, D_y = D \\ &\Leftrightarrow (C \times D)_{(x,y)} = C_x \times D_y = C \times D \Leftrightarrow (x, y) \in \operatorname{rint}(C \times D).\end{aligned}$$

(2) *Let $f : C \to E'$ be an affine function. Then, $f(C)$ and $f^{-1}(D)$ are convex and*

$$f^{-1}(D)_x = C_x \cap f^{-1}(D_{f(x)}) \text{ for each } x \in f^{-1}(D)\ (\subset C).$$

In particular, $C_x \subset f^{-1}(f(C)_{f(x)})$ (i.e., $f(C_x) \subset f(C)_{f(x)}$) for each $x \in C$. When f is injective, $f(C_x) = f(C)_{f(x)}$ for each $x \in C$.

Sketch of Proof It is easy to see $f(f^{-1}(D)_x) \subset D_{f(x)}$, i.e., $f^{-1}(D)_x \subset f^{-1}(D_{f(x)})$. Also, $f^{-1}(D)_x \subset C_x$ because $f^{-1}(D) \subset C$. Hence, $f^{-1}(D)_x \subset C_x \cap f^{-1}(D_{f(x)})$. To prove the converse inclusion, for each $y \in f^{-1}(D_{f(x)}) \cap C_x$, choose $\delta > 0$ so that $(1 + \delta)f(x) - \delta f(y) \in D$ and $(1 + \delta)x - \delta y \in C$. Then, $(1 + \delta)x - \delta y \in f^{-1}(D)$.

Remark 3.2.5 In (2) above, $f(C_x) \neq f(C)_{f(x)}$ in general. For instance, let $C = \{(s, t) \in \mathbb{R}^2 \mid |s| \leq t \leq 1\} \subset \mathbb{R}^2$. Then, $\operatorname{pr}_1(C) = [-1, 1]$, $\operatorname{pr}_1(C_{\mathbf{0}}) = \operatorname{pr}_1(\{\mathbf{0}\}) = \{0\}$, and $\operatorname{pr}_1(C)_0 = \operatorname{pr}_1(C)$.

3.3 The Hahn–Banach Extension Theorem

We here prove the Hahn–Banach Extension Theorem and present a relationship between the sublinear functionals and the convex sets.

Let E be a linear space. A functional $p : E \to \mathbb{R}$ is **sublinear** if it satisfies the following conditions:

(SL_1) $p(x+y) \leqslant p(x)+p(y)$ for each $x, y \in E$, and
(SL_2) $p(tx) = tp(x)$ for each $x \in E$ and $t > 0$.

Note that if $p : E \to \mathbb{R}$ is sublinear then $p(\mathbf{0}) = 0$ and $-p(-x) \leqslant p(x)$. For each $x, y \in E$ and $t \in \mathbf{I}$,

$$p((1-t)x + ty) \leqslant (1-t)p(x) + tp(y).$$

When $p : E \to \mathbb{R}$ is a non-negative sublinear functional, $p^{-1}([0, r)) = rp^{-1}([0, 1))$ and $p^{-1}([0, r]) = rp^{-1}(\mathbf{I})$ are convex for each $r > 0$.

In the HAHN–BANACH EXTENSION THEOREM below, no topological concepts appear (even in the proof). Nevertheless, this theorem is very important in the study of topological linear spaces.

Theorem 3.3.1 (HAHN–BANACH EXTENSION) *Let $p : E \to \mathbb{R}$ be a sublinear functional of a linear space E and F be a linear subspace of E. If $f : F \to \mathbb{R}$ is a linear functional such that $f(x) \leqslant p(x)$ for every $x \in F$, then f extends to a linear functional $\tilde{f} : E \to \mathbb{R}$ such that $\tilde{f}(x) \leqslant p(x)$ for every $x \in E$.*

Proof Let $\mathcal{F}$ be the collection of all linear functionals $f' : F' \to \mathbb{R}$ of a linear subspace $F' \subset E$ such that $F \subset F'$, $f'|F = f$, and $f'(x) \leqslant p(x)$ for every $x \in F'$. For $f', f'' \in \mathcal{F}$, we define $f' \leqslant f''$ if f'' is an extension of f'. Then, $\mathcal{F} = (\mathcal{F}, \leqslant)$ is an inductive ordered set, i.e., every totally ordered subset of $\mathcal{F}$ is upper bounded. By Zorn's lemma, $\mathcal{F}$ has a maximal element $f_0 : F_0 \to \mathbb{R}$. It suffices to show that $F_0 = E$.

Assume that $F_0 \neq E$. Taking $x_1 \in E \setminus F_0$, we have a linear subspace $F_1 = F_0 + \mathbb{R}x_1 \supsetneqq F_0$. We show that f_0 has a linear extension $f_1 : F_1 \to \mathbb{R}$ in $\mathcal{F}$, which contradicts the maximality of f_0. By assigning x_1 to $\alpha \in \mathbb{R}$, f_1 can be defined, i.e., $f_1(x + tx_1) = f_0(x) + t\alpha$ for $x \in F_0$ and $t \in \mathbb{R}$. In order that $f_1 \in \mathcal{F}$, we have to choose α so that for every $x \in F_0$ and $t > 0$,

$$f_0(x) + t\alpha \leqslant p(x + tx_1) \quad \text{and} \quad f_0(x) - t\alpha \leqslant p(x - tx_1).$$

Dividing by t, we obtain the following equivalent condition:

$$f_0(y) - p(y - x_1) \leqslant \alpha \leqslant p(y + x_1) - f_0(y) \quad \text{for every } y \in F_0.$$

Hence, such an $\alpha \in \mathbb{R}$ exists if

$$\sup\{f_0(y) - p(y - x_1) \mid y \in F_0\} \leqslant \inf\{p(y + x_1) - f_0(y) \mid y \in F_0\}.$$

This inequality can be proved as follows: For each $y, y' \in F_0$,

$$f_0(y) + f_0(y') = f_0(y + y') \leqslant p(y + y') \leqslant p(y - x_1) + p(y' + x_1),$$

hence $f_0(y) - p(y - x_1) \leqslant p(y' + x_1) - f_0(y')$. Thus, we have the result. □

Let F be a flat in a linear space E and $A \subset F$. The following set is called the **core** of A in F:

$$\operatorname{core}_F A \underset{\text{def}}{=} \big\{x \in A \;\big|\; \forall y \in F,\ \exists \delta > 0 \ \text{ such that } \ |t| \leqslant \delta \ \Rightarrow\ (1-t)x + ty \in A\big\},$$

where $|t| \leqslant \delta$ can be replaced by $-\delta \leqslant t \leqslant 0$ (or $0 \leqslant t \leqslant \delta$). Each point of $\operatorname{core}_F A$ is called a **core point** of A in F. In the case that A is convex,

$$\begin{aligned} x \in \operatorname{core}_F A &\Leftrightarrow \forall y \in F,\ \exists \delta > 0 \ \text{ such that } \ (1+\delta)x - \delta y \in A \\ &\Leftrightarrow \forall y \in F,\ \exists \delta > 0 \ \text{ such that } \ (1-\delta)x + \delta y \in A. \end{aligned}$$

When $F = E$, we can omit the phrase "in E" and simply write core A by removing the subscript E. By definition, $A \subset B \subset F$ implies $\operatorname{core}_F A \subset \operatorname{core}_F B$. We also have the following fact:

Fact *For each $A \subset F$, $\operatorname{core}_F A \neq \emptyset$ implies $\operatorname{fl} A = F$.*

Indeed, $\operatorname{fl} A \subset F$ because $A \subset F$. To see the converse inclusion, let $x \in \operatorname{core}_F A$. For each $y \in F$, we have $\delta > 0$ such that $z = (1+\delta)x - \delta y \in A$. Then, $y = \delta^{-1}(1+\delta)x - \delta^{-1}z \in \operatorname{fl} A$. Consequently, $\operatorname{fl} A = F$.

The following is easily obtained:

Proposition 3.3.2 *For every non-empty convex set $A \subset E$, $\operatorname{core}_{\operatorname{fl} A} A = \operatorname{rint} A$, which is also convex. Hence, $\operatorname{core} A \neq \emptyset$ implies $\operatorname{core} A = \operatorname{rint} A$ and $\operatorname{core}\operatorname{core} A = \operatorname{core} A$.*

Sketch of Proof By definition, $\operatorname{core}_{\operatorname{fl} A} A \subset \operatorname{rint} A$. To see that $\operatorname{rint} A \subset \operatorname{core}_{\operatorname{fl} A} A$, for each $x \in \operatorname{rint} A$ and $y \in \operatorname{fl} A$, we need to find some $s > 0$ such that $(1+s)x - sy \in A$. This can be done using the same proof of the inclusion $C \cap \operatorname{fl} C_x \subset C_x$ in Proposition 3.2.8.

Remark 3.3.1 When A is a non-empty finite-dimensional convex set, $\operatorname{core}_F A \neq \emptyset$ if and only if $F = \operatorname{fl} A$ according to Propositions 3.3.2 and 3.2.10. However, this does not hold for an infinite-dimensional convex set. For example, consider the convex set $\mathbf{I}_f^{\mathbb{N}}$ in $\mathbb{R}^{\mathbb{N}}$. Then, $\mathbb{R}_f^{\mathbb{N}} = \operatorname{fl} \mathbf{I}_f^{\mathbb{N}}$ and $\operatorname{core}_{\mathbb{R}_f^{\mathbb{N}}} \mathbf{I}_f^{\mathbb{N}} = \operatorname{rint} \mathbf{I}_f^{\mathbb{N}} = \emptyset$.

With regard to convex sets defined by a non-negative sublinear functional, we have the following proposition:

Proposition 3.3.3 *Let $p : E \to \mathbb{R}$ be a non-negative sublinear functional of a linear subspace E. Then,*

$$p^{-1}([0,1)) = \operatorname{core} p^{-1}([0,1)) = \operatorname{core} p^{-1}(\mathbf{I}).$$

Proof Let $x \in p^{-1}([0,1))$, i.e., $p(x) < 1$. For each $y \in E$, choose $\delta > 0$ so that $\delta p(x-y) < 1 - p(x)$. Then,

$$0 \leqslant p((1+\delta)x - \delta y) = p(x + \delta(x-y)) \leqslant p(x) + \delta p(x-y) < 1,$$

which means $x \in \operatorname{core} p^{-1}([0,1))$. Thus, we have

$$p^{-1}([0,1)) \subset \operatorname{core} p^{-1}([0,1)) \subset \operatorname{core} p^{-1}(\mathbf{I}).$$

If $p(x) \geqslant 1$, then $x \notin \operatorname{core} p^{-1}(\mathbf{I})$ because

$$p((1+t)x - t\mathbf{0}) = (1+t)p(x) > 1 \ \text{ for any } t > 0.$$

Therefore, $\operatorname{core} p^{-1}(\mathbf{I}) \subset p^{-1}([0,1))$. □

For each $A \subset E$ with $\mathbf{0} \in \operatorname{core} A$, the **Minkowski functional** $p_A : E \to \mathbb{R}_+$ can be defined as follows:

$$p_A(x) = \inf\left\{s > 0 \;\middle|\; x \in sA\right\} = \inf\left\{s > 0 \;\middle|\; s^{-1}x \in A\right\}.$$

Then, for each $x \in E$ and $t > 0$,

$$\begin{aligned} p_A(tx) &= \inf\left\{s > 0 \;\middle|\; s^{-1}tx \in A\right\} = \inf\left\{ts > 0 \;\middle|\; (ts)^{-1}tx \in A\right\} \\ &= t \inf\left\{s > 0 \;\middle|\; s^{-1}x \in A\right\} = tp_A(x), \end{aligned}$$

i.e., p_A satisfies (SL_2). In the above, $p_A(tx) = p_{t^{-1}A}(x)$. Then, it follows that $p_{t^{-1}A} = tp_A$ for each $t > 0$. Replacing t by t^{-1}, we have

$$p_{tA} = t^{-1}p_A \ \text{ for each } t > 0.$$

When $A \subset E$ is convex, the Minkowski functional p_A has the following desirable properties:

Proposition 3.3.4 *Let $A \subset E$ be a convex set with $\mathbf{0} \in \operatorname{core} A$. Then, the Minkowski functional p_A is sublinear and*

$$\operatorname{rint} A = \operatorname{core} A = p_A^{-1}([0,1)) \subset A \subset p_A^{-1}(\mathbf{I}) = \operatorname{rcl} A,$$

so $\partial A = p_A^{-1}(1)$. Moreover, the following equivalence holds:

$$p_A(x) = 0 \ \Leftrightarrow \ \mathbb{R}_+ x \subset A.$$

In order that p_A is a norm on E, it is necessary and sufficient that $\mathbb{R}_+ x \not\subset A$ if $x \neq \mathbf{0}$ and $tA \subset A$ if $|t| < 1$.

Proof First, we prove that p_A is sublinear. As already observed, p_A satisfies (SL_2). To show that p_A satisfies (SL_1), let $x, y \in E$. Since A is convex, we have

$$s^{-1}x,\ t^{-1}y \in A \ \Rightarrow\ (s+t)^{-1}(x+y) = \frac{s}{s+t}s^{-1}x + \frac{t}{s+t}t^{-1}y \in A,$$

which implies that $p_A(x+y) \leqslant p_A(x) + p_A(y)$.

The equality $\operatorname{rint} A = \operatorname{core} A$ was obtained in Proposition 3.3.2. The following easily follows from the definitions:

$$\operatorname{core} A \subset p_A^{-1}([0,1)) \subset A \subset p_A^{-1}(\mathbf{I}) \subset \operatorname{rcl} A.$$

By Propositions 3.3.2 and 3.3.3, we have

$$\operatorname{core} A = \operatorname{core}\operatorname{core} A \subset \operatorname{core} p_A^{-1}([0,1)) = p_A^{-1}([0,1)) \subset \operatorname{core} A.$$

Hence, $\operatorname{core} A = p_A^{-1}([0,1))$. To prove $p_A^{-1}(\mathbf{I}) = \operatorname{rcl} A$, it remains to show that $\operatorname{rcl} A \subset p_A^{-1}(\mathbf{I})$. Let $x \in \operatorname{rcl} A$. Since $\mathbf{0} \in \operatorname{rint} A$, it follows from Proposition 3.2.6 that $s^{-1}x \in \operatorname{rint} A \subset A$ for each $s > 1$, which implies that $p_A(x) \leqslant 1$, i.e., $x \in p_A^{-1}(\mathbf{I})$.

By definition, $p_A(x) = 0$ if and only if $tx \in A$ for an arbitrarily large $t > 0$, which means that $\mathbb{R}_+ x \subset A$ because A is convex.

Because p_A is sublinear, p_A is a norm if and only if $p_A(x) \neq 0$ and $p_A(x) = p_A(-x)$ for every $x \in E \setminus \{\mathbf{0}\}$. Because $p_A(x) \neq 0$ if and only if $\mathbb{R}_+ x \not\subset A$, it remains to show that $p_A(x) = p_A(-x)$ for every $x \in E \setminus \{\mathbf{0}\}$ if and only if $tA \subset A$ whenever $|t| < 1$.

Assume that $p_A(x) = p_A(-x)$ for each $x \in E$. If $x \in A$ and $|t| < 1$ then $p_A(tx) = p_A(|t|x) = |t| p_A(x) < 1$, which implies that $tx \in A$. Hence, $tA \subset A$ whenever $|t| < 1$.

Conversely, assume that $tA \subset A$ whenever $|t| < 1$. For each $s > p_A(x)$, $r^{-1}x \in A$ for some $0 < r < s$, and we have $s^{-1}(-x) = (-s^{-1}r)r^{-1}x \in A$, hence $p_A(-x) \leqslant p_A(x)$. Replacing x with $-x$, we have $p_A(x) \leqslant p_A(-x)$. Therefore, $p_A(x) = p_A(-x)$. □

When the Minkowski functional p_A is a norm on E, we call it the **Minkowski norm**. In this case, $\operatorname{rcl} A$, $\operatorname{rint} A$, and ∂A are the unit closed ball, the unit open ball, and the unit sphere, respectively, of the normed linear space $E = (E, p_A)$. Then, $\operatorname{rcl} A$ and $\operatorname{rint} A$ are symmetric about $\mathbf{0}$, i.e., $\operatorname{rcl} A = -\operatorname{rcl} A$ and $\operatorname{rint} A = -\operatorname{rint} A$. We should note that a convex set $A \subset E$ is symmetric about $\mathbf{0}$ if and only if $tA \subset A$ whenever $|t| \leqslant 1$ (in the next section, A is said to be circled).

A subset $W \subset E$ is called a **wedge** if $x + y \in W$ for each $x, y \in W$ and $tx \in W$ for each $x \in W$, $t \geqslant 0$, or, equivalently, W is convex and $tW \subset W$ for every $t \geqslant 0$. Note that if $A \subset E$ is convex then $\mathbb{R}_+ A$ is a wedge. For a wedge $W \subset E$, the following statements are true:

(1) $\mathbf{0} \in \operatorname{core} W \Leftrightarrow W = E$;
(2) $W \neq E,\ x \in \operatorname{core} W \Rightarrow -x \notin W$.

Using the Hahn–Banach Extension Theorem, we can prove the following separation theorem:

Theorem 3.3.5 (SEPARATION) *Let A and B be non-empty convex sets in E such that* $\operatorname{core} A \neq \emptyset$ *and* $(\operatorname{core} A) \cap B = \emptyset$. *Then, there exists a linear functional $f : E \to \mathbb{R}$ such that $f(x) < f(y)$ for every $x \in \operatorname{core} A$ and $y \in B$, and* $\sup f(A) \leqslant \inf f(B)$.

Proof Recall that core $A = \operatorname{rint} A$ (Proposition 3.3.2). For a linear functional $f : E \to \mathbb{R}$, if $f(x) < f(y)$ for every $x \in \operatorname{core} A$ and $y \in B$, then $\sup f(A) \leqslant \inf f(B)$. Indeed, let $x \in A$, $y \in B$, $v \in \operatorname{core} A$, and $0 \leqslant t < 1$. Since $(1-t)v + tx \in \operatorname{core} A$ by Proposition 3.2.6, we have

$$(1-t)f(v) + tf(x) = f((1-t)v + tx) < f(y),$$

where the left side tends to $f(x)$ as $t \to 1$, and hence $f(x) \leqslant f(y)$.

Note that $W = \mathbb{R}_+(A - B)$ is a wedge. Moreover, $(\operatorname{core} A) - B \subset \operatorname{core} W$. Indeed, let $x \in \operatorname{core} A$ and $y \in B$. For each $z \in E$, choose $\delta > 0$ so that $(1+\delta)x - \delta(y+z) \in A$. Then,

$$(1+\delta)(x-y) - \delta z = (1+\delta)x - \delta(y+z) - y \in A - B \subset W.$$

Therefore, it suffices to construct a linear functional $f : E \to \mathbb{R}$ such that $f(\operatorname{core} W) \subset (-\infty, 0)$.

Now, we will show that $W \cap (B - \operatorname{core} A) = \emptyset$. Assume that there exist $x_0 \in A$, $x_1 \in \operatorname{core} A$, $y_0, y_1 \in B$, and $t_0 \geqslant 0$ such that $t_0(x_0 - y_0) = y_1 - x_1$. Note that $\operatorname{rint}\langle x_0, x_1\rangle \subset \operatorname{rint} A = \operatorname{core} A$ by Proposition 3.2.6. Hence,

$$\frac{t_0}{t_0+1}x_0 + \frac{1}{t_0+1}x_1 = \frac{t_0}{t_0+1}y_0 + \frac{1}{t_0+1}y_1 \in (\operatorname{core} A) \cap B,$$

which contradicts the fact that $(\operatorname{core} A) \cap B = \emptyset$.

Take $v_0 \in (\operatorname{core} A) - B \subset \operatorname{core} W$. Then, note that $-v_0 \notin W$. For each $x \in E$, we have $\delta > 0$ such that $(1+\delta)v_0 - \delta(-x) \in W$, which implies $x + \delta^{-1}(1+\delta)v_0 \in W$. Then, we can define $p : E \to \mathbb{R}$ by

$$p(x) = \inf\left\{t \geqslant 0 \;\middle|\; x + tv_0 \in W\right\}.$$

Because W is a wedge, we see that p is sublinear. Since $-v_0 \notin W$, it follows that $p(s(-v_0)) = s$ and $p(sv_0) = 0$ for every $s \geqslant 0$. Applying the Hahn–Banach Extension Theorem 3.3.1, we can obtain a linear functional $f : E \to \mathbb{R}$ such that $f(s(-v_0)) = s$ for each $s \in \mathbb{R}$ and $f(x) \leqslant p(x)$ for every $x \in E$ (see Fig. 3.3). For each $z \in \operatorname{core} W$, we have $\delta > 0$ such that $(1+\delta)z - \delta(z+v_0) \in W$, i.e., $z - \delta v_0 \in W$. Accordingly, $(z - \delta v_0) + tv_0 \in W$ for every $t \geqslant 0$, which means $p(z - \delta v_0) = 0$. Thus, we have

$$f(z) < f(z) + \delta = f(z - \delta v_0) \leqslant p(z - \delta v_0) = 0. \qquad \square$$

Remark 3.3.2 Using the Hahn–Banach Extension Theorem, we have just proved the Separation Theorem. Conversely, the Hahn–Banach Extension Theorem can be derived from the Separation Theorem. Indeed, under the assumption of the Hahn–Banach Extension Theorem 3.3.1, we define

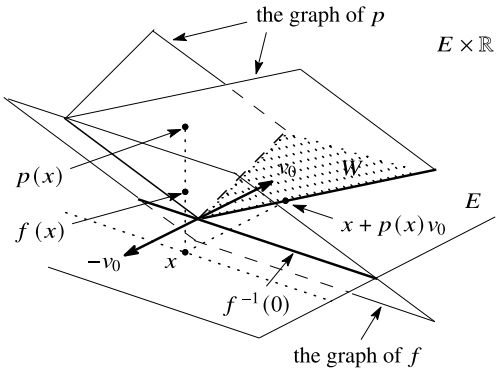


Fig. 3.3 The graphs of p and f

$$A = \big\{(x,t) \in E \times \mathbb{R} \,\big|\, t > p(x)\big\} \text{ and } B = \big\{(x, f(x)) \in E \times \mathbb{R} \,\big|\, x \in F\big\},$$

where $B = \mathrm{Gr}(f)$ is the graph of f. Then, A and B are disjoint convex sets in $E \times \mathbb{R}$. It is straightforward to show that $\mathrm{core}\, A = A \neq \emptyset$. By the Separation Theorem 3.3.5, we have a linear functional $\varphi : E \times \mathbb{R} \to \mathbb{R}$ such that $A \subset \varphi^{-1}((-\infty, r])$ and $B \subset \varphi^{-1}([r, \infty))$ for some $r \in \mathbb{R}$. Then, $r \leqslant 0$ because $0 = \varphi(\mathbf{0}, 0) \in \varphi(B)$. If $\varphi(z) < 0$ for some $z \in B$, then $\varphi(tz) = t\varphi(z) < r$ for sufficiently large $t > 0$. This is a contradiction because $tz \in B$. If $\varphi(z) > 0$ for some $z \in B$, then $-z \in B$ and $\varphi(-z) = -\varphi(z) < 0$, which is a contradiction. Therefore, $B \subset \varphi^{-1}(0)$. Note that $\varphi(\mathbf{0}, 1) < 0$ because $(0, 1) \in A$. Since $\varphi(x, t) = \varphi(x, 0) + t\varphi(0, 1)$ for each $x \in E$, we have $\varphi(\{x\} \times \mathbb{R}) = \mathbb{R}$. Observe that $(\{x\} \times \mathbb{R}) \cap \varphi^{-1}(0)$ is a singleton. Then, f extends to the linear functional $\tilde{f} : E \to \mathbb{R}$ whose graph is $\varphi^{-1}(0)$, i.e., $(x, \tilde{f}(x)) \in \varphi^{-1}(0)$ for each $x \in E$. Since $\varphi^{-1}(0) \subset (E \times \mathbb{R}) \setminus A$, it follows that $\tilde{f}(x) \leqslant p(x)$ for every $x \in E$.

The Separation Theorem 3.3.5 can also be obtained as a corollary of the following two theorems, where we do not use the Hahn–Banach Extension Theorem 3.3.1.

Theorem 3.3.6 *For each pair of disjoint non-empty convex sets $A, B \subset E$, there exists a pair of disjoint convex sets $\widetilde{A}, \widetilde{B} \subset E$ such that $A \subset \widetilde{A}$, $B \subset \widetilde{B}$, and $\widetilde{A} \cup \widetilde{B} = E$.*

Proof Let $\mathcal{P}$ be the collection of pairs (C, D) of disjoint convex sets such that $A \subset C$ and $B \subset D$. For $(C, D), (C', D') \in \mathcal{P}$, we define $(C, D) \leqslant (C', D')$ if $C \subset C'$ and $D \subset D'$. Then, it is easy to see that $\mathcal{P} = (\mathcal{P}, \leqslant)$ is an inductive ordered set. Due to Zorn's Lemma, $\mathcal{P}$ has a maximal element $(\widetilde{A}, \widetilde{B})$.

To show that $\widetilde{A} \cup \widetilde{B} = E$, assume the contrary, i.e., there exists a point $v_0 \in E \setminus (\widetilde{A} \cup \widetilde{B})$. By the maximality of $(\widetilde{A}, \widetilde{B})$, we can obtain two points

$$x \in \widetilde{A} \cap \langle \widetilde{B} \cup \{v_0\} \rangle \text{ and } y \in \widetilde{B} \cap \langle \widetilde{A} \cup \{v_0\} \rangle.$$

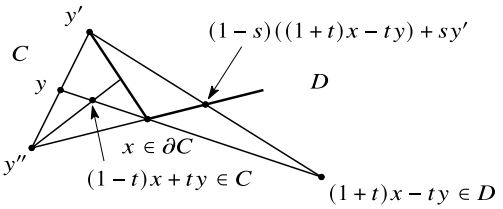


Fig. 3.4 $\partial C \subset \partial D$

Then, $x \in \langle v_0, y_1 \rangle$ for some $y_1 \in \widetilde{B}$ and $y \in \langle v_0, x_1 \rangle$ for some $x_1 \in \widetilde{A}$. Note that $x \in \operatorname{rint}\langle v_0, y_1 \rangle$ and $y \in \operatorname{rint}\langle v_0, x_1 \rangle$. Consider the triangle $\langle v_0, x_1, y_1 \rangle$. It is easy to see that $\langle x_1, x \rangle$ and $\langle y_1, y \rangle$ meet at a point v_1. Since $\langle x_1, x \rangle \subset \widetilde{A}$ and $\langle y_1, y \rangle \subset \widetilde{B}$, it follows that $v_1 \in \widetilde{A} \cap \widetilde{B}$, which is a contradiction. □

Theorem 3.3.7 *For each pair of disjoint non-empty convex sets $C, D \subset E$ with $C \cup D = E$, $\operatorname{rcl} C \cap \operatorname{rcl} D$ is a hyperplane of E if $\emptyset \neq \operatorname{rcl} C \cap \operatorname{rcl} D \subsetneqq E$.*

Proof First, we prove that $\operatorname{rcl} C \cap \operatorname{rcl} D = \partial C = \partial D \neq \emptyset$. To prove that $\partial C \subset \partial D$, let $x \in \partial C$. It suffices to find $y \in C$ such that

$$\begin{aligned} &1 - t)x + ty \in C \ \text{ for } 0 < t \leqslant 1 \text{ and} \\ &(1 + t)x - ty \in E \setminus C = D \ \text{ for } t > 0. \end{aligned}$$

To this end, take $y', y'' \in C$ such that $(1 - t)x + ty' \in C$ for $0 < t \leqslant 1$ and $(1 + t)x - ty'' \notin C$ for $t > 0$. Then, $y = \frac{1}{2}y' + \frac{1}{2}y'' \in C$ is the desired point. Indeed, for each $0 < t \leqslant 1$,

$$\begin{aligned} (1 - t)x + ty &= (1 - t)x + \tfrac{1}{2}ty' + \tfrac{1}{2}ty'' \\ &= (1 - \tfrac{1}{2}t)\left(\frac{1 - t}{1 - \frac{1}{2}t}x + \frac{\frac{1}{2}t}{1 - \frac{1}{2}t}y'\right) + \tfrac{1}{2}ty'' \in C. \end{aligned}$$

Moreover, note that

$$\begin{aligned} &(1 - s)((1 + t)x - ty) + sy' \\ &\qquad = (1 - s)(1 + t)x - \tfrac{1}{2}(1 - s)ty' - \tfrac{1}{2}(1 - s)ty'' + sy'. \end{aligned}$$

For each $t > 0$, let $s = t/(2 + t) \in (0, 1)$. Then, $(1 - s)t = 2s$. Therefore, we have

$$(1 - s)((1 + t)x - ty) + sy' = (1 + s)x - sy'' \notin C,$$

which means that $(1 + t)x - ty \notin C$ (Fig. 3.4). Similarly, we have $\partial D \subset \partial C$. Hence, $\partial C = \partial D$. Since $\operatorname{rint} C \cap \operatorname{rint} D = \emptyset$, it follows that $\operatorname{rcl} C \cap \operatorname{rcl} D = \partial C = \partial D$.

Next, we will show that ∂C is a flat. To see that $\partial C \neq \emptyset$, let $u \in C$ and $v \in D$. Since $w = 2u - v \in D$ implies $u = (v + w)/2 \in D$ because D is convex, we

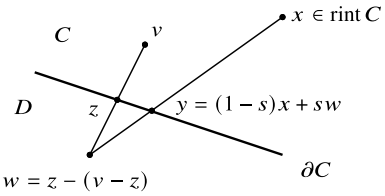


Fig. 3.5 The case $x \in \operatorname{rint} C$

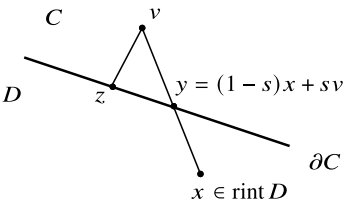


Fig. 3.6 The case $x \in \operatorname{rint} D$

have $w \in C$. Then, $v = 2u - w \in \operatorname{fl} C \cap D = \operatorname{fl} C \setminus C$. Hence, C is not a flat, which means $\partial C \neq \emptyset$ by Proposition 3.2.9 (vi).

We now verify that $x' = (1+t)x - ty \in \partial C$ for each $x, y \in \partial C$ and $t > 0$. If $x' \notin \partial C$, then $x' \in \operatorname{rint} C$ or $x' \in \operatorname{rint} D$. In this case, $x \in \operatorname{rint}\langle x', y\rangle \subset \operatorname{rint} C$ or $x \in \operatorname{rint}\langle x', y\rangle \subset \operatorname{rint} D$ by Proposition 3.2.6. This is a contradiction, so $x' \in \partial C$.

All that remains is to show that ∂C is a hyperplane if $\partial C \neq E$. To this end, let $v \in E \setminus \partial C$. It suffices to show that $E = \operatorname{fl}(\partial C \cup \{v\})$. Without loss of generality, we may assume that $v \in \operatorname{rint} C$. On the other hand, we can take $z \in \partial C$ (we saw $\partial C \neq \emptyset$). Then, $w = 2z - v \in \operatorname{rint} D$. If not, then $w \in \operatorname{rcl} C$, from which, using Proposition 3.2.6, we have $z = (v+w)/2 \in \operatorname{rint}\langle v, w\rangle \subset \operatorname{rint} C$, which is a contradiction.

For each $x \in E \setminus \partial C$, $x \in \operatorname{rint} C$ or $x \in \operatorname{rint} D$. When $x \in \operatorname{rint} C$, let

$$s = \sup\left\{t \in \mathbf{I} \;\middle|\; (1-t)x + tw \in C\right\}.$$

Refer to Fig. 3.5. Then, $y = (1-s)x + sw \in \partial C$, which implies that

$$x = \frac{1}{1-s}y - \frac{2s}{1-s}z + \frac{s}{1-s}v \in \operatorname{fl}(\partial C \cup \{v\}).$$

In the case that $x \in \operatorname{rint} D$, let

$$s = \sup\left\{t \in \mathbf{I} \;\middle|\; (1-t)x + tv \in D\right\}.$$

Now, refer to Fig. 3.6. Then, $y = (1-s)x + sv \in \partial D = \partial C$, which implies that

$$x = \frac{1}{1-s}y + \frac{-s}{1-s}v \in \operatorname{fl}(\partial C \cup \{v\}).$$

Consequently, it follows that $E = \operatorname{fl}(\partial C \cup \{v\})$. □

Remark 3.3.3 In the above, the condition $\operatorname{rcl} C \cap \operatorname{rcl} D \neq E$ is necessary. For example, define the convex set C in the linear space $\mathbb{R}_f^{\mathbb{N}}$ as follows:

$$C = \left\{ x \in \mathbb{R}_f^{\mathbb{N}} \;\middle|\; n = \max\{i \mid x(i) \neq 0\} \Rightarrow x(n) > 0 \right\}.$$

Let $D = \mathbb{R}_f^{\mathbb{N}} \setminus C = (-C) \setminus \{\mathbf{0}\}$. Then, D is also convex. As is easily observed, $\operatorname{rcl} C = \operatorname{rcl} D = \mathbb{R}_f^{\mathbb{N}}$, hence $\operatorname{rcl} C \cap \operatorname{rcl} D = \mathbb{R}_f^{\mathbb{N}}$.

The Separation Theorem 3.3.5 can also be obtained as a corollary of Theorems 3.3.6 and 3.3.7. In fact, let $A, B \subset E$ be non-empty convex sets with $\operatorname{core} A \neq \emptyset$ and $(\operatorname{core} A) \cap B = \emptyset$. Then, $\operatorname{core} A = \operatorname{rint} A$ is convex. We apply Theorem 3.3.6 to obtain disjoint non-empty convex sets C and D such that $\operatorname{core} A \subset C$, $B \subset D$, and $C \cup D = E$. Observe that $\operatorname{core} A \cap \operatorname{rcl} D = \emptyset$, hence $\operatorname{rcl} D \neq E$. It follows from Theorem 3.3.7 that $\operatorname{rcl} C \cap \operatorname{rcl} D$ is a hyperplane. Then, we have a linear functional $f : E \to \mathbb{R}$ such that $\operatorname{rcl} C \cap \operatorname{rcl} D = f^{-1}(s)$ for some $s \in \mathbb{R}$ (Proposition 3.1.3 (1)). Since $\operatorname{core} A \subset E \setminus f^{-1}(s)$, we have $\operatorname{core} A \subset f^{-1}((s, \infty))$ or $\operatorname{core} A \subset f^{-1}((-\infty, s))$. If $\operatorname{core} A \subset f^{-1}((s, \infty))$, by replacing f and s by $-f$ and $-s$, it can be assumed that $\operatorname{core} A \subset f^{-1}((-\infty, s))$.

We now show that $\operatorname{rcl} C \subset f^{-1}((-\infty, s])$. Let $x \in \operatorname{core} A$ $(\subset \operatorname{rint} C)$. Then, $x \in \operatorname{rint} C$ and $f(x) < s$. If $f(y) > s$ for some $y \in \operatorname{rcl} C$, we have $z \in \operatorname{rint}\langle x, y\rangle \cap f^{-1}(s)$. Because $z \in \operatorname{rcl} D$, $\operatorname{rint}\langle w, z\rangle \subset D$ for some $w \in D$. On the other hand, $z \in \operatorname{rint}\langle x, y\rangle \subset \operatorname{rint} C$ (Proposition 3.2.6). Because $\operatorname{rint} C = \operatorname{core} C$, $\langle v, z\rangle \subset C = E \setminus D$ for some $v \in \operatorname{rint}\langle w, z\rangle$, which is a contradiction.

Since $C \subset f^{-1}((-\infty, s])$, it follows that $D \supset f^{-1}((s, \infty))$. Observe that $\operatorname{rint} D \supset f^{-1}((s, \infty))$. So, we have $x \in \operatorname{rint} D$ such that $f(x) > s$. Likewise for $\operatorname{rcl} D$, we can show that $\operatorname{rcl} D \subset f^{-1}([s, \infty))$. Since $\operatorname{core} A \subset f^{-1}((-\infty, s))$ and $B \subset f^{-1}([s, \infty))$, we have the desired result.

3.4 Topological Linear Spaces

A **topological linear space** E is a linear space with a topology such that the algebraic operations of addition $(x, y) \mapsto x + y$ and scalar multiplication $(x, t) \mapsto tx$ are continuous.[9] Every linear space E has such a topology. In fact, E has a Hamel basis B. As a linear subspace of the product space $\mathbb{R}^B$, we have a topological linear space $\mathbb{R}_f^B$ that is linearly isomorphic to E by the linear isomorphism $\varphi : \mathbb{R}_f^B \to E$ defined by $\varphi(x) = \sum_{v \in B} x(v)v$. Then, φ induces a topology that makes E a topological linear space. In the next section, it will be seen that if E is finite-dimensional, then such a topology is unique. However, an infinite-dimensional linear space has various topologies for which the algebraic operations are continuous.

[9] Here, we only consider linear spaces over $\mathbb{R}$. Recall that topological spaces are assumed to be **Hausdorff**. For topological linear spaces (more generally for topological groups), it suffices to assume axiom T_0, which implies **regularity** (Proposition 3.4.2 and its footnote). The continuity of scalar multiplication implies the continuity of the operation $x \mapsto -x$ because $(-1)x = -x$.

In the following proposition, we present the basic properties of a neighborhood basis at $\mathbf{0}$ in a topological linear space.

Proposition 3.4.1 *Let E be a topological linear space and $\mathcal{U}$ be a neighborhood basis at $\mathbf{0}$ in E. Then, $\mathcal{U}$ has the following properties:*

(i) *For each $U, V \in \mathcal{U}$, there is some $W \in \mathcal{U}$ such that $W \subset U \cap V$.*
(ii) *For each $U \in \mathcal{U}$, there is some $V \in \mathcal{U}$ such that $V + V \subset U$.*
(iii) *For each $U \in \mathcal{U}$, there is some $V \in \mathcal{U}$ such that $[-1, 1]V \subset U$.*
(iv) *For each $x \in E$ and $U \in \mathcal{U}$, there is some $a > 0$ such that $x \in aU$.*
(v) $\bigcap \mathcal{U} = \{\mathbf{0}\}$.

Conversely, let E be a linear space with $\mathcal{U}$ a non-empty collection of subsets satisfying these conditions. Then, E has a topology such that addition and scalar multiplication are continuous and $\mathcal{U}$ is a neighborhood basis at $\mathbf{0}$.

Proof Property (i) is trivial; (ii) comes from the continuity of addition at $(\mathbf{0}, \mathbf{0}) \in E \times E$; (iii) is obtained by the continuity of scalar multiplication at each $(\mathbf{0}, t) \in E \times [-1, 1]$ and the compactness of $[-1, 1]$; (iv) follows from the continuity of scalar multiplication at $(x, 0) \in E \times \mathbb{R}$; (v) comes from axiom T_1.

Given $\mathcal{U}$ with these properties, an open set in E is defined as a subset $W \subset E$ satisfying the condition that, for each $x \in W$, there is some $U \in \mathcal{U}$ such that $x + U \subset W$. (Verify the axioms of open sets, i.e., every intersection of finitely many open sets is open; every union of open sets is open.)

For each $x \in E$ and $U \in \mathcal{U}$, $x + U$ is a neighborhood of x in this topology.[10] Indeed, let

$$W = \{y \in E \mid \exists V \in \mathcal{U} \text{ such that } y + V \subset x + U\}.$$

Then, $x \in W \subset x + U$ because of (v). For each $y \in W$, we have $V \in \mathcal{U}$ such that $y + V \subset x + U$. Take $V' \in \mathcal{U}$ so that $V' + V' \subset V$ as in (ii). Then, $y + V' \subset W$ because $(y + y') + V' \subset y + V \subset x + U$ for every $y' \in V'$. Therefore, W is open in E, so $x + U$ is a neighborhood of x in E. By the definition of the topology, $\{x + U \mid U \in \mathcal{U}\}$ is a neighborhood basis at x. In particular, $\mathcal{U}$ is a neighborhood basis at $\mathbf{0}$.

Since $\{x + U \mid U \in \mathcal{U}\}$ is a neighborhood basis at x, the continuity of addition follows from (ii). Using (iii), we can show that the operation $x \mapsto -x$ is continuous.

For scalar multiplication, let $x \in E$, $\alpha \in \mathbb{R}$, and $U \in \mathcal{U}$. Because of the continuity of $x \mapsto -x$, it can be assumed that $\alpha \geqslant 0$. Then, we can write $\alpha = n + t$, where $n \in \omega$ and $0 \leqslant t < 1$. Using (ii) inductively, we can find $V_1 \supset \cdots \supset V_n \supset V_{n+1}$ in $\mathcal{U}$ such that

$$V_1 + \cdots + V_n + (V_{n+1} + V_{n+1}) \subset U.$$

By (iii), we have $W \in \mathcal{U}$ such that $[-1, 1]W \subset V_{n+1}$. Then, $x \in rW$ for some $r > 0$ by (iv). Choose $\delta > 0$ so that $\delta < \min\{1/r, 1 - t\}$. Let $y \in x + W$ and $|\alpha - \beta| < \delta$. Then, we can write $\beta = n + s$, where $t - \delta < s < t + \delta$. It follows that

$$\begin{aligned} \beta y - \alpha x &= (n + s)y - (n + t)x = n(y - x) + s(y - x) + (s - t)x \\ &\in nW + [-1, 1]W + \delta[-1, 1](rW) \\ &\subset nV_{n+1} + V_{n+1} + V_{n+1} \subset \underbrace{V_{n+1} + \cdots + V_{n+1}}_{n+2} \subset U, \end{aligned}$$

[10] If E is a topological linear space, then $x + U$ is a neighborhood of $x \in E$ for any neighborhood U of $\mathbf{0}$.

hence $\beta y \in \alpha x + U$.[11]

To see the Hausdorffness, let $x \neq y \in E$. By (v), we have $U \in \mathcal{U}$ such that $x - y \notin U$. By (ii) and (iii), we can find $V \in \mathcal{U}$ such that $V - V \subset U$. Then, $x + V$ and $y + V$ are neighborhoods of x and y, respectively. Observe that $(x + V) \cap (y + V) = \emptyset$. □

It is said that $A \subset E$ is **circled** if $tA \subset A$ for every $t \in [-1, 1]$. It should be noted that a circled set is defined in an arbitrary linear space. When E is a topological linear space, the closure of a circled set A is also circled.

Indeed, let $x \in \mathrm{cl}\, A$ and $t \in [-1, 1]$. If $t = 0$, then $tx = \mathbf{0} \in A \subset \mathrm{cl}\, A$. When $t \neq 0$, for each neighborhood U of tx in E, since $t^{-1}U$ is a neighborhood of $t^{-1}tx = x$, $t^{-1}U \cap A \neq \emptyset$, which implies that $U \cap tA \neq \emptyset$. Because $tA \subset A$, $U \cap A \neq \emptyset$. Thus, we have $tx \in \mathrm{cl}\, A$.

In the above (iii), $[-1, 1]V$ is a circled neighborhood of $\mathbf{0} \in E$, where if V is open in E, then $[-1, 1]V = \bigcup_{0<|t|\leqslant 1} tV$ is also open in E. Hence, (iii) is equivalent to the following condition:

(iii)′ $\mathbf{0} \in E$ has a neighborhood basis consisting of circled (open) sets.

A **topological group** G is a group with a topology such that the algebraic operations of multiplication $(x, y) \mapsto xy$ and taking inverses $x \mapsto x^{-1}$ are both continuous.[12] Then, G is **homogeneous**, that is, for each distinct $x_0, x_1 \in G$, there is a homeomorphism $h : G \to G$ such that $h(x_0) = x_1$. Such an h can be defined by $h(x) = x_0 x^{-1} x_1$, where not only $h(x_0) = x_1$ but also $h(x_1) = x_0$. Every topological linear space is a topological group with respect to addition, so it is homogeneous.

Proposition 3.4.2 *Every topological group G has a closed neighborhood basis at each $g \in G$, i.e., it is regular.*[13] *For a topological linear space E, $\mathbf{0} \in E$ has a circled closed neighborhood basis.*

Sketch of Proof Each neighborhood U of the unit $\mathbf{1} \in G$ contains a neighborhood V of $\mathbf{1}$ such that $V^{-1}V \subset U$. For each $x \in \mathrm{cl}\, V$, we have $y \in Vx \cap V$. Consequently, $x \in V^{-1}y \subset V^{-1}V \subset U$, so we have $\mathrm{cl}\, V \subset U$. For the additional statement, recall that if V is circled, then $\mathrm{cl}\, V$ is also circled.

Concerning a subgroup of a topological group, we have the following:

Proposition 3.4.3 *Let G be a topological group and H be a subgroup of G.*

(1) *If H is open in G, then H is closed in G.*
(2) *The closure* $\mathrm{cl}\, H$ *of H is a subgroup of G.*

Sketch of Proof (1): For each $x \in G \setminus H$, Hx is an open neighborhood of x in G and $Hx \subset G \setminus H$.

(2): For each $x, y \in \mathrm{cl}\, H$, show that $x^{-1}y \in \mathrm{cl}\, H$, i.e., each neighborhood W of $x^{-1}y$ meets H. To this end, choose neighborhoods U and V of x and y, respectively, so that $U^{-1}V \subset W$.

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[11] It should be noted that, in general, $2V \subset V + V$ but $V + V \not\subset 2V$.

[12] These two operations are continuous if and only if the operation $(x, y) \to x^{-1}y$ is continuous.

[13] A topological group G is assumed to be **Hausdorff**, but it suffices to assume axiom T_0. In fact, axiom T_0 implies T_1 for a topological group G because of the homogeneity of G.

The following is a directed consequence of the above (1):

Corollary 3.4.4 *A connected topological group G has no open subgroups except for G itself.*

For a topological linear space, we have the following:

Proposition 3.4.5 *Let E be a topological linear space and F a linear subspace.*

(1) *If $F \neq E$, then F is not open in E.*
(2) *The closure* cl F *of F is also a linear subspace of E.*

Sketch of Proof Since every topological linear space is path-connected, (1) follows from Corollary 3.4.4. To show (2), we can apply Proposition 3.4.3 (2). So it suffices to show that $tx \in \mathrm{cl}\, F$ if $x \in \mathrm{cl}\, F$ and $t \in \mathbb{R}$. This is an easy exercise.

The continuity of linear functionals is characterized as follows:

Proposition 3.4.6 *Let E be a topological linear space. For a linear functional $f : E \to \mathbb{R}$ with $f(E) \neq \{0\}$, the following are equivalent:*

(a) *f is continuous;*
(b) *$f^{-1}(0)$ is closed in E;*
(c) *$f^{-1}(0)$ is not dense in E;*
(d) *$f(V)$ is bounded for some neighborhood V of $\mathbf{0} \in E$.*

Proof The implication (a) $\Rightarrow$ (b) is obvious, and (b) $\Rightarrow$ (c) follows from $f(E) \neq \{0\}$ (i.e., $f^{-1}(0) \neq E$).

(c) $\Rightarrow$ (d): We have $x \in E$ and a circled neighborhood V of $\mathbf{0} \in E$ such that $(x + V) \cap f^{-1}(0) = \emptyset$. Then, $f(V)$ is bounded. Indeed, if $f(V)$ is unbounded, then there is some $z \in V$ such that $|f(z)| > |f(x)|$. In this case, $f(tz) = tf(z) = -f(x)$ for some $t \in [-1, 1]$, which implies that $-f(x) \in f(V)$. It follows that $0 \in f(x) + f(V) = f(x + V)$, which contradicts the fact that $(x + V) \cap f^{-1}(0) = \emptyset$.

(d) $\Rightarrow$ (a): For each $\varepsilon > 0$, we have $n \in \mathbb{N}$ such that $f(V) \subset (-n\varepsilon, n\varepsilon)$. Then, $n^{-1}V$ is a neighborhood of $\mathbf{0}$ in E and $f(n^{-1}V) \subset (-\varepsilon, \varepsilon)$. Therefore, f is continuous at $\mathbf{0} \in E$. Since f is linear, it follows that f is continuous at every point of E. □

Proposition 3.4.7 *Let E be a topological linear space and $A, B \subset E$.*

(1) *If B is open in E then $A + B$ is open in E.*
(2) *If A is compact and B is closed in E then $A + B$ is also closed in E.*

Sketch of Proof (1): Note that $A + B = \bigcup_{x \in A}(x + B)$.

(2): To show that $E \setminus (A + B)$ is open in E, let $z \in E \setminus (A + B)$. For each $x \in A$, because $z - x \in E \setminus B$, we have open neighborhoods U_x, V_x of x, z in E such that $V_x - U_x \subset E \setminus B$. Since A is compact, $A \subset \bigcup_{i=1}^{n} U_{x_i}$ for some $x_1, \ldots, x_n \in A$. Then, $V = \bigcap_{i=1}^{n} V_{x_i}$ is an open neighborhood of z in E. We can show that $V \cap (A + B) = \emptyset$, i.e., $V \subset E \setminus (A + B)$.

Remark 3.4.1 In (2) above, we cannot assert that $A + B$ is closed in E even if both A and B are closed and convex in E. For example, $A = \mathbb{R} \times \{0\}$ and $B = \{(x, y) \in \mathbb{R}^2 \mid x > 0,\ y \geqslant x^{-1}\}$ are closed convex sets in $\mathbb{R}^2$, but $A + B = \mathbb{R} \times (0, \infty)$ is not closed in $\mathbb{R}^2$.

Concerning the quotient space of a topological linear space, we have the following:

Proposition 3.4.8 *Let F be a closed linear subspace of a topological linear space E. Then, the quotient linear space E/F with the quotient topology is also a topological linear space, and the quotient map $q : E \to E/F$ (i.e., $q(x) = x + F \in E/F$) is open, where if $\mathcal{U}$ is a neighborhood basis at $\mathbf{0}$ in E, then $q(\mathcal{U}) = \{q(U) \mid U \in \mathcal{U}\}$ is a neighborhood basis $\mathbf{0}$ in E/F.*

Sketch of Proof Apply Proposition 3.4.7(1) to show that the quotient map $q : E \to E/F$ is open. Then, in the diagrams below, $q \times q$ and $q \times \mathrm{id}_{\mathbb{R}}$ are open, so they are quotient maps:

$$\begin{array}{ccc} E \times E & \longrightarrow & E \\ {\scriptstyle q\times q}\downarrow & & \downarrow{\scriptstyle q} \\ E/F \times E/F & \longrightarrow & E/F, \end{array} \qquad \begin{array}{ccc} E \times \mathbb{R} & \longrightarrow & E \\ {\scriptstyle q\times \mathrm{id}_{\mathbb{R}}}\downarrow & & \downarrow{\scriptstyle q} \\ E/F \times \mathbb{R} & \longrightarrow & E/F. \end{array}$$

Accordingly, the continuity of addition and scalar multiplication are clear. Note that E/F is Hausdorff if and only if F is closed in E.

For convex sets in a topological linear space, we have the following:

Proposition 3.4.9 *For each non-empty convex set C in a topological linear space E, the following holds:*

(i) $\mathrm{cl}\, C$ *is convex and* $\mathrm{rcl}\, C \subset \mathrm{cl}\, C$, *hence* $\mathrm{rcl}\, C = C$ *if C is closed in E.*
(ii) $\mathrm{int}_F C = \emptyset$ *for any flat F with* $\mathrm{fl}\, C \subsetneqq F$.
(iii) $\mathrm{int}_{\mathrm{fl}\, C} C \neq \emptyset$ *implies* $\mathrm{int}_{\mathrm{fl}\, C} C = \mathrm{core}_{\mathrm{fl}\, C} C = \mathrm{rint}\, C$.
(iv) $\mathrm{int}_F C$ *is convex for any flat F containing C. In particular,* $\mathrm{int}\, C$ *is convex.*

Proof By the definition and the continuity of algebraic operations, we can easily obtain (i). For (ii), observe that $\mathrm{int}_F C \subset \mathrm{core}_F C$. If $\mathrm{int}_F C \neq \emptyset$ then $\mathrm{fl}\, C = F$ by the Fact stated just before Proposition 3.3.2.

(iii): Due to Proposition 3.3.2, $\mathrm{core}_{\mathrm{fl}\, C} C = \mathrm{rint}\, C$. Since $\mathrm{int}_{\mathrm{fl}\, C} C \subset \mathrm{core}_{\mathrm{fl}\, C} C$, we show $\mathrm{rint}\, C \subset \mathrm{int}_{\mathrm{fl}\, C} C$. Without loss of generality, we may assume that $\mathbf{0} \in \mathrm{int}_{\mathrm{fl}\, C} C$. Then, for each $x \in \mathrm{rint}\, C$, we can find $0 < s < 1$ such that $x \in sC$. Since $(1 - s)C$ is a neighborhood of $\mathbf{0} = x - x$ in $\mathrm{fl}\, C$, we have a neighborhood U of x in $\mathrm{fl}\, C$ such that $U - x \subset (1 - s)C$. Then, it follows that $U \subset (1 - s)C + sC = C$. Hence, $x \in \mathrm{int}_{\mathrm{fl}\, C} C$.

(iv): If $\mathrm{int}_F C \neq \emptyset$, then $F = \mathrm{fl}\, C$ by (ii) and $\mathrm{int}_{\mathrm{fl}\, C} C = \mathrm{rint}\, C$ by (iii), which is convex by Proposition 3.3.2. □

Remark 3.4.2 In the above, we cannot assert any one of $\operatorname{cl} C = \operatorname{rcl} C$, $\operatorname{int}_{\operatorname{fl} C} C = \operatorname{core}_{\operatorname{fl} C} C$, or $\operatorname{int}_{\operatorname{fl} C} C \neq \emptyset$. For example, $[-1, 1]_f^{\mathbb{N}}$ is a convex set in $\mathbb{R}^{\mathbb{N}}$ such that $\operatorname{rcl}[-1, 1]_f^{\mathbb{N}} = [-1, 1]_f^{\mathbb{N}}$ but $\operatorname{cl}[-1, 1]_f^{\mathbb{N}} = [-1, 1]^{\mathbb{N}}$. Note that $\operatorname{fl}[-1, 1]_f^{\mathbb{N}} = \mathbb{R}_f^{\mathbb{N}}$. Regarding $[-1, 1]_f^{\mathbb{N}}$ as a convex set in $\mathbb{R}_f^{\mathbb{N}}$, we have

$$\operatorname{int}_{\mathbb{R}_f^{\mathbb{N}}}[-1, 1]_f^{\mathbb{N}} = \emptyset \quad \text{but} \quad \operatorname{core}_{\mathbb{R}_f^{\mathbb{N}}}[-1, 1]_f^{\mathbb{N}} = \operatorname{rint}[-1, 1]_f^{\mathbb{N}} = (-1, 1)_f^{\mathbb{N}}.$$

By Proposition 3.4.9 (i), if A is a subset of a topological linear space E, then $\operatorname{cl}\langle A\rangle$ is the smallest closed convex set containing A, which is called the **closed convex hull** of A.

Remark 3.4.3 In general, $\langle A\rangle$ is not closed in E even if A is compact. For example, let $A = \{a_n \mid n \in \omega\} \subset \ell_1$, where $a_0(i) = 2^{-i}$ for every $i \in \mathbb{N}$ and, for each $n \in \mathbb{N}$, $a_n(i) = 2^{-i}$ if $i \leqslant n$ and $a_n(i) = 0$ if $i > n$. Then, A is compact and $\langle A\rangle = \bigcup_{n\in\mathbb{N}}\langle a_0, a_1, \ldots, a_n\rangle$. For each $n \in \mathbb{N}$, let

$$x_n = 2^{-n}a_0 + 2^{-1}a_1 + \cdots + 2^{-n}a_n \in \langle a_0, a_1, \ldots, a_n\rangle.$$

Then, $x_n(i) = 2^{-2i+1}$ if $i \leqslant n$ and $x_n(i) = 2^{-n-i}$ if $i > n$. Hence, $(x_n)_{n\in\mathbb{N}}$ converges to $x_0 \in \ell_1$, where $x_0(i) = 2^{-2i+1}$ for each $i \in \mathbb{N}$. However, $x_0 \notin \langle A\rangle$. Otherwise, $x_0 \in \langle a_0, a_1, \ldots, a_n\rangle$ for some $n \in \mathbb{N}$, where we can write

$$x_0 = \sum_{i=0}^{n} z(i+1)a_i, \ z \in \Delta^n.$$

Then, we have the following:

$$z(1)a_0(n+1) = x_0(n+1) = 2^{-2n-1} = 2^{-n}a_0(n+1) \ \text{ and}$$
$$z(1)a_0(n+2) = x_0(n+2) = 2^{-2n-3} = 2^{-n-1}a_0(n+2),$$

hence $z(1) = 2^{-n}$ and $z(1) = 2^{-n-1}$. This is a contradiction. Therefore, $\langle A\rangle$ is not closed in ℓ_1.

The following is a topological version of the Separation Theorem 3.3.5:

Theorem 3.4.10 (SEPARATION) *Let A and B be non-empty convex sets in a topological linear space E such that* $\operatorname{int} A \neq \emptyset$ *and* $(\operatorname{int} A) \cap B = \emptyset$. *Then, there is a continuous linear functional $f : E \to \mathbb{R}$ such that $f(x) < f(y)$ for each $x \in \operatorname{int} A$ and $y \in B$, and* $\sup f(A) \leqslant \inf f(B)$.

Proof First, $\operatorname{int} A \neq \emptyset$ implies $\operatorname{core} A = \operatorname{int} A \neq \emptyset$ by Proposition 3.4.9 (iii). Then, by the Separation Theorem 3.3.5, we have a linear functional $f : E \to \mathbb{R}$ such that $f(x) < f(y)$ for every $x \in \operatorname{int} A$ and $y \in B$, and $\sup f(A) \leqslant \inf f(B)$. Note that $B - \operatorname{int} A$ is open in E and $f(z) > 0$ for every $z \in B - \operatorname{int} A$. Thus, $f^{-1}(0)$ is not dense in E. Therefore, f is continuous by Proposition 3.4.6. □

A topological linear space E is **locally convex** if $\mathbf{0} \in E$ has a neighborhood basis consisting of convex sets. For each convex neighborhood V of $\mathbf{0}$ in E, $\operatorname{int} V$ is an open convex neighborhood of $\mathbf{0}$ by Proposition 3.4.9 (iv). Hence, every locally convex topological linear space E has a neighborhood basis of $\mathbf{0}$ consisting of open convex sets, equivalently, open convex sets make up an open basis for E. Note that if V is a convex set in E with $\mathbf{0} \in V$, then $V \cap (-V)$ is a circled convex set.

Indeed, $V \cap (-V)$ is convex as an intersection of two convex sets. For each $x \in V \cap (-V)$ and $t \in \mathbf{I}$, $tx = (1-t)\mathbf{0} + tx \in V \cap (-V)$ and $(-t)x = t(-x) \in V \cap (-V)$. Hence, $t(V \cap (-V)) \subset V \cap (-V)$ for every $t \in [-1, 1]$.

Consequently, the following holds:

- *Every locally convex topological linear space has a neighborhood basis of* $\mathbf{0}$ *consisting of circled open convex sets.*

We obtain the following from Proposition 3.4.8:

Proposition 3.4.11 *For each locally convex topological space E and each closed linear subspace $F \subset E$, the quotient linear space E/F is also locally convex.* □

For locally convex topological linear spaces, we have the following separation theorem:

Theorem 3.4.12 (STRONG SEPARATION) *Let A and B be disjoint non-empty closed convex sets in a locally convex topological linear space E. If at least one of A and B is compact, then there is a continuous linear functional $f : E \to \mathbb{R}$ such that* $\sup f(A) < \inf f(B)$.

Proof By Proposition 3.4.7 (2), $B - A$ is closed in E. Since $A \cap B = \emptyset$, it follows that $\mathbf{0} \notin B - A$. Choose an open convex neighborhood U of $\mathbf{0}$ so that $U \cap (B - A) = \emptyset$. By the Separation Theorem 3.4.10, we have a non-trivial continuous linear functional $f : E \to \mathbb{R}$ such that $\sup f(U) \leqslant \inf f(B - A)$. Then, $\sup f(A) + \sup f(U) \leqslant \inf f(B)$, where $\sup f(U) > 0$ by the non-triviality of f. Thus, we have the result. □

As a particular case, we have the following:

Corollary 3.4.13 *Let E be a locally convex topological linear space. For each pair of distinct points $x, y \in E$, there exists a continuous linear functional $f : E \to \mathbb{R}$ such that $f(x) \neq f(y)$.* □

Concerning the continuity of sublinear functionals, we have the following:

Proposition 3.4.14 *Let $p : E \to \mathbb{R}$ be a non-negative sublinear functional of a topological linear space E. Then, p is continuous if and only if $p^{-1}([0, 1))$ is a neighborhood of $\mathbf{0} \in E$.*

Proof The "only if" part follows from $p^{-1}([0,1)) = p^{-1}((-1,1))$. To see the "if" part, let $\varepsilon > 0$. Since $p^{-1}([0,\varepsilon)) = \varepsilon p^{-1}([0,1))$ is a neighborhood of $\mathbf{0} \in E$, each $x \in E$ has the following neighborhood:

$$U = \big(x + p^{-1}([0,\varepsilon))\big) \cap \big(x - p^{-1}([0,\varepsilon))\big).$$

For each $y \in U$, since $p(y-x) < \varepsilon$ and $p(x-y) < \varepsilon$, it follows that

$$p(x) - \varepsilon < p(x) - p(x-y) \leqslant p(y) \leqslant p(y-x) + p(x) < p(x) + \varepsilon,$$

which means that p is continuous at x. □

For each convex set $C \subset E$ with $\mathbf{0} \in \operatorname{int} C$, we have $\operatorname{int} C = \operatorname{core} C = p_C^{-1}([0,1))$ by Propositions 3.3.4 and 3.4.9 (iii). Then, the following is obtained from Proposition 3.4.14.

Corollary 3.4.15 *Let E be a topological linear space. For each convex set $C \subset E$ with $\mathbf{0} \in \operatorname{int} C$, the Minkowski functional $p_C : E \to \mathbb{R}$ is continuous. Moreover,*

$$p_C^{-1}([0,1)) = \operatorname{int} C = \operatorname{rint} C \quad \textit{and} \quad p_C^{-1}(\mathbf{I}) = \operatorname{cl} C = \operatorname{rcl} C,$$

hence $p_C^{-1}(1) = \operatorname{bd} C = \partial C$. □

The boundedness is a metric concept, but it can be extended to subsets of a topological linear space E. A subset $A \subset E$ is **topologically bounded**[14] provided that, for each neighborhood U of $\mathbf{0} \in E$, there exists some $r > 0$ such that $A \subset rU$. The following can be easily shown:

Proposition 3.4.16 (Some Properties of Topologically Bounded Sets) *Let E be a topological linear space and $A, B \subset E$.*

(1) *If A is topologically bounded in E, then so is every subset $B \subset A$.*
(2) *Let F be a linear subspace of E and $A \subset F$. Then, A is topologically bounded in F if and only if A is topologically bounded in E.*
(3) *If A and B are topologically bounded in E and $s, t \in \mathbb{R}$, then $sA + tB$ is also topologically bounded. In particular, if A is topologically bounded, $s \in \mathbb{R}$, and $x \in E$, sA and $A + x$ are topologically bounded.*
(4) *If A is topologically bounded, $A \neq \emptyset$, and $f : \operatorname{fl} A \to E'$ is an affine map of the flat hull $\operatorname{fl} A$ into another topological linear space E', then $f(A)$ is topologically bounded in E'.*
(5) *Every compact set in E is topologically bounded in E.*
(6) *When $E = (E, \|\cdot\|)$ is a normed linear space and $A \neq \emptyset$, A is topologically bounded in E if and only if A is bounded in the metric sense, i.e.,* $\operatorname{diam} A < \infty$.

[14] Usually, we say simply "*bounded*" but here add "*topologically*" in order to distinguish the metric sense. It should be noted that every metrizable space has an admissible bounded metric.

Proof (1) and (2) are trivial.

(3): For each neighborhood U of $\mathbf{0} \in E$, take a circled neighborhood V of $\mathbf{0}$ so that $V + V \subset U$ (Proposition 3.4.1 (ii) and (iii)′, which is equivalent to (iii)) and choose $r > 0$ so that $A, B \subset rV$. Let $r_0 = r \max\{|s|, |t|\} > 0$. Then, $sA + tB \subset r|s|V + r|t|V \subset r_0 U$.

(4): Let $v \in A$. Then, $F = \mathrm{fl}\, A - v$ is a linear subspace. By (2) and (3) above, $A - v$ is topologically bounded in F. Let $g : F \to E'$ be a linear map defined by $g(x) = f(x + v) - f(v)$ for each $x \in F$. Note that $f(x) = g(x - v) + f(v)$ for each $x \in A$. For each neighborhood U of $\mathbf{0}$ in E', there is a neighborhood V of $\mathbf{0}$ in F such that $g(V) \subset U$. Then, we have $r > 0$ such that $A - v \subset rV$, which implies $g(A - v) \subset g(rV) = rg(V) \subset rU$. Hence, $g(A - v)$ is topologically bounded in E'. Again by (2), $f(A) = g(A - v) + f(v)$ is also topologically bounded in E'.

(5): Each neighborhood U of $\mathbf{0} \in E$ contains a circled open neighborhood V of $\mathbf{0}$ ((iii)′ equiv., (iii)). Since $A \subset E = \bigcup_{r>0} rV$ and A is compact, we have $r_1, \ldots, r_n > 0$ such that $A \subset \bigcup_{i=1}^n r_i V$. Let $r_0 = \max\{r_1, \ldots, r_n\} > 0$. Because V is circled, $A \subset \bigcup_{i=1}^n r_i V \subset r_0 V \subset r_0 U$.

(6): If A is topologically bounded, then there is $r > 0$ such that $A \subset r\mathrm{B}(\mathbf{0}, 1) = \mathrm{B}(\mathbf{0}, r)$, which implies $\operatorname{diam} A < 2r$. Conversely, if $\operatorname{diam} A < \infty$, for each neighborhood U of $\mathbf{0} \in E$, choose $\delta > 0$ so that $\overline{\mathrm{B}}(\mathbf{0}, \delta) \subset U$. Take $v \in A$ and let $r = (\|v\| + \operatorname{diam} A)/\delta > 0$. For each $x \in A$, $\|x\| \leqslant \|v\| + \operatorname{diam} A = \delta r$. Hence, $A \subset \overline{\mathrm{B}}(\mathbf{0}, \delta r) = r\overline{\mathrm{B}}(\mathbf{0}, \delta) \subset rU$. □

Applying Minkowski functionals, we can show the following:

Theorem 3.4.17 *Let C and D be topologically bounded closed convex sets in a topological linear space E with $\operatorname{int} C \neq \emptyset$ and $\operatorname{int} D \neq \emptyset$. Then, C and D are homeomorphic to each other by a homeomorphism of E onto itself, hence*

$$(C, \operatorname{bd} C) \approx (D, \operatorname{bd} D) \quad \textit{and} \quad \operatorname{int} C \approx \operatorname{int} D.$$

Proof By virtue of Proposition 3.4.16 (3), without loss of generality, we may assume that $\mathbf{0} \in \operatorname{int} C \cap \operatorname{int} D$. Let p_C and p_D be the Minkowski functionals for C and D, respectively. By the topological boundedness of C and D, it is easy to see that $p_C(x),\ p_D(x) > 0$ for every $x \in E \setminus \{\mathbf{0}\}$. Then, we can define maps $\varphi, \psi : E \to E$ as follows: $\varphi(\mathbf{0}) = \psi(\mathbf{0}) = \mathbf{0}$,

$$\varphi(x) = \frac{p_C(x)}{p_D(x)} x \quad \text{and} \quad \psi(x) = \frac{p_D(x)}{p_C(x)} x \quad \text{for each } x \in E \setminus \{\mathbf{0}\}.$$

It follows from the continuity of p_C and p_D (Corollary 3.4.15) that φ and ψ are continuous at each $x \in E \setminus \{\mathbf{0}\}$.

To verify the continuity of φ at $\mathbf{0} \in E$, let U be a neighborhood of $\mathbf{0} \in E$. Since D is topologically bounded and C is a neighborhood of $\mathbf{0}$, there is an $r > 0$ such that $D \subset rC$. Then, $p_C(x) \leqslant r p_D(x)$ for every $x \in E$. Choose a circled neighborhood V of $\mathbf{0} \in E$ so that $rV \subset U$. Then, $\varphi(V) \subset U$. Indeed, for each $x \in V \setminus \{\mathbf{0}\}$,

$$\varphi(x) = \frac{p_C(x)}{p_D(x)} x \in \frac{p_C(x)}{p_D(x)} V \subset rV \subset U.$$

Similarly, ψ is continuous at $\mathbf{0} \in E$.

For each $x \in E \setminus \{\mathbf{0}\}$, since $\varphi(x) \neq \mathbf{0}$,

$$\psi\varphi(x) = \frac{p_D(\varphi(x))}{p_C(\varphi(x))}\varphi(x) = \frac{\dfrac{p_C(x)}{p_D(x)}p_D(x)}{\dfrac{p_C(x)}{p_D(x)}p_C(x)} \cdot \frac{p_C(x)}{p_D(x)}x = x.$$

Hence, $\psi\varphi = \mathrm{id}$. Similarly, $\varphi\psi = \mathrm{id}$. Therefore, φ is a homeomorphism with $\varphi^{-1} = \psi$. Moreover, observe that $\varphi(C) \subset D$ and $\psi(D) \subset C$, hence $\varphi(C) = D$. Thus, we have the result. □

The norm of a normed linear space E is the Minkowski functional for the unit closed ball $\mathbf{B}_E$ of E. Since $\mathrm{bd}\,\mathbf{B}_E$ is the unit sphere $\mathbf{S}_E$ of E, we have the following:

Corollary 3.4.18 *Let $E = (E, \|\cdot\|)$ be a normed linear space. For every bounded closed convex set $C \subset E$ with $\mathrm{int}\, C \neq \emptyset$, the pair $(C, \mathrm{bd}\, C)$ is homeomorphic to the pair $(\mathbf{B}_E, \mathbf{S}_E)$ of the unit closed ball and the unit sphere of E.* □

It is easy to see that every normed linear space $E = (E, \|\cdot\|)$ is homeomorphic to the unit open ball $\mathrm{B}(\mathbf{0}, 1) = \mathbf{B}_E \setminus \mathbf{S}_E$ of E.

In fact, the following are homeomorphisms (each of them is the inverse of the other):

$$E \ni x \mapsto \frac{1}{1 + \|x\|}x \in \mathrm{B}(\mathbf{0}, 1); \quad \mathrm{B}(\mathbf{0}, 1) \ni y \mapsto \frac{1}{1 - \|y\|}y \in E.$$

By applying the Minkowski functional, this can be extended as follows:

Theorem 3.4.19 *Every non-empty open convex set V in a topological linear space E is homeomorphic to E itself.*

Proof Without loss of generality, it can be assumed that $\mathbf{0} \in \mathrm{int}\, V = V$. Then, we have $V = \mathrm{int}\, V = p_V^{-1}([0, 1))$ by Corollary 3.4.15. Using the Minkowski functional p_V, we can define maps $\varphi : V \to E$ and $\psi : E \to V$ as follows:

$$\varphi(x) = \frac{1}{1 - p_V(x)}x \quad \text{for } x \in V; \quad \psi(y) = \frac{1}{1 + p_V(y)}y \quad \text{for } y \in E.$$

Observe that $\psi\varphi = \mathrm{id}_V$ and $\varphi\psi = \mathrm{id}_E$. This means that φ is a homeomorphism with $\psi = \varphi^{-1}$. □

3.5 Finite Dimensionality

Here, we prove that every finite-dimensional linear space has a unique topology that is compatible with the algebraic operations, and that a topological linear space is finite-dimensional if and only if it is locally compact.

First, we show the following proposition:

Proposition 3.5.1 *Every finite-dimensional flat F in an arbitrary linear space E has a unique (Hausdorff) topology such that the following operation is continuous:*

$$F \times F \times \mathbb{R} \ni (x, y, t) \mapsto (1-t)x + ty \in F.$$

With respect to this topology, every affine bijection $f : \mathbb{R}^n \to F$ is a homeomorphism, where $n = \dim F$. Then, F is affinely homeomorphic to $\mathbb{R}^n$. Moreover, if E is a topological linear space, then F is closed in E.

Proof As mentioned at the beginning of Sect. 3.4, E has a topology that makes E a topological linear space. With respect to the topology of F inherited from this topology, the above operation is continuous.

Note that there exists an affine bijection $f : \mathbb{R}^n \to F$, where $\dim F = n$. We shall show that any affine bijection $f : \mathbb{R}^n \to F$ is a homeomorphism with respect to any other topology of F such that the above operation is continuous, which implies that such a topology is unique and F is affinely homeomorphic to $\mathbb{R}^n$.

Since f is affine, we have

$$f(z) = \left(1 - \sum_{i=1}^{n} z(i)\right) f(\mathbf{0}) + \sum_{i=1}^{n} z(i) f(\mathbf{e}_i) \quad \text{for each } z \in \mathbb{R}^n.$$

Note that the following function is continuous:

$$\mathbb{R}^n \ni z \mapsto \left(1 - \sum_{i=1}^{n} z(i), z(1), \ldots, z(n)\right) \in \mathrm{fl}\, \Delta^n \subset \mathbb{R}^{n+1}.$$

Then, the continuity of f follows from the claim:

Claim Given $v_1, \ldots, v_k \in F$, $k \leqslant n$, the following function is continuous:

$$\varphi_k : \mathrm{fl}\, \Delta^{k-1} \ni z \mapsto \sum_{i=1}^{k} z(i) v_i \in F.$$

Since $\mathrm{fl}\, \Delta^0 = \Delta^0 = \{\mathbf{e}_1\}$ is a singleton, the continuity of φ_1 is obvious. Assuming the continuity of φ_k, we shall show the continuity of φ_{k+1}. Let $\psi : \mathrm{fl}\, \Delta^{k-1} \times \mathbb{R} \to \mathrm{fl}\, \Delta^k$ be the map defined by $\psi(z, t) = ((1-t)z, t)$. Observe that

$$\varphi_{k+1}\psi(z, t) = (1-t) \sum_{i=1}^{k} z(i) v_i + t v_{k+1} = (1-t)\varphi_k(z) + t v_{k+1}.$$

From the property of the topology of F and the continuity of φ_k, it follows that $\varphi_{k+1}\psi$ is continuous. For each $i = 1, \ldots, k+1$, let $p_i = \mathrm{pr}_i | \mathrm{fl}\, \Delta^k : \mathrm{fl}\, \Delta^k \to \mathbb{R}$ be the restriction of the projection onto the i-th factor. Note that the following restriction of ψ is a homeomorphism:

$$\psi|\operatorname{fl}\Delta^{k-1}\times(\mathbb{R}\setminus\{1\}):\operatorname{fl}\Delta^{k-1}\times(\mathbb{R}\setminus\{1\})\to\operatorname{fl}\Delta^{k}\setminus p_{k+1}^{-1}(1).$$

Hence, $\varphi_{k+1}|\operatorname{fl}\Delta^k\setminus p_{k+1}^{-1}(1)$ is continuous. Replacing the $(k+1)$-th coordinates with the i-th coordinates, we can see the continuity of $\varphi_{k+1}|\operatorname{fl}\Delta^k\setminus p_i^{-1}(1)$. Since $\operatorname{fl}\Delta^k=\bigcup_{i=1}^{k+1}(\operatorname{fl}\Delta^k\setminus p_i^{-1}(1))$, it follows that φ_{k+1} is continuous. Thus, the claim can be obtained by induction.

It remains to show the openness of f. On the contrary, assume that f is not open. Then, we have $x\in\mathbb{R}^n$ and $\varepsilon>0$ such that $f(\mathrm{B}(x,\varepsilon))$ is not a neighborhood of $f(x)$ in F. Since $\operatorname{bd}\mathrm{B}(x,\varepsilon)$ is a bounded closed set of $\mathbb{R}^n$, it is compact, hence $f(\operatorname{bd}\mathrm{B}(x,\varepsilon))$ is closed in F. Then, $F\setminus f(\operatorname{bd}\mathrm{B}(x,\varepsilon))$ is a neighborhood of $f(x)$ in F. Using the compactness of $\mathbf{I}$, we can find an open neighborhood U of $f(x)$ in F such that

$$(1-t)f(x)+tU\subset F\setminus f(\operatorname{bd}\mathrm{B}(x,\varepsilon))\ \text{ for every } t\in\mathbf{I}.$$

Then, $U\cap f(\operatorname{bd}\mathrm{B}(x,\varepsilon))=\emptyset$. Since $f(\mathrm{B}(x,\varepsilon))$ is not a neighborhood of $f(x)$, it follows that $U\not\subset f(\mathrm{B}(x,\varepsilon))$, and so we can take a point $y\in U\setminus f(\overline{\mathrm{B}}(x,\varepsilon))$. Now, we define a linear path $g:\mathbf{I}\to\mathbb{R}^n$ by $g(t)=(1-t)x+tf^{-1}(y)$. Since f is affine and $y\in U$, it follows that

$$fg(t)=(1-t)f(x)+ty\in F\setminus f(\operatorname{bd}\mathrm{B}(x,\varepsilon))\ \text{ for every } t\in\mathbf{I}.$$

Since f is a bijection, we have

$$g(\mathbf{I})\subset\mathbb{R}^n\setminus\operatorname{bd}\mathrm{B}(x,\varepsilon)=\mathrm{B}(x,\varepsilon)\cup(\mathbb{R}^n\setminus\overline{\mathrm{B}}(x,\varepsilon)).$$

Then, $g(0)=x\in\mathrm{B}(x,\varepsilon)$ and $g(1)=f^{-1}(y)\in\mathbb{R}^n\setminus\overline{\mathrm{B}}(x,\varepsilon)$, which contradicts the connectedness of $\mathbf{I}$. Thus, f is open.

In the case when E is a topological linear space, to prove that F is closed in E, take a point $x\in E\setminus F$ and consider the flat $F_x=\operatorname{fl}(F\cup\{x\})$. It is easy to construct an affine bijection $f:\mathbb{R}^{n+1}\to F_x$ such that $f(\mathbb{R}^n\times\{0\})=F$. As we saw in the above, f is a homeomorphism, hence F is closed in F_x. Since $F_x\setminus F$ is open in F_x, we have an open set U in E such that $U\cap F_x=F_x\setminus F$. Then, U is a neighborhood of x in E and $U\subset E\setminus F$. Therefore, $E\setminus F$ is open in E, that is, F is closed in E. □

If a linear space E has a topology such that the operation

$$E\times E\times\mathbb{R}\ni(x,y,t)\mapsto(1-t)x+ty\in E$$

is continuous, then scalar multiplication and addition are also continuous with this topology because they can be written as follows:

$$E\times\mathbb{R}\ni(x,t)\mapsto tx=(1-t)\mathbf{0}+tx\in E;$$
$$E\times E\ni(x,y)\mapsto x+y=2\left(\tfrac{1}{2}x+\tfrac{1}{2}y\right)\in E.$$

Then, the following is obtained by Proposition 3.5.1:

Corollary 3.5.2 *Every finite-dimensional linear space E has a unique (Hausdorff) topology compatible with the algebraic operations (addition and scalar multiplication), and then it is linearly homeomorphic to $\mathbb{R}^n$, where $n = \dim E$.* □

Moreover, we have the following:

Corollary 3.5.3 *Let E be a topological linear space and F a finite-dimensional flat in another topological linear space. Then, every affine function $f : F \to E$ is continuous, and if f is injective then f is a closed embedding.*

Proof By Proposition 3.5.1, F can be replaced with $\mathbb{R}^n$, where $n = \dim F$. Then, we can write

$$f(x) = \left(1 - \sum_{i=1}^{n} x(i)\right) f(\mathbf{0}) + \sum_{i=1}^{n} x(i) f(\mathbf{e}_i) \quad \text{for each } x \in \mathbb{R}^n,$$

where $\mathbf{e}_1, \ldots, \mathbf{e}_n$ is the canonical orthonormal basis for $\mathbb{R}^n$. Thus, the continuity of f is obvious. Since $f(\mathbb{R}^n)$ is a finite-dimensional flat in E, $f(\mathbb{R}^n)$ is closed in E by Proposition 3.5.1. If f is injective then $f : \mathbb{R}^n \to f(\mathbb{R}^n)$ is an affine bijection, which is a homeomorphism by Proposition 3.5.1. Hence, f is a closed embedding. □

Combining Proposition 3.2.3 and Corollary 3.5.3, we have

Corollary 3.5.4 *Let E be a topological linear space and C a finite-dimensional convex set in another topological linear space. Then, every affine function $f : C \to E$ is continuous. Moreover, if f is injective, then f is an embedding.* □

For finite-dimensional convex sets in a linear space, we have the following:

Proposition 3.5.5 *Let C be a non-empty finite-dimensional convex set in an arbitrary linear space E. Then,* $\operatorname{rint} C = \operatorname{int}_{\mathrm{fl}\, C} C \neq \emptyset$, $\operatorname{rcl} C = \operatorname{cl}_{\mathrm{fl}\, C} C$, *and* $\partial C = \operatorname{bd}_{\mathrm{fl}\, C} C$ *with respect to the unique topology for* $\mathrm{fl}\, C$ *as in Proposition 3.5.1.*

Proof We may assume that E is a topological linear space. By Proposition 3.4.9 (iii), it suffices to show that $\operatorname{int}_{\mathrm{fl}\, C} C \neq \emptyset$. We have affinely independent $v_0, v_1, \ldots, v_n \in C$ with $\mathrm{fl}\, C = \mathrm{fl}\{v_0, v_1, \ldots, v_n\}$, where $n = \dim C$. We have an affine bijection $f : \mathbb{R}^n \to \mathrm{fl}\, C$ such that $f(\mathbf{0}) = v_0$, $f(\mathbf{e}_1) = v_1$, …, $f(\mathbf{e}_n) = v_n$. Then, f is a homeomorphism by Proposition 3.5.1, hence

$$\operatorname{int}_{\mathrm{fl}\, C} C \supset \operatorname{int}_{\mathrm{fl}\, C} \langle v_0, v_1, \ldots, v_n \rangle = f(\operatorname{int}_{\mathbb{R}^n} \langle \mathbf{0}, \mathbf{e}_1, \ldots, \mathbf{e}_n \rangle) \neq \emptyset.$$

In the above, $f : \mathbb{R}^n \to \mathrm{fl}\, C$ is an affine homeomorphism and $f^{-1}(C)$ is a convex set in $\mathbb{R}^n$. To see $\operatorname{rcl} C = \operatorname{cl}_{\mathrm{fl}\, C} C$, identifying C with $f^{-1}(C)$, we may assume that $C \subset \mathbb{R}^n = \mathrm{fl}\, C$, where $\operatorname{int} C \neq \emptyset$ as seen in the above. Take $v \in \operatorname{int} C$. Then, $\mathbf{0} \in \operatorname{int} C - v = \operatorname{int}(C - v)$, which implies

$$\operatorname{rcl} C - v = \operatorname{rcl}(C - v) = \operatorname{cl}(C - v) = \operatorname{cl} C - v$$

by Corollary 3.4.15. Thus, we have $\operatorname{rcl} C = \operatorname{cl} C$. □

Note that every compact set in a topological linear space is topologically bounded and closed. For an n-dimensional non-empty convex set C in a linear space, the flat hull $\operatorname{fl} C$ is affinely isomorphic to $\mathbb{R}^n$. Combining Propositions 3.5.1 and 3.5.5 with Corollary 3.4.18, we have the following:

Corollary 3.5.6 *Let $C \neq \emptyset$ be an n-dimensional compact convex set in an arbitrary topological linear space E. Then, the pair $(C, \partial C)$ is homeomorphic to the pair $(\mathbf{B}^n, \mathbf{S}^{n-1})$ of the unit closed n-ball and the unit $(n-1)$-sphere.* □

It should be noted that every bounded closed set in Euclidean space $\mathbb{R}^n$ is compact. More generally, we can prove the following:

Proposition 3.5.7 *Let E be an arbitrary topological linear space and $\emptyset \neq A \subset E$ with $\dim \operatorname{fl} A < \infty$. Then, A is compact if and only if A is topologically bounded and closed in E.*

Sketch of Proof The "only if" part is Proposition 3.4.16 (5). The "if" part can be reduced to the case of $\mathbb{R}^n$ by Propositions 3.4.16 (4) and 3.5.1.

The following convex version of Proposition 3.5.1 is non-trivial.

Proposition 3.5.8 *Let $C \neq \emptyset$ be an n-dimensional convex set in an arbitrary linear space E. If (1) C is the convex hull of a finite set*[15] *or (2) $C = \operatorname{rint} C$, then C has a unique (Hausdorff) topology such that the following operation is continuous:*

$$C \times C \times \mathbf{I} \ni (x, y, t) \mapsto (1-t)x + ty \in C.$$

Moreover, in case (1), $\operatorname{rcl} C = C$ and $(C, \partial C) \approx (\mathbf{B}^n, \mathbf{S}^{n-1})$; in case (2), $C \approx \mathbb{R}^n$.

Proof Like Proposition 3.5.1, it suffices to see the uniqueness and the additional statement. To this end, suppose that C has such a topology, but it is unknown whether this is induced from a topology of $\operatorname{fl} C$ or not.

Case (1): Let $C = \langle v_1, \ldots, v_k \rangle$ and define $f : \Delta^{k-1} \to C$ by $f(z) = \sum_{i=1}^{k} z(i) v_i$. In the same way as for the claim in the proof of Proposition 3.5.1, we can see that the continuity of the operation above induces the continuity of f. Since Δ^{k-1} is compact, f is a closed map, hence it is quotient. Thus, the topology of C is unique and C is compact with respect to this topology. Giving any topology on E so that E is a topological linear space, we have $\operatorname{rcl} C = C$ by Proposition 3.4.9(i) and $(C, \partial C) \approx (\mathbf{B}^n, \mathbf{S}^{n-1})$ by Corollary 3.5.6.

Case (2): Let $f : \mathbb{R}^n \to \operatorname{fl} C$ be an affine bijection, where $n = \dim \operatorname{fl} C = \dim C$. Since $D = f^{-1}(C)$ is an n-dimensional convex set in $\mathbb{R}^n$, $D = \operatorname{rint} D = \operatorname{int} D$ is open

[15] In this case, C is called a **cell** or a (**convex**) **linear cell** (cf. Sect. 4.1).

in $\mathbb{R}^n$ by Proposition 3.5.5, hence $D \approx \mathbb{R}^n$ by Proposition 3.4.19. Then, it suffices to show that $f|D : D \to C$ is a homeomorphism. For each $x \in D$, choose $\delta > 0$ so that $x + \delta \mathbf{B}^n = \overline{\mathrm{B}}(x, \delta) \subset D$. Let $v_0 = x - \delta \hat{\Delta}^{n-1}$, where $\hat{\Delta}^{n-1}$ is the barycenter of the standard $(n-1)$-simplex $\Delta^{n-1} = \langle \mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_n \rangle \subset \mathbb{R}^n$. For each $i = 1, \ldots, n$, let $v_i = x + \delta \mathbf{e}_i$. Then, $v_0, v_1, \ldots, v_n$ are affinely independent and

$$x \in \mathrm{int}_{\mathbb{R}^n} \langle v_0, v_1, \ldots, v_n \rangle \subset x + \delta \mathbf{B}^n \subset D,$$

hence $\langle v_0, v_1, \ldots, v_n \rangle$ is a neighborhood of x in D. On the other hand, we have the affine homeomorphism $\varphi : \Delta^n \to \langle v_0, v_1, \ldots, v_n \rangle$ defined by $\varphi(z) = \sum_{i=0}^n z(i+1) v_i$. Since $f\varphi(z) = \sum_{i=0}^n z(i+1) f(v_i)$, the continuity of the operation above implies that of $f\varphi$, hence $f|\langle v_0, v_1, \ldots, v_n \rangle$ is continuous at x. Then, it follows that $f|D$ is continuous at x.

Since D is open in $\mathbb{R}^n$, we can apply the same argument as in the proof of Proposition 3.5.1 to prove that $f|D : D \to C$ is open. Consequently, $f|D : D \to C$ is a homeomorphism. □

Remark 3.5.1 For an arbitrary finite-dimensional convex set C, Proposition 3.5.8 does not hold in general. For example, let

$$C = \{\mathbf{0}\} \cup \{(x, y) \in (0, 1]^2 \mid x \geqslant y\} \subset \mathbb{R}^2.$$

Then, C is a convex set that has a finer topology than the usual one such that the operation in Proposition 3.5.8 is continuous. Such a topology is generated by open sets in the usual topology and the following sets:

$$D_r = \{\mathbf{0}\} \cup (\mathrm{B}((0, r), r) \cap C), \ r > 0.$$

Note that this topology induces the same relative topology on $C \setminus \{\mathbf{0}\}$ as the usual topology does. Since $D_{\varepsilon/\sqrt{2}} \subset \mathrm{B}(\mathbf{0}, \varepsilon)$ for each $\varepsilon > 0$, $\{D_r \mid r > 0\}$ is a neighborhood basis at $\mathbf{0} \in C$ with respect to this topology.

We shall show the continuity of the following operation:

$$C \times C \times \mathbf{I} \ni (p, q, t) \mapsto (1-t)p + tq \in C.$$

When $(1-t)p + tq \neq \mathbf{0}$, the continuity at $(p, q, t) \in C \times C \times \mathbf{I}$ follows from the continuity with respect to the usual topology. The continuity at $(\mathbf{0}, \mathbf{0}, t)$ follows from the convexity of D_r, $r > 0$.

To see the continuity at $(\mathbf{0}, q, 0)$ $(q \neq \mathbf{0})$, let $q = (x, y)$, where $0 < y \leqslant x \leqslant 1$. Choose $s > 0$ so that $q \in D_s$ (i.e., $s > (x^2 + y^2)/2y$). For each $0 < r < \min\{1, s\}$, let $0 \leqslant t \leqslant r/2s$, $p' \in D_{r/2}$, and $q' \in D_s$ (cf. Fig. 3.7). Observe that

$$\frac{1-t}{r/s - t}(r/s)p' \in \frac{1-t}{r/s - t}(r/s)D_{r/2} \subset \frac{1}{r/2s}(r/s)D_{r/2} = D_r$$

and $(r/s)q' \in (r/s)D_s \subset D_r$. Since D_r is convex, it follows that

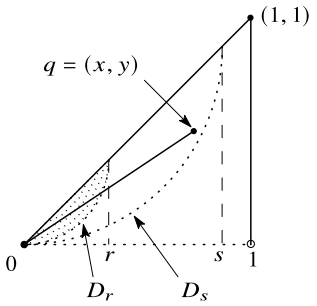


Fig. 3.7 The continuity of the operation at $(\mathbf{0}, q, 0) \in C \times C \times \mathbf{I}$

$$(1-t)p' + tq' = \left(1 - \frac{t}{r/s}\right)\frac{1-t}{r/s - t}(r/s)p' + \frac{t}{r/s}(r/s)q' \in D_r.$$

Thus, the operation is continuous at $(\mathbf{0}, q, 0)$. The continuity at $(p, \mathbf{0}, 1)$ $(p \neq \mathbf{0})$ is the same.

A subset A of a topological linear space E is **totally bounded** provided, for each neighborhood U of $\mathbf{0} \in E$, there exists some finite set $M \subset E$ such that $A \subset M + U$. In this definition, M can be taken as a subset of A.

Indeed, for each neighborhood U of $\mathbf{0} \in E$, we have a circled neighborhood V such that $V + V \subset U$. Then, $A \subset M + V$ for some finite set $M \subset E$, where it can be assumed that $(x + V) \cap A \neq \emptyset$ for every $x \in M$. For each $x \in M$, choose $a_x \in A$ so that $a_x \in x + V$. Then, $x \in a_x - V = a_x + V$. It follows that $A \subset \bigcup_{x \in M}(x + V) \subset \bigcup_{x \in M}(a_x + V + V) \subset \bigcup_{x \in M}(a_x + U)$.

If $A \subset E$ is totally bounded and $B \subset A$, then B is also totally bounded. It is easy to see that every compact subset of E is totally bounded and every totally bounded subset of E is topologically bounded. In other words, we have

$$\boxed{\text{compact}} \Rightarrow \boxed{\text{totally bounded}} \Rightarrow \boxed{\text{topologically bounded}}$$

For topological linear spaces, the finite dimensionality can be simply characterized as follows:

Theorem 3.5.9 *Let E be a topological linear space. The following are equivalent:*

(a) *E is finite-dimensional;*
(b) *E is locally compact;*
(c) *$\mathbf{0} \in E$ has a totally bounded neighborhood in E.*

Proof Since each n-dimensional topological linear space is linearly homeomorphic to $\mathbb{R}^n$ (Corollary 3.5.2), we have (a) $\Rightarrow$ (b). Since every compact subset of E is totally bounded, the implication (b) $\Rightarrow$ (c) follows.

(c) $\Rightarrow$ (a): Let U be a totally bounded neighborhood of $\mathbf{0} \in E$. By Proposition 3.4.1, we have a circled neighborhood V of $\mathbf{0}$ such that $V + V \subset U$. Then, V is also totally bounded. First, we show the following:

Claim For each closed linear subspace $F \subsetneqq E$, there is some $x \in U$ such that $(x + V) \cap F = \emptyset$.

Contrary to the claim, suppose that $(x + V) \cap F \neq \emptyset$ for every $x \in U$. Since $V = -V$, it follows that $U \subset F + V$, so we have $V + V \subset F + V$. If $(n-1)V \subset F + V$, then

$$nV \subset (n-1)V + V \subset F + V + V \subset F + F + V = F + V.$$

By induction, we have $nV \subset F + V$ for every $n \in \mathbb{N}$, which implies that $V \subset \bigcap_{n\in\mathbb{N}}(F + n^{-1}V)$.

Take $z \in E \setminus F$. Since F is closed in E, we have a circled neighborhood W of $\mathbf{0} \in E$ such that $W \subset V$ and $(z + W) \cap F = \emptyset$. The total boundedness of V implies the topological boundedness, hence $V \subset mW$ for some $m \in \mathbb{N}$. On the other hand, $k^{-1}z \in V$ for some $k \in \mathbb{N}$. Since $k^{-1}z \in V \subset F + (km)^{-1}V$, it follows that $z \in F + m^{-1}V \subset F + W$. This contradicts the fact that $(z + W) \cap F = \emptyset$. Thus, we have the claim.

Now, assume that E is infinite-dimensional. Let $v_1 \in U \setminus \{\mathbf{0}\}$ and $F_1 = \mathbb{R}v_1$. Then, F_1 is closed in E (Proposition 3.5.1) and $F_1 \neq E$. Applying the claim above, we have $v_2 \in U$ such that $(v_2 + V) \cap F_1 = \emptyset$. Note that $v_2 \notin v_1 + V$. Let $F_2 = \mathbb{R}v_1 + \mathbb{R}v_2$. Since F_2 is closed in E (Proposition 3.5.1) and $F_2 \neq E$, we can again apply the claim to find $v_3 \in U$ such that $(v_3 + V) \cap F_2 = \emptyset$. Then, note that $v_3 \notin v_i + V$ for $i = 1, 2$. By induction, we have $v_n \in U$, $n \in \mathbb{N}$, such that $v_n \notin v_i + V$ for $i < n$. Then, $\{v_n \mid n \in \mathbb{N}\}$ is not totally bounded. This is a contradiction. Consequently, E is finite-dimensional. □

As a direct consequence of Theorem 3.5.9, we have the following:

Corollary 3.5.10 *Every infinite-dimensional topological linear space is not locally compact.* □

3.6 Metrizability and Normability

In this section, we prove metrization and normability theorems for topological linear spaces. The metrizability of a topological linear space has the following very simple characterization:

Theorem 3.6.1 *A topological linear space E is metrizable if and only if $\mathbf{0} \in E$ has a countable neighborhood basis.*

In a more general setting, we will prove a stronger result. A metric d on a group G is said to be **left** (resp. **right**) **invariant** if $d(x, y) = d(zx, zy)$ (resp. $d(x, y) = d(xz, yz)$) for each $x, y, z \in G$; equivalently, $d(x, y) = d(x^{-1}y, \mathbf{1})$ (resp. $d(x, y) = d(xy^{-1}, \mathbf{1})$) for each $x, y \in G$. When both of two metrics d and d' on a group G are left (or right) invariant, they are uniformly equivalent to each other if and only if they induce the same topology. It is said that d is **invariant** if it is left and right invariant. Every invariant metric d on a group G induces the topology on G that makes G a topological group. In fact,

$$d(x, y) = d(x^{-1}xy^{-1}, x^{-1}yy^{-1}) = d(y^{-1}, x^{-1}) = d(x^{-1}, y^{-1}) \text{ and}$$
$$d(xy, x'y') \leqslant d(xy, x'y) + d(x'y, x'y') = d(x, x') + d(y, y').$$

It is easy to verify that a left (or right) invariant metric d on a group G is invariant if $d(x, y) = d(x^{-1}, y^{-1})$ for each $x, y \in G$. Theorem 3.6.1 comes from the following:

Theorem 3.6.2 *For a topological group G, the following are equivalent:*

(a) *G is metrizable;*
(b) *the unit $\mathbf{1} \in G$ has a countable neighborhood basis;*
(c) *G has an admissible bounded left invariant (right invariant) metric.*

Proof Since the implications (a) $\Rightarrow$ (b) and (c) $\Rightarrow$ (a) are obvious, it suffices to show the implication (b) $\Rightarrow$ (c).

(b) $\Rightarrow$ (c):[16] We will construct a bounded left invariant metric $\rho \in \mathrm{Metr}^B(G)$. Then, a bounded right invariant metric $\rho' \in \mathrm{Metr}^B(G)$ can be defined by $\rho'(x, y) = \rho(x^{-1}, y^{-1})$. By (b), we can find an open neighborhood basis $\{V_n \mid n \in \mathbb{N}\}$ at $\mathbf{1} \in G$ such that $V_n^{-1} = V_n$ and $V_{n+1}V_{n+1}V_{n+1} \subset V_n$ for each $n \in \mathbb{N}$. [17] Let $V_0 = G$, and define

$$p(x) = \inf\{2^{-i} \mid x \in V_i\} \in \mathbf{I} \text{ for each } x \in G.$$

Since $V_n = V_n^{-1}$ for each $n \in \mathbb{N}$, it follows that $p(x) = p(x^{-1})$ for every $x \in G$. Note that $\bigcap_{n\in\omega} V_n = \{\mathbf{1}\}$.[18] Hence, for every $x \in G$,

$$p(x) = 0 \Leftrightarrow x = \mathbf{1}.$$

By induction on n, we will prove the following:

---

[16] The idea of the proof is the same as that of Theorem 2.4.1 (b) $\Rightarrow$ (a).

[17] Note that $\{xV_n \mid n \in \mathbb{N}\}$ is an open neighborhood basis at $x \in G$. For each $x, y \in G$ and $n \in \mathbb{N}$, $xV_{n+1} \cap yV_{n+1} \neq \emptyset$ implies $yV_{n+1} \subset xV_n$. Indeed, $xu = yv$ for some $u, v \in V_{n+1}$, hence $yV_{n+1} = xuv^{-1}V_{n+1} \subset xV_n$. Thus, the metrizability of G can be obtained by the Frink Metrization Theorem 2.4.1. On the other hand, $\mathcal{V}_n = \{xV_n \mid x \in G\} \in \mathrm{cov}(G)$ and $\mathrm{st}\,\mathcal{V}_{n+1} \prec \mathcal{V}_n$. Indeed, $\mathrm{st}(xV_{n+1}, \mathcal{V}_{n+1}) \subset xV_n$. Thus, the metrizability of G can also be obtained by Corollary 2.4.3.

[18] It is assumed that G is Hausdorff.

$$(*)\quad p(x_0^{-1}x_n) \leqslant 2\sum_{i=1}^{n} p(x_{i-1}^{-1}x_i) \text{ for each } x_0, x_1, \ldots, x_n \in G.^{19}$$

The case $n = 1$ is obvious. Assume $(*)$ for $m < n$. If $\sum_{i=1}^{n} p(x_{i-1}^{-1}x_i) = 0$ or $\sum_{i=1}^{n} p(x_{i-1}^{-1}x_i) \geqslant \frac{1}{2}$, it is trivial. When $2^{-k-1} \leqslant \sum_{i=1}^{n} p(x_{i-1}^{-1}x_i) < 2^{-k}$ for some $k \in \mathbb{N}$, choose $1 \leqslant m \leqslant n$ so that

$$\sum_{i=1}^{m-1} p(x_{i-1}^{-1}x_i) < 2^{-k-1} \quad \text{and} \quad \sum_{i=m+1}^{n} p(x_{i-1}^{-1}x_i) < 2^{-k-1}.$$

Note that $p(x_{m-1}^{-1}x_m) < 2^{-k}$. By the inductive assumption, $p(x_0^{-1}x_{m-1}) < 2^{-k}$ and $p(x_m^{-1}x_n) < 2^{-k}$. Then, $x_0^{-1}x_{m-1},\ x_{m-1}^{-1}x_m,\ x_m^{-1}x_n \in V_{k+1}$. Since $V_{k+1}V_{k+1}V_{k+1} \subset V_k$, it follows that $x_0^{-1}x_n \in V_k$, hence

$$p(x_0^{-1}x_n) \leqslant 2^{-k} \leqslant 2\sum_{i=1}^{n} p(x_{i-1}^{-1}x_i).$$

Now, we can define a metric ρ on G as follows:

$$\rho(x, y) = \inf\Big\{ \sum_{i=1}^{n} p(x_{i-1}^{-1}x_i) \;\Big|\; n \in \mathbb{N},\ x_i \in G,\ x_0 = x,\ x_n = y \Big\}.$$

By the definition, ρ is left invariant. Moreover, ρ is bounded because $\rho(x, y) \leqslant p(x^{-1}y) \leqslant 1$. Then, $x^{-1}y \in V_n$ implies $\rho(x, y) \leqslant p(x^{-1}y) \leqslant 2^{-n} < 2^{-n+1}$, which means $xV_n \subset \mathrm{B}_\rho(x, 2^{-n+1})$. On the other hand, if $\rho(x, y) < 2^{-n}$ then $p(x^{-1}y) \leqslant 2\rho(x, y) < 2^{-n+1}$ by $(*)$, which implies $x^{-1}y \in V_n$. Thus, $\mathrm{B}_\rho(x, 2^{-n}) \subset xV_n$. Therefore, ρ is admissible. □

In the above proof, a right invariant metric $\rho \in \mathrm{Metr}^B(G)$ can be directly defined as follows:

$$\rho(x, y) = \inf\Big\{ \sum_{i=1}^{n} p(x_{i-1}x_i^{-1}) \;\Big|\; n \in \mathbb{N},\ x_i \in G,\ x_0 = x,\ x_n = y \Big\}.$$

Every metrizable topological linear space E has an admissible (bounded) metric ρ that is not only invariant but also satisfies the following:

$$(\sharp)\quad |t| \leqslant 1 \ \Rightarrow\ \rho(tx, \mathbf{0}) \leqslant \rho(x, \mathbf{0}).$$

To verify this, let us recall how to define the metric ρ in the above proof. Taking a neighborhood basis $\{V_n \mid n \in \mathbb{N}\}$ at $\mathbf{0} \in E$ so that $V_n = -V_n$ and $V_{n+1} + V_{n+1} + V_{n+1} \subset V_n$ for each $n \in \mathbb{N}$, we define the admissible invariant metric ρ as follows:

[19] For each $x, y \in G$, let $\delta(x, y) = p(x^{-1}y)$. Then, this inequality is simply the one given in the Sketch of the direct proof for Corollary 2.4.3.

$$\rho(x, y) = \inf\Big\{ \sum_{i=1}^{n} p(x_i - x_{i-1}) \;\Big|\; n \in \mathbb{N},\ x_i \in E,\ x_0 = x,\ x_n = y \Big\},$$

where $p(x) = \inf\{2^{-i} \mid x \in V_i\}$. Since E is a topological linear space, the condition that $V_n = -V_n$ can be replaced by a stronger condition that V_n is circled, i.e., $tV_n \subset V_n$ for $t \in [-1, 1]$. Then, $p(tx) \leqslant p(x)$ for each $x \in E$ and $t \in [-1, 1]$, which implies that $\rho(tx, \mathbf{0}) \leqslant \rho(x, \mathbf{0})$ for each $x \in E$ and $t \in [-1, 1]$.

In general, any invariant metric d on a linear space E satisfies the following:

($\natural$) $\forall x \in X, \forall n \in \mathbb{Z}, d(nx, \mathbf{0}) \leqslant |n| d(x, \mathbf{0})$.

Indeed, ($\natural$) is obvious if $n = 0, 1$. For $n \in \mathbb{N}$, observe that

$$d((n+1)x, \mathbf{0}) \leqslant d((n+1)x, x) + d(x, \mathbf{0}) = d(nx, \mathbf{0}) + d(x, \mathbf{0}).$$

Then, by induction, ($\natural$) can be shown for $n \in \omega$. In addition, $d(-nx, \mathbf{0}) = nd(-x, \mathbf{0}) = nd(x, \mathbf{0})$ for each $x \in E$ and $n \in \mathbb{N}$. Thus, ($\natural$) holds for every $n \in \mathbb{Z}$.

By this ($\natural$), for an invariant metric d on E, the above ($\sharp$) implies the following stronger condition:

($\sharp$)$'$ $\forall x \in X, \forall t \geqslant 0, d(tx, \mathbf{0}) \leqslant \lceil t \rceil d(x, \mathbf{0}) \quad \big(d(-tx, \mathbf{0}) \leqslant \lceil t \rceil d(x, \mathbf{0})\big)$,

where $\lceil t \rceil$ denotes the least integer $\geqslant t$.[20]

Write $t = n + s, n \in \omega, 0 < s < 1$, where $n + 1 = \lceil t \rceil$. Then, it follows that

$$d(tx, \mathbf{0}) \leqslant d((n+s)x, sx) + d(sx, \mathbf{0}) \leqslant d(nx, \mathbf{0}) + d(x, \mathbf{0}) \leqslant (n+1) d(x, \mathbf{0}).$$

Let d be an invariant metric on a linear space E. Addition on a linear space E is continuous with respect to d.

Indeed, for each $x, x', y, y' \in E$,

$$d(x + y, x' + y') \leqslant d(x + y, x' + y) + d(x' + y, x' + y') = d(x, x') + d(y, y').$$

On the other hand, scalar multiplication on E is continuous with respect to d if and only if d satisfies the following three conditions:

(i) $t_n \to 0 \ \Rightarrow\ \forall x \in E,\ d(t_n x, \mathbf{0}) \to 0$;
(ii) $d(x_n, \mathbf{0}) \to 0 \ \Rightarrow\ \forall t \in \mathbb{R},\ d(tx_n, \mathbf{0}) \to 0$;
(iii) $t_n \to 0,\ d(x_n, \mathbf{0}) \to 0 \ \Rightarrow\ d(t_n x_n, \mathbf{0}) \to 0$.

Indeed, the "only if" part is trivial. To show the "if" part, observe that

$$t_n x_n - tx = (t_n - t)(x_n - x) + t(x_n - x) + (t_n - t)x.$$

[20] $\lfloor t \rfloor$ denotes the greatest integer $\leqslant t$. Then, $\lceil t \rceil - 1 \leqslant \lfloor t \rfloor \leqslant \lceil t \rceil \leqslant \lfloor t \rfloor + 1$ for every $t \in \mathbb{R}$, where $\lfloor t \rfloor = \lceil t \rceil$ if and only if $t \in \mathbb{Z}$.

Since d is invariant, it follows that

$$\begin{aligned} d(t_n x_n, tx) &= d((t_n - t)(x_n - x) + t(x_n - x) + (t_n - t)x, \mathbf{0}) \\ &\leqslant d((t_n - t)(x_n - x), \mathbf{0}) + d(t(x_n - x), \mathbf{0}) + d((t_n - t)x, \mathbf{0}). \end{aligned}$$

Thus, the above three conditions imply the continuity of scalar multiplication on E with respect to d.

It should be remarked that ($\sharp$) and ($\sharp$)$'$ imply (iii) and (ii), respectively.

An invariant metric d on E satisfying these three conditions is called a **linear metric**. In particular, an invariant metric d on E satisfying ($\sharp$) and the above (i) is a linear metric. A linear space with a linear metric is called a **metric linear space**. Evidently, every metric linear space is a metrizable topological linear space. Conversely, we have the following fact:

Fact *Every admissible invariant metric for a metrizable topological linear space is a linear metric.*

For subsets of a metric linear space, the total boundedness coincides with that in the metric sense. On the other hand, the topological boundedness does not coincide with the metric boundedness. In fact, according to Theorem 3.6.2, every metrizable topological linear space E has an admissible bounded invariant metric. Moreover, given any admissible invariant metric d for E, the following are admissible bounded invariant metrics:

$$\min\big\{1,\ d(x, y)\big\},\ \frac{d(x, y)}{1 + d(x, y)}.$$

Let d be a linear metric on E with condition ($\sharp$) (i.e., an invariant metric satisfying ($\sharp$) and (i)). Then, the functional $E \ni x \mapsto d(x, \mathbf{0}) \in \mathbb{R}$ is called an ***F*-norm**. In other words, a functional $\|\cdot\| : E \to \mathbb{R}$ on a linear space E is called an ***F*-norm** if it satisfies the following conditions:

(F_1) $\|x\| \geqslant 0$ for every $x \in E$;
(F_2) $\|x\| = 0 \ \Rightarrow\ x = \mathbf{0}$;
(F_3) $|t| \leqslant 1 \ \Rightarrow\ \|tx\| \leqslant \|x\|$ for every $x \in E$;
(F_4) $\|x + y\| \leqslant \|x\| + \|y\|$ for every $x, y \in E$;
(F_5) $t_n \to 0 \ \Rightarrow\ \|t_n x\| \to 0$ for every $x \in E$.

Here, (F_3) and (F_5) correspond to ($\sharp$) and (i), respectively.[21] Since $\|\mathbf{0}\| = 0$ by (F_5), the converse of (F_2) is true, i.e., $\|x\| = 0 \Leftrightarrow x = \mathbf{0}$. Moreover, ($F_3$) implies that $\|-x\| = \|x\|$ for every $x \in E$.

[21] In Köthe's book, an F-norm is called an (F)-norm, where the following condition is added in the definition:

(F_6) $\|x_n\| \to 0 \ \Rightarrow\ \|tx_n\| \to 0$ for every $t \in \mathbb{R}$.

This (F_6) corresponds to (ii), which comes from (F_1), (F_3), and (F_4). For this reason, we do not include (F_6) in the definition. In Köthe's book, the naming of conditions (F_5) and (F_6) is reversed.

A linear space E given an F-norm $\|\cdot\|$ is called an ***F*-normed linear space**. Every norm is an F-norm, hence every normed linear space is an F-normed space. An F-norm $\|\cdot\|$ induces an invariant metric $d(x, y) = \|x - y\|$, which is a linear metric because it satisfies (♯) and (i). Then, every F-normed linear space is a metric linear space. An F-norm on a topological linear space E is said to be **admissible** if it induces the topology for E. As we saw above, if E is metrizable, then E has an admissible invariant metric ρ satisfying (♯), which induces the F-norm. Therefore, we have the following:

Theorem 3.6.3 *A topological linear space has an admissible F-norm if and only if it is metrizable.* □

For each metrizable topological linear space, there exists an F-norm with the following partially stronger condition than (F_3):

(F_3^*) $x \neq \mathbf{0}$, $|t| < 1 \Rightarrow \|tx\| < \|x\|$,

which implies that $\|sx\| < \|tx\|$ for each $x \neq \mathbf{0}$ and $0 < s < t$.

Since (F_3^*) does not state the case $t = -1$ of (F_3), it cannot be said that (F_3^*) is stronger than (F_3). But, the case $t = -1$ of (F_3) can be derived from (F_3^*), (F_4), and (F_5) as follows: We may assume that $x \neq \mathbf{0}$. From (F_4) and (F_3^*), it follows that

$$\|-x\| = \|(n^{-1} - 1)x + n^{-1}x\| \leqslant \|(n^{-1} - 1)x\| + \|n^{-1}x\| < \|x\| + \|n^{-1}x\|.$$

Since $n^{-1} \to 0$, we have $\|-x\| \leqslant \|x\|$ by (F_5).

The following proposition guarantees the existence of an F-norm satisfying (F_3^*):

Proposition 3.6.4 *Every (completely) metrizable topological linear space E has an admissible invariant (complete) metric d such that $d(tx, \mathbf{0}) < d(x, \mathbf{0})$ if $x \neq \mathbf{0}$ and $|t| < 1$, which induces an admissible F-norm satisfying (F_3^*). If an admissible invariant metric ρ for E is given, d can be chosen so that $d \geqslant \rho$ (hence, if ρ is complete, then so is d).*[22] *Moreover, if ρ is bounded, d can be chosen to be bounded.*

Proof Given an admissible (bounded) invariant metric ρ for E, we define $d_1(x, y) = \sup_{0<s\leqslant 1} \rho(sx, sy)$. Then, d_1 is an invariant metric on E with $d_1 \geqslant \rho$ (if ρ is bounded then so is d_1). For each $\varepsilon > 0$, since the scalar multiplication $E \times \mathbb{R} \ni (x, s) \mapsto sx \in E$ is continuous at $(\mathbf{0}, s)$ and $\mathbf{I}$ is compact, we can find $\delta > 0$ such that $\rho(x, \mathbf{0}) < \delta$ implies $\rho(sx, \mathbf{0}) < \varepsilon$ for every $s \in \mathbf{I}$, hence $\rho(x, y) < \delta$ implies $d_1(x, y) = \sup_{0<s\leqslant 1} \rho(sx, sy) \leqslant \varepsilon$. Thus, d_1 is uniformly equivalent to ρ. In particular, d_1 is admissible. For $r > 0$, we define an admissible invariant metric d_r for E by $d_r(x, y) = d_1(rx, ry)$ $(= \sup_{0<s\leqslant r} \rho(sx, sy))$. Observe that $d_r(tx, \mathbf{0}) \leqslant d_r(x, \mathbf{0})$ for each $x \in E$ and $t \in \mathbf{I}$.

Now, let $\mathbb{Q} \cap (0, 1] = \{r_n \mid n \in \mathbb{N}\}$, where $r_1 = 1$. We define $d(x, y) = \sum_{n\in\mathbb{N}} 2^{-n+1} d_{r_n}(x, y)$. Then, d is an invariant metric on E and

[22] Since every d-Cauchy sequence is ρ-Cauchy, if ρ is complete, then so is d. However, as will be shown in Theorem 3.6.6, for a completely metrizable topological linear space E, every admissible invariant metric is complete.

$$\rho(x, y) \leqslant d_1(x, y) \leqslant d(x, y) \leqslant 2d_1(x, y),$$

hence d is admissible (if ρ is bounded then so is d). It also follows that $d(tx, \mathbf{0}) \leqslant d(x, \mathbf{0})$ for each $x \in E$ and $t \in \mathbf{I}$. It remains to show that $d(tx, \mathbf{0}) \neq d(x, \mathbf{0})$ for each $x \in E \setminus \{\mathbf{0}\}$ and $0 < t < 1$. Since $\mathbb{Q} \cap (0, 1)$ is dense in $(0, 1)$, it suffices to show that $d(tx, \mathbf{0}) \neq d(x, \mathbf{0})$ for each $x \in E \setminus \{\mathbf{0}\}$ and $t \in \mathbb{Q} \cap (0, 1)$. Assume that there exists some $x \in E \setminus \{\mathbf{0}\}$ and $t \in \mathbb{Q} \cap (0, 1)$ such that $d(tx, \mathbf{0}) = d(x, \mathbf{0})$. Note that $d_r(tx, \mathbf{0}) = d_r(x, \mathbf{0})$ for each $r \in \mathbb{Q} \cap (0, 1)$. Then, it follows that

$$d_t(x, \mathbf{0}) = d_t(tx, \mathbf{0}) = d_{t^2}(x, \mathbf{0}) = d_{t^2}(tx, \mathbf{0}) = d_{t^3}(x, \mathbf{0}) = d_{t^3}(tx, \mathbf{0}) = \cdots,$$

so $d_t(x, \mathbf{0}) = d_{t^{n+1}}(x, \mathbf{0}) = d_t(t^n x, \mathbf{0})$ for every $n \in \mathbb{N}$. Since $\lim_{n\to\infty} t^n = 0$, it follows that $d_t(x, \mathbf{0}) = \lim_{n\to\infty} d_t(t^n x, \mathbf{0}) = 0$, hence $x = \mathbf{0}$, which is a contradiction. □

The topological linear space $\mathbb{R}^{\mathbb{N}} = \boldsymbol{s}$ (the space of sequences) has the following admissible F-norms:

$$\sup_{i\in\mathbb{N}} \min\left\{1/i, |x(i)|\right\}, \ \sum_{i\in\mathbb{N}} \min\left\{2^{-i}, |x(i)|\right\}, \ \sum_{i\in\mathbb{N}} \frac{2^{-i}|x(i)|}{1 + |x(i)|}, \ \ldots.$$

The first two do not satisfy condition (F_3^*), but the third does.

We now consider the completion of metric linear spaces (cf. Corollary 2.3.11).

Proposition 3.6.5 *Let G be a topological group such that the topology is induced by an invariant metric d. The completion $\widetilde{G} = (\widetilde{G}, \tilde{d})$ of (G, d) is a group such that G is its subgroup and $\tilde{d}$ is invariant. Similarly, the completion of a metric (F-normed or normed) linear space E is a metric (F-normed or normed) linear space containing E as a linear subspace.*

Proof We define the algebraic operations on $\widetilde{G}$ as follows: For each $x, y \in \widetilde{G}$, choose sequences $(x_i)_{i\in\mathbb{N}}$ and $(y_i)_{i\in\mathbb{N}}$ in G so as to converge to x and y, respectively. Since d is invariant, $(x_i y_i)_{i\in\mathbb{N}}$ and (x_i^{-1}) are Cauchy sequences in G. Then, define xy and x^{-1} as the limits of $(x_i y_i)_{i\in\mathbb{N}}$ and $(x_i^{-1})_{i\in\mathbb{N}}$, respectively. It is easily verified that these are well-defined. Since $\tilde{d}(x, y) = \lim_{i\to\infty} d(x_i, y_i)$, it is also easy to see that $\tilde{d}$ is invariant, which implies the continuity of the algebraic operations $(x, y) \mapsto xy$ and $x \mapsto x^{-1}$.

For the completion $\widetilde{E}$ of a metric linear space E, we can define not only addition but also scalar multiplication in the same way. To see the continuity of scalar multiplication, let $x \in \widetilde{E}$ and $t \in \mathbb{R}$. Choose a sequence $(x_i)_{i\in\mathbb{N}}$ in E so as to converge to x. For each $\varepsilon > 0$, we can choose $\delta_0 > 0$ (depending on t) so that

$$z \in E, \ d(z, \mathbf{0}) < \delta_0, \ |t - t'| < \delta_0 \Rightarrow d(t'z, \mathbf{0}) < \varepsilon/4.$$

Then, we have $n_0 \in \mathbb{N}$ such that $d(x_n, x_{n_0}) < \delta_0$ for every $n \geqslant n_0$. Choose $\delta_1 > 0$ so that $\delta_1 < \delta_0$ and

$$|s| < \delta_1 \Rightarrow d(sx_{n_0}, \mathbf{0}) < \varepsilon/4.$$

Now, let $x' \in \widetilde{E}$ and $t' \in \mathbb{R}$ such that $\tilde{d}(x, x') < \delta_0$ and $|t - t'| < \delta_1$. Take a sequence $(x_i')_{i\in\mathbb{N}}$ in E so as to converge to x' and choose $n_1 \in \mathbb{N}$ so that $n_1 \geqslant n_0$ and $d(x_n, x_n') < \delta_0$ for every $n \geqslant n_1$. Then, for every $n \geqslant n_1$, it follows that

$$\begin{aligned} d(tx_n, t'x_n') &\leqslant d(tx_n, tx_{n_0}) + d(tx_{n_0}, t'x_{n_0}) + d(t'x_{n_0}, t'x_n) + d(t'x_n, t'x_n') \\ &= d(t(x_n - x_{n_0}), \mathbf{0}) + d((t - t')x_{n_0}, \mathbf{0}) \\ &\qquad + d(t'(x_{n_0} - x_n), \mathbf{0}) + d(t'(x_n - x_n'), \mathbf{0}) \\ &< \varepsilon/4 + \varepsilon/4 + \varepsilon/4 + \varepsilon/4 = \varepsilon. \end{aligned}$$

When E is an F-normed (or normed) linear space, it is easy to see that the F-norm (or norm) for E naturally extends to $\widetilde{E}$. □

Concerning the completeness of admissible invariant metrics, we have the following:

Theorem 3.6.6 *Let G be a completely metrizable topological group. Every admissible invariant metric for G is complete. In particular, a metric linear space is complete if it is absolutely G_δ (i.e., completely metrizable).*

Proof Let d be an admissible invariant metric for G and $\widetilde{G}$ be the completion of (G, d). Note that $\widetilde{G}$ is a topological group by Proposition 3.6.5. It suffices to show that $\widetilde{G} = G$. Since G is completely metrizable, G is a dense G_δ-set in $\widetilde{G}$ (Theorem 2.5.2), hence we can write $\widetilde{G} \setminus G = \bigcup_{n\in\mathbb{N}} F_n$, where each F_n is a nowhere dense closed set in $\widetilde{G}$. Assume $\widetilde{G} \setminus G \neq \emptyset$ and take $x_0 \in \widetilde{G} \setminus G$. Since $x_0 x \in \widetilde{G} \setminus G$ for every $x \in G$, it follows that $G \subset \bigcup_{n\in\mathbb{N}} x_0^{-1} F_n$, where each $x_0^{-1} F_n$ is also a nowhere dense closed set in $\widetilde{G}$. Then, we have

$$\widetilde{G} = \bigcup_{n\in\mathbb{N}} F_n \cup \bigcup_{n\in\mathbb{N}} x_0^{-1} F_n,$$

which is a countable union of nowhere dense closed sets. This contradicts the complete metrizability of $\widetilde{G}$ (the Baire Category Theorem 2.5.1). □

Corollary 3.6.7 *Let G be a metrizable topological group. Every completely metrizable Abelian subgroup H of G is closed in G. Hence, in a metrizable topological linear space, every completely metrizable linear subspace is closed.*

Proof By Theorem 3.6.2, G has an admissible left invariant metric d. Because H is an Abelian subgroup of G, the restriction of d on H is an admissible invariant metric for H, which is complete by Theorem 3.6.6. Hence, H is closed in G. □

It is said that an F-norm (or an F-normed space) is **complete** if the metric induced by the F-norm is complete. It should be noted that every metrizable topological linear space has an admissible F-norm (Theorem 3.6.3) and that every admissible F-norm

for a completely metrizable topological linear space is complete (Theorem 3.6.6). A completely metrizable topological linear space (or a complete F-normed linear space) is called an ***F*-space**. A **Fréchet space** is a locally convex F-space, that is, a completely metrizable locally convex topological linear space. Every Banach space is a Fréchet space, but the converse does not hold. In fact, $\boldsymbol{s} = \mathbb{R}^{\mathbb{N}}$ is a Fréchet space but it is not normable (Proposition 1.2.2).

Concerning the quotient of an F-normed (or normed) linear space, we have the following:

Proposition 3.6.8 *Let $E = (E, \|\cdot\|)$ be an F-normed (or normed) linear space and F a closed linear subspace of E. Then, the quotient space E/F has the admissible F-norm (or norm) $|||\xi||| = \inf_{x\in\xi} \|x\|$, where if $\|\cdot\|$ is complete then so is $|||\cdot|||$. Hence, if E is (completely) metrizable or (completely) normable then so is E/F.*

Proof It is easy to see that $|||\cdot|||$ is an F-norm (or a norm). It should be noted that the closedness of F is necessary for the condition (F_2). Let $q : E \to E/F$ be the natural linear surjection, i.e., $q(x) = x + F$. Then, for each $\varepsilon > 0$,

$$\big\{q(x) \;\big|\; \|x\| < \varepsilon\big\} = \big\{\xi \in E/F \;\big|\; |||\xi||| < \varepsilon\big\},$$

which means that $q : E \to (E/F, |||\cdot|||)$ is open and continuous, so it is a quotient map. Then, $|||\cdot|||$ induces the quotient topology, i.e., $|||\cdot|||$ is admissible for the quotient topology.

We should remark the following fact:

Fact $|||\xi - \xi'||| = \inf\big\{\|x - x'\| \;\big|\; x' \in \xi'\big\}$ *for each* $x \in \xi$.

Indeed, the left side is not greater than the right side by definition. For each $x, y \in \xi$ and $y' \in \xi'$,

$$\|y - y'\| = \|x - (y' + x - y)\| \geqslant \inf\big\{\|x - x'\| \;\big|\; x' \in \xi'\big\}$$

because $y' + x - y \in \xi'$. Thus, the left side is not less than the right side.

We will show that if $\|\cdot\|$ is complete then so is $|||\cdot|||$. To see the completeness of $|||\cdot|||$, it suffices to prove that each Cauchy sequence $(\xi_i)_{i\in\mathbb{N}}$ in E/F contains a convergent subsequence. Then, by replacing $(\xi_i)_{i\in\mathbb{N}}$ with its subsequence, we may assume that $|||\xi_i - \xi_{i+1}||| < 2^{-i}$ for each $i \in \mathbb{N}$. Using the Fact above, we can inductively choose $x_i \in \xi_i$ so that $\|x_i - x_{i+1}\| < 2^{-i}$. Then, $(x_i)_{i\in\mathbb{N}}$ is a Cauchy sequence in E, which converges to some $x \in E$. It follows that $(\xi_i)_{i\in\mathbb{N}}$ converges to some $x + F$. □

In the above, E/F is called the **quotient *F*-normed** (or **normed**) **linear space** with the F-norm (or norm) $|||\cdot|||$, which is called the **quotient *F*-norm** (or **norm**). Recall that E/F is locally convex if so is E (Proposition 3.4.11). If E is a Banach space, a Fréchet space, or an F-space, then so is E/F for any closed linear subspace F of E.

Recall that $A \subset E$ is topologically bounded if, for each neighborhood U of $\mathbf{0} \in E$, there exists some $r \in \mathbb{R}$ such that $A \subset rU$.

Theorem 3.6.9 *A topological linear space E is normable if and only if there is a topologically bounded convex neighborhood of $\mathbf{0} \in E$.*

Proof The "only if" part is trivial. To see the "if" part, let V be a topologically bounded convex neighborhood of $\mathbf{0} \in E$. Then, $W = V \cap (-V)$ is a topologically bounded circled convex neighborhood of $\mathbf{0} \in E$. Hence, the Minkowski functional p_W is a norm on E by Proposition 3.3.4. By Corollary 3.4.15,

$$\left\{x \in E \;\middle|\; p_W(x) < \varepsilon\right\} = \varepsilon p_W^{-1}([0,1)) = \varepsilon \operatorname{int} W \quad \text{for each } \varepsilon > 0.$$

For each neighborhood U of $\mathbf{0} \in E$, we can choose $r > 0$ such that $W \subset rU$. Then,

$$\left\{x \in E \;\middle|\; p_W(x) < r^{-1}\right\} = r^{-1} \operatorname{int} W \subset r^{-1} W \subset U,$$

hence p_W induces the topology for E. □

For the local convexity, we have the following:

Theorem 3.6.10 *A topological linear space E is (metrizable and) locally convex if and only if E is linearly homeomorphic to a linear subspace of the (countable) product $\prod_{\lambda\in\Lambda} E_\lambda$ of normed linear spaces E_λ.*

Proof As is easily observed, the product of locally convex topological linear spaces is locally convex, and so is any linear subspace of a locally convex topological linear space. Moreover, a countable product of metrizable spaces is metrizable. Then, the "if" part follows.

We show the "only if" part. By the local convexity, E has a neighborhood basis $\{V_\lambda \mid \lambda \in \Lambda\}$ of $\mathbf{0} \in E$ consisting of circled closed convex sets (cf. Proposition 3.4.2), where $\operatorname{card} \Lambda = \aleph_0$ if E is metrizable (Theorem 3.6.1). For each $\lambda \in \Lambda$, let F_λ be a maximal linear subspace of E contained in V_λ. (The existence of F_λ is guaranteed by Zorn's Lemma.) Then, F_λ is closed in E (Proposition 3.4.5 (2)). Let $q_\lambda : E \to E/F_\lambda$ be the natural linear surjection, where we do not give the quotient topology to E/F_λ but we want to define a norm on E/F_λ.

Observe that $q_\lambda(V_\lambda)$ is a circled convex set in E/F_λ and $\mathbf{0} \in \operatorname{core} q_\lambda(V_\lambda)$. Moreover, $\mathbb{R}_+\xi \not\subset q_\lambda(V_\lambda)$ for each $\xi \in (E/F_\lambda) \setminus \{\mathbf{0}\}$. Indeed, take $x \in E \setminus F_\lambda$ so that $q_\lambda(x) = \xi$. By the maximality of F_λ, $\mathbb{R}x + F_\lambda \not\subset V_\lambda$, i.e., $tx + y \notin V_\lambda$ for some $t \in \mathbb{R}$ and $y \in F_\lambda$, where we can take $t > 0$ because V_λ is circled. For each $z \in F_\lambda$,

$$tx + y = \tfrac{1}{2}(2tx + z) + \tfrac{1}{2}(2y - z).$$

Since $2y - z \in F_\lambda \subset V_\lambda$, we have $2tx + z \notin V_\lambda$. Then, $2t\xi = q_\lambda(2tx) \notin q_\lambda(V_\lambda)$.

By Proposition 3.3.4, the Minkowski functional $p_\lambda = p_{q_\lambda(V_\lambda)} : E/F_\lambda \to \mathbb{R}$ for $q_\lambda(V_\lambda)$ is a norm. Thus, we have a normed linear space $E_\lambda = (E/F_\lambda, p_\lambda)$. Observe

$$\begin{aligned}\mathbf{0} \in \operatorname{int} V_\lambda = \operatorname{core} V_\lambda &\subset q_\lambda^{-1}(\operatorname{core} q_\lambda(V_\lambda)) \\ &= q_\lambda^{-1}\big(p_{q_\lambda(V_\lambda)}^{-1}([0,1))\big) = (p_\lambda q_\lambda)^{-1}([0,1)).\end{aligned}$$

By Proposition 3.4.14, the sublinear functional $p_\lambda q_\lambda : E \to \mathbb{R}$ is continuous, which implies that $q_\lambda : E \to E_\lambda$ is continuous.

Let $h : E \to \prod_{\lambda\in\Lambda} E_\lambda$ be the linear map[23] defined by $h(x) = (q_\lambda(x))_{\lambda\in\Lambda}$. If $x \neq \mathbf{0} \in E$ then $x \notin V_\lambda$ (so $x \notin F_\lambda$) for some $\lambda \in \Lambda$, which implies $q_\lambda(x) \neq \mathbf{0}$, hence $h(x) \neq \mathbf{0}$. Thus, h is a continuous linear injection. To see that h is an embedding, it suffices to show that

$$h(V_\lambda) \supset h(E) \cap \mathrm{pr}_\lambda^{-1}(p_\lambda^{-1}([0, 1/2))) \quad \text{for each } \lambda \in \Lambda.$$

If $p_\lambda(\mathrm{pr}_\lambda h(x)) < 1/2$, then $q_\lambda(2x) = \mathrm{pr}_\lambda h(2x) \in p_\lambda^{-1}([0, 1)) \subset q_\lambda(V_\lambda)$, hence $2x - y \in F_\lambda$ for some $y \in V_\lambda$. Then, it follows that $x = \frac{1}{2}(2x - y) + \frac{1}{2}y \in V_\lambda$, so $h(x) \in h(V_\lambda)$. This completes the proof. □

Combining Theorem 3.6.10 with Corollary 3.6.7, we have the following:

Corollary 3.6.11 *A topological linear space E is a Fréchet space if and only if E is linearly homeomorphic to a closed linear subspace of a countable product $\prod_{i\in\mathbb{N}} E_i$ of Banach spaces E_i.* □

3.7 The Closed Graph and Open Mapping Theorems

This section is devoted to two very important theorems, the Closed Graph Theorem and the Open Mapping Theorem. They are proved using the Baire Category Theorem 2.5.1. First, we prove the CLOSED GRAPH THEOREM:

Theorem 3.7.1 (CLOSED GRAPH) *Let E and F be completely metrizable topological linear spaces and $f : E \to F$ be a linear function. If the graph of f is closed in $E \times F$, then f is continuous.*

Proof It suffices to show the continuity of f at $\mathbf{0} \in E$. Let d and ρ be admissible complete invariant metrics for E and F, respectively (cf. Proposition 3.6.4).

First, we show that for each $\varepsilon > 0$, there is some $\delta(\varepsilon) > 0$ such that $\mathrm{B}_d(\mathbf{0}, \delta(\varepsilon)) \subset \mathrm{cl}\, f^{-1}(\mathrm{B}_\rho(\mathbf{0}, \varepsilon))$. Since $F = \bigcup_{n\in\mathbb{N}} n\mathrm{B}_\rho(\mathbf{0}, \varepsilon/2)$ and f is linear, it follows that $E = \bigcup_{n\in\mathbb{N}} nf^{-1}(\mathrm{B}_\rho(\mathbf{0}, \varepsilon/2))$. By the Baire Category Theorem 2.5.1, $\mathrm{int}\,\mathrm{cl}\, nf^{-1}(\mathrm{B}_\rho(\mathbf{0}, \varepsilon/2)) \neq \emptyset$ for some $n \in \mathbb{N}$, which implies that $\mathrm{int}\,\mathrm{cl}\, f^{-1}(\mathrm{B}_\rho(\mathbf{0}, \varepsilon/2)) \neq \emptyset$. Let $z \in \mathrm{int}\,\mathrm{cl}\, f^{-1}(\mathrm{B}_\rho(\mathbf{0}, \varepsilon/2))$ and choose $\delta(\varepsilon) > 0$ so that

$$z + \mathrm{B}_d(\mathbf{0}, \delta(\varepsilon)) = \mathrm{B}_d(z, \delta(\varepsilon)) \subset \mathrm{cl}\, f^{-1}(\mathrm{B}_\rho(\mathbf{0}, \varepsilon/2)).$$

Then, it follows that

$$\mathrm{B}_d(\mathbf{0}, \delta(\varepsilon)) \subset \mathrm{cl}\, f^{-1}(\mathrm{B}_\rho(\mathbf{0}, \varepsilon/2)) - z \subset \mathrm{cl}\, f^{-1}(\mathrm{B}_\rho(\mathbf{0}, \varepsilon)).$$

[23] That is, a continuous linear function.

The second inclusion can be proved as follows: For each $y \in \operatorname{cl} f^{-1}(\mathrm{B}_\rho(\mathbf{0}, \varepsilon/2))$ and $\eta > 0$, we have $y', z' \in f^{-1}(\mathrm{B}_\rho(\mathbf{0}, \varepsilon/2))$ such that $d(y, y'), d(z, z') < \eta/2$, which implies $d(y - z, y' - z') < \eta$. Observe that

$$\rho(f(y' - z'), \mathbf{0}) = \rho(f(y'), f(z')) \leqslant \rho(f(y'), \mathbf{0}) + \rho(f(z'), \mathbf{0}) < \varepsilon,$$

which means $y' - z' \in f^{-1}(\mathrm{B}_\rho(\mathbf{0}, \varepsilon))$. Therefore, $y - z \in \operatorname{cl} f^{-1}(\mathrm{B}_\rho(\mathbf{0}, \varepsilon))$.

Now, for each $\varepsilon > 0$ and $x \in \mathrm{B}_d(\mathbf{0}, \delta(\varepsilon/2))$, we can inductively choose $x_n \in E$, $n \in \mathbb{N}$, so that $x_n \in f^{-1}(\mathrm{B}_\rho(\mathbf{0}, 2^{-n}\varepsilon))$ and

$$d\big(x, \sum_{i=1}^{n} x_i\big) = d\big(x - \textstyle\sum_{i=1}^{n} x_i,\ \mathbf{0}\big) < \min\big\{2^{-n}, \delta(2^{-n-1}\varepsilon)\big\}.$$

Indeed, if $x_1, \ldots, x_{n-1}$ have been chosen, then

$$x - \sum_{i=1}^{n-1} x_i \in \mathrm{B}_d(\mathbf{0}, \delta(2^{-n}\varepsilon)) \subset \operatorname{cl} f^{-1}(\mathrm{B}_\rho(\mathbf{0}, 2^{-n}\varepsilon)),$$

hence we can choose $x_n \in f^{-1}(\mathrm{B}_\rho(\mathbf{0}, 2^{-n}\varepsilon))$ so that

$$d\big(x, \sum_{i=1}^{n} x_i\big) = d\big(x - \textstyle\sum_{i=1}^{n-1} x_i,\ x_n\big) < \min\big\{2^{-n}, \delta(2^{-n-1}\varepsilon)\big\}.$$

Since $\rho(f(x_n), \mathbf{0}) < 2^{-n}\varepsilon$ for each $n \in \mathbb{N}$, it follows that $(f(\sum_{i=1}^{n} x_i))_{n\in\mathbb{N}}$ is a Cauchy sequence, which converges to some $y \in F$. For each $n \in \mathbb{N}$,

$$\rho\big(f(\sum_{i=1}^{n} x_i),\ \mathbf{0}\big) \leqslant \sum_{i=1}^{n} \rho(f(x_i),\ \mathbf{0}) < \sum_{i=1}^{n} 2^{-i}\varepsilon < \varepsilon,$$

hence $y \in \overline{\mathrm{B}}_\rho(\mathbf{0}, \varepsilon)$. On the other hand, $\big(\sum_{i=1}^{n} x_i\big)_{n\in\mathbb{N}}$ converges to x. Since the graph of f is closed in $E \times F$, the point (x, y) belongs to the graph of f, which means $f(x) = y \in \overline{\mathrm{B}}_\rho(\mathbf{0}, \varepsilon)$. Thus, we have $f(\mathrm{B}_d(\mathbf{0}, \delta(\varepsilon/2))) \subset \overline{\mathrm{B}}_\rho(\mathbf{0}, \varepsilon)$. Therefore, f is continuous. □

Corollary 3.7.2 *Let E and F be completely metrizable topological linear spaces. Then, every continuous linear isomorphism $f : E \to F$ is a homeomorphism.*

Proof In general, the continuity of f implies the closedness of the graph of f in $E \times F$. By changing coordinates, the graph of f can be regarded as the graph of f^{-1}. Then, it follows that the graph of f^{-1} is closed in $F \times E$, which implies the continuity of f^{-1} by Theorem 3.7.1. □

Next, we prove the OPEN MAPPING THEOREM:[24]

[24] This is also known as the BANACH–SCHAUDER THEOREM.

Theorem 3.7.3 (OPEN MAPPING) *Let E and F be completely metrizable topological linear spaces. Then, every continuous linear surjection $f : E \to F$ is open.*

Proof Since $f^{-1}(\mathbf{0})$ is a closed linear subspace of E, the quotient linear space $E/f^{-1}(\mathbf{0})$ is completely metrizable by Proposition 3.6.8. Then, f induces the continuous linear isomorphism $\tilde{f} : E/f^{-1}(\mathbf{0}) \to F$. By Corollary 3.7.2, $\tilde{f}$ is a homeomorphism. Note that the quotient map $q : E \to E/f^{-1}(\mathbf{0})$ is open. Indeed, for every open set U in E, $q^{-1}(q(U)) = U + f^{-1}(\mathbf{0})$ is open in E, which means that $q(U)$ is open in $E/f^{-1}(\mathbf{0})$. Hence, $f = \tilde{f}q$ is also open. □

Note In the above, the Closed Graph Theorem is first proved and then the Open Mapping Theorem is obtained as a corollary of the Closed Graph Theorem. Conversely, we can directly prove the Open Mapping Theorem and then obtain the Closed Graph Theorem as a corollary of the Open Mapping Theorem.

Proof (*Open Mapping Theorem* 3.7.3) Let d and ρ be admissible complete invariant metrics for E and F, respectively.

First, we show that for each $\varepsilon > 0$, there is some $\delta(\varepsilon) > 0$ such that $\mathrm{B}_\rho(\mathbf{0}, \delta(\varepsilon)) \subset \mathrm{cl}\, f(\mathrm{B}_d(\mathbf{0}, \varepsilon))$. Since $E = \bigcup_{n\in\mathbb{N}} n\mathrm{B}_d(\mathbf{0}, \varepsilon/2)$, we have

$$F = f(E) = \bigcup_{n\in\mathbb{N}} nf(\mathrm{B}_d(\mathbf{0}, \varepsilon/2)).$$

By the Baire Category Theorem 2.5.1, $\mathrm{int}\,\mathrm{cl}\, nf(\mathrm{B}_d(\mathbf{0}, \varepsilon/2)) \neq \emptyset$ for some $n \in \mathbb{N}$, which implies that $\mathrm{int}\,\mathrm{cl}\, f(\mathrm{B}_d(\mathbf{0}, \varepsilon/2)) \neq \emptyset$. Let $z \in \mathrm{int}\,\mathrm{cl}\, f(\mathrm{B}_d(\mathbf{0}, \varepsilon/2))$ and choose $\delta(\varepsilon) > 0$ so that

$$z + \mathrm{B}_\rho(\mathbf{0}, \delta(\varepsilon)) = \mathrm{B}_\rho(z, \delta(\varepsilon)) \subset \mathrm{cl}\, f(\mathrm{B}_d(\mathbf{0}, \varepsilon/2)).$$

Then, it follows that

$$\mathrm{B}_\rho(\mathbf{0}, \delta(\varepsilon)) \subset \mathrm{cl}\, f(\mathrm{B}_d(\mathbf{0}, \varepsilon/2)) - z \subset \mathrm{cl}\, f(\mathrm{B}_d(\mathbf{0}, \varepsilon)),$$

where the second inclusion can be seen as follows: For $y \in \mathrm{cl}\, f(\mathrm{B}_d(\mathbf{0}, \varepsilon/2))$ and $\eta > 0$, choose $y', z' \in \mathrm{B}_d(\mathbf{0}, \varepsilon/2)$ so that $\rho(y, f(y')), \rho(z, f(z')) < \eta/2$. Then,

$$\rho(y - z, f(y' - z')) < \eta \quad \text{and} \quad d(y' - z', \mathbf{0}) = d(y', z') < \varepsilon,$$

hence $y - z \in \mathrm{cl}\, f(\mathrm{B}_d(\mathbf{0}, \varepsilon))$.

Next, we prove that $\mathrm{cl}\, f(\mathrm{B}_d(\mathbf{0}, \varepsilon/2)) \subset f(\overline{\mathrm{B}}_d(\mathbf{0}, \varepsilon))$ for each $\varepsilon > 0$. For each $y \in \mathrm{cl}\, f(\mathrm{B}_d(\mathbf{0}, \varepsilon/2))$, choose $x_1 \in \mathrm{B}_d(\mathbf{0}, \varepsilon/2)$ so that

$$\rho(y, f(x_1)) < \min\{2^{-1}, \delta(2^{-2}\varepsilon)\}.$$

By induction, we can choose $x_n \in \mathrm{B}_d(\mathbf{0}, 2^{-n}\varepsilon)$, $n \in \mathbb{N}$, so that

$$\rho\big(y, f\big(\sum_{i=1}^{n} x_i\big)\big) = \rho\big(y - \textstyle\sum_{i=1}^{n} f(x_i),\ \mathbf{0}\big) < \min\{2^{-n}, \delta(2^{-n-1}\varepsilon)\}.$$

Indeed, if $x_1, \ldots, x_{n-1}$ have been chosen, then

$$y - \sum_{i=1}^{n-1} f(x_i) \in \mathrm{B}_\rho(\mathbf{0}, \delta(2^{-n}\varepsilon)) \subset \mathrm{cl}\, f(\mathrm{B}_d(\mathbf{0}, 2^{-n}\varepsilon)),$$

hence we can choose $x_n \in \mathrm{B}_d(\mathbf{0}, 2^{-n}\varepsilon)$ so that

$$\rho\big(y, f\big(\sum_{i=1}^{n} x_i\big)\big) = \rho\big(y - \sum_{i=1}^{n-1} f(x_i),\ f(x_n)\big) < \min\{2^{-n}, \delta(2^{-n-1}\varepsilon)\}.$$

Since $(\sum_{i=1}^{n} x_i)_{n\in\mathbb{N}}$ is a Cauchy sequence in E, it converges to some $x \in E$. On the other hand, $(f(\sum_{i=1}^{n} x_i))_{n\in\mathbb{N}}$ converges to y. By the continuity of f, we have $f(x) = y$. For each $n \in \mathbb{N}$,

$$d\big(\sum_{i=1}^{n} x_i,\ \mathbf{0}\big) \leqslant \sum_{i=1}^{n} d(x_i, \mathbf{0}) < \sum_{i=1}^{n} 2^{-i}\varepsilon < \varepsilon,$$

hence $x \in \overline{\mathrm{B}}_d(\mathbf{0}, \varepsilon)$. Thus, it follows that $\mathrm{cl}\, f(\mathrm{B}_d(\mathbf{0}, \varepsilon/2)) \subset f(\overline{\mathrm{B}}_d(\mathbf{0}, \varepsilon))$.

To see that f is open, let U be an open set in E. For each $x \in U$, choose $\varepsilon > 0$ so that $\overline{\mathrm{B}}_d(\mathbf{0}, \varepsilon) \subset -x + U$. Since

$$\mathrm{B}_\rho(\mathbf{0}, \delta(\varepsilon/2)) \subset \mathrm{cl}\, f(\mathrm{B}_d(\mathbf{0}, \varepsilon/2)) \subset f(\overline{\mathrm{B}}_d(\mathbf{0}, \varepsilon)) \subset -f(x) + f(U),$$

it follows that $\mathrm{B}_\rho(f(x), \delta(\varepsilon/2)) \subset f(U)$. Hence, $f(U)$ is open in F. □

Now, using the Open Mapping Theorem, we will prove the Closed Graph Theorem.

Proof (Closed Graph Theorem 3.7.1) The product space $E \times F$ is a completely metrizable topological linear space. The graph G of f is a linear subspace of $E \times F$ that is completely metrizable because it is closed in $E \times F$. Since $p = \mathrm{pr}_E|G : G \to E$ is a homeomorphism by the Open Mapping Theorem, $f = \mathrm{pr}_F \circ p^{-1}$ is continuous. □

Remark 3.7.1 In both the Closed Graph Theorem and the Open Mapping Theorem, the completeness is essential. Let $E = (\ell_1, \|\cdot\|_2)$, where $\ell_1 \subset \ell_2$ as sets and $\|\cdot\|_2$ is the norm inherited from ℓ_2. Then, E is not completely metrizable. Indeed, if so, it would be closed in ℓ_2 by Corollary 3.6.7, but E is dense in ℓ_2 and $E \neq \ell_2$. The linear bijection $f = \mathrm{id} : \ell_1 \to E$ is continuous, but is not a homeomorphism, so it is not an open map. It follows from the continuity of f that the graph of f is closed in $\ell_1 \times E$, hence the graph of f^{-1} is closed in $E \times \ell_1$. However, $f^{-1} : E \to \ell_1$ is not continuous.

3.8 Continuous Selections

In this section, we prove the Michael Selection Theorem. Let X and Y be spaces and $\varphi : X \to \mathfrak{P}_0(Y)$ $(= \mathfrak{P}(Y) \setminus \{\emptyset\})$ be a set-valued function. A (**continuous**) **selection** for φ is a map $f : X \to Y$ such that $f(x) \in \varphi(x)$ for each $x \in X$. For a topological linear space E, let $\mathrm{Conv}(E)$ denote the set of all non-empty convex sets in E. In this section, we will discuss the problem of when a convex-valued function $\varphi : X \to \mathrm{Conv}(E)$ has a selection.

In Sect. 2.12, we introduced the Vietoris topology on the power set $\mathfrak{P}(Y)$, which is generated by the sets V^+, V^-, where V is an open set in Y. Then, $\emptyset$ is an isolated point of $\mathfrak{P}(Y)$. It is said that $\varphi : X \to \mathfrak{P}_0(Y)$ is **lower semi-continuous** (**l.s.c.**) (resp.

upper semi-continuous (u.s.c.)) if $\varphi^{-1}(V^-)$ (resp. $\varphi^{-1}(V^+)$) is open in X for each open set V in Y. Observe that

$$\varphi^{-1}(V^-) = \left\{x \in X \;\middle|\; \varphi(x) \cap V \neq \emptyset\right\} \left(\text{resp. } \varphi^{-1}(V^+) = \left\{x \in X \;\middle|\; \varphi(x) \subset V\right\}\right).$$

Thus, φ is l.s.c. (resp. u.s.c.) if and only if for each $x_0 \in X$ and each open set V in Y and $x_0 \in X$ such that $\varphi(x_0) \cap V \neq \emptyset$ (resp. $\varphi(x_0) \subset V$), there exists a neighborhood U of x_0 in X such that $\varphi(x) \cap V \neq \emptyset$ (resp. $\varphi(x) \subset V$) for every $x \in U$. Moreover, if $\varphi : X \to \mathfrak{P}_0(Y)$ is l.s.c. and u.s.c., then φ is **continuous** (with respect to the Vietoris topology).

Note Recall that $f : X \to \mathbb{R}$ is l.s.c. (resp. u.s.c.) if $f^{-1}((t, \infty))$ (resp. $f^{-1}((-\infty, t))$) is open in X for each $t \in \mathbb{R}$ (cf. Sect. 2.7), where the topology of $\mathbb{R}$ is generated by the half open intervals $(t, \infty), (-\infty, t), t \in \mathbb{R}$. Comparing $U^+ = \{A \in \mathfrak{P}_0(Y) \mid A \subset U\}$ with $(-\infty, t) = \{s \in \mathbb{R} \mid s < t\}$, we can perceive the similarity between the upper semi-continuity of set-valued functions and that of real-valued functions. However, the lower semi-continuity of set-valued functions does not look like that of real-valued functions because $\subset$ is not a total order on $\mathfrak{P}_0(Y)$. As for the lower semi-continuity, rather compare $(Y \setminus U)^+ = \mathfrak{P}_0(Y) \setminus U^-$ and $(-\infty, t] = \mathbb{R} \setminus (t, \infty)$.

The following is easily shown:

Proposition 3.8.1 *For a surjection $g : Y \to X$, the set-valued function $g^{-1} : X \to \mathfrak{P}_0(Y)$ is l.s.c. (resp. u.s.c.) if and only if g is open (resp. closed).*

Sketch of Proof Observe that, for $V \subset Y$, $\{x \in X \mid g^{-1}(x) \cap V \neq \emptyset\} = g(V)$ and

$$\{x \in X \mid g^{-1}(x) \subset V\} = X \setminus \{x \in X \mid g^{-1}(x) \cap (Y \setminus V) \neq \emptyset\} = X \setminus g(Y \setminus V).$$

Because of the following proposition, we consider the selection problem for l.s.c. set-valued functions.

Proposition 3.8.2 *Let $\varphi : X \to \mathfrak{P}_0(Y)$ be a set-valued function. Assume that, for each $x_0 \in X$ and $y_0 \in \varphi(x_0)$, there exists a neighborhood U of x_0 in X and a selection $f : U \to Y$ for $\varphi|U$ such that $f(x_0) = y_0$. Then, φ is l.s.c.*

Proof Let V be an open set in Y and $x_0 \in X$ such that $\varphi(x_0) \cap V \neq \emptyset$. Take any $y_0 \in \varphi(x_0) \cap V$. From the assumption, there is a neighborhood U of x_0 in X with a selection $f : U \to Y$ for $\varphi|U$ such that $f(x_0) = y_0$. Then, $f^{-1}(V)$ is a neighborhood of x_0 in X and $f(x) \in \varphi(x) \cap V$ for each $x \in f^{-1}(V)$. □

Lemma 3.8.3 *Let $\varphi, \psi : X \to \mathfrak{P}_0(Y)$ be set-valued functions such that $\operatorname{cl} \varphi(x) = \operatorname{cl} \psi(x)$ for each $x \in X$. If φ is l.s.c. then so is ψ.*

Sketch of Proof. This follows from the fact that, for each open set V in Y and $B \subset Y$, $V \cap B \neq \emptyset$ if and only if $V \cap \operatorname{cl} B \neq \emptyset$.

Lemma 3.8.4 *Let $\varphi : X \to \mathfrak{P}_0(Y)$ be l.s.c., A be a closed set in X, and $f : A \to Y$ be a selection for $\varphi|A$. Define $\psi : X \to \mathfrak{P}_0(Y)$ by*

$$\psi(x) = \begin{cases} \{f(x)\} & \text{if } x \in A, \\ \varphi(x) & \text{otherwise.} \end{cases}$$

Then, ψ is also l.s.c.

Proof For each open set V in Y, $f^{-1}(V)$ is open in A and

$$f^{-1}(V) \subset \big\{x \in X \;\big|\; \varphi(x) \cap V \neq \emptyset\big\},$$

where the latter set is open in X because φ is l.s.c. Then, we can choose an open set U in X so that $f^{-1}(V) = U \cap A$ and $U \subset \{x \in X \mid \varphi(x) \cap V \neq \emptyset\}$. Observe that

$$\big\{x \in X \;\big|\; \psi(x) \cap V \neq \emptyset\big\} = U \cup \Big(\big\{x \in X \;\big|\; \varphi(x) \cap V \neq \emptyset\big\} \setminus A\Big).$$

Thus, it follows that ψ is l.s.c. □

For each $W \subset Y^2$ and $y_0 \in Y$, we denote

$$W(y_0) = \big\{y \in Y \;\big|\; (y_0, y) \in W\big\}.$$

If W is a neighborhood of the diagonal $\Delta_Y = \{(y, y) \mid y \in Y\}$ in Y^2, then $W(y_0)$ is a neighborhood of y_0 in Y.

Lemma 3.8.5 *Let $\varphi : X \to \mathfrak{P}_0(Y)$ be l.s.c., $f : X \to Y$ a map, and W an open neighborhood of Δ_Y in Y^2. Suppose that $\varphi(x) \cap W(f(x)) \neq \emptyset$ for each $x \in X$. Then, the set-valued function $\psi : X \to \mathfrak{P}_0(Y)$ defined by $\psi(x) = \varphi(x) \cap W(f(x))$ is l.s.c.*

Proof Let V be an open set in Y and $x_0 \in X$ such that $\psi(x_0) \cap V \neq \emptyset$. Then, we have $y_0 \in \varphi(x_0) \cap W(f(x_0)) \cap V$. Since $(f(x_0), y_0) \in W$, there are open sets V_1 and V_2 in Y such that $(f(x_0), y_0) \in V_1 \times V_2 \subset W$. Then, the following U is an open neighborhood of x_0 in X:

$$U = f^{-1}(V_1) \cap \big\{x \in X \;\big|\; \varphi(x) \cap V_2 \cap V \neq \emptyset\big\}.$$

For each $x \in U$, we have $y \in \varphi(x) \cap V_2 \cap V$. Since $(f(x), y) \in V_1 \times V_2 \subset W$, it follows that $y \in \varphi(x) \cap W(f(x)) \cap V$, hence $\psi(x) \cap V \neq \emptyset$. Therefore, ψ is l.s.c. □

Let E be a linear space. The set of all non-empty convex sets in E is denoted by $\mathrm{Conv}(E)$. Recall that $\langle A \rangle$ denotes the convex hull of $A \subset E$.

Lemma 3.8.6 *Let E be a topological linear space and $\varphi : X \to \mathfrak{P}_0(E)$ be an l.s.c. set-valued function. Define a convex-valued function $\psi : X \to \mathrm{Conv}(E)$ by $\psi(x) = \langle \varphi(x) \rangle$ for each $x \in X$. Then, ψ is also l.s.c.*

Proof Let V be an open set in E and $x_0 \in X$ such that $\psi(x_0) \cap V \neq \emptyset$. Choose any $y_0 = \sum_{i=1}^{n} t_i y_i \in \psi(x_0) \cap V$, where $y_1, \ldots, y_n \in \varphi(x_0)$ and $t_1, \ldots, t_n \geqslant 0$ with $\sum_{i=1}^{n} t_i = 1$. Then, each y_i has an open neighborhood V_i such that $t_1 V_1 + \cdots + t_n V_n \subset V$. Since φ is l.s.c., the following is an open neighborhood of x_0 in X:

$$U = \bigcap_{i=1}^{n} \{x \in X \mid \varphi(x) \cap V_i \neq \emptyset\}.$$

For each $x \in U$, let $z_i \in \varphi(x) \cap V_i$, $i = 1, \ldots, n$. Then, $\sum_{i=1}^{n} t_i z_i \in \psi(x) \cap V$, hence $\psi(x) \cap V \neq \emptyset$. Therefore, ψ is l.s.c. □

Lemma 3.8.7 *Let X be paracompact, E be a topological linear space, and $\varphi : X \to \mathrm{Conv}(E)$ be an l.s.c. convex-valued function. Then, for each convex open neighborhood V of $\mathbf{0}$ in E, there exists a map $f : X \to E$ such that $f(x) \in \varphi(x) + V$ for each $x \in X$.*

Proof For each $y \in E$, let

$$U_y = \{x \in X \mid \varphi(x) \cap (y - V) \neq \emptyset\}.$$

Since φ is l.s.c., we have $\mathcal{U} = \{U_y \mid y \in E\} \in \mathrm{cov}(X)$. From paracompactness, X has a locally finite partition of unity $(f_\lambda)_{\lambda \in \Lambda}$ subordinated to $\mathcal{U}$. For each $\lambda \in \Lambda$, choose $y_\lambda \in E$ so that $\mathrm{supp}\, f_\lambda \subset U_{y_\lambda}$. We define a map $f : X \to E$ by $f(x) = \sum_{\lambda \in \Lambda} f_\lambda(x) y_\lambda$. If $f_\lambda(x) \neq 0$ then $x \in U_{y_\lambda}$, which means that $\varphi(x) \cap (y_\lambda - V) \neq \emptyset$, i.e., $y_\lambda \in \varphi(x) + V$. Since each $\varphi(x) + V$ is convex, $f(x) \in \varphi(x) + V$. □

Now, we can prove the following:

Theorem 3.8.8 (MICHAEL SELECTION) *Let X be a paracompact space and $E = (E, d)$ be a locally convex metric linear space.*[25] *Then, an l.s.c. convex-valued function $\varphi : X \to \mathrm{Conv}(E)$ admits a selection if each $\varphi(x)$ is d-complete. Moreover, if A is a closed set in X then each selection $f : A \to E$ for $\varphi|A$ can extend to a selection $\tilde{f} : X \to E$ for φ.*

Proof Let $\{V_i \mid i \in \mathbb{N}\}$ be a neighborhood basis of $\mathbf{0}$ in E such that each V_i is a circled open convex set with $\mathrm{diam}\, V_i < 2^{-(i+1)}$. By induction, we construct maps $f_i : X \to E$, $i \in \mathbb{N}$, so that, for each $x \in X$ and $i \in \mathbb{N}$,

(1) $f_i(x) \in \varphi(x) + V_i$ and
(2) $d(f_{i+1}(x), f_i(x)) < 2^{-i}$.

The existence of f_1 is guaranteed by Lemma 3.8.7. Assume we have maps $f_1, \ldots, f_n$ satisfying (1) and (2). Define $\psi : X \to \mathrm{Conv}(E)$ by

$$\psi(x) = \varphi(x) \cap (f_n(x) + V_n) \quad \text{for each } x \in X.$$

[25] Recall that a metric linear space is a linear space with a linear metric (cf., Sect. 3.6).

Since V_n is symmetric, we have $\psi(x) \neq \emptyset$ by (1). Consider the open neighborhood $W = \{(x, y) \in E^2 \mid y - x \in V_n\}$ of Δ_E in E^2. Then, $W(f_n(x)) = f_n(x) + V_n$. By Lemma 3.8.5, ψ is l.s.c. We can apply Lemma 3.8.7 to obtain a map $f_{n+1} : X \to E$ such that

$$f_{n+1}(x) \in \psi(x) + V_{n+1} \quad \text{for each } x \in X.$$

Then, as is easily observed, f_{n+1} satisfies (1) and (2). Thus, we have the desired sequence of maps f_i, $i \in \mathbb{N}$.

Using maps $f_i : X \to E$, $i \in \mathbb{N}$, we will define a selection $f : X \to E$ for φ. For each $x \in X$ and $i \in \mathbb{N}$, we have $x_i \in \varphi(x)$ such that $d(f_i(x), x_i) < 2^{-(i+1)}$ by (1). Then, $(x_i)_{i\in\mathbb{N}}$ is Cauchy in $\varphi(x)$. Since $\varphi(x)$ is complete, $(x_i)_{i\in\mathbb{N}}$ converges to $f(x) \in \varphi(x)$. Thus, we have $f : X \to E$. Note that $(f_i)_{i\in\mathbb{N}}$ uniformly converges to f, so f is continuous. Hence, f is a selection for φ.

For the additional statement, apply Lemma 3.8.4. □

Concerning factors of a metric linear space, we have the following:

Corollary 3.8.9 (BARTLE–GRAVES–MICHAEL) *Let E be a locally convex metric linear space and F be a linear subspace of E that is complete (so a Fréchet space). Then, $E \approx F \times E/F$. In particular, $E \approx \mathbb{R} \times G$ for some metric linear space G whenever $E \neq \{\mathbf{0}\}$.*

Proof Note that the quotient space E/F is metrizable (Proposition 3.6.8) and the natural map $g : E \to E/F$ is open, hence $g^{-1} : E/F \to \mathrm{Conv}(E)$ is l.s.c. by Proposition 3.8.1. Since $g^{-1}g(x) = x + F$ is complete for each $x \in E$, we apply the Michael Selection Theorem 3.8.8 to obtain a map $f : E/F \to E$ that is a selection for g^{-1}, i.e., $gf = \mathrm{id}$. Then, $x - fg(x) \in F$ for each $x \in E$. Hence, a homeomorphism $h : E \to F \times E/F$ can be defined by

$$h(x) = (x - fg(x), g(x)) \quad \text{for each } x \in E.$$

In fact, $h^{-1}(y, z) = y + f(z)$ for each $(y, z) \in F \times E/F$. □

By combining the Michael Selection Theorem 3.8.8 and the Open Mapping Theorem 3.7.3, the BARTLE–GRAVES THEOREM can be obtained as a corollary:

Theorem 3.8.10 (BARTLE–GRAVES) *Let E and F be Fréchet spaces and $f : E \to F$ be a continuous linear surjection. Then, there is a map $g : F \to E$ such that $fg = \mathrm{id}$. Therefore, $E \approx F \times f^{-1}(\mathbf{0})$ by the homeomorphism h defined as follows:*

$$h(x) = (f(x), x - gf(x)) \quad \textit{for each } x \in E. \qquad \square$$

We show that each Banach space is a (topological) factor of $\ell_1(\Gamma)$. To this end, we need the following:

Theorem 3.8.11 (BANACH–MAZUR, KLEE) *For every Banach space E, there is a continuous linear surjection $q : \ell_1(\Gamma) \to E$, where* $\mathrm{card}\,\Gamma = \mathrm{dens}\,E$.

Proof The unit closed ball $\mathbf{B}_E$ of E has a dense set $\{\mathbf{e}_\gamma \mid \gamma \in \Gamma\}$. Since $\sum_{\gamma\in\Gamma} |x(\gamma)| = \|x\| < \infty$ for each $x \in \ell_1(\Gamma)$ and E is complete, we can define a linear map $q : \ell_1(\Gamma) \to E$ as follows:[26]

$$q(x) = \sum_{\gamma\in\Gamma} x(\gamma)\mathbf{e}_\gamma = \lim_{F\in\mathrm{Fin}(\Gamma)} \sum_{\gamma\in F} x(\gamma)\mathbf{e}_\gamma \quad \text{for each } x \in E.$$

Since $\|q(x)\| \leqslant \sum_{\gamma\in\Gamma} |x(\gamma)| = \|x\|$, it follows that q is continuous.

To see that q is surjective, it suffices to show $\mathbf{B}_E \subset q(\ell_1(\Gamma))$. For each $y \in \mathbf{B}_E$, we can inductively choose $\mathbf{e}_{\gamma_i}$, $i \in \mathbb{N}$, so that $\gamma_i \neq \gamma_j$ if $i \neq j$, and

$$\|y - \mathbf{e}_{\gamma_1}\| < 2^{-1}, \ \|y - \mathbf{e}_{\gamma_1} - 2^{-1}\mathbf{e}_{\gamma_2}\| < 2^{-2},$$
$$\|y - \mathbf{e}_{\gamma_1} - 2^{-1}\mathbf{e}_{\gamma_2} - 2^{-2}\mathbf{e}_{\gamma_3}\| < 2^{-3}, \ \ldots.$$

We have $x \in \ell_1(\Gamma)$ defined by

$$x(\gamma) = \begin{cases} 2^{1-i} & \text{if } \gamma = \gamma_i, \\ 0 & \text{otherwise.} \end{cases}$$

Then, it follows that $y = \sum_{i=1}^{\infty} 2^{1-i}\mathbf{e}_{\gamma_i} = q(x)$. This completes the proof. □

As a combination of the Bartle–Graves Theorem 3.8.10 and Theorem 3.8.11, we have the following:

Corollary 3.8.12 *For any Banach space E, there exists a Banach space F such that $E \times F \approx \ell_1(\Gamma)$, where* $\operatorname{card}\Gamma = \operatorname{dens} E$. □

In the Michael Selection Theorem 3.8.8, the paracompactness of X is necessary. Actually, we have the following characterization:

Theorem 3.8.13 *A space X is paracompact if and only if the following holds for any Banach space E: if $\varphi : X \to \mathrm{Conv}(E)$ is l.s.c. and each $\varphi(x)$ is closed, then φ has a selection.*

Proof Since the "only if" part is simply Theorem 3.8.8, it suffices to prove the "if" part. For each $\mathcal{U} \in \mathrm{cov}(X)$, we define $\varphi : X \to \mathfrak{P}_0(\ell_1(\mathcal{U}))$ as follows:

$$\varphi(x) = \big\{z \in \ell_1(\mathcal{U}) \ \big|\ \|z\| = 1, \ \forall U \in \mathcal{U}, \ z(U) \geqslant 0, z(U) = 0 \ \text{ if } x \notin U\big\}.$$

Clearly, each $\varphi(x)$ is a closed convex set.

To see that φ is l.s.c., let W be an open set in $\ell_1(\mathcal{U})$ and $z \in \varphi(x) \cap W$. Choose $\delta > 0$ so that $\mathrm{B}(z, 2\delta) \subset W$. Then, we have $V_1, \ldots, V_n \in \mathcal{U}[x]$ such that $\sum_{i=1}^{n} z(V_i) > 1 - \delta$, where $\bigcap_{i=1}^{n} V_i$ is a neighborhood of x in X. We define $z' \in \ell_1(\mathcal{U})$ as follows:

[26] See the definition of $\sum_{\gamma\in\Gamma} x(\gamma)$ just before Proposition 1.2.5.

$$z'(V_i) = \frac{z(V_i)}{\sum_{j=1}^{n} z(V_j)} \text{ and } z'(U) = 0 \text{ for } U \neq V_1, \ldots, V_n.$$

It is easy to see that $z' \in \varphi(x') \cap W$ for every $x' \in \bigcap_{i=1}^{n} V_i$. Thus, φ is l.s.c.

By the assumption, φ has a selection $f : X \to \ell_1(\mathcal{U})$. For each $U \in \mathcal{U}$, let $f_U : X \to \mathbf{I}$ be the map defined by $f_U(x) = f(x)(U)$ for $x \in X$. Then, $(f_U)_{U \in \mathcal{U}}$ is a partition of unity such that $f_U^{-1}((0, 1]) \subset U$ for every $U \in \mathcal{U}$. The result follows from Theorem 2.7.5. □

Remark 3.8.1 Let $g, h : X \to \mathbb{R}$ be real-valued functions on a paracompact space X such that g is u.s.c., h is l.s.c., and $g(x) \leqslant h(x)$ for each $x \in X$. We define the convex-valued function $\varphi : X \to \mathrm{Conv}(\mathbb{R})$ by $\varphi(x) = [g(x), h(x)]$ for each $x \in X$. Then, φ is l.s.c. Indeed, for each open set V in $\mathbb{R}$, let $\varphi(x) \cap V \neq \emptyset$. Take $y \in \varphi(x) \cap V$ and $a < y < b$ so that $[a, b] \subset V$. Since g is u.s.c. and h is l.s.c., x has a neighborhood U in X such that $x' \in U$ implies $g(x') < b$ and $h(x') > a$. Since $g(x') \leqslant h(x')$, it follows that

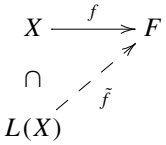
$$\varphi(x') \cap V \supset [g(x'), h(x')] \cap [a, b] = [\max\{a, g(x')\}, \min\{b, h(x')\}] \neq \emptyset.$$

Now, we can apply the Michael Selection Theorem 3.8.8 to obtain a map $f : X \to \mathbb{R}$ such that $g(x) \leqslant f(x) \leqslant h(x)$ for each $x \in X$. This is analogous to Theorem 2.7.8.

3.9 Free Topological Linear Spaces

The **free topological linear space** over a space X is a topological linear space $L(X)$ that contains X as a subspace and has the following extension property:

(LE) For an arbitrary topological linear space F, every map $f : X \to F$ of X uniquely extends to a linear map[27] $\tilde{f} : L(X) \to F$.



If such a space $L(X)$ exists, then it is uniquely determined up to linear homeomorphism, that is, if E is a topological linear space that contains X and has the property (LE), then E is linearly homeomorphic to $L(X)$.

Indeed, there exist linear maps $\varphi : L(X) \to E$ and $\psi : E \to L(X)$ such that $\varphi|X = \psi|X = \mathrm{id}_X$. Since $\mathrm{id}_{L(X)}$ is a linear map extending id_X, it follows from the uniqueness that

[27] I.e., a continuous linear function.

$\psi\varphi = \mathrm{id}_{L(X)}$. Similarly, we have $\varphi\psi = \mathrm{id}_E$. Therefore, φ is a linear homeomorphism with $\psi = \varphi^{-1}$.

In this section, we will show that $L(X)$ actually exists for every Tychonoff space X (see Theorem 3.9.3) and give basic information on free topological linear spaces, which will be used in Section 7.12 to construct a metrizable linear space that is not an absolute extensor for metrizable spaces (an AE).

We start from proving the following:

Proposition 3.9.1 *Let X be a Tychonoff space. Then,*

(1) *X is a Hamel basis for $L(X)$;*
(2) *X is closed in $L(X)$.*

Proof (1): First, let F be the linear span of X. Applying (LE), we have a linear map $r : L(X) \to F$ such that $r|X = \mathrm{id}_X$. Since $r : L(X) \to L(X)$ is a linear map extending id_X, we have $r = \mathrm{id}_{L(X)}$, which implies $F = L(X)$, that is, $L(X)$ is generated by X.

To see that X is linearly independent in $L(X)$, let $x_1, \ldots, x_n \in X$, where $x_i \neq x_j$ if $i \neq j$. For each $i = 1, \ldots, n$, there is a map $f_i : X \to \mathbf{I}$ such that $f_i(x_i) = 1$ and $f_i(x_j) = 0$ for $j \neq i$. Let $f : X \to \mathbb{R}^n$ be the map defined by $f(x) = (f_1(x), \ldots, f_n(x))$. Then, by (LE), f extends to a linear map $\tilde{f} : L(X) \to \mathbb{R}^n$, where $\tilde{f}(x_i) = f(x_i) = \mathbf{e}_i$ for each $i = 1, \ldots, n$. Since $\mathbf{e}_1, \ldots, \mathbf{e}_n$ is linearly independent in $\mathbb{R}^n$, it follows that $x_1, \ldots, x_n \in X$ is linearly independent in $L(X)$.

(2): Let $z = \sum_{i=1}^n t_i x_i \in L(X) \setminus X$, where $t_i \in \mathbb{R} \setminus \{0\}$, $x_i \in X$, and $x_i \neq x_j$ if $i \neq j$. We have to find a neighborhood of z missing X. When $\sum_{i=1}^n t_i \neq 1$, property (LE) provides us a linear map $f : L(X) \to \mathbb{R}$ such that $f(X) = \{1\}$. Then, $f^{-1}(\mathbb{R} \setminus \{1\})$ is an open neighborhood of z in $L(X)$ that misses X. We will show the case when $\sum_{i=1}^n t_i = 1$, where $n > 1$ because $z \notin X$. Let $V_1, \ldots, V_n$ be pairwise disjoint open sets in X with $x_i \in V_i$. Since X is Tychonoff, we can take maps $g_i : X \to \mathbf{I}$, $i = 1, \ldots, n$, so that $g_i(x_i) = 1$ and $g_i(X \setminus V_i) = \{0\}$.

When $t_i < 0$ for some $i = 1, \ldots, n$, we define a map $f : X \to \mathbb{R}$ as follows:

$$f(x) = \sum_{i=1}^n \frac{t_i}{|t_i|} g_i(x) \quad \text{for each } x \in X.$$

Then, $f(x) = 0$ if $x \in X \setminus \bigcup_{i=1}^n V_i$, and $f(x) = (t_i/|t_i|)g_i(x) \leqslant g_i(x)$ if $x \in V_i$. Hence, $f(x) \leqslant 1$ for any $x \in X$. By (LE), f can be extended to a linear map $\tilde{f} : L(X) \to \mathbb{R}$. Observe that

$$\tilde{f}(z) = \sum_{i=1}^n t_i f(x_i) = \sum_{i=1}^n t_i \frac{t_i}{|t_i|} g_i(x_i) = \sum_{i=1}^n |t_i| > 1.$$

Hence, $\tilde{f}^{-1}((1, \infty))$ is an open neighborhood of z in $L(X)$ missing X.

When $t_i > 0$ for every $i = 1, \ldots, n$, since $t_i < 1$, take $t_i < t_i' < 1$ and define $W_i = g_i^{-1}((t_i', 1])$ for each $i = 1, \ldots, n$. Then, each W_i is an open neighborhood of x_i in X.

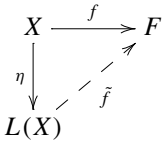
Since X is Tychonoff, we have maps $h_i : X \to \mathbf{I}, i = 1, \ldots, n$, so that $h_i(x_i) = 1$ and $h_i(X \setminus W_i) = \{0\}$. Then, we define a map $h : X \to \mathbb{R}$ by $h(x) = 1 - \sum_{i=1}^{n} h_i(x)$ for each $x \in X$. By (LE), g_i and h can be extended to linear maps $\tilde{g}_i : L(X) \to \mathbb{R}$ and $\tilde{h} : L(X) \to \mathbb{R}$. Then, as is easily seen, the following U is an open neighborhood of z in $L(X)$:

$$U = \bigcap_{i=1}^{n} \tilde{g}_i^{-1}((-\infty, t_i')) \cap \tilde{h}^{-1}((-\infty, 1)).$$

If $x \in W_i$, then $\tilde{g}_i(x) = g_i(x) > t_i'$. If $x \in X \setminus \bigcup_{i=1}^{n} W_i$, then $\tilde{h}(x) = h(x) = 1$. Therefore, $U \cap X = \emptyset$. □

In the definition of a free topological linear space $L(X)$, specify a map $\eta : X \to L(X)$ instead of assuming $X \subset L(X)$ and replace the property (LE) with the following universality:

(LE′) For each map $f : X \to F$ of X to an arbitrary topological linear space F, there exists a unique linear map $\tilde{f} : L(X) \to F$ such that $\tilde{f}\eta = f$.



Then, we have the following:

Proposition 3.9.2 *Let X be a Tychonoff space and $\eta : X \to L(X)$ a map to a topological linear space $L(X)$ satisfying condition* (LE′)*. Then, η is a closed embedding and $\eta(X) \subset L(X)$ satisfies* (LE)*.*

Proof To see that η is injective, let $x \neq y \in X$. Then, there is a map $f : X \to \mathbf{I}$ with $f(x) = 0$ and $f(y) = 1$. By (LE′), we have a linear map $\tilde{f} : L(X) \to \mathbb{R}$ such that $\tilde{f}\eta = f$. Then, observe $\eta(x) \neq \eta(y)$.

To show that $\eta : X \to \eta(X)$ is open, let U be an open set in X. For each $x \in U$, there is a map $g : X \to \mathbf{I}$ such that $g(x) = 0$ and $g(X \setminus U) = 1$. By (LE′), we have a linear map $\tilde{g} : L(X) \to \mathbb{R}$ such that $\tilde{g}\eta = g$. Then, $V = \tilde{g}^{-1}((-1/2, 1/2))$ is an open neighborhood $\eta(x)$ in $L(X)$. Since $\eta^{-1}(V) = g^{-1}([0, 1/2)) \subset U$, it follows that $V \cap \eta(X) \subset \eta(U)$, hence $\eta(U)$ is a neighborhood of $\eta(x)$ in $\eta(X)$. This means that $\eta(U)$ is open in $\eta(X)$. Thus, $\eta : X \to \eta(X)$ is open. Consequently, $\eta : X \to L(X)$ is an embedding.

It is trivial to see that $\eta(X) \subset L(X)$ satisfies (LE). Due to Proposition 3.9.1(2), $\eta(X)$ is closed in $L(X)$, which means that η is a closed embedding. □

In Proposition 3.9.2, X can be identified with the subspace $\eta(X) \subset L(X)$ because η is an embedding. Thus, the **free topological linear space** over X can be defined

as $(L(X), \eta)$ satisfying (LE′). It should be noted that the uniqueness of $\tilde{f}$ in (LE′) is not used to prove that η is an embedding. In addition, the following can be easily shown:

- *The linear map $\tilde{f}$ in* (LE′) *is unique if and only if $L(X)$ is generated by $\eta(X)$.*

The "if" part is trivial. For the "only if" part, refer to the proof of Proposition 3.9.1(1).

Remark 3.9.1 In the definition of the free topological linear space $L(X)$ over X, we required $L(X)$ to contain X as a subspace but it is enough to suppose the continuity of the inclusion $X \subset L(X)$.

Now, we prove the following existence theorem:

Theorem 3.9.3 (EXISTENCE) *For every Tychonoff space X, there exists the free topological linear space $L(X)$ over X.*

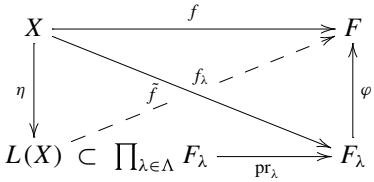
Proof First of all, we will construct a collection $\mathcal{F} = \{f_\lambda : X \to F_\lambda \mid \lambda \in \Lambda\}$ of maps into topological linear spaces F_λ so that, for an arbitrary topological linear space F and each map $f : X \to F$, there exist $\lambda \in \Lambda$ and a linear embedding $\varphi : F_\lambda \to F$ such that $\varphi f_\lambda = f$. For each $A \subset X$, let $\mathfrak{T}_A$ be the collection of topologies $\mathcal{T}$ on $\mathbb{R}^A_f$ such that $(\mathbb{R}^A_f, \mathcal{T})$ is a topological linear space. Then, the following is a desired collection:

$$\mathcal{F} = \bigcup_{A \subset X} \bigcup_{\mathcal{T} \in \mathfrak{T}_A} \mathrm{C}(X, (\mathbb{R}^A_f, \mathcal{T})).$$

Indeed, for an arbitrary topological linear space F and each map $f : X \to F$, let $B \subset f(X)$ be a basis of the linear span F_X of $f(X)$, that is, a maximal linearly independent subset of $f(X)$ (by Zorn's Lemma). Then, there is a subset $A \subset X$ such that $f|A : A \to B$ is a bijection (by Axiom of Choice). Note that $\mathbb{R}^A_f$ is linearly isomorphic to F_X, i.e., there is a linear isomorphism $\varphi : \mathbb{R}^A_f \to F_X$. We can take $\mathcal{T} \in \mathfrak{T}_A$ such that $\varphi : (\mathbb{R}^A_f, \mathcal{T}) \to F_X$ is a homeomorphism. Accordingly, we have $g = \varphi^{-1} f \in \mathrm{C}(X, (\mathbb{R}^A_f, \mathcal{T}))$ and a linear embedding $\varphi : \mathbb{R}^A_f \to F$ such that $f = \varphi g$.

Now, we consider the product space $\prod_{\lambda \in \Lambda} F_\lambda$, which is a topological linear space. Let $\eta : X \to \prod_{\lambda \in \Lambda} F_\lambda$ be a map defined by $\eta(x) = (f_\lambda(x))_{\lambda \in \Lambda}$. We define $L(X)$ as the linear span of $\eta(X)$ in $\prod_{\lambda \in \Lambda} F_\lambda$. Then, $(L(X), \eta)$ satisfies the condition (LE′) in Proposition 3.9.2. In fact, for an arbitrary topological linear space F and each map $f : X \to F$, there exists $\lambda \in \Lambda$ and a linear embedding $\varphi : F_\lambda \to F$ such that $\varphi f_\lambda = f$. Consequently, we have a linear map $\tilde{f} = \varphi \mathrm{pr}_\lambda | L(X) : L(X) \to F$ and

$$\tilde{f}\eta(x) = \varphi \mathrm{pr}_\lambda \eta(x) = \varphi f_\lambda(x) = f(x) \quad \text{for every } x \in X.$$



Because $L(X)$ is generated by $\eta(X)$, a linear map $\tilde{f} : L(X) \to F$ is uniquely determined by the condition that $\tilde{f}\eta = f$. Due to Proposition 3.9.2, η is an embedding. As noted after the proof of Proposition 3.9.2, identifying X with $\eta(X)$, we can consider $L(X)$ as the free topological linear space over X. □

Let X and Y be Tychonoff spaces. For each map $f : X \to Y$, we have a unique linear map $f_\natural : L(X) \to L(Y)$ that is an extension of f by (LE), that is, the following diagram commutes:

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ \cap & & \cap \\ L(X) & \xrightarrow[f_\natural]{} & L(Y) \end{array}$$

This extension is functorial, i.e., $(gf)_\natural = g_\natural f_\natural$ for every pair of maps $f : X \to Y$ and $g : Y \to Z$, and $\mathrm{id}_{L(X)} = (\mathrm{id}_X)_\natural$. Accordingly, we have a covariant functor from the category of Tychonoff spaces into the category of topological linear spaces. Consequently, every homeomorphism $f : X \to Y$ extends to a linear homeomorphism $f_\natural : L(X) \to L(Y)$.

Now, we will investigate some topological properties of the free topological linear space $L(X)$ over X. Since X is a Hamel basis for $L(X)$ (Proposition 3.9.1(1)), each $z \in L(X) \setminus \{\mathbf{0}\}$ can be uniquely represented as follows:

$$z = \sum_{i=1}^{n} t_i x_i, \ x_i \in X, \ t_i \in \mathbb{R} \setminus \{0\},$$

where $x_i \neq x_j$ if $i \neq j$. Such a representation is called the **irreducible representation** of z. The set $\{x_1, \ldots, x_n\}$ is called the **support** of z and is denoted by $\operatorname{supp} z$. For simplicity, we write $\nu(z) = \operatorname{card} \operatorname{supp} z \in \mathbb{N}$. Notice the following:

- *When we write* $\operatorname{supp} z = \{x_1, \ldots, x_{\nu(z)}\}$, *the condition that* $x_i \neq x_j$ *for* $i \neq j$ *is implicit in this description.*

For convenience, let $\operatorname{supp} \mathbf{0} = \emptyset$ and $\nu(\mathbf{0}) = 0$. Using this notation, we represent $z \in L(X)$ as follows:

$$z = \sum_{x \in X} z(x)x = \sum_{x \in \operatorname{supp} z} z(x)x,$$

where $z(x) \in \mathbb{R}$ is defined by

$$z(x) = \begin{cases} t_i & \text{if } x = x_i, \\ 0 & \text{otherwise.} \end{cases}$$

For each $n \in \mathbb{N}$, we define

$$L_n(X) \underset{\text{def}}{=} \left\{ z \in L(X) \;\middle|\; \nu(z) \leqslant n \right\} \quad \text{and}$$
$$S_n(X) \underset{\text{def}}{=} \left\{ z \in L(X) \;\middle|\; \nu(z) = n \right\} = L_n(X) \setminus L_{n-1}(X),$$

where $L_0(X) = \{\mathbf{0}\}$. Then, we have the following:

Proposition 3.9.4 *Let X be a Tychonoff space. Then, $L_n(X)$ is closed in $L(X)$ for each $n \in \omega$. As a result, $S_n(X)$ is open in $L_n(X)$ for each $n \in \mathbb{N}$.*

Proof For $z \in L(X) \setminus L_n(X)$, let $\operatorname{supp} z = \{x_1, \ldots, x_m\}$, where $m = \nu(z) > n$. Each x_i has an open neighborhood U_i in X such that $U_i \cap U_j = \emptyset$ if $i \neq j$. Taking maps $f_i : X \to \mathbf{I}$, $i = 1, \ldots, m$, so that $f(x_i) = 1$ and $f_i(X \setminus U_i) = \{0\}$, we define a map $f : X \to \mathbf{I}^m \subset \mathbb{R}^m$ by $f(x) = (f_1(x), \ldots, f_m(x))$. Then, $f(x_i) = \mathbf{e}_i$, $f(U_i) \subset \mathbf{I}\mathbf{e}_i$ for each $i = 1, \ldots, m$, and $f(X \setminus \bigcup_{i=1}^m U_i) = \{\mathbf{0}\}$. By (LE), f extends to a linear map $\tilde{f} : L(X) \to \mathbb{R}^m$. Then,

$$\tilde{f}(z) = \sum_{i=1}^m z(x_i) f(x_i) = \sum_{i=1}^m z(x_i)\mathbf{e}_i = (z(x_1), \ldots, z(x_m)) \in (\mathbb{R} \setminus \{0\})^m.$$

Hence, $\tilde{f}^{-1}((\mathbb{R} \setminus \{0\})^m)$ is an open neighborhood of z in $L(X)$. Then, it suffices to show that $z' \in \tilde{f}^{-1}((\mathbb{R} \setminus \{0\})^m)$ implies $z' \notin L_n(X)$, that is, $\nu(z') > n$. For each $i = 1, \ldots, m$,

$$\operatorname{pr}_i \tilde{f}(z') = \sum_{x \in X} z'(x) f_i(x) = \sum_{x \in U_i} z'(x) f_i(x).$$

Since $\operatorname{pr}_i \tilde{f}(z') \neq 0$, it follows that $z'(x) \neq 0$ for some $x \in U_i$, that is, $\operatorname{supp} z' \cap U_i \neq \emptyset$. This means that $\nu(z') = \operatorname{card} \operatorname{supp} z' \geqslant m > n$. Thus, we have the result. □

Remark 3.9.2 In Proposition 3.9.4, the case $n = 0$ means that $\{\mathbf{0}\}$ is closed in $L(X)$. This condition is contained in the fact that $L(X)$ is topological linear space.

For each $n \in \mathbb{N}$, we define a surjective map $\gamma_n : X^n \times \mathbb{R}^n \to L_n(X)$ as follows:

$$\gamma_n(x, \alpha) = \sum_{i=1}^n \alpha(i) x(i) \quad \text{for each } (x, \alpha) \in X^n \times \mathbb{R}^n.$$

As is easily observed, we have

$$\gamma_n^{-1}(S_n(X)) = \left\{ (x, \alpha) \in X^n \times (\mathbb{R} \setminus \{0\})^n \;\middle|\; x(i) \neq x(j) \;\text{ for } i \neq j \right\}.$$

The n-th symmetric group is denoted by $\mathfrak{S}_n$.[28] Then, we have the following:

[28] I.e., $\mathfrak{S}_n$ is the group of all bijections of $\{1, \ldots, n\}$ onto itself (i.e., permutations of $\{1, \ldots, n\}$).

Lemma 3.9.5 *Let $V_1, \dots, V_n$ be pairwise disjoint open sets in a Tychonoff space X and $(a_i, b_i) \subset \mathbb{R} \setminus \{0\}$, $i = 1, \dots, n$. Then, $U = \gamma_n(\prod_{i=1}^n V_i \times \prod_{i=1}^n (a_i, b_i))$ is open in $S_n(X)$ and*

$$\gamma_n^{-1}(U) = \bigcup_{\theta \in \mathfrak{S}_n} \left(\prod_{i=1}^n V_{\theta(i)} \times \prod_{i=1}^n (a_{\theta(i)}, b_{\theta(i)}) \right)$$
$$\left(= \bigoplus_{\theta \in \mathfrak{S}_n} \left(\prod_{i=1}^n V_{\theta(i)} \times \prod_{i=1}^n (a_{\theta(i)}, b_{\theta(i)}) \right) \right),$$

where each $\gamma_n | \prod_{i=1}^n V_{\theta(i)} \times \prod_{i=1}^n (a_{\theta(i)}, b_{\theta(i)})$ is a homeomorphism onto U.

Proof To show that U is open in $S_n(X)$, let $z = \sum_{i=1}^n z(x_i)x_i \in U$, where $x_i \in V_i$ and $z(x_i) \in (a_i, b_i)$ for each $i = 1, \dots, n$. As in the proof of Proposition 3.9.4, taking maps $f_i : X \to \mathbf{I}$, $i = 1, \dots, n$, so that $f_i(x_i) = 1$ and $f_i(X \setminus V_i) = \{0\}$, we define a map $f : X \to \mathbb{R}^n$ by $f(x) = (f_1(x), \dots, f_n(x))$. Then, $f(x_i) = \mathbf{e}_i$, $f(V_i) \subset \mathbf{I}\mathbf{e}_i$ for each $i = 1, \dots, n$. By (LE), f extends to a linear map $\tilde{f} : L(X) \to \mathbb{R}^n$. Observe that

$$\tilde{f}(z) = \sum_{i=1}^n z(x_i) f(x_i) = \sum_{i=1}^n z(x_i)\mathbf{e}_i = (z(x_1), \dots, z(x_n)) \in \prod_{i=1}^n (a_i, b_i).$$

For each $i = 1, \dots, n$, choose $c_i < d_i$ so that $z(x_i) \in (c_i, d_i) \subset [c_i, d_i] \subset (a_i, b_i)$. Then, $\tilde{f}^{-1}(\prod_{i=1}^n (c_i, d_i))$ is an open neighborhood of z in $L(X)$.

Choose $0 < s < 1$ so that $a_i < c_i/s$ and $d_i/s < b_i$ for every $i = 1, \dots, n$.[29] Each x_i has an open neighborhood $W_i \subset V_i$ such that $f_i(W_i) \subset (s, 1]$. As in the above, taking maps $g_i : X \to \mathbf{I}$, $i = 1, \dots, n$, so that $g_i(x_i) = 1$ and $g_i(X \setminus W_i) = \{0\}$, we define a map $g : X \to \mathbb{R}^n$ by $g(x) = (g_1(x), \dots, g_n(x))$. By (LE), g extends to a linear map $\tilde{g} : L(X) \to \mathbb{R}^n$. Since $\tilde{g}(z) = \tilde{f}(z) \in (\mathbb{R} \setminus \{0\})^n$, $\tilde{g}^{-1}((\mathbb{R} \setminus \{0\})^n)$ is an open neighborhood of z in $L(X)$. The following is an open neighborhood of z in $S_n(X)$:

$$U_z = S_n(X) \cap \tilde{f}^{-1}\big(\textstyle\prod_{i=1}^n (c_i, d_i)\big) \cap \tilde{g}^{-1}\big((\mathbb{R} \setminus \{0\})^n\big).$$

To see $U_z \subset U$, let $z' \in U_z$. For each $i = 1, \dots, n$,

$$\mathrm{pr}_i \tilde{g}(z') = \sum_{x \in X} z'(x) g_i(x) = \sum_{x \in W_i} z'(x) g_i(x).$$

Since $\mathrm{pr}_i \tilde{g}(z') \neq 0$, it follows that $z'(x_i') \neq 0$ for some $x_i' \in W_i$. Because $\nu(z') = n$, $\operatorname{supp} z' \cap W_i = \{x_i'\}$ and $\operatorname{supp} z' = \{x_1', \dots, x_n'\}$. Thus, we have $z' = \sum_{i=1}^n z'(x_i')x_i'$ and

$$\tilde{f}(z') = \sum_{i=1}^n z'(x_i') f_i(x_i') = (z(x_1') f_1(x_1'), \dots, z(x_n') f_n(x_n')) \in \prod_{i=1}^n (c_i, d_i).$$

[29] Note that $c/s > c$ if $c > 0$, and $c/s < c$ if $c < 0$.

Note that $s < f_i(x_i') \leqslant 1$ because $x_i' \in W_i$. Then, it follows that $z'(x_i') \in (a_i, b_i)$ for each $i = 1, \ldots, n$. Hence,

$$z' = \sum_{i=1}^{n} z'(x_i')x_i' \in \gamma_n\big(\textstyle\prod_{i=1}^n W_i \times \prod_{i=1}^n (a_i, b_i)\big) \subset U.$$

Consequently, $U_z \subset U$. Thus, U is a neighborhood of every $z \in U$ in $S_n(X)$, which means that U is open in $S_n(X)$.

Evidently, each $(x, \alpha) \in \gamma_n^{-1}(S_n(X))$ has a neighborhood basis consisting of sets $\prod_{i=1}^n V_i \times \prod_{i=1}^n (a_i, b_i)$, where $V_1, \ldots, V_n$ are pairwise disjoint open sets in X with $x(i) \in V_i$, and $\alpha(i) \in (a_i, b_i) \subset \mathbb{R} \setminus \{0\}$. As just proved above, $\gamma_n(\prod_{i=1}^n V_i \times \prod_{i=1}^n (a_i, b_i))$ is open in $S_n(X)$. Therefore, $\gamma_n | \gamma_n^{-1}(S_n(X))$ is an open map.

The following equality is trivial:

$$\gamma_n^{-1}(U) = \bigcup_{\theta \in \mathfrak{S}_n} \left(\prod_{i=1}^{n} V_{\theta(i)} \times \prod_{i=1}^{n} (a_{\theta(i)}, b_{\theta(i)}) \right).$$

Since X is a Hamel basis for $L(X)$, it follows that $\gamma_n | \prod_{i=1}^n V_{\theta(i)} \times \prod_{i=1}^n (a_{\theta(i)}, b_{\theta(i)})$ is injective. Thus, $\gamma_n | \prod_{i=1}^n V_{\theta(i)} \times \prod_{i=1}^n (a_{\theta(i)}, b_{\theta(i)})$ is a homeomorphism onto U as it is a bijective open map. □

Remark 3.9.3 Observe that Lemma 3.9.5 is valid even if each (a_i, b_i) can be replaced by $\mathbb{R} \setminus \{0\}$. In other words, for pairwise disjoint open sets $V_1, \ldots, V_n$ in X, $U = \gamma_n(\prod_{i=1}^n V_i \times (\mathbb{R} \setminus \{0\})^n)$ is open in $S_n(X)$ and

$$\gamma_n^{-1}(U) = \bigcup_{\theta \in \mathfrak{S}_n} \prod_{i=1}^{n} V_{\theta(i)} \times (\mathbb{R} \setminus \{0\})^n \left(= \bigoplus_{\theta \in \mathfrak{S}_n} \prod_{i=1}^{n} V_{\theta(i)} \times (\mathbb{R} \setminus \{0\})^n \right),$$

where each $\gamma_n | \prod_{i=1}^n V_{\theta(i)} \times (\mathbb{R} \setminus \{0\})^n$ is a homeomorphism onto U.

In general, a map $p : \widetilde{X} \to X$ is called a **covering projection** if each $x \in X$ has an open neighborhood U **evenly covered** by p, that is, there are disjoint open sets U_λ, $\lambda \in \Lambda$, such that $p^{-1}(U) = \bigcup_{\lambda \in \Lambda} U_\lambda$ and each $p | U_\lambda$ is a homeomorphism onto U. It should be noted that every covering projection $p : \widetilde{X} \to X$ is an open map.

Indeed, let $\mathcal{U}$ be an open cover of X such that each $U \in \mathcal{U}$ is evenly covered by p. Then, $p | p^{-1}(U)$ is an open map. Since $p^{-1}(\mathcal{U})$ is an open cover of $\widetilde{X}$, it is easy to see that p is an open map.

Therefore, Lemma 3.9.5 means the following:

Proposition 3.9.6 *Let X be a Tychonoff space. For each $n \in \mathbb{N}$, $\gamma_n | \gamma_n^{-1}(S_n(X))$ is a covering projection over $S_n(X)$ such that* $\operatorname{card} \gamma_n^{-1}(z) = n!$ *for each $z \in S_n(X)$. Moreover, each $z \in S_n(X)$ has an open neighborhood in $S_n(X)$ that is homeomorphic to an open set in $X^n \times \mathbb{R}^n$.* □

As a corollary, we have the following:

Corollary 3.9.7 *Let X be a Tychonoff space. Then, the following hold:*

(1) *If X is infinite, then $w(S_n(X)) = w(X)$ for each $n \in \mathbb{N}$.*
(2) *If X is locally compact, then so is each $S_n(X)$.*
(3) *If X is metrizable, then so is each $S_n(X)$.*

Sketch of Proof (1) and (2) are direct consequences of Proposition 3.9.6. For (3), to apply the Nagata–Smirnov Metrization Theorem (cf. 2.3.4), take a σ-locally finite open basis $\mathcal{B} = \bigcup_{i\in\mathbb{N}} \mathcal{B}_i$ for $\gamma_n^{-1}(S_n(X))$, where $\mathcal{B}_i$ is locally finite in $\gamma_n^{-1}(S_n(X))$. It is easy to see that $\gamma_n(\mathcal{B})$ is an open basis for $S_n(X)$. To verify the local finiteness of $\gamma_n(\mathcal{B}_i)$, let $z \in S_n(X)$. By Proposition 3.9.6, there exist an open neighborhood V of z in $S_n(X)$ and disjoint open sets V_y, $y \in \gamma_n^{-1}(z)$ such that $\gamma_n^{-1}(V) = \bigcup_{y\in\gamma_n^{-1}(z)} V_y$ and each $\gamma_n|V_y$ is a homeomorphism of V_y onto V. Then, each $y \in \gamma_n^{-1}(z)$ has an open neighborhood W_y contained in V_y such that $\mathcal{B}_i[W_y]$ is finite. Since $\gamma_n^{-1}(z)$ is finite, $W = \bigcap_{y\in\gamma_n^{-1}(z)} \gamma_n(W_y)$ is an open neighborhood of z in $S_n(X)$ and $\gamma_n(\mathcal{B}_i)[W]$ is finite.

The following is easy:

Proposition 3.9.8 *Let X be a Tychonoff space. Each $L_n(X)$ is nowhere dense in $L_{n+1}(X)$ if X has at least $n+1$ points.*

Sketch of Proof Since $L_n(X)$ is closed in $L_{n+1}(X)$ by Proposition 3.9.4, it suffices to show $\mathrm{int}_{L_{n+1}(X)} L_n(X) = \emptyset$. Let $z = \sum_{i=1}^{\nu(z)} z(x_i)x_i \in L_n(X)$, where $\nu(z) \leqslant n$. We can take points $x_{\nu(z)+1}, x_{\nu(z)+2}, \ldots, x_{n+1} \in X$ so that $x_i \neq x_j$ if $1 \leqslant i < j \leqslant n+1$. For any neighborhood U of z in $L_{n+1}(X)$, we can choose $\varepsilon > 0$ sufficiently small so that

$$z' = \gamma_{n+1}((x_1, \ldots, x_{n+1}), (z(x_1), \ldots, z(x_{\nu(z)}), \varepsilon, \ldots, \varepsilon)) \in U$$

because γ_{n+1} is continuous at $((x_1, \ldots, x_{n+1}), (z(x_1), \ldots, z(x_{\nu(z)}), 0, \ldots, 0))$. Then, $z' \in U \setminus L_n(X)$, which means $U \setminus L_n(X) \neq \emptyset$.

Choosing a point $* \in X$ as a base point, we consider a pointed space $X = (X, *)$. Then, we can define the direct limit $X^\infty = \varinjlim X^n$ of the closed tower $X \subset X^2 \subset X^3 \subset \cdots$, where each X^n is identified with $X^n \times \{(*, *, \ldots)\} \subset X_f^{\mathbb{N}}$ (see where $X_f^{\mathbb{N}}$ is defined just before Proposition 2.8.6). Moreover, identifying each $X^n \times \mathbb{R}^n$ with

$$\big(X^n \times \{(*, *, \ldots)\}\big) \times \big(\mathbb{R}^n \times \{(0, 0, \ldots)\}\big) \subset X_f^{\mathbb{N}} \times \mathbb{R}_f^{\mathbb{N}},$$

we have the direct limit $\varinjlim(X^n \times \mathbb{R}^n)$ of the following closed tower:

$$X \times \mathbb{R} \subset X^2 \times \mathbb{R}^2 \subset X^3 \times \mathbb{R}^3 \subset \cdots,$$

that is, $\varinjlim(X^n \times \mathbb{R}^n)$ is the space $\bigcup_{n\in\mathbb{N}}(X^n \times \mathbb{R}^n) = X_f^{\mathbb{N}} \times \mathbb{R}_f^{\mathbb{N}}$ with the weak topology with respect to the above tower.[30] Since $\gamma_{n+1}|X^n \times \mathbb{R}^n = \gamma_n$ for each $n \in \mathbb{N}$, we have a map $\gamma_\infty : \varinjlim(X^n \times \mathbb{R}^n) \to L(X)$, which is surjective because

[30] Due to Proposition 2.8.5, $\varinjlim(X^n \times \mathbb{R}^n) = \varinjlim X^n \times \varinjlim \mathbb{R}^n$ if X is locally compact. But, as seen in Remark 2.8.5, $\varinjlim(X^n \times \mathbb{R}^n) \neq \varinjlim X^n \times \varinjlim \mathbb{R}^n$ in general.

$$\gamma_\infty\big(\varinjlim(X^n \times \mathbb{R}^n)\big) = \bigcup_{n\in\mathbb{N}} \gamma_n(X^n \times \mathbb{R}^n) = \bigcup_{n\in\mathbb{N}} L_n(X) = L(X).$$

As is easily observed, for each $z \in L(X)$,

$$\gamma_\infty^{-1}(z) = \Big\{(y_1, \dots, y_m; s_1, \dots, s_m) \;\Big|\; m \geqslant \nu(z),\ \forall x \in X,\ z(x) = \sum_{y_i = x} s_i\Big\}$$
$$\in \bigcup_{n\in\mathbb{N}} (X^n \times \mathbb{R}^n) = X_f^{\mathbb{N}} \times \mathbb{R}_f^{\mathbb{N}}.$$

In the proof of the Existence Theorem 3.9.3, the free topological linear space $L(X)$ was constructed as a linear subspace of a product of topological linear spaces. However, we need a more concrete construction of $L(X)$ to study its topological structure and properties. As is seen from Proposition 3.9.1 (1), $L(X)$ can be obtained by topologizing the following linear space:

$$\mathbb{R}_f^X = \Big\{z \in \mathbb{R}^X \;\Big|\; z(x) = 0 \ \text{ except for finitely many } x \in X\Big\}.$$

In fact, as discussed just before Proposition 3.9.4, regarding $z \in L(X)$ as the function $z \in \mathbb{R}_f^X$ $(X \ni x \mapsto z(x) \in \mathbb{R})$ with finite support, we gave the following representation:

$$z = \sum_{x\in X} z(x)x = \sum_{x\in \operatorname{supp}(z)} z(x)x,$$

where each $x \in X$ is identified with $\delta_x \in \mathbb{R}_f^X$ defined by $\delta_x(x) = 1$ and $\delta_x(y) = 0$ for $y \neq x$. Then, for $z \in \mathbb{R}_f^X$, we have

$$\operatorname{supp} z = z^{-1}(\mathbb{R} \setminus \{0\}) \ \text{ and } \ \nu(z) = \operatorname{card} z^{-1}(\mathbb{R} \setminus \{0\}).$$

In the following, we regard $L(X)$ as the linear space $\mathbb{R}_f^X$ topologized so that the algebraic operations are continuous and (LE) is satisfied, where X is identified with $\{\delta_x \mid x \in X\}$.[31]

Let $L_\infty(X)$ be the space $\mathbb{R}_f^X$ with the quotient topology induced by the surjection $\gamma_\infty : \varinjlim(X^n \times \mathbb{R}^n) \to \mathbb{R}_f^X$, i.e., $\gamma_\infty : \varinjlim(X^n \times \mathbb{R}^n) \to L_\infty(X)$ is a quotient map. We regard X as a subspace of $\varinjlim(X^n \times \mathbb{R}^n)$ by the natural identification

$$X = X \times \{1\} \subset X \times \mathbb{R} \subset \varinjlim(X^n \times \mathbb{R}^n).$$

[31] With respect to the product topology of $\mathbb{R}_f^X$, $X \ni x \mapsto \delta_x \in \mathbb{R}_f^X \subset \mathbb{R}^X$ is not continuous at a non-isolated point $x \in X$. Indeed, $\mathrm{pr}_x^{-1}((0, \infty))$ is an open neighborhood of δ_x in $\mathbb{R}^X$ but any neighborhood U of x in X contains $y \neq x$ and $\delta_y \notin \mathrm{pr}_x^{-1}((0, \infty))$ because $\mathrm{pr}_x(\delta_y) = \delta_y(x) = 0$.

Then, the inclusion $X \subset L_\infty(X)$ is continuous because it is just the restriction of $\gamma_\infty : \varinjlim(X^n \times \mathbb{R}^n) \to L_\infty(X)$. On the other hand, $\mathrm{id} : L_\infty(X) \to L(X)$ is also continuous because $\gamma_\infty : \varinjlim(X^n \times \mathbb{R}^n) \to L(X)$ is continuous. In particular, $L_\infty(X)$ is Hausdorff.

If $L_\infty(X)$ is a topological linear space, then $\mathrm{id} : L(X) \to L_\infty(X)$ is also continuous by the property (LE) of $L(X)$, which means $L_\infty(X) = L(X)$ as topological linear spaces, that is, $\gamma_\infty : \varinjlim(X^n \times \mathbb{R}^n) \to L(X)$ is a quotient map.

Note For an arbitrary topological linear space F, every map $f : X \to F$ uniquely extends to a linear function $\tilde{f} : L_\infty(X) \to F$, which is defined as follows:

$$\tilde{f}\gamma_\infty(x, \alpha) = \sum_{i=1}^{n} \alpha(i) f(x(i)) \quad \text{for } (x, \alpha) \in X^n \times \mathbb{R}^n.$$

Evidently, every $\tilde{f}\gamma_\infty | X^n \times \mathbb{R}^n = \tilde{f}\gamma_n$ is continuous. Hence, $\tilde{f}\gamma_\infty : \varinjlim(X^n \times \mathbb{R}^n) \to F$ is continuous, which means that $\tilde{f}$ is continuous. Thus, $L_\infty(X)$ satisfies (LE′). If $L_\infty(X)$ is a topological linear space, then $L_\infty(X)$ is a free topological linear space over X (cf. Proposition 3.9.2 and Remark 3.9.1), which means $L_\infty(X) = L(X)$ as topological linear spaces by the uniqueness of $L(X)$.

The continuity of scalar multiplication is trivial:

Proposition 3.9.9 *For every Tychonoff space X, the scalar multiplication $\mathfrak{m} : L_\infty(X) \times \mathbb{R} \to L_\infty(X)$ is continuous.*

Proof The following diagram is commutative:

$$\begin{array}{ccc} \varinjlim(X^n \times \mathbb{R}^n) \times \mathbb{R} & \xrightarrow{\bar{\mathfrak{m}}} & \varinjlim(X^n \times \mathbb{R}^n) \\ \downarrow{\scriptstyle \gamma_\infty \times \mathrm{id}} & & \downarrow{\scriptstyle \gamma_\infty} \\ L_\infty(X) \times \mathbb{R} & \xrightarrow[\mathfrak{m}]{} & L_\infty(X), \end{array}$$

where the top $\bar{\mathfrak{m}}$ is a map induced by the following maps:

$$(X^n \times \mathbb{R}^n) \times \mathbb{R} \ni ((x, \alpha), t) \mapsto (x, t\alpha) \in X^n \times \mathbb{R}^n, \ n \in \mathbb{N}.$$

Since $\mathbb{R}$ is locally compact, $\gamma_\infty \times \mathrm{id}$ is a quotient map by Proposition 2.1.9. Thus, the continuity of $\mathfrak{m}$ follows from that of $\mathfrak{m}(\gamma_\infty \times \mathrm{id}) = \gamma_\infty \bar{\mathfrak{m}}$. □

One might expect the continuity of addition $\mathfrak{a} : L_\infty(X) \times L_\infty(X) \to L_\infty(X)$ can be proved in the same way as the above. However, it is unknown whether $\gamma_\infty \times \gamma_\infty$ is a quotient map or not.

In fact, there is a quotient map p such that $p \times p$ is not quotient. Consider the quotient map $q : \mathbb{N} \times \mathbf{I} \to X = (\mathbb{N} \times \mathbf{I})/(\mathbb{N} \times \{0\})$ observed in Remark 2.1.1, where $q \times \mathrm{id}_\mathbb{Q}$ is not a quotient map. Then, $p = q \oplus \mathrm{id}_\mathbb{Q} : (\mathbb{N} \times \mathbf{I}) \oplus \mathbb{Q} \to X \oplus \mathbb{Q}$ is a quotient map. But, as is easily observed, $p \times p$ is not a quotient map.

Thus, a different approach is needed. Under the assumption that X *is the direct limit of a tower of compact spaces*, we will show the continuity of $\mathfrak{a}$. We begin by showing the following:

Proposition 3.9.10 *Let* $X = \varinjlim X_n$ *with a base point* $* \in X_1$, *where* $X_1 \subset X_2 \subset \cdots$ *is a tower of locally compact spaces. Then,* $X^\infty = \varinjlim X^n = \varinjlim X_n^n$.

Proof Observe that $\varinjlim X^n = \varinjlim X_n^n = X_f^{\mathbb{N}}$ as sets. The continuity of $\mathrm{id} : \varinjlim X_n^n \to \varinjlim X^n$ is obvious. To show that a closed set A in $\varinjlim X_n^n$ is closed in $\varinjlim X^n$, we have to verify that $A \cap X^n$ is closed in X^n for each $n \in \mathbb{N}$. Note that $X^n = (\varinjlim X_k)^n = \varinjlim X_k^n$ by Proposition 2.8.5. For each $k \geqslant n$, since $A \cap X_k^k$ is closed in X_k^k and $X_k^n \subset X_k^k$, it follows that $(A \cap X^n) \cap X_k^n = (A \cap X_k^k) \cap X_k^n$ is closed in X_k^n. For $k < n$, since $A \cap X_n^n$ is closed in X_n^n and $X_k^n \subset X_n^n$, it follows that $(A \cap X^n) \cap X_k^n = (A \cap X_n^n) \cap X_k^n$ is closed in X_k^n. Hence, $A \cap X^n$ is closed in X^n. □

When X is the direct limit of a tower $X_1 \subset X_2 \subset \cdots$ of locally compact spaces, $X_1 \times \mathbb{R} \subset X_2 \times \mathbb{R} \subset \cdots$ is a tower of locally compact spaces with $X \times \mathbb{R} = \varinjlim(X_n \times \mathbb{R})$. We can apply Propositions 3.9.10 and 2.8.5 to obtain

$$\begin{aligned}\varinjlim(X^n \times \mathbb{R}^n) &= \varinjlim(X \times \mathbb{R})^n = \varinjlim(X_n \times \mathbb{R})^n \\ &= \varinjlim(X_n^n \times \mathbb{R}^n) = \varinjlim X_n^n \times \varinjlim \mathbb{R}^n = X^\infty \times \mathbb{R}^\infty,\end{aligned}$$

where we identify $X^n \times \mathbb{R}^n = (X \times \mathbb{R})^n$ (or $X_n^n \times \mathbb{R}^n = (X_n \times \mathbb{R})^n$) by natural identification. Thus, we have

Corollary 3.9.11 *Let* $X = \varinjlim X_n$ *with a base point* $* \in X_1$, *where* $X_1 \subset X_2 \subset \cdots$ *is a tower of locally compact spaces. Then,* $\varinjlim(X^n \times \mathbb{R}^n) = X^\infty \times \mathbb{R}^\infty$. □

When X is the direct limit of a tower $X_1 \subset X_2 \subset \cdots$ of compact spaces, we can apply Proposition 2.8.5 to obtain

$$X \times \mathbb{R} = \varinjlim X_n \times \varinjlim[-n, n] = \varinjlim(X_n \times [-n, n]),$$

where each $X_n \times [-n, n]$ is compact. Applying Proposition 3.9.10 as above, we can obtain

$$\varinjlim(X^n \times \mathbb{R}^n) = \varinjlim(X \times \mathbb{R})^n = \varinjlim(X_n \times [-n, n])^n = \varinjlim(X_n^n \times [-n, n]^n),$$

where we identify $X^n \times \mathbb{R}^n = (X \times \mathbb{R})^n$ and $X_n^n \times [-n, n]^n = (X \times [-n, n])^n$.

For each $n \in \mathbb{N}$, let $\hat{L}_n$ be a compact subspace of $L_n(X)$ defined as follows:

$$\hat{L}_n = \gamma_\infty(X_n^n \times [-n, n]^n) \subset L_n(X) = \gamma_\infty(X^n \times \mathbb{R}^n) \subset L_\infty(X),$$

where $\bigcup_{n \in \mathbb{N}} \hat{L}_n = L_\infty(X)$. For each $n \in \mathbb{N}$, let

$$\hat{\gamma}_n = \gamma_\infty | X_n^n \times [-n, n]^n : X_n^n \times [-n, n]^n \to \hat{L}_n.$$

Since $X_n^n \times [-n, n]^n$ is compact, $\hat{\gamma}_n$ is a quotient map.[32] Then, we have the following:

Lemma 3.9.12 *Let $X = \varinjlim X_n$ with a base point $* \in X_1$, where $X_1 \subset X_2 \subset \cdots$ is a tower of compact spaces. The quotient topology of $L_\infty(X)$ induced by $\gamma_\infty : \varinjlim(X^n \times \mathbb{R}^n) \to L_\infty(X)$ coincides with the weak topology with respect to the tower $\hat{L}_1 \subset \hat{L}_2 \subset \cdots$, that is, $L_\infty(X) = \varinjlim \hat{L}_n$.*

Proof Let $A \subset L_\infty(X) = \mathbb{R}_f^X$. For each $n \in \mathbb{N}$,

$$\gamma_\infty^{-1}(A) \cap (X_n^n \times [-n, n]^n) = \gamma_\infty^{-1}(A \cap \hat{L}_n) \cap (X_n^n \times [-n, n]^n) = \hat{\gamma}_n^{-1}(A \cap \hat{L}_n).$$

Then, the following equivalences hold:

$$\begin{aligned}
A \text{ is closed in } L_\infty(X) &\Leftrightarrow \gamma_\infty^{-1}(A) \text{ is closed in } \varinjlim(X^n \times \mathbb{R}^n) \\
&\qquad = \varinjlim(X_n^n \times [-n, n]^n) \\
&\Leftrightarrow \forall n \in \mathbb{N}, \gamma_\infty^{-1}(A) \cap (X_n^n \times [-n, n]^n) \\
&\qquad \text{is closed in } X_n^n \times [-n, n]^n \\
&\Leftrightarrow \forall n \in \mathbb{N}, \hat{\gamma}_n^{-1}(A \cap \hat{L}_n) \text{ is closed in } X_n^n \times [-n, n]^n \\
&\Leftrightarrow \forall n \in \mathbb{N}, A \cap \hat{L}_n \text{ is closed in } \hat{L}_n \\
&\Leftrightarrow A \text{ is closed in } \varinjlim \hat{L}_n.
\end{aligned}$$

Thus, the topologies of $L_\infty(X)$ and $\varinjlim \hat{L}_n$ coincide. □

Now, we can prove the continuity of the addition $\mathfrak{a} : L_\infty(X) \times L_\infty(X) \to L_\infty(X)$ as follows: By virtue of Lemma 3.9.12 and Proposition 2.8.5,

$$L_\infty(X) \times L_\infty(X) = \varinjlim \hat{L}_n \times \varinjlim \hat{L}_n = \varinjlim(\hat{L}_n \times \hat{L}_n).$$

So, to show the continuity of $\mathfrak{a}$, it suffices to verify the continuity of restrictions $\mathfrak{a}_n = \mathfrak{a} | \hat{L}_n \times \hat{L}_n$, $n \in \mathbb{N}$, where $\mathfrak{a}_n(\hat{L}_n \times \hat{L}_n) \subset \hat{L}_{2n}$. The following diagram is commutative:

$$\begin{array}{ccc}
(X_n^n \times [-n, n]^n) \times (X_n^n \times [-n, n]^n) & \xrightarrow{\bar{\mathfrak{a}}_n} & X_{2n}^{2n} \times [-2n, 2n]^{2n} \\
\downarrow{\scriptstyle \hat{\gamma}_n \times \hat{\gamma}_n} & & \downarrow{\scriptstyle \hat{\gamma}_{2n}} \\
\hat{L}_n \times \hat{L}_n & \xrightarrow[\mathfrak{a}_n]{} & \hat{L}_{2n}
\end{array}$$

[32] Note that $X_n^n \times [-n, n]^n$ is not saturated with respect to γ_∞, that is, $\gamma_\infty^{-1}(\gamma_\infty(X_n^n \times [-n, n]^n)) \neq X_n^n \times [-n, n]^n$.

where $\bar{\mathfrak{a}}_n((x,\alpha),(x',\alpha')) = ((x,x'),(\alpha,\alpha'))$, and the left-hand map $\hat{\gamma}_n \times \hat{\gamma}_n$ is quotient because it is surjective and closed. Hence, $\mathfrak{a}_n$ is continuous.

Note In the above, $\bar{\mathfrak{a}}_n$ does not induce

$$\bar{\mathfrak{a}} : \varinjlim(X_n^n \times [-n,n]^n) \times \varinjlim(X_n^n \times [-n,n]^n) \to \varinjlim(X_{2n}^{2n} \times [-2n,2n]^{2n}).$$

However, to make it so, we can change the definition of $\bar{\mathfrak{a}}_n$ as follows:

$$\bar{\mathfrak{a}}_n((x,\alpha),(x',\alpha')) = \big((x(1),x'(1),\dots,x(n),x'(n)),\\ (\alpha(1),\alpha'(1),\dots,\alpha(n),\alpha'(n)\big).$$

We can conclude that $L_\infty(X)$ is a topological linear space. Summarizing the above results on $L_\infty(X)$, we have

Theorem 3.9.13 *Let $X = \varinjlim X_n$ be the direct limit of a tower of compact spaces. Then, $\gamma_\infty : \varinjlim(X^n \times \mathbb{R}^n) \to L(X)$ is a quotient map and $L(X) = \varinjlim \gamma_n(X_n^n \times [-n,n]^n)$ is the direct limit of a tower of compact spaces. In particular, when X itself is compact, $L(X) = \varinjlim \gamma_n(X^n \times [-n,n]^n)$.* □

It should be noted that if $X = \varinjlim X_n$ is the direct limit of a tower of compacta (i.e., compact metrizable spaces), then $L(X)$ is also the direct limit of a tower of compacta. Indeed, each $\hat{L}_n$ is compact metrizable because $\hat{\gamma}_n : X_n^n \times [-n,n]^n \to \hat{L}_n$ is a perfect map (see Proposition 2.4.4 (1)). Thus, we have the following corollary (cf. Corollary 2.8.10 and Theorem 2.2.8):

Corollary 3.9.14 *Let $X = \varinjlim X_n$ be the direct limit of a tower of compacta. Then, $L(X) = \varinjlim \gamma_n(X_n^n \times [-n,n]^n)$ is the direct limit of a tower of compacta, which is a perfectly normal and paracompact k-space, hence it is hereditarily paracompact (so hereditarily normal = completely normal).* □

In Sect. 7.12, we will construct a metrizable linear space that is not an absolute extensor for metrizable spaces. In the construction, the free topological linear space $L(X)$ over a compactum X has an important role. In Sect. 7.11, we will study more topological and geometrical structures of $L(X)$.

Notes for Chap. 3

There are lots of good textbooks for studying topological linear spaces. The following classical book by Köthe is still a very good source on this subject. The textbook by Kelley and Namioka is also recommended by many people. Besides these two books, Day's textbook is a good reference for normed linear spaces, as is Valentine's book for convex sets. Concerning non-locally convex F-spaces and Roberts' example (a compact convex set with no extreme points), one can refer to the book by Kalton, Peck, and Roberts.

- G. Köthe, *Topological Vector Spaces, I*, GMW **159** (Springer–Verlag, New York, 1969), English translation of the 1966 original German edition, GMW 107, published by Springer–Verlag
- J.L. Kelley and I. Namioka, *Linear Topological Spaces*, GTM **36** (Springer–Verlag, New York, 1976), Reprint of the 1963 original edition published by Van Nostrand Co.
- M.M. Day, *Normed Linear Spaces, 3rd edition*, EMG **21** (Springer–Verlag, Berlin, 1973)
- F.A. Valentine, *Convex Sets* (R.E. Krieger Publ. Co., New York, 1976), Reprint of the 1964 original edition published by McGraw–Hill Inc.
- N.J. Kalton, N.T. Peck and J.W. Roberts, *An F-space Sampler*, London Math. Soc. Lect. Note Ser. **89** (Cambridge Univ. Press, Cambridge, 1984)

For a systematic and comprehensive study on continuous selections, refer to the following book by Repovš and Semenov, which is written in instructive style.

- D. Repovš and P.V. Semenov, *Continuous Selections of Multivalued Mappings*, MIA **455** (Kluwer Acad. Publ., Dordrecht, 1998)

Proposition 3.5.1 (or Corollary 3.5.2) is well known but Proposition 3.5.8 and the example in Remark 3.5.1 have not been known. In Theorem 3.6.4, the construction of a metric d from d_1 is due to Eidelheit and Mazur [2].

The results of Sect. 3.8 are contained in the first part of Michael's paper [6], which consists of three parts. The second and third parts of [6] treated the finite-dimensional case, which is deeply related with the concept discussed in Sect. 6.11 (also refer to [7]) and will not be treated in this book. The zero-dimensional case will be treated in Sect. 7.2. For more information about this section, see the above book by Repovš and Semenov.

In the definition of the free topological linear spaces over a space X, replacing the term "topological linear space" with "locally convex topological linear space," we can define the **free locally convex topological linear space** over a space X, which is a locally convex topological linear space with X a Hamel basis. The free locally convex topological linear space over X was introduced and studied by Markov [5]. Graev [4] defined the free locally convex topological linear space over a pointed space X with a base point x_0, which is a locally convex topological linear space with $\mathbf{0} = x_0$ and $X \setminus \{x_0\}$ a Hamel basis. Thus, initially, free locally convex topological linear spaces have been investigated. Recently, non-locally convex topological linear spaces have been gathering attention. The paper [3] by Gabriyelyan and Morris was published after publication of the previous edition of this book, in which Theorem 3.9.13 was proved, but Cauty already implicitly proved in [1] when X is a compactum.

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Chapter 4
Simplicial Complexes and Polyhedra



In this chapter, we introduce and demonstrate the basic concepts and properties of simplicial complexes, where the local finiteness is not assumed. The importance and usefulness of simplicial complexes lie in the fact that they can be used to approximate and explore (topological) spaces.

A polyhedron is the underlying space of a simplicial complex, which has two typical topologies, the so-called Whitehead (or weak) topology and the metric topology. We prove J.H.C. Whitehead's Theorem on small subdivisions and its metric topology version. The paracompactness of the Whitehead topology is shown. We also demonstrate basic properties of simplicial maps and PL (piecewise linear) maps. A map between polyhedra is PL if it is simplicial with respect to some triangulations but the converse is not true in general. We will give a characterization of such a PL map. Regular neighborhoods are very useful and basic tools in PL Topology, on which basic results will be shown for arbitrary polyhedra. We also show that a simplicial complex K is a combinatorial manifold if and only if the polyhedron $|K|$ is a PL manifold. It is shown that every paracompact space can be represented as the limit of an inverse system of polyhedra (with the weak topology) and that every completely metrizable space can be represented as the limit of an inverse sequence of locally finite-dimensional polyhedra with the metric topology, where the bonding maps of the inverse system (sequence) are PL. We show that a compactum X cannot be represented as the limit of any inverse sequence of finite simplicial complexes with simplicial bonding maps if $\dim X > 0$ and X contains no arcs. In addition, we give a proof of the Whitehead–Milnor Theorem on the homotopy type of simplicial complexes. It is also proved that a map between polyhedra is a homotopy equivalence if it induces isomorphisms between their homotopy groups.

This chapter is based on Chaps. 2 and 3. In particular, we employ the theory of convex sets and the related concepts discussed in Sects. 3.1, 3.2, and 3.5.

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4.1 Simplexes and Cells

Throughout this section, let E and E' be linear spaces. The convex hull $C = \langle A \rangle$ of a non-empty finite subset $A \subset E$ is called a **cell** (or a **linear cell**),[1] where the dimension of C is defined as the dimension of the flat hull $\mathrm{fl}\, C$, i.e., $\dim C = \dim \mathrm{fl}\, A < \mathrm{card}\, A < \infty$ (cf. Sect. 3.2). Recall that $\langle v_1, \ldots, v_n \rangle$ denotes $\langle \{v_1, \ldots, v_n\} \rangle$. For each $n \in \mathbb{N}$, when $v_1, \ldots, v_n$ are affinely independent in E, $\sigma = \langle v_1, \ldots, v_n \rangle$ is called a **simplex**, each v_i is called a **vertex** of σ, and $\dim \sigma = n - 1$. Every simplex is a cell. An n-dimensional cell and an n-dimensional simplex are simply called an ***n*-cell** and an ***n*-simplex**, respectively. If $\langle v_1, \ldots, v_{n+1} \rangle$ is an n-simplex, then $v_1, \ldots, v_{n+1}$ are affinely independent, hence they are vertices of the simplex. If $\dim \langle A \rangle = n$ and $\mathrm{card}\, A = n + 1$, then A is affinely independent and $\langle A \rangle$ is an n-simplex. The 0- and 1-cells are the same as 0- and 1-simplexes, which are singletons and line segments, respectively. The affine image of a cell (i.e., the image $f(C)$ of an affine function $f : C \to E'$) is always a cell and every cell is the affine image of some simplex. But, the affine image of a simplex σ is, in general, not a simplex. When $\sigma = \langle v_1, \ldots, v_{n+1} \rangle$ is an n-simplex and $f : \sigma \to E'$ is affine, $f(\sigma) = \langle f(v_1), \ldots, f(v_{n+1}) \rangle$ but $f(\sigma)$ is not necessarily a simplex if $\dim f(\sigma) < n$.

The *radial* interior $\mathrm{rint}\, C$ and the *radial* boundary ∂C of a cell C are just called the **interior** and the **boundary** of C, which are defined without topology, that is,

$$\mathrm{rint}\, C = \left\{ x \in C \;\middle|\; \forall y \in C,\ \exists \delta > 0 \ \text{ such that } \ (1+\delta)x - \delta y \in C \right\}$$

and $\partial C = C \setminus \mathrm{rint}\, C$. Then, $\mathrm{rint}\, C \neq \emptyset$ by Proposition 3.2.10. It should be noted that C itself is equal to the radial closure defined as follows:

$$\mathrm{rcl}\, C = \left\{ x \in E \;\middle|\; \exists y \in C \ \text{ such that } \ 0 < \forall t < 1,\ (1+t)x - ty \in C \right\}.$$

In fact, $\mathrm{rcl}\, \mathrm{rint}\, C = \mathrm{rcl}\, C = C$ (cf. Propositions 3.2.9 (iii) and 3.5.8). For $x \neq y \in E$,

$$\partial \langle x, y \rangle = \{x, y\},\ \ \mathrm{rint} \langle x, y \rangle = \langle x, y \rangle \setminus \{x, y\}.$$

Then, we can also write as follows:

$$\mathrm{rint}\, C = \left\{ x \in C \;\middle|\; \forall y \in C,\ \exists z \in C \ \text{ such that } \ x \in \mathrm{rint} \langle y, z \rangle \right\}.$$

According to Proposition 3.5.1, the flat hull $\mathrm{fl}\, C$ has a unique topology such that the following operation is continuous:

$$\mathrm{fl}\, C \times \mathrm{fl}\, C \times \mathbb{R} \ni (x, y, t) \mapsto (1-t)x + ty \in \mathrm{fl}\, C,$$

and $\mathrm{fl}\, C$ is affinely homeomorphic to $\mathbb{R}^n$, where $n = \dim C = \dim \mathrm{fl}\, C$. With respect to this topology, as is shown in Proposition 3.5.5,

[1] More precisely, it is called a **convex linear cell**.

$$\operatorname{rint} C = \operatorname{int}_{\operatorname{fl} C} C \quad \text{and} \quad \partial C = \operatorname{bd}_{\operatorname{fl} C} C.$$

Moreover, we have $(C, \partial C) \approx (\mathbf{B}^n, \mathbf{S}^{n-1})$ by Corollary 3.5.6. For every cell (or every simplex) C, we always consider the topology to be inherited from this unique topology of $\operatorname{fl} C$.

In fact, as seen in Proposition 3.5.8, C itself has a unique topology such that the following operation is continuous:

$$C \times C \times \mathbf{I} \ni (x, y, t) \mapsto (1-t)x + ty \in C.$$

For the standard n-simplex $\Delta^n \subset \mathbb{R}^{n+1}$,

$$\operatorname{rint} \Delta^n = \left\{ z \in \Delta^n \;\middle|\; z(i) > 0 \;\text{ for every }\; i = 1, \ldots, n+1 \right\} \text{ and}$$
$$\partial \Delta^n = \left\{ z \in \Delta^n \;\middle|\; z(i) = 0 \;\text{ for some }\; i = 1, \ldots, n+1 \right\}.$$

For an n-simplex $\sigma = \langle v_1, \ldots, v_{n+1} \rangle \subset E$, we have the *natural* affine homeomorphism $\delta_\sigma : \operatorname{fl} \Delta^n \to \operatorname{fl} \sigma$ defined as follows:

$$\delta_\sigma(z) = \sum_{i=1}^{n+1} z(i) v_i \quad \text{for each } z \in \operatorname{fl} \Delta^n.$$

Then, $\sigma = \delta_\sigma(\Delta^n)$, $\operatorname{rint} \sigma = \delta_\sigma(\operatorname{rint} \Delta^n)$, and $\partial \sigma = \delta_\sigma(\partial \Delta^n)$. The **barycenter** $\hat{\sigma}$ of σ is defined as follows:

$$\hat{\sigma} = \delta_\sigma\left(\frac{1}{n+1}, \ldots, \frac{1}{n+1} \right) = \sum_{i=1}^{n+1} \frac{1}{n+1} \cdot v_i.$$

The affine homeomorphism δ_σ is not unique because it depends on how the vertices $v_1, \ldots, v_{n+1}$ are ordered but the barycenter $\hat{\sigma}$ is uniquely determined independent of the order of the vertices.

When $v_1, \ldots, v_{n+1}$ are not affinely independent, the map defined in the same way as δ_σ is not a homeomorphism, but we do have the following result:

Proposition 4.1.1 *For every finite subset $A = \{v_1, \ldots, v_n\} \subset E$,*

$$\operatorname{rint} \langle A \rangle = \left\{ \textstyle\sum_{i=1}^n z(i) v_i \;\middle|\; z \in \operatorname{rint} \Delta^{n-1} \right\}.$$

Proof Take $z_0 \in \operatorname{rint} \Delta^{n-1}$ and let $x_0 = \sum_{i=1}^n z_0(i) v_i \in \langle A \rangle$. For each $x \in \operatorname{rint} \langle A \rangle$, we have $x_1 \in \langle A \rangle$ and $0 < \delta < 1$ such that $x = (1-\delta)x_0 + \delta x_1$. Write $x_1 = \sum_{i=1}^n z_1(i) v_i$, $z_1 \in \Delta^{n-1}$. Then, $z = (1-\delta)z_0 + \delta z_1 \in \operatorname{rint} \Delta^{n-1}$ (Proposition 3.2.6) and

$$x = (1-\delta) \sum_{i=1}^n z_0(i) v_i + \delta \sum_{i=1}^n z_1(i) v_i = \sum_{i=1}^n z(i) v_i.$$

Conversely, for each $z \in \operatorname{rint} \Delta^{n-1}$, we show $\sum_{i=1}^n z(i)v_i \in \operatorname{rint}\langle A\rangle$. Each $y \in \langle A\rangle$ can be written as $y = \sum_{i=1}^n z_0(i)v_i$ for some $z_0 \in \Delta^{n-1}$. Then, we have $z_1 \in \Delta^{n-1}$ and $0 < \delta < 1$ such that $z = (1-\delta)z_0 + \delta z_1$. Let $y_1 = \sum_{i=1}^n z_1(i)v_i \in \langle A\rangle$. It follows that

$$\sum_{i=1}^n z(i)v_i = (1-\delta)\sum_{i=1}^n z_0(i)v_i + \delta\sum_{i=1}^n z_1(i)v_i = (1-\delta)y + \delta y_1.$$

This means that $\sum_{i=1}^n z(i)v_i \in \operatorname{rint}\langle A\rangle$. □

Proposition 4.1.2 *Each cell $C \subset E$ has the smallest finite set $C^{(0)}$ such that $\langle C^{(0)}\rangle = C$, i.e., $\langle A\rangle = C \Rightarrow C^{(0)} \subset A$. Additionally, $C^{(0)} \subset \partial C$ if $\dim C > 0$.*

Proof By the definition of a cell, we can easily find a minimal finite set $\{v_1, \ldots, v_n\} \subset C$ such that $\langle v_1, \ldots, v_n\rangle = C$, i.e., $\langle B\rangle \neq C$ if $B \subsetneqq \{v_1, \ldots, v_n\}$, where $v_i \neq v_j$ if $i \neq j$. Then, it suffices to show that $\{v_1, \ldots, v_n\}$ is the minimum, i.e.,

$$\langle A\rangle = C \;\Rightarrow\; \{v_1, \ldots, v_n\} \subset A. \tag{♯}$$

Before showing this, we prove the following:

$$v_1 = \sum_{i=1}^n z(i)v_i,\ z \in \Delta^{n-1} \;\Rightarrow\; z(1) = 1 \quad (\text{i.e., } z = \mathbf{e}_1). \tag{♮}$$

Indeed, when $z(1) \neq 1$, i.e., $0 \leqslant z(1) < 1$, since $(1 - z(1))v_1 = \sum_{i=2}^n z(i)v_i$, we have

$$v_1 = \sum_{i=2}^n \frac{z(i)}{1-z(1)}v_i \quad \text{and} \quad \sum_{i=2}^n \frac{z(i)}{1-z(1)} = 1.$$

Then, it follows that $\langle v_2, \ldots, v_n\rangle = C$, which contradicts the choice of $\{v_1, \ldots, v_n\}$.

To see that (♯), on the contrary, suppose that $\langle A\rangle = C$ but $\{v_1, \ldots, v_n\} \not\subset A$. Then, we may assume $v_1 \notin A$. We can write

$$v_1 = \sum_{i=1}^m z(i)x_i,\ x_1, \ldots, x_m \in A,\ z \in \operatorname{rint} \Delta^{m-1},$$

where $x_i \neq x_j$ if $i \neq j$. Since $v_1 \notin A$, we have $m \geqslant 2$. Define

$$v' = (z(1) + \varepsilon)x_1 + (z(2) - \varepsilon)x_2 + \sum_{i=3}^m z(i)x_i,$$

$$v'' = (z(1) - \varepsilon)x_1 + (z(2) + \varepsilon)x_2 + \sum_{i=3}^m z(i)x_i,$$

where $\varepsilon > 0$ is chosen so that $z(1) \pm \varepsilon,\ z(2) \pm \varepsilon \in (0, 1)$. Then, $v' \neq v''$ because $v' - v'' = 2\varepsilon(x_1 - x_2) \neq \mathbf{0}$. Since $v', v'' \in C = \langle v_1, \ldots, v_n \rangle$, they can be written

$$v' = \sum_{i=1}^{n} z'(i)v_i,\ \ v'' = \sum_{i=1}^{n} z''(i)v_i,\ \ z', z'' \in \Delta^{n-1},$$

from which it follows that

$$v_1 = \frac{1}{2}(v' + v'') = \sum_{i=1}^{n} \frac{z'(i) + z''(i)}{2} v_i.$$

By (♮), we have $z'(1) + z''(1) = 2$. Since $z'(1), z''(1) \in \mathbf{I}$, it follows that $z'(1) = z''(1) = 1$. Hence, $v' = v'' = v_1$, which is a contradiction. Thus, $C^{(0)} = \{v_1, \ldots, v_n\}$ is the smallest finite set such that $\langle C^{(0)} \rangle = C$.

To see the additional statement, let $C^{(0)} = \{v_1, \ldots, v_n\}$, where $v_i \neq v_j$ if $i \neq j$. When $\dim C > 0$, equiv., $n > 1$, assume that some v_i is not contained in ∂C. Changing indexes if necessary, we may assume that $v_n \in \operatorname{rint} C$. By Proposition 4.1.1, we can write $v_n = \sum_{i=1}^{n} z(i)v_i$, $z \in \operatorname{rint} \Delta^{n-1}$. Since $0 < z(n) < 1$, it follows that $v_n = \sum_{i=1}^{n-1} (1 - z(n))^{-1} z(i)v_i$, where $\sum_{i=1}^{n-1} (1 - z(n))^{-1} z(i) = 1$. Hence, we have $C = \langle C^{(0)} \setminus \{v_n\} \rangle$, which contradicts the minimality of $C^{(0)}$. Therefore, every v_i is contained in ∂C, that is, $C^{(0)} \subset \partial C$. □

In Proposition 4.1.2, each point of $C^{(0)}$ is called a **vertex** of C; namely, $C^{(0)}$ is the set of vertices of C. Note that if $\sigma = \langle v_1, \ldots, v_{n+1} \rangle$ is an n-simplex then $\sigma^{(0)} = \{v_1, \ldots, v_{n+1}\}$, i.e., $\operatorname{card} \sigma^{(0)} = \dim \sigma + 1$. Thus, the above definition is consistent with the definition of a vertex of a simplex, and we have the following:

Corollary 4.1.3 *A cell $C \subset E$ is a simplex if and only if $C^{(0)}$ is affinely independent, equiv.,* $\operatorname{card} C^{(0)} = \dim C + 1$. □

It is said that two simplexes σ and τ are **joinable** (or σ is **joinable** to τ) if $\sigma \cap \tau = \emptyset$ and $\sigma^{(0)} \cup \tau^{(0)}$ is affinely independent. Then, $\langle \sigma^{(0)} \cup \tau^{(0)} \rangle$ is a simplex called the **join** of σ and τ and is denoted by $\sigma\tau$, where

$$\dim \sigma\tau = \dim \sigma + \dim \tau + 1.$$

The following observation is trivial:

- *Let σ_1, σ_2, and σ_3 be simplexes. If σ_1 is joinable σ_2 and $\sigma_1\sigma_2$ is joinable to σ_3, then σ_2 is joinable to σ_3, σ_1 is joinable to $\sigma_2\sigma_3$, and $\sigma_1(\sigma_2\sigma_3) = (\sigma_1\sigma_2)\sigma_3$.*

  Indeed, the assumption implies that $\sigma_i \cap \sigma_j = \emptyset$ for $i \neq j$, and $\sigma_1^{(0)} \cup \sigma^{(0)} \cup \sigma_3^{(0)}$ are affinely independent. Then, it follows that $\sigma_1(\sigma_2\sigma_3) = (\sigma_1\sigma_2)\sigma_3 = \langle \sigma_1^{(0)} \cup \sigma^{(0)} \cup \sigma_3^{(0)} \rangle$.

In the above, $\sigma_1(\sigma_2\sigma_3) = (\sigma_1\sigma_2)\sigma_3$ is simply written as $\sigma_1\sigma_2\sigma_3$. For finitely many simplexes $\sigma_1, \ldots, \sigma_k$ such that $\sigma_1 \ldots \sigma_i$ is joinable to σ_{i+1} for each $i = 1, \ldots, k - 1$

(equiv., $\sigma_1, \ldots, \sigma_k$ are pairwise disjoint and $\sigma_1^{(0)} \cup \cdots \cup \sigma_k^{(0)}$ is affinely independent), we inductively define $\sigma_1 \cdots \sigma_{k-1}\sigma_k = (\sigma_1 \cdots \sigma_{k-1})\sigma_k$, which is equal to $\langle \sigma_1^{(0)} \cup \cdots \cup \sigma_k^{(0)} \rangle$. When $\sigma = \{v\}$, the join $\{v\}\tau$ is simply denoted by $v\tau$. For distinct two points $u, v \in E$, we can just write $\langle u, v \rangle = uv$. Moreover, an n-simplex σ with vertices $v_1, \ldots, v_{n+1}$ can be denoted as follows:

$$\sigma = \langle v_1, \ldots, v_{n+1} \rangle = v_1 \cdots v_{n+1}.$$

For convenience, we define $\sigma\emptyset = \emptyset\sigma = \sigma$ and $\emptyset\emptyset = \emptyset$.

The face of a cell C at $x \in C$ is defined as in Sect. 3.2, that is,

$$C_x = \big\{ y \in C \;\big|\; \exists \delta > 0 \;\text{ such that }\; (1+\delta)x - \delta y \in C \big\}.$$

Recall $\operatorname{rint} C = \{x \in C \mid C_x = C\}$, hence $\partial C = \{x \in C \mid C_x \neq C\}$. Moreover, $x \in \operatorname{rint} C_x$ and $C_x = C_y$ for every $y \in \operatorname{rint} C_x$ (Proposition 3.2.9 (viii), (x)). Recall that an extreme point of C is a point $x \in C$ such that $C_x = \{x\}$.

Proposition 4.1.4 *For each cell $C \subset E$ and $x \in C$, the following hold:*

(1) *C_x is a cell with $C_x^{(0)} = C^{(0)} \cap C_x$;*
(2) *x is a vertex of C if and only if it is an extreme point of C, i.e.,*

$$x \in C^{(0)} \;\Leftrightarrow\; C_x = \{x\}.$$

Proof (1): To see that C_x is a cell, it suffices to show that $C_x = \langle C^{(0)} \cap C_x \rangle$. Since C_x is convex (Proposition 3.2.8), we have $\langle C^{(0)} \cap C_x \rangle \subset C_x$. Each $y \in C_x$ can be written as

$$y = \sum_{i=1}^{n} z(i) v_i, \;\; v_1, \ldots, v_n \in C^{(0)}, \;\; z \in \operatorname{rint} \Delta^{n-1}.$$

We can choose $\delta \in (0, 1)$ so that $(1+\delta)x - \delta y \in C$. For each $i = 1, \ldots, n$, let

$$x_i = (1 - \delta + \delta z(i))x + \sum_{j \neq i} \delta z(j) v_j \in C.$$

Then, it follows that

$$\Big(1 + \frac{1}{2}\delta z(i)\Big)x - \frac{1}{2}\delta z(i) v_i = \frac{1}{2}\big((1+\delta)x - \delta y\big) + \frac{1}{2}x_i \in C,$$

which means that $v_i \in C_x$. Hence, $y \in \langle C^{(0)} \cap C_x \rangle$.

Since $C_x = \langle C^{(0)} \cap C_x \rangle$, it follows that $C_x^{(0)} \subset C^{(0)} \cap C_x$ and

$$\begin{aligned} \langle C_x^{(0)} \cup (C^{(0)} \setminus C_x) \rangle &= \langle C_x \cup (C^{(0)} \setminus C_x) \rangle \\ &= \big\langle (C^{(0)} \cap C_x) \cup (C^{(0)} \setminus C_x) \big\rangle = \langle C^{(0)} \rangle = C. \end{aligned}$$

The latter implies $C^{(0)} \cap C_x \subset C_x^{(0)}$. Hence, $C^{(0)} \cap C_x = C_x^{(0)}$.

(2): If $C_x = \{x\}$, then $x \in C_x^{(0)} \subset C^{(0)}$ by (1). Conversely, if $x \in C^{(0)}$, then $x \in C^{(0)} \cap C_x = C_x^{(0)}$ by (1). Since $x \in \operatorname{rint} C_x$ (Proposition 3.2.9 (viii)), we have $C_x = \{x\}$ by Proposition 4.1.2. □

A cell D is called a **face** of a cell C if $D = C_x$ for some $x \in C$, where we write

$$D \leqslant C \quad \text{or} \quad C \geqslant D.$$

According to Proposition 4.1.4 (1), each cell C has only a finite number of faces because $C^{(0)}$ is finite. We regard each vertex v of a cell C as a face of C because $\{v\} = C_v$ by Proposition 4.1.4 (2). Frequently, a vertex v is implicitly identified with the singleton $\{v\}$. When $D \leqslant C$ and $D \neq C$, D is called a **proper face** of C and is denoted as follows:

$$D < C \quad \text{or} \quad C > D.$$

An n-dimensional face is called an ***n*-face**, where a 0-face is just a vertex.

A face of a simplex σ is also a simplex (cf. Proposition 4.1.4 (1)) and $\langle v_1, \ldots, v_k \rangle \leqslant \sigma$ for any $v_1, \ldots, v_k \in \sigma^{(0)}$. In fact, we have the following:

- *A simplex τ is a face of σ if and only if $\tau^{(0)} \subset \sigma^{(0)}$.*

  The "if" part can be shown as follows: Assume that $\tau^{(0)} \subset \sigma^{(0)}$, which implies $\tau \subset \sigma$. Take any $x \in \operatorname{rint} \tau$. Then, $\tau \subset \sigma_x$ by the definition of σ_x. For each $y \in \sigma \setminus \tau$, we have distinct $v_1, \ldots, v_n \in \sigma^{(0)}$ and $z \in \operatorname{rint} \Delta^{n-1}$ such that $y = \sum_{i=1}^{n} z(i) v_i$ and $v_1 \notin \tau^{(0)}$. Then, $\tau \cap \operatorname{rint}\langle y, y' \rangle = \emptyset$ for any $y' \in \sigma$, hence $y \notin \sigma_x$. Thus, we have $\tau = \sigma_x$.

If $\dim \sigma = n$, then σ has $2^{n+1} - 1$ faces, and hence it has $2^{n+1} - 2$ proper faces. Note that σ is the join $\sigma_0 \sigma_1$ of any two disjoint faces σ_0 and σ_1 with $\sigma^{(0)} = \sigma_0^{(0)} \cup \sigma_1^{(0)}$, where σ_i is called the **opposite face** of σ to σ_{1-i} $(i = 0, 1)$. Moreover, the following can be easily shown:

- *If $\tau_i \leqslant \sigma_i$, $i = 0, 1$, then $\tau_0 \tau_1 \leqslant \sigma = \sigma_0 \sigma_1$;*
- *For each $\tau \leqslant \sigma_0 \sigma_1$, $\tau = (\tau \cap \sigma_0)(\tau \cap \sigma_1)$, where $\tau \cap \sigma_i \leqslant \sigma_i$ if $\tau \cap \sigma_i \neq \emptyset$.*

In the following, we will list basic properties of cells:

Proposition 4.1.5 (Basic Properties of Cells) *For cells $C, D \subset E$, the following hold:*

(i) $\emptyset \neq A \subset C^{(0)} \Rightarrow \langle A \rangle \subset C$, $\langle A \rangle^{(0)} = A$;
(ii) *If $A \subset C^{(0)}$ is not a singleton, then $C^{(0)} \cap \operatorname{rint}\langle A \rangle = \emptyset$;*
(iii) $D \leqslant C \Rightarrow D = C \cap \operatorname{fl} D$ *and* $D^{(0)} = C^{(0)} \cap D = C^{(0)} \cap \operatorname{fl} D$;
(iv) $D \leqslant C \Rightarrow D_x = C_x$ *for each* $x \in D$, *hence* $D' \leqslant C$ *for each* $D' \leqslant D$;
(v) $D \leqslant C \Rightarrow D = C_x$ *for each* $x \in \operatorname{rint} D$;
(vi) $D,\ D' \leqslant C$, $\operatorname{rint} D \cap \operatorname{rint} D' \neq \emptyset \Rightarrow D = D'$;
(vii) $D,\ D' \leqslant C$, $D \cap \operatorname{rint} D' \neq \emptyset \Rightarrow D' \leqslant D$, *hence* $D' \subset D \Leftrightarrow D' \leqslant D$;
(viii) $D < C \Rightarrow \dim D < \dim C$;
(ix) $\partial C = \bigcup \{D \mid D < C\} = \bigcup \{D \mid D < C,\ \dim D = n - 1\}$.

Sketch of Proof As a combination of Propositions 3.2.8 and 4.1.4 (1), we have (iii). For (iv), (v), and (vi), refer to Propositions 3.2.9 (ix), (x), and (xi), respectively. And (vii) is a combination of (iv) and (v), where (vi) can also be derived from (vii). Since $D < C \Rightarrow \mathrm{fl}\, D \subsetneqq \mathrm{fl}\, C$ by (iii), we have (viii). For the first equality of (ix), recall $\partial C = \{x \in C \mid C_x \neq C\}$. The second equality can be obtained by Proposition 4.1.6 below. It remains to prove (i) and (ii).

(i): Note that $\langle A\rangle^{(0)} \subset A \subset C^{(0)}$. Let $B = (C^{(0)} \setminus A) \cup \langle A\rangle^{(0)} \subset C^{(0)}$. Since $A \subset \langle A\rangle \subset \langle B\rangle$, we have $C^{(0)} \subset \langle B\rangle \subset C$, so $\langle B\rangle = C$. By Proposition 4.1.2, $B = C^{(0)}$, which means $\langle A\rangle^{(0)} = A$.

(ii): By (i) and Proposition 4.1.2, $A = \langle A\rangle^{(0)} \subset \partial\langle A\rangle$. If $\mathrm{rint}\langle A\rangle$ contains some $v \in C^{(0)}$, then $v \in \langle A \setminus \{v\}\rangle$ by Proposition 4.1.1, which implies $\langle C^{(0)}\rangle = \langle C^{(0)} \setminus \{v\}\rangle$, which contradicts the definition of $C^{(0)}$.

Proposition 4.1.6 *Let $C \subset E$ be an n-cell and D a proper k-face of C. Then, the following hold:*

(i) *D is a maximal proper face of C if and only if $k = n - 1$*[2]*;*
(ii) *If $k < n - 1$, then there exist faces $D = D_k < D_{k+1} < \cdots < D_n = C$ such that $\dim D_i = i$ for $k \leqslant i \leqslant n$.*

Proof (i): By Proposition 4.1.5 (viii), we have $k < n$. Then, the "if" part is obvious. To see the "only if" part, assume $\dim D = k < n - 1$. Because of the maximality of D, $\mathrm{rint}\, D$ misses any other proper face of C by Proposition 4.1.5 (vii). Then, $\partial C \setminus \mathrm{rint}\, D = \bigcup\{D' \mid D \neq D' < C\}$ is a compact set in the flat $\mathrm{fl}\, C$ with a unique topology (Proposition 3.5.1). Let $x_0 \in \mathrm{rint}\, D$. Since $\mathrm{fl}\, C$ is affinely homeomorphic to $\mathbb{R}^n$, x_0 has a convex neighborhood V in $\mathrm{fl}\, C$ missing $\partial C \setminus \mathrm{rint}\, D$, i.e., $V \cap \partial C \subset \mathrm{rint}\, D$. Since $C = \mathrm{rcl}\,\mathrm{rint}\, C = \mathrm{cl}_{\mathrm{fl}\, C}\,\mathrm{int}_{\mathrm{fl}\, C}\, C$ by Propositions 3.2.9 (iii), (iv), and 3.5.5, we can take $y \in V \cap \mathrm{rint}\, C$. Then, $\mathrm{fl}(D \cup \{y\})$ is nowhere dense in $\mathrm{fl}\, C$ because

$$\dim \mathrm{fl}(D \cup \{y\}) \leqslant \dim \mathrm{fl}\, D + 1 \leqslant n - 1 < n = \dim \mathrm{fl}\, C.$$

Note that $V \setminus C \neq \emptyset$ because $\partial C = C \setminus \mathrm{rint}\, C = \mathrm{bd}_{\mathrm{fl}\, C}\, C$ by Proposition 3.5.5. Hence, we can also take $z \in V \setminus (C \cup \mathrm{fl}(D \cup \{y\}))$. If $v \in \langle y, z\rangle \cap D$, then $z \in \mathrm{fl}\{y, v\} \subset \mathrm{fl}(D \cup \{y\})$. Therefore, $\langle y, z\rangle \cap D = \emptyset$. Since $\langle y, z\rangle \cap (\partial C \setminus D) \subset V \cap \partial C \setminus D = \emptyset$, it follows that $\langle y, z\rangle \cap \partial C = \emptyset$, that is, $\langle y, z\rangle \subset \mathrm{int}_{\mathrm{fl}\, C}\, C \cup (\mathrm{fl}\, C \setminus C)$, where $y \in \mathrm{int}_{\mathrm{fl}\, C}\, C$ and $z \in \mathrm{fl}\, C \setminus C$. This contradicts the connectivity of $\langle y, z\rangle$.

(ii) Let $D_n = C$. Since D is not a maximal proper face of D_n by (i), there is a proper face $D' < D_n$ with $D < D'$. If $\dim D' = n - 1$, then D' is a maximal proper face of D_n. Otherwise, there is a proper face $D'' < D_n$ with $D' < D''$. Repeating this operation finitely many times, we can obtain a maximal proper face $D_{n-1} < D_n$ such that $D < D_{n-1}$, where $\dim D_{n-1} = n - 1$ by (i). When $k = n - 2$, $D = D_{n-2} < D_{n-1} < D_n$ is the desired sequence of faces. When $k < n - 2$, as the same way as above, we can obtain a maximal proper face $D_{n-2} < D_{n-1}$ such that $D < D_{n-2}$. By the downward induction, it is easy to obtain the desired sequence of faces of C. □

[2] The relations $\subset$ and $\leqslant$ induce the same order among proper faces of C by Proposition 4.1.5 (vii); "maximal" is used with respect to this order.

Using affine functionals, we characterize cells as follows:

Proposition 4.1.7 *Let $\emptyset \neq C \subset E$ be non-degenerate. In order for C to be a cell, it is necessary and sufficient that* $\dim \mathrm{fl}\, C < \infty$, $x + \mathbb{R}_+(y - x) \not\subset C$ *for each pair of distinct points $x, y \in C$, and there are finitely many non-constant affine functionals $f_1, \ldots, f_k : \mathrm{fl}\, C \to \mathbb{R}$ such that $C = \bigcap_{i=1}^k f_i^{-1}(\mathbb{R}_+)$.*

Proof (*Necessity*) Let $n = \dim C$. By virtue of Proposition 4.1.5 (ix), we can write $\partial C = \bigcup_{i=1}^k D_i$, where each D_i is an $(n-1)$-face of C. By using Proposition 3.1.3 (1), we can find an affine functional $f_i : \mathrm{fl}\, C \to \mathbb{R}$ such that $\mathrm{fl}\, D_i = f_i^{-1}(0)$. Then, $C \subset f_i^{-1}(\mathbb{R}_+)$ or $C \subset f_i^{-1}(-\mathbb{R}_+)$. Replacing f_i with $-f_i$ if $C \subset f_i^{-1}(-\mathbb{R}_+)$, we may assume that $C \subset f_i(\mathbb{R}_+)$ for every $i = 1, \ldots, k$. Thus, we have $C \subset \bigcap_{i=1}^k f_i^{-1}(\mathbb{R}_+)$. Suppose that there is a point $z \in \bigcap_{i=1}^k f_i^{-1}(\mathbb{R}_+) \setminus C$. By taking $y \in \mathrm{rint}\, C$, we have $x \in \mathrm{rint}\langle y, z\rangle \cap \partial C$. Then, x is contained in some D_i. Since $f_i(x) = 0$ and $f_i(y) > 0$, it follows that $f_i(z) < 0$, which is a contradiction. Therefore, $C = \bigcap_{i=1}^k f_i^{-1}(\mathbb{R}_+)$.

(*Sufficiency*) First, note that $C = \bigcap_{i=1}^k f_i^{-1}(\mathbb{R}_+)$ is convex, where we may assume that $C \neq \bigcap_{i\neq j} f_i^{-1}(\mathbb{R}_+)$ for any $j = 1, \ldots, n$. Then, $C \cap f_j^{-1}(0) \neq \emptyset$ for every $j = 1, \ldots, k$. Indeed, $C \neq \bigcap_{i\neq j} f_i^{-1}(\mathbb{R}_+)$ means that $f_j(x) < 0$ for some $x \in \bigcap_{i\neq j} f_i^{-1}(\mathbb{R}_+)$. Take any point $y \in C$. Since $f_j(y) \geqslant 0$, we have $z \in \langle x, y\rangle$ such that $f_j(z) = 0$. If $i \neq j$, then $f_i(z) \geqslant 0$ because $f_i(x), f_i(y) \geqslant 0$. Therefore,

$$z \in f_j^{-1}(0) \cap \bigcap_{i\neq j}^{k} f_i^{-1}(\mathbb{R}_+) = C \cap f_j^{-1}(0).$$

Next, we show that $\mathrm{rint}\, C = \bigcap_{i=1}^k f_i^{-1}((0,\infty))$. Since $\dim \mathrm{fl}\, C < \infty$, each f_i is continuous (Corollary 3.5.3). Hence, $\bigcap_{i=1}^k f_i^{-1}((0,\infty)) \subset \mathrm{int}_{\mathrm{fl}\, C} C = \mathrm{rint}\, C$ by Proposition 3.5.5. Each $x \in \mathrm{int}_{\mathrm{fl}\, C} C$ has a neighborhood V in $\mathrm{fl}\, C$ such that $V \subset C$. Assume that $f_i(x) = 0$ for some $i = 1, \ldots, k$. As observed in the above, $f_i(y) < 0$ for some $y \in \bigcap_{j\neq i} f_j^{-1}(\mathbb{R}_+)$. Choose $0 < t < 1$ so small that $z = (1-t)x + ty \in V$. Then, it follows that $f_i(z) < 0$, which contradicts $z \in V \subset C$. Therefore, $f_i(x) > 0$ for every $i = 1, \ldots, k$, that is, $x \in \bigcap_{i=1}^k f_i^{-1}((0,\infty))$.

Now, by induction on $n = \dim \mathrm{fl}\, C$, we show that C is a cell. For each $i = 1, \ldots, k$, let $D_i = C \cap f_i^{-1}(0) \neq \emptyset$. Then, as observed above, $\partial C = \bigcup_{i=1}^k D_i$. Since $\mathrm{fl}\, D_i \subset f_i^{-1}(0)$ and $\dim f_i^{-1}(0) = n - 1$ (Proposition 3.1.3(2)), each $D_i = \bigcap_{j\neq i} (f_j | \mathrm{fl}\, D_i)^{-1}(\mathbb{R}_+)$ is a cell by the inductive assumption. Thus, we have a finite set $A = \bigcup_{i=1}^k D_i^{(0)} \subset \partial C$. Consequently, $\partial C = \bigcup_{i=1}^k D_i \subset \langle A\rangle \subset C$. Take any point $v \in A \subset \partial C$. For each $x \in \mathrm{rint}\, C$, $v + \mathbb{R}_+(x - v) \not\subset C$, hence there is a $y \in \partial C$ such that $x \in \langle v, y\rangle$. Then, $x \in \langle A\rangle$ because $v, y \in \langle A\rangle$. Therefore, $C = \langle A\rangle$ is a cell. □

Proposition 4.1.8 *Let C be a cell represented as $C = \bigcap_{i=1}^k f_i^{-1}(\mathbb{R}_+)$ for some non-constant affine functionals $f_1, \ldots, f_k : \mathrm{fl}\, C \to \mathbb{R}$. Then, the following hold:*

(i) $\operatorname{rint} C = \bigcap_{i=1}^{k} f_i^{-1}((0,\infty))$;
(ii) $D = f_{i_0}^{-1}(0) \cap C \neq \emptyset \ \Rightarrow\ D \leqslant C$;
(iii) $f_{i_0}^{-1}(0) \cap C = \emptyset \ \Rightarrow\ C = \bigcap_{i \neq i_0} f_i^{-1}(\mathbb{R}_+)$;
(iv) $C \neq \bigcap_{i \neq i_0} f_i^{-1}(\mathbb{R}_+) \ \Rightarrow\ D = f_{i_0}^{-1}(0) \cap C < C$ *and* $\operatorname{fl} D = f_{i_0}^{-1}(0)$.

Proof In the proof of Theorem 4.1.7 (Sufficiency), (i) and (iii) were proved. For (ii), refer to Proposition 3.2.11 (2). For (iv), as a combination of (ii) and (iii), we have $C \neq \bigcap_{i \neq i_0} f_i^{-1}(\mathbb{R}_+) \Rightarrow D = f_{i_0}^{-1}(0) \cap C < C$. To see $\operatorname{fl} D = f_{i_0}^{-1}(0)$, let $F = f_{i_0}^{-1}(0)$. Observe that

$$F \cap \bigcap_{i \neq i_0} f_i^{-1}((0,\infty)) \subset \operatorname{int}_F D \subset \operatorname{core}_F D.$$

According Fact immediatly preceding Proposition 3.3.2, it suffices to show that $F \cap \bigcap_{i \neq i_0} f_i^{-1}((0,\infty)) \neq \emptyset$. If this does not hold true, then

$$\bigcap_{i \neq i_0} f_i^{-1}((0,\infty)) \subset f_{i_0}^{-1}((-\infty,0)) \cup f_{i_0}^{-1}((0,\infty)) \ \text{ and}$$

$$f_{i_0}^{-1}((0,\infty)) \cap \bigcap_{i \neq i_0} f_i^{-1}((0,\infty)) = \bigcap_{i=1}^{k} f_i^{-1}((0,\infty)) = \operatorname{rint} C \neq \emptyset \ \text{ by (i).}$$

Since $\bigcap_{i \neq i_0} f_i^{-1}((0,\infty))$ is connected, it follows that $\bigcap_{i \neq i_0} f_i^{-1}((0,\infty)) \subset f_{i_0}^{-1}((0,\infty))$. Take $v \in \operatorname{rint} C$. If $f_{i_0}(x) < 0$ for some $x \in \bigcap_{i \neq i_0} f_i^{-1}(\mathbb{R}_+)$, then $f_{i_0}((1-t)x + tv) < 0$ for a sufficiently small $t > 0$ but $(1-t)x + tv \in \bigcap_{i \neq i_0} f_i^{-1}((0,\infty)) \subset f_{i_0}^{-1}((0,\infty))$, which is a contradiction. Thus, we have $\bigcap_{i \neq i_0} f_i^{-1}(\mathbb{R}_+) \subset f_{i_0}^{-1}(\mathbb{R}_+)$, which contradicts $C \neq \bigcap_{i \neq i_0} f_i^{-1}(\mathbb{R}_+)$. □

Proposition 4.1.9 (Additional Results for Cells)

(1) *For each cell $C \subset E$ and each flat $F \subset E$ with $C \cap F \neq \emptyset$, the intersection $C \cap F$ is also a cell.*
(2) *For every two cells $C, D \subset E$ with $C \cap D \neq \emptyset$, the intersection $C \cap D$ is also a cell and $(C \cap D)_x = C_x \cap D_x$ for each $x \in C \cap D$. In addition,*

$$\operatorname{rint} C \cap \operatorname{rint} D \neq \emptyset \ \Rightarrow\ \operatorname{rint}(C \cap D) = \operatorname{rint} C \cap \operatorname{rint} D.$$

(3) *For a cell $C \subset E$ and an affine map $f : C \to E'$, if $D \subset E'$ is a cell and $D \cap f(C) \neq \emptyset$, then $f^{-1}(D)$ is a cell and $f^{-1}(D)_x = C_x \cap f^{-1}(D_{f(x)})$ for each $x \in f^{-1}(D)$. When f is injective, $f(C_x) = f(C)_{f(x)}$ for each $x \in C$.*

Sketch of Proof For the above three items, apply the characterization 4.1.7 (cf. Proposition 3.2.3 for (3)). The statements about faces in (2) and (3) are the same as Propositions 3.2.9 (xii) and 3.2.11 (2), respectively. The statement about the radial interior in (2) is Proposition 3.2.9 (vii).

(4) *For every two cells $C, D \subset E$ and their faces $C' \leqslant C$, $D' \leqslant D$ with $C' \cap D' \neq \emptyset$, $C' \cap D' \leqslant C \cap D$.*

Sketch of Proof Let $x \in \operatorname{rint}(C' \cap D')$. Using Proposition 3.2.9 (ix) and (2) above, we have

$$C' \cap D' = (C' \cap D')_x = C'_x \cap D'_x = C_x \cap D_x = (C \cap D)_x.$$

(5) *For every two cells $C \subset E$ and $D \subset E'$, the product $C \times D$ is also a cell in $E \times E'$ with*

$$\operatorname{rint}(C \times D) = \operatorname{rint} C \times \operatorname{rint} D,\ (C \times D)^{(0)} = C^{(0)} \times D^{(0)},\ \textit{and}$$
$$(C \times D)_{(x,y)} = C_x \times D_y \ \textit{for each}\ (x, y) \in C \times D.$$

Sketch of Proof Note that $C \times D = \langle C^{(0)} \times D^{(0)} \rangle$, which is a cell. In addition, see Proposition 3.2.11 (1). To see $(C \times D)^{(0)} = C^{(0)} \times D^{(0)}$, use Proposition 4.1.4 (2).

4.2 Cell Complexes and Simplicial Complexes

Throughout this section, as in the previous section, let E be a linear space. A collection K of cells in E is called a **cell complex**[3] if K satisfies the following two conditions:

(C_1) If $C \in K$ and $D \leqslant C$, then $D \in K$;
(C_2) For each $C, D \in K$ with $C \cap D \neq \emptyset$, $C \cap D \leqslant C$ (and $C \cap D \leqslant D$).[4]

Under condition (C_1), condition (C_2) is equivalent to each of the following:

(C'_2) For each $C, D \in K$, $C \cap \operatorname{rint} D \neq \emptyset$ implies $D \leqslant C$;
(C''_2) For each $C, D \in K$, $C \neq D$ implies $\operatorname{rint} C \cap \operatorname{rint} D = \emptyset$ (equiv., $\operatorname{rint} C \cap \operatorname{rint} D \neq \emptyset$ implies $C = D$).

Sketch of Proof The implication (C'_2) $\Rightarrow$ (C''_2) is easy. Since $C \cap \operatorname{rint} D \neq \emptyset$ means $(C \cap D) \cap \operatorname{rint} D \neq \emptyset$, we can apply Proposition 4.1.5 (vii) to show (C_2) $\Rightarrow$ (C'_2). To see (C''_2) $\Rightarrow$ (C_2), assume $C \cap D \neq \emptyset$ and take a point $x \in \operatorname{rint}(C \cap D)$. Since $C_x, D_x \in K$ by (C_1) and $x \in \operatorname{rint} C_x \cap \operatorname{rint} D_x$, we have $C_x = D_x$ by (C''_2). It follows from Proposition 4.1.9 (2) that

$$C \cap D = (C \cap D)_x = C_x \cap D_x = C_x \leqslant C.$$

A subcollection L of a cell complex K is called a **subcomplex** if L is also a cell complex, equivalently L satisfies (C_1). As is easily observed, unions and intersections of subcomplexes of K are also subcomplexes of K. For each cell C, we define

$$F(C) \underset{\text{def}}{=} \{D \mid D \leqslant C\} \ \text{ and } \ \partial F(C) \underset{\text{def}}{=} \{D \mid D < C\},$$

which are cell complexes (cf. Proposition 4.1.5 (vi) or (vii)). Namely, $F(C)$ is the cell complex consisting of all faces of C and $\partial F(C)$ is that of all proper faces of C,

[3] More precisely, it is called a (**convex**) **linear cell complex**.
[4] Any vertex v of a cell is identified with the singleton $\{v\}$.

where $\partial F(C)$ is a subcomplex of $F(C)$. Condition (C_1) means that $F(C) \subset K$ for each $C \in K$. The next fact follows from Proposition 4.1.5 (iii) and (C_2') (or (C_2)):

Fact *For each* $C, D \in K$, $D \leqslant C \Leftrightarrow D^{(0)} \subset C^{(0)} \Leftrightarrow D \subset C$.

This fact means that the face relation $\leqslant$ induces the same order on K as the subset relation $\subset$.

A cell complex consisting of simplexes is called a **simplicial complex**. For a simplex σ, $F(\sigma)$ and $\partial F(\sigma)$ are simplicial complexes. Every subcomplex of a simplicial complex is a simplicial complex.

Let K be a cell (simplicial) complex. We call $K^{(0)} = \bigcup_{C \in K} C^{(0)}$ the **set of vertices**. It is said that K is **finite**, **infinite**, or **countable** according to card K (equivalently card $K^{(0)}$). When card K is infinite, we have card K = card $K^{(0)}$.

Indeed, $K \ni C \mapsto C^{(0)} \in \mathrm{Fin}(K^{(0)})$ is an injection by Proposition 4.1.2 (or the above Fact). Then, card $K^{(0)}$ is also infinite, hence it follows that

$$\mathrm{card}\, K \leqslant \mathrm{card}\, \mathrm{Fin}(K^{(0)}) = \mathrm{card}\, K^{(0)} \leqslant \mathrm{card}\, K.$$

The following should be noticed:

- *For each* $C \in K$, $C \cap K^{(0)} = C^{(0)}$.
- *For each* $C, D \in K$ *with* $C \cap D \neq \emptyset$, $(C \cap D)^{(0)} = C^{(0)} \cap D^{(0)}$.

The **dimension** of K is $\dim K = \sup_{C \in K} \dim C$, where we consider $\dim K = -1$ if $K = \emptyset$. If $\dim K = \infty$, K is said to be **infinite-dimensional** (abbrev. **i.d.**). When $\dim K < \infty$, K is **finite-dimensional** (abbrev. **f.d.**). It is said that K is ***n*-dimensional** if $\dim K = n$. Note that every cell complex K with $\dim K \leqslant 1$ is simplicial. For each $n \in \omega$, the ***n*-skeleton** of K is defined as the following subcomplex:

$$K^{(n)} \underset{\text{def}}{=} \left\{ C \in K \;\middle|\; \dim C \leqslant n \right\} \subset K.$$

The 0-skeleton $K^{(0)}$ is the set of vertices. For each n-cell C (i.e., $\dim C = n$), $\partial F(C) = F(C)^{(n-1)}$. Obviously $\dim K \leqslant n$ if and only if $K = K^{(n)}$. It is convenient to define $K(n) = K^{(n)} \setminus K^{(n-1)}$, where $K^{-1} = \emptyset$. Namely, the set of n-cells of K is denoted by $K(n)$.

The **polyhedron** $|K|$ of K is defined as follows:

$$|K| \underset{\text{def}}{=} \bigcup K = \bigcup_{C \in K} C = \bigcup_{C \in K} \mathrm{rint}\, C \; (\subset E).$$

Recall that each cell $C \in K$ is given the unique topology, as mentioned in the previous section, and C with this topology is homeomorphic to the unit closed n-ball $\mathbf{B}^n$ if $\dim C = n$ (Proposition 3.5.8). We equip $|K|$ with the weak topology with respect to K (a collection of compacta), that is, the topology of $|K|$ is defined as follows:

$$U \subset |K| \text{ is open in } |K| \underset{\text{def}}{\Leftrightarrow} \forall C \in K,\ U \cap C \text{ is open in } C$$

$$\Big(\text{equiv. } A \subset |K| \text{ is closed in } |K| \underset{\text{def}}{\Leftrightarrow} \forall C \in K,\ A \cap C \text{ is closed in } C\Big),$$

which can be also defined as the quotient topology induced by the natural surjection $q : \bigoplus_{C\in K} C \to |K|$ (cf. the definition of the weak topology and its remark in Sect. 1.1). This topology is called the **Whitehead** (or **weak**) **topology**. Then, $K^{(0)}$ is discrete in $|K|$. Each $C \in K$ is a closed subspace of $|K|$ because $C \cap D \leqslant D$ for any $D \in K$ with $C \cap D \neq \emptyset$. The following fact is useful:

Fact *For an arbitrary space X, each $f : |K| \to X$ is continuous if and only if $f|C$ is continuous for every $C \in K$.*

Remark 4.2.1 If V is a neighborhood of $x \in |K|$, then $V \cap C$ is a neighborhood x in C for every $C \in K[x]$. However, the converse does not hold. For example, let K be the 2-dimensional simplicial complex in $\mathbb{R}^{\mathbb{N}}_f$ defined as follows:

$$K = \big\{\mathbf{0}, \mathbf{e}_i, \langle \mathbf{0}, \mathbf{e}_i\rangle, \langle \mathbf{e}_1, \mathbf{e}_{i+1}\rangle, \langle \mathbf{0}, \mathbf{e}_1, \mathbf{e}_{i+1}\rangle \ \big|\ i \in \mathbb{N}\big\}.$$

We define $V = \bigcup_{i\in\mathbb{N}} \langle \mathbf{0}, 2^{-i}\mathbf{e}_1, 2^{-i}\mathbf{e}_{i+1}\rangle \subset |K|$. For each simplex $\sigma \in K[\mathbf{0}]$, $V \cap \sigma$ is a neighborhood of $\mathbf{0}$ in σ. Nevertheless, V is not a neighborhood of $\mathbf{0}$ in $|K|$. Indeed,

$$(\mathrm{int}_{|K|}\, V) \cap \langle \mathbf{0}, \mathbf{e}_1\rangle \subset \bigcap_{n\in\mathbb{N}} \langle \mathbf{0}, 2^{-i}\mathbf{e}_1, 2^{-i}\mathbf{e}_{i+1}\rangle = \{\mathbf{0}\},$$

which implies $(\mathrm{int}_{|K|}\, V) \cap \langle \mathbf{0}, \mathbf{e}_1\rangle = \emptyset$. Thus, V is not a neighborhood of $\mathbf{0}$ in $|K|$.

Due to (C''_2), each $x \in |K|$ is contained in the (radial) interior of the unique cell $c_K(x) \in K$, which is called the **carrier** of x in K. In other words, $c_K(x)$ is the smallest cell of K containing x. If $x \in C \in K$, then $C_x = c_K(x)$. A cell $C \in K$ is said to be **principal** if C is not a proper face of any cell of K, that is, it is a maximal cell of K. A cell $C \in K$ is principal if and only if $\mathrm{int}_{|K|}\, C \neq \emptyset$. In general, $\mathrm{int}_{|K|}\, C \neq \mathrm{rint}\, C$ even if $C \in K$ is principal. If $\dim K = n$, then every n-cell of K is principal.

Proposition 4.2.1 *For every subcomplex L of a cell complex K, $|L|$ is a closed subspace of $|K|$.*

Proof It is easy to see that $A \cap |L|$ is closed in $|L|$ for each closed set A in $|K|$. So, we will show that every closed set A in $|L|$ is closed in $|K|$. For each $C \in K$,

$$A \cap C = A \cap C \cap |L| = \bigcup_{D\in L} A \cap C \cap D = \bigcup_{D\in L\cap F(C)} A \cap D.$$

For each $D \in L \cap F(C)$, $A \cap D$ is closed in D and D is a closed subspace of C, hence $A \cap D$ is closed in C. Because $L \cap F(C)$ is finite, $A \cap C$ is closed in C. Therefore, A is closed in $|K|$. □

Proposition 4.2.2 *The polyhedron $|K|$ is a perfectly normal k-space.*

Proof By definition, it is obvious that $|K|$ is T_1. To see that $|K|$ is perfectly normal, it suffices to show that for any disjoint closed sets $A, B \subset |K|$, there exists a map $f : |K| \to \mathbf{I}$ such that $f^{-1}(0) = A$ and $f^{-1}(1) = B$. We will inductively construct maps $f_n : |K^{(n)}| \to \mathbf{I}$, $n \in \omega$, so that $f_n^{-1}(0) = A \cap |K^{(n)}|$, $f_n^{-1}(1) = B \cap |K^{(n)}|$, and $f_{n+1}||K^{(n)}| = f_n$. Then, the map f can be defined by $f||K^{(n)}| = f_n$ for each $n \in \omega$.

Because $K^{(0)}$ is discrete, f_0 can be easily constructed. Assume that $f^{(n-1)}$ has been constructed. For each n-cell $C \in K$, we apply the Tietze–Urysohn Extension Theorem 2.2.2 to obtain a map $g_C : C \to \mathbf{I}$ such that $g_C|\partial C = f_{n-1}|\partial C$, $g_C(A \cap C) = \{0\}$, and $g_C(B \cap C) = \{1\}$. On the other hand, since $C \in K$ is metrizable (so perfectly normal), there is a map $h_C : C \to \mathbf{I}$ with

$$h_C^{-1}(0) = (A \cap C) \cup (B \cap C) \cup \partial C.$$

We define a map $f_C : C \to \mathbf{I}$ by

$$f_C(x) = (1 - h_C(x))g_C(x) + h_C(x)/2.$$

Then, $f_C|\partial C = f_{n-1}|\partial C$, $f_C^{-1}(0) = A \cap C$ and $f_C^{-1}(1) = B \cap C$. Hence, f_n can be defined by $f_n||K^{(n-1)}| = f_{n-1}$ and $f_n|C = f_C$ for every n-cell C of K.

To see that $|K|$ is a k-space, just observe the following implications for $A \subset |K|$:

$$\begin{aligned} A \text{ is closed in } |K| &\Rightarrow A \cap H \text{ is closed in every compact set } H \text{ in } |K| \\ &\Rightarrow \forall C \in K,\ A \cap C \text{ is closed in } C \Leftrightarrow A \text{ is closed in } |K| \end{aligned}$$ □

Concerning compact sets in $|K|$, we have the following:

Proposition 4.2.3 *Let K be a cell complex. Every compact set $A \subset |K|$ is contained in $|L|$ for some finite subcomplex $L \subset K$. Consequently, $|K|$ is compact if and only if K is finite.*

Proof It suffices to show that $K_A = \{D \in K \mid A \cap \operatorname{rint} D \neq \emptyset\}$ is finite. For each $D \in K_A$, take $x_D \in A \cap \operatorname{rint} D$. Since $C \cap \{x_D \mid D \in K_A\}$ is finite for each $C \in K$ by (C_2'), any subset of $\{x_D \mid D \in K_A\}$ is closed in $|K|$, hence $\{x_D \mid D \in K_A\}$ is discrete in $|K|$, so in A. Since A is compact, it follows that K_A is finite. □

Now, the following can be proved:

Proposition 4.2.4 *Let K and L be cell complexes with $|L| \subset |K|$. Then, $|L|$ is a closed subspace of $|K|$ if and only if each $D \in L$ is covered by only finitely many cells of K and each $C \in K$ contains interior points of only finitely many cells of L.*

Proof (*The "only if" Part*) Since each cell $D \in L$ is a compact set in $|K|$, it is covered by only finitely many cells of K by Proposition 4.2.3. For each $C \in K$,

$C \cap |L|$ is a compact set in $|L|$. Then, as seen in the proof of Proposition 4.2.3, $L_C = \{D \in L \mid C \cap \operatorname{rint} D \neq \emptyset\}$ is finite.

(*The "if" Part*) It suffices to show that the inclusion $|L| \subset |K|$ is a closed map. First, it should be noted that for each $C \in K$ and $D \in L$, $C \cap D$ is a closed subspace of both C and D. Indeed, $C \cap D$ is a cell by Proposition 4.1.9 (2), which has a unique topology.

Let A be a closed set in $|K|$. For each $D \in L$, there are $C_1, \ldots, C_n \in K$ such that $D \subset \bigcup_{i=1}^n C_i$. Since each $A \cap C_i$ is closed in C_i, $A \cap C_i \cap D$ is closed in $C_i \cap D$, and hence it is closed in D. Then, $A \cap D = \bigcup_{i=1}^n (A \cap C_i \cap D)$ is closed in D. Therefore, $A \cap |L|$ is closed in $|L|$. Thus, the inclusion $|L| \subset |K|$ is continuous.

Now, let B be a closed set in $|L|$. For each $C \in K$, we can write

$$\left\{D \in L \;\middle|\; C \cap \operatorname{rint} D \neq \emptyset\right\} = \left\{D_1, \ldots, D_m\right\}.$$

Since each $B \cap D_i$ is closed in D_i, $B \cap C \cap D_i$ is closed in $C \cap D_i$, and hence it is closed in C. Then, $B \cap C = \bigcup_{i=1}^m (B \cap C \cap D_i)$ is closed in C. It follows that B is closed in $|K|$. Thus, the inclusion $|L| \subset |K|$ is a closed map. □

When $|K| = |L|$ as sets in the above Proposition 4.2.4, we can see conditions that $|K|$ and $|L|$ have the same topology. The following is easily shown:

Proposition 4.2.5 *Let K_1 and K_2 be cell complexes with $|K_1| = |K_2|$ as sets. Then, the following are equivalent:*

(a) $|K_1| = |K_2|$ *as spaces;*
(b) *Each $C \in K_i$ is covered by only finitely many cells of K_{3-i} for each $i = 1, 2$;*
(c) *Each $C \in K_i$ contains interior points of only finitely many cells of K_{3-i} for each $i = 1, 2$.* □

The following subcomplex of K is called the **star** at $C \in K$:

$$\operatorname{St}(C, K) \underset{\text{def}}{=} \left\{D' \;\middle|\; \exists D \in K \text{ such that } C \leqslant D,\ D' \leqslant D\right\} = \bigcup_{C \leqslant D \in K} F(D).$$

Evidently, $\operatorname{St}(C, K) = F(C)$ (i.e., $|\operatorname{St}(C, K)| = C$) if and only if C is principal in K. Even if similar notations are used, we must not confuse $\operatorname{St}(C, K)$ with $\operatorname{st}(C, K)$.[5] As is easily observed, there are the following relations:

$$|\operatorname{St}(C, K)| = \operatorname{st}(\operatorname{rint} C, K) \subset \operatorname{st}(C, K),$$

where $|\operatorname{St}(C, K)| \neq \operatorname{st}(C, K)$ in general. Moreover,

$$|\operatorname{St}(v, K)| = \operatorname{st}(v, K) \text{ for each } v \in K^{(0)},$$
$$\operatorname{st}(x, K) = |\operatorname{St}(c_K(x), K)| \text{ for each } x \in |K|.$$

[5] Due to definition in Sect. 2.4, $\operatorname{st}(C, K) = \bigcup K[C]$, where $K[C] = \{D \in K \mid C \cap D \neq \emptyset\}$.

The following subset of $|K|$ is called the **open star** at $x \in |K|$ (with respect to K):

$$O_K(x) \underset{\text{def}}{=} |K| \setminus |K \setminus K[x]| = \bigcup_{C \in K[x]} \operatorname{rint} C = \bigcup_{c_K(x) \leqslant C \in K} \operatorname{rint} C.$$

Since $K \setminus K[x] = \{C \in K \mid x \notin C\}$ is a subcomplex of K, the open star $O_K(x)$ is an open neighborhood of x in $|K|$ (cf. Proposition 4.2.1). On the other hand, the star $\operatorname{st}(x, K)$ $(= |\operatorname{St}(c_K(x), K)|)$ is a closed neighborhood of x in $|K|$ because $O_K(x) \subset \operatorname{st}(x, K)$. Observe the following equivalences:

$$\begin{aligned} &y \in O_K(x) \Leftrightarrow c_K(x) \leqslant c_K(y) \Leftrightarrow c_K(x)^{(0)} \subset c_K(y)^{(0)} \\ &\Leftrightarrow \forall v \in c_K(x)^{(0)},\ c_K(y) \in K[v] \Leftrightarrow \forall v \in c_K(x)^{(0)},\ y \in O_K(v). \end{aligned}$$

Therefore, we have

$$O_K(x) = \bigcap_{v \in c_K(x)^{(0)}} O_K(v).$$

Then, $|K|$ has the following open and closed covers:

$$\mathcal{O}_K \underset{\text{def}}{=} \left\{O_K(v) \;\middle|\; v \in K^{(0)}\right\};\ \mathcal{S}_K \underset{\text{def}}{=} \left\{|\operatorname{St}(v, K)| \;\middle|\; v \in K^{(0)}\right\},$$

where it should be noted that $\mathcal{O}_K^{\mathrm{cl}} = \mathcal{S}_K$.

For $A \subset |K|$, we define the following subcomplexes of K:

$$\begin{aligned} N(A, K) &\underset{\text{def}}{=} \bigcup_{C \in K[A]} F(C) = \left\{D \in K \;\middle|\; \exists C \in K[A] \text{ such that } D \leqslant C\right\}; \\ C(A, K) &\underset{\text{def}}{=} K \setminus K[A] = \left\{D \in K \;\middle|\; D \cap A = \emptyset\right\} \text{ and} \\ B(A, K) &\underset{\text{def}}{=} N(A, K) \cap C(A, K). \end{aligned}$$

It should be noted that $K = N(A, K) \cup C(A, K)$ and $|N(A, K)| = \bigcup_{x \in A} \operatorname{st}(x, K)$ is a neighborhood of A in $|K|$, which is called the **polyhedral neighborhood** of A with respect to K. For simplicity, $N(\{x\}, K)$, $C(\{x\}, K)$, and $B(\{x\}, K)$ are just written $N(x, K)$, $C(x, K)$, and $B(x, K)$, respectively. For a subcomplex L of K, $N(|L|, K)$, $C(|L|, K)$, and $B(|L|, K)$ are simply written as $N(L, K)$, $C(L, K)$, and $B(L, K)$, respectively. Observe that $N(v, K) = \operatorname{St}(v, K)$ for any vertex $v \in K^{(0)}$. For a cell $C \in K$, $N(C, K) \supset \operatorname{St}(C, K)$, but in general $N(C, K) = \operatorname{St}(C, K)$ is not always true.

A cell complex K is said to be **locally finite**, **locally countable**, or **locally finite-dimensional** (abbrev. **l.f.d.**) according to the case that the star $\operatorname{St}(v, K)$ at every $v \in K^{(0)}$ is finite, countable, or finite-dimensional, respectively. Every locally finite cell complex is l.f.d. Note that $K[O_K(v)] = K[v]$ and $\operatorname{St}(v, K) = \bigcup_{C \in K[v]} F(C)$ for every $v \in K^{(0)}$. Then, we have the following:

Proposition 4.2.6 *A cell complex K is locally finite (or locally countable) if and only if K is locally finite (or locally countable) as a collection of subsets in the polyhedron $|K|$ (as a space).* □

For two simplicial complexes K and L in the same linear space, they are **joinable** (or K is **joinable** to L) if they satisfy the following conditions:

(J_1) Every $\sigma \in K$ and $\tau \in L$ are joinable;
(J_2) $\sigma\tau \cap \sigma'\tau' = (\sigma \cap \sigma')(\tau \cap \tau')$ for each $\sigma, \sigma' \in K$ and $\tau, \tau' \in L$.[6]

Observe that (J_1) implies $|K| \cap |L| = \emptyset$. Under condition (J_1), the above (J_2) is equivalent to the following:

(J_2') For $\sigma, \sigma' \in K$ and $\tau, \tau' \in L$, $\sigma \neq \sigma'$ or $\tau \neq \tau' \Rightarrow \operatorname{rint}\sigma\tau \cap \operatorname{rint}\sigma'\tau' = \emptyset$.

The equivalence (J_2) $\Leftrightarrow$ (J_2') can be proved as follows: For (J_2) $\Rightarrow$ (J_2'), we may show the case that $\sigma\tau \cap \sigma'\tau' \neq \emptyset$. Then, by ($J_2$), $\sigma \cap \sigma' \neq \emptyset$ or $\tau \cap \tau' \neq \emptyset$, and $\sigma\tau \cap \sigma'\tau' = (\sigma \cap \sigma')(\tau \cap \tau')$ is a common face of $\sigma\tau$ and $\sigma'\tau'$. Since $\sigma \neq \sigma'$ or $\tau \neq \tau'$, we have $\sigma\tau \cap \sigma'\tau' < \sigma\tau$ or $\sigma\tau \cap \sigma'\tau' < \sigma'\tau'$. In either case, $\operatorname{rint}\sigma\tau \cap \operatorname{rint}\sigma'\tau' = \emptyset$.

For (J_2') $\Rightarrow$ (J_2), since $(\sigma \cap \sigma')(\tau \cap \tau') \subset \sigma\tau \cap \sigma'\tau'$ is obvious, it suffices to show that $\sigma\tau \cap \sigma'\tau' \subset (\sigma \cap \sigma')(\tau \cap \tau')$. We may assume that $\sigma\tau \cap \sigma'\tau' \neq \emptyset$. Let $x \in \sigma$, $y \in \tau$, $x' \in \sigma'$, and $y' \in \tau'$ such that $\langle x, y\rangle \cap \langle x', y'\rangle \neq \emptyset$. Then, we may show that $\langle x, y\rangle \cap \langle x', y'\rangle \subset (\sigma \cap \sigma')(\tau \cap \tau')$. When $\sigma_x = \sigma'_{x'}$ and $\tau_y = \tau'_{y'}$,

$$\langle x, y\rangle \cap \langle x', y'\rangle \subset \sigma_x\tau_y \leqslant (\sigma \cap \sigma')(\tau \cap \tau').$$

When $\sigma_x \neq \sigma'_{x'}$ or $\tau_y \neq \tau'_{y'}$, since $\operatorname{rint}\sigma_x\tau_y \cap \operatorname{rint}\sigma'_{x'}\tau'_{y'} = \emptyset$ by (J_2'), it follows that $\operatorname{rint}\langle x, y\rangle \cap \operatorname{rint}\langle x', y'\rangle = \emptyset$. If $x \in \operatorname{rint}\langle x', y'\rangle$, then $\operatorname{rint}\sigma_x\tau'_{y'} \cap \operatorname{rint}\sigma'_{x'}\tau'_{y'} \supset \operatorname{rint}\langle x, y'\rangle \neq \emptyset$, which implies $\sigma_x = \sigma'_{x'}$ by (J_2'). Then, $y' \in \operatorname{fl}\{x, x'\} \subset \operatorname{fl}\sigma_x$, which is a contradiction because σ_x and $\tau'_{y'}$ are joinable. Hence, $x \notin \operatorname{rint}\langle x', y'\rangle$. Similarly, we have $y \notin \operatorname{rint}\langle x', y'\rangle$. Consequently, $\langle x, y\rangle \cap \operatorname{rint}\langle x', y'\rangle = \emptyset$. In the same way, we have $\langle x', y'\rangle \cap \operatorname{rint}\langle x, y\rangle = \emptyset$. Note that $x \neq y'$ and $y \neq x'$ because $|K| \cap |L| = \emptyset$, Therefore, $x = x'$ or $y = y'$. If $x = x'$, then

$$\langle x, y\rangle \cap \langle x', y'\rangle = \{x\} \subset \sigma \cap \sigma' \subset (\sigma \cap \sigma')(\tau \cap \tau').$$

The same is true for $y = y'$.

When K and L are joinable, we can obtain the following simplicial complex:

$$\begin{aligned} K * L &\underset{\text{def}}{=} \left\{\sigma\tau \;\middle|\; \sigma \in K \cup \{\emptyset\},\ \tau \in L \cup \{\emptyset\}\right\} \setminus \{\emptyset\} \\ &= K \cup L \cup \left\{\sigma\tau \;\middle|\; \sigma \in K,\ \tau \in L\right\}, \end{aligned}$$

which is called the **join** of K and L. When $K = \{v_0\}$, we write $v_0 * L$, which is called a **simplicial cone** over L.

[6] For definition, see Sect. 4.1. Recall $\sigma\emptyset = \emptyset\sigma = \sigma$ for any simplex σ and $\emptyset\emptyset = \emptyset$.

When K and L sit in linear spaces E and F respectively, they are regarded as simplicial complexes in the linear space $E \times F \times \mathbb{R}$ by the following identification:

$$E = E \times \{\mathbf{0}\} \times \{0\} \quad \text{and} \quad F = \{\mathbf{0}\} \times F \times \{1\}.$$

Then, K and L are joinable and the join $K * L$ can be defined for an arbitrary pair of simplicial complexes K and L.

Proposition 4.2.7 *Let K_i, $i = 1, 2, 3$, be simplicial complexes in the same linear space. If K_1 is joinable to K_2 and $K_1 * K_2$ is joinable to K_3 (hence K_2 is joinable to K_3), then, K_1 is joinable to $K_2 * K_3$ and $(K_1 * K_2) * K_3 = K_1 * (K_2 * K_3)$.*

Proof To see that K_1 is joinable to $K_2 * K_3$, we have to check the conditions (J_1) and (J_2). The condition (J_1) follows from the observation after the definition of the join of simplexes (see Sect. 4.1). To verify (J_2), let $\sigma_i, \tau_i \in K_i$, $i = 1, 2, 3$. Then, it follows that

$$\begin{aligned}\sigma_1(\sigma_2\sigma_3) \cap \tau_1(\tau_2\tau_3) &= (\sigma_1\sigma_2)\sigma_3 \cap (\tau_1\tau_2)\tau_3 \\ &= (\sigma_1\sigma_2 \cap \tau_1\tau_2)(\sigma_3 \cap \tau_3) = \big((\sigma_1 \cap \tau_1)(\sigma_2 \cap \tau_2)\big)(\sigma_3 \cap \tau_3) \\ &= (\sigma_1 \cap \tau_1)\big((\sigma_2 \cap \tau_2)(\sigma_3 \cap \tau_3)\big) = (\sigma_1 \cap \tau_1)(\sigma_2\sigma_3 \cap \tau_2\tau_3).\end{aligned}$$

Thus, (J_2) holds. Hence, K_1 is joinable to $K_2 * K_3$. The equality $(K_1 * K_2) * K_3 = K_1 * (K_2 * K_3)$ is trivial. □

Let K be a simplicial complex. Two simplexes $\sigma, \tau \in K$ are **joinable** in K (or σ is **joinable** to τ in K) if they are joinable and the join $\sigma\tau$ belongs to K. Two subcomplexes K_1, K_2 of K are **joinable** in K (or K_1 is **joinable** to K_2 in K) if they are joinable and the join $K_1 * K_2$ is a subcomplex of K. If $\sigma, \tau \in K$ are joinable in K, then $F(\sigma)$ and $F(\tau)$ are joinable in K and the following hold:

$$F(\sigma\tau) = F(\sigma) * F(\tau) \quad \text{and} \quad \partial F(\sigma\tau) = \big(F(\sigma) * \partial F(\tau)\big) \cup \big(\partial F(\sigma) * F(\tau)\big).$$

When K is a simplicial complex, the following subcomplex of K is called the **link** of $\sigma \in K$:

$$\begin{aligned}\mathrm{Lk}(\sigma, K) &\underset{\text{def}}{=} \mathrm{St}(\sigma, K) \setminus K[\sigma] = \big\{\tau \in \mathrm{St}(\sigma, K) \;\big|\; \tau \cap \sigma = \emptyset\big\} \\ &= \big\{\tau \in K \;\big|\; \tau \text{ is joinable to } \sigma \text{ in } K\big\}.\end{aligned}$$

Note that $\mathrm{Lk}(\sigma, K) = \emptyset$ if and only if σ is principal in K. For each non-principal simplex $\sigma \in K$, we have

$$\mathrm{St}(\sigma, K) = F(\sigma) * \mathrm{Lk}(\sigma, K) \quad \text{and} \quad |\mathrm{St}(\sigma, K)| = \bigcup_{\tau \in \mathrm{Lk}(\sigma, K)} \sigma\tau.$$

In the case when K is a simplicial complex,

$$O_K(v) = |\operatorname{St}(v, K)| \setminus |\operatorname{Lk}(v, K)| \quad \text{for each } v \in K^{(0)}.$$

This equality can be seen as follows:

$$\begin{aligned} O_K(v) &= \bigcup_{\sigma \in K[v]} \operatorname{rint} \sigma = \bigcup_{\tau \in \operatorname{Lk}(v,K)} \operatorname{rint}(v\tau) \cup \{v\} \\ &= \bigcup_{\tau \in \operatorname{Lk}(v,K)} (v\tau \setminus \tau) \cup \{v\} = |\operatorname{St}(v, K)| \setminus |\operatorname{Lk}(v, K)|. \end{aligned}$$

Proposition 4.2.8 *Let K be a simplicial complex and $v_1, \ldots, v_n \in K^{(0)}$. Then, $\langle v_1, \ldots, v_n \rangle \in K$ if and only if $\bigcap_{i=1}^n O_K(v_i) \neq \emptyset$.*

Proof If $\sigma = \langle v_1, \ldots, v_n \rangle \in K$, then $\bigcap_{i=1}^n O_K(v_i) \supset \operatorname{rint} \sigma \neq \emptyset$. Thus, we have the "only if" part. To prove the "if" part, assume that $\bigcap_{i=1}^n O_K(v_i)$ contains a point x. Then, $v_1, \ldots, v_n \in c_K(x)^{(0)}$. Hence, $\langle v_1, \ldots, v_n \rangle \in K$. □

Proposition 4.2.9 *Let K be a simplicial complex and L a subcomplex of K. Then, $O_K||L| = O_L$.*

Proof Since $O_L(v) = O_K(v) \cap |L|$ for each $v \in L^{(0)}$, we have $O_L \subset O_K||L|$. To prove $O_K||L| \subset O_L$, let $v \in K^{(0)}$ with $O_K(v) \cap |L| \neq \emptyset$. We have a simplex $\sigma \in L$ such that $O_K(v) \cap \sigma \neq \emptyset$, which means that $v \in \sigma^{(0)} \subset L^{(0)}$. Then, $O_K(v) \cap |L| = O_L(v)$. Therefore, $O_K||L| = O_L$. □

Note Related to the above proposition, it should be noted that $\mathcal{S}_K||L| \neq \mathcal{S}_L$ in general. Indeed, let $K = F(\sigma)$ and $L = \partial F(\sigma)$, where $\sigma = \langle v_0, v_1, v_2 \rangle$ is a 2-simplex. Then, $|\operatorname{St}(v_0, K)| \cap |L| = \partial\sigma$ but is not equal to any one of $|\operatorname{St}(v_i, L)|$, $i = 0, 1, 2$, which is the union of two 1-faces of σ.

A **full complex** (or **full simplicial complex**) is a simplicial complex K such that $K^{(0)}$ is affinely independent and $\langle v_1, \ldots, v_n \rangle \in K$ for all finitely many distinct vertices $v_1, \ldots, v_n \in K^{(0)}$, where $|K| = \langle K^{(0)} \rangle$. For any simplex σ, $F(\sigma)$ is a finite full complex with $|F(\sigma)| = \sigma$. Given an affinely independent set V in E, we have the following full complex:

$$\mathrm{F}_V = \big\{ \langle A \rangle \;\big|\; A \in \operatorname{Fin}(V) \big\},$$

where $(\mathrm{F}_V)^{(0)} = V$ and $|\mathrm{F}_V| = \langle V \rangle$. Note that $\mathrm{F}_{\sigma^{(0)}} = F(\sigma)$. In the case when V is infinite, $|\mathrm{F}_V| = \langle V \rangle$ might be considered as an infinite-dimensional simplex. An infinite full complex has no principal simplexes. For a simplicial complex K, if $K^{(0)}$ is affinely independent, then K is a subcomplex of the full complex $\mathrm{F}_{K^{(0)}}$.

On the other hand, it is said that a subcomplex L of a simplicial complex K is **full in** K or a **full subcomplex** of K if $\sigma \in L$ for any $\sigma \in K$ with $\sigma^{(0)} \subset L^{(0)}$, that is, L is a maximal subcomplex of K such that the set of vertices is $L^{(0)}$. The n-skeleton $K^{(n)}$ is not full in K unless $K^{(n)} = K$.

Note Although the same word *full* is used, *full subcomplex* and *full complex* are different concepts. The former is used in the relative sense, but the latter is in the absolute sense. It should be noted that a full subcomplex of a simplicial complex K is not always a full complex unless K is a full complex.

We have the following characterization of a full subcomplex:

Proposition 4.2.10 *For a subcomplex L of a simplicial complex K, the following are equivalent:*

(a) *L is full in K, that is, L is a full subcomplex of K;*
(b) *For $\sigma \in K$, $\sigma \cap |L| \neq \emptyset$ implies $\sigma \cap |L| \leqslant \sigma$;*
(c) *There does not exist $\sigma \in K \setminus L$ such that $\sigma \cap |L| = \partial\sigma \neq \emptyset$.*[7]

Proof The implication (b) $\Rightarrow$ (c) is obvious. So, we show (c) $\Rightarrow$ (a) $\Rightarrow$ (b).

(c) $\Rightarrow$ (a): If L is not full in K, then there exists a simplex $\tau \in K \setminus L$ with $\tau^{(0)} \subset L^{(0)}$. Let σ be a minimal face of τ such that $\sigma \notin L$. Then, $\dim \sigma \geqslant 1$ and any proper face of σ belongs to L because of the minimality of σ. Hence, $\partial\sigma \subset |L|$, which means that $\sigma \cap |L| = \partial\sigma$. This contradicts condition (c).

(a) $\Rightarrow$ (b): For $\sigma \in K$ with $\sigma \cap |L| \neq \emptyset$, $\sigma' = \langle \sigma^{(0)} \cap |L| \rangle$ is a face of σ. Then, $\sigma' \in L$ because L is full in K. For $\tau \in L[\sigma]$, since $(\tau \cap \sigma)^{(0)} \subset \sigma^{(0)} \cap |L| = \sigma'^{(0)}$, we have $\tau \cap \sigma \leqslant \sigma'$. Then,

$$\sigma \cap |L| = \bigcup_{\tau \in L[\sigma]} (\tau \cap \sigma) \subset \sigma' = \langle \sigma^{(0)} \cap |L| \rangle \subset \sigma \cap |L|.$$

Consequently, $\sigma \cap |L| = \sigma' \leqslant \sigma$. □

Proposition 4.2.11 (Some Topological Properties of Polyhedra) *For a cell complex K, the following hold:*

(1) *$|K|$ is separable if and only if K is countable.*

Sketch of Proof Each $C \in K$ has a countable dense set D_C. If K is countable, then $D = \bigcup_{C \in K} D_C$ is a countable dense set in $|K|$.

For a countable subset $A \subset |K|$, the following is countable:

$$\{C \in K \mid \exists D \in K \text{ such that } C \leqslant D,\ A \cap \operatorname{rint} D \neq \emptyset\}.$$

If K is uncountable, K has a cell $C_0 \in K$ such that $A \cap \operatorname{rint} C = \emptyset$ if $C_0 \leqslant C \in K$. Take $x \in \operatorname{rint} C_0$. Since $O_K(x) \cap A = \emptyset$, A is not dense in $|K|$.

(2) *The following are equivalent:*

(a) *K is locally finite*;
(b) *$|K|$ is locally compact*;
(c) *$|K|$ is metrizable*;
(d) *$|K|$ is first countable*.

[7] $\partial\sigma \neq \emptyset \Leftrightarrow \dim \sigma \geqslant 1$.

Sketch of Proof The implication (a) $\Rightarrow$ (b) follows from Proposition 4.2.3 and (c) $\Rightarrow$ (d) is obvious. Since a space is metrizable if it is a locally finite union of metrizable closed subspaces (Proposition 2.4.4 (2)), we have (a) $\Rightarrow$ (c) by Proposition 4.2.6.

(b) $\Rightarrow$ (a): By Proposition 4.2.3, we have a finite subcomplex $L \subset K$ such that $|L|$ is a neighborhood of v in $|K|$. For each $C \in K[v]$, $|L| \cap C$ is a neighborhood of v in C. Since $|L| \cap \operatorname{rint} C \neq \emptyset$, some $D \in L$ meets $\operatorname{rint} C$, which implies $C \leqslant D$. Hence, $\operatorname{St}(v, K) \subset L$.

(d) $\Rightarrow$ (a): Assume that $\operatorname{St}(v, K)$ is infinite for some $v \in K^{(0)}$. Then, v is a common vertex of distinct cells A_n, $n \in \mathbb{N}$. Let $\{U_n \mid n \in \mathbb{N}\}$ be a neighborhood basis at v in $|K|$. For each $n \in \mathbb{N}$, choose $a_n \in \operatorname{rint} A_n \cap U_n$. Then, $|K| \setminus \{a_n \mid n \in \mathbb{N}\}$ is an open neighborhood of v in $|K|$ but it does not contain any U_n, which is a contradiction.

Note There is another proof of (a) $\Rightarrow$ (c): When K is a simplicial complex, the implication (a) $\Rightarrow$ (c) will be reproved in Theorem 4.6.4, where an actual metric will be given for $|K|$. Due to Theorem 4.3.6, every cell complex has a simplicial subdivision. Evidently, every subdivision of a locally finite cell complex is also locally finite. Thus, we have (a) $\Rightarrow$ (c).

(3) *The following are equivalent*:

(a) *K is countable and locally finite*;
(b) *$|K|$ is second countable*;
(c) *$|K|$ is separable and metrizable*.

Sketch of Proof Combine (1) and (2) above.

(4) *In general*, $w(|K|) \neq \operatorname{dens}|K|$ (i.e., $w(|K|) > \operatorname{dens}|K|$).

Sketch of Proof Let $K = \{v_i \mid i \in \omega\} \cup \{\langle v_0, v_i\rangle \mid i \in \mathbb{N}\}$. Then, $\operatorname{dens}|K| = \aleph_0$ by (1) above. However, $|K|$ is not first countable by (2), hence it is not second countable, that is, $w(|K|) > \aleph_0$.

(5) *Let $f : X \to |K|$ be a map of a metrizable (more generally, first countable) space X. Then, each $x \in X$ has a neighborhood U_x in X such that $f(U_x) \subset |K_x|$ for some finite subcomplex K_x of K.*

Sketch of Proof If $x \in X$ does not have such a neighborhood, then we can find a sequence $(x_i)_{i\in\mathbb{N}}$ in X such that $\lim_{i\to\infty} x_i = x$ and $c_K(f(x_i)) \neq c_K(f(x_j))$ if $i \neq j$. Then, $\{f(x), f(x_i) \mid i \in \mathbb{N}\} \subset |K|$ is compact. This contradicts Proposition 4.2.3.

(6) *The following are equivalent*:

(a) *$|K|$ is connected*;
(b) *$|K|$ is path-connected*;
(c) *For every two distinct vertices $v, v' \in K^{(0)}$, there exist vertices $v_1, \ldots, v_n \in K^{(0)}$ such that $v_1 = v$, $v_n = v'$, and $\langle v_n, v_{n+1}\rangle \in K$, where $v_i \neq v_j$ if $i \neq j$.*

Sketch of Proof Since (c) $\Rightarrow$ (b) $\Rightarrow$ (a) are trivial. we have to show (a) $\Rightarrow$ (c). For each $C, C' \in K$, define $C \sim C'$ if $C = C'$ or there exist $v \in C^{(0)}$ and $v' \in C'^{(0)}$ which satisfy condition (c). Then, $\sim$ is an equivalent relation on K. Each equivalence class L of K is a subcomplex of K and $|L|$ is clopen in $|K|$. Hence, $K = L$ satisfies (c).

It is said that a cell complex K is **connected** if K satisfies (c) in (6) above.

The following is easy, but will serve to confirm the details in later discussions:

Proposition 4.2.12 *Let K be a cell complex.*

(1) *For a cell $C \in K$ and a subcomplex L of K, $C \in L$ if and only if $C \subset |L|$. Consequently, $L = \{C \in K \mid C \subset |L|\}$.*
(2) *For subcomplexes $L_1, L_2 \subset K$, $L_1 \subset L_2$ if and only if $|L_1| \subset |L_2|$. In particular, $L_1 = L_2$ if and only if $|L_1| = |L_2|$.*

Sketch of Proof The "only if" parts of both (1) and (2) are obvious. The "if" part of (2) easily follows from (1). For the "if" part of (1), take $D \in L$ so that $\operatorname{rint} C \cap \operatorname{rint} D \neq \emptyset$ and apply condition (C_2'').

4.3 Subdivisions of Cell Complexes

A **subdivision** of a cell complex K is a cell complex K' satisfying the following two conditions:

(S_1) Each cell of K is covered by finitely many cells of K';
(S_2) Each $C' \in K'$ is contained in some $C \in K$, i.e., $K' \prec K$.

Then, it is also said that K' **subdivides** K or K is **subdivided** by K'. Evidently, if K' is a subdivision of K and K'' is a subdivision of K', then K'' is a subdivision of K. A simplicial complex K' subdividing K is called a **simplicial subdivision** and is denoted as follows:

$$K' \lhd K \quad \text{or} \quad K \rhd K'.$$

As can be seen from Proposition 4.2.5, condition (S_1) guarantees the following:

- *For every subdivision K' of a cell complex K, $|K'| = |K|$ as spaces.*

Related to condition (S_2), we have the following:

Lemma 4.3.1 *Let K' be a subdivision of a cell complex K. For cells $C \in K'$ and $D \in K$, the following are equivalent:*

(a) *D is the smallest cell of K containing C;*
(b) $\operatorname{rint} C \subset \operatorname{rint} D$*;*
(c) $\operatorname{rint} C \cap \operatorname{rint} D \neq \emptyset$ *(i.e., some $x \in \operatorname{rint} C$ is contained in* $\operatorname{rint} D$*).*

Proof The implication (b) $\Rightarrow$ (c) is trivial because $\operatorname{rint} C \neq \emptyset$.

(a) $\Rightarrow$ (b): For each $x \in \operatorname{rint} C$, $C = C_x \subset D_x \leqslant D$. Since $D_x = D$ by (a), it follows that $x \in \operatorname{rint} D$. Hence, $\operatorname{rint} C \subset \operatorname{rint} D$.

(c) $\Rightarrow$ (a): Let D' be the smallest cell of K containing C. Since (a) $\Rightarrow$ (b) is valid, we have $\operatorname{rint} C \subset \operatorname{rint} D'$. Then, it follows from (c) that $\operatorname{rint} D \cap \operatorname{rint} D' \neq \emptyset$, which means $D = D'$ by (C_2''). □

It should be noted that condition (S_1) can be strengthened as follows:

Proposition 4.3.2 *Let K' be a subdivision of a cell complex K. For each $C \in K$, there are finitely many $D_1, \ldots, D_k \in K'$ such that $C = \bigcup_{i=1}^{k} \operatorname{rint} D_i = \bigcup_{i=1}^{k} D_i$.*

Proof Because of condition (S_1), we can find finitely many $D_1, \ldots, D_k \in K'$ such that $C \subset \bigcup_{i=1}^{k} \operatorname{rint} D_i \subset \bigcup_{i=1}^{k} D_i$ and each $\operatorname{rint} D_i$ meets C. For each $i = 1, \ldots, k$, let $C_i \in K$ be the smallest cell containing D_i. Then, $\operatorname{rint} D_i \subset \operatorname{rint} C_i$ by Lemma 4.3.1. Since $C \cap \operatorname{rint} C_i \neq \emptyset$, we have $C_i \leqslant C$ by (C_2'). Hence, $\bigcup_{i=1}^{k} D_i \subset \bigcup_{i=1}^{k} C_i \subset C$. □

Concerning the open star of subdivision, we have the following:

Proposition 4.3.3 *Let K' be a subdivision of a cell complex K. Then, $O_{K'}(v') \subset O_K(v)$ for each $v' \in K'^{(0)}$ and $v \in c_K(v')^{(0)}$. Consequently, $O_{K'} \prec O_K$.*

Proof For each $C \in K'$ with $v' \in C^{(0)}$, let $D \in K$ be the smallest cell containing C. Since $D \cap \operatorname{rint} c_K(v') \neq \emptyset$, we have $c_K(v') \leqslant D$ by (C_2'), hence $v \in c_K(v')^{(0)} \subset D^{(0)}$. On the other hand, $\operatorname{rint} C \subset \operatorname{rint} D \subset O_K(v)$ by Lemma 4.3.1. Therefore, $O_{K'}(v') \subset O_K(v)$. □

Restricting a subdivision, we can obtain a subdivision of a subcomplex:

Proposition 4.3.4 *Let K be a cell complex and K' a subdivision of K. Then, every subcomplex L of K is subdivided by the subcomplex $L' = \{C' \in K' \mid C' \subset |L|\}$ of K'. Moreover, $L' = K' \wedge L$.*

Proof Obviously, L' is a subcomplex of K'. For each $C' \in L'$, let $C \in K$ be the smallest cell containing C'. On the other hand, we have $D \in L$ such that $D \cap \operatorname{rint} C' \neq \emptyset$. Since $\operatorname{rint} C' \subset \operatorname{rint} C$ by Lemma 4.3.1, $D \cap \operatorname{rint} C \neq \emptyset$, which implies $C \leqslant D$ by (C_2'), hence $C \in L$. Thus, we have $L' \prec L$.

For each $C \in L$, we have finitely many $D_1', \ldots, D_k' \in K'$ such that $C = \bigcup_{i=1}^{k} D_i'$ by Proposition 4.3.2, where $D_i' \in L'$ because $D_i' \subset |L|$. Thus, C is covered by finitely many cells of L'.

Finally, we will show that $L' = K' \wedge L$. Each $C' \in L'$ is contained in some $D \in L$ because $L' \prec L$. Then, $C' = C' \cap D \in K' \wedge L$. Thus, we have $L' \subset K' \wedge L$. Conversely, for each $C' \in K'$, let $C \in K$ be the smallest cell containing C'. For each $D \in L$ with $C' \cap D \neq \emptyset$, since $C \cap D \leqslant C$, we can apply Proposition 4.1.9 (4) to obtain $C' \cap D = C' \cap (C \cap D) \leqslant C' \cap C = C'$. Hence, $C' \cap D \in K'$. Since $C' \cap D \subset D \subset |L|$, it follows that $K' \wedge L \subset L'$. □

The subdivision L' of L in Proposition 4.3.4 above is called the **induced subdivision** by K', where it is said that K' **induces** L' or L' is **induced** by K'. Due to Proposition 4.2.12 (2), the induced subdivision L' is characterized as the only subcomplex of K' such that $|L'| = |L|$. The pair (K', L') of subdivisions is called a **subdivision** of a pair (K, L) of a cell complex and a subcomplex. Obviously, if K' is simplicial, then so is L'. Then, (K', L') is called a **simplicial subdivision** of (K, L), which is denoted as follows:

$$(K', L') \lhd (K, L) \quad \text{or} \quad (K, L) \rhd (K', L').$$

Proposition 4.3.5 *Let K be a simplicial complex and L a full subcomplex of K. Every simplicial subdivision K' of K induces a subdivision L' of L being a full subcomplex of K' and $L' = K'||L|$, i.e., $K'||L| = \{\sigma \in K' \mid \sigma \subset |L|\}$.*

Proof Applying Proposition 4.2.10, we show that L' is a full subcomplex of K'. Let $\sigma' \in K'$ with $\sigma' \cap |L'| \neq \emptyset$. Then, σ' is contained in some $\sigma \in K$. Since $\sigma \cap |L| \neq \emptyset$, we have $\sigma \cap |L| \leqslant \sigma$ by Proposition 4.2.10. Then, by Proposition 4.1.9 (4),

$$\sigma' \cap |L'| = \sigma' \cap (\sigma \cap |L|) \leqslant \sigma' \cap \sigma = \sigma'.$$

Thus, L' is a full subcomplex of K' and $K'||L| \subset L'$. Since $L' \subset K'||L|$, we have $L' = K'||L|$. □

Remark 4.3.1 Proposition 4.3.5 above does not hold for a subcomplex L of K which is not full in K. Namely, $L' \neq K'||L|$ in general. For example, let $K = F(\Delta^3)$ and $L = \partial F(\Delta^2)$, where we regard $\Delta^2 \subset \Delta^3$. Take $v \in \operatorname{rint} \Delta^3$ and let $K' = v * \partial F(\Delta^3) \lhd K$. Then, $\Delta^2 \in K'$ but $\Delta^2 \cap |L| = \partial \Delta^2 \notin L$.

With regard to simplicial subdivisions, we have the following:

Theorem 4.3.6 *Let K be a cell complex with an order on $K^{(0)}$ such that there exists the maximum $v_C \in C^{(0)}$ for each cell $C \in K$ (e.g., a total order on $K^{(0)}$ satisfies this condition). Then, the following is a simplicial subdivision with $K'^{(0)} = K^{(0)}$:*

$$K' = \left\{ \langle v_{C_1}, \ldots, v_{C_n} \rangle \;\middle|\; C_1 > \cdots > C_n \in K \right\}.$$

Proof First of all, it should be remarked that the representation of simplexes of K' in the above is not unique. In fact, $v_{C_i} = v_{C_j}$ is possible for $i < j$. Then, in the above definition of K', by replacing $C_1 > \cdots > C_n \in K$ with its subsequence, we can add the following condition:

(i) $v_{C_1} > \cdots > v_{C_n}$.

Even if this condition is added, the representation of simplexes of K' in the above is not unique. It might be happen that $v_{C_i} = v_C$ for some $C \neq C_i$ with $C_{i-1} > C > C_{i+1}$. Then, each C_i can be replaced with the smallest cell of K containing $v_{C_i}, \ldots, v_{C_n}$, where the condition $C_1 > \cdots > C_n$ is preserved by Fact following the definition of $F(C)$ in Sect. 4.2. Thus, in the definition of K', we can add the following second condition:

(ii) Each C_i is the smallest cell of K containing $v_{C_i}, \ldots, v_{C_n}$.

It should be noted that (ii) implies (i). Indeed, if $v_{C_i} = v_{C_{i+1}}$, then C_{i+1} contains $v_{C_i}, \ldots, v_{C_n}$, which contradicts the minimality of C_i. Moreover, (ii) implies that $v_{C_1}, \ldots, v_{C_n}$ are affinely independent. Indeed, for each $i = 1, \ldots, n-1$, we have $v_{C_i} \notin C_{i+1}^{(0)}$ because $v_{C_i} > v_{C_{i+1}} = \max C_{i+1}^{(0)}$. Since $C_{i+1}^{(0)} = C_i^{(0)} \cap \operatorname{fl} C_{i+1}$ by Proposition 4.1.5 (iii), it follows that $v_{C_i} \notin \operatorname{fl} C_{i+1} \supset \operatorname{fl}\{v_{C_{i+1}}, \ldots, v_{C_n}\}$. Hence, $v_{C_1}, \ldots, v_{C_n}$ are affinely independent. Consequently, K' is a collection of simplexes.

To show that K' is a simplicial subdivision of K, we have to verify the conditions (C_1), (C_2''), (S_1), and (S_2), where (C_1) and (S_2) are obvious. To see (C_2'') and (S_1), it suffices to show the following:

Claim *For each point $x \in |K|$, there exists a unique sequence $C_1 > \cdots > C_n \in K$ such that $x \in \operatorname{rint}\langle v_{C_1}, \ldots, v_{C_n}\rangle$ and each C_i is the smallest cell of K containing $v_{C_i}, \ldots, v_{C_n}$, i.e., (ii) holds.*

Obviously, this claim implies (C_2''). Since $\langle v_{C_1}, \ldots, v_{C_n}\rangle \subset C_1$ for $C_1 > \cdots > C_n \in K$ and each cell of K has only finitely many faces, (S_1) follows from this claim.

To prove the claim, for each $C \in K \setminus K^{(0)}$ and $x \in \operatorname{rint} C$, define $\pi_C(x) \in \partial C$ as the unique point such that $x \in \operatorname{rint}\langle v_C, \pi_C(x)\rangle$. Such a point exists because C is a compact convex set in $\operatorname{fl} C$ with a unique topology as in Proposition 3.5.1 and such a point is unique by Proposition 3.2.6.

(*Uniqueness*) Let $C_1 > \cdots > C_n \in K$ be a sequence as in the claim. Since $x \in \operatorname{rint}\langle v_{C_1}, \ldots, v_{C_n}\rangle$, we have $\langle v_{C_1}, \ldots, v_{C_n}\rangle \subset (C_1)_x$ by the definition of $(C_1)_x$. From the minimality of C_1, it follows that $C_1 = (C_1)_x$, which means $x \in \operatorname{rint} C_1$ (Proposition 3.2.9 (viii)). Hence, $C_1 = c_K(x)$. When $n > 1$, let $x_1 = x$ and $x_2 = \pi_{C_1}(x_1)$. Then, $x_2 \in \operatorname{rint}\langle v_{C_2}, \ldots, v_{C_n}\rangle$. By the same argument for x, we can see $x_2 \in \operatorname{rint} C_2$ and $C_2 = c_K(x_2)$. Now, we can inductively define $x_i = \pi_{C_{i-1}}(x_{i-1})$, $i = 2, \ldots, n$, and it follows that $x_i \in \operatorname{rint}\langle v_{C_i}, \ldots, v_{C_n}\rangle$ and $C_i = c_K(x_i)$, where $x_n = v_{C_n} \in C_n = \{v_{C_n}\}$. By the same argument as x, we have $C_i = c_K(x_i)$. Thus, a finite sequence $C_1 > \cdots > C_n \in K$ in the claim is uniquely determined.

(*Existence*) For each $x \in |K|$, let $x_1 = x$ and $C_1 = c_K(x_1) \in K$. If $\dim C_1 = 0$, then $C_1 = \{x_1\} = \{v_{C_1}\}$ and $x = x_1 \in \operatorname{rint}\langle v_{C_1}\rangle$. Otherwise, we define $x_2 = \pi_{C_1}(x_1) \in \partial C_1$. Then, $x_2 \in \operatorname{rint}\langle v_{C_1}, x_2\rangle$ and let $C_2 = c_K(x_2) < C_1$. For any $C_1 > C \geqslant C_2$, we have $v_{C_1} \notin C^{(0)}$ because $x_1 \notin C$, $x_2 \in C$, and $x_1 \in \langle v_{C_1}, x_2\rangle$. If $\dim C_2 = 0$, then $C_2 = \{x_2\} = \{v_{C_2}\}$ and $x = x_1 \in \operatorname{rint}\langle v_{C_1}, v_{C_2}\rangle$.

When x_i and $C_i = c_K(x_i)$ have been defined and $C_i \neq \{x_i\}$ (i.e., $\dim C_i > 0$), we define $x_{i+1} = \pi_{C_i}(x_i) \in \partial C_i$. Then, $x_i \in \operatorname{rint}\langle v_{C_i}, x_{i+1}\rangle$ and let $C_{i+1} = c_K(x_{i+1}) < C_i$. By induction, we can obtain a finite sequence $x = x_1, x_2, \ldots, x_n$ and cells $C_1 > C_2 > \cdots > C_n = \{x_n\}$ such that $C_i = c_K(x_i)$ and $x_i \in \langle v_{C_i}, x_{i+1}\rangle$. Similarly to v_{C_1}, we have $v_{C_i} \notin C^{(0)}$ for any $C_i > C \geqslant C_{i+1}$ $(i < n)$. By Proposition 4.1.1, we have

$$x = x_1 \in \operatorname{rint}\langle v_{C_1}, x_2\rangle \subset \operatorname{rint}\langle v_{C_1}, v_{C_2}, x_3\rangle \subset \cdots \subset \operatorname{rint}\langle v_{C_1}, v_{C_2}, \ldots, v_{C_n}\rangle.$$

Moreover, using the fact that $v_{C_i} \notin C^{(0)}$ for $C_i > C \geqslant C_{i+1}$ $(i < n)$, we can see that each C_i is the smallest cell of K containing $v_{C_i}, \ldots, v_{C_n}$.

This can be shown by a downward induction on i as follows: For $i = n$, there is nothing to do because $C_n = \{v_{C_n}\}$. Let $i < n$ and assume that C_{i+1} is the smallest cell containing $v_{C_{i+1}}, \ldots, v_{C_n}$. To see the minimality of C_i, let $v_{C_i}, \ldots, v_{C_n} \in D \in K$ and let $C = D \cap C_i$. Then, $v_{C_{i+1}}, \ldots, v_{C_n} \in C \leqslant C_i$, which implies $C_{i+1} \leqslant C$ by the minimality of C_{i+1}. If $C_i \neq C$, then $C_i > C \geqslant C_{i+1}$, and hence $v_{C_i} \notin C^{(0)} = D^{(0)} \cap C_i^{(0)}$, which is a contradiction. Thus, $C_i = C \leqslant D$.

The proof is completed. □

Remark 4.3.2 In Theorem 4.3.6 above, it should be noted that if a simplicial complex L is a subcomplex of K, then L is also a subcomplex of K'.

A cell complex K with such an order on $K^{(0)}$ as in Theorem 4.3.6 is called an **ordered cell complex**. If K is a simplicial complex, $\sigma^{(0)}$ has the maximum for each simplex $\sigma \in K$ if and only if $\sigma^{(0)}$ is totally ordered for each $\sigma \in K$. Thus, an **ordered simplicial complex** is a simplicial complex K with an order on $K^{(0)}$ such that $\sigma^{(0)}$ is totally ordered for each $\sigma \in K$.

We show that two cell complexes have a common subdivision if they have the same polyhedron, that is:

Proposition 4.3.7 *For every two cell complexes K_1 and K_2, if $|K_1| = |K_2|$ as spaces, then $K_1 \wedge K_2$ is a common subdivision of K_1 and K_2.*

In the above, recall

$$K_1 \wedge K_2 = \left\{ C \cap D \;\middle|\; C \in K_1,\ D \in K_2,\ C \cap D \neq \emptyset \right\}.$$

Then, if L is a subcomplex of both K_1 and K_2, then $L \subset K_1 \wedge K_2$. Due to Proposition 4.2.5 and the definition of a subdivision of a cell complex, the above Proposition 4.3.7 is easily derived from the following proposition:

Proposition 4.3.8 *For every two cell complexes K_1 and K_2 in the same linear space, let $K = K_1 \wedge K_2$. Then, the following hold:*

(1) *K is a cell complex with $|K| = |K_1| \cap |K_2|$ (as sets), where if L is a subcomplex of both K_1 and K_2, then L is also a subcomplex of K. Moreover,*

$$K = \left\{ C \cap D \;\middle|\; C \in K_1,\ D \in K_2,\ \operatorname{rint} C \cap \operatorname{rint} D \neq \emptyset \right\}. \qquad (\sharp)$$

(2) *$|K|$ is a closed subspace of both $|K_1|$ and $|K_2|$ if and only if each cell of K_i, $i = 1, 2$, contains interior points of only finitely many cells of K_{3-i}.*

Proof (1): For each $C \in K_1$ and $D \in K_2$ with $C \cap D \neq \emptyset$, $C \cap D$ is a cell and $(C \cap D)_x = C_x \cap D_x$ for each $x \in C \cap D$ by Proposition 4.1.9 (2). Thus, K satisfies (C_1). Here, if $\operatorname{rint} C \cap \operatorname{rint} D = \emptyset$, take $x \in \operatorname{rint}(C \cap D)$. Then, $x \in \operatorname{rint} C_x \cap \operatorname{rint} D_x$ $(\neq \emptyset)$ by Proposition 3.2.9 (viii) and $C_x \cap D_x = (C \cap D)_x = C \cap D$ by Proposition 4.1.9 (2) (Proposition 3.2.9 (xii)). Hence, $(\sharp)$ is easy to see.

To verify (C_2''), let $C, C' \in K_1$ and $D, D' \in K_2$ with $\operatorname{rint} C \cap \operatorname{rint} D \neq \emptyset$ and $\operatorname{rint} C' \cap \operatorname{rint} D' \neq \emptyset$. Assume that $\operatorname{rint}(C \cap D) \cap \operatorname{rint}(C' \cap D') \neq \emptyset$. By Proposition 4.1.9 (2), we have $\operatorname{rint} C \cap \operatorname{rint} D \cap \operatorname{rint} C' \cap \operatorname{rint} D' \neq \emptyset$. Since K_1 and K_2 satisfy (C_2''), it follows that $C = C'$ and $D = D'$, so $C \cap D = C' \cap D'$. The additional statement is obvious from the definition of K.

(2): Assume that $|K|$ is a closed subspace of $|K_i|$ $(i = 1, 2)$. By Proposition 4.2.4, each cell $C \in K_i$ contains interior points of only finitely many $D_1, \ldots, D_n \in K$. For

each $j = 1, \dots, n$, there is a unique $D'_j \in K_{3-i}$ such that $\operatorname{rint} D_j \subset \operatorname{rint} D'_j$. Then, C does not contain interior points of cells of $K_{3-i} \setminus \{D'_1, \dots, D'_n\}$.

Conversely, assume that each cell $C \in K_i$ contains interior points of only finitely many cells of K_{3-i}. Since C does not contain interior point of cells of K_i except faces of C, it follows from ($\sharp$) and Proposition 4.1.9 (2) that C contains interior points of only finitely many cells of K. Moreover, $K \prec K_i$, that is, each cell of K is contained in a cell of K_i. Hence, $|K|$ is a closed subspace of $|K_i|$ by Proposition 4.2.4. □

Combining Proposition 4.3.7 and Theorem 4.3.6 with Remark 4.3.2, we have:

Corollary 4.3.9 *Let K_1 and K_2 be cell complexes such that $|K_1| = |K_2|$ as spaces. Then, K_1 and K_2 have a common simplicial subdivision K (i.e., $K \lhd K_1 \wedge K_2$), where if L is a common subcomplex of both K_1 and K_2 which is simplicial, then K can be taken so that $L \subset K$ (i.e., L is a subcomplex of K).* □

Remark 4.3.3 In Proposition 4.3.7, if L_1 and L_2 are subcomplexes of K_1 and K_2, then $L_1 \wedge L_2$ is a subcomplex of $K_1 \wedge K_2$. Hence, for pairs (K_1, L_1) and (K_2, L_2) of cell complexes and subcomplexes, if $(|K_1|, |L_1|) = (|K_2|, |L_2|)$ as pairs of spaces, then $(K_1 \wedge K_2, L_1 \wedge L_2)$ is their common subdivision.

Now, we introduce a derived subdivision of a cell complex, which is a simplicial subdivision. The following lemma is trivial but useful to construct simplicial subdivisions:

Lemma 4.3.10 *Let C be a cell with $u_0 \in \operatorname{rint} C$ and L a simplicial subdivision of the cell complex $\partial F(C)$, i.e., $L \lhd \partial F(C)$. Then, u_0 and L are joinable, and the simplicial cone $u_0 * L$ over L is a simplicial subdivision of $F(C)$, i.e., $u_0 * L \lhd F(C)$, where L is a full subcomplex of $u_0 * L$.* □

Using Lemma 4.3.10 above, we prove the following:

Proposition 4.3.11 *Let K be a cell complex and L' a simplicial subdivision of a subcomplex L of K. Given $v_C \in \operatorname{rint} C$ for each $C \in K \setminus L$, there exists a simplicial subdivision K' of K such that L' is a full subcomplex of K' and*

$$K'^{(0)} = L'^{(0)} \cup K^{(0)} \cup \left\{ v_C \;\middle|\; C \in K \setminus L \right\}.$$

Proof For each $n \in \omega$, let $K_n = L \cup K^{(n)}$. Then, each K_n is a subcomplex of K and $K = \bigcup_{n \in \mathbb{N}} K_n$. Note that $K'_0 = L' \cup K^{(0)} \lhd K_0$ with $K'^{(0)}_0 = L'^{(0)} \cup K^{(0)}$ and L' is full in K'_0. Suppose that $K'_{n-1} \lhd K_{n-1}$ was constructed so that L' is a full subcomplex of K'_{n-1} and

$$K'^{(0)}_{n-1} = K'^{(0)}_0 \cup \left\{ v_C \;\middle|\; C \in K^{(n-1)} \setminus L \right\}.$$

Then, K'_{n-1} induces the subdivision $K_{\partial C} \lhd \partial F(C)$ for each $C \in K^{(n)} \setminus K_{n-1}$ (Proposition 4.3.4). By Lemma 4.3.10, we have

$$K_C = K_{\partial C} \cup \{v_C\} \cup \left\{ v_C \sigma \;\middle|\; \sigma \in K_{\partial C} \right\} \lhd F(C),$$

where $K_{\partial C}$ is full in K_C. We define a simplicial complex

$$K'_n = K'_{n-1} \cup \bigcup_{C \in K^{(n)} \setminus K_{n-1}} K_C.$$

As is easily observed, $K'_n \lhd K_n$, L' is a full subcomplex in K'_n, and

$$K'^{(0)}_n = K'^{(0)}_0 \cup \{v_C \mid C \in K^{(n)} \setminus L\}.$$

Thus, we can inductively construct $K'_n \lhd K_n$, $n \in \mathbb{N}$, with the above conditions. Then, $K' = \bigcup_{n\in\mathbb{N}} K'_n$ is the desired simplicial subdivision. □

When $L = \emptyset$ in Proposition 4.3.11 above, the obtained subdivision K' is called a **derived subdivision** of K, which is simplicial. As can be seen from the above proof, K' can also be defined as follows:

$$K' = \left\{\langle v_{C_1}, \ldots, v_{C_n}\rangle \;\middle|\; C_1 < \cdots < C_n \in K\right\},$$

and K' is an ordered simplicial complex with the natural order on $K'^{(0)}$ defined as follows: $v_C \leqslant v_D$ if $C \leqslant D$. Any derived subdivision of K induces a derived subdivision of every subcomplex of K. If K is simplicial and $v_\sigma = \hat{\sigma}$ for each $\sigma \in K$, the derived subdivision K' of K is called the **barycentric subdivision** of K, which is denoted by $\mathrm{Sd}\, K$. The barycentric subdivision of K induces the barycentric subdivision of each subcomplex of K.

When L is simplicial and $L' = L$ in Proposition 4.3.11, we call K' a **derived subdivision** of K **relative to** L, where

$$\begin{aligned} K' = L \cup \left\{\langle v_{C_1}, \ldots, v_{C_n}\rangle \;\middle|\; C_1 < \cdots < C_n \in K \setminus L\right\} \\ \cup \left\{\langle v_1, \ldots, v_m, v_{C_1}, \ldots, v_{C_n}\rangle \;\middle|\; \langle v_1, \ldots, v_m\rangle \in L, \right. \\ \left. C_1 < \cdots < C_n \in K \setminus L,\ \langle v_1, \ldots, v_m\rangle < C_1\right\}. \end{aligned}$$

If K is simplicial (which implies that L is also simplicial) and $v_\sigma = \hat{\sigma}$ for each $\sigma \in K \setminus L$, the derived subdivision of K relative to L is called the **barycentric subdivision** of K **relative to** L, which is denoted by $\mathrm{Sd}_L\, K$ (cf. Fig. 4.1). Concerning derived subdivisions, we have the following:

Proposition 4.3.12 *Let K' be a derived subdivision of a cell complex K. For each subcomplex L of K, the subdivision L' of L induced from K' is a full subcomplex of K'. If L is simplicial and K' is a derived subdivision of K relative to L, then L is a full subcomplex of K'. In particular, when K is a simplicial complex,* $\mathrm{Sd}\, L$ *and* L *are full subcomplexes of* $\mathrm{Sd}\, K$ *and* $\mathrm{Sd}_L\, K$*, respectively.* □

Note The above second assertion is just the case $L' = L$ of Proposition 4.3.11.

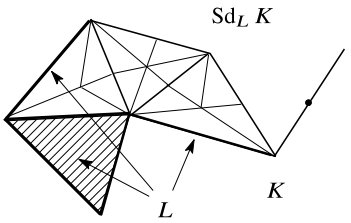


Fig. 4.1 Definition of $\mathrm{Sd}_L\, K$

4.4 Combinatorial Equivalence and Triangulation of Polyhedra

In this section, we will discuss equivalence relations among cell complexes and triangulation of polyhedra.

Let K_1 and K_2 be cell complexes. It is said that K_1 is **congruent** to K_2, or K_1 and K_2 are **congruent**, written as $K_1 \equiv K_2$, if there exists a bijection $\eta : K_1 \to K_2$ such that

$$C, D \in K_1,\ D \leqslant C \ \Leftrightarrow\ \eta(D) \leqslant \eta(C) \quad \big(\text{equiv., } D < C \ \Leftrightarrow\ \eta(D) < \eta(C)\big).$$

It should be noted that η is a function between two collections of cells but not a map from $|K_1|$ to $|K_2|$. In fact, η is an order isomorphism with respect to the face relation $\leqslant$.[8] Hence, $K_1 \equiv K_2$ means that ordered sets $(K_1, \leqslant)$ and $(K_2, \leqslant)$ are order isomorphic. Thus, $\equiv$ is an equivalence relation among cell complexes. Observe that $\eta(K_1^{(n)}) = K_2^{(n)}$ for every $n \in \omega$. In particular, $\eta|K_1^{(0)} : K_1^{(0)} \to K_2^{(0)}$ is a bijection. For each $C \in K_1$, $\eta(C)^{(0)} = \eta(C^{(0)})$ and $\dim \eta(C) = \dim C$. Since a cell $C \in K_1$ is a simplex if and only if $\mathrm{card}\, C^{(0)} = \dim C + 1$, it follows that C is a simplex if and only if $\eta(C)$ is a simplex. Therefore, if $K_1 \equiv K_2$ and K_1 is a simplicial complex, then so is K_2.

If K_1' and K_2' are derived subdivisions of K_1 and K_2 respectively, then $K_1 \equiv K_2$ implies $K_1' \equiv K_2'$. In particular, every derived subdivision K' of a simplicial complex K is congruent to $\mathrm{Sd}\, K$, i.e., $K' \equiv \mathrm{Sd}\, K$. Moreover, by using Theorem 4.3.6, it can be shown that every pair of congruent cell complexes K_1 and K_2 has simplicial subdivisions $K_1' \lhd K_1$ and $K_2' \lhd K_2$ such that $K_1'^{(0)} = K_1^{(0)}$, $K_2'^{(0)} = K_2^{(0)}$, and $K_1' \equiv K_2'$.

> Indeed, we can give orders on $K_1^{(0)}$ and $K_2^{(0)}$ so that K_1 and K_2 are ordered cell complexes and $\eta|K_1^{(0)} : K_1^{(0)} \to K_2^{(0)}$ is an order isomorphism. Using these orders, we can apply Theorem 4.3.6 to obtain $K_1' \lhd K_1$ and $K_2' \lhd K_2$ such that $K_i'^{(0)} = K_i^{(0)}$, $i = 1, 2$, and $K_1' \equiv K_2'$.

For pairs (K_1, L_1) and (K_2, L_2) of cell complexes and subcomplexes, when the bijection $\eta : K_1 \to K_2$ in the definition of $K_1 \equiv K_2$ satisfies $\eta(L_1) = L_2$, it is said

[8] Cf. Fact following the definition of $F(C)$ in Sect. 4.2.

that (K_1, L_1) is **congruent** to (K_2, L_2), or (K_1, L_1) and (K_2, L_2) are **congruent**, written as $(K_1, L_1) \equiv (K_2, L_2)$. Then, $\equiv$ is an equivalence relation among pairs of cell complexes. As in the above, by Theorem 4.3.6, if $(K_1, L_1) \equiv (K_2, L_2)$, then there are simplicial subdivisions $(K_1', L_1') \lhd (K_1, L_1)$ and $(K_2', L_2') \lhd (K_2, L_2)$ such that $K_1'^{(0)} = K_1^{(0)}$, $K_2'^{(0)} = K_2^{(0)}$, and $(K_1', L_1') \equiv (K_2', L_2')$, where $L_i' \lhd L_i$ is induced from $K_i' \lhd K_i$.

When $L_1 = L_2 = L$ and $\eta : K_1 \to K_2$ satisfies $\eta|L = \mathrm{id}$, it is said that K_1 is **congruent** to K_2 **relative to** L, or K_1 and K_2 are **congruent relative to** L, written as $K_1 \equiv K_2$ rel. L. Moreover, if $K_1 \equiv K_2$ rel. L, then there exist simplicial subdivisions $K_1' \lhd K_1$ and $K_2' \lhd K_2$ such that $K_1'^{(0)} = K_1^{(0)}$, $K_2'^{(0)} = K_2^{(0)}$, K_1' and K_2' induce the same simplicial subdivision $L' \lhd L$, and $K_1' \equiv K_2'$ rel. L', where if L is simplicial, then $L' = L$. Moreover, every derived subdivision K' of a simplicial complex K relative to L is congruent to $\mathrm{Sd}_L K$ relative to L, i.e., $K' \equiv \mathrm{Sd}_L K$ rel. L.

It is said that K_1 is **combinatorially equivalent** to K_2, or K_1 and K_2 are **combinatorially equivalent**, denoted by $K_1 \cong K_2$ if they have subdivisions which are congruent. As we just saw above, subdivisions can be simplicial. Thus, we can define as follows:

$$K_1 \cong K_2 \underset{\text{def}}{\Leftrightarrow} \exists K_1' \lhd K_2,\ \exists K_2' \lhd K_2 \ \text{ such that } \ K_1' \equiv K_2'.$$

Due to Proposition 4.3.7, the following holds:

$$|K_1| = |K_2| \ \text{ (as spaces)} \ \Rightarrow \ K_1 \cong K_2.$$

For pairs (K_1, L_1) and (K_2, L_2), it is said that (K_1, L_1) is **combinatorially equivalent** to (K_2, L_2) or (K_1, L_1) and (K_2, L_2) are **combinatorially equivalent**, denoted by $(K_1, L_1) \cong (K_2, L_2)$ if they have (simplicial) subdivisions which are congruent, that is,

$$(K_1, L_1) \cong (K_2, L_2) \underset{\text{def}}{\Leftrightarrow} \exists (K_1', L_1') \lhd (K_1, L_1),\ \exists (K_2', L_2') \lhd (K_2, L_2) \ \text{ such that } \ (K_1', L_1') \equiv (K_2', L_2').$$

In the above, when $|L_1| = |L_2|$ (as spaces), $L_1' = L_2' = L'$, and $K_1' \equiv K_2'$ rel. L' (note that $L' \lhd L_1 \wedge L_2$), we say that K_1 is **combinatorially equivalent** to K_2 **relative to** $L_1 \wedge L_2$, or K_1 and K_2 are **combinatorially equivalent relative to** $L_1 \wedge L_2$, and write $K_1 \cong K_2$ rel. $L_1 \wedge L_2$. Namely,

$$K_1 \cong K_2 \ \text{ rel. } L_1 \wedge L_2 \underset{\text{def}}{\Leftrightarrow} \exists K_1' \lhd K_1,\ \exists K_2' \lhd K_2,\ \exists L' \lhd L_1 \wedge L_2 \ \text{ such that } \ L' \subset K_1', K_2' \ \text{ and } \ K_1' \equiv K_2' \ \text{ rel. } L',$$

where $L_1 \wedge L_2 = L$ if $L_1 = L_2 = L$. By the observation in Remark 4.3.3, we have the following implication:

$$(|K_1|, |L_1|) = (|K_2|, |L_2|) \ \Rightarrow \ K_1 \cong K_2 \ \text{ rel. } L_1 \wedge L_2.$$

For the relations $\equiv$ and $\cong$ defined above, the following implications are obvious:

$$K_1 \equiv K_2 \Rightarrow K_1 \cong K_2; \quad K_1 \equiv K_2 \text{ rel. } L \Rightarrow K_1 \cong K_2 \text{ rel. } L;$$
$$K_1 \cong K_2 \text{ rel. } L_1 \wedge L_2 \Rightarrow (K_1, L_1) \cong (K_2, L_2);$$
$$(K_1, L_1) \equiv (K_2, L_2) \Rightarrow (K_1, L_1) \cong (K_2, L_2).$$

Compared with $\equiv$, it is non-trivial that $\cong$ is an equivalence relation among cell complexes (or pairs of cell complexes). In fact, the transitivity of $\cong$ needs to be proved, which will be done in Proposition 4.4.4.

For two cells C and D, there are many cases that $F(C) \equiv F(D)$ even if C and D are not affinely homeomorphic. In these cases, given a subdivision K of $F(C)$, it is not clear to construct a subdivision L of $F(D)$ such that $K \equiv L$. Concerning subdivisions of congruent simplicial complexes, we show the following:

Proposition 4.4.1 *Let K_1 and K_2 be simplicial complexes such that $K_1 \equiv K_2$. For each subdivision K_1' of K_1, there is a subdivision K_2' of K_2 such that $K_1' \equiv K_2'$. When K_1 and K_2 have a common subcomplex L and $K_1 \equiv K_2$ rel. L, if L' is a subdivision of L induced by K_1', then the above K_2' satisfies that $L' \subset K_2'$ and $K_1' \equiv K_2'$ rel. L'.*

Proof Note that K_1 is a simplicial complex but K_1' is not necessarily simplicial. Since $K_1 \equiv K_2$, we have a bijection $\eta : K_1 \to K_2$ such that

$$\forall \sigma, \forall \tau \in K_1, \ \tau \leqslant \sigma \ \Leftrightarrow \ \eta(\tau) \leqslant \eta(\sigma).$$

As observed in the definition of congruence, $\eta|K_1^{(0)} : K_1^{(0)} \to K_2^{(0)}$ is a bijection and $\eta(\sigma)^{(0)} = \eta(\sigma^{(0)})$ for each simplex $\sigma \in K$, where $\sigma^{(0)}$ and $\eta(\sigma)^{(0)}$ are affinely independent. We can easily obtain a homeomorphism $\varphi : |K_1| \to |K_2|$ such that $\varphi|K_1^{(0)} = \eta|K_1^{(0)}$ and $\varphi|\sigma : \sigma \to \varphi(\sigma) = \eta(\sigma)$ is an affine homeomorphism for every $\sigma \in K_1$.[9] For each simplex $\sigma \in K_1$, $K_\sigma' = \{C \in K_1' \mid C \subset \sigma\}$ is a subdivision of $F(\sigma)$ by Proposition 4.3.4. Since $\varphi|\sigma$ is an affine homeomorphism, $\varphi(C)$ is a cell for each $C \in K_1'$. Observe that $\varphi(K_\sigma') = \{\varphi(C) \mid C \in K_1', \ C \subset \sigma\}$ is a subdivision of $F(\varphi(\sigma)) = F(\eta(\sigma))$. Then, it is straightforward to show that $K_2' = \varphi(K_1') = \bigcup_{\sigma \in K_1} \varphi(K_\sigma')$ is a subdivision of K_2 and $K_1' \equiv K_2'$.

When $K_1 \equiv K_2$ rel. L, since $L \subset K_1$ and $\eta|L = \mathrm{id}$, we have $\varphi||L| = \mathrm{id}$ by the definition of φ. Then, it follows that $L' = \varphi(L') \subset K_2'$ and $K_1' \equiv K_2'$ rel. L'. □

For cell complexes, we have the following:

Proposition 4.4.2 *Let K_1 and K_2 be cell complexes such that $K_1 \equiv K_2$. For each subdivision K_1' of K_1, there is a subdivision K_2' of K_2 such that $K_1' \cong K_2'$. When K_1 and K_2 have a common subcomplex L and $K_1 \equiv K_2$ rel. L, the above K_2' satisfies that $K_1' \cong K_2'$ rel. $L_1' \wedge L_2'$, where L_i' is the subdivision of L induced by K_i', $i = 1, 2$.*

[9] In Sect. 4.10, such a homeomorphism φ will be called a **simplicial homeomorphism** (with respect to K_1 and K_2).

Proof Because of similarity, we give a proof for the relative case only. As observed just after the definition of congruence, applying Theorem 4.3.6, we can take simplicial subdivisions $K_1'' \lhd K_1$ and $K_2'' \lhd K_2$ such that K_1'' and K_2'' induce the same subdivision $L'' \lhd L$ and $K_1'' \equiv K_2''$ rel. L''. Since $L_1' \wedge L'' \subset K_1' \wedge K_1''$, we can apply Proposition 4.4.1 to obtain a subdivision K_2' of K_2'' such that $L_1' \wedge L'' \subset K_2'$ and $K_2' \equiv K_1' \wedge K_1''$ rel. $L_1' \wedge L''$, where $L_2' = L_1' \wedge L''$ is a subdivision of L induced by K_2'. Hence, $K_1' \cong K_2'$ rel. $L_1' \wedge L_2'$. □

The following can be proved by a suitable modification of the above proof, but we leave the details to the reader.

Proposition 4.4.3 *Let (K_1, L_1) and (K_2, L_2) be pairs of cell complexes and subcomplexes such that $(K_1, L_1) \equiv (K_2, L_2)$. For each subdivision (K_1', L_1') of (K_1, L_1), there is a subdivision (K_2', L_2') of (K_2, L_2) such that $(K_1', L_1') \cong (K_2', L_2')$, where (K_2', L_2') can be taken so that $(K_1', L_1') \equiv (K_2', L_2')$ in the case when K_1 and K_2 are simplicial complexes.*

Now, we can prove the following:

Proposition 4.4.4 *The combinatorial equivalence $\cong$ is an equivalence relation among cell complexes (or pairs of cell complexes). Given a cell complex L, $\cong$ rel. L is an equivalence relation among cell complexes containing L as a subcomplex.*

Proof It is obvious that $\cong$ is reflective and symmetric. To see that $\cong$ is transitive, let $K_1 \cong K_2$ and $K_2 \cong K_3$. Then, we have $K_1' \lhd K_1$, $K_2' \lhd K_2$, $K_2'' \lhd K_2$, $K_3'' \lhd K_3$ such that $K_1' \equiv K_2'$ and $K_2'' \equiv K_3''$. Since $|K_2'| = |K_2''|$, we apply Corollary 4.3.9 to obtain $K_2''' \lhd K_2' \wedge K_2''$. By Proposition 4.4.1, K_1 and K_3 have subdivisions K_1''' and K_3''' such that $K_1''' \equiv K_2''' \equiv K_3'''$. Thus, we have $K_1 \cong K_3$. (It is easy to adjust the above proof to pairs. The details are left to the reader.)

In the above, when $K_1 \cong K_2$ rel. L and $K_2 \cong K_3$ rel. L, we have $K_1' \equiv K_2'$ rel. L' and $K_2'' \equiv K_3''$ rel. L'', where $L \rhd L' \subset K_1'$, K_2' and $L \rhd L'' \subset K_2''$, K_3''. Then, the above $K_2''' \lhd K_2' \wedge K_2''$ induces $L''' \lhd L' \wedge L''$. The above K_1''' and K_3''' obtained from Proposition 4.4.1 satisfy that $L''' \subset K_i'''$ and $K_i''' \equiv K_2'''$ rel. L''', $i = 1, 3$. Hence, $K_1 \cong K_3$ rel. L. □

A **polyhedron** (resp. **topological polyhedron**) is defined as a space P such that $P = |K|$ (resp. $P \approx |K|$) for some cell complex K. According to Theorem 4.3.6,

- P is a polyhedron (resp. topological polyhedron) if and only if $P = |K|$ (resp. $P \approx |K|$) for some *simplicial* complex K.

Such a simplicial complex K is called a **triangulation** (resp. **topological triangulation**) of P, where it is also said that P is **triangulated** (resp. **topologically triangulated**) by K or K **triangulates** (resp. **topologically triangulates**) P. Note that $\mathbf{B}^2$ is a topological polyhedron but not a polyhedron, a polyhedron $\mathbf{I}^2$ can be topologically triangulated by $F(\Delta^2)$ but any triangulation of $\mathbf{I}^2$ cannot be congruent to $F(\Delta^2)$.

By Propositions 4.2.2, 4.2.11 (2), (3), and (4), we have the following:

Proposition 4.4.5 (Additional Topological Properties of Polyhedra) *For a (topological) polyhedron P, the following hold:*

(1) *P is a perfectly normal k-space.*
(2) *The following three conditions are equivalent:*

(a) *P is metrizable;* (b) *P is locally compact;* (c) *P is first countable.*

(3) *P is separable metrizable if and only if P is second countable.*
(4) *In general, $w(P) \neq \operatorname{dens} P$, i.e., $w(P) > \operatorname{dens} P$.* □

For a polyhedron P, the **dimension** of P, $\dim P$, is defined by the dimension of a triangulation K of P (or a cell complex K with $|K| = P$), i.e., $\dim P = \dim K$. This is well-defined, that is, $\dim P$ does not depend on K with $|K| = P$. In other words, for cell complexes K_1 and K_2, $|K_1| = |K_2|$ implies $\dim K_1 = \dim K_2$.

Indeed, let $C \in K_1$ with $\dim C = \dim K$. Then, $C \subset \bigcup_{i=1}^{n} D_i$ for some $D_1, \ldots, D_n \in K_1$ by Proposition 4.2.5. By Baire Category Theorem 2.5.1, some $C \cap D_i$ has an interior point in C, which implies $\dim(C \cap D_i) = \dim C$. Then, it follows that

$$\dim K_1 = \dim C = \dim(C \cap D_i) \leqslant \dim D_i \leqslant \dim K_2.$$

Similarly, $\dim K_2 \leqslant \dim K_1$. Consequently, $\dim K_1 = \dim K_2$.

It can be said that P is **finite-dimensional** (abbrev. **f.d.**), **infinite-dimensional** (abbrev. **i.f.d.**), or **locally finite-dimensional** (abbrev. **l.f.d.**) according to whether a triangulation K of P (or a cell complex K with $|K| = P$) is.

Due to Theorem 4.3.6 and Proposition 4.3.7, *every two triangulations of a polyhedron are combinatorial equivalent*, that is:

Theorem 4.4.6 *Let P be a polyhedron. If P is defined by two cell complexes K_1 and K_2, then P has a triangulation which is a common subdivision of K_1 and K_2. Namely, $P = |K_1| = |K_2| \Rightarrow K_1 \cong K_2$.* □

Remark 4.4.1 It is natural to ask if the above Theorem 4.4.6 is also valid for a topological polyhedron, that is, the implication $|K_1| \approx |K_2| \Rightarrow K_1 \cong K_2$ holds or not. This implication is called **Hauptvermutung** (the *fundamental conjecture*). It had taken a long time to find finite simplicial complexes K and L such that $K \ncong L$ but $|K| \approx |L|$. It is known that this conjecture does not hold even if $|K|$ and $|L|$ are n-manifolds, i.e., there exists an n-manifold that has topological triangulations $K \ncong L$. Refer to Notes at the end of this chapter.

A subspace Q of a polyhedron P is called a **subpolyhedron** of P if there exists a pair (K, L) of a cell complex and a subcomplex such that $P = |K|$ and $Q = |L|$. To discuss triangulations of subpolyhedra, using derived subdivisions, we prove the following theorem:

Theorem 4.4.7 *Let K, K_0, and L be simplicial complexes such that $|L|$ is a closed subspace of $|K|$ and $K_0 \subset K \cap L$, i.e., K_0 is a subcomplex of both K and L. Then,*

K has a simplicial subdivision K' such that K_0 is a full subcomplex of K' and $L' = \{\tau \in K' \mid \tau \subset |L|\}$ is a subcomplex of K' subdividing L.[10]

Proof The proof will be proceeded in two steps. The first is the case that $L \setminus K_0$ is finite. The second general case will be proved by applying the first case.

Step 1: Under the assumption that $L \setminus K_0$ is finite, we will show the theorem by induction on $n = \mathrm{card}(L \setminus K_0)$. In the case $n = 0$, a derived subdivision K' of K relative to $L = K_0$ is the desired one.

In the case $n > 0$, let σ be a maximal dimensional simplex of $L \setminus K_0$. Since σ is a principal simplex of L, $L_1 = L \setminus \{\sigma\}$ is a subcomplex of L with $K_0 \subset L_1$ and $\mathrm{card}(L_1 \setminus K_0) = n - 1$, where $|L_1|$ is a closed subspace of $|K|$ because it is a closed subspace of $|L|$. By the inductive assumption, we have $K' \lhd K$ such that K_0 is a full subcomplex of K' and $L'_1 = \{\tau \in K' \mid \tau \subset |L_1|\}$ is a subcomplex of K' subdividing L_1. Let $L'_{\partial\sigma} \lhd \partial F(\sigma)$ be the subdivision induced by L'_1, where $L'_{\partial\sigma} \subset L'_1 \subset K'$. According to the additional statement of Proposition 4.3.4,

$$L'_{\partial\sigma} = \left\{\tau \in L'_1 \;\middle|\; \tau \subset \partial\sigma\right\} = L'_1 \wedge \partial F(\sigma).$$

Applying Proposition 4.3.8 (1), we have the following cell complex:

$$L'_\sigma = K' \wedge F(\sigma) = \left\{\tau \cap \sigma' \;\middle|\; \tau \in K',\ \sigma' \in F(\sigma),\ \mathrm{rint}\,\tau \cap \mathrm{rint}\,\sigma' \neq \emptyset\right\},$$

where $L'_{\partial\sigma} \subset L'_\sigma$ because $L'_1 \subset K'$. Since $|L|$ is a subspace of $|K|$, σ is a compact subspace of $|K| = |K'|$. Hence, σ is covered by only finitely many simplexes of K', which means that σ is covered by only finitely many cells of L'_σ. Thus, L'_σ is a subdivision of $F(\sigma)$. Since $L'_{\partial\sigma}$ is a subcomplex of K' and $\partial\sigma = |L'_{\partial\sigma}|$, if $\tau \in K'$ and $(\mathrm{rint}\,\tau) \cap \partial\sigma \neq \emptyset$, then $\tau \in L'_{\partial\sigma}$, which means $\tau \subset \partial\sigma$. Thus, we have

$$L'_\sigma \setminus L'_{\partial\sigma} = \left\{\tau \cap \sigma \;\middle|\; \tau \in K',\ \mathrm{rint}\,\tau \cap \mathrm{rint}\,\sigma \neq \emptyset\right\}.$$

In the above, for each $C \in L'_\sigma \setminus L'_{\partial\sigma}$, there is a unique $\tau_C \in K'$ such that $C = \tau_C \cap \sigma$ and $\mathrm{rint}\,\tau_C \cap \mathrm{rint}\,\sigma \neq \emptyset$, where $\mathrm{rint}\,C = \mathrm{rint}\,\tau_C \cap \mathrm{rint}\,\sigma$.

Indeed, for $\tau, \tau' \in K'$, assume that $\mathrm{rint}\,\tau \cap \mathrm{rint}\,\sigma \neq \emptyset$, $\mathrm{rint}\,\tau' \cap \mathrm{rint}\,\sigma \neq \emptyset$, and $\tau \cap \sigma = \tau' \cap \sigma$. Then, by Proposition 4.1.9 (2),

$$\mathrm{rint}\,\tau \cap \mathrm{rint}\,\sigma = \mathrm{rint}(\tau \cap \sigma) = \mathrm{rint}(\tau' \cap \sigma) = \mathrm{rint}\,\tau' \cap \mathrm{rint}\,\sigma.$$

Hence, $\mathrm{rint}\,\tau \cap \mathrm{rint}\,\tau' \neq \emptyset$, which means that $\tau = \tau'$.

Now, consider the subcomplex $K'_1 = \{\tau \in K' \mid \tau \cap \mathrm{rint}\,\sigma = \emptyset\}$ of K'. Since $\partial\sigma \subset |L_1| = |L| \setminus \mathrm{rint}\,\sigma$, it follows that $L'_{\partial\sigma} \subset L'_1 \subset K'_1$, and hence

$$\sigma \cap |K'_1| = (\sigma \setminus \mathrm{rint}\,\sigma) \cap |K'_1| = |L'_{\partial\sigma}| = \partial\sigma.$$

[10] Note that L' is defined exactly as L' in Proposition 4.3.4, but here L is not necessarily a subcomplex of K.

For each $\tau \in K' \setminus K'_1 = K'[\operatorname{rint}\sigma]$, choose $v_\tau \in \operatorname{rint}\tau$ so that

$$\operatorname{rint}\tau \cap \operatorname{rint}\sigma \neq \emptyset \;\Rightarrow\; v_\tau \in \operatorname{rint}\tau \cap \operatorname{rint}\sigma,$$

which means that for each $C \in L'_\sigma \setminus L'_{\partial\sigma}$, $\tau_C \in K' \setminus K'_1$ and $v_{\tau_C} \in \operatorname{rint}\tau_C \cap \operatorname{rint}\sigma = \operatorname{rint} C$. Using these points v_τ, $\tau \in K' \setminus K'_1$, we define K'' as a derived subdivision of K' relative to K'_1 (cf. Fig. 4.2). Then, $K_0 \cup L'_1 \subset K''$. Since K_0 is full in K'_1 and K'_1 is full in K'', it follows that K_0 is full in K''.

On the other hand, for each cell $C \in L'_\sigma \setminus L'_{\partial\sigma}$, let $v_C = v_{\tau_C} \in \operatorname{rint} C$ (cf. Fig. 4.3). Then, for $C, C' \in L'_\sigma \setminus L'_{\partial\sigma}$, the following equivalence holds:

$$C' \leqslant C \;\Leftrightarrow\; \tau_{C'} \leqslant \tau_C.$$

Indeed, let $x \in \operatorname{rint} C' = \operatorname{rint}\tau_{C'} \cap \operatorname{rint}\sigma$. Then, $C'_x = C'$, $(\tau_{C'})_x = \tau_{C'}$, and $\sigma_x = \sigma$. If $C' \leqslant C$, then $C' = C_x$, so

$$\tau_{C'} \cap \sigma = C' = C_x = (\tau_C \cap \sigma)_x = (\tau_C)_x \cap \sigma_x = (\tau_C)_x \cap \sigma,$$

where $x \in \operatorname{rint}(\tau_C)_x \cap \operatorname{rint}\sigma$. By the uniqueness of $\tau_{C'}$, we have $\tau_{C'} = (\tau_C)_x$, which means $\tau_{C'} \leqslant \tau_C$. Conversely, if $\tau_{C'} \leqslant \tau_C$, then $\tau_{C'} = (\tau_C)_x$, so

$$C' = \tau_{C'} \cap \sigma = (\tau_C)_x \cap \sigma_x = (\tau_C \cap \sigma)_x = C_x,$$

which means that $C' \leqslant C$.

Using these points v_C, $C \in L'_\sigma \setminus L'_{\partial\sigma}$, we have the derived subdivision L''_σ of L'_σ relative to $L'_{\partial\sigma}$. Then, $L''_\sigma \lhd F(\sigma)$ and a subcomplex of K''. By the way of construction of derived subdivisions (defined after the proof of Proposition 4.3.11), L''_σ is simply the subcomplex of K'' consisting of simplexes with vertices in σ, that is, $L''_\sigma = \{\tau \in K'' \mid \tau \subset \sigma\}$.

We can verify $L''_\sigma \subset K''$ as follows: First, $L'_{\partial\sigma} \subset K'_1 \subset K''$. For $C_1 < \cdots < C_n \in L'_\sigma \setminus L'_{\partial\sigma}$, since $\tau_{C_1} < \cdots < \tau_{C_n} \in K' \setminus K'_1$, it follows that

$$\langle v_{C_1}, \ldots, v_{C_n}\rangle = \langle v_{\tau_{C_1}}, \ldots, v_{\tau_{C_n}}\rangle \in K''.$$

Let $\langle v_1, \ldots, v_m\rangle \in L'_{\partial\sigma}$ and $C_1 < \cdots < C_n \in L'_\sigma \setminus L'_{\partial\sigma}$ with $\langle v_1, \ldots, v_m\rangle < C_1$. Since $v_1, \ldots, v_m \in {L'_{\partial\sigma}}^{(0)} \cap \tau_{C_1} \subset \tau_{C_1}^{(0)}$, we have $\langle v_1, \ldots, v_m\rangle < \tau_{C_1}$, and hence

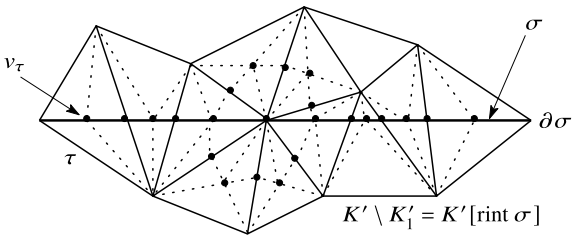


Fig. 4.2 Definition of $K'' \lhd K'$

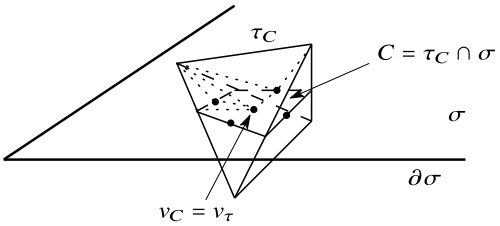


Fig. 4.3 Subdivision $L''_\sigma \lhd L'_\sigma$

$$\langle v_1, \ldots, v_m, v_{C_1}, \ldots, v_{C_n} \rangle = \langle v_1, \ldots, v_m, v_{\tau_{C_1}}, \ldots, v_{\tau_{C_n}} \rangle \in K''.$$

Thus, we have a subcomplex $L'' = L'_1 \cup L''_\sigma$ of K'', where $L'' \lhd L$ because $L'_1 \lhd L_1 = L \setminus \{\sigma\}$ and $L''_\sigma \lhd F(\sigma)$.

Step 2: Using Step 1, we will prove the general case. Since $|L|$ is a closed subspace of $|K|$, each simplex of L is covered by finitely many simplexes of K (so it contains interior points of only finitely many simplexes of K) and each simplex of K contains interior points of only finitely many simplexes of L by Proposition 4.2.4. Then, $K \wedge L$ is a cell complex and $|L| = |K \wedge L|$ $(= |K| \cap |L|)$ as spaces by Proposition 4.3.8 (1) and (2). Note that $K_0 \subset K \wedge L$ and $K \wedge L$ has a simplicial subdivision with the same vertices as $K \wedge L$ (Theorem 4.3.6). By replacing L with such a simplicial subdivision of $K \wedge L$, it can be assumed that $L \prec K$, that is, each simplex of L is contained in some simplex of K, where each simplex $\sigma \in K$ contains only finitely many simplexes of L (because it contains interior points of only finitely many simplexes of L) and $K^{(0)} \cap |L| \subset L^{(0)}$.

Indeed, let $v \in K^{(0)}$ and $\tau \in L$ with $v \in \tau$. Since $L \prec K$, we have $\sigma \in K$ such that $\tau \subset \sigma$. Then, $\tau_v \subset \sigma_v = \{v\}$, which means $v \in \tau^{(0)}$.

Then, $L \cup K^{(0)}$ is a simplicial complex and $|L \cup K^{(0)}|$ is a closed subspace of $|K|$. Hence, replacing K_0 and L with $K_0 \cup K^{(0)}$ and $L \cup K^{(0)}$, respectively, we may assume that $K^{(0)} \subset K_0 \subset L$.

Now, let $K_n = K_0 \cup K^{(n)}$ for each $n \in \mathbb{N}$ and $L_n = \{\tau \in L \mid \tau \subset |K_n|\}$ for each $n \in \omega$. Then, $L_0 = K_0$ and $\bigcup_{n\in\omega} L_n = L$ because $L \prec K$. Moreover, each L_n is a subcomplex of L with $|L_n| = |L| \cap |K_n|$.

Indeed, $|L_n| \subset |L| \cap |K_n|$ by the definition. To see $|L| \cap |K_n| \subset |L_n|$, let $x \in |L| \cap |K_n|$. We can take $\tau \in L$ and $\sigma \in K_n$ such that $x \in \operatorname{rint}\tau$ and $x \in \operatorname{rint}\sigma$. Since $L \prec K$, we also have $\sigma' \in K$ such that $\tau \subset \sigma'$. Since $x \in \operatorname{rint}\sigma'_x$, it follows that $\operatorname{rint}\sigma \cap \operatorname{rint}\sigma'_x \neq \emptyset$, which implies $\sigma'_x = \sigma$. Since $x \in \operatorname{rint}\tau$, we have

$$\tau = \tau_x = (\tau \cap \sigma')_x = \tau_x \cap \sigma'_x = \tau \cap \sigma \subset \sigma \subset |K_n|,$$

so $\tau \in L_n$. Therefore, $x \in |L_n|$.

We will inductively construct simplicial subdivisions $K'_n \lhd K_n$, $n \in \omega$, so as to satisfy the following conditions:

(1) $K'_{n-1} \subset K'_n$, K_0 is a full subcomplex of K'_n, and
(2) $L'_n = \{\tau \in K'_n \mid \tau \subset |L_n|\} \lhd L_n$,

where $K'_0 = K_0$ and

$$L'_0 = \left\{\tau \in K'_0 \;\middle|\; \tau \subset |L_0|\right\} = \left\{\tau \in K_0 \;\middle|\; \tau \subset |K_0|\right\} = K_0 = L_0.$$

Then, $K' = \bigcup_{n\in\omega} K'_n$ is the desired one. Indeed, K_0 is a full subcomplex of K' and

$$L' = \left\{\tau \in K' \;\middle|\; \tau \subset |L|\right\} = \bigcup_{n\in\omega} L'_n \lhd \bigcup_{n\in\omega} L_n = L.$$

Suppose that K'_{n-1} has been defined so as to satisfy conditions (1) and (2). For each $\sigma \in K_n \setminus K_{n-1} = K(n) \setminus K_0$,[11] we have the following simplicial complex:

$$K'_{\partial\sigma} = \left\{\tau \in K'_{n-1} \;\middle|\; \tau \subset \partial\sigma\right\},$$

which is a simplicial subdivision of $\partial F(\sigma)$ because $K'_{n-1} \lhd K_{n-1}$ and $\partial F(\sigma) \subset K_{n-1}$. Since $L'_{n-1} \subset K'_{n-1}$ and $L'_{n-1} \lhd L_{n-1}$, the following is a subcomplex of both $K'_{\partial\sigma}$ and L'_{n-1}:

$$L'_{\partial\sigma} = \left\{\tau \in K'_{\partial\sigma} \;\middle|\; \tau \subset |L|\right\} = \left\{\tau \in L'_{n-1} \;\middle|\; \tau \subset \partial\sigma\right\},$$

where $|L'_{\partial\sigma}| = \partial\sigma \cap |L|$. Applying Lemma 4.3.10 to $\hat{\sigma} \in \operatorname{rint}\sigma$ and $K'_{\partial\sigma} \lhd \partial F(\sigma)$, we can obtain $K_\sigma \lhd F(\sigma)$ such that $K'_{\partial\sigma}$ is a full subcomplex of K_σ. By Proposition 4.3.8 (1) and (2), $K_\sigma \wedge L$ is a cell complex and $|K_\sigma \wedge L| = \sigma \cap |L|$ as spaces. Then, $L'_{\partial\sigma}$ is a subcomplex of $K_\sigma \wedge L$. Using Theorem 4.3.6, we can construct $L_\sigma \lhd K_\sigma \wedge L$ with $L_\sigma^{(0)} = (K_\sigma \wedge L)^{(0)}$. It should be observed that $L'_{\partial\sigma} = \{\tau \in L_\sigma \mid \tau \subset \partial\sigma\}$. Then, $K'_{\partial\sigma} \cup L_\sigma$ is a simplicial complex with

$$|K'_{\partial\sigma} \cup L_\sigma| = |K'_{\partial\sigma}| \cup |L_\sigma| = |K'_{\partial\sigma}| \cup |K_\sigma \wedge L| = \partial\sigma \cup (\sigma \cap |L|).$$

Now, for each $\sigma \in K_n \setminus K_{n-1}$, applying Step 1 to K_σ, $K'_{\partial\sigma}$, and $K'_{\partial\sigma} \cup L_\sigma$, we can obtain $K'_\sigma \lhd K_\sigma$ such that $K'_{\partial\sigma}$ is a full subcomplex of K'_σ and

$$\left\{\tau \in K'_\sigma \;\middle|\; \tau \subset |K'_{\partial\sigma} \cup L_\sigma|\right\} = K'_{\partial\sigma} \cup L'_\sigma \lhd K'_{\partial\sigma} \cup L_\sigma,$$

where $L'_\sigma = \{\tau \in K'_\sigma \mid \tau \subset |L_\sigma|\}$. Then, note $L'_\sigma \lhd L_\sigma$. The desired K'_n can be defined as follows:

$$K'_n = K'_{n-1} \cup \bigcup_{\sigma\in K_n\setminus K_{n-1}} K'_\sigma \lhd K_n.$$

Indeed, condition (1) is obvious. For (2), observe that

[11] Recall that $K(n) = K^{(n)} \setminus K^{(n-1)}$ is the collection of all n-simplexes of K.

$$L'_n = L'_{n-1} \cup \bigcup_{\sigma \in K_n \setminus K_{n-1}} L'_\sigma \lhd L_n.$$

The proof is completed. □

By applying Theorem 4.4.7, the following theorem can be easily proved:

Theorem 4.4.8 *Let P and Q be polyhedra with $Q \subset P$ as sets. Then, the following are equivalent:*

(a) *Q is a subpolyhedron of P;*
(b) *Q is a closed subspace of P;*
(c) *Every triangulation K of P has a simplicial subdivision K' such that Q is triangulated by a full subcomplex of K';*
(d) *Every triangulation L of Q has a simplicial subdivision L' which is a full subcomplex of some triangulation of P.*

Sketch of Proof The implications (c) ⇒ (a) and (d) ⇒ (a) are obvious. By Propositions 4.2.1 and 4.3.12, we have (a) ⇒ (b) and (a) ⇒ (c), respectively. The implication (b) ⇒ (d) can be obtained by applying Theorem 4.4.7 together with Proposition 4.3.12.

Concerning the intersection and the union of two polyhedra, the following holds:

Theorem 4.4.9 *Let P_1 and P_2 be polyhedra in the same linear space and suppose that $P_1 \cup P_2$ is equipped with a topology such that P_1 and P_2 are closed subspaces of $P_1 \cup P_2$, hence $P_1 \cap P_2$ is a subspace of both P_1 and P_2. Then, $P_1 \cap P_2$ is a subpolyhedron of P_1 and P_2. Moreover, $P_1 \cup P_2$ is a polyhedron, and P_1, P_2 and $P_1 \cap P_2$ are subpolyhedra of $P_1 \cup P_2$.*

Proof For each $i = 1, 2$, let K_i be a triangulation of P_i. Then, each simplex $\sigma \in K_i$ contains interior points of only finitely many simplexes of K_{3-i}. Indeed, since P_{3-i} is closed in $P_1 \cup P_2$, it follows that $\sigma \cap P_{3-i}$ is a compact set in $P_{3-i} = |K_{3-i}|$, which contains interior points of only finitely many simplexes of K_{3-i} (cf. the proof of Proposition 4.2.3). Thus, it follows from Proposition 4.3.8 (2) that $P_1 \cap P_2 = |K_1 \wedge K_2|$ as spaces, which implies that $P_1 \cap P_2$ is a polyhedron. Moreover, by Theorem 4.4.8, $P_1 \cap P_2$ is a subpolyhedron of both P_1 and P_2. Then, we can replace K_i, $i = 1, 2$, so that $P_1 \cap P_2 = |L_i|$ for some subcomplex L_i of K_i. Due to Corollary 4.3.9, L_1 and L_2 have a common simplicial subdivision L. Applying Proposition 4.3.11, we can obtain $K'_i \lhd K_i, i = 1, 2$, such that L is a subcomplex of K'_i. Since $L = K'_1 \cap K'_2$ and $|L| = |K'_1| \cap |K'_2|$, it is easy to see that $K = K'_1 \cup K'_2$ is a simplicial complex. Hence, $P_1 \cup P_2 = |K|$ is a polyhedron. Since K'_1, K'_2, and L are subcomplexes of K, $P_1 = |K'_1|$, $P_2 = |K'_2|$, and $P_1 \cap P_2 = |L|$ are subpolyhedra of $P_1 \cup P_2$. □

Remark 4.4.2 It is not practical that a subpolyhedron of a *topological* polyhedron P is defined as a subspace Q of P such that $(P, Q) \approx (|K|, |L|)$ for some pair of a cell complex K and its subcomplex L. To see this, suppose that Q_1 and Q_2 are subpolyhedra of P in this sense. Then, we have homeomorphisms $h_1 : (P, Q_1) \to$

$(|K_1|, |L_1|)$ and $h_2 : (P, Q_2) \to (|K_2|, |L_2|)$, but K_1 and K_2 are different complexes in general, and even worse, it may happen that $K_1 \ncong K_2$ (cf. Remark 4.4.1). Even in the case when $K_1 = K_2 = K$, the homeomorphisms $h_1 : P \to |K|$ and $h_2 : P \to |K|$ can be different.

4.5 Product Complexes and Homotopy Extension

By virtue of Proposition 4.1.9 (5), the **product cell complex** of two cell complexes K (in E) and L (in F) can be defined as the following cell complex (in $E \times F$):

$$K \times_c L = \left\{ C \times D \;\middle|\; C \in K,\ D \in L \right\}.$$

Note the following facts:

- $|K \times_c L| = |K| \times |L|$ as sets;
- $\mathrm{St}(C \times D, K \times_c L) = \mathrm{St}(C, K) \times_c \mathrm{St}(D, L)$ for each $C \in K$ and $D \in L$ (cf. Proposition 4.1.9 (5));
- $c_{K \times_c L}(x, y) = c_K(x) \times c_L(y)$ for each $(x, y) \in |K| \times |L|$ (cf. Proposition 4.1.9 (5));
- $O_{K \times_c L}(x, y) = O_K(x) \times O_L(y)$ as sets for each $(x, y) \in |K| \times |L|$.

The projections $\mathrm{pr}_1 : |K \times_c L| \to |K|$ and $\mathrm{pr}_2 : |K \times_c L| \to |L|$ are continuous because $\mathrm{pr}_i | C \times D$, $i = 1, 2$, are continuous. Hence, the identity $\mathrm{id} : |K \times_c L| \to |K| \times |L|$ is continuous. When K and L are finite, $|K \times_c L| = |K| \times |L|$ as spaces because $|K \times_c L|$ is compact by Proposition 4.2.3. More generally, we can prove the following:

Theorem 4.5.1 *For each pair of cell complexes K and L, $|K \times_c L| = |K| \times |L|$ as spaces if* (1) *both K and L are locally countable or* (2) *one of K or L is locally finite.*

Proof *Case* (1): Since $\mathrm{id} : |K \times_c L| \to |K| \times |L|$ is continuous, it suffices to show the continuity of $\mathrm{id} : |K| \times |L| \to |K \times_c L|$ at each $(x, y) \in |K| \times |L|$. Choose $(v, u) \in K^{(0)} \times L^{(0)}$ so that $(x, y) \in O_K(v) \times O_L(u)$. Then,

$$O_{K \times_c L}(v, u) = O_K(v) \times O_L(u) \ \text{ as sets,}$$

where the left side is an open neighborhood of (x, y) in $|K \times_c L|$ and the right side is in $|K| \times |L|$. Replacing K and L by $\mathrm{St}(v, K)$ and $\mathrm{St}(u, L)$, we can assume that both K and L are countable. Then, K and L have towers $K_1 \subset K_2 \subset \cdots$ and $L_1 \subset L_2 \subset \cdots$ of finite subcomplexes such that $x \in |K_1|$, $y \in |L_1|$, $K = \bigcup_{n \in \mathbb{N}} K_n$, and $L = \bigcup_{n \in \mathbb{N}} L_n$. Observe that $|K|$, $|L|$, and $|K \times_c L|$ have the weak topologies determined by $\{|K_n| \mid n \in \mathbb{N}\}$, $\{|L_n| \mid n \in \mathbb{N}\}$, and $\{|K_n \times_c L_n| \mid n \in \mathbb{N}\}$, respectively. Thus, $|K| = \varinjlim |K_n|$, $|L| = \varinjlim |L_n|$, and $|K \times_c L| = \varinjlim |K_n \times_c L_n|$. For

each $n \in \mathbb{N}$, $|K_n \times_c L_n| = |K_n| \times |L_n|$ as spaces because K_n and L_n are finite. Applying Proposition 2.8.5, we have

$$\begin{aligned} |K| \times |L| &= \varinjlim |K_n| \times \varinjlim |L_n| = \varinjlim (|K_n| \times |L_n|) \\ &= \varinjlim |K_n \times_c L_n| = |K \times_c L| \ \text{ as spaces.} \end{aligned}$$

Case (2): We may assume that L is locally finite, which means that $|L|$ is locally compact. The polyhedra $|K|$ and $|L|$ can be regarded as quotient spaces of $\bigoplus_{C\in K} C$ and $\bigoplus_{D\in L} D$ with the natural quotient maps $q_K : \bigoplus_{C\in K} C \to |K|$ and $q_L : \bigoplus_{D\in L} D \to |L|$, respectively (cf. Sect. 4.2). For simplicity, let $\widetilde{K} = \bigoplus_{C\in K} C$ and $\widetilde{L} = \bigoplus_{D\in L} D$. Then, as is easily observed,

$$\widetilde{K} \times \widetilde{L} = \bigoplus_{C\in K, D\in L} (C \times D) = \bigoplus_{C\times D \in K\times_c L} (C \times D)$$

and $|K \times_c L|$ is regarded as a quotient space of $|K| \times |L|$ whose quotient map is $q_K \times q_L : \widetilde{K} \times \widetilde{L} \to |K| \times |L|$. By Proposition 2.1.9, the following are quotient maps:

$$q_K \times \mathrm{id}_{|K|} : \widetilde{K} \times |L| \to |K| \times |L| \ \text{ and } \ \mathrm{id}_{\widetilde{K}} \times q_L : \widetilde{K} \times \widetilde{L} \to \widetilde{K} \times |L|.$$

Consequently, $q_K \times q_L = (q_K \times \mathrm{id}_{|K|})(\mathrm{id}_{\widetilde{K}} \times q_L)$ is also quotient, which means $|K \times_c L| = |K| \times |L|$ as spaces. □

For simplicity, we denote $F(\mathbf{I}) = I$, i.e., $I = \{0, 1, \mathbf{I}\}$, which is the cell complex with $|I| = \mathbf{I}$. It follows from Theorem 4.5.1 that $|K| \times \mathbf{I} = |K \times_c I|$ as spaces for any cell complex K. Due to the following proposition, the conditions given by Theorem 4.5.1 are essential.

Proposition 4.5.2 *There exist* 1*-dimensional simplicial complexes* K *and* L *with* $\mathrm{card}\, K^{(0)} = 2^{\aleph_0}$ *and* $\mathrm{card}\, L^{(0)} = \aleph_0$ *such that* K *is not locally countable,* L *is not locally finite, and* $|K \times_c L| \neq |K| \times |L|$ *as spaces.*

Proof (*Example*) We define K and L in the linear spaces $\mathbb{R}^{\mathbb{N}^{\mathbb{N}}}$ and $\mathbb{R}^{\mathbb{N}}$ as follows:

$$K = \{\mathbf{0}, \mathbf{e}_a, \langle \mathbf{0}, \mathbf{e}_a \rangle \mid a \in \mathbb{N}^{\mathbb{N}}\} \ \text{ and } \ L = \{\mathbf{0}, \mathbf{e}_n, \langle \mathbf{0}, \mathbf{e}_n \rangle \mid n \in \mathbb{N}\},$$

where $\mathbf{e}_a \in \mathbb{R}^{\mathbb{N}^{\mathbb{N}}}$ and $\mathbf{e}_n \in \mathbb{R}^{\mathbb{N}}$ are the unit vectors (i.e., $\mathbf{e}_\gamma(\gamma) = 1$ and $\mathbf{e}_\gamma(\gamma') = 0$ for $\gamma' \neq \gamma$). For each $a \in \mathbb{N}^{\mathbb{N}}$ and $n \in \mathbb{N}$, let

$$y_{a,n} = a(n)^{-1}(\mathbf{e}_a, \mathbf{e}_n) \in \langle \mathbf{0}, \mathbf{e}_a \rangle \times \langle \mathbf{0}, \mathbf{e}_n \rangle \subset \mathbb{R}^{\mathbb{N}^{\mathbb{N}}} \times \mathbb{R}^{\mathbb{N}}.$$

Then, $Y = \{y_{a,n} \mid a \in \mathbb{N}^{\mathbb{N}}, n \in \mathbb{N}\}$ is closed in $|K \times_c L|$, where it should be noted that $(\mathbf{0}, \mathbf{0}) \notin Y$.

To see that Y is not closed in $|K| \times |L|$, we may show that $(\mathbf{0}, \mathbf{0}) \in \mathrm{cl}_{|K|\times|L|} Y$. Let U be a neighborhood of $\mathbf{0}$ in $|K|$ and V a neighborhood of $\mathbf{0}$ in $|L|$. For each

$a \in \mathbb{N}^{\mathbb{N}}$, choose $\varepsilon_a > 0$ so that $[0, \varepsilon_a]\mathbf{e}_a \subset U$. For each $n \in \mathbb{N}$, choose $\delta_n > 0$ so that $[0, \delta_n]\mathbf{e}_n \subset V$. Then, we have $a_0 \in \mathbb{N}^{\mathbb{N}}$ such that $a_0(n)^{-1} < \min\{\delta_n, n^{-1}\}$ for each $n \in \mathbb{N}$. Choose $n_0 \in \mathbb{N}$ so that $n_0^{-1} < \varepsilon_{a_0}$. Since $a_0(n_0)^{-1} < n_0^{-1} < \varepsilon_{a_0}$ and $a_0(n_0)^{-1} < \delta_{n_0}$, it follows that $y_{a_0,n_0} \in U \times V$, so $(U \times V) \cap Y \neq \emptyset$. Thus, we have $(\mathbf{0}, \mathbf{0}) \in \mathrm{cl}_{|K|\times|L|} Y$. □

For simplicial complexes K and L, the product complex $K \times_c L$ is not simplicial but does have a simplicial subdivision with the same vertices by Theorem 4.3.6, which can be obtained by giving an order on $(K \times_c L)^{(0)} = K^{(0)} \times L^{(0)}$ so that $K \times_c L$ is an ordered cell complex. If K and L are ordered simplicial complexes, we can define an order on $K^{(0)} \times L^{(0)}$ as follows:

$$(u, v) \leqslant (u', v') \underset{\text{def}}{\Leftrightarrow} u \leqslant u' \text{ and } v \leqslant v'.$$

By this order, $K \times_c L$ is an ordered cell complex. Using this order, we can define the simplicial subdivision $K \times_s L$ of $K \times_c L$, which is called the **product simplicial complex** of K and L (Fig. 4.4). In fact, $K \times_s L$ can be written as follows:

$$K \times_s L = \Big\{\langle (u_1, v_1), \ldots, (u_k, v_k)\rangle \;\Big|\; \exists \sigma \in K, \exists \tau \in L \text{ such that } u_1 \leqslant \cdots \leqslant u_k \in \sigma^{(0)},\ v_1 \leqslant \cdots \leqslant v_k \in \tau^{(0)}\Big\}.$$

Indeed, let $C_1 < \cdots < C_k \in K \times_c L$, where $C_i = \sigma_i \times \tau_i$, $\sigma_i \in K$, $\tau_i \in L$. For each $i = 1, \ldots, n$, let $(u_i, v_i) = \max C_i^{(0)}$. Then, $u_i = \max \sigma_i^{(0)}$ and $v_i = \max \tau_i^{(0)}$. Since $\sigma_i^{(0)} \subset \sigma_{i+1}^{(0)}$ and $\tau_i^{(0)} \subset \tau_{i+1}^{(0)}$, we have $u_1 \leqslant \cdots \leqslant u_k \in \sigma_k^{(0)}$ and $v_1 \leqslant \cdots \leqslant v_k \in \tau_k^{(0)}$.

Conversely, let $u_1 \leqslant \cdots \leqslant u_k \in \sigma^{(0)}$ and $v_1 \leqslant \cdots \leqslant v_k \in \tau^{(0)}$, where $(u_i, v_i) \neq (u_{i+1}, v_{i+1})$ for each $i = 1, \ldots, k-1$. For each $i = 1, \ldots, n$, let $\sigma_i = \langle u_1 \ldots, u_i\rangle$, $\tau_i = \langle v_1 \ldots, v_i\rangle$, and $C_i = \sigma_i \times \tau_i \in K \times_c L$. Then, $(u_i, v_i) = \max C_i^{(0)}$ and $C_i \neq C_{i+1}$ because $(u_i, v_i) \neq (u_{i+1}, v_{i+1})$. Since $\sigma_i \leqslant \sigma_{i+1}$ and $\tau_i \leqslant \tau_{i+1}$, we have $C_1 < \cdots < C_k$.

For simplicial complexes K and L, since the barycentric subdivisions $\mathrm{Sd}\, K$ and $\mathrm{Sd}\, L$ are ordered simplicial complexes, we can define the product simplicial complex $\mathrm{Sd}\, K \times_s \mathrm{Sd}\, L$, that consists of simplexes

$$\langle (\hat{\sigma}_1, \hat{\tau}_1), \ldots, (\hat{\sigma}_n, \hat{\tau}_n)\rangle,\ \sigma_1 \leqslant \cdots \leqslant \sigma_n \in K,\ \tau_1 \leqslant \cdots \leqslant \tau_n \in L.$$

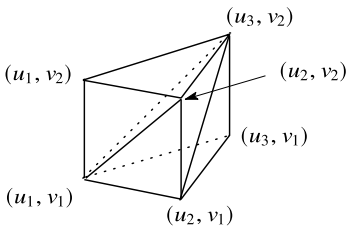


Fig. 4.4 A product simplicial complex

On the other hand, by giving $v_{\sigma\times\tau} \in \mathrm{rint}(\sigma \times \tau)$ for each $\sigma \times \tau \in K \times_c L$, we can define a derived subdivision of the product cell complex $K \times_c L$, which consists of simplexes

$$\langle v_{\sigma_1\times\tau_1}, \ldots, v_{\sigma_n\times\tau_n}\rangle,\ \sigma_1 \times \tau_1 < \cdots < \sigma_n \times \tau_n \in K \times_c L.$$

When $v_{\sigma\times\tau} = (\hat{\sigma}, \hat{\tau})$ for each $\sigma \times \tau \in K \times_c L$, the derived subdivision of $K \times_c L$ is just $\mathrm{Sd}\, K \times_s \mathrm{Sd}\, L$.

We now consider the following HOMOTOPY EXTENSION THEOREM, which will be used very often:

Theorem 4.5.3 (HOMOTOPY EXTENSION) *Let L be a subcomplex of a cell complex K and $h : |L| \times \mathbf{I} \to X$ a homotopy into an arbitrary space X. If h_0 extends to a map $f : |K| \to X$, then h extends to a homotopy $\bar{h} : |K| \times \mathbf{I} \to X$ with $\bar{h}_0 = f$. Moreover, if h is a $\mathcal{U}$-homotopy for an open cover $\mathcal{U}$ of X, then $\bar{h}$ can be taken as a $\mathcal{U}$-homotopy.*

Proof Let $h : |L| \times \mathbf{I} \to X$ be a $\mathcal{U}$-homotopy such that h_0 extends to a map $f : |K| \to X$. For each $n \in \omega$, we define

$$K_n = L \cup K^{(n)} \ \text{ and } \ P_n = (K \times_c \{0\}) \cup (K_n \times_c I).$$

Then, K_n and P_n are subcomplexes of K and $K \times_c I$, respectively. Observe that each $|P_n|$ is a closed subspace of $|K| \times \mathbf{I}$ that contains $|L| \times \mathbf{I}$ as a subspace (cf. Theorem 4.5.1), and $|K| \times \mathbf{I} = \bigcup_{n\in\omega} |P_n|$ has the weak topology with respect to the tower $|P_0| \subset |P_1| \subset |P_2| \subset \cdots$.

We can define the map $g_0 : |P_0| \to X$ as follows:

$$g_0(x,t) = \begin{cases} h(x,t) & \text{for } (x,t) \in |L| \times \mathbf{I}, \\ f(x) & \text{for } (x,t) \in (|K| \times \{0\}) \cup (|K^{(0)}| \setminus |L|) \times \mathbf{I}. \end{cases}$$

It is obvious that $g_0||K_0| \times \mathbf{I}$ is a $\mathcal{U}$-homotopy. Suppose that maps $g_i : |P_i| \to X$, $i < n$, were defined so that $g_i||P_{i-1}| = g_{i-1}$ and $g_i||K_i| \times \mathbf{I}$ is a $\mathcal{U}$-homotopy. Let $C \in K_n \setminus K_{n-1}$ (i.e., an n-cell of $K \setminus L$). By taking $v_C \in \mathrm{rint}\, C$, each $x \in C$ can be written as

$$x = (1-s)y + sv_C,\ y \in \partial C,\ s \in \mathbf{I}.$$

We can choose $0 < \delta_C < 1$ so that $\{g_{n-1}(C(y)) \mid y \in \partial C\} \prec \mathcal{U}$, where

$$C(y) = \{y\} \times \mathbf{I} \cup \{((1-s)y + sv_C,\ 0) \mid 0 \leqslant s \leqslant 2\delta_C\}.$$

See the left side of Fig. 4.5. Then, we can define a map

$$r_C : C \times \mathbf{I} \to C \times \{0\} \cup \partial C \times \mathbf{I} \subset |P_{n-1}|$$

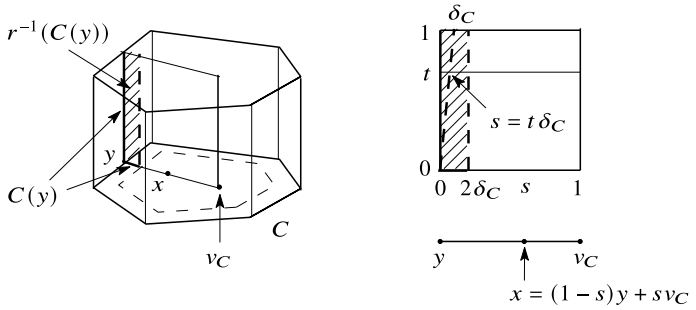


Fig. 4.5 $r_C : C \times \mathbf{I} \to C \times \{0\} \cup \partial C \times \mathbf{I}$

as follows:

$$r_C((1-s)y + sv_C,\ t) = \begin{cases} \big((1-s)y + sv_C,\ 0\big) & \text{if } 2\delta_C \leqslant s \leqslant 1, \\ \left(\left(1 - \dfrac{2(s - t\delta_C)}{2 - t}\right)y + \dfrac{2(s - t\delta_C)}{2 - t}v_C,\ 0\right) & \text{if } t\delta_C \leqslant s \leqslant 2\delta_C, \\ \big(y,\ t - \delta_C^{-1}s\big) & \text{if } 0 \leqslant s \leqslant t\delta_C. \end{cases}$$

Refer to Fig. 4.5. For each $x \in C$, $r_C(\{x\} \times \mathbf{I}) = \{(x, 0)\}$ or $r_C(\{x\} \times \mathbf{I}) \subset C(y)$, where $y \in \partial C$ with $x \in \langle y, v_C \rangle$. It follows that

$$\{g_{n-1}(r_C(\{x\} \times \mathbf{I})) \mid x \in C\} \prec \mathcal{U}.$$

Observe that $r_C | C \times \{0\} \cup \partial C \times \mathbf{I} = \mathrm{id}$. We can extend g_{n-1} to a map $g_n : |P_n| \to X$ defined by $g_n | C \times \mathbf{I} = g_{n-1} r_C$ for each n-cell $C \in K \setminus L$. Then, $g_n || K_n | \times \mathbf{I}$ is a $\mathcal{U}$-homotopy.

By induction, we can obtain maps $g_n : |P_n| \to X$, $n \in \omega$, such that $g_n || P_{n-1}| = g_{n-1}$ and $g_n || K_n | \times \mathbf{I}$ is a $\mathcal{U}$-homotopy. The desired $\mathcal{U}$-homotopy $\bar{h} : |K| \times \mathbf{I} \to X$ can be defined by $\bar{h} || P_n | = g_n$ for each $n \in \omega$. □

In the above proof, assume that K is locally finite and both f and h are proper. For each compact set $A \subset X$,

$$g_0^{-1}(A) = f^{-1}(A) \times \{0\} \cup h^{-1}(A) \cup \bigcup_{v \in (K^{(0)} \setminus |L|) \cap f^{-1}(A)} \{v\} \times \mathbf{I},$$

which is compact. Since $K \times_c I$ is locally finite, the following is finite:

$$S = \big\{C \in K \setminus K_0 \;\big|\; C \times \mathbf{I} \cap g_0^{-1}(A) \neq \emptyset\big\}.$$

Thus, we have a compact set $\bigcup_{C \in S} C \times \mathbf{I}$ in $|K| \times \mathbf{I}$, where

$$\bar{h}^{-1}(A) \cap |P_0| = g_0^{-1}(A) \subset \bigcup_{C \in S} C \times \mathbf{I}.$$

For each $(x,t) \in \bar{h}^{-1}(A) \setminus |P_0|$, let $C = c_K(x)$. Since $(x,t) \notin |P_0|$, it follows that $t > 0$ and $C \in K \setminus K_0$. By the definition of $\bar{h}$, $\bar{h}(x,t) = g_0(x',t')$ for some $(x',t') \in C_0 \times \mathbf{I}$. Since $(x',t') \in (C_0 \times \mathbf{I}) \cap g_0^{-1}(A)$, it follows that $C_0 \in S$. Hence, $(x,t) \in C_0 \times \mathbf{I} \subset \bigcup_{C \in S} C \times \mathbf{I}$. Thus, we have $\bar{h}^{-1}(A) \subset \bigcup_{C \in S} C \times \mathbf{I}$, so $\bar{h}^{-1}(A)$ is compact. As a consequence, we have the following proper version of the Homotopy Extension Theorem 4.5.3:

Theorem 4.5.4 (PROPER HOMOTOPY EXTENSION) *Let L be a subcomplex of a locally finite cell complex K and $h : |L| \times \mathbf{I} \to X$ a proper homotopy into an arbitrary space X. If h_0 extends to a proper map $f : |K| \to X$, then h extends to a proper homotopy $\bar{h} : |K| \times \mathbf{I} \to X$ with $\bar{h}_0 = f$. Moreover, if h is a proper $\mathcal{U}$-homotopy for an open cover $\mathcal{U}$ of X, then $\bar{h}$ can be taken as a proper $\mathcal{U}$-homotopy.* □

For a cell complex L, two maps $f, g : X \to |L|$ are said to be **contiguous** (with respect to L) if f, g are L-close, that is, for each $x \in X$, there is some $C \in L$ such that $f(x), g(x) \in C$. Then, we can define a function $h : X \to |L|$ as follows:

$$h(x,t) = (1-t)f(x) + tg(x) \quad \text{for each } (x,t) \in X \times \mathbf{I}.$$

This function is not necessarily continuous, but it is called the **straight-line homotopy**. When the straight-line homotopy is continuous, we have $f \simeq_L g$. In the rest of this section, we will consider when h is continuous and give an example that h is not continuous. First of all, we prove the following:

Lemma 4.5.5 *Let L be a finite cell complex and X an arbitrary space. If two maps $f, g : X \to |L|$ are contiguous (with respect to L), then the straight-line homotopy h from f to g is continuous.*

Proof We may assume that $|L| \neq \emptyset$. Note that the flat $\mathrm{fl}\,|L| = \mathrm{fl}\,L^{(0)}$ is finite-dimensional. Due to Proposition 3.5.1, $\mathrm{fl}\,|L|$ has a unique topology for which the following operation is continuous:

$$\mathrm{fl}\,|L| \times \mathrm{fl}\,|L| \times \mathbb{R} \ni (y,z,t) \mapsto (1-t)y + tz \in \mathrm{fl}\,|L|.$$

Hence, $h : X \times \mathbf{I} \to \mathrm{fl}\,|L|$ is continuous. The topology of $|L|$ is equal to the relative topology with respect to this topology.

> Indeed, due to Proposition 3.5.8, each $D \in L$ has a unique topology for which the following operation is continuous:
>
> $$D \times D \times I \ni (y,z,t) \mapsto (1-t)y + tz \in D.$$
>
> Hence, the inclusion $|L| \subset \mathrm{fl}\,|L|$ is continuous. By the compactness of $|L|$, this inclusion is a closed embedding.

Consequently, $h : X \times \mathbf{I} \to |L|$ is continuous. □

Using the above lemma, we have the following:

Proposition 4.5.6 *Let L be a cell complex and $f, g : X \to |L|$ be maps which are contiguous with respect to L (i.e., L-close). In the following cases, $f \simeq_L g$ by the straight-line homotopy h (equiv., h is continuous):*

(1) *L is locally finite;*
(2) *X is a k-space (= compactly generated).*

Proof (1): For each $x \in X$, let

$$V_x = f^{-1}(O_L(f(x)) \cap g^{-1}(O_L(g(x)) \subset X \text{ and}$$
$$L_x = \mathrm{St}(c_L(f(x)), L) \cap \mathrm{St}(c_L(g(x)), L) \subset L.$$

Then, V_x is an open neighborhood of x in X and L_x is a finite subcomplex of L because L is locally finite. For each $y \in V_x$, we have $C \in L$ such that $f(y), g(y) \in C$. Since $f(y) \in O_L(f(x))$ and $g(y) \in O_L(g(x))$, it follows that $c_L(f(x)) \leqslant c_L(f(y)) \leqslant C$ and $c_L(g(x)) \leqslant c_L(g(y)) \leqslant C$, which implies $C \in L_x$. Hence,

$$h(y, t) = (1 - t)f(y) + tg(y) \in C \subset |L_x|.$$

Thus, $h|V_x \times \mathbf{I}$ is continuous by Lemma 4.5.5. Therefore, h is continuous.

(2): Since X is a k-space, $X \times \mathbf{I}$ is also a k-space. Hence, it suffices to show that for every compact set A in X, $h|A \times \mathbf{I}$ is continuous. According to Proposition 4.2.3, $f(A) \cup g(A) \subset |L_0|$ for some finite subcomplex L_0 of L. For each pair $(D_1, D_2) \in L^2$, let $L(D_1, D_2)$ be the subcomplex of L consisting of faces of the smallest cell of L containing $D_1 \cup D_2$, where $L(D_1, D_2) = \emptyset$ if L has no cells containing $D_1 \cup D_2$. Then, $L_1 = \bigcup_{(D_1,D_2)\in L_0^2} L(D_1, D_2)$ is a finite subcomplex of L with $L_0 \subset L_1$. Since f and g are contiguous, it follows that $h(A \times \mathbf{I}) \subset |L_1|$. Hence, $h|A \times \mathbf{I}$ is continuous by Lemma 4.5.5. □

According to Proposition 4.5.6 (2), in the cases that X is a polyhedron or locally compact or metrizable, the straight-line homotopy is continuous.

Remark 4.5.1 It will be shown in Sect. 4.15 that every two contiguous maps defined on an arbitrary space are homotopic, where the homotopy is not always given by the straight-line homotopy. In fact, there are some cases where the straight-line homotopy is not continuous. Such an example can be obtained by reforming the example given in Proposition 4.5.2.

Let $\Gamma = \mathbb{N}^{\mathbb{N}} \cup \mathbb{N}$. We define $K = \{\mathbf{0}, \mathbf{e}_\gamma, \langle \mathbf{0}, \mathbf{e}_\gamma \rangle \mid \gamma \in \Gamma\}$ and

$$L = K \cup \left\{ \langle \mathbf{e}_\gamma, \mathbf{e}_{\gamma'} \rangle,\ \langle \mathbf{0}, \mathbf{e}_\gamma, \mathbf{e}_{\gamma'} \rangle \;\middle|\; \gamma \neq \gamma' \in \Gamma \right\},$$

where $\mathbf{e}_\gamma \in \mathbb{R}^\Gamma$ is the unit vector in $\mathbb{R}^\Gamma$ (i.e., $\mathbf{e}_\gamma(\gamma) = 1$ and $\mathbf{e}_\gamma(\gamma') = 0$ for $\gamma' \neq \gamma$). Then, $K \subset L$ are simplicial complexes in $\mathbb{R}^\Gamma$ with $\dim K = 1$ and $\dim L = 2$. Let $f, g : |K|^2 \to |L|$ be maps defined by $f(x, y) = x$ and $g(x, y) = y$ for each

$(x, y) \in |K|^2$, where $|K|^2 \neq |K \times_c K|$ as spaces (see the proof of Proposition 4.5.2). Evidently, these maps f, g are contiguous. We can define $h : |K|^2 \times \mathbf{I} \to |L|$ as follows:

$$h(x, y, t) = (1 - t) f(x, y) + tg(x, y) = (1 - t)x + ty.$$

We will prove that h is not continuous at $(\mathbf{0}, \mathbf{0}, 1/2) \in |K|^2 \times \mathbf{I}$. For each $a \in \mathbb{N}^{\mathbb{N}}$ and $n \in \mathbb{N}$, let

$$v_{a,n} = \frac{1}{2} a(n)^{-1} \mathbf{e}_a + \frac{1}{2} a(n)^{-1} \mathbf{e}_n \in \langle \mathbf{0}, \mathbf{e}_a, \mathbf{e}_n \rangle \subset |L|.$$

As is easily observed, $U = |L| \setminus \{v_{a,n} \mid a \in \mathbb{N}^{\mathbb{N}}, n \in \mathbb{N}\}$ is an open neighborhood of $\mathbf{0} = h(\mathbf{0}, \mathbf{0}, 1/2)$ in $|L|$. Then, $h(W^2 \times \{1/2\}) \not\subset U$ for any neighborhood W of $\mathbf{0}$ in $|K|$. Indeed, for each $\gamma \in \Gamma$, we can choose $\delta_\gamma > 0$ so that $[0, \delta_\gamma] \mathbf{e}_\gamma \subset W$. Then, we have $a_0 \in \mathbb{N}^{\mathbb{N}}$ such that $a_0(n) > \max\{n, \delta_n^{-1}\}$ for every $n \in \mathbb{N}$. Take $n_0 \in \mathbb{N}$ so that $n_0 > \delta_{a_0}^{-1}$. Since $a_0(n_0)^{-1} < n_0^{-1} < \delta_{a_0}$, we have $x_0 = a_0(n_0)^{-1} \mathbf{e}_{a_0} \in W$. On the other hand, since $a_0(n_0)^{-1} < \delta_{n_0}$, we have $y_0 = a_0(n_0)^{-1} \mathbf{e}_{n_0} \in W$. Then,

$$h(x_0, y_0, 1/2) = \frac{1}{2} a_0(n_0)^{-1} \mathbf{e}_{a_0} + \frac{1}{2} a_0(n_0)^{-1} \mathbf{e}_{n_0} = v_{a_{n_0}, n_0} \notin U.$$

Therefore, h is not continuous at $(\mathbf{0}, \mathbf{0}, 1/2)$.

4.6 The Metric Topology of Polyhedra

As shown in Proposition 4.2.11 (2), for a non-locally finite cell complex K, the polyhedron $|K|$ is non-metrizable. In this section, when K is a *simplicial complex*, we introduce a natural metric on the polyhedron $|K|$ that induces the same topology as the Whitehead topology if K is locally finite.

Each point $x \in |K|$ has a unique representation

$$x = \sum_{i=1}^{n+1} z(i) v_i, \ z \in \operatorname{rint} \Delta^n, \ c_K(x) = \langle v_1, \ldots, v_{n+1} \rangle,$$

where $v_i \neq v_j$ if $i \neq j$. For each $v \in K^{(0)}$, we define a map $\beta_v^K : |K| \to \mathbf{I}$ as follows:

$$\beta_v^K(x) = \begin{cases} z(i) & \text{if } v = v_i, \ i = 1, \ldots, n+1, \\ 0 & \text{otherwise.} \end{cases}$$

Then, maps $\beta_v^K : |K| \to \mathbf{I}$, $v \in K^{(0)}$, are affine on each simplex of K. It follows from the definition that

$$\sum_{v\in K^{(0)}} \beta_v^K(x) = 1 \quad \text{for each } x \in |K|,$$

where $\{v \in K^{(0)} \mid \beta_v^K(x) \neq 0\} = c_K(x)^{(0)}$ (i.e., $\beta_v^K(x) > 0 \Leftrightarrow v \in c_K(x)^{(0)}$). Namely, $(\beta_v^K)_{v\in K^{(0)}}$ is a *partition of unity* on $|K|$ with $\operatorname{supp}\beta_v^K = |\operatorname{St}(v, K)|$. In fact, each $(\beta_v^K)^{-1}((0, 1])$ is nothing but the open star $O_K(v)$, that is,

$$(\beta_v^K)^{-1}((0, 1]) = O_K(v) \quad \text{for every } v \in K^{(0)}.$$

Thus, each $x \in |K|$ is uniquely represented in the form

$$x = \sum_{v\in K^{(0)}} \beta_v^K(x)v,$$

where $\beta_v^K(x)$ is called the **barycentric coordinate** of x at v with respect to K. The injection $\beta^K : |K| \to \ell_1(K^{(0)})$ defined by $\beta^K(x)(v) = \beta_v^K(x)$ is called the **canonical representation** of K. Observe that $\beta^K(v) = \mathbf{e}_v$ is the unit vector of $\ell_1(K^{(0)})$. For each $\sigma \in K$, the restriction $\beta^K|\sigma$ is affine. It should be noted that $\beta^K(K) = \{\beta^K(\sigma) \mid \sigma \in K\}$ is a simplicial complex in the Banach space $\ell_1(K^{(0)})$, which is congruent to K.

Now, we can define a metric ρ_K on $|K|$ as follows:

$$\rho_K(x, y) = \|\beta^K(x) - \beta^K(y)\|_1 = \sum_{v\in K^{(0)}} |\beta_v^K(x) - \beta_v^K(y)|,$$

where $\|\cdot\|_1$ is the norm for $\ell_1(K^{(0)})$. Note that $\rho_K(v, v') = 2$ for each pair of distinct vertices $v, v' \in K^{(0)}$. The topology on $|K|$ induced by this metric ρ_K is called the **metric topology**. The space $|K|$ with this topology is denoted by $|K|_{\mathrm{m}}$. This space is homeomorphic to the subspace $\beta^K(|K|) = |\beta^K(K)|$ of the Banach space $\ell_1(K^{(0)})$ because β^K is an isometry. The space $|K|_{\mathrm{m}}$ (or the metric space $(|K|, \rho_K)$) is called a **metric polyhedron**. Note that $\ell_1(K^{(0)}) \subset \mathbb{R}^{K^{(0)}}$, and that the topology of $\beta^K(|K|)$ inherited from $\ell_1(K^{(0)})$ coincides with the one inherited from the product space $\mathbb{R}^{K^{(0)}}$ because $\beta^K(|K|)$ is contained in the unit sphere of $\ell_1(K^{(0)})$ (cf. Proposition 1.2.6). Hence, the metric topology on $|K|$ is the coarsest topology such that all $\beta_v^K : |K| \to \mathbf{I}$, $v \in K^{(0)}$, are continuous. Then, we have the following:

Fact *For an arbitrary space X, each $f : X \to |K|_{\mathrm{m}}$ is continuous if and only if $\beta_v^K f$ is continuous for every $v \in K^{(0)}$.*

Since every $\beta_v^K : |K| \to \mathbf{I}$ is continuous (with respect to the Whitehead topology), the identity $\mathrm{id}_{|K|} : |K| \to |K|_{\mathrm{m}}$ is continuous, hence the Whitehead topology and the metric topology are identical on each simplex of K.

The open star $O_K(v)$ at each $v \in K^{(0)}$ is open in $|K|_{\mathrm{m}}$ because it is just $(\beta_v^K)^{-1}((0, 1])$ $(= |K| \setminus (\beta_v^K)^{-1}(0))$. Hence, $\mathcal{O}_K \in \operatorname{cov}(|K|_{\mathrm{m}})$. For each $x \in |K|$, we have

$$O_K(x) = \bigcup_{\sigma \in K[x]} \operatorname{rint} \sigma = \bigcap_{v \in c_K(x)^{(0)}} O_K(v) \subset |\operatorname{St}(c_K(x), K)|.$$

Hence, the open star $O_K(x)$ is an open neighborhood of x in $|K|_{\mathrm{m}}$. For each $0 < r \leqslant 1$, the following is also an open neighborhood of x in $|K|_{\mathrm{m}}$:

$$O_K(x, r) = (1 - r)x + r O_K(x) = \big\{(1 - r)x + ry \;\big|\; y \in O_K(x)\big\}.$$

Then, note that $O_K(x, r) \subset \mathrm{B}_{\rho_K}(x, 2r)$. Indeed, for each $y \in O_K(x)$, since $c_K(x) \leqslant c_K(y)$ and β_v^K is affine on $c_K(y)$, it follows that

$$\begin{aligned} \rho_K((1 - r)x + ry, x) &= \sum_{v \in K^{(0)}} \left|\beta_v^K((1 - r)x + ry) - \beta_v^K(x)\right| \\ &= \sum_{v \in K^{(0)}} r \cdot \left|\beta_v^K(y) - \beta_v^K(x)\right| = r\rho_K(x, y) < 2r. \end{aligned}$$

In the above, when $c_K(x)$ is not principal, for arbitrarily small $\varepsilon > 0$, we can take $v_0 \in \operatorname{Lk}(c_K(x), K)^{(0)}$ and $y \in \operatorname{rint} v_0 c_K(x) \subset O_K(x)$ so that $0 < \beta_v^K(y) < \beta_v^K(x)$ for every $v \in c_K(x)^{(0)}$ and $\sum_{v \in c_K(x)^{(0)}} \beta_v^K(y) < \varepsilon/2$. Then,

$$\begin{aligned} \rho_K(x, y) &= \sum_{v \in c_K(x)^{(0)}} \big(\beta_v^K(x) - \beta_v^K(y)\big) + \sum_{v \in \operatorname{Lk}(c_K(x), K)^{(0)}} \beta_v^K(y) \\ &= 1 - \sum_{v \in c_K(x)^{(0)}} \beta_v^K(y) + 1 - \sum_{v \in c_K(x)^{(0)}} \beta_v^K(y) > 2 - \varepsilon. \end{aligned}$$

As a consequence, we have the following:

Proposition 4.6.1 *Let K be a simplicial complex and $x \in |K|$. Then, $\{O_K(x, r) \mid 0 < r \leqslant 1\}$ is an open neighborhood basis of x in $|K|_{\mathrm{m}}$.* □

The following proposition can be easily proved:

Proposition 4.6.2 *For every subcomplex L of a simplicial complex K, the metric ρ_L is the restriction of ρ_K and $|L|_{\mathrm{m}}$ is a closed subspace of $|K|_{\mathrm{m}}$.* □

Moreover, we have:

Proposition 4.6.3 *For a simplicial complex K and each $x \in |K|$, the closure of $O_K(x)$ in $|K|_{\mathrm{m}}$ coincides with $|\operatorname{St}(c_K(x), K)|$. In particular, for a vertex $v \in K^{(0)}$, $|\operatorname{St}(v, K)|$ is the closure of $O_K(v)$ in $|K|_{\mathrm{m}}$.*

Proof According to Proposition 4.6.2, $|\operatorname{St}(c_K(x), K)|$ is closed in $|K|_{\mathrm{m}}$. Then, it suffices to show that $(1 - t)y + tx \in O_K(x)$ for each $y \in |\operatorname{St}(c_K(x), K)|$ and $0 < t \leqslant 1$. For each $v \in c_K(x)^{(0)}$, since $\beta_v^K(x) > 0$, it follows that

$$\beta_v^K((1-t)y+tx) = (1-t)\beta_v^K(y) + t\beta_v^K(x) > 0,$$
$$\text{i.e., } (1-t)y+tx \in (\beta_v^K)^{-1}((0,1]) = O_K(v).$$

Hence, $(1-t)y+tx \in \bigcap_{v\in c_K(x)^{(0)}} O_K(v) = O_K(x)$. □

Thus, with respect to the metric topology as well as the Whitehead topology, O_K is an open cover of $|K|_{\mathrm{m}}$ with $\mathcal{S}_K = O_K^{\mathrm{cl}}$ and $(\beta_v^K)_{v\in K^{(0)}}$ is a *partition of unity* on $|K|_{\mathrm{m}}$ with $\operatorname{supp}\beta_v^K = |\operatorname{St}(v,K)|$.

For a finite simplicial complex K, since $|K|$ is compact (Proposition 4.2.3), it follows that $\mathrm{id}_{|K|} : |K| \to |K|_{\mathrm{m}}$ is a homeomorphism, that is, the metric topology of $|K|$ coincides with the Whitehead topology. More generally, we can prove the following theorem (cf. Proposition 4.2.11 (2)):

Theorem 4.6.4 *For a simplicial complex K, the metric topology of $|K|$ coincides with the Whitehead topology (i.e., $|K|_{\mathrm{m}} = |K|$ as spaces) if and only if K is locally finite.*

Proof If $|K|_{\mathrm{m}} = |K|$ as spaces, then $|K|$ is metrizable, so K is locally finite by Proposition 4.2.11 (2). To show the converse, let $\varphi = \mathrm{id} : |K| \to |K|_{\mathrm{m}}$, which is a continuous bijection. Assume that K is locally finite, that is, $\operatorname{St}(v,K)$ is finite for each $v \in K^{(0)}$. Then, each $\varphi|\operatorname{St}(v,K)|$ is a homeomorphism, so $\varphi|O_K(v)$ is also a homeomorphism. Since O_K is an open cover of both $|K|_{\mathrm{m}}$ and $|K|$, it follows that φ is a homeomorphism. □

The complete metrizability of $|K|_{\mathrm{m}}$ can be characterized as follows:

Theorem 4.6.5 *For a simplicial complex K, the following are equivalent:*

(a) $|K|_{\mathrm{m}}$ *is completely metrizable;*
(b) K *contains no infinite full complexes as subcomplexes;*
(c) ρ_K *is complete.*

Proof Since (c) ⇒ (a) is obvious, we show (a) ⇒ (b) ⇒ (c).

(a) ⇒ (b): Assume that K contains an infinite full complex as a subcomplex. Then, we have a countably infinite full complex $L \subset K$. Because $|K|_{\mathrm{m}}$ is completely metrizable, its closed subspace $|L|_{\mathrm{m}}$ is also completely metrizable. However, $|L|_{\mathrm{m}}$ is a union of countably many simplexes that have no interior points. This contradicts the Baire Category Theorem 2.5.1. Therefore, K contains no infinite full complexes.

(b) ⇒ (c): Let $(x_i)_{i\in\mathbb{N}}$ be a ρ_K-Cauchy sequence in $|K|_{\mathrm{m}}$. Then, $(\beta^K(x_i))_{i\in\mathbb{N}}$ is Cauchy in $\ell_1(K^{(0)})$ because $\beta^K : (|K|,\rho_K) \to \ell_1(K^{(0)})$ is an isometry. Then, we have $z = \lim_{i\to\infty}\beta^K(x_i) \in \ell_1(K^{(0)})$. Observe that

$$\sum_{v\in K^{(0)}} z(v) = \|z\|_1 = \lim_{i\to\infty}\|\beta^K(x_i)\|_1 = 1.$$

Let $V = \{v \in K^{(0)} \mid z(v) > 0\}$. For each $S \in \operatorname{Fin}(V)$, we can choose $i_0 \in \mathbb{N}$ so that

$$\left\|z - \beta^K(x_{i_0})\right\|_1 < \min\left\{z(v) \mid v \in S\right\}.$$

If there exists $v_0 \in S \setminus c_K(x_{i_0})^{(0)}$, then

$$\left\|z - \beta^K(x_{i_0})\right\|_1 \geqslant \left|z(v_0) - \beta^K_{v_0}(x_{i_0})\right| = z(v_0) \geqslant \min\left\{z(v) \mid v \in S\right\},$$

which is a contradiction. Hence, $S \subset c_K(x_{i_0})^{(0)}$, which means $\langle S\rangle \leqslant c_K(x_{i_0})$. Then, it follows that $\langle S\rangle \in K$. Consequently, the full complex F_V with $\mathrm{F}_V^{(0)} = V$ is a subcomplex of K. Since V is finite by (b), we have $\langle V\rangle \in K$. Then, $(x_i)_{i\in\mathbb{N}}$ converges to

$$x = \sum_{v\in V} z(v)v \in \langle V\rangle \subset |K|.$$

Indeed, since $\beta^K(x) = z$, it follows that $\rho_K(x, x_i) = \|z - \beta^K(x_i)\|_1 \to 0\ (i \to \infty)$. Therefore, ρ_K is complete. □

Remark 4.6.1 Obviously, every locally finite-dimensional simplicial complex K contains no infinite full complexes as subcomplexes. However, the converse does not hold. For example, take simplexes σ_n, $n \in \mathbb{N}$, so that $\dim \sigma_n = n$ and $\sigma_i \cap \sigma_j = \{v_0\}$ if $i \neq j$, where v_0 is a vertex of every σ_n. Then, $K = \bigcup_{n\in\mathbb{N}} F(\sigma_n)$ is a simplicial complex which contains no infinite full complexes as subcomplexes, but it is not locally finite-dimensional.

In Proposition 4.5.6, we considered cases where contiguous two maps between cell complexes are homotopic by the straight-line homotopy. Moreover, as seen in Remark 4.5.1, the straight-line homotopy is not always continuous. However, using the metric topology, the straight-line homotopy is always continuous, that is:

Proposition 4.6.6 *Let L be a simplicial complex and X an arbitrary space. If two maps $f, g : X \to |L|_{\mathrm{m}}$ are contiguous (with respect to L) then $f \simeq_L g$ by the straight-line homotopy $h : X \times \mathbf{I} \to |L|_{\mathrm{m}}$, that is,*

$$h(x,t) = (1-t)f(x) + tg(x) \ \textit{for each}\ (x,t) \in X \times \mathbf{I}.$$

Proof It suffices to verify the continuity of h. This follows from the continuity of $\beta^L_v h : X \times \mathbf{I} \to \mathbf{I}$, $v \in L^{(0)}$, where

$$\beta^L_v h(x,t) = (1-t)\beta^K_v f(x) + t\beta^K_v g(x).$$ □

As seen in Proposition 4.5.2, in general $|K \times_s L| \neq |K| \times |L|$ as spaces for ordered simplicial complexes K and L. However, concerning the metric topology, always $|K \times_s L|_{\mathrm{m}} = |K|_{\mathrm{m}} \times |L|_{\mathrm{m}}$, that is:

Theorem 4.6.7 *For each pair of ordered simplicial complexes K and L,*

$$|K \times_s L|_{\mathrm{m}} = |K|_{\mathrm{m}} \times |L|_{\mathrm{m}} \ \textit{as spaces.}$$

Proof To see the continuity of $\mathrm{id} : |K \times_s L|_{\mathrm{m}} \to |K|_{\mathrm{m}} \times |L|_{\mathrm{m}}$, it suffices to show the continuity of the projections $\mathrm{pr}_1 : |K \times_s L|_{\mathrm{m}} \to |K|_{\mathrm{m}}$ and $\mathrm{pr}_2 : |K \times_s L|_{\mathrm{m}} \to |L|_{\mathrm{m}}$. For each $(x, y) \in |K \times_s L| = |K| \times |L|$, observe that

$$\begin{aligned}(x, y) &= \sum_{(u,v)\in(K\times_s L)^{(0)}} \beta^{K\times_s L}_{(u,v)}(x, y)\cdot(u, v)\\ &= \left(\sum_{(u,v)\in K^{(0)}\times L^{(0)}} \beta^{K\times_s L}_{(u,v)}(x, y)u, \sum_{(u,v)\in K^{(0)}\times L^{(0)}} \beta^{K\times_s L}_{(u,v)}(x, y)v \right).\end{aligned}$$

Hence, we have the following:

$$\begin{aligned}\beta^K_u(x) &= \sum_{v\in L^{(0)}} \beta^{K\times_s L}_{(u,v)}(x, y) \ \text{ for each } u \in K^{(0)} \text{ and}\\ \beta^L_v(y) &= \sum_{u\in K^{(0)}} \beta^{K\times_s L}_{(u,v)}(x, y) \ \text{ for each } v \in L^{(0)}.\end{aligned}$$

Then, it is easy to see that each $\beta^K_u \mathrm{pr}_1$ and $\beta^L_v \mathrm{pr}_2$ are continuous, which means that pr_1 and pr_2 are continuous.

Indeed, for each $(x, y), (x', y') \in |K \times_s L|_{\mathrm{m}}$,

$$\begin{aligned}\left|\beta^K_u \mathrm{pr}_1(x, y) - \beta^K_u \mathrm{pr}_1(x', y')\right| &= \left|\beta^K_u(x) - \beta^K_u(x')\right|\\ &= \left| \sum_{v\in L^{(0)}} \beta^{K\times_s L}_{(u,v)}(x, y) - \sum_{v\in L^{(0)}} \beta^{K\times_s L}_{(u,v)}(x', y') \right|\\ &\leqslant \sum_{v\in L^{(0)}} \left|\beta^{K\times_s L}_{(u,v)}(x, y) - \beta^{K\times_s L}_{(u,v)}(x', y')\right|\\ &\leqslant \left\|\beta^{K\times_s L}(x, y) - \beta^{K\times_s L}(x', y')\right\|_1 = \rho_{K\times_s L}((x, y), (x', y')).\end{aligned}$$

Thus, $\beta^K_u \mathrm{pr}_1$ is continuous. The continuity of $\beta^L_v \mathrm{pr}_2$ is similarly shown.

To prove the continuity of $\mathrm{id} : |K|_{\mathrm{m}} \times |L|_{\mathrm{m}} \to |K \times_s L|_{\mathrm{m}}$ at $(x, y) \in |K|_{\mathrm{m}} \times |L|_{\mathrm{m}}$, we need to derive the data for $(\beta^{K\times_s L}_{(u,v)}(x, y))_{(u,v)\in K^{(0)}\times L^{(0)}}$ from $(\beta^K_u(x))_{u\in K^{(0)}}$ and $(\beta^L_v(y))_{v\in L^{(0)}}$. Note that the carrier $c_{K\times_s L}(x, y)$ is contained in $c_K(x) \times c_L(y)$. Let

$$\begin{aligned}c_K(x) &= \langle u_1, \ldots, u_n\rangle,\ u_1 < \cdots < u_n, \text{ and}\\ c_L(y) &= \langle v_1, \ldots, v_m\rangle,\ v_1 < \cdots < v_m,\end{aligned}$$

and define $0 = a_0 < a_1 < \cdots < a_n = 1$ and $0 = b_0 < b_1 < \cdots < b_m = 1$ as follows:

$$a_k = \sum_{i=1}^{k} \beta^K_{u_i}(x) \ \text{ and } \ b_k = \sum_{i=1}^{k} \beta^L_{v_i}(y).$$

We define $0 = c_0 < c_1 < \cdots < c_\ell = 1$ by putting these a_i and b_j in order from smallest to largest, where $\max\{n, m\} \leqslant \ell < m + n$. It should be noted that $\sum_{k=1}^{\ell}(c_k - c_{k-1}) = c_\ell - c_0 = 1$. For each $k = 1, \ldots, \ell$, let

$$a_{i(k)-1} < c_k \leqslant a_{i(k)} \text{ and } b_{j(k)-1} < c_k \leqslant b_{j(k)},$$

and define $(\bar{u}_k, \bar{v}_k) = (u_{i(k)}, v_{j(k)})$. Then, $\langle(\bar{u}_1, \bar{v}_1), \ldots, (\bar{u}_\ell, \bar{v}_\ell)\rangle \in K \times_s L$. Moreover, observe that

$$\begin{aligned}
&\sum_{k=1}^{\ell}(c_k - c_{k-1})(\bar{u}_k, \bar{v}_k) \\
&\quad = \left(\sum_{i=1}^{n}\sum_{i(k)=i}(c_k - c_{k-1})u_{i(k)}, \sum_{j=1}^{m}\sum_{j(k)=j}(c_k - c_{k-1})v_{j(k)}\right) \\
&\quad = \left(\sum_{i=1}^{n}(a_i - a_{i-1})u_i, \sum_{j=1}^{m}(b_j - b_{j-1})v_j\right) \\
&\quad = \left(\sum_{i=1}^{n}\beta^K_{u_i}(x)u_i, \sum_{j=1}^{m}\beta^L_{v_j}(y)v_j\right) = (x, y),
\end{aligned}$$

which means that

$$c_{K\times_s L}(x, y) = \langle(\bar{u}_1, \bar{v}_1), \ldots, (\bar{u}_\ell, \bar{v}_\ell)\rangle \text{ and } \beta^{K\times_s L}_{(\bar{u}_k, \bar{v}_k)}(x, y) = c_k - c_{k-1}.$$

For each $\varepsilon > 0$, choose $\delta > 0$ so that $2\delta < c_k - c_{k-1}$ for every $k = 1, \ldots, \ell$ and $2(\ell + mn + 1)\delta < \varepsilon$. Then, $\beta^K_{u_i}(x) > 2\delta$ for $i = 1, \ldots, n$ and $\beta^L_{v_j}(y) > 2\delta$ for $j = 1, \ldots, m$. Now, let $(x', y') \in |K| \times |L|$ with $\rho_K(x, x') < \delta$ and $\rho_L(y, y') < \delta$. To show that $\rho_{K\times_s L}((x, y), (x', y')) < \varepsilon$, let

$$\begin{aligned}
c_K(x') &= \langle u'_1, \cdots, u'_{n'}\rangle,\ u'_1 < \cdots < u'_{n'}, \text{ and} \\
c_L(y') &= \langle v'_1, \cdots, v'_{m'}\rangle,\ v'_1 < \cdots < v'_{m'}.
\end{aligned}$$

In the same way as (x, y), define $0 = a'_0 < a'_1 < \cdots < a'_{n'} = 1$ and $0 = b'_0 < b'_1 < \cdots < b'_{m'} = 1$ as follows:

$$a'_k = \sum_{i=1}^{k}\beta^K_{u'_i}(x') \text{ and } b'_k = \sum_{i=1}^{k}\beta^L_{v'_i}(y'),$$

and then define $0 = c'_0 < c'_1 < \cdots < c'_{\ell'} = 1$ by putting these a'_i and b'_j in order from smallest to largest. For each $k = 1, \ldots, \ell'$, let

$$a'_{i'(k)-1} < c'_k \leqslant a'_{i'(k)} \text{ and } b'_{j'(k)-1} < c'_k \leqslant b'_{j'(k)},$$

and define $(\bar{u}'_k, \bar{v}'_k) = (u'_{i'(k)}, v'_{j'(k)})$. Then, we have

$$c_{K\times_s L}(x', y') = \langle(\bar{u}'_1, \bar{v}'_1), \ldots, (\bar{u}'_{\ell'}, \bar{v}'_{\ell'})\rangle \text{ and } \beta^{K\times_s L}_{(\bar{u}'_k, \bar{v}'_k)}(x', y') = c'_k - c'_{k-1}.$$

For each $i = 1, \ldots, n$,

$$\beta^K_{u_i}(x') \geqslant \beta^K_{u_i}(x) - |\beta^K_{u_i}(x) - \beta^K_{u_i}(x')| > 2\delta - \rho_K(x, x') > \delta > 0,$$

which implies that $u_1 < \cdots < u_n$ is a subsequence of $u'_1 < \cdots < u'_{n'}$, that is, we can take $1 \leqslant p(1) < \cdots < p(n) \leqslant n'$ to write $u_i = u'_{p(i)}$. Then, we have

$$|a'_{p(k)} - a_k| < \delta \text{ and } |a'_{p(k)-1} - a_{k-1}| < \delta.$$

Indeed, these can be shown as follows:

$$\begin{aligned} |a'_{p(k)} - a_k| &\leqslant \sum_{i=1}^{k} |\beta^K_{u_i}(x) - \beta^K_{u_i}(x')| \\ &\quad + \sum_{u\notin\{u_1,\ldots,u_n\}} \beta^K_u(x') \leqslant \rho_K(x, x') < \delta \text{ and} \\ |a'_{p(k)-1} - a_{k-1}| &\leqslant \sum_{i=1}^{k-1} |\beta^K_{u_i}(x) - \beta^K_{u_i}(x')| \\ &\quad + \sum_{u\notin\{u_1,\ldots,u_n\}} \beta^K_u(x') \leqslant \rho_K(x, x') < \delta. \end{aligned}$$

Similarly, we can take $1 \leqslant q(1) < \cdots < q(m) \leqslant m'$ to write $v_j = v'_{q(j)}$. Then,

$$|b'_{q(k)} - b_k| < \delta \text{ and } |b'_{q(k)-1} - b_{k-1}| < \delta.$$

On the other hand, for each $k = 1, \ldots, \ell$, because $a_{i(k)-1} < c_k \leqslant a_{i(k)}$ and $b_{j(k)-1} < c_k \leqslant b_{j(k)}$, we have

$$c_k = \min\{a_{i(k)}, b_{j(k)}\} \text{ and } c_{k-1} = \max\{a_{i(k)-1}, b_{j(k)-1}\}.$$

Then, it follows that

$$b'_{q(j(k))-1} < b_{j(k)-1} + \delta \leqslant c_{k-1} + \delta < c_k - \delta \leqslant a_{i(k)} - \delta < a'_{p(i(k))}.$$

Similarly, $a'_{p(i(k))-1} < b'_{q(j(k))}$. Hence, there is some $r(k) = 1, \ldots, \ell'$ such that

$$c'_{r(k)-1} = \max\{a'_{p(i(k))-1},\ b'_{q(j(k))-1}\} < c'_{r(k)} = \min\{a'_{p(i(k))},\ b'_{q(j(k))}\}.$$

This means that $p(i(k)) = i'(r(k))$ and $q(j(k)) = j'(r(k))$, which implies that

$$\begin{aligned}(\bar{u}'_{r(k)}, \bar{v}'_{r(k)}) &= (u'_{i'(r(k))}, v'_{j'(r(k))}) = (u'_{p(i(k))}, v'_{q(j(k))}) \\ &= (u_{i(k)}, v_{j(k)}) = (\bar{u}_k, \bar{v}_k)\end{aligned}$$

and $\beta^{K\times,L}_{(\bar{u}_k,\bar{v}_k)}(x', y') = c'_{r(k)} - c'_{r(k)-1}$. Observe that

$$\begin{aligned}c_k - \delta &= \min\left\{a_{i(k)} - \delta, b_{j(k)} - \delta\right\} \\ &< c'_{r(k)} = \min\left\{a'_{p(i(k))}, b'_{q(j(k))}\right\} \\ &< c_k + \delta = \min\left\{a_{i(k)} + \delta, b_{j(k)} + \delta\right\} \text{ and} \\ c_{k-1} - \delta &= \max\left\{a_{i(k)-1} - \delta, b_{j(k)-1} - \delta\right\} \\ &< c'_{r(k)-1} = \max\left\{a'_{p(i(k))-1}, b'_{q(j(k))-1}\right\} \\ &< c_{k-1} + \delta = \max\left\{a_{i(k)-1} + \delta, b_{j(k)-1} + \delta\right\}.\end{aligned}$$

Moreover, it should be noted that for each $r = 1, \ldots, \ell'$,

$$\begin{aligned}\sum_{i'(r)=i} (c'_r - c'_{r-1}) &= a'_i - a'_{i-1} = \beta^K_{u'_i}(x'), \\ \sum_{j'(r)=j} (c'_r - c'_{r-1}) &= b'_j - b'_{j-1} = \beta^L_{v'_j}(y') \text{ and}\end{aligned}$$

$$r \in \{r(1), \ldots, r(\ell)\} \Leftrightarrow \begin{cases}\exists i = 1, \ldots, n,\ \exists j = 1, \ldots, m, \text{ such that} \\ i'(r) = p(i),\ j'(r) = q(j),\ a_i \neq b_{j-1},\ b_j \neq a_{i-1}.\end{cases}$$

The last equivalence can be proved as follows:

($\Rightarrow$): Recall that for each $k = 1, \ldots, \ell$, $p(i(k)) = i'(r(k))$ and $q(j(k)) = j'(r(k))$, where $b_{j(k)-1} < c_k \leqslant a_{i(k)}$ and $a_{i(k)-1} < c_k \leqslant b_{j(k)}$. Hence, $a_{i(k)} \neq b_{j(k)-1}$ and $b_{j(k)} \neq a_{i(k)-1}$.

($\Leftarrow$): Take i, j as in the right-hand condition. If $a_i < b_{j-1}$, then $a_i + \delta < b_{j-1} - \delta$ but

$$b_{j-1} - \delta < b'_{q(j-1)} \leqslant b'_{q(j)-1} = b'_{j'(r)-1} < c'_r \leqslant a'_{i'(r)} = a'_{p(i)} < a_i + \delta,$$

which is a contradiction. Hence, $b_{j-1} < a_i$. Similarly, we have $a_{i-1} < b_j$. Consequently, $\max\{a_{i-1}, b_{j-1}\} < \min\{a_i, b_j\}$. Choose $k = 1, \ldots, \ell$ so that $c_k = \min\{a_i, b_j\}$. Then, $a_{i-1} < c_k \leqslant a_i$ and $b_{j-1} < c_k \leqslant b_j$, which means $i = i(k)$ and $j = j(k)$. Thus, it follows that

$$c'_r = \min\{a'_{i'(r)}, b'_{j'(r)}\} = \min\{a'_{p(i)}, b'_{q(j)}\} = \min\{a'_{p(i(k))}, b'_{q(j(k))}\},$$

which implies that $r = r(k)$.

By the last equivalence, we have

$$\begin{aligned}&\left\{1, \ldots, \ell'\right\} \setminus \left\{r(1), \ldots, r(\ell)\right\} \\ &\qquad = \left\{r \in R \;\middle|\; i'(r) \notin \{p(1), \ldots, p(n)\}\right\} \\ &\qquad\quad \cup \left\{r \in R \;\middle|\; j'(r) \notin \{q(1), \ldots, q(m)\}\right\} \cup R_1 \cup R_2,\end{aligned}$$

where $R = \{1, \ldots, \ell'\}$,

$$R_1 = \Big\{ r \in R \;\Big|\; \exists i = 1, \ldots, n,\ \exists j = 1, \ldots, m,\ \text{such that} \\ i'(r) = p(i),\ j'(r) = q(j),\ a_i = b_{j-1} \Big\},\ \text{and}$$
$$R_2 = \Big\{ r \in R \;\Big|\; \exists i = 1, \ldots, n,\ \exists j = 1, \ldots, m,\ \text{such that} \\ i'(r) = p(i),\ j'(r) = q(j),\ b_j = a_{i-1} \Big\}.$$

Then, it follows that

$$\sum_{r \notin \{r(1), \ldots, r(\ell)\}} (c'_r - c'_{r-1}) \leqslant \sum_{i \notin \{p(1), \ldots, p(n)\}} \sum_{i'(r) = i} (c'_r - c'_{r-1}) \\ + \sum_{j \notin \{q(1), \ldots, q(m)\}} \sum_{j'(r) = j} (c'_r - c'_{r-1}) \\ + \sum_{r \in R_1 \cup R_2} (c'_r - c'_{r-1}),$$

where we have the following:

$$\sum_{i \notin \{p(1), \ldots, p(n)\}} \sum_{i'(r) = i} (c'_r - c'_{r-1}) = \sum_{i \notin \{p(1), \ldots, p(n)\}} \beta^K_{u'_i}(x') \\ = \sum_{u \notin \{u_1, \ldots, u_n\}} \beta^K_u(x') \leqslant \rho_K(x, x') < \delta,$$
$$\sum_{j \notin \{q(1), \ldots, q(m)\}} \sum_{j'(r) = j} (c'_r - c'_{r-1}) = \sum_{j \notin \{q(1), \ldots, q(m)\}} \beta^L_{v'_j}(y') \\ = \sum_{v \notin \{v_1, \ldots, v_m\}} \beta^L_v(y') \leqslant \rho_L(y, y') < \delta.$$

For each $r \in R_1$, take $i = 1, \ldots, n$ and $j = 1, \ldots, m$ so that

$$i'(r) = p(i),\ j'(r) = q(j),\ a_i = b_{j-1}.$$

Since $|a_i - a'_{p(i)}| < \delta$ and $|b_{j-1} - b'_{q(j)-1}| < \delta$, we have $|a'_{p(i)} - b'_{q(j)-1}| < 2\delta$. Then, it follows that

$$\begin{aligned} c'_r - c'_{r-1} &= \min\{a'_{i'(r)}, b'_{j'(r)}\} - \max\{a'_{i'(r)-1}, b'_{j'(r)-1}\} \\ &= \min\{a'_{p(i)}, b'_{q(j)}\} - \max\{a'_{p(i)-1}, b'_{q(j)-1}\} \\ &\leqslant a'_{p(i)} - b'_{q(j)-1} < 2\delta. \end{aligned}$$

For each $r \in R_2$, we can similarly obtain $c'_r - c'_{r-1} < 2\delta$. Since the correspondence $r \mapsto (i'(r), j'(r))$ is injective, it is easy to see that $\operatorname{card}(R_1 \cup R_2) \leqslant mn$. Thus, we have $\sum_{r \in R_1 \cup R_2} (c'_r - c'_{r-1}) \leqslant 2\delta \cdot mn$. Therefore,

$$\sum_{r\notin\{r(1),\dots,r(\ell)\}}(c'_r - c'_{r-1}) < \delta + \delta + 2mn\delta = 2(mn+1)\delta.$$

Consequently, we have the following estimation:

$$\begin{aligned}\rho_{K\times_s L}((x,y),(x',y')) &= \sum_{k=1}^{\ell}\left|(c_k - c_{k-1}) - (c'_{r(k)} - c'_{r(k)-1})\right| \\ &\qquad + \sum_{r\notin\{r(1),\dots,r(\ell)\}}(c'_r - c'_{r-1}) \\ &< \sum_{k=1}^{\ell}|c_k - c'_{r(k)}| + \sum_{k=1}^{\ell}|c_{k-1} - c'_{r(k)-1}| + 2(mn+1)\delta \\ &< 2(\ell + mn + 1)\delta < \varepsilon.\end{aligned}$$

This completes the proof. □

Concerning simplicial subdivisions, we have the following result:

Proposition 4.6.8 *Given simplicial complexes $K \rhd K'$, $\rho_K(x,y) \leqslant \rho_{K'}(x,y)$ for each $x, y \in |K| = |K'|$, hence* $\mathrm{id} : |K'|_{\mathrm{m}} \to |K|_{\mathrm{m}}$ *is continuous.*

Proof For each $x \in |K| = |K'|$ and $v \in K^{(0)}$, we have

$$\beta_v^K(x) = \beta_v^K\big(\textstyle\sum_{w\in K'^{(0)}}\beta_w^{K'}(x)w\big) = \displaystyle\sum_{w\in K'^{(0)}}\beta_w^{K'}(x)\beta_v^K(w).$$

Then, for each $x, y \in |K| = |K'|$,

$$\begin{aligned}\rho_K(x,y) &= \sum_{v\in K^{(0)}}\left|\beta_v^K(x) - \beta_v^K(y)\right| \\ &\leqslant \sum_{v\in K^{(0)}}\sum_{w\in K'^{(0)}}\left|\beta_w^{K'}(x) - \beta_w^{K'}(y)\right|\beta_v^K(w) \\ &= \sum_{w\in K'^{(0)}}\left|\beta_w^{K'}(x) - \beta_w^{K'}(y)\right| = \rho_{K'}(x,y).\end{aligned}$$

□

Remark 4.6.2 In Proposition 4.6.8 above, $\mathrm{id} : |K|_{\mathrm{m}} \to |K'|_{\mathrm{m}}$ is not necessarily continuous. Namely, in contrast to the Whitehead topology, $|K'|_{\mathrm{m}} \neq |K|_{\mathrm{m}}$ for some subdivision $K' \lhd K$. Such an example can be defined in ℓ_1 as follows:

$$K = \left\{\mathbf{0},\ \mathbf{e}_i,\ \langle\mathbf{0},\mathbf{e}_i\rangle \;\middle|\; i\in\mathbb{N}\right\} \text{ and}$$
$$K' = \left\{\mathbf{0},\ 2^{-i}\mathbf{e}_i,\ \mathbf{e}_i,\ \langle\mathbf{0},2^{-i}\mathbf{e}_i\rangle,\ \langle 2^{-i}\mathbf{e}_i,\mathbf{e}_i\rangle \;\middle|\; i\in\mathbb{N}\right\} \lhd K,$$

where each $\mathbf{e}_i \in \ell_1$ is defined by $\mathbf{e}_i(i) = 1$ and $\mathbf{e}_i(j) = 0$ for $j \neq i$. Then, $|K|_{\mathrm{m}}$ is simply the hedgehog $J(\mathbb{N})$. The set $\{2^{-i}\mathbf{e}_i \mid i\in\mathbb{N}\}$ is closed in $|K'|_{\mathrm{m}}$ but $\lim_{i\to\infty}2^{-i}\mathbf{e}_i = \mathbf{0}$ in $|K|_{\mathrm{m}}$. See Fig. 4.6.

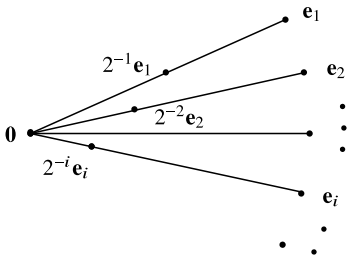


Fig. 4.6 $|K'|_{\mathrm{m}} \neq |K|_{\mathrm{m}}$

By using the example in the above remark, the following can be easily shown:

Proposition 4.6.9 *Every non-locally finite simplicial complex K has a simplicial subdivision K' such that $|K'|_{\mathrm{m}} \neq |K|_{\mathrm{m}}$.*

Sketch of Proof For some vertex $v \in K^{(0)}$, $\mathrm{St}(v, K)$ is infinite. Taking infinite distinct vertices $v_i \in \mathrm{Lk}(v, K)^{(0)}$, $i \in \mathbb{N}$, we can construct a subcomplex L of K which is congruent to the example in Remark 4.6.2. We can apply Proposition 4.3.11 to extend $L' \lhd L$ to $K' \lhd K$.

We can generalize Proposition 4.2.11 (1) as follows:

Proposition 4.6.10 *For each infinite cell complex K,* $\operatorname{dens}|K| = \operatorname{card} K^{(0)}$. *If K is simplicial, then* $\operatorname{dens}|K| = \operatorname{dens}|K|_{\mathrm{m}} = \operatorname{card} K^{(0)}$.

Sketch of Proof As in Proposition 4.2.11 (1), we can construct a dense set D in $|K|$ with $\operatorname{card} D = \operatorname{card} K = \operatorname{card} K^{(0)}$, hence $\operatorname{dens}|K| \leqslant \operatorname{card} K^{(0)}$. On the other hand, let K' be a derived subdivision of K. Then, $\{O_{K'}(v) \mid v \in K^{(0)}\}$ is a pairwise disjoint collection of open sets in $|K|$, hence $\operatorname{card} K^{(0)} \leqslant c(|K|) \leqslant \operatorname{dens}|K|$.

If K is simplicial, the above D is also dense in $|K|_{\mathrm{m}}$. Moreover, $\{O_{\mathrm{Sd}\,K}(v) \mid v \in K^{(0)}\}$ is a pairwise disjoint collection of open sets in $|K|_{\mathrm{m}}$.

It should be remarked that $w(|K|_{\mathrm{m}}) = \operatorname{dens}|K|_{\mathrm{m}} = \operatorname{card} K^{(0)}$ for every infinite simplicial complex K (see Sect. 1.1) but, as we saw in Proposition 4.2.11 (4), $w(|K|) \neq \operatorname{card} K^{(0)}$ in general, where $w(|K|) \geqslant \operatorname{dens}|K| = \operatorname{card} K^{(0)}$.

There is another admissible metric on $|K|_{\mathrm{m}}$ that is widely adopted because it allows for easy estimates of distances.

Proposition 4.6.11 (Another Metric on $|K|$) *Let K be a simplicial complex.*

(1) *The following metric is admissible for $|K|_{\mathrm{m}}$:*

$$\rho_K^{\infty}(x, y) = \left\|\beta^K(x) - \beta^K(y)\right\|_{\infty} = \sup_{v \in K^{(0)}} \left|\beta_v^K(x) - \beta_v^K(y)\right|,$$

where $\|\cdot\|_{\infty}$ is the norm for $\ell_{\infty}(K^{(0)})$.

Sketch of Proof Evidently, $\rho_K^{\infty}(x, y) \leqslant \rho_K(x, y)$ for each $x, y \in |K|$. For the continuity of $\mathrm{id} : (|K|, \rho_K^{\infty}) \to (|K|, \rho_K)$, see Proposition 1.2.6.

(2) *The inequality* $\rho_K^\infty(x, y) \leqslant \rho_{K'}^\infty(x, y)$ *does not hold for some* $K' \lhd K$. *Namely, Proposition* 4.6.8 *is not valid for the metric* ρ_K^∞.

Example. Consider the following simplicial complexes in $\mathbb{R}$:

$$K = \{0, \pm 1, \langle 0, \pm 1\rangle\},\ K' = \{0, \pm 1/2, \pm 1, \langle 0, \pm 1/2\rangle, \langle \pm 1/2, \pm 1\rangle\}.$$

Then, $K' \lhd K$ but $\rho_K^\infty(-3/4, 3/4) = 3/4 > \rho_{K'}^\infty(-3/4, 3/4) = 1/2$.

(3) *For a simplicial complex* K, *the following are equivalent*:

(a) K *is finite-dimensional*;
(b) ρ_K *is uniformly equivalent to* ρ_K^∞;
(c) ρ_K^∞ *is complete*.

Sketch of Proof (a) $\Rightarrow$ (b) and (c): For each $x, y \in |K|$,

$$\rho_K^\infty(x, y) \leqslant \rho_K(x, y) \leqslant 2(\dim K + 1)\rho_K^\infty(x, y).$$

Then, applying Theorem 4.6.5, we have (c).

(b) or (c) $\Rightarrow$ (a): If $\dim K = \infty$, then we can inductively choose n-simplexes $\sigma_n \in K, n \in \mathbb{N}$, so that $\sigma_n \cap \sigma_m = \emptyset$ if $n \neq m$. Then, for any $n < m$, $\rho_K(\hat{\sigma}_n, \hat{\sigma}_m) = 2$ and $\rho_K^\infty(\hat{\sigma}_n, \hat{\sigma}_m) = 1/(n + 1)$. This is contrary to both (b) and (c).

Remark 4.6.3 In the same way as (1) above, we can see that the following metric is also admissible for $|K|_{\mathrm{m}}$:

$$d(x, y) = \|\beta^K(x) - \beta^K(x)\|_2 = \sqrt{\sum_{v \in K^{(0)}} \left(\beta_v^K(x) - \beta_v^K(y)\right)^2},$$

where $\|\cdot\|_2$ is the norm for Hilbert space $\ell_2(K^{(0)})$.

4.7 Admissible Subdivisions

Due to Proposition 4.6.9, by subdividing a non-locally finite simplicial complex K, the metric topology of $|K|$ may be changed in general. In this section, we discuss simplicial subdivisions preserving the metric topology. A simplicial subdivision K' of a simplicial complex K is called an **admissible subdivision** of K if the metric $\rho_{K'}$ is admissible for $|K|_{\mathrm{m}}$; equivalently, $|K'|_{\mathrm{m}} = |K|_{\mathrm{m}}$ as spaces.[12]

For an arbitrary simplicial complex K, the barycentric subdivision is admissible, that is:

Theorem 4.7.1 *For each simplicial complex* K, *the barycentric subdivision* $\mathrm{Sd}\, K$ *is admissible, i.e.,* $|\mathrm{Sd}\, K|_{\mathrm{m}} = |K|_{\mathrm{m}}$ *as spaces.*

[12] In [15], such a subdivision was introduced and studied under the name **proper subdivision**.

Proof When K is finite, the result follows from Proposition 4.6.8 and the compactness of $|\operatorname{Sd} K|_{\mathrm{m}}$. We may assume that K is infinite. By Proposition 4.6.8, it suffices to show that $\mathrm{id} : |K|_{\mathrm{m}} \to |\operatorname{Sd} K|_{\mathrm{m}}$ is continuous. Let $x \in |K|$ and $k = \dim c_K(x)$. For each $\varepsilon > 0$, choose $\delta > 0$ so that $\delta < (2k+3)^{-1}\varepsilon$ and

$$\beta_v^K(x) \neq \beta_{v'}^K(x) \;\Rightarrow\; \delta < |\beta_v^K(x) - \beta_{v'}^K(x)|/2.$$

The last condition implies that $\delta < \beta_v^K(x)/2$ for each $v \in c_K(x)^{(0)}$ because $\beta_{v'}^K(x) = 0$ for some $v' \in K^{(0)}$. For each $y \in |K|$ with $\rho_K(x, y) < \delta$, the following hold:

$$\begin{gathered}\beta_v^K(x) > \beta_{v'}^K(x) \;\Rightarrow\; \beta_v^K(y) > \beta_{v'}^K(y);\\ \beta_v^K(x) = 0 \;\Leftrightarrow\; \beta_v^K(y) < \delta.\end{gathered}$$

Note that the first implication above holds even if $\beta_{v'}^K(x) = 0$, hence $c_K(x) \leqslant c_K(y)$. The first implication is equivalent to the following:

$$\beta_v^K(y) \leqslant \beta_{v'}^K(y) \;\Rightarrow\; \beta_v^K(x) \leqslant \beta_{v'}^K(x).$$

Hence, we can write

$$\begin{gathered}c_K(x) = \langle v_0, \dots, v_k\rangle,\ c_K(y) = \langle v_0, \dots, v_n\rangle,\ k \leqslant n,\\ \beta_{v_0}^K(x) \geqslant \cdots \geqslant \beta_{v_k}^K(x) > 0 \ \text{ and } \ \beta_{v_0}^K(y) \geqslant \cdots \geqslant \beta_{v_n}^K(y) > 0.\end{gathered}$$

For each $i = 0, \dots, n$, let $\sigma_i = \langle v_0, \dots, v_i\rangle$. Then, $\sigma_0 < \sigma_1 < \cdots < \sigma_n$, $\sigma_k = c_K(x)$, and $\sigma_n = c_K(y)$. Observe that

$$\begin{aligned}x &= \beta_{v_k}^K(x)v_k + \beta_{v_{k-1}}^K(x)v_{k-1} + \cdots + \beta_{v_1}^K(x)v_1 + \beta_{v_0}^K(x)v_0\\ &= (k+1)\beta_{v_k}^K(x)\hat{\sigma}_k + k\big(\beta_{v_{k-1}}^K(x) - \beta_{v_k}^K(x)\big)\hat{\sigma}_{k-1} +\\ &\qquad \cdots + 2\big(\beta_{v_1}^K(x) - \beta_{v_2}^K(x)\big)\hat{\sigma}_1 + \big(\beta_{v_0}^K(x) - \beta_{v_1}^K(x)\big)\hat{\sigma}_0.\end{aligned}$$

Thus, we have $x \in \langle \hat{\sigma}_0, \dots, \hat{\sigma}_k\rangle \in \operatorname{Sd} K$, $\beta_{\hat{\sigma}_k}^{\operatorname{Sd} K}(x) = (k+1)\beta_{v_k}^K(x)$ and

$$\beta_{\hat{\sigma}_i}^{\operatorname{Sd} K}(x) = (i+1)\big(\beta_{v_i}^K(x) - \beta_{v_{i+1}}^K(x)\big) \ \text{ for } i = 0, \dots, k-1.$$

Similarly, $y \in \langle \hat{\sigma}_0, \dots, \hat{\sigma}_n\rangle \in \operatorname{Sd} K$, $\beta_{\hat{\sigma}_n}^{\operatorname{Sd} K}(y) = (n+1)\beta_{v_n}^K(y)$ and

$$\beta_{\hat{\sigma}_i}^{\operatorname{Sd} K}(y) = (i+1)\big(\beta_{v_i}^K(y) - \beta_{v_{i+1}}^K(y)\big) \ \text{ for } i = 0, \dots, n-1.$$

Then, it follows that

$$\begin{aligned}\rho_{\mathrm{Sd}\,K}(x,y) &= \sum_{i=0}^{k}\left|\beta^{\mathrm{Sd}\,K}_{\hat{\sigma}_i}(x)-\beta^{\mathrm{Sd}\,K}_{\hat{\sigma}_i}(y)\right| + \sum_{i=k+1}^{n}\beta^{\mathrm{Sd}\,K}_{\hat{\sigma}_i}(y)\\ &\leqslant \sum_{i=0}^{k}(2i+1)\left|\beta^{K}_{v_i}(x)-\beta^{K}_{v_i}(y)\right| + (k+1)\beta^{K}_{v_{k+1}}(y)\\ &\qquad + (k+2)\beta^{K}_{v_{k+1}}(y) + \sum_{i=k+2}^{n}\beta^{K}_{v_i}(y)\\ &\leqslant (2k+3)\sum_{i=0}^{n}\left|\beta^{K}_{v_i}(x)-\beta^{K}_{v_i}(y)\right|\\ &= (2k+3)\rho_K(x,y) < (2k+3)\delta < \varepsilon.\end{aligned}$$

Thus, $\mathrm{id} : |K|_{\mathrm{m}} \to |\mathrm{Sd}\,K|_{\mathrm{m}}$ is continuous. □

By Proposition 4.6.8, Theorem 4.7.1 can be generalized as follows:

Corollary 4.7.2 *For each simplicial complex K with a subcomplex L, $|\mathrm{Sd}_L\,K|_{\mathrm{m}} = |K|_{\mathrm{m}}$ as spaces.* □

Now, using open stars, we can characterize admissible subdivisions as follows:

Lemma 4.7.3 *A simplicial subdivision K' of a simplicial complex K is admissible if and only if the open star $O_{K'}(v)$ of each vertex $v \in K'^{(0)}$ is open in $|K|_{\mathrm{m}}$.*

Proof The "only if" part follows from the fact that the open star $O_{K'}(v)$ is open in $|K'|_{\mathrm{m}}$. To see the "if" part, it suffices to show that $\mathrm{id} : |K|_{\mathrm{m}} \to |K'|_{\mathrm{m}}$ is continuous at each $x \in |K|$ (Proposition 4.6.8). Let U be a neighborhood of x in $|K'|_{\mathrm{m}}$. We can find $0 < r < 1$ such that $O_{K'}(x,r) \subset U$ by Proposition 4.6.1. On the other hand, $O_{K'}(x) = \bigcap_{v \in c_{K'}(x)^{(0)}} O_{K'}(v)$ is open in $|K|_{\mathrm{m}}$ because so is each $O_{K'}(v)$. Again by Proposition 4.6.1, we can find $0 < s < 1$ such that $O_K(x,s) \subset O_{K'}(x)$. Then, it follows that

$$\begin{aligned}O_K(x,rs) &= (1-rs)x + rs\,O_K(x) = (1-r)x + r\big((1-s)x + s\,O_K(x)\big)\\ &= (1-r)x + r\,O_K(x,s) \subset (1-r)x + r\,O_{K'}(x) = O_{K'}(x,r) \subset U,\end{aligned}$$

which means that U is a neighborhood of x in $|K|_{\mathrm{m}}$. □

Let K be a simplicial complex. Recall that $N(\{x\}, K)$, $C(\{x\}, K)$, and $B(\{x\}, K)$ are just written $N(x, K)$, $C(x, K)$, and $B(x, K)$, respectively.[13] For each $x \in |K|$,

$$\begin{gathered}N(x,K) = \mathrm{St}(c_K(x), K),\ C(x,K) = K \setminus K[x],\\ B(x,K) = N(x,K) \cap C(x,K) = N(x,K) \setminus K[x] \text{ and}\\ O_K(x) = |N(x,K)| \setminus |B(x,K)| \subset |N(x,K)| = \mathrm{st}(x,K).\end{gathered}$$

[13] For $A \subset |K|$, $N(A,K)$, $C(A,K)$, and $B(A,K)$ were defined in Sect. 4.2.

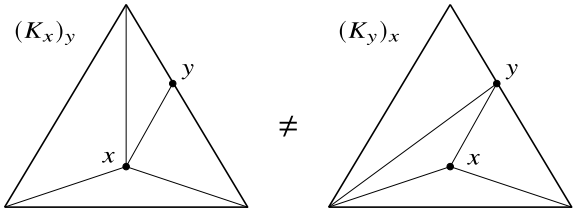


Fig. 4.7 Example of $(K_x)_y \neq (K_y)_x$

For $v \in K^{(0)}$, $N(v, K) = \mathrm{St}(v, K)$ and $B(v, K) = \mathrm{Lk}(v, K)$.[14] For each $\sigma \in K$,

$$N(\sigma, K) = \bigcup_{v \in \sigma^{(0)}} \mathrm{St}(v, K), \ |N(\sigma, K)| = \mathrm{st}(\sigma, K), \ N(\hat{\sigma}, K) = \mathrm{St}(\sigma, K), \text{ and}$$
$$|N(\hat{\sigma}, K)| = |\mathrm{St}(\sigma, K)| = \mathrm{st}(\hat{\sigma}, K) = \mathrm{st}(\mathrm{rint}\,\sigma, K).$$

Moreover, note that each $x \in |K|$ is joinable to each simplex $\sigma \in B(x, K)$, where the join $x\sigma$ is contained in $|N(x, K)|$ $(= \mathrm{st}(x, K) = |\mathrm{St}(c_K(x), K)|)$.

Now, take $A \subset |K| \setminus K^{(0)}$ so that $O_K(x) \cap O_K(x') = \emptyset$ if $x \neq x' \in A$, that is, K has no simplex containing more than one point of A.[15]

By the definition of the open star and condition (C_2''), we have the following equivalence:

$$O_K(x) \cap O_K(x') \neq \emptyset \ \Leftrightarrow \ \exists \sigma \in K[x] \cap K[x'] \ \text{(i.e., } x, x' \in \sigma).$$

Then, the simplicial subdivision K_A of K can be defined as follows:

$$K_A = C(A, K) \cup A \cup \left\{ x\sigma \;\middle|\; x \in A, \ \sigma \in B(x, K) \right\}.$$

Observe that $K_A^{(0)} = A \cup K^{(0)}$, $C(A, K_A) = C(A, K)$, and $O_{K_A}(x) = O_K(x)$ for each $x \in A$. When $A = \{x\}$, we simply write $K_{\{x\}} = K_x$. It should be noted that $K_x = K$ if $x \in K^{(0)}$. The operation $K \to K_x$ (or K_x itself) is called the **starring** of K at x. A subdivision obtained by finite starrings is known as a **stellar subdivision**. In general, $(K_x)_y \neq (K_y)_x$ for distinct two points $x, y \in |K|$ (see Fig. 4.7).

Lemma 4.7.4 *The starring K_w of a simplicial complex K at $w \in |K| \setminus K^{(0)}$ is an admissible subdivision of K.*

Proof Due to Proposition 4.6.8, all we have to show is that $\mathrm{id} : |K|_{\mathrm{m}} \to |K_w|_{\mathrm{m}}$ is continuous. We need to prove that $\beta_v^{K_w} : |K|_{\mathrm{m}} \to \mathbf{I}$ is continuous for every $v \in K_w^{(0)}$. By using the barycentric coordinates with respect to K_w, each point $x \in |K|$ can be written as follows:

[14] For $x \in |K| \setminus K^{(0)}$, $B(x, K) = \mathrm{St}(c_K(x), K) \cap C(x, K) \supsetneqq \mathrm{Lk}(c_K(x), K)$.

[15] Note that this implies that A is discrete in $|K|$.

$$x = \beta_w^{K_w}(x)w + \sum_{v \in K^{(0)}} \beta_v^{K_w}(x)v.$$

Since $\beta_v^K(v) = 1$ and $\beta_v^K(u) = 0$ for each $u \in K^{(0)} \setminus \{v\}$, it follows that

$$\beta_v^K(x) = \beta_w^{K_w}(x)\beta_v^K(w) + \beta_v^{K_w}(x)$$
$$\big(\text{i.e., } \beta_v^{K_w}(x) = \beta_v^K(x) - \beta_w^{K_w}(x)\beta_v^K(w)\big).$$

Then, it is enough to verify the continuity of $\beta_w^{K_w} : |K|_{\mathrm{m}} \to \mathbf{I}$.

We shall show that

$$\beta_w^{K_w} = \min_{v \in c_K(w)^{(0)}} \beta_v^K(w)^{-1}\beta_v^K : |K|_{\mathrm{m}} \to \mathbf{I},$$

which implies the continuity of $\beta_w^{K_w}$. For each $x \in |K|$, if $c_K(w) \not\leqslant c_K(x)$, then $\beta_w^{K_w}(x) = 0$ and $\min_{v \in c_K(w)^{(0)}} \beta_v^K(w)^{-1}\beta_v^K(x) = 0$ because $\beta_v^K(x) = 0$ if $v \notin c_K(x)^{(0)}$. When $c_K(w) \leqslant c_K(x)$, take $v_0 \in c_K(w)^{(0)}$ so that

$$\beta_{v_0}^K(w)^{-1}\beta_{v_0}^K(x) = \min_{v \in c_K(w)^{(0)}} \beta_v^K(w)^{-1}\beta_v^K(x) \in (0, 1].$$

Let σ be the opposite face of $c_K(x)$ to v_0. Observe that

$$\frac{\beta_{v_0}^K(x)}{\beta_{v_0}^K(w)} + \sum_{v \in \sigma^{(0)}} \left(\beta_v^K(x) - \frac{\beta_{v_0}^K(x)}{\beta_{v_0}^K(w)}\beta_v^K(w)\right)$$
$$= \frac{\beta_{v_0}^K(x)}{\beta_{v_0}^K(w)} + \big(1 - \beta_{v_0}^K(x)\big) - \frac{\beta_{v_0}^K(x)}{\beta_{v_0}^K(w)}\big(1 - \beta_{v_0}^K(w)\big) = 1.$$

Then, we have

$$\frac{\beta_{v_0}^K(x)}{\beta_{v_0}^K(w)}w + \sum_{v \in \sigma^{(0)}} \left(\beta_v^K(x) - \frac{\beta_{v_0}^K(x)}{\beta_{v_0}^K(w)}\beta_v^K(w)\right)v \in w\sigma \in K_w.$$

This point is equal to x. Indeed,

$$\frac{\beta_{v_0}^K(x)}{\beta_{v_0}^K(w)}w + \sum_{v \in K^{(0)} \setminus \{v_0\}} \left(\beta_v^K(x) - \frac{\beta_{v_0}^K(x)}{\beta_{v_0}^K(w)}\beta_v^K(w)\right)v$$
$$= \sum_{v \in K^{(0)}} \frac{\beta_{v_0}^K(x)}{\beta_{v_0}^K(w)}\beta_v^K(w)v + \sum_{v \in K^{(0)} \setminus \{v_0\}} \left(\beta_v^K(x) - \frac{\beta_{v_0}^K(x)}{\beta_{v_0}^K(w)}\beta_v^K(w)\right)v$$
$$= \frac{\beta_{v_0}^K(x)}{\beta_{v_0}^K(w)}\beta_{v_0}^K(w)v_0 + \sum_{v \in \sigma^{(0)}} \beta_v^K(x)v = \sum_{v \in K^{(0)}} \beta_v^K(x)v = x.$$

Therefore, $\beta_w^{K_w}(x) = \beta_{v_0}^K(x)/\beta_{v_0}^K(w)$. □

For a subcomplex L of K, we define maps $\delta_L^K, \beta_L^K : |K| \to \mathbb{R}$ as follows:

$$\delta_L^K(x) = \rho_K(x, |L|) \text{ and } \beta_L^K(x) = \sum_{v \in L^{(0)}} \beta_v^K(x) \text{ for each } x \in |K|.$$

Then, $\mathrm{N}_{\rho_K}(|L|, t) = (\delta_L^K)^{-1}([0, t))$ and $\overline{\mathrm{N}}_{\rho_K}(|L|, t) = (\delta_L^K)^{-1}([0, t])$ for each $t > 0$. For convenience, let $\delta_\emptyset^K, \beta_\emptyset^K : |K| \to \mathbb{R}$ be the constant maps with $\delta_\emptyset^K(|K|) = \{2\}$ and $\beta_\emptyset^K(|K|) = \{0\}$. For each $\sigma \in K$, $\delta_{F(\sigma)}^K$ and $\beta_{F(\sigma)}^K$ are just written as δ_σ^K and β_σ^K, respectively. When $\sigma_L = \sigma \cap |L| \in L$, we have $\beta_L^K|\sigma = \beta_{\sigma_L}^K|\sigma$ by the definition.

Lemma 4.7.5 *For a full subcomplex L of K, $\delta_L^K = 2(1 - \beta_L^K) = 2\beta_{C(L,K)}^K$, which is affine on each simplex of K.*

Proof It is easy to see that $\beta_{C(L,K)}^K(x) = 1 - \beta_L^K(x)$ for each $x \in |K|$.

Indeed, $\beta_L^K(x) + \beta_{C(L,K)}^K(x) = \sum_{v \in L^{(0)}} \beta_v^K(x) + \sum_{v \in K^{(0)} \setminus L^{(0)}} \beta_v^K(x) = \sum_{v \in K^{(0)}} \beta_v^K(x) = 1.$

When $x \in |C(L, K)|$, since $c_K(x) \cap |L| = \emptyset$,

$$\rho_K(x, y) = \sum_{v \in K^{(0)} \setminus L^{(0)}} \beta_v^K(x) + \sum_{v \in L^{(0)}} \beta_v^K(y) = 2 \text{ for every } y \in |L|.$$

Hence, $\delta_L^K(x) = \rho_K(x, |L|) = 2 = 2(1 - \beta_L^K(x))$ for each $x \in |C(L, K)|$.

When $x \in |K| \setminus |C(L, K)|$, for every $y \in |L|$,

$$\begin{aligned}\rho_K(x, y) &\geqslant \sum_{v \in L^{(0)}} \left(\beta_v^K(y) - \beta_v^K(x)\right) + \sum_{v \in K^{(0)} \setminus L^{(0)}} \beta_v^K(x) \\ &= \sum_{v \in L^{(0)}} \beta_v^K(y) - \sum_{v \in L^{(0)}} \beta_v^K(x) + 1 - \sum_{v \in L^{(0)}} \beta_v^K(x) \\ &= 2\big(1 - \beta_L^K(x)\big).\end{aligned}$$

Since L is a full subcomplex of K, $c_K(x)_L = c_K(x) \cap |L| \leqslant c_K(x)$ by Proposition 4.2.10. Hence, we can define a map $r_L : |K| \setminus |C(K, L)| \to |L|$ as follows:

$$r_L(x) = \sum_{v \in L^{(0)}} \frac{\beta_v^K(x)}{\beta_L^K(x)} v = \sum_{v \in c_K(x)_L^{(0)}} \frac{\beta_v^K(x)}{\beta_{c_K(x)_L}^K(x)} v \in c_K(x)_L.$$

Then, it follows that

$$\begin{aligned}\rho_K(x, r_L(x)) &= \sum_{v \in K^{(0)} \setminus L^{(0)}} \beta_v^K(x) + \sum_{v \in L^{(0)}} \left(\frac{\beta_v^K(x)}{\beta_L^K(x)} - \beta_v^K(x)\right) \\ &= \beta_{C(L,K)}^K(x) + 1 - \beta_L^K(x) = 2(1 - \beta_L^K(x)).\end{aligned}$$

Thus, $\delta_L^K(x) = \rho_K(x, |L|) = 2(1 - \beta_L^K(x))$ for each $x \in |K| \setminus |C(L, K)|$. □

For a space X with $A \subset X$, a map $r : X \to A$ is called a **retraction** if $r|A = \mathrm{id}$, where A is called a **retract** of X.[16] As is easily shown, every retract of X is closed in X. From the above proof, we have the following:

Proposition 4.7.6 *For a simplicial complex K and a full subcomplex L of K, there exists a retraction $r_L : |K| \setminus |C(L, K)| \to |L|$ such that $r_L(x) \in c_K(x)$ and $\rho_K(x, r_L(x)) = \rho_K(x, |L|)$ for each $x \in |K| \setminus |C(L, K)|$.* □

By using β_L^K, a full subcomplex L of a simplicial complex K can be characterized as follows:

Lemma 4.7.7 *Let K be a simplicial complex. A subcomplex L of K is full in K (i.e., L is a full subcomplex of K) if and only if $|L| = (\beta_L^K)^{-1}(1)$ $(= (\beta_{C(L,K)}^K)^{-1}(0))$.*

Proof (*The "only if" Part*) Evidently, $|L| \subset (\beta_L^K)^{-1}(1)$. To see $(\beta_L^K)^{-1}(1) \subset |L|$, let $x \in (\beta_L^K)^{-1}(1) = (\beta_{C(L,K)}^K)^{-1}(0)$. Then, it follows that $\beta_v^K(x) = 0$ for every $v \in K^{(0)} \setminus L^{(0)}$, which means that $c_K(x)^{(0)} \subset L^{(0)}$. Since L is full in K, we have $c_K(x) \in L$, so $x \in |L|$.

(*The "if" Part*) Let $\sigma \in K$ with $\sigma^{(0)} \subset L^{(0)}$. Since $\beta_L^K(\sigma^{(0)}) = \{1\}$ and $\beta_L^K|\sigma$ is affine, we have $\beta_L^K(\sigma) = \{1\}$. Hence, $\sigma \subset (\beta_L^K)^{-1}(1) = |L|$, which means $\sigma \in L$. Thus, L is full in K. □

Lemma 4.7.8 *Let K' and K'' be simplicial subdivisions of a simplicial complex K such that $K'^{(0)}$ and $K''^{(0)}$ are discrete in $|K|_{\mathrm{m}}$. Then, K' and K'' have a common simplicial subdivision K''' such that $K'''^{(0)}$ is discrete in $|K|_{\mathrm{m}}$.*

Proof Here, we use the following admissible metric on $|K|_{\mathrm{m}}$:

$$d(x, y) = \sqrt{\sum_{v \in K^{(0)}} \left(\beta_v^K(x) - \beta_v^K(y)\right)^2} = \|\beta^K(x) - \beta^K(y)\|_2,$$

where $\|\cdot\|_2$ is the norm for Hilbert space $\ell_2(K^{(0)})$ (cf. Remark 4.6.3). Then, $\mathrm{diam}_d\, \sigma = \mathrm{diam}_d\, \sigma^{(0)} = \sqrt{2}$ for each $\sigma \in K \setminus K^{(0)}$. For each $\sigma \in K$ and $x \in |K|$, there exists the nearest point $p_\sigma(x) \in \sigma$ to x, i.e., $d(x, p_\sigma(x)) = d(x, \sigma)$. Then, $p_\sigma : |K|_{\mathrm{m}} \to \sigma$ is a map which is affine on each simplex of K and $p_\sigma|\sigma = \mathrm{id}$.

To see this fact, it suffices to show that $p_\sigma(x)$ can be represented as follows:

$$p_\sigma(x) = \sum_{v \in \sigma^{(0)}} \left(\beta_v^K(x) + \alpha_\sigma(x)\right)v,\ \alpha_\sigma(x) = \frac{1 - \sum_{v \in \sigma^{(0)}} \beta_v^K(x)}{\dim \sigma + 1}.$$

For $x \in \sigma$, this is obvious, where $\alpha_\sigma(x) = 0$. When $x \notin \sigma$ (i.e., $\alpha_\sigma(x) \neq 0$), let $y = \sum_{v \in \sigma^{(0)}} \left(\beta_v^K(x) + \alpha_\sigma(x)\right)v \in \sigma$, (i.e., $\beta_v^K(y) = \beta_v^K(x) + \alpha_\sigma(x), v \in \sigma^{(0)}$). In $\ell_2(K^{(0)})$, the line segment $\beta^K(xy) = \beta^K(x)\beta^K(y)$ is perpendicular to $\beta^K(\sigma)$. Indeed, for each $u \in \sigma^{(0)}$,

[16] Theory of retracts is the subject of Chap. 6.

$$\begin{aligned}
&\langle \beta^K(y) - \beta^K(x), \beta^K(y) - \beta^K(u) \rangle \\
&\quad = \sum_{v \in \sigma^{(0)}} \big(\beta_v^K(y) - \beta_v^K(x)\big)\big(\beta_v^K(y) - \beta_v^K(u)\big) \\
&\quad = \alpha_\sigma(x)(\beta_u^K(x) + \alpha_\sigma(x) - 1) + \sum_{v \in \sigma^{(0)} \setminus \{u\}} \alpha_\sigma(x)(\beta_v^K(x) + \alpha_\sigma(x)) \\
&\quad = \alpha_\sigma(x)\Big(\sum_{v \in \sigma^{(0)}} \beta_v^K(x) + (1 + \dim \sigma)\alpha_\sigma(x) - 1 \Big) = 0.
\end{aligned}$$

Then, $\beta^K(y)$ is the nearest point of $\beta^K(\sigma)$ to $\beta^K(x)$. Since β^K is an isometry, y is the nearest point of σ to x, which means $p_\sigma(x) = y$.

By Proposition 4.3.8, we have the following cell complex L, which is a common subdivision of K' and K'':

$$L = K' \wedge K'' = \big\{ \sigma' \cap \sigma'' \;\big|\; \sigma' \in K',\ \sigma'' \in K'' \text{ such that } \sigma' \cap \sigma'' \neq \emptyset \big\}.$$

Due to Theorem 4.3.6, L has a simplicial subdivision K''' such that $K'''^{(0)} = L^{(0)}$. Then, it suffices to show that $L^{(0)}$ is discrete in $|K|_{\mathrm{m}}$.

Let $x_0 \in |K|_{\mathrm{m}}$. Since $c_K(x_0)$ is compact and $K'^{(0)} \cup K''^{(0)}$ is discrete in $|K|_{\mathrm{m}}$, we have

$$\delta_0 = \mathrm{dist}_{\rho_K}\big(c_K(x_0), (K'^{(0)} \cup K''^{(0)}) \setminus c_K(x_0)\big) > 0.$$

Then, it follows that

$$\mathrm{N}_{\rho_K}(c_K(x_0), \delta_0) \cap \big(|C(c_K(x_0), K')| \cup |C(c_K(x_0), K'')|\big) = \emptyset.$$

Indeed, let $\tau = \langle v_1, \ldots, v_{n+1} \rangle \in C(c_K(x_0), K') \cup C(c_K(x_0), K'')$, $n = \dim \tau$, and $x = \sum_{i=1}^{n+1} z(i) v_i \in \tau$, $z \in \Delta^n$. Then, by Lemma 4.7.5, we have

$$\begin{aligned}
\rho_K(x, c_K(x_0)) &= 2\beta^K_{C(c_K(x_0),K)}(x) = 2\sum_{i=1}^{n+1} z(i)\beta^K_{C(c_K(x_0),K)}(v_i) \\
&= \sum_{i=1}^{n+1} z(i)\rho_K(v_i, c_K(x_0)) \geqslant \sum_{i=1}^{n+1} z(i)\delta_0 = \delta_0.
\end{aligned}$$

Since d is admissible for $|K|_{\mathrm{m}}$ and $c_K(x_0)$ contains only finitely many simplexes of K' and K'', we can find $\delta > 0$ so that

(i) $\mathrm{B}_d(x_0, \delta) \subset O_K(x_0)$,
(ii) $\mathrm{N}_d(c_K(x_0), \delta) \subset \mathrm{N}_{\rho_K}(c_K(x_0), \delta_0)$,
(iii) $\delta < d(u, v)$ for any distinct $u, v \in (L^{(0)} \cap c_K(x_0)) \cup \{x_0\}$, and
(iv) $\delta < \mathrm{dist}_d(\sigma', \sigma'')$ for any disjoint $\sigma' \in K'$, $\sigma'' \in K''$ with $\sigma', \sigma'' \subset c_K(x_0)$.

Then, (ii) implies that

(ii′) $\mathrm{N}_d(c_K(x_0), \delta) \cap \big(|C(c_K(x_0), K')| \cup |C(c_K(x_0), K'')|\big) = \emptyset$.

Combining this (ii′) with (iii), we have

(iii′) $\mathrm{B}_d(x_0, \delta) \cap \big((K'^{(0)} \cup K''^{(0)}) \setminus \{x_0\}\big) = \emptyset$,

Now, we will show that

$$d(x_0, v) \geqslant \delta^2/2\sqrt{2} \text{ for each } v \in (L^{(0)} \cap \mathrm{B}_d(x_0, \delta)) \setminus \{x_0\}.$$

Since $v \notin c_K(x_0) \cup K'^{(0)} \cup K''^{(0)}$ by (iii) and (iii′), there exist $\sigma' \in K' \setminus K'^{(0)}$ and $\sigma'' \in K'' \setminus K''^{(0)}$ such that

$$\{v\} = \sigma' \cap \sigma'' = \operatorname{rint}\sigma' \cap \operatorname{rint}\sigma''.$$

Then, $\sigma'_0 = \sigma' \cap c_K(x_0) \neq \emptyset$ and $\sigma''_0 = \sigma'' \cap c_K(x_0) \neq \emptyset$ by (ii′). Moreover, due to (i) and the definition of $O_K(x_0)$ (cf. Sect. 4.2),

$$v \in \mathrm{B}_d(x_0, \delta) \subset O_K(x_0) = \bigcup_{c_K(x_0) \leqslant \sigma \in K} \operatorname{rint}\sigma,$$

from which it follows that $c_K(x_0) \leqslant c_K(v)$. By Proposition 4.1.9 (4),

$$\sigma'_0 = \sigma' \cap c_K(x_0) \leqslant \sigma' \cap c_K(v) = \sigma'.$$

Since $v \in \sigma' \setminus \sigma'_0$, it follows that $\sigma'_0 < \sigma'$. Similarly, we have $\sigma''_0 < \sigma''$. Let σ'_1 and σ''_1 be the faces of σ' and σ'' that are opposite to σ'_0 and σ''_0, respectively. In other words, $\sigma' = \sigma'_0\sigma'_1$ and $\sigma'' = \sigma''_0\sigma''_1$. See Fig. 4.8. Then, we can write

$$\begin{gathered} v = (1-t')y' + t'z' = (1-t'')y'' + t''z'', \\ y' \in \sigma'_0,\ z' \in \sigma'_1,\ y'' \in \sigma''_0,\ z'' \in \sigma''_1,\ t', t'' \in (0,1). \end{gathered}$$

Let $u = p_{c_K(x_0)}(v) \in c_K(x_0)$. Since $\sigma' \cap \sigma'' = \{v\}$ and $v \notin c_K(x_0)$, it follows that $\sigma'_0 \cap \sigma''_0 = \emptyset$. Hence, by (iv),

$$d(y', u) + d(y'', u) \geqslant d(y', y'') \geqslant \operatorname{dist}_d(\sigma'_0, \sigma''_0) > \delta.$$

Then, $d(y', u) \geqslant \delta/2$ or $d(y'', u) \geqslant \delta/2$. We may assume that $d(y', u) \geqslant \delta/2$.

Now, let $x' = p_{c_K(x_0)}(z') \in c_K(x_0)$, where $z' \in \sigma'_1 \in C(c_K(x_0), K')$. Then, the line segment $\langle x', z' \rangle$ is upright on $c_K(x_0)$. Hence, by (i′),

$$d(z', x') = d(z', c_K(x_0)) \geqslant \operatorname{dist}_d(\sigma'_1, c_K(x_0)) > \delta.$$

Since $c_K(x_0)$ is isometric to the standard simplex of Euclidean space, we have $d(x', y') \leqslant \operatorname{diam}_d c_K(x_0) = \sqrt{2}$. Moreover, the right triangle $x'y'z'$ is similar to the right triangle $uy'v$. Then, it follows that

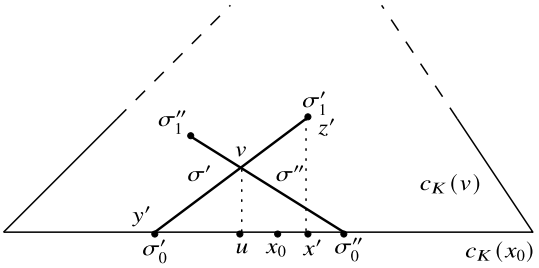


Fig. 4.8 Estimation of $d(x_0, v)$

$$d(x_0, v) \geqslant d(u, v) = \frac{d(x', z')}{d(x', y')} \cdot d(u, y') \geqslant \delta^2/2\sqrt{2}.$$

This completes the proof. □

Now, we can characterize admissible subdivisions as follows:

Theorem 4.7.9 *A simplicial subdivision K' of a simplicial complex K is admissible if and only if $K'^{(0)}$ is discrete in $|K|_{\mathrm{m}}$.*

Proof Since $K'^{(0)}$ is discrete in $|K'|_{\mathrm{m}}$, it suffices to show the "if" part. By virtue of Proposition 4.6.8, we need only show the continuity of $\mathrm{id} : |K|_{\mathrm{m}} \to |K'|_{\mathrm{m}}$ at each $w \in |K|$. Due to Proposition 4.7.4, the starring K_w at w is an admissible subdivision, i.e., $|K|_{\mathrm{m}} = |K_w|_{\mathrm{m}}$, where $w \in K_w^{(0)}$. By Lemma 4.7.8, there is a common simplicial subdivision K'' of K_w and K' such that $K''^{(0)}$ is discrete in $|K|_{\mathrm{m}}$. Then, $\mathrm{id} : |K''|_{\mathrm{m}} \to |K'|_{\mathrm{m}}$ is continuous. Thus, it suffices to show the continuity of $\mathrm{id} : |K_w|_{\mathrm{m}} \to |K''|_{\mathrm{m}}$ at w. Replacing K and K' with K_w and K'', respectively, we may show the continuity of $\mathrm{id} : |K|_{\mathrm{m}} \to |K'|_{\mathrm{m}}$ at each $w \in K^{(0)}$.

For each $x \in |K|$, observe that

$$\begin{aligned}\rho_K(x, w) &= \sum_{v \in K^{(0)}} \left|\beta_v^K(x) - \beta_v^K(w)\right| \\ &= 1 - \beta_w^K(x) + \sum_{v \in K^{(0)} \setminus \{w\}} \beta_v^K(x) = 2\big(1 - \beta_w^K(x)\big).\end{aligned}$$

By the same reasoning, we have $\rho_{K'}(x, w) = 2\big(1 - \beta_w^{K'}(x)\big)$.

Let $\delta = \mathrm{dist}_{\rho_K}(w, K'^{(0)} \setminus \{w\}) > 0$. For every $v \in K'^{(0)} \setminus \{w\}$, $\beta_w^K(v) \leqslant 1 - \delta/2$ because $2\big(1 - \beta_w^K(v)\big) = \rho_K(v, w) \geqslant \delta$. For each $x \in |K|$,

$$\begin{aligned}\beta_w^K(x) &= \sum_{v \in K'^{(0)}} \beta_v^{K'}(x)\beta_w^K(v) \leqslant \beta_w^{K'}(x) + \sum_{v \in K'^{(0)} \setminus \{w\}} \beta_v^{K'}(x)(1 - \delta/2) \\ &= \beta_w^{K'}(x) + \big(1 - \beta_w^{K'}(x)\big)(1 - \delta/2) = 1 - \delta/2 + \beta_w^{K'}(x)\delta/2.\end{aligned}$$

Then, it follows that

$$\rho_{K'}(x, w) = 2\big(1 - \beta_w^{K'}(x)\big) \leqslant 4\big(1 - \beta_w^{K}(x)\big)/\delta = 2\rho_K(x, w)/\delta.$$

Therefore, for each $\varepsilon > 0$, $\rho_K(x, w) < \delta\varepsilon/2$ implies $\rho_{K'}(x, w) < \varepsilon$. □

Combining Theorem 4.7.9 with Lemma 4.7.8, we have the following:

Corollary 4.7.10 *Every two admissible subdivisions of a simplicial complex K have an admissible common subdivision.* □

A metric space $P = (P, d)$ is called a **metric polyhedron** if P has a triangulation K such that ρ_K is admissible for P, that is, $P = |K|_{\mathrm{m}}$ as spaces, where K is called an **admissible triangulation** of P. For example, $(|K|, \rho_K^{\infty})$ is a metric polyhedron (cf. Proposition 4.6.11 (1)). For simplicial complexes K and L, $(|K| \times |L|, d)$ is a metric polyhedron if d is an admissible metric for the product space $|K|_{\mathrm{m}} \times |L|_{\mathrm{m}}$ such as the following metrics:

$$\rho_K(x, x') + \rho_L(y, y'), \ \max\big\{\rho_K(x, x'), \ \rho_L(y, y')\big\}.$$

The following is unknown:

Common Admissible Subdivision Problem
Do any two admissible triangulations of a metric polyhedron P have a common admissible subdivision? In other words, for two simplicial complexes K_1 and K_2, if $|K_1|_{\mathrm{m}} = |K_2|_{\mathrm{m}}$ as spaces (i.e., ρ_{K_1} and ρ_{K_2} give the same topology), do they have a common admissible subdivision?

Corollary 4.7.10 is the affirmative answer to the above problem in the case when K_1 and K_2 are admissible subdivisions of some simplicial complex K.

4.8 Small Subdivisions and Paracompactness of Polyhedra

In this section, we give a proof of the J.H.C. Whitehead Theorem that every simplicial complex K has an arbitrarily small simplicial subdivision, that is, for any $\mathcal{U} \in \mathrm{cov}(|K|)$, there is $K' \lhd K$ such that $\mathcal{S}_{K'} \prec \mathcal{U}$. Applying this theorem, we can prove that $|K|$ is paracompact.

Let K be a simplicial complex and L a subcomplex of K. We inductively define the ***n*-th barycentric subdivision** $\mathrm{Sd}^n K$ of K and the ***n*-th barycentric subdivision** $\mathrm{Sd}_L^n K$ of K **relative to** L as follows:

$$\mathrm{Sd}^n K = \mathrm{Sd}(\mathrm{Sd}^{n-1} K) \ \text{ and } \ \mathrm{Sd}_L^n K = \mathrm{Sd}_L(\mathrm{Sd}_L^{n-1} K),$$

where $\mathrm{Sd}^0 K = \mathrm{Sd}^0_L K = K$. In the following, we will show that if K is finite then the size of $\mathrm{Sd}^m K$ becomes smaller as m gets larger. The following is a special case of Proposition 3.2.2 (the direct proof is rather easy):

Lemma 4.8.1 *For a cell C in a normed linear space $E = (E, \|\cdot\|)$, the following hold:*

(i) $\|x - y\| \leqslant \sup_{v \in C^{(0)}} \|x - v\|$ *for each* $x \in E$ *and* $y \in C$;
(ii) $\operatorname{diam} C = \operatorname{diam} C^{(0)}$. □

Proposition 4.8.2 *For every simplicial complex K in a normed linear space E,* $\operatorname{mesh} K = \operatorname{mesh} K^{(1)}$.

Proof By Lemma 4.8.1(ii), we have $\operatorname{diam} \sigma = \max_{v,u \in \sigma^{(0)}} \operatorname{diam} \langle u, v \rangle$ for each $\sigma \in K$. Then, it follows that $\operatorname{mesh} K = \operatorname{mesh} K^{(1)}$. □

Lemma 4.8.3 *Let $f : \sigma \to E$ be an affine map of an n-simplex σ into a normed linear space E. Then,*

$$\operatorname{mesh} f(\mathrm{Sd}\, F(\sigma)) \leqslant \frac{n}{n+1} \cdot \operatorname{diam} f(\sigma).$$

Proof Note that $f(\sigma)$ is a cell in E and $f(\sigma)^{(0)} \subset f(\sigma^{(0)})$. Let $\sigma'' < \sigma' \leqslant \sigma$. It follows from Lemma 4.8.1(i), (ii) that

$$\begin{aligned}
\|f(\hat{\sigma}') - f(\hat{\sigma}'')\| &\leqslant \max_{v \in \sigma''^{(0)}} \|f(\hat{\sigma}') - f(v)\| \\
&\leqslant \max_{v \in \sigma''^{(0)}} \sum_{u \in \sigma'^{(0)}} \frac{1}{\dim \sigma' + 1} \|f(u) - f(v)\| \\
&= \max_{v \in \sigma''^{(0)}} \sum_{u \in \sigma'^{(0)} \setminus \{v\}} \frac{1}{\dim \sigma' + 1} \|f(u) - f(v)\| \\
&\leqslant \frac{\dim \sigma'}{\dim \sigma' + 1} \cdot \operatorname{diam} f(\sigma') \leqslant \frac{n}{n+1} \cdot \operatorname{diam} f(\sigma).
\end{aligned}$$

Then, we have the result by Proposition 4.8.2. □

The following is the special case of Lemma 4.8.3 when $f = \mathrm{id}_\sigma$:

Lemma 4.8.4 *For every n-simplex σ in a normed linear space E,*

$$\operatorname{mesh} \mathrm{Sd}\, F(\sigma) \leqslant \frac{n}{n+1} \cdot \operatorname{diam} \sigma.$$ □

For each simplicial complex K, as can be seen from the definition of ρ_K, the canonical representation $\beta^K : (|K|, \rho_K) \to \ell_1(K^{(0)})$ is an isometry. Hence, we can regard $|K| \subset \ell_1(K^{(0)})$, and ρ_K is induced from the norm of $\ell_1(K^{(0)})$. Moreover, for an n-simplex $\sigma \in K$, $\rho_K(v, \hat{\sigma}) = 2n/(n+1)$ for every $v \in \sigma^{(0)}$. Thus, the following can be obtained from Proposition 4.8.2 and Lemma 4.8.4:

Proposition 4.8.5 *Let K be a simplicial complex. For any simplicial subdivision $K' \lhd K$, $\mathrm{mesh}_{\rho_K} K' = \mathrm{mesh}_{\rho_K} K'^{(1)}$. If $\dim K < \infty$, then*

$$\mathrm{mesh}_{\rho_K} \mathrm{Sd}\, K' \leqslant \frac{\dim K}{\dim K + 1} \cdot \mathrm{mesh}_{\rho_K} K'.$$

In case $K' = K$ and $\dim K = m$, we have $\mathrm{mesh}_{\rho_K} \mathrm{Sd}\, K = 2m/(m+1)$. □

Using Proposition 4.8.5 inductively, we have the following:

Theorem 4.8.6 *For every finite-dimensional simplicial complex K,*

$$\mathrm{mesh}_{\rho_K} \mathrm{Sd}^n K \leqslant 2\left(\frac{\dim K}{\dim K + 1}\right)^n \quad \textit{for every } n \in \mathbb{N}.$$

Hence, $\lim_{n\to\infty} \mathrm{mesh}_{\rho_K} \mathrm{Sd}^n K = 0$. □

Corollary 4.8.7 *For a finite simplicial complex K, each open cover $\mathcal{U}$ of $|K|_{\mathrm{m}}$ $(= |K|)$ is refined by $\mathcal{S}_{\mathrm{Sd}^n K}$ for some $n \in \mathbb{N}$.*

Proof Since $|K|_{\mathrm{m}} = |K|$ is compact (Proposition 4.2.3 and Theorem 4.6.4) and $\dim K < \infty$, we can choose $n \in \mathbb{N}$ so large that $\mathrm{mesh}_{\rho_K} \mathrm{Sd}^n K$ is less than the Lebesgue number of $\mathcal{U}$. Then, n is the desired one. □

For infinite-dimensional simplicial complexes, we have the following:

Proposition 4.8.8 *If K is an infinite-dimensional simplicial complex, then*

$$\mathrm{mesh}_{\rho_K} \mathrm{Sd}^n K = 2 \ (= \mathrm{mesh}_{\rho_K} K) \quad \textit{for every } n \in \mathbb{N}.$$

Proof By induction on $n \in \mathbb{N}$, we shall show the following:

$(\star)_n$ for each $m \in \mathbb{N}$ and $\varepsilon > 0$, there exist a vertex $v \in K^{(0)}$ and an m-simplex $\sigma \in \mathrm{Lk}(v, \mathrm{Sd}^n K)$ such that $\beta_v^K(u) < \varepsilon$ for every $u \in \sigma^{(0)}$.

If $(\star)_n$ has been shown, then $\tau = v\sigma \in \mathrm{Sd}^n K$ and

$$\mathrm{diam}\, \tau \geqslant \max_{u \in \sigma^{(0)}} \rho_K(v, u) = \max_{u \in \sigma^{(0)}} 2(1 - \beta_v^K(u)) > 2(1 - \varepsilon).$$

Thus, it would follow that $\mathrm{mesh}_{\rho_K} \mathrm{Sd}^n K = 2$.

To see $(\star)_1$, for each $m \in \mathbb{N}$ and $\varepsilon > 0$, choose $k \in \mathbb{N}$ so that $(k+2)^{-1} < \varepsilon$. Since $\dim K = \infty$, K has simplexes $\sigma_1 < \cdots < \sigma_{m+1}$ with $\dim \sigma_i = k + i$. Let $v \in \sigma_1^{(0)}$. Then, we have an m-simplex $\sigma = \langle \hat{\sigma}_1, \ldots, \hat{\sigma}_{m+1} \rangle \in \mathrm{Lk}(v, \mathrm{Sd}\, K)$. Observe that $\beta_v^K(\hat{\sigma}_i) = (k + i + 1)^{-1} < \varepsilon$ for every $i = 1, \ldots, m+1$.

Now, we prove the implication $(\star)_n \Rightarrow (\star)_{n+1}$. For each $m \in \mathbb{N}$ and $\varepsilon > 0$, choose $k \in \mathbb{N}$ so that $(k+2)^{-1} < \varepsilon/2$. By $(\star)_n$, we have $v \in K^{(0)}$ and a $(k+m)$-simplex $\sigma = \langle v_1, \ldots, v_{k+m+1} \rangle \in \mathrm{Lk}(v, \mathrm{Sd}^n K)$ such that

$$\beta_v^K(v_i) < \varepsilon/2 \ \text{ for every } i = 1, \dots, k+m+1.$$

For each $i = 1, \dots, m+1$, let $\sigma_i = \langle v, v_1, \dots, v_{k+i} \rangle \in \mathrm{Sd}^n K$. Since $v < \sigma_1 < \cdots < \sigma_{m+1}$, we have an m-simplex $\sigma = \langle \hat{\sigma}_1, \dots, \hat{\sigma}_{m+1} \rangle \in \mathrm{Lk}(v, \mathrm{Sd}^{n+1} K)$. Then, it follows that

$$\begin{aligned} \beta_v^K(\hat{\sigma}_i) &= \frac{1}{k+i+1}\left(1 + \sum_{j=1}^{k+i} \beta_v^K(v_j)\right) \\ &< \frac{1}{k+i+1}\left(1 + \frac{(k+i)\varepsilon}{2}\right) < \frac{1}{k+2} + \frac{\varepsilon}{2} < \varepsilon. \end{aligned}$$

This completes the proof. □

Remark 4.8.1 For the metric ρ_K^∞, we have the following:

- If $\dim K < \infty$, then $\mathrm{mesh}_{\rho_K^\infty} \mathrm{Sd}^n K \leqslant \left(\dfrac{\dim K}{\dim K + 1}\right)^n$ for every $n \in \mathbb{N}$;
- If $\dim K = \infty$, then $\mathrm{mesh}_{\rho_K^\infty} \mathrm{Sd}^n K = 1$ $(= \mathrm{mesh}_{\rho_K^\infty} K)$ for every $n \in \mathbb{N}$.

To construct small subdivisions of infinite simplicial complexes, we need the following two lemmas:

Lemma 4.8.9 *Let K be a finite simplicial complex and L a subcomplex of K. Given an open neighborhood $U(v)$ of $|\mathrm{St}(v, L)|$ in $|K|_{\mathrm{m}}$ $(= |K|)$ for every $v \in L^{(0)}$, there exists a simplicial subdivision K' of K such that $L \subset K'$ and $|\mathrm{St}(v, K')| \subset U(v)$ for every $v \in L^{(0)}$.*

Proof Replacing K by $\mathrm{Sd}_L K$, we may assume that L is full in K. Since $|\mathrm{St}(v, L)|$ and $|K| \setminus U(v)$ are disjoint closed sets in the metric compactum $(|K|, \rho_K)$, we have

$$\delta = \min_{v \in L^{(0)}} \mathrm{dist}_{\rho_K}(|\mathrm{St}(v, L)|,\ |K| \setminus U(v)) > 0.$$

For each $\sigma \in K \setminus L$, choose $v(\sigma) \in \mathrm{rint}\, \sigma$ so that

$$\sigma \cap |L| \neq \emptyset \ \Rightarrow\ \rho_K(v(\sigma), \sigma \cap |L|) < \delta.$$

Using the points $v(\sigma)$, we define K' as a derived subdivision of K relative to L, which is as desired. See Fig. 4.9.

To see that $|\mathrm{St}(v, K')| \subset U(v)$ for each $v \in L^{(0)}$, let $x \in |\mathrm{St}(v, K')|$. Then, we have $\sigma_0 = \langle v_1, \dots, v_m \rangle \in \mathrm{St}(v, L)$ and $\sigma_1 < \cdots < \sigma_n \in K \setminus L$ such that

$$v_1 = v,\ \sigma_0 < \sigma_1 \ \text{ and } \ x \in \langle v_1, \dots, v_m, v(\sigma_1), \dots, v(\sigma_n) \rangle \subset \sigma_n,$$

where we can write

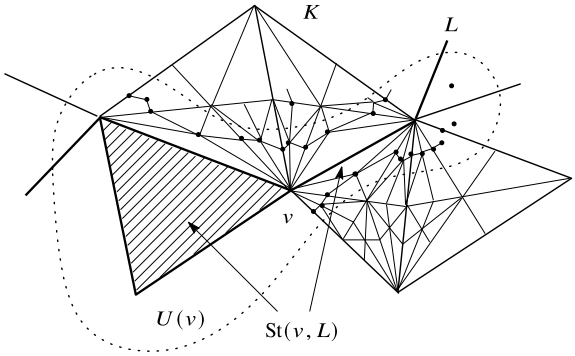


Fig. 4.9 Small subdivision

$$x = \sum_{i=1}^{m} z(i)v_i + \sum_{j=1}^{n} z(j+m)v(\sigma_j) \quad \text{for some} \quad z \in \Delta^{m+n-1}.$$

For each $j = 1, \ldots, n$, since $\sigma_j \cap |L| \neq \emptyset$, we can choose $u_j \in \sigma_j \cap |L|$ so that $\rho_K(v(\sigma_j), u_j) < \delta$. Since L is full in K, $\sigma_i \cap |L| = \langle \sigma_i^{(0)} \cap |L| \rangle \in L$ for each $i = 1, \ldots, n$ and $\sigma_1 \cap |L| \leqslant \cdots \leqslant \sigma_n \cap |L|$, from which we have

$$y = \sum_{i=1}^{m} z(i)v_i + \sum_{j=1}^{n} z(j+m)u_j \in \sigma_n \cap |L| \subset |\operatorname{St}(v, L)|.$$

Since $\beta^K|\sigma_n$ is affine, it follows that

$$\begin{aligned}
\rho_K(x, y) &= \left\| \beta^K(x) - \beta^K(y) \right\|_1 \\
&= \left\| \sum_{j=1}^{n} z(j+m)\beta^K(v(\sigma_j)) - \sum_{j=1}^{n} z(j+m)\beta^K(u_j) \right\|_1 \\
&\leqslant \sum_{j=1}^{n} z(j+m) \left\| \beta^K(v(\sigma_j)) - \beta^K(u_j) \right\|_1 \\
&= \sum_{j=1}^{n} z(j+m)\rho_K(v(\sigma_j), u_j) < \delta.
\end{aligned}$$

Thus, $\rho_K(x, |\operatorname{St}(v, L)|) < \delta$, which means that $x \in U(v)$. □

Lemma 4.8.10 *Let K be a simplicial complex and L a subcomplex of K.*

(1) $\operatorname{St}(\hat{\sigma}, \operatorname{Sd}_L K) = \operatorname{St}(\hat{\sigma}, \operatorname{Sd} K)$ *for each* $\sigma \in C(L, K)$;[17]

[17] $C(L, K)$ was defined in Sect. 4.2.

(2) $\mathrm{Sd}^n C(L,K) \subset C(L, \mathrm{Sd}_L^n K)$ *for every* $n \in \mathbb{N}$;
(3) $\mathrm{St}(v, \mathrm{Sd}_L^n K) = \mathrm{St}(v, \mathrm{Sd}^n K)$ *for every* $n \in \mathbb{N}$ *and* $v \in (\mathrm{Sd}^n C(L,K))^{(0)}$.

Proof (1): Since $\sigma \in C(L,K)$ has no face in L, we have $\mathrm{St}(\hat{\sigma}, \mathrm{Sd}_L K) = \mathrm{St}(\hat{\sigma}, \mathrm{Sd} K)$.

According to the definition (cf. Sect. 4.4), for $\sigma \in K \setminus L$, $\mathrm{St}(\hat{\sigma}, \mathrm{Sd}_L K)$ consists of all faces of the following simplexes:

$$\begin{aligned} &\langle \hat{\sigma}_1, \dots, \hat{\sigma}_n \rangle, \sigma_1 < \cdots < \sigma_n \in K \setminus L, \sigma_k = \sigma \ \text{ for some } k = 1, \dots, n; \\ &\langle v_1, \dots, v_m, \hat{\sigma}_1, \dots, \hat{\sigma}_n \rangle, \sigma_1 < \cdots < \sigma_n \in K \setminus L, \langle v_1, \dots, v_m \rangle \in L, \\ &\qquad\qquad \langle v_1, \dots, v_m \rangle < \sigma_1, \ \sigma_k = \sigma \ \text{ for some } k = 1, \dots, n. \end{aligned}$$

(2): By (1), we have

$$\mathrm{Sd}\, C(L,K) \subset \bigcup_{\sigma \in C(L,K)} \mathrm{St}(\hat{\sigma}, \mathrm{Sd} K) = \bigcup_{\sigma \in C(L,K)} \mathrm{St}(\hat{\sigma}, \mathrm{Sd}_L K) \subset \mathrm{Sd}_L K.$$

Hence, $\mathrm{Sd}\, C(L,K) \subset C(L, \mathrm{Sd}_L K)$, that is, (2) holds for $n = 1$.

Now, assume that (2) holds for $n = k-1$ $(k > 1)$, that is, $\mathrm{Sd}^{k-1} C(L,K) \subset C(L, \mathrm{Sd}_L^{k-1} K)$. Applying (2) for $n = 1$ to $\mathrm{Sd}_L^{k-1} K$ and L, we have

$$\mathrm{Sd}^k C(L,K) \subset \mathrm{Sd}\, C(L, \mathrm{Sd}_L^{k-1} K) \subset C(L, \mathrm{Sd}_L(\mathrm{Sd}_L^{k-1} K)) = C(L, \mathrm{Sd}_L^k K).$$

By induction, (2) holds for every $n \in \mathbb{N}$.

(3): Evidently, (1) above means that (3) holds for $n = 1$. Assume that (3) holds for $n = k-1$ $(k > 1)$ and let $v \in (\mathrm{Sd}^k C(L,K))^{(0)}$. Then, $v = \hat{\sigma}$ for some $\sigma \in \mathrm{Sd}^{k-1} C(L,K)$, where $\sigma \in C(L, \mathrm{Sd}_L^{k-1} K)$ by (2). Applying (1), we have

$$\mathrm{St}(v, \mathrm{Sd}_L^k K) = \mathrm{St}(v, \mathrm{Sd}_L \mathrm{Sd}_L^{k-1} K) = \mathrm{St}(v, \mathrm{Sd}\, \mathrm{Sd}_L^{k-1} K).$$

Take any $v_0 \in \sigma^{(0)}$. Applying (3) for $n = k-1$ to $v_0 \in (\mathrm{Sd}^{k-1} C(L,K))^{(0)}$, we have $\mathrm{St}(v_0, \mathrm{Sd}_L^{k-1} K) = \mathrm{St}(v_0, \mathrm{Sd}^{k-1} K)$. Since $v_0 \in \sigma^{(0)}$ and $v = \hat{\sigma}$, it follows that $\mathrm{St}(v, \mathrm{Sd}\, \mathrm{Sd}_L^{k-1} K) = \mathrm{St}(v, \mathrm{Sd}^k K)$.

To see this, let $\tau \in (\mathrm{Sd}\, \mathrm{Sd}_L^{k-1} K)[v]$. Since $v = \hat{\sigma}$, we can write $\tau = \langle \hat{\sigma}_1, \dots, \hat{\sigma}_m \rangle$, $\sigma_1 < \cdots < \sigma_m \in \mathrm{Sd}_L^{k-1} K$, where $\sigma = \sigma_i$ for some i. Since $v_0 \in \sigma^{(0)} = \sigma_i^{(0)} \subset \sigma_m^{(0)}$, it follows that $\sigma_m \in \mathrm{St}(v_0, \mathrm{Sd}_L^{k-1} K) = \mathrm{St}(v_0, \mathrm{Sd}^{k-1} K) \subset \mathrm{Sd}^{k-1} K$. Hence, $\tau = \langle \hat{\sigma}_1, \dots, \hat{\sigma}_m \rangle \in \mathrm{Sd}\, \mathrm{Sd}^{k-1} K = \mathrm{Sd}^k K$. Thus, we have $(\mathrm{Sd}\, \mathrm{Sd}_L^{k-1} K)[v] \subset (\mathrm{Sd}^k K)[v]$, which means $\mathrm{St}(v, \mathrm{Sd}\, \mathrm{Sd}_L^{k-1} K) \subset \mathrm{St}(v, \mathrm{Sd}^k K)$. The converse inclusion $\mathrm{St}(v, \mathrm{Sd}^k K) \subset \mathrm{St}(v, \mathrm{Sd}\, \mathrm{Sd}_L^{k-1} K)$ can be proved similarly.

Consequently, $\mathrm{St}(v, \mathrm{Sd}_L^k K) = \mathrm{St}(v, \mathrm{Sd}^k K)$. By induction, (3) holds for every $n \in \mathbb{N}$. □

We can apply Lemmas 4.8.9 and 4.8.10 above to strengthen Corollary 4.8.7 as follows:

Proposition 4.8.11 *Let K be a finite simplicial complex, L a subcomplex of K, and $\mathcal{U}$ an open cover of $|K|_{\mathrm{m}}$ $(= |K|)$ such that $\mathcal{S}_L \prec \mathcal{U}$. Then, K has a simplicial subdivision K' such that $L \subset K'$ and $\mathcal{S}_{K'} \prec \mathcal{U}$. Moreover, if $U(v) \in \mathcal{U}$ are given for all $v \in L^{(0)}$ so that $|\operatorname{St}(v, L)| \subset U(v)$, then $|\operatorname{St}(v, K')| \subset U(v)$ for all $v \in L^{(0)}$.*

Proof As in the additional statement, we assume that $|\operatorname{St}(v, L)| \subset U(v) \in \mathcal{U}$ for each $v \in L^{(0)}$. By Lemma 4.8.9, we have $K' \lhd K$ such that $L \subset K'$ and $|\operatorname{St}(v, K')| \subset U(v)$ for each $v \in L^{(0)}$. By Corollary 4.8.7, $\mathcal{S}_{\operatorname{Sd}^m K'} \prec \mathcal{U}$ for some $m \in \mathbb{N}$. Then, $\operatorname{Sd}_L^m K'$ is the desired subdivision of K. Indeed, let $v \in (\operatorname{Sd}_L^m K')^{(0)}$. If $v \in L^{(0)}$, then

$$|\operatorname{St}(v, \operatorname{Sd}_L^m K')| \subset |\operatorname{St}(v, K')| \subset U(v) \in \mathcal{U}.$$

When $v \in |C(L, K')|$, since $\operatorname{Sd}_L^m C(L, K') \subset C(L, \operatorname{Sd}_L^m K') \subset \operatorname{Sd}_L^m K'$ by Lemma 4.8.10 (2), it follows that $v \in (\operatorname{Sd}_L^m C(L, K'))^{(0)}$. Since $\operatorname{St}(v, \operatorname{Sd}_L^m K') = \operatorname{St}(v, \operatorname{Sd}^m K')$ by Lemma 4.8.10 (3) and $\mathcal{S}_{\operatorname{Sd}^m K'} \prec \mathcal{U}$, $|\operatorname{St}(v, \operatorname{Sd}_L^m K')|$ is contained in some member of $\mathcal{U}$. When $v \notin L^{(0)} \cup |C(L, K')|$, let $\sigma = c_{K'}(v)$. Then, $v_0 \in \sigma^{(0)} \cap L^{(0)}$ because $\sigma \cap |L| \neq \emptyset$ and $L \subset K'$. Each $\tau \in (\operatorname{Sd}_L^m K')[v]$ is contained in some $\sigma' \in K'$. Since $v \in \sigma' \cap \operatorname{rint} \sigma$, it follows that $\sigma \leqslant \sigma'$, and hence $v_0 \in \sigma'$. Thus, $\tau \subset \sigma' \subset |\operatorname{St}(v_0, K')| \subset U(v_0)$. Therefore, $|\operatorname{St}(v, \operatorname{Sd}_L^m K')| \subset U(v_0) \in \mathcal{U}$. □

For a simplicial complex K, recall that $K(n) = K^{(n)} \setminus K^{(n-1)}$ denotes the collection of all n-simplexes of K. Now, we can prove WHITEHEAD'S THEOREM on small subdivisions:

Theorem 4.8.12 (J.H.C. WHITEHEAD) *Let K be an arbitrary simplicial complex. For any open cover $\mathcal{U}$ of $|K|$, K has a simplicial subdivision K' such that $\mathcal{S}_{K'} \prec \mathcal{U}$. When L is a subcomplex of K such that $\mathcal{S}_L \prec \mathcal{U}$, K' can be taken so that $L \subset K'$ and $\mathcal{S}_{K'} \prec \mathcal{U}$.*

Proof Let $K_0 = L \cup K^{(0)}$. For each $v \in K_0^{(0)}$, we have $U(v) \in \mathcal{U}$ such that $|\operatorname{St}(v, K_0)| \subset U(v)$ because $\operatorname{St}(v, K_0) = \operatorname{St}(v, L)$ if $v \in L^{(0)}$, and $\operatorname{St}(v, K_0) = \{v\}$ if $v \in K^{(0)} \setminus L$. By induction, we shall construct subdivisions $K_n \lhd L \cup K^{(n)}$ and choose $U(v) \in \mathcal{U}$ for each $v \in K_n^{(0)} \setminus K_{n-1}^{(0)}$ so that $K_{n-1} \subset K_n$ and $|\operatorname{St}(v, K_n)| \subset U(v)$ for every $v \in K_n^{(0)} = \bigcup_{i \leqslant n} K_i^{(0)}$. Then, $K' = \bigcup_{n \in \omega} K_n$ would be the desired subdivision of K. Indeed, each $v \in K'^{(0)}$ belongs to some K_n, hence

$$|\operatorname{St}(v, K')| = \bigcup_{i \geqslant n} |\operatorname{St}(v, K_i)| \subset U(v) \in \mathcal{U}.$$

Now, assume that $K_{n-1} \lhd L \cup K^{(n-1)}$ has been constructed and $U(v) \in \mathcal{U}$ has been chosen for every $v \in K_{n-1}^{(0)}$ such that $|\operatorname{St}(v, K_{n-1})| \subset U(v)$. For each $\sigma \in K(n) \setminus L$, let

$$K_{\partial\sigma} = \{\tau \in K_{n-1} \mid \tau \subset \partial\sigma\} \lhd \partial F(\sigma).$$

Applying Lemma 4.3.10, we have

$$K_\sigma = K_{\partial\sigma} \cup \{\hat{\sigma}\} \cup \left\{\hat{\sigma}\tau \;\middle|\; \tau \in K_{\partial\sigma}\right\} \lhd F(\sigma).$$

By Proposition 4.8.11, we have $K'_\sigma \lhd K_\sigma$ such that $K_{\partial\sigma} \subset K'_\sigma$, $\mathcal{S}_{K'_\sigma} \prec \mathcal{U}$, and $|\operatorname{St}(v, K'_\sigma)| \subset U(v)$ for every $v \in K^{(0)}_{\partial\sigma}$. Then, we define

$$K_n = \bigcup_{\sigma \in K(n)} K'_\sigma \cup K_{n-1} \lhd L \cup K^{(n)}.$$

For each $v \in K^{(0)}_n \setminus K^{(0)}_{n-1}$, we have $\sigma \in K(n) \setminus L$ such that $v \in \operatorname{rint}\sigma$. Then, $\operatorname{St}(v, K_n) = \operatorname{St}(v, K'_\sigma)$. Therefore, we can choose $U(v) \in \mathcal{U}$ so that $|\operatorname{St}(v, K_n)| \subset U(v)$. On the other hand, for every $v \in K^{(0)}_{n-1}$,

$$|\operatorname{St}(v, K_n)| = \bigcup_{v \in \sigma \in K(n) \setminus L} |\operatorname{St}(v, K'_\sigma)| \cup |\operatorname{St}(v, K_{n-1})| \subset U(v).$$

This completes the proof. □

By Theorem 4.8.12 above, we have the following:

Corollary 4.8.13 *Every polyhedron is paracompact.*

Proof It suffices to show that $|K|$ is paracompact for any simplicial complex K. For any open cover $\mathcal{U}$ of $|K|$, K has a simplicial subdivision K' such that $\mathcal{S}_{K'} \prec \mathcal{U}$ (Theorem 4.8.12). Then, $(\beta_v^{K'})_{v \in K'^{(0)}}$ is a partition of unity on $|K'| = |K|$ subordinated to $\mathcal{U}$. Hence, $|K|$ is paracompact by Theorem 2.7.5. □

4.9 Small Admissible Subdivisions

In this section, we prove the metric topology version of Whitehead's Theorem 4.8.12, that is, for each simplicial complex K and for any $\mathcal{U} \in \operatorname{cov}(|K|_{\mathrm{m}})$, there is an admissible subdivision $K' \lhd K$ such that $\mathcal{S}_{K'} \prec \mathcal{U}$. It should be remarked that $\operatorname{mesh}_{\rho_K} \operatorname{Sd}^n K = 2 \,(= \operatorname{mesh}_{\rho_K} K)$ for every $n \in \mathbb{N}$ if $\dim K = \infty$ (Proposition 4.8.8).

In the following, for simplicity, we just write $N(L, K)$, $C(L, K)$, and $B(L, K)$ instead of $N(|L|, K)$, $C(|L|, K)$, and $B(|L|, K)$, respectively, where L is a cell complex with $|L| \subset |K|$ which is not necessarily a subcomplex of K.[18] In other words, these can be defined as follows:

[18] For $A \subset |K|$, $N(A, K)$, $C(A, K)$, and $B(A, K)$ were defined in Sect. 4.2.

$$N(L,K)=\bigcup_{\tau\in L}N(\tau,K),\ C(L,K)=\bigcap_{\tau\in L}C(\tau,K),\ \text{and}$$
$$B(L,K)=N(L,K)\cap C(L,K).$$

Similarly, we also write $K[L]$ instead of $K[|L|]$. To prove the metric topology version of Whitehead's Theorem 4.8.12 on small subdivisions, we need several lemmas. In the following lemmas, let K be a simplicial complex.

Lemma 4.9.1 *For a subcomplex L of K, every simplicial subdivision B' of $B(L,K)$ extends to a simplicial subdivision N' of $N(L,K)$ such that $L\cup B'\subset N'$ and $N'^{(0)}=L^{(0)}\cup B'^{(0)}$.*

Proof Let $L_1=\{\sigma\in K\mid \sigma^{(0)}\subset L^{(0)}\}$. Then, $L\subset L_1$, $L^{(0)}=L_1^{(0)}$, and L_1 is a full subcomplex of K. Moreover, $N(L,K)=N(L_1,K)$ and $B(L,K)=B(L_1,K)$. By replacing L with L_1, we can assume that L is a full subcomplex of K.

For each $\tau\in B'$, let $c_K(\hat{\tau})$ be the carrier of the barycenter of τ in K. Then, $c_K(\hat{\tau})\in B(L,K)$ and $\mathrm{Lk}(c_K(\hat{\tau}),K)\cap L\neq\emptyset$. For each $\sigma\in\mathrm{Lk}(c_K(\hat{\tau}),K)\cap L$, we have $\sigma\tau\subset\sigma c_K(\hat{\tau})\in K$. Then, we can define

$$N'=L\cup B'\cup\{\sigma\tau\mid\sigma\in\mathrm{Lk}(c_K(\hat{\tau}),K)\cap L,\ \tau\in B'\}.$$

Obviously, $N'^{(0)}=L^{(0)}\cup B'^{(0)}$. For each $x\in|N(L,K)|\setminus|L\cup B'|$, since L is full in K, we have $\sigma=c_K(x)\cap|L|\in L$. Let σ' be the opposite face of $c_K(x)$ from σ. Then, $\sigma'\in B(L,K)$. Since B' is a subdivision of $B(L,K)$, we have $\tau\in B'$ such that $c_K(\hat{\tau})=\sigma'$ and $x\in\sigma\tau$. Thus, N' is a subdivision of $N(L,K)$. □

The following is easy to see:

Lemma 4.9.2 *For $\sigma\in\mathrm{Sd}\,K$ and $\tau\in K$, the following are equivalent:*

(a) *τ is the smallest simplex of K containing σ;*
(b) *$\hat{\sigma}\in\mathrm{rint}\,\tau$;*
(c) *$\hat{\tau}\in\sigma^{(0)}$ and every vertex of σ is the barycenter of a face of τ.*

Proof The equivalence (a) ⇔ (b) is easily shown by Lemma 4.3.1 and the implication (c) ⇒ (b) is trivial. To see (b) ⇒ (c), take $\tau_1>\cdots>\tau_k\in K$ so that $\sigma=\langle\hat{\tau}_1,\ldots,\hat{\tau}_n\rangle$. Then, $\hat{\sigma}\in\mathrm{rint}\,\tau_1\cap\mathrm{rint}\,\tau$, which means that $\tau_1=\tau$. □

Lemma 4.9.3 *For each $\tau\in K$,*

$$|N(\mathrm{rint}\,\tau,\mathrm{Sd}^2K)|\setminus\partial\tau\subset O_{\mathrm{Sd}\,K}(\hat{\tau}).$$

Proof For each $\sigma\in\mathrm{Sd}^2K$ with $\sigma\cap\mathrm{rint}\,\tau\neq\emptyset$, we may show that $\sigma\setminus\partial\tau\subset O_{\mathrm{Sd}\,K}(\hat{\tau})$. First, note $\sigma\cap\tau\leqslant\sigma$ by Proposition 4.2.10 because $\mathrm{Sd}^2F(\tau)$ is full in Sd^2K (Proposition 4.3.5). Moreover, $(\sigma\cap\tau)^{(0)}\cap\mathrm{rint}\,\tau\neq\emptyset$. Otherwise,

$$(\sigma\cap\tau)^{(0)}\subset(\mathrm{Sd}^2F(\tau))^{(0)}\cap\partial\tau=(\mathrm{Sd}^2\,\partial F(\tau))^{(0)}.$$

Since $\mathrm{Sd}^2\,\partial F(\tau)$ is a full subcomplex of $\mathrm{Sd}^2\,F(\tau)$ by Proposition 4.3.12, it follows that $\sigma\cap\tau\in\mathrm{Sd}^2\,\partial F(\tau)$, which means $\sigma\cap\tau\subset\partial\tau$. This is a contradiction because

$$(\sigma\cap\tau)\cap\operatorname{rint}\tau=\sigma\cap\operatorname{rint}\tau\neq\emptyset.$$

Since $\sigma\in\mathrm{Sd}^2\,K$, there are $\varrho_1>\cdots>\varrho_n\in\mathrm{Sd}\,K$ such that $\sigma=\langle\hat{\varrho}_1,\dots,\hat{\varrho}_n\rangle$. Then, $\sigma\subset\varrho_1$. Let ϱ_1' be the opposite face of ϱ_1 to $\hat{\tau}$. As seen in the above, some $\hat{\varrho}_i$ is contained in $\operatorname{rint}\tau$. Then, we have $k=\max\{i\mid\hat{\varrho}_i\in\operatorname{rint}\tau\}$. Due to Lemma 4.9.2, $\hat{\tau}\in\varrho_k^{(0)}$ and the other vertex of ϱ_k is the barycenter of some proper face of τ. For $i\leqslant k$, $\hat{\varrho}_i\in\operatorname{rint}\varrho_i$ and $\hat{\tau}\in\varrho_k^{(0)}\subset\varrho_i^{(0)}$. Let $\sigma'=\langle\hat{\varrho}_{k+1},\dots,\hat{\varrho}_n\rangle<\sigma$, where $\sigma'=\varrho_{k+1}=\emptyset$ if $k=n$. Then, it follows that[19]

$$\sigma\setminus\sigma'=\bigcup_{i=1}^{k}\big(\langle\hat{\varrho}_i,\dots,\hat{\varrho}_n\rangle\setminus\langle\hat{\varrho}_{i+1},\dots,\hat{\varrho}_n\rangle\big)\subset\bigcup_{i=1}^{k}\operatorname{rint}\varrho_i\subset O_{\mathrm{Sd}\,K}(\hat{\tau}).$$

Since $\sigma'\subset\varrho_{k+1}$ is contained in some face of τ, we have $\sigma\setminus\partial\tau\subset O_{\mathrm{Sd}\,K}(\hat{\tau})$. □

For a full subcomplex L of a simplicial complex K and $0<\varepsilon<2$, we will define a subdivision K_L^{ε} of K such that

$$L\cup C(L,K)\subset K_L^{\varepsilon}\ \text{ and }\ |N(L,K_L^{\varepsilon})|=\overline{\mathrm{N}}_{\rho_K}(|L|,\varepsilon).$$

First, for each $\sigma\in K[L]\setminus L$, let $\sigma_L=\sigma\cap|L|$. Then, $\sigma_L\in L$ and $\sigma_L<\sigma$ because L is full in K. We can write $\sigma=\sigma_L\sigma^L$, where σ^L is the opposite face of σ to σ_L. Since $\delta_L^K|\sigma=\delta_{\sigma_L}^K|\sigma:\sigma\to\mathbb{R}$ is affine (Lemma 4.7.5), $(\delta_L^K)^{-1}(\varepsilon)$ divides σ into the following two cells:

$$\sigma_L^{\varepsilon}=\sigma\cap(\delta_L^K)^{-1}([0,\varepsilon]),\ \ \sigma_{\varepsilon}^L=\sigma\cap(\delta_L^K)^{-1}([\varepsilon,2]).$$

Then, $\sigma_L^{\varepsilon}\cap\sigma_{\varepsilon}^L=\sigma\cap(\delta_L^K)^{-1}(\varepsilon)$, which is denoted by $\sigma_L(\varepsilon)$.

Observe that $\delta_L^K((1-t)x+ty)=2t$ for each $x\in\sigma_L$, $y\in\sigma^L$, and $t\in\mathbf{I}$. Hence,

$$\sigma_L(\varepsilon)=\sigma\cap(\delta_L^K)^{-1}(\varepsilon)=\big\{(1-\varepsilon/2)x+(\varepsilon/2)y\;\big|\;x\in\sigma_L,\ y\in\sigma^L\big\}.$$

By using Proposition 4.1.8 (or 4.1.9 (3)), the following can be easily seen:

$$\begin{gathered}\operatorname{rint}\sigma_L^{\varepsilon}=\operatorname{rint}\sigma\cap(\delta_L^K)^{-1}((0,\varepsilon)),\ \ \operatorname{rint}\sigma_{\varepsilon}^L=\operatorname{rint}\sigma\cap(\delta_L^K)^{-1}((\varepsilon,2)),\\ \operatorname{rint}\sigma_L(\varepsilon)=\operatorname{rint}\sigma\cap(\delta_L^K)^{-1}(\varepsilon),\\ F(\sigma_L^{\varepsilon})=F(\sigma_L(\varepsilon))\cup F(\sigma_L)\cup\{\tau_L^{\varepsilon}\mid\tau\in F(\sigma)[L]\setminus L\},\\ F(\sigma_{\varepsilon}^L)=F(\sigma_L(\varepsilon))\cup F(\sigma^L)\cup\{\tau_{\varepsilon}^L\mid\tau\in F(\sigma)[L]\setminus L\},\\ F(\sigma_L(\varepsilon))=\{\tau_L(\varepsilon)\mid\tau\in F(\sigma)[L]\setminus L\}.\end{gathered}$$

[19] For the first $\subset$, refer to Proposition 3.2.6.

The subdivision K_L^ε of K can be defined as follows:

$$K_L^\varepsilon \underset{\text{def}}{=} L \cup C(L, K) \cup \left\{\sigma_L^\varepsilon, \sigma_\varepsilon^L, \sigma_L(\varepsilon) \;\middle|\; \sigma \in K[L] \setminus L\right\},$$

where it should be noted that $K \setminus (L \cup C(L, K)) = K[L] \setminus L$.

To see that K_L^ε is a cell complex, verify (C_1) and (C_2'') for K_L^ε (cf. Proposition 4.1.9 (3)).

Observe that $K_L^\varepsilon[L] \setminus L = \{\sigma_L^\varepsilon \mid \sigma \in K[L] \setminus L\}$. Hence, we have

$$N(L, K_L^\varepsilon) = L \cup \left\{\sigma_L^\varepsilon, \sigma_L(\varepsilon) \;\middle|\; \sigma \in K[L] \setminus L\right\} \text{ and}$$
$$|N(L, K_L^\varepsilon)| = (\delta_L^K)^{-1}([0, \varepsilon]) = \overline{\mathrm{N}}_{\rho_K}(|L|, \varepsilon).$$

We have also the following:

$$|B(L, K_L^\varepsilon)| = (\delta_L^K)^{-1}(\varepsilon) = \overline{\mathrm{N}}_{\rho_K}(|L|, \varepsilon) \setminus \mathrm{N}_{\rho_K}(|L|, \varepsilon),$$
$$|C(L, K_L^\varepsilon)| = (\delta_L^K)^{-1}([\varepsilon, 2]) = |K| \setminus \mathrm{N}_{\rho_K}(|L|, \varepsilon) = \overline{\mathrm{N}}_{\rho_K}(|C(L, K)|, 2 - \varepsilon).$$

Additionally, $(K_L^\varepsilon)^{(0)}$ consists of the vertices $K^{(0)}$ and the points:

$$v_w = (1 - \varepsilon/2)v + (\varepsilon/2)w, \;\; v \in L^{(0)}, \;\; w \in \mathrm{Lk}(v, K)^{(0)} \setminus L.$$

Then, $(K_L^\varepsilon)^{(0)}$ is discrete in $|K|_{\mathrm{m}}$.

In fact, $\rho_K(u, v) \geqslant \min\{\varepsilon, 2 - \varepsilon\}$ for any $u \neq v \in (K_L^\varepsilon)^{(0)}$. This can be shown as follows: It is obvious that $\rho_K(u, v) = 2$ if $u \neq v \in K^{(0)}$. For $u \in K^{(0)}$, $v, v' \in L^{(0)}$, $w \in \mathrm{Lk}(v, K)^{(0)} \setminus L$, and $w' \in \mathrm{Lk}(v', K)^{(0)} \setminus L$, we have

$$\rho_K(v_w, u) = \begin{cases} 2 & \text{if } u \neq v, w, \\ \varepsilon & \text{if } u = v, \\ 2 - \varepsilon & \text{if } u = w, \end{cases} \quad \text{and} \quad \rho_K(v_w, v'_{w'}) = \begin{cases} 2 & \text{if } v \neq v', w \neq w', \\ \varepsilon & \text{if } v = v', w \neq w', \\ 2 - \varepsilon & \text{if } v \neq v', w = w'. \end{cases}$$

Using the above subdivision K_L^ε of K, we can prove the following:

Lemma 4.9.4 *Let L be a finite full subcomplex of K and $\mathcal{U}$ an open cover of $|L|$ in $|K|_{\mathrm{m}}$ such that $L \prec \mathcal{U}$. Then, there is an admissible subdivision $K_L \lhd K$ with $0 < \varepsilon_L < 2$ such that*

$$L \cup C(L, K) \subset K_L, \;\; N(L, K_L) \prec \mathcal{U}, \;\; \textit{and} \;\; |N(L, K_L)| = \overline{\mathrm{N}}_{\rho_K}(|L|, \varepsilon_L).$$

Proof For each $\tau \in L$, choose $U_\tau \in \mathcal{U}$ and $0 < \varepsilon_\tau < 2$ so that

$$(\delta_\tau^K)^{-1}([0, \varepsilon_\tau]) = \overline{\mathrm{N}}_{\rho_K}(\tau, \varepsilon_\tau) \subset U_\tau.$$

Since L is finite, we have $\varepsilon_L = \min\{\varepsilon_\tau \mid \tau \in L\} > 0$. Then, it follows that

$$\sigma_L^{\varepsilon_L} = \sigma \cap (\delta_{\sigma_L}^K)^{-1}([0, \varepsilon_L]) \subset U_{\sigma_L} \;\text{ for each } \sigma \in K[L] \setminus L.$$

Using Theorem 4.3.6, we have a simplicial subdivision $K_L \lhd K_L^{\varepsilon_L}$ with $K_L^{(0)} = (K_L^{\varepsilon_L})^{(0)}$. Since $K_L^{(0)}$ is discrete in $|K|_{\mathrm{m}}$, K_L is an admissible subdivision of K by Theorem 4.7.9. The subcomplex $L \cup C(L, K)$ of $K_L^{\varepsilon_L}$ is simplicial, and hence it is also a subcomplex of K_L. Observe that

$$N(L, K_L) \prec K_L[L] \prec L \cup \left\{\sigma_L^{\varepsilon_L} \;\middle|\; \sigma \in K[L] \setminus L\right\} \prec \mathcal{U} \text{ and}$$
$$|N(L, K_L)| = |N(L, K_L^{\varepsilon_L})| = \overline{\mathrm{N}}_{\rho_K}(|L|, t_L).$$

For the first equality of the above second line, the inclusion $|N(L, K_L)| \supset |N(L, K_L^{\varepsilon_L})|$ can be verified as follows: For each $\sigma \in K[L] \setminus L$, we may show that $\sigma_L^{\varepsilon_L} \subset |N(L, K_L)|$. Since $|N(L, K_L)|$ is closed in $|K_L|$, it suffices to show $\operatorname{rint} \sigma_L^{\varepsilon_L} \subset |N(L, K_L)|$. For each $x \in \operatorname{rint} \sigma_L^{\varepsilon_L}$, since K_L is a subdivision of $K_L^{\varepsilon_L}$, we have $\tau \in K_L$ such that $x \in \operatorname{rint} \tau \subset \operatorname{rint} \sigma_L^{\varepsilon_L}$. Since $K_L^{(0)} = (K_L^{\varepsilon_L})^{(0)}$, it follows that $\tau^{(0)} \subset (\sigma_L^{\varepsilon_L})^{(0)} = \sigma_L^{(0)} \cup \sigma_L(\varepsilon_L)^{(0)}$. Then, $\tau^{(0)} \cap \sigma_L^{(0)} \neq \emptyset$ because $\tau^{(0)} \not\subset \sigma_L(\varepsilon_L)^{(0)}$. Thus, $\tau \cap |L| \neq \emptyset$, which implies $x \in \tau \subset |N(L, K_L)|$.

Therefore, K_L is the desired one. □

Now, we will prove the following metric topology version of Whitehead's Theorem 4.8.12 on small subdivisions:

Theorem 4.9.5 (HENDERSON–SAKAI) *Let K be an arbitrary simplicial complex. For any open cover $\mathcal{U}$ of $|K|_{\mathrm{m}}$, K has an admissible subdivision K' such that $\mathcal{S}_{K'} \prec \mathcal{U}$.*

Proof It suffices to construct an admissible subdivision K' of K such that $K' \prec \mathcal{U}$. Indeed, every open cover of $|K|_{\mathrm{m}}$ has the open star-refinement and $K' \prec \mathcal{U}$ implies $\mathcal{S}_{K'} \prec \operatorname{st} \mathcal{U}$. We will inductively construct admissible subdivisions K_n of K, $n \in \omega$, so as to satisfy the following conditions:

(1) $K_n \lhd K_{n-1}$,
(2) $K_n||K^{(n-1)}| = K_{n-1}||K^{(n-1)}|$,
(3) $K_n[K^{(n)}] \prec \mathcal{U}$,
(4) $K_n^{(0)} \setminus K_{n-1}^{(0))} \subset |C(K^{(n-1)}, K_{n-1})|$,
(5) $|C(K^{(n-1)}, K_n)| = |C(K^{(n-1)}, K_{n-1})|$,

where $K_{-1} = \mathrm{Sd}^2 K$, $K^{(-1)} = \emptyset$, and $C(\emptyset, \mathrm{Sd}^2 K) = \mathrm{Sd}^2 K$.[20] Due to Proposition 4.3.5, $K_n||K^{(n-1)}|$ and $K_{n-1}||K^{(n-1)}|$ in (2) are simplicial subdivisions of $K^{(n-1)}$ induced by K_n and K_{n-1}, respectively.

It should be noted that the subdivision of each n-skeleton $K^{(n)}$ induced by $K_{-1} = \mathrm{Sd}^2 K$ is a full subcomplex of K_{-1}. For each $i \in \omega$, since $K_i \lhd K_{-1}$, K_i induces a subdivision of $K^{(n)}$ which is a full subcomplex of K_i by Proposition 4.3.5.

So, (2) means that K_n and K_{n-1} induce the same subdivision of $K^{(n-1)}$. Hence, (2) guarantees that $K' = \bigcup_{n\in\mathbb{N}} K_n||K^{(n)}|$ is a simplicial subdivision of K. Moreover, we have $K' \prec \mathcal{U}$ by (3). To see that K' is admissible, let $x \in |K|$ and choose $n \in \omega$

[20] Here, recall the convention on notation mentioned just before Lemma 4.9.1.

so that $x \in |K^{(n)}|$. Since K_n is admissible, $K'^{(0)}||K^{(n)}| = K_n^{(0)}||K^{(n)}|$ is discrete in $|K|_{\mathrm{m}}$ by (2). Hence, x has a neighborhood U in $|K|_{\mathrm{m}}$ which meets $K'^{(0)}||K^{(n)}|$ at no more than one point. Observe that

$$K'^{(0)} \setminus |K^{(n)}| \subset \bigcup_{i>n} K_i^{(0)} \setminus K_n^{(0)} \subset \bigcup_{i>n} \left(K_i^{(0)} \setminus K_{i-1}^{(0)}\right),$$

which is contained in $|C(K^{(n)}, K_n)|$. Indeed, by (4), $K_i^{(0)} \setminus K_{i-1}^{(0)} \subset |C(K^{(i-1)}, K_{i-1})|$ for each $i > n$, and by (5),

$$|C(K^{(i)}, K_i)| \subset |C(K^{(i-1)}, K_i)| = |C(K^{(i-1)}, K_{i-1})| \quad \text{for each } i > n.$$

Since $|C(K^{(n)}, K_n)|$ is closed in $|K_n|_{\mathrm{m}}$ and K_n is an admissible subdivision of K, $V = U \setminus |C(K^{(n)}, K_n)|$ is a neighborhood of x in $|K|_{\mathrm{m}}$, which meets $K'^{(0)}$ at no more than one point. Therefore, $K'^{(0)}$ is discrete in $|K|_{\mathrm{m}}$, which means that K' is an admissible subdivision of K according to Theorem 4.7.9.

(*The First Step*) We will construct K_0. For each vertex $v \in K^{(0)}$, we apply Lemma 4.9.4 to obtain a simplicial subdivision $N_v \lhd N(v, \mathrm{Sd}^2 K)$ $(= \mathrm{St}(v, \mathrm{Sd}^2 K))$ such that

$$B(v, \mathrm{Sd}^2 K) \subset N_v, \ \ N(v, N_v) \prec \mathcal{U},$$

and $N_v^{(0)}$ is discrete in $|K|_{\mathrm{m}}$. The collection $\{|N_v| \mid v \in K^{(0)}\}$ is discrete in $|K|_{\mathrm{m}}$ because $|N_v| \subset O_{\mathrm{Sd}\, K}(v)$ for each $v \in K^{(0)}$ by Lemma 4.9.3, each $O_{\mathrm{Sd}\, K}(v)$ is open in $|K|_{\mathrm{m}}$, $O_{\mathrm{Sd}\, K}(v) \cap O_{\mathrm{Sd}\, K}(v') = \emptyset$ if $v \neq v' \in K^{(0)}$, and $\bigcup_{v \in K^{(0)}} |N_v| = |N(K^{(0)}, \mathrm{Sd}^2 K)|$ is closed in $|K|_{\mathrm{m}}$. Observe that

$$|N_v| \cap |C(K^{(0)}, \mathrm{Sd}^2 K)| = |B(v, \mathrm{Sd}^2 K)| \ \text{ and}$$
$$N_v \cap C(K^{(0)}, \mathrm{Sd}^2 K) = B(v, \mathrm{Sd}^2 K).$$

Then, the desired $K_0 \lhd \mathrm{Sd}^2 K$ can be defined as follows:

$$\begin{aligned} K_0 &\underset{\text{def}}{=} \bigcup_{v \in K^{(0)}} N_v \cup C(K^{(0)}, \mathrm{Sd}^2 K) \\ &\lhd \bigcup_{v \in K^{(0)}} N(v, \mathrm{Sd}^2 K) \cup C(K^{(0)}, \mathrm{Sd}^2 K) = \mathrm{Sd}^2 K. \end{aligned}$$

Indeed, $\bigcup_{v \in K^{(0)}} N_v^{(0)}$ is discrete in $|K|_{\mathrm{m}}$ as the discrete union of discrete sets in $|K|_{\mathrm{m}}$. Since $(\mathrm{Sd}^2 K)^{(0)}$ is discrete in $|K|_{\mathrm{m}}$, $K_0^{(0)} = \bigcup_{v \in K^{(0)}} N_v^{(0)} \cup (\mathrm{Sd}^2 K)^{(0)}$ is discrete in $|K|_{\mathrm{m}}$, which means that K_0 is an admissible subdivision of K by Theorem 4.7.9. Observe that

$$N(K^{(0)}, K_0) = \bigcup_{v \in K^{(0)}} N(v, K_0) = \bigcup_{v \in K^{(0)}} N(v, N_v) \prec \mathcal{U}.$$

Namely, K_0 satisfies condition (3). Other conditions are obvious.

(*The Inductive Step*) Assume that K_{n-1} has been obtained. For each n-simplex $\tau \in K(n)$ $(= K^{(n)} \setminus K^{(n-1)})$, we define

$$\tau^* = \tau \cap |C(K^{(n-1)}, K_{n-1})| = |F(\tau) \wedge C(K^{(n-1)}, K_{n-1})|,$$

where the last equality comes from Proposition 4.3.8. Due to Propositions 4.3.4 and 4.3.5,

$$F(\tau) \wedge K_{n-1} = \left\{\sigma \in K_{n-1} \;\middle|\; \sigma \subset \tau\right\} = K_{n-1}|\tau.$$

Then, it follows that

$$\begin{aligned} F(\tau) \wedge C(K^{(n-1)}, K_{n-1}) &= \left\{\sigma \in C(K^{(n-1)}, K_{n-1}) \;\middle|\; \sigma \subset \tau\right\} \\ &= \left\{\sigma \in K_{n-1} \;\middle|\; \sigma \subset \tau^*\right\} = K_{n-1}|\tau^*, \end{aligned}$$

which is a subcomplex of $K_{n-1}|\tau$ triangulating τ^*. It should be noted that $K_{n-1}|\tau^*$ is a full subcomplex of K_{n-1}. Moreover, the following collection is discrete in $|K|_{\mathrm{m}}$:

$$\mathcal{N} = \left\{|N(\tau^*, C(K^{(n-1)}, K_{n-1}))| \;\middle|\; \tau \in K(n)\right\}.$$

Indeed, for each $\tau \in K(n)$,

$$\begin{aligned} |N(\tau^*, C(K^{(n-1)}, K_{n-1}))| &= |N(\tau^*, K_{n-1})| \cap |C(K^{(n-1)}, K_{n-1})| \\ &\subset |N(\operatorname{rint}\tau, \operatorname{Sd}^2 K)| \setminus |K^{(n-1)}| \subset O_{\operatorname{Sd} K}(\hat{\tau}), \end{aligned}$$

where the last inclusion comes from Lemma 4.9.3. Each $O_{\operatorname{Sd} K}(\hat{\tau})$ is open in $|K|_{\mathrm{m}}$ and

$$\tau \neq \tau' \in K(n) \;\Rightarrow\; O_{\operatorname{Sd} K}(\hat{\tau}) \cap O_{\operatorname{Sd} K}(\hat{\tau}') = \emptyset.$$

Moreover, $\bigcup \mathcal{N}$ is closed in $|K|_{\mathrm{m}}$ because it is triangulated by the following subcomplex of K_{n-1}:

$$\bigcup_{\tau \in K(n)} N(\tau^*, C(K^{(n-1)}, K_{n-1}).$$

By Corollary 4.8.7, we can choose $n(\tau) \in \omega$ so that $\operatorname{Sd}^{n(\tau)}(K_{n-1}|\tau^*) \prec \mathcal{U}$. To avoid notational complications, we write

$$\begin{aligned} L_\tau = \operatorname{Sd}^{n(\tau)}(K_{n-1}|\tau^*),\ D_\tau = B(\tau^*, C(K^{(n-1)}, K_{n-1})),\ \text{and} \\ M_\tau = \operatorname{Sd}_{D_\tau}^{n(\tau)} N(\tau^*, C(K^{(n-1)}, K_{n-1})). \end{aligned}$$

Then, $|L_\tau| = \tau^*$ and $L_\tau \prec \mathcal{U}$. Moreover, M_τ is an admissible subdivision of $N(\tau^*, C(K^{(n-1)}, K_{n-1}))$ by Corollary 4.7.2, hence $|M_\tau|_{\mathrm{m}}$ is a closed subspace of $|K_{n-1}|_{\mathrm{m}} = |K|_{\mathrm{m}}$. Since $\tau^* \cap |D_\tau| = \emptyset$ and $K_{n-1}|\tau^*$ is a full subcomplex of

$N(\tau^*, C(K^{(n-1)}, K_{n-1}))$ $(\subset K_{n-1})$, it follows from Proposition 4.3.5 that L_τ is a full subcomplex of M_τ subdividing $K_{n-1}|\tau^*$. We can apply Lemma 4.9.4 to obtain a simplicial subdivision $N_\tau \lhd M_\tau$ such that

$$L_\tau \cup C(L_\tau, M_\tau) \subset N_\tau, \ N(L_\tau, N_\tau) \prec \mathcal{U},$$

and $N_\tau^{(0)}$ is discrete in $|M_\tau|_{\mathrm{m}}$. It should be noted that $\{|N_\tau| \mid \tau \in K(n)\} = \mathcal{N}$ is discrete in $|K|_{\mathrm{m}}$ and $D_\tau \subset C(L_\tau, N_\tau)$ $(D_\tau \neq B(L_\tau, N_\tau)$ in general). Observe that

$$\begin{aligned} |N_\tau| \cap |C(K^{(n)}, K_{n-1})| &= |N(\tau^*, C(K^{(n-1)}, K_{n-1}))| \cap |C(\tau^*, C(K^{(n-1)}, K_{n-1}))| \\ &= |B(\tau^*, C(K^{(n-1)}, K_{n-1}))| = |D_\tau|. \end{aligned}$$

Since D_τ is a subcomplex of both N_τ and $C(K^{(n)}, K_{n-1})$, we can define a simplicial subdivision $C' \lhd C(K^{(n-1)}, K_{n-1})$ as follows:

$$C' = C(K^{(n)}, K_{n-1}) \cup \bigcup_{\tau \in K(n)} N_\tau.$$

Then, $\bigcup_{\tau \in K(n)} N_\tau^{(0)}$ is discrete in $|K|_{\mathrm{m}}$ as a discrete union of discrete sets. Since $C(K^{(n)}, K_{n-1})^{(0)}$ is also discrete in $|K|_{\mathrm{m}}$, it follows that $C'^{(0)}$ is discrete in $|K|_{\mathrm{m}}$.

Let B' be the simplicial subdivision of $B(K^{(n-1)}, K_{n-1})$ induced by C'. According to Lemma 4.9.1, B' extends to a simplicial subdivision N' of $N(K^{(n-1)}, K_{n-1})$ such that

$$B' \cup (K^{(n-1)}||K_{n-1}|) \subset N' \ \text{ and } \ N'^{(0)} = N(K^{(n-1)}, K_{n-1})^{(0)} \cup B'^{(0)}.$$

Then, $K_n = C' \cup N'$ is the desired simplicial subdivision of K_{n-1}. Indeed, as is easily observed,

$$K_n^{(0)} = C'^{(0)} \cup N'^{(0)} = C'^{(0)} \cup K_{n-1}^{(0)},$$

which is discrete in $|K|_{\mathrm{m}}$ because $C'^{(0)}$ and $K_{n-1}^{(0)}$ are discrete in $|K|_{\mathrm{m}}$. This means that K_n is an admissible subdivision of K by Theorem 4.7.9. We also have

$$\begin{aligned} K_n||K^{(n-1)}| = N'||K^{(n-1)}| = K_{n-1}||K^{(n-1)}| \ \text{ and} \\ K_n^{(0)} \setminus K_{n-1}^{(0)} \subset C'^{(0)} \subset |C(K^{(n-1)}, K_{n-1})|, \end{aligned}$$

that is, K_n satisfies (2) and (4). Since $C(K^{(n-1)}, K_n) = C' \lhd C(K^{(n-1)}, K_{n-1})$, K_n satisfies (5) too. Moreover, $K_n[K^{(n)}] \prec \mathcal{U}$ because

$$\begin{aligned} K_n[K^{(n-1)}] \prec K_{n-1}[K^{(n-1)}] \prec \mathcal{U} \ \text{ and} \\ K_n[K^{(n)}] \setminus K_n[K^{(n-1)}] \subset \bigcup_{\tau \in K(n)} N(L_\tau, N_\tau) \prec \mathcal{U}. \end{aligned}$$

Thus, K_n also satisfies (3). This completes the proof. □

4.10 Simplicial Maps

In this section, we define simplicial maps and prove their basic properties. We will also prove the Simplicial Approximation Theorem 4.10.11.

Let K and L be simplicial complexes. A function $f : |K| \to |L|$ is called a **simplicial map** from K to L (or with respect to K and L) if $f|\sigma$ is affine and $f(\sigma) \in L$ for each $\sigma \in K$, where $\dim f(\sigma) \leqslant \dim \sigma$. Then, f is continuous (Corollary 3.5.4), $f(K^{(0)}) \subset L^{(0)}$ and $f(K) = \{f(\sigma) \mid \sigma \in K\}$ is a subcomplex of L, where $f(\tau) \leqslant f(\sigma)$ if $\tau \leqslant \sigma$. For each $\sigma \in K$, if $\sigma^{(0)} = \{v_1, \ldots, v_{n+1}\}$, then $f(\sigma)^{(0)} = \{f(v_1), \ldots, f(v_{n+1})\}$ and

$$f\big(\textstyle\sum_{i=1}^{n+1} z(i)v_i\big) = \displaystyle\sum_{i=1}^{n+1} z(i) f(v_i) \quad \text{for each } z \in \Delta^n,$$

where it is possible that $f(v_i) = f(v_j)$ for some $i \neq j$. Hence,

$$f(x) = \sum_{v \in K^{(0)}} \beta_v^K(x) f(v) \quad \text{for each } x \in |K|;$$
$$\beta_u^L(f(x)) = \sum_{v \in f^{-1}(u) \cap K^{(0)}} \beta_v^K(x) \quad \text{for each } u \in L^{(0)}.$$

Although a simplicial map f from K to L is actually a (continuous) function from $|K|$ to $|L|$, we may write $f : K \to L$ because it induces a function between the collections K and L. Evidently, the composition of simplicial maps and the restriction of a simplicial map to a subcomplex are also simplicial.

As seen in the above, by restricting a simplicial map $f : K \to L$ on $K^{(0)}$, we can obtain a function $f_0 : K^{(0)} \to L^{(0)}$. It is helpful if we can know which function $f_0 : K^{(0)} \to L^{(0)}$ extends to a simplicial map $f : K \to L$. The following proposition characterizes such a function:

Proposition 4.10.1 *Let K and L be simplicial complexes. For a function $f_0 : K^{(0)} \to L^{(0)}$, the following are equivalent:*

(a) *f_0 extends to a simplicial map $f : K \to L$;*
(b) *$\langle f_0(\sigma^{(0)})\rangle \in L$ for each $\sigma \in K$;*
(c) *$\bigcap_{v \in \sigma^{(0)}} O_L(f_0(v)) \neq \emptyset$ for each $\sigma \in K$.*

In this case, the simplicial extension f of f_0 is unique.

Proof The implication (a) $\Rightarrow$ (b) follows from the definition of a simplicial map, where $\langle f_0(\sigma^{(0)})\rangle = f(\sigma)$ for each $\sigma \in K$. By Proposition 4.2.8, we have (b) $\Leftrightarrow$ (c). It remains to show (b) $\Rightarrow$ (a). For each $\sigma \in K$, the function $f_0|\sigma^{(0)}$ uniquely extends to an affine map $f_\sigma : \sigma \to \langle f_0(\sigma^{(0)})\rangle \subset |L|$, where $f_\sigma(\sigma) = \langle f_0(\sigma^{(0)})\rangle \in L$ by (b). Since $f_\sigma|\sigma \cap \tau = f_\tau|\sigma \cap \tau$ for each $\sigma, \tau \in K$, we can define $f : |K| \to |L|$ by $f|\sigma = f_\sigma$. Then, f is simplicial with respect to K and L. The uniqueness of f follows from the uniqueness of f_σ, $\sigma \in K$. □

In the following, we list basic properties of a simplicial map, which can be easily proved (the proofs are left to the reader):

Proposition 4.10.2 (Basic Properties of a Simplicial Map) *Let K and L be simplicial complexes. A simplicial map $f : K \to L$ has the following properties:*

(i) $\operatorname{rint} f(\sigma) = f(\operatorname{rint} \sigma)$ *for each* $\sigma \in K$.
(ii) $c_L(f(x)) = f(c_K(x))$ *for each* $x \in |K|$.
(iii) $\rho_L(f(x), f(y)) \leqslant \rho_K(x, y)$ *for each* $x, y \in |K|$, *i.e., f is Lipschitz (1-Lipschitz); when f is injective, $\rho_L(f(x), f(y)) = \rho_K(x, y)$, i.e., f is an isometry.*
(iv) *For each subcomplex K_0 of K, the image $f(K_0) = \{f(\sigma) \mid \sigma \in K_0\}$ is a subcomplex of L and $|f(K_0)| = f(|K_0|)$.*
(v) *For each subcomplex L_0 of L, the inverse image $f^{-1}(L_0) = \{\sigma \in K \mid f(\sigma) \in L_0\}$ is a subcomplex of K and $|f^{-1}(L_0)| = f^{-1}(|L_0|)$.*

In (i) above, it should be noted that $\partial f(\sigma) \neq f(\partial\sigma)$ in general. Due to (iii), for each simplicial map $f : K \to L$, $f : |K|_{\mathrm{m}} \to |L|_{\mathrm{m}}$ is also continuous.

It is also easy to show the following:

Proposition 4.10.3 *For a simplicial map $f : K \to L$, the following are equivalent:*

(a) *$f : |K| \to |L|$ is a homeomorphism;*
(b) *$f : |K| \to |L|$ is a bijection;*
(c) *f induces a bijection $f : K \to L$ between collections of simplexes.*

Sketch of Proof The implication (a) $\Rightarrow$ (b) is obvious. In (b) $\Rightarrow$ (c), to see that $f : K \to L$ is surjective, let $\tau \in L$ and take $y \in \operatorname{rint} \tau$. Let $\sigma = c_K(f^{-1}(y)) \in K$. Then, $f(\sigma) = f(c_K(f^{-1}(y))) = c_L(y) = \tau$. To see (c) $\Rightarrow$ (a), use Proposition 4.10.2 (ii) and observe that $\beta^L_{f(v)}(f(x)) = \beta^K_v(x)$ for each $x \in |K|$ and $v \in K^{(0)}$.

A bijective simplicial map is called a **simplicial isomorphism**, that is, a simplicial isomorphism is a simplicial map $f : K \to L$ such that $f : |K| \to |L|$ is bijective (i.e., it satisfies (b) above). For each simplicial isomorphism $f : K \to L$, the inverse $f^{-1} : L \to K$ is also a simplicial isomorphism.

Indeed, if a bijection $f : |K| \to |L|$ is simplicial with respect to K and L, then the inverse $f^{-1} : |L| \to |K|$ is also simplicial with respect to L and K.

Due to Proposition 4.10.3 above, a simplicial isomorphism $f : K \to L$ is a homeomorphism from $|K|$ to $|L|$, so it is also called a **simplicial homeomorphism**. Then, the induced bijection $f : K \to L$ (in (c) above) satisfies the following:

$$\sigma, \tau \in K,\ \tau \leqslant \sigma \ \Leftrightarrow\ f(\tau) \leqslant f(\sigma).$$

Therefore, K and L are congruent. Conversely, if K and L are congruent, then there is a bijection $\eta : K \to L$ such that

$$\sigma, \tau \in K,\ \tau \leqslant \sigma \ \Leftrightarrow\ \eta(\tau) \leqslant \eta(\sigma).$$

Due to Proposition 4.10.1, $\eta|K^{(0)} : K^{(0)} \to L^{(0)}$ extends to a simplicial map $f : K \to L$, which is a simplicial isomorphism by Proposition 4.10.3. Thus, we have the following:

Proposition 4.10.4 *Two simplicial complexes K_1 and K_2 are congruent, i.e., $K_1 \equiv K_2$ if and only if there exists some simplicial isomorphism (simplicial homeomorphism) $f : K_1 \to K_2$. When K_1 and K_2 have a common subcomplex L, $K_1 \equiv K_2$ rel. L if and only if there exists some simplicial homeomorphism $f : K_1 \to K_2$ such that $f|L = \mathrm{id}$.* □

By virtue of Proposition 4.10.4, speaking of simplicial complexes alone, we can use the term *simplicially isomorphic* instead of *congruent*. Namely, it is said that simplicial complexes K_1 and K_2 are **simplicially isomorphic** (relative to L) or K_1 is **simplicially isomorphic** to K_2 (relative to L) if $K_1 \equiv K_2$ (rel. L).

The following is also easy:

Proposition 4.10.5 *For a simplicial map $f : K \to L$, the following are equivalent:*

(a) *$f : |K| \to |L|$ is a closed embedding;*
(b) *$f : |K| \to |L|$ is an embedding;*
(c) *$f : |K| \to |L|$ is an injection;*
(d) *$f|K^{(0)} : K^{(0)} \to L^{(0)}$ is an injection;*
(e) *f induces an injection $f : K \to L$ between collections of simplexes.*

Sketch of Proof The implications (a) ⇒ (b) ⇒ (c) ⇒ (d) and (e) ⇒ (d) are trivial. Because every simplex is determined by its vertices, (d) ⇒ (e) is also trivial. Since $f(K)$ is a subcomplex of L, the image $f(|K|) = |f(K)|$ is closed in $|L|$ (Proposition 4.10.2 (iv)). Hence, we have (b) ⇒ (a). To see (d) ⇒ (b), use Proposition 4.10.2 (ii) and observe that $\beta^L_{f(v)}(f(x)) = \beta^K_v(x)$ for each $x \in |K|$ and $v \in K^{(0)}$ if $f|K^{(0)}$ is an injection.

An injective simplicial map is called a **simplicial embedding**. It should be noted that Proposition 4.10.3 does not have the condition corresponding to (d) in Proposition 4.10.5. In fact, $f : |K| \to |L|$ need not be surjective even if $f|K^{(0)} : K^{(0)} \to L^{(0)}$ is bijective.

We obtain the following from Proposition 4.10.2 (iii):

Proposition 4.10.6 *For each simplicial embedding (resp. a simplicial isomorphism) $f : K \to L$, $f : (|K|, \rho_K) \to (|L|, \rho_L)$ is a closed isometric embedding (resp. an isometric homeomorphism), hence $f : |K|_{\mathrm{m}} \to |L|_{\mathrm{m}}$ is a closed embedding (resp. a homeomorphism).* □

With regard to subdivisions of simplicial maps, we have the following:

Proposition 4.10.7 *Let K and L be simplicial complexes and $f : K \to L$ a simplicial map. For each simplicial subdivision $L' \lhd L$, there exists a simplicial subdivision $K' \lhd K$ such that $f : K' \to L'$ is simplicial.*

Proof We define

$$K_0 = \left\{\sigma \cap f^{-1}(\tau) \;\middle|\; \sigma \in K,\ \tau \in L',\ \tau \subset f(\sigma)\right\}.$$

Then, K_0 is a cell complex subdividing K. Indeed, let $\sigma \in K$ and $\tau \in L'$ with $\tau \subset f(\sigma)$ and $x \in \sigma \cap f^{-1}(\tau)$. By Proposition 4.1.9 (3), $\sigma \cap f^{-1}(\tau) = (f|\sigma)^{-1}(\tau)$ is a cell with $(\sigma \cap f^{-1}(\tau))_x = \sigma_x \cap f^{-1}(\tau_{f(x)})$. Since $\operatorname{rint} f(\sigma_x) \cap \operatorname{rint} \tau_{f(x)} \neq \emptyset$, we have $\tau_{f(x)} \subset f(\sigma_x)$ because $L' \lhd L$. Thus, K_0 satisfies (C_1). To show (C_2''), let $\sigma, \sigma' \in K$ and $\tau, \tau' \in L'$ with $\tau \subset f(\sigma)$, $\tau' \subset f(\sigma')$, and

$$x \in \operatorname{rint}(\sigma \cap f^{-1}(\tau)) \cap \operatorname{rint}(\sigma' \cap f^{-1}(\tau')) \left(\subset \sigma \cap \sigma' \cap f^{-1}(\tau \cap \tau')\right).$$

Since $\operatorname{rint} \sigma_x \cap \operatorname{rint} \sigma'_x \neq \emptyset$ and $\operatorname{rint} \tau_{f(x)} \cap \operatorname{rint} \tau'_{f(x)} \neq \emptyset$ (cf. Proposition 3.2.9 (viii)), we have $\sigma_x = \sigma'_x$ and $\tau_{f(x)} = \tau'_{f(x)}$. Then, from Proposition 4.1.9 (3), it follows that

$$\begin{aligned}\sigma \cap f^{-1}(\tau) &= (\sigma \cap f^{-1}(\tau))_x = \sigma_x \cap f^{-1}(\tau_{f(x)}) \\ &= \sigma'_x \cap f^{-1}(\tau'_{f(x)}) = (\sigma' \cap f^{-1}(\tau'))_x = \sigma' \cap f^{-1}(\tau').\end{aligned}$$

Therefore, K_0 satisfies (C_2'').

Let $\sigma \in K$ and $\tau \in L'$ with $\tau \subset f(\sigma)$. Then, $f(\sigma \cap f^{-1}(\tau)) = f(\sigma) \cap \tau = \tau$. For each $x \in \sigma \cap f^{-1}(\tau)$, since $\tau_{f(x)} \subset f(\sigma_x)$ as seen in the verification of (C_1) above, it follows that

$$f((\sigma \cap f^{-1}(\tau))_x) = f(\sigma_x \cap f^{-1}(\tau_{f(x)})) = f(\sigma_x) \cap \tau_{f(x)} = \tau_{f(x)}.$$

In particular, if $v \in (\sigma \cap f^{-1}(\tau))^{(0)}$, then $f(v) \in \tau^{(0)}$. Thus, $f(K_0^{(0)}) \subset L'^{(0)}$.

Applying Theorem 4.3.6, we can obtain a simplicial subdivision $K' \lhd K_0$ with $K'^{(0)} = K_0^{(0)}$. Then, $f(K'^{(0)}) \subset L'^{(0)}$. For each simplex $\sigma' \in K'$, we have $\sigma \in K$ and $\tau \in L'$ such that $\tau \subset f(\sigma)$, $\sigma' \subset \sigma \cap f^{-1}(\tau)$, and $\sigma'^{(0)} \subset (\sigma \cap f^{-1}(\tau))^{(0)}$. Since $f|\sigma'$ is affine and $f(\sigma'^{(0)}) \subset \tau^{(0)}$, it follows that $f(\sigma') \leqslant \tau$, hence $f(\sigma') \in L'$. Thus, $f : K' \to L'$ is simplicial. □

Remark 4.10.1 When K and L are cell complexes, in the same way as a simplicial map, we can define a **cellular map** from K to L as a map $f : |K| \to |L|$ such that $f|C$ is affine and $f(C) \in L$ for each $C \in K$.[21] Notice that the simplicial map $f : K \to L$ in the proof of Proposition 4.10.7 is a cellular map from K_0 to L. Although a simplicial map can be easily defined by using Proposition 4.10.1, it is not easy to define a cellular map directly. So, such a cellular map seems to be impractical. In fact, for any quadrilateral cell D, $F(D) \cong F(\mathbf{I}^2)$ but an injective affine image of $\mathbf{I}^2$ must be a parallelogram.

[21] In fact, this definition is employed in the textbook by C.P. Rourke and B.J. Sanderson, p. 16 (cf. Notes for this chapter).

Note For CW complexes X and Y, a **cellular map** $f : X \to Y$ is defined as a map such that $f(X^n) \subset Y^n$ for each $n \in \mathbb{N}$. Any cell complex K defines a CW complex $X = |K|$ with $X^n = |K^{(n)}|$, $n \in \omega$. Evidently, a cellular map between cell complexes is cellular as a map between CW complexes but the converse is not true.

Now, let $f : |K| \to |L|$ be a map, where K and L are cell complexes. Recall the **graph** $\mathrm{Gr}(f)$ of f is defined in Sect. 1.1 as follows:

$$\mathrm{Gr}(f) \underset{\text{def}}{=} \big\{(x, f(x)) \;\big|\; x \in |K|\big\} \subset |K| \times |L|.$$

Then, $\mathrm{Gr}(f)$ is closed in $|K| \times |L|$ and $\mathrm{pr}_{|K|} | \mathrm{Gr}(f) : \mathrm{Gr}(f) \to |K|$ is a homeomorphism whose inverse is the natural injection $i_f : |K| \to \mathrm{Gr}(f)$ defined by $i_f(x) = (x, f(x))$ for each $x \in |K|$. As observed in Proposition 4.5.2, $|K| \times |L| \neq |K \times_c L|$ as spaces in general. However, we have the following:

Lemma 4.10.8 *Let $f : |K| \to |L|$ be a map, where K and L are cell complexes. The graph $\mathrm{Gr}(f)$ of f is closed in $|K \times_c L|$ and its topology is equal to the one inherited from $|K \times_c L|$, that is, $\mathrm{Gr}(f)$ is a closed subspace of $|K \times_c L|$.*

Proof Since $\mathrm{id} : |K \times_c L| \to |K| \times |L|$ is continuous, it suffices to show that $A \subset \mathrm{Gr}(f)$ is closed in $|K| \times |L|$ if $A \cap (C \times D)$ is closed in $C \times D$ for each $C \in K$ and $D \in L$. Because $\mathrm{pr}_{|K|} | \mathrm{Gr}(f) : \mathrm{Gr}(f) \to |K|$ is a homeomorphism, we may show that $\mathrm{pr}_{|K|}(A)$ is closed in $|K|$. For each $C \in K$, since $f(C)$ is compact, we have $D_1, \ldots, D_k \in L$ such that $f(C) \subset \bigcup_{i=1}^{k} D_i$. Since $\mathrm{pr}_{|K|}^{-1}(C) \cap \mathrm{Gr}(f) = (C \times f(C)) \cap \mathrm{Gr}(f)$, it follows that

$$A \cap \mathrm{pr}_{|K|}^{-1}(C) = A \cap (C \times f(C)) = \bigcup_{i=1}^{k} A \cap (C \times D_i).$$

Each $A \cap (C \times D_i)$ is compact as a closed subset of $C \times D_i$. Hence, $A \cap \mathrm{pr}_{|K|}^{-1}(C)$ is compact. Thus, $\mathrm{pr}_{|K|}(A) \cap C = \mathrm{pr}_{|K|}(A \cap \mathrm{pr}_{|K|}^{-1}(C))$ is compact, which implies that $\mathrm{pr}_{|K|}(A) \cap C$ is closed in C. Therefore, $\mathrm{pr}_{|K|}(A)$ is closed in $|K|$. □

For simplicial complexes K and L, the product complex $K \times_c L$ is not simplicial. However, in Sect. 4.5, we defined the product simplicial complex $K \times_s L$ for ordered simplicial complexes K and L.

Theorem 4.10.9 *Let $f : K \to L$ be a simplicial map between ordered simplicial complexes K and L. If $f|K^{(0)} : K^{(0)} \to L^{(0)}$ is order preserving (i.e., $\forall u, v \in K^{(0)}$, $u \leqslant v \Rightarrow f(u) \leqslant f(v)$), then the graph $\mathrm{Gr}(f)$ is triangulated by a subcomplex of the product simplicial complex $K \times_s L$.*

Proof For each $\sigma \in K$, let $\sigma = \langle v_1, \ldots, v_n \rangle$ with $v_1 < \cdots < v_n$. Then, $f(\sigma) = \langle f(v_1), \ldots, f(v_n) \rangle$ and $f(v_1) \leqslant \cdots \leqslant f(v_n)$. Hence,

$$i_f(\sigma) = \langle (v_1, f(v_1)), \ldots, (v_n, f(v_n)) \rangle \in K \times_s L,$$

where $i_f : |K| \to \mathrm{Gr}(f)$ is the natural injection. It follows from Proposition 4.1.9 (3) that $i_f(K) = \{i_f(\sigma) \mid \sigma \in K\}$ is a subcomplex of $K \times_s L$, where $|i_f(K)| = i_f(|K|) = \mathrm{Gr}(f)$. □

Let K and L be simplicial complexes. A simplicial map $g : K \to L$ is called a **simplicial approximation** of a map $f : |K| \to |L|$ if each $g(x)$ is contained in the carrier $c_L(f(x))$ of $f(x)$ in L. Then, $f \simeq_L g$ by Proposition 4.5.6, which is realized by the straight-line homotopy. A **simplicial approximation** $g : K \to L$ of a map $f : |K|_{\mathrm{m}} \to |L|_{\mathrm{m}}$ is similarly defined. Then, $f \simeq_L g$ by Proposition 4.6.6, which is also realized by the straight-line homotopy.

Lemma 4.10.10 *A simplicial map $g : K \to L$ is a simplicial approximation of a map $f : |K| \to |L|$ (or $f : |K|_{\mathrm{m}} \to |L|_{\mathrm{m}}$) if and only if $f(O_K(v)) \subset O_L(g(v))$ for every $v \in K^{(0)}$.*

Proof First, assume that $g : K \to L$ is a simplicial approximation of f and let $v \in K^{(0)}$. For each $x \in O_K(v)$, we have $v \in c_K(x)^{(0)}$. Observe that $g(x) \in c_L(f(x)) \cap \operatorname{rint} g(c_K(x))$. Then, $g(c_K(x)) \leqslant c_L(f(x))$, hence $g(v) \in g(c_K(x))^{(0)} \subset c_L(f(x))^{(0)}$. Therefore, $f(x) \in \operatorname{rint} c_L(f(x)) \subset O_L(g(v))$.

Conversely, assume that $f(O_K(v)) \subset O_L(g(v))$ for every $v \in K^{(0)}$ and let $x \in |K|$. For every $v \in c_K(x)^{(0)}$, since $x \in \operatorname{rint} c_K(x) \subset O_K(v)$, we have $f(x) \in f(O_K(v)) \subset O_L(g(v))$, which means that $g(v) \in c_L(f(x))^{(0)}$. Thus, $g(c_K(x)) \leqslant c_L(f(x))$. Therefore, $g(x) \in c_L(f(x))$. □

As an application of Theorem 4.8.12, we have the following:

Theorem 4.10.11 (Simplicial Approximation) *Let K and L be simplicial complexes. Then, each map $f : |K| \to |L|$ has a simplicial approximation $g : K' \to L$ for some $K' \lhd K$.*

Proof By Theorem 4.8.12, we have $K' \lhd K$ such that $\mathcal{S}_{K'} \prec f^{-1}(\mathcal{O}_L)$. Let $g' : K'^{(0)} \to L^{(0)}$ be a function such that $|\operatorname{St}(v, K')| \subset f^{-1}(O_L(g'(v)))$. For each $\sigma \in K'$,

$$\sigma \subset \bigcap_{v \in \sigma^{(0)}} |\operatorname{St}(v, K')| \subset \bigcap_{v \in \sigma^{(0)}} f^{-1}(O_L(g'(v))),$$

hence $\bigcap_{v \in \sigma^{(0)}} O_L(g'(v)) \neq \emptyset$. Due to Proposition 4.10.1, g' induces a simplicial map $g : K' \to L$. By Lemma 4.10.10, g is a simplicial approximation of f. □

Remark 4.10.2 In the above theorem, when K is finite, we can take $K' = \mathrm{Sd}^n K$ for some $n \in \mathbb{N}$. Applying Theorem 4.8.6, we can choose such an $n \in \mathbb{N}$ so that $2 \operatorname{mesh}_{\rho_K} \mathrm{Sd}^n K$ is a Lebegue number of $f^{-1}(\mathcal{O}_L)$.

We can easily generalize Theorem 4.10.11 above as follows:

Theorem 4.10.12 *Let K and $L_1, \ldots, L_n$ be simplicial complexes. For any maps $f_i : |K| \to |L_i|$, $i = 1, \ldots, n$, there exist a simplicial subdivision K' of K and simplicial maps $g_i : K' \to L_i$, $i = 1, \ldots, n$, such that each g_i is a simplicial approximation of f_i.* □

The following metric versions of the above results can be proved by the same arguments using Theorem 4.9.5 instead of Whitehead's Theorem 4.8.12:

Theorem 4.10.13 (ADMISSIBLE SIMPLICIAL APPROXIMATION) *Let K and L be simplicial complexes. Then, each map $f : |K|_{\mathrm{m}} \to |L|_{\mathrm{m}}$ has a simplicial approximation $g : K' \to L$ for some admissible subdivision $K' \lhd K$, where $g : |K|_{\mathrm{m}} \to |L|_{\mathrm{m}}$ is continuous.* □

Theorem 4.10.14 *Let K and $L_1, \ldots, L_n$ be simplicial complexes. For any maps $f_i : |K|_{\mathrm{m}} \to |L_i|_{\mathrm{m}}$, $i = 1, \ldots, n$, there exist an admissible subdivision K' of K and simplicial maps $g_i : K' \to L_i$, $i = 1, \ldots, n$, such that each g_i is a simplicial approximation of f_i, where each $g_i : |K|_{\mathrm{m}} \to |L|_{\mathrm{m}}$ is continuous.* □

Let (K_1, L_1) and (K_2, L_2) be pairs of simplicial complexes and subcomplexes. A **simplicial map** $f : (K_1, L_1) \to (K_2, L_2)$ from (K_1, L_1) to (K_2, L_2) (or with respect to (K_1, L_1) to (K_2, L_2)) is a map $f : (|K_1|, |L_1|) \to (|K_2|, |L_2|)$ such that $f : K_1 \to K_2$ is simplicial, where $f||L_1| : L_1 \to L_2$ is also simplicial as a consequence. A simplicial map $g : (K_1, L_1) \to (K_2, L_2)$ is called a **simplicial approximation** of a map $f : (|K_1|, |L_1|) \to (|K_2|, |L_2|)$ if $g : K_1 \to K_2$ is a simplicial approximation of $f : |K_1| \to |K_2|$. Then, it follows that $g||L_1| : L_1 \to L_2$ is a simplicial approximation of $f||L_1| : |L_1| \to |L_2|$. The Simplicial Approximation Theorem 4.10.11 is valid for pairs:

Theorem 4.10.15 (SIMPLICIAL APPROXIMATION FOR PAIRS) *Let (K_1, L_1) and (K_2, L_2) be pairs of simplicial complexes. For each map $f : (|K_1|, |L_1|) \to (|K_2|, |L_2|)$, there is a simplicial subdivision $K_1' \lhd K_1$ and a simplicial approximation $g : (K_1', L_1') \to (K_2, L_2)$, where $L_1' \lhd L_1$ induced by K_1'.*

Sketch of Proof Observe that $\mathrm{St}(v, L_1') \subset \mathrm{St}(v, K_1')$ and $O_{L_2}(u) = O_{K_2}(u) \cap |L_2|$.[22] As in the proof of Theorem 4.10.11, take $K_1' \lhd K_1$ so that $\mathcal{S}_{K_1'} \prec f^{-1}(O_{K_2})$. Then, it follows that $\mathcal{S}_{L_1'} \prec f^{-1}(O_{K_2})||L_1| = (f||L_1|)^{-1}(O_{L_2}) \setminus \{\emptyset\}$. Hence, we have $g' : ({K_1'}^{(0)}, {L_1'}^{(0)}) \to ({K_2}^{(0)}, {L_2}^{(0)})$ so that $|\mathrm{St}(v, K_1')| \subset f^{-1}(O_{K_2}(g'(v)))$ for $v \in {K_1'}^{(0)}$ and $|\mathrm{St}(v, L_1')| \subset f^{-1}(O_{L_2}(g'(v)))$ for $v \in {L_1'}^{(0)}$.

A **simplicial isomorphism** or a **simplicial homeomorphism** $f : (K_1, L_1) \to (K_2, L_2)$ is a simplicial isomorphism $f : K_1 \to K_2$ such that $f(|L_1|) = |L_2|$. In other words, it is a simplicial map $f : (K_1, L_1) \to (K_2, L_2)$ such that $f : |K_1| \to |K_2|$ is a bijection and $f(|L_1|) = |L_2|$, where $f||L_1| : L_1 \to L_2$ is also a simplicial isomorphism as a consequence. Then, the inverse $f^{-1} : (K_2, L_2) \to (K_1, L_1)$ is also a simplicial isomorphism. It is said that (K_1, L_1) and (K_2, L_2) are **simplicially isomorphic** or (K_1, L_1) is **simplicially isomorphic** to (K_2, L_2) if there exists a simplicial isomorphism $f : (K_1, L_1) \to (K_2, L_2)$. A **simplicial embedding** $f : (K_1, L_1) \to (K_2, L_2)$ is a simplicial embedding $f : K_1 \to K_2$ with $f(|L_1|) \subset |L_2|$

[22] It should be remarked that $\mathrm{St}(v, L_1') \neq \mathrm{St}(v, K_1') \cap L_1'$ in general. Indeed, if $\sigma = v\tau \in K_1' \setminus L_1'$, $v \in {L_1'}^{(0)}$, and $\tau \in L_1'$, then $\tau \in \mathrm{St}(v, K_1')$ but $\tau \notin \mathrm{St}(v, L_1')$.

(i.e., a simplicial map $f : (K_1, L_1) \to (K_2, L_2)$ such that $f : |K_1| \to |K_2|$ is an injection), where $f||L_1| : L_1 \to L_2$ is also simplicial embedding.

As is easily observed, Proposition 4.10.4 is valid for pairs of simplicial complexes, that is:

Proposition 4.10.16 *Two pairs of simplicial complexes (K_1, L_1) and (K_2, L_2) are congruent (i.e., $(K_1, L_1) \equiv (K_2, L_2)$) if and only if they are simplicially isomorphic, that is, there exists a simplicial isomorphism (simplicial homeomorphism) $f : (K_1, L_1) \to (K_2, L_2)$.* □

4.11 Piecewise-Linear Maps

Let P and Q be polyhedra (in linear spaces E and F, respectively). A map $f : P \to Q$ is said to be **piecewise linear** (**PL**) if there is a cell (or simplicial) complex K such that $|K| = P$ and f is affine on each cell (or simplex) $C \in K$, i.e., $f|C : C \to F$ is affine. When K_1 is a cell complex and $P = |K_1|$, the above K can be taken as a subdivision of K_1.

Indeed, since $|K| = |K_1|$, we have a common subdivision K' of K and K_1 (Proposition 4.3.7). Then, f is affine on each cell $C \in K'$ because C is contained in some cell $D \in K$.

According to the following theorem, a PL map can be characterized as a map whose graph is a polyhedron:

Theorem 4.11.1 *Let P and Q be polyhedra. A map $f : P \to Q$ is PL if and only if the graph $\mathrm{Gr}(f)$ of f is a polyhedron, where if K and L are cell complexes with $|K| = P$ and $|L| = Q$, then $\mathrm{Gr}(f)$ is a subpolyhedron of $|K \times_c L|$.*[23]

Proof (*The "only if" Part*) Let K and L are cell complexes with $|K| = P$ and $|L| = Q$. Since f is PL, K has a subdivision K' such that f is affine on each cell in K'. Then, the natural injection $i_f : |K| \to \mathrm{Gr}(f)$ is affine on each cell in K'. Using Proposition 4.1.9 (3), we can see that $i_f(K') = \{i_f(C) \mid C \in K'\}$ is a cell complex. Since $i_f|C : C \to i_f(C)$ is a homeomorphism for each $C \in K'$, it follows that $i_f : |K| = |K'| \to |i_f(K')|$ is a homeomorphism. Recall that $\mathrm{pr}_{|K|}|\mathrm{Gr}(f)$ is a homeomorphism. Hence, $\mathrm{id} = i_f \mathrm{pr}_{|K|}|\mathrm{Gr}(f) \to |i_f(K')|$ is a homeomorphism, i.e., $\mathrm{Gr}(f) = |i_f(K')|$ as spaces. Thus, $\mathrm{Gr}(f)$ is a polyhedron. Due to Lemma 4.10.8, $\mathrm{Gr}(f)$ is a closed subspace of $|K \times_c L|$. Consequently, $\mathrm{Gr}(f)$ is a subpolyhedron of $|K \times_c L|$ by Theorem 4.4.8.

(*The "if" Part*) Let M, K, and L be cell complexes with $|M| = \mathrm{Gr}(f)$, $|K| = P$, and $|L| = Q$. Then, $\mathrm{pr}_{|K|}(M) = \{\mathrm{pr}_{|K|}(D) \mid D \in M\}$ is a cell complex by Proposition 4.1.9 (3). Since $\mathrm{pr}_{|K|}|\mathrm{Gr}(f) : \mathrm{Gr}(f) = |M| \to |\mathrm{pr}_{|K|}(M)|$ is a homeomorphism, it follows that $|\mathrm{pr}_{|K|}(M)| = |K|$ as spaces. By Proposition 4.3.7, K and

[23] Note that $P \times Q \neq |K \times_c L|$ as spaces. Moreover, we cannot apply Theorem 4.10.9 because f is not necessarily simplicial with respect to some $K' \lhd K$ and $L' \lhd L$.

$\mathrm{pr}_{|K|}(M)$ have a common subdivision K'. For each $C \in K'$, $i_f(C)$ is a cell contained in some cell in M. Note that $\mathrm{pr}_{|K|}|i_f(C)$ is an affine homeomorphism, hence so is $i_f|C$. Since $\mathrm{pr}_{|L|}|i_f(C)$ is affine, $f|C = \mathrm{pr}_{|L|}(i_f|C)$ is affine. Thus, f is affine on each cell in K', that is, f is PL. □

In the definition of a PL map $f : |K| \to |L|$, we did not require a subdivision K' of K to satisfy that the image of each cell of K' is contained in a cell of L, i.e., $f(K') \prec L$. However, we can take such a subdivision, that is:

Theorem 4.11.2 *Let K and L be cell complexes. Then, a function $f : |K| \to |L|$ is a PL map if and only if there is a subdivision (or a simplicial subdivision) K' of K such that for each cell (or each simplex) $C \in K'$, $f(C)$ is contained in some $D \in L$ and $f|C : C \to D$ is affine.*

Proof (*The "if" Part*) By the uniqueness of the topology of cells (Proposition 3.5.8 (1)) and the continuity of affine functions on finite-dimensional convex sets (Corollary 3.5.4), $f|C : C \to D$ is continuous in the condition. From the definition of topology of $|K'|$, f is continuous. Hence, f is a PL map by the definition of a PL map.

(*The "only if" Part*) Replacing K with a subdivision, we may assume that $f|C$ is affine for each $C \in K$. According to Theorem 4.11.1, $\mathrm{Gr}(f)$ is a subpolyhedron of $|K \times_c L|$. By Theorem 4.4.8, $\mathrm{Gr}(f)$ is triangulated by a subcomplex M of some simplicial subdivision of $K \times_c L$. Note that $\mathrm{pr}_{|K|}|\tau : \tau \to \mathrm{pr}_{|K|}(\tau)$ is an affine homeomorphism for each $\tau \in M$. Then, $K' = \{\mathrm{pr}_{|K|}(\tau) \mid \tau \in M\}$ is a simplicial subdivision of K. For each $\sigma \in K'$, $i_f(\sigma) \in M$ is contained in a cell $C \times D \in K \times_c L$. Hence, $f(\sigma) = \mathrm{pr}_{|L|} i_f(\sigma)$ is contained in $D \in L$. Consequently, $f(K') \prec L$. □

Remark 4.11.1 In the definition of a PL map, the continuity of f is necessary. For example, let $K = \{0,\ 1,\ \mathbf{I}\}$ and $L = \{0, 1/n, [1/(n+1), 1/n] \mid n \in \mathbb{N}\}$, where $|K| = \mathbf{I}$ (as spaces) and $|L| = \mathbf{I}$ (as sets) but $|L| \approx \{0\} \cup [1, \infty)$ ($\subset \mathbb{R}$). Then, $f = \mathrm{id}_{\mathbf{I}} : |K| \to |L| \subset \mathbb{R}$ is affine but non-continuous.

By the above characterization, the following can be easily proved:

Proposition 4.11.3 *The composition of PL maps is also PL.* □

For a subpolyhedron P_0 of a polyhedron P, the inclusion $P_0 \subset P$ is a PL map. Then, the following is a direct consequence of the above proposition:

Corollary 4.11.4 *Let $f : P \to Q$ be a PL map between polyhedra. Then, for each subpolyhedron P_0 of P, the restriction $f|P_0$ is also PL.* □

Remark 4.11.2 (1) It should be remarked that a PL map was not defined as a map $f : |K| \to |L|$ which is simplicial with respect to some simplicial subdivisions $K' \lhd K$ and $L' \lhd L$. Such a definition depends on cell complexes K and L, that is, when K_1 and L_1 are cell complexes with $|K| = |K_1|$ and $|L| = |L_1|$ as spaces, there need not exist $K_1' \lhd K_1$ and $L_1' \lhd L_1$ for which $f : K_1' \to L_1'$ is simplicial. For example, let $K = \{n,\ [n, n+1] \mid n \in \omega\}$ and $I = \{0,\ 1,\ \mathbf{I}\}$ be the natural triangulations of

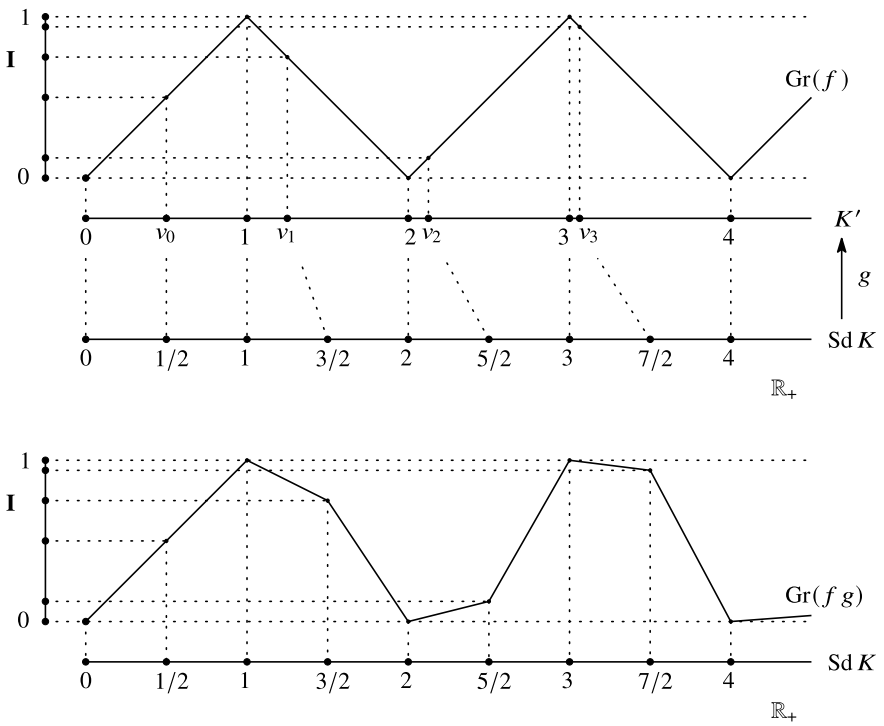


Fig. 4.10 The composition of PL maps

$[0, \infty)$ and $\mathbf{I}$, respectively, and $f : K \to I$ be a simplicial map defined by $f(2n) = 0$ and $f(2n+1) = 1$ for each $n \in \omega$. Consider the following simplicial subdivision of K:

$$K' = \big\{n,\ v_n,\ [n, v_n],\ [v_n, n+1] \ \big|\ n \in \omega\big\},$$

where $v_n = n + 2^{-(n+1)}$ (cf. the upper graph of Fig. 4.10). Then, there do not exist $K'' \lhd K'$ and $I' \lhd I$ such that $f : K'' \to I'$ is simplicial. Indeed, for such K'' and I', we have

$$\big\{2^{-(2n+1)} \ \big|\ n \in \omega\big\} \subset \{f(v_n) \mid n \in \omega\} \subset f(K'^{(0)}) \subset f(K''^{(0)}) \subset I'^{(0)},$$

which implies $I'^{(0)}$ is infinite. This contradicts the fact that $|I'| = \mathbf{I}$ is compact.

(2) Alternatively, we might define a PL map as a map $f : P \to Q$ between polyhedra P and Q which is simplicial for *some* triangulations K and L of P and Q, respectively. However, if we adopted such a definition, we could not assert that the composition fg of PL maps f and g is PL, even if g is a simplicial isomorphism for some triangulation. For example, let K, K', I and $f : K \to I$ be as in (1) and let $g : \mathrm{Sd}\, K \to K'$ be a simplicial isomorphism defined by $g(n) = n$ and

$g(n+1/2) = v_n$ for each $n \in \omega$. Then, $fg : [0,\infty) \to \mathbf{I}$ is not PL in this sense. To see this, suppose $fg : K'' \to I'$ is simplicial for some triangulations K'' and I' of $[0,\infty)$ and $\mathbf{I}$, respectively. Then, $n+1/2 \in K''^{(0)}$ for every $n \in \omega$ because fg is not affine on any interval containing $n+1/2$ as an interior point (see Fig. 4.10). Thus, we have

$$\begin{aligned}\left\{2^{-(2n+1)} \;\middle|\; n \in \omega\right\} &\subset \{f(v_n) \mid n \in \omega\} = \{fg(n+1/2) \mid n \in \omega\} \\ &\subset fg(K''^{(0)}) \subset I'^{(0)},\end{aligned}$$

which implies $I'^{(0)}$ is infinite. This is a contradiction as above.

Concerning the preimage of a subpolyhedron by a PL map, we have:

Proposition 4.11.5 *Let $f : P \to Q$ be a PL map between polyhedra. For every subpolyhedron Q_0 of Q, the preimage $f^{-1}(Q_0)$ is a subpolyhedron of P.*

Proof Let L be a triangulation of Q such that Q_0 is triangulated by a subcomplex L_0 of L. Take a triangulation K of P such that f is affine on each simplex of K and $f(K) \prec L$ ($K \prec f^{-1}(L)$), and define

$$K' = K \wedge f^{-1}(L) = \left\{\sigma \cap f^{-1}(\tau) \;\middle|\; \sigma \in K,\ \tau \in L,\ \sigma \cap f^{-1}(\tau) \neq \emptyset\right\}.$$

In the same way as Proposition 4.10.7, we can show that K' is a cell complex subdividing K.[24] Then, the following is similarly a cell complex:

$$K'_0 = K \wedge f^{-1}(L_0) = \left\{\sigma \cap f^{-1}(\tau) \;\middle|\; \sigma \in K,\ \tau \in L_0,\ \sigma \cap f^{-1}(\tau) \neq \emptyset\right\},$$

where $K'_0 \subset K'$ and $|K'_0| = f^{-1}(Q_0)$. Hence, $f^{-1}(Q_0)$ is a subpolyhedron of P. □

Remark 4.11.3 (1) As for the image of a PL map, there is no proposition corresponding to the above. In fact, for a PL map $f : P \to Q$ between polyhedra, the image $f(P)$ is not necessarily a subpolyhedron of Q even if $f : P \to Q$ is an injective PL map. For example, let $f : \mathbb{R}_+ \to \mathbf{I}^2$ be the PL map defined as follows:

$$f(t) = \begin{cases} (t, 0) & \text{if } 0 \leqslant t \leqslant 1, \\ (1, t-1) & \text{if } 1 \leqslant t \leqslant 2, \\ (2^{-n}(2n+2-t), 2^{-n+1}) & \text{if } 2n \leqslant t \leqslant 2n+1,\ n \in \mathbb{N}, \\ (2^{-n}, 2^{-n}(2n+3-t)) & \text{if } 2n+1 \leqslant t \leqslant 2n+2,\ n \in \mathbb{N}. \end{cases}$$

Then, f is injective but $f(\mathbb{R}_+)$ is not a subpolyhedron of I^2. Any triangulation of $f(\mathbb{R}_+)$ would contain $f(\omega)$ as a part of the set of vertices (the 0-skeleton). See Fig. 4.11.

[24] For each $\sigma \in K$, $f(\sigma)$ is covered by finitely many simplexes of L. Hence, σ is covered by finitely many cells of K'.

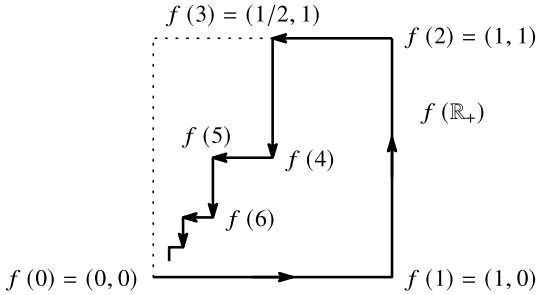


Fig. 4.11 The image of the PL map f

(2) Furthermore, even if $f(P)$ is a subpolyhedron of Q, the image $f(P_0)$ of a subpolyhedron P_0 of P is not necessarily a subpolyhedron of Q. For example, consider the simplicial map $f : K \to I$ in Remark 4.11.2 (see Fig. 4.10), where K and I are the natural triangulations of $\mathbb{R}_+$ and $\mathbf{I}$, respectively. Then, $P_0 = \{v_n \mid n \in \omega\}$ is a subpolyhedron of $\mathbb{R}_+$. But $f(P_0) = \{2^{-(2n+1)}, 1 - 2^{-(2n+2)} \mid n \in \omega\}$ is not a subpolyhedron of $\mathbf{I} = |I|$ because any subpolyhedron of $\mathbf{I}$ is compact.

A function $f : P \to Q$ is called a **tight PL map** if it has the following property:

- For *any* triangulation K of P, there exist a simplicial subdivision $K' \lhd K$ and a triangulation L of Q such that $f : K' \to L$ is simplicial.

The following is obvious:

Fact *Every tight PL map $f : P \to Q$ is a PL map and the image $f(P)$ is a subpolyhedron of Q.*

A tight PL map $f : |K| \to |L|$ can be also defined as a PL map with the following property:

- For *any* subdivision K' of K, there exist simplicial subdivisions $K'' \lhd K'$ and $L' \lhd L$ such that $f : K'' \to L'$ is simplicial.

As the example $f : K \to I$ in Remark 4.11.2 (1), in general, $f : |K| \to |L|$ is not a tight PL map even if $f : K \to L$ is simplicial (cf. the above Fact). We can show that every proper PL map is tight. To this end, we need the following:

Lemma 4.11.6 *Let $C_1, \ldots, C_n$ be cells contained in a cell C and K_0 be a triangulation of ∂C such that $C_i \cap \partial C$ is triangulated by a subcomplex of K_0 if $C_i \cap \partial C \neq \emptyset$. Then, C has a triangulation K such that $K_0 \subset K$ and $C_1, \ldots, C_n$ are triangulated by subcomplexes of K.*

Proof It suffices to prove the case $n = 1$. Indeed, assume that C has triangulations $K_1, \ldots, K_n$ such that $K_0 \subset K_i$ and C_i is triangulated by a subcomplex of K_i. By Corollary 4.3.9, we have a common simplicial subdivision K of $K_1, \ldots, K_n$ with $K_0 \subset K$, which is the desired triangulation of C.

The case $n = 1$ can be shown as follows: Consider the cell complex $F(C_1)$ and its subcomplex $L_1 = \{D \in F(C_1) \mid D \subset \partial C\}$. Then, $|L_1| = C_1 \cap \partial C$. Indeed, $|L_1| \subset C_1 \cap \partial C$ by the definition. To see the converse inclusion, for each $x \in C_1 \cap \partial C$, take $D \leqslant C_1$ with $x \in \operatorname{rint} D$. Then, $D = D_x \subset C_x$. Since $x \notin \operatorname{rint} C$ implies $C_x < C$, we have $D \subset \partial C$, that is, $D \in L_1$. Hence, $x \in |L_1|$. Note that K_0 gives the simplicial subdivision L_1' of L_1. We can apply Proposition 4.3.11 to obtain a simplicial subdivision K_1 of $F(C_1)$ with $L_1' \subset K_1$. Then, $K_1 \cup K_0$ is a triangulation of $C_1 \cup \partial C$. On the other hand, $F(C)$ has a simplicial subdivision K with $K_0 \subset K$ by Proposition 4.3.11. Applying Theorem 4.4.7, we can obtain a simplicial subdivision K_1' of K such that $K_0 \subset K_1'$ and $K_1 \cup K_0$ is subdivided by a subcomplex of K_1'. Then, $C_1 = |K_1|$ is triangulated by a subcomplex of K_1'. □

A tight PL map can be characterized as follows:

Theorem 4.11.7 *Let P and Q be polyhedra. For a PL map $f : P \to Q$, the following are equivalent:*

(a) *f is tight PL;*
(b) *Q has a triangulation L such that* $\operatorname{cl} f^{-1}(\operatorname{rint} \tau)$ *is compact for any simplex $\tau \in L \setminus L^{(0)}$;*
(c) *$Q_0 = \{y \in Q \mid f^{-1}(y)$ is non-compact$\}$ is discrete in Q and* $\operatorname{cl} f^{-1}(A \setminus Q_0)$ *is compact for any compact set A in Q.*

Proof (a) $\Rightarrow$ (b): Since f is tight PL, it is simplicial with respect to some triangulations K and L of P and Q, respectively. Suppose there is some $\tau \in L \setminus L^{(0)}$ such that $\operatorname{cl} f^{-1}(\operatorname{rint} \tau)$ is not compact. Then, since $f^{-1}(\operatorname{rint} \tau)$ cannot be covered by finitely many simplexes of K, we have distinct simplexes $\sigma_i \in K$, $i \in \mathbb{N}$, such that $\operatorname{rint} \sigma_i \cap f^{-1}(\operatorname{rint} \tau) \neq \emptyset$, where $f(\sigma_i) \in L$ because $f : K \to L$ is simplicial. Since $\operatorname{rint} f(\sigma_i) \cap \operatorname{rint} \tau \neq \emptyset$, it follows that $f(\sigma_i) = \tau$ for every $i \in \mathbb{N}$. Because $\dim \tau > 0$ and $f(\operatorname{rint} \sigma_i) = \operatorname{rint} \tau$, we can inductively choose $x_i \in \operatorname{rint} \sigma_i$, $i \in \mathbb{N}$, so that $f(x_i) \neq f(x_j)$ if $j < i$. Let K' be a derived subdivision of K defined by using points $v_\sigma \in \operatorname{rint} \sigma$, $\sigma \in K$, which are given as follows:

$$v_\sigma = \begin{cases} x_i & \text{if } \sigma = \sigma_i, i \in \mathbb{N}, \\ \hat{\sigma} & \text{otherwise.} \end{cases}$$

Because of tightness of f, there are simplicial subdivisions $K'' \lhd K'$ and $L' \lhd L$ such that $f : K'' \to L'$ is simplicial. Then, $f(x_i) \in L'^{(0)}$ because $x_i \in K'^{(0)} \subset K''^{(0)}$. This contradicts the fact that τ contains only finitely many simplexes of L'. Consequently, $\operatorname{cl} f^{-1}(\operatorname{rint} \tau)$ is compact for each $\tau \in L \setminus L^{(0)}$.

(b) $\Rightarrow$ (c): Let L be a triangulation of Q such that $\operatorname{cl} f^{-1}(\operatorname{rint} \tau)$ is compact for each simplex $\tau \in L \setminus L^{(0)}$. Then, it follows that $Q_0 \subset L^{(0)}$, which implies that Q_0 is discrete in Q. For any compact set A in Q, there is a finite subcomplex L_0 of L such that $A \subset |L_0|$. Since $A \subset \bigcup_{\tau \in L_0} \operatorname{rint} \tau$, it follows that

$$f^{-1}(A \setminus Q_0) \subset \bigcup_{v \in L_0^{(0)} \setminus Q_0} f^{-1}(v) \cup \bigcup_{\tau \in L_0 \setminus L^{(0)}} \operatorname{cl} f^{-1}(\operatorname{rint} \tau),$$

where the right-hand set is closed in P because L_0 is finite. Since $f^{-1}(v)$ is compact for each $v \in L_0^{(0)} \setminus Q_0$ and $\operatorname{cl} f^{-1}(\operatorname{rint} \tau)$ is compact for each $\tau \in L_0 \setminus L^{(0)}$, it follows that $\operatorname{cl} f^{-1}(A \setminus Q_0)$ is compact.

(c) $\Rightarrow$ (b): Due to Theorem 4.4.8, Q_0 is a 0-dimensional subpolyhedron of Q, that is, Q_0 is triangulated by a subcomplex of a triangulation L of Q, which means that $Q_0 \subset L^{(0)}$. For each $\tau \in L \setminus L^{(0)}$, $\operatorname{cl} f^{-1}(\tau \setminus Q_0)$ is compact, whose closed subset $\operatorname{cl} f^{-1}(\operatorname{rint} \tau)$ is also compact.

(b) $\Rightarrow$ (a): Let L be a triangulation of Q as in (b) and K be any triangulation of P. Then, we may show that f is simplicial with respect to some simplicial subdivisions $K' \lhd K$ and $L' \lhd L$. Since f is PL, replacing K with a simplicial subdivision, we can assume that $f|\sigma$ is affine for each $\sigma \in K$, where $f(\sigma)$ is a cell. For each $\tau \in L$, we define

$$S_\tau = \left\{ \sigma \in K \;\middle|\; \operatorname{rint} \sigma \cap f^{-1}(\operatorname{rint} \tau) \neq \emptyset \right\}.$$

When $\dim \tau > 0$, S_τ is finite because $\operatorname{cl} f^{-1}(\operatorname{rint} \tau)$ is compact. Let $L_0 = L^{(0)}$. We will inductively construct simplicial subdivisions $L_n \lhd L^{(n)}$, $n \in \mathbb{N}$, so as to satisfy the following:

$(*)_n$ $L_{n-1} \subset L_n$ and for each $\tau \in L^{(n)}$ and $\sigma \in S_\tau$, the cell $f(\sigma) \cap \tau$ is triangulated by some subcomplex of L_n.

Suppose that $L_{n-1} \lhd L^{(n-1)}$ has been constructed so as to satisfy $(*)_{n-1}$. Let $\tau \in L(n) = L^{(n)} \setminus L^{(n-1)}$ and $\sigma \in S_\tau$. Then, we have

$$f(\sigma) \cap \partial\tau = \bigcup_{\tau' < \tau} \bigcup_{\sigma' \in F(\sigma) \cap S_{\tau'}} (f(\sigma') \cap \tau'). \tag{$\sharp$}$$

Indeed, the inclusion $\supset$ is obvious. To see the inclusion $\subset$, for each $y \in f(\sigma) \cap \partial\tau$, take $x \in \sigma$ so that $f(x) = y$. Then, $\tau' = \tau_y < \tau$ and $\sigma' = \sigma_x \leqslant \sigma$. Since $x \in \operatorname{rint} \sigma' \cap f^{-1}(\operatorname{rint} \tau')$, it follows that $\sigma' \in S_{\tau'}$ and $y \in f(\sigma') \cap \tau'$, which means that y belongs to the right-hand set.

For each $\tau' < \tau$ and $\sigma' \in F(\sigma) \cap S_{\tau'}$, $f(\sigma') \cap \tau'$ is triangulated by some subcomplex of L_{n-1} by $(*)_{n-1}$. Hence, $f(\sigma) \cap \partial\tau$ is also triangulated by some subcomplex of L_{n-1}. For each $\tau \in L(n)$, let $L_{\partial\tau}$ be the subdivision $\partial F(\tau)$ induced by L_{n-1}. We can apply Lemma 4.11.6 to obtain a triangulation L_τ of τ such that $L_{\partial\tau} \subset L_\tau$ and $f(\sigma) \cap \tau$ is triangulated by a subcomplex of L_τ for each $\sigma \in S_\tau$. Then, $L_n \lhd L^{(n)}$ is defined as follows:

$$L_n = L_{n-1} \cup \bigcup_{\tau \in L(n)} L_\tau,$$

which satisfies $(*)_n$ and $L_{n-1} \subset L_n$.

Now, we have a simplicial subdivision $L' = \bigcup_{n\in\omega} L_n \lhd L$. Then, $f(\sigma)$ is triangulated by a subcomplex of L' for each $\sigma \in K$.

This is obtained from the following, which can be shown in the same way as (♯) above:

$$f(\sigma) = \bigcup_{\tau\in L[f(\sigma)]} \bigcup_{\sigma'\in F(\sigma)\cap S_\tau} (f(\sigma') \cap \tau).$$

Then, the following is a cell complex subdividing K:

$$K_0 = \left\{\sigma \cap f^{-1}(\tau) \;\middle|\; \sigma \in K,\ \tau \in L',\ \tau \subset f(\sigma)\right\}.$$

To see that K_0 satisfies (C_1), let $\sigma \in K$, $\tau \in L'$, $\tau \subset f(\sigma)$, and $x \in \sigma \cap f^{-1}(\tau)$. By Proposition 4.1.9 (3), we have

$$(\sigma \cap f^{-1}(\tau))_x = (f|\sigma)^{-1}(\tau)_x = \sigma_x \cap (f|\sigma)^{-1}(\tau_{f(x)}) = \sigma_x \cap f^{-1}(\tau_{f(x)}).$$

Since $f(\sigma_x)$ is triangulated by some subcomplex of L', there is some $\tau' \in L'$ such that $f(x) \in \tau' \subset f(\sigma_x)$. Since $\tau' \cap \operatorname{rint} \tau_{f(x)} \neq \emptyset$, it follows that $\tau_{f(x)} \leqslant \tau'$, which implies $\tau_{f(x)} \subset f(\sigma_x)$. Therefore, $(\sigma \cap f^{-1}(\tau))_x \in K_0$. For (C_2''), we can imitate the proof of Proposition 4.10.7.

Finally, take a simplicial subdivision $K' \lhd K_0$ so that $K'^{(0)} = K_0^{(0)}$. Then, in the same way as Proposition 4.10.7, we can show that $f : K' \to L'$ is simplicial. (The details are left to the reader.) □

By the above characterization, the following are trivial:

Corollary 4.11.8 *Every proper PL map is a tight PL map. In particular, for every subpolyhedron Q of a polyhedron P, the inclusion $Q \subset P$ is a tight PL map.* □

Remark 4.11.4 The converse of the above corollary is not true, that is, we have a non-proper tight PL map. Let $v_0 * L$ be the cone over an infinite 0-dimensional simplicial complex $L = \{v_i \mid i \in \mathbb{N}\}$, where $v_i \neq v_j$ if $i \neq j$. We define a map $f : \mathbf{I} \times \mathbb{N} \to |v_0 * L|$ as follows:

$$f(t, n) = (1 - t)v_0 + t v_n \quad \text{for each } (t, n) \in \mathbf{I} \times \mathbb{N}.$$

Then, f is a tight PL map but $f^{-1}(v_0)$ is not compact.

By applying Proposition 4.10.7, the following can be easily proved (the proof is left to the reader):

Proposition 4.11.9 *The composition of tight PL maps is a tight PL map.*

The following is a combination of Corollary 4.11.8 and Proposition 4.11.9 (cf. Fact preceding Lemma 4.11.6):

Proposition 4.11.10 *Let $f : P \to Q$ be a tight PL map between polyhedra. For every subpolyhedron P_0 of P, the restriction $f|P_0 : P_0 \to Q$ is a tight PL map and the image of $f(P_0)$ is a subpolyhedron of Q.* □

As mentioned in Remark 4.11.3, the above Proposition 4.11.10 does not hold for a PL map in general.

The following is a combination of Whitehead's Theorem 4.8.12 on small subdivisions and the Simplicial Approximation Theorem 4.10.11:

Theorem 4.11.11 (PL APPROXIMATION) *Let K and L be simplicial complexes and $f : |K| \to |L|$ be a map. For each open cover $\mathcal{U}$ of $|L|$, there is a PL map $g : |K| \to |L|$ that is $\mathcal{U}$-homotopic to f.* □

Remark 4.11.5 Recall that for each simplicial map $f : K \to L$ between simplicial complexes K and L, both $f : |K| \to |L|$ and $f : |K|_{\mathrm{m}} \to |L|_{\mathrm{m}}$ are continuous (Proposition 4.10.2). However, for a PL map $f : |K| \to |L|$, $f : |K|_{\mathrm{m}} \to |L|_{\mathrm{m}}$ is not necessarily continuous even if f is bijective. Indeed, consider the example of $K' \lhd K$ with $|K'|_{\mathrm{m}} \neq |K|_{\mathrm{m}}$ which was given in Remark 4.6.2 (Fig. 4.6). Let $L = K$ and define a simplicial map $f : K' \to \mathrm{Sd}\, K$ by $f(\mathbf{0}) = \mathbf{0}$, $f(\mathbf{e}_i) = \mathbf{e}_i$, and $f(2^{-i}\mathbf{e}_i) = 2^{-1}\mathbf{e}_i$, where $2^{-1}\mathbf{e}_i$ is the barycenter of $\langle \mathbf{0}, \mathbf{e}_i \rangle$. Then, $f : |K| \to |L| = |K|$ is a PL map, $f : |K|_{\mathrm{m}} \to |L|_{\mathrm{m}} = |K|_{\mathrm{m}}$ is not continuous, because $2^{-i}\mathbf{e}_i \to \mathbf{0}$ in $|K|_{\mathrm{m}}$ but $f(2^{-i}\mathbf{e}_i) = 2^{-1}\mathbf{e}_i \not\to f(\mathbf{0}) = \mathbf{0}$ in $|L|_{\mathrm{m}}$.

Let K and L be simplicial complexes. A map $f : |K|_{\mathrm{m}} \to |L|_{\mathrm{m}}$ is defined to be an **admissible PL map** if there exists an admissible subdivision $K' \lhd K$ such that f is affine on each simplex of K'.

Note This definition depends on simplicial complexes K and L. If the Common Admissible Subdivision Problem mentioned at the end of Sect. 4.7 is affirmative, being admissible PL is independent from the choice of admissible triangulations, so the above can be regarded as the definition of an admissible PL map between metric polyhedra. In fact, when $|K_1|_{\mathrm{m}} = |K_2|_{\mathrm{m}}$ and $|L_1|_{\mathrm{m}} = |L_2|_{\mathrm{m}}$ as spaces, let $f : |K_1|_{\mathrm{m}} = |K_2|_{\mathrm{m}} \to |L_1|_{\mathrm{m}} = |L_2|_{\mathrm{m}}$ be a map and assume that K_1 has an admissible subdivision K_1' such that $f|\sigma$ is affine for each $\sigma \in K_1'$. If we can show that K_2 also has such an admissible subdivision K_2'. then we can define an admissible PL map between metric polyhedra.

As a combination of Theorem 4.9.5 (the metric topology version of Whitehead's Theorem 4.8.12), the Admissible Simplicial Approximation Theorem 4.10.13, and Proposition 4.6.6, we have the following version of the PL Approximation Theorem 4.11.11:

Theorem 4.11.12 (ADMISSIBLE PL APPROXIMATION) *Let K and L be simplicial complexes. For each open cover $\mathcal{U}$ of $|L|_{\mathrm{m}}$, every map $f : |K|_{\mathrm{m}} \to |L|_{\mathrm{m}}$ is $\mathcal{U}$-homotopic to an admissible PL map $g : |K|_{\mathrm{m}} \to |L|_{\mathrm{m}}$, where $g : K' \to L'$ is simplicial with respect to admissible subdivisions $K' \lhd K$ and $L' \lhd L$.* □

Remark 4.11.6 Due to Proposition 4.5.6 (2), any contiguous PL maps $f, g : |K| \to |L|$ are homotopic by the straight-line homotopy $h : |K| \times \mathbf{I} \to |L|$. Then, each $h_t : |K| \to |L|$ is PL, but $h : |K \times_c I| \to |L|$ is not PL in general. In fact, by Proposition

4.3.7, K has a subdivision K' such that both $f|C$ and $g|C$ are affine for each $C \in K'$. Then, $h_t|C$ is affine by definition. As an example of the straight-line homotopy h being non-PL, consider the affine maps $f, g : \mathbf{I} \to \mathbf{I}^2$ defined by $f(s) = (s, 0)$ and $g(s) = (0, s)$. In this case, the straight-line homotopy h is defined by $h(s, t) = ((1 - t)s, ts)$. Note that

$$h((1-t)(0,0) + t(1,1)) = h(t,t) = (t - t^2, t^2) \text{ for each } t \in \mathbf{I}.$$

Any cell complex K with $|K| = \mathbf{I}^2$ has a cell C such that $A = C \cap \langle (0,0), (1,1) \rangle$ is a non-degenerate line segment. Then, $h|C$ is not affine.

A **piecewise linear** (**PL**) **homeomorphism** can be literally defined as a homeomorphism being PL. As is easily derived from Theorem 4.11.1, the inverse of a PL homeomorphism is also a PL homeomorphism. By Proposition 4.11.3 the composition of PL homeomorphisms is a PL homeomorphism. Every PL homeomorphism $f : P \to Q$ between polyhedra is tight PL (Corollary 4.11.8). Hence, f is a simplicial isomorphism (= bijective simplicial map) with respect to some triangulations K and L of P and Q, respectively (cf. the definition of a simplicial isomorphism in Sect. 4.10). Obviously, every simplicial isomorphism is a PL homomorphism. Thus, we have the following characterization:

Proposition 4.11.13 *Let P and Q be polyhedra. Then, $f : P \to Q$ is a PL homeomorphism if and only if f is a simplicial isomorphism with respect to some triangulations K and L of P and Q, respectively.* □

The above can be adopted as the definition of a PL homeomorphism but if that is the case, it would be non-trivial that the composition of PL homeomorphisms is a PL homeomorphism.

Two polyhedra P and Q are said to be **PL homeomorphic** (or P is **PL homeomorphic** to Q) if there exists a PL homeomorphism $f : P \to Q$, which is denoted as follows:

$$P \overset{\text{PL}}{\approx} Q.$$

By the above arguments, we have the following characterization:

Theorem 4.11.14 *For polyhedra P and Q, the following are equivalent:*

(a) *P is PL homeomorphic to Q;*
(b) *P and Q have simplicially isomorphic (congruent) triangulations;*
(c) *P and Q are combinatorially equivalent triangulations;*
(d) *Any triangulations of P and Q are combinatorially equivalent.* □

This can be restated as follows:

Theorem 4.11.15 *For cell complexes K and L,*

$$|K| \overset{PL}{\approx} |L| \;\Leftrightarrow\; K \cong L \;\Leftrightarrow\; \exists K' \lhd K,\ \exists L' \lhd L \text{ such that } K' \equiv L'. \qquad \square$$

Remark 4.11.7 Every PL bijection between compact polyhedra is a PL homeomorphism. However, a PL bijection $f : |K| \to |L|$ is, in general, not a PL homeomorphism. For example, define $f : \mathbb{R}_+ \to \partial \mathbf{I}^2$ as follows:

$$f(t) = \begin{cases} (t, 0) & \text{if } t \in \mathbf{I} = [0, 1], \\ (1, t-1) & \text{if } t \in [1, 2], \\ (3-t, 1) & \text{if } t \in [2, 3], \\ (0, 2^{-n+2}(n+2-t)) & \text{if } t \in [n, n+1], n \geqslant 3. \end{cases}$$

Then, f is a PL bijection that is not a PL homeomorphism.

Remark 4.11.8 For cell complexes K and L, the following implications hold:

$$|K| = |L| \underset{(4.3.7)}{\Rightarrow} K \cong L \underset{(4.11.14)}{\Leftrightarrow} |K| \overset{\text{PL}}{\approx} |L| \Rightarrow |K| \approx |L|.$$

Although it goes without saying that the converse of the first implication does not hold, the converse of the last does not either. The converse of the second implication is called **Hauptvermutung** (*the fundamental conjecture*). Refer to Remark 4.4.1.

An embedding $f : P \to Q$ is called a **PL embedding** if $f(P)$ is a subpolyhedron of Q and $f : P \to f(P)$ is a PL homeomorphism. Note that every PL embedding is a closed embedding because every subpolyhedron is closed. If a PL map is a closed embedding, then it is proper, hence it is a tight PL map by Corollary 4.11.8. Thus, we have the following:

Theorem 4.11.16 *Let P and Q be polyhedra. For a map $f : P \to Q$, the following are equivalent:*

(a) *f is a PL embedding;*
(b) *f is PL and a closed embedding;*
(c) *f is injective and simplicial with respect to some triangulations of P and Q.* □

Every injective PL map of a compact polyhedron is a PL embedding. As observed in Remark 4.11.7, an injective PL map is not always a PL embedding.

Let (P_1, Q_1) and (P_2, Q_2) be pairs of polyhedra and subpolyhedra. We call $f : (P_1, Q_1) \to (P_2, Q_2)$ a **PL map** if $f : P_1 \to P_2$ and $f|Q_1 : Q_1 \to Q_2$ are PL maps, where the condition for $f|Q_1$ can be omitted by virtue of Corollary 4.11.4. According to Proposition 4.11.10, if f is a tight PL, then $f|Q_1$ is also a tight PL map. In this case, $f : (P_1, Q_1) \to (P_2, Q_2)$ is called a **tight PL map**. We also call $f : (P_1, Q_1) \to (P_2, Q_2)$ a **PL homeomorphism** if $f : P_1 \to P_2$ and $f|Q_1 : Q_1 \to Q_2$ are PL homeomorphism, where the condition for $f|Q_1$ can be replaced by the condition $f(Q_1) = Q_2$. It is said that (P_1, Q_1) and (P_2, Q_2) are **PL homeomorphic** (or (P_1, Q_1) is **PL homeomorphic** to (P_2, Q_2)) if there exists a PL homeomorphism $f : (P_1, Q_1) \to (P_2, Q_2)$, which is denoted by

$$(P_1, Q_1) \overset{\text{PL}}{\approx} (P_2, Q_2).$$

In the case $Q_1 = Q_2 = Q$ (i.e., Q is a subpolyhedron of both P_1 and P_2), when there exists a PL homeomorphism $f : P_1 \to P_2$ such that $f|Q = \mathrm{id}$, it is said that P_1 and P_2 are **PL homeomorphic relative to** Q (or P_1 is **PL homeomorphic** to P_2 **relative to** Q), denoted by

$$P_1 \overset{\text{PL}}{\approx} P_2 \ \text{ rel. } Q.$$

For pairs (K_1, L_1) and (K_2, L_2) of cell complexes and subcomplexes,

$$(K_1, L_1) \cong (K_2, L_2) \quad \Leftrightarrow \quad (|K_1|, |L_1|) \overset{\text{PL}}{\approx} (|K_2|, |L_2|).$$

When $|L_1| = |L_2|$ as spaces, the following equivalence holds:

$$K_1 \cong K_2 \ \text{ rel. } L_1 \wedge L_2 \quad \Leftrightarrow \quad |K_1| \overset{\text{PL}}{\approx} |K_2| \ \text{ rel. } |L_1| = |L_2|.$$

4.12 Derived Neighborhoods and Regular Neighborhoods

This section is devoted to derived neighborhoods and regular neighborhoods, which are important tools in PL Topology.

For a simplicial complex K and its full subcomplex L, let K' be a derived subdivision of K with $\mathrm{Sd}\, L \subset K'$, that is, $\mathrm{Sd}\, L$ is the subdivision of L induced from K'. We call $N(\mathrm{Sd}\, L, K')$ a **derived neighborhood** of L in K. Recall that K' is defined by choosing $v_\sigma \in \operatorname{rint} \sigma$ for each $K \setminus L$ and $v_\sigma = \hat{\sigma}$ for each $\sigma \in L$ (cf. Sect. 4.3). Then, we can define

$$N(\mathrm{Sd}\, L, K') = \left\{ \langle v_{\sigma_1}, \ldots, v_{\sigma_n} \rangle \;\middle|\; \sigma_1 < \cdots < \sigma_n \in K[L] \right\}.$$

We have a simplicial isomorphism $h : K' \to \mathrm{Sd}\, K$ defined by $h(v_\sigma) = \hat{\sigma}$ for each $\sigma \in K$. Then, $h||L| = \mathrm{id}$ and $h(N(\mathrm{Sd}\, L, K')) = N(\mathrm{Sd}\, L, \mathrm{Sd}\, K)$. Thus,

$$N(\mathrm{Sd}\, L, K') \equiv N(\mathrm{Sd}\, L, \mathrm{Sd}\, K) \ \text{ rel. } \mathrm{Sd}\, L.$$

Consequently, a derived neighborhood of L in K is *simplicially unique*.

Remark 4.12.1 Since $C(L, K)$ is not involved in defining a derived neighborhood of L in K, $N(\mathrm{Sd}\, L, K')$ does not change even if we replace v_σ with $\hat{\sigma}$ for every $\sigma \in C(L, K)$, that is, for K', we require not only $\mathrm{Sd}\, L \subset K'$ but also $\mathrm{Sd}\, C(L, K) \subset K'$. In this case, there exists a simplicial isomorphism $h : K' \to \mathrm{Sd}\, K$ such that $h||L| \cup |C(L, K)| = \mathrm{id}$ and $h(N(\mathrm{Sd}\, L, K')) = N(\mathrm{Sd}\, L, \mathrm{Sd}\, K)$.

Let K be a simplicial complex and L a subcomplex of K. We will use the maps $\delta_L^K, \beta_L^K : |K| \to \mathbb{R}$ defined in Sect. 4.7. Recall they are defined as follows:

$$\delta_L^K(x) = \rho_K(x, |L|) \text{ and } \beta_L^K(x) = \sum_{v \in L^{(0)}} \beta_v^K(x) \text{ for each } x \in |K|.$$

Then, for each $\varepsilon > 0$,

$$\mathrm{N}_{\rho_K}(|L|, \varepsilon) = (\delta_L^K)^{-1}([0, \varepsilon)) \text{ and } \overline{\mathrm{N}}_{\rho_K}(|L|, \varepsilon) = (\delta_L^K)^{-1}([0, \varepsilon])$$

The following were shown in Sect. 4.7:

- *When L is full in K, $\delta_L^K = 2(1 - \beta_L^K) = 2\beta_{C(L,K)}^K$, which is affine on each simplex of K* — Lemma 4.7.5.
- *When L is full in K, there is a retraction $r_L : |K| \setminus |C(K, L)| \to |L|$ such that $r(x) \in c_K(x) \cap |L|$ and $\rho_K(x, r_L(x)) = \rho_K(x, |L|)$ for each $x \in |K| \setminus |C(L, K)|$* — Proposition 4.7.6.
- *L is full in K if and only if $|L| = (\beta_L^K)^{-1}(1) = (\beta_{C(L,K)}^K)^{-1}(0)$* — Lemma 4.7.7.

For each full subcomplex L of K and $0 < \varepsilon < 2$, we defined in Sect. 4.9 a cell complex K_L^ε subdividing K such that

$$L \cup C(L, K) \subset K_L^\varepsilon \text{ and } |N(L, K_L^\varepsilon)| = \overline{\mathrm{N}}_{\rho_K}(|L|, \varepsilon).$$

We recall the construction of K_L^ε. Each $\sigma \in K[L] \setminus L$ can be written as $\sigma = \sigma_L \sigma^L$, where $\sigma_L = \sigma \cap |L| \in L$ and $\sigma^L \in C(L, K)$ is the opposite face of σ to σ_L. For $0 < \varepsilon < 2$, $(\delta_L^K)^{-1}(\varepsilon)$ divides σ into the following two cells:

$$\sigma_L^\varepsilon = \sigma \cap (\delta_L^K)^{-1}([0, \varepsilon]), \ \sigma_\varepsilon^L = \sigma \cap (\delta_L^K)^{-1}([\varepsilon, 2]),$$

where $\sigma_L^\varepsilon \cap \sigma_\varepsilon^L = \sigma \cap (\delta_L^K)^{-1}(\varepsilon)$ is denoted by $\sigma_L(\varepsilon)$. Then, we define

$$K_L^\varepsilon \underset{\text{def}}{=} L \cup C(L, K) \cup \left\{ \sigma_L^\varepsilon, \sigma_\varepsilon^L, \sigma_L(\varepsilon) \;\middle|\; \sigma \in K[L] \setminus L \right\}.$$

We call $N(L, K_L^\varepsilon)$ the ***ε*-neighborhood** of L in K, which is denoted by $N_\varepsilon(L, K)$, i.e.,

$$N_\varepsilon(L, K) \underset{\text{def}}{=} L \cup \left\{ \sigma_L^\varepsilon, \sigma_L(\varepsilon) \;\middle|\; \sigma \in K[L] \setminus L \right\}.$$

Note Let $I = \{0, 1, \mathbf{I}\}$ be the natural triangulation of $\mathbf{I}$. For each $0 < a < 1$,[25]

$$I_a = \{0, a, 1, [0, a], [a, 1]\} \lhd I.$$

Then, $\beta_{C(L,K)}^K = 2^{-1}\delta_L^K : K \to I$ is a simplicial map. Observe that

[25] I_a is a starring of I at a, which was defined just before Lemma 4.7.4.

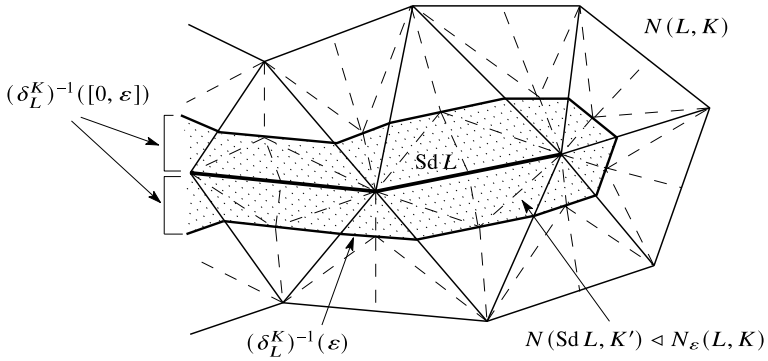


Fig. 4.12 $N(\mathrm{Sd}\,L, K') \lhd N_\varepsilon(L, K)$

$$\begin{aligned} K_L^\varepsilon &= K \wedge (\beta^K_{C(L,K)})^{-1}(I_{\varepsilon/2}) \\ &= \left\{\sigma \cap (\beta^K_{C(L,K)})^{-1}(\tau) \,\middle|\, \sigma \in K,\ \tau \in I_{\varepsilon/2},\ \sigma \cap (\beta^K_{C(L,K)})^{-1}(\tau) \neq \emptyset\right\}; \\ N_\varepsilon(L,K) &= \left\{\sigma \cap (\beta^K_{C(L,K)})^{-1}(\tau) \,\middle|\, \sigma \in K[L],\ \tau \leqslant [0,\varepsilon/2],\right. \\ &\qquad\qquad \left. \sigma \cap (\beta^K_{C(L,K)})^{-1}(\tau) \neq \emptyset\right\}. \end{aligned}$$

In fact, for $\sigma \in K$ and $\tau \in I_{\varepsilon/2}$,

$$\sigma \cap (\beta^K_{C(L,K)})^{-1}(\tau) = \begin{cases} \sigma & \text{if } \sigma \in L, 0 \in \tau, \text{ or } \sigma \in C(L,K), 1 \in \tau, \\ \emptyset & \text{if } \sigma \in L, 0 \notin \tau, \text{ or } \sigma \in C(L,K), 1 \notin \tau, \\ \sigma^L & \text{if } \sigma \in K[L] \setminus L, \tau = 1, \\ \sigma^L_\varepsilon & \text{if } \sigma \in K[L] \setminus L, \tau = [\varepsilon/2, 1], \\ \sigma_L(\varepsilon) & \text{if } \sigma \in K[L] \setminus L, \tau = \varepsilon/2, \\ \sigma^\varepsilon_L & \text{if } \sigma \in K[L] \setminus L, \tau = [0, \varepsilon/2], \\ \sigma_L & \text{if } \sigma \in K[L] \setminus L, \tau = 0. \end{cases}$$

Note that $K \setminus (L \cup C(L,K)) = K[L] \setminus L$. Choosing $v_\sigma \in \mathrm{rint}\,\sigma \cap (\delta_L^K)^{-1}(\varepsilon)$ for each $\sigma \in K[L] \setminus L$ and letting $v_\sigma = \hat{\sigma}$ for $\sigma \in L \cup C(L,K)$, we can obtain a derived subdivision K' of K such that

$$\mathrm{Sd}\,L \cup \mathrm{Sd}\,C(L,K) \subset K' \lhd K_L^\varepsilon \ \text{ and } \ N(\mathrm{Sd}\,L, K') \lhd N_\varepsilon(L,K).$$

Then, $N(\mathrm{Sd}\,L, K')$ is a derived neighborhood of L in K with $|N(\mathrm{Sd}\,L, K')| = |N_\varepsilon(L,K)| = (\delta_L^K)^{-1}([0,\varepsilon])$ (see Fig. 4.12). As observed in Remark 4.12.1, it should be noted that there exists a simplicial isomorphism $h : K' \to \mathrm{Sd}\,K$ such that $h(N(\mathrm{Sd}\,L, K')) = N(\mathrm{Sd}\,L, \mathrm{Sd}\,K)$ and $h|\,\mathrm{Sd}\,L \cup \mathrm{Sd}\,C(L,K) = \mathrm{id}$.

Let X be a space with $A \subset X$. A retraction map $r : X \to A$ is called a **strong deformation retraction** if $r \simeq \mathrm{id}$ rel. A in X, where A is called a **strong deformation retract** of X.[26]

[26] In Chap. 6, we will discuss retracts and strong deformation retracts.

Proposition 4.12.1 *Let K be a simplicial complex with a full subcomplex L and K' a derived subdivision of K such that* $\mathrm{Sd}\, L \subset K'$. *Then, there is a simplicial retraction* $r : N(\mathrm{Sd}\, L, K') \to \mathrm{Sd}\, L$ *such that* $r \simeq_K \mathrm{id}$ *rel.* $|L|$ *in* $|N(\mathrm{Sd}\, L, K')|$ *(hence* $|L|$ *is a strong deformation retract of* $|N(\mathrm{Sd}\, L, K')|$*) and*

$$r^{-1}(O_L(v)) \subset O_K(v) \cap |N(\mathrm{Sd}\, L, K')| \ \text{ for each } v \in L^{(0)}.$$

Proof Since all derived neighborhoods are congruent (simplicially isomorphic), we can assume

$$|N(\mathrm{Sd}\, L, K')| = |N_1(L, K)| = (\delta_L^K)^{-1}([0, 1]),$$

where $K' \lhd K_L^1$ is defined by choosing $v_\sigma \in \mathrm{rint}\, \sigma \cap (\delta_L^K)^{-1}(1)$ for each $\sigma \in K[L] \setminus L$ and $v_\sigma = \hat{\sigma}$ for each $\sigma \in L \cup C(L, K)$. Then, we can write

$$N(\mathrm{Sd}\, L, K') = \left\{ \langle v_{\sigma_1}, \ldots, v_{\sigma_n} \rangle \;\middle|\; \sigma_1 < \cdots < \sigma_n \in K[L] \right\}.$$

Since L is full in K, we have $\sigma_L = \sigma \cap |L| \leqslant \sigma$ for each $\sigma \in K[L]$. Observe that $\sigma' < \sigma \in K[L]$ implies $\sigma'_L \leqslant \sigma_L$. Hence, there exists a simplicial map $r : N(\mathrm{Sd}\, L, K') \to \mathrm{Sd}\, L$ such that $r(v_\sigma) = \hat{\sigma}_L = v_{\sigma_L}$ for each $\sigma \in K[L] \setminus L$ and $r||L| = \mathrm{id}$, that is, r is a simplicial retraction.

For each $x \in |N(\mathrm{Sd}\, L, K')|$, the carrier $c_{K'}(x) \in K'$ is written as follows:

$$c_{K'}(x) = \langle v_{\sigma_1}, \ldots, v_{\sigma_n} \rangle, \ \sigma_1 < \cdots < \sigma_n \in K[L].$$

Then, observe that $c_K(x) = \sigma_n$ and

$$\begin{aligned} c_{\mathrm{Sd}\, L}(r(x)) = r(c_{K'}(x)) &= \langle v_{\sigma_{1L}}, \ldots, v_{\sigma_{nL}} \rangle \\ &\subset c_L(r(x)) = \sigma_{nL} = \sigma_n \cap |L| \leqslant \sigma_n. \end{aligned}$$

Hence, $x, r(x) \in \sigma_n = c_K(x)$. For $v \in L^{(0)}$, $\beta_v^L(r(x)) > 0$ implies $\beta_v^K(x) > 0$ because $v \in c_L(r(x))^{(0)} \subset \sigma_n^{(0)} = c_K(x)^{(0)}$, which means that

$$r^{-1}(O_L(v)) \subset O_K(v) \cap |N(\mathrm{Sd}\, L, K')|.$$

Moreover, for each $x \in |N(\mathrm{Sd}\, L, K')| = (\delta_L^K)^{-1}([0, 1])$, $\delta_L^K(x) \leqslant 1$ and $\delta_L^K(r(x)) = 0$. Since $r(x) \in c_K(x) \in K$ and $\delta_L^K | c_K(x)$ is affine, it follows that

$$\delta_L^K((1 - t)x + tr(x)) = (1 - t)\delta_L^K(x) \leqslant 1 \ \text{ for each } t \in \mathbf{I}.$$

This means that the homotopy tracks of the straight-line homotopy from $\mathrm{id}_{|N(\mathrm{Sd}\, L, K')|}$ to r are contained in $|N(\mathrm{Sd}\, L, K')|$. Hence, $r \simeq_K \mathrm{id}$ rel. $|L|$ in $|N(\mathrm{Sd}\, L, K')|$. □

Remark 4.12.2 In Proposition 4.12.1, $\simeq_K$ cannot be replaced by $\simeq_{K'}$ (see Fig. 4.13). In the above proof, $\mathrm{id}_{|N(\mathrm{Sd}\, L, K')|} \simeq_K r$ by the straight-line homotopy h, which

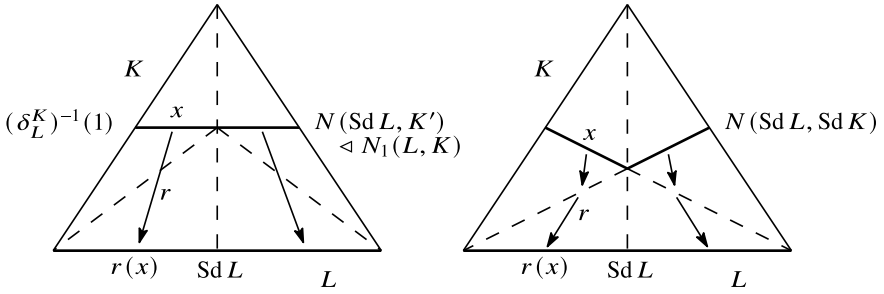


Fig. 4.13 $r : N(\mathrm{Sd}\,L, K') \to \mathrm{Sd}\,L$ and $r : N(\mathrm{Sd}\,L, \mathrm{Sd}\,K) \to \mathrm{Sd}\,L$

has the property that each h_t is PL. Since all derived neighborhoods are congruent (simplicially isomorphic), any derived neighborhood $N(\mathrm{Sd}\,L, K')$ satisfies this property, that is, $r \simeq_K \mathrm{id}$ rel. $|L|$ in $|N(\mathrm{Sd}\,L, K')|$ by a homotopy h such that each h_t is PL, but it is unknown whether we can take the straight-line homotopy as h or not. For example, if we use $\mathrm{Sd}\,K$ instead of K', then it would be bothersome to find out whether the line segment from $x \in |N(\mathrm{Sd}\,L, K')|$ to $r(x) \in |\mathrm{Sd}\,L|$ is contained in $|N(\mathrm{Sd}\,L, \mathrm{Sd}\,K)|$ or not. It should be noted that the image of the line segment from $x \in |N(\mathrm{Sd}\,L, K')|$ to $r(x) \in |\mathrm{Sd}\,L|$ by the simplicial isomorphism from $N(\mathrm{Sd}\,L, K')$ onto $N(\mathrm{Sd}\,L, \mathrm{Sd}\,K)$ is a broken line segment (cf. Fig. 4.13).

Note There are some variations in defining a derived neighborhood of L in K. If an arbitrary derived subdivision of K is used, then the uniqueness of a derived neighborhood of L in K is a little loose. When a derived subdivision K' of K relative to L is used, the notation is simpler and the uniqueness can be written as

$$N(L, K') \equiv N(L, \mathrm{Sd}_L\, K) \ \text{ rel. } L.$$

Moreover, in relation to the ε-neighborhood of L in K, we can take a derived subdivision K' of K relative to $L \cup C(L, K)$ such that $N(L, K') \lhd N_\varepsilon(L, K)$ and $K' \lhd K_L^\varepsilon$. Then, there exists a simplicial isomorphism $h : K' \to \mathrm{Sd}_{L\cup C(L,K)}\, K$ such that

$$h(N(L, K')) = N(L, \mathrm{Sd}_{L\cup C(L,K)}\, K) \ \text{ and } \ h|L \cup C(L, K) = \mathrm{id},$$

where $N(L, \mathrm{Sd}_{L\cup C(L,K)}\, K) = N(L, \mathrm{Sd}_L\, K)$. However, in order that we can naturally define a simplicial retraction of $N(\mathrm{Sd}\,L, K')$ as in Proposition 4.12.1 while preserving the uniqueness of $N(\mathrm{Sd}\,L, K')$, we used a derived subdivision K' of K with $\mathrm{Sd}\,L \subset K'$.

Since $N(\mathrm{Sd}\,L, \mathrm{Sd}\,K) \neq \mathrm{Sd}\,N(L, K)$ in general, the following is non-trivial:

Proposition 4.12.2 *For a simplicial complex K with a full subcomplex L, there exists a PL homeomorphism $h : |K| \to |K|$ such that $h||L| \cup |C(L, K)| = \mathrm{id}$ and $h(|N(\mathrm{Sd}\,L, \mathrm{Sd}\,K)|) = |N(\mathrm{Sd}^2\,L, \mathrm{Sd}^2\,K)|$. Consequently,*

$$N(\mathrm{Sd}\,L, \mathrm{Sd}\,K) \cong N(\mathrm{Sd}^2\,L, \mathrm{Sd}^2\,K) \ \textit{rel.}\ \mathrm{Sd}^2\,L.$$

Proof Let K_1 and K_2 be derived subdivisions of K such that

$$\operatorname{Sd} L \cup \operatorname{Sd} C(L, K) \subset K_i \lhd K_L^{1/i} \text{ and } N(\operatorname{Sd} L, K_i) \lhd N_{1/i}(L, K) \text{ for } i = 1, 2.$$

For each $\sigma \in K_1[L] \setminus \operatorname{Sd} L$, since $\operatorname{rint}\sigma \cap (\delta_L^K)^{-1}(1/2) \neq \emptyset$, we can choose $v_\sigma \in \operatorname{rint}\sigma \cap (\delta_L^K)^{-1}(1/2)$ to define a derived subdivision K' of K_1 such that

$$\operatorname{Sd}^2 L \cup \operatorname{Sd} C(\operatorname{Sd} L, K_1) \subset K' \text{ and}$$
$$|N(\operatorname{Sd}^2 L, K')| = |N_{1/2}(L, K)| = \overline{\mathrm{N}}_{\rho_K}(|N|, 1/2) = |N(\operatorname{Sd} L, K_2)|,$$

where it should be noted that $\operatorname{Sd}^2 C(L, K) \subset \operatorname{Sd} C(\operatorname{Sd} L, K_1) \subset \operatorname{Sd} K_1$. Then, we have three simplicial isomorphisms $h_1 : \operatorname{Sd} K \to K_2, h_2 : \operatorname{Sd}^2 K \to \operatorname{Sd} K_2$, and $h_3 : \operatorname{Sd} K_2 \to K'$ such that

$$h_1|\operatorname{Sd} L \cup \operatorname{Sd} C(L, K) = \mathrm{id},\ h_1(N(\operatorname{Sd} L, \operatorname{Sd} K)) = N(\operatorname{Sd} L, K_2),$$
$$h_2|\operatorname{Sd}^2 L \cup \operatorname{Sd} C(\operatorname{Sd} L, K_1) = \mathrm{id},\ h_2(N(\operatorname{Sd}^2 L, \operatorname{Sd} K_2)) = N(\operatorname{Sd}^2 L, K'),$$
$$h_3|\operatorname{Sd}^2 L \cup \operatorname{Sd}^2 C(L, K) = \mathrm{id}, \text{ and } h_3(N(\operatorname{Sd}^2 L, \operatorname{Sd}^2 K)) = N(\operatorname{Sd}^2 L, \operatorname{Sd} K_2).$$

Then, the composition $h = h_3^{-1}h_2^{-1}h_1 : |K| \to |K|$ is the desired PL homeomorphism. Indeed, $h||L| \cup |C(L, K)| = \mathrm{id}$ and

$$\begin{aligned} h(|N(\operatorname{Sd} L, \operatorname{Sd} K)|) &= h_3^{-1}h_2^{-1}(|N(\operatorname{Sd} L, K_2)|) = h_3^{-1}h_2^{-1}(|N(\operatorname{Sd}^2 L, K')|) \\ &= h_3^{-1}(|N(\operatorname{Sd}^2 L, \operatorname{Sd} K_2)|) = |N(\operatorname{Sd}^2 L, \operatorname{Sd}^2 K)|. \end{aligned}$$ □

Remark 4.12.3 When $\dim L < \infty$, the above argument can be made a little simpler. Let $0 < \varepsilon < 2/(n+2)$, where $n = \dim L$. Then,

$$|N_\varepsilon(L, K)| = \overline{\mathrm{N}}_{\rho_K}(|L|, \varepsilon) \subset |K| \setminus |C(\operatorname{Sd} L, \operatorname{Sd} K)|.$$

Indeed, for each $\sigma \in K \setminus K[L]$, $\rho_K(\hat{\sigma}, |L|) = 2 > \varepsilon$. For each $\sigma \in K[L] \setminus L$, let $\sigma_L = \sigma \cap |L| \neq \emptyset$. Then, $\sigma_L < \sigma$ because L is full in K. We can write $\sigma = \sigma_L\sigma^L$, where $\sigma^L < \sigma$ is the opposite face to σ_L. Observe that

$$\begin{aligned} \rho_K(\hat{\sigma}, |L|) = \delta_L^K(\hat{\sigma}) = 2\beta^K_{C(L,K)}(\hat{\sigma}) &= 2\frac{\dim \sigma^L + 1}{\dim \sigma + 1} \\ &= \frac{2\dim \sigma^L + 2}{\dim \sigma_L + \dim \sigma^L + 2} \geqslant \frac{2}{\dim \sigma_L + 2} \geqslant \frac{2}{n+2}. \end{aligned}$$

Thus, $\rho_K(x, |L|) \geqslant 2/(n+2) > \varepsilon$ for every $x \in |C(\operatorname{Sd} L, \operatorname{Sd} K)|$, because δ_L^K is affine on each simplex of K. Hence, $\overline{\mathrm{N}}_{\rho_K}(|L|, \varepsilon)$ does not meet with $|C(\operatorname{Sd} L, \operatorname{Sd} K)|$.

Since $\operatorname{rint}\tau \cap (\delta_L^K)^{-1}(\varepsilon) \neq \emptyset$ for each $\tau \in (\operatorname{Sd} K)[L] \setminus \operatorname{Sd} L$, we have a derived subdivision K_1 of $\operatorname{Sd} K$ such that $\operatorname{Sd}^2 L \subset K_1$ and $|N(\operatorname{Sd}^2 L, K_1)| = |N_\varepsilon(L, K)|$, where $K_1 \equiv \operatorname{Sd}^2 K$ rel. $\operatorname{Sd}^2 L$. Moreover, as in the above proof, we can also define a derived subdivision K_2 of $\operatorname{Sd} K$ such that $\operatorname{Sd} L \subset K_2$ and $|N(\operatorname{Sd} L, K_2)| = |N_\varepsilon(L, K)|$. Since $|N(\operatorname{Sd} L, K_1)| = |N(\operatorname{Sd}^2 L, K_2)|$ (as spaces), it follows that

$N(\mathrm{Sd}\, L, K_1) \cong N(\mathrm{Sd}^2 L, K_2)$ rel. $\mathrm{Sd}^2 L$. Furthermore, $N(\mathrm{Sd}^2 L, \mathrm{Sd}^2 K) \equiv N(\mathrm{Sd}^2 L, K_1)$ rel. $\mathrm{Sd}^2 L$ and $N(\mathrm{Sd}\, L, \mathrm{Sd}\, K) \equiv N(\mathrm{Sd}\, L, K_2)$ rel. $\mathrm{Sd}\, L$. Consequently, we have $N(\mathrm{Sd}\, L, \mathrm{Sd}\, K) \cong N(\mathrm{Sd}^2 L, \mathrm{Sd}^2 K)$ rel. $\mathrm{Sd}^2 L$.

When $\dim L = \infty$, the above argument cannot be applied. Indeed, if there are $\sigma_n \in L$, $v_n \in K^{(0)} \setminus L$, $n \in \mathbb{N}$, such that $\dim \sigma_n \geqslant n$ and $\tau_n = v_n \sigma_n \in K$, then we cannot take $0 < \varepsilon < 2$ as above because

$$\rho_K(\hat{\tau}_n, |L|) = 2\beta^K_{C(L,K)}(\hat{\tau}_n) = \frac{2}{\dim \tau_n + 1} = \frac{2}{\dim \sigma_n + 2} \leqslant \frac{2}{n+2} \to 0 \ \text{ as } n \to \infty.$$

Let K be a simplicial complex and L a full subcomplex of K. When K is locally finite and L is finite, every neighborhood of $|L|$ in $|K|$ contains some ε-neighborhood of L in K. However, when K is not locally finite, even if L is finite, this does not necessarily hold. Such an example can be easily constructed by referring to the example in Remark 4.6.2 (cf. Remark 4.12.3). So we will introduce a more general concept. Suppose that for every $\sigma \in K[L] \setminus L$, a hyperplane θ_σ of $\mathrm{fl}\, \sigma$ is chosen so as to satisfy the following two conditions:

(Sl_1) $\theta_\sigma \cap \sigma \neq \emptyset$ and $\theta_\sigma \cap \sigma \cap (|L| \cup |C(L, K)|) = \emptyset$;
(Sl_2) $\sigma, \sigma' \in K[L] \setminus L, \sigma' < \sigma \ \Rightarrow\ \theta_{\sigma'} = \theta_\sigma \cap \mathrm{fl}\, \sigma'$.

Then, each θ_σ cuts σ into two cells $\theta_-(\sigma)$ and $\theta_+(\sigma)$ with $\theta_-(\sigma) \cap |L| \neq \emptyset$ and $\theta_+(\sigma) \cap |L| = \emptyset$, where $\theta_-(\sigma) \cap \theta_+(\sigma) = \theta_\sigma \cap \sigma$ is denoted by $\theta(\sigma)$. The following cell complex is a subdivision of K:

$$K^\theta \underset{\text{def}}{=} C(L, K) \cup L \cup \big\{\theta_-(\sigma), \theta_+(\sigma), \theta(\sigma) \,\big|\, \sigma \in K[L] \setminus L\big\}.$$

For $\sigma \in K[L] \setminus L$, if σ meets $\sigma' \in C(L, K)$, then $\theta_+(\sigma) \cap \sigma' = \sigma \cap \sigma'$ and $\theta_-(\sigma) \cap \sigma' = \emptyset$; if σ meets $\sigma' \in L$, then $\theta_-(\sigma) \cap \sigma' = \sigma \cap \sigma'$ and $\theta_+(\sigma) \cap \sigma' = \emptyset$. For $\sigma_1, \sigma_2 \in K[L] \setminus L$, $\theta_-(\sigma_1) \cap \theta_+(\sigma_2) = \theta_+(\sigma_1) \cap \theta_-(\sigma_2) = \emptyset$ or $\theta_-(\sigma_1) \cap \theta_+(\sigma_2) = \theta_+(\sigma_1) \cap \theta_-(\sigma_2) = \theta(\sigma_1 \cap \sigma_2)$. If $\theta_-(\sigma_1) \cap \theta_-(\sigma_2) \neq \emptyset$ and $\theta_+(\sigma_1) \cap \theta_+(\sigma_2) \neq \emptyset$, then $\theta_-(\sigma_1) \cap \theta_-(\sigma_2) = \theta_-(\sigma_1 \cap \sigma_2)$ and $\theta_+(\sigma_1) \cap \theta_+(\sigma_2) = \theta_+(\sigma_1 \cap \sigma_2)$. If $\theta_-(\sigma_1) \cap \theta_-(\sigma_2) \neq \emptyset$ and $\theta_+(\sigma_1) \cap \theta_+(\sigma_2) = \emptyset$, then $\theta_-(\sigma_1) \cap \theta_-(\sigma_2) = \sigma_1 \cap \sigma_2 \in L$. If $\theta_-(\sigma_1) \cap \theta_-(\sigma_2) = \emptyset$ and $\theta_+(\sigma_1) \cap \theta_+(\sigma_2) \neq \emptyset$, then $\theta_+(\sigma_1) \cap \theta_+(\sigma_2) = \sigma_1 \cap \sigma_2 \in C(L, K)$.

We call $\theta = (\theta_\sigma)_{\sigma \in K[L] \setminus L}$ a **slicer** of K along L and the following subcomplex of K^θ a **sliced neighborhood** of L in K defined by θ:

$$N_\theta(L, K) \underset{\text{def}}{=} N(L, K^\theta) = L \cup \big\{\theta_-(\sigma), \theta(\sigma) \,\big|\, \sigma \in K[L] \setminus L\big\}.$$

For example, the ε-neighborhood $N_\varepsilon(L, K)$ is a sliced neighborhood of L in K defined by $(\varphi_\sigma^{-1}(\varepsilon))_{\sigma \in K[L] \setminus L}$, where $\varphi_\sigma : \mathrm{fl}\, \sigma \to \mathbb{R}$ is the unique affine extension of the following:

$$\delta_L^K|\sigma = 2\beta_{C(L,K)}^K|\sigma = 2\sum_{v\in\sigma^{(0)}\setminus L^{(0)}} \beta_v^K|\sigma : \sigma \to \mathbb{R}.$$

Similar to the ε-neighborhood $N_\varepsilon(L, K)$, a sliced neighborhood $N_\theta(L, K)$ can be subdivided by a derived neighborhood $N(\mathrm{Sd}\, L, K')$. In fact, choosing $v_\sigma \in \mathrm{rint}\,\theta(\sigma)$ for each $\sigma \in K[L] \setminus L$ (let $v_\sigma = \hat{\sigma}$ for $\sigma \in L \cup C(L, K)$), we can obtain a derived subdivision K' of K such that $\mathrm{Sd}\, L \cup \mathrm{Sd}\, C(L, K) \subset K' \lhd K^\theta$ and $N(\mathrm{Sd}\, L, K') \lhd N_\theta(L, K)$.

Due to the following lemma, a small sliced neighborhood is subdivided by a sliced neighborhood in a simplicial subdivision.

Lemma 4.12.3 *Let K be a simplicial complex with a full subcomplex L, and K' a simplicial subdivision of K with a subcomplex $L' \lhd L$. If $\theta = (\theta_\sigma)_{\sigma\in K[L]\setminus L}$ is a slicer of K along L such that $|N_\theta(L, K)| \subset |K| \setminus |C(L', K')|$, then there exists a slicer $\theta' = (\theta'_\tau)_{\tau\in K'[L]\setminus L'}$ of K' along L' such that $K'^{\theta'}$ is a subdivision of K^θ and $|N_{\theta'}(L', K')| = |N_\theta(L, K)|$.*

Proof Each $\tau \in K'[L] \setminus L'$ is contained in the carrier $c_K(\hat{\tau})$ of the barycenter $\hat{\tau}$ of τ and $c_K(\hat{\tau}) \in K[L] \setminus L$. Let $\theta'_\tau = \theta_{c_K(\hat{\tau})} \cap \mathrm{fl}\,\tau$. Then, θ'_τ satisfies (Sl_1) because

$$\theta'_\tau \cap \tau \cap (|L'| \cup |C(L', K')|) \subset \theta_{c_K(\hat{\tau})} \cap c_K(\hat{\tau}) \cap (|L| \cup |C(L', K')|) = \emptyset.$$

To see that $\theta' = (\theta'_\tau)_{\tau\in K'[L]\setminus L'}$ satisfies (Sl_2), let $\tau' < \tau \in K'[L] \setminus L$. Since $c_K(\hat{\tau}') \leqslant c_K(\hat{\tau})$, it follows that

$$\begin{aligned}\theta'_{\tau'} = \theta_{c_K(\hat{\tau}')} \cap \mathrm{fl}\,\tau' &= \theta_{c_K(\hat{\tau})} \cap \mathrm{fl}\, c_K(\hat{\tau}') \cap \mathrm{fl}\,\tau' \\ &= \theta_{c_K(\hat{\tau})} \cap \mathrm{fl}\,\tau \cap \mathrm{fl}\,\tau' = \theta'_\tau \cap \mathrm{fl}\,\tau'.\end{aligned}$$

Thus, $\theta' = (\theta'_\tau)_{\tau\in K'[L]\setminus L'}$ is a slicer of K' along L'. By the definition of θ', it is easy to see that $K'^{\theta'}$ is a subdivision of K^θ and $|N_{\theta'}(L', K')| = |N_\theta(L, K)|$. □

By the following lemma, we can take an arbitrarily small sliced neighborhood:

Lemma 4.12.4 *Let K be a simplicial complex and L a full subcomplex of K. For each open neighborhood U of $|L|$ in $|K|$, $\mathrm{Sd}\, K$ has a slicer $\theta = (\theta_\tau)_{\tau\in(\mathrm{Sd}\, K)[L]\setminus \mathrm{Sd}\, L}$ along $\mathrm{Sd}\, L$ such that $|N_\theta(\mathrm{Sd}\, L, \mathrm{Sd}\, K)| \subset U$.*

Proof Each $\tau \in (\mathrm{Sd}\, K)[L] \setminus \mathrm{Sd}\, L$ can be written as

$$\begin{aligned}\tau = \langle \hat{\sigma}'_1, \ldots, \hat{\sigma}'_k, \hat{\sigma}_1, \ldots, \hat{\sigma}_m\rangle,\ \sigma'_1 < \cdots < \sigma'_k \in L,& \\ \sigma_1 < \cdots < \sigma_m \in K \setminus L,\ \sigma'_k < \sigma_1,& \quad (\natural)\end{aligned}$$

where $k, m \in \mathbb{N}$. Choosing $0 < t_\tau < 1$ for each $\tau \in K[L] \setminus L$, we will define an affine functional $\psi_\tau : \mathrm{fl}\,\tau \to \mathbb{R}$ as the unique affine extension of $\sum_{i=1}^m t_{\sigma_i}^{-1} \beta_{\hat{\sigma}_i}^{\mathrm{Sd}\, K}|\tau : \tau \to \mathbb{R}$, where τ is represented as in $(\natural)$. Then, $\theta = (\psi_\tau^{-1}(1))_{\tau\in(\mathrm{Sd}\, K)[L]\setminus \mathrm{Sd}\, L}$ is a slicer of $\mathrm{Sd}\, K$ along $\mathrm{Sd}\, L$. Indeed,

$$\tau \cap |L| = \langle \hat{\sigma}'_1, \dots, \hat{\sigma}'_k \rangle = \bigcap_{i=1}^{m} \big(\beta^{\mathrm{Sd}\,K}_{\hat{\sigma}_i}\big)^{-1}(0) \cap \tau \subset \psi_\tau^{-1}(0) \ \text{ and}$$

$$x \in \tau \cap |C(\mathrm{Sd}\,L, \mathrm{Sd}\,K)| = \langle \hat{\sigma}_1, \dots, \hat{\sigma}_m \rangle$$

$$\Rightarrow\ \psi_\tau(x) = \sum_{i=1}^{m} t_{\sigma_i}^{-1} \beta^{\mathrm{Sd}\,K}_{\hat{\sigma}_i}(x) \geqslant \min\left\{t_{\sigma_1}^{-1}, \dots, t_{\sigma_m}^{-1}\right\} > 1.$$

This shows that θ satisfies (Sl_1). By the definition, it is easy to see that θ satisfies (Sl_2). For each $\tau \in (\mathrm{Sd}\,K)[L] \setminus \mathrm{Sd}\,L$, $\theta_-(\tau) = \psi_\tau^{-1}([0,1]) \cap \tau$ because $\tau \subset \psi_\tau^{-1}(\mathbb{R}_+)$ and $\tau \cap |L| \subset \psi_\tau^{-1}((-\infty, 1])$. Hence, θ defines the following sliced neighborhood of $\mathrm{Sd}\,L$ in $\mathrm{Sd}\,K$:

$$\big|N_\theta(\mathrm{Sd}\,L, \mathrm{Sd}\,K)\big| = |L| \cup \bigcup_{\tau \in (\mathrm{Sd}\,K)[L] \setminus \mathrm{Sd}\,L} \big(\psi_\tau^{-1}([0,1]) \cap \tau\big).$$

In order that $|N_\theta(\mathrm{Sd}\,L, \mathrm{Sd}\,K)| \subset U$, it suffices to define ψ_τ, $\tau \in (\mathrm{Sd}\,K)[L] \setminus \mathrm{Sd}\,L$, so that $\psi_\tau^{-1}([0,1]) \cap \tau \subset U$. Namely, we may find $0 < t_\sigma < 1$, $\sigma \in K[L] \setminus L$, so that when τ is represented as in (♮),

$$x \in \tau,\ \sum_{i=1}^{m} t_{\sigma_i}^{-1} \beta^{\mathrm{Sd}\,K}_{\hat{\sigma}_i}(x) \leqslant 1\ \Rightarrow\ x \in U. \tag{$*$}$$

It should be noticed that $K[L] = \bigcup_{n \in \mathbb{N}} K^{(n)}[L]$ and $(\mathrm{Sd}\,K)[L] = \bigcup_{n \in \mathbb{N}} (\mathrm{Sd}\,K^{(n)})[L]$. Thus, by induction on $n \in \mathbb{N}$, we may choose $0 < t_\sigma < 1$, $\sigma \in K^{(n)}[L] \setminus L$, so that $(*)$ is satisfied for each $\tau \in (\mathrm{Sd}\,K^{(n)})[L] \setminus \mathrm{Sd}\,L$.

(*The First Step*) Each $\sigma \in K^{(1)}[L] \setminus L$ has exactly one vertex $v_\sigma \in L^{(0)}$ because L is full in K. Choose $0 < t_\sigma < 1$ so that

$$\left\{(1-t)v_\sigma + t\hat{\sigma} \;\middle|\; 0 \leqslant t \leqslant t_\sigma\right\} \subset U.$$

For each $\tau \in (\mathrm{Sd}\,K^{(1)})[L] \setminus \mathrm{Sd}\,L$, we have a unique $\sigma \in K^{(1)}[L] \setminus L$ such that $\tau = \langle v_\sigma, \hat{\sigma} \rangle$. To see the condition $(*)$, take $x \in \tau$. If $t_\sigma^{-1} \beta^{\mathrm{Sd}\,K}_{\hat{\sigma}}(x) \leqslant 1$, i.e., $0 \leqslant \beta^{\mathrm{Sd}\,K}_{\hat{\sigma}}(x) \leqslant t_\sigma$, then

$$x = (1 - \beta^{\mathrm{Sd}\,K}_{\hat{\sigma}}(x))v_\sigma + \beta^{\mathrm{Sd}\,K}_{\hat{\sigma}}(x)\hat{\sigma} \in U.$$

(*The Inductive Step*) Assume that $0 < t_\sigma < 1, \sigma \in K^{(n)}[L] \setminus L$, have been chosen so that $(*)$ is satisfied for each $\tau \in (\mathrm{Sd}\,K^{(n)})[L] \setminus \mathrm{Sd}\,L$. Let $\sigma \in K^{(n+1)}[L] \setminus L$ and $\sigma \notin K^{(n)}[L] \setminus L$, that is, $\sigma \in K^{(n+1)}[L] \setminus (K^{(n)} \cup L)$. Since $\partial F(\sigma) \subset K^{(n)}$, we can define

$$M_\sigma = \big(\sigma \cap |L|\big) \cup \bigcup_{\tau \in (\mathrm{Sd}\,\partial F(\sigma))[L] \setminus \mathrm{Sd}\,L} \big(\psi_\tau^{-1}([0,1]) \cap \tau\big) \subset \sigma.$$

Then, $M_\sigma \subset U$ because $(*)$ is satisfied for each $\tau \in (\mathrm{Sd}\, K^{(n)})[L] \setminus \mathrm{Sd}\, L$. Hence, we can choose $0 < t_\sigma < 1$ so that

$$t_\sigma \leqslant \min\left\{t_{\sigma'} \mid \sigma' < \sigma\right\} \text{ and } \overline{\mathrm{N}}_{\rho_{\mathrm{Sd}\, K}}(M_\sigma, 2t_\sigma) \subset U.$$

For each $\tau \in (\mathrm{Sd}\, K^{(n+1)})[L] \setminus \mathrm{Sd}\, L$, we have to show that $(*)$ holds. Since $(*)$ holds for $\tau \in (\mathrm{Sd}\, K^{(n)})[L] \setminus \mathrm{Sd}\, L$ by the inductive assumption, we may assume that $\tau \notin \mathrm{Sd}\, K^{(n)}$. We write τ as in $(\natural)$, where $\dim \sigma_m = n+1$ because $\tau \notin \mathrm{Sd}\, K^{(n)}$. Let

$$\tau_0 = \begin{cases} \langle \hat{\sigma}'_1, \ldots, \hat{\sigma}'_k \rangle & \text{if } m = 1, \\ \langle \hat{\sigma}'_1, \ldots, \hat{\sigma}'_k, \hat{\sigma}_1, \ldots, \hat{\sigma}_{m-1} \rangle & \text{if } m > 1. \end{cases}$$

Then $\tau_0 < \tau$ and $\tau_0 \in (\mathrm{Sd}\, \partial F(\sigma_m))[L]$. To see the condition $(*)$, take $x \in \tau$ and assume that $\sum_{i=1}^m t_{\sigma_i}^{-1} \beta_{\hat{\sigma}_i}^{\mathrm{Sd}\, K}(x) \leqslant 1$. We define $y \in \tau_0$ as follows:

$$y = \begin{cases} \left(\beta_{\hat{\sigma}'_1}^{\mathrm{Sd}\, K}(x) + \beta_{\hat{\sigma}'_1}^{\mathrm{Sd}\, K}(x)\right)\hat{\sigma}'_1 + \displaystyle\sum_{i=2}^{k} \beta_{\hat{\sigma}'_i}^{\mathrm{Sd}\, K}(x)\hat{\sigma}'_i & \text{if } m = 1, \\ \left(\beta_{\hat{\sigma}_m}^{\mathrm{Sd}\, K}(x) + \beta_{\hat{\sigma}'_1}^{\mathrm{Sd}\, K}(x)\right)\hat{\sigma}'_1 + \displaystyle\sum_{i=2}^{k} \beta_{\hat{\sigma}'_i}^{\mathrm{Sd}\, K}(x)\hat{\sigma}'_i + \sum_{j=1}^{m-1} \beta_{\hat{\sigma}_j}^{\mathrm{Sd}\, K}(x)\hat{\sigma}_j & \text{if } m > 1. \end{cases}$$

If $m > 1$, then $\tau_0 \in \mathrm{Sd}\, K^{(n)}[L] \setminus \mathrm{Sd}\, L$ and

$$\begin{aligned} \psi(y) = \sum_{i=1}^{m-1} t_{\sigma_i}^{-1} \beta_{\hat{\sigma}_i}^{\mathrm{Sd}\, K}(y) &= t_{\sigma_1}^{-1}\left(\beta_{\hat{\sigma}_m}^{\mathrm{Sd}\, K}(x) + \beta_{\hat{\sigma}'_1}^{\mathrm{Sd}\, K}(x)\right) + \sum_{i=2}^{m-1} t_{\sigma_i}^{-1} \beta_{\hat{\sigma}_i}^{\mathrm{Sd}\, K}(x) \\ &\leqslant \sum_{i=1}^{m} t_{\sigma_i}^{-1} \beta_{\hat{\sigma}_i}^{\mathrm{Sd}\, K}(x) \leqslant 1, \end{aligned}$$

which implies $y \in M_{\sigma_m}$. Moreover,

$$\begin{aligned} \rho_{\mathrm{Sd}\, K}(x, y) &= \sum_{i=1}^{k} \left|\beta_{\hat{\sigma}'_i}^{\mathrm{Sd}\, K}(x) - \beta_{\hat{\sigma}'_i}^{\mathrm{Sd}\, K}(y)\right| + \sum_{i=1}^{m-1} \left|\beta_{\hat{\sigma}_i}^{\mathrm{Sd}\, K}(x) - \beta_{\hat{\sigma}_i}^{\mathrm{Sd}\, K}(y)\right| + \beta_{\hat{\sigma}_m}^{\mathrm{Sd}\, K}(x) \\ &= 2\beta_{\hat{\sigma}_m}^{\mathrm{Sd}\, K}(x) \leqslant 2t_{\sigma_m} \sum_{i=1}^{m} t_{\sigma_i}^{-1} \beta_{\hat{\sigma}_i}^{\mathrm{Sd}\, K}(x) \leqslant 2t_{\sigma_m}. \end{aligned}$$

Hence, $x \in \overline{\mathrm{N}}_{\rho_{\mathrm{Sd}\, K}}(M_{\sigma_m}, 2t_{\sigma_m}) \subset U$. When $m = 1$, since $\tau_0 \in \mathrm{Sd}\, L$, $y \in \tau_0 \subset \sigma \cap |L| \subset M_{\sigma_m}$, and

$$\rho_{\mathrm{Sd}\,K}(x,y)=\sum_{i=1}^{k}\left|\beta^{\mathrm{Sd}\,K}_{\hat{\sigma}'_i}(x)-\beta^{\mathrm{Sd}\,K}_{\hat{\sigma}'_i}(y)\right|+\beta^{\mathrm{Sd}\,K}_{\hat{\sigma}_1}(x)=2\beta^{\mathrm{Sd}\,K}_{\hat{\sigma}_1}(x)$$
$$\leqslant 2t_{\sigma_1}\left(t_{\sigma_1}^{-1}\beta^{\mathrm{Sd}\,K}_{\hat{\sigma}_1}(x)\right)\leqslant 2t_{\sigma_1}=2t_{\sigma_m},$$

it follows that $x\in\overline{\mathrm{N}}_{\rho_{\mathrm{Sd}\,K}}(M_{\sigma_m},2t_{\sigma_m})\subset U$. Consequently, $(*)$ holds for every $\tau\in(\mathrm{Sd}\,K^{(n+1)})[L]\setminus\mathrm{Sd}\,L$. Thus, the inductive step is completed. □

Remark 4.12.4 In Lemma 4.12.4 above, replacing $(\mathrm{Sd}\,K,\mathrm{Sd}\,L)$ with $(\mathrm{Sd}_L\,K,L)$, we can assert that $\mathrm{Sd}_L\,K$ has a slicer θ along L such that $|N_\theta(L,\mathrm{Sd}_L\,K)|\subset U$.

As mentioned after the definition of a sliced neighborhood, a sliced neighborhood $N_\theta(L,K)$ can be subdivided by a derived neighborhood $N(\mathrm{Sd}\,L,K')$. Hence, the following is a direct consequence of Lemma 4.12.4:

Proposition 4.12.5 *Let K be a simplicial complex with a full subcomplex L. For any neighborhood U of $|L|$ in $|K|$, there exists a derived subdivision K' of* $\mathrm{Sd}\,K$ *such that* $\mathrm{Sd}^2\,L\cup\mathrm{Sd}\,C(\mathrm{Sd}\,L,\mathrm{Sd}\,K)\subset K'$ *and* $|N(\mathrm{Sd}^2\,L,K')|\subset U$. □

Let P be a polyhedron with a subpolyhedron $Q\subset P$. A subpolyhedron $N\subset P$ is called a **regular neighborhood** of Q in P if there exist a triangulation K of P, a full subcomplex $L\subset K$ with $|L|=Q$, and a derived subdivision K' of K with $\mathrm{Sd}\,L\subset K'$ such that $N=|N(\mathrm{Sd}\,L,K')|$, where Theorem 4.4.8 guarantees the existence of such a pair (K,L). The following are easy:

- *For any simplicial complex K, any full subcomplex L of K, and $0<\varepsilon<2$, $\overline{\mathrm{N}}_{\rho_K}(|L|,\varepsilon)=|N_\varepsilon(L,K)|$ is a regular neighborhood of $|L|$ in $|K|$.*
- *For any simplicial complex K and any subcomplex L of K, $|N(\mathrm{Sd}^2\,L,\mathrm{Sd}^2\,K)|$ is a regular neighborhood of $|L|$ in $|K|$ (cf. Proposition 4.3.12).*

Concerning regular neighborhoods, we have the following as a direct consequence of Proposition 4.12.1 and Remark 4.12.2:

Proposition 4.12.6 *Let P be a polyhedron. Every subpolyhedron Q of P is a strong deformation retract of any regular neighborhood N of Q in P. More precisely, there exists a homotopy $h:N\times\mathbf{I}\to N$ such that $h_1:N\to Q$ is a PL retraction, $h_0=\mathrm{id}$, $h_t|Q=\mathrm{id}$, and h_t is PL for each $t\in\mathbf{I}$.* □

The following is a direct consequence of Proposition 4.12.5:

Proposition 4.12.7 *Let P be a polyhedron with a subpolyhedron $Q\subset P$. Every neighborhood U of Q in P contains some regular neighborhood of Q.* □

For regular neighborhoods, the following uniqueness holds:

Theorem 4.12.8 (Uniqueness of Regular Neighborhoods) *Let P be a polyhedron. If N_1 and N_2 are regular neighborhoods of a subpolyhedron Q in P, then there is a PL homeomorphism $h:P\to P$ such that $h(N_1)=N_2$ and $h|Q=\mathrm{id}$.*

Proof Let (K_i, L_i), $i = 1, 2$, are pairs of simplicial complexes and their full subcomplexes and let K_i' be a derived subdivision of K_i such that $P = |K_i|$, $Q = |L_i|$, $\mathrm{Sd}\, L_i \subset K_i'$, and $N_i = |N(\mathrm{Sd}\, L_i, K_i')|$. By Proposition 4.3.7, P has a triangulation K that is a common subdivision of $\mathrm{Sd}\, K_1$ and $\mathrm{Sd}\, K_2$. Let L be a subcomplex of K with $|L| = Q$, which is a common subdivision of $\mathrm{Sd}\, L_1$ and $\mathrm{Sd}\, L_2$ induced by K. Then, L is full in K (Proposition 4.3.5).

We can apply Lemma 4.12.4 to obtain a slicer θ_i of $\mathrm{Sd}\, K_i$ along $\mathrm{Sd}\, L_i$ such that

$$|N_{\theta_i}(\mathrm{Sd}\, L_i, \mathrm{Sd}\, K_i)| \subset |K| \setminus |C(L, K)|.$$

Then, we have a derived subdivision K_i'' of $\mathrm{Sd}\, K_i$ such that

$$\mathrm{Sd}^2 L_i \subset K_i'' \lhd (\mathrm{Sd}\, K_i)^{\theta_i} \quad \text{and} \quad N(\mathrm{Sd}^2 L_i, K_i'') \lhd N_{\theta_i}(\mathrm{Sd}\, L_i, \mathrm{Sd}\, K_i).$$

Since $K_i' \equiv \mathrm{Sd}\, K_i$ rel. $\mathrm{Sd}\, L_i$ and $K_i'' \equiv \mathrm{Sd}^2 K_i$ rel. $\mathrm{Sd}^2 L_i$, we can apply Proposition 4.12.2 to obtain a PL homeomorphism $h_i : |K_i'| \to |K_i''|$ such that $h_i||L_i| = \mathrm{id}$ and

$$h_i(|N(\mathrm{Sd}\, L_i, K_i')|) = |N(\mathrm{Sd}^2 L_i, K_i'')| = |N_{\theta_i}(\mathrm{Sd}\, L_i, \mathrm{Sd}\, K_i)|.$$

Due to Lemma 4.12.3, there is a slicer θ_i' of K along L such that $|N_{\theta_i'}(L, K)| = |N_{\theta_i}(\mathrm{Sd}\, L_i, \mathrm{Sd}\, K_i)|$. Since $N_{\theta_i'}(L, K)$ is subdivided by a derived neighborhood of L in K and a derived neighborhood of L in K is simplicially unique, we have a PL homeomorphism $h_0 : |K| \to |K|$ such that $h||L| = |L|$ and

$$h_0(|N_{\theta_1'}(L, K)|) = |N_{\theta_2'}(L, K)| = |N_{\theta_2}(\mathrm{Sd}\, L_2, \mathrm{Sd}\, K_2)|.$$

Then, $h = h_2^{-1} h_0 h_1 : |K| \to |K|$ is the desired PL homeomorphism. Indeed, $h||L| = \mathrm{id}$ and

$$\begin{aligned} h(|N(\mathrm{Sd}\, L_1, K_1')|) &= h_2^{-1} h_0 h_1(|N(\mathrm{Sd}\, L_1, K_1')|) = h_2^{-1} h_0(|N_{\theta_1}(\mathrm{Sd}\, L_1, \mathrm{Sd}\, K_1)|) \\ &= h_2^{-1}(|N_{\theta_2}(\mathrm{Sd}\, L_2, \mathrm{Sd}\, K_2)| = |N(\mathrm{Sd}\, L_2, K_2')|. \end{aligned}$$

□

A subpolyhedron A of P is said to be **PL bi-collared** (resp. **PL collared**) in P if there is a PL embedding $k : A \times [-1, 1] \to P$ (resp. $k : A \times \mathbf{I} \to P$) such that $k(x, 0) = x$ for every $x \in A$ and $k(A \times (-1, 1))$ (resp. $k(A \times [0, 1))$) is open in P (i.e., $k|A \times (-1, 1)$ (resp. $k|A \times [0, 1)$) is an open embedding in P), where k is called a **PL bi-collar** (resp. a **PL collar**) in P.

Theorem 4.12.9 *Let P be a polyhedron and let N be a regular neighborhood of a subpolyhedron Q in P. The boundary* bd N *is PL bi-collared in P (and PL collared in N). Furthermore, we have a homeomorphism $h :$* bd $N \times \mathbb{R}_+ \to N \setminus Q$ *such that $h(x, 0) = x$ for every $x \in$* bd N, *so* bd N *is a strong deformation retract of $N \setminus Q$.*

Proof We have a triangulation K of P, a full subcomplex L with $|L| = Q$, and a derived subdivision $K' \lhd K$ such that $\mathrm{Sd}\, L \subset K'$ and $|N(\mathrm{Sd}\, L, K')| = N$. Since a

derived neighborhood is simplicially unique, we may assume that $|N(\mathrm{Sd}\, L, K')| = |N_1(L, K)|$, that is, $N = |N_1(L, K)| = \overline{\mathrm{N}}_{\rho_K}(|L|, 1)$, where K' is defined by choosing $v_\sigma \in \mathrm{rint}\,\sigma$ for each $\sigma \in K$ so that $v_\sigma \in \mathrm{rint}\,\sigma \cap (\delta_L^K)^{-1}(1)$ if $\sigma \in K[L] \setminus L$. Then, observe that $\mathrm{bd}\, N = (\delta_L^K)^{-1}(1)$.

We define a subcomplex $\partial N_1(L, K)$ of $N_1(L, K)$ as follows:

$$\partial N_1(L, K) = \left\{\sigma \cap (\delta_L^K)^{-1}(1) \;\middle|\; \sigma \in K[L] \setminus L\right\},$$

where $|\partial N_1(L, K)| = \mathrm{bd}\, N = (\delta_L^K)^{-1}(1)$. Let $J = \{\pm 1, [-1, 1]\}$ be the natural triangulation of $[-1, 1]$ and define $M = \partial N_1(L, K) \times_c J$. Then,

$$|M| = \mathrm{bd}\, N \times [-1, 1] = (\delta_L^K)^{-1}(1) \times [-1, 1].$$

For each $C \in M$, we define $v_C \in \mathrm{rint}\, C$ as follows:

$$v_C = \begin{cases} (v_\sigma, 0) & \text{if } C = (\sigma \cap (\delta_L^K)^{-1}(1)) \times [-1, 1], \\ (v_\sigma, \pm 1) & \text{if } C = (\sigma \cap (\delta_L^K)^{-1}(1)) \times \{\pm 1\}, \end{cases}$$

where $\sigma \in K[L] \setminus L$. Let M' be the derived subdivision of M defined by using these points.

Now, consider the following cell complex:

$$B = \left\{\sigma \cap (\delta_L^K)^{-1}(\tau) \;\middle|\; \sigma \in K[L] \setminus L,\ \tau \leqslant [1/2, 3/2]\right\}.$$

Then, $|B| = (\delta_L^K)^{-1}([1/2, 3/2])$, which is a subpolyhedron of P. For each $D \in B$, choose $v_D \in \mathrm{rint}\, D$ so that $v_D = v_\sigma \in \mathrm{rint}\,\sigma \cap (\delta_L^K)^{-1}(1)$ if $D = \sigma \cap (\delta_L^K)^{-1}([1/2, 3/2])$, $\sigma \in K[L] \setminus L$. Let B' be a derived subdivision of B defined by using these points.

We have a bijection $\eta : M \to B$ defined by

$$\begin{gathered} \eta((\sigma \cap (\delta_L^K)^{-1}(1)) \times [-1, 1]) = \sigma \cap (\delta_L^K)^{-1}([1/2, 3/2]), \\ \eta((\sigma \cap (\delta_L^K)^{-1}(1)) \times \{1\}) = \sigma \cap (\delta_L^K)^{-1}(1/2), \text{ and} \\ \eta((\sigma \cap (\delta_L^K)^{-1}(1)) \times \{-1\}) = \sigma \cap (\delta_L^K)^{-1}(3/2), \end{gathered}$$

where $\sigma \in K[L] \setminus L$. Observe that for $C, C' \in M$,

$$C' \leqslant C \;\Leftrightarrow\; \eta(C') \leqslant \eta(C).$$

Hence, we have a simplicial isomorphism $k : M' \to B'$ such that $k(v_C) = v_{\eta(C)}$ for each $C \in M$. When $C = (\sigma \cap (\delta_L^K)^{-1}(1)) \times [-1, 1]$, $\sigma \in K[L] \setminus L$, we have $v_C = (v_\sigma, 0)$ and $v_{\eta(C)} = v_\sigma$ because $\eta(C) = \sigma \cap (\delta_L^K)^{-1}([1/2, 3/2])$. Since k is simplicial, it follows that $k(x, 0) = x$ for every $x \in \mathrm{bd}\, N$. Thus, $k : \mathrm{bd}\, N \times [-1, 1] \to P$ is a PL bi-collar, so $\mathrm{bd}\, N$ is PL bi-collared in P. Moreover, observe that

$$k(\operatorname{bd} N \times [0,1]) = (\delta_L^K)^{-1}([1/2,1]) \subset (\delta_L^K)^{-1}([0,1]) = N.$$

Hence, the restriction $k|\operatorname{bd} N \times [0,1] : \operatorname{bd} N \times [0,1] \to N$ is a PL collar in N.

To see the additional statement, consider $1/2 = 2^{-1} > 2^{-2} > \cdots > 0$. Then,

$$N \setminus Q = (\delta_L^K)^{-1}((0,1]) = \bigcup_{n\in\mathbb{N}} (\delta_L^K)^{-1}([2^{-n}, 2^{-(n-1)}]).$$

In the above, we obtained a homeomorphism

$$k_1 = k|(\delta_L^K)^{-1}(1) \times \mathbf{I} : (\delta_L^K)^{-1}(1) \times \mathbf{I} \to (\delta_L^K)^{-1}([1/2,1]),$$

where $k_1(x,0) = x$ for every $x \in (\delta_L^K)^{-1}(1)$ and $k_1((\delta_L^K)^{-1}(1) \times \{1\}) = (\delta_L^K)^{-1}(1/2)$. In the same way, we can define homeomorphisms

$$k_n : (\delta_L^K)^{-1}(2^{-(n-1)}) \times \mathbf{I} \to (\delta_L^K)^{-1}([2^{-n}, 2^{-(n-1)}]),\ n > 1,$$

such that $k_n(x,0) = x$ for every $x \in (\delta_L^K)^{-1}(2^{-(n-1)})$ and $k_n((\delta_L^K)^{-1}(2^{-(n-1)}) \times \{1\}) = (\delta_L^K)^{-1}(2^{-n})$. By induction, homeomorphisms

$$h_n : \big(\operatorname{bd} N \times [n-1,n], \operatorname{bd} N \times \{n\}\big) \to \big((\delta_L^K)^{-1}([2^{-n}, 2^{-(n-1)}]), (\delta_L^K)^{-1}(2^{-n})\big),\ n \in \mathbb{N},$$

can be defined as follows:

$$h_n(x,t) = k_n(h_{n-1}(x,n-1), t-n+1) \ \text{ for each } (x,t) \in \operatorname{bd} N \times [n-1,n],$$

where $h_0(x,0) = x$, so $h_1(x,t) = k_1(x,t)$. Then, for every $x \in \operatorname{bd} N$ and $n \in \mathbb{N}$,

$$h_n(x,n-1) = k_n(h_{n-1}(x,n-1),0) = h_{n-1}(x,n-1).$$

So, a homeomorphism $h : \operatorname{bd} N \times \mathbb{R}_+ \to N \setminus Q$ can be defined by

$$h|\operatorname{bd} N \times [n-1,n] = h_n,\ n \in \mathbb{N}.$$

Then, $h(x,0) = k_1(x,0) = x$ for every $x \in \operatorname{bd} N$. □

Remark 4.12.5 In the above, the homeomorphism h extends to a homeomorphism

$$\tilde{h} : \operatorname{bd} N \times \mathbb{R} \to (\delta_L^K)^{-1}((0,2)) = |N(L,K)| \setminus |L \cup C(L,K)|$$

such that $\tilde{h}(x,0) = x$ for every $x \in \operatorname{bd} N$.

Indeed, replacing L with $C(L,K)$ in the above proof, we can obtain a homeomorphism

$$h' : (\delta_{C(L,K)}^K)^{-1}(1) \times \mathbb{R}_+ \to (\delta_{C(L,K)}^K)^{-1}((0,1])$$

such that $h'(x,0) = x$ for every $x \in (\delta^K_{C(L,K)})^{-1}(1)$. Since $\delta^K_{C(L,K)} = 2 - \delta^K_L$, we have $(\delta^K_{C(L,K)})^{-1}(1) = (\delta^K_L)^{-1}(1)$, $(\delta^K_{C(L,K)})^{-1}((0,1]) = (\delta^K_L)^{-1}([1,2))$. Then, by combining h and h', the desired homeomorphism $\tilde{h}$ can be easily constructed.

4.13 PL n-Manifolds and Combinatorial n-Manifolds

A separable polyhedron M is called a **PL n-manifold** (or an **n-dimensional PL manifold**) if each $x \in M$ has a neighborhood that is PL homeomorphic to $\mathbf{I}^n$. Then, M is an n-dimensional polyhedron, that is, $M = |K|$ for some n-dimensional simplicial (or cell) complex K. Obviously, every PL n-manifold M is an n-manifold (cf. Sect. 1.1). PL n-manifolds are the main subjects in PL Topology. A countable simplicial complex K is called a **combinatorial n-manifold** (or an **n-dimensional combinatorial manifold**) if $\mathrm{St}(v,K) \cong F(\Delta^n)$ for every $v \in K^{(0)}$. Then, K is an n-dimensional complex. In this section, we give some fundamental results on PL n-manifolds and combinatorial n-manifolds. Among them, we show that a simplicial complex K is a combinatorial n-manifold if and only if $|K|$ is a PL n-manifold, i.e., K is a triangulation of a PL manifold (Theorem 4.13.7).

First of all, we show the following:

Proposition 4.13.1 *For every $n \in \omega$, $F(\Delta^n)$ and $\partial F(\Delta^{n+1})$ are combinatorial n-manifolds.*

Proof Since the proposition is obvious for $F(\Delta^n)$, we show the one for $\partial F(\Delta^{n+1})$. For each vertex $v \in \partial F(\Delta^{n+1})^{(0)}$, let $\sigma < \Delta^{n+1}$ be the opposite face to v, that is, $\Delta^{n+1} = v\sigma$. Then, it follows that

$$\begin{aligned}\mathrm{St}(v, \partial F(\Delta^{n+1})) &= \partial F(\Delta^{n+1}) \setminus \{\sigma\} = v * \partial F(\sigma) \\ &\equiv \hat{\sigma} * \partial F(\sigma) \cong F(\sigma) \equiv F(\Delta^n),\end{aligned}$$

where $\hat{\sigma}$ is the barycenter of σ. Hence, $\partial F(\Delta^{n+1})$ is a combinatorial n-manifold. □

Let C be an n-cell in $\mathbb{R}^n$ with $\mathbf{0} \in \mathrm{int}\, C$. For $r > 0$, $rC = \{rx \mid x \in C\}$ is an n-cell and $\mathbf{0} \in \mathrm{int}\, rC$. For each proper face $D < C$, $\mathbf{0}D = \{tx \mid t \in \mathbf{I}, x \in D\}$ is a cell with $(\mathbf{0}D)^{(0)} = \{\mathbf{0}\} \cup D^{(0)}$. Then, $F(C)$ has the following subdivision:[27]

$$F(C)_{\mathbf{0}} = \left\{\mathbf{0}D \;\middle|\; D < C\right\} \cup \partial F(C) \cup \{\mathbf{0}\}.$$

Note that $\mathbf{0}r\,D = \{tx \mid 0 \leqslant t \leqslant r, x \in D\}$ for $r > 0$ and $D < C$. Using this notation, we can show the following:

[27] When C is a simplex, $F(C)_{\mathbf{0}}$ is just the starring of $F(C)$ at $\mathbf{0}$, which is defined in Sect. 4.7.

Proposition 4.13.2 *Every two n-cells C and D are PL homeomorphic to each other, that is,*

$$(F(C), \partial F(C)) \cong (F(D), \partial F(D)).$$

Proof Although C and D are contained in some linear spaces, $\mathrm{fl}\, C$ and $\mathrm{fl}\, D$ are affinely homeomorphic to $\mathbb{R}^n$. Hence, it can be assumed that $C, D \subset \mathbb{R}^n$. Moreover, using translation, we may assume that $\mathbf{0} \in \operatorname{int} C \cap \operatorname{int} D$. Since C and D are compact, $C \subset rD$ and $D \subset rC$ for sufficiently large $r > 0$. Due to Proposition 4.3.8 (1), the following are cell complexes:

$$S_C = F(C)_{\mathbf{0}} \wedge F(rD)_{\mathbf{0}},\ L_C = \partial F(C) \wedge F(rD)_{\mathbf{0}},$$
$$S_D = F(rC)_{\mathbf{0}} \wedge F(D)_{\mathbf{0}},\ \text{and}\ L_D = F(rC)_{\mathbf{0}} \wedge \partial F(D),$$

which subdivide $F(C), \partial F(C), F(D)$, and $\partial F(D)$. As is easily observed, for $C' < C$ and $D' < D$,

$$C' \cap \mathbf{0}rD' \neq \emptyset \ \Leftrightarrow\ \mathbf{0}C' \cap \mathbf{0}D' \setminus \{\mathbf{0}\} \neq \emptyset \ \Leftrightarrow\ \mathbf{0}rC' \cap D' \neq \emptyset.$$

Both sides obviously imply the middle. To see that the middle implies the right, assume that there are $s, t \in (0, 1]$, $x \in C'$, and $y \in D'$ such that $sx = ty$. Then, $(s/t)x = y \in D \subset rC$. Since $rx \in \partial rC$ and $\mathbf{0} \in \operatorname{rint} rC$, it follows that $r'x \notin rC$ for any $r' > r$. Hence, $s/t \leqslant r$, which implies that $(s/t)x = y \in \mathbf{0}rC' \cap D'$. Similarly the middle implies the left.

Then, we can define a bijection $\eta : S_C \to S_D$ as follows:

$$\eta(\mathbf{0}C' \cap \mathbf{0}rD') = \mathbf{0}rC' \cap \mathbf{0}D' \ \text{ for } C' < C,\ D' < D, \text{ and}$$
$$\eta(C' \cap \mathbf{0}rD') = \mathbf{0}rC' \cap D' \ \text{ for } C' < C,\ D' < D \text{ with } C' \cap \mathbf{0}rD' \neq \emptyset.$$

Then, we have $\eta(L_C) = L_D$. The following can be easily verified:

$$C', C'' \in S_C,\ C'' < C' \ \Leftrightarrow\ \eta(C'') < \eta(C'),$$

so $(S_C, L_C) \equiv (S_D, L_D)$. Consequently, $(F(C), \partial F(C)) \cong (F(D), \partial F(D))$. □

The following seems almost obvious but some proof is needed:

Lemma 4.13.3 *Let σ and τ be simplexes in $\mathbb{R}^n$ joinable to $\mathbf{0}$ such that $\mathbf{0}\tau \subset \mathbf{0}\sigma$. If $r\tau \cap \mathbf{0}\sigma = \emptyset$ for some $r > 0$, then $r\tau$ is a simplex joinable to $\mathbf{0}$ and $\sigma \cap \mathbf{0}r\tau$ is a simplex with $\dim \sigma \cap \mathbf{0}r\tau = \dim \tau$, where $\mathbf{0}r\tau = r\mathbf{0}\tau$ and $\mathbf{0}(\sigma \cap \mathbf{0}r\tau) = \mathbf{0}\sigma \cap \mathbf{0}r\tau$.*

Proof It is obvious that $r\tau$ is a simplex joinable to $\mathbf{0}$. But we verify that $\sigma \cap \mathbf{0}r\tau$ is a simplex. For each vertex $v \in \tau^{(0)}$, $\sigma \cap \mathbf{0}rv$ is a singleton because σ is joinable to $\mathbf{0}$. Then, $\sigma \cap \mathbf{0}rv = \{t_v v\}$ for some $0 < t_v < r$. If $\sum_{v \in \tau^{(0)}} s_v(t_v v) = \mathbf{0}$ and $\sum_{v \in \tau^{(0)}} s_v = 0$, then

$$\sum_{v \in \tau^{(0)}} s_v(t_v v) + \Big(\sum_{v \in \tau^{(0)}} s_v(1 - t_v) \Big)\mathbf{0} = \mathbf{0} \ \text{ and } \ \sum_{v \in \tau^{(0)}} s_v t_v + \Big(\sum_{v \in \tau^{(0)}} s_v(1 - t_v) \Big) = 0.$$

Since $\tau^{(0)} \cup \{\mathbf{0}\}$ is affinely independent, it follows that $s_v t_v = 0$ for all $v \in \tau^{(0)}$, which implies that $s_v = 0$ for all $v \in \tau^{(0)}$ because $t_v > 0$. Hence, $\{t_v v \mid v \in \tau^{(0)}\}$ is affinely independent. Thus, $\langle \{t_v v \mid v \in \tau^{(0)}\} \rangle$ is a simplex.

We have to show $\sigma \cap \mathbf{0}r\tau = \langle \{t_v v \mid v \in \tau^{(0)}\} \rangle$, where $\supset$ is obvious. To verify $\subset$, let $y \in \sigma \cap \mathbf{0}r\tau$. Then, we have $x \in \tau$ and $0 < t < r$ such that $y = tx$. Write $x = \sum_{v \in \tau^{(0)}} s_v v$, $\sum_{v \in \tau^{(0)}} s_v = 1$, where $s_v \geqslant 0$ for each $v \in \tau^{(0)}$. Then,

$$y = t \sum_{v \in \tau^{(0)}} s_v v = \alpha t \sum_{v \in \tau^{(0)}} \frac{s_v}{\alpha t_v}(t_v v), \text{ where } \alpha = \sum_{v \in \tau^{(0)}} \frac{s_v}{t_v}.$$

Since $\sum_{v \in \tau^{(0)}} s_v/\alpha t_v = \alpha^{-1} \sum_{v \in \tau^{(0)}} s_v/t_v = 1$ and $t_v v \in \sigma$ for every $v \in \tau^{(0)}$, it follows that $\sum_{v \in \tau^{(0)}} (s_v/\alpha t_v)(t_v v) \in \sigma$, which means $\alpha t = 1$ because σ is joinable to $\mathbf{0}$. Hence, $y \in \langle \{t_v v \mid v \in \tau^{(0)}\} \rangle$. □

Using the above lemma, we prove the following:

Lemma 4.13.4 *Let K be a locally finite simplicial complex and $K' \lhd K$. For each $v \in K^{(0)}$,*

$$\big(\mathrm{St}(v, K'), \mathrm{Lk}(v, K')\big) \cong \big(\mathrm{St}(v, K), \mathrm{Lk}(v, K)\big).$$

Proof Since $\mathrm{St}(v, K)$ is finite, we may assume that $|\mathrm{St}(v, K)| \subset \mathbb{R}^n$ and $v = \mathbf{0}$. For sufficiently large $r > 0$,

$$|\mathrm{St}(\mathbf{0}, K)| \subset r O_{K'}(v) = r|\mathrm{St}(\mathbf{0}, K')| \setminus r|\mathrm{Lk}(\mathbf{0}, K')|.$$

Indeed, since $O_{K'}(v)$ is an open neighborhood of $\mathbf{0}$ in $|K'|$, $0|\mathrm{St}(\mathbf{0}, K)| = \{\mathbf{0}\} \subset O_{K'}(v)$, and $|\mathrm{St}(\mathbf{0}, K)|$ is compact, we have $r > 0$ such that $1/r|\mathrm{St}(\mathbf{0}, K)| \subset O_{K'}(v)$.

For each $\tau \in \mathrm{Lk}(\mathbf{0}, K')$, we have a unique $\sigma_\tau \in \mathrm{Lk}(\mathbf{0}, K)$ such that $\mathbf{0}\sigma_\tau$ is the smallest simplex containing $\mathbf{0}\tau$, i.e., $\mathbf{0}\sigma_\tau = c_K(2^{-1}\hat{\tau})$. Then, $\sigma_\tau \cap \mathbf{0}r\tau$ is a simplex with $\dim \sigma_\tau \cap \mathbf{0}r\tau = \dim \tau$ by Lemma 4.13.3. As is easily observed, the following is a simplicial subdivision of $\mathrm{Lk}(\mathbf{0}, K)$:

$$L_0 = \big\{\sigma_\tau \cap \mathbf{0}r\tau \;\big|\; \tau \in \mathrm{Lk}(\mathbf{0}, K')\big\},$$

which is congruent to $\mathrm{Lk}(\mathbf{0}, K')$ by the following correspondence:

$$\mathrm{Lk}(\mathbf{0}, K') \ni \tau \mapsto \sigma_\tau \cap \mathbf{0}r\tau \in L_0.$$

Thus, we have a simplicial subdivision

$$S_0 = \bigcup_{\tau \in \mathrm{Lk}(\mathbf{0}, K')} F(\mathbf{0}(\sigma_\tau \cap \mathbf{0}r\tau)) \lhd \mathrm{St}(\mathbf{0}, K),$$

where $(S_0, L_0) \equiv (\mathrm{St}(\mathbf{0}, K'), \mathrm{Lk}(\mathbf{0}, K'))$. Consequently, we have the result. □

The above lemma can be generalized as follows:

Proposition 4.13.5 *Let K and L be locally finite simplicial complexes. For $v \in K^{(0)}$ and $u \in L^{(0)}$, if $(K, v) \cong (L, u)$, then*

$$\big(\mathrm{St}(v, K), \mathrm{Lk}(v, K)\big) \cong \big(\mathrm{St}(u, L), \mathrm{Lk}(u, L)\big).$$

Proof Take subdivisions $K' \triangleleft K$ and $L' \triangleleft L$ so that $(K', v) \equiv (L', u)$. Then,

$$\big(\mathrm{St}(v, K'), \mathrm{Lk}(v, K')\big) \equiv \big(\mathrm{St}(u, L'), \mathrm{Lk}(u, L')\big).$$

Applying Lemma 4.13.4, we have

$$\begin{aligned}\big(\mathrm{St}(v, K), \mathrm{Lk}(v, K)\big) &\cong \big(\mathrm{St}(v, K'), \mathrm{Lk}(v, K')\big)\\ &\equiv \big(\mathrm{St}(u, L'), \mathrm{Lk}(u, L')\big) \cong \big(\mathrm{St}(u, L), \mathrm{Lk}(u, L)\big). \qquad \square\end{aligned}$$

By combining the following proposition with Theorem 4.11.14, a simplicial complex is a combinatorial n-manifold if it is PL homeomorphic to $\mathbf{I}^n$.

Proposition 4.13.6 *Let K be a simplicial complex with a PL homeomorphism $h : |K| \to \Delta^n$. For each vertex $v \in K^{(0)}$,*

$$\big(\mathrm{St}(v, K), \mathrm{Lk}(v, K)\big) \cong \begin{cases} \big(F(\Delta^n), \partial F(\Delta^n)\big) & \text{if } h(v) \in \mathrm{rint}\, \Delta^n, \\ \big(F(\Delta^n), F(\Delta^{n-1})\big) & \text{if } h(v) \in \partial \Delta^n, \end{cases}$$

where $\Delta^{n-1} = \langle \mathbf{e}_1, \ldots, \mathbf{e}_n \rangle \subset \Delta^n = \langle \mathbf{e}_1, \ldots, \mathbf{e}_n, \mathbf{e}_{n+1} \rangle$.

Proof When $h(v) \in \mathrm{rint}\, \Delta^n$, let L be the starring $F(\Delta^n)_{h(v)}$ of $F(\Delta^n)$ at $h(v)$, i.e.,

$$L = \big\{ h(v)\sigma \;\big|\; \sigma \in \partial F(\Delta^n) \big\} \cup \partial F(\Delta^n) \cup \{h(v)\}.$$

Then, $|L| = \Delta^n$, $h(v) \in L^{(0)}$, $L = \mathrm{St}(h(v), L)$, and $\partial F(\Delta^n) = \mathrm{Lk}(h(v), L)$. Since $(K, v) \cong (L, h(v))$ by the pair version of Theorem 4.11.15 noted at the end of Sect. 4.11, we can apply Proposition 4.13.5 to obtain

$$\big(\mathrm{St}(v, K), \mathrm{Lk}(v, K)\big) \cong \big(\mathrm{St}(h(v), L), \mathrm{Lk}(h(v), L)\big) \cong \big(F(\Delta^n), \partial F(\Delta^n)\big).$$

When $h(v) \in \partial \Delta^n$, it is easy to see that $(\Delta^n, h(v))$ is PL homeomorphic to $(\Delta^n, \mathbf{e}_{n+1})$. Then, it follows that $(K, v) \cong (F(\Delta^n), \mathbf{e}_{n+1})$. So, we can apply Proposition 4.13.5 to obtain

$$\begin{aligned}\big(\mathrm{St}(v, K), \mathrm{Lk}(v, K)\big) &\cong \big(\mathrm{St}(\mathbf{e}_{n+1}, F(\Delta^n)), \mathrm{Lk}(\mathbf{e}_{n+1}, F(\Delta^n))\big)\\ &= \big(F(\Delta^n), F(\Delta^{n-1})\big). \qquad \square\end{aligned}$$

Now, we can prove the following characterization:

Theorem 4.13.7 *A simplicial complex K is a combinatorial n-manifold if and only if $|K|$ is a PL n-manifold.*

Proof (*The "only if" Part*) Each $x \in |K|$ is contained in the open star $O_K(v)$ of some vertex $v \in K^{(0)}$. Then, $|\operatorname{St}(v, K)|$ is neighborhood of x in $|K|$. Since $\operatorname{St}(v, K) \cong F(\Delta^n)$, $|\operatorname{St}(v, K)|$ is PL homeomorphic to $\mathbf{I}^n$ by Theorem 4.11.14 and Proposition 4.13.2. Therefore, $|K|$ is a PL n-manifold.

(*The "if" Part*) Each $v \in K^{(0)}$ has a neighborhood which is PL homeomorphic to $\mathbf{I}^n$. Applying Proposition 4.4.7, we have a simplicial subdivision $K' \lhd K$ and a subcomplex L of K' such that $|L|$ is a neighborhood of v in $|K|$ being PL homeomorphic to $\mathbf{I}^n$. Since $v \in \operatorname{int}_{|K'|}|L|$, it follows that $\operatorname{St}(v, K') = \operatorname{St}(v, L)$. Due to Proposition 4.13.2, $|L|$ is PL homeomorphic to Δ^n. Combining Lemma 4.13.4 and Proposition 4.13.6, we have $\operatorname{St}(v, K) \cong \operatorname{St}(v, K') = \operatorname{St}(v, L) \cong F(\Delta^n)$. □

Evidently, a polyhedron is a PL n-manifold if it is PL homeomorphic to a PL n-manifold. Hence, we have the following:

Corollary 4.13.8 *A simplicial complex is a combinatorial n-manifold if it is combinatorially equivalent to a combinatorial n-manifold.* □

Let K and L be joinable simplicial complexes. Then, each pair of simplicial subdivisions $K' \lhd K$ and $L' \lhd L$ is also joinable and $K' * L' \lhd K * L$. Hence, the following is trivial:

Lemma 4.13.9 *For joinable simplicial complexes K_i and L_i, $i = 1, 2$,*

$$K_1 \cong K_2,\ L_1 \cong L_2 \ \Rightarrow\ K_1 * L_1 \cong K_2 * L_2,$$

where any PL homeomorphisms $f : |K_1| \to |K_2|$ and $g : |L_1| \to |L_2|$ extend to a PL homeomorphism $h : |K_1 * L_1| \to |K_2 * L_2|$. □

For joinable simplexes, we have the following:

Proposition 4.13.10 *Let σ and τ be joinable simplexes with $\dim \sigma = n$ and $\dim \tau = m$. Then, the following hold:*

(i) $\big(F(\sigma) * F(\tau),\ F(\sigma) * \partial F(\tau) \cup \partial F(\sigma) * F(\tau)\big)$
$= \big(F(\sigma\tau),\ \partial F(\sigma\tau)\big) \equiv \big(F(\Delta^{n+m+1}),\ \partial F(\Delta^{n+m+1})\big)$;

(ii) $\big(F(\sigma) * \partial F(\tau),\ \partial F(\sigma) * \partial F(\tau)\big) \cong \big(F(\Delta^{n+m}),\ \partial F(\Delta^{n+m})\big)$,

where $F(\sigma) * \emptyset = F(\sigma)$, $\emptyset * F(\tau) = F(\tau)$, and $\emptyset * \emptyset = \emptyset$.

Proof Since the join $\sigma\tau$ is defined as the simplex $\langle \sigma^{(0)} \cup \tau^{(0)} \rangle$, (i) is trivial. Before proving (ii), consider particular triangulations of n-cells in the Banach space ℓ_1. We identify $\mathbb{R}^n = \mathbb{R}^n \times \{(0, 0, \dots)\} \subset \ell_1$.

For each $i \in \mathbb{N}$, let $\mathbf{e}_i \in \ell_1$ be the unit vector defined by $\mathbf{e}_i(i) = 1$ and $\mathbf{e}_i(j) = 0$ if $j \neq i$. Using 0-dimensional simplicial complexes $A_n = \{\mathbf{e}_n, -\mathbf{e}_n\}$, $n \in \mathbb{N}$, we

inductively define simplicial complexes K_n, $n \in \mathbb{N}$, by $K_1 = A_1$ and $K_n = K_{n-1} * A_n$ for $n > 1$. Then, it is easy to see

$$K_n^{(0)} = \{\mathbf{e}_1, -\mathbf{e}_1, \ldots, \mathbf{e}_n, -\mathbf{e}_n\} \text{ and } |K_n| = \mathbb{R}^n \cap \mathbf{S}_{\ell_1}.$$

Since each K_n is joinable to $\mathbf{0}$, we have the simplicial cone $\widetilde{K}_n = \mathbf{0} * K_n$, where

$$|\widetilde{K}_n| = |\mathbf{0} * K_n| = \bigcup_{\sigma \in K_n} \mathbf{0}\sigma = \mathbb{R}^n \cap \mathbf{B}_{\ell_1} = \langle \mathbf{e}_1, -\mathbf{e}_1, \ldots, \mathbf{e}_n, -\mathbf{e}_n \rangle.$$

For the last equality, the inclusion $\supset$ comes from the convexity of $\mathbb{R}^n \cap \mathbf{B}_{\ell_1}$. Since the right-hand side set contains $\mathbf{0}$ and all vertices of K_n, the inclusion $\subset$ holds.

Thus, $|\widetilde{K}_n|$ is an n-cell, and by Proposition 3.2.5, we have

$$\partial |\widetilde{K}_n| = \partial(\mathbb{R}^n \cap \mathbf{B}_{\ell_1}) = \mathbb{R}^n \cap \mathbf{S}_{\ell_1} = |K_n|.$$

Since $\widetilde{K}_n$ is joinable to A_{n+1}, we apply Proposition 4.2.7 to obtain

$$\widetilde{K}_n * A_{n+1} = (\mathbf{0} * K_n) * A_{n+1} = \mathbf{0} * (K_n * A_{n+1}) = \mathbf{0} * K_{n+1} = \widetilde{K}_{n+1}.$$

For each $n \in \mathbb{N}$, let $s_n : \ell_1^f \to \ell_1^f$ be the linear map defined by $s_n(\mathbf{e}_i) = \mathbf{e}_{n+i}$ for each $i \in \mathbb{N}$, where ℓ_1^f is the linear span of $\{e_i \mid i \in \mathbb{N}\}$ in ℓ_1. Then, $s_n(A_m) = A_{n+m}$ for each $n, m \in \mathbb{N}$. Moreover, $\widetilde{K}_n$ is joinable to $s_n(K_m)$ and $K_m \equiv s_n(K_m)$. By induction on $m \in \mathbb{N}$, we will show the following:

$$\big(\widetilde{K}_n * s_n(K_m),\, K_n * s_n(K_m)\big) = \big(\widetilde{K}_{n+m},\, K_{n+m}\big). \tag{$*$}$$

For $m = 1$, we have $\widetilde{K}_n * s_n(K_1) = \widetilde{K}_n * s_n(A_1) = \widetilde{K}_n * A_{n+1} = \widetilde{K}_{n+1}$ and similarly $K_n * s_n(K_1) = K_{n+1}$. For $m > 1$, assume that $(*)$ holds for $m - 1$. Then, by Proposition 4.2.7,

$$\begin{aligned}\widetilde{K}_n * s_n(K_m) &= \widetilde{K}_n * s_n(K_{m-1} * A_m) = \widetilde{K}_n * \big(s_n(K_{m-1}) * s_n(A_m)\big) \\ &= \big(\widetilde{K}_n * s_n(K_{m-1})\big) * s_n(A_m) = \widetilde{K}_{n+m-1} * A_{n+m} = \widetilde{K}_{n+m}\end{aligned}$$

and similarly $K_n * s_n(K_m) = K_{n+m}$. Thus, $(*)$ is proved.

Now, to see (ii), let σ and τ be joinable simplexes with $\dim \sigma = n$ and $\dim \tau = m$. By Proposition 4.13.2, $(F(\sigma), \partial F(\sigma)) \cong (\widetilde{K}_n, K_n)$ and $\partial F(\tau) \cong K_m \equiv s_n(K_m)$. Then, using Lemma 4.13.9, $(*)$, and Proposition 4.13.2, we have

$$\begin{aligned}\big(F(\sigma) * \partial F(\tau), \partial F(\sigma) * \partial F(\tau)\big) &\cong \big(\widetilde{K}_n * s_n(K_m),\, K_n * s_n(K_m)\big) \\ &= \big(\widetilde{K}_{n+m},\, K_{n+m}\big) \cong \big(F(\Delta^{n+m}), \partial F(\Delta^{n+m})\big),\end{aligned}$$

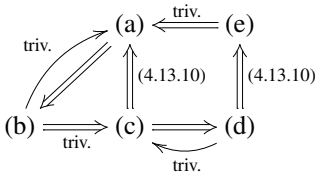
which concludes (ii). □

Now, we have the following characterization of combinatorial manifolds:

Theorem 4.13.11 *For a simplicial complex K, the following are equivalent:*

(a) *K is a combinatorial n-manifold;*
(b) $(\mathrm{St}(v, K), \mathrm{Lk}(v, K)) \cong (F(\Delta^n), \partial F(\Delta^n))$ *or* $(F(\Delta^n), F(\Delta^{n-1}))$ *for each vertex $v \in K^{(0)}$;*
(c) $\mathrm{Lk}(v, K) \cong \partial F(\Delta^n)$ *or* $F(\Delta^{n-1})$ *for each vertex $v \in K^{(0)}$;*
(d) $\mathrm{Lk}(\sigma, K) \cong \partial F(\Delta^{n-k})$ *or* $F(\Delta^{n-k-1})$ *for each k-simplex $\sigma \in K$;*
(e) $\mathrm{St}(\sigma, K) \cong F(\Delta^n)$ *for each simplex $\sigma \in K$.*

Proof The implications (e) $\Rightarrow$ (a) and (b) $\Rightarrow$ (c) are obvious ((d) $\Rightarrow$ (c) and (b) $\Rightarrow$ (a) are obvious too). Since $\mathrm{St}(\sigma, K) = \mathrm{Lk}(\sigma, K) * F(\sigma)$ for each i-simplex $\sigma \in K$, the implications (c) $\Rightarrow$ (a) and (d) $\Rightarrow$ (e) come from Proposition 4.13.10 (cf. Lemma 4.13.9). Then, it suffices to show the implications (a) $\Rightarrow$ (b) and (c) $\Rightarrow$ (d).



(a) $\Rightarrow$ (b): For each $v \in K^{(0)}$, let $S = \mathrm{St}(v, K)$. Then, $v \in S^{(0)}$ and

$$(\mathrm{St}(v, S), \mathrm{Lk}(v, S)) = (\mathrm{St}(v, K), \mathrm{Lk}(v, K)).$$

Since $S \cong F(\Delta^n)$ by (a), there exists a PL homeomorphism $h : |S| \to \Delta^n$. Applying Proposition 4.13.6, we can obtain

$$\big(\mathrm{St}(v, S), \mathrm{Lk}(v, S)\big) \cong \begin{cases} \big(F(\Delta^n), \partial F(\Delta^n)\big) & \text{if } h(v) \in \mathrm{rint}\, \Delta^n, \\ \big(F(\Delta^n), F(\Delta^{n-1})\big) & \text{if } h(v) \in \partial \Delta^n. \end{cases}$$

(c) $\Rightarrow$ (d): By induction on n, we show this implication on n. The case $n = 0$ is obvious. Assuming the implication (c) $\Rightarrow$ (d) for $n - 1$, we show the case n. The assumption means the theorem holds for $n - 1$. For $k > 0$, each k-simplex σ can be written as $\sigma = v\tau$, where $v \in \sigma^{(0)}$ and τ is the opposite face of σ to v, i.e., $\tau^{(0)} = \sigma^{(0)} \setminus \{v\}$. Then, as is easily observed, $\mathrm{Lk}(\sigma, K) = \mathrm{Lk}(\tau, \mathrm{Lk}(v, K))$. Due to Proposition 4.13.1 and Corollary 4.13.8, (c) implies that $\mathrm{Lk}(v, K)$ is a combinatorial $(n - 1)$-manifold. Since τ is an $(k - 1)$-simplex, it follows from the inductive assumption that

$$\mathrm{Lk}(\sigma, K) = \mathrm{Lk}(\tau, \mathrm{Lk}(v, K)) \cong \partial F(\Delta^{n-k}) \ \text{ or } \ F(\Delta^{n-k-1}).$$

Hence, (c) $\Rightarrow$ (d) is valid for n. Thus, we have the implication (c) $\Rightarrow$ (d) by induction. □

As a corollary, we have the following:

Corollary 4.13.12 *Let K be a combinatorial n-manifold.*

(1) *Every $\sigma \in K$ is a face of an n-simplex of K. Equivalently, every k-simplex $\sigma \in K$ is joinable to some $(n-k-1)$-simplex of K in K.*
(2) *Each $(n-1)$-simplex $\sigma \in K$ is a common face of distinct two n-simplexes of K or a face of exactly one n-simplex of K. Equivalently, each $(n-1)$-simplex $\sigma \in K$ is joinable to one or two vertices in K.* □

Since a PL n-manifold M is an n-manifold, the interior $\operatorname{Int} M$ and the boundary ∂M of M can be defined like an n-manifold (cf. Sect. 1.1). Actually, $x \in M$ is called an **interior point** of M if it has a neighborhood N with a PL homeomorphism $h : N \to \mathbf{I}^n$ such that $h(x) \in (0,1)^n$. The set $\operatorname{Int} M$ of all interior points of M is called the **interior** of M, which is open in M. We call $x \in M$ a **boundary point** of M if it is not an interior point and the set ∂M of all boundary points is called the **boundary** of M. Then, ∂M is closed in M and it can be proved that $x \in M$ *is a boundary point if and only if it has a neighborhood N with a PL homeomorphic $h : N \to \mathbf{I}^n$ such that $h(x) \in \partial \mathbf{I}^n$.*[28] A PL n-manifold M with $\partial M = \emptyset$ is called a **PL n-manifold without boundary**. A compact PL n-manifold without boundary is called a **closed PL n-manifold**.

Remark 4.13.1 The n-cube $\mathbf{I}^n$ is a PL n-manifold by definition. Even n-cell C is a PL n-manifold because it is PL homeomorphic to $\mathbf{I}^n$ by Proposition 4.13.2. Then, the notation ∂C may have two meanings: one is the radial boundary as a convex set defined in Sect. 3.2 and the other is the boundary as a PL n-manifold. For $C = \mathbf{I}^n$, the two meanings clearly coincide. For a general n-cell C, Proposition 4.13.2 guarantees that $(C, \partial C)$ is PL homeomorphic to $(\mathbf{I}^n, \partial \mathbf{I}^n)$, where ∂ means the radial boundary. Therefore, the two meanings coincide for any n-cell C.

Theorem 4.13.13 *The boundary ∂M of each PL n-manifold M is a PL $(n-1)$-manifold without boundary.*

Proof Each $x \in \partial M$ has a neighborhood N with a PL homeomorphism $h : N \to \mathbf{I}^n$ such that $h(x) \in \partial \mathbf{I}^n$. We may assume that

$$h(x) = (1/2, \ldots, 1/2, 0) \in [1/4, 3/4]^{n-1} \times \{0\}.$$

Because $h(\operatorname{int} N)$ is an open neighborhood of $h(x)$ in $\mathbf{I}^n$, we can choose $0 < a < 1/2 < b < 1$ and $c > 0$ so that $[a,b]^{n-1} \times [0,c] \subset h(\operatorname{int} N)$. Then, note that

$$h^{-1}([a,b]^{n-1} \times (0,c]) \subset h^{-1}((0,1)^n) \subset \operatorname{Int} M.$$

On the other hand, $h^{-1}([a,b]^{n-1} \times \{0\}) \subset \partial M$ because N is a neighborhood of each $y \in h^{-1}([a,b]^{n-1} \times \{0\})$ and $h(y) \in \partial \mathbf{I}^n$. Thus, we have

[28] The "only if" part is trivial, but the "if" part is non-trivial and can be proved by using the Invariance of Domain (Theorem 5.1.10) in the same way as Footnote 7 of Sect. 1.1.

$$h^{-1}([a,b]^{n-1} \times [0,c]) \cap \partial M = h^{-1}([a,b]^{n-1} \times \{0\}).$$

Therefore, $N' = h^{-1}([a,b]^{n-1} \times \{0\})$ is a neighborhood of x in ∂M. Let $p : [a,b]^{n-1} \times \{0\} \to [a,b]^{n-1}$ be the projection. Then, $h' = ph|N' : N' \to [a,b]^{n-1}$ is a PL homeomorphism and $h'(x) = ph(x) \in (a,b)^{n-1}$, which means $x \in \operatorname{Int} \partial M$. Consequently, ∂M is an $(n-1)$-manifold without boundary. □

For every combinatorial manifold K, $|K|$ is a PL manifold by Theorem 4.13.7. Then, we can consider the interior $\operatorname{Int}|K|$ and the boundary $\partial|K|$. The following details condition (d) of Theorem 4.13.11:

Proposition 4.13.14 *Let K be a combinatorial n-manifold and $k < n$. For a k-simplex $\sigma \in K$, the following hold:*

(i) $\operatorname{Lk}(\sigma, K) \cong \partial F(\Delta^{n-k})$ *if and only if* $\operatorname{rint}\sigma \subset \operatorname{Int}|K|$;
(ii) $\operatorname{Lk}(\sigma, K) \cong F(\Delta^{n-k-1})$ *if and only if* $\sigma \subset \partial|K|$.

Moreover, $\operatorname{rint}\sigma \subset \operatorname{Int}|K|$ *for every n-simplex $\sigma \in K$.*

Proof Let $\sigma \in K$ be a k-simplex. Then, $\operatorname{Lk}(\sigma, K) \cong \partial F(\Delta^{n-k})$ or $F(\Delta^{n-k-1})$ by Theorem 4.13.11. On the other hand, $\operatorname{rint}\sigma \subset \operatorname{Int}|K|$ and $\sigma \subset \partial|K|$ cannot occur simultaneously. Thus, it is enough to show the "only if" part of (i) and (ii).

(*The "only if" Part of* (i)) Suppose that $\operatorname{Lk}(\sigma, K) \cong \partial F(\Delta^{n-k})$. Since $\operatorname{St}(\sigma, K) = F(\sigma) * \operatorname{Lk}(\sigma, K)$, by Lemma 4.13.9, we have a PL homeomorphism

$$h : \big(|\operatorname{St}(\sigma, K)|, |\partial F(\sigma) * \operatorname{Lk}(\sigma, K)|\big) \to \big(|F(\Delta^k) * \partial F(\Delta^{n-k})|, |\partial F(\sigma) * \partial F(\Delta^{n-k})|\big),$$

On the other hand, by Proposition 4.13.10 (ii), we have a PL homeomorphism

$$g : \big(|F(\Delta^k) * \partial F(\Delta^{n-k})|, |\partial F(\Delta^k) * \partial F(\Delta^{n-k})|\big) \to \big(\Delta^n, \partial\Delta^n\big),$$

Let $K_{\hat{\sigma}}$ be the starring of K at $\hat{\sigma}$.[29] Observe that

$$\begin{aligned} \operatorname{rint}\sigma \subset O(\hat{\sigma}, K_{\hat{\sigma}}) &= |\operatorname{St}(\hat{\sigma}, K)| \setminus |\operatorname{Lk}(\hat{\sigma}, K)| \\ &= |\operatorname{St}(\sigma, K)| \setminus |\partial F(\sigma) * \operatorname{Lk}(\sigma, K)|. \end{aligned}$$

Since $O(\hat{\sigma}, K_{\hat{\sigma}})$ is a neighborhood of each point of $\operatorname{rint}\sigma$ and

$$gh(O(\hat{\sigma}, K_{\hat{\sigma}})) = \Delta^n \setminus \partial\Delta^n = \operatorname{rint}\Delta^n \approx (0,1)^n,$$

it follow that $\operatorname{rint}\sigma \subset \operatorname{Int} M$.

(*The "only if" Part of* (ii)) Suppose that $\operatorname{Lk}(\sigma, K) \cong F(\Delta^{n-k-1})$. In the same way as above, we have PL homeomorphisms

[29] The starring was defined just before Lemma 4.7.4.

$$h : \big(|\operatorname{St}(\sigma, K)|, \sigma\big) \to \big(|F(\Delta^k) * F(\Delta^{n-k-1})|, \Delta^k\big),$$
$$g : \big(|F(\Delta^k) * F(\Delta^{n-k-1})|, \Delta^k\big) \to \big(\Delta^n, \Delta^k\big).$$

Since $|\operatorname{St}(\sigma, K)|$ is a neighborhood of each point of $\operatorname{rint}\sigma$ and $gh(\sigma) = \Delta^k \subset \partial\Delta^n$, it follows that $\operatorname{rint}\sigma \subset \partial|K|$. Hence, $\sigma \subset \partial|K|$ because $\partial|K|$ is closed in $|K|$.

(*Additional Statement*) Every n-simplex $\sigma \in K$ is principal, which implies $\operatorname{rint}\sigma$ is open in $|K|$. Since $(\sigma, \partial\sigma)$ is PL homeomorphic to $(\mathbf{I}^n, \partial\mathbf{I}^n)$ (Proposition 4.13.2), the statement follows from the definition of $\operatorname{Int}|K|$. □

Let K be a combinatorial n-manifold. The following ∂K is called the **boundary** of K:

$$\partial K = \big\{\sigma \in K^{(n-1)} \;\big|\; \operatorname{Lk}(\sigma, K) \cong F(\Delta^{n-\dim\sigma-1})\big\},$$

where if $\dim\sigma = n$, then $\operatorname{Lk}(\sigma, K) = \emptyset$ because σ is principal.

When $\partial K = \emptyset$, K is called a **combinatorial *n*-manifold without boundary**. A **closed combinatorial *n*-manifold** is a finite combinatorial n-manifold without boundary.

Remark 4.13.2 For an n-simplex σ, the notation $\partial F(\sigma)$ may have two meanings: one is $\partial F(\sigma) = \{\tau \mid \tau < \sigma\}$ defined in Sect. 4.2 and the other is the boundary of a combinatorial manifold $F(\sigma)$. It should be noticed that these two meanings coincide.

Due to Corollary 4.13.12 (2), the following fact is trivial:

- *For a combinatorial n-manifold K, an $(n-1)$-simplex $\sigma \in K$ belongs to the boundary ∂K if and only if σ is a face of only one n-simplex of K; equivalently, σ is joinable to only one vertex in K.*

Concerning of the boundary of a combinatorial n-manifold, we have the following:

Theorem 4.13.15 *For a combinatorial n-manifold K with $\partial K \neq \emptyset$, ∂K is a combinatorial $(n-1)$-manifold without boundary and $|\partial K| = \partial|K|$.*

Proof It follows from Proposition 4.13.14 that ∂K is a subcomplex of K with $|\partial K| = \partial|K|$. Since $M = \partial|K|$ is a PL $(n-1)$-manifold without boundary by Proposition 4.13.13, ∂K is a combinatorial $(n-1)$-manifold by Theorem 4.13.7. Replacing K with ∂K, we have $|\partial\partial K| = \partial|\partial K| = \partial M = \emptyset$, which implies $\partial\partial K = \emptyset$. □

In the rest of this section, using the Invariance of Domain proved later (Theorem 5.1.10), we will show the following:

Theorem 4.13.16 *Let M be a PL n-manifold and P a subpolyhedron of M. Then, a regular neighborhood N of P in M is a PL n-manifold with boundary. If $P \subset \operatorname{Int} M = M \setminus \partial M$, then $\partial N = \operatorname{bd}_M N$.*

Proof Let K be a triangulation of M (i.e., $M = |K|$) such that $P = |L|$ for some full subcomplex L of K and $N = |N(L, K')|$ for some derived subdivision K' of K relative to $L \cup C(L, K)$. Then, K is a combinatorial n-manifold by Theorem 4.13.7. By virtue of the uniqueness of regular neighborhoods (Theorem 4.12.8), we may assume that $N(L, K') \lhd N_\varepsilon(L, K)$, $0 < \varepsilon < 1$.

As is easily observed,

$$N = |N_\varepsilon(L, K)| = \bigcup_{v \in L^{(0)}} O_K(v) \cap |N_\varepsilon(L, K)|,$$

where $O_K(v)$ is the open star at v. For each $x \in N$, choose $v \in L^{(0)}$ so that $x \in O_K(v)$. Then, $|\operatorname{St}(v, K)| \cap N$ is a neighborhood of x in N and

$$\big|\operatorname{St}(v, K)\big| \cap N = \big|\operatorname{St}(v, K)\big| \cap \big|N_\varepsilon(L, K)\big| = \big|\operatorname{St}(v, K_L^\varepsilon)\big|.$$

It should be remarked that K_L^ε is not a simplicial subdivision of K nor $|\operatorname{St}(v, K')| \neq |\operatorname{St}(v, K_L^\varepsilon)|$. However, due to Theorem 4.3.6, giving an order on $(K_L^\varepsilon)^{(0)}$ so that $C^{(0)}$ has the maximum v_C for each $C \in K_L^\varepsilon$, we can define a simplicial subdivision of K_L^ε with the same set of vertices as K_L^ε. Giving such an order on $(K_L^\varepsilon)^{(0)}$ satisfying the additional condition that v is the maximum of $C^{(0)}$ for every $C \in K_L^\varepsilon[v]$ $(\subset \operatorname{St}(v, K_L^\varepsilon))$, we apply Theorem 4.3.6 to define a simplicial subdivision $K^v \lhd K_L^\varepsilon$. Then, it follows that $|\operatorname{St}(v, K^v)| = |\operatorname{St}(v, K_L^\varepsilon)|$. Since $\operatorname{St}(v, K^v) \cong \operatorname{St}(v, K) \cong F(\Delta^n)$ by Lemma 4.13.4, $|\operatorname{St}(v, K_L^\varepsilon)|$ is PL homeomorphic to $\mathbf{I}^n$. Therefore, $N = |N(L, K')|$ is a PL n-manifold.

In the above, assume that $P \subset \operatorname{Int} M$. For each $v \in L^{(0)}$ $(\subset P)$, the following follows from Theorem 4.13.11:

$$O_K(v) = |\operatorname{St}(v, K)| \setminus |\operatorname{Lk}(v, K)| \approx \Delta^n \setminus \partial\Delta^n = \operatorname{rint} \Delta^n.$$

Then, we have $O_K(v) \subset \operatorname{Int} M$. Hence, $N \subset \bigcup_{v \in L^{(0)}} O_K(v) \subset \operatorname{Int} M$. Each $x \in N$ has a polyhedral neighborhood B in M with a PL homeomorphism $h : B \to \mathbf{I}^n$ such that $h(x) \in (0, 1)^n$.

If $x \in \operatorname{int}_M N$, then the above B can be taken so small that $B \subset \operatorname{int}_M N$, which implies $x \in \operatorname{Int} N$. Conversely, if $x \in \operatorname{Int} N$, then there is a polyhedral neighborhood D in N with a PL homeomorphism $g : D \to \mathbf{I}^n$ such that $g(x) \in (0, 1)^n$, where D can be taken so small that $D \subset B$. By virtue of the Invariance of Domain (Theorem 5.1.10), $hg^{-1}((0, 1)^n)$ is open in $(0, 1)^n$, which implies that D is a neighborhood of x in B. Therefore, $x \in \operatorname{int}_M N$. Thus, we have $\operatorname{int}_M N = \operatorname{Int} N$, which means

$$\partial N = N \setminus \operatorname{Int} N = N \setminus \operatorname{int}_M N = \operatorname{bd}_M N. \qquad \square$$

4.14 PL Embedding and Relative PL Approximation Theorems

In this section, we first prove the General Position Lemma, which is very important in PL Topology and Dimension Theory. Applying this lemma, we show the PL Embedding Approximation Theorem, that is, for every n-dimensional separable locally compact polyhedron P, each proper map $f : P \to \mathbb{R}^{2n+1}$ can be approximated by PL embeddings. Besides, using derived (or regular) neighborhoods, we prove a relative version of the PL Approximation Theorem 4.11.11.

We will prove the GENERAL POSITION LEMMA in a slightly strengthened form as follows:

Lemma 4.14.1 (GENERAL POSITION) *Let U_i, $i \in \mathbb{N}$, be non-empty open sets in $\mathbb{R}^n$ and $A \subset \mathbb{R}^n$ with* $\operatorname{card} A \leqslant \aleph_0$. *Then, there exist distinct $v_i \in U_i \setminus A$, $i \in \mathbb{N}$, such that $S \subset A \cup \{v_i \mid i \in \mathbb{N}\}$ is affinely independent whenever* $\operatorname{card} S \leqslant n+1$ *and $S \cap A$ is affinely independent (in particular, $S \subset \{v_i \mid i \in \mathbb{N}\}$ is affinely independent if* $\operatorname{card} S \leqslant n+1$*).*[30]

Proof Suppose that distinct $v_1 \in U_1 \setminus A, \ldots, v_k \in U_k \setminus A$ have been chosen so that $S \subset A \cup \{v_i \mid i \leqslant k\}$ is affinely independent whenever $\operatorname{card} S \leqslant n+1$ and $S \cap A$ is affinely independent. Using the Baire Category Theorem 2.5.1 and the fact that every $(n-1)$-dimensional flat (= hyperplane) of $\mathbb{R}^n$ is nowhere dense in $\mathbb{R}^n$, we can find a point

$$v_{k+1} \in U_{k+1} \setminus \bigcup \left\{ \operatorname{fl} S \;\middle|\; S \subset A \cup \{v_i \mid i \leqslant k\}, 0 < \operatorname{card} S \leqslant n \right\}.$$

Then, $v_{k+1} \in U_{k+1} \setminus (A \cup \{v_i \mid i \leqslant k\})$ and $S \subset A \cup \{v_i \mid i \leqslant k+1\}$ is affinely independent whenever $\operatorname{card} S \leqslant n+1$ and $S \cap A$ is affinely independent. By induction, we can obtain $v_i \in U_i \setminus A$, $i \in \mathbb{N}$, with the desired property. □

Applying the General Position Lemma 4.14.1, we can prove the following PL EMBEDDING APPROXIMATION THEOREM:

Theorem 4.14.2 (PL EMBEDDING APPROXIMATION FOR $\mathbb{R}^n$) *Let P be a finite-dimensional separable locally compact polyhedron with Q a subpolyhedron and $n \geqslant 2 \dim P + 1$. If $f : P \to \mathbb{R}^n$ is a proper map and $f|Q$ is a PL embedding, then for any open cover $\mathcal{U}$ of $f(P)$ in $\mathbb{R}^n$, there is a PL embedding $g : P \to \mathbb{R}^n$ such that $f|Q = g|Q$ and $f \simeq^p_{\mathcal{U}} g$ rel. Q.*

Proof Since f is a closed map by Proposition 2.1.12, $f(P)$ is closed in $\mathbb{R}^n$. Then, $\mathcal{U} \cup \{\mathbb{R}^n \setminus f(P)\} \in \operatorname{cov}(\mathbb{R}^n)$. Take a locally finite open refinement $\mathcal{V} \in \operatorname{cov}(\mathbb{R}^n)$ of $\mathcal{U} \cup \{\mathbb{R}^n \setminus f(P)\}$ such that each $V \in \mathcal{V}$ has the compact closure in $\mathbb{R}^n$, where it should be noted that $\mathcal{V}[f(P)] \prec \mathcal{U}$. Then, we can find an open refinement $\mathcal{B}$ of $\mathcal{V}$ consisting of open balls in $\mathbb{R}^n$. Let $\mathcal{W} \in \operatorname{cov}(\mathbb{R}^n)$ be a locally finite open refinement

[30] We consider the empty set to be affinely independent (see the definition in Sect. 3.1).

of $\mathcal{B}$. Then, the convex hull $\langle W\rangle$ of each $W \in \mathcal{W}$ is contained in some ball of $\mathcal{B}$, hence it is contained in some member of $\mathcal{V}$. By Theorem 4.4.8 and Whitehead's Theorem 4.8.12 on small subdivision, P has a triangulation K such that $K \prec f^{-1}(\mathcal{W})$ (i.e., for each $\tau \in K$, $f(\tau)$ is contained in some $W \in \mathcal{W}$), Q is triangulated by a full subcomplex L of K, and for each $\tau \in L$, $f|\tau : \tau \to f(\tau)$ is an affine homeomorphism (so $f(\tau)$ is a simplex with $f(\tau)^{(0)} = f(\tau^{(0)})$). Then, $2\dim K + 1 \leqslant n$, and K is countable because P is separable (Proposition 4.2.11 (1)).

For each $v \in K^{(0)} \setminus L^{(0)}$, $W_v = \bigcap \mathcal{W}[f(v)]$ is open in $\mathbb{R}^n$ because $\mathcal{W}$ is locally finite. Applying the General Position Lemma 4.14.1, we can obtain an injection $g_0 : K^{(0)} \to \mathbb{R}^n$ such that

(1) $g_0|L^{(0)} = f|L^{(0)}$,
(2) $g_0(v) \in W_v$ for each $v \in K^{(0)} \setminus L^{(0)}$,
(3) $S \subset g_0(K^{(0)})$ with $\operatorname{card} S \leqslant n+1$ is affinely independent whenever $S \cap g_0(L^{(0)})$ is affinely independent.

From (3) above, it follows that $S \subset g_0(K^{(0)} \setminus L^{(0)})$ is affinely independent if $\operatorname{card} S \leqslant n+1$.

Then, g_0 is uniquely extended to a PL map $g : P = |K| \to \mathbb{R}^n$ so that $g|\sigma$ is affine for each $\sigma \in K$. For each $\tau \in L$, $g|\tau = f|\tau$ because $g_0|\tau^{(0)} = f|\tau^{(0)}$ by (1) and $f|\tau$ is affine. Hence, $g|Q = f|Q$. Moreover, for each $\sigma \in K$, $g|\sigma : \sigma \to g(\sigma)$ is an affine homeomorphism. Indeed, by (3), if $\sigma \cap |L| = \emptyset$, then $g(\sigma^{(0)})$ is affinely independent. When $\sigma \cap |L| \neq \emptyset$, since L is full in K, we have $\tau = \sigma \cap |L| \in L$ is a face of σ and $\tau^{(0)} = \sigma^{(0)} \cap L^{(0)}$. Then, $g(\sigma^{(0)}) \cap g(L^{(0)}) = g(\tau^{(0)}) = f(\tau^{(0)})$ is affinely independent and

$$\operatorname{card} g(\sigma^{(0)}) = \operatorname{card} \sigma^{(0)} \leqslant \dim K < 2\dim K + 2 \leqslant n+1.$$

Hence, by (3), $g(\sigma^{(0)})$ is affinely independent, which means that $g(\sigma)$ is a simplex.

For each $x \in |K|$, $f(c_K(x))$ is contained in some $W \in \mathcal{W}$ because $K \prec f^{-1}(\mathcal{W})$. For each $v \in c_K(x)^{(0)} \setminus L^{(0)}$, $g(v) \in W_v = \bigcap \mathcal{W}[f(v)] \subset W$ because $f(v) \in W$. Then, it follows that $g(c_K(x)^{(0)}) \subset W$. Hence, $f(c_K(x)) \cup g(c_K(x)) \subset \langle W\rangle$, which implies that $\langle f(x), g(x)\rangle \subset \langle W\rangle$. Since $\langle W\rangle$ is contained in some member of $\mathcal{V}$, it follows that $f \simeq_{\mathcal{V}} g$ rel. Q by the straight-line homotopy. Hence, $f \simeq_{\mathcal{U}} g$ rel. Q because $\mathcal{V}[f(P)] \prec \mathcal{U}$. Since f is proper, it follows from Proposition 2.1.19 (1) that g is also proper, hence it is a closed map by Proposition 2.1.12. Moreover, the homotopy is also proper by Proposition 2.1.19 (2), that is, $f \simeq^p_{\mathcal{U}} g$ rel. Q. All that remains is to show that g is an injection.

Let $x, y \in P$ and assume that $g(x) = g(y)$. When $x, y \in Q$, since $g|Q = f|Q$ is an embedding, we have $x = y$. We will consider the case when $x \notin Q$ or $y \notin Q$. When $c_K(x) \cap Q = \emptyset$, let $S = g(c_K(x)^{(0)}) \cup g(c_K(y)^{(0)})$. Then, $\operatorname{card} S \leqslant 2(\dim K + 1) \leqslant n+1$. Since $c_K(y) \cap |L| = \emptyset$ or $c_K(y) \cap |L|$ is a simplex of L (because L is full in K), it follows that $S \cap g(L^{(0)}) = g(c_K(y)^{(0)} \cap L^{(0)}) = f(c_K(y)^{(0)} \cap L^{(0)})$ is affinely independent. Thus, by (3), S is affinely independent, and hence $\langle S\rangle$ is a simplex, where $g(c_K(x))$ and $g(c_K(y))$ are its faces. Since

$g|c_K(x) : c_K(x) \to g(c_K(x))$ and $g|c_K(y) : c_K(y) \to g(c_K(y))$ are affine homeomorphisms, it follows that $g(x) \in \operatorname{rint} g(c_K(x))$ and $g(y) \in \operatorname{rint} g(c_K(y))$. Hence, $g(c_K(x)) = g(c_K(y))$ because $g(x) = g(y)$. Since g_0 is an injection, $c_K(x)^{(0)} = c_K(y)^{(0)}$, which means that $c_K(x) = c_K(y)$. Moreover, $g|c_K(x) = g|c_K(y)$ is injective, we have $x = y$. When $c_K(y) \cap Q = \emptyset$, it similarly follows that $x = y$.

Finally, we consider the case when $c_K(x) \cap Q \neq \emptyset$ and $c_K(y) \cap Q \neq \emptyset$ under the assumption that $x \notin Q$ or $y \notin Q$. Then, we may assume that $x \notin Q$ without loss of generality. Since L is full in K, $\sigma = c_K(x) \cap |L|$ and $\tau = c_K(y) \cap |L|$ are simplexes of L. Take a maximal subset $\{v_1, \ldots, v_k\} \subset \sigma^{(0)} \cup \tau^{(0)}$ such that $\{g(v_1), \ldots, g(v_k)\}$ is affinely independent, where $v_i \neq v_j$ if $i \neq j$. Then, we have $g(\sigma) \cup g(\tau) \subset \mathrm{fl}\{g(v_1), \ldots, g(v_k)\}$. Note that $(c_K(x)^{(0)} \cup c_K(y)^{(0)}) \setminus L^{(0)} \neq \emptyset$ because $x \notin |L|$. We write this set as $\{w_1, \ldots, w_\ell\}$ such that

$$c_K(x)^{(0)} \setminus L^{(0)} = \{w_i \mid i \leqslant b\} \text{ and } c_K(y)^{(0)} \setminus L^{(0)} = \{w_j \mid j > a\},$$

where $0 \leqslant a \leqslant b$, $1 \leqslant b \leqslant \ell$, and $w_i \neq w_j$ if $i \neq j$. Observe that

$$k + \ell \leqslant (\dim c_K(x) + 1) + (\dim c_K(y) + 1) \leqslant 2(\dim K + 1) \leqslant n + 1.$$

Hence, $\{g(v_1), \ldots, g(v_k), g(w_1), \ldots, g(w_\ell)\}$ is affinely independent by (3). Write $c_K(x) = \sigma\sigma'$, where $\sigma' \neq \emptyset$ because $x \notin |L|$. Then, x can be uniquely written as

$$x = (1 - s)x' + sx'', \ x' \in \sigma, \ x'' \in \sigma', \ 0 < s < 1.$$

Since $g(x') \in g(\sigma) \subset \mathrm{fl}\{g(v_1), \ldots, g(v_k)\}$, we can write

$$g(x') = \sum_{i=1}^{k} s_i g(v_i), \ \sum_{i=1}^{k} s_i = 1.$$

On the other hand, $x'' \in \sigma' = \langle w_1, \ldots, w_b \rangle$ can be written as $x'' = \sum_{j=1}^{b} s'_j w_j$, $\sum_{j=1}^{b} s'_j = 1$, $s'_j \geqslant 0$. Since $g|\sigma'$ is affine, we have $g(x'') = \sum_{j=1}^{b} s'_j g(w_j)$. Then, it follows that

$$g(x) = \sum_{i=1}^{k} (1 - s)s_i g(v_i) + \sum_{j=1}^{b} ss'_j g(w_j).$$

If $y \in Q$, then $g(x) = g(y) \in g(\tau) \subset \mathrm{fl}\{g(v_1), \ldots, g(v_k)\}$, which implies that $ss'_j = 0$ for each $j = 1, \ldots, b$. Since $s \neq 0$, it follows that $s'_j = 0$ for each $j = 1, \ldots, b$, which contradicts $\sum_{j=1}^{b} s'_j = 1$. Therefore, $y \notin Q$ and $a < \ell$. Then, similarly to x, we can uniquely write

$$y = (1 - t)y' + ty'', \ y' \in \tau, \ y'' \in \tau', \ 0 < t < 1,$$

where $c_K(y) = \tau\tau'$, $\tau' \neq \emptyset$. Moreover, similarly to x' and x'', we can write

$$g(y') = \sum_{i=1}^{k} t_i g(v_i),\ \sum_{i=1}^{k} t_i = 1,\ \text{and}\ y'' = \sum_{j=a+1}^{\ell} t'_j w_j,\ \sum_{j=a+1}^{\ell} t'_j = 1,\ t'_j \geqslant 0.$$

Then, in the same way as $g(x)$, we have

$$g(y) = \sum_{i=1}^{k} (1-t) t_i g(v_i) + \sum_{j=a+1}^{\ell} t t'_j g(w_j).$$

Since $g(x) = g(y)$ and $\{g(v_1), \ldots, g(v_k), g(w_1), \ldots, g(w_\ell)\}$ is affinely independent, it follows that $(1-s)s_i = (1-t)t_i$ for each $i = 1, \ldots, k$; $a < b$ and

$$j \leqslant a \ \Rightarrow\ s s'_j = 0;\ j > b \ \Rightarrow\ t t'_j = 0;\ a < j \leqslant b \ \Rightarrow\ s s'_j = t t'_j.$$

Because $\sum_{j=1}^{b} s'_j = \sum_{j=a+1}^{\ell} t'_j = 1$, we have $s = \sum_{j=1}^{b} s s'_j = \sum_{j=a+1}^{\ell} t t'_j = t$. Hence, $s_i = t_i$ for each $i = 1, \ldots, k$, and

$$j \leqslant a \ \Rightarrow\ s'_j = 0;\ j > b \ \Rightarrow\ t'_j = 0;\ a < j \leqslant b \ \Rightarrow\ s'_j = t'_j.$$

Thus, it follows that

$$f(x') = g(x') = \sum_{i=1}^{k} s_i g(v_i) = \sum_{i=1}^{k} t_i g(v_i) = g(y') = f(y') \ \text{ and}$$
$$x'' = \sum_{j=1}^{b} s'_j w_j = \sum_{j=a+1}^{\ell} t'_j w_j = y''.$$

Then, $x' = y'$ because $f|Q$ is an embedding. Consequently, $x = (1-s)x' + s x'' = (1-t)y' + t y'' = y$. □

Note the following fact:

- *Every locally compact separable metrizable space X has a proper map $f : X \to \mathbb{R}_+$.*[31]

  Indeed, if X is non-compact, then there is a compact tower $X_1 \subset X_2 \subset \cdots$ such that $\bigcup_{n\in\mathbb{N}} X_n = X$ and $X_n \subset \operatorname{int} X_{n+1}$ for each $n \in \mathbb{N}$. Connecting Urysohn maps, we can construct a proper map $f : X \to \mathbb{R}_+$.

Since a polyhedron is metrizable if and only if it is locally compact (Proposition 4.2.11 (2)), we have the following corollary:

Corollary 4.14.3 *Every n-dimensional locally compact separable polyhedron can be embedded into $\mathbb{R}^{2n+1}$ as a subpolyhedron.* □

[31] According to the Urysohn Metrization Theorem (Corollary 2.3.5), the term "separable metrizable" can be replaced by "regular and second countable."

Remark 4.14.1 In Theorem 4.14.2 and Corollary 4.14.3, as is easily observed, $\mathbb{R}^n$ can be replaced by $\mathbf{I}^n$, where the considered polyhedron must be compact.

The PL Approximation Theorem 4.11.11 can be proved by using derived (or regular) neighborhoods but the following stronger version can be proved:

Theorem 4.14.4 (RELATIVE PL APPROXIMATION) *Let P and Q be polyhedra, P_0 a subpolyhedron of P, and $f : P \to Q$ a map such that $f|P_0$ is PL. For each open cover $\mathcal{U}$ of Q, there is a PL map $g : P \to Q$ such that $f|P_0 = g|P_0$ and $f \simeq_{\mathcal{U}} g$ rel. P_0.*

Proof Let L be a triangulation of Q. Identifying $v \in L^{(0)}$ with the unit vector $\mathbf{e}_v \in \ell_1^f(L^{(0)})$ ($\mathbf{e}_v(v) = 1$ and $\mathbf{e}_v(v') = 0$ for $v' \in L^{(0)} \setminus \{v\}$), we regard L as a subcomplex of the full complex $\mathrm{F}_{L^{(0)}}$. For simplicity, we write $F = \mathrm{F}_{L^{(0)}}$. Thus, we can regard Q as a subspace of the convex polyhedron $|F| = \langle L^{(0)} \rangle$ in $\ell_1(L^{(0)})$.[32] Applying Proposition 4.12.1 to $(\mathrm{Sd}\, F, \mathrm{Sd}\, L)$, we have a simplicial retraction $r : N(\mathrm{Sd}^2 L, \mathrm{Sd}^2 F) \to \mathrm{Sd}^2 L$ (as a map, $r : |N(\mathrm{Sd}^2 L, \mathrm{Sd}^2 F)| \to Q$). Since $|\mathrm{Sd}^2 F|_{\mathrm{m}} = |\mathrm{Sd}\, F|_{\mathrm{m}} = |F|_{\mathrm{m}}$ as spaces, it follows that $|N(\mathrm{Sd}^2 L, \mathrm{Sd}^2 F)|$ is a neighborhood of Q in $|F|_{\mathrm{m}}$. Then, we can take an open cover $\mathcal{V}$ of Q in $|F|_{\mathrm{m}}$ so that each $V \in \mathcal{V}$ is convex and contained in $|N(\mathrm{Sd}^2 L, \mathrm{Sd}^2 F)|$, and $\mathcal{V} \prec r^{-1}(\mathcal{U})$. Note that $\mathcal{V}$ is also an open cover of Q in $|F|$. By Theorem 4.4.8 and Whitehead's Theorem 4.8.12 on small subdivision, P has a triangulation K such that $K \prec f^{-1}(\mathcal{V})$ and P_0 is triangulated by a subcomplex K_0 of K.

Since $f|P_0$ is PL, we can assume that $f|\sigma$ is affine for each $\sigma \in K_0$ by replacing K with its subdivision if necessary. Because $|F|$ is convex, we can define a PL map $\varphi : |K| \to |F|$ so that $\varphi|K^{(0)} = f|K^{(0)}$ and $\varphi|\sigma$ is affine for each $\sigma \in K$. Then, it follows that $\varphi||K_0| = f||K_0|$, i.e., $\varphi|P_0 = f|P_0$. For each $\sigma \in K$, we can take $V_\sigma \in \mathcal{V}$ so that $f(\sigma) \subset V_\sigma$. Then, since V_σ is convex, it follows that

$$\varphi(\sigma) = \langle f(\sigma^{(0)}) \rangle \subset V_\sigma \subset |N(\mathrm{Sd}^2 L, \mathrm{Sd}^2 F)|.$$

Thus, we have a PL map $g = r\varphi : P \to Q$ (Proposition 4.11.3), where $g|P_0 = r\varphi|P_0 = f|P_0$. By the same reasoning as $g = r\varphi$, we can also define a homotopy $h : P \times \mathbf{I} \to Q$ as follows:

$$h(x, t) = r((1 - t)f(x) + t\varphi(x)) \quad \text{for each } (x, t) \in P \times \mathbf{I}.$$

Then, $h_0 = f$, $h_1 = r\varphi = g$, and $h_t|P_0 = f|P_0$ for each $t \in \mathbf{I}$. For each $\sigma \in K$, $h(\sigma \times \mathbf{I}) \subset r(V_\sigma)$ and $r(V_\sigma)$ is contained in some $U \in \mathcal{U}$. Hence, h is a $\mathcal{U}$-homotopy. Consequently, we have $f \simeq_{\mathcal{U}} g$ rel. P_0. □

Combining the above Relative PL Approximation Theorem 4.14.4 with Proposition 2.1.19 (2), we have the following:

[32] Note that $|F|_{\mathrm{m}}$ is a subspace of the Banach space $\ell_1(L^{(0)})$. In fact, the metric ρ_F is defined by the norm of $\ell_1(L^{(0)})$ but $|F| \neq |F|_{\mathrm{m}}$ as spaces.

Theorem 4.14.5 (Relative Proper PL Approximation) *Let P and Q be locally compact polyhedra, P_0 a subpolyhedron of P, and $f : P \to Q$ a proper map such that $f|P_0$ is PL. For each open cover $\mathcal{U}$ of Q, there is a proper PL map $g : P \to Q$ such that $f|P_0 = g|P_0$ and $f \simeq^p_{\mathcal{U}} g$ rel. P_0.* □

Using the Relative Proper PL Approximation Theorem 4.14.5 and the General Position Lemma 4.14.1, we can prove the following PL Embedding Approximation Theorem:

Theorem 4.14.6 (PL Embedding Approximation for Products) *Let P and Q be finite-dimensional locally compact separable polyhedra, P_0 a subpolyhedron of P, and $n \geqslant 2 \dim P + 1$. If $f : P \to Q \times \mathbf{I}^n$ is a proper map and $f|P_0$ is a PL embedding, then for any $\mathcal{U} \in \mathrm{cov}(Q \times \mathbf{I}^n)$, there exists a PL embedding $h : P \to Q \times \mathbf{I}^n$ such that $h|P_0 = f|P_0$ and $h \simeq^p_{\mathcal{U}} f$ rel. P_0.*

Proof We assume that $\mathbf{I}^n$ inherits the Euclidean metric from $\mathbb{R}^n$. Then, for each $\varepsilon > 0$, the ε-neighborhood $\mathrm{B}(z, \varepsilon)$ of $z \in \mathbf{I}^n$ is convex. Since $\mathbf{I}^n$ is compact and Q is locally compact, it is easy to find a compact neighborhood N_y of each $y \in Q$ and a finite open cover $\mathcal{O}_y$ of $\mathbf{I}^n$ such that $\{N_y \times O \mid O \in \mathcal{O}_y\} \prec \mathcal{U}$. Then, there exists a locally finite open cover $\mathcal{V} \in \mathrm{cov}(Q)$ such that $\mathrm{st}\,\mathcal{V} \prec \{\mathrm{int}\, N_y \mid y \in Q\}$. For each $V \in \mathcal{V}$, choose $y \in Q$ so that $\mathrm{st}(V, \mathcal{V}) \subset \mathrm{int}\, N_y$ and let $\delta_V > 0$ be a Lebesgue number for $\mathcal{O}_y$. Then, for each $V \in \mathcal{V}$, $\mathrm{cl}\,\mathrm{st}(V, \mathcal{V})$ is compact and

$$\left\{ \mathrm{cl}\,\mathrm{st}(V, \mathcal{V}) \times \mathrm{B}(z, \delta_V/3) \;\middle|\; z \in \mathbf{I}^n \right\} \prec \mathcal{U}.$$

Note that $\mathrm{pr}_Q f : P \to Q$ is a proper map and $\mathrm{pr}_Q f|P_0$ is a PL map. Applying the Relative Proper PL Approximation Theorem 4.14.5, we have a proper PL map $k : P \to Q$ such that $k|P_0 = \mathrm{pr}_Q f|P_0$ and $k \simeq^p_{\mathcal{V}} \mathrm{pr}_Q f$ rel. P_0. Let

$$\mathcal{W} = \left\{ V \times \mathrm{B}(z, \delta_V/6) \;\middle|\; V \in \mathcal{V},\ z \in \mathbf{I}^n \right\} \in \mathrm{cov}(Q \times \mathbf{I}^n).$$

By Theorem 4.4.8 and Whitehead's Theorem 4.8.12 on small subdivisions, we have a triangulation K of P such that P_0 is triangulated by a full subcomplex K_0 of K, $K \prec f^{-1}(\mathcal{W})$, $k|\sigma$ is affine for each $\sigma \in K$, and $\mathrm{pr}_{\mathbf{I}^n} f|\tau$ is affine for each $\tau \in K_0$, which means that $f|\tau$ is affine for each $\tau \in K_0$.

For each $v \in K^{(0)} \setminus K_0^{(0)}$, since $\mathcal{V}$ is locally finite, we have

$$\delta_v = \min \left\{ \delta_V/6 \;\middle|\; V \in \mathcal{V}[\mathrm{pr}_Q f(v)] \right\} > 0.$$

Applying Lemma 4.14.1, we can obtain a function $g_0 : K^{(0)} \to \mathbf{I}^n$ such that

(1) $g_0|K_0^{(0)} = \mathrm{pr}_{\mathbf{I}^n} f|K_0^{(0)}$,
(2) $g_0|K^{(0)} \setminus K_0^{(0)}$ is an injection into $\mathbf{I}^n \setminus \mathrm{pr}_{\mathbf{I}^n} f(K_0^{(0)})$,
(3) $\|g_0(v) - \mathrm{pr}_{\mathbf{I}^n} f(v)\| < \delta_v$ for each $v \in K^{(0)} \setminus K_0^{(0)}$, and
(4) $S \subset g_0(K^{(0)})$ with $\mathrm{card}\, S \leqslant n+1$ is affinely independent whenever $S \cap g_0(K_0^{(0)})$ is affinely independent.

The above implies that $S \subset g_0(K^{(0)} \setminus K_0^{(0)})$ is affinely independent if card $S \leqslant n+1$.

Then, g_0 uniquely extends to a PL map $g : P = |K| \to \mathbf{I}^n$ so that $g|\sigma$ is affine for each $\sigma \in K$. Since $\mathrm{pr}_{\mathbf{I}^n} f|\tau$ is affine for each $\tau \in K_0$, it follows from (1) that $g|P_0 = \mathrm{pr}_{\mathbf{I}^n} f|P_0$.

Define an injection $h_0 : K^{(0)} \to Q \times \mathbf{I}^n$ by $h_0(v) = (k(v), g_0(v))$ for each $v \in K^{(0)}$. By (1), for each $v \in K_0^{(0)}$,

$$h_0(v) = (k(v), \mathrm{pr}_{\mathbf{I}^n} f(v)) = (\mathrm{pr}_Q f(v), \mathrm{pr}_{\mathbf{I}^n} f(v)) = f(v).$$

Since $f|P_0$ is an injection, h_0 is also an injection by (2). For each $\sigma \in K$, $k(\sigma)$ is convex because $k|\sigma$ affine, and $h_0(\sigma^{(0)}) \subset k(\sigma) \times \mathbf{I}^n$. Hence, h_0 uniquely extends to a PL map $h : P = |K| \to Q \times \mathbf{I}^n$ such that $h|\sigma$ is affine for each $\sigma \in K$. Since $k|\sigma$ and $g|\sigma$ are affine for each $\sigma \in K$, it follows that $\mathrm{pr}_Q h = k$ and $\mathrm{pr}_{\mathbf{I}^n} h = g$, which means that $h(x) = (k(x), g(x))$ for each $x \in P$. Then, $h|P_0 = f|P_0$ because $k|P_0 = \mathrm{pr}_Q f|P_0$ and $g|P_0 = \mathrm{pr}_{\mathbf{I}^n} f|P_0$. Moreover, since k is proper, it follows that h is proper, so it is a closed map.

To see that h is injective, recall that $h_0 = h|K^{(0)}$ is an injection. If h_0 satisfy (3) in the proof of the Theorem 4.14.2, then the last part (the last two paragraphs) of the proof can be applied to this case. Thus, it suffices to show the following:

(5) $S \subset h_0(K^{(0)})$ with card $S \leqslant n+1$ is affinely independent whenever $S \cap h_0(K_0^{(0)})$ is affinely independent.

Here, the affine independence means the one in the linear space $F \times \mathbb{R}^n$, where F is a linear space containing Q.

To show (5), let $S = \{h_0(v_1), \ldots, h_0(v_k)\} \subset h_0(K^{(0)})$ such that $S \cap h_0(K_0^{(0)}) = \{h_0(v_1), \ldots, h_0(v_a)\}$ is affinely independent, where $0 \leqslant a \leqslant k \leqslant n+1$, $v_i \neq v_j$ if $i \neq j$. In the case when $a = 0$, as noted at (4), $\{g_0(v_1), \ldots, g_0(v_k)\}$ is affinely independent, which implies that $S = \{h_0(v_1), \ldots, h_0(v_k)\}$ is affinely independent because $h_0(v_i) = (k(v_i), g_0(v_i))$. When $a > 0$, reordering $v_1, \ldots, v_a$ if necessary, we may assume that there exists $1 \leqslant b \leqslant a$ such that $\{g_0(v_1), \ldots, g_0(v_b)\}$ is affinely independent and $g_0(v_i) \in \mathrm{fl}\{g_0(v_1), \ldots, g_0(v_b)\}$ for $b < i \leqslant a$, where $g_0(v_i) \neq g_0(v_j)$ if $1 \leqslant i < j \leqslant b$. For each $b < i \leqslant a$, we can write

$$g_0(v_i) = \sum_{j=1}^{b} s_{i,j} g_0(v_j), \quad \sum_{j=1}^{b} s_{i,j} = 1.$$

To verify that S is affinely independent, suppose $\sum_{i=1}^{k} t_i h_0(v_i) = \mathbf{0}$ and $\sum_{i=1}^{k} t_i = 0$. Observe that

$$\sum_{i=b+1}^{a} t_i g_0(v_i) = \sum_{i=b+1}^{a} t_i \left(\sum_{j=1}^{b} s_{i,j} g_0(v_j) \right) = \sum_{j=1}^{b} \left(\sum_{i=b+1}^{a} t_i s_{i,j} \right) g_0(v_j).$$

Then, we have

$$\sum_{j=1}^{b}\left(t_j+\sum_{i=b+1}^{a}t_i s_{i,j}\right)g_0(v_j)+\sum_{i=a+1}^{k}t_i g_0(v_i)=\sum_{i=1}^{k}t_i g_0(v_i)=\mathbf{0}\ \text{ and}$$
$$\sum_{j=1}^{b}\left(t_j+\sum_{i=b+1}^{a}t_i s_{i,j}\right)+\sum_{i=a+1}^{k}t_i=\sum_{i=1}^{k}t_i=0.$$

Moreover, the following is affinely independent (cf. (1) and (2)):

$$\big\{g_0(v_1),\dots,g_0(v_b),g_0(v_{a+1}),\dots,g_0(v_k)\big\}\cap g_0(K_0^{(0)})=\big\{g_0(v_1),\dots,g_0(v_b)\big\}.$$

Then, by (4), the following is also affinely independent:

$$\big\{g_0(v_1),\dots,g_0(v_b),g_0(v_{a+1}),\dots,g_0(v_k)\big\}.$$

Hence, $t_j+\sum_{i=b+1}^{a}t_i s_{i,j}=0$ for every $j=1,\dots,b$, and $t_i=0$ for every $i=a+1,\dots,k$. Thus, we have $\sum_{i=1}^{a}t_i h_0(v_i)=\mathbf{0}$ and $\sum_{i=1}^{a}t_i=0$. Now, by the assumption that $S\cap h_0(K_0^{(0)})=\{h_0(v_1),\dots,h_0(v_a)\}$ is affinely independent, we have $t_i=0$ for every $i=1,\dots,a$. Therefore, S is affinely independent.

Finally, we will show that $h\simeq^p_{\mathcal{U}}f$. Let $\varphi:P\times\mathbf{I}\to Q$ be a proper homotopy from k to $\mathrm{pr}_Q f$ realizing $k\simeq^p_{\mathcal{V}}\mathrm{pr}_Q f$ rel. P_0. Define a homotopy $\psi:P\times\mathbf{I}\to Q\times\mathbf{I}^n$ as follows:

$$\psi(x,t)=(\varphi(x,t),(1-t)g(x)+t\mathrm{pr}_{\mathbf{I}^n}f(x))\ \text{ for each } (x,t)\in P\times\mathbf{I}.$$

Then, $\psi_0=h$, $\psi_1=f$, and $\psi_t|P_0=\psi_0|P_0$ for every $t\in\mathbf{I}$ because $g|P_0=\mathrm{pr}_{\mathbf{I}^n}f|P_0$. Since φ is proper, ψ is also proper. To see that ψ is a $\mathcal{U}$-homotopy, let $x\in P$. Since $K\prec f^{-1}(\mathcal{W})$, we can take $V\in\mathcal{V}$ and $z\in\mathbf{I}^n$ such that $f(c_K(x))\subset V\times\mathrm{B}(z,\delta_V/6)$. For each $v\in c_K(x)^{(0)}\setminus K_0^{(0)}$, $\delta_v\leqslant\delta_V/6$ because $\mathrm{pr}_Q f(v)\in V$. Hence, by (1) and (3), $\|g_0(v)-\mathrm{pr}_{\mathbf{I}^n}f(v)\|<\delta_V/6$ for each $v\in c_K(x)^{(0)}$. Moreover, we have

$$y=\sum_{v\in c_K(x)^{(0)}}\beta_v^K(x)\mathrm{pr}_{\mathbf{I}^n}f(v)\in\big\langle\mathrm{pr}_{\mathbf{I}^n}f(c_K(x)^{(0)})\big\rangle$$
$$\subset\big\langle\mathrm{pr}_{\mathbf{I}^n}f(c_K(x))\big\rangle\subset\mathrm{B}(z,\delta_V/6).$$

Then, it follows that

$$\begin{aligned}\|g(x)-z\| &\leqslant \|g(x)-y\|+\|y-z\| \\ &< \left\| \sum_{v\in c_K(x)^{(0)}} \beta_v^K(x) g_0(v) - \sum_{v\in c_K(x)^{(0)}} \beta_v^K(x)\mathrm{pr}_{\mathbf{I}^n} f(v) \right\| + \delta_V/6 \\ &\leqslant \sum_{v\in c_K(x)^{(0)}} \beta_v^K(x)\|g_0(v)-\mathrm{pr}_{\mathbf{I}^n} f(v)\| + \delta_V/6 < \delta_V/3.\end{aligned}$$

On the other hand, $\mathrm{pr}_{\mathbf{I}^n} f(x) \in \mathrm{pr}_{\mathbf{I}^n} f(c_K(x)) \subset \mathrm{B}(z, \delta_V/6)$. Thus, we have

$$(1-t)g(x) + t\mathrm{pr}_{\mathbf{I}^n} f(x) \in \mathrm{B}(z, \delta_V/3) \text{ for every } t \in \mathbf{I}.$$

Since φ is a $\mathcal{V}$-homotopy, $\varphi(\{x\}\times\mathbf{I})$ is contained in some $V' \in \mathcal{V}$, where $V' \subset \mathrm{st}(V,\mathcal{V})$ because $\mathrm{pr}_Q f(x) \in V' \cap V$. Hence,

$$\psi(\{x\}\times\mathbf{I}) \subset \mathrm{st}(V,\mathcal{V}) \times \mathrm{B}(z, \delta_V/3),$$

which is contained in some $U \in \mathcal{U}$. Consequently, ψ is a $\mathcal{U}$-homotopy. □

Now, we can generalize Theorem 4.14.2 to PL manifolds as follows:

Theorem 4.14.7 (PL Embedding Approximation for PL Manifolds) *Let P be a compact polyhedron with a subpolyhedron $P_0 \subset P$ and let M be a PL n-manifold. Suppose that $n \geqslant 2\dim P + 1$ and $f : P \to M$ is a map such that $f|P_0$ is a PL embedding. Then, for each open cover $\mathcal{U}$ of M, there exists a PL embedding $h : P \to M$ such that $h|P_0 = f|P_0$ and $h \simeq_{\mathcal{U}} f$ rel. P_0.*

Proof Take subpolyhedra $C_1, \ldots, C_k \subset M$ so that $f(P) \subset \bigcup_{i=1}^k \mathrm{int}\, C_i$ and each C_i is PL homeomorphic to $\mathbf{I}^n$. Then, we have PL homeomorphisms $\varphi_i : \mathbf{I}^n \to C_i$, $i = 1, \ldots, k$. We can take subpolyhedra $P_0 \subset P_1 \subset \cdots \subset P_k = P$ such that $f(\mathrm{cl}(P_i \setminus P_{i-1})) \subset \mathrm{int}\, C_i$ for each $i = 1, \ldots, k$.

Indeed, P has a triangulation K whose simplex contained in some $f^{-1}(\mathrm{int}\, C_i)$. Let K_i be the subcomplex of K consisting of all simplexes contained in $f^{-1}(\mathrm{int}\, C_i)$. For $i = 1, \ldots, k$, define $P_i = P_0 \cup |K_1| \cup \cdots \cup |K_i|$. Then, $\mathrm{cl}(P_i \setminus P_{i-1}) \subset |K_i| \subset f^{-1}(\mathrm{int}\, C_i)$.

By Proposition 2.7.9 (1), we can take $d \in \mathrm{Metr}(M)$ such that $\{\overline{\mathrm{B}}_d(x,1) \mid x \in X\} \prec \mathcal{U}$. By induction, we will construct maps $f_i : P \to M$, $i = 1, \ldots, k$, so that $f_i|P_i$ is a PL embedding, $f_i|P_{i-1} = f_{i-1}|P_{i-1}$, $f_i \simeq_{1/k} f_{i-1}$ rel. P_{i-1}, and $f_i(\mathrm{cl}(P_j \setminus P_{j-1})) \subset \mathrm{int}\, C_j$ for each $j > i$, where $f_0 = f$. Then, $h = f_k : P = P_k \to M$ is the desired PL embedding. Indeed, $h|P_0 = f_k|P_0 = f_0|P_0 = f|P_0$ and

$$h = f_k \simeq_{1/k} f_{k-1} \simeq_{1/k} \cdots \simeq_{1/k} f_0 = f \text{ rel. } P_0,$$

which implies that $h \simeq_{\mathcal{U}} f$ rel. P_0.

Suppose f_{i-1} has been constructed. Choose $\varepsilon > 0$ so that $\varepsilon < 1/k$ and

$$\mathrm{N}(f_{i-1}(\mathrm{cl}(P_j \setminus P_{j-1})), \varepsilon) \subset \mathrm{int}\, C_j \text{ for each } j > i.$$

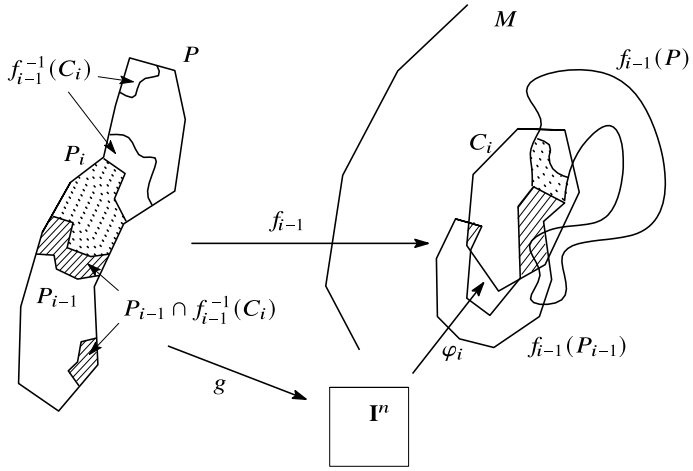


Fig. 4.14 $P_i \setminus \operatorname{int} f_{i-1}^{-1}(C_i)$ and $f_{i-1}(P_{i-1}) \setminus \operatorname{int} C_i$

By the uniform continuity of φ_i, we have $\delta > 0$ such that

$$x, y \in \mathbf{I}^n, \|x - y\| < \delta \ \Rightarrow\ d(\varphi_i(x), \varphi_i(y)) < \varepsilon.$$

Since $f_{i-1}(\operatorname{cl}(P_i \setminus P_{i-1})) \subset C_i$, it follows that

$$P_i \cap f_{i-1}^{-1}(C_i) = (P_{i-1} \cap f_{i-1}^{-1}(C_i)) \cup \operatorname{cl}(P_i \setminus P_{i-1})$$

is a subpolyhedron of P and $P_{i-1} \cap f_{i-1}^{-1}(C_i)$ is its subpolyhedron. Consider the following map:

$$\varphi_i^{-1} f_{i-1} | P_i \cap f_{i-1}^{-1}(C_i) : P_i \cap f_{i-1}^{-1}(C_i) \to \mathbf{I}^n,$$

where $\varphi_i^{-1} f_{i-1} | P_{i-1} \cap f_{i-1}^{-1}(C_i)$ is a PL embedding. Applying the PL Embedding Approximation Theorem 4.14.6 (the case Q is a singleton), we can obtain a PL embedding $g : P_i \cap f_{i-1}^{-1}(C_i) \to \mathbf{I}^n$ such that

$$g | P_{i-1} \cap f_{i-1}^{-1}(C_i) = \varphi_i^{-1} f_{i-1} | P_{i-1} \cap f_{i-1}^{-1}(C_i) \ \text{ and}$$
$$g \simeq_\delta \varphi_i^{-1} f_{i-1} \ \text{ rel. } P_{i-1} \cap f_{i-1}^{-1}(C_i).$$

Refer to Fig. 4.14. Then, $\varphi_i g$ extends to a PL embedding $f' : P_i \to M$ defined by $f' | P_{i-1} = f_{i-1} | P_{i-1}$. Observe that $f' \simeq_\varepsilon f_{i-1} | P_i$ rel. P_{i-1}. Applying the Homotopy Extension Theorem 4.5.3, we can extend f' to a map $f_i : P \to M$ such that $f_i \simeq_\varepsilon f_{i-1}$ rel. P_{i-1}. Then, $f_i | P_i = f'$ is a PL embedding, $f_i | P_{i-1} = f_{i-1} | P_{i-1}$, $f_i \simeq_{1/k} f_{i-1}$ rel. P_{i-1}, and $f_i(\operatorname{cl}(P_j \setminus P_{j-1})) \subset \operatorname{int} C_j$ for each $j > i$. □

4.15 The Nerves of Open Covers

In this section, we introduce the concept of nerve, which is a method of approximating a space with a simplicial complex.

Let Γ be an arbitrary set. Recall $\mathrm{Fin}(\Gamma)$ is the collection of all non-empty finite subsets of Γ. An **abstract complex** $\mathcal{K}$ over Γ is a subcollection $\mathcal{K} \subset \mathrm{Fin}(\Gamma)$ satisfying the following condition:

(AC) If $A \in \mathcal{K}$ and $\emptyset \neq B \subset A$, then $B \in \mathcal{K}$.

A subcollection $\mathcal{L} \subset \mathcal{K}$ satisfying (AC) is called a **subcomplex** of $\mathcal{K}$. In particular, $\mathrm{Fin}(\Gamma)$ is an abstract complex and every abstract complex $\mathcal{K}$ over Γ is a subcomplex of $\mathrm{Fin}(\Gamma)$. For each $n \in \omega$, the ***n*-skeleton** $\mathcal{K}^{(n)}$ of $\mathcal{K}$ is defined by

$$\mathcal{K}^{(n)} = \left\{ A \in \mathcal{K} \;\middle|\; \mathrm{card}\, A \leqslant n+1 \right\} = \mathcal{K} \cap \mathrm{Fin}_{n+1}(\Gamma),$$

where we regard $\mathcal{K}^{(0)} \subset \Gamma$, and so $\mathrm{Fin}(\Gamma)^{(0)} = \Gamma$. Each $\mathcal{K}^{(n)}$ is a subcomplex of $\mathcal{K}$. When $\mathcal{K} = \mathcal{K}^{(n)}$, we say that $\mathcal{K}$ is **at most *n*-dimensional** and write $\dim \mathcal{K} \leqslant n$. It is said that $\mathcal{K}$ is ***n*-dimensional** (written as $\dim \mathcal{K} = n$) if $\dim \mathcal{K} \leqslant n$ and $\dim \mathcal{K} \nleqslant n-1$. Note that every abstract complex $\mathcal{K}$ over Γ with $\dim \mathcal{K} \leqslant n$ is a subcomplex of $\mathrm{Fin}(\Gamma)^{(n)}$.

Let $\mathcal{K}$ and $\mathcal{L}$ be abstract complexes. It is said that $\mathcal{K}$ and $\mathcal{L}$ are **congruent** or $\mathcal{K}$ is **congruent** to $\mathcal{L}$, denoted by $\mathcal{K} \equiv \mathcal{L}$, if there exists a bijection $\varphi : \mathcal{K} \to \mathcal{L}$ such that for each $A, B \in \mathcal{K}$, $A \subset B$ if and only if $\varphi(A) \subset \varphi(B)$.

For any simplicial complex K, observe that $\{\sigma^{(0)} \mid \sigma \in K\} \subset \mathrm{Fin}(K^{(0)})$ is an abstract complex over $K^{(0)}$, which is called the **abstract complex defined by** K. When an abstract complex $\mathcal{K}$ is congruent to the abstract complex defined by a simplicial complex K, that is, $\mathcal{K} \equiv \{\sigma^{(0)} \mid \sigma \in K\}$, it is said that $\mathcal{K}$ is **realized by** K or K **realizes** $\mathcal{K}$, and K is called a **simplicial complex realizing** $\mathcal{K}$. Evidently, if $\mathcal{K}$ is realized by K, then every subcomplex of $\mathcal{K}$ is realized by a subcomplex of K. Every abstract complex $\mathcal{K}$ over Γ can be realized by some simplicial complex. In fact, consider the linear space $\mathbb{R}^{\Gamma}_{f}$. For each $\gamma \in \Gamma$, let $\mathbf{e}_{\gamma} \in \mathbb{R}^{\Gamma}_{f}$ be the unit vector defined by $\mathbf{e}_{\gamma}(\gamma) = 1$ and $\mathbf{e}_{\gamma}(\gamma') = 0$ if $\gamma \neq \gamma'$, where $V = \{\mathbf{e}_{\gamma} \mid \gamma \in \Gamma\}$ is the canonical Hamel basis for $\mathbb{R}^{\Gamma}_{f}$. Then, we have the following simplicial complex:

$$K = \left\{ \langle \mathbf{e}_{\gamma_1}, \ldots, \mathbf{e}_{\gamma_n} \rangle \;\middle|\; \{\gamma_1, \ldots, \gamma_n\} \in \mathcal{K} \right\},$$

where $K^{(0)} \subset V$. By the bijection corresponding each $\gamma \in \mathcal{K}^{(0)}$ to $\mathbf{e}_{\gamma} \in K^{(0)}$, $\mathcal{K}$ is congruent to the abstract complex defined by K, that is, $\mathcal{K}$ is realized by K. Here, K is a subcomplex of the full complex F_V (cf. the definition of F_V in Sect. 4.2), $\mathrm{Fin}(V)$ is the abstract complex defined by F_V and $\mathrm{Fin}(\Gamma)$ is realized by F_V.

Remark 4.15.1 (1) In the above construction of a simplicial complex K realizing $\mathcal{K}$, instead of $\{\mathbf{e}_{\gamma} \mid \gamma \in \Gamma\} \subset \mathbb{R}^{\Gamma}_{f}$, take a set $V = \{v_{\gamma} \mid \gamma \in \Gamma\}$ in a linear space E so that $v_{\gamma} \neq v_{\gamma'}$ if $\gamma \neq \gamma' \in \Gamma$, and the following condition is satisfied:

(♯) $\{v_\gamma \mid \gamma \in A \cup A'\}$ is affinely independent in E for each $A, A' \in \mathcal{K}$.

For simplicity, we write $\{v_\gamma \mid \gamma \in A\} = V_A$ for each $A \in \mathrm{Fin}(\Gamma)$. Then, by the condition (♯), $\langle V_A \rangle$ is a simplex in E if $A \in \mathcal{K}$. However, for any two $A, A' \in \mathcal{K}$, $\langle V_{A\cup A'} \rangle$ is also a simplex in E. Since $\langle V_A \rangle$ and $\langle V_{A'} \rangle$ are faces of $\langle V_{A\cup A'} \rangle$, we have $\langle V_A \rangle \cap \langle V_{A'} \rangle = \langle V_{A\cap A'} \rangle$. Hence, the following is a simplicial complex realizing $\mathcal{K}$:

$$K = \big\{\langle V_A \rangle \;\big|\; A \in \mathcal{K}\big\} = \big\{\langle v_{\gamma_1}, \ldots, v_{\gamma_n} \rangle \;\big|\; \{\gamma_1, \ldots, \gamma_n\} \in \mathcal{K}\big\}.$$

(2) Due to the General Position Lemma 4.14.1, there exists a countable (discrete) set V in $\mathbb{R}^{2n+1}$ such that any $2n+2$ points of V are affinely independent. Then, $\{\langle A \rangle \mid A \in \mathrm{Fin}(V)^{(n)}\}$ is a simplicial complex, by which the abstract complex $\mathrm{Fin}(V)^{(n)}$ is defined. For any abstract complex $\mathcal{K}$ over a countable set with $\dim \mathcal{K} \leqslant n$, we can construct as (1) a simplicial complex K in $\mathbb{R}^{2n+1}$ realizing $\mathcal{K}$.

(3) For a simplicial complex K, the following is an abstract complex:

$$\big\{\{\sigma_1, \cdots, \sigma_n\} \;\big|\; \sigma_1 < \cdots < \sigma_n \in K\big\},$$

which is realized by the barycentric subdivision $\mathrm{Sd}\, K$.

Now, consider two abstract complexes $\mathcal{K}$ and $\mathcal{L}$ over Γ and Λ, respectively. Let K and L be the simplicial complexes realizing $\mathcal{K}$ and $\mathcal{L}$ with $K^{(0)} = \{v_\gamma \mid \gamma \in \Gamma\}$ and $L^{(0)} = \{v_\lambda \mid \lambda \in \Lambda\}$, respectively. Suppose that a function $\varphi : \mathcal{K}^{(0)} \to \mathcal{L}^{(0)}$ satisfies the following condition:

(∗) $A \in \mathcal{K}$ implies $\varphi(A) \in \mathcal{L}$.

Then, φ defines a function $f_0 : K^{(0)} \to L^{(0)}$ by $f_0(v_\gamma) = v_{\varphi(\gamma)}$ for each $\gamma \in \Gamma$. By Proposition 4.10.1 and (∗), f_0 extends to a simplicial map $f : K \to L$, which is called the **simplicial map induced by** φ. Conversely, for any simplicial map $f : K \to L$, we have a function $\varphi : \mathcal{K}^{(0)} \to \mathcal{L}^{(0)}$ such that $f(v_\gamma) = v_{\varphi(\gamma)}$ for each $\gamma \in \Gamma$. Due to Proposition 4.10.1, φ satisfies (∗). Then, $f|K_0$ is defined by this φ, that is, f is induced by φ. A function $\varphi : \mathcal{K}^{(0)} \to \mathcal{L}^{(0)}$ satisfying (∗) is also called a **simplicial map** from $\mathcal{K}$ to $\mathcal{L}$, and is written as $\varphi : \mathcal{K} \to \mathcal{L}$. Actually, (∗) means that φ induces a function from $\mathcal{K}$ to $\mathcal{L}$.

If a bijection $\varphi : K^{(0)} \to L^{(0)}$ satisfies the condition that $A \in \mathcal{K}$ if and only if $\varphi(A) \in \mathcal{L}$ (i.e., both φ and φ^{-1} satisfy (∗)), then φ induces the simplicial isomorphism $f : K \to L$, and any simplicial isomorphism $f : K \to L$ is induced by such a bijection $\varphi : K^{(0)} \to L^{(0)}$. Such a bijection $\varphi : K^{(0)} \to L^{(0)}$ is also called a **simplicial isomorphism** from $\mathcal{K}$ to $\mathcal{L}$. It is said that $\mathcal{K}$ and $\mathcal{L}$ are **simplicially isomorphic** or $\mathcal{K}$ is **simplicially isomorphic** to $\mathcal{L}$ if there is a simplicial isomorphism from $\mathcal{K}$ to $\mathcal{L}$. Clearly, $\mathcal{K}$ is simplicially isomorphic to $\mathcal{L}$ if and only if $\mathcal{K}$ is congruent to $\mathcal{L}$, i.e., $\mathcal{K} \equiv \mathcal{L}$.

For any open cover $\mathcal{U}$ of a space X, we define the abstract complex $\mathfrak{N}(\mathcal{U})$ over $\mathcal{U}$ (or $\mathcal{U} \setminus \{\emptyset\}$) as follows:

$$\mathfrak{N}(\mathcal{U}) = \big\{\{U_1, \cdots, U_n\} \in \mathrm{Fin}(\mathcal{U}) \;\big|\; U_1 \cap \cdots \cap U_n \neq \emptyset\big\}.$$

The **nerve** $N(\mathcal{U})$ of $\mathcal{U}$ is defined as a simplicial complex realizing $\mathfrak{N}(\mathcal{U})$, that is, $N(\mathcal{U})$ is a simplicial complex with vertices v_U, $U \in \mathcal{U}$, such that $v_U \neq v_{U'}$ if $U \neq U'$, and

$$\langle v_{U_1}, \cdots, v_{U_n} \rangle \in N(\mathcal{U}) \ \Leftrightarrow \ U_1 \cap \cdots \cap U_n \neq \emptyset.$$

Here, it is useful to identify each open set $U \in \mathcal{U}$ with its corresponding vertex $v_U \in N(\mathcal{U})^{(0)}$, that is, each $U \in \mathcal{U}$ is regarded as a vertex of $N(\mathcal{U})$. Then, $N(\mathcal{U})^{(0)} = \mathcal{U}$ and

$$N(\mathcal{U}) = \left\{ \langle U_1, \cdots, U_n \rangle \;\middle|\; U_1, \ldots, U_n \in \mathcal{U},\ U_1 \cap \cdots \cap U_n \neq \emptyset \right\}.$$

A map $p : X \to |N(\mathcal{U})|$ (or $p : X \to |N(\mathcal{U})|_{\mathrm{m}}$) is called a **canonical map** for $\mathcal{U}$ if

$$p^{-1}(O_{N(\mathcal{U})}(U)) \subset U \ \text{ for each } \ U \in \mathcal{U} \setminus \{\emptyset\}.$$

Evidently, if $p : X \to |N(\mathcal{U})|$ is a canonical map, then $p : X \to |N(\mathcal{U})|_{\mathrm{m}}$ is also a canonical map. For each compact set $A \subset |N(\mathcal{U})|$, there are finitely many $U_1, \ldots, U_n \in \mathcal{U}$ such that $A \subset \bigcup_{i=1}^{n} O_{N(\mathcal{U})}(U_i)$ because $O_{N(\mathcal{U})}$ is an open cover of $|N(\mathcal{U})|$. Hence, we have the following:

Proposition 4.15.1 *Let $\mathcal{U}$ be an open cover of a locally compact space X such that each $U \in \mathcal{U}$ has the compact closure in X. Then, every canonical map $p : X \to |N(\mathcal{U})|$ is proper.* □

Due to Proposition 4.2.8, every simplicial complex K can be regarded as the nerve $N(O_K)$ of the open cover O_K of $|K|$ (or $|K|_{\mathrm{m}}$) by identifying each $v \in K^{(0)}$ with $O_K(v) \in O_K$. The identity $\mathrm{id} : |K| \to |K|$ is a canonical map for O_K, where it should be noted that $\mathrm{id} : |K| \to |K|_{\mathrm{m}}$ and $\mathrm{id} : |K|_{\mathrm{m}} \to |K|_{\mathrm{m}}$ are also canonical maps for O_K.

We have the following characterization of canonical maps:

Proposition 4.15.2 *For an open cover $\mathcal{U}$ of X, a map $p : X \to |N(\mathcal{U})|$ (or $p : X \to |N(\mathcal{U})|_{\mathrm{m}}$) is a canonical map if and only if*

$$c_{N(\mathcal{U})}(p(x))^{(0)} \subset \mathcal{U}[x] \ \textit{ for each } x \in X,$$

where $c_{N(\mathcal{U})}(p(x)) \in N(\mathcal{U})$ is the carrier of $p(x)$. If $\mathcal{U}[x]$ is finite, this condition is equivalent to $p(x) \in \langle \mathcal{U}[x] \rangle \in N(\mathcal{U})$. Cf. Fig. 4.15.

Proof To prove the "if" part, let $U \in \mathcal{U}$ and $x \in p^{-1}(O_{N(\mathcal{U})}(U))$. From the condition, it follows that $U \in c_{N(\mathcal{U})}(p(x))^{(0)} \subset \mathcal{U}[x]$, which means that $x \in U$. Therefore, $p^{-1}(O_{N(\mathcal{U})}(U)) \subset U$.

To show the "only if" part, let $x \in X$ and $U \in c_{N(\mathcal{U})}(p(x))^{(0)}$. Observe that $p(x) \in \operatorname{rint} c_{N(\mathcal{U})}(p(x)) \subset O_{N(\mathcal{U})}(U)$, hence $x \in p^{-1}(O_{N(\mathcal{U})}(U)) \subset U$. Thus, we have $c_{N(\mathcal{U})}(p(x))^{(0)} \subset \mathcal{U}[x]$. □

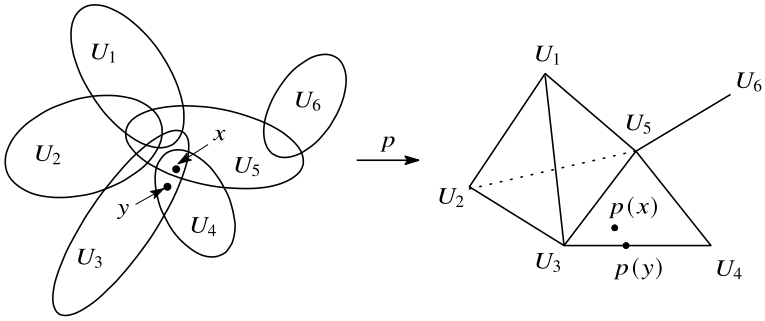


Fig. 4.15 A canonical map

Remark 4.15.2 For a subspace A of X and $\mathcal{U} \in \mathrm{cov}(X)$, we have $\mathcal{U}|A \in \mathrm{cov}(A)$. The nerve $N(\mathcal{U}|A)$ cannot be regarded as a subcomplex of $N(\mathcal{U})$ in general. Indeed, there may be cases where $U \cap A = U' \cap A$ even if $U \neq U' \in \mathcal{U}$. Assume that $U_1 \cap A \neq U_2 \cap A$ if $U_1 \neq U_2 \in \mathcal{U}[A]$. Then, by identifying each $U \cap A \in \mathcal{U}|A$ with $U \in \mathcal{U}[A]$ ($\subset \mathcal{U} \setminus \{\emptyset\}$), the nerve $N(\mathcal{U}|A)$ can be regarded as the following subcomplex of the nerve $N(\mathcal{U})$:[33]

$$\left\{\langle U_1, \ldots, U_n\rangle \in N(\mathcal{U}) \;\middle|\; U_1 \cap \cdots \cap U_n \cap A \neq \emptyset\right\}.$$

In this case, for each canonical map $p : X \to |N(\mathcal{U})|$ (or $p : X \to |N(\mathcal{U})|_{\mathrm{m}}$), the restriction $p|A : A \to |N(\mathcal{U}|A)|$ (or $p|A : A \to |N(\mathcal{U}|A)|_{\mathrm{m}}$) is a canonical map for $\mathcal{U}|A$. Indeed, for each $x \in A$, $c_{N(\mathcal{U})}(p(x))^{(0)} \subset \mathcal{U}[x] = (\mathcal{U}|A)[x]$ (Proposition 4.15.2), so $c_{N(\mathcal{U})}(p(x)) \in N(\mathcal{U}|A)$. Therefore, $p(A) \subset |N(\mathcal{U}|A)|$.

Proposition 4.15.2 yields the following:

Corollary 4.15.3 *Let $\mathcal{U}$ be an open cover of X. Then, any two canonical maps $p, q : X \to |N(\mathcal{U})|$ (or $p, q : X \to |N(\mathcal{U})|_{\mathrm{m}}$) are contiguous.* □

Given an open refinement $\mathcal{V}$ of $\mathcal{U} \in \mathrm{cov}(X)$, for each $V \in \mathcal{V} = N(\mathcal{V})^{(0)}$, choosing $\varphi(V) \in \mathcal{U}$ so that $V \subset \varphi(V)$, we can define a simplicial map $\varphi : N(\mathcal{V}) \to N(\mathcal{U})$, which is called a **refining simplicial map**. Then, we have the following:

Proposition 4.15.4 *Let $\mathcal{U}$ and $\mathcal{V}$ be open covers of X with $\mathcal{V} \prec \mathcal{U}$.*

(1) *Any two refining simplicial maps $\varphi, \psi : N(\mathcal{V}) \to N(\mathcal{U})$ are contiguous.*
(2) *If $\varphi : N(\mathcal{V}) \to N(\mathcal{U})$ is a refining simplicial map and $p : X \to |N(\mathcal{V})|$ (or $p : X \to |N(\mathcal{V})|_{\mathrm{m}}$) is a canonical map for $\mathcal{V}$, then $\varphi p : X \to |N(\mathcal{U})|$ (or $\varphi p : X \to |N(\mathcal{U})|_{\mathrm{m}}$) is also a canonical map for $\mathcal{U}$.*

[33] This is not equal to $N(\mathcal{U}[A])$.

Sketch of Proof To see (1), let $\sigma = \langle V_1, \dots, V_n \rangle \in N(\mathcal{V})$. Then, $\varphi(\sigma)$ and $\psi(\sigma)$ are faces of the same simplex $\tau \in N(\mathcal{U})$ with $\tau^{(0)} = \{\varphi(V_i), \psi(V_i) \mid i = 1, \dots, n\}$ because $\bigcap_{i=1}^n \varphi(V_i) \cap \bigcap_{i=1}^n \psi(V_i) \supset \bigcap_{i=1}^n V_i \neq \emptyset$.

For (2), take $x \in X$. Then, $c_{N(\mathcal{U})}(\varphi p(x))^{(0)} = \varphi\big(c_{N(\mathcal{V})}(p(x))^{(0)}\big) \subset \varphi(\mathcal{V}[x]) \subset \mathcal{U}[x]$.

Recall an open cover $\mathcal{U} \in \mathrm{cov}(X)$ is called a **numerable cover** if there exists a partition of unity $(f_U)_{U \in \mathcal{U}}$ such that $f_U^{-1}((0,1]) \subset U$ for each $U \in \mathcal{U}$. As shown in Theorem 2.7.2, for a numerable cover $\mathcal{U} \in \mathrm{cov}(X)$, X has a locally finite partition of unity $(f_U)_{U \in \mathcal{U}}$ such that $\mathrm{supp}\, f_U^{-1} \subset U$ for each $U \in \mathcal{U}$. It was also shown in Theorem 2.7.2 that $\mathcal{U} \in \mathrm{cov}(X)$ *is numerable if and only if it is a normal cover*, that is, there exists a sequence of open star-refinements:

$$\mathcal{U} \overset{*}{\succ} \mathcal{U}_1 \overset{*}{\succ} \mathcal{U}_2 \overset{*}{\succ} \cdots .$$

The numerability of an open cover characterizes the existence of a canonical map, that is:

Theorem 4.15.5 *For an open cover $\mathcal{U}$ of a space X, the following are equivalent:*

(a) *There exists a canonical map $p : X \to |N(\mathcal{U})|$;*
(b) *There exists a canonical map $p : X \to |N(\mathcal{U})|_{\mathrm{m}}$;*
(c) *$\mathcal{U}$ is numerable (= normal).*

Proof The implication (a) $\Rightarrow$ (b) is trivial because if $p : X \to |N(\mathcal{U})|$ is a canonical map, then $p : X \to |N(\mathcal{U})|_{\mathrm{m}}$ is also a canonical map.

(b) $\Rightarrow$ (c): Let $p : X \to |N(\mathcal{U})|_{\mathrm{m}}$ be a canonical map. Since $O_{N(\mathcal{U})}$ is a numerable open cover of $|N(\mathcal{U})|_{\mathrm{m}}$, $|N(\mathcal{U})|_{\mathrm{m}}$ has a partition of unity $(f_U)_{U \in \mathcal{U}}$ such that $f_U^{-1}((0,1]) \subset O_{N(\mathcal{U})}(U)$ for each $U \in \mathcal{U}$. Then, $(f_U p)_{U \in \mathcal{U}}$ is a partition of unity on X and for each $U \in \mathcal{U}$,

$$(f_U p)^{-1}((0,1]) = p^{-1} f_U^{-1}((0,1]) \subset p^{-1}(O_{N(\mathcal{U})}(U)) \subset U,$$

which means that $\mathcal{U}$ is numerable.

(c) $\Rightarrow$ (b), (a): We have a locally finite partition of unity $(f_U)_{U \in \mathcal{U}}$ on X such that $\mathrm{supp}\, f_U \subset U$ for each $U \in \mathcal{U}$. We can define a map $p : X \to |N(\mathcal{U})|_{\mathrm{m}}$ as follows:

$$p(x) = \sum_{U \in \mathcal{U}} f_U(x) \cdot U \quad \big(\text{i.e., } \beta_U^{N(\mathcal{U})}(p(x)) = f_U(x)\big).$$

Observe that $p^{-1}(O_{N(\mathcal{U})}(U)) \subset \mathrm{supp}\, f_U \subset U$ for each $U \in \mathcal{U}$. Then, $p : X \to |N(\mathcal{U})|_{\mathrm{m}}$ is a canonical map for $\mathcal{U}$.

Each $x \in X$ has a neighborhood V_x such that $\mathcal{U}_x = \{U \in \mathcal{U} \mid \mathrm{supp}\, f_U \cap V_x \neq \emptyset\}$ is finite. Let $K_x = \{\sigma \in N(\mathcal{U}) \mid \sigma^{(0)} \subset \mathcal{U}_x\}$, which is the full subcomplex of $N(\mathcal{U})$ with $K_x^{(0)} = \mathcal{U}_x$. Then, $p(V_x) \subset |K_x|$. It follows that $p : X \to |N(\mathcal{U})|$ is continuous, so it is also a canonical map. □

Since every open cover of a paracompact space is numerable (Theorem 2.7.5) and a locally finite open cover of a normal space is numerable (Theorem 2.7.7), we have the following corollary:

Corollary 4.15.6 *Let $\mathcal{U}$ be an open cover of a normal space X. If X is paracompact or $\mathcal{U}$ is locally finite, then there exists a canonical map $p : X \to |N(\mathcal{U})|$.* □

Applying Corollary 4.15.6, we will prove the following:

Theorem 4.15.7 *For every simplicial complex K, the identity $\varphi = \mathrm{id} : |K| \to |K|_{\mathrm{m}}$ is a homotopy equivalence with a homotopy inverse $\psi : |K|_{\mathrm{m}} \to |K|$ such that $\psi\varphi \simeq_K \mathrm{id}$ and $\varphi\psi \simeq_K \mathrm{id}$, where $\psi\varphi \simeq_{O_K} \mathrm{id}$ and $\varphi\psi \simeq_{O_K} \mathrm{id}$ are also valid. These homotopies are realized by the straight-line homotopy.*

Proof Consider K as the nerve $N(O_K)$ of the open cover $O_K \in \mathrm{cov}(|K|_{\mathrm{m}})$, where each vertex $v \in K^{(0)}$ is identified with the open star $O_K(v) \in O_K$. By virtue of Corollary 4.15.6, we have a canonical map $\psi : |K|_{\mathrm{m}} \to |N(O_K)| = |K|$. Then, $\psi\varphi, \mathrm{id} : |K| \to |K|$ are contiguous and $\varphi\psi, \mathrm{id} : |K|_{\mathrm{m}} \to |K|_{\mathrm{m}}$ are also contiguous by Corollary 4.15.3. Due to Propositions 4.5.6 and 4.6.6, $\psi\varphi \simeq_K \mathrm{id}$ and $\varphi\psi \simeq_K \mathrm{id}$ by the straight-line homotopy defined as

$$h(x,t) = (1-t)\psi\varphi(x) + tx = (1-t)\varphi\psi(x) + tx.$$

Since $c_K(\psi(x)) \leqslant c_K(x)$ for each $x \in |K|$, each $h(\{x\} \times \mathbf{I})$ is contained in not only $c_K(x)$ but also $O_K(v)$ for any $v \in c_K(\psi(x))^{(0)}$. □

Remark 4.15.3 In the above proof, let L be a subcomplex of a simplicial complex K. Then, $O_L = O_K||L|$ according to Proposition 4.2.9. As noted in Remark 4.15.2, identifying $O_L(v)$ with $O_K(v)$ for each $v \in L^{(0)}$, the nerve $N(O_L)$ can be regarded as a subcomplex of the nerve $N(O_K)$, where the pair (K, L) can be identified with the pair $(N(O_K), N(O_L))$. Moreover, the restriction $\psi||L| : |L|_{\mathrm{m}} \to |N(O_L)| = |L|$ is also a canonical map for O_L, which is a homotopy inverse of $\varphi||L| = \mathrm{id} : |L| \to |L|_{\mathrm{m}}$. In this case, $(\psi||L|)(\varphi||L|) \simeq_L \mathrm{id}$ and $(\varphi||L|)(\psi||L|) \simeq_L \mathrm{id}$ are given by the straight-line homotopies, that is, the restrictions of the homotopies $\psi\varphi \simeq_K \mathrm{id}$ and $\varphi\psi \simeq_K \mathrm{id}$, respectively.

We can now generalize Proposition 4.5.6 as follows:

Proposition 4.15.8 *Let K be a simplicial complex and X an arbitrary space. If two maps $f, g : X \to |K|$ are contiguous then $f \simeq_K g$.*

Proof Let $\varphi = \mathrm{id} : |K| \to |K|_{\mathrm{m}}$. By virtue of Theorem 4.15.7, we have a map $\psi : |K|_{\mathrm{m}} \to |K|$ such that $\psi\varphi \simeq_K \mathrm{id}$ by the straight-line homotopy $h : |K| \times \mathbf{I} \to |K|$, where $c_K(\psi(x)) \leqslant c_K(x)$ and $h(\{x\} \times \mathbf{I}) \subset c_K(x)$ for each $x \in |K|$ (see the proof of Theorem 4.15.7). On the other hand, by Proposition 4.6.6, we have $\varphi f \simeq_K \varphi g$, which is realized by the straight-line homotopy $h' : X \times \mathbf{I} \to |K|_{\mathrm{m}}$. For each $x \in X$, choose $\sigma_x \in K$ so that $h'(\{x\} \times \mathbf{I}) \subset \sigma_x$. Observe that $\psi h'(\{x\} \times \mathbf{I}) \subset \psi(\sigma_x) \subset \sigma_x, h(\{f(x)\} \times \mathbf{I}) \subset c_K(f(x)) \subset \sigma_x$, and $h(\{g(x)\} \times \mathbf{I}) \subset c_K(g(x)) \subset \sigma_x$. Then, by connecting three homotopies $h(f \times \mathrm{id}_{\mathbf{I}})$, $\psi h'$, and $h(g \times \mathrm{id}_{\mathbf{I}})$, we can get a K-homotopy from f to g, hence $f \simeq_K g$. □

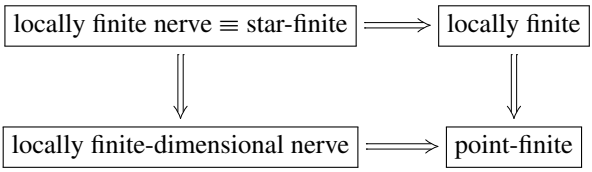
Combining Corollary 4.15.3 with Proposition 4.15.8 (or 4.6.6), we have the following corollary:

Corollary 4.15.9 *Let $\mathcal{U}$ be an open cover of a space X. Then, $p \simeq_{N(\mathcal{U})} q$ for any two canonical maps $p, q : X \to |N(\mathcal{U})|$ (or $p, q : X \to |N(\mathcal{U})|_{\mathrm{m}}$).* □

An open cover $\mathcal{U}$ of a space X is said to be **star-finite** if $\mathcal{U}[U]$ is finite for each $U \in \mathcal{U}$, which is equivalent to the condition that the nerve $N(\mathcal{U})$ is locally finite. In fact, $\mathrm{St}(U, N(\mathcal{U}))^{(0)} = \mathcal{U}[U]$ for each $U \in \mathcal{U} = N(\mathcal{U})^{(0)}$. Thus, the star-finiteness of an open cover characterizes the local finiteness of its nerve. On the other hand, the nerve $N(\mathcal{U})$ is locally finite-dimensional (l.f.d.) if and only if $\sup_{x \in U} \operatorname{card} \mathcal{U}[x] < \infty$ for each $U \in \mathcal{U}$. In this case, we have

$$\sup_{x \in U} \operatorname{card} \mathcal{U}[x] = \dim \mathrm{St}(U, N(\mathcal{U})) + 1.$$

Note that every star-finite open cover is locally finite and its nerve is locally finite-dimensional, and that if an open cover is locally finite or its nerve is locally finite-dimensional then it is point-finite, that is, we have the following implications:



In the above, the converse implications do not hold and there are no connections between the local finiteness of an open cover and the local finite dimensionality of its nerve. In fact, $\mathcal{U} = \{\mathbb{R}, (n, \infty) \mid n \in \mathbb{N}\}$ is a locally finite open cover of $\mathbb{R}$ but the nerve $N(\mathcal{U})$ is not locally finite-dimensional. This example also shows that the converse implication in the top row of the figure does not hold. Since the cover is point-finite, the converse in the bottom row does not hold either. On the other hand, let $X = (\mathbb{N} \times \mathbf{I})/(\mathbb{N} \times \{0\})$ be the quotient space (or $X = J(\mathbb{N})$ the hedgehog (cf. Sect. 1.2)). Let $U_0 = X$ and, for each $n \in \mathbb{N}$, let $U_n = \{n\} \times (0, 1]$. Then, $\mathcal{U} = \{U_n \mid n \in \omega\}$ is an open cover with $\dim N(\mathcal{U}) = 1$, which is not locally finite in X. This shows that the converse implication on the left side of the figure does not hold. Since the cover is point-finite, the converse on the right side does not hold either.

Theorem 4.15.10 *Every numerable cover of a space X has a locally finite σ-discrete open refinement with the locally finite-dimensional nerve.*

Proof It suffices to show that, for a simplicial complex K, the open cover $O_K \in \mathrm{cov}(|K|)$ has a locally finite σ-discrete open refinement $\mathcal{V}$ with the locally finite-dimensional nerve. Indeed, every numerable cover $\mathcal{U}$ of X has a canonical map $p : X \to |N(\mathcal{U})|$ by Theorem 4.15.5. When $K = N(\mathcal{U})$,

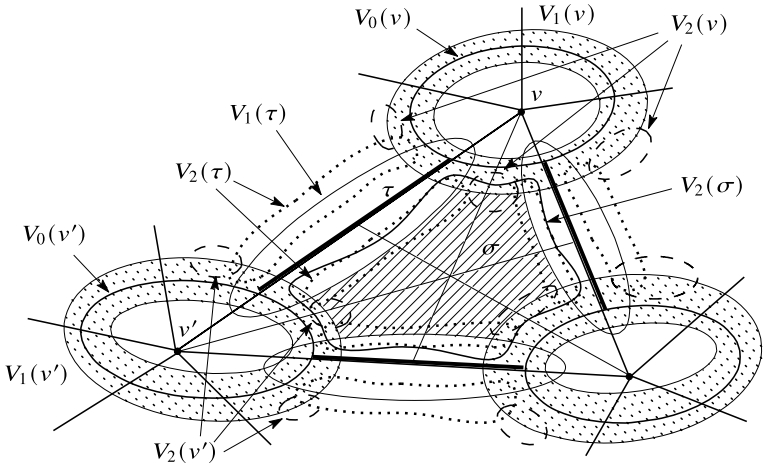


Fig. 4.16 $\mathcal{V} = \{V_n(\sigma) \mid n \in \omega,\ \sigma \in K^{(n)}\}$

$$p^{-1}(\mathcal{V}) \prec p^{-1}(\mathcal{O}_K) = p^{-1}(\mathcal{O}_{N(\mathcal{U})}) \prec \mathcal{U},$$

$p^{-1}(\mathcal{V})$ is locally finite σ-discrete. Moreover, $N(p^{-1}(\mathcal{V}))$ is l.f.d. because

$$\sup_{x \in p^{-1}(V)} \operatorname{card} p^{-1}(\mathcal{V})[x] \leqslant \sup_{p(x) \in V} \operatorname{card} \mathcal{V}[p(x)] < \infty \ \text{ for each } V \in \mathcal{V}.$$

We will construct an open collection $\mathcal{V} = \{V_n(\sigma) \mid \sigma \in K^{(n)},\ n \in \omega\}$ satisfying the following conditions:

(1) $\{V_n(\sigma) \mid \sigma \in K^{(n)},\ \dim \sigma = i\}$ is discrete in $|K|$ for each $i \leqslant n$;
(2) $\operatorname{cl} V_n(\sigma) \subset O_{\operatorname{Sd} K}(\hat{\sigma}) \setminus \operatorname{cl} V_{n-2}$ for each $\sigma \in K$ and $n \geqslant \dim \sigma$;
(3) $|K^{(n)}| \cup \operatorname{cl} V_{n-1} \subset V_n$,

where

$$V_n = \bigcup \{V_i(\sigma) \mid \sigma \in K^{(i)},\ i \leqslant n\} \quad (V_{-1} = V_{-2} = \emptyset).$$

Then, $\mathcal{V} \in \operatorname{cov}(|K|)$ by (3), $\mathcal{V} \prec \mathcal{O}_{\operatorname{Sd} K} \prec \mathcal{O}_K$ by (2), and $\mathcal{V}$ is σ-discrete in $|K|$ by (1). Moreover, we can see that $\mathcal{V}$ is locally finite and $N(\mathcal{V})$ is l.f.d. (Fig. 4.16).

The local finiteness of $\mathcal{V}$ can be shown as follows: Each $x \in |K|$ is contained in some $V_n \setminus \operatorname{cl} V_{n-2}$ by (3). From (2), it follows that

$$(V_n \setminus \operatorname{cl} V_{n-2}) \cap V_i(\sigma) = \emptyset \ \text{ for } i > n+1 \text{ and } i < n-1.$$

By (1), x has a neighborhood W in $V_n \setminus \operatorname{cl} V_{n-2}$ that meets at most one member of $\{V_i(\sigma) \mid \dim \sigma = k\}$ for each $i = n-1, n, n+1$ and each $k \leqslant i$. Consequently, W meets at most $3n+3$ of the open sets $V_i(\sigma)$.

To prove that $N(\mathcal{V})$ is l.f.d., let $n \in \omega$ and $\sigma \in K^{(n)}$. Due to (2), for each $x \in V_n(\sigma)$,

$$\left\{i \in \omega \mid \exists \tau \in K \text{ such that } x \in V_i(\tau)\right\} = \{n-1, n\} \text{ or } \{n\} \text{ or } \{n, n+1\}.$$

For each $i \in \omega$, $V_i(\tau)$ is defined only if $\dim \tau \leqslant i$, so x is contained in at most $i+1$ of the open sets $V_i(\tau)$ by (1). Therefore, each $x \in V_n(\sigma)$ is contained in at most $2n+3$ of the open sets $V_i(\tau)$, that is, $\dim \mathrm{St}(V_n(\sigma), N(\mathcal{V})) \leqslant 2n+2$.

Now, let us construct $\mathcal{V}$. First, for each $v \in K^{(0)}$, choose an open set $V_0(v)$ in $|K|$ so that $v \in V_0(v) \subset \mathrm{cl}\, V_0(v) \subset O_{\mathrm{Sd}\, K}(v)$. Each $x \in |K|$ is contained in the open star $O_{\mathrm{Sd}\, K}(\hat{\sigma})$ for some $\sigma \in K$, which meets only finitely many $\mathrm{cl}\, V_0(v)$ because

$$O_{\mathrm{Sd}\, K}(\hat{\sigma}) \cap O_{\mathrm{Sd}\, K}(v) \neq \emptyset \;\Leftrightarrow\; \langle v, \hat{\sigma}\rangle \in \mathrm{Sd}\, K \;\Leftrightarrow\; v \in \sigma^{(0)}.$$

Then, $\{\mathrm{cl}\, V_0(v) \mid v \in K^{(0)}\}$ is a pairwise disjoint locally finite collection of closed sets in $|K|$, which means that it is discrete in $|K|$. Therefore, $\{V_0(v) \mid v \in K^{(0)}\}$ is discrete in $|K|$.

Suppose that $V_i(\sigma)$ has been defined for $i \leqslant n-1$ and $\sigma \in K^{(i)}$ so as to satisfy (1), (2), and (3). Let V'_{n-1} be an open set in $|K|$ such that

$$|K^{(n-1)}| \cup \mathrm{cl}\, V_{n-2} \subset V'_{n-1} \subset \mathrm{cl}\, V'_{n-1} \subset V_{n-1}.$$

For each n-simplex $\sigma \in K$, since $|K^{(n-1)}| \subset V_{n-1}$, it follows that

$$\sigma \setminus V_{n-1} = \mathrm{rint}\, \sigma \setminus V_{n-1} \subset O_{\mathrm{Sd}\, K}(\hat{\sigma}) \setminus \mathrm{cl}\, V'_{n-1} \subset O_{\mathrm{Sd}\, K}(\hat{\sigma}) \setminus \mathrm{cl}\, V_{n-2}.$$

Then, we have an open set $V_n(\sigma)$ in $|K|$ such that

$$\sigma \setminus V_{n-1} \subset V_n(\sigma) \subset \mathrm{cl}\, V_n(\sigma) \subset O_{\mathrm{Sd}\, K}(\hat{\sigma}) \setminus \mathrm{cl}\, V'_{n-1}.$$

Each $x \in |K^{(n-1)}|$ is contained in the open set V'_{n-1}, which misses $\mathrm{cl}\, V_n(\sigma)$ for every n-simplex $\sigma \in K$. On the other hand, each $x \in |K| \setminus |K^{(n-1)}|$ is contained in the open star $O_{\mathrm{Sd}\, K}(\hat{\tau})$ for some $\tau \in K \setminus K^{(n-1)}$, which meets only finitely many $\mathrm{cl}\, V_n(\sigma)$ (where $\dim \sigma = n$) because

$$O_{\mathrm{Sd}\, K}(\hat{\tau}) \cap O_{\mathrm{Sd}\, K}(\hat{\sigma}) \neq \emptyset \;\Leftrightarrow\; \langle \hat{\sigma}, \hat{\tau}\rangle \in \mathrm{Sd}\, K \;\Leftrightarrow\; \sigma \leqslant \tau.$$

Then, $\{\mathrm{cl}\, V_n(\sigma) \mid \sigma \in K^{(n)},\ \dim \sigma = n\}$ is a pairwise disjoint locally finite collection of closed sets in $|K|$, so it is discrete in $|K|$. Thus, $\{V_n(\sigma) \mid \sigma \in K^{(n)},\ \dim \sigma = n\}$ is discrete in $|K|$.[34]

[34] Note that $|K|$ is paracompact by Corollary 4.8.13, so it is collectionwise normal by Theorem 2.6.1. Observe that $\{\sigma \setminus V_{n-1} \mid \sigma \in K^{(n)},\ \dim \sigma = n\}$ is discrete in $|K|$. Then, we can obtain $\{\mathrm{cl}\, V_n(\sigma) \mid \sigma \in K^{(n)},\ \dim \sigma = n\}$ without taking V'_{n-1}.

For each $\sigma \in K^{(n-1)}$, observe that

$$\operatorname{cl} V_{n-1}(\sigma) \setminus V_{n-1} \subset O_{\operatorname{Sd} K}(\hat{\sigma}) \setminus \operatorname{cl} V'_{n-1} \subset O_{\operatorname{Sd} K}(\hat{\sigma}) \setminus \operatorname{cl} V_{n-2}.$$

Then, we have an open set $V_n(\sigma)$ in $|K|$ such that

$$\operatorname{cl} V_{n-1}(\sigma) \setminus V_{n-1} \subset V_n(\sigma) \subset \operatorname{cl} V_n(\sigma) \subset O_{\operatorname{Sd} K}(\hat{\sigma}) \setminus \operatorname{cl} V'_{n-1}.$$

By the same approach as above, we can see that $\{V_n(\sigma) \mid \sigma \in K^{(n)},\ \dim \sigma = i\}$ is discrete in $|K|$ for each $i < n$.

Since $\{V_i(\sigma) \mid \sigma \in K^{(i)},\ i \leqslant n-1\}$ is locally finite in $|K|$, we have

$$\operatorname{cl} V_{n-1} = \bigcup \left\{ \operatorname{cl} V_i(\sigma) \;\middle|\; \sigma \in K^{(i)},\ i \leqslant n-1 \right\}.$$

Note that $\operatorname{cl} V_i(\sigma) \subset \operatorname{cl} V_{n-2} \subset V_{n-1}$ for $i \leqslant n-2$ and $\sigma \in K^{(i)}$. Hence,

$$\operatorname{bd} V_{n-1} \subset \bigcup_{\sigma \in K^{(n-1)}} \operatorname{cl} V_{n-1}(\sigma) \setminus V_{n-1} \subset \bigcup_{\sigma \in K^{(n-1)}} V_n(\sigma).$$

Then, it follows that $|K^{(n)}| \cup \operatorname{cl} V_{n-1} \subset V_n$. Thus, we have obtained $V_n(\sigma)$ for every $\sigma \in K^{(n)}$ such that conditions (1), (2), and (3) are satisfied. The proof is completed by induction. □

Since every open cover of a paracompact space is numerable, the following is a direct consequence of the above:

Theorem 4.15.11 *Every open cover of a paracompact space X has a locally finite σ-discrete open refinement with the locally finite-dimensional nerve.* □

When X is a locally compact paracompact space, each $\mathcal{U} \in \operatorname{cov}(X)$ has a locally finite open refinement $\mathcal{V}$ such that $\operatorname{cl} V$ is compact for each $V \in \mathcal{V}$. Then, as is easily observed, $\mathcal{V}$ is star-finite, so the nerve $N(\mathcal{V})$ is locally finite. Thus, we have the following theorem:

Theorem 4.15.12 *Every open cover of a locally compact paracompact space has a star-finite open refinement whose nerve is locally finite.* □

In addition, we can show that:

Theorem 4.15.13 *Every open cover of a regular Lindelöf space has a countable star-finite open refinement whose nerve is countable and locally finite.*

Proof Due to Corollary 2.6.5, a regular Lindelöf space X is paracompact. Then, each $\mathcal{U} \in \operatorname{cov}(X)$ has a countable locally finite open refinement $\mathcal{V} = \{V_i \mid i \in \mathbb{N}\}$. Indeed, $\mathcal{U}$ has a locally finite open refinement $\mathcal{V}'$. Since X is Lindelöf, the desired $\mathcal{V}$ can be taken as a countable subcover of $\mathcal{V}'$. According to Lemma 2.7.6, $\mathcal{V}$ has a

strong open shrinking $\{V_{i,1} \mid i \in \mathbb{N}\}$. We can inductively choose open sets $V_{i,j}$ in X so that $\mathrm{cl}\, V_{i,j-1} \subset V_{i,j} \subset \mathrm{cl}\, V_{i,j} \subset V_i$. For each $i, j \in \mathbb{N}$, let

$$W_{i,j} = V_{i,j} \setminus \bigcup_{i'+j'<i+j-1} \mathrm{cl}\, V_{i',j'}.$$

Then, $\mathcal{W} = \{W_{i,j} \mid i, j \in \mathbb{N}\} \in \mathrm{cov}(X)$ is a star-finite countable open refinement of $\mathcal{U}$. □

The Nerves of Indexed Open Covers

For an indexed open cover $(U_\gamma)_{\gamma\in\Gamma}$ of a space X, we similarly define the abstract complex $\mathfrak{N}((U_\gamma)_{\gamma\in\Gamma})$ over Γ as follows:

$$\mathfrak{N}((U_\gamma)_{\gamma\in\Gamma}) = \left\{\{\gamma_1, \cdots, \gamma_n\} \in \mathrm{Fin}(\Gamma) \;\middle|\; U_{\gamma_1} \cap \cdots \cap U_{\gamma_n} \neq \emptyset\right\}.$$

The **nerve** $N((U_\gamma)_{\gamma\in\Gamma})$ of $(U_\gamma)_{\gamma\in\Gamma}$ is defined as a simplicial complex realizing $\mathfrak{N}((U_\gamma)_{\gamma\in\Gamma})$, that is, $N((U_\gamma)_{\gamma\in\Gamma})$ is a simplicial complex with vertices v_γ, $\gamma \in \Gamma$, such that $v_\gamma \neq v_{\gamma'}$ if $\gamma \neq \gamma'$, and

$$\langle v_{\gamma_1}, \cdots, v_{\gamma_n}\rangle \in N((U_\gamma)_{\gamma\in\Gamma}) \;\Leftrightarrow\; U_{\gamma_1} \cap \cdots \cap U_{\gamma_n} \neq \emptyset.$$

For $A \subset X$, the nerve $N((U_\gamma \cap A)_{\gamma\in\Gamma})$ is always a subcomplex of $N((U_\gamma)_{\gamma\in\Gamma})$. As observed in Remark 4.15.2, for an open cover $\mathcal{U}$ of X, $N(\mathcal{U}|A)$ cannot be regarded as a subcomplex of $N(\mathcal{U})$.

Every open cover $\mathcal{U}$ of X can be regarded as an indexed open cover such that each member of $\mathcal{U}$ is indexed by itself. Then, $N(\mathcal{U}) = N((U)_{U\in\mathcal{U}})$. For $A \subset X$, the nerve $N((U \cap A)_{U\in\mathcal{U}})$ is a subcomplex of $N(\mathcal{U})$ but $N(\mathcal{U}|A)$ is not, where the former is indexed by $\mathcal{U}$ and the latter is indexed by $\mathcal{U}|A$. Moreover, given an indexed open cover $(U_\gamma)_{\gamma\in\Gamma}$ of X, we have an open cover $\mathcal{U} = \{U_\gamma \mid \gamma \in \Gamma\} \in \mathrm{cov}(X)$. However, it should be noted that $N((U_\gamma)_{\gamma\in\Gamma}) \not\equiv N(\mathcal{U})$ in general. Indeed, if $U_\gamma = U_{\gamma'}$ for some $\gamma \neq \gamma' \in \Gamma$, then $v_{U_\gamma} = v_{U_{\gamma'}}$ in $N(\mathcal{U})$ but $v_\gamma \neq v_{\gamma'}$ in $N((U_\gamma)_{\gamma\in\Gamma})$.

For an indexed open cover $(U_\gamma)_{\gamma\in\Gamma}$ of X, a map $p : X \to |N((U_\gamma)_{\gamma\in\Gamma})|$ (or $p : X \to |N((U_\gamma)_{\gamma\in\Gamma})|_{\mathrm{m}}$) is called a **canonical map** if

$$p^{-1}(O_{N((U_\gamma)_{\gamma\in\Gamma})}(v_\gamma)) \subset U_\gamma \quad \text{for each } \gamma \in \Gamma \text{ with } U_\gamma \neq \emptyset.$$

Then, for any $A \subset X$, $p|A : A \to |N((U_\gamma \cap A)_{\gamma\in\Gamma})|$ (or $p|A : A \to |N((U_\gamma \cap A)_{\gamma\in\Gamma})|_{\mathrm{m}}$) is a canonical map. Evidently, if $p : X \to |N((U_\gamma)_{\gamma\in\Gamma})|$ is a canonical map, then $p : X \to |N((U_\gamma)_{\gamma\in\Gamma})|_{\mathrm{m}}$ is also a canonical map.

4.16 Inverse Systems of Polyhedra

In this section, we will show that every paracompact space can be represented as the limit of an inverse system of simplicial complexes with the weak topology.

Let $\Lambda = (\Lambda, \leqslant)$ be a directed set.[35] An **inverse system** $(X_\lambda, f_{\lambda,\lambda'})_{\lambda\in\Lambda}$ is an indexed family of spaces X_λ and maps $f_{\lambda,\lambda'} : X_{\lambda'} \to X_\lambda$, $\lambda < \lambda' \in \Lambda$, with the following condition:

$$\lambda < \lambda' < \lambda'' \;\Rightarrow\; f_{\lambda,\lambda'} f_{\lambda',\lambda''} = f_{\lambda,\lambda''},$$

[35] Cf. Footnote 17 of Sect. 1.2.

where $f_{\lambda,\lambda'}$ are called the **bonding maps**. For convenience, let $f_{\lambda,\lambda} = \mathrm{id}_{X_\lambda}$. The (**inverse**) **limit** $\varprojlim(X_\lambda, f_{\lambda,\lambda'})_{\lambda\in\Lambda}$ is defined as the following subspace of the product space $\prod_{\lambda\in\Lambda} X_\lambda$:

$$\varprojlim(X_\lambda, f_{\lambda,\lambda'})_{\lambda\in\Lambda} \underset{\text{def}}{=} \left\{x \in \prod_{\lambda\in\Lambda} X_\lambda \;\middle|\; x(\lambda) = f_{\lambda,\lambda'}(x(\lambda')) \text{ if } \lambda < \lambda' \in \Lambda\right\},$$

which is closed in $\prod_{\lambda\in\Lambda} X_\lambda$. For each $\lambda \in \Lambda$, the restriction $p_\lambda : \varprojlim(X_\lambda, f_{\lambda,\lambda'})_{\lambda\in\Lambda} \to X_\lambda$ of the projection $\mathrm{pr}_\lambda : \prod_{\lambda\in\Lambda} X_\lambda \to X_\lambda$ is called the (**inverse limit**) **projection** of $\varprojlim(X_\lambda, f_{\lambda,\lambda'})_{\lambda\in\Lambda}$ to X_λ. Note that $f_{\lambda,\lambda'} p_{\lambda'} = p_\lambda$ for each $\lambda < \lambda' \in \Lambda$. When each X_λ is a subspace of Y and each bonding map $f_{\lambda,\lambda'}$ is an inclusion, $(X_\lambda, f_{\lambda,\lambda'})_{\lambda\in\Lambda}$ is called a **nested system** in Y. A nested system and its limit are simply written as $(X_\lambda)_{\lambda\in\Lambda}$ and $\varprojlim(X_\lambda)_{\lambda\in\Lambda}$ (or $\varprojlim X_\lambda$). The following is easy:

Proposition 4.16.1 *For every nested system $(X_\lambda)_{\lambda\in\Lambda}$ in a space Y, there exists a homeomorphism $h : \bigcap_{\lambda\in\Lambda} X_\lambda \to \varprojlim(X_\lambda)_{\lambda\in\Lambda}$ such that $p_\lambda h : \bigcap_{\lambda\in\Lambda} X_\lambda \to X_\lambda$ is the inclusion.*

Sketch of Proof The diagonal map $h : \bigcap_{\lambda\in\Lambda} X_\lambda \to \varprojlim(X_\lambda)_{\lambda\in\Lambda}$ defined by $h(x)(\lambda) = x$, $\lambda \in \Lambda$, is the desired homeomorphism.

An inverse system $(X_i, f_{i,j})_{i\in\mathbb{N}}$ indexed by $\mathbb{N}$ is called an **inverse sequence**. If $f_i = f_{i,i+1} : X_{i+1} \to X_i$ is given for each $i \in \mathbb{N}$, then $f_{i,j}$ can be defined for every $i < j$ as follows:

$$f_{i,j} = f_i \cdots f_{j-1} : X_j \to X_i.$$

For this reason, an inverse sequence is represented as $(X_i, f_i)_{i\in\mathbb{N}}$, which can be written as follows:

$$X_1 \xleftarrow{f_1} X_2 \xleftarrow{f_2} X_3 \xleftarrow{f_3} \cdots.$$

The limit of an inverse sequence is defined as follows:

$$\varprojlim(X_i, f_i) \underset{\text{def}}{=} \left\{x \in \prod_{i\in\mathbb{N}} X_i \;\middle|\; x(i) = f_i(x(i+1)) \text{ for each } i \in \mathbb{N}\right\},$$

When the bonding maps are evidently known, we simply write $\varprojlim(X_i, f_i) = \varprojlim X_i$. A nested system $X_1 \supset X_2 \supset \cdots$ indexed by $\mathbb{N}$ is called a **nested sequence**.

The following easily follows from the Tychonoff Theorem 2.1.1:

Proposition 4.16.2 *The limit of an inverse system $(X_\lambda, f_{\lambda,\lambda'})_{\lambda\in\Lambda}$ of compact spaces is compact, where $\varprojlim(X_\lambda, f_{\lambda,\lambda'})_{\lambda\in\Lambda} \neq \emptyset$ if $X_\lambda \neq \emptyset$ for every $\lambda \in \Lambda$.*

To see the additional statement, for each $\lambda \in \Lambda$, define

$$X^*_\lambda = \left\{x \in \prod_{\lambda\in\Lambda} X_\lambda \;\middle|\; x(\delta) = f_{\delta,\delta'}(x(\delta')) \text{ for each } \delta < \delta' \leqslant \lambda\right\}.$$

Then, $(X^*_\lambda)_{\lambda\in\Lambda}$ is a nested system in the compact product space $\prod_{\lambda\in\Lambda} X_\lambda$, Since $(X^*_\lambda)_{\lambda\in\Lambda}$ has the f.i.p., it follows that $\varprojlim(X_\lambda, f_{\lambda,\lambda'})_{\lambda\in\Lambda} = \bigcap_{\lambda\in\Lambda} X^*_\lambda \neq \emptyset$.

Remark 4.16.1 It is possible that $\varprojlim(X_i, f_i) = \emptyset$ even if $X_i \neq \emptyset$ for every $i \in \mathbb{N}$. For example, the following nested sequence has the empty limit:

$$[1, \infty) \supset [2, \infty) \supset [3, \infty) \supset \cdots,$$

where the bonding maps are the inclusions.

A subset Δ of a directed set $\Lambda = (\Lambda, \leqslant)$ is said to be **cofinal** in Λ if for each $\lambda \in \Lambda$, Δ contains some $\delta \geqslant \lambda$. Evidently, every cofinal subset of Λ is a directed set with the preorder inherited from Λ.

Proposition 4.16.3 *Let $(X_\lambda, f_{\lambda,\lambda'})_{\lambda\in\Lambda}$ be an inverse system. For any cofinal subset Δ of Λ, there exists a homeomorphism*

$$h : \varprojlim(X_\lambda, f_{\lambda,\lambda'})_{\lambda\in\Lambda} \to \varprojlim(X_\delta, f_{\delta,\delta'})_{\delta\in\Delta}$$

such that $q_\delta h = p_\delta$ for each $\delta \in \Delta$, where $p_\lambda : \varprojlim(X_\lambda, f_{\lambda,\lambda'})_{\lambda\in\Lambda} \to X_\lambda$ and $q_\delta : \varprojlim(X_\delta, f_{\delta,\delta'})_{\delta\in\Delta} \to X_\delta$ are the inverse limit projections.

Sketch of Proof The desired homeomorphism h is a restriction of the projection $\mathrm{pr}_\Delta : \prod_{\lambda\in\Lambda} X_\lambda \to \prod_{\delta\in\Delta} X_\delta$, that is, $h(x)(\delta) = x(\delta)$, $\delta \in \Delta$. The inverse of h can be defined as follows: Let $y \in \varprojlim(X_\delta, f_{\delta,\delta'})_{\delta\in\Delta}$, For each $\lambda \in \Lambda$, choose $\delta \in \Delta$ so that $\delta \geqslant \lambda$ and let $x(\lambda) = f_{\lambda,\delta}(y(\delta))$, where $x(\lambda)$ is determined independently of how $\delta \in \Delta$ is chosen. Indeed, for $\delta' \in \Delta$ such that $\delta' \geqslant \lambda$, choose $\delta'' \in \Delta$ so that $\delta'' \geqslant \delta, \delta'$. Then,

$$f_{\lambda,\delta'}(y(\delta')) = f_{\lambda,\delta'} f_{\delta',\delta''}(y(\delta'')) = f_{\lambda,\delta''}(y(\delta'')) = f_{\lambda,\delta} f_{\delta,\delta''}(y(\delta'')) = f_{\lambda,\delta}(y(\delta)).$$

Since the inverse limit $\varprojlim(X_\lambda, f_{\lambda,\lambda'})_{\lambda\in\Lambda}$ is a subspace of the product space $\prod_{\lambda\in\Lambda} X_\lambda$, the following can be easily obtained:

Proposition 4.16.4 *Let $(X_\lambda, f_{\lambda,\lambda'})_{\lambda\in\Lambda}$ be an inverse system. For any cofinal subset $\Delta \subset \Lambda$, the following $\mathcal{B}$ is an open basis for $\varprojlim(X_\lambda, f_{\lambda,\lambda'})_{\lambda\in\Lambda}$:*

$$\mathcal{B} = \left\{ p_\delta^{-1}(V) \;\middle|\; \delta \in \Delta, V \text{ is open in } X_\delta \right\},$$

and hence the following $\mathcal{V}$ is a neighborhood basis of $x \in \varprojlim(X_\lambda, f_{\lambda,\lambda'})_{\lambda\in\Lambda}$:

$$\mathcal{V} = \left\{ p_\delta^{-1}(V) \;\middle|\; \delta \in \Delta, V \text{ is a neighborhood of } x(\delta) \text{ in } X_\delta \right\}.$$

Indeed, for each open set U in $\varprojlim(X_\lambda, f_{\lambda,\lambda'})_{\lambda\in\Lambda}$ and $x \in U$, there are $\lambda_1, \ldots, \lambda_n \in \Lambda$ and open sets U_i in X_{λ_i}, $i = 1, \ldots, n$, such that $x \in \bigcap_{i=1}^n p_{\lambda_i}^{-1}(U_i) \subset U$. We have $\delta \in \Delta$ such that $\delta \geqslant \lambda_i$ for every $i = 1, \ldots, n$. Then, $V = \bigcap_{i=1}^n f_{\lambda_i,\delta}^{-1}(U_i)$ is an open set in X_δ and $x \in p_\delta^{-1}(V) = \bigcap_{i=1}^n p_{\lambda_i}^{-1}(U_i) \subset U$.

Concerning subspaces, we have the following:

Proposition 4.16.5 *Let* $X = \varprojlim(X_\lambda, f_{\lambda,\lambda'})_{\lambda\in\Lambda}$ *be the limit of an inverse system* $(X_\lambda, f_{\lambda,\lambda'})_{\lambda\in\Lambda}$ *with the projections* $p_\lambda : X \to X_\lambda$, $\lambda \in \Lambda$.

(1) *For each* $\lambda \in \Lambda$, *let* A_λ *be a subspace of* X_λ *such that* $f_{\lambda,\lambda'}(A_{\lambda'}) \subset A_\lambda$ *for each* $\lambda' > \lambda$. *Then,* $A = \varprojlim(A_\lambda, f_{\lambda,\lambda'}|A_{\lambda'})_{\lambda\in\Lambda}$ *is a subspace of* X *and* $p_\lambda|A$, $\lambda \in \Lambda$, *are the inverse limit projections.*

(2) *For every closed subspace* A *of* X, $A = \varprojlim(p_\lambda(A), f_{\lambda,\lambda'}|p_{\lambda'}(A))_{\lambda\in\Lambda}$ *and* $p_\lambda|A$, $\lambda \in \Lambda$, *are the inverse limit projections.*

Sketch of Proof For (2), $A \subset \varprojlim(p_\lambda(A), f_{\lambda,\lambda'}|p_{\lambda'}(A))_{\lambda\in\Lambda}$ is trivial. Conversely, let $x \in \varprojlim(p_\lambda(A), f_{\lambda,\lambda'}|p_{\lambda'}(A))_{\lambda\in\Lambda}$. For each neighborhood U of x in X, there are $\lambda_0 \in \Lambda$ and an open neighborhood V of $x(\lambda_0)$ in X_{λ_0} such that $p_{\lambda_0}^{-1}(V) \subset U$ (Proposition 4.16.4). Since $x(\lambda_0) = p_{\lambda_0}(x) \in p_{\lambda_0}(A)$, it follows that $p_{\lambda_0}^{-1}(V) \cap A \neq \emptyset$, which implies $U \cap A \neq \emptyset$. Hence, $x \in \mathrm{cl}\, A = A$.

Remark 4.16.2 Evidently, if $(X_\lambda, f_{\lambda,\lambda'})_{\lambda\in\Lambda}$ is a nested system, then Proposition 4.16.5 (2) is valid for an arbitrary subspace A of X. However, in general, it is necessary to assume the closedness of A. In fact, if every X_λ is finite then the inverse limit X is compact (Proposition 4.16.2). For $A \subset X$, since each $p_\lambda(A)$ is finite, the inverse limit $\varprojlim(p_\lambda(A), f_{\lambda,\lambda'}|p_{\lambda'}(A))_{\lambda\in\Lambda}$ is also compact. Then, $A \neq \varprojlim(p_\lambda(A), f_{\lambda,\lambda'}|p_{\lambda'}(A))_{\lambda\in\Lambda}$ unless A is closed in X.

Proposition 4.16.6 *The limit of an inverse system* $(X_\lambda, f_{\lambda,\lambda'})_{\lambda\in\Lambda}$ *of locally compact spaces and proper maps is locally compact and the inverse limit projections* $p_\lambda : \varprojlim(X_\lambda, f_{\lambda,\lambda'})_{\lambda\in\Lambda} \to X_\lambda$, $\lambda \in \Lambda$, *are proper.*

Proof If p_{λ_0} is proper for some $\lambda_0 \in \Lambda$, then $X = \varprojlim(X_\lambda, f_{\lambda,\lambda'})_{\lambda\in\Lambda}$ is locally compact because X_{λ_0} is locally compact. Hence, we may show that $p_\lambda : X \to X_\lambda$ is proper for each $\lambda \in \Lambda$. Let $\Delta = \{\delta \in \Lambda \mid \delta \geqslant \lambda\}$. Then, Δ is cofinal in Λ. By Proposition 4.16.3, we have a homeomorphism

$$h : \varprojlim(X_\lambda, f_{\lambda,\lambda'})_{\lambda\in\Lambda} \to \varprojlim(X_\delta, f_{\delta,\delta'})_{\delta\in\Delta}$$

such that $q_\delta h = p_\delta$, where $q_\delta : \varprojlim(X_\delta, f_{\delta,\delta'})_{\delta\in\Delta} \to X_\delta$ is the inverse limit projection. Thus, it is enough to show that $q_\lambda : \varprojlim(X_\delta, f_{\delta,\delta'})_{\delta\in\Delta} \to X_\lambda$ is proper.

Let A be a compact set in X_λ. For each $\delta \in \Delta$ (i.e., $\delta \geqslant \lambda$), since $q_\lambda = f_{\lambda,\delta} q_\delta$, we have $f_{\lambda,\delta} q_\delta(q_\lambda^{-1}(A)) \subset A$, i.e., $q_\delta(q_\lambda^{-1}(A)) \subset f_{\lambda,\delta}^{-1}(A)$, where $f_{\lambda,\delta}^{-1}(A)$ is compact because $f_{\lambda,\delta}$ is proper. Hence, $\varprojlim(f_{\lambda,\delta}^{-1}(A), f_{\delta,\delta'}|f_{\lambda,\delta'}^{-1}(A))_{\delta\in\Delta}$ is a compact subspace of $\varprojlim(X_\delta, f_{\delta,\delta'})_{\delta\in\Delta}$ by Propositions 4.16.2 and 4.16.5 (1). Since $q_\lambda^{-1}(A)$ is closed in $\varprojlim(X_\delta, f_{\delta,\delta'})_{\delta\in\Delta}$, it follows from Proposition 4.16.5 (2) that

$$q_\lambda^{-1}(A) = \varprojlim(q_\delta(q_\lambda^{-1}(A)), f_{\delta,\delta'}|q_{\delta'}(q_\lambda^{-1}(A)))_{\delta\in\Delta},$$

which is also closed in $\varprojlim(f_{\lambda,\delta}^{-1}(A), f_{\delta,\delta'}|f_{\lambda,\delta'}^{-1}(A))_{\delta\in\Delta}$. So $q_\lambda^{-1}(A)$ is compact. □

Let $(X_\lambda, f_{\lambda,\lambda'})_{\lambda\in\Lambda}$ be an inverse system with the limit $X = \varprojlim(X_\lambda, f_{\lambda,\lambda'})_{\lambda\in\Lambda}$. Then, the inverse limit projections $p_\lambda : X \to X_\lambda$, $\lambda \in \Lambda$, satisfy the following:

(inv-1) $f_{\lambda,\lambda'} p_{\lambda'} = p_\lambda$ for each $\lambda < \lambda' \in \Lambda$;
(inv-2) Given maps $g_\lambda : Y \to X_\lambda$, $\lambda \in \Lambda$, such that $f_{\lambda,\lambda'} g_{\lambda'} = g_\lambda$ for each $\lambda < \lambda' \in \Lambda$, there exists a unique map $g : Y \to X$ such that $p_\lambda g = g_\lambda$ for each $\lambda \in \Lambda$.

The above two conditions characterize the inverse limit, that is, more formally:

Theorem 4.16.7 *A space X is homeomorphic to the limit $\varprojlim(X_\lambda, f_{\lambda,\lambda'})_{\lambda\in\Lambda}$ of an inverse system $(X_\lambda, f_{\lambda,\lambda'})_{\lambda\in\Lambda}$ if and only if there exist maps $q_\lambda : X \to X_\lambda$, $\lambda \in \Lambda$, which satisfy conditions* (inv-1) *and* (inv-2). *In this case, there exists a unique homeomorphism $h : X \to \varprojlim(X_\lambda, f_{\lambda,\lambda'})_{\lambda\in\Lambda}$ such that $p_\lambda h = q_\lambda$ for each $\lambda \in \Lambda$.*

Proof (*The "only if" Part*) If there is a homeomorphism $h : X \to \varprojlim(X_\lambda, f_{\lambda,\lambda'})_{\lambda\in\Lambda}$, then the maps $q_\lambda = p_\lambda h : X \to X_\lambda$, $\lambda \in \Lambda$, satisfy conditions (inv-1) and (inv-2), where $p_\lambda : \varprojlim(X_\lambda, f_{\lambda,\lambda'})_{\lambda\in\Lambda} \to X_\lambda$ is the inverse limit projection.

(*The "if" Part*) Because q_λ, $\lambda \in \Lambda$, satisfy condition (inv-1), we apply condition (inv-2) for p_λ, $\lambda \in \Lambda$ to obtain a unique map $h : X \to \varprojlim(X_\lambda, f_{\lambda,\lambda'})_{\lambda\in\Lambda}$ such that $p_\lambda h = q_\lambda$ for each $\lambda \in \Lambda$. Similarly, we apply condition (inv-2) for q_λ, $\lambda \in \Lambda$, to obtain a map $g : \varprojlim(X_\lambda, f_{\lambda,\lambda'})_{\lambda\in\Lambda} \to X$ such that $q_\lambda g = p_\lambda$ for each $\lambda \in \Lambda$. Since $p_\lambda hg = p_\lambda$ and $q_\lambda gh = q_\lambda$ for each $\lambda \in \Lambda$, $hg = \mathrm{id}$ and $gh = \mathrm{id}$ by the uniqueness in condition (inv-2). Therefore, h is a homeomorphism with $h^{-1} = g$. □

The following theorem shows the importance of paracompact spaces:

Theorem 4.16.8 *Every paracompact space X is homeomorphic to the limit $\varprojlim(X_\gamma, \varphi_{\gamma,\gamma'})_{\gamma\in\Gamma}$ of an inverse system of polyhedra and PL maps, where $\dim(X_\gamma) \leqslant w(X)$.*

Proof Let $\mathrm{Part}(X)$ be the collection of all locally finite partitions of unity on X. For each $\boldsymbol{\alpha} = (\alpha_\lambda)_{\lambda\in\Lambda(\boldsymbol{\alpha})} \in \mathrm{Part}(X)$, let $\mathcal{U}_{\boldsymbol{\alpha}} = \{\alpha_\lambda^{-1}((0,1]) \mid \lambda \in \Lambda(\boldsymbol{\alpha})\}$. Then, every open cover of X is refined by $\mathcal{U}_{\boldsymbol{\alpha}}$ for some $\boldsymbol{\alpha} \in \mathrm{Part}(X)$. Indeed, since X is paracompact, every open cover of X has a locally finite open refinement $\mathcal{U}$. By Theorem 2.7.7, there exists a partition of unity $\boldsymbol{\alpha} = (\alpha_\lambda)_{\lambda\in\Lambda(\boldsymbol{\alpha})}$ such that $\operatorname{supp}\alpha_\lambda \subset U$ for every $\lambda \in \Lambda(\boldsymbol{\alpha})$. Thus, we have $\mathcal{U}_{\boldsymbol{\alpha}} \prec \mathcal{U}$.

Let $F(X) = (\mathrm{Fin}(\mathrm{Part}(X)), \subset)$, which is a directed set. For each $F \in F(X)$, let

$$S_F = \left\{ v \in \prod_{\boldsymbol{\alpha}\in F} \Lambda(\boldsymbol{\alpha}) \;\middle|\; W(v) \neq \emptyset \right\}, \text{ where } W(v) = \bigcap_{\boldsymbol{\alpha}\in F} \alpha_{v(\boldsymbol{\alpha})}^{-1}((0,1]).$$

Then, we have the following abstract complex:

$$\mathcal{K}_F = \left\{ A \in \mathrm{Fin}(S_F) \;\middle|\; \textstyle\bigcap_{v\in A} W(v) \neq \emptyset \right\}.$$

Let K_F be the simplicial complex defined by $\mathcal{K}_F$, that is, $K_F^{(0)} = S_F$ and

$$\langle v_1, \dots, v_n \rangle \in K_F \Leftrightarrow W(v_1) \cap \cdots \cap W(v_n) \neq \emptyset.$$

When $F = \{\boldsymbol{\alpha}_1, \dots, \boldsymbol{\alpha}_k\}$, we have $\{W(v) \mid v \in S_F\} = \mathcal{U}_{\boldsymbol{\alpha}_1} \wedge \cdots \wedge \mathcal{U}_{\boldsymbol{\alpha}_k}$ but it might be possible that $W(v) = W(v')$ even if $v \neq v'$. It should be noted that K_F is not necessarily the nerve of $\mathcal{U}_{\boldsymbol{\alpha}_1} \wedge \cdots \wedge \mathcal{U}_{\boldsymbol{\alpha}_k}$, although the definition is similar.

For each $v \in S_F = K_F^{(0)}$, we define $\alpha_v = \prod_{\boldsymbol{\alpha} \in F} \alpha_{v(\boldsymbol{\alpha})} : X \to \mathbf{I}$. Evidently,

$$\alpha_v^{-1}((0,1]) = \bigcap_{\boldsymbol{\alpha} \in F} \alpha_{v(\boldsymbol{\alpha})}^{-1}((0,1]) = W(v).$$

Moreover, $(\alpha_v)_{v \in S_F}$ is a locally finite partition of unity. Indeed, let $x \in X$ and $F = \{\boldsymbol{\alpha}_1, \dots, \boldsymbol{\alpha}_k\}$, where $\boldsymbol{\alpha}_i \neq \boldsymbol{\alpha}_j$ if $i \neq j$. Since $\boldsymbol{\alpha}_i = (\alpha_\lambda)_{\lambda \in \Lambda(\boldsymbol{\alpha}_i)}$ is locally finite, $\alpha_\lambda(x) = 0$ except for finitely many $\lambda \in \Lambda(\boldsymbol{\alpha}_i)$. Then, it follows that

$$\begin{aligned} \sum_{v \in S_F} \alpha_v(x) &= \sum_{v \in S_F} \prod_{i=1}^{k} \alpha_{v(\boldsymbol{\alpha}_i)}(x) = \sum_{\substack{(\lambda_1, \dots, \lambda_k) \\ \in \prod_{i=1}^{k} \Lambda(\boldsymbol{\alpha}_i)}} \prod_{i=1}^{k} \alpha_{\lambda_i}(x) \\ &= \prod_{i=1}^{k} \Big(\sum_{\lambda \in \Lambda(\boldsymbol{\alpha}_i)} \alpha_\lambda(x) \Big) = 1. \end{aligned}$$

The local finiteness of $(\alpha_v)_{v \in S_F}$ easily follows from that of $\boldsymbol{\alpha}_1, \dots, \boldsymbol{\alpha}_k$.

Now, we define a map $\varphi_F : X \to |K_F|$ as follows:

$$\varphi_F(x) = \sum_{v \in K_F^{(0)}} \alpha_v(x) v \quad \Big(\text{i.e., } \forall v \in K_F^{(0)},\ \beta_v^{K_F} \varphi_F(x) = \alpha_v(x) \Big).$$

Observe that for each $v \in K_F^{(0)}$,

$$\varphi_F^{-1}(O_{K_F}(v)) = \varphi_F^{-1}((\beta_v^{K_F})^{-1}((0,1])) = \alpha_v^{-1}((0,1]) = W(v). \qquad (*)$$

For each $F \subset F' \in F(X)$, we have a simplicial map $\varphi_{F,F'} : K_{F'} \to K_F$ defined by $\varphi_{F,F'}(v') = v'|F \in K_F^{(0)} = S_F$ for each $v' \in K_{F'}^{(0)} = S_{F'}$, where $\varphi_{F,F} = \mathrm{id}$. As is easily observed, $\varphi_{F,F'}\varphi_{F',F''} = \varphi_{F,F''}$ for $F \subset F' \subset F'' \in F(X)$.

Indeed, $\varphi_{F,F'}\varphi_{F',F''}(v'') = (v''|F')|F = v''|F = \varphi_{F,F''}(v'')$ for each $v'' \in K_{F''}^{(0)}$.

Thus, $(|K_F|, \varphi_{F,F'})_{F \in F(X)}$ is an inverse system. Moreover, $\varphi_{F,F'}\varphi_{F'} = \varphi_F$ for $F \subsetneqq F' \in F(X)$. Indeed, for each $x \in X$,

$$\begin{aligned}\varphi_{F,F'}\varphi_{F'}(x) &= \varphi_{F,F'}\Big(\textstyle\sum_{v'\in K^{(0)}_{F'}} \alpha_{v'}(x)v'\Big) = \sum_{v'\in K^{(0)}_{F'}} \alpha_{v'}(x)\varphi_{F,F'}(v') \\ &= \sum_{v'\in K^{(0)}_{F'}} \alpha_{v'}(x)v'|F = \sum_{v\in K^{(0)}_{F}} \sum_{\substack{v'\in K^{(0)}_{F'}\\ v'|F=v}} \alpha_{v'}(x)v'|F \\ &= \sum_{v\in K^{(0)}_{F}} \alpha_{v}(x)v = \varphi_F(x).\end{aligned}$$

To show the first equality of the last line, let $F' \setminus F = \{\boldsymbol{\alpha}_1, \dots, \boldsymbol{\alpha}_k\}$, where $\boldsymbol{\alpha}_i \neq \boldsymbol{\alpha}_j$ if $i \neq j$. Then, it follows that

$$\begin{aligned}\sum_{\substack{v'\in K^{(0)}_{F'}\\ v'|F=v}} \alpha_{v'}(x) &= \sum_{\substack{v'\in K^{(0)}_{F'}\\ v'|F=v}} \prod_{\boldsymbol{\alpha}\in F'} \alpha_{v'(\boldsymbol{\alpha})}(x) = \sum_{\substack{(\lambda_1,\dots,\lambda_k)\\ \in\prod_{i=1}^k \Lambda(\boldsymbol{\alpha}_i)}} \alpha_v(x) \prod_{i=1}^k \alpha_{\lambda_i}(x) \\ &= \alpha_v(x) \prod_{i=1}^k \Big(\sum_{\lambda\in\Lambda(\boldsymbol{\alpha}_i)} \alpha_\lambda(x) \Big) = \alpha_v(x).\end{aligned}$$

For each $F \in F(X)$, let $\mathfrak{N}_F(X)$ be the collection of all polyhedral neighborhoods $N = |N(\varphi_F(X), K'_F)|$ of $\varphi_F(X)$ with respect to $K'_F \lhd K_F$,[36] that is,

$$\mathfrak{N}_F(X) = \big\{ N \;\big|\; \exists K'_F \lhd K_F \;\text{ such that }\; N = |N(\varphi_F(X), K'_F)| \big\}.$$

We define $\Gamma(X) = \bigcup_{F\in F(X)} \{F\} \times \mathfrak{N}_F(X)$, which is a directed set with the following preorder $\leqslant$:

$$(F, N) \leqslant (F', N') \underset{\text{def}}{\Leftrightarrow} F \subset F',\ \varphi_{F,F'}(N') \subset N.$$

For each $\gamma = (F, N) \in \Gamma(X)$, let $X_\gamma = N$ and $\varphi_\gamma = \varphi_F : X \to X_\gamma\ (= N \subset |K_F|)$. For each $\gamma \leqslant \gamma' = (F', N') \in \Gamma(X)$, because $\varphi_{F,F'}(X_{\gamma'}) = \varphi_{F,F'}(N') \subset N = X_\gamma$, we have a PL map $\varphi_{\gamma,\gamma'} = \varphi_{F,F'}|X_{\gamma'} : X_{\gamma'} \to X_\gamma$. Since $(|K_F|, \varphi_{F,F'})_{F\in F(X)}$ is an inverse system, $(X_\gamma, \varphi_{\gamma,\gamma'})_{\gamma\in\Gamma(X)}$ is an inverse system by the definition of $\Gamma(X)$. Moreover, as is easily observed, $\varphi_\gamma = \varphi_{\gamma,\gamma'}\varphi_{\gamma'}$ for $\gamma \leqslant \gamma' \in \Gamma(X)$. Hence, by (inv-2), we have a unique map $h : X \to \varprojlim(X_\gamma, \varphi_{\gamma,\gamma'})_{\gamma\in\Gamma(X)}$ such that $p_\gamma h = \varphi_\gamma$ for each $\gamma \in \Gamma(X)$, where each $p_\gamma : \varprojlim(X_\gamma, \varphi_{\gamma,\gamma'})_{\gamma\in\Gamma(X)} \to X_\gamma$ is the inverse limit projection. To see that h is homeomorphism, we have to show that h is bijective and open.

(*Injective*) For each $x_1 \neq x_2 \in X$, consider $\{X \setminus \{x_2\}, X \setminus \{x_1\}\} \in \mathrm{cov}(X)$. Since X is paracompact, we can take $\boldsymbol{\alpha} = (\alpha_\lambda)_{\lambda\in\Lambda(\boldsymbol{\alpha})} \in \mathrm{Part}(X)$ so that

$$\mathcal{U}_{\boldsymbol{\alpha}} = \big\{ \alpha_\lambda^{-1}((0,1]) \;\big|\; \lambda \in \Lambda(\boldsymbol{\alpha}) \big\} \prec \big\{ X \setminus \{x_2\}, X \setminus \{x_1\} \big\}.$$

[36] For $A \subset |K|$, $N(A, K)$ was defined in Sect. 4.2.

Choose $\lambda_0 \in \Lambda(\boldsymbol{\alpha})$ so that $x_1 \in \alpha_{\lambda_0}^{-1}((0,1])$, which means $\alpha_{\lambda_0}^{-1}((0,1]) \subset X \setminus \{x_2\}$. Then, $\alpha_{\lambda_0}(x_1) > 0$ and $\alpha_{\lambda_0}(x_2) = 0$. Let $F = \{\boldsymbol{\alpha}\} \in F(X) = \mathrm{Fin}(\mathrm{Part}(X))$ and $N = |N(\varphi_F(X), K_F)| \in \mathfrak{N}_F(X)$. Thus, we have $\gamma = (F, N) \in \Gamma(X)$, where $X_\gamma = N$. Then, we show $p_\gamma h(x_1) \neq p_\gamma h(x_2)$, which implies $h(x_1) \neq h(x_2)$. We can take $v_0 \in K_F^{(0)} = S_F$ so that $v_0(\boldsymbol{\alpha}) = \lambda_0 \in \Lambda(\boldsymbol{\alpha})$. Then, it follows that

$$\beta_{v_0}^{K_F} p_\gamma h(x_1) = \beta_{v_0}^{K_F} \varphi_\gamma(x_1) = \beta_{v_0}^{K_F} \varphi_F(x_1) = \alpha_{v_0}(x_1) > 0 \text{ and}$$
$$\beta_{v_0}^{K_F} p_\gamma h(x_2) = \beta_{v_0}^{K_F} \varphi_\gamma(x_2) = \beta_{v_0}^{K_F} \varphi_F(x_2) = \alpha_{v_0}(x_2) = 0.$$

(*Surjective*) For simplicity, we write $\varprojlim(X_\gamma, \varphi_{\gamma,\gamma'})_{\gamma \in \Gamma(X)} = L$. Suppose that $h(X) \neq L$. For each $y \in L \setminus h(X)$, let $\mathcal{V}$ be an open neighborhood basis of y in L. By regularity of L, we have $\bigcap_{V \in \mathcal{V}} \mathrm{cl}\, V = \{y\}$, which means

$$\{h^{-1}(L \setminus \mathrm{cl}\, V) \mid V \in \mathcal{V}\} \in \mathrm{cov}(X).$$

Then, we can take $\boldsymbol{\alpha} = (\alpha_\lambda)_{\lambda \in \Lambda(\boldsymbol{\alpha})} \in \mathrm{Part}(X)$ so that

$$\mathcal{U}_{\boldsymbol{\alpha}} = \left\{\alpha_\lambda^{-1}((0,1]) \;\middle|\; \lambda \in \Lambda(\boldsymbol{\alpha})\right\} \prec \left\{h^{-1}(L \setminus \mathrm{cl}\, V) \;\middle|\; V \in \mathcal{V}\right\}.$$

Let $F = \{\boldsymbol{\alpha}\} \in F(X)$ and $\gamma_0 = (F, |N(\varphi_F(X), K_F)|) \in \Gamma(X)$. Since $y(\gamma_0) \in X_{\gamma_0} \subset |K_F)|$, we can choose $v_0 \in K_F^{(0)} = S_F$ so that $p_{\gamma_0}(y) = y(\gamma_0) \in O_{K_F}(v_0)$, and let $\lambda_0 = v_0(\boldsymbol{\alpha}) \in \Lambda(\boldsymbol{\alpha})$. Then, $\alpha_{v_0(\boldsymbol{\alpha})}^{-1}((0,1]) = \alpha_{\lambda_0}^{-1}((0,1]) \subset h^{-1}(L \setminus \mathrm{cl}\, V_0)$ for some $V_0 \in \mathcal{V}$. Since p_{γ_0} is continuous, there is $V \in \mathcal{V}$ such that $V \subset V_0$ and $p_{\gamma_0}(V) \subset O_{K_F}(v_0)$. From (*), it follows that

$$h^{-1}(V) \subset h^{-1} p_{\gamma_0}^{-1}(p_{\gamma_0}(V)) \subset \varphi_{\gamma_0}^{-1}(O_{K_F}(v_0)) = \alpha_{v_0}^{-1}((0,1])$$
$$\subset h^{-1}(L \setminus \mathrm{cl}\, V_0) \subset h^{-1}(L \setminus V) = X \setminus h^{-1}(V),$$

which means $h^{-1}(V) = \emptyset$, that is, $V \cap h(X) = \emptyset$.

By Proposition 4.16.4, we can take $\gamma_* = (F_*, N_*) \in \Gamma(X)$ and an open neighborhood V_* of $y(\gamma_*)$ in X_{γ_*} such that $p_{\gamma_*}^{-1}(V_*) \subset V$. Hence, $p_{\gamma_*}^{-1}(V_*) \cap h(X) = \emptyset$, which implies $V_* \cap \varphi_{\gamma_*}(X) = V_* \cap p_{\gamma_*} h(X) = \emptyset$. Recall that $X_{\gamma_*} = N_* = |N(\varphi_{\gamma_*}(X), K'_{F_*})|$ for some $K'_{F_*} \lhd K_{F_*}$. Let W be an open set in $|K_{F_*}|$ such that $V_* = W \cap X_{\gamma_*}$. Since $\{W, |K_{F_*}| \setminus \{y(\gamma_*)\}\} \in \mathrm{cov}(|K_{F_*}|)$, applying Whitehead's Theorem 4.8.12 on small subdivisions, we can obtain $K''_{F_*} \lhd K'_{F_*}$ such that $\mathcal{S}_{K''_{F_*}} \prec \{W, |K_{F_*}| \setminus \{y(\gamma_*)\}\}$. Observe that $|N(\varphi_{\gamma_*}(X), K''_{F_*})| \subset |K_{F_*}| \setminus \{y(\gamma_*)\}$. Let $\gamma'_* = (F_*, |N(\varphi_{\gamma_*}(X), K''_{F_*})|) \in \Gamma(X)$. Then, $\gamma_* \leqslant \gamma'_*$ and $\varphi_{\gamma_*,\gamma'_*} = \varphi_{F_*,F_*}|X_{\gamma'_*} = \mathrm{id}$ is the inclusion. Hence, $\varphi_{\gamma_*} = \varphi_{\gamma_*,\gamma'_*}\varphi_{\gamma'_*} = \varphi_{\gamma'_*}$. Thus, it follows that

$$y(\gamma'_*) = \varphi_{\gamma_*,\gamma'_*}(y(\gamma'_*)) = y(\gamma_*) \notin |N(\varphi_{\gamma_*}(X), K''_{F_*})| = |N(\varphi_{\gamma'_*}(X), K''_{F_*})| = X_{\gamma'_*},$$

which is a contradiction.

(*Open*) For each open neighborhood U of $x \in X$, it suffices to prove that $h(U)$ is a neighborhood of $h(x)$ in $\varprojlim(X_\gamma, \varphi_{\gamma,\gamma'})_{\gamma\in\Gamma(X)}$. Since $\{U, X \setminus \{x\}\} \in \mathrm{cov}(X)$, we have $\boldsymbol{\alpha} = (\alpha_\lambda)_{\lambda\in\Lambda(\boldsymbol{\alpha})} \in \mathrm{Part}(X)$ such that

$$\mathcal{U}_{\boldsymbol{\alpha}} = \left\{\alpha_\lambda^{-1}((0,1]) \;\middle|\; \lambda \in \Lambda(\boldsymbol{\alpha})\right\} \prec \left\{U, X \setminus \{x\}\right\}.$$

Take $\lambda_0 \in \Lambda(X)$ so that $x \in \alpha_{\lambda_0}^{-1}((0,1])$. Then, $\alpha_{\lambda_0}^{-1}((0,1]) \subset U$. Let $F = \{\boldsymbol{\alpha}\} \in F(X)$, $\gamma_0 = (F, |N(\varphi_F(X), K_F)|) \in \Gamma(X)$, and $v_0 \in K_F^{(0)}$ such that $v_0(\boldsymbol{\alpha}) = \lambda_0$. Since $p_{\gamma_0} h = \varphi_{\gamma_0} = \varphi_F$, it follows from (*) that

$$h^{-1}\big(p_{\gamma_0}^{-1}(O_{K_F}(v_0))\big) = \varphi_F^{-1}(O_{K_F}(v_0)) = \alpha_{v_0}^{-1}((0,1]) = \alpha_{\lambda_0}^{-1}((0,1]).$$

Hence, $x \in h^{-1}(p_{\gamma_0}^{-1}(O_{K_F}(v_0))) \subset U$. Because h is bijective, we have

$$h(x) \in p_{\gamma_0}^{-1}(O_{K_F}(v_0)) \subset h(U).$$

Therefore, $h(U)$ is a neighborhood of $h(x)$ in $\varprojlim(X_\gamma, \varphi_{\gamma,\gamma'})_{\gamma\in\Gamma(X)}$. □

In the above, if X is compact, then $\mathrm{Part}(X)$ can be replaced with the set $\mathrm{Part}_f(X)$ consisting of all finite partitions of unity on X. Then, each K_F is finite, so each X_γ is compact. Thus, we have the following:

Theorem 4.16.9 *Every compact space X can be represented as the limit of inverse system of compact polyhedra and PL maps.* □

The direct proof of the above theorem is rather easy.

Sketch of Proof In fact, by Theorem 2.1.3, we can regard X as a closed set in $\mathbf{I}^{\mathrm{C}(X,\mathbf{I})}$. For each finite subset $F \subset \mathrm{C}(X, \mathbf{I})$, let $\mathrm{pr}_F : \mathbf{I}^{\mathrm{C}(X,\mathbf{I})} \to \mathbf{I}^F$ be the projection and let $\mathfrak{N}_F$ be the collection of all polyhedral neighborhoods of $\mathrm{pr}_F(X)$ in $\mathbf{I}^F$ (i.e., neighborhoods of $\mathrm{pr}_F(X)$ in $\mathbf{I}^F$ being subpolyhedra of $\mathbf{I}^F$). Let $\Gamma(X) = \bigcup_{F\in\mathrm{Fin}(\mathrm{C}(X,\mathbf{I}))} \mathfrak{N}_F$, which is a directed set with the following preorder:

$$N \in \mathfrak{N}_F,\ N' \in \mathfrak{N}_{F'},\ N \leqslant N' \underset{\text{def}}{\Leftrightarrow} F \subset F',\ \mathrm{pr}_F^{-1}(N) \supset \mathrm{pr}_{F'}^{-1}(N').$$

For $N \leqslant N' \in \Gamma(X)$, let $\varphi_{N,N'} : N' \to N$ be the restriction of the projection of $\mathbf{I}^{F'}$ onto $\mathbf{I}^F$. For each $N \in \Gamma(X)$, let $\varphi_N = \mathrm{pr}_F | X : X \to N$. Then, $(N, \varphi_{N,N'})_{N\in\Gamma(X)}$ is an inverse system. We can uniquely define a map $h : X \to \varprojlim(N, \varphi_{N,N'})_{N\in\Gamma(X)}$ such that $p_N h = \varphi_N$ for each $N \in \Gamma(X)$. It is easy to show that h is a continuous bijection, that is, a homeomorphism.

Expansion in the Homotopy Category

We can consider inverse systems in the homotopy category HTop of topological spaces, whose morphisms are homotopy classes of maps. Then, arguments will be simple as follows:

(1) For a space X, $\mathrm{cov}(X)$ is a directed set with the preorder $\leqslant$ defined as follows:

$$\mathcal{U} \leqslant \mathcal{V} \underset{\text{def}}{\Leftrightarrow} \mathcal{U} \succ \mathcal{V}\ \ (\mathcal{V} \prec \mathcal{U}).$$

For each $\mathcal{U} \leqslant \mathcal{V} \in \mathrm{cov}(X)$, we have a refining simplicial map $\varphi_{\mathcal{U},\mathcal{V}} : N(\mathcal{V}) \to N(\mathcal{U})$ defined by $V \subset \varphi_{\mathcal{U},\mathcal{V}}(V) \in \mathcal{U} = N(\mathcal{U})^{(0)}$ for each $V \in \mathcal{V} = N(\mathcal{V})^{(0)}$ (cf. the definition immediately

preceding Proposition 4.15.4). A refining simplicial map is not unique but it is homotopically unique, that is, $\varphi_{\mathcal{U},\mathcal{V}} \simeq \varphi'_{\mathcal{U},\mathcal{V}}$ for any other refining simplicial map $\varphi'_{\mathcal{U},\mathcal{V}} : N(\mathcal{V}) \to N(\mathcal{U})$ (Proposition 4.15.4 (1)). In particular, $\varphi_{\mathcal{U},\mathcal{U}} \simeq \mathrm{id}$. Thus, it follows that

$$[\varphi_{\mathcal{U},\mathcal{V}}][\varphi_{\mathcal{V},\mathcal{W}}] = [\varphi_{\mathcal{U},\mathcal{W}}] \text{ for } \mathcal{U} \leqslant \mathcal{V} \leqslant \mathcal{W},$$

but as pointed out previously, $\varphi_{\mathcal{U},\mathcal{V}}\varphi_{\mathcal{V},\mathcal{W}} = \varphi_{\mathcal{U},\mathcal{W}}$ does not always hold in general. Thus, $(|N(\mathcal{U})|, [\varphi_{\mathcal{U},\mathcal{V}}])_{\mathcal{U}\in \mathrm{cov}(X)}$ is an inverse system in the homotopy category HTop. Moreover, let $f : X \to Y$ be a map. Then, $f^{-1}(\mathcal{U}) \in \mathrm{cov}(X)$ for each $\mathcal{U} \in \mathrm{cov}(Y)$, and simplicial maps $f_{\mathcal{U}} : N(f^{-1}(\mathcal{U})) \to N(\mathcal{U}), \mathcal{U} \in \mathrm{cov}(Y)$, can be defined by $f_{\mathcal{U}}(f^{-1}(U)) = U$ for each $U \in \mathcal{U} = N(\mathcal{U})^{(0)}$. Then, the following holds:

$$[f_{\mathcal{U}}][\varphi_{f^{-1}(\mathcal{U}),f^{-1}(\mathcal{V})}] = [\varphi_{\mathcal{U},\mathcal{V}}][f_{\mathcal{V}}] \text{ for each } \mathcal{U} \leqslant \mathcal{V} \in \mathrm{cov}(Y).$$

Indeed, by Proposition 4.15.4 (1), a refining simplicial map $\varphi_{f^{-1}(\mathcal{U}),f^{-1}(\mathcal{V})} : f^{-1}(\mathcal{V}) \to f^{-1}(\mathcal{U})$ can be replaced with the one defined as follows:

$$\varphi_{f^{-1}(\mathcal{U}),f^{-1}(\mathcal{V})}(f^{-1}(V)) = f^{-1}(\varphi_{\mathcal{U},\mathcal{V}}(V)) \text{ for each } V \in \mathcal{V}.$$

Then, $f_{\mathcal{U}}\varphi_{f^{-1}(\mathcal{U}),f^{-1}(\mathcal{V})} = \varphi_{\mathcal{U},\mathcal{V}}f_{\mathcal{V}}$.

(2) Let $\mathrm{cov}^*(X)$ denote the collection of all numerable (= normal) covers of X. It should be noted that $\mathrm{cov}^*(X) = \mathrm{cov}(X)$ if X is paracompact. Since $\mathrm{cov}^*(X)$ is a directed subset of $\mathrm{cov}(X)$ (Corollary 2.7.3), we have an inverse system $(|N(\mathcal{U})|, [\varphi_{\mathcal{U},\mathcal{V}}])_{\mathcal{U}\in \mathrm{cov}^*(X)}$. By Theorem 4.15.5, we have canonical maps $p_{\mathcal{U}} : X \to |N(\mathcal{U})|, \mathcal{U} \in \mathrm{cov}^*(X)$. Then, by Corollary 4.15.3 and Proposition 4.15.4 (2),

$$[p_{\mathcal{U}}] = [\varphi_{\mathcal{U},\mathcal{V}}][p_{\mathcal{V}}] \text{ for } \mathcal{U} \leqslant \mathcal{V} \in \mathrm{cov}^*(X).$$

For each map $f : X \to Y$, $\mathcal{U} \in \mathrm{cov}^*(X)$ implies $f^{-1}(\mathcal{U}) \in \mathrm{cov}^*(Y)$ (Corollary 2.7.4) and

$$[f_{\mathcal{U}}][p_{f^{-1}(\mathcal{U})}] = [p_{\mathcal{U}}][f] \text{ for each } \mathcal{U} \in \mathrm{cov}^*(Y).$$

Indeed, for each $x \in X$, let $U \in \mathcal{U}$ such that $f^{-1}(U) \in c_{N(f^{-1}(\mathcal{U}))}(p_{f^{-1}(\mathcal{U})}(x))^{(0)}$. By Proposition 4.15.2, $f^{-1}(U) \in f^{-1}(\mathcal{U})[x]$, which means $f_{\mathcal{U}}(f^{-1}(U)) = U \in \mathcal{U}[f(x)]$. Thus, we have

$$f_{\mathcal{U}}(c_{N(f^{-1}(\mathcal{U}))}(p_{f^{-1}(\mathcal{U})}(x))^{(0)}) \subset \mathcal{U}[f(x)].$$

Since $c_{N(\mathcal{U})}(p_{\mathcal{U}}f(x))^{(0)} \subset \mathcal{U}[f(x)]$ by Proposition 4.15.2, $f_{\mathcal{U}}(c_{N(f^{-1}(\mathcal{U}))}(p_{f^{-1}(\mathcal{U})}(x)))$ and $c_{N(\mathcal{U})}(p_{\mathcal{U}}f(x))$ are faces of the same simplex of $N(\mathcal{U})$. Hence, $f_{\mathcal{U}}p_{f^{-1}(\mathcal{U})}$ and $p_{\mathcal{U}}f$ are contiguous, which implies that $f_{\mathcal{U}}p_{f^{-1}(\mathcal{U})} \simeq p_{\mathcal{U}}f$.

Let K be a simplicial complex. As mentioned just before Proposition 4.15.2, K can be regarded as the nerve $N(\mathcal{O}_K)$ by identifying $v \in K^{(0)}$ with $O_K(v) \in \mathcal{O}_K$ (cf. Proposition 4.2.8), where $\mathrm{id} : |K| \to |K| = |N(\mathcal{O}_K)|$ is a canonical map for $\mathcal{O}_K$. For a map $f : X \to |K|$, we have a simplicial map $f_{\mathcal{O}_K} : N(f^{-1}(\mathcal{O}_K)) \to N(\mathcal{O}_K) \equiv K$, where $f^{-1}(\mathcal{O}_K) \in \mathrm{cov}^*(X)$. Then, $[f_{\mathcal{O}_K}][p_{f^{-1}(\mathcal{O}_K)}] = [f]$. Thus, the following holds:

(i) *For each map $f : X \to |K|$, there exist $\mathcal{U} \in \mathrm{cov}^*(X)$ and a map $g : |N(\mathcal{U})| \to |K|$ such that $[f] = [g][p_{\mathcal{U}}]$.*

Moreover, the following is also known:

(ii) *For each locally finite $\mathcal{U} \in \mathrm{cov}^*(X)$ and two maps $f, g : |N(\mathcal{U})| \to |K|$, if $fp_{\mathcal{U}} \simeq gp_{\mathcal{U}}$, then $f\varphi_{\mathcal{U},\mathcal{V}} \simeq g\varphi_{\mathcal{U},\mathcal{V}}$ for some refinement $\mathcal{V} \in \mathrm{cov}^*(X)$ of $\mathcal{U}$.*

4.17 The Inverse Limits of Metric Polyhedra

In this section, we show that every complete metrizable space can be represented as the limit of an inverse *sequence* of locally finite-dimensional simplicial complexes with the metric topology and PL maps, where for some compactum, PL bonding maps cannot be replaced by simplicial maps.

Inverse sequences are easier to handle than inverse systems. In particular, it is true for metrizable spaces. Since the inverse limit of an inverse sequence $(X_i, f_i)_{i\in\mathbb{N}}$ is a closed subspace of the countable product $\prod_{i\in\mathbb{N}} X_i$, we have the following:

Proposition 4.17.1 *The limit of an inverse sequence $(X_i, f_i)_{i\in\mathbb{N}}$ of metrizable (resp. completely metrizable) spaces is also metrizable (resp. completely metrizable). So, the limit of an inverse sequence of compacta is also a compactum.*[37] □

For inverse sequences, inductive arguments can be available. By induction, we can show the following:

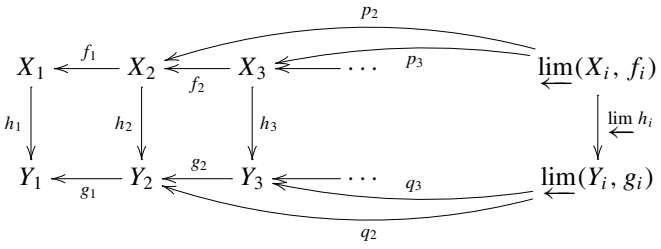
Proposition 4.17.2 *For an inverse sequence $(X_i, f_i)_{i\in\mathbb{N}}$, if every bonding map f_i is surjective, then the inverse limit projection $p_n : \varprojlim(X_i, f_i) \to X_n$ is also surjective for each $n \in \mathbb{N}$.*

Proof For each $x \in X_n$ and each $i \leqslant n$, define $x_i = f_{i,n}(x)$. For each $i \geqslant n$, inductively choose $x_{i+1} \in f_i^{-1}(x_i)$. Then, $(x_i)_{i\in\mathbb{N}} \in \varprojlim(X_i, f_i)$ and $p_n((x_i)_{i\in\mathbb{N}}) = x_n = x$. □

Let $(X_i, f_i)_{i\in\mathbb{N}}$ and $(Y_i, g_i)_{i\in\mathbb{N}}$ be inverse sequences. Given maps $h_i : X_i \to Y_i$, $i \in \mathbb{N}$, such that $h_i f_i = g_i h_{i+1}$ for every $i \in \mathbb{N}$, we can define a map $\varprojlim h_i : \varprojlim(X_i, f_i) \to \varprojlim(Y_i, g_i)$ as follows:

$$(\varprojlim h_i)((x_i)_{i\in\mathbb{N}}) = (h_i(x_i))_{i\in\mathbb{N}}.$$

Then, $q_i \varprojlim h_i = h_i p_i$ for every $i \in \mathbb{N}$, where $p_i : \varprojlim(X_i, f_i) \to X_i$ and $q_i : \varprojlim(Y_i, g_i) \to Y_i$ are the projections.



[37] Cf. Proposition 4.16.2.

For an inverse sequence $(X_i, f_i)_{i\in\mathbb{N}}$ and an increasing sequence $n(1) < n(2) < \cdots \in \mathbb{N}$, we have the inverse sequence $(X_{n(i)}, f_{n(i),n(i+1)})_{i\in\mathbb{N}}$:

$$X_{n(1)} \xleftarrow{f_{n(1),n(2)}} X_{n(2)} \xleftarrow{f_{n(2),n(3)}} X_{n(3)} \longleftarrow \cdots,$$

which is called a **subsequence** of $(X_i, f_i)_{i\in\mathbb{N}}$. Evidently, $\{n(i) \mid i \in \mathbb{N}\}$ is cofinal in $\mathbb{N}$. For inverse sequences, Propositions 4.16.3 and 4.16.4 can be restated as follows:

Proposition 4.17.3 *Let $(X_i, f_i)_{i\in\mathbb{N}}$ be an inverse sequence. For any increasing sequence $n(1) < n(2) < \cdots \in \mathbb{N}$, there exists a homeomorphism*

$$h : \varprojlim(X_i, f_i) \to \varprojlim(X_{n(i)}, f_{n(i),n(i+1)})$$

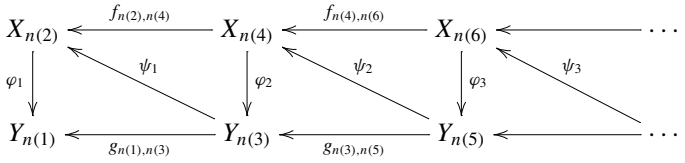
such that $q_i h = p_{n(i)}$ for each $i \in \mathbb{N}$, where $p_i : \varprojlim(X_i, f_i) \to X_i$ and $q_i : \varprojlim(X_{n(i)}, f_{n(i),n(i+1)}) \to X_{n(i)}$ are the projections. □

Proposition 4.17.4 *Let $(X_i, f_i)_{i\in\mathbb{N}}$ be an inverse sequence. For any increasing sequence $n(1) < n(2) < \cdots \in \mathbb{N}$, the following $\mathcal{B}$ is an open basis for $\varprojlim(X_i, f_i)$:*

$$\mathcal{B} = \left\{ p_{n(i)}^{-1}(V) \;\middle|\; i \in \mathbb{N}, V \text{ is open in } X_{n(i)} \right\}.$$ □

The following can be easily proved by using Proposition 4.17.3:

Corollary 4.17.5 *Let $(X_i, f_i)_{i\in\mathbb{N}}$ and $(Y_i, g_i)_{i\in\mathbb{N}}$ be inverse sequences. Suppose that there exists an increasing sequence $n(1) < n(2) < \cdots \in \mathbb{N}$ and maps $\varphi_i : X_{n(2i)} \to Y_{n(2i-1)}$ and $\psi_i : Y_{n(2i+1)} \to X_{n(2i)}$ such that $\psi_i\varphi_{i+1} = f_{n(2i),n(2i+2)}$ and $\varphi_i\psi_i = g_{n(2i-1),n(2i+1)}$, that is, the following diagram is commutative:*



Then, $\varprojlim(X_i, f_i)$ is homeomorphic to $\varprojlim(Y_i, g_i)$.

Sketch of Proof Consider the following inverse sequence:

$$Y_{n(1)} \xleftarrow{\varphi_1} X_{n(2)} \xleftarrow{\psi_1} Y_{n(3)} \xleftarrow{\varphi_2} X_{n(4)} \xleftarrow{\psi_2} \cdots,$$

whose limit is homeomorphic to the inverse limits of the upper and lower sequences in the above diagram.

Restricting the natural homeomorphism from $\prod_{i\in\mathbb{N}} X_i \times \prod_{i\in\mathbb{N}} Y_i$ onto $\prod_{i\in\mathbb{N}} (X_i \times Y_i)$, we can state the following:

Proposition 4.17.6 *For inverse sequences* $(X_i, f_i)_{i\in\mathbb{N}}$ *and* $(Y_i, g_i)_{i\in\mathbb{N}}$, *the product* $\varprojlim(X_i, f_i) \times \varprojlim(Y_i, g_i)$ *is homeomorphic to the inverse limit* $\varprojlim(X_i \times Y_i, f_i \times g_i)$. □

In Theorem 4.16.8, we saw that every paracompact space (hence every metrizable space) is represented as the limit of an inverse system of polyhedra (with the weak topology). However, that system seems to be complicated and difficult to handle. For the remainder of this section, we show that every completely metrizable space is represented as the limit of an inverse *sequence* of metric polyhedra. This can be stated in the following theorem:

Theorem 4.17.7 *Every completely metrizable space* X *is homeomorphic to the limit* $\varprojlim(|K_i|_{\mathrm{m}}, f_i)$ *of an inverse sequence of metric polyhedra and maps such that each* K_i *is locally finite-dimensional,* $\operatorname{card} K_i \leqslant \aleph_0 w(X)$, *and each* $f_i : K_{i+1} \to K_i'$ *is simplicial for some admissible subdivision* K_i' *of* K_i.[38]

Proof Because this theorem is obvious if X is finite, we may assume that $w(X) \geqslant \aleph_0$. Let d be an admissible complete metric for X. By induction, we will construct a refining sequence of open covers of X:

$$\mathcal{U}_1 \succ \mathcal{U}_2^{\mathrm{cl}} \succ \mathcal{U}_2 \succ \mathcal{U}_3^{\mathrm{cl}} \succ \mathcal{U}_3 \succ \cdots$$

with admissible subdivisions K_i' of the nerves $K_i = N(\mathcal{U}_i)$ and simplicial maps $f_i : K_{i+1} \to K_i'$ satisfying the following conditions:

(1) $\mathcal{U}_i$ is locally finite;
(2) $K_i = N(\mathcal{U}_i)$ is locally finite-dimensional;
(3) $\operatorname{mesh}_d \mathcal{U}_i < 2^{-i}$;
(4) $\operatorname{cl} U \subset \varphi_i^{-1}(O_{K_i'}(f_i(U)))$ for each $U \in K_{i+1}^{(0)} = \mathcal{U}_{i+1}$;
(5) $\operatorname{mesh}_{\rho_j} f_{j,i}(\mathcal{S}_{K_i}) < 2^{-(i-j)}$ for $j < i$,

where $\varphi_i : X \to |K_i|_{\mathrm{m}}$ is a canonical map for $\mathcal{U}_i$ and $\rho_j = \rho_{K_j}$. By (1) and (2), we can obtain $\operatorname{card} K_i \leqslant \aleph_0 w(X) = w(X)$.

First, by Theorem 4.15.11, we have $\mathcal{U}_1 \in \operatorname{cov}(X)$ satisfying (1)–(3). Next, assume that $\mathcal{U}_1, \ldots, \mathcal{U}_i$ and $f_1, \ldots, f_{i-1}$ have been defined so as to satisfy (1)–(5). By Theorem 4.9.5, we can find an admissible subdivision K_i' of K_i such that

$$\operatorname{mesh}_{\rho_i} \mathcal{S}_{K_i'} < 1/2 \quad \text{and} \quad \operatorname{mesh}_{\rho_j} f_{j,i}(\mathcal{S}_{K_i'}) < 2^{-(i+1-j)} \quad \text{for all } j < i.$$

Note that $O_{K_i'}$ is an open cover of $|K_i'|_{\mathrm{m}} = |K_i|_{\mathrm{m}}$ (cf. Lemma 4.7.3). Let $\varphi_i : X \to |K_i|_{\mathrm{m}}$ be a canonical map. Then,

$$\varphi_i^{-1}(O_{K_i'}) \in \operatorname{cov}(X) \quad \text{and} \quad \varphi_i^{-1}(O_{K_i'}) \prec \varphi_i^{-1}(O_{K_i}) \prec \mathcal{U}_i.$$

[38] The admissibility of K_i' guarantees the continuity of $f_i : |K_{i+1}|_{\mathrm{m}} \to |K_i|_{\mathrm{m}}$. Such a map f_i is called an admissible PL map , which was defined before Theorem 4.11.12.

By the regularity of X and Theorem 4.15.11, we have $\mathcal{U}_{i+1} \in \mathrm{cov}(X)$ satisfying (1)–(3) and $\mathcal{U}_{i+1}^{\mathrm{cl}} \prec \varphi_i^{-1}(\mathcal{O}_{K_i'})$. Then, there is a simplicial map $f_i : K_{i+1} \to K_i'$ satisfying (4), that is,

$$\mathrm{cl}\, U \subset \varphi_i^{-1}(O_{K_i'}(f_i(U))) \quad \text{for each } U \in K_{i+1}^{(0)} = \mathcal{U}_{i+1}.$$

Indeed, for each $\langle U_1, \dots, U_k \rangle \in K_{i+1}$, $\bigcap_{j=1}^k \varphi_i^{-1}(O_{K_i'}(f_i(U_j))) \neq \emptyset$, hence $\bigcap_{j=1}^k O_{K_i'}(f_i(U_j)) \neq \emptyset$, which implies that $\langle f_i(U_1), \dots, f_i(U_k) \rangle \in K_i'$. Since $f_i(\mathcal{S}_{K_{i+1}}) \prec \mathcal{S}_{K_i'}$, (5) holds for $j < i+1$, that is,

$$\mathrm{mesh}_{\rho_j}\, f_{j,i+1}(\mathcal{S}_{K_{i+1}}) < 2^{-(i+1-j)} \quad \text{for each } j < i+1.$$

We now construct a homeomorphism $h : X \to \varprojlim(|K_i|_{\mathrm{m}}, f_i)$. For each $x \in X$ and $i \in \mathbb{N}$, let $\tau_i(x) = \langle \mathcal{U}_i[x] \rangle \in K_i$. We will show that $f_i(\tau_{i+1}(x)) \subset \tau_i(x)$. Let $\sigma \in K_i$ be the smallest simplex containing $f_i(\tau_{i+1}(x)) \in K_i'$. Then,

$$\bigcup_{V \in \mathcal{U}_{i+1}[x]} c_{K_i}(f_i(V))^{(0)} = \sigma^{(0)}.$$

Indeed, for each $V \in \mathcal{U}_{i+1}[x] = \tau_{i+1}(x)^{(0)}$, we have $c_{K_i}(f_i(V)) \leqslant \sigma$, that is, $c_{K_i}(f_i(V))^{(0)} \subset \sigma^{(0)}$. If $\bigcup_{V \in \mathcal{U}_{i+1}[x]} c_{K_i}(f_i(V))^{(0)} \subsetneqq \sigma^{(0)}$ then $\tau_{i+1}(x) = \langle \mathcal{U}_{i+1}[x] \rangle$ is contained in a proper face of σ, which contradicts the minimality of σ.

For each $V \in \tau_{i+1}(x)^{(0)} = \mathcal{U}_{i+1}[x]$ and $U \in c_{K_i}(f_i(V))^{(0)}$, we have $O_{K_i'}(f_i(V)) \subset O_{K_i}(U)$ by Proposition 4.3.3. Thus,

$$\bigcap_{V \in \mathcal{U}_{i+1}[x]} O_{K_i'}(f_i(V)) \subset \bigcap_{U \in \sigma^{(0)}} O_{K_i}(U).$$

Then, it follows that

$$\begin{aligned} x \in \bigcap_{V \in \mathcal{U}_{i+1}[x]} V &\subset \bigcap_{V \in \mathcal{U}_{i+1}[x]} \varphi_i^{-1}\big(O_{K_i'}(f_i(V))\big) \\ &\subset \bigcap_{U \in \sigma^{(0)}} \varphi_i^{-1}\big(O_{K_i}(U)\big) \subset \bigcap_{U \in \sigma^{(0)}} U. \end{aligned}$$

Hence, $\sigma^{(0)} \subset \mathcal{U}_i[x]$, that is, $\sigma \leqslant \tau_i(x)$, so $f_i(\tau_{i+1}(x)) \subset \tau_i(x)$. Now, we have the following nested sequence of compacta:

$$\tau_i(x) = f_{i,i}(\tau_i(x)) \supset f_{i,i+1}(\tau_{i+1}(x)) \supset f_{i,i+2}(\tau_{i+2}(x)) \supset \cdots.$$

Since $\bigcap_{j \geqslant i} f_{i,j}(\tau_j(x)) \neq \emptyset$ and $\lim_{k \to \infty} \mathrm{diam}_{\rho_i}\, f_{i,i+k}(\tau_{i+k}(x)) = 0$ by (5), it follows that $\bigcap_{j \geqslant i} f_{i,j}(\tau_j(x))$ is a singleton. Thus, we have $h_i : X \to |K_i|$ such that

$$\{h_i(x)\} = \bigcap_{j \geqslant i} f_{i,j}(\tau_j(x)) \text{ for each } x \in X.$$

Observe that $f_i h_{i+1} = h_i$. Thus, we have $h : X \to \varprojlim(|K_i|_{\mathrm{m}}, f_i)$ such that $p_i h = h_i$ for each $i \in \mathbb{N}$, where $p_i : \varprojlim(|K_i|_{\mathrm{m}}, f_i) \to |K_i|_{\mathrm{m}}$ is the projection.

We now verify the continuity of h. For each $i \in \mathbb{N}$, $x \in X$, and $\varepsilon > 0$, we can choose $j > i$ so that $\mathrm{mesh}_{\rho_i} f_{i,j}(\mathcal{S}_{K_j}) < \varepsilon$ by (5). Then, $\bigcap \mathcal{U}_j[x]$ is an open neighborhood of x in X. Since $y \in \bigcap \mathcal{U}_j[x]$ implies $\tau_j(x) \leqslant \tau_j(y)$, we have $h_j(y) \in \tau_j(y) \in \mathrm{St}(\tau_j(x), K_j)$, hence

$$h_i(\textstyle\bigcap \mathcal{U}_j[x]) = f_{i,j} h_j(\bigcap \mathcal{U}_j[x]) \subset f_{i,j}(|\,\mathrm{St}(\tau_j(x), K_j)|).$$

Then, it follows that

$$\mathrm{diam}_{\rho_i} h_i(\textstyle\bigcap \mathcal{U}_j[x]) \leqslant \mathrm{diam}_{\rho_i} f_{i,j}(|\,\mathrm{St}(\tau_j(x), K_j)|) \leqslant \mathrm{mesh}_{\rho_i} f_{i,j}(\mathcal{S}_{K_j}) < \varepsilon.$$

Thus, each $h_i : X \to |K_i|_{\mathrm{m}}$ is continuous, so h is continuous.

Next, we show that h is injective. For $x \neq x' \in X$, we can choose $i \in \mathbb{N}$ so that $\mathrm{mesh}_d \mathcal{U}_i < d(x, x')$ by (3). Then, it follows that

$$\tau_i(x)^{(0)} \cap \tau_i(x')^{(0)} = \mathcal{U}_i[x] \cap \mathcal{U}_i[x'] = \emptyset,$$

which means $\tau_i(x) \cap \tau_i(x') = \emptyset$, hence $h_i(x) \neq h_i(x')$, so $h(x) \neq h(x')$. Thus, h is injective.

To prove that h is surjective, let $y \in \varprojlim(|K_i|_{\mathrm{m}}, f_i)$. For each $i \in \mathbb{N}$, let $\tau_i = c_{K_i}(p_i(y)) \in K_i$ be the carrier of $p_i(y)$. Then, we have an open set $W_i = \bigcup_{U \in \tau_i^{(0)}} U \subset X$. Since $\bigcap_{U \in \tau_i^{(0)}} U \neq \emptyset$, it follows that $\mathrm{diam}_d W_i \leqslant 2\,\mathrm{mesh}_d \mathcal{U}_i$, hence $\lim_{i \to \infty} \mathrm{diam}_d W_i = 0$ by (3). For each $U \in \tau_{i+1}^{(0)}$, choose $V \in \tau_i^{(0)}$ so that $f_i(U) \in O_{K_i}(V)$. Since $f_i(U) \in K_i'^{(0)}$ and $K_i' \lhd K_i$, it follows that $O_{K_i'}(f_i(U)) \subset O_{K_i}(V)$, hence by (4),

$$\mathrm{cl}\, U \subset \varphi_i^{-1}(O_{K_i'}(f_i(U))) \subset \varphi_i^{-1}(O_{K_i}(V)) \subset V.$$

Therefore, $\mathrm{cl}\, W_{i+1} \subset W_i$. By the completeness of X, we have $z \in X$ such that $\{z\} = \bigcap_{i \in \mathbb{N}} \mathrm{cl}\, W_i = \bigcap_{i \in \mathbb{N}} W_i$. For each $i \in \mathbb{N}$, z is contained in some $U \in \tau_i^{(0)}$, so $\tau_i^{(0)} \cap \tau_i(z)^{(0)} \neq \emptyset$, that is, $\tau_i \cap \tau_i(z) \neq \emptyset$. It follows from the definition of h_i that $h_i(z) = p_i(y)$ for each $i \in \mathbb{N}$, which means $h(z) = y$.

To conclude that h is a homeomorphism, it remains to show that h is an open map. Let V be a neighborhood of x in X. By (3), $\mathrm{st}^2(x, \mathcal{U}_i) \subset V$ for some $i \in \mathbb{N}$. Since $h_i(x) \in \tau_i(x)$, we have $U \in \tau_i(x)^{(0)} = \mathcal{U}_i[x]$ such that $h_i(x) \in O_{K_i}(U)$, which means $h(x) \in p_i^{-1}(O_{K_i}(U))$. For each $y \in p_i^{-1}(O_{K_i}(U))$, let $z = h^{-1}(y)$. Then, U is a vertex of the carrier $\tau_i \in K_i$ of $p_i(y)$. It follows from the above argument that $z \in \mathrm{st}(U, \mathcal{U}_i) \subset V$, so $y = h(z) \in h(V)$. Thus, we have $p_i^{-1}(O_{K_i}(U)) \subset h(V)$. Hence, $h(V)$ is a neighborhood of $h(x)$ in $\varprojlim(|K_i|_{\mathrm{m}}, f_i)$. This shows that h is open. □

In the above proof, if X is compact, then each $\mathcal{U}_i$ can be finite, so each K_i is finite. When X is separable and completely metrizable, each $\mathcal{U}_i$ can be countable and star-finite by virtue of Theorem 4.15.13, which means that each K_i is locally finite. Thus, we have the following:

Corollary 4.17.8 *Every compactum (resp. separable completely metrizable space) is homeomorphic to the inverse limit of an inverse sequence of compact (resp. separable locally compact) polyhedra and PL maps.* □

Note The compact case of the above Corollary 4.17.8 can be easily proved as follows: By virtue of Corollary 2.3.9, we may assume that X is a closed set in the Hilbert cube $\mathbf{I}^{\mathbb{N}}$. For each $i \in \mathbb{N}$, let $p_i : \mathbf{I}^{\mathbb{N}} \to \mathbf{I}^i$ be the projection defined by $p_i(x) = (x(1), \ldots, x(i))$. Construct compact polyhedra $P_i \subset \mathbf{I}^i$ so that $p_i(X) \subset \mathrm{int}_{\mathbf{I}^i} P_i$, $P_{i+1} \subset P_i \times \mathbf{I}$ (hence $p_{i+1}^{-1}(P_{i+1}) \subset p_i^{-1}(P_i)$) and $\bigcap_{i\in\mathbb{N}} p_i^{-1}(P_i) = X$. For each $i \in \mathbb{N}$, let $f_i : P_{i+1} \to P_i$ be the restriction of the projection of $P_i \times \mathbf{I}$ onto P_i. Then, $X = \bigcap_{i\in\mathbb{N}} p_i^{-1}(P_i) \approx \varprojlim(P_i, f_i)$ by the homeomorphism induced by $p_i|X : X \to P_i$.

In the case that X is separable and locally compact metrizable, not only can each K_i be locally finite, but each f_i can also be proper. Indeed, we can take $\mathcal{U}_i$ such that each $U \in \mathcal{U}_i$ has the compact closure, hence $\mathcal{U}_i[U]$ is finite for each $U \in \mathcal{U}_i$, which means that $K_i = N(\mathcal{U}_i)$ is locally finite, hence $|K_i|_{\mathrm{m}} = |K_i|$ is locally compact. For each $U \in K_{i+1}^{(0)} = \mathcal{U}_{i+1}$ and $V \in K_i^{(0)} = \mathcal{U}_i$, if $f_i(U) \in O_{K_i}(V)$ then

$$\mathrm{cl}\, U \subset \varphi_i^{-1}(O_{K_i'}(f_i(U))) \subset \varphi_i^{-1}(O_{K_i}(V)) \subset V.$$

Since V contains only finitely many members of $\mathcal{U}_{i+1}$, $f_i^{-1}(f_i(U))$ contains only finitely many vertices of K_{i+1}. Hence, f_i is proper.

Corollary 4.17.9 *Every separable locally compact metrizable space is homeomorphic to the inverse limit of an inverse sequence of separable locally compact polyhedra and proper PL maps.* □

Remark 4.17.1 The following fact is useful: In the proof of Theorem 4.17.7, we needed the following condition that is derived from condition (5):

$(*)$ $\lim_{k\to\infty} \mathrm{mesh}_{\rho_i}\, f_{i,i+k}(\mathcal{S}_{K_{i+k}}) = 0.$

In Theorem 4.17.7, if $\dim K_i < \infty$ for each $i \in \mathbb{N}$, we can take $K_i' = \mathrm{Sd}^{n_i} K_i$ for some $n_i \in \mathbb{N}$. Indeed, choose $n_i \in \mathbb{N}$ so that

$$\lim_{i\to\infty} \left(\frac{\dim K_i}{\dim K_i + 1} \right)^{n_i} = 0.$$

Then, condition $(*)$ holds because the following inequality is obtained from Lemma 4.8.3:

$$\begin{aligned}\operatorname{mesh}_{\rho_i} f_{i,i+k}(\operatorname{Sd}^{n_{i+k}} K_{i+k}) &\leqslant \left(\frac{\dim K_{i+k}}{\dim K_{i+k}+1}\right)^{n_{i+k}} \operatorname{mesh}_{\rho_i} f_{i,i+k}(K_{i+k}) \\ &\leqslant \left(\frac{\dim K_{i+k}}{\dim K_{i+k}+1}\right)^{n_{i+k}} \operatorname{mesh}_{\rho_i} \operatorname{Sd}^{n_i} K_i \\ &\leqslant 2\left(\frac{\dim K_i}{\dim K_i+1}\right)^{n_i}\left(\frac{\dim K_{i+k}}{\dim K_{i+k}+1}\right)^{n_{i+k}}.\end{aligned}$$

When $\dim K_i \leqslant n$ for every $i \in \mathbb{N}$, we can take $K_i' = \operatorname{Sd} K_i$. Indeed, condition $(*)$ is satisfied because the following inequality follows from Lemma 4.8.3:

$$\begin{aligned}\operatorname{mesh}_{\rho_i} f_{i,i+k}(\operatorname{Sd} K_{i+k}) &\leqslant \frac{n}{n+1} \operatorname{mesh}_{\rho_i} f_{i,i+k}(K_{i+k}) \\ &\leqslant \frac{n}{n+1} \operatorname{mesh}_{\rho_i} f_{i,i+k-1}(\operatorname{Sd} K_{i+k-1}) \\ &\leqslant \left(\frac{n}{n+1}\right)^2 \operatorname{mesh}_{\rho_i} f_{i,i+k-1}(K_{i+k-1}) \\ &\leqslant \cdots \leqslant \left(\frac{n}{n+1}\right)^k \operatorname{mesh}_{\rho_i} \operatorname{Sd} K_i \leqslant 2\left(\frac{n}{n+1}\right)^{k+1}.\end{aligned}$$

Remark 4.17.2 In Theorem 4.17.7, unlike Theorem 4.16.8, each bonding map f_i cannot taken simplicial with respect to K_{i+1} and K_i even if X is a compactum. Such an example is a compactum X with $\dim X > 0$ which contains no arcs.[39] The pseudo-arc is such a compactum.[40] In fact, using a theorem (Theorem 5.4.2) presented in Chap. 5, we have the following theorem:

Theorem 4.17.10 (MARDEŠIĆ) *Let $X = \varprojlim(|K_i|_{\mathrm{m}}, f_i)$, where each K_i is a finite simplicial complex and each $f_i : K_{i+1} \to K_i$ is a simplicial map. If $\dim X > 0$, then X contains an arc.*

Proof For each $i \in \mathbb{N}$, let $p_i : X \to |K_i|$ be the inverse limit projection. Let $X_0 = \varprojlim(K_i^{(0)}, f_i|K_i^{(0)}) \subset X$. Since $\dim X_0 \leqslant 0$ (which will be shown in Theorem 5.4.2), we can take a point $x_0 \in X \setminus X_0$. Then, there exists $i_0 \in \mathbb{N}$ such that $p_j(x_0) \notin K_j^{(0)}$ for every $j \geqslant i_0$. Otherwise, for each $i \in \mathbb{N}$, there exists $j > i$ such that $p_j(x_0) \in K_j^{(0)}$. Then, $p_i(x_0) \in K_i^{(0)}$ because $f_{i,j} = f_i \cdots f_{j-1} : K_j \to K_i$ is simplicial and $p_i = f_{i,j} p_j$. Thus, we have $x_0 \in X_0$, which is a contradiction.

For each $i \geqslant i_0$, let $\sigma_i = c_{K_i}(p_i(x_0)) \in K_i$ be the carrier of $p_i(x_0)$ in K_i. Then, $\dim \sigma_i > 0$. Since $f_i(\sigma_{i+1}) \in K_i$ and $p_i(x_0) = f_i p_{i+1}(x_0) \in f_i(\operatorname{rint} \sigma_{i+1}) = \operatorname{rint} f_i(\sigma_{i+1})$, it follows that $f_i(\sigma_{i+1}) = \sigma_i$. Let τ_{i_0} be a 1-face of σ_{i_0}. For each $i \geqslant i_0$, we can inductively choose 1-face τ_i of σ_i so that $f_i(\tau_{i+1}) = \tau_i$. Thus, we can obtain an inverse sequence $(\tau_i, f_i|\tau_{i+1})_{i \geqslant i_0}$. Then, $X_1 = \varprojlim(\tau_i, f_i|\tau_{i+1})_{i \geqslant i_0}$ is a subspace of X by Proposition 4.16.5(1). Since each $f_i|\tau_{i+1}$ is an affine homeomorphism, we have

[39] For the definition of dimension of a space, refer to the preface to Chap. 5.

[40] The **pseudo-arc** is the topologically unique non-degenerate hereditarily indecomposable arc-like continuum. Refer to Notes for this chapter.

affine homeomorphisms $h_i : \mathbf{I} \to \tau_i$, $i \geqslant i_0$, such that $h_i = (f_i|\tau_{i+1})h_{i+1}$. These h_i, $i \geqslant i_0$, define a homeomorphism $h : \mathbf{I} \to X_1$, which means that X_1 is an arc. □

4.18 The Mapping Cylinders

The **mapping cylinder** M_f of a map $f : X \to Y$ is defined as the following adjunction space:

$$M_f = Y \cup_{f \circ \mathrm{pr}_X | X \times \{0\}} X \times \mathbf{I},$$

that is, M_f is obtained from the topological sum $Y \oplus (X \times \mathbf{I})$ by identifying the points $(x, 0) \in X \times \{0\}$ with the points $f(x) \in Y$, where M_f is Hausdorff.

Indeed, $Y \cup X \times (0, t/2)$ and $X \times (t/2, 1]$ are disjoint open neighborhoods of $y \in Y$ and $(x, t) \in X \times (0, 1]$ in M_f, respectively. Every pair of distinct points of $X \times (0, 1]$ has disjoint open neighborhoods in $X \times (0, 1]$, which are open in M_f. Every pair of distinct points of Y has disjoint open neighborhoods V and V' in Y. Then, the following are disjoint open neighborhoods of y and y' in M_f:

$$V \cup (f^{-1}(V) \times (0, 1]), \ V' \cup (f^{-1}(V') \times (0, 1]).$$

The identification map (the quotient map) is denoted by $q_f : Y \oplus (X \times \mathbf{I}) \to M_f$. The **collapsing** of M_f is the map $c_f : M_f \to Y$ defined by

$$c_f|Y = \mathrm{id} \ \text{ and } \ c_f|X \times (0, 1] = f\mathrm{pr}_X.$$

Then, c_f is a retraction and Y is a retract of M_f. Let $i_f : X \to M_f$ denote the natural embedding defined by $i_f(x) = (x, 1)$. Then,

$$c_f i_f = f \ \text{ and } \ c_f \simeq \mathrm{id} \ \text{ rel. } Y \text{ in } M_f,$$

which is realized by the homotopy $h^f : M_f \times \mathbf{I} \to M_f$ defined by $h^f_0 = c_f$, $h^f_t|Y = \mathrm{id}$ for each $t \in \mathbf{I}$, and

$$h^f_t(x, s) = (x, st) \ \text{ for each } (x, s) \in X \times (0, 1] \text{ and } t \in (0, 1].$$

Thus, c_f is a strong deformation retraction and Y is a strong deformation retract of M_f.[41] It should be noted that $h^f(c_f^{-1}(y) \times \mathbf{I}) = c_f^{-1}(y)$ for every $y \in Y$.

For each proper map $f : X \to Y$, the collapsing $c_f : M_f \to Y$ and the homotopy $h^f : M_f \times \mathbf{I} \to M_f$ are proper, which means that $c_f \simeq^p \mathrm{id}$ rel. Y in M_f.

Indeed, for each compact set $A \subset Y$, since $f^{-1}(A)$ is compact, the following is compact:

$$c_f^{-1}(A) = A \cup q_f(f^{-1}(A) \times \mathbf{I}).$$

[41] The definition was given just before Proposition 4.12.1 (see Footnote 26).

For each $(x, u) \in X \times (0, 1]$,

$$(h^f)^{-1}(x, u) = \{(x, s, t) \mid st = u\} \subset \{x\} \times [u, 1]^2$$
$$\subset \big(\{f(x)\} \cup (\{x\} \times (0, 1])\big) \times \mathbf{I} \subset c_f^{-1}(f(x)) \times \mathbf{I} = c_f^{-1}(c_f(x, u)) \times \mathbf{I}.$$

Hence, for each compact set $C \subset M_f$, $(h^f)^{-1}(C) \subset c_f^{-1}(c_f(C)) \times \mathbf{I}$. Thus, $(h^f)^{-1}(C)$ is compact because so is $c_f^{-1}(c_f(C)) \times \mathbf{I}$.

When X and Y are metrizable, the mapping cylinder M_f of a proper (perfect) map $f : X \to Y$ is metrizable because it is a perfect image of the metrizable space $Y \oplus (X \times \mathbf{I})$ (Proposition 2.4.4 (1)).

When Y is a singleton, the mapping cylinder M_f is homeomorphic to the quotient space $(X \times \mathbf{I})/(X \times \{0\})$, which is called the **cone** over X. We regard the mapping cylinder M_{id_X} of the identity map id_X as the product space $X \times \mathbf{I}$.

If X is a *closed subspace* of Y, the mapping cylinder M_i of the inclusion $i : X \subset Y$ can be regarded as the subspace $(Y \times \{0\}) \cup (X \times \mathbf{I})$ of the product space $Y \times \mathbf{I}$. However, if Y is perfectly normal and X is not closed in Y then the mapping cylinder M_i of the inclusion $i : X \subset Y$ cannot be regarded as the subspace of $Y \times \mathbf{I}$. Indeed, let $y \in (\mathrm{cl}_Y X) \setminus X$. Then, we have a map $k : Y \to \mathbf{I}$ such that $\{y\} = k^{-1}(0)$. Observe that the graph $G = \{(x, k(x)) \mid x \in X\}$ of the map $k|X$ is closed in M_i but is not closed in the subspace $(Y \times \{0\}) \cup (X \times \mathbf{I})$ of $Y \times \mathbf{I}$.

For spaces X and Y with $A \subset X \cap Y$, when there exist homotopy equivalences $f : X \to Y$ and $g : Y \to X$ such that $f|A = g|A = \mathrm{id}$, $gf \simeq \mathrm{id}_X$ rel. A, and $fg \simeq \mathrm{id}_Y$ rel. A, it is said that X and Y are **homotopy equivalent relative to** A (or X is **homotopy equivalent** to Y **relative to** A), denoted as follows:

$$X \simeq Y \ \text{ rel. } A.$$

In the above, when f and g are proper, $gf \simeq^p \mathrm{id}_X$ rel. A, and $fg \simeq^p \mathrm{id}_Y$ rel. A, it is said that X and Y are **proper homotopy equivalent relative to** A (or X is **proper homotopy equivalent** to Y **relative to** A), denoted as follows:

$$X \simeq^p Y \ \text{ rel. } A.$$

As is easily observed, the relation $\simeq$ rel. A ($\simeq^p$ rel. A) is an equivalence relation on the spaces containing A as a subspace.

Using the mapping cylinders M_f and M_g, we can characterize $f \simeq g$ as follows:

Theorem 4.18.1 *For maps $f, g : X \to Y$, the following are equivalent:*

(a) $f \simeq g$;
(b) $M_f \simeq M_g$ rel. $Y \cup (X \times \{1\})$;
(c) *There is a map $\varphi : M_f \to M_g$ such that $\varphi|Y \cup (X \times \{1\}) = \mathrm{id}$.*

Proof The implication (b) $\Rightarrow$ (c) is obvious. Using the above homotopy h^f of M_f, we can see (c) $\Rightarrow$ (a) as follows:

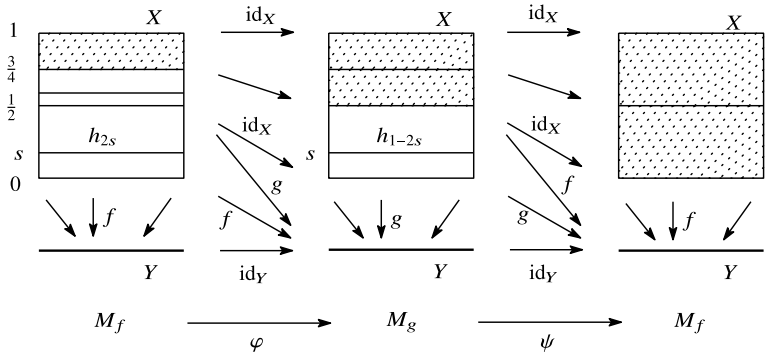


Fig. 4.17 The mapping cylinders of homotopic maps

$$f = c_f \circ i_f = c_g \circ \varphi \circ h_0^f \circ i_f \simeq c_g \circ \varphi \circ h_1^f \circ i_f = c_g \circ \varphi \circ i_f = c_g \circ i_g = g.$$

(a) $\Rightarrow$ (b): Let $h : X \times \mathbf{I} \to Y$ be a homotopy from f to g. We define maps $\varphi : M_f \to M_g$ and $\psi : M_g \to M_f$ as follows: $\varphi|Y = \psi|Y = \mathrm{id}$ and

$$\varphi(x,s) = \begin{cases} (x, 2s-1) & \text{for } (x,s) \in X \times (1/2, 1], \\ h_{2s}(x) & \text{for } (x,s) \in X \times (0, 1/2]; \end{cases}$$

$$\psi(x,s) = \begin{cases} (x, 2s-1) & \text{for } (x,s) \in X \times (1/2, 1], \\ h_{1-2s}(x) & \text{for } (x,s) \in X \times (0, 1/2]. \end{cases}$$

Observe that $\psi\varphi|Y = \mathrm{id}$ and

$$\psi\varphi(x,s) = \begin{cases} (x, 4s-3) & \text{for } (x,s) \in X \times (3/4, 1], \\ h_{3-4s}(x) & \text{for } (x,s) \in X \times (1/2, 3/4], \\ h_{2s}(x) & \text{for } (x,s) \in X \times (0, 1/2]. \end{cases}$$

See Fig. 4.17.

We define an auxiliary map $\theta : M_f \to M_f$ as follows: $\theta|Y = \mathrm{id}$ and

$$\theta(x,s) = \begin{cases} (x, 4s-3) & \text{for } (x,s) \in X \times (3/4, 1], \\ f(x) & \text{for } (x,s) \in X \times (0, 3/4]. \end{cases}$$

Then $\psi\varphi \simeq \theta \simeq \mathrm{id}_{M_f}$ rel. $Y \cup (X \times \{1\})$ by the homotopies $\xi : M_f \times \mathbf{I} \to M_f$ and $\zeta : M_f \times \mathbf{I} \to M_f$ defined as follows: $\xi_t|Y = \zeta_t|Y = \mathrm{id}$ and

$$\xi_t(x,s) = \begin{cases} (x, 4s-3) & \text{for } (x,s) \in X \times (3/4, 1], \\ h_{(3-4s)t}(x) & \text{for } (x,s) \in X \times (1/2, 3/4], \\ h_{2st}(x) & \text{for } (x,s) \in X \times (0, 1/2]; \end{cases}$$

$$\zeta_t(x,s) = \begin{cases} \left(x, \dfrac{4s-3t}{4-3t}\right) & \text{for } (x,s) \in X \times (3t/4, 1], \\ f(x) & \text{for } (x,s) \in X \times (0, 3t/4]. \end{cases}$$

Similarly, we can see that $\varphi\psi \simeq \mathrm{id}$ rel. $Y \cup (X \times \{1\})$. □

Transcribing the above proof for proper maps f and g, we have the following characterization of $f \simeq^p g$, that is, the proper version of Theorem 4.18.1:

Theorem 4.18.2 *For proper maps $f, g : X \to Y$, the following are equivalent:*

(a) $f \simeq^p g$;
(b) $M_f \simeq^p M_g$ rel. $Y \cup (X \times \{1\})$;
(c) *There is a proper map $\varphi : M_f \to M_g$ such that $\varphi | Y \cup (X \times \{1\}) = \mathrm{id}$.*

Sketch of Proof The implication (b) ⇒ (c) is obvious. In the proof (c) ⇒(a), since c_g, φ, and hf are proper, it follows that $c_g \varphi h f (i_f \times \mathrm{id}_{\mathbf{I}})$ is a proper homotopy. In the proof (a) ⇒ (b), h is a proper homotopy. Then, show that φ, ψ, and θ are proper maps and ξ and ζ are proper homotopies.

Concerning the composition of maps, we have the following:

Theorem 4.18.3 *For maps $f : X \to Y$ and $g : Y \to Z$,*

$$M_g \cup_{i_g} M_f \simeq M_{gf} \quad \textit{rel. } Z \cup (X \times \{1\}).$$

Proof We define maps $\varphi : M_g \cup_{i_g} M_f \to M_{gf}$ and $\psi : M_{gf} \to M_g \cup_{i_g} M_f$ as follows: $\varphi | M_g = c_g$, $\varphi | X \times (0, 1] = \mathrm{id}$ and $\psi | Z = \mathrm{id}$,

$$\psi(x,s) = \begin{cases} (x, 2s-1) \in M_f & \text{for } (x,s) \in X \times (1/2, 1], \\ (f(x), 2s) \in M_g & \text{for } (x,s) \in X \times (0, 1/2]. \end{cases}$$

Observe that $\psi\varphi | M_g = c_g$, $\psi\varphi | X \times (0, 1] = \psi | X \times (0, 1]$, and $\varphi\psi | Z = \mathrm{id}$,

$$\varphi\psi(x,s) = \begin{cases} (x, 2s-1) & \text{for } (x,s) \in X \times (1/2, 1], \\ gf(x) & \text{for } (x,s) \in X \times (0, 1/2]. \end{cases}$$

See Figs. 4.18 and 4.19.

It is easy to show that

$$\psi\varphi \simeq \mathrm{id} \ \text{ rel. } Z \cup (X \times \{1\}) \ \text{ and } \ \varphi\psi \simeq \mathrm{id} \ \text{ rel. } Z \cup (X \times \{1\}).$$

In fact, these are realized by the homotopies

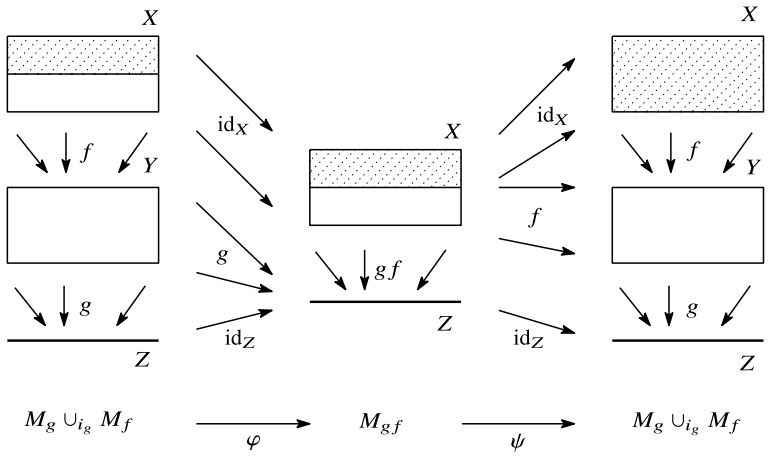


Fig. 4.18 The mapping cylinder of the composition of maps (1)

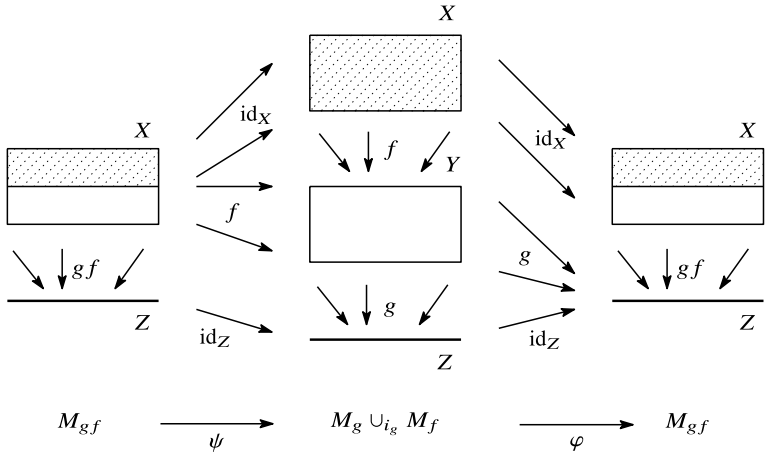


Fig. 4.19 The mapping cylinder of the composition of maps (2)

$$\xi : (M_g \cup_{i_g} M_f) \times \mathbf{I} \to M_g \cup_{i_g} M_f \quad \text{and} \quad \zeta : M_{gf} \times \mathbf{I} \to M_{gf}$$

defined as follows: $\xi_t | M_g = h^g_{1-t}$, $\zeta_t | Z = \mathrm{id}$, and

$$\xi_t(x,s) = \begin{cases} \left(x, \dfrac{2s-t}{2-t}\right) & \text{for } (x,s) \in X \times (t/2, 1], \\ (f(x), 2s+1-t) & \text{for } (x,s) \in X \times (0, t/2]; \end{cases}$$

$$\zeta_t(x,s) = \begin{cases} \left(x, \dfrac{2s-t}{2-t}\right) & \text{for } (x,s) \in X \times (t/2, 1], \\ gf(x) & \text{for } (x,s) \in X \times (0, t/2]. \end{cases}$$

Thus, we have the result. □

It is easy to obtain the following proper version of Theorem 4.18.3 by transcribing the above proof:

Theorem 4.18.4 *For proper maps $f : X \to Y$ and $g : Y \to Z$,*

$$M_g \cup_{i_g} M_f \simeq^p M_{gf} \text{ rel. } Z \cup (X \times \{1\}).$$

Sketch of Proof In the above proof, assuming that f and g are proper, show that φ and ψ are proper maps and that ξ and ζ are proper homotopies.

Let K and L be simplicial complexes in the linear spaces E and F, respectively. Consider the linear product space $E \times F \times \mathbb{R}$ and regard K and L as being contained in $E \times \{\mathbf{0}\} \times \{1\}$ and $\{\mathbf{0}\} \times F \times \{0\}$, respectively. Assume that K is an *ordered simplicial complex*. The **simplicial mapping cylinder** Z_f of a simplicial map $f : K \to L$ is defined as the following simplicial complex:

$$Z_f = K \cup L \cup \big\{ \langle f(v_1), \cdots, f(v_{m'}), v_m, \cdots, v_n \rangle \;\big|\; \langle v_1, \cdots, v_n \rangle \in K,\ v_1 < \cdots < v_n,\ 1 \leqslant m' \leqslant m \leqslant n \big\}.$$

The **collapsing** of Z_f is the simplicial map $\bar{c}_f : Z_f \to L$ defined by $\bar{c}_f(v) = f(v)$ for each $v \in K^{(0)}$ and $\bar{c}_f(u) = u$ for each $u \in L^{(0)}$. Then, $f = \bar{c}_f | |K|$. If $L = \{v\}$ is a singleton, Z_f is the simplicial cone $v * K$ over K, which was defined just before Proposition 4.2.7.

Remark 4.18.1 Note that the simplicial mapping cylinder changes if the order on $K^{(0)}$ is changed. Moreover, in general, $|Z_f| \not\approx M_f$. For instance, let K and L be the standard triangulations of a 2-simplex $\langle v_1, v_2, v_3 \rangle$ and a 1-simplex $\langle u_1, u_2 \rangle$, respectively. Consider the simplicial map $f : K \to L$ defined by $f(v_1) = f(v_2) = u_1$ and $f(v_3) = u_2$. Let Z_f be the simplicial mapping cylinder defined by the order $v_1 < v_2 < v_3$ and Z'_f be the one defined by the order $v_3 <' v_2 <' v_1$. Then, we have not only $Z_f \not\equiv Z'_f$ but also $|Z_f| \not\approx |Z'_f|$ (Fig. 4.20). Evidently, $|Z_f| \not\approx M_f$.

Because K is assumed to be an ordered simplicial complex, we can consider the product simplicial complex $K \times_s I$ of K and the ordered simplicial complex $I = \{\mathbf{I}, 0, 1\}$ with the natural order, i.e.,

$$K \times_s I = K \times_c \{0, 1\} \cup \big\{ \langle (v_1, 0), \cdots, (v_{m'}, 0), (v_m, 1), \cdots, (v_n, 1) \rangle \;\big|\; \langle v_1, \cdots, v_n \rangle \in K,\ v_1 < \cdots < v_n,\ 1 \leqslant m' \leqslant m \leqslant n \big\},$$

where $K \times_c \{0, 1\} = \{\sigma \times \{i\} \mid \sigma \in K, \quad i = 0, 1\}$. Then, $|K \times_s I| = |K| \times \mathbf{I}$ according to Theorem 4.5.1. We define the simplicial map $\bar{q}_f : L \oplus (K \times_s I) \to Z_f$ by $\bar{q}_f | L^{(0)} = \mathrm{id}$,

$$\bar{q}_f(v, 0) = f(v) \text{ and } \bar{q}_f(v, 1) = v \text{ for each } v \in K^{(0)}.$$

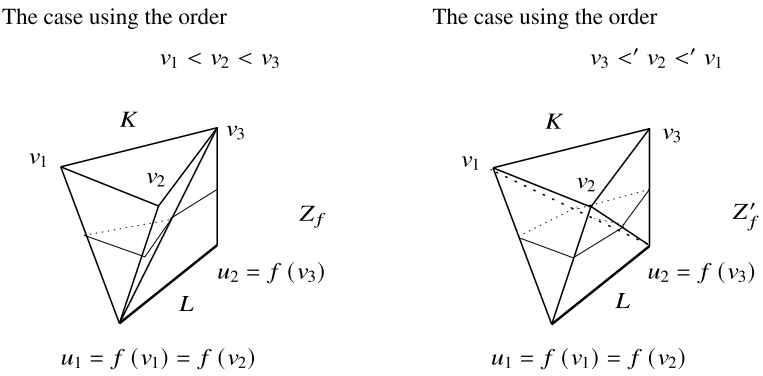


Fig. 4.20 The simplicial mapping cylinders

Then, $\bar{q}_f : |L| \oplus (|K| \times \mathbf{I}) \to |Z_f|$ is a quotient map. Indeed, let $A \subset |Z_f|$ and assume that $\bar{q}_f^{-1}(A)$ is closed in $|L| \oplus (|K| \times \mathbf{I})$. Observe that

$$Z_f = L \cup \left\{ \bar{q}_f(\tau) \;\middle|\; \tau \in K \times_s I \right\}.$$

For each $\sigma \in L$, $A \cap \sigma = \bar{q}_f^{-1}(A) \cap \sigma$ is closed in σ. For each $\tau \in K \times_s I$, since $\bar{q}_f|\tau$ is a quotient map and $(\bar{q}_f|\tau)^{-1}(A \cap \bar{q}_f(\tau)) = \bar{q}_f^{-1}(A) \cap \tau$ is closed in τ, it follows that $A \cap \bar{q}_f(\tau)$ is closed in $\bar{q}_f(\tau)$. As a consequence, A is closed in $|Z_f|$. Since $\bar{q}_f||L| = \mathrm{id}$ and $\bar{q}_f||K| \times \{0\} = f \circ \mathrm{pr}_{|K|}$, we have the map $p_f : M_f \to |Z_f|$ such that $p_f \circ q_f = \bar{q}_f$. Then, p_f is a quotient map. See Fig. 4.21.

For each $(x, s) \in |K| \times \mathbf{I}$, let $c_K(x) = \langle v_1, \cdots, v_n \rangle \in K$ be the carrier of x and write $x = \sum_{i=1}^{n} t_i v_i$, where $v_1 < \cdots < v_n$ and $t_i = \beta_{v_i}^K(x) > 0$. Choose $m = 1, \cdots, n$ so that

$$\sum_{i=m+1}^{n} t_i \leqslant s \leqslant \sum_{i=m}^{n} t_i \quad \left(\text{i.e., } \sum_{i=1}^{m-1} t_i \leqslant 1 - s \leqslant \sum_{i=1}^{m} t_i \right),$$

where $\sum_{i=n+1}^{n} t_i = 0$. Then, it follows that

$$\begin{aligned}(x, s) = \sum_{i=1}^{m-1} t_i (v_i, 0) + \left(\sum_{i=m}^{n} t_i - s \right)(v_m, 0) \\ + \left(s - \sum_{i=m+1}^{n} t_i \right)(v_m, 1) + \sum_{i=m+1}^{n} t_i (v_i, 1) \\ \in \langle (v_1, 0), \cdots, (v_m, 0), (v_m, 1), \cdots, (v_n, 1) \rangle \in K \times_s I.\end{aligned}$$

By the definition of $\bar{q}_f$, we have

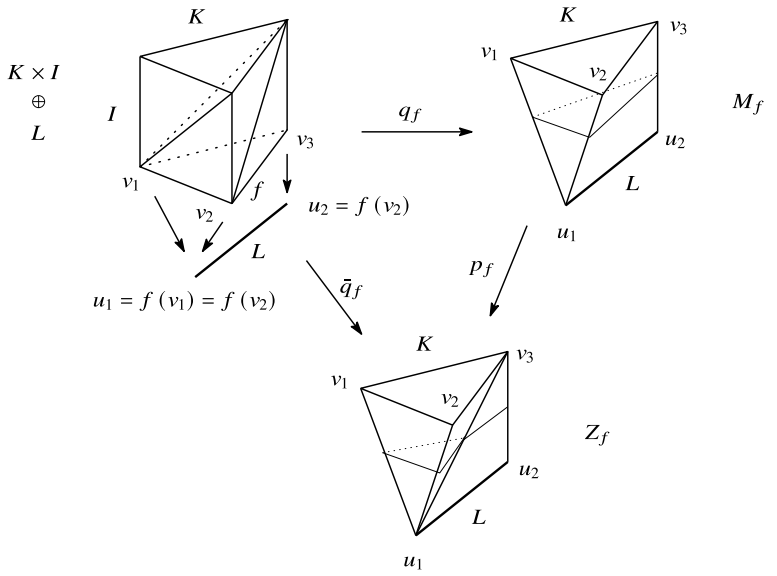


Fig. 4.21 The mapping cylinder and the simplicial mapping cylinder

$$\bar{q}_f(x,s) = \sum_{i=1}^{m-1} t_i f(v_i) + \left(\sum_{i=m}^{n} t_i - s\right) f(v_m) + \left(s - \sum_{i=m+1}^{n} t_i\right) v_m + \sum_{i=m+1}^{n} t_i v_i$$
$$\in \langle f(v_1), \cdots, f(v_m), v_m, \cdots, v_n \rangle \in Z_f.$$

If $\bar{q}_f(x,s) = \bar{q}_f(x',s')$, then

$$s = \sum_{v \in K^{(0)}} \beta_v^{Z_f}(\bar{q}_f(x,s)) = \sum_{v \in K^{(0)}} \beta_v^{Z_f}(\bar{q}_f(x',s')) = s'.$$

In addition, if $\bar{q}_f(x,s) = \bar{q}_f(x',s)$, then $\bar{q}_f(x,s') = \bar{q}_f(x',s')$ for every $s' \in [0,s]$ because

$$\sum_{i=m'+1}^{n} t_i \leqslant s' \leqslant \sum_{i=m'}^{n} t_i \quad \text{for } m' \geqslant m.$$

Therefore, we can define a homotopy $\bar{h}^f : |Z_f| \times \mathbf{I} \to |Z_f|$ by $\bar{h}_t^f || L| = \mathrm{id}$ and $\bar{h}_t^f(\bar{q}_f(x,s)) = \bar{q}_f(x,st)$ on $|Z_f| \setminus |L| \subset \bar{q}_f(|K| \times \mathbf{I})$. Thus, we have

$$\bar{c}_f = \bar{h}_0^f \simeq \bar{h}_1^f = \mathrm{id}_{|Z_f|} \text{ rel. } |L|,$$

where $\bar{h}^f(\bar{c}_f^{-1}(y) \times \mathbf{I}) = \bar{c}_f^{-1}(y)$ for every $y \in |L|$. Moreover, the following diagram commutes:

$$\begin{array}{ccc} M_f \times \mathbf{I} & \xrightarrow{h^f} & M_f \\ {\scriptstyle p_f \times \mathrm{id}}\downarrow & & \downarrow{\scriptstyle p_f} \\ |Z_f| \times \mathbf{I} & \xrightarrow[\bar{h}^f]{} & |Z_f|. \end{array}$$

For each ordered simplicial complex K and its simplicial subdivision K', we define the simplicial subdivision $I(K', K)$ of the product simplicial complex $K \times_s I$ as follows:

$$\begin{aligned} I(K', K) \underset{\text{def}}{=} & (K' \times \{0\}) \cup (K \times \{1\}) \\ & \cup \Big\{ \langle \sigma \times \{0\} \cup \{v_m, \cdots, v_n\} \times \{1\} \rangle \;\Big|\; \sigma \in K',\ 1 \leqslant m \leqslant n, \\ & \qquad \langle v_1, \cdots, v_n \rangle \in K,\ v_1 < \cdots < v_n,\ \sigma \subset \langle v_1, \cdots, v_m \rangle \Big\}, \end{aligned}$$

where $K' \times \{0\} = \{\sigma \times \{0\} \mid \sigma \in K'\}$ and $K \times \{1\} = \{\tau \times \{1\} \mid \tau \in K\}$. For any simplicial map $f : K' \to L$, given an order on $K'^{(0)}$ so that K' is an ordered simplicial complex, we have the simplicial mapping cylinder Z_f. Identifying $K' \subset Z_f$ with $K' \times \{0\} \subset I(K', K)$, we have the simplicial complex

$$Z_f^K = Z_f \cup_{K' = K' \times \{0\}} I(K', K),$$

from which it follows that $|Z_f^K| = |Z_f| \cup_{\mathrm{pr}_{|K|}||K| \times \{0\}} |K| \times \mathbf{I}$.

Theorem 4.18.5 *Let K and K' be ordered simplicial complexes such that $K' \lhd K$ and let L be a simplicial complex. For any simplicial approximation $f : K' \to L$ of a map $g : |K| \to |L|$,*

$$|Z_f^K| \simeq M_g \quad \textit{rel.}\ |L| \cup (|K| \times \{1\}).$$

Proof Since $f \simeq g$, we may assume that $f = g$ by virtue of Theorem 4.18.1. We define maps $\varphi : |Z_f^K| \to M_f$ and $\psi : M_f \to |Z_f^K|$ as follows: $\varphi||Z_f| = \bar{c}_f, \varphi||K| \times (0, 1] = \mathrm{id}$, and $\psi||L| = \mathrm{id}$,

$$\psi(x, s) = \begin{cases} (x, 2s - 1) & \text{for } (x, s) \in |K| \times (\frac{1}{2}, 1], \\ \bar{q}_f(x, 2s) & \text{for } (x, s) \in |K| \times (0, \frac{1}{2}]. \end{cases}$$

Refer to Figs. 4.22 and 4.23, and observe the following:

$$\begin{gathered} \varphi||L| \cup (|K| \times \{1\}) = \psi||L| \cup (|K| \times \{1\}) = \mathrm{id}, \\ \psi\varphi||Z_f| = \bar{c}_f = \bar{h}_0^f \ \text{ and } \ \psi\varphi||K| \times (0, 1] = \psi||K| \times (0, 1]. \end{gathered}$$

We now define the homotopies $\xi : |Z_f^K| \times \mathbf{I} \to |Z_f^K|$ and $\zeta : M_f \times \mathbf{I} \to M_f$ as follows: $\xi_t||Z_f| = \bar{h}_{1-t}^f$, $\zeta_t||L| = \mathrm{id}$, and

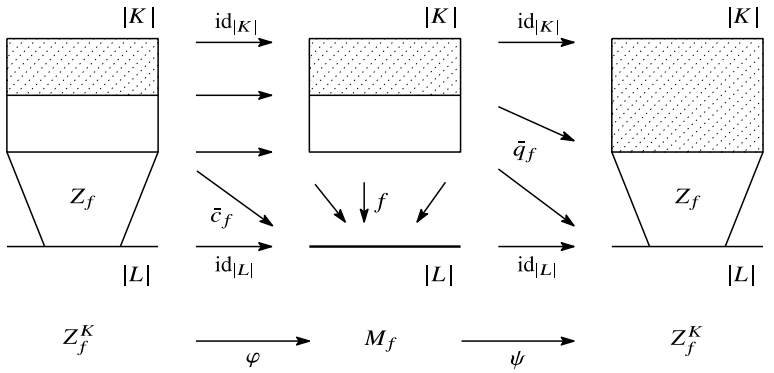


Fig. 4.22 The map $\psi\varphi : |Z_f^K| \to |Z_f^K|$

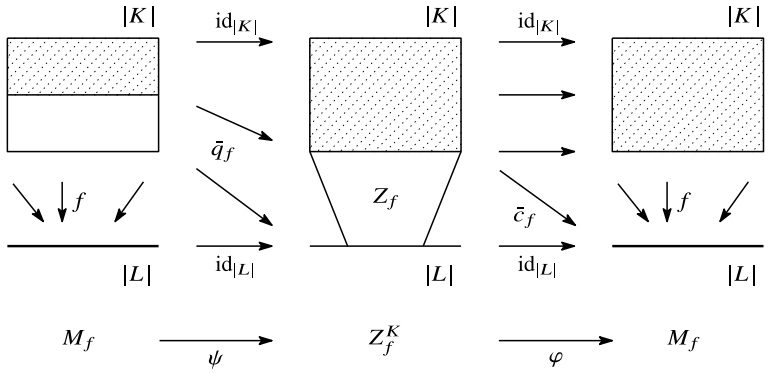


Fig. 4.23 The map $\varphi\psi : M_f \to M_f$

$$\xi_t(x,s) = \begin{cases} \left(x, \dfrac{2s-t}{2-t}\right) & \text{for } (x,s) \in |K| \times (\tfrac{1}{2}t, 1], \\ \bar{q}_f(x, 2s+1-t) & \text{for } (x,s) \in |K| \times (0, \tfrac{1}{2}t]; \end{cases}$$

$$\zeta_t(x,s) = \begin{cases} \left(x, \dfrac{2s-t}{2-t}\right) & \text{for } (x,s) \in |K| \times (\tfrac{1}{2}t, 1], \\ f(x) & \text{for } (x,s) \in |K| \times (0, \tfrac{1}{2}t]. \end{cases}$$

By these homotopies, we have

$$\psi\varphi = \xi_1 \simeq \xi_0 = \mathrm{id} \ \text{ rel. } |L| \cup (|K| \times \{1\}) \text{ and}$$
$$\varphi\psi = \zeta_1 \simeq \zeta_0 = \mathrm{id} \ \text{ rel. } |L| \cup (|K| \times \{1\}).$$

This completes the proof. □

When K and L are locally finite simplicial complexes, any simplicial approximation $f : K' \to L$ of a proper map $g : |K| \to |L|$ is also proper. The following proper version of Theorem 4.18.5 can be proved by transcribing the above proof:

Theorem 4.18.6 *Let K and K' be locally finite ordered simplicial complexes such that $K' \lhd K$ and let L be a locally finite simplicial complex. For any simplicial approximation $f : K' \to L$ of a proper map $g : |K| \to |L|$,*

$$|Z_f^K| \simeq^p M_g \quad \textit{rel.}\ |L| \cup (|K| \times \{1\}). \qquad \square$$

In Theorems 4.18.5 and 4.18.6, consider the case that $f = g$, $K' = K$, and $f : K \to L$ is simplicial. Then, $|Z_f| \simeq^p |Z_f^K|$ rel. $|L|$. Indeed, let $r : |Z_f^K| \to |Z_f|$ be the proper map defined by $r||Z_f| = \mathrm{id}$ and $r||K| \times \mathbf{I} = \mathrm{pr}_{|K|}$. As is easily observed, $r \simeq^p \mathrm{id}$ rel. $|Z_f|$ in $|Z_f^K|$. Thus, we have the following corollary:

Corollary 4.18.7 *Let K be an ordered simplicial complex and L be a simplicial complex. Then, $|Z_f| \simeq M_f$ rel. $|L|$ for each simplicial map $f : K \to L$, where $|Z_f| \simeq^p M_f$ rel. $|L|$ if K and L are locally finite and f is proper.* $\square$

4.19 The Homotopy Type of Simplicial Complexes

In this section, we apply the mapping cylinder and the simplicial mapping cylinder to prove the Whitehead–Milnor Theorem on the homotopy type of simplicial complexes.

A space X is said to be **homotopy dominated by** Y (or Y **homotopy dominates** X) if there are maps $f : X \to Y$ and $g : Y \to X$ such that $gf \simeq \mathrm{id}$, where g is called a **homotopy domination**. It is easy to see that X has the homotopy type of the singleton $\{0\}$ if and only if X is homotopy dominated by $\{0\}$. The **homotopy type of a simplicial complex** K means the homotopy type of the polyhedron $|K|$ (or $|K|_{\mathrm{m}}$) (cf. Theorem 4.15.7). We say that X is **homotopy dominated by a simplicial complex** K if X is homotopy dominated by $|K|$ (or $|K|_{\mathrm{m}}$). Applying Theorems 4.18.1, 4.18.3, and 4.18.5, we can prove the following theorem called the WHITEHEAD–MILNOR THEOREM:

Theorem 4.19.1 (J.H.C. WHITEHEAD; MILNOR) *If a space X is homotopy dominated by a simplicial complex K, then X has the homotopy type of a simplicial complex L with* $\operatorname{card} L^{(0)} = \operatorname{dens} |L| \leqslant \operatorname{dens} X$. *When X is separable, X has the homotopy type of a countable simplicial complex.*

Proof We may assume that X is infinite, so $\operatorname{dens} X \geqslant \aleph_0$. Let $f : X \to |K|$ and $g : |K| \to X$ be maps such that $gf \simeq \mathrm{id}$. Then, we may assume that $\operatorname{card} K \leqslant \operatorname{dens} X$. Indeed, take a dense set D in X with $\operatorname{card} D = \operatorname{dens} X$ and define

$$K_0 = \left\{\tau \in K \;\middle|\; \exists x \in D \ \text{ such that } \ \tau \leqslant c_K(f(x))\right\},$$

where $c_K(f(x))$ is the carrier of $f(x)$. Observe that card $K_0 \leqslant$ card D and $f(X) = f(\mathrm{cl}\, D) \subset \mathrm{cl}\, f(D) \subset |K_0|$. Because $(g||K_0|)f \simeq \mathrm{id}$, we can replace K with K_0.

By the Simplicial Approximation Theorem 4.10.11, $fg : |K| \to |K|$ has a simplicial approximation $\varphi : K' \to K$, where K' is a subdivision of K. Giving orders on $K'^{(0)}$ and $K^{(0)}$ so that K' and K are ordered simplicial complexes, we obtain the simplicial complex Z_φ^K with card $Z_\varphi^K \leqslant \mathrm{dens}\, X$.

For each $n \in \mathbb{Z}$, let L_n be a copy of Z_φ^K. Identifying K of L_{n+1} with $K \times \{1\}$ of L_n for each $n \in \mathbb{Z}$, we have a simplicial complex $L = \bigcup_{n\in\mathbb{Z}} L_n$, where card $L \leqslant \mathrm{dens}\, X$. For each $n \in \mathbb{Z}$, let M_{2n-1} and M_{2n} be copies of M_f and M_g, respectively. Identifying $X \times \{1\} \subset M_{2n-1}$ with $X \subset M_{2n}$ and $|K| \times \{1\} \subset M_{2n}$ with $|K| \subset M_{2n+1}$, let $M = \bigcup_{n\in\mathbb{Z}} M_n$.

By Theorems 4.18.5 and 4.18.3, we have

$$|L_n| = |Z_\varphi^K| \simeq M_{fg} \simeq M_f \cup_{i_f} M_g = M_{2n-1} \cup M_{2n} \ \text{ rel. } |K| \cup (|K| \times \{1\}),$$

which implies that $|L| \simeq M$. By Theorems 4.18.3 and 4.18.1,

$$M_{2n} \cup M_{2n+1} = M_g \cup_{i_g} M_f \simeq M_{gf} \simeq M_{\mathrm{id}_X} \text{ rel. } X \cup (X \times \{1\}).$$

Regarding M_{id_X} as $X \times [n, n+1]$, we have $M \simeq X \times \mathbb{R}$. Thus, it follows that $X \simeq X \times \mathbb{R} \simeq |L|$. See Fig. 4.24. □

In the above proof, if $\dim K < \infty$ then $\dim L < \infty$. Therefore, we have the following corollary:

Corollary 4.19.2 *A space X homotopy dominated by a finite-dimensional simplicial complex has the homotopy type of a finite-dimensional simplicial complex K with* card $K^{(0)} = \mathrm{dens}\, |K| \leqslant \mathrm{dens}\, X$. □

Applying product simplicial complexes, we can prove the following theorem:

Theorem 4.19.3 *Every simplicial complex has the homotopy type of a locally finite-dimensional simplicial complex with the same density. In addition, every countable simplicial complex has the homotopy type of a countable locally finite simplicial complex.*

Proof For each $n \in \omega$, let $L_n = \{i,\ [i, i+1] \mid i \geqslant n\}$ be the ordered simplicial complex with the natural order. For each simplicial complex K, assuming that K is an ordered simplicial complex, we define a locally finite-dimensional simplicial complex $S_K = \bigcup_{n\in\omega} K^{(n)} \times_s L_n$ (see Fig. 4.25). By Theorem 4.5.1, we have

$$|S_K| = \bigcup_{n\in\omega} |K^{(n)}| \times [n, \infty) \subset |K| \times [0, \infty).$$

To prove that $|K| \simeq |S_K|$, it suffices to show that $|K| \times [0, \infty) \simeq |S_K|$ because $|K| \simeq |K| \times [0, \infty)$. For each $n \in \omega$, let

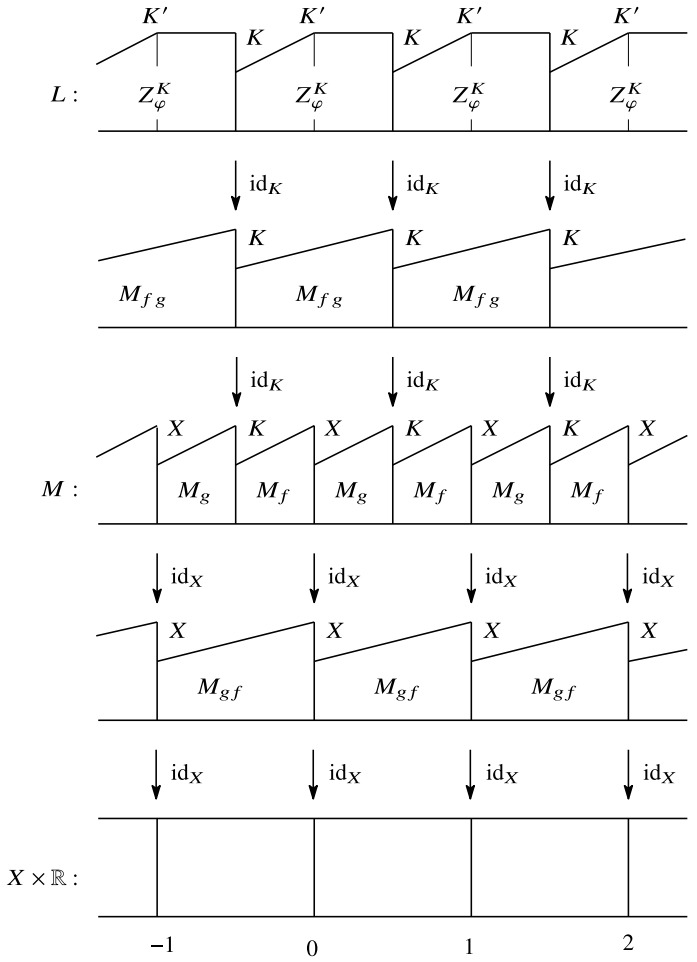


Fig. 4.24 $X \times \mathbb{R} \simeq |L|$

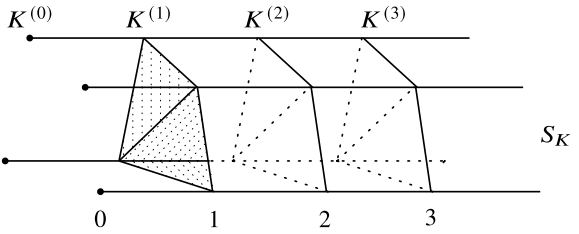


Fig. 4.25 The locally finite-dimensional simplicial complex S_K

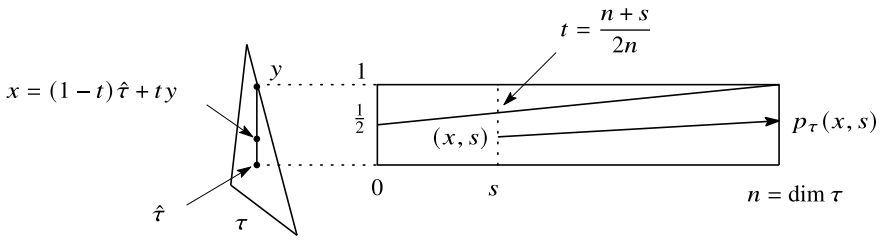


Fig. 4.26 The map p_τ

$$T_n = |K^{(n)}| \times [0, n] \cup |S_K|.$$

Then, $|S_K| = T_0 \subset T_1 \subset T_2 \subset \cdots$ and $|K| \times [0, \infty) = \varinjlim T_n$ (cf. Sect. 2.8). For each n-simplex $\tau \in K$, we have a map

$$p_\tau : \tau \times [0, n] \to \partial\tau \times [0, n] \cup \tau \times \{n\} = \tau \times [0, n] \cap T_{n-1}$$

defined as follows: $p_\tau(\hat{\tau}, s) = (\hat{\tau}, n)$ and

$$p_\tau(x, s) = \begin{cases} (y, s + 2n(1-t)) & \text{if } \dfrac{n+s}{2n} \leqslant t \leqslant 1, \\ \left(\left(1 - \dfrac{2n}{n+s}t\right)\hat{\tau} + \dfrac{2n}{n+s}ty, n\right) & \text{if } 0 < t \leqslant \dfrac{n+s}{2n}, \end{cases}$$

where $x = (1-t)\hat{\tau} + ty \in \tau$, $y \in \partial\tau$, $0 < t \leqslant 1$ (see Fig. 4.26).

For each $n \in \mathbb{N}$, let $p_n : T_n \to T_{n-1}$ be the map defined by $p_n|T_{n-1} = \mathrm{id}$ and $p_n|\tau \times [0, n] = p_\tau$ for each n-simplex $\tau \in K$. Then, we can define a map $r : |K| \times [0, \infty) \to |S_K|$ by $r|T_n = p_1 \cdots p_n$ for each $n \in \mathbb{N}$. Note that $r||S_K| = \mathrm{id}$ and $r(\tau \times [0, \infty)) \subset \tau \times [0, \infty)$ for each $\tau \in K$. Observe that $r \simeq \mathrm{id}$ rel. $|S_K|$ in $|K| \times [0, \infty)$ by the straight-line homotopy (i.e., $t \mapsto (1-t)r(x, s) + t(x, s)$). Therefore, we have $|K| \times [0, \infty) \simeq |S_K|$.

When K is countable, K has a tower $K_0 \subset K_1 \subset K_2 \subset \cdots$ of finite subcomplexes such that each $K_{n+1} \setminus K_n$ consists of one simplex and $K = \bigcup_{n\in\omega} K_n$. In the above, replacing $K^{(n)}$ with K_n, we define a countable locally finite simplicial complex $T_K = \bigcup_{n\in\omega} K_n \times_s L_n$. We can similarly prove that $|K| \simeq |T_K|$. □

By Theorems 4.19.1 and 4.19.3, we have the following corollary:

Corollary 4.19.4 *A space X homotopy dominated by a simplicial complex has the homotopy type of a locally finite-dimensional simplicial complex K with* $\operatorname{card} K^{(0)} = \operatorname{dens}|K| \leqslant \operatorname{dens} X$. *If X is separable, X has the homotopy type of a countable locally finite simplicial complex.* □

It is said that X is **proper homotopy dominated** by Y (or Y **proper homotopy dominates** X) if there are proper maps $f : X \to Y$ and $g : Y \to X$ such



Fig. 4.27 $\mathbf{I} \times [0, 1) \approx (0, 1) \times [0, 1)$

that $gf \simeq^p \mathrm{id}$, where $g : Y \to X$ is called a **proper homotopy domination**. For a locally compact space, we can show the following proper homotopy version of the Whitehead–Milnor Theorem 4.19.1:

Theorem 4.19.5 *Let X be a locally compact space. If X is proper homotopy dominated by a locally finite simplicial complex K, then $X \times \mathbb{R}_+$ has the proper homotopy type of a locally finite simplicial complex L.*

Proof To prove the theorem, we can perform the same proof as Theorem 4.19.1. For the reader's convenience, we give its outline below:

> Let $f : X \to |K|$ and $g : |K| \to X$ be proper maps such that $gf \simeq^p \mathrm{id}$. By the Simplicial Approximation Theorem 4.10.11, fg has a simplicial approximation $\varphi : K' \to K$ for some $K' \lhd K$. Since fg is proper, φ is also proper. Giving orders on $K'^{(0)}$ and $K^{(0)}$ so that K' and K are ordered simplicial complexes, we can obtain the simplicial complex Z_φ^K. Since K is locally finite and φ is proper, Z_φ^K is locally finite. For each $n \in \mathbb{Z}$, let L_n be a copy of Z_φ^K. Identifying $K \subset L_{n+1}$ with $K \times \{1\} \subset L_n$ for each $n \in \mathbb{Z}$, we have a locally finite simplicial complex $L = \bigcup_{n\in\mathbb{Z}} L_n$. For each $n \in \mathbb{Z}$, let M_{2n-1} and M_{2n} be copies of M_f and M_g, respectively. Identifying $X \times \{1\} \subset M_{2n-1}$ and $|K| \times \{1\} \subset M_{2n}$ with $X \subset M_{2n}$ and $|K| \subset M_{2n+1}$ respectively, we define $M = \bigcup_{n\in\mathbb{Z}} M_n$. Then, by using Theorems 4.18.2, 4.18.4, and 4.18.6, we can prove that $|L| \simeq^p |M| \simeq^p X \times \mathbb{R}$. Refer to Fig. 4.24.

As is easily observed (see Fig. 4.27), we have

$$\mathbf{I} \times \mathbb{R}_+ \approx \mathbf{I} \times [0, 1) \approx (0, 1) \times [0, 1) \approx \mathbb{R} \times \mathbb{R}_+.$$

Then, it follows that

$$X \times \mathbb{R}_+ \simeq^p X \times \mathbf{I} \times \mathbb{R}_+ \approx X \times \mathbb{R} \times \mathbb{R}_+ \simeq^p |L| \times \mathbb{R}_+ = |L \times_s R_+|,$$

where R_+ is the natural triangulation of $\mathbb{R}_+$, i.e., $R_+ = \{[n, n + 1], n \mid n \in \omega\}$. Thus, $X \times \mathbb{R}_+$ has the proper homotopy type of $L \times_s R_+$, which is a locally finite simplicial complex. □

Remark 4.19.1 In Theorem 4.19.5, if X is connected, then L is countable because L is also connected (i.e., $|L|$ is connected).

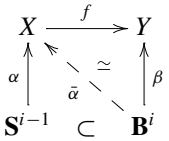
Remark 4.19.2 Relating to Theorem 4.19.1 and Corollary 4.19.4, the question arises whether a compact space homotopy dominated by a simplicial complex has the homotopy type of a finite simplicial complex. This is equivalent to asking whether a compact space having the homotopy type of a simplicial complex has the homotopy type of a finite simplicial complex. As will be seen in Sect. 6, every ANR has the homotopy type of a simplicial complex and every simplicial complex has the homotopy type of an ANR. Thus, the question is equivalent to asking whether a compact ANR has the homotopy type of a finite simplicial complex. The positive answer has been expected and it is called the Borsuk conjecture, which was posed by Borsuk in 1954 and proved by J. West in 1975 (see Notes at the end of this chapter). This conjecture is discussed in Chapter 4 of the author's second book [TIDM].

4.20 Weak Homotopy Equivalences

Let $n \in \omega$. A map $f : X \to Y$ is called an ***n*-equivalence** if it satisfies the following condition $(\pi)_i$ for each $i = 0, \ldots, n$:

$(\pi)_i$ For each map $\alpha : \mathbf{S}^{i-1} \to X$, if $f\alpha$ extends to a map $\beta : \mathbf{B}^i \to Y$, then α extends to a map $\bar{\alpha} : \mathbf{B}^i \to X$ such that $f\bar{\alpha} \simeq \beta$ rel. $\mathbf{S}^{i-1}$,

where $\mathbf{B}^0 = \{0\}$ and $\mathbf{S}^{-1} = \emptyset$.



When $f : X \to Y$ is an n-equivalence for every $n \in \omega$, we call f a **weak homotopy equivalence**. For convenience, a weak homotopy equivalence is sometimes called an **∞-equivalence**.

In the Appendix (Sect. 8.1), we will give a characterization of n-equivalences in the framework of homotopy groups. In particular, it will be shown that a map is a weak homotopy equivalence if and only if it induces a bijection between the sets of path-components and an isomorphism between their homotopy groups in every dimension. The following proposition is an immediate consequence of this characterization, but we will give a direct proof.

Proposition 4.20.1 *Every homotopy equivalence $f : X \to Y$ is a weak homotopy equivalence.*

Proof Let $g : Y \to X$ be a homotopy inverse of f. Then, there are homotopies $h : X \times \mathbf{I} \to X$ and $k : Y \times \mathbf{I} \to Y$ such that $h_0 = \mathrm{id}_X, h_1 = gf, k_0 = \mathrm{id}_Y$, and $k_1 = fg$. For each pair of maps $\alpha : \mathbf{S}^{i-1} \to X$ and $\beta : \mathbf{B}^i \to Y$ with $f\alpha = \beta|\mathbf{S}^{i-1}$, we can extend α to the map $\bar{\alpha} : \mathbf{B}^i \to X$ defined as follows:

$$\bar{\alpha}(x) = \begin{cases} g\beta(8x) & \text{if } 0 \leqslant \|x\| \leqslant 1/8, \\ gfh(\alpha(\|x\|^{-1}x), 8\|x\| - 1) & \text{if } 1/8 \leqslant \|x\| \leqslant 1/4, \\ gk(f\alpha(\|x\|^{-1}x), 2 - 4\|x\|) & \text{if } 1/4 \leqslant \|x\| \leqslant 1/2, \\ h(\alpha(\|x\|^{-1}x), 2 - 2\|x\|) & \text{if } 1/2 \leqslant \|x\| \leqslant 1. \end{cases}$$

It remains to show that $f\bar{\alpha} \simeq \beta$ rel. $\mathbf{S}^{i-1}$.

We define an auxiliary map $\gamma : \mathbf{B}^i \to Y$ as follows:

$$\gamma(x) = \begin{cases} \beta(8x) & \text{if } 0 \leqslant \|x\| \leqslant 1/8, \\ fh(\alpha(\|x\|^{-1}x), 8\|x\| - 1) & \text{if } 1/8 \leqslant \|x\| \leqslant 1/4, \\ k(f\alpha(\|x\|^{-1}x), 2 - 4\|x\|) & \text{if } 1/4 \leqslant \|x\| \leqslant 1/2, \\ k(f\alpha(\|x\|^{-1}x), 4\|x\| - 2) & \text{if } 1/2 \leqslant \|x\| \leqslant 3/4, \\ fh(\alpha(\|x\|^{-1}x), 4 - 4\|x\|) & \text{if } 3/4 \leqslant \|x\| \leqslant 1. \end{cases}$$

Then, $\gamma \simeq f\bar{\alpha}$ rel. $\mathbf{S}^{i-1}$ by the homotopy $\varphi : \mathbf{B}^i \times \mathbf{I} \to Y$ defined as follows:

$$\varphi_t(x) = \begin{cases} k_t(\beta(8x)) & \text{if } 0 \leqslant \|x\| \leqslant 1/8, \\ k_t(fh(\alpha(\|x\|^{-1}x), 8\|x\| - 1)) & \text{if } 1/8 \leqslant \|x\| \leqslant 1/4, \\ k_t k(f\alpha(\|x\|^{-1}x), 2 - 4\|x\|) & \text{if } 1/4 \leqslant \|x\| \leqslant 1/2, \\ k(f\alpha(\|x\|^{-1}x), 4\|x\| - 2 + t) & \text{if } 1/2 \leqslant \|x\| \leqslant (3 - t)/4, \\ fh(\alpha(\|x\|^{-1}x), (4 - 4\|x\|)/(1 + t)) & \text{if } (3 - t)/4 \leqslant \|x\| \leqslant 1. \end{cases}$$

On the other hand, let $\gamma', \gamma'' : \mathbf{B}^i \to Y$ be the maps defined as follows:

$$\gamma'(x) = \begin{cases} \beta(8x) & \text{if } 0 \leqslant \|x\| \leqslant 1/8, \\ fh(\alpha(\|x\|^{-1}x), 8\|x\| - 1) & \text{if } 1/8 \leqslant \|x\| \leqslant 1/4, \\ fgf\alpha(\|x\|^{-1}x) & \text{if } 1/4 \leqslant \|x\| \leqslant 3/4, \\ fh(\alpha(\|x\|^{-1}x), 4 - 4\|x\|) & \text{if } 3/4 \leqslant \|x\| \leqslant 1; \end{cases}$$

$$\gamma''(x) = \begin{cases} \beta(8x) & \text{if } 0 \leqslant \|x\| \leqslant 1/8, \\ f\alpha(\|x\|^{-1}x) & \text{if } 1/8 \leqslant \|x\| \leqslant 1. \end{cases}$$

It is then easy to obtain the homotopies $\gamma \simeq \gamma' \simeq \gamma'' \simeq \beta$ rel. $\mathbf{S}^{i-1}$. □

In this section, we will show the converse of Proposition 4.20.1 when X and Y are polyhedra. Namely, we will prove that every weak homotopy equivalence between polyhedra is a homotopy equivalence.

A **path-component** of a space X is a maximal path-connected subset of X. The set of path-components of X is denoted by $\pi_0(X)$. Every map $f : X \to Y$ induces the function $f_\sharp : \pi_0(X) \to \pi_0(Y)$, which sends the path-component of $x \in X$ to the path-component of $f(x) \in Y$. The following propositions are easily proved:

Proposition 4.20.2 (1) *A map $f : X \to Y$ is a 0-equivalence if and only if every path-component of Y meets $f(X)$, i.e., $f_\sharp : \pi_0(X) \to \pi_0(Y)$ is a surjection.*

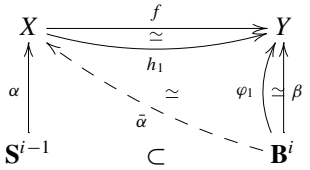
(2) *Every 1-equivalence $f : X \to Y$ induces the bijection $f_\sharp : \pi_0(X) \to \pi_0(Y)$ and the surjection $f_\sharp : [(\mathbf{I}, \partial\mathbf{I}), (X, x_0)] \to [(\mathbf{I}, \partial\mathbf{I}), (Y, f(x_0))]$ for every $x_0 \in X$.*[42] □

Proposition 4.20.3 *Let $n \in \omega \cup \{\infty\}$. The composition of n-equivalences is also an n-equivalence.* □

The following proposition, similar to Proposition 4.20.1, is an immediate consequence of the characterization of an n-equivalence in the framework of homotopy groups, which will be discussed in the Appendix (Sect. 8.1). However, here we will give a direct proof.

Proposition 4.20.4 *Let $n \in \omega \cup \{\infty\}$. If a map $f : X \to Y$ is homotopic to an n-equivalence, then f is also an n-equivalence.*

Proof For each $i < n + 1$,[43] let $\alpha : \mathbf{S}^{i-1} \to X$ and $\beta : \mathbf{B}^i \to Y$ be maps such that $f\alpha = \beta|\mathbf{S}^{i-1}$. By the assumption, we have a homotopy $h : X \times \mathbf{I} \to Y$ such that $h_0 = f$ and h_1 is an n-equivalence. According to the Homotopy Extension Theorem 4.5.3, there is a homotopy $\varphi : \mathbf{B}^i \times \mathbf{I} \to Y$ such that $\varphi_0 = \beta$ and $\varphi|\mathbf{S}^{i-1} \times \mathbf{I} = h(\alpha \times \mathrm{id}_\mathbf{I})$. Since $h_1\alpha$ extends to the map $\varphi_1 : \mathbf{B}^i \to Y$, α extends to a map $\bar{\alpha} : \mathbf{B}^i \to X$ such that $h_1\bar{\alpha} \simeq \varphi_1$ rel. $\mathbf{S}^{i-1}$.



Let $\psi : \mathbf{B}^i \times \mathbf{I} \to Y$ be a homotopy such that $\psi_0 = h_1\bar{\alpha}$, $\psi_1 = \varphi_1$, and $\psi_t|\mathbf{S}^{i-1} = h_1\alpha$ for every $t \in \mathbf{I}$. We define a homotopy $k : \partial(\mathbf{B}^i \times \mathbf{I}) \times \mathbf{I} \to Y$ as follows:

$$k_t(x, s) = \begin{cases} \varphi_{1-t}(x) & \text{if } s = 1, \\ h_{1-t}\bar{\alpha}(x) & \text{if } s = 0, \\ h_{1-t}\alpha(x) & \text{if } x \in \mathbf{S}^{i-1}. \end{cases}$$

Refer to Fig. 4.28. Then, ψ is an extension of k_0. Hence, we can apply the Homotopy Extension Theorem 4.5.3 to extend k_1 to a map $\psi' : \mathbf{B}^i \times \mathbf{I} \to Y$, which is a homotopy from $f\bar{\alpha}$ to β with $\psi'_s|\mathbf{S}^{i-1} = f\alpha$ for every $s \in \mathbf{I}$, i.e., $f\bar{\alpha} \simeq \beta$ rel. $\mathbf{S}^{i-1}$. Therefore, f is an n-equivalence. □

When f is the inclusion, we have the following:

[42] That is, $f_\sharp : \pi_1(X, x_0) \to \pi_1(Y, f(x_0))$ is an epimorphism for every $x_0 \in X$.

[43] We use the convention that $\infty + 1 = \infty$.

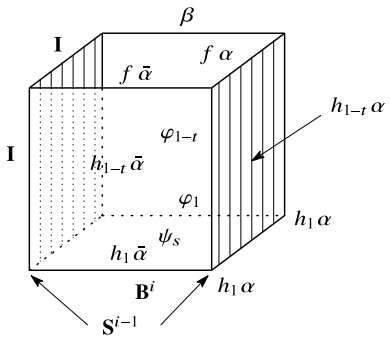
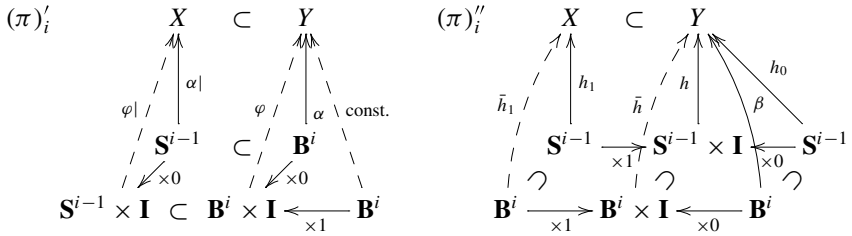


Fig. 4.28 The homotopy k

Lemma 4.20.5 *For the inclusion $X \subset Y$ and $i \geqslant 1$, each of the following is equivalent to condition $(\pi)_i$:*

$(\pi)'_i$ *Every map $\alpha : \mathbf{B}^i \to Y$ with $\alpha(\mathbf{S}^{i-1}) \subset X$ is null-homotopic by a homotopy $\varphi : \mathbf{B}^i \times \mathbf{I} \to Y$ with $\varphi(\mathbf{S}^{i-1} \times \mathbf{I}) \subset X$, i.e., $[(\mathbf{B}^i, \mathbf{S}^{i-1}), (Y, X)] = \{0\}$;*[44]

$(\pi)''_i$ *For a homotopy $h : \mathbf{S}^{i-1} \times \mathbf{I} \to Y$ with $h_1(\mathbf{S}^{i-1}) \subset X$, if h_0 extends to a map $\beta : \mathbf{B}^i \to Y$, then h extends to a homotopy $\bar{h} : \mathbf{B}^i \times \mathbf{I} \to Y$ such that $\bar{h}_0 = \beta$ and $\bar{h}_1(\mathbf{B}^i) \subset X$.*



Proof $(\pi)_i \Rightarrow (\pi)'_i$: From condition $(\pi)_i$, we have a homotopy $h : \mathbf{B}^i \times \mathbf{I} \to Y$ such that $h_0 = \alpha$, $h_1(\mathbf{B}^i) \subset X$ and $h_t|\mathbf{S}^{i-1} = \alpha|\mathbf{S}^{i-1}$ for every $t \in \mathbf{I}$. Because $\mathbf{B}^i \simeq \{\mathbf{0}\}$,[45] we have $h_1 \simeq 0$ in X. Then, the homotopy φ in $(\pi)'_i$ is easily defined.

$(\pi)'_i \Rightarrow (\pi)''_i$: We have a homeomorphism $\psi : \mathbf{B}^i \times \mathbf{I} \to \mathbf{B}^i \times \mathbf{I}$ such that

$$\psi\big((\mathbf{S}^{i-1} \times \mathbf{I}) \cup (\mathbf{B}^i \times \{0\})\big) = \mathbf{B}^i \times \{0\},$$
$$\psi(\mathbf{S}^{i-1} \times \{1\}) = \mathbf{S}^{i-1} \times \{0\} \text{ and}$$
$$\psi(\mathbf{B}^i \times \{1\}) = (\mathbf{S}^{i-1} \times \mathbf{I}) \cup (\mathbf{B}^i \times \{1\}).$$

[44] In terms of the homotopy groups, this means that $\pi_i(Y, X, x_0) = 0$ for each $x_0 \in X$. In Proposition 8.2.3, we will give another proof of the equivalence between this condition and $(\pi)_i$.

[45] I.e., $\mathbf{B}^i$ is contractible (see the definition in Sect. 6.1).

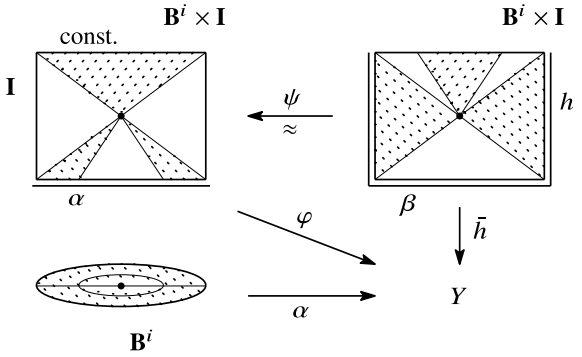


Fig. 4.29 The construction of $\bar{h}$

Let $\beta : \mathbf{B}^i \to Y$ be an extension of h_0 and define a map $\alpha : \mathbf{B}^i \to Y$ as follows:

$$\alpha(x) = \begin{cases} h\psi^{-1}(x, 0) & \text{if } (x, 0) \in \psi(\mathbf{S}^{i-1} \times \mathbf{I}), \\ \beta \mathrm{pr}_{\mathbf{B}^i} \psi^{-1}(x, 0) & \text{if } (x, 0) \in \psi(\mathbf{B}^i \times \{0\}). \end{cases}$$

Since $\alpha(\mathbf{S}^{i-1}) = h_1(\mathbf{S}^{i-1}) \subset X$, we can apply $(\pi)'_i$ to obtain a homotopy $\varphi : \mathbf{B}^i \times \mathbf{I} \to Y$ such that $\varphi_0 = \alpha, \varphi(\mathbf{S}^{i-1} \times \mathbf{I}) \subset X$, and φ_1 is constant. Then, $\bar{h} = \varphi\psi : \mathbf{B}^i \times \mathbf{I} \to Y$ is the desired extension of h. See Fig. 4.29.

$(\pi)''_i \Rightarrow (\pi)_i$: Given a map $\beta : \mathbf{B}^i \to Y$ such that $\alpha = \beta|\mathbf{S}^{i-1} : \mathbf{S}^{i-1} \to X$, let $h : \mathbf{S}^{i-1} \times \mathbf{I} \to Y$ be the constant homotopy defined by $h_t = \alpha$ for each $t \in \mathbf{I}$. Due to condition $(\pi)''_i$, h extends to a homotopy $\bar{h} : \mathbf{B}^i \times \mathbf{I} \to Y$ such that $\bar{h}_0 = \beta$ and $\bar{h}_1(\mathbf{B}^i) \subset X$. Then, $\bar{h}_1|\mathbf{S}^{i-1} = \alpha$ and $\bar{h}_1 \simeq \beta$ rel. $\mathbf{S}^{i-1}$ in Y. □

Using this lemma, we can prove the following proposition:

Proposition 4.20.6 *Let X be a subspace of a space Y such that the inclusion $X \subset Y$ is an n-equivalence, where $n \in \omega \cup \{\infty\}$. Given a simplicial complex K and a subcomplex $L \subset K$ with $K \setminus L \subset K^{(n)}$ ($K^{(\infty)}$ means K), if a map $f : |L| \to X$ extends to a map $f' : |K| \to Y$, then f extends to a map $f'' : |K| \to X$ such that $f' \simeq f''$ rel. $|L|$ in Y.*

Proof According to Theorem 4.5.1,

$$|L \cup K^{(i)}| \times \mathbf{I} = |(L \cup K^{(i)}) \times_c I| \text{ for each } i \in \omega,$$

where $I = \{\mathbf{I}, 0, 1\}$. Applying condition $(\pi)''_i$ in Lemma 4.20.5 simplex-wise, we can inductively construct homotopies $h^{(i)} : |L \cup K^{(i)}| \times \mathbf{I} \to Y, i \in \omega$, such that

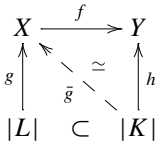
$$h^{(i)}_0 = f'||L \cup K^{(i)}|, \ h^{(i)}_1(|L \cup K^{(i)}|) \subset X \text{ and}$$
$$h^{(i)}||L \cup K^{(i-1)}| \times \mathbf{I} = h^{(i-1)},$$

where $h^{(-1)} : |L| \times \mathbf{I} \to Y$ is the constant homotopy defined by $h_t^{(-1)} = f||L| = f'||L|$ for every $t \in \mathbf{I}$. Because $|K| \times \mathbf{I} = |K \times_c I|$, we can define a homotopy $h : |K| \times \mathbf{I} \to Y$ by $h||L \cup K^{(i)}| \times \mathbf{I} = h^{(i)}$ for each $i \in \omega$. Hence, $f'' = h_1$ is the desired map. □

In the following proposition, we identify X with the subspace $X \times \{1\}$ of the mapping cylinder M_f of $f : X \to Y$.

Proposition 4.20.7 *Let $n \in \omega \cup \{\infty\}$. For a map $f : X \to Y$, the following are equivalent:*

(a) *f is an n-equivalence;*
(b) *The inclusion $i_f : X \subset M_f$ is an n-equivalence;*[46]
(c) *Given a simplicial complex K and a subcomplex $L \subset K$ with $K \setminus L \subset K^{(n)}$ ($K^{(\infty)}$ means K), for a map $g : |L| \to X$, if fg extends to a map $h : |K| \to Y$ then g extends to a map $\bar{g} : |K| \to X$ such that $f\bar{g} \simeq h$ rel. $|L|$.*



Proof Since $(\mathbf{B}^i, \mathbf{S}^{i-1}) \approx (|F(\Delta^i)|, |\partial F(\Delta^i)|)$, the implication (c) $\Rightarrow$ (a) is trivial. Since the collapsing $c_f : M_f \to Y$ is a homotopy equivalence and the inclusion $i_Y : Y \subset M_f$ is its homotopy inverse, the map $i_Y f = i_Y c_f i_f : X \to Y \subset M_f$ is homotopic to $i_f : X \to M_f$. By virtue of Propositions 4.20.1, 4.20.3, and 4.20.4, $f = c_f i_Y f : X \to Y$ is an n-equivalence if and only if i_f is. Thus, we have the equivalence (a) $\Leftrightarrow$ (b). It remains to show the implication (b) $\Rightarrow$ (c).

(b) $\Rightarrow$ (c): Let K be a simplicial complex and $L \subset K$ be a subcomplex such that $\dim \sigma \leqslant n$ for every $\sigma \in K \setminus L$. Let $g : |L| \to X$ and $h : |K| \to Y$ be maps such that $fg = h||L|$. The homotopy $h^f : M_f \times \mathbf{I} \to M_f$ from $i_Y c_f$ to id induces a homotopy $\varphi : |L| \times \mathbf{I} \to M_f$ from $i_Y c_f i_f g = i_Y fg = i_Y h||L|$ to $i_f g$, which is defined by $\varphi_t = h_t^f i_f g$. Then, $c_f \varphi_t = c_f h_t^f i_f g = c_f i_f g = fg$ for each $t \in \mathbf{I}$. According to the Homotopy Extension Theorem 4.5.3, φ extends to a homotopy $\bar{\varphi} : |K| \times \mathbf{I} \to M_f$ with $\bar{\varphi}_0 = i_Y h$, so $\bar{\varphi}_1$ is an extension of $i_f g$. By (b) and Proposition 4.20.6, g extends to a map $\bar{g} : |K| \to X$ such that $\bar{g} \simeq \bar{\varphi}_1$ rel. $|L|$ in M_f, hence $f\bar{g} = c_f \bar{g} \simeq c_f \bar{\varphi}_1$ rel. $|L|$ in Y. On the other hand, $c_f \bar{\varphi}_1 \simeq c_f \bar{\varphi}_0 = c_f i_Y h = h$ rel. $|L|$ in Y because $c_f \bar{\varphi}_t||L| = c_f \varphi_t = fg$ for each $t \in \mathbf{I}$. Thus, we have $f\bar{g} \simeq h$ rel. $|L|$ in Y. □

For polyhedra, we have the following theorem:

Theorem 4.20.8 *Let K and L be simplicial complexes and $n \in \mathbb{N}$. If $\dim K \leqslant n-1$ and $\dim L \leqslant n$, then every n-equivalence $f : |K| \to |L|$ is a homotopy equivalence.*

[46] According to Lemma 4.20.5 with Footnote 44, this condition means that $\pi_i(M_f, X, x_0) = 0$ for each $x_0 \in X$ and $i < n+1$ (where $\infty + 1 = \infty$).

Proof According to the Simplicial Approximation Theorem 4.10.11, f has a simplicial approximation $g : K' \to L$ for some $K' \lhd K$. Then, $M_f \simeq |Z_g^K|$ rel. $|L| \cup |K|$ by Theorem 4.18.5, where $|K|$ is identified with $|K| \times \{1\}$ in M_f and $|Z_g^K|$. Since the inclusion $|K| \subset M_f$ is an n-equivalence by (b) in Proposition 4.20.7, it follows that the inclusion $|K| \subset |Z_g^K|$ is also an n-equivalence. Note that $\dim Z_g^K \leqslant n$. We can now apply condition (c) of Proposition 4.20.7 to obtain a map $\varphi : |Z_g^K| \to |K|$ such that $\varphi||K| = \mathrm{id}$ and $\varphi \simeq \mathrm{id}$ rel. $|K|$ in $|Z_g^K|$, which means that the inclusion $|K| \subset |Z_g^K|$ is a homotopy equivalence, hence the inclusion $|K| \subset M_f$ is also a homotopy equivalence. Since the collapsing $c_f : M_f \to |L|$ is a homotopy equivalence, it follows that $f = c_f||K|$ is a homotopy equivalence. □

If $n = \infty$, $K^{(\infty)} = K$ in Propositions 4.20.6 and 4.20.7, so the above argument is valid without the dimensional assumption. Thus, we have the following theorem:

Theorem 4.20.9 *For simplicial complexes K and L, every weak homotopy equivalence $f : |K| \to |L|$ is a homotopy equivalence.* □

Remark 4.20.1 It should be noted that there are connected polyhedra X and Y such that $X \not\simeq Y$ but $\pi_n(X) \cong \pi_n(Y)$ for every $n \in \mathbb{N}$. For example, $X = \mathbf{S}^2 \times \mathbb{R}\mathrm{P}^3$ and $\mathbb{R}\mathrm{P}^2 \times \mathbf{S}^3$ are not homotopy equivalent but they have the isomorphic homotopy groups.[47]

Combining the above with Theorem 4.19.1 and Proposition 4.20.1, we have the following corollary:

Corollary 4.20.10 *Let X and Y be spaces that are homotopy dominated by simplicial complexes. Then, a map $f : X \to Y$ is a homotopy equivalence if and only if f is a weak homotopy equivalence.* □

Notes for Chap. 4

There are no good textbooks for studying *non-locally finite* simplicial complexes or *infinite-dimensional* simplicial complexes. For the study of Piecewise-Linear (PL) Topology, the following can be recommended:

- C.P. Rourke and B.J. Sanderson, *Introduction to Piecewise-Linear Topology*, Springer Study Edition (Springer-Verlag, Berlin, 1982)

The following classical lecture notes are still excellent resources for PL Topology:

- J.F.P. Hudson, *Piecewise-Linear Topology* (W.A. Benjamin, Inc., New York, 1969)
- J. Stallings, *Lectures on Polyhedral Topology* (Tata Institute, Bombay, 1967)
- E.C. Zeeman, *Seminar on Combinatorial Topology* (Institute des Hautes Etude Sci., Paris, 1963)

[47] Here, $\mathbb{R}\mathrm{P}^2$ is the real projective plane and $\mathbb{R}\mathrm{P}^3$ is the 3-dimensional real projective space. For example, refer to Hatcher's book: *Algebraic Topology* (cf. Notes for Chap. 8).

Stellar subdivisions are discussed in Volume I of the following:

- L.C. Glaser, *Geometrical Combinatorial Topology, Vol. I, Vol. II* (Van Nostrand Reinhold Co., New York, 1970, 1972)

Concerning the history of polyhedra and simplicial complexes, an article [9] by Burde and Zieschang gives interesting information. For an overview of PL Topology, Bryant's article [8] is helpful. More survey articles can be found in the references of those articles. In the following, we give some information on the results in this chapter but it is restricted. For more information, refer to the articles [8, 9].

The example of Proposition 4.5.2 was given by C.H. Dowker [11]. The German terminology "Hauptvermutung" was introduced by Kneser [19], but it was claimed by Poincaré [28, 29], and formulated in 1908 as a conjecture by Steinitz [34] and Tietze [35]. In 1961, Hauptvermutung for polyhedra was disproved by J. Milnor [24]. In 1969, Kirby and Siebenmann [17] demonstrated that Hauptvermutung does not hold for n-manifolds for $n \geqslant 5$. Nevertheless, it was discovered in [32] that for any two countable simplicial complexes K and L, $|K| \simeq |L|$ implies $K \times_s \mathrm{F}(\mathbb{N}) \cong L \times_s \mathrm{F}(\mathbb{N})$, where $\mathrm{F}(\mathbb{N})$ is the countable infinite full complex. The completions of the metrics ρ_K and ρ_K^∞ on $|K|$ are discussed in [31].

A characterization of admissible subdivisions in Lemma 4.7.3 can be found in [15] and another in Theorem 4.7.9 in [25]. At the end of Sect. 4.7, we posed the Common Admissible Subdivision Problem, which arose during discussions with A. Yamashita.

Theorem 4.8.12 was established by J.H.C. Whitehead [38]. Theorem 4.9.5 was established by D.W. Henderson [15]. However, his proof is valid only for locally finite-dimensional complexes, and the proof presented here was the one given in [33], whose some gaps were corrected by A. Yamashita.

Remark 4.11.1 and the proof (b) $\Rightarrow$ (a) of Theorem 4.11.7 are due to Yamashita (personal communication). The name "tight PL map" was first introduced in the present book.

The nerves of open covers were introduced by P. Alexandroff [2, 3] and canonical maps by K. Kuratowski [20]. Theorems 4.15.7 and 4.15.11 were given by C.H. Dowker in [10, 11], respectively. In Theorem 4.15.7, it can be asserted that $\mathrm{id} : |K| \to |K|_{\mathrm{m}}$ is a **fine homotopy equivalence**, which will be defined in Sect. 6.7. This can be obtained by combining Theorems 4.15.7 and 4.9.5. For an alternative proof, refer to [30].

By H. Freudenthal [14], it was proved that every compactum X is the inverse limit of compact polyhedra, i.e., the limit of an inverse sequence of compact polyhedra (cf. Theorem 5.2.7 and Notes for Chap. 5). This was generalized to non-metrizable compact spaces (the limit of an inverse system of compact polyhedra) in the book by Eilenberg and Steenrod [12, Chap. X, Theorem 10.1]. M.D. Alder [1] generalized to paracompact spaces as Theorem 4.16.8 (cf. [4, 27]). For "Expansion in the Homotopy Category" at the end of Sect. 4.16, refer to the paper by K. Morita [26], where (i) and (ii) were proved.

In [16, 7.2], Theorem 4.17.7 was proved under the assumption that every open cover of X has an open refinement whose nerve is finite-dimensional. In fact, this is proved in a more general setting (for a complete uniform space). Theorem 4.17.10 is due to S. Mardešić [21]. The pseudo-arc in Footnote 40 was first constructed by Knaster [18]. In [5, 6], Bing showed that it is homogeneous and topologically unique, respectively.

The countable case of Theorem 4.19.3 was proved in the proof of [39, Theorem 13]. The simplicial mapping cylinder defined in Sect. 4.19 is different from the mapping cylinder of a simplicial map in [38]. Note that our collapsing is simplicial. Theorem 4.19.1 was first established by J.H.C. Whitehead [40] in the separable case and extended by J. Milnor [23] to the general case. The mapping cylinder technique used in the proof of Theorem 4.19.1 was essentially invented by Mather [22] (cf. [13]). The Borsuk conjecture in Remark 4.19.2 was posed in [7] and its proof was announced in [36] and given in [37].

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Chapter 5
Dimensions of Spaces



For a collection $\mathcal{A}$ of subsets of a space X, the **order** of $\mathcal{A}$ is defined as follows: when $n = \sup\{\text{card}\,\mathcal{A}[x] \mid x \in X\}$ is finite, we define $\text{ord}\,\mathcal{A} = n$; otherwise, we write $\text{ord}\,\mathcal{A} = \infty$. For an open cover $\mathcal{U} \in \text{cov}(X)$, $\text{ord}\,\mathcal{U} = \dim N(\mathcal{U}) + 1$, where $N(\mathcal{U})$ is the nerve of $\mathcal{U}$. The (**covering**) **dimension** of X, $\dim X$, is defined as follows: $\dim X \leqslant n$ if each *finite* open cover of X has a *finite* open refinement $\mathcal{U}$ with $\text{ord}\,\mathcal{U} \leqslant n + 1$, and then $\dim X = n$ if $\dim X \leqslant n$ and $\dim X \not\leqslant n$. By $\dim X = -1$, we mean that $X = \emptyset$. We say that X is ***n*-dimensional** if $\dim X = n$ and that X is **finite-dimensional** (**f.d.**) ($\dim X < \infty$) if $\dim X \leqslant n$ for some $n \in \omega$. Otherwise, X is said to be **infinite-dimensional** (**i.d.**) ($\dim X = \infty$). The dimension is a topological invariant (i.e., $\dim X = \dim Y$ if $X \approx Y$).

This chapter is devoted to lectures on Dimension Theory, where metrizable spaces are our main concern. Fundamental theorems are proved and various characterizations of dimension are given. We introduce inductively defined dimensions, $\text{ind}\,X$ and $\text{Ind}\,X$, which coincide with $\dim X$ for any separable metrizable space X. We construct a non-separable completely metrizable space X such that $\dim X = \text{Ind}\,X = 1$ but $\text{ind}\,X = 0$, and a non-metrizable compact space X such that $\dim X = 1$ but $\text{ind}\,X = \text{Ind}\,X \geqslant 2$. Moreover, several types of infinite dimensions are discussed. It is shown that every n-dimensional separable metrizable space has n-dimensional metrizable compactification and it can be embedded into $\mathbb{R}^{2n+1}$. We construct universal spaces such as the Nöbeling spaces and the Menger compacta, and also an infinite-dimensional compactum any non-empty subspace of which is either zero-dimensional or infinite-dimensional. In this context, we discuss the Brouwer Fixed Point Theorem and the characterization of the Cantor set. In addition, we show that every compactum is a continuous image of the Cantor set μ^0 and that every locally connected continuum (a Peano continuum) is a continuous image of $\mathbf{I} = [0, 1]$.

We will use the results in Chaps. 2 and 4. In particular, we will need the combinatorial techniques treated in Chap. 4. Also, the concept of the nerves of open covers is very important in Dimension Theory.

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5.1 The Brouwer Fixed Point Theorem

It is said that a space X has the **fixed point property** if any map $f : X \to X$ has a fixed point, i.e., $f(x) = x$ for some $x \in X$. In this section, we prove the following BROUWER FIXED POINT THEOREM:

Theorem 5.1.1 (BROUWER FIXED POINT) *For every $n \in \mathbb{N}$, the n-cube $\mathbf{I}^n$ has the fixed point property.*

To prove this theorem, we need two lemmas. Let K be a simplicial complex and K' a simplicial subdivision of K. A simplicial map $h : K' \to K$ is called a **Sperner map** if for each $v \in K'^{(0)}$, $h(v)$ is a vertex of the carrier $c_K(v)^{(0)}$ of v in K, equivalently $v \in O_K(h(v))$. In other words, h is a simplicial approximation of $\mathrm{id}_{|K|}$.

Indeed, for each $x \in |K'| = |K|$, $c_{K'}(x) \subset c_K(x)$. Since $c_K(v) \leqslant c_K(x)$ for every $v \in c_{K'}(x)^{(0)}$, it follows that $h(c_{K'}(x)^{(0)}) \subset c_K(x)^{(0)}$, hence $h(x) \in h(c_{K'}(x)) \leqslant c_K(x)$.

Lemma 5.1.2 (SPERNER) *Let τ be an n-simplex, and K' a subdivision of $F(\tau)$, where $F(\tau)$ is the natural triangulation of τ. If $h : K' \to F(\tau)$ is a Sperner map, then the number of n-simplexes $\tau' \in K'$ such that $h(\tau') = \tau$ is odd; hence, there exists such an n-simplex $\tau' \in K'$.*

Proof We prove the lemma by induction with respect to n. The case $n = 0$ is obvious. Assume the lemma has been established for any $(n-1)$-simplex. Let σ be an $(n-1)$-face of τ. Then, $h(\sigma) \subset \sigma$. The natural triangulation $F(\sigma)$ of σ is a subcomplex of $F(\tau)$. Let L' be the subdivision of $F(\sigma)$ induced by K'. As is easily observed, $h|\sigma : L' \to F(\sigma)$ is also a Sperner map. Let a be the number of $(n-1)$-simplexes $\sigma' \in L'$ such that $h(\sigma') = \sigma$. Then, a is odd by the inductive assumption. Let $\mathcal{S}$ be the set of all $(n-1)$-simplexes $\sigma' \in K'$ such that $h(\sigma') = \sigma$. For each n-simplex $\tau' \in K'$, let $b(\tau')$ denote the number of faces σ' of τ' that belong to $\mathcal{S}$, i.e., $h(\sigma') = \sigma$. Then, it follows that

$$b(\tau') = \begin{cases} 2 & \text{if } h(\tau') = \sigma; \\ 1 & \text{if } h(\tau') = \tau; \\ 0 & \text{otherwise.} \end{cases}$$

Let c be the number of n-simplexes $\tau' \in K'$ such that $h(\tau') = \tau$. Then,

$$\sum_{\tau' \in K' \setminus K'^{(n-1)}} b(\tau') - c \quad \text{is even.}$$

On the other hand, a is equal to the number of $(n-1)$-simplexes σ' of $\mathcal{S}$ such that $\sigma' \subset \sigma$. Each $\sigma' \in \mathcal{S}$ is a face of exactly two n-simplexes (resp. exactly one n-simplex) of K' if and only if $\sigma' \not\subset \sigma$ (resp. $\sigma' \subset \sigma$). Hence,

$$\sum_{\tau' \in K' \setminus K'^{(n-1)}} b(\tau') - a \quad \text{is even.}$$

Therefore, $a - c$ is also even. Recall that a is odd. Thus, c is also odd. □

Lemma 5.1.3 *Let $\tau = \langle v_1, \ldots, v_{n+1} \rangle$ be an n-simplex and $F_1, \ldots, F_{n+1}$ be closed sets in τ. If $\langle v_{i(1)}, \ldots, v_{i(m)} \rangle \subset F_{i(1)} \cup \cdots \cup F_{i(m)}$ for each $1 \leqslant i(1) < \cdots < i(m) \leqslant n+1$, then $F_1 \cap \cdots \cap F_{n+1} \neq \emptyset$.*

Proof Assume that $F_1 \cap \cdots \cap F_{n+1} = \emptyset$. Then,

$$\mathcal{U} = \{\tau \setminus F_1, \ldots, \tau \setminus F_{n+1}\} \in \mathrm{cov}(\tau).$$

Let K' be a simplicial subdivision of $F(\tau)$ that refines $\mathcal{U}$. For each $v \in K'^{(0)}$, choose a vertex v_i of the carrier of v in $F(\tau)$ so that $v \in F_i$, and let $h(v) = v_i$. Then, we have a Sperner map $h : K' \to F(\tau)$. By Lemma 5.1.2, there is a simplex $\tau' \in K'$ such that $h(\tau') = \tau$. Write $\tau' = \langle v'_1, \ldots, v'_{n+1} \rangle$ so that $h(v'_i) = v_i$. By the definition of h, $v'_i \in F_i$ for each $i = 1, \ldots, n+1$. Thus, τ' is not contained in any $\tau \setminus F_i$, which is a contradiction. □

Proof (*Theorem* 5.1.1) It suffices to show that any map $f : \Delta^n \to \Delta^n$ has a fixed point, where $\Delta^n \subset \mathbb{R}^{n+1}$ is the standard n-simplex. For each $i = 1, \ldots, n+1$, let

$$F_i = \{x \in \Delta^n \mid \mathrm{pr}_i(f(x)) \leqslant \mathrm{pr}_i(x)\},$$

where $\mathrm{pr}_i : \mathbb{R}^{n+1} \to \mathbb{R}$ is the projection onto the i-th factor. Then, F_i is closed in Δ^n. Moreover, each face $\sigma = \langle \mathbf{e}_{i(1)}, \ldots, \mathbf{e}_{i(m)} \rangle \leqslant \Delta^n$ is contained in $F_{i(1)} \cup \cdots \cup F_{i(m)}$, where $\{\mathbf{e}_1, \ldots, \mathbf{e}_{n+1}\}$ is the canonical orthonormal basis for $\mathbb{R}^{n+1}$. In fact, if $x \in \sigma$ then

$$\sum_{j=1}^{m} \mathrm{pr}_{i(j)}(f(x)) \leqslant 1 = \sum_{j=1}^{m} \mathrm{pr}_{i(j)}(x),$$

which implies that $\mathrm{pr}_{i(j)}(f(x)) \leqslant \mathrm{pr}_{i(j)}(x)$ for some $j = 1, \ldots, m$. By Lemma 5.1.3, we have a point $a \in F_1 \cap \cdots \cap F_{n+1}$. Since $0 \leqslant \mathrm{pr}_i(f(a)) \leqslant \mathrm{pr}_i(a)$ for each $i = 1, \ldots, n+1$ and

$$\sum_{i=1}^{n+1} \mathrm{pr}_i(f(a)) = 1 = \sum_{i=1}^{n+1} \mathrm{pr}_i(a),$$

it follows that $\mathrm{pr}_i(f(a)) = \mathrm{pr}_i(a)$ for each $i = 1, \ldots, n+1$, which means that $f(a) = a$. □

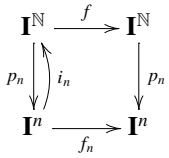
The following is the infinite-dimensional version of Theorem 5.1.1:

Corollary 5.1.4 *The Hilbert cube $\mathbf{I}^{\mathbb{N}}$ has the fixed point property.*

Proof For each $n \in \mathbb{N}$, let $p_n : \mathbf{I}^{\mathbb{N}} \to \mathbf{I}^n$ be the projection onto the first n factors and $i_n : \mathbf{I}^n \to \mathbf{I}^{\mathbb{N}}$ the natural injection defined by

$$i_n(x) = (x(1), \ldots, x(n), 0, 0, \ldots).$$

For each map $f : \mathbf{I}^{\mathbb{N}} \to \mathbf{I}^{\mathbb{N}}$, consider the map $f_n = p_n f i_n : \mathbf{I}^n \to \mathbf{I}^n$.



By the Brouwer Fixed Point Theorem 5.1.1, f_n has a fixed point. We define

$$K_n = \left\{x \in \mathbf{I}^{\mathbb{N}} \;\middle|\; p_n f(x) = p_n(x)\right\},$$

which is closed in $\mathbf{I}^{\mathbb{N}}$ and $K_n \supset K_{n+1}$ for each $n \in \mathbb{N}$. Moreover, $K_n \neq \emptyset$. Indeed, if $y \in \mathbf{I}^n$ is a fixed point of f_n, then $p_n f(i_n(y)) = f_n(y) = y = p_n(i_n(y))$, i.e., $i_n(y) \in K_n$. By compactness, we have $a \in \bigcap_{n\in\mathbb{N}} K_n$. Since $p_n f(a) = p_n(a)$ for every $n \in \mathbb{N}$, we have $f(a) = a$. □

As another corollary of the Brouwer Fixed Point Theorem 5.1.1, we have the following NO RETRACTION THEOREM:

Corollary 5.1.5 (NO RETRACTION) *There does not exist any map $r : \mathbf{B}^n \to \mathbf{S}^{n-1}$ with $r|\mathbf{S}^{n-1} = \mathrm{id}$.*[1]

Proof Suppose that there exists a map $r : \mathbf{B}^n \to \mathbf{S}^{n-1}$ with $r|\mathbf{S}^{n-1} = \mathrm{id}$. We define a map $f : \mathbf{B}^n \to \mathbf{B}^n$ by $f(x) = -r(x)$. Then, f has no fixed points, which contradicts the Brouwer Fixed Point Theorem 5.1.1. □

Remark 5.1.1 It should be noted that the Brouwer Fixed Point Theorem 5.1.1 can be derived from the No Retraction Theorem 5.1.5. Indeed, if there is a map $f : \mathbf{B}^n \to \mathbf{B}^n$ without fixed points, then we have a map $r : \mathbf{B}^n \to \mathbf{S}^{n-1}$ such that $x \in \langle f(x), r(x)\rangle$ for each $x \in \mathbf{B}^n$, which implies that $r|\mathbf{S}^{n-1} = \mathrm{id}$ (see Fig. 5.1).

In fact, such a map r can be defined as follows:

$$r(x) = (1 + \alpha(x))x - \alpha(x) f(x),$$

where $\alpha(x) \geqslant 0$ can be obtained by solving the equation

$$\alpha(x)^2 \|x - f(x)\|^2 + 2\alpha(x)\langle x - f(x), x\rangle + \|x\|^2 - 1 = 0,$$

where $\langle y, z\rangle = \sum_{i=1}^{n} y(i)z(i)$ is the inner product.

Therefore, the No Retraction Theorem 5.1.5 implies that $\mathbf{I}^n \approx \mathbf{B}^n$ has the fixed point property. Thus, the Brouwer Fixed Point Theorem 5.1.1 and the No Retraction Theorem 5.1.5 are equivalent.

[1] Such a map r is called a **retraction**, which will be discussed in Chapter 6.

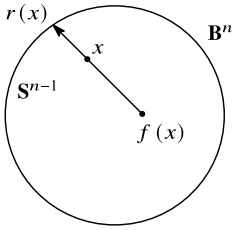


Fig. 5.1 The construction of r

Note In Algebraic Topology, the homotopy groups or the homology groups are used to prove the No Retraction Theorem 5.1.5, and then the Brouwer Fixed Point Theorem 5.1.1 is proved as in Remark 5.1.1.

Using the Tietze–Urysohn Extension Theorem 2.2.2, we have the following extension theorem:

Theorem 5.1.6[2] *Let A be a closed set in a normal space X and $n \in \mathbb{N}$.*

(1) *Every map $f : A \to \mathbf{B}^n$ extends over X.*
(2) *Every map $f : A \to \mathbf{S}^{n-1}$ extends over a neighborhood of A in X.*

Sketch of Proof (1): By the coordinate-wise application of the Tietze–Urysohn Extension Theorem 2.2.2, each map $f : A \to \mathbf{I}^n \approx \mathbf{B}^n$ can be extended over X.

(2): Let $f : A \to \mathbf{S}^{n-1}$ be a map. By (1), f extends to a map $\tilde{f} : X \to \mathbf{B}^n$. Then, $W = \tilde{f}^{-1}(\mathbf{B}^n \setminus \{0\})$ is an open neighborhood of A in X. Let $r : \mathbf{B}^n \setminus \{0\} \to \mathbf{S}^{n-1}$ be the radial projection, i.e., $r(x) = x/\|x\|$. Then, $r\tilde{f}|W : W \to \mathbf{S}^{n-1}$ is an extension of f.

Applying Theorem 5.1.6, the Simplicial Approximation Theorem 4.10.11, and the Homotopy Extension Theorem 4.5.3, we prove the following:

Lemma 5.1.7 *Let A be a closed set in $\mathbf{S}^{n-1}$. Then, every map $f : A \to \mathbf{S}^{n-1}$ extends to a map $\tilde{f} : \mathbf{S}^{n-1} \to \mathbf{S}^{n-1}$.*

Proof Since $\mathbf{S}^{n-1} \approx \partial\Delta^n$, we may replace $\mathbf{S}^{n-1}$ with $\partial\Delta^n$. Let A be a closed set in $\partial\Delta^n$. Due to Theorem 5.1.6(2), every map $f : A \to \partial\Delta^n$ extends to a map $f' : U \to \partial\Delta^n$ for some neighborhood U of A in $\partial\Delta^n$. By Theorem 4.8.6, we can take $K \lhd \partial F(\Delta^n)$ so that $|N(A, K)| \subset U$. According to the Simplicial Approximation Theorem 4.10.11, $f'||N(A, K)|$ has a simplicial approximation $g : N' \to \partial F(\Delta^n)$ for some $N' \lhd N(A, K)$. By Proposition 4.3.11, N' is a subcomplex of some $K' \lhd K$. Since $\dim K' = n - 1$ and any n vertices of $\partial F(\Delta^n)$ span a simplex of $\partial F(\Delta^n)$, g extends to a simplicial map $\tilde{g} : K' \to \partial F(\Delta^n)$. Since $f'||N(A, K)| \simeq g$, we can apply the Homotopy Extension Theorem 4.5.3 to extend $f'||N(A, K)|$ to a map $\tilde{f} : |K| = \partial\Delta^n \to \partial\Delta^n$. □

[2] This theorem can be stated as: $\mathbf{B}^n$ *is an AE for normal spaces and* $\mathbf{S}^{n-1}$ *is an ANE for normal spaces*. For an AE (ANE) for normal spaces, refer to the preface to Chap. 6.

Using Theorem 5.1.6, Lemma 5.1.7, and the No Retraction Theorem 5.1.5, we can obtain the following characterization of boundary points of a closed set X in Euclidean space $\mathbb{R}^n$:

Theorem 5.1.8 *Let X be a closed subset of Euclidean space $\mathbb{R}^n$. For a point $x \in X$, $x \in \operatorname{bd} X$ if and only if each neighborhood U of x in X contains a neighborhood V of x in X such that every map $f : X \setminus V \to \mathbf{S}^{n-1}$ extends to a map $\tilde{f} : X \to \mathbf{S}^{n-1}$.*

Proof (*The "only if" Part*) For each neighborhood U of x in X, choose $\varepsilon > 0$ so that $\overline{\mathrm{B}}(x, \varepsilon) \cap X \subset U$. Since $x \in \operatorname{bd} X$, we have $z \in \mathrm{B}(x, \varepsilon) \setminus X$. Let $\delta = \|x - z\| > 0$. Then, $z \in \partial\mathrm{B}(x, \delta) \approx \mathbf{S}^{n-1}$ and $X \cap \overline{\mathrm{B}}(x, \delta) \subset U$. Choose $0 < \lambda < \delta$ so that $\overline{\mathrm{B}}(z, \lambda) \cap X = \emptyset$. Since $\partial\mathrm{B}(x, \delta) \setminus \mathrm{B}(z, \lambda) \approx \mathbf{B}^{n-1}$, we have a retraction $r : \overline{\mathrm{B}}(x, \delta) \to \partial\mathrm{B}(x, \delta) \setminus \mathrm{B}(z, \lambda)$ by Theorem 5.1.6(1), which can be extended to a retraction

$$r' : X \cup \overline{\mathrm{B}}(x, \delta) \to (X \setminus \mathrm{B}(x, \delta)) \cup (\partial\mathrm{B}(x, \delta) \setminus \mathrm{B}(z, \lambda)).$$

Then, $V = X \cap \mathrm{B}(x, \delta)$ is the desired neighborhood of x in X. Indeed, by Lemma 5.1.7, every map $f : X \setminus V = X \setminus \mathrm{B}(x, \delta) \to \mathbf{S}^{n-1}$ can be extended to a map $f' : (X \setminus V) \cup \partial\mathrm{B}(x, \delta) \to \mathbf{S}^{n-1}$. The map $f'r'|X : X \to \mathbf{S}^{n-1}$ is an extension of f.

(*The "if" Part*) Assume that $x \in \operatorname{int} X$. Then, $\overline{\mathrm{B}}(x, \delta) \subset X$ for some $\delta > 0$. By the condition, $\mathrm{B}(x, \delta)$ contains a neighborhood V of x such that every map $f : X \setminus V \to \mathbf{S}^{n-1}$ extends to a map $\tilde{f} : X \to \mathbf{S}^{n-1}$. We have the radial projection

$$r : \mathbb{R}^n \setminus \{x\} \to \partial\mathrm{B}(x, \delta) \approx \mathbf{S}^{n-1},$$

which is a retraction. Then, $r|X \setminus V$ extends to a retraction $\tilde{r} : X \to \partial\mathrm{B}(x, \delta)$. Since $\overline{\mathrm{B}}(x, \delta) \subset X$, it follows that $\partial\mathrm{B}(x, \delta)$ is a retract of $\overline{\mathrm{B}}(x, \delta)$, which contradicts the No Retraction Theorem 5.1.5 because $(\overline{\mathrm{B}}(x, \delta), \partial\mathrm{B}(x, \delta)) \approx (\mathbf{B}^n, \mathbf{S}^{n-1})$. Thus, we have $x \in \operatorname{bd} X$. □

Applying Theorem 5.1.8, we prove the following:

Theorem 5.1.9 *For any open set $U \subset \mathbb{R}^n$, every continuous injection $f : U \to \mathbb{R}^n$ is an open embedding.*

Proof Let $x \in U$ and take any compact neighborhood C of x in $\mathbb{R}^n$ with $C \subset U$. Then, $f|C : C \to f(C)$ is a homeomorphism. We show that $f(x) \in \operatorname{int} f(C)$. On the contrary, assume that $f(x) \in \operatorname{bd} f(C)$. For each neighborhood V of x in C, since $f(V)$ is a neighborhood of $f(x)$ in $f(C)$, we can apply Theorem 5.1.8 to find a neighborhood W of $f(x)$ in $f(C)$ such that $W \subset f(V)$ and every map $g : f(C) \setminus W \to \mathbf{S}^{n-1}$ extends to a map $\tilde{g} : f(C) \to \mathbf{S}^{n-1}$. Then, $f^{-1}(W)$ is a neighborhood of x in C with $f^{-1}(W) \subset V$. For every map $h : C \setminus f^{-1}(W) \to \mathbf{S}^{n-1}$, $hf^{-1} : f(C) \setminus W \to \mathbf{S}^{n-1}$ extends to a map $\tilde{g} : f(C) \to \mathbf{S}^{n-1}$. Thus, we have a map $\tilde{g}f : C \to \mathbf{S}^{n-1}$ such that

$$\tilde{g} f | C \setminus f^{-1}(W) = (\tilde{g} | f(C) \setminus W)(f | C \setminus f^{-1}(W))$$
$$= h f^{-1} f | C \setminus f^{-1}(W) = h.$$

Due to Theorem 5.1.8, this means that $x \in \operatorname{bd} C$, which is a contradiction. Therefore, $f(C)$ is a neighborhood of x in $\mathbb{R}^n$ and $f(C) \subset f(U)$. Consequently, $f(U)$ is open in $\mathbb{R}^n$ and $f : U \to f(U)$ is a homeomorphism, hence f is an open embedding. □

Applying Theorem 5.1.9, we can easily obtain BROUWER'S THEOREM ON INVARIANCE OF DOMAIN, simply called the INVARIANCE OF DOMAIN:

Theorem 5.1.10 (INVARIANCE OF DOMAIN) *For each pair of subsets $X, Y \subset \mathbb{R}^n$, $X \approx Y$ implies* $\operatorname{int} X \approx \operatorname{int} Y$. *In fact,* $h(\operatorname{int} X) = \operatorname{int} Y$ *for any homeomorphism* $h : X \to Y$.

Sketch of Proof Just apply Theorem 5.1.9 to $h | \operatorname{int} X$ and $h^{-1} | \operatorname{int} Y$.

5.2 Characterizations of Dimension

Recall that we define $\dim X \leqslant n$ if each *finite* open cover of X has a *finite* open refinement $\mathcal{U}$ with $\operatorname{ord} \mathcal{U} \leqslant n + 1$. The following lemma shows that the refinement $\mathcal{U}$ in this definition need not be finite.

Lemma 5.2.1 *Let $\mathcal{U}$ be an open cover of a space X and $\mathcal{V}$ an open refinement of $\mathcal{U}$. Then, $\mathcal{U}$ has an open refinement $\mathcal{W} = \{W_U \mid U \in \mathcal{U}\}$ such that $W_U \subset U$ for each $U \in \mathcal{U}$ and* $\operatorname{card} \mathcal{W}[x] \leqslant \operatorname{card} \mathcal{V}[x]$ *for each $x \in X$, which implies that* $\operatorname{ord} \mathcal{W} \leqslant \operatorname{ord} \mathcal{V}$ *and if $\mathcal{U}$ is (locally) finite (or σ-discrete) then so is $\mathcal{W}$.*

Proof Let $\varphi : \mathcal{V} \to \mathcal{U}$ be a function such that $V \subset \varphi(V)$ for each $V \in \mathcal{V}$. For each $U \in \mathcal{U}$, define

$$W_U = \bigcup \varphi^{-1}(U) = \bigcup \big\{ V \in \mathcal{V} \,\big|\, \varphi(V) = U \big\}.$$

Then, $\mathcal{W} = \{W_U \mid U \in \mathcal{U}\}$ is the desired refinement of $\mathcal{U}$. □

The following is a finite case of the Open Cover Shrinking Lemma 2.7.6, which is easily proved directly.

Lemma 5.2.2 *Every finite open cover $\{U_1, \ldots, U_n\}$ of a normal space X has a strong open shrinking $\{V_1, \ldots, V_n\}$, where* $\{\operatorname{cl} V_1, \ldots, \operatorname{cl} V_n\}$ *is a closed shrinking of* $\{U_1, \ldots, U_n\}$.

Sketch of Proof Using the normality of X, V_i can be inductively chosen so that

$$\operatorname{cl} V_i \subset U_i \quad \text{and} \quad V_1 \cup \cdots \cup V_i \cup U_{i+1} \cup \cdots \cup U_n = X.$$

Let $\mathcal{A} = \{A_\lambda \mid \lambda \in \Lambda\}$ be a collection of subsets of a space X. A **swelling** of $\mathcal{A}$ in X is a collection $\mathcal{B} = \{B_\lambda \mid \lambda \in \Lambda\}$ of subsets of X such that $A_\lambda \subset B_\lambda$ for each $\lambda \in \Lambda$ and for every finitely many $\lambda_1, \ldots, \lambda_n \in \Lambda$,

$$A_{\lambda_1} \cap \cdots \cap A_{\lambda_n} \neq \emptyset \Leftrightarrow B_{\lambda_1} \cap \cdots \cap B_{\lambda_n} \neq \emptyset.$$

A swelling $\mathcal{B}$ is said to be **open** or **closed** depending on whether every member of $\mathcal{B}$ is open or closed. A closed swelling $\mathcal{B}$ is said to be **strong** if each B_λ is a neighborhood of A_λ. From the definition, we have $\operatorname{ord} \mathcal{A} = \operatorname{ord} \mathcal{B}$ for every swelling $\mathcal{B}$ of $\mathcal{A}$. The following is called the SWELLING LEMMA:

Lemma 5.2.3 (SWELLING) *Every finite closed collection $\{A_1, \ldots, A_n\}$ of a normal space X has a strong closed swelling $\{B_1, \ldots, B_n\}$. Moreover, if open neighborhoods U_i of A_i, $i = 1, \ldots, n$, are given, then the swelling can be taken so that $B_i \subset U_i$.*

Proof Let C_1 be the union of all intersections $A_{i_1} \cap \cdots \cap A_{i_k}$ disjoint from A_1. Since X is normal, A_1 has a closed neighborhood B_1 in X such that $B_1 \subset (X \setminus C_1) \cap U_1$. Then, $\{B_1, A_2, \ldots, A_n\}$ is a swelling of $\{A_1, \ldots, A_n\}$.

Assume that for each $i < j \leqslant n$, A_i has a closed neighborhood B_i in X such that $B_i \subset U_i$ and $\{B_1, \ldots, B_{j-1}, A_j, \ldots, A_n\}$ is a swelling of $\{A_1, \ldots, A_n\}$. Applying the above argument to $\{A_j, \ldots, A_n, B_1, \ldots, B_{j-1}\}$ and A_j, we have a closed neighborhood B_j of A_j in X such that $B_j \subset U_j$ and $\{B_1, \ldots, B_j, A_{j+1}, \ldots, A_n\}$ is a swelling of $\{A_1, \ldots, A_n\}$. By induction, we can obtain a strong closed swelling $\{B_1, \ldots, B_n\}$ of $\{A_1, \ldots, A_n\}$. □

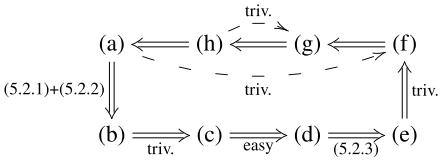
We now prove the following characterizations of dimension:

Theorem 5.2.4 *For $n \in \omega$ and a normal space X, the following are equivalent:*

(a) $\dim X \leqslant n$;
(b) *Every finite open cover $\{U_1, \ldots, U_k\}$ of X has a strong open shrinking $\mathcal{V} = \{V_1, \ldots, V_k\}$ with $\operatorname{ord} \mathcal{V} \leqslant n + 1$.*
(c) *Every open cover $\{U_1, \ldots, U_{n+2}\}$ of X has a strong open shrinking $\{V_1, \ldots, V_{n+2}\}$ with $V_1 \cap \cdots \cap V_{n+2} = \emptyset$.*
(d) *Every open cover $\{U_1, \ldots, U_{n+2}\}$ of X has a closed shrinking $\{A_1, \ldots, A_{n+2}\}$ with $A_1 \cap \cdots \cap A_{n+2} = \emptyset$.*
(e) *Every open cover $\{U_1, \ldots, U_{n+2}\}$ of X has an open shrinking $\{V_1, \ldots, V_{n+2}\}$ with $V_1 \cap \cdots \cap V_{n+2} = \emptyset$.*
(f) *Every open cover $\{U_1, \ldots, U_{n+2}\}$ of X has an open refinement $\mathcal{V}$ with $\operatorname{ord} \mathcal{V} \leqslant n + 1$.*
(g) *Each map $f : A \to \mathbf{S}^n$ of any closed set A in X extends over X.*
(h) *For every $k \geqslant n$, each map $f : A \to \mathbf{S}^k$ of any closed set A in X extends over X.*

Proof The implications (b) ⇒ (c), (d), and (e) ⇒ (f) are trivial. (Additionally, (a) ⇒ (f) and (h) ⇒ (g) are also trivial). Since using (c) twice yields (d), we have (c) ⇒ (d). By Lemmas 5.2.1 and 5.2.2, we have (a) ⇒ (b). The implication (d) ⇒ (e)

follows from the Swelling Lemma 5.2.3. We will prove the implications (f) $\Rightarrow$ (g) $\Rightarrow$ (h) $\Rightarrow$ (a). See the following diagram of implications:



(f) $\Rightarrow$ (g): Let Δ^{n+1} be the standard $(n+1)$-simplex and $K = \partial F(\Delta^{n+1})$ (i.e., the simplicial complex consisting of all proper faces of Δ^{n+1}). Then, $|K| = \partial\Delta^{n+1} \approx \mathbf{S}^n$. To extend a given map $f : A \to \mathbf{S}^n$ over X, we consider $\mathbf{S}^n = |K|$. By Theorem 5.1.6 (2), $f : A \to |K|$ is extended to a map $\tilde{f} : \operatorname{cl} W \to |K|$, where W is an open neighborhood of A in X. Note that $\operatorname{card} K^{(0)} = n+2$. By (f), X has a finite open cover $\mathcal{V}$ such that $\operatorname{ord}\mathcal{V} \leqslant n+1$ and

$$\mathcal{V} \prec \left\{ \tilde{f}^{-1}(O_K(v)) \cup (X \setminus \operatorname{cl} W) \;\middle|\; v \in K^{(0)} \right\}.$$

We have a function $\varphi : \mathcal{V} \to K^{(0)}$ such that

$$V \subset \tilde{f}^{-1}(O_K(\varphi(V))) \cup (X \setminus \operatorname{cl} W) \quad \text{for each } V \in \mathcal{V},$$

which defines a simplicial map $\varphi : N(\mathcal{V}) \to K$ because every $n+1$ vertices span a simplex of K and each simplex of $N(\mathcal{V})$ has at most $n+1$ vertices. Since $\mathcal{V}$ is finite, there is a canonical map $g : X \to |N(\mathcal{V})|$ for $N(\mathcal{V})$ by Corollary 4.15.6. For each $x \in \operatorname{cl} W$, $\tilde{f}(x)$ and $\varphi g(x)$ are contained in the same simplex of K. In fact, let $\tau \in K$ be the carrier of $\tilde{f}(x)$, i.e., $\tilde{f}(x) \in \operatorname{rint}\tau$. Then, for each $V \in \mathcal{V}[x]$,

$$x \in V \cap \operatorname{cl} W \subset \tilde{f}^{-1}(O_K(\varphi(V))),$$

hence $\tilde{f}(x) \in O_K(\varphi(V))$. Thus, we have $\tilde{f}(x) \in \bigcap_{V\in\mathcal{V}[x]} O_K(\varphi(V))$, which implies that $\varphi(V) \in \tau^{(0)}$ for each $V \in \mathcal{V}[x]$, i.e., $\langle\varphi(\mathcal{V}[x])\rangle \leqslant \tau$. On the other hand, $g(x) \in \langle\mathcal{V}[x]\rangle$, which implies

$$\varphi g(x) \in \varphi(\langle\mathcal{V}[x]\rangle) = \langle\varphi(\mathcal{V}[x])\rangle \leqslant \tau,$$

so $\varphi g(x), \tilde{f}(x) \in \tau$. Thus, we can define a map $h : X \times \{0\} \cup \operatorname{cl} W \times \mathbf{I} \to |K|$ as follows:

$$h(x,0) = \varphi g(x) \quad \text{for } x \in X \text{ and}$$
$$h(x,t) = (1-t)\varphi g(x) + t\tilde{f}(x) \quad \text{for } (x,t) \in \operatorname{cl} W \times \mathbf{I}.$$

Let $k : X \to \mathbf{I}$ be a Urysohn map with $X \setminus W \subset k^{-1}(0)$ and $A \subset k^{-1}(1)$. Then, an extension $f^* : X \to |K|$ of f can be defined by $f^*(x) = h(x, k(x))$ for $x \in X$.

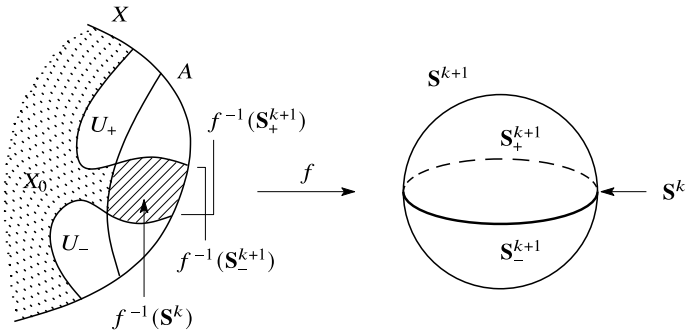


Fig. 5.2 Extending a map $f : A \to \mathbf{S}^{k+1}$

(g) $\Rightarrow$ (h): By induction on $k \geqslant n$, we show that each map $f : A \to \mathbf{S}^{k+1}$ of any closed set A in X extends over X. Let

$$\mathbf{S}^{k+1}_+ = \mathbf{S}^{k+1} \cap (\mathbb{R}^{k+1} \times \mathbb{R}_+) \text{ and } \mathbf{S}^{k+1}_- = -\mathbf{S}^{k+1}_+,$$

where we identify $\mathbf{S}^k = \mathbf{S}^k \times \{0\} = \mathbf{S}^{k+1}_+ \cap \mathbf{S}^{k+1}_- \subset \mathbf{S}^{k+1}$. We have disjoint open sets U_+ and U_- in X such that

$$U_+ \cap A = A \setminus f^{-1}(\mathbf{S}^{k+1}_-) \text{ and } U_- \cap A = A \setminus f^{-1}(\mathbf{S}^{k+1}_+).$$

In fact, by Theorem 5.1.6 (2), f extends to a map $f' : U \to \mathbf{S}^{k+1}$ of an open neighborhood of A in X. Then, $U_\pm = f'^{-1}(\mathbf{S}^{k+1} \setminus \mathbf{S}^{k+1}_\mp)$ are the desired open sets.

Now, let $X_0 = X \setminus (U_+ \cup U_-)$ and $A_0 = A \cap X_0 = f^{-1}(\mathbf{S}^k)$ (see Fig. 5.2). Since $f|A_0 : A_0 \to \mathbf{S}^k$ extends over X by the inductive assumption, $f|A_0$ extends to a map $f_0 : X_0 \to \mathbf{S}^k$. Then, f_0 has the following extensions:

$$f_+ : X_0 \cup f^{-1}(\mathbf{S}^{k+1}_+) \to \mathbf{S}^{k+1}_+ \text{ and } f_- : X_0 \cup f^{-1}(\mathbf{S}^{k+1}_-) \to \mathbf{S}^{k+1}_-$$

defined by $f_\pm | f^{-1}(\mathbf{S}^{k+1}_\pm) = f | f^{-1}(\mathbf{S}^{k+1}_\pm)$. Let

$$X_+ = X_0 \cup U_+ = X \setminus U_- \text{ and } X_- = X_0 \cup U_- = X \setminus U_+,$$

which are closed in X, and hence they are normal. Note that $X_0 \cup f^{-1}(\mathbf{S}^{k+1}_\pm)$ is closed in $X_\pm$. Since $\mathbf{S}^{k+1}_\pm \approx \mathbf{B}^{k+1}$, $f_\pm$ extend to maps $\tilde{f}_\pm : X_\pm \to \mathbf{S}^{k+1}_\pm$ by Theorem 5.1.6 (1). Since $X = X_+ \cup X_-$, $X_0 = X_+ \cap X_-$, and $\tilde{f}_\pm | X_0 = f_\pm | X_0 = f_0$, the desired extension $\tilde{f} : X \to \mathbf{S}^{k+1}$ of f can be defined by $\tilde{f} | X_\pm = \tilde{f}_\pm$.

(h) $\Rightarrow$ (a): For each finite open cover $\mathcal{U}$ of X, let $K = N(\mathcal{U})$ be the nerve of $\mathcal{U}$ with $f : X \to |K|$ a canonical map (cf. Corollary 4.15.6). If $f(X) \subset |K^{(n)}|$, $f^{-1}(\mathcal{O}_{K^{(n)}}) \in \mathrm{cov}(X)$ is a finite open refinement of $\mathcal{U}$ and

$$\operatorname{ord} f^{-1}(\mathcal{O}_{K^{(n)}}) \leqslant \operatorname{ord} \mathcal{O}_{K^{(n)}} = \dim K^{(n)} + 1 \leqslant n + 1.$$

Otherwise, choose $m > n$ so that $f(X) \subset |K^{(m)}|$ but $f(X) \not\subset |K^{(m-1)}|$. Let $\tau_1, \ldots, \tau_k$ be the m-simplexes of K. Since $\partial\tau_i \approx \mathbf{S}^{m-1}$ and $m - 1 \geqslant n$, we have maps $f_i : X \to \partial\tau_i$ such that $f_i | f^{-1}(\partial\tau_i) = f | f^{-1}(\partial\tau_i)$ by (h). Let $f' : X \to |K|$ be the map defined by

$$f' | f^{-1}(|K^{(m-1)}|) = f | f^{-1}(|K^{(m-1)}|) \text{ and}$$
$$f' | f^{-1}(\tau_i) = f_i | f^{-1}(\tau_i) \text{ for each } i = 1, \ldots, k$$

Then, $f'(X) \subset |K^{(m-1)}|$. Since $c_K(f'(x)) \leqslant c_K(f(x)) \leqslant \langle \mathcal{U}[x] \rangle$ for each $x \in X$, f' is still a canonical map. By the downward induction on $m \geqslant n$, we can obtain a canonical map $f : X \to |K|$ such that $f(X) \subset |K^{(n)}|$. This completes the proof. □

Remark 5.2.1 In the above proof of (h) ⇒ (a), instead of a finite open cover $\mathcal{U}$ of X, let us take a locally finite open cover $\mathcal{U}$ whose nerve $N(\mathcal{U})$ is *locally finite-dimensional* (l.f.d.). It can be shown that $\mathcal{U}$ has a locally finite open refinement $\mathcal{V}$ with $\operatorname{ord} \mathcal{V} \leqslant n + 1$ (i.e., $\dim N(\mathcal{V}) \leqslant n$).

Indeed, let $K = N(\mathcal{U})$, where K might be infinite-dimensional but l.f.d. By Corollary 4.15.6, we have a canonical map $f : X \to |K|$ such that each $x \in X$ has a neighborhood V_x in X with $f(V_x) \subset |K_x|$ for some finite subcomplex K_x of K. Now, inductively define the following subcomplexes of K:

$$\begin{aligned} K_i &= K \setminus \left\{ \tau \in K \;\middle|\; \dim \tau > n, \tau \text{ is principal in } K_{i-1} \right\} \\ &= K^{(n)} \cup \left\{ \tau \in K_{i-1} \;\middle|\; \tau \text{ is not principal in } K_{i-1} \right\}, \; i \in \mathbb{N}, \end{aligned}$$

where $K_0 = K$. Then, $K^{(n)} = \bigcap_{i \in \mathbb{N}} K_i$ because K is l.f.d. We will inductively construct canonical maps $f_i : X \to |K|, i \in \mathbb{N}$, such that

$$f_i | f_{i-1}^{-1}(|K_i|) = f_{i-1} | f_{i-1}^{-1}(|K_i|), \; f_i(X) \subset |K_i| \text{ and}$$
$$f_i(V_x) \subset |K_x| \text{ for each } x \in X,$$

where $f_0 = f$. Suppose f_{i-1} have been constructed. For each $\tau \in K_{i-1} \setminus K_i$, since $\dim \tau > n$, we can apply (h) to obtain an extension $f_\tau : X \to \partial\tau$ of $f_{i-1} | f_{i-1}^{-1}(\partial\tau)$. We can define $f_i : X \to |K|$ as follows:

$$f_i | f_{i-1}^{-1}(|K_i|) = f_{i-1} | f_{i-1}^{-1}(|K_i|) \text{ and}$$
$$f_i | f_{i-1}^{-1}(\tau) = f_\tau | f_{i-1}^{-1}(\tau) \text{ for each } \tau \in K_{i-1} \setminus K_i.$$

Then, $f_i(X) \subset |K_i|$. Since $f_i(f_{i-1}^{-1}(\tau)) \subset \partial\tau \subset \tau$ for each $\tau \in K_{i-1} \setminus K_i$, it follows that $f_i(V_x) \subset |K_x|$ for each $x \in X$, so f_i is continuous because each K_x is finite. Moreover, $c_K(f_i(x)) \leqslant c_K(f_{i-1}(x))$ for each $x \in X$, hence f_i is also a canonical map.

For each $x \in X$, since K_x is finite, $K_x^{(n)} = K_x \cap K_{i(x)}$ for some $i(x) \in \mathbb{N}$. For every $i \geqslant i(x)$, because $K_x \cap K_i = K_x \cap K_{i(x)}$, we have $f_i | V_x = f_{i(x)} | V_x$. Therefore, we can define a map $\tilde{f} : X \to |K^{(n)}|$ by $\tilde{f} | V_x = f_{i(x)} | V_x$ for each $x \in X$. Then, $\tilde{f}^{-1}(\mathcal{O}_{K^{(n)}}) \in \operatorname{cov}(X)$ is an open refinement of $\mathcal{U}$ with $\operatorname{ord} \leqslant n + 1$. By applying Lemma 5.2.1, we can obtain the desired refinement $\mathcal{V}$ of $\mathcal{U}$.

When X is paracompact, since every open cover of X has a locally finite (and σ-discrete) open refinement with the l.f.d. nerve by Theorem 4.15.11, if $\dim X \leqslant n$, then an arbitrary open cover of X has a (locally finite σ-discrete) open refinement $\mathcal{V}$ with $\operatorname{ord}\mathcal{V} \leqslant n+1$ by the above remark. Since the converse obviously holds, we have the following characterization:

Theorem 5.2.5 *For $n \in \omega$ and a paracompact space X, $\dim X \leqslant n$ if and only if an arbitrary open cover of X has a (locally finite σ-discrete) open refinement $\mathcal{V}$ with $\operatorname{ord}\mathcal{V} \leqslant n+1$.* □

Instead of Theorem 4.15.11, we can use Theorem 4.15.12 to obtain the following corollary:

Corollary 5.2.6 *Let X be regular Lindelöf and $n \in \omega$. Then, $\dim X \leqslant n$ if and only if an arbitrary open cover of X has a (star-finite and countable) open refinement $\mathcal{V}$ with $\operatorname{ord}\mathcal{V} \leqslant n+1$.* □

In the proof of Theorem 4.17.7, if $\dim X \leqslant n$, we can apply Theorem 5.2.5 (Corollary 5.2.6) to obtain $\mathcal{U}_i$ with $\operatorname{ord}\mathcal{U}_i \leqslant n+1$, namely, $\dim K_i \leqslant n$. By Remark 4.17.1, we have the following version of Theorem 4.17.7 (Corollaries 4.17.8 and 4.17.9):

Theorem 5.2.7 *Every completely metrizable space X with $\dim X \leqslant n < \infty$ is homeomorphic to the inverse limit of an inverse sequence $(|K_i|_{\mathrm{m}}, f_i)_{i\in\mathbb{N}}$ of metric polyhedra and PL maps such that $\dim K_i \leqslant n$, $\operatorname{card} K_i \leqslant \aleph_0 w(X)$, and $f_i : K_{i+1} \to \operatorname{Sd} K_i$ is simplicial. Moreover, if X is compact metrizable (resp. separable and completely metrizable), then each $|K_i|_{\mathrm{m}} = |K_i|$ is compact (resp. locally compact). If X is separable and locally compact metrizable, each $|K_i|_{\mathrm{m}} = |K_i|$ is locally compact and each f_i is proper.* □

Remark 5.2.2 Theorem 5.2.7 does not hold for a non-metrizable space X, that is, Theorem 4.16.8 has no version like the above even if X is compact. In fact, there exists a one-dimensional compact (non-metrizable) space which cannot be represented as the limit of an inverse system of polyhedra with $\dim \leqslant 1$ (refer to Notes for this chapter).

Applying Theorem 5.2.4, we will prove two types of sum theorems. The first one is the COUNTABLE SUM THEOREM:

Theorem 5.2.8 (COUNTABLE SUM) *Let $X = \bigcup_{i\in\mathbb{N}} F_i$ be normal and $n \in \omega$, where each F_i is closed in X. If $\dim F_i \leqslant n$ for every $i \in \mathbb{N}$, then $\dim X \leqslant n$.*

Proof Let $\{U_1, \ldots, U_{n+2}\} \in \operatorname{cov}(X)$. By induction on $i \in \mathbb{N}$, we can define $\mathcal{U}_i = \{U_{i,1}, \ldots, U_{i,n+2}\} \in \operatorname{cov}(X)$ so that

$$\operatorname{cl} U_{i,j} \subset U_{i-1,j} \quad \text{and} \quad U_{i,1} \cap \cdots \cap U_{i,n+2} \cap F_i = \emptyset,$$

where $U_{0,j} = U_j$. Indeed, assume that $\mathcal{U}_{i-1}$ has been defined, where $F_0 = \emptyset$. By Theorem 5.2.4, there is $\{V_{i,1}, \ldots, V_{i,n+2}\} \in \mathrm{cov}(F_i)$ such that

$$V_{i,j} \subset U_{i-1,j} \text{ and } V_{i,1} \cap \cdots \cap V_{i,n+2} = \emptyset.$$

Let $W_{i,j} = V_{i,j} \cup (U_{i-1,j} \setminus F_i)$. Then, $\{W_{i,1}, \ldots, W_{i,n+2}\} \in \mathrm{cov}(X)$. By Lemma 5.2.2, we have $\mathcal{U}_i = \{U_{i,1}, \ldots, U_{i,n+2}\} \in \mathrm{cov}(X)$ such that $\mathrm{cl}\, U_{i,j} \subset W_{i,j}$. Observe that $\mathcal{U}_i$ is as desired.

For each $j = 1, \ldots, n+2$, let $A_j = \bigcap_{i\in\mathbb{N}} U_{i,j}$. Observe that $A_j = \bigcap_{i\in\mathbb{N}} \mathrm{cl}\, U_{i,j}$ is closed in X. Since

$$A_1 \cap \cdots \cap A_{n+2} \cap F_i \subset U_{i,1} \cap \cdots \cap U_{i,n+2} \cap F_i = \emptyset,$$

we have $A_1 \cap \cdots \cap A_{n+2} = \emptyset$. For each $x \in X$, $\{i \in \mathbb{N} \mid x \in U_{i,j}\}$ is infinite for some j. Then, $x \in \bigcap_{i\in\mathbb{N}} U_{i,j} = A_j$. Hence, $X = A_1 \cup \cdots \cup A_{n+2}$. According to Theorem 5.2.4, we have $\dim X \leqslant n$. □

The second is the LOCALLY FINITE SUM THEOREM:

Theorem 5.2.9 (LOCALLY FINITE SUM) *Let X be normal and $n \in \omega$. If X has a locally finite closed cover $\{F_\lambda \mid \lambda \in \Lambda\}$ such that $\dim F_\lambda \leqslant n$ for each $\lambda \in \Lambda$, then $\dim X \leqslant n$.*

Proof We may assume that $\Lambda = (\Lambda, \leqslant)$ is a well-ordered set, and $F_{\min\Lambda} = \emptyset$. Let $\{U_1, \ldots, U_{n+2}\} \in \mathrm{cov}(X)$. Using transfinite induction, we will define $\mathcal{U}_\lambda = \{U_{\lambda,1}, \ldots, U_{\lambda,n+2}\} \in \mathrm{cov}(X)$ so that

$$U_{\min\Lambda,j} = U_j,\ U_{\lambda,1} \cap \cdots \cap U_{\lambda,n+2} \cap F_\lambda = \emptyset, \text{ and}$$
$$\mu < \lambda \Rightarrow U_{\lambda,j} \subset U_{\mu,j},\ U_{\mu,j} \setminus U_{\lambda,j} \subset \bigcup_{\mu\leqslant\nu\leqslant\lambda} F_\nu.$$

Suppose that $\mathcal{U}_\mu$ has been obtained for $\mu < \lambda$. Let $U'_{\lambda,j} = \bigcap_{\mu<\lambda} U_{\mu,j}$. Then, $\{U'_{\lambda,1}, \ldots, U'_{\lambda,n+2}\} \in \mathrm{cov}(X)$. Indeed, if there exists $\mu_0 = \max\{\mu \in \Lambda \mid \mu < \lambda\}$, then $U'_{\lambda,j} = U_{\mu_0,j}$ for each $j = 1, \ldots, n+2$. Otherwise, for each $x \in X$, choose an open neighborhood U of x in X that meets only finitely many F_μ. Then, there exists $\mu_1 < \lambda$ such that $U \cap F_\mu = \emptyset$ for $\mu_1 \leqslant \mu < \lambda$. If $x \in U_{\mu_1,j} \in \mathcal{U}_{\mu_1}$, then $U \cap U_{\mu_1,j} \subset U_{\mu,j}$ for $\mu_1 \leqslant \mu < \lambda$ because

$$(U \cap U_{\mu_1,j}) \setminus U_{\mu,j} \subset \bigcup_{\mu_1\leqslant\nu\leqslant\mu} (U \cap F_\nu) = \emptyset.$$

It follows that

$$x \in U \cap U_{\mu_1,j} \subset \bigcap_{\mu_1\leqslant\mu<\lambda} U_{\mu,j} = \bigcap_{\mu<\lambda} U_{\mu,j} = U'_{\lambda,j}.$$

Thus, $X = \bigcup_{j=1}^{n+2} U'_{\lambda,j}$. Moreover, for each $j = 1, \ldots, n+2$, applying the argument above to $x \in U'_{\lambda,j}$, we can see that $x \in U \cap U_{\mu_1,j} \subset U'_{\lambda,j}$ as above. This shows that $U'_{\lambda,j}$ is open in X.

We apply Theorem 5.2.4 to obtain $\{V_{\lambda,1}, \ldots, V_{\lambda,n+2}\} \in \mathrm{cov}(F_\lambda)$ such that $V_{\lambda,j} \subset U'_{\lambda,j}$ and $V_{\lambda,1} \cap \cdots \cap V_{\lambda,n+2} = \emptyset$. Now, let $U_{\lambda,j} = V_{\lambda,j} \cup (U'_{\lambda,j} \setminus F_\lambda)$. Then, $\{U_{\lambda,1}, \ldots, U_{\lambda,n+2}\} \in \mathrm{cov}(X)$ is the desired open cover. In fact, if $\mu < \lambda$ then

$$U_{\mu,j} \setminus U_{\lambda,j} \subset F_\lambda \cup \left((U_{\mu,j} \setminus F_\lambda) \setminus (U'_{\lambda,j} \setminus F_\lambda)\right) \subset F_\lambda \cup \left(U_{\mu,j} \setminus \bigcap_{\nu<\lambda} U_{\nu,j}\right)$$

$$\subset F_\lambda \cup \bigcup_{\mu<\nu<\lambda} (U_{\mu,j} \setminus U_{\nu,j}) \subset \bigcup_{\mu \leqslant \nu \leqslant \lambda} F_\nu.$$

The proofs of the other properties are easy.

For each $j = 1, \ldots, n+2$, let $U^*_j = \bigcap_{\lambda\in\Lambda} U_{\lambda,j}$. Then, similar to the above, $\{U^*_1, \ldots, U^*_{n+2}\} \in \mathrm{cov}(X)$. Clearly, $U^*_j \subset U_j$ and $U^*_1 \cap \cdots \cap U^*_{n+2} = \emptyset$. Therefore, $\dim X \leqslant n$ by Theorem 5.2.4. □

Remark 5.2.3 As is easily observed, the following holds:

- $\dim Y \leqslant \dim X$ for every *closed* set Y in X.

Thus, in Theorem 5.2.8 (resp. 5.2.9), if $\dim F_i = n$ (resp. $\dim F_\lambda = n$) for some $i \in \mathbb{N}$ (resp. $\lambda \in \Lambda$), then $\dim X = n$.

The following corollary is a combination of Theorems 5.2.8 and 5.2.9:

Corollary 5.2.10 *Let X be a normal space and $n \in \omega$. If X has a σ-locally finite closed cover $\{F_\lambda \mid \lambda \in \Lambda\}$ such that $\dim F_\lambda \leqslant n$ for each $\lambda \in \Lambda$, then $\dim X \leqslant n$.* □

The next corollary follows from Theorem 5.2.9:

Corollary 5.2.11 *Let X be a paracompact space and $n \in \omega$. If each point of X has a closed neighborhood with $\dim \leqslant n$, then $\dim X \leqslant n$.*

Proof Since X is paracompact and each point of X has a closed neighborhood with $\dim \leqslant n$, X has a locally finite open cover $\mathcal{U}$ such that $\dim \mathrm{cl}\, U \leqslant n$ for each $U \in \mathcal{U}$. Then, $\mathcal{U}^{\mathrm{cl}}$ is also locally finite in X, hence $\dim X \leqslant n$ by Theorem 5.2.9. □

Remark 5.2.4 Corollary 5.2.11 can also be proved by applying Michael's Theorem on local properties of closed sets (Corollary 2.6.8) and Theorem 5.2.8 (or 5.2.9).

Now, we can prove the following theorem:

Theorem 5.2.12 *For each $n \in \mathbb{N}$, $\dim \mathbf{B}^n = \dim \mathbb{R}^n = n$.*

Proof For any $\mathcal{U} \in \mathrm{cov}(\Delta^n)$, Δ^n has a triangulation K such that $O_K \prec \mathcal{S}_K \prec \mathcal{U}$ (Corollary 4.8.7). Since $\mathrm{ord}\, O_K = \dim K + 1 = n + 1$ and $|K| = \Delta^n \approx \mathbf{B}^n$, it follows that $\dim \mathbf{B}^n \leqslant n$. If $\dim \mathbf{B}^n \leqslant n - 1$, then we apply Theorem 5.2.4 to obtain a map $r : \mathbf{B}^n \to \mathbf{S}^{n-1}$ such that $r|\mathbf{S}^{n-1} = \mathrm{id}$, which contradicts the No Retraction Theorem 5.1.5. Consequently, we have $\dim \mathbf{B}^n = n$.

Since $\mathbb{R}^n = \bigcup_{k\in\mathbb{N}} k\mathbf{B}^n$, we have $\dim \mathbb{R}^n = n$ by the Countable Sum Theorem 5.2.8 and Remark 5.2.3. □

Using simplicial complexes, we can characterize the dimension of paracompact spaces as follows:

Theorem 5.2.13 *Let X be paracompact and $n \in \omega$. Then, $\dim X \leqslant n$ if and only if, for every simplicial complex K, each map $f : X \to |K|$ (or $f : X \to |K|_{\mathrm{m}}$) is contiguous to a map $g : X \to |K^{(n)}|$ (or $g : X \to |K^{(n)}|_{\mathrm{m}}$). In this case, each $g(x)$ is contained in the carrier $c_K(f(x)) \in K$ of $f(x)$.*

Proof (*The "if" Part*) Each (finite) open cover $\mathcal{U}$ of X has an open star-refinement $\mathcal{V}$. Let $K = N(\mathcal{V})$ be the nerve of $\mathcal{V}$. A canonical map $f : X \to |K|$ (or $f : X \to |K|_{\mathrm{m}}$) is contiguous to a map $g : X \to |K^{(n)}|$ (or $g : X \to |K^{(n)}|_{\mathrm{m}}$). Then, $g^{-1}(O_{K^{(n)}}) \in \mathrm{cov}(X)$ with

$$\mathrm{ord}\, g^{-1}(O_{K^{(n)}}) \leqslant \mathrm{ord}\, O_{K^{(n)}} = \dim K^{(n)} + 1 \leqslant n + 1.$$

Let $V \in \mathcal{V} = K^{(0)}$ and $x \in g^{-1}(O_{K^{(n)}}(V))$. We have $\sigma \in K$ such that $f(x), g(x) \in \sigma$. Then, $c_K(f(x)) \leqslant \sigma$ and $V \in \sigma^{(0)}$. Since f is canonical, we have $c_K(f(x))^{(0)} \subset \mathcal{V}[x]$ (Proposition 4.15.2). It follows that $V \cap V' \neq \emptyset$ and $x \in V'$ for any $V' \in c_K(f(x))^{(0)}$, which implies $x \in \mathrm{st}(V, \mathcal{V})$. Thus, $g^{-1}(O_{K^{(n)}}(V)) \subset \mathrm{st}(V, \mathcal{V})$, which means $g^{-1}(O_{K^{(n)}}) \prec \mathcal{U}$. Therefore, $\dim X \leqslant n$.

(*The "only if" Part*) Let $f : X \to |K|$ (or $f : X \to |K|_{\mathrm{m}}$) be a map. Because $\dim X \leqslant n$, X has an open cover $\mathcal{U} \prec f^{-1}(O_K)$ with $\mathrm{ord}\, \mathcal{U} \leqslant n + 1$ by Theorem 5.2.5. Let $L = N(\mathcal{U})$ be the nerve of $\mathcal{U}$ with $\varphi : X \to |L|$ (or $\varphi : X \to |L|_{\mathrm{m}}$) a canonical map. Then, we have a function $\psi : L^{(0)} = \mathcal{U} \to K^{(0)}$ such that $U \subset f^{-1}(O_K(\psi(U)))$, i.e., $f(U) \subset O_K(\psi(U))$. By Proposition 4.10.1, $\psi : L^{(0)} \to K^{(0)}$ induces the simplicial map $\psi : L \to K$. Since $\dim L \leqslant n$, it follows that $\psi\varphi(X) \subset \psi(|L|) \subset |K^{(n)}|$. Thus, we have a map $g = \psi\varphi : X \to |K^{(n)}|$ (or $g = \psi\varphi : X \to |K^{(n)}|_{\mathrm{m}}$).

We will show that $g(x) \in c_K(f(x))$ for every $x \in X$. For each $x \in X$, $\varphi(x) \in \langle \mathcal{U}[x] \rangle \in L$ because φ is canonical. Then, $g(x) = \psi\varphi(x) \in \psi(\langle \mathcal{U}[x] \rangle) \in K$. For each $U \in \mathcal{U}[x]$, $f(x) \in f(U) \subset O_K(\psi(U))$, which means $\psi(U) \in c_K(f(x))^{(0)}$. Hence, $\psi(\langle \mathcal{U}[x] \rangle) \leqslant c_K(f(x))$. Thus, $g(x) \in c_K(f(x))$ for every $x \in X$. □

As a corollary of Theorems 5.2.12 and 5.2.13, we have the following:

Corollary 5.2.14 *For any simplicial complex K,*

$$\dim K = \dim |K| = \dim |K|_{\mathrm{m}}.$$

Proof An n-simplex $\tau \in K$ is closed in both $|K|$ and $|K|_{\mathrm{m}}$, and $\dim \tau = n$ by Theorem 5.2.12. By the definition of dimension, $\dim |K| \geqslant \dim \tau$ and $\dim |K|_{\mathrm{m}} \geqslant \dim \tau$. On the other hand, combining Theorem 5.2.13 with the Simplicial Approximation Theorems 4.10.11 and 4.10.13, we arrive at $\dim |K| \leqslant \dim K$ and $\dim |K|_{\mathrm{m}} \leqslant \dim K$. □

5.3 Partitions and Essential Maps

In the previous section, we characterized $\dim \leqslant n$. Here, we will characterize $\dim \geqslant n$. We first show that the existence of a homotopically non-trivial map of X to $\mathbf{S}^n$ implies $\dim X \geqslant n$, that is,

Proposition 5.3.1 *For a normal space X, if there exists a map $f : X \to \mathbf{S}^n$ that is not null-homotopic, then* $\dim X \geqslant n$.

Proof Define $\mathbf{S}^n_+$ and $\mathbf{S}^n_-$ as in the proof of Theorem 5.2.4 (f) $\Rightarrow$ (e) and identify $\mathbf{S}^{n-1} = \mathbf{S}^n_+ \cap \mathbf{S}^n_- \subset \mathbf{S}^n$. If $\dim X \leqslant n-1$ then $f|f^{-1}(\mathbf{S}^{n-1})$ extends to a map $f' : X \to \mathbf{S}^{n-1}$ by Theorem 5.2.4. We can define a map $g : X \to \mathbf{S}^n$ as follows:

$$g|f^{-1}(\mathbf{S}^n_+) = f'|f^{-1}(\mathbf{S}^n_+) \quad \text{and} \quad g|f^{-1}(\mathbf{S}^n_-) = f|f^{-1}(\mathbf{S}^n_-).$$

Then, $g \simeq f$ rel. $f^{-1}(\mathbf{S}^n_-)$. Indeed, we have a homeomorphism $\varphi : \mathbf{S}^n_+ \to \mathbf{B}^n$ with $\varphi|\mathbf{S}^{n-1} = \mathrm{id}$. Then, $\varphi f|f^{-1}(\mathbf{S}^n_+) \simeq \varphi f'|f^{-1}(\mathbf{S}^n_+)$ rel. $f^{-1}(\mathbf{S}^{n-1})$ in $\mathbf{B}^n$, which is realized by the straight-line homotopy. Hence, $f|f^{-1}(\mathbf{S}^n_+) \simeq f'|f^{-1}(\mathbf{S}^n_+)$ rel. $f^{-1}(\mathbf{S}^{n-1})$ in $\mathbf{S}^n_+$, which implies $g \simeq f$ rel. $f^{-1}(\mathbf{S}^n_-)$. Since $g(X) \subset \mathbf{S}^n_- \approx \mathbf{B}^n$, it follows that $f \simeq g \simeq 0$. This is a contradiction. □

Remark 5.3.1 The converse of Proposition 5.3.1 does not hold. In fact, if a space X (even if $\dim X \geqslant n$) is homotopy equivalent to a singleton,[3] then every map $f : X \to \mathbf{S}^n$ is null-homotopic.

Let A and B be disjoint closed sets in a space X. A closed set C in X is called a **partition** between A and B in X if there exist disjoint open sets U and V in X such that $A \subset U$, $B \subset V$, and $X \setminus C = U \cup V$. Concerning partitions, we have the following:

Proposition 5.3.2 *Let X be a normal space and $n \in \omega$. Suppose that for every disjoint closed sets A and B in X, there exists a partition L between A and B in X such that* $\dim L \leqslant n-1$. *Then,* $\dim X \leqslant n$.

Proof Each finite open cover $\mathcal{U} = \{U_1, \ldots, U_k\} \in \mathrm{cov}(X)$ has a closed shrinking $\{A_1, \ldots, A_k\}$ by Lemma 5.2.2. By the assumption, for each $i = 1, \ldots, k$, we have a partition L_i between A_i and $X \setminus U_i$ such that $\dim L_i \leqslant n-1$. Choose an open set W_i

[3] I.e., X is contractible (which will be defined just before Theorem 6.1.5).

in X so that $A_i \subset W_i \subset \operatorname{cl} W_i \subset U_i$ and $\operatorname{bd} W_i \subset L_i$. Then, $\{W_1, \ldots, W_k\} \in \operatorname{cov}(X)$ because $X = \bigcup_{i=1}^k A_i$. Since $L = \bigcup_{i=1}^k L_i$ is normal, we have $\dim L \leqslant n-1$ by the Countable Sum Theorem 5.2.8 (or the Locally Finite Sum Theorem 5.2.9). Due to Theorem 5.2.4, the cover $\{L \cap U_i \mid i = 1, \ldots, k\} \in \operatorname{cov}(L)$ has a closed shrinking $C = \{C_1, \ldots, C_k\}$ with $\operatorname{ord} C \leqslant n$. By the Swelling Lemma 5.2.3, C has a strong closed swelling $\{D_1', \ldots, D_k'\}$ in X. For each $i = 1, \ldots, k$, choose a closed set D_i in X so that $C_i \subset \operatorname{int} D_i \subset D_i \subset D_i' \cap U_i$. Thus, we have a strong closed swelling $\mathcal{D} = \{D_1, \ldots, D_k\}$ of C in X such that $D_i \subset U_i$ for each $i = 1, \ldots, k$. Then, $\operatorname{ord} \mathcal{D} \leqslant n$ and $V = \bigcup_{i=1}^k \operatorname{int} D_i$ is an open neighborhood of L in X. For each $i = 1, \ldots, k$, let

$$Z_i = \operatorname{cl} W_i \setminus (V \cup W_1 \cup \cdots \cup W_{i-1}) \subset U_i.$$

Then, $\mathcal{F} = \{D_i,\ Z_i \mid i = 1, \ldots, k\} \prec \mathcal{U}$ is a closed cover of X. Indeed,

$$\bigcup_{i=1}^k D_i \cup \bigcup_{i=1}^k Z_i \supset \bigcup_{i=1}^k (V \cup Z_i) \supset \bigcup_{i=1}^k (W_i \setminus (W_1 \cup \cdots \cup W_{i-1})) = \bigcup_{i=1}^k W_i = X.$$

For each $j < i \leqslant k$,

$$\begin{aligned} Z_i \cap Z_j &\subset (\operatorname{cl} W_i \setminus (V \cup W_j)) \cap \operatorname{cl} W_j \subset (X \setminus V) \cap (X \setminus W_j) \cap \operatorname{cl} W_j \\ &= (X \setminus V) \cap \operatorname{bd} W_j \subset (X \setminus V) \cap L_j = \emptyset. \end{aligned}$$

Since $\operatorname{ord} \mathcal{D} \leqslant n$, it follows that $\operatorname{ord} \mathcal{F} \leqslant n+1$. Hence, $\dim X \leqslant n$ by Theorem 5.2.4. □

A family $(A_\gamma, B_\gamma)_{\gamma \in \Gamma}$ of pairs of disjoint closed sets in X is **inessential** in X if there are partitions L_γ between A_γ and B_γ with $\bigcap_{\gamma \in \Gamma} L_\gamma = \emptyset$. Note that if one of A_γ or B_γ is empty for some $\gamma \in \Gamma$, then (A_γ, B_γ) is inessential. If $(A_\gamma, B_\gamma)_{\gamma \in \Gamma}$ is not inessential in X (i.e., $\bigcap_{\gamma \in \Gamma} L_\gamma \neq \emptyset$ for any partitions L_γ between A_γ and B_γ), it is said to be **essential** in X.

A map $f : X \to \mathbf{I}^n$ is said to be **essential** if every map $g : X \to \mathbf{I}^n$ with $g|f^{-1}(\partial \mathbf{I}^n) = f|f^{-1}(\partial \mathbf{I}^n)$ is surjective, where it should be noted that g is also essential. It is said that f is **inessential** if it is not essential, i.e., there is a map $g : X \to \mathbf{I}^n$ such that $g|f^{-1}(\partial \mathbf{I}^n) = f|f^{-1}(\partial \mathbf{I}^n)$ and $g(X) \neq \mathbf{I}^n$.

Lemma 5.3.3 *For two maps $f, g : X \to \mathbf{B}^n$, if $f(x) \neq g(x)$ for any $x \in X$, then there is a map $h : X \to \mathbf{S}^{n-1}$ such that $h|f^{-1}(\mathbf{S}^{n-1}) = f|f^{-1}(\mathbf{S}^{n-1})$.*

Proof In the same way as for the map r in the remark for the No Retraction Theorem 5.1.5, we can obtain a map $h : X \to \mathbf{S}^{n-1}$ such that $f(x) \in \langle h(x), g(x) \rangle$ for each $x \in X$, which implies $h|f^{-1}(\mathbf{S}^{n-1}) = f|f^{-1}(\mathbf{S}^{n-1})$. □

For a map $f : X \to \mathbf{B}^n$ with $f(X) \neq \mathbf{B}^n$, by taking g as a constant map, the following can be obtained as a special case of Lemma 5.3.3:

Lemma 5.3.4 *If a map $f : X \to \mathbf{B}^n$ is not surjective, then there is a map $h : X \to \mathbf{S}^{n-1}$ such that $h|f^{-1}(\mathbf{S}^{n-1}) = f|f^{-1}(\mathbf{S}^{n-1})$.* □

Thus, we have

Proposition 5.3.5 *A map $f : X \to \mathbf{I}^n$ is inessential if and only if there is a map $g : X \to \partial\mathbf{I}^n$ such that $g|f^{-1}(\partial\mathbf{I}^n) = f|f^{-1}(\partial\mathbf{I}^n)$.* □

Using Lemma 5.3.4, we will show the following:

Proposition 5.3.6 *Let X be a normal space and $h : X \times \mathbf{I} \to \mathbf{I}^n$ be a homotopy such that h_0 is essential and $h(h_0^{-1}(\partial\mathbf{I}^n) \times \mathbf{I}) \subset \partial\mathbf{I}^n$. Then, h_1 is also essential, hence it is surjective.*

Proof Let $h_0 = f$ and assume that h_1 is inessential. By Lemma 5.3.4, there is a map $g : X \to \partial\mathbf{I}^n$ such that $g|h_1^{-1}(\partial\mathbf{I}^n) = h_1|h_1^{-1}(\partial\mathbf{I}^n)$. Then, $f^{-1}(\partial\mathbf{I}^n) \subset h_1^{-1}(\partial\mathbf{I}^n)$ and $h_1 \simeq g$ rel. $h_1^{-1}(\partial\mathbf{I}^n)$ by the straight-line homotopy:

$$(1-t)h_1(x) + tg(x) \quad \text{for each } (x,t) \in X \times \mathbf{I}.$$

Connecting this to h, we obtain a homotopy $\varphi : X \times \mathbf{I} \to \mathbf{I}^n$ such that

$$\varphi(f^{-1}(\partial\mathbf{I}^n) \times I) \subset \partial\mathbf{I}^n, \ \varphi_0 = f \ \text{ and } \ \varphi_1 = g.$$

Then, $A = \mathrm{pr}_X(\varphi^{-1}([1/3, 2/3]^n))$ is a closed set in X. Observe that

$$\varphi^{-1}([1/3, 2/3]^n) \cap (f^{-1}(\partial\mathbf{I}^n) \times \mathbf{I}) = \emptyset,$$

which implies $A \cap f^{-1}(\partial\mathbf{I}^n) = \emptyset$. Taking a Urysohn map $k : X \to \mathbf{I}$ with $k(f^{-1}(\partial\mathbf{I}^n)) = 0$ and $k(A) = 1$, we define a map $g' : X \to \mathbf{I}^n$ as follows:

$$g'(x) = \varphi(x, k(x)) \quad \text{for each } x \in X.$$

Then, $g'|f^{-1}(\partial\mathbf{I}^n) = f|f^{-1}(\partial\mathbf{I}^n)$ but $g'(X) \neq \mathbf{I}^n$. In fact, $g'(A) = g(A) \subset \partial\mathbf{I}^n$ and

$$\begin{aligned} g'(X \setminus A) &\subset \varphi((X \setminus A) \times \mathbf{I}) \subset \varphi\big((X \times \mathbf{I}) \setminus \varphi^{-1}([1/3, 2/3]^n)\big) \\ &\subset \mathbf{I}^n \setminus [1/3, 2/3]^n. \end{aligned}$$

This is a contradiction because $h_0 = f$ is essential. □

Essential maps can be characterized as follows:

Theorem 5.3.7 *Let X be a normal space. For a map $f : X \to \mathbf{I}^n$, the following are equivalent:*

(a) *f is essential.*
(b) *For each map $g : X \to \mathbf{I}^n$, there is some $x \in X$ such that $f(x) = g(x)$.*

(c) $(f^{-1}(\mathrm{pr}_i^{-1}(0)), f^{-1}(\mathrm{pr}_i^{-1}(1)))_{i=1}^n$ *is essential in* X.

Proof The implication (a) $\Rightarrow$ (b) follows from Lemma 5.3.3.

(b) $\Rightarrow$ (c): Assume that $(f^{-1}(\mathrm{pr}_i^{-1}(0)), f^{-1}(\mathrm{pr}_i^{-1}(1)))_{i=1}^n$ is inessential, that is, there are partitions L_i between $f^{-1}(\mathrm{pr}_i^{-1}(0))$ and $f^{-1}(\mathrm{pr}_i^{-1}(1))$ such that $\bigcap_{i=1}^n L_i = \emptyset$. Then, we have disjoint open sets U_i and V_i in X such that

$$X \setminus L_i = U_i \cup V_i, \ \ f^{-1}(\mathrm{pr}_i^{-1}(0)) \subset U_i \ \text{ and } \ f^{-1}(\mathrm{pr}_i^{-1}(1)) \subset V_i.$$

Applying Lemma 5.2.2 to the open cover $\{X \setminus L_i \mid i = 1, \ldots, n\}$ of X, we have a closed cover $\{F_i \mid i = 1, \ldots, n\}$ of X such that $F_i \subset X \setminus L_i = U_i \cup V_i$, where we may assume that

$$f^{-1}(\mathrm{pr}_i^{-1}(0)) \cup f^{-1}(\mathrm{pr}_i^{-1}(1)) \subset F_i.$$

Each $U_i \cap F_i = F_i \setminus V_i$ and $V_i \cap F_i = F_i \setminus U_i$ are disjoint closed sets in X. Using Urysohn maps for $U_i \cap F_i$ and $V_i \cap F_i$, we can define a map $g : X \to \mathbf{I}^n$ such that $\mathrm{pr}_i g(U_i \cap F_i) = 1$ and $\mathrm{pr}_i g(V_i \cap F_i) = 0$. Observe that

$$\bigcup_{i=1}^n \big((U_i \cap F_i) \cup (V_i \cap F_i)\big) = \bigcup_{i=1}^n \big((U_i \cup V_i) \cap F_i\big) = \bigcup_{i=1}^n F_i = X,$$
$$(\mathrm{pr}_i f)^{-1}(1) \subset V_i \subset X \setminus U_i \ \text{ and } \ (\mathrm{pr}_i f)^{-1}(0) \subset U_i \subset X \setminus V_i.$$

It follows that $g(x) \neq f(x)$ for any $x \in X$.

(c) $\Rightarrow$ (a): Suppose that f is inessential. Then, there is a map $h : X \to \partial\mathbf{I}^n$ with $h|f^{-1}(\partial\mathbf{I}^n) = f|f^{-1}(\partial\mathbf{I}^n)$ by Lemma 5.3.4. Note that

$$f^{-1}(\mathrm{pr}_i^{-1}(0)) \subset h^{-1}(\mathrm{pr}_i^{-1}(0)) \ \text{ and } \ f^{-1}(\mathrm{pr}_i^{-1}(1)) \subset h^{-1}(\mathrm{pr}_i^{-1}(1)).$$

Each $h^{-1}(\mathrm{pr}_i^{-1}(1/2))$ is a partition between $f^{-1}(\mathrm{pr}_i^{-1}(0))$ and $f^{-1}(\mathrm{pr}_i^{-1}(1))$, and then

$$\bigcap_{i=1}^n h^{-1}(\mathrm{pr}_i^{-1}(1/2)) = h^{-1}(1/2, \ldots, 1/2) = \emptyset.$$

Thus, $(f^{-1}(\mathrm{pr}_i^{-1}(0)), f^{-1}(\mathrm{pr}_i^{-1}(1)))_{i=1}^n$ is inessential. □

The Brouwer Fixed Point Theorem 5.1.1 means that the identity map of $\mathbf{I}^n$ satisfies condition (b) in Theorem 5.3.7, hence we have the following corollary:

Corollary 5.3.8 *The family* $(\mathrm{pr}_i^{-1}(0), \mathrm{pr}_i^{-1}(1))_{i=1}^n$ *is essential in* $\mathbf{I}^n$. □

Remark 5.3.2 Due to Theorem 5.3.7, Corollary 5.3.8 is equivalent to the Brouwer Fixed Point Theorem 5.1.1. Moreover, Corollary 5.3.8 is valid even if $n = \infty$. In fact, $(\mathrm{pr}_i^{-1}(0), \mathrm{pr}_i^{-1}(1))_{i\in\mathbb{N}}$ is essential in $\mathbf{I}^{\mathbb{N}}$. This will be shown in the proof of Theorem 5.6.1.

Using essential families and essential maps, we can also characterize dimension as follows:

Theorem 5.3.9 (EILENBERG–OTTO; ALEXANDROFF) *Let X be a normal space and $n \in \mathbb{N}$. Then, the following are equivalent:*

(a) $\dim X \geqslant n$;
(b) *X has an essential map $f : X \to \mathbf{I}^n$;*
(c) *X has an essential family of n pairs of disjoint closed sets.*

Proof The implication (b) $\Rightarrow$ (c) follows from Theorem 5.3.7. For an essential map $f : X \to \mathbf{I}^n$, $f|f^{-1}(\partial \mathbf{I}^n) : f^{-1}(\partial \mathbf{I}^n) \to \partial \mathbf{I}^n$ cannot extend to any map from X to $\partial \mathbf{I}^n$, which means $\dim X \geqslant n$ by Theorem 5.2.4. Thus, we have also (b) $\Rightarrow$ (a). The implications (a) $\Rightarrow$ (b) and (c) $\Rightarrow$ (b) remain to be proved.

(a) $\Rightarrow$ (b): By Theorem 5.2.4, there exists a map $f' : A \to \partial \mathbf{I}^n$ of a closed set A in X that cannot extend over X. Nevertheless, f' can be extended to a map $f : X \to \mathbf{I}^n$ by Theorem 5.1.6(1). If there is a map $g : X \to \mathbf{I}^n$ such that $g(x) \neq f(x)$ for any $x \in X$, then we have a map $h : X \to \partial \mathbf{I}^n$ such that $h|f^{-1}(\partial \mathbf{I}^n) = f|f^{-1}(\partial \mathbf{I}^n)$ by Lemma 5.3.3. This is a contradiction because h is an extension of f'. Therefore, for each map $g : X \to \mathbf{I}^n$, there is some $x \in X$ such that $f(x) = g(x)$. By Theorem 5.3.7, f is essential.

(c) $\Rightarrow$ (b): Let $(A_i, B_i)_{i=1}^n$ be an essential family of n pairs of disjoint closed sets in X. Using Urysohn maps for A_i and B_i, we can define a map $f : X \to \mathbf{I}^n$ so that $\mathrm{pr}_i f(A_i) = 0$ and $\mathrm{pr}_i f(B_i) = 1$ for each $i = 1, \ldots, n$. Since $A_i \subset f^{-1}(\mathrm{pr}_i^{-1}(0))$ and $B_i \subset f^{-1}(\mathrm{pr}_i^{-1}(1))$, it follows that $(f^{-1}(\mathrm{pr}_i^{-1}(0)), f^{-1}(\mathrm{pr}_i^{-1}(1)))_{i=1}^n$ is essential, which means that f is essential according to Theorem 5.3.7. □

Equivalences (a) $\Leftrightarrow$ (b) and (a) $\Leftrightarrow$ (c) are called the ALEXANDROFF CHARACTERIZATION and the EILENBERG–OTTO CHARACTERIZATION of dimension, respectively. Using Theorem 5.3.9, we can easily show the following corollary:

Corollary 5.3.10 *Let X be a non-degenerate normal space. If $\dim X = 0$, then X is disconnected. Equivalently, if X is connected, then $\dim X > 0$.* □

5.4 Dimension of Metrizable Spaces

In this section, we will give characterizations of dimension for metrizable spaces and prove some fundamental theorems. For metric spaces, the following characterization can be established:

Theorem 5.4.1 *Let $X = (X, d)$ be a metric space. Then, $\dim X \leqslant n$ if and only if X has a sequence $\mathcal{U}_1 \succ \mathcal{U}_2 \succ \cdots$ of (locally finite σ-discrete) open covers such that $\operatorname{ord} \mathcal{U}_i \leqslant n + 1$ and $\lim_{i \to \infty} \operatorname{mesh} \mathcal{U}_i = 0$.*

Proof (*The "only if" Part*) When $\dim X \leqslant n$, using the "only if" of Theorem 5.2.5, we can inductively construct locally finite σ-discrete open covers $\mathcal{U}_1 \succ \mathcal{U}_2 \succ \cdots$ of X such that $\operatorname{ord} \mathcal{U}_i \leqslant n+1$ and $\lim_{i\to\infty} \operatorname{mesh} \mathcal{U}_i = 0$.

(*The "if" Part*) Since $\mathcal{U}_{i+1} \prec \mathcal{U}_i$, we can take a function $\varphi_i : \mathcal{U}_{i+1} \to \mathcal{U}_i$ such that $U \subset \varphi_i(U)$ for each $U \in \mathcal{U}_{i+1}$. For each $j > i$, let $\varphi_{i,j} = \varphi_i \circ \cdots \circ \varphi_{j-1} : \mathcal{U}_j \to \mathcal{U}_i$ and $\varphi_{i,i} = \mathrm{id}_{\mathcal{U}_i}$. Then, it follows that $U \subset \varphi_{i,j}(U)$ for each $U \in \mathcal{U}_j$.

Let $\mathcal{W}$ be a finite open cover of X. For each $i \in \mathbb{N}$, we have the following open set in X:

$$X_i = \bigcup \big\{ U \in \mathcal{U}_i \;\big|\; \operatorname{st}(U, \mathcal{U}_i) \subset W \ \text{ for some } W \in \mathcal{W} \big\},$$

where it does not necessarily follow that $X_i \subset X_{i+1}$. Since $\lim_{i\to\infty} \operatorname{mesh} \mathcal{U}_i = 0$, it follows that $X = \bigcup_{i\in\mathbb{N}} X_i$. Moreover, let

$$\mathcal{U}'_i = \mathcal{U}_i[X_i] \ \text{ and } \ \mathcal{U}''_i = \mathcal{U}'_i \setminus \mathcal{U}_i[\textstyle\bigcup_{j<i} X_j] = \mathcal{U}_i[X_i] \setminus \bigcup_{j<i} \mathcal{U}_i[X_j],$$

where $X_0 = \emptyset$. For each $U \in \mathcal{U}'_i$, since $\varphi_{i,i}(U) \cap X_i = U \cap X_i \neq \emptyset$, we can define

$$m(U) = \min \big\{ j \leqslant i \;\big|\; \varphi_{j,i}(U) \cap X_j \neq \emptyset \big\}.$$

Since $\mathcal{U}'_1 = \mathcal{U}''_1$ and $\varphi_{m(U),i}(U) \subset \varphi_{m(U)-1,i}(U) \subset \cdots \subset \varphi_{1,i}(U)$ if $m(U) > 1$, we have $\varphi_{m(U),i}(U) \in \mathcal{U}''_{m(U)}$.

Now, for each $U \in \mathcal{U}''_i$, we define

$$V_U = \bigcup_{j\geqslant i} \bigcup \big\{ U' \cap X_j \;\big|\; U' \in \mathcal{U}'_j,\ m(U') = i,\ \varphi_{i,j}(U') = U \big\} \subset U.$$

For the above last inclusion, just observe that $U' \subset \varphi_{i,j}(U') = U$.

Since $U \cap X_i \neq \emptyset$, there is $U' \in \mathcal{U}_i$ such that $U \cap U' \neq \emptyset$ and $\operatorname{st}(U', \mathcal{U}_i) \subset W$ for some $W \in \mathcal{W}$. Then, it follows that $V_U \subset U \subset W$. Thus, we have the following open collection:

$$\mathcal{V} = \big\{ V_U \;\big|\; U \in \textstyle\bigcup_{i\in\mathbb{N}} \mathcal{U}''_i \big\} \prec \mathcal{W}.$$

If $\mathcal{V} \in \operatorname{cov}(X)$ and $\operatorname{ord} \mathcal{V} \leqslant n+1$, the proof is complete. Hence, it suffices to show that $0 < \operatorname{card} \mathcal{V}[x] \leqslant n+1$ for each $x \in X$.

For each $x \in X$, there uniquely exists $j(x) \in \mathbb{N}$ such that $x \in X_{j(x)} \setminus \bigcup_{j<j(x)} X_j$. For each $U \in \mathcal{U}_{j(x)}[x]$ $(\subset \mathcal{U}_{j(x)}[X_{j(x)}] = \mathcal{U}'_{j(x)})$, as observed in the above,

$$U_0 = \varphi_{m(U),j(x)}(U) \in \mathcal{U}''_{m(U)} \ \text{ and } \ x \in U \cap X_{j(x)} \subset V_{U_0} \in \mathcal{V}.$$

Hence, $\mathcal{V}[x] \neq \emptyset$, i.e., $\operatorname{card} \mathcal{V}[x] > 0$. Moreover, a function $\psi : \mathcal{U}_{j(x)}[x] \to \mathcal{V}[x]$ can be defined as follows:

$$\psi(U) = V_{U_0} = V_{\varphi_{m(U),j(x)}(U)} \ \text{ for each } U \in \mathcal{U}_{j(x)}[x].$$

Then, ψ is surjective. Indeed, let $V_U \in \mathcal{V}[x]$, where $U \in \mathcal{U}_i''$. By the definition of V_U, we have $k \geqslant i$ and $U' \in \mathcal{U}_k'$ such that $m(U') = i$, $\varphi_{i,k}(U') = U$, and $x \in U' \cap X_k$. Since $x \in V_U \subset U$ and $U \cap \bigcup_{j<i} X_j = \emptyset$, it follows that $x \notin \bigcup_{j<i} X_j$, which implies $i \leqslant j(x)$. Moreover, $k \geqslant j(x)$ because $x \in X_k$. Since $x \in U' \subset \varphi_{j(x),k}(U')$, we have

$$U_1 = \varphi_{j(x),k}(U') \in \mathcal{U}_{j(x)}[x] \subset \mathcal{U}_{j(x)}[X_{j(x)}] = \mathcal{U}'_{j(x)}.$$

Then, observe that $m(U_1) = m(U') = i$ and

$$\varphi_{m(U_1),j(x)}(U_1) = \varphi_{i,j(x)}(\varphi_{j(x),k}(U')) = \varphi_{i,k}(U') = U.$$

Therefore, $\psi(U_1) = V_U$. Consequently,

$$\operatorname{card} \mathcal{V}[x] \leqslant \operatorname{card} \mathcal{U}_{j(x)}[x] \leqslant \operatorname{ord} \mathcal{U}_{j(x)} \leqslant n+1.$$

The proof is complete. □

Applying Theorem 5.4.1, we can show that the inverse limit preserves the dimension, that is,

Theorem 5.4.2 *Let $X = \varprojlim(X_i, f_i)$ be the limit of an inverse sequence $(X_i, f_i)_{i\in\mathbb{N}}$ of metrizable spaces. If $\dim X_i \leqslant n$ for infinitely many $i \in \mathbb{N}$, then $\dim X \leqslant n$.*

Proof By Proposition 4.17.3, we may assume that $\dim X_i \leqslant n$ for every $i \in \mathbb{N}$. Recall that X is the following subspace of the product space $\prod_{i\in\mathbb{N}} X_i$:

$$X = \left\{x \in \textstyle\prod_{i\in\mathbb{N}} X_i \;\middle|\; x(i) = f_i(x(i+1)) \text{ for every } i \in \mathbb{N}\right\}.$$

We define $d \in \operatorname{Metr}(X)$ as follows:

$$d(x, y) \underset{\text{def}}{=} \sup_{i\in\mathbb{N}} \min\{d_i(x(i), y(i)), 2^{-i}\},$$

where $d_i \in \operatorname{Metr}(X_i)$. For each $i \in \mathbb{N}$, we can inductively choose $\mathcal{V}_i \in \operatorname{cov}(X_i)$ so that $\operatorname{ord} \mathcal{V}_i \leqslant n+1$,

$$\mathcal{V}_i \prec (f_j \dots f_{i-1})^{-1}(\mathcal{V}_j) \text{ and } \operatorname{mesh} f_j \dots f_{i-1}(\mathcal{V}_i) < 2^{-i} \text{ for } j < i.$$

Let $\mathcal{U}_i = p_i^{-1}(\mathcal{V}_i) \in \operatorname{cov}(X)$, where $p_i = \operatorname{pr}_i | X : X \to X_i$ is the inverse limit projection. Then, $\mathcal{U}_1 \succ \mathcal{U}_2 \succ \cdots$, $\operatorname{ord} \mathcal{U}_i \leqslant n+1$, and $\operatorname{mesh} \mathcal{U}_i \leqslant 2^{-i}$. Therefore, $\dim X \leqslant n$ by Theorem 5.4.1. □

When X is compact, Theorem 5.4.2 is valid even if X is non-metrizable, that is,

Theorem 5.4.3 *Let $X = \varprojlim(X_\lambda, f_{\lambda,\lambda'})_{\lambda\in\Lambda}$ be the inverse limit of an inverse system of compact spaces. If $\Delta = \{\lambda \in \Lambda \mid \dim X_\lambda \leqslant n\}$ is cofinal in Λ, then $\dim X \leqslant n$.*

Proof Let $\mathcal{U} \in \mathrm{cov}(X)$. Since X is compact, we can apply Proposition 4.16.4 to find $\lambda_1, \ldots, \lambda_k \in \Lambda$ and open sets U_i in X_{λ_i}, $i = 1, \ldots, k$, such that $\{p_{\lambda_i}^{-1}(U_i) \mid i = 1, \ldots, k\} \prec \mathcal{U}$ and $X = \bigcup_{i=1}^{k} p_{\lambda_i}^{-1}(U_i)$, where $p_\lambda : X \to X_\lambda$ is the inverse limit projection. Choose $\delta_0 \in \Delta$ so that $\delta_0 \geqslant \lambda_i$, $i = 1, \ldots, k$, and define

$$\mathcal{U}_0 = \left\{ f_{\lambda_i, \delta_0}^{-1}(U_i) \;\middle|\; i = 1, \ldots, k \right\} \cup \left\{ X_{\delta_0} \setminus p_{\delta_0}(X) \right\} \in \mathrm{cov}(X_{\delta_0}).$$

Since $\dim X_{\delta_0} \leqslant n$, $\mathcal{U}_0$ has a finite open refinement $\mathcal{V}$ with $\mathrm{ord}\, \mathcal{V} \leqslant n+1$. Then, $p_{\delta_0}^{-1}(\mathcal{V})$ is an open refinement of $\mathcal{U}$ with $\mathrm{ord}\, p_{\delta_0}^{-1}(\mathcal{V}) \leqslant n+1$. Therefore, $\dim X \leqslant n$. □

Remark 5.4.1 Theorem 5.4.3 is valid for the inverse limit of an inverse system of paracompact spaces and *perfect maps* (cf. Notes for this chapter).

As mentioned in Remark 5.2.3, if Y is a *closed* set in X, then $\dim Y \leqslant \dim X$. This is not true in the case when Y is not closed in X. In fact, there exists a zero-dimensional compact space X containing a subspace Y with $\dim Y > 0$. Such a space will be constructed in Sect. 5.5 (cf. Theorem 5.5.5). However, when X is metrizable, the following SUBSET THEOREM holds, which is a trivial consequence of Theorem 5.4.1:

Theorem 5.4.4 (SUBSET) *For an arbitrary subset Y of a metrizable space X,* $\dim Y \leqslant \dim X$. □

We can apply Theorem 5.4.1 to prove the following DIMENSION-PRESERVING COMPLETION THEOREM:

Theorem 5.4.5 (DIMENSION-PRESERVING COMPLETION) *Every n-dimensional metrizable space X can be embedded in an n-dimensional completely metrizable space as a dense set.*

Proof We can regard X as a dense subset of a complete metric space $Y = (Y, d)$ (Corollary 2.3.11). Applying Theorem 5.4.1, we can obtain a sequence $\mathcal{U}_1 \succ \mathcal{U}_2 \succ \cdots \in \mathrm{cov}(X)$ such that $\mathrm{ord}\, \mathcal{U}_i \leqslant n+1$ and $\mathrm{mesh}_d\, \mathcal{U}_i \to 0$ as $i \to \infty$. For each $i \in \mathbb{N}$, we can inductively take a collection $\widetilde{\mathcal{U}}_i$ of open sets in Y such that $\widetilde{\mathcal{U}}_i | X = \mathcal{U}_i$ and $\widetilde{\mathcal{U}}_i \prec \widetilde{\mathcal{U}}_{i-1}$, where $\widetilde{\mathcal{U}}_{-1} = \{Y\}$.

> Indeed, for each $U \in \mathcal{U}_i$, choose an open set U' in Y and $U'' \in \widetilde{\mathcal{U}}_{i-1}$ so that $U = U' \cap X$ and $U \subset U''$, and define $\widetilde{U} = U' \cap U''$. Then, $\widetilde{\mathcal{U}}_i = \{\widetilde{U} \mid U \in \mathcal{U}_i\}$ is the desired one.

Then, $\mathrm{ord}\, \widetilde{\mathcal{U}}_i = \mathrm{ord}\, \mathcal{U}_i \leqslant n+1$ and $\mathrm{mesh}_d\, \widetilde{\mathcal{U}}_i = \mathrm{mesh}_d\, \mathcal{U}_i$. For each $i \in \mathbb{N}$, $G_i = \bigcup \widetilde{\mathcal{U}}_i$ is an open set in Y. Thus, we have a G_δ-set $\widetilde{X} = \bigcap_{i \in \mathbb{N}} G_i$ in Y and X is dense in $\widetilde{X}$. According to Theorem 2.5.3(2), $\widetilde{X}$ is completely metrizable. Using Theorems 5.4.1 and 5.4.4, we can obtain $\dim \widetilde{X} \leqslant n$ and $\dim \widetilde{X} \geqslant n$, that is, $\dim \widetilde{X} = n$. □

Recall that a **clopen set** in a space X is both closed and open in X. A **clopen basis** for X is an open basis consisting of clopen sets. For metrizable spaces, we characterize the zero-dimensionality as follows:

Theorem 5.4.6 *For a metrizable space X ($\neq \emptyset$), $\dim X = 0$ if and only if X has a σ-locally finite clopen basis.*

Proof (*The "only if" Part*) Assume that $\dim X = 0$ and let $d \in \mathrm{Metr}(X)$. By Theorem 5.4.1, X has a sequence of locally finite open covers $\mathcal{B}_1 \succ \mathcal{B}_2 \succ \cdots$ such that $\operatorname{ord} \mathcal{B}_i = 1$ and $\lim_{i\to\infty} \operatorname{mesh} \mathcal{B}_i = 0$. Note that each $B \in \mathcal{B}_i$ is clopen in X because $B = X \setminus \bigcup\{B' \in \mathcal{B}_i \mid B' \neq B\}$. It is easy to see that $\mathcal{B} = \bigcup_{i\in\mathbb{N}} \mathcal{B}_i$ is a σ-locally finite clopen basis for X.

(*The "if" Part*) Let $\mathcal{B} = \bigcup_{i\in\mathbb{N}} \mathcal{B}_i$ be a σ-locally finite clopen basis for X, where each $\mathcal{B}_i$ is locally finite. Let $\{U_1, U_2\} \in \mathrm{cov}(X)$. For each $i \in \mathbb{N}$, let

$$V_{2i-1} = \bigcup \{B \in \mathcal{B}_i \mid B \subset U_1\} \text{ and } V_{2i} = \bigcup \{B \in \mathcal{B}_i \mid B \subset U_2\}.$$

Because V_i is clopen, we have an open set $W_i = V_i \setminus \bigcup_{j<i} V_j$ in X. Then, $\mathcal{W} = \{W_i \mid i \in \mathbb{N}\}$ is an open refinement of $\{U_1, U_2\}$. Indeed, $W_{2i-1} \subset U_1$, $W_{2i} \subset U_2$, and

$$\bigcup_{i\in\mathbb{N}} W_i = \bigcup_{i\in\mathbb{N}} V_i = \bigcup_{i\in\mathbb{N}} V_{2i-1} \cup \bigcup_{i\in\mathbb{N}} V_{2i} = U_1 \cup U_2 = X.$$

Since $\operatorname{ord} \mathcal{W} \leqslant 1$ by definition, we have $\dim X \leqslant 0$ by Theorem 5.2.4. □

Using the above characterization, we can easily show that $\dim \mathbb{Q} = \dim(\mathbb{R} \setminus \mathbb{Q}) = 0$ and $\dim \mu^0 = 0$, where μ^0 is the Cantor (ternary) set. The following theorem can also be easily proved by applying this characterization.

Theorem 5.4.7 *The countable product of zero-dimensional metrizable spaces is zero-dimensional.* □

The following is called the PARTITION EXTENSION LEMMA, which is simple but very useful in Dimension Theory:

Lemma 5.4.8 (PARTITION EXTENSION) *Let A, B be closed and U, V be open sets in a metrizable space X such that $A \subset U$ and $B \subset V$ and $\mathrm{cl}\, U \cap \mathrm{cl}\, V = \emptyset$. For any subspace Y of X, if Y has a partition S between $Y \cap \mathrm{cl}\, U$ and $Y \cap \mathrm{cl}\, V$, then X has a partition L between A and B with $Y \cap L \subset S$.*

Proof Let U' and V' be disjoint open sets in Y such that $Y \setminus S = U' \cup V'$, $Y \cap \mathrm{cl}\, U \subset U'$, and $Y \cap \mathrm{cl}\, V \subset V'$. From $U \cap V' = \emptyset$, it follows that $A \cap \mathrm{cl}\, V' = \emptyset$. Then,

$$(A \cup U') \cap \mathrm{cl}(B \cup V') = (A \cup U') \cap (B \cup \mathrm{cl}\, V') = \emptyset.$$

Similarly, we have $(B \cup V') \cap \mathrm{cl}(A \cup U') = \emptyset$. Take a metric $d \in \mathrm{Metr}(X)$ and define

$$U'' = \{x \in X \mid d(x, A \cup U') < d(x, B \cup V')\} \text{ and}$$
$$V'' = \{x \in X \mid d(x, B \cup V') < d(x, A \cup U')\}.$$

Then, U'' and V'' are disjoint open sets in X, $A \cup U' \subset U''$, and $B \cup V' \subset V''$. Hence, $L = X \setminus (U'' \cup V'')$ is the desired partition. □

Remark 5.4.2 In Lemma 5.4.8, it does not suffice to assume that S is a partition between $A \cap Y$ and $B \cap Y$ in Y. In fact, $A = [-1, 0] \times \{0\}$ and $B = [0, 1] \times \{1\}$ are disjoint closed sets in $X = \mathbb{R}^2$. Let $Y = \mathbb{R}^2 \setminus \{\mathbf{0}, \mathbf{e}_2\} \subset X$, where $\mathbf{e}_2 = (0, 1) \in \mathbb{R}^2$. Then, $S = (\{0\} \times \mathbb{R}) \cap Y$ is a partition between $A \cap Y$ and $B \cap Y$ in Y but X has no partition L between A and B such that $Y \cap L \subset S$.

Using partitions, we can characterize the dimension for metrizable spaces as in the following theorem:

Theorem 5.4.9 *Let X be metrizable and $n \in \omega$. Then,* $\dim X \leqslant n$ *if and only if, for any pair of disjoint closed sets A and B in X, there is a partition L in X between A and B with* $\dim L \leqslant n - 1$.

Proof The "if" part is Proposition 5.3.2. So, we will prove the "only if" part. Let $d \in \mathrm{Metr}(X)$ such that $\mathrm{dist}(A, B) > 1$. (Such a metric can be obtained by a metric for X and a Urysohn map for A and B.) Using Theorem 5.2.5 (cf. Theorem 5.4.1), we can construct a sequence $(\mathcal{U}_i)_{i \in \mathbb{N}}$ of locally finite open covers of X such that $\operatorname{ord} \mathcal{U}_i \leqslant n + 1$, $\operatorname{mesh} \mathcal{U}_i < i^{-1}$, and $\mathcal{U}_{i+1}^{\mathrm{cl}} \prec \mathcal{U}_i$. Let A_0 and B_0 be open neighborhoods of A and B in X, respectively, such that $\mathrm{dist}(A_0, B_0) > 1$. We inductively define sets A_i and B_i, $i \in \mathbb{N}$, as follows:

$$A_i = X \setminus \bigcup \left\{ \operatorname{cl} U \;\middle|\; U \in \mathcal{U}_i[B_{i-1}] \right\} \quad \text{and}$$
$$B_i = X \setminus \bigcup \left\{ \operatorname{cl} U \;\middle|\; U \in \mathcal{U}_i \setminus \mathcal{U}_i[B_{i-1}] \right\}.$$

Then, $A_i \cap B_i = \emptyset$. Because of the local finiteness of $\mathcal{U}_i^{\mathrm{cl}}$, A_i and B_i are open in X. Note that $B_{i-1} \cap U = \emptyset$ if and only if $B_{i-1} \cap \operatorname{cl} U = \emptyset$ for each $U \in \mathcal{U}_i$. Then, it follows that $B_{i-1} \subset B_i$ for each $i \in \mathbb{N}$. Hence, $\mathcal{U}_i[B_{i-1}] \subset \mathcal{U}_i[B_i]$. We also have $\mathcal{U}_i[B_i] \subset \mathcal{U}_i[B_{i-1}]$. Indeed, each $U \in \mathcal{U}_i[B_i]$ contains some point of B_i, where that point does not belong to any member of $\mathcal{U}_i \setminus \mathcal{U}_i[B_{i-1}]$. This means $U \in \mathcal{U}_i[B_{i-1}]$. Therefore, $\mathcal{U}_i[B_i] = \mathcal{U}_i[B_{i-1}]$ for each $i \in \mathbb{N}$.

We will show that $\operatorname{cl} A_{i-1} \subset A_i$ for each $i \in \mathbb{N}$. This follows from the fact that $\operatorname{cl} A_{i-1} \cap \operatorname{cl} U = \emptyset$ for each $U \in \mathcal{U}_i[B_{i-1}]$. This fact can be shown as follows: The case $i = 1$ follows from $\operatorname{mesh} \mathcal{U}_1 < 1$ and $\mathrm{dist}(A_0, B_0) > 1$. When $i > 1$, for each $U \in \mathcal{U}_i[B_{i-1}]$, $\operatorname{cl} U$ is contained in some $V \in \mathcal{U}_{i-1}$. Since $V \in \mathcal{U}_{i-1}[B_{i-1}] = \mathcal{U}_{i-1}[B_{i-2}]$, it follows that $A_{i-1} \cap V = \emptyset$, and hence $\operatorname{cl} A_{i-1} \cap V = \emptyset$, which implies $\operatorname{cl} A_{i-1} \cap \operatorname{cl} U = \emptyset$.

For each $i \in \mathbb{N}$, let $L_i = X \setminus (A_i \cup B_i)$ and let $L = \bigcap_{i \in \mathbb{N}} L_i$. Then, L is a partition between A and B. Indeed,

$$X \setminus L = \left(\bigcup_{i \in \mathbb{N}} A_i \right) \cup \left(\bigcup_{i \in \mathbb{N}} B_i \right), \quad A \subset \bigcup_{i \in \mathbb{N}} A_i, \quad B \subset \bigcup_{i \in \mathbb{N}} B_i, \quad \text{and}$$

$$\left(\bigcup_{i\in\mathbb{N}} A_i\right)\cap\left(\bigcup_{i\in\mathbb{N}} B_i\right)=\bigcup_{i,j\in\mathbb{N}}(A_i\cap B_j)\subset\bigcup_{i,j\in\mathbb{N}}(A_{\max\{i,j\}}\cap B_{\max\{i,j\}})$$
$$=\bigcup_{i\in\mathbb{N}}(A_i\cap B_i)=\emptyset.$$

For each $i\in\mathbb{N}$, we have

$$\mathcal{W}_i=\left\{U\cap L\;\middle|\; U\in\mathcal{U}_i[B_{i-1}]\right\}\in\operatorname{cov}(L).$$

Indeed, each $x\in L$ is not contained in A_{i+1}, so $x\in\operatorname{cl}V$ for some $V\in\mathcal{U}_{i+1}[B_i]$. Choose $U\in\mathcal{U}_i$ so that $\operatorname{cl}V\subset U$. Then, $U\in\mathcal{U}_i[B_i]=\mathcal{U}_i[B_{i-1}]$, hence $x\in U\cap L\in\mathcal{W}_i$. Therefore, $\mathcal{W}_i\in\operatorname{cov}(L)$. Note that $\operatorname{mesh}\mathcal{W}_i\leqslant\operatorname{mesh}\mathcal{U}_i<i^{-1}$. Moreover, $\mathcal{W}_{i+1}\prec\mathcal{W}_i$ because each $V\in\mathcal{U}_{i+1}[B_i]$ is contained in some $U\in\mathcal{U}_i$, where $U\in\mathcal{U}_i[B_i]=\mathcal{U}_i[B_{i-1}]$.

We will show that $\operatorname{ord}\mathcal{W}_i\leqslant n$. Suppose that there exist $n+1$ distinct $U_1,\dots,U_{n+1}\in\mathcal{U}_i[B_{i-1}]$ that contain a common point $x\in L$. Since $x\notin B_i$, $x\in\operatorname{cl}U_{n+2}$ for some $U_{n+2}\in\mathcal{U}_i\setminus\mathcal{U}_i[B_{i-1}]$. Since $\bigcap_{j=1}^{n+1}U_j$ is a neighborhood of x, it follows that $\bigcap_{j=1}^{n+2}U_j\neq\emptyset$, which is contrary to $\operatorname{ord}\mathcal{U}_i\leqslant n+1$. Therefore, we have $\dim L\leqslant n-1$ by Theorem 5.4.1. □

The following lemma is very useful, and will be also used several times in Chapter 6:

Lemma 5.4.10 (Canonical Open Set Extension) *Let X be metrizable and $\emptyset\neq A\subset X$. Then, there exists an operation extending each open set U in A to an open set $E(U)$ in X satisfying the following properties:*

(i) $E(\emptyset)=\emptyset$, $E(A)=X$;
(ii) $E(U)\cap A=U$;
(iii) $U\subset V\Rightarrow E(U)\subset E(V)$;
(iv) $E(U\cap V)=E(U)\cap E(V)$;
(v) $\operatorname{cl}_A U=A\cap\operatorname{cl}_X E(U)$, *so* $\operatorname{bd}_A U=A\cap\operatorname{bd}_X E(U)$.

Proof Using $d\in\operatorname{Metr}(X)$, for each open set U in A, we define

$$E(U)=\left\{x\in X\;\middle|\; d(x,U)<d(x,A\setminus U)\right\},$$

where $d(x,\emptyset)=\infty$. Then, $E(U)$ is an open set in X. Conditions (i), (ii), and (iii) are obvious. For (iv), it suffices to show $E(U)\cap E(V)\subset E(U\cap V)$ because the converse inclusion follows from (iii). For each $x\in E(U)\cap E(V)$, we prove

$$(*)\;\; d(x,U\cap V)<d(x,A\setminus(U\cap V))=\min\left\{d(x,A\setminus U),\; d(x,A\setminus V)\right\}.$$

Then, without loss of generality, we may assume that

$$d(x,A\setminus(U\cap V))=d(x,A\setminus U)\leqslant d(x,A\setminus V).$$

Since $x \in E(U)$, $d(x, y) < d(x, A \setminus U)$ for some $y \in U$. Then, it follows that $d(x, y) < d(x, A \setminus V)$, which implies that $y \in V$. Hence, $y \in U \cap V$, so we have the inequality $(*)$.

To see (v), assume that $\mathrm{cl}_A U \neq A \cap \mathrm{cl}_X E(U)$. Because $\mathrm{cl}_A U = A \cap \mathrm{cl}_X U \subset A \cap \mathrm{cl}_X E(U)$, there exists $x \in A \cap \mathrm{cl}_X E(U)$ such that $x \notin \mathrm{cl}_A U$. Since $x \in \mathrm{cl}_X E(U)$ and $d(x, U) > 0$, we have $y \in E(U)$ such that $d(x, y) < d(x, U)/2$, that is, $d(x, y) < d(x, U) - d(x, y)$. Then,

$$d(y, A \setminus U) \leqslant d(y, x) < d(x, U) - d(x, y) \leqslant d(y, U),$$

which contradicts the fact that $y \in E(U)$. □

The real line $\mathbb{R}$ is one-dimensional and it can be decomposed into two zero-dimensional subsets $\mathbb{Q}$ and $\mathbb{R} \setminus \mathbb{Q}$. Using Lemma 5.4.10, we can generalize this to the following DECOMPOSITION THEOREM:

Theorem 5.4.11 (DECOMPOSITION) *Let X be metrizable and $n \in \omega$. Then,* $\dim X \leqslant n$ *if and only if X is covered by $n+1$ subsets $X_1, \ldots, X_{n+1}$ with* $\dim X_i \leqslant 0$.

Proof (*The "if" Part*) Let $\mathcal{U}$ be a finite open cover of X. Since $\dim X_i \leqslant 0$, X_i has a finite open cover $\mathcal{V}_i$ such that $\mathcal{V}_i \prec \mathcal{U}$ and $\mathrm{ord}\, \mathcal{V}_i \leqslant 1$. For each $V \in \mathcal{V}_i$, choose $U_V \in \mathcal{U}$ such that $V \subset U_V$ and let $E_i(V)$ be an open set in X obtained by applying Lemma 5.4.10, where $A = X_i$. Thus, we have an open set $\widetilde{V} = U_V \cap E_i(V)$ in X such that $\widetilde{V} \cap X_i = V$. If $V \neq V' \in \mathcal{V}_i$, then $\widetilde{V} \cap \widetilde{V}' = \emptyset$. Indeed, since $\mathrm{ord}\, \mathcal{V}_i = 1$, we have $V \cap V' = \emptyset$, which implies $E_i(V) \cap E_i(V') = \emptyset$. Hence, $\widetilde{\mathcal{V}}_i = \{\widetilde{V} \mid V \in \mathcal{V}_i\}$ is a pairwise disjoint open collection in X that covers X_i and refines $\mathcal{U}$. Observe that $\widetilde{\mathcal{V}} = \bigcup_{i=1}^{n+1} \widetilde{\mathcal{V}}_i$ is an open refinement of $\mathcal{U}$ with $\mathrm{ord}\, \mathcal{V} \leqslant n+1$. Therefore, $\dim X \leqslant n$.

(*The "only if" Part*) This part can be easily obtained by induction once the following proposition has been proved. □

Proposition 5.4.12 *Let X be metrizable and* $\dim X \leqslant n < \infty$. *Then, $X = Y \cup Z$ for some $Y, Z \subset X$ with* $\dim Y \leqslant n-1$ *and* $\dim Z \leqslant 0$.

Proof Assume that X is a metric space. For each $i \in \mathbb{N}$, let $\mathcal{U}_i$ be a locally finite open cover of X with mesh $\mathcal{U}_i < i^{-1}$. By normality (Lemma 2.7.6), $\mathcal{U}_i$ has a closed shrinking $\{F_U \mid U \in \mathcal{U}_i\}$ (i.e., $X = \bigcup_{U \in \mathcal{U}_i} F_U$ and $F_U \subset U$ for all $U \in \mathcal{U}_i$). For each $U \in \mathcal{U}_i$, we apply Theorem 5.4.9 to obtain an open set B_U such that

$$F_U \subset B_U \subset \mathrm{cl}\, B_U \subset U \quad \text{and} \quad \dim \mathrm{bd}\, B_U \leqslant n-1.$$

It is easy to see that $\mathcal{B} = \{B_U \mid U \in \mathcal{U}_i, i \in \mathbb{N}\}$ is a σ-locally finite open basis for X. Let

$$Y = \bigcup \{\mathrm{bd}\, B \mid B \in \mathcal{B}\} \quad \text{and} \quad Z = X \setminus Y.$$

According to Corollary 5.2.10, $\dim Y \leqslant n-1$. Since $\{B \cap Z \mid B \in \mathcal{B}\}$ is a σ-locally finite clopen basis for Z, we have $\dim Z \leqslant 0$ by Theorem 5.4.6. □

In the above, we considered the following conditions for a metrizable space X:

(a) $\dim X \leqslant n$;
(b) *X has a σ-locally finite open basis $\mathcal{B}$ such that* $\dim \mathrm{bd}\, B \leqslant n-1$ *for every* $B \in \mathcal{B}$;
(c) $X = Y \cup Z$ *for some* $Y, Z \subset X$ *with* $\dim Y \leqslant n-1$ *and* $\dim Z \leqslant 0$;
(d) *X is covered by $n+1$ subsets $X_1, \ldots, X_{n+1}$ with* $\dim X_i \leqslant 0$.

In the proof of Proposition 5.4.12, we proved (a) $\Rightarrow$ (b) $\Rightarrow$ (c). The implication (d) $\Rightarrow$ (a) is just the "if" part of the Decomposition Theorem 5.4.11. As observed in the proof of the "only if" part, (d) can be obtained by inductive use of (c), that is, (c) $\Rightarrow$ (d). Thus, the above four conditions are equivalent to each other for a metrizable space X. In particular, we have the following characterization, which is a generalization of Theorem 5.4.6:

Theorem 5.4.13 *Let X be metrizable and $n \in \omega$. Then,* $\dim X \leqslant n$ *if and only if X has a σ-locally finite open basis $\mathcal{B}$ such that* $\dim \mathrm{bd}\, B \leqslant n-1$ *for each* $B \in \mathcal{B}$. □

As a corollary of the Decomposition Theorem 5.4.11, we have the following ADDITION THEOREM:

Theorem 5.4.14 (ADDITION) *For any two subspaces X and Y of a metrizable space,*

$$\dim X \cup Y \leqslant \dim X + \dim Y + 1.$$ □

The following PRODUCT THEOREM can be proved by using Theorem 5.4.13:

Theorem 5.4.15 (PRODUCT) *For any metrizable spaces X and Y,*

$$\dim X \times Y \leqslant \dim X + \dim Y.$$

Proof If $\dim X = \infty$ or $\dim Y = \infty$, the theorem is obvious.

When $\dim X, \dim Y < \infty$, we prove the theorem by induction on $\dim X + \dim Y$. The case $\dim X = \dim Y = 0$ is a special case of Theorem 5.4.7. Assume the theorem is true for any two spaces X and Y with $\dim X + \dim Y < k$. Now, let $\dim X = m$, $\dim Y = n$, and $m + n = k$. According to Theorem 5.4.13, X and Y have σ-locally finite open bases $\mathcal{B}_X$ and $\mathcal{B}_Y$ such that $\dim \mathrm{bd}\, B \leqslant m-1$ for each $B \in \mathcal{B}_X$ and $\dim \mathrm{bd}\, B \leqslant n-1$ for each $B \in \mathcal{B}_Y$. Then,

$$\mathcal{B} = \left\{ B_1 \times B_2 \;\middle|\; B_1 \in \mathcal{B}_X \text{ and } B_2 \in \mathcal{B}_Y \right\}$$

is a σ-locally finite open basis for $X \times Y$. For each $B_1 \in \mathcal{B}_X$ and $B_2 \in \mathcal{B}_Y$,

$$\mathrm{bd}(B_1 \times B_2) = (\mathrm{bd}\, B_1 \times \mathrm{cl}\, B_2) \cup (\mathrm{cl}\, B_1 \times \mathrm{bd}\, B_2).$$

Hence, $\dim \mathrm{bd}(B_1 \times B_2) \leqslant m + n - 1$ by the inductive assumption and Theorem 5.2.8 or 5.2.9. Then, we have $\dim X \times Y \leqslant m + n$ by Theorem 5.4.13. □

Remark 5.4.3 In Theorem 5.4.15, the equality $\dim X \times Y = \dim X + \dim Y$ does not hold in general. In fact, there exists a separable metrizable space X such that $\dim X^2 \neq 2 \dim X$. Such a space will be constructed in Theorem 5.12.1. However, if one of X or Y is a locally compact polyhedron or a metric polyhedron (cf. Sect. 4.6), the equality does hold. This will be proved in Theorem 7.9.8.

5.5 Inductive Dimensions

In this section, we introduce two types of dimension defined by induction. First, the **large inductive dimension** of a *normal* space X, $\operatorname{Ind} X$, can be defined as follows: $\operatorname{Ind} X = -1$ if $X = \emptyset$; $\operatorname{Ind} X \leqslant n$ if each closed set $A \subset X$ has an arbitrarily small *open* neighborhood V with $\operatorname{Ind} \operatorname{bd} V \leqslant n-1$, that is, any neighborhood of A contains such a neighborhood V. Then, we define $\operatorname{Ind} X = n$ if $\operatorname{Ind} X \leqslant n$ and $\operatorname{Ind} X \not\leqslant n-1$. We write $\operatorname{Ind} X < \infty$ if $\operatorname{Ind} X \leqslant n$ for some $n \in \mathbb{N}$, and otherwise $\operatorname{Ind} X = \infty$. Observe the following:

- If Y is a *closed* set in X, then $\operatorname{Ind} Y \leqslant \operatorname{Ind} X$.

For an open set G and a closed set F in X,

$$\operatorname{bd} \operatorname{cl} G = \operatorname{cl} G \setminus \operatorname{int} \operatorname{cl} G \subset \operatorname{cl} G \setminus G = \operatorname{bd} G \quad \text{and}$$
$$\operatorname{bd} \operatorname{int} F = \operatorname{cl} \operatorname{int} F \setminus \operatorname{int} F \subset F \setminus \operatorname{int} F = \operatorname{bd} F.$$

For every normal space X, $\operatorname{Ind} X \leqslant n$ if and only if each closed set A in X has an arbitrarily small *closed* neighborhood V with $\operatorname{Ind} \operatorname{bd} V \leqslant n-1$. As is easily observed, the following holds:

- *For every normal space* X, $\operatorname{Ind} X \leqslant n$ *if and only if, for any two disjoint closed sets* A *and* B *in* X, *there is a partition* L *between* A *and* B *with* $\operatorname{Ind} L \leqslant n-1$.

The next theorem follows, by induction, from Proposition 5.3.2 and Theorem 5.4.9:

Theorem 5.5.1 (1) *For every normal space* X, $\dim X \leqslant \operatorname{Ind} X$.
(2) *For every metrizable space* X, $\dim X = \operatorname{Ind} X$. □

Remark 5.5.1 (2) above does not hold even if X is compact. Such an example will be seen in Theorem 5.5.6.

Concerning zero-dimensionality, we have the following:

Theorem 5.5.2 *For every normal space* X, $\dim X = 0$ *if and only if* $\operatorname{Ind} X = 0$.

Proof The "if" part comes from $\dim X \leqslant \operatorname{Ind} X$ (Theorem 5.5.1 (1)). To see the "only if" part, assume $\dim X = 0$. Let A be a closed set in X and U an open neighborhood of A in X. The open cover $\{X \setminus A, U\} \in \operatorname{cov}(X)$ has a finite open refinement $\mathcal{V}$ with $\operatorname{ord} \mathcal{V} \leqslant 1$. Then, $\operatorname{st}(A, \mathcal{V}) \subset U$. Since $\mathcal{V}$ is pairwise disjoint, $\operatorname{st}(A, \mathcal{V}) \subset U$ is clopen in X, which means $\operatorname{bd} \operatorname{st}(A, \mathcal{V}) \subset U = \emptyset$. Therefore, $\operatorname{Ind} X = 0$. □

Next, the **small inductive dimension** of a *regular* space X, $\operatorname{ind} X$, is defined as follows: $\operatorname{ind} X = -1$ if $X = \emptyset$; $\operatorname{ind} X \leqslant n$ if each point $x \in X$ has an arbitrarily small *open* neighborhood V with $\operatorname{ind} \operatorname{bd} V \leqslant n-1$. Then, we define $\operatorname{ind} X = n$ if $\operatorname{ind} X \leqslant n$ and $\operatorname{ind} X \not\leqslant n-1$. We write $\operatorname{ind} X < \infty$ if $\operatorname{ind} X \leqslant n$ for some $n \in \mathbb{N}$, and otherwise $\operatorname{ind} X = \infty$. The following are easily observed:

- $\operatorname{ind} X \leqslant n$ if and only if X has an open basis $\mathcal{B}$ such that $\operatorname{ind} \operatorname{bd} B \leqslant n-1$ for every $B \in \mathcal{B}$.
- $\operatorname{ind} Y \leqslant \operatorname{ind} X$ for an *arbitrary* subset $Y \subset X$.

For every regular space X, $\operatorname{ind} X \leqslant n$ if and only if each point x of X has an arbitrarily small *closed* neighborhood V with $\operatorname{ind} \operatorname{bd} V \leqslant n-1$. By induction, we can deduce the following from the definition and Theorem 5.4.13:

Theorem 5.5.3 (1) *For every space* X, $\operatorname{ind} X \leqslant \operatorname{Ind} X$.
(2) *For every metrizable space* X, $\operatorname{ind} X \leqslant \dim X$ $(= \operatorname{Ind} X)$. □

Remark 5.2.2 (2) above does not hold even if X is compact. Such an example will be seen in Theorem 5.5.6. When X is a normal space, both cases $\operatorname{ind} X < \dim X$ and $\dim X < \operatorname{ind} X$ are possible (cf. Remark 5.5.4).

One might expect that the equality $\operatorname{ind} X = \dim X$ $(= \operatorname{Ind} X)$ holds for an arbitrary metrizable space X. However, there exists a completely metrizable space X such that $\operatorname{ind} X \neq \operatorname{Ind} X$. Before constructing such a space, we first prove the following COINCIDENCE THEOREM:

Theorem 5.5.4 (COINCIDENCE) *For every separable metrizable space* X,

$$\dim X = \operatorname{Ind} X = \operatorname{ind} X.$$

Proof By virtue of Theorem 5.5.3, it is enough to show that $\dim X \leqslant \operatorname{ind} X$ when $\operatorname{ind} X < \infty$. We will prove this by induction on $\operatorname{ind} X$. Assume that $\dim X \leqslant \operatorname{ind} X$ for every separable metrizable space X with $\operatorname{ind} X < n$. Now, let $\operatorname{ind} X = n$. Then, X has an open basis $\mathcal{B}$ such that $\operatorname{ind} \operatorname{bd} B \leqslant n-1$ for every $B \in \mathcal{B}$. Since X is separable metrizable, X has a countable open basis $\{V_i \mid i \in \mathbb{N}\}$. We define

$$P = \left\{(i, j) \in \mathbb{N}^2 \;\middle|\; V_i \subset B \subset V_j \;\text{ for some }\; B \in \mathcal{B}\right\}.$$

For each $p = (i, j) \in P$, choose $B_p \in \mathcal{B}$ so that $V_i \subset B_p \subset V_j$. Then, $\{B_p \mid p \in P\}$ is a countable open basis for X such that $\dim \operatorname{bd} B_p \leqslant \operatorname{ind} \operatorname{bd} B_p \leqslant n-1$ for each $p \in P$. By Theorem 5.4.13, we have $\dim X \leqslant n$. □

As mentioned above, the Coincidence Theorem 5.5.4 does not hold for non-separable spaces. In fact, we have the following:

Theorem 5.5.5 *There exists a completely metrizable space* Z *such that* $\operatorname{ind} Z = 0$ *but* $\operatorname{Ind} Z = \dim Z = 1$. *Furthermore,* Z *has a zero-dimensional compactification.*

Proof (*Example*) Let $\Omega = [0, \omega_1)$ be the space of all countable ordinals with the order topology. Note that the space $\overline{\Omega} = [0, \omega_1]$ is compact and zero-dimensional. In fact, for each open cover $\mathcal{U}$ of $\overline{\Omega}$, we can inductively choose $\omega_1 = \alpha_0 > \alpha_1 > \alpha_2 > \cdots$ so that each $(\alpha_i, \alpha_{i-1}]$ is contained in some member of $\mathcal{U}$. Since Ω is well ordered, some α_n is equal to 0. Thus, $\mathcal{U}$ has a finite open refinement $\{0\} \cup \{(\alpha_i, \alpha_{i-1}] \mid i = 1, \ldots, n\}$, which is pairwise disjoint.

Our space is constructed as a subspace of the product $\Omega^{\mathbb{N}}$. Let L be the subset of Ω consisting of infinite limit ordinals and $S = \Omega \setminus L$. For each $k \in \mathbb{N}$, let $S_k = \{\alpha + k \mid \alpha \in L\}$. We define

$$Z = \Big\{ z \in \Omega^{\mathbb{N}} \;\Big|\; z(k) \in L \Rightarrow z(k+1) = z(k) + k, \\ z(k+j) \in S_k \;\text{ for } j > 1 \Big\}.$$

By definition, we have $\operatorname{ind} \Omega^{\mathbb{N}} = 0$, so $\operatorname{ind} Z = 0$. On the other hand, we can write $Z = S^{\mathbb{N}} \cup \bigcup_{k \in \mathbb{N}} Z_k$, where each Z_k is a closed set in Z defined as follows:

$$Z_k = \Big\{ z \in Z \;\Big|\; z(k) \in L \Big\} \subset S^{k-1} \times L \times S_k^{\mathbb{N}}.$$

Since S is a discrete space, it follows from Theorem 5.4.7 that $\dim S^{\mathbb{N}} = 0$. As is easily seen, each Z_k is homeomorphic to $S^{k-1} \times S_k^{\mathbb{N}}$ via the following correspondence:

$$Z_k \ni z \mapsto (z(1), \ldots, z(k-1), z(k+1), z(k+2), \ldots) \in S^{k-1} \times S_k^{\mathbb{N}},$$

where $z(k+1) = z(k) + k$. Then, it follows that $\dim Z_k = 0$ for each $k \in \mathbb{N}$.

(*Neighborhood Bases*) For each $\alpha \in L$, choose $\xi_1(\alpha) < \xi_2(\alpha) < \cdots < \alpha$ so that $\sup_{i \in \mathbb{N}} \xi_i(\alpha) = \alpha$. Each $z \in Z$ has the neighborhood basis $\{U_n(z) \mid n \in \mathbb{N}\}$ defined as follows:

$$U_n(z) = \begin{cases} \Big\{ x \in Z \;\Big|\; x(i) = z(i) \;\text{ for } i \leqslant n \Big\} & \text{if } z \in S^{\mathbb{N}}, \\ \Big\{ x \in Z \;\Big|\; x(i) = z(i) \;\text{ for } k \neq i \leqslant k+n, \\ \qquad\qquad \text{and } \; \xi_n(z(k)) < x(k) \leqslant z(k) \Big\} & \text{if } z \in Z_k. \end{cases}$$

Note that each $U_n(z)$ is clopen in Z, but $\{U_n(z) \mid z \in S^{\mathbb{N}}\}$ is not locally finite at the point $(\omega, \omega+1, \omega+1, \ldots)$ in Z (cf. Theorem 5.4.6). The following statements can be easily proved:

(1) If $z, z' \in S^{\mathbb{N}}$ or $z, z' \in Z_k$, then $U_n(z) \cap U_n(z') \neq \emptyset \Rightarrow U_n(z) = U_n(z')$.
(2) If $z \in Z_k$, $z' \in Z_{k'}$, and $k < k' < n + k$, then $U_n(z) \cap U_{n'}(z') = \emptyset$ for every $n' \in \mathbb{N}$.
(3) If $z \in S^{\mathbb{N}}$, $z' \in Z_k$, and $n < k$, then $U_n(z) \cap U_{n'}(z') \neq \emptyset \Rightarrow U_{n'}(z') \subset U_n(z)$.

Furthermore, we have the following:

(4) For any $z \in S^{\mathbb{N}}$ and $n \in \mathbb{N}$, there exists some $m > n$ such that $U_m(z) \cap U_m(z') = \emptyset$ for every $z' \in \bigcup_{k \leqslant n} Z_k$.

Indeed, if $z(n+1) \notin S_k$ for any $k \leqslant n$, then $U_{n+1}(z) \cap U_{n+1}(z') = \emptyset$ for every $z' \in \bigcup_{k \leqslant n} Z_k$. If $z(n+1) \in S_k$ for some $k \leqslant n$, then $U_m(z) \cap U_m(z') = \emptyset$ for every $m > n$ and $z' \in \bigcup_{k \neq j \leqslant n} Z_j$. On the other hand, because $z(k+1) \in S$, we can write $z(k+1) = \alpha + r$, where $\alpha \in L \cup \{0\}$ and $r \in \mathbb{N}$. If $\alpha = 0$ or $r \neq k$, then $U_m(z) \cap U_m(z') = \emptyset$ for every $m > k$ and $z' \in Z_k$. When $\alpha \in L$ and $r = k$, choose $m > k$ so that $z(k) \notin (\xi_m(\alpha), \alpha]$. Then, it follows that $U_m(z) \cap U_m(z') = \emptyset$ for every $z' \in Z_k$.

(*Metrizability*) To prove the metrizability, by the Frink Metrization Theorem 2.4.1 it suffices to show that, for each $z \in Z$ and $n \in \mathbb{N}$, there exists $m \in \mathbb{N}$ so that $U_m(z) \cap U_m(z') \neq \emptyset$ implies $U_m(z') \subset U_n(z)$.

When $z \in Z_k$ for some $k \in \mathbb{N}$, if $z' \in \bigcup_{k'<k} Z_{k'}$ or $z' \in \bigcup_{k<k'<n+2k} Z_{k'}$ then $U_{n+k}(z) \cap U_{n+k}(z') = \emptyset$ by (2). Assume $U_{n+k}(z) \cap U_{n+k}(z') \neq \emptyset$. If $z' \in S^{\mathbb{N}} \cup \bigcup_{k'>n+k} Z_{k'}$, then $U_{n+k}(z') \subset U_n(z)$ by definition. If $z' \in Z_k$, then $U_{n+k}(z') = U_{n+k}(z) \subset U_n(z)$ by (1). Thus, we have

$$U_{n+k}(z) \cap U_{n+k}(z') \neq \emptyset \Rightarrow U_{n+k}(z') \subset U_n(z).$$

For $z \in S^{\mathbb{N}}$, we can choose $m > n$ by (4) such that $U_m(z) \cap U_m(z') = \emptyset$ for every $z' \in \bigcup_{k \leqslant n} Z_k$. Assume $U_m(z) \cap U_m(z') \neq \emptyset$. Then, $z' \in S^{\mathbb{N}}$ or $z' \in Z_k$ for some $k > n$. If $z' \in S^{\mathbb{N}}$, then $U_m(z') = U_m(z) \subset U_n(z)$ by (1). If $z' \in Z_k$ for some $k > n$, then $U_m(z') \subset U_n(z)$ by (3). Thus, we have

$$U_m(z) \cap U_m(z') \neq \emptyset \Rightarrow U_m(z') \subset U_n(z).$$

(*Complete Metrizability*) Because of Theorem 2.5.6, to show the complete metrizability of Z, it is enough to prove that Z is a G_δ-set in the compact space $\overline{\Omega}^{\mathbb{N}}$. For each $z \in Z$ and $n \in \mathbb{N}$, we extend $U_n(z)$ to a neighborhood of z in $\overline{\Omega}^{\mathbb{N}}$ as follows:

$$\widetilde{U}_n(z) = \begin{cases} \left\{ x \in \overline{\Omega}^{\mathbb{N}} \;\middle|\; x(i) = z(i) \;\text{ for } i \leqslant n \right\} & \text{for } z \in S^{\mathbb{N}}, \\ \left\{ x \in \overline{\Omega}^{\mathbb{N}} \;\middle|\; x(i) = z(i) \;\text{ for } k \neq i \leqslant k+n, \right. \\ \qquad\qquad \left. \text{and } \; \xi_n(z(k)) < x(k) \leqslant z(k) \right\} & \text{for } z \in Z_k. \end{cases}$$

Then, each $W_n = \bigcup_{z \in Z} \widetilde{U}_n(z)$ is an open neighborhood of Z in $\overline{\Omega}^{\mathbb{N}}$. Since $x(n) \neq \omega_1$ for each $x \in W_n$, it follows that $\bigcap_{n \in \mathbb{N}} W_n \subset \Omega^{\mathbb{N}}$. We can also show that $Z = \bigcap_{n \in \mathbb{N}} W_n$. Indeed, if $x \in \bigcap_{n \in \mathbb{N}} W_n \setminus S^{\mathbb{N}}$, then $x(k) \in L$ for some $k \in \mathbb{N}$. For $n > k$, choose $z_n \in Z$ so that $x \in \widetilde{U}_n(z_n)$. Since $x(k) \in L$ and $k < n$, it follows that $z_n \notin S^{\mathbb{N}} \cup \bigcup_{k' \neq k} Z_{k'}$, i.e., $z_n \in Z_k$. Then, $x(k+i) = z_n(k+i) \in S_k$ for each $0 < i \leqslant n$ and $\xi_n(z_n(k)) < x(k) \leqslant z_n(k)$. Moreover, $x(i) = z_{k+1}(i) \in S$ for each $i < k$. Since $x(k+1) = z_n(k+1) = z_n(k) + k$, every $z_n(k)$ is identical, say $z(k)$. Since $z(k) = \sup_{n \in \mathbb{N}} \xi_n(z(k))$, we have $x(k) = z(k)$. Then, $x(k+1) = z(k) + k = x(k) + k$. Taking $n \in \mathbb{N}$ arbitrarily large, we can see that $x(i) \in S_k$ for any $i > k$. Hence, $x \in Z_k \subset Z$.

(*One-Dimensionality*) It has been shown that Z is metrizable and each Z_k is closed in Z. Then, applying the Countable Sum Theorem 5.2.8 and the Addition Theorem 5.4.14, we have $\dim Z \leqslant 1$.

To see that $\dim Z > 0$, assume $\dim Z = 0$. Let $\mathcal{W} = \{W_\alpha \mid \alpha \in \Omega\} \in \mathrm{cov}(Z)$, where $W_\alpha = \{z \in Z \mid 0 \leqslant z(2) \leqslant \alpha\}$. By the assumption, $\mathcal{W}$ has an open refinement $\mathcal{V}$ with $\mathrm{ord}\,\mathcal{V} \leqslant 1$. Then, $\mathcal{V}$ is discrete in Z. Here, we call $s \in S^n$ **regular** if there exist $f : \bigoplus_{i\in\mathbb{N}} S^i \to S$ and $V \in \mathcal{V}$ such that $R(s; f) \subset V$, where

$$R(s; f) = \Big\{x \in S^{\mathbb{N}} \;\Big|\; x(i) = s(i) \;\text{ for } i \leqslant n \text{ and } x(n+i) \geqslant f(x(n), \ldots, x(n+i-1)) \;\text{ for } i \in \mathbb{N}\Big\}.$$

Otherwise, s is said to be **irregular**.

First, we verify the following fact:

(5) Every $s \in S$ is irregular.

For each $f : \bigoplus_{i\in\mathbb{N}} S^i \to S$ and $\alpha \in \Omega$, define $s_f^\alpha \in S^{\mathbb{N}}$ as follows: $s_f^\alpha(1) = s$, $s_f^\alpha(2) = \max\{f(s), \alpha + 1\}$, and $s_f^\alpha(i+1) = f(s_f^\alpha(1), \ldots, s_f^\alpha(i))$ for $i \geqslant 2$. Then, $s_f^\alpha \in R(s; f) \setminus W_\alpha$. Hence, $R(s; f)$ is not contained in any $V \in \mathcal{V}$.

Next, we show the following fact:

(6) If $s \in S^n$ is irregular, then $(s, t) \in S^{n+1}$ is irregular for some $t \in S$.

Suppose that (s, t) is regular for every $t \in S$, that is, there are $f_t : \bigoplus_{i\in\mathbb{N}} S^i \to S$ and $V_t \in \mathcal{V}$ such that $R(s, t; f_t) \subset V_t$. When there exist $a \in S$ and $V \in \mathcal{V}$ such that $V_t = V$ for $t \geqslant \max\{a, f_a(a)\}$, we define $f : \bigoplus_{i\in\mathbb{N}} S^i \to S$ by

$$f(t) = \max\{a, f_a(a)\} \;\text{ for } t \in S \text{ and}$$
$$f(t_1, \ldots, t_i) = f_{t_2}(t_2, \ldots, t_i) \;\text{ for } (t_1, \ldots, t_i) \in S^i,\ i \geqslant 2.$$

For $x \in R(s; f)$, let $t = x(n+1)$. Then, $x \in R(s, t; f_t)$ because

$$\begin{aligned} x(n+1+i) &\geqslant f(x(n), \ldots, x(n+i)) \\ &= f_t(x(n+1), \ldots, x(n+1+(i-1))) \;\text{ for } i \in \mathbb{N}. \end{aligned}$$

Moreover, $t = x(n+1) \geqslant f(x(n)) = \max\{a, f_a(a)\} \geqslant a$. Hence,

$$R(s; f) \subset \bigcup_{t\geqslant a} R(s, t; f_t) \subset V,$$

which contradicts the irregularity of s. Therefore, we can obtain an increasing sequence $a_1 < a_2 < \cdots$ in S such that $V_{a_i} \neq V_{a_{i+1}}$ and $a_{i+1} \geqslant f_{a_i}(a_i)$. Let $\alpha = \sup_{i\in\mathbb{N}} a_i \in L$ and $b_0 = \alpha + n + 1$. For each $j \in \mathbb{N}$, we can inductively choose $b_j \in S_{n+1}$ so that

$$b_j \geqslant \sup_{i\in\mathbb{N}} f_{a_i}(a_i, b_0, \ldots, b_{j-1}).$$

Then, we have

$$z = (s(1), \dots, s(n), \alpha, b_0, b_1, b_2, \dots) \in Z_{n+1} \text{ and}$$
$$z_i = (s(1), \dots, s(n), a_i, b_0, b_1, b_2, \dots) \in R(s, a_i; f_{a_i}) \subset V_{a_i},$$

where $\lim_{i\to\infty} z_i = z$. This contradicts the discreteness of $\mathcal{V}$ because $V_{a_i} \neq V_{a_{i+1}}$.

By (5) and (6), we obtain $s \in S^{\mathbb{N}}$ such that each $(s(1), \dots, s(n)) \in S^n$ is irregular. Then, s is contained in some $V \in \mathcal{V}$, from which $U_n(s) \subset V$ for some $n \in \mathbb{N}$, which implies that $(s(1), \dots, s(n))$ is regular. This is a contradiction.

(*Zero-Dimensional Compactification*) Finally, we show that $\mathrm{cl}_{\overline{\Omega}^{\mathbb{N}}} Z$ is a zero-dimensional compactification of Z. It suffices to show that $\dim \overline{\Omega}^{\mathbb{N}} = 0$. Because $\overline{\Omega}^{\mathbb{N}}$ is compact, each open cover $\mathcal{U}$ of $\overline{\Omega}^{\mathbb{N}}$ has a finite refinement

$$\left\{ p_{m_i}^{-1}\left(\prod_{j=1}^{m_i} [\alpha_{i,j}, \beta_{i,j}] \right) \;\middle|\; i = 1, \dots, n \right\},$$

where $p_k : \overline{\Omega}^{\mathbb{N}} \to \overline{\Omega}^k$ is the projection onto the first k factors. We write

$$\{\alpha_{i,j}, \beta_{i,j} \mid i = 1, \dots, n;\ j = 1, \dots, m_i\} = \{\gamma_k \mid k = 1, \dots, \ell\},$$

where $\gamma_k < \gamma_{k+1}$ for each $k = 1, \dots, \ell - 1$. Note that $\gamma_1 = 0$ and $\gamma_\ell = \omega_1$. Then, $\mathcal{U}$ has the following pairwise disjoint open refinement:

$$\left\{ p_m^{-1}\left(\prod_{j=1}^{m} (\gamma_{k_j - 1}, \gamma_{k_j}] \right) \;\middle|\; k_j = 1, \dots, \ell \right\},$$

where $m = \max\{m_1, \dots, m_n\}$ and $(\gamma_0, \gamma_1] = \{0\}$. Therefore, $\dim \overline{\Omega}^{\mathbb{N}} = 0$. This completes the proof. □

Remark 5.5.3 According to Theorem 5.5.5, there exists a zero-dimensional compact space that contains a one-dimensional metrizable subspace. Thus, the Subset Theorem 5.4.4 does not hold for non-metrizable compact spaces.

We will provide a non-metrizable compact space X with $\dim X < \operatorname{ind} X = \operatorname{Ind} X$.

Theorem 5.5.6 (LOKUCIEVSKI) *There exists a compact space X such that*

$$\dim X = 1 < \operatorname{ind} X = \operatorname{Ind} X = 2$$

and $X = Y_1 \cup Y_2$ for some closed sets Y_i with $\operatorname{ind} Y_i = \operatorname{Ind} Y_i = 1$, $i = 1, 2$.

Proof (*Example*) Consider the Cantor ternary set μ^0 ($\subset \mathbf{I}$) and the long segment $\overline{\mathbb{L}}$.[4] Note that each point of μ^0 is written as $\sum_{i\in\mathbb{N}} 2x_i/3^i$ for a unique sequence $(x_i)_{i\in\mathbb{N}} \in \mathbf{2}^{\mathbb{N}}$ ($= \{0, 1\}^{\mathbb{N}}$). Using this expression, we define a map $\varphi : \mu^0 \to \mathbf{I}$ by $\varphi(\sum_{i\in\mathbb{N}} 2x_i/3^i) = \sum_{i\in\mathbb{N}} x_i/2^i$. Observe that $\operatorname{card} \varphi^{-1}(t) = 1$ or 2 for each $t \in \mathbf{I}$. Let

[4] For μ^0 and $\overline{\mathbb{L}}$, see Sects. 1.1 and 2.10, respectively.

$$D = \left\{ i/2^n \;\middle|\; n \in \mathbb{N},\ i = 1, 2, \ldots, 2^i - 1 \right\}.$$

Then, D is a countable dense set in $\mathbf{I}$ and $D = \{t \in \mathbf{I} \mid \operatorname{card} \varphi^{-1}(t) = 2\}$. Identifying μ^0 with $\{\omega_1\} \times \mu^0$, we define a compact space Y as the adjunction space $\mathbf{I} \cup_\varphi (\overline{\mathbb{L}} \times \mu^0)$.

Take two disjoint copies Y_1, Y_2 of Y. The counterparts of $\mathbf{I}$ and D in Y_i are denoted by $\mathbf{I}_i$ and D_i, respectively. Let $h : \mathbf{I}_2 \to \mathbf{I}_1$ be the homeomorphism defined as follows:

$$h(t) = \begin{cases} \sqrt{2}t & \text{if } t \leqslant 1/2, \\ (2 - \sqrt{2})t + \sqrt{2} - 1 & \text{if } t \geqslant 1/2. \end{cases}$$

Then, $h(D_2) \cap D_1 = \emptyset$ because $h(t)$ is irrational for every rational $0 < t < 1$. The compact space X is defined as the adjunction space $Y_1 \cup_h Y_2$.

To show that $\operatorname{Ind} X \geqslant 2$, the following fact is essential:

(1) For an open set U in $\overline{\mathbb{L}} \times \mu^0$ with $U \cap (\{\omega_1\} \times \mu^0) \neq \emptyset$, let $t_U = \sup\{t \in \mu^0 \mid (\omega_1, t) \in U\} \in \mu^0$. If $t_U \notin \varphi^{-1}(D) \cup \{1\}$, at least one of the following holds:

  (i) There exists $x' \in \mathbb{L}$ such that $(x', \omega_1]_{\overline{\mathbb{L}}} \times \{t_U\} \subset \operatorname{bd} U$.

  (ii) There exist $t' \in (t_U, 1]$ and a cofinal subset $L' \subset \mathbb{L}$ such that $L' \times ([t_U, t'] \cap \mu^0) \subset U$.

To see this, take $(\omega_1, t_i) \in U$, $i \in \mathbb{N}$, so that $t_U = \lim_{i\to\infty} t_i$. Since U is open in $\overline{\mathbb{L}} \times \mu^0$, we have ordinals $\alpha_i < \omega_1$, $i \in \mathbb{N}$, such that $(\alpha_i, \omega_1]_{\overline{\mathbb{L}}} \times \{t_i\} \subset U$. Let $\alpha_0 = \sup_{i\in\mathbb{N}} \alpha_i < \omega_1$. Then, $(\alpha_0, \omega_1]_{\overline{\mathbb{L}}} \times \{t_U\} \subset \operatorname{cl} U$. If (i) does not hold, $\mathbb{L}$ has a cofinal subset L_0 such that $(x, t_U) \in U$ for any $x \in L_0$. For each $x \in L_0$, since U is an open neighborhood of (x, t_U) in $\overline{\mathbb{L}} \times \mu^0$, we can take $t_x \in (t_U, 1]$ so that $\{x\} \times ([t_U, t_x] \cap \mu^0) \subset U$. For each $k \in \mathbb{N}$, let $L_k = \{x \in L_0 \mid t_x - t_U > 2^{-k}\}$. Since $\bigcup_{k\in\mathbb{N}} L_k = L_0$, some L_k is cofinal in $\mathbb{L}$. Then, (ii) holds with $t' = t_U + 2^{-k}$ and $L' = L_k$.

We can derive the following from (1):

(2) For an open set U in $Y = \mathbf{I} \cup_\varphi (\overline{\mathbb{L}} \times \mu^0)$ with $U \cap \mathbf{I} \neq \emptyset$, let $s_U = \sup(U \cap \mathbf{I})$. If $s_U \in [0, 1) \setminus D$, then $\operatorname{ind} \operatorname{bd} U \geqslant 1$.

Let $q : \mathbf{I} \oplus (\overline{\mathbb{L}} \times \mu^0) \to Y$ be the quotient map and let $U' = q^{-1}(U) \cap (\overline{\mathbb{L}} \times \mu^0)$. Then, $U' \cap (\{\omega_1\} \times \mu^0) \neq \emptyset$ and $\varphi(t_{U'}) = s_U$. Moreover, $s_U \in [0, 1) \setminus D$ implies $t_{U'} \notin \varphi^{-1}(D) \cup \{1\}$. Thus, we can apply (1) to U'. When (i) holds, there exists $x' \in \mathbb{L}$ such that $(x', \omega_1]_{\overline{\mathbb{L}}} \times \{t_{U'}\} \subset \operatorname{bd} U'$, and hence $(x', \omega_1)_{\overline{\mathbb{L}}} \times \{t_{U'}\} \subset \operatorname{bd} U$. When (ii) holds, there exist $t' \in (t_{U'}, 1]$ and a cofinal subset $L' \subset \mathbb{L}$ such that $L' \times ([t_{U'}, t'] \cap \mu^0) \subset U'$, where t' can be taken in μ^0.

> Indeed, if $t' \notin \mu^0$, then t' can be replaced by $t'' = \sup((0, t') \cap \mu^0) \in \mu^0$ because $[t_{U'}, t'] \cap \mu^0 = [t_{U'}, t''] \cap \mu^0$ and $t_{U'} < t''$.

Since $\varphi^{-1}(s_U) = \{t_{U'}\}$, and hence $[s_U, \varphi(t')] \subset \operatorname{bd} U$. In both cases, $\operatorname{bd} U$ contains an arc, which implies $\operatorname{ind} \operatorname{bd} U \geqslant 1$.

Now, we will show that $\operatorname{ind} Y = \operatorname{Ind} Y = 1$. Since Y contains $\mathbf{I}$, it follows that $\operatorname{ind} Y \geqslant 1$. To see $\operatorname{ind} Y \leqslant 1$, observe that $x \in Y \setminus \mathbf{I} = \mathbb{L} \times \mu^0$ has a neighborhood basis in Y consisting of open sets with boundary homeomorphic to $\mu^0 \oplus \mu^0$, and $x \in \mathbf{I}$ has a neighborhood basis in Y consisting of open sets with boundary homeomorphic to $\mu^0 \oplus \mathbf{2}$ or $\mu^0 \oplus \{0\}$ if $x \notin \partial\mathbf{I}$ or $\in \partial\mathbf{I}$, respectively.

Indeed, for each $a \in D$, write $\varphi^{-1}(a) = \{a_-, a_+\}$, where $a_- = \min \varphi^{-1}(a)$ and $a_+ = \max \varphi^{-1}(a)$. Then, it follows that

$$a < b \in D \;\Rightarrow\; a_+ < b_- \;\text{ and }\; [a_+, b_-] \cap \mu^0 = (a_-, b_+) \cap \mu^0.$$

It is convenient to write $0_- = -1$, $0_+ = 0$, $1_- = 1$, and $1_+ = 2$. The above implication holds even if D is replaced by $D \cup \mathbf{2}$. Consider the following types of open sets in Y:

$$U_1 = (s,t)_{\overline{\mathbb{L}}} \times ([a_+, b_-] \cap \mu^0),\ s < t \in \mathbb{L},\ a < b \in D \cup \mathbf{2};$$
$$U_2 = (a,b) \cup ((\alpha, \omega_1)_{\overline{\mathbb{L}}} \times ([a_+, b_-] \cap \mu^0),\ a < b \in D,\ \alpha < \omega_1;$$
$$U_3 = [0,a) \cup ((\alpha, \omega_1)_{\overline{\mathbb{L}}} \times ([0, a_-] \cap \mu^0),\ a \in D,\ \alpha < \omega_1;$$
$$U_4 = (a,1] \cup ((\alpha, \omega_1)_{\overline{\mathbb{L}}} \times ([a_+, 1] \cap \mu^0),\ a \in D,\ \alpha < \omega_1.$$

Then, they have the following boundaries in Y:

$$\operatorname{bd} U_1 = \{s,t\} \times ([a_+, b_-] \cap \mu^0) \approx \mu^0 \oplus \mu^0;$$
$$\operatorname{bd} U_2 = \{a,b\} \cup (\{\alpha\} \times ([a_+, b_-] \cap \mu^0)) \approx \mathbf{2} \oplus \mu^0;$$
$$\operatorname{bd} U_3 = \{a\} \cup (\{\alpha\} \times ([0, a_-] \cap \mu^0)) \approx \{0\} \oplus \mu^0;$$
$$\operatorname{bd} U_4 = \{a\} \cup (\{\alpha\} \times ([a_+, 1] \cap \mu^0)) \approx \{0\} \oplus \mu^0.$$

To see $\operatorname{Ind} Y \leqslant 1$, let A be a closed set in Y and V an open set in Y such that $A \subset V$. Each $x \in A$ has a neighborhood U_x such that $U_x \subset V$ and $\operatorname{bd} U_x$ is a zero-dimensional compact metrizable space. Since A is compact, we have finitely many $x_1, \ldots, x_k \in A$ such that $A \subset U = \bigcup_{i=1}^k U_{x_i} \subset V$. Then, $\operatorname{bd} U$ is contained in $B = \bigcup_{i=1}^k \operatorname{bd} U_{x_i}$. By the Countable Sum Theorem 5.2.8 and Theorem 5.5.2, $\operatorname{Ind} B = \dim B = 0$. Since $\operatorname{bd} U$ is a closed set in B, we have $\operatorname{Ind} \operatorname{bd} U \leqslant 0$. Hence, $\operatorname{Ind} Y \leqslant 1$. By Theorem 5.5.3 (1), $\operatorname{ind} Y = \operatorname{Ind} Y = 1$.

We can regard $X = Y_1 \cup Y_2$ and $\mathbf{I} = Y_1 \cap Y_2$ $(= \mathbf{I}_1 = h(\mathbf{I}_2))$, where each $x \in \mathbf{I}_2$ is identified with $h(x) \in \mathbf{I}_1 = \mathbf{I}$. As seen in the above, $\operatorname{Ind} Y_i = \operatorname{ind} Y_i = 1$, $i = 1, 2$. Then, by Theorem 5.5.1 (1) and the Countable Sum Theorem 5.2.8, we have $\dim X \leqslant 1$. Since X contains $\mathbf{I}$ as a closed set, it follows that $\dim X = 1$.

To see that $\operatorname{Ind} X \geqslant 2$, let $x \in [0, 3/4) \subset \mathbf{I}$ $(\subset X)$. For any open neighborhood U of x in X disjoint from $[3/4, 1]$, let $s_U = \sup(U \cap \mathbf{I})$. Since $D \cap h(D) = \emptyset$, either $s_U \in [0,1) \setminus D$ or $s_U \in [0,1) \setminus h(D)$ holds. Then, applying (2), we have $\operatorname{ind} \operatorname{bd}_{Y_i}(U \cap Y_i) \geqslant 1$ for some $i \in \{1,2\}$. Since $\operatorname{bd}_{Y_i}(U \cap Y_i) \subset \operatorname{bd} U$ for $i = 1, 2$, it follows that $\operatorname{ind} \operatorname{bd} U \geqslant 1$. Thus, we have $\operatorname{Ind} X \geqslant \operatorname{ind} X \geqslant 2$.

Applying Proposition 5.5.7 (2), we have

$$\operatorname{Ind} X \leqslant \operatorname{Ind} Y_1 + \operatorname{Ind} Y_2 = 2.$$

Therefore, $\operatorname{ind} X = \operatorname{Ind} X = 2$. □

Remark 5.5.4 More generally, the following are known (cf. Notes for this chapter):

(1) For any $m \leqslant n$, there exists a compact space X such that $\dim X = m$ and $\operatorname{ind} X = \operatorname{Ind} X = n$.
(2) For $0 \leqslant l \leqslant n$ and $0 < m \leqslant n$, there exists a (collectionwise) normal space X such that $\operatorname{ind} X = l$, $\dim X = m$, and $\operatorname{Ind} X = n$.

Proposition 5.5.7 *Let $X = Y \cup Z$ be a normal space, where $Y, Z \neq \emptyset$. Then, the following holds:*

(1) *If Y is normal, Z is closed in X, and* $\operatorname{Ind} Z = 0$, *then* $\operatorname{Ind} X \leqslant \operatorname{Ind} Y$.
(2) *If Y and Z are closed in X, then* $\operatorname{Ind} X \leqslant \operatorname{Ind} Y + \operatorname{Ind} Z$.

Proof (1): Let $\operatorname{Ind} Y = n$ and we will show that $\operatorname{Ind} X \leqslant n$. Let A and B be disjoint closed sets in X. Since $\operatorname{Ind} Z = 0$, there are disjoint open sets U_1' and V_1' in Z such that $A \cap Z \subset U_1'$, $B \cap Z \subset V_1'$, and $Z = U_1' \cup V_1'$. Then, U_1' and V_1' are also closed in Z, hence in X. Since $A \cup U_1'$ and $B \cup V_1'$ are disjoint closed sets in X, there are disjoint open sets U_1 and V_1 in X such that $A \cup U_1' \subset U_1$, $B \cup V_1' \subset V_1$, and $\operatorname{cl} U_1 \cap \operatorname{cl} V_1 = \emptyset$. Note that $Z \subset U_1 \cup V_1$. Then, there is a partition L between $Y \cap \operatorname{cl} U_1$ and $Y \cap \operatorname{cl} V_1$ in Y and $\operatorname{Ind} L \leqslant n - 1$. Let U_2 and V_2 be disjoint open set in Y such that $Y \setminus L = U_2 \cup V_2$, $Y \cap \operatorname{cl} U_1 \subset U_2$, and $Y \cap \operatorname{cl} V_1 \subset V_2$. Observe that $L \subset X \setminus (U_1 \cup V_1) \subset X \setminus Z \subset Y$. Then, L is closed in $X \setminus (U_1 \cup V_1)$ because it is closed in Y. Hence, L is closed in X. Note that $U_2 \setminus Z$ and $V_2 \setminus Z$ are open in X because they are open in $Y \setminus Z = X \setminus Z$. Thus, we have open sets $U = U_1 \cup (U_2 \setminus Z)$ and $V = V_1 \cup (V_2 \setminus Z)$ in X. Then, $A \subset U$ and $B \subset V$. Observe that

$$U_1 \cap (V_2 \setminus Z) \subset (U_1 \cap Y) \cap V_2 \subset U_2 \cap V_2 = \emptyset.$$

Similarly, $V_1 \cap (U_2 \setminus Z) = \emptyset$. Hence, $U \cap V = \emptyset$. Since $Z \subset U_1 \cup V_1 \subset U \cup V$, it follows that

$$\begin{aligned} X \setminus (U \cup V) &= Y \setminus (U \cup V) = (Y \setminus (U_1 \cup V_1)) \cap (Y \setminus (U_2 \cup V_2)) \\ &= (Y \setminus (U_1 \cup V_1)) \cap L = L. \end{aligned}$$

Therefore, L is a partition in X between A and B.

(2): In (1), the case when $\operatorname{Ind} Z = 0$ has been proved. Assuming that (2) holds if $\operatorname{Ind} Z \leqslant n - 1$, we may show the case when $\operatorname{Ind} Z = n$. For each disjoint closed sets A and B in X, Z has a partition L_1 between $A \cap Z$ and $B \cap Z$ such that $\operatorname{Ind} L_1 \leqslant n - 1$, and then Z has disjoint open sets U_1 and V_1 such that $Z \setminus L_1 = U_1 \cup V_1$, $A \cap Z \subset U_1$, and $B \cap Z \subset V_1$. It should be noted that L_1 is closed in X, $Z \setminus U_1$ and

$Z \setminus V_1$ are closed in X. Since $A \subset X \setminus (Z \setminus U_1)$ and $B \subset X \setminus (Z \setminus V_1)$, we can take disjoint open sets U_0 and V_0 in X such that

$$A \subset U_0 \subset X \setminus (Z \setminus U_1) \text{ and } B \subset V_0 \subset X \setminus (Z \setminus V_1).$$

Then, $U_0 \cap Z \subset U_1$, $V_0 \cap Z \subset V_1$, and $U_0 \cup V_0 \subset X \setminus L_1$. Since $U_1 \setminus Y$ and $V_1 \setminus Y$ are open in X, we have open sets $U = U_0 \cup (U_1 \setminus Y)$ and $V = V_0 \cup (V_1 \setminus Y)$ in X such that $A \subset U$ and $B \subset V$. Then, $L = L_1 \cup (Y \setminus (U_0 \cup V_0))$ is closed in X and

$$\begin{aligned} X \setminus (U \cup V) &= ((Z \setminus Y) \setminus (U \cup V)) \cup (Y \setminus (U \cup V)) \\ &= ((Z \setminus Y) \setminus (U_1 \cup V_1)) \cup (Y \setminus (U_0 \cup V_0)) \\ &= ((Z \setminus Y) \setminus (Z \setminus L_1)) \cup (Y \setminus (U_0 \cup V_0)) \\ &= (L_1 \setminus Y) \cup (Y \setminus (U_0 \cup V_0)) \\ &= L_1 \cup (Y \setminus (U_0 \cup V_0)) = L, \end{aligned}$$

which means that L is a partition between A and B in X. Since $\operatorname{Ind} L_1 \leqslant n - 1$ and $Y \setminus (U_0 \cup V_0)$ is closed in X, it follows from the assumption that

$$\operatorname{Ind} L \leqslant \operatorname{Ind}(Y \setminus (U_0 \cup V_0)) + \operatorname{Ind} L_1 \leqslant \operatorname{Ind} Y + \operatorname{Ind} Z - 1.$$

Therefore, $\operatorname{Ind} X \leqslant \operatorname{Ind} Y + \operatorname{Ind} Z$. □

Remark 5.5.5 Due to Theorem 5.5.6, unlike dim, the (Finite) Sum Theorem does not hold for Ind nor ind.

5.6 Infinite Dimensions

In this section, several types of infinite dimensions are defined and discussed. According to Theorem 5.3.9, a normal space X is infinite-dimensional if and only if X has an essential family of n pairs of disjoint closed sets for any $n \in \mathbb{N}$. By this characterization, we can divide infinite dimensions into two types. A *normal* space X is said to be **strongly infinite-dimensional** (**s.i.d.**) if X has an infinite essential family of pairs of disjoint closed sets. Obviously, if X is s.i.d. then $\dim X = \infty$. A *infinite-dimensional normal* space X is **weakly infinite-dimensional** (**w.i.d.**) if X is not s.i.d.,[5] that is, for every family $(A_i, B_i)_{i \in \mathbb{N}}$ of pairs of disjoint closed sets in X, there are partitions L_i between A_i and B_i such that $\bigcap_{i \in \mathbb{N}} L_i = \emptyset$.

[5] In many articles, the infinite dimensionality is not assumed, i.e., w.i.d. = not s.i.d., so every f.d. normal space is w.i.d. However, here we assume the infinite dimensionality because we discuss the difference among infinite-dimensional spaces.

Theorem 5.6.1 *The Hilbert cube* $\mathbf{I}^{\mathbb{N}}$ *is strongly infinite-dimensional.*

Proof For each $i \in \mathbb{N}$, let

$$A_i = \left\{x \in \mathbf{I}^{\mathbb{N}} \;\middle|\; x(i) = 0\right\} \text{ and } B_i = \left\{x \in \mathbf{I}^{\mathbb{N}} \;\middle|\; x(i) = 1\right\}.$$

Then, $(A_i, B_i)_{i\in\mathbb{N}}$ is essential in $\mathbf{I}^{\mathbb{N}}$. Indeed, for each $i \in \mathbb{N}$, let L_i be a partition between A_i and B_i. For each $n \in \mathbb{N}$, let $j_n : \mathbf{I}^n \to \mathbf{I}^{\mathbb{N}}$ be the natural injection defined by

$$j_n(x) = (x(1), \ldots, x(n), 0, 0, \ldots).$$

Then, for each $i \leqslant n$, $j_n^{-1}(L_i)$ is a partition between

$$j_n^{-1}(A_i) = \left\{x \in \mathbf{I}^n \;\middle|\; x(i) = 0\right\} \text{ and } j_n^{-1}(B_i) = \left\{x \in \mathbf{I}^n \;\middle|\; x(i) = 1\right\}.$$

Since $(j_n^{-1}(A_i), j_n^{-1}(B_i))_{i=1}^n$ is essential in $\mathbf{I}^n$ (Corollary 5.3.8), it follows that $\bigcap_{i=1}^n j_n^{-1}(L_i) \neq \emptyset$, so $\bigcap_{i=1}^n L_i \neq \emptyset$. Because $\mathbf{I}^{\mathbb{N}}$ is compact, we can conclude that $\bigcap_{i\in\mathbb{N}} L_i \neq \emptyset$. □

By definition, a normal space is strongly infinite-dimensional if it contains an s.i.d. closed subspace. Then, it follows from Theorem 5.6.1 that every normal space containing a copy of $\mathbf{I}^{\mathbb{N}}$ is s.i.d. For example, ℓ_1, ℓ_2, and $\mathbb{R}^{\mathbb{N}}$ are s.i.d.[6] Moreover, rint $\boldsymbol{Q} = \bigcup_{n\in\mathbb{N}}[-1 + 2^{-n}, 1 - 2^{-n}]^{\mathbb{N}}$ and $\mathbf{I}^{\mathbb{N}} \setminus (0, 1)^{\mathbb{N}}$ are also s.i.d.[7]

A *normal* space X is said to be **countable-dimensional** (**c.d.**) if X is a countable union of f.d. normal subspaces, where it should be noted that subspaces of normal spaces need not be normal (cf. Sect. 2.9). Recall that an f.d. metrizable space is a finite union of zero-dimensional subspaces by the Decomposition Theorem 5.4.11. Hence, we have the following:

Proposition 5.6.2 *A metrizable space is countable-dimensional if and only if it is a countable union of zero-dimensional subspaces.*

By the following theorem, any strongly infinite-dimensional metrizable space is not countable-dimensional.

Theorem 5.6.3 *A countable-dimensional metrizable space* X *with* $\dim X = \infty$ *is weakly infinite-dimensional.*

Proof Let $(A_i, B_i)_{i\in\mathbb{N}}$ be a family of pairs of disjoint closed sets in X. We can write $X = \bigcup_{i\in\mathbb{N}} X_i$, where $\dim X_i = 0$. From Theorem 5.3.9 and the Partition Extension Lemma 5.4.8, it follows that for each $i \in \mathbb{N}$, X has a partition L_i between A_i and B_i such that $L_i \cap X_i = \emptyset$. Then, we have

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[6] It is known that $\ell_1 \approx \ell_2 \approx \mathbb{R}^{\mathbb{N}}$, where the latter homeomorphy was proved by R.D. Anderson. Refer to Notes for Chap. 2.

[7] Since rint $\boldsymbol{Q}$ and $\mathbf{I}^{\mathbb{N}} \setminus (0, 1)^{\mathbb{N}}$ are not completely metrizable, they are not homeomorphic to $\mathbb{R}^{\mathbb{N}}$, but it is known that rint $\boldsymbol{Q} \approx \mathbf{I}^{\mathbb{N}} \setminus (0, 1)^{\mathbb{N}}$.

$$\bigcap_{i\in\mathbb{N}} L_i = \left(\bigcap_{i\in\mathbb{N}} L_i\right) \cap \left(\bigcup_{i\in\mathbb{N}} X_i\right) = \bigcup_{i\in\mathbb{N}} \left(\left(\bigcap_{j\in\mathbb{N}} L_j\right) \cap X_i\right)$$
$$\subset \bigcup_{i\in\mathbb{N}} (L_i \cap X_i) = \emptyset.$$

Therefore, X is w.i.d. □

The space $\bigoplus_{n\in\mathbb{N}} \mathbf{I}^n$ and its one-point compactification are c.d., and hence they are w.i.d. by Theorem 5.6.3. The following space is also c.d. (so w.i.d.):

$$\mathbf{I}^{\mathbb{N}}_f = \left\{x \in \mathbf{I}^{\mathbb{N}} \;\middle|\; x(i) = 0 \text{ except for finitely many } i\right\}.$$

As is easily seen, any subspace of a c.d. metrizable space is also c.d. However, an infinite-dimensional subspace of a w.i.d. metrizable space need not be w.i.d. In fact, there exists a w.i.d. compactum that has an s.i.d. subspace. Such a compactum will be constructed in Theorem 5.12.5. Notice that this w.i.d compactum is not c.d., as its s.i.d. subspace is not c.d. by Theorem 5.6.3.

Now, we introduce a strong version of countable dimensionality. A *normal* space X is **strongly countable-dimensional** (**s.c.d.**) if X is a countable union of f.d. *closed* subspaces. The space $\bigoplus_{n\in\mathbb{N}} \mathbf{I}^n$, its one-point compactification, and the space $\mathbf{I}^{\mathbb{N}}_f$ are s.c.d. Every s.c.d. space is c.d. but the converse does not hold. Let ν_ω be the subspace of the Hilbert cube $\mathbf{I}^{\mathbb{N}}$ defined as follows:

$$\nu_\omega = \left\{x \in \mathbf{I}^{\mathbb{N}} \;\middle|\; x(i) \in \mathbf{I} \setminus \mathbb{Q} \text{ except for finitely many } i\right\}.$$

Theorem 5.6.4 *The space ν_ω is countable-dimensional but not strongly countable-dimensional.*

Proof Since ν_ω is a countable union of subspaces

$$\left\{x \in \mathbf{I}^{\mathbb{N}} \;\middle|\; x(i) \in \mathbf{I} \setminus \mathbb{Q} \text{ for } i \geqslant n\right\} \approx \mathbf{I}^n \times (\mathbf{I} \setminus \mathbb{Q})^{\mathbb{N}},$$

it follows that ν_ω is c.d. Moreover, $\dim \nu_\omega = \infty$ because $\mathbf{I}^n \times \{0\} \subset \nu_\omega$ for any $n \in \mathbb{N}$.

Assume that ν_ω is s.c.d., that is, $\nu_\omega = \bigcup_{n\in\mathbb{N}} F_n$, where each F_n is f.d. and closed in ν_ω. Consider the subspace $(\mathbf{I} \setminus \mathbb{Q})^{\mathbb{N}} \subset \nu_\omega$. Since $(\mathbf{I} \setminus \mathbb{Q})^{\mathbb{N}}$ is completely metrizable, at least one $F_n \cap (\mathbf{I} \setminus \mathbb{Q})^{\mathbb{N}}$ has the non-empty interior in $(\mathbf{I} \setminus \mathbb{Q})^{\mathbb{N}}$ by the Baire Category Theorem 2.5.1. Then, we have a non-empty open set U in ν_ω such that $U \cap (\mathbf{I} \setminus \mathbb{Q})^{\mathbb{N}} \subset F_n \cap (\mathbf{I} \setminus \mathbb{Q})^{\mathbb{N}}$. Since U contains a copy of every n-cube $\mathbf{I}^n$, it follows that $\dim U = \infty$, hence $U \setminus F_n \neq \emptyset$ because $\dim F_n < \infty$. Since $(\mathbf{I} \setminus \mathbb{Q})^{\mathbb{N}}$ is dense in ν_ω, we have

$$\left(U \cap (\mathbf{I} \setminus \mathbb{Q})^{\mathbb{N}}\right) \setminus \left(F_n \cap (\mathbf{I} \setminus \mathbb{Q})^{\mathbb{N}}\right) = (U \setminus F_n) \cap (\mathbf{I} \setminus \mathbb{Q})^{\mathbb{N}} \neq \emptyset,$$

which is a contradiction. Therefore, ν_ω is not s.c.d. □

A collection $\mathcal{A}$ of subsets of X is **locally countable** if each $x \in X$ has a neighborhood U that meets only countably many members of $\mathcal{A}$, i.e., $\operatorname{card} \mathcal{A}[U] \leqslant \aleph_0$. In the following, we give a list of basic properties of (strong) countable dimension. Since their proofs are easy, they are left as an exercise for the reader.

Proposition 5.6.5 (Basic Properties of (Strong) Countable Dimension)

(1) *If a normal space X is a countable union of countable-dimensional subspaces, then X is countable-dimensional.*
(2) *If a normal space X is a countable union of strongly countable-dimensional closed subspaces, then X is strongly countable-dimensional.*
(3) *Every closed subspace of a (strongly) countable-dimensional space is (strongly) countable-dimensional. For a metrizable space, this is valid for a non-closed subspace, that is, every subspace of a (strongly) countable-dimensional metrizable space is (strongly) countable-dimensional.*

Sketch of Proof For the above three items, just apply the definition.

(4) *A paracompact space X is (strongly) countable-dimensional if each point $x \in X$ has a (strongly) countable-dimensional neighborhood.*

Sketch of Proof Let $\mathcal{P}$ be the property of closed sets in X being c.d. (or s.c.d.). Apply Michael's Theorem on local properties (Corollary 2.6.8). To show (F-3), use the Locally Finite Sum Theorem 5.2.9.

(5) *If a paracompact space X has a locally countable union of countable-dimensional subspaces, then X is countable-dimensional.*
(6) *If a paracompact space X has a locally countable union of strongly countable-dimensional closed subspaces, then X is strongly countable-dimensional.*

Sketch of Proof For the above two items, let $\mathcal{A}$ be a locally countable (closed) cover of X such that each $A \in \mathcal{A}$ is c.d. (s.c.d.). Each $x \in X$ has an open neighborhood V_x in X such that $\mathcal{A}[V_x]$ is countable. Then, $\operatorname{st}(V_x, \mathcal{A}) = \bigcup \mathcal{A}[V_x]$ is a c.d. (s.c.d.) neighborhood of x in X.

By using Theorem 5.5.1 (2) (or 5.4.9), the Decomposition Theorem 5.4.11, and the definition of (strong) countable dimension, the following can be proved (the details are left to the reader):

Theorem 5.6.6 *Every finite-dimensional metrizable space X contains an n-dimensional closed set for any $n \leqslant \dim X$. Moreover, every (strongly) countable-dimensional metrizable space X with $\dim X = \infty$ contains an n-dimensional (closed) set for any $n \in \omega$.* □

However, the above is not true for an infinite-dimensional space. Namely, there exists an infinite-dimensional compactum such that every subset with $\dim \not\leqslant 0$ is infinite-dimensional. Such a space is called a **hereditarily infinite-dimensional** (**h.i.d.**) space. We will construct an h.i.d. compactum in Theorem 5.12.8.

Next, we introduce infinite-dimensional versions of inductive dimensions. By transfinite induction on ordinals α, the **large transfinite inductive dimension** of a *normal* space X, $\mathrm{trInd}\, X$, and the **small transfinite inductive dimension** of a *regular* space X, $\mathrm{trind}\, X$, are defined as follows: After defining $\mathrm{trInd}\, X = \mathrm{trind}\, X = -1$ if $X = \emptyset$, we define $\mathrm{trInd}\, X \leqslant \alpha$ (resp. $\mathrm{trind}\, X \leqslant \alpha$) if each closed set $A \subset X$ (resp. $x \in X$) has an arbitrarily small open neighborhood V with $\mathrm{trInd}\, \mathrm{bd}\, V < \alpha$ (resp. $\mathrm{trind}\, \mathrm{bd}\, V < \alpha$). Then, we define $\mathrm{trInd}\, X = \alpha$ (resp. $\mathrm{trind}\, X = \alpha$) if $\mathrm{trInd}\, X \leqslant \alpha$ (resp. $\mathrm{trind}\, X \leqslant \alpha$) and $\mathrm{trInd}\, X \not\leqslant \beta$ (resp. $\mathrm{trind}\, X \not\leqslant \beta$) for any $\beta < \alpha$.

It should be noted that $\mathrm{trInd}\, X < \omega$ (resp. $\mathrm{trind}\, X < \omega$) if and only if $\mathrm{Ind}\, X < \infty$ (resp. $\mathrm{ind}\, X < \infty$), and that, in this case, $\mathrm{trInd}\, X = \mathrm{Ind}\, X$ (resp. $\mathrm{trind}\, X = \mathrm{ind}\, X$). Moreover, using transfinite induction, we can show the following:

Lemma 5.6.7 *Let X be a normal (resp. regular) space. If* $\mathrm{trInd}\, X = \alpha$ *(resp.* $\mathrm{trind}\, X = \alpha$*) and $\beta < \alpha$, then X has a closed set Y such that* $\mathrm{trInd}\, Y = \beta$ *(resp.* $\mathrm{trind}\, Y = \beta$*).*

Proof Because of the similarity, we prove the lemma only for trInd. Assume that the lemma holds for any ordinal $< \alpha$ and let $\beta < \alpha$. Since $\mathrm{trInd}\, X \not\leqslant \beta$, there is a closed set A in X with its neighborhood U such that $\mathrm{trInd}\, \mathrm{bd}\, V \not< \beta$ for any open neighborhood V of A contained in U. On the other hand, since $\mathrm{trInd}\, X \leqslant \alpha$, A has an open neighborhood V contained in U such that $\mathrm{trInd}\, \mathrm{bd}\, V < \alpha$. If $\beta = \mathrm{trInd}\, \mathrm{bd}\, V$, then $\mathrm{bd}\, V$ is the desired Y. When $\beta < \mathrm{trInd}\, \mathrm{bd}\, V$, by the inductive assumption, $\mathrm{bd}\, V$ has a closed set Y with $\mathrm{trInd}\, Y = \beta$. □

It is said that a *normal* (resp. *regular*) space X has **large** (resp. **small**) **transfinite inductive dimension** (abbrev. trInd (or trind)) if $\mathrm{trInd}\, X \leqslant \alpha$ (or $\mathrm{trind}\, X \leqslant \alpha$) for some ordinal α.

Proposition 5.6.8 (Basic Properties of trInd and trind)

(1) *If a normal space X has* trInd*, then X has* trind *and* $\mathrm{trind}\, X \leqslant \mathrm{trInd}\, X$.
(2) *If a regular space X has* trind*, then every subspace A of X also has* trind*, where* $\mathrm{trind}\, A \leqslant \mathrm{trind}\, X$.
(3) *If a normal space X has* trInd*, then every* closed *subspace A of X has* trInd*, where* $\mathrm{trInd}\, A \leqslant \mathrm{trInd}\, X$.

Statements (1) – (3) above are easily proved by the definitions.

(4) *If a normal space X has no* trInd*, then X has a closed set A with an open neighborhood U such that the boundary of each neighborhood of A contained in U has no* trInd.

Sketch of Proof Let $\mathcal{P}$ be the collection of pairs (A, U) of closed sets A in X and open sets U in X with $A \subset U$. Suppose that for each $(A, U) \in \mathcal{P}$, A has a neighborhood $V_{(A,U)}$ in X such that $V_{(A,U)} \subset U$ and $\mathrm{bd}\, V_{(A,U)}$ has trInd. Then, $V'_{(A,U)} = \mathrm{int}\, V_{(A,U)}$ is an open neighborhood of A contained in U and $\mathrm{bd}\, V'_{(A,U)} \subset \mathrm{bd}\, V_{(A,U}$, where $\mathrm{bd}\, V'_{(A,U)}$ has trInd by (3). Take an ordinal α so that $\alpha > \sup_{(A,U)\in\mathcal{P}} \mathrm{trInd}\, \mathrm{bd}\, V_{(A,U)}$. Then, $\mathrm{Ind}\, X \leqslant \alpha$, so X has trInd.

(5) *If a regular space X has no* trind, *then X has a point $x \in X$ with an open neighborhood U such that the boundary of each neighborhood of x contained in U has no trind.*

Sketch of Proof In the proof of (4), replace the closed sets A in X with points $x \in X$.

We now prove that the converse of (1) does not hold, that is,

Theorem 5.6.9 *The strongly countable-dimensional space $\bigoplus_{n\in\mathbb{N}} \mathbf{I}^n$ has no* trInd *but* trind $\bigoplus_{n\in\mathbb{N}} \mathbf{I}^n = \omega$.

Proof Each point of $\bigoplus_{n\in\mathbb{N}} \mathbf{I}^n$ is contained in some $\mathbf{I}^n$, hence trind $\bigoplus_{n\in\mathbb{N}} \mathbf{I}^n \leqslant \omega$. Because ind $\bigoplus_{n\in\mathbb{N}} \mathbf{I}^n = \infty$, we have trind $\bigoplus_{n\in\mathbb{N}} \mathbf{I}^n = \omega$.

On the other hand, assume that $\bigoplus_{n\in\mathbb{N}} \mathbf{I}^n$ has trInd, i.e., trInd $\bigoplus_{n\in\mathbb{N}} \mathbf{I}^n = \alpha$ for some ordinal α. Then, $\alpha \geqslant \omega$ because dim $\bigoplus_{n\in\mathbb{N}} \mathbf{I}^n = \infty$. By Lemma 5.6.7, $\bigoplus_{n\in\mathbb{N}} \mathbf{I}^n$ contains a closed set X with trInd $X = \omega$. For each $n \in \mathbb{N}$, let $X_n = X \cap \mathbf{I}^n$. Then, each X_n is finite-dimensional, but $\sup_{n\in\mathbb{N}} \dim X_n = \infty$ because $X = \bigoplus_{n\in\mathbb{N}} X_n$. By Theorem 5.4.9, we have disjoint closed sets A_n and B_n in X_n such that $\dim L \geqslant \dim X_n - 1$ for any partition L between A_n and B_n in X_n. Then, $A = \bigoplus_{n\in\mathbb{N}} A_n$ and $B = \bigoplus_{n\in\mathbb{N}} B_n$ are disjoint closed sets in X. Since trInd $X = \omega$, we have a partition L in X between A and B such that trInd $L < \omega$, i.e., $\dim L < \infty$. Choose $n \in \mathbb{N}$ so that $\dim X_n > \dim L + 1$. Then, $X_n \cap L$ is a partition in X_n between A_n and B_n and $\dim X_n \cap L \leqslant \dim L < \dim X_n - 1$. This is a contradiction. Therefore, $\bigoplus_{n\in\mathbb{N}} \mathbf{I}^n$ has no trInd. □

Theorem 5.6.9 above also shows that the converse of the following theorem does not hold.

Theorem 5.6.10 *A metrizable space is countable-dimensional if it has* trInd.

Proof This can be proved by transfinite induction. Assume that all metrizable spaces with trInd $< \alpha$ are c.d. and let X be a metrizable space with trInd $X = \alpha$. By the analogy of Proposition 5.4.12, we can construct a σ-locally finite basis $\mathcal{B}$ for X such that trInd bd $B < \alpha$ for each $B \in \mathcal{B}$. Let

$$Y = \bigcup \{ \text{bd } B \mid B \in \mathcal{B} \} \quad \text{and} \quad Z = X \setminus Y.$$

Then, $\dim Z \leqslant 0$ by Theorem 5.4.6. On the other hand, by the assumption, bd B is c.d. for all $B \in \mathcal{B}$. Then, Y is also c.d. by Proposition 5.6.5 (5) and (1). Therefore, X is c.d. □

The following theorem can be proved in a similar manner (cf. the proof of Theorem 5.5.4).

Theorem 5.6.11 *A separable metrizable space is countable-dimensional if it has* trind. □

Remark 5.6.1 In Theorem 5.6.11, it is unknown whether the separability is necessary or not, that is, the existence of a metrizable space that has trind but is not c.d. is unknown.

As we saw in Theorem 5.6.9, the converse of Theorem 5.6.10 is not true in general, but it is true for compacta. Namely, the following theorem holds:

Theorem 5.6.12 *A compactum has* trInd *if and only if it is countable-dimensional.*

Proof It is enough to prove the "if" part. Let X be a compactum and $X = \bigcup_{n\in\mathbb{N}} A_n$, where $\dim A_n \leqslant 0$ for each $n \in \mathbb{N}$. Suppose that X has no trInd. Then, by Proposition 5.6.8 (4), X has a closed set A with an open neighborhood U such that the boundary of each neighborhood of A contained in U has no trInd. Since $\dim A_1 \leqslant 0$, we can use the Partition Extension Lemma 5.4.8 to find a closed neighborhood V_1 of A contained in U such that $\mathrm{bd}\, V_1 \cap A_1 = \emptyset$. Then, $X_1 = \mathrm{bd}\, V_1$ has no trInd and $X_1 \cap A_1 = \emptyset$. By the same argument, we have a closed set $X_2 \subset X_1$ that misses A_2 and has no trInd. Thus, by induction, we can obtain closed sets $X_1 \supset X_2 \supset \cdots$ such that each X_n has no trInd and $X_n \cap A_n = \emptyset$. Then,

$$\bigcap_{n\in\mathbb{N}} X_n = \bigcap_{n\in\mathbb{N}} X_n \cap \bigcup_{n\in\mathbb{N}} A_n \subset \bigcup_{n\in\mathbb{N}} (X_n \cap A_n) = \emptyset,$$

which contradicts the compactness of X. □

Although $\bigoplus_{n\in\mathbb{N}} \mathbf{I}^n$ has no trInd (Theorem 5.6.9), the one-point compactification of $\bigoplus_{n\in\mathbb{N}} \mathbf{I}^n$ has trInd by Theorem 5.6.12. Thus, even if a space X has trInd, it does not imply that a subspace A of X has trInd, that is, Theorem 5.6.8 (3) does not hold without the closedness of A.

Theorem 5.6.13 *A completely metrizable space has* trind *if it is countable-dimensional.*

Proof Let $X = (X, d)$ be a complete metric space and $X = \bigcup_{n\in\mathbb{N}} A_n$, where $\dim A_n \leqslant 0$ for each $n \in \mathbb{N}$. Suppose that X has no trind. Then, by Proposition 5.6.8(5), X has a point a with an open neighborhood U such that the boundary of each neighborhood of a contained in U has no trind, where we may assume that $\mathrm{diam}\, U < 2^{-1}$. In the same way as for Theorem 5.6.12, we can inductively obtain non-empty closed sets $X_1 \supset X_2 \supset \cdots$ such that $X_n \cap A_n = \emptyset$ and $\mathrm{diam}\, X_n < 2^{-n}$ for each $n \in \mathbb{N}$. Then,

$$\bigcap_{n\in\mathbb{N}} X_n = \bigcap_{n\in\mathbb{N}} X_n \cap \bigcup_{n\in\mathbb{N}} A_n \subset \bigcup_{n\in\mathbb{N}} (X_n \cap A_n) = \emptyset.$$

However, the completeness of X implies $\bigcap_{n\in\mathbb{N}} X_n \neq \emptyset$. This is a contradiction. □

Combining Theorems 5.6.11 and 5.6.13, we have the following corollary:

Corollary 5.6.14 *A separable completely metrizable space has* trind *if and only if it is countable-dimensional.* □

The next theorem shows that the above "if" part (Theorem 5.6.13) does not hold without the completeness.

Theorem 5.6.15 *The strongly countable-dimensional space* $\mathbf{I}_f^{\mathbb{N}}$ *has no* trind.

To prove this theorem, we need the following two lemmas:

Lemma 5.6.16 *Let X be a subspace of a metrizable space M. Then, each open set U in M contains an open set U' in M such that $X \cap U' = X \cap U$ and $X \cap \mathrm{cl}_M U' = \mathrm{cl}_X(X \cap U')$, hence $X \cap \mathrm{bd}_M U' = \mathrm{bd}_X(X \cap U')$.*

Proof For each open set U in M, let $U' = U \cap E(X \cap U)$, where $E(X \cap U)$ is an open set in M by applying the Canonical Open Set Extension Lemma 5.4.10 to $X \subset M$. Then, $X \cap U' = U \cap E(X \cap U) \cap X = X \cap U$. Moreover,

$$\begin{aligned} X \cap \mathrm{cl}_M U' &\subset X \cap \mathrm{cl}_M E(X \cap U) = \mathrm{cl}_X(X \cap U) \\ &= \mathrm{cl}_X(X \cap U') = X \cap \mathrm{cl}_M(X \cap U') \subset X \cap \mathrm{cl}_M U'. \end{aligned}$$

Hence, $X \cap \mathrm{cl}_M U' = \mathrm{cl}_X(X \cap U')$. □

Lemma 5.6.17 *Let M be a separable metrizable space and $X \subset M$ with* trind $X \leqslant \alpha$. *Then, X is contained in some G_δ-set X^* in M with* trind $X^* \leqslant \alpha$.

Proof Assuming that the lemma is true for any ordinal $< \alpha$, we will show the lemma for α. Take $d \in \mathrm{Metr}(M)$. For each $i \in \mathbb{N}$, applying Lemma 5.6.16, we can find a countable open collection $\mathcal{U}_i$ in M such that $X \subset X_i = \bigcup \mathcal{U}_i$, $\mathrm{mesh}\, \mathcal{U}_i < 1/i$, and trind $X \cap \mathrm{bd}_M U < \alpha$ for each $U \in \mathcal{U}_i$, where $X \cap \mathrm{bd}_M U = \mathrm{bd}_X(X \cap U)$ for each $U \in \mathcal{U}_i$. By the inductive assumption, for each $U \in \mathcal{U}_i$, there is a G_δ-set G_U in M such that $X \cap \mathrm{bd}_M U \subset G_U$ and trind $G_U < \alpha$. Then,

$$\begin{aligned} X^* &= \bigcap_{i \in \mathbb{N}} X_i \cap \bigcap_{i \in \mathbb{N}} \bigcap_{U \in \mathcal{U}_i} (G_U \cup (M \setminus \mathrm{bd}_M U)) \\ &= \bigcap_{i \in \mathbb{N}} X_i \setminus \bigcup_{i \in \mathbb{N}} \bigcup_{U \in \mathcal{U}_i} (\mathrm{bd}_M U \setminus G_U) \end{aligned}$$

is a G_δ-set in M and $X \subset X^*$. For any $i \in \mathbb{N}$, every $x \in X^*$ is contained in some $U \in \mathcal{U}_i$. Then, $\mathrm{diam}\, X^* \cap U < 1/i$ and

$$\begin{aligned} \mathrm{bd}_{X^*}(X^* \cap U) &= \mathrm{cl}_{X^*}(X^* \cap U) \setminus U \\ &\subset (X^* \cap \mathrm{cl}_M U) \setminus U = X^* \cap \mathrm{bd}_M U \subset G_U, \end{aligned}$$

which implies trind $\mathrm{bd}_{X^*}(X^* \cap U) < \alpha$. Thus, each point $x \in X^*$ has an arbitrarily small neighborhood V with trind $\mathrm{bd}_{X^*} V < \alpha$. Hence, trind $X^* \leqslant \alpha$. □

Proof (*Theorem* 5.6.15) Assume that $\mathbf{I}_f^{\mathbb{N}}$ has trind. According to Lemma 5.6.17, $\mathbf{I}_f^{\mathbb{N}}$ is contained in some G_δ-set G in $\mathbf{I}^{\mathbb{N}}$ that also has trind. Then, G is c.d. by Theorem 5.6.11. We show that G contains a copy of $\mathbf{I}^{\mathbb{N}}$, hence G is s.i.d., which contradicts Theorem 5.6.3. Thus, we obtain the desired result.

Let $G = \bigcap_{k\in\mathbb{N}} U_k$, where U_k is open in $\mathbf{I}^{\mathbb{N}}$. Note that $0 = (0, 0, \dots) \in \mathbf{I}_f^{\mathbb{N}} \subset G \subset U_1$. Choose $n_1 \in \mathbb{N}$ and $a_1, \dots, a_{n_1} \in (0, 1)$ so that

$$\left\{x \in \mathbf{I}^{\mathbb{N}} \;\middle|\; x(i) \leqslant a_i \;\text{ for }\; i = 1, \dots, n_1\right\} \subset U_1.$$

Note that $\prod_{i=1}^{n_1}[0, a_i] \times \{0\} \subset \mathbf{I}_f^{\mathbb{N}} \subset U_2$. According to the Wallace Theorem 2.1.2, we can choose $n_2 \in \mathbb{N}$ with $n_2 > n_1$ and $a_{n_1+1}, \dots, a_{n_2} \in (0, 1)$ so that

$$\left\{x \in \mathbf{I}^{\mathbb{N}} \;\middle|\; x(i) \leqslant a_i \;\text{ for }\; i = 1, \dots, n_2\right\} \subset U_2.$$

By induction, we can obtain an increasing sequence n_i of natural numbers and a sequence $a_i \in (0, 1)$ such that

$$\left\{x \in \mathbf{I}^{\mathbb{N}} \;\middle|\; x(i) \leqslant a_i \;\text{ for } i = 1, \dots, n_k\right\} \subset U_k \;\text{ for each } k \in \mathbb{N}.$$

Then, $G = \bigcap_{k\in\mathbb{N}} U_k$ contains $\prod_{i\in\mathbb{N}}[0, a_i] \approx \mathbf{I}^{\mathbb{N}}$. □

Remark 5.6.2 There exists a slightly stronger version of the weak infinite dimension. We say that an *infinite-dimensional normal* space X is **weakly infinite-dimensional in the sense of Smirnov** (**S-w.i.d.**) if for every family $(A_i, B_i)_{i\in\mathbb{N}}$ of pairs of disjoint closed sets in X, there are partitions L_i between A_i and B_i such that $\bigcap_{i=1}^{n} L_i = \emptyset$ for some $n \in \mathbb{N}$. To distinguish w.i.d. from S-w.i.d. the term "**weakly infinite-dimensional in the sense of Alexandroff** (**A-w.i.d.**)" is used. Obviously, every S-w.i.d. space is (A-)w.i.d. For compact spaces, the converse is also true, that is, the two notions of weak infinite dimension are equivalent. It was shown in [38] that the Stone–Čech compactification of a normal space X is w.i.d. if and only if X is S-w.i.d.[8]

5.7 Dimension-Preserving Compactification Theorems

In Theorem 5.4.5, we proved that every finite-dimensional metric space has a dimension-preserving completion, that is, every n-dimensional metrizable space can be embedded in an n-dimensional completely metrizable space as a dense set. In this section, we present similar results for metrizable compactification, that is, every separable metrizable space has a dimension-preserving metrizable compactification, that

[8] Refer to Problem 6.1.E in Engelking's book: *Theory of Dimensions, Finite and Infinite* (cf. Notes for this chapter).

is, every n-dimensional separable metrizable space has an n-dimensional metrizable compactification and that every c.d. (resp. s.c.d.) separable completely metrizable space has a c.d. (resp. s.c.d.) metrizable compactification.

Recall that every separable metrizable space X has a metrizable compactification. Indeed, embedding X into the Hilbert cube $\mathbf{I}^{\mathbb{N}}$ (Corollary 2.3.9), the closure of X in $\mathbf{I}^{\mathbb{N}}$ is a metrizable compactification of X.

Note Here is an alternative direct proof of Corollary 2.3.9:

Proof (*Corollary* 2.3.9) Let $X = (X, d)$ be a separable metric space with $\{a_i \mid i \in \mathbb{N}\}$ a countable dense set. For each $i \in \mathbb{N}$, we define a map $f_i : X \to \mathbf{I}$ by $f_i(x) = \min\{1, d(x, a_i)\}$ for each $x \in X$. Then, the map $f : X \to \mathbf{I}^{\mathbb{N}}$ defined by $f(x) = (f_i(x))_{i\in\mathbb{N}}$ is an embedding. Indeed, for $x \neq y \in X$, choose a_i so that $d(x, a_i) < \min\{1, d(x, y)/2\}$. Then, it follows that

$$f_i(x) = d(x, a_i) < d(x, y)/2 < d(x, y) - d(x, a_i) \leqslant d(y, a_i) = f_i(y).$$

Thus, f is injective. If f is not an embedding, then there are $x, x_n \in X$, $n \in \mathbb{N}$, and $0 < \delta < 1$ such that $\lim_{n\to\infty} f(x_n) = f(x)$ but $d(x_n, x) \geqslant \delta$ for all $n \in \mathbb{N}$. Choose a_i so that $d(x, a_i) < \delta/3$. Then, we have $f_i(x) < 1$. If $n \in \mathbb{N}$ is sufficiently large, then $f_i(x_n) < 1$ holds, and hence

$$\begin{aligned} f_i(x_n) - f_i(x) &= d(x_n, a_i) - d(x, a_i) \\ &\geqslant d(x_n, x) - 2d(x, a_i) > \delta - 2\delta/3 = \delta/3, \end{aligned}$$

which contradicts $\lim_{n\to\infty} f_i(x_n) = f_i(x)$. Therefore, f is an embedding. □

A metric space $X = (X, d)$ or a metric d is said to be **totally bounded** provided that, for any $\varepsilon > 0$, there is a finite set $A \subset X$ such that $d(x, A) < \varepsilon$ for every $x \in X$, i.e., $X = \bigcup_{a\in A} \mathrm{B}_d(a, \varepsilon)$. It is now easy to show that X is totally bounded if and only if, for any $\varepsilon > 0$, X has a finite open cover $\mathcal{U}$ with mesh $\mathcal{U} < \varepsilon$. Then, every compact metric space is totally bounded. As is easily seen, any subspace of a totally bounded metric space X is also totally bounded with respect to the metric inherited from X.

Theorem 5.7.1 *A metrizable space is separable if and only if it has an admissible totally bounded metric.*

Proof If a metrizable space X is separable, then X can be embedded in the Hilbert cube $\mathbf{I}^{\mathbb{N}}$. Restricting a metric for $\mathbf{I}^{\mathbb{N}}$, we can obtain an admissible totally bounded metric on X.

Conversely, if X has an admissible totally bounded metric d, then X has finite subsets A_i, $i \in \mathbb{N}$, so that $d(x, A_i) < 2^{-i}$ for every $x \in X$. Then, $A = \bigcup_{i\in\mathbb{N}} A_i$ is a countable dense subset of X. Hence, X is separable. □

Now, we can prove the following Dimension-Preserving Compactification Theorem:

Theorem 5.7.2 (Dimension-Preserving Compactification) *Every n-dimensional separable metrizable space has an n-dimensional metrizable compactification.*

Proof Let X be a separable metrizable space with $\dim X = n$. By Theorem 5.7.1, X has an admissible totally bounded metric d. For each $i \in \mathbb{N}$, X has a finite open cover $\mathcal{U}_i = \{U_{i,j} \mid j = 1, \ldots, m_i\}$ such that $\operatorname{ord} \mathcal{U}_i \leqslant n+1$, $\operatorname{mesh}_d \mathcal{U}_i < 2^{-i}$, and mesh $f_{i',j'}(\mathcal{U}_i) < 2^{-i}$ for $i' < i$ and $j' \leqslant m_{i'}$, where $f_{i,j} : X \to \mathbf{I}$ is the map defined by

$$f_{i,j}(x) = \frac{d(x, X \setminus U_{i,j})}{\sum_{k=1}^{m_i} d(x, X \setminus U_{i,k})}.$$

For each $i \in \mathbb{N}$, we define a map $f_i : X \to \mathbf{I}^{m_i}$ by

$$f_i(x) = (f_{i,1}(x), \ldots, f_{i,m_i}(x)).$$

Then, the map $f : X \to \prod_{i\in\mathbb{N}} \mathbf{I}^{m_i}$ defined by $f(x) = (f_i(x))_{i\in\mathbb{N}}$ is an embedding. Indeed, $\bigcup_{i\in\mathbb{N}} \mathcal{U}_i = \{U_{i,j} \mid i \in \mathbb{N},\ j \leqslant m_i\}$ is an open basis for X. Since $x \in U_{i,j}$ if and only if $f_{i,j}(x) > 0$, it follows that f is injective, and

$$f(U_{i,j}) = f(X) \cap \left\{ z \in \textstyle\prod_{i\in\mathbb{N}} \mathbf{I}^{m_i} \;\middle|\; z(i)(j) > 0 \right\}.$$

The closure $\widetilde{X}$ of $f(X)$ in $\prod_{i\in\mathbb{N}} \mathbf{I}^{m_i}$ is a metrizable compactification of X. Let ρ be the admissible metric for $\prod_{i\in\mathbb{N}} \mathbf{I}^{m_i}$ defined by $\rho(z, z') = \sup_{i\in\mathbb{N}} 2^{-i} \rho_i(z(i), z'(i))$, where ρ_i is the metric for $\mathbf{I}^{m_i}$ defined by

$$\rho_i(x, y) = \max \left\{ |x(j) - y(j)| \;\middle|\; j = 1, \ldots, m_i \right\} \text{ for } x, y \in \mathbf{I}^{m_i}.$$

For each $i \in \mathbb{N}$ and $j \leqslant m_i$, let $W_{i,j} = \{z \in \widetilde{X} \mid z(i)(j) > 0\}$. Then, $W_{i,j} \cap f(X) = f(U_{i,j})$ is dense in $W_{i,j}$. For $i' < i$,

$$\operatorname{diam}_{\rho_{i'}} f_{i'}(U_{i,j}) = \max \left\{ \operatorname{diam} f_{i',j'}(U_{i,j}) \;\middle|\; j' \leqslant m_{i'} \right\} < 2^{-i}.$$

Thus, it follows that $\operatorname{diam}_\rho W_{i,j} = \operatorname{diam}_\rho f(U_{i,j}) \leqslant 2^{-i}$. For each $z \in \widetilde{X}$, we have $x_n \in X$, $n \in \mathbb{N}$, such that $f(x_n) \to z$ $(n \to \infty)$. Note that $\sum_{j=1}^{m_i} f_{i,j}(x_n) = 1$. For each $i \in \mathbb{N}$, we can find $j \leqslant m_i$ such that $f_{i,j}(x_n) \geqslant 1/m_i$ for infinitely many $n \in \mathbb{N}$. Because $f_{i,j}(x_n) \to z(i)(j)$ $(n \to \infty)$, we have $z(i)(j) \geqslant 1/m_i$, i.e., $z \in W_{i,j}$. Therefore, $\mathcal{W}_i = \{W_{i,j} \mid j = 1, \ldots, m_i\} \in \operatorname{cov}(\widetilde{X})$ with $\operatorname{mesh}_\rho \mathcal{W}_i \leqslant 2^{-i}$. Since $W_{i,j} \cap f(X) = f(U_{i,j})$ and $f(X)$ is dense in $\widetilde{X}$, it follows that $\operatorname{ord} \mathcal{W}_i = \operatorname{ord} f(\mathcal{U}_i) = \operatorname{ord} \mathcal{U}_i \leqslant n+1$. Since $\widetilde{X}$ is compact, we can find $i_1 < i_2 < \cdots$ in $\mathbb{N}$ so that $\mathcal{W}_{i_1} \succ \mathcal{W}_{i_2} \succ \cdots$. Then, $\dim \widetilde{X} \leqslant n$ by Theorem 5.4.1. On the other hand, $\dim X \leqslant \dim \widetilde{X}$ by the Subset Theorem 5.4.4. Thus, we have $\dim \widetilde{X} = n$. □

In the above proof, suppose now that X is a closed subset of a separable metrizable space Y and d is an admissible totally bounded metric for Y. Then, applying the Canonical Open Set Extension Lemma 5.4.10, we can easily construct open covers $\mathcal{V}_i = \{V_{i,j} \mid j = 1, \ldots, m_i\}$ of Y such that $\operatorname{ord} \mathcal{V}_i[X] \leqslant n+1$, $\operatorname{mesh}_d \mathcal{V}_i < 2^{-i}$, and mesh $g_{i',j'}(\mathcal{V}_i) < 2^{-i}$ for $i' < i$ and $j' \leqslant m_{i'}$, where $g_{i,j} : Y \to \mathbf{I}$ is the map defined by

$$g_{i,j}(y) = \frac{d(y, Y \setminus V_{i,j})}{\sum_{k=1}^{m_i} d(y, Y \setminus V_{i,k})}.$$

As for f in the above proof, using maps $g_{i,j}$, we can define an embedding $g : Y \to \prod_{i\in\mathbb{N}} \mathbf{I}^{m_i}$. The closure $\widetilde{Y}$ of $g(Y)$ in $\prod_{i\in\mathbb{N}} \mathbf{I}^{m_i}$ is a metrizable compactification of Y such that $\dim \mathrm{cl}_{\widetilde{Y}} X = \dim X$. Furthermore, we can strengthen this as follows:

Theorem 5.7.3 *Let X be a separable metrizable space and $X_1, X_2, \ldots$ be closed sets in X. Then, there exists a metrizable compactification $\widetilde{X}$ of X such that* $\dim \mathrm{cl}_{\widetilde{X}} X_i = \dim X_i$.

Sketch of Proof Assume that $\dim X_i = n_i < \infty$. Let d be an admissible totally bounded metric for X. Applying the Canonical Open Set Extension Lemma 5.4.10, construct open covers $\mathcal{U}_{i,j} = \{U_{i,j,k} \mid k = 1, \ldots, m(i, j)\}$ of X so that $\mathrm{ord}\, \mathcal{U}_{i,j}[X_i] \leqslant n_i + 1$, $\mathrm{mesh}_d\, \mathcal{U}_{i,j} < 2^{-i-j}$, and $\mathrm{mesh}\, f_{i',j',k'}(\mathcal{U}_{i,j}) < 2^{-i-j}$ for $i' + j' < i + j$ and $k' \leqslant m(i', j')$, where $f_{i,j,k} : X \to \mathbf{I}$ is the map defined by

$$f_{i,j,k}(x) = \frac{d(x, X \setminus U_{i,j,k})}{\sum_{l=1}^{m(i,j)} d(x, X \setminus U_{i,j,l})}.$$

As above, we can now use these maps $f_{i,j,k}$ to define an embedding

$$f : X \to \prod_{n\in\mathbb{N}} \prod_{i+j=n+1} \mathbf{I}^{m(i,j)}.$$

The desired compactification of X is obtained as the closure $\widetilde{X}$ of $f(X)$ in the compactum $\prod_{n\in\mathbb{N}} \prod_{i+j=n+1} \mathbf{I}^{m(i,j)}$.

Next, we show the following compactification theorem:

Theorem 5.7.4 (Compactification with the S.C.D. Remainder) *Every separable completely metrizable space X has a metrizable compactification γX such that the remainder $\gamma X \setminus X$ is a countable union of finite-dimensional compact sets, hence it is strongly countable-dimensional.*

Proof We may assume that X is a subset of a compact metric space $Z = (Z, d)$ with $\mathrm{diam}\, Z \leqslant 1$. Since X is completely metrizable, we can write $X = \bigcap_{i\in\mathbb{N}} G_i$, where $G_1 \supset G_2 \supset \cdots$ are open in Z. Since each G_i is totally bounded, G_i has a finite open cover $\mathcal{U}_i$ with $\mathrm{mesh}\, \mathcal{U}_i < 2^{-i}$. We can write $\bigcup_{i\in\mathbb{N}} \mathcal{U}_i = \{U_n \mid n \in \mathbb{N}\}$. Let $f : Z \to \mathbf{I}^{\mathbb{N}}$ be a map defined by

$$f(z)(n) = d(z, Z \setminus U_n), \ n \in \mathbb{N}.$$

Then, $f|X$ is an embedding. In fact, if $x \neq y \in X$, there exists some U_n such that $x \in U_n$ but $y \notin U_n$. Then, $f(x)(n) \neq 0 = f(y)(n)$, which implies that $f(x) \neq f(y)$. For each $x \in X$ and each neighborhood U of x in X, choose $n \in \mathbb{N}$ so that $x \in U_n \cap X \subset U$. Then, $f(U_n \cap X) = f(X) \cap \mathrm{pr}_n^{-1}((0, 1])$, where $\mathrm{pr}_n : \mathbf{I}^{\mathbb{N}} \to \mathbf{I}$ is the projection onto the n-th factor. Therefore, $f(U)$ is a neighborhood of $f(x)$ in $f(X)$.

Let γX be the closure of $f(X)$ in $\mathbf{I}^{\mathbb{N}}$. Identifying X with $f(X)$, γX is a compactification of X. Note that $\gamma X \subset f(Z)$. If $f(z)(n) > 0$ for infinitely many $n \in \mathbb{N}$, then z is contained in infinitely many G_i, which implies that $z \in \bigcap_{i\in\mathbb{N}} G_i = X$. Thus, we have $f(Z) \setminus f(X) \subset \mathbf{I}^{\mathbb{N}}_f$, hence $\gamma X \setminus X \subset \mathbf{I}^{\mathbb{N}}_f$. Since X is completely metrizable, X is G_δ in γX, hence $\gamma X \setminus X$ is F_σ in γX. Consequently, $\gamma X \setminus X$ is a countable union of f.d. compact sets. □

Now, we prove the following theorem of COMPACTIFICATION PRESERVING THE (STRONG) COUNTABLE DIMENSIONALITY:

Theorem 5.7.5 (COMPACTIFICATION PRESERVING THE (S.)C.D.) *Every (strongly) countable-dimensional separable completely metrizable space has a (strongly) countable-dimensional metrizable compactification.*

Proof The c.d. case is a direct consequence of Theorem 5.7.4. To prove the s.c.d. case, let X be a separable completely metrizable space with $X = \bigcup_{i\in\mathbb{N}} X_i$, where each X_i is closed in X, $\dim X_i < \infty$, and $X_1 \subset X_2 \subset \cdots$. By Theorem 5.7.3, X has a metrizable compactification Y such that $\dim \mathrm{cl}_Y X_i = \dim X_i$. By the complete metrizability of X, we can write $X = \bigcap_{i\in\mathbb{N}} U_i$, where each U_i is open in Y and $Y = U_1 \supset U_2 \supset \cdots$. Let $Z = \bigcup_{i\in\mathbb{N}}(U_i \cap \mathrm{cl}_Y X_i)$. Then, $X = \bigcup_{i\in\mathbb{N}} X_i \subset Z$. Since each $U_i \cap \mathrm{cl}_Y X_i$ is an F_σ-set in Y, Z is a countable union of f.d. compact sets.

We show that $Y \setminus Z = \bigcup_{i\in\mathbb{N}}((Y \setminus \mathrm{cl}_Y X_i) \setminus U_{i+1})$, which is an F_σ-set in Y. For each $y \in Y \setminus Z$, let $i_0 = \max\{i \in \mathbb{N} \mid y \in U_i\}$. Then, $y \in U_{i_0} \setminus U_{i_0+1}$, which implies $y \notin \mathrm{cl}_Y X_{i_0}$ because $y \notin Z$. Hence, $y \in (Y \setminus \mathrm{cl}_Y X_{i_0}) \setminus U_{i_0+1}$. On the other hand, for each $z \in Z$, we have i_1 such that $z \in U_{i_1} \cap \mathrm{cl}_Y X_{i_1}$. For $i \geqslant i_1$, $z \notin (Y \setminus \mathrm{cl}_Y X_i) \setminus U_{i+1}$ because $z \in \mathrm{cl}_Y X_{i_1} \subset \mathrm{cl}_Y X_i$. For $i < i_1$, $z \notin (Y \setminus \mathrm{cl}_Y X_i) \setminus U_{i+1}$ because $z \in U_{i_1} \subset U_{i+1}$. Thus, Z is a G_δ-set in a compactum Y, hence it is completely metrizable.

Now, applying Theorem 5.7.4, we have a metrizable compactification $\widetilde{Z}$ of Z such that $\widetilde{Z} \setminus Z$ is a countable union of f.d. compact sets. Then, $\widetilde{Z}$ is a compactification of X and it is a countable union of f.d. compact sets, hence it is s.c.d. □

5.8 Embeddings into Euclidean Spaces

Recall that every separable metrizable space X can be embedded into the Hilbert cube $\mathbf{I}^{\mathbb{N}}$ and hence into $\mathbb{R}^{\mathbb{N}}$ (Corollary 2.3.9; cf. Note in the second paragraph of Sect. 5.7). As a finite-dimensional version of this result, we prove the following EMBEDDING THEOREM:

Theorem 5.8.1 (EMBEDDING INTO $\mathbf{I}^{2n+1}$) *Every separable metrizable space with* $\dim \leqslant n$ *can be embedded in* $\mathbf{I}^{2n+1}$*, and can hence be embedded in the Euclidean space* $\mathbb{R}^{2n+1}$.

Remark 5.8.1 In Theorem 5.8.1, the cube $\mathbf{I}^{2n+1}$ cannot be replaced by any cube of dimension $< 2n + 1$. In fact, there exist n-dimensional compact polyhedra that cannot be embedded into $\mathbf{I}^{2n}$. Refer to Fig. 5.3.

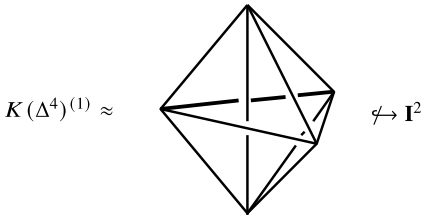


Fig. 5.3 A one-dimensional polyhedron that cannot be embedded in $\mathbf{I}^2$

To prove the Embedding Theorem 5.8.1, we introduce a new notion. Now, let $X = (X, d)$ be a *compact* metric space. Given $\varepsilon > 0$, a map $f : X \to Y$ is called an ***ε*-map** if $\operatorname{diam} f^{-1}(y) < \varepsilon$ for each $y \in Y$. As is easily observed, a map $f : X \to Y$ is an embedding if and only if $f : X \to Y$ is an ε-map for every $\varepsilon > 0$.

Lemma 5.8.2 *Let $f : X \to Y$ be an ε-map from a compact metric space $X = (X, d)$ to a metric space $Y = (Y, \rho)$. Then, there is some $\delta > 0$ such that any map $g : X \to Y$ with $\rho(f, g) < \delta$ is an ε-map.*

Proof Since f is a closed map, each $y \in Y$ has an open neighborhood V_y in Y such that $\operatorname{diam} f^{-1}(V_y) < \varepsilon$. Indeed, for each $y \in Y$, $f^{-1}(y)$ has an open neighborhood U_y in X with $\operatorname{diam} U_y < \varepsilon$. Then, $V_y = Y \setminus f(X \setminus U_y)$ is an open neighborhood of y in X such that $f^{-1}(V_y) \subset U_y$. Since X is compact, we can choose $\delta > 0$ so that each $\mathrm{B}_\rho(f(x), 2\delta)$ is contained in some V_y, hence $\operatorname{diam} f^{-1}(\mathrm{B}_\rho(f(x), 2\delta)) < \varepsilon$. Let $g : X \to Y$ be a map with $\rho_\infty(f, g) < \delta$.[9] For $y \in Y$ and $x, x' \in g^{-1}(y)$,

$$\rho(f(x), f(x')) \leqslant \rho(f(x), g(x)) + \rho(f(x'), g(x')) < 2\delta,$$

which implies that $g^{-1}(y) \subset f^{-1}(\mathrm{B}_\rho(f(x), 2\delta))$. Therefore, $\operatorname{diam} g^{-1}(y) < \varepsilon$, that is, g is an ε-map. □

For spaces X and Y, let $\mathrm{Emb}(X, Y)$ denote the subset of $\mathrm{C}(X, Y)$ consisting of all closed embeddings.

Theorem 5.8.3 *Let $X = (X, d)$ be a compact metric space and $Y = (Y, \rho)$ a complete metric space. Assume that for each $\varepsilon > 0$ and $\delta > 0$, every map $f : X \to Y$ is δ-close to an ε-map. Then, every map $f : X \to Y$ can be approximated by an embedding, that is,* $\mathrm{Emb}(X, Y)$ *is dense in the space* $\mathrm{C}(X, Y)$ *with the sup-metric.*

Proof For each $n \in \mathbb{N}$, let G_n be the set of all 2^{-n}-maps from X to Y. Then, G_n is open and dense in the space $\mathrm{C}(X, Y)$ by Lemma 5.8.2 and the assumption. By the Baire Category Theorem 2.5.1, $\mathrm{Emb}(X, Y) = \bigcap_{n\in\mathbb{N}} G_n$ is also dense in $\mathrm{C}(X, Y)$, hence so is the set of embeddings of X into Y. □

[9] Recall $\rho_\infty(f, g) = \sup_{x\in X} \rho(f(x), g(x))$, which was defined in Sect. 1.1.

Since every separable metrizable space has a metrizable compactification with the same dimension by the Dimension-Preserving Compactification Theorem 5.7.2, the Embedding Theorem 5.8.1 can be obtained as a corollary of the next theorem, which can be proved by applying the General Position Lemma 4.14.1 together with Theorem 5.8.3:

Theorem 5.8.4 (EMBEDDING APPROXIMATION) *Let X be a compact metric space with* $\dim X \leqslant n$. *Then, every map $f : X \to \mathbf{I}^{2n+1}$ can be approximated by embeddings, that is, for each $\varepsilon > 0$, there is an embedding $h : X \to \mathbf{I}^{2n+1}$ that is ε-close to f. In particular, every compact metrizable space with* $\dim \leqslant n$ *can be embedded in $\mathbf{I}^{2n+1}$.*

Proof Let d be the restriction of the Euclidean metric on $\mathbf{I}^{2n+1}$. Because of Theorem 5.8.3, it is enough to show that for each $\varepsilon > 0$ and $\delta > 0$, every map $f : X \to \mathbf{I}^{2n+1}$ is δ-close to an ε-map. We have a finite open cover $\mathcal{U}$ of X such that $\operatorname{ord} \mathcal{U} \leqslant n+1$, $\operatorname{mesh} \mathcal{U} < \varepsilon$, and $\operatorname{mesh} f(\mathcal{U}) < \delta/2$. Let $K = N(\mathcal{U})$ be the nerve of $\mathcal{U}$. A canonical map $\varphi : X \to |K|$ is an ε-map because $\varphi^{-1}(O_K) \prec \mathcal{U}$. Applying the General Position Lemma 4.14.1, we can obtain an injection $g_0 : K^{(0)} = \mathcal{U} \to \mathbf{I}^{2n+1}$ such that $d(g_0(U), f(U)) < \delta/2$ for every $U \in \mathcal{U}$ and every $2n+2$ points of $g_0(K^{(0)})$ are affinely independent. Then, g_0 is uniquely extended to a map $g : |K| \to \mathbf{I}^{2n+1}$ such that g is affine on each simplex of K_1. It is easy to see that g is also an injection (so an embedding) (cf. the proof of Theorem 4.14.2). Hence, $h = g\varphi : X \to \mathbf{I}^{2n+1}$ is an ε-map. For each $x \in X$, let $\mathcal{U}[x] = \{U_1, \ldots, U_k\}$. Then, $\langle U_1, \ldots, U_k \rangle \in K$ and

$$d(g(U_i), f(x)) \leqslant d(g(U_i), f(U_i)) + \operatorname{diam} f(U_i) < \delta.$$

Since $\mathrm{B}(f(x), \delta)$ is convex, it follows that

$$g\varphi(x) \in g(\langle U_1, \ldots, U_k \rangle) = \langle g(U_1), \ldots, g(U_k) \rangle \subset \mathrm{B}(f(x), \delta).$$

Therefore, $h = g\varphi$ is δ-close to f. □

We want to generalize the Embedding Approximation Theorem 5.8.4 to the non-compact case. To do so, it is necessary to extend Theorem 5.8.3 to the non-compact case. Given $\mathcal{U} \in \operatorname{cov}(X)$, a map $f : X \to Y$ is called a **$\mathcal{U}$-map** if $f^{-1}(\mathcal{V}) \prec \mathcal{U}$ for some $\mathcal{V} \in \operatorname{cov}(Y)$. By $\mathrm{C}_{\mathcal{U}}(X, Y)$, we denote the subset of $\mathrm{C}(X, Y)$ consisting of all $\mathcal{U}$-maps. In the case that X is a compact metric space, let $\varepsilon > 0$ be a Lebesgue number for $\mathcal{U} \in \operatorname{cov}(X)$. Then, every ε-map is a $\mathcal{U}$-map.

> Indeed, as seen in the proof of Lemma 5.8.2, each $y \in Y$ has an open neighborhood V_y in Y such that $\operatorname{diam} f^{-1}(V_y) < \varepsilon$. Then, we have $\mathcal{V} = \{V_y \mid y \in Y\} \in \operatorname{cov}(Y)$. It follows that $f^{-1}(\mathcal{V}) \prec \mathcal{U}$.

Conversely, given $\varepsilon > 0$, let $\mathcal{U} = \{\mathrm{B}(x, \varepsilon/3) \mid x \in X\}$. Then, it is easy to see that every $\mathcal{U}$-map $f : X \to Y$ is an ε-map.

When Y is paracompact, the limitation topology can be defined as the topology on $\mathrm{C}(X, Y)$ in which $\{\mathcal{V}(f) \mid \mathcal{V} \in \operatorname{cov}(Y)\}$ is a neighborhood basis of each $f \in \mathrm{C}(X, Y)$,[10] where

[10] Cf. Sect. 2.11. The limitation topology is Hausdorff.

$$\mathcal{V}(f) = \left\{ g \in \mathrm{C}(X, Y) \;\middle|\; g \text{ is } \mathcal{V}\text{-close to } f \right\}.$$

To extend Theorem 5.8.3, we need the following two lemmas.

Lemma 5.8.5 *Let Y be a paracompact space. For each $\mathcal{U} \in \mathrm{cov}(X)$, $\mathrm{C}_{\mathcal{U}}(X, Y)$ is open in the space $\mathrm{C}(X, Y)$ with the limitation topology.*

Proof For each $f \in \mathrm{C}_{\mathcal{U}}(X, Y)$, $f^{-1}(\mathcal{V}) \prec \mathcal{U}$ for some $\mathcal{V} \in \mathrm{cov}(Y)$. Let $\mathcal{W} \in \mathrm{cov}(Y)$ such that $\mathrm{st}\,\mathcal{W} \prec \mathcal{V}$. For each $g \in \mathcal{W}(f)$, $f(g^{-1}(\mathcal{W})) \prec \mathrm{st}\,\mathcal{W} \prec \mathcal{V}$, so $g^{-1}(\mathcal{W}) \prec f^{-1}(\mathcal{V}) \prec \mathcal{U}$, which implies $g \in \mathrm{C}_{\mathcal{U}}(X, Y)$. □

Lemma 5.8.6 *Let X be a complete metric space and Y a metrizable space. Then, $\mathrm{Emb}(X, Y) = \bigcap_{n\in\mathbb{N}} \mathrm{C}_{\mathcal{U}_n}(X, Y)$, where $\mathcal{U}_n \in \mathrm{cov}(X)$ with $\mathrm{mesh}\,\mathcal{U}_n < 2^{-n}$. Thus, for each completely metrizable space X and each metrizable space Y, $\mathrm{Emb}(X, Y)$ is a G_δ-set in the space $\mathrm{C}(X, Y)$ with the limitation topology.*

Proof Obviously, $\mathrm{Emb}(X, Y) \subset \bigcap_{n\in\mathbb{N}} \mathrm{C}_{\mathcal{U}_n}(X, Y)$. Every $f \in \bigcap_{n\in\mathbb{N}} \mathrm{C}_{\mathcal{U}_n}(X, Y)$ is injective. For $x_n \in X$, $n \in \mathbb{N}$, if $(f(x_n))_{n\in\mathbb{N}}$ is convergent, then $(x_n)_{n\in\mathbb{N}}$ is Cauchy, so it is convergent. This means that f is closed, hence $f \in \mathrm{Emb}(X, Y)$. Thus, we have $\mathrm{Emb}(X, Y) = \bigcap_{n\in\mathbb{N}} \mathrm{C}_{\mathcal{U}_n}(X, Y)$. □

When Y is also completely metrizable, the space $\mathrm{C}(X, Y)$ with the limitation topology is a Baire space by Theorem 2.11.4. Then, using Lemmas 5.8.5 and 5.8.6, we can extend Theorem 5.8.3 as follows:

Theorem 5.8.7 *Let X and Y be completely metrizable spaces. Suppose that, for each $\mathcal{U} \in \mathrm{cov}(X)$, $\mathrm{C}_{\mathcal{U}}(X, Y)$ is dense in the space $\mathrm{C}(X, Y)$ with the limitation topology. Then, $\mathrm{Emb}(X, Y)$ is also dense in $\mathrm{C}(X, Y)$. In other words, if every map $f : X \to Y$ is approximated by $\mathcal{U}$-maps for each $\mathcal{U} \in \mathrm{cov}(X)$, then every map $f : X \to Y$ is approximated by closed embeddings.* □

We now consider the case that X and Y are locally compact metrizable. Let $\mathrm{C}^P(X, Y)$ be the subspace of the space $\mathrm{C}(X, Y)$ with the limitation topology consisting of all proper maps.[11] The space $\mathrm{C}^P(X, Y)$ is a Baire space by Theorem 2.11.7. It should be noted that $\mathrm{Emb}(X, Y) \subset \mathrm{C}^P(X, Y)$. Moreover, if X is non-compact, then any constant map of X to Y is not proper, which implies that $\mathrm{Emb}(X, Y)$ is not dense in the space $\mathrm{C}(X, Y)$ with the limitation topology because $\mathrm{C}^P(X, Y)$ is clopen in $\mathrm{C}(X, Y)$ due to Proposition 2.11.6. For an open cover $\mathcal{U} \in \mathrm{cov}(X)$ consisting of open sets with the compact closure, we have $\mathrm{C}_{\mathcal{U}}(X, Y) \subset \mathrm{C}^P(X, Y)$.

Indeed, for each $f \in \mathrm{C}_{\mathcal{U}}(X, Y)$, let $\mathcal{V}$ be a locally finite open cover of Y such that $f^{-1}(\mathcal{V}) \prec \mathcal{U}$. Each compact set A in Y meets only finitely many $V_1, \ldots, V_n \in \mathcal{V}$, where each $f^{-1}(V_i)$ is contained in some $U_i \in \mathcal{U}$. Then, $f^{-1}(A) \subset \bigcup_{i=1}^n \mathrm{cl}\, U_i$. Since $\bigcup_{i=1}^n \mathrm{cl}\, U_i$ is compact, $f^{-1}(A)$ is also compact.

The following theorem is the locally compact version of Theorem 5.8.7:

[11] In this case, a proper map coincides with a perfect map (Proposition 2.1.13).

Theorem 5.8.8 *Let X and Y be locally compact metrizable spaces. If $C_{\mathcal{U}}(X, Y)$ is dense in the space $C^P(X, Y)$ with the limitation topology for each open cover $\mathcal{U}$ of X consisting of open sets with the compact closure, then $\mathrm{Emb}(X, Y)$ is also dense in $C^P(X, Y)$.* □

Now, we show the following locally compact version of the Embedding Approximation Theorem 5.8.4:

Theorem 5.8.9 (EMBEDDING APPROXIMATION) *Let X be a locally compact separable metrizable space with $\dim X \leqslant n$. Then, $\mathrm{Emb}(X, \mathbb{R}^{2n+1})$ is dense in the space $C^P(X, \mathbb{R}^{2n+1})$ with the limitation topology, that is, for each open cover $\mathcal{V}$ of $\mathbb{R}^{2n+1}$, every proper map $f : X \to \mathbb{R}^{2n+1}$ is $\mathcal{V}$-close to a closed embedding $h : X \to \mathbb{R}^{2n+1}$.*

Proof We prove the case when X is non-compact because the compact case follows immediately from Theorem 5.8.4. By Theorem 5.8.8, it suffices to show that for each $\mathcal{U} \in \mathrm{cov}(X)$ and $\mathcal{V} \in \mathrm{cov}(\mathbb{R}^{2n+1})$, every proper map $f : X \to \mathbb{R}^{2n+1}$ is $\mathcal{V}$-close to some $\mathcal{U}$-map $h : X \to \mathbb{R}^{2n+1}$. Replacing $\mathcal{V}$ with a refinement, we may assume that $\mathcal{V}$ is locally finite and $\mathrm{cl}\, V$ is compact for each $V \in \mathcal{V}$.

We can find a countable locally finite open cover $\mathcal{W}$ of $\mathbb{R}^{2n+1}$ such that for each $y \in \mathbb{R}^{2n+1}$, the convex hull $\langle \mathrm{st}(y, \mathcal{W})\rangle$ is contained in some member of $\mathcal{V}$, that is, $\{\langle \mathrm{st}(y, \mathcal{W})\rangle \mid y \in \mathbb{R}^{2n+1}\} \prec \mathcal{V}$. Then, $\mathrm{cl}\, W$ is compact for each $W \in \mathcal{W}$. By replacing $\mathcal{U}$ with a refinement, we may assume that $\emptyset \notin \mathcal{U} \prec f^{-1}(\mathcal{W})$, $\mathcal{U}$ is countable and locally finite in X, and $\mathrm{ord}\, \mathcal{U} \leqslant n + 1$ (cf. Corollary 5.2.6). Write $\mathcal{U} = \{U_i \mid i \in \mathbb{N}\}$ so that $U_i \neq U_j$ if $i \neq j$ and choose $W_i \in \mathcal{W}$, $i \in \mathbb{N}$, so that $f(U_i) \subset W_i$, where it should be noticed that $i \neq j$ does not imply $W_i \neq W_j$. For each $W \in \mathcal{W}$, $\{i \in \mathbb{N} \mid W = W_i\}$ is finite (or empty). Indeed,

$$\{i \in \mathbb{N} \mid W = W_i\} \subset \{i \in \mathbb{N} \mid f(U_i) \subset W\} \subset \{i \in \mathbb{N} \mid U_i \cap f^{-1}(\mathrm{cl}\, W) \neq \emptyset\},$$

where the last set is finite because $\mathcal{U}$ is locally finite and $f^{-1}(\mathrm{cl}\, W)$ is compact (f is proper). Since $\mathcal{W}$ is locally finite in $\mathbb{R}^{2n+1}$, it follows that the indexed collection $(W_i)_{i\in\mathbb{N}}$ is locally finite in $\mathbb{R}^{2n+1}$ (cf. Sect. 1.1).

Let $K = N(\mathcal{U})$ be the nerve of $\mathcal{U}$ with $\varphi : X \to |K|$ a canonical map. Then, $\dim K \leqslant n$ and φ is a $\mathcal{U}$-map because $\varphi^{-1}(O_K(U)) \subset U$ for each $U \in \mathcal{U} = K^{(0)}$. We will construct a closed embedding $g : |K| \to \mathbb{R}^{2n+1}$ such that $h = g\varphi : X \to \mathbb{R}^{2n+1}$ is $\mathcal{V}$-close f. Then, as is easily observed, h is also a $\mathcal{U}$-map, which will complete the proof.

By the General Position Lemma 4.14.1, we have points $v_i \in \mathbb{R}^{2n+1}$, $i \in \mathbb{N}$, such that $v_i \in W_i$ and every $2n + 2$ points $v_{i_1}, \ldots, v_{i_{2n+2}}$ are affinely independent, where it should be noted that $\{v_i \mid i \in \mathbb{N}\}$ is discrete in $\mathbb{R}^{2n+1}$ because $(W_i)_{i\in\mathbb{N}}$ is locally finite in $\mathbb{R}^{2n+1}$. Then, we have a PL-map $g : |K| \to \mathbb{R}^{2n+1}$ such that $g(U_i) = v_i \in W_i$ for each $U_i \in K^{(0)} = \mathcal{U}$ and $g|\sigma$ is affine on each simplex $\sigma \in K$. For each pair of simplexes $\sigma, \tau \in K$, since $\mathrm{card}(\sigma^{(0)} \cup \tau^{(0)}) \leqslant 2n + 2$, $g(\sigma^{(0)} \cup \tau^{(0)})$ is affinely independent, which implies that $g|\sigma \cup \tau$ is an embedding. Hence, g is injective.

We will show that $h = g\varphi : X \to Y$ is $\mathcal{V}$-close to f. For each $x \in X$, take the carrier $\sigma \in K$ of $\varphi(x)$ and let $\sigma^{(0)} = \{U_{i_1}, \ldots, U_{i_k}\}$. Since $x \in \bigcap_{j=1}^{k} U_{i_j}$, we have

$f(x) \in \bigcap_{j=1}^{k} W_{i_j}$, which implies $v_{i_1}, \ldots, v_{i_k} \in \bigcup_{j=1}^{k} W_{i_j} \subset \mathrm{st}(f(x), \mathcal{W})$. Then, it follows that

$$h(x) = g(\varphi(x)) \in g(\sigma) = \langle g(U_{i_1}), \ldots, g(U_{i_k}) \rangle = \langle v_{i_1}, \ldots, v_{i_k} \rangle \subset \langle \mathrm{st}(f(x), \mathcal{W}) \rangle.$$

Recall that $\langle \mathrm{st}(f(x), \mathcal{W}) \rangle$ is contained in some $V \in \mathcal{V}$. Then, $h(x), f(x) \in V$. Thus, h is $\mathcal{V}$-close to f. Notice that, using the above argument, we can show that $\{g(\sigma) \mid \sigma \in K\} \prec \mathcal{V}$, where each $V \in \mathcal{V}$ contains only a finite number of $g(\sigma)$, $\sigma \in K$, because $g(K^{(0)}) = \{v_i \mid i \in \mathbb{N}\}$ is discrete in $\mathbb{R}^{2n+1}$ and $\mathrm{cl}\, V$ is compact.

It remains to show that g is a closed embedding. Since g is an injective map, by Proposition 2.1.12, it is enough to show that g is proper. Let C be a compact set in $\mathbb{R}^{2n+1}$. Since $\mathcal{V}$ is locally finite, it follows that $\mathcal{V}[C]$ is finite. For each $\sigma \in K$ with $\sigma \cap f^{-1}(C) \neq \emptyset$, $g(\sigma)$ is contained in some $V \in \mathcal{V}$, where $V \in \mathcal{V}[C]$. As observed in the above, each $V \in \mathcal{V}$ contains only finite number of $g(\sigma)$, $\sigma \in K$. Hence, the closed set $f^{-1}(C)$ in $|K|$ is covered by finitely many simplexes of K, which means that $f^{-1}(C)$ is compact. □

Remark 5.8.2 In the Embedding Approximation Theorem 5.8.9, it should be noticed that $\mathrm{C}^P(X, \mathbb{R}^{2n+1}) \neq \emptyset$. Indeed, as observed just before Corollary 4.14.3, there exists a proper map $f : X \to \mathbb{R}_+$, where we can regard $f \in \mathrm{C}^P(X, \mathbb{R}^{2n+1})$ by identifying $\mathbb{R}_+$ with $\{\mathbf{0}\} \times \mathbb{R}_+ \subset \mathbb{R}^{2n} \times \mathbb{R} = \mathbb{R}^{2n+1}$. Hence, every locally compact separable metrizable space X with $\dim X \leqslant n$ can be embedded into $\mathbb{R}^{2n+1}$ as a closed set. Moreover, it should be also noticed that a map $f : X \to \mathbb{R}^{2n+1}$ cannot be approximated by closed embeddings if f is not proper. Indeed, $A = f^{-1}(C)$ is not compact for some compact set C in $\mathbb{R}^{2n+1}$. If $h : X \to \mathbb{R}^{2n+1}$ is a closed embedding, then $h(A)$ is closed in $\mathbb{R}^{2n+1}$ and non-compact, which means that $h(A)$ is unbounded. Then, $\sup_{x \in A} d(h(x), f(x)) \geqslant \sup_{x \in A} d(f(x), C) = \infty$, where d is the Euclidean metric on $\mathbb{R}^{2n+1}$.

We now show the following proposition:

Proposition 5.8.10 *Let X be an arbitrary space and $n \in \omega$. Suppose that for each $\mathcal{U} \in \mathrm{cov}(X)$, there exist a paracompact space Y with $\dim Y \leqslant n$ and a $\mathcal{U}$-map $f : X \to Y$. Then, $\dim X \leqslant n$.*

Proof For each finite open cover $\mathcal{U}$ of X, we have a paracompact space Y with $\dim Y \leqslant n$ and a $\mathcal{U}$-map $f : X \to Y$. Then, by Theorem 5.2.5, we have $\mathcal{V} \in \mathrm{cov}(Y)$ such that $f^{-1}(\mathcal{V}) \prec \mathcal{U}$ and $\mathrm{ord}\, \mathcal{V} \leqslant n + 1$. Since $\mathrm{ord}\, f^{-1}(\mathcal{V}) \leqslant n + 1$, $\dim X \leqslant n$ can be obtained by Theorem 5.2.4. □

When X is a metric compactum, the following can be easily derived from Theorem 5.8.10:

Proposition 5.8.11 *Let X be a metric compactum and $n \in \omega$. Suppose that for each $\varepsilon > 0$, there exist a space Y with $\dim Y \leqslant n$ and an ε-map $f : X \to Y$. Then, $\dim X \leqslant n$.* □

The n-dimensionality of subsets of $\mathbb{R}^n$ can be characterized as follows:

Theorem 5.8.12 *(n-DIMENSIONALITY OF $X \subset \mathbb{R}^n$) For a subset $X \subset \mathbb{R}^n$, $\dim X = n$ if and only if X contains an interior point. Equivalently, $\dim X < n$ if and only if $\mathbb{R}^n \setminus X$ is dense in $\mathbb{R}^n$.*

Proof For $X \subset \mathbb{R}^n$, we have $\dim X \leqslant \dim \mathbb{R}^n = n$. If X contains an interior point x, then $\overline{\mathrm{B}}(x, \delta) \subset X$ for some $\delta > 0$. Since $\overline{\mathrm{B}}(x, \delta) \approx \mathbf{B}^n$, it follows that $\dim X \geqslant n$, hence $\dim X = n$.

Conversely, assume that X contains no interior point, equivalently $\mathbb{R}^n \setminus X$ is dense in $\mathbb{R}^n$. Then, $\mathbb{R}^n \setminus X$ contains a countable dense set D. Since $X \subset \mathbb{R}^n \setminus D$, it suffices to show that $\dim(\mathbb{R}^n \setminus D) \leqslant n - 1$. Consider the following subset of $\mathbb{R}^n$:

$$\mathbb{R}^n \setminus \mathbb{Q}^n = \left\{ x \in \mathbb{R}^n \;\middle|\; x(i) \in \mathbb{R} \setminus \mathbb{Q} \;\text{ for at least one } i \right\}.$$

Since $\{t\} \times \mathbb{R}^{n-1}$ is an $(n-1)$-dimensional closed subspace of $\mathbb{R}^n \setminus \mathbb{Q}^n$ for any $t \in \mathbb{R} \setminus \mathbb{Q}$, it follows $\dim(\mathbb{R}^n \setminus \mathbb{Q}^n) \geqslant n - 1$. As is easily observed, $\mathbb{R}^n \setminus \mathbb{Q}^n$ is a union of the following n subsets:

$$\mathbb{Q}^n_k = \left\{ x \in \mathbb{R}^n \;\middle|\; x(i) \in \mathbb{Q} \;\text{ for exactly } k\, i \right\},\; k = 0, 1, \ldots, n-1.$$

Each $\mathbb{Q}^n_k$ is a union of finitely many copies of the subspace $\mathbb{Q}^k \times (\mathbb{R} \setminus \mathbb{Q})^{n-k} \subset \mathbb{R}^n$. Since $\dim(\mathbb{R} \setminus \mathbb{Q})^{n-k} = 0$, it follows that $\mathbb{Q}^n_k$ is a countable union of zero-dimensional closed sets in $\mathbb{Q}^n_k$, which implies $\dim \mathbb{Q}^n_k = 0$ by the Countable Sum Theorem 5.2.8. Then, $\dim(\mathbb{R}^n \setminus \mathbb{Q}^n) \leqslant n - 1$ by the Decomposition Theorem 5.4.11. Hence, the following theorem completes the proof. □

A space X is **countable dense homogeneous** if for and two countable dense sets $A, B \subset X$, there is a homeomorphism $h : X \to X$ such that $h(A) = B$. We show that $\mathbb{R}^n$ is countable dense homogeneous, that is,

Theorem 5.8.13 *(COUNTABLE DENSE HOMOGENEITY OF $\mathbb{R}^n$) For any two countable dense sets $A, B \subset \mathbb{R}^n$, there is a homeomorphism $h : \mathbb{R}^n \to \mathbb{R}^n$ such that $h(A) = B$.*

To prove the above theorem, we need the following lemma:

Lemma 5.8.14 *Let A and B be dense sets in $\mathbb{R}$. Every order isomorphism $f : A \to B$ extends to a homeomorphism $\tilde{f} : \mathbb{R} \to \mathbb{R}$, where A and B have the natural order inherited from $\mathbb{R}$.*

Proof For each $x \in \mathbb{R}$, let $A_x = A \cap (-\infty, x]$. Since A is dense in $\mathbb{R}$, A_x has an upper bound in A, hence $f(A_x)$ has an upper bound in $B = f(A)$. In particular, $f(A_x)$ has an upper bound in $\mathbb{R}$. Thus, we can define $\tilde{f}(x) : \mathbb{R} \to \mathbb{R}$ by $\tilde{f}(x) = \sup f(A_x)$ for each $x \in \mathbb{R}$. If $x \in A$, then

$$f(x) = f(\max A_x) = \max f(A_x) = \sup f(A_x) = \tilde{f}(x),$$

which means that $\tilde{f}|A = f$. Then, $\tilde{f}$ is an order-preserving injection, i.e., $x < x' \in \mathbb{R} \Rightarrow \tilde{f}(x) < \tilde{f}(x')$. Indeed, take $a, a' \in A$ so that $x < a < a' < x'$. Then, it is easy to see that $\tilde{f}(x) \leqslant f(a) < f(a') \leqslant \tilde{f}(x')$. Now, consider the inverse $g = f^{-1} : B \to A$. As above, we can define an order-preserving injection $\tilde{g} : \mathbb{R} \to \mathbb{R}$ with $\tilde{g}|B = g$. Then, $\tilde{g}\tilde{f} : \mathbb{R} \to \mathbb{R}$ is an order-preserving injection with $\tilde{g}\tilde{f}|A = \mathrm{id}_A$. Since A is dense in $\mathbb{R}$, we have $\tilde{g}\tilde{f} = \mathrm{id}_{\mathbb{R}}$. Therefore, $\tilde{f}$ is an order isomorphism. Since the topology of $\mathbb{R}$ is induced by the natural order, $\tilde{f}$ is also a homeomorphism. □

Proof (*Theorem* 5.8.13) Let $P = \{x - y \mid x, y \in A \cup B\}$. Then, P is a countable subset of $\mathbb{R}^n$ with $\mathbf{0} \in P$. Applying the General Position Lemma 4.14.1, we can obtain affinely independent $v_1, \ldots, v_n \in \mathbb{R}^n \setminus P$ such that

- $S \subset P \cup \{v_1, \ldots, v_n\}$ is affinely independent whenever card $S \leqslant n + 1$ and $S \cap P$ is affinely independent.

Let $k : \mathbb{R}^n \to \mathbb{R}^n$ be the (unique) linear homeomorphism such that $k(v_i) = \mathbf{e}_i$ for each $i = 1, \ldots, n$. Let $A' = k(A)$ and $B' = k(B)$. Then, $P' = \{x' - y' \mid x', y' \in A' \cup B'\} = k(P)$. Replacing A, B, and P with A', B', and P', respectively, we can assume $\mathbf{e}_1, \ldots, \mathbf{e}_n \notin P$ and

- $S \subset P \cup \{\mathbf{e}_1, \ldots, \mathbf{e}_n\}$ is affinely independent whenever card $S \leqslant n + 1$ and $S \cap P$ is affinely independent.

By this assumption, we can assert that $x(i) \neq y(i)$ for each $x \neq y \in A \cup B$ and $i = 1, \ldots, n$. Namely, for each $i = 1, \ldots, n$, the restriction $\mathrm{pr}_i|A \cup B$ of the projection $\mathrm{pr}_i : \mathbb{R}^n \to \mathbb{R}$ is injective. Indeed, let

$$S = \{\mathbf{0}, x - y\} \cup \{\mathbf{e}_j \mid j \neq i\} \subset P \cup \{\mathbf{e}_1, \ldots, \mathbf{e}_n\}.$$

Then, since card $S \leqslant n + 1$ and $S \cap P = \{\mathbf{0}, x - y\}$ is affinely independent, S is affinely independent. Hence, $x - y \notin \mathrm{fl}\{\mathbf{0}, \mathbf{e}_j \mid j \neq i\} = \mathrm{pr}_i^{-1}(0)$, which means that $x(i) \neq y(i)$.

For each $i = 1, \ldots, n$, $\mathrm{pr}_i|A : A \to \mathrm{pr}_i(A)$ and $\mathrm{pr}_i|B : B \to \mathrm{pr}_i(B)$ are bijections. We want to define a coordinate-wise order-preserving bijection $f : A \to B$, that is, each $\mathrm{pr}_i f(\mathrm{pr}_i|A)^{-1} : \mathrm{pr}_i(A) \to \mathrm{pr}_i(B)$ is order-preserving (i.e., an order isomorphism), which extends to a homeomorphism $h_i : \mathbb{R} \to \mathbb{R}$ by Lemma 5.8.14. Then, we can extend f to the homeomorphism $h : \mathbb{R}^n \to \mathbb{R}^n$ defined by $h(x) = (h_1(x(1)), \ldots, h_n(x(n)))$. To define f, we show the following claim:

Claim Let $F \subset A$ and $F' \subset B$ be finite subsets with a coordinate-wise order-preserving bijection $g : F \to F'$.

(1) For each $x \in A \setminus F$, there is $y \in B \setminus F'$ such that g extends to a coordinate-wise order-preserving bijection $g : F \cup \{x\} \to F' \cup \{y\}$.
(2) For each $y \in B \setminus F'$, there is $x \in B \setminus F$ such that g extends to a coordinate-wise order-preserving bijection $g : F \cup \{x\} \to F' \cup \{y\}$.

Because of similarity, we may show only (1). For each $i = 1, \ldots, n$, $g_i = \mathrm{pr}_i g(\mathrm{pr}_i|F)^{-1} : \mathrm{pr}_i(F) \to \mathrm{pr}_i(F')$ is an order isomorphism. The connected component of $\mathbb{R} \setminus \mathrm{pr}_i(F)$ containing $x(i)$ is an open interval (p_i, q_i), where $p_i, q_i \in \mathrm{pr}_i(F) \cup \{\pm\infty\}$. Then, we have an open set $U = \prod_{i=1}^n (g_i(p_i), g_i(q_i))$ in $\mathbb{R}^n$, where $g_i(\pm\infty) = \pm\infty$. Since B is dense in $\mathbb{R}^n$, we can take $y \in U \cap B$, which satisfies the required condition.

Now, we write $A = \{a_j \mid j \in \mathbb{N}\}$ and $B = \{b_j \mid j \in \mathbb{N}\}$, where $a_j \neq a_{j'}, b_j \neq b_{j'}$ if $j \neq j'$. Let $F_1 = \{a_1\}$ and $F_1' = \{b_1\}$, where the unique bijection $f_1 : F_1 \to F_1'$ is coordinate-wise order-preserving. Applying the above claim inductively, we can easily construct finite subsets $F_1 \subset F_2 \subset \cdots \subset A$ and $F_1' \subset F_2' \subset \cdots \subset B$ with coordinate-wise order-preserving bijections $f_j : F_j \to F_j'$, $j \in \mathbb{N}$, such that $a_1, \ldots, a_j \in F_j$, $b_1, \ldots, b_j \in F_j'$, and $f_{j+1}|F_j = f_j$ for each $j \in \mathbb{N}$. Since $\bigcup_{j\in\mathbb{N}} F_j = A$, $\bigcup_{j\in\mathbb{N}} F_j' = B$, and $f_{j+1}|F_j = f_j$, we can define $f : A \to B$ by $f|F_j = f_j$ for each $j \in \mathbb{N}$. Then, f is evidently a coordinate-wise order-preserving bijection. □

Corollary 5.8.15 *For a non-empty open set $U \subset \mathbb{R}^n$, if U is not dense in $\mathbb{R}^n$, then* $\dim \mathrm{bd}\, U = n - 1$.

Proof Since $\mathrm{int}\,\mathrm{bd}\, U = \emptyset$, we have $\dim \mathrm{bd}\, U \leqslant n - 1$ by Theorem 5.8.12. Thus, it suffices to show that $\dim \mathrm{bd}\, U \not< n - 1$.

First, consider the case that U is bounded. Assume $\dim \mathrm{bd}\, U = \mathrm{ind}\, \mathrm{bd}\, U < n - 1$. Let $p \in U$. For every $\varepsilon > 0$, $p + \varepsilon U$ is a neighborhood of p in $\mathbb{R}^n$, where $\mathrm{ind}\, \mathrm{bd}(p + \varepsilon U) < n - 1$ because $\mathrm{bd}(p + \varepsilon U) \approx \mathrm{bd}\, U$. By the homogeneity of $\mathbb{R}^n$, every point of $\mathbb{R}^n$ has an arbitrarily small neighborhood whose boundary has $\mathrm{ind} < n - 1$, which means $\dim \mathbb{R}^n = \mathrm{ind}\, \mathbb{R}^n \leqslant n - 1$. This is a contradiction.

In the case when U is unbounded, $\mathrm{int}(\mathbb{R}^n \setminus U) \neq \emptyset$ because U is not dense in $\mathbb{R}^n$. Without loss of generality, we may assume that $\mathbf{0} \in \mathrm{int}(\mathbb{R}^n \setminus U)$. Choose $\delta > 0$ so that $\delta \mathbf{B}^n \subset \mathrm{int}(\mathbb{R}^n \setminus U)$. Then, there exists a homeomorphism $\varphi : \mathbb{R}^n \setminus \{\mathbf{0}\} \to \mathbb{R}^n \setminus \{\mathbf{0}\}$ such that $\varphi(\delta\, \mathrm{int}\, \mathbf{B}^n \setminus \{\mathbf{0}\}) = \mathbb{R}^n \setminus \mathbf{B}^n$ and $\varphi(\mathbb{R}^n \setminus \delta \mathbf{B}^n) = \mathbf{B}^n \setminus \{\mathbf{0}\}$. Indeed, φ can be defined by $\varphi(x) = \|x\|^{-2}\delta x$, where $\varphi^{-1} = \varphi$ (because $\varphi^2 = \mathrm{id}$). Since $\varphi(U) \subset \mathrm{int}\, \mathbf{B}^n \setminus \{\mathbf{0}\}$ is bounded open in $\mathbb{R}^n$, it follows from the bounded case that $\dim \mathrm{bd}_{\mathbb{R}^n} \varphi(U) = n - 1$.

When $\mathbf{0} \notin \mathrm{bd}_{\mathbb{R}^n} \varphi(U)$, $\dim \mathrm{bd}_{\mathbb{R}^n\setminus\{\mathbf{0}\}} \varphi(U) = n - 1$ because $\mathrm{bd}_{\mathbb{R}^n\setminus\{\mathbf{0}\}} \varphi(U) = \mathrm{bd}_{\mathbb{R}^n} \varphi(U)$. When $\mathbf{0} \in \mathrm{bd}_{\mathbb{R}^n} \varphi(U)$, note that $\mathrm{bd}_{\mathbb{R}^n\setminus\{\mathbf{0}\}} \varphi(U) = \mathrm{bd}_{\mathbb{R}^n} \varphi(U) \setminus \{\mathbf{0}\}$ is F_σ in $\mathbb{R}^n$, that is, $\mathrm{bd}_{\mathbb{R}^n\setminus\{\mathbf{0}\}} \varphi(U) = \bigcup_{i\in\mathbb{N}} F_i$, where each F_i is closed in $\mathbb{R}^n$. Since $\mathrm{bd}_{\mathbb{R}^n} \varphi(U) = \{\mathbf{0}\} \cup \bigcup_{i\in\mathbb{N}} F_i$, $\dim F_i \leqslant n - 1$ for every $i \in \mathbb{N}$ and $\dim F_{i_0} = n - 1$ for some $i_0 \in \mathbb{N}$ or $\dim\{\mathbf{0}\} = n - 1$ (i.e., $n = 1$) by the Countable Sum Theorem 5.2.8. If $\dim F_i < n - 1$ for every $i \in \mathbb{N}$, then $n = 1$ and hence $\mathrm{bd}_{\mathbb{R}^n} \varphi(U) = \{\mathbf{0}\}$. This is a contradiction because $\varphi(U) \subset \mathrm{int}\, \mathbf{B}^n \setminus \{\mathbf{0}\}$. Hence, $\dim F_{i_0} = n - 1$ for some $i_0 \in \mathbb{N}$, which implies that $\dim \mathrm{bd}_{\mathbb{R}^n\setminus\{\mathbf{0}\}} \varphi(U) = n - 1$. Thus, we have

$$\dim \mathrm{bd}_{\mathbb{R}^n} U = \dim \mathrm{bd}_{\mathbb{R}^n\setminus\{\mathbf{0}\}} U = \dim \mathrm{bd}_{\mathbb{R}^n\setminus\{\mathbf{0}\}} \varphi(U) = n - 1. \qquad \square$$

5.9 Universal Spaces

Given a class $\mathfrak{C}$ of spaces, a space $Y \in \mathfrak{C}$ is called a **universal space** for $\mathfrak{C}$ if every space $X \in \mathfrak{C}$ can be embedded into Y. The Hilbert cubes $\mathbf{I}^{\mathbb{N}}$ and $\mathbb{R}^{\mathbb{N}}$ are universal spaces for separable metrizable spaces (Corollary 2.3.9) and a countable power $J(\Gamma)^{\mathbb{N}}$ of the hedgehog is the universal space for metrizable spaces with weight $\leqslant \operatorname{card}\Gamma$ (Corollary 2.3.8).[12]

In this section, we show the existence of universal spaces for metrizable spaces with $\dim \leqslant n$, and for countable-dimensional and strongly countable-dimensional metrizable spaces.

First, we will show that the space $\mathbf{I}_f^{\mathbb{N}}$ is also a universal space for strongly countable-dimensional separable metrizable spaces.

Lemma 5.9.1 *Let X be a separable metrizable space and $X_0 \subset X_1$ be closed sets in X with $\dim X_1 \leqslant n$. Then, there exists a map $f : X \to \mathbf{I}^{2n+2}$ such that $X_0 = f^{-1}(\mathbf{0})$ and $f|X_1 \setminus X_0$ is an embedding.*

Proof Applying the Tietze–Urysohn Extension Theorem 2.2.2 coordinate-wise, we can extend an embedding of X_1 into $\mathbf{I}^{2n+1}$ obtained by Theorem 5.8.1 to a map $h : X \to \mathbf{I}^{2n+1}$. Let $g : X \to \mathbf{I}$ be a map with $g^{-1}(0) = X_0$. We define a map $f : X \to \mathbf{I}^{2n+2} = \mathbf{I}^{2n+1} \times \mathbf{I}$ by $f(x) = (g(x)h(x), g(x))$. Then, $f^{-1}(\mathbf{0}) = X_0$. It is easy to prove that $f|X_1 \setminus X_0$ is injective. To see that $f|X_1 \setminus X_0$ is an embedding, let $x, x_i \in X_1 \setminus X_0, i \in \mathbb{N}$, and assume that $f(x) = \lim_{i\to\infty} f(x_i)$. Since $g(x_i) \to g(x)$ and $g(x_i), g(x) > 0$, we have $g(x_i)^{-1} \to g(x)^{-1}$, which implies that $h(x_i) \to h(x)$ in $\mathbf{I}^{2n+1}$, hence $x_i \to x$ in X. Therefore, $f|X_1 \setminus X_0$ is an embedding. □

Theorem 5.9.2 *The space $\mathbf{I}_f^{\mathbb{N}}$ is a universal space for strongly countable-dimensional separable metrizable spaces.*

Proof Let X be an s.c.d. separable metric space. We can write $X = \bigcup_{k\in\mathbb{N}} X_k$, where $X_1 \subsetneqq X_2 \subsetneqq \cdots$ are closed in X and $\dim X_k = n_k < \infty$. By Theorem 5.7.3, X has a metrizable compactification Y such that $\dim \operatorname{cl}_Y X_k = \dim X_k = n_k$. By Lemma 5.9.1, we have maps $f_k : Y \to \mathbf{I}^{2n_k+2}$ $(k \in \mathbb{N})$ such that $f_k^{-1}(\mathbf{0}) = \operatorname{cl}_Y X_{k-1}$ and $f_k|\operatorname{cl}_Y X_k \setminus \operatorname{cl}_Y X_{k-1}$ is an embedding for each $k \in \mathbb{N}$, where $X_0 = \emptyset$. We define a map $f : Y \to \prod_{k\in\mathbb{N}} \mathbf{I}^{2n_k+2} = \mathbf{I}^{\mathbb{N}}$ by $f(x) = (f_k(x))_{k\in\mathbb{N}}$. Then, $f|\bigcup_{k\in\mathbb{N}} \operatorname{cl}_Y X_k$ is injective. By definition, $f(\bigcup_{k\in\mathbb{N}} \operatorname{cl}_Y X_k) \subset \mathbf{I}_f^{\mathbb{N}}$. For $y \in Y$, if $f(y) \in \mathbf{I}_f^{\mathbb{N}}$ then $f_{k+1}(y) = 0$ for some $k \in \mathbb{N}$, which means that $y \in \operatorname{cl}_Y X_k$. Then, it follows that

$$f(A) \cap \mathbf{I}_f^{\mathbb{N}} = f\big(A \cap \textstyle\bigcup_{k\in\mathbb{N}} \operatorname{cl}_Y X_k\big) \quad \text{for each } A \subset Y.$$

Since f is a closed map, the restriction $f|\bigcup_{k\in\mathbb{N}} \operatorname{cl}_Y X_k : \bigcup_{k\in\mathbb{N}} \operatorname{cl}_Y X_k \to \mathbf{I}_f^{\mathbb{N}}$ is also a closed map. Therefore, $f|\bigcup_{k\in\mathbb{N}} \operatorname{cl}_Y X_k$ is an embedding, hence so is $f|X$. □

[12] Usually, the phrase "the class of" is omitted.

For each $n \in \omega$, let

$$\begin{aligned}\nu_n &= \left\{x \in \mathbf{I}^{\mathbb{N}} \;\middle|\; x(i) \in \mathbf{I} \setminus \mathbb{Q} \text{ except for } n\, i\right\}\\ &= \left\{x \in \mathbf{I}^{\mathbb{N}} \;\middle|\; x(i) \in \mathbb{Q} \text{ for at most } n\, i\right\}.\end{aligned}$$

Then, $\nu_0 = (\mathbf{I} \setminus \mathbb{Q})^{\mathbb{N}} \subset \nu_1 \subset \nu_2 \subset \cdots \subset \nu_\omega = \bigcup_{n\in\omega} \nu_n$.[13] Recall that ν_ω is c.d. but not s.c.d. (Theorem 5.6.4). We will show that ν_n is a universal space for separable metrizable spaces with $\dim \leqslant n$ and that ν_ω is the universal space for c.d. separable metrizable spaces. To avoid restricting ourselves to separable spaces, we construct non-separable analogs to ν_n and ν_ω.

Let Γ be an infinite set. Recall that the hedgehog $J(\Gamma)$ is the closed subspace of $\ell_1(\Gamma)$ defined as

$$\begin{aligned}J(\Gamma) &= \left\{x \in \ell_1(\Gamma) \;\middle|\; x(\gamma) \in \mathbf{I} \text{ for all } \gamma \in \Gamma \text{ and } x(\gamma) \neq 0 \text{ for at most one } \gamma \in \Gamma\right\}\\ &= \bigcup_{\gamma\in\Gamma} \langle \mathbf{0}, \mathbf{e}_\gamma\rangle = \bigcup_{\gamma\in\Gamma} \mathbf{I}\mathbf{e}_\gamma \subset \ell_1(\Gamma).\end{aligned}$$

Then, $\dim J(\Gamma) = 1$. Let

$$\begin{aligned}P(\Gamma) &= \left\{x \in J(\Gamma) \;\middle|\; x(\gamma) \in (\mathbf{I} \setminus \mathbb{Q}) \cup \{0\} \text{ for all } \gamma \in \Gamma\right\}\\ &= \{\mathbf{0}\} \cup \bigcup_{\gamma\in\Gamma} (\mathbf{I} \setminus \mathbb{Q})\mathbf{e}_\gamma.\end{aligned}$$

Observe that $P(\Gamma) = \{\mathbf{0}\} \cup \bigcup_{i\in\mathbb{N}} P_i$, where $P_i = P(\Gamma) \setminus \mathrm{B}(\mathbf{0}, 1/i)$. Each P_i is the discrete union of zero-dimensional closed sets in $P(\Gamma)$ that are homeomorphic to $\mathbf{I} \setminus \mathbb{Q}$, hence $\dim P_i = 0$ by the Locally Finite Sum Theorem 5.2.9. Then, $\dim P(\Gamma) = 0$ by the Countable Sum Theorem 5.2.8. Now, we define

$$\nu_\omega(\Gamma) = \left\{z \in J(\Gamma)^{\mathbb{N}} \;\middle|\; z(i) \in P(\Gamma) \text{ except for finitely many } i\right\}.$$

Observe that $\nu_\omega(\Gamma)$ is a countable union of subspaces that are homeomorphic to $J(\Gamma)^n \times P(\Gamma)^{\mathbb{N}}$. Since $\dim J(\Gamma)^n \times P(\Gamma)^{\mathbb{N}} \leqslant n$ (Product Theorems 5.4.15 and 5.4.7) and $J(\Gamma)^n$ contains a copy of $\mathbf{I}^n$, we have $\dim J(\Gamma)^n \times P(\Gamma)^{\mathbb{N}} = n$. Therefore, it follows that $\nu_\omega(\Gamma)$ is c.d. For each $n \in \omega$, we define

$$\nu_n(\Gamma) = \left\{z \in J(\Gamma)^{\mathbb{N}} \;\middle|\; z(i) \in P(\Gamma) \text{ except for } n\, i\right\}.$$

Then, $\nu_0(\Gamma) = P(\Gamma)^{\mathbb{N}} \subset \nu_1(\Gamma) \subset \nu_2(\Gamma) \subset \cdots \subset \nu_\omega(\Gamma) = \bigcup_{n\in\omega} \nu_n(\Gamma)$.

[13] Recall that ν^0 denotes the space $\mathbb{R} \setminus \mathbb{Q}$. Then, $\nu_0 \subsetneqq \nu^0$ but $\nu_0 \approx ((-1, 1) \setminus \mathbb{Q})^{\mathbb{N}} \approx \nu^0$.

Theorem 5.9.3 *For each $n \in \omega$, $\dim \nu_n(\Gamma) = \dim \nu_n = n$.*

Proof We only give a proof of $\dim \nu_n(\Gamma) = n$ because $\dim \nu_n = n$ is similar and simpler.

We already proved that $\dim \nu_0(\Gamma) = \dim P(\Gamma)^{\mathbb{N}} = 0$. Assuming that $\dim \nu_{n-1}(\Gamma) = n - 1 \geqslant 0$, we now prove that $\dim \nu_n(\Gamma) = n$. We can write

$$\nu_n(\Gamma) = \nu_0(\Gamma) \cup \bigcup_{i \in \mathbb{N}} \bigcup_{q \in (0,1] \cap \mathbb{Q}} \bigcup_{\gamma \in \Gamma} \nu_n(i, q, \gamma),$$

where $\nu_n(i, q, \gamma)$ is a closed set in $\nu_n(\Gamma)$ defined as follows:

$$\nu_n(i, q, \gamma) = \{z \in \nu_n(\Gamma) \mid z(i)(\gamma) = q\}.$$

Since $\{\nu_n(i, q, \gamma) \mid \gamma \in \Gamma\}$ is discrete in $\nu_n(\Gamma)$ and $\nu_n(i, q, \gamma) \approx \nu_{n-1}(\Gamma)$, $\bigcup_{\gamma \in \Gamma} \nu_n(i, q, \gamma)$ is an $(n-1)$-dimensional closed set in $\nu_n(\Gamma)$ by the Locally Finite Sum Theorem 5.2.9. Then, $\dim \nu_n(\Gamma) \leqslant n$ by the Countable Sum Theorem 5.2.8 and the Addition Theorem 5.4.14. Since $\nu_n(\Gamma)$ contains an n-dimensional subspace $J(\Gamma)^n \times P(\Gamma)^{\mathbb{N}}$, we have $\dim \nu_n(\Gamma) \geqslant n$, hence $\dim \nu_n(\Gamma) = n$. The result follows by induction. □

We will show that the space $\nu_n(\Gamma)$ is a universal space for metrizable spaces with $\dim \leqslant n$ and weight $\leqslant \operatorname{card} \Gamma$, and that the space $\nu_\omega(\Gamma)$ is a universal space for c.d. metrizable spaces with weight $\leqslant \operatorname{card} \Gamma$.

Lemma 5.9.4 *Let X be a metrizable space and $X_0, X_1, \cdots \subset X$ with $\dim X_n \leqslant 0$. Suppose that $L_0 = \emptyset$, $L_1, \ldots, L_{m-1}$ are closed sets in X satisfying the following condition:*

(∗) *No $x \in X_n$ are contained in $n + 1$ sets L_i.*

Then, for each pair (A, B) of disjoint closed sets in X, there exists a partition L_m in X between A and B that does not violate the condition (∗).

Proof Let $C_0 = X_0$. For $n < m$, define

$$C_n = \bigcup \{X_n \cap \textstyle\bigcap_{j=1}^{n} L_{i_j} \mid 0 \leqslant i_1 < i_2 < \cdots < i_n < m\}.$$

Then, $C_i \cap C_j = \emptyset$ for $i \neq j$ by (∗). Let $D = \bigcup_{i=0}^{m-1} C_i$. For each $n < m - 1$, $\bigcup_{i=n+1}^{m-1} C_i$ is contained in the closed set

$$F = \bigcup \{\textstyle\bigcap_{j=1}^{n+1} L_{i_j} \mid 0 \leqslant i_1 < i_2 < \cdots < i_{n+1} < m\}.$$

Note that $F \cap \bigcup_{i=0}^{n} X_i = \emptyset$ by (∗). For this reason, $\bigcup_{i=0}^{n} C_i = D \setminus F$ is open in D. Therefore, each $C_n = \bigcup_{i=0}^{n} C_i \setminus \bigcup_{i=0}^{n-1} C_i$ is an F_σ-set in D. It follows from the Subset Theorem 5.4.4 and the Countable Sum Theorem 5.2.8 that $\dim D \leqslant 0$.

Using Theorem 5.3.9 and the Partition Extension Lemma 5.4.8, we obtain a partition L_m between A and B such that $L_m \cap D = \emptyset$. Condition $(*)$ is trivial for $n \geqslant m$. For $n < m$, if $x \in X_n$ is contained in n sets L_i $(i < m)$, then $x \in C_n \subset D$, which implies $x \notin L_m$. Therefore, condition $(*)$ is satisfied. □

Lemma 5.9.5 *Let X be a metrizable space and $X_0, X_1, \cdots \subset X$ with $\dim X_n \leqslant 0$ and let $a < b \in \mathbb{R}$. Then, for any sequence $(A_i, B_i)_{i\in\mathbb{N}}$ of pairs of disjoint closed sets in X, there exist maps $f_i : X \to [a, b]$, $i \in \mathbb{N}$, such that $A_i = f_i^{-1}(a)$, $B_i = f_i^{-1}(b)$, and*

$$\operatorname{card}\left\{i \in \mathbb{N} \;\middle|\; f_i(x) \in (a, b) \cap \mathbb{Q}\right\} \leqslant n \;\text{ for } x \in X_n.$$

Proof Let $\{q_j \mid j \in \mathbb{N}\} = (a, b) \cap \mathbb{Q}$, where $q_i \neq q_j$ if $i \neq j$. For each $j \in \mathbb{N}$, let

$$\delta_j = \min\left\{q_i - a, b - q_i, |q_i - q_{i'}| \;\middle|\; i, i' \leqslant j,\ i \neq i'\right\},$$

where it should be noted that $\delta_j \geqslant \delta_{j+1}$. We also define $a_j = q_j - 2^{-j-1}\delta_j$ and $b_j = q_j + 2^{-j-1}\delta_j$. For each $i \in \mathbb{N}$, let $f_{i,0} : X \to [a, b]$ be a map with $A_i = f_{i,0}^{-1}(a)$ and $B_i = f_{i,0}^{-1}(b)$. We construct maps $f_{i,j} : X \to [a, b]$, $i, j \in \mathbb{N}$, so as to satisfy the following conditions:

(1) $A_i = f_{i,j}^{-1}(a)$ and $B_i = f_{i,j}^{-1}(b)$;
(2) $f_{i,j}(x) \neq f_{i,j-1}(x) \Rightarrow f_{i,j-1}(x), f_{i,j}(x) \in (a_j, b_j)$
$\big($i.e., $f_{i,j}|f_{i,j-1}^{-1}([a, a_j] \cup [b_j, b]) = f_{i,j-1}|f_{i,j-1}^{-1}([a, a_j] \cup [b_j, b])$
and $f_{i,j}|f_{i,j-1}^{-1}((a_j, b_j)) \subset (a_j, b_j)\big)$;
(3) No $x \in X_n$ are contained in $n + 1$ sets $f_{i,j}^{-1}(q_j)$.

For each $(i, j) \in \mathbb{N}^2$, let

$$k(i, j) = \frac{1}{2}(i + j - 2)(i + j - 1) + j \in \mathbb{N}.$$

Then, (i, j) is the $k(i, j)$-th element of $\mathbb{N}^2$ in the ordering

$$(1, 1), (2, 1), (1, 2), (3, 1), (2, 2), (1, 3), \ldots.$$

By induction on $k(i, j)$, we construct maps $f_{i,j}$ satisfying conditions (1), (2), and (3) above. Assume that $f_{i',j'}$ have been defined for $k(i', j') < m$. We will define $f_{i,j}$ for $k(i, j) = m$. Applying Lemma 5.9.4 to $L_0 = \emptyset$, $L_{k(i',j')} = f_{i',j'}^{-1}(q_{j'})$, $k(i', j') < m$, $A = f_{i,j-1}^{-1}([a, a_j])$, and $B = f_{i,j-1}^{-1}([b_j, b])$, we obtain a partition L_m in X between A and B such that

$(*)$ No $x \in X_n$ are contained in $n + 1$ sets L_i.

Then, we can easily obtain a map $f_{i,j} : X \to [a, b]$ such that $L_m = f_{i,j}^{-1}(q_j)$, $f_{i,j}|A \cup B = f_{i,j-1}|A \cup B$, and $f_{i,j}(X \setminus (A \cup B)) \subset (a_j, b_j)$ for which conditions (1), (2), and (3) are satisfied.

Since $b_j - a_j = 2^{-j}\delta_j$, it follows from (2) that $|f_{i,j}(x) - f_{i,j-1}(x)| < 2^{-j}\delta_j$ for each $x \in X$. Then, $(f_{i,j})_{j\in\mathbb{N}}$ uniformly converges to a map $f_i : X \to [a, b]$ and $|f_{i,j}(x) - f_i(x)| \leqslant 2^{-j}\delta_j$. For each $x \in A_i$, $f_i(x) = \lim_{j\to\infty} f_{i,j}(x) = a$ by (1). For each $x \in X \setminus A_i$, we have $k = \min\{j \in \mathbb{N} \mid f_{i,0}(x) > a_j\}$ because $f_{i,0}(x) > a = \inf_{j\in\mathbb{N}} a_j$. Then, $f_{i,0}(x) = f_{i,1}(x) = \cdots = f_{i,k-1}(x) > a_k$ and $f_{i,k}(x) > a_k = q_k - 2^{-k-1}\delta_k$, hence

$$f_i(x) \geqslant f_{i,k}(x) - 2^{-k}\delta_k > q_k - \delta_k \geqslant q_k - (q_k - a) = a.$$

Therefore, $A_i = f_i^{-1}(a)$. Similarly, we have $B_i = f_i^{-1}(b)$.

For each $x \in X_n$, let

$$M = \left\{i \in \mathbb{N} \;\middle|\; f_{i,j}(x) = q_j \text{ for some } j \in \mathbb{N}\right\}.$$

Then, M has at most n elements by (3). For $i \in \mathbb{N} \setminus M$ and $j \in \mathbb{N}$, let $K = \{k > j \mid f_{i,k}(x) \neq f_{i,j}(x)\}$. If $K = \emptyset$, then $f_i(x) = f_{i,j}(x) \neq q_j$ because $i \notin M$. Otherwise, let $k = \min K > j \geqslant 1$. Since $f_{i,k-1}(x) = f_{i,j}(x) \neq f_{i,k}(x)$, we have $a_k < f_{i,k}(x) < b_k$ by (2). Then, $|f_{i,k}(x) - q_j| \geqslant \delta_k - 2^{-k-1}\delta_k$. On the other hand, $|f_i(x) - f_{i,k}(x)| < 2^{-k+1}\delta_k$. Therefore,

$$\begin{aligned} |f_i(x) - q_j| &\geqslant |f_{i,k}(x) - q_j| - |f_i(x) - f_{i,k}(x)| \\ &> \delta_k - 2^{-k-1}\delta_k - 2^{-k+1}\delta_k > \delta_k/4 > 0. \end{aligned}$$

Thus, $\operatorname{card}\{i \mid f_i(x) \in (a, b) \cap \mathbb{Q}\} \leqslant n$ for $x \in X_n$. □

Proposition 5.9.6 *Let X be a metrizable space and Γ be an infinite set with $w(X) \leqslant \operatorname{card}\Gamma$. For each sequence $X_0, X_1, \cdots \subset X$ with $\dim X_n \leqslant 0$, there exists an embedding $h : X \to J(\Gamma)^{\mathbb{N}}$ such that $h(X_n) \subset \nu_n(\Gamma)$.*

Proof By Corollary 2.3.2, X has an open basis $\mathcal{B} = \bigcup_{i\in\mathbb{N}} \mathcal{B}_i$, where each $\mathcal{B}_i$ is discrete in X. Then, as is easily observed, $\operatorname{card}\mathcal{B}_i \leqslant w(X) \leqslant \operatorname{card}\Gamma$, hence we have $\Gamma_i \subset \Gamma, i \in \mathbb{N}$, such that $\operatorname{card}\mathcal{B}_i = \operatorname{card}\Gamma_i$ and $\Gamma_i \cap \Gamma_j = \emptyset$ if $i \neq j$. For each $i \in \mathbb{N}$, we write $\mathcal{B}_i = \{B_\gamma \mid \gamma \in \Gamma_i\}$, where $B_\gamma \neq B_{\gamma'}$ if $\gamma \neq \gamma'$. We apply Lemma 5.9.5 to obtain maps $f_i : X \to [0, 1], i \in \mathbb{N}$, such that $f_i^{-1}(0) = X \setminus \bigcup_{\gamma\in\Gamma_i} B_\gamma$, $f_i^{-1}(1) = \emptyset$, and $\operatorname{card}\{i \in \mathbb{N} \mid f_i(x) \in (0, 1) \cap \mathbb{Q}\} \leqslant n$ for $x \in X_n$. We define $h_i : X \to J(\Gamma)$ by

$$h_i(x)(\gamma) = \begin{cases} f_i(x) & \text{if } x \in B_\gamma,\ \gamma \in \Gamma_i, \\ 0 & \text{otherwise.} \end{cases}$$

In other words, $h_i(x) = f_i(x)\mathbf{e}_\gamma$ for $x \in B_\gamma, \gamma \in \Gamma_i$, and $h_i(x) = \mathbf{0}$ for $x \in f_i^{-1}(0)$. The desired embedding $h : X \to J(\Gamma)^{\mathbb{N}}$ can be defined by $h(x) = (h_i(x))_{i\in\mathbb{N}}$. Indeed, if $x \neq y \in X$, then $x \in B_\gamma$ and $y \notin B_\gamma$ for some $\gamma \in \Gamma_i$. Then, $h_i(x)(\gamma) = f_i(x) > 0 = h_i(y)(\gamma)$. Thus, h is injective. For each $\gamma \in \Gamma_i$, $U_\gamma = \{z \in J(\Gamma)^{\mathbb{N}} \mid z(i)(\gamma) > 0\}$ is open in $J(\Gamma)^{\mathbb{N}}$. Observe that $h(B_\gamma) = U_\gamma \cap h(X)$. Therefore, h is an embedding of X into $J(\Gamma)^{\mathbb{N}}$. For $x \in X_n$,

$$\operatorname{card}\{i \in \mathbb{N} \mid h_i(x) \notin P(\Gamma)\} = \operatorname{card}\{i \in \mathbb{N} \mid f_i(x) \in \mathbb{Q} \setminus \{0\}\} \leqslant n.$$

Then, it follows that $h(X_n) \subset \nu_n(\Gamma)$. □

Theorem 5.9.7 *Let Γ be an infinite set. The space $\nu_n(\Gamma)$ is a universal space for metrizable spaces X with $w(X) \leqslant \operatorname{card}\Gamma$ and $\dim X \leqslant n$, and the space $\nu_\omega(\Gamma)$ is a universal space for countable-dimensional metrizable spaces X with $w(X) \leqslant \operatorname{card}\Gamma$.*

Proof We can write $X = \bigcup_{i\in\omega} X_i$, where $\dim X_i \leqslant 0$ and $X_i = \emptyset$ for $i > n$ if $\dim X = n$. The theorem follows from Proposition 5.9.6. □

In the proof of Proposition 5.9.6, when X is a separable metrizable space, we can take a $\mathcal{B}_i$ with only one element. Then, replacing $\mathbf{I}$ with $[a, 1]$ where $a \in \mathbf{I} \setminus \mathbb{Q}$, the map $h : X \to [a, 1]^{\mathbb{N}} \subset \mathbf{I}^{\mathbb{N}}$ defined by $h(x) = (f_i(x))_{i\in\mathbb{N}}$ is an embedding such that $h(X_n) \subset \nu_n$. Similar to Theorem 5.9.7, we have the following separable version:

Theorem 5.9.8 *The space ν_n is a universal space for separable metrizable spaces with $\dim \leqslant n$ and the space ν_ω is a universal space for countable-dimensional separable metrizable spaces.* □

Next, recall that $\mathbf{I}_f^{\mathbb{N}}$ is s.c.d. We now define

$$K_\omega = \bigcup_{n\in\mathbb{N}} \nu_n \times \left((0, 1]^n \times \{\mathbf{0}\}\right) \subset \nu_\omega \times \mathbf{I}_f^{\mathbb{N}}.$$

For an infinite set Γ, we define

$$K_\omega(\Gamma) = \bigcup_{n\in\mathbb{N}} \nu_n(\Gamma) \times \left((0, 1]^n \times \{\mathbf{0}\}\right) \subset \nu_\omega(\Gamma) \times \mathbf{I}_f^{\mathbb{N}} \subset J(\Gamma)^{\mathbb{N}} \times \mathbf{I}^{\mathbb{N}}.$$

Then, K_ω is separable and $w(K_\omega(\Gamma)) = \operatorname{card}\Gamma$.[14] Moreover, $K_\omega(\Gamma)$ and K_ω are s.c.d. Indeed, for $(x, y) \in K_\omega(\Gamma)$,

$$(x, y) \in \bigcup_{i=1}^{n} \nu_i(\Gamma) \times \left((0, 1]^i \times \{\mathbf{0}\}\right) \Leftrightarrow y(n + 1) = 0.$$

Hence, $\bigcup_{i=1}^{n} \nu_i(\Gamma) \times ((0, 1]^i \times \{\mathbf{0}\})$ is a closed set in $K_\omega(\Gamma)$, which is finite-dimensional by the Product Theorem 5.4.15 and the Addition Theorem 5.4.14.

Theorem 5.9.9 *Let Γ be an infinite set. The space $K_\omega(\Gamma)$ is a universal space for strongly countable-dimensional metrizable spaces X with $w(X) \leqslant \operatorname{card}\Gamma$.*

---

[14] Our notation is different from the usual one. In the literature for Dimension Theory, our K_ω is represented by $K_\omega(\aleph_0)$ and K_ω stands for $\mathbf{I}_f^{\mathbb{N}}$.

Proof We can write $X = \bigcup_{i\in\mathbb{N}} F_i$, where each F_i is closed in X, $\dim F_i \leqslant i-1$, and $F_i \subset F_{i+1}$ for each $i \in \mathbb{N}$. By the Decomposition Theorem 5.4.11, we have a sequence $X_1, X_2, \cdots \subset X$ such that $\dim X_n \leqslant 0$ and

$$F_1 = X_1,\ F_2 \setminus F_1 = X_2 \cup X_3,\ F_3 \setminus F_2 = X_4 \cup X_5 \cup X_6,\ \ldots,$$

i.e., $F_i \setminus F_{i-1} = \bigcup_{n=k(i-1)+1}^{k(i)} X_n$, where $F_0 = \emptyset$ and $k(i) = \frac{1}{2}i(i+1)$. We apply Proposition 5.9.6 to obtain an embedding $h : X \to J(\Gamma)^{\mathbb{N}}$ such that $h(X_n) \subset \nu_n(\Gamma)$ for each $n \in \mathbb{N}$. For each $i \in \mathbb{N}$, let $f_i : X \to \mathbf{I}$ be a map with $f_i^{-1}(0) = F_{i-1}$, and define a map $f : X \to \mathbf{I}^{\mathbb{N}}$ as follows:

$$f(x) = (f_1(x), f_2(x), f_2(x), f_3(x), f_3(x), f_3(x), \ldots),$$

where each $f_i(x)$ appears i times, i.e., $\mathrm{pr}_n f = f_i$ for $k(i-1)+1 \leqslant n \leqslant k(i)$. Now, we can define the embedding $g : X \to J(\Gamma)^{\mathbb{N}} \times \mathbf{I}^{\mathbb{N}}$ by $g(x) = (h(x), f(x))$. For each $x \in X$, choose $i \in \mathbb{N}$ and $k(i-1)+1 \leqslant n \leqslant k(i)$ so that $x \in X_n \subset F_i \setminus F_{i-1}$. Then, $h(x) \in h(X_n) \subset \nu_n(\Gamma) \subset \nu_{k(i)}(\Gamma)$. Since $x \in F_i \setminus F_{i-1}$, it follows that $f_j(x) > 0$ for $j \leqslant i$ and $f_j(x) = 0$ for $j \geqslant i+1$, i.e., $\mathrm{pr}_j f(x) > 0$ for $j \leqslant k(i)$ and $\mathrm{pr}_j f(x) = 0$ for $j \geqslant k(i)+1$. Therefore, $f(x) \in (0,1]^{k(i)} \times \{\mathbf{0}\} \subset \mathbf{I}^{\mathbb{N}}$. Thus, we have

$$g(x) = (h(x), f(x)) \in \nu_{k(i)}(\Gamma) \times \left((0,1]^{k(i)} \times \{\mathbf{0}\}\right) \subset K_\omega(\Gamma).$$

Consequently, X can be embedded into $K_\omega(\Gamma)$. □

Similarly, we can obtain the following separable version:

Theorem 5.9.10 *The space K_ω is a universal space for strongly countable-dimensional separable metrizable spaces.* □

5.10 Nöbeling Spaces and Menger Compacta

In this section, we shall construct two universal spaces for separable metrizable spaces with $\dim \leqslant n$, which are named the n-dimensional Nöbeling space and the n-dimensional Menger compactum.

In the previous section, we defined the universal space ν_n. In the definition of ν_n, we replace $\mathbf{I}^{\mathbb{N}}$ with $\mathbb{R}^{2n+1}$ to define the ***n*-dimensional Nöbeling space** ν^n, that is,

$$\begin{aligned}\nu^n &= \left\{x \in \mathbb{R}^{2n+1} \;\middle|\; x(i) \in \mathbb{R} \setminus \mathbb{Q} \text{ for at least } n+1\, i\right\}\\ &= \left\{x \in \mathbb{R}^{2n+1} \;\middle|\; x(i) \in \mathbb{Q} \text{ for at most } n\, i\right\},\end{aligned}$$

which is the n-dimensional version of the space of irrationals $\nu^0 = \mathbb{R} \setminus \mathbb{Q}$.[15] Note that ν^n, $n \in \omega$, are different from ν_n defined in Sect. 5.9. In the proof of Theorem 5.8.12, we defined the following zero-dimensional subsets of $\mathbb{R}^n$:

$$\mathbb{Q}_k^n = \left\{ x \in \mathbb{R}^n \;\middle|\; x(i) \in \mathbb{Q} \text{ for exactly } k\, i \right\},\ k = 0, 1, \ldots, n-1.$$

Since $\nu^n = \mathbb{Q}_0^{2n+1} \cup \cdots \cup \mathbb{Q}_n^{2n+1}$, we have $\dim \nu^n = n$ by the Decomposition Theorem 5.4.11. Observe that

$$\mathbb{R}^{2n+1} \setminus \nu^n = \left\{ x \in \mathbb{R}^{2n+1} \;\middle|\; x(i) \in \mathbb{Q} \text{ for at least } n+1\, i \right\},$$

which is a countable union of n-dimensional flats that are closed in $\mathbb{R}^{2n+1}$. Then, ν^n is a G_δ-set in $\mathbb{R}^{2n+1}$, hence it is completely metrizable. Thus, we have the following proposition:

Proposition 5.10.1 *The space ν^n is a separable completely metrizable space with* $\dim \nu^n = n$. □

Moreover, ν^n has the following additional property:

Proposition 5.10.2 *Each point of ν^n has an arbitrarily small neighborhood that is homeomorphic to ν^n. In fact, $\nu^n \cap \prod_{i=1}^{2n+1}(a_i, b_i) \approx \nu^n$ for each $a_i < b_i \in \mathbb{Q}$, $i = 1, \ldots, 2n+1$.*

Proof Let $\varphi : \mathbb{R} \to (-1, 1)$ be the homeomorphism defined by

$$\varphi(t) = \frac{t}{1+|t|}, \text{ where } \varphi^{-1}(s) = \frac{s}{1-|s|}.$$

We define a homeomorphism $h : \mathbb{R}^{2n+1} \to \prod_{i=1}^{2n+1}(a_i, b_i)$ as follows:

$$h(x) = (h_1(x(1)), \ldots, h_{2n+1}(x(2n+1))),$$

where $h_i : \mathbb{R} \to (a_i, b_i)$ is the homeomorphism defined by

$$h_i(t) = \frac{b_i - a_i}{2}(\varphi(t) + 1) + a_i.$$

Since $h_i(\mathbb{Q}) = \mathbb{Q} \cap (a_i, b_i)$, we have $h(\nu^n) = \nu^n \cap \prod_{i=1}^{2n+1}(a_i, b_i)$. □

We will show the universality of ν^n.

Theorem 5.10.3 *The n-dimensional Nöbeling space ν^n is a universal space for separable metrizable spaces with* $\dim \leqslant n$.

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[15] As the definition of ν_n in Sect. 5.9, the first condition of the definition of ν^n can be expressed as "$x(i) \in \mathbb{R} \setminus \mathbb{Q}$ except for $n\, i$."

According to the Dimension-Preserving Compactification Theorem 5.7.2, every n-dimensional separable metrizable space X has an n-dimensional metrizable compactification. Theorem 5.10.3 comes from the following proposition:

Proposition 5.10.4 *For each locally compact separable metrizable space X with $\dim X \leqslant n$ and $\mathcal{U} \in \mathrm{cov}(\mathbb{R}^{2n+1})$, every proper map $f : X \to \mathbb{R}^{2n+1}$ is $\mathcal{U}$-close to a closed embedding $g : X \to \mathbb{R}^{2n+1}$ with $g(X) \subset \nu^n$.*

This can be shown by a modification of the proof of the Embedding Approximation Theorem 5.8.9 (or 5.8.4). To this end, we need the following generalization of Theorem 5.8.8:

Lemma 5.10.5 *Let X and Y be locally compact metrizable spaces and $Y_0 = \bigcap_{n\in\mathbb{N}} G_n \subset Y$, where each G_n is open in Y (hence Y_0 is a G_δ-set in Y). Suppose that for each $n \in \mathbb{N}$ and each open cover $\mathcal{U}$ of X consisting of open sets with the compact closure, $\mathrm{C}_{\mathcal{U}}(X, Y) \cap \mathrm{C}(X, G_n)$ is dense in the space $\mathrm{C}^P(X, Y)$ with the limitation topology.*[16] *Then, $\mathrm{Emb}(X, Y) \cap \mathrm{C}(X, Y_0)$ is dense in $\mathrm{C}^P(X, Y)$. In particular, $\mathrm{Emb}(X, Y)$ is dense in $\mathrm{C}^P(X, Y)$.*

Proof Let d be a complete admissible metric for X and take $\mathcal{U}_n \in \mathrm{cov}(X)$, $n \in \mathbb{N}$, so that $\mathrm{mesh}_d\, \mathcal{U}_n < 2^{-n}$ and $\mathrm{cl}\, U$ is compact for each $U \in \mathcal{U}$. By Lemma 5.8.6, we have

$$\begin{aligned}\mathrm{Emb}(X, Y) \cap \mathrm{C}(X, Y_0) &= \bigcap_{n\in\mathbb{N}} \mathrm{C}_{\mathcal{U}_n}(X, Y) \cap \bigcap_{n\in\mathbb{N}} \mathrm{C}(X, G_n)\\ &= \bigcap_{n\in\mathbb{N}} \big(\mathrm{C}_{\mathcal{U}_n}(X, Y) \cap \mathrm{C}(X, G_n)\big).\end{aligned}$$

By the assumption, each $\mathrm{C}_{\mathcal{U}_n}(X, Y) \cap \mathrm{C}(X, G_n)$ is dense in $\mathrm{C}^P(X, Y)$. Moreover, it is open in $\mathrm{C}^P(X, Y)$ because $\mathrm{C}_{\mathcal{U}_n}(X, Y)$ is open in $\mathrm{C}^P(X, Y)$ by Lemma 5.8.5 and $f(X)$ is closed in Y for each $f \in \mathrm{C}_{\mathcal{U}_n}(X, Y) \subset \mathrm{C}^P(X, Y)$ (Proposition 2.1.12). Since $\mathrm{C}^P(X, Y)$ is a Baire space by Theorem 2.11.7, we have the desired result. □

Proof (*Proposition* 5.10.4) According to the definition of ν^n, we can write

$$\nu^n = \mathbb{R}^{2n+1} \setminus \bigcup_{i\in\mathbb{N}} H_i = \bigcap_{i\in\mathbb{N}} (\mathbb{R}^{2n+1} \setminus H_i),$$

where each H_i is an n-dimensional flat.

In fact, each H_i can be written as $p^{-1}_{i_1,\dots,i_{n+1}}(y)$ for some $1 \leqslant i_1 < \cdots < i_{n+1} \leqslant 2n+1$ and $y \in \mathbb{Q}^{n+1}$, where $p_{i_1,\dots,i_{n+1}} : \mathbb{R}^{2n+1} \to \mathbb{R}^{n+1}$ is the projection defined by $p_{i_1,\dots,i_{n+1}}(x) = (x(i_1), \dots, x(i_{n+1}))$.

[16] Just before Theorem 5.8.8, we observed that $\mathrm{C}_U(X, Y) \subset \mathrm{C}^P(X, Y)$.

Because of Lemma 5.10.5, it suffices to show that, for each n-dimensional flat H in $\mathbb{R}^{2n+1}$ and each $\mathcal{U} \in \mathrm{cov}(X)$ consisting of open sets with the compact closure, $\mathrm{C}_{\mathcal{U}}(X, \mathbb{R}^{2n+1}) \cap \mathrm{C}(X, \mathbb{R}^{2n+1} \setminus H)$ is dense in the space $\mathrm{C}^P(X, \mathbb{R}^{2n+1})$ with the limitation topology.

In the proof of Theorem 5.8.9, we can choose $v_i \in \mathbb{R}^{2n+1}$, $i \in \mathbb{N}$, to satisfy the additional condition that the flat hull of every $n+1$ points $v_{i_1}, \ldots, v_{i_{n+1}}$ misses H (i.e., $\mathrm{fl}\{v_{i_1}, \ldots, v_{i_{n+1}}\} \cap H = \emptyset$). Thus, we can obtain the PL embedding $g : |K| \to \mathbb{R}^{2n+1}$ such that $g(|K|) \cap H = \emptyset$. The map $g\varphi : X \to \mathbb{R}^{2n+1} \setminus H$ belongs to $\mathrm{C}_{\mathcal{U}}(X, \mathbb{R}^{2n+1}) \cap \mathrm{C}(X, \mathbb{R}^{2n+1} \setminus H)$ and is $\mathcal{V}$-close to f. □

Remark 5.10.1 It is known that if X is a separable completely metrizable space with $\dim X \leqslant n$, then every map $f : X \to \nu^n$ can be approximated by closed embeddings $h : X \to \nu^n$. Refer to Remark 5.8.2.

Before defining the n-dimensional Menger compactum, let us recall the construction of the Cantor (ternary) set μ^0. We can geometrically describe $\mu^0 \subset \mathbf{I}$ as follows: For each $k \in \mathbb{N}$, let

$$R_k = \bigcup_{m=0}^{3^{k-1}-1} \left(\frac{m}{3^{k-1}} + \frac{1}{3^k}, \frac{m}{3^{k-1}} + \frac{2}{3^k} \right) \subset \mathbf{I}.$$

Then, $\mu^0 = \bigcap_{k\in\mathbb{N}} (\mathbf{I} \setminus R_k) = \mathbf{I} \setminus \bigcup_{k\in\mathbb{N}} R_k$ (cf. Fig. 5.4). Observe that

$$\bigcap_{i=1}^{k} (\mathbf{I} \setminus R_i) = [0, 3^{-k}] + V_k^0, \quad \text{where} \quad V_k^0 = \left\{ \sum_{i=1}^{k} \frac{2x(i)}{3^i} \;\middle|\; x \in \mathbf{2}^k \right\}.$$

Moreover, $\{3^{-k}\mu^0 + v \mid v \in V_k^0\}$ is an open cover of μ^0 with $\mathrm{ord} = 1$, where

$$\mu^0 \approx 3^{-k}\mu^0 + v = \mu^0 \cap ([0, 3^{-k}] + v) = \mu^0 \cap ((-3^{-k-1}, 3^{-k} + 3^{-k-1}) + v).$$

As the n-dimensional version of μ^0, the ***n*-dimensional Menger compactum** μ^n is defined as follows: For each $k \in \mathbb{N}$, let

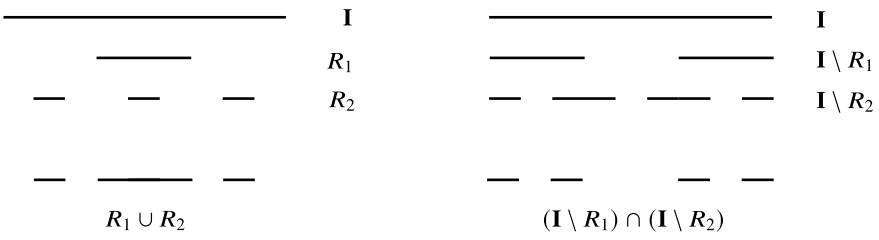


Fig. 5.4 R_k and $\mathbf{I} \setminus R_k$

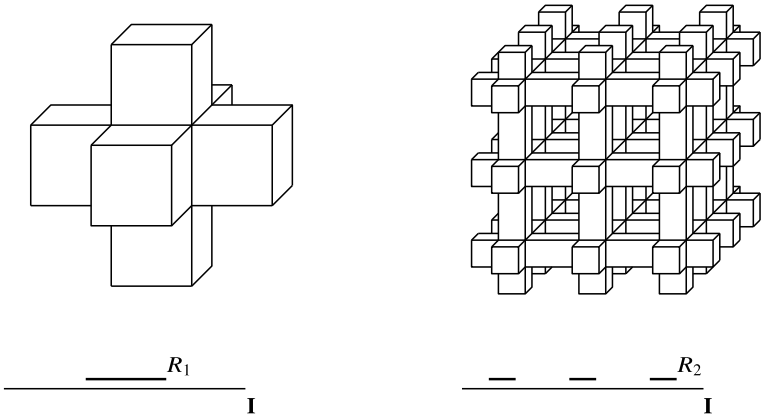


Fig. 5.5 $\mathbf{I}^3 \setminus M_1^3$ and $\mathbf{I}^3 \setminus M_2^3$

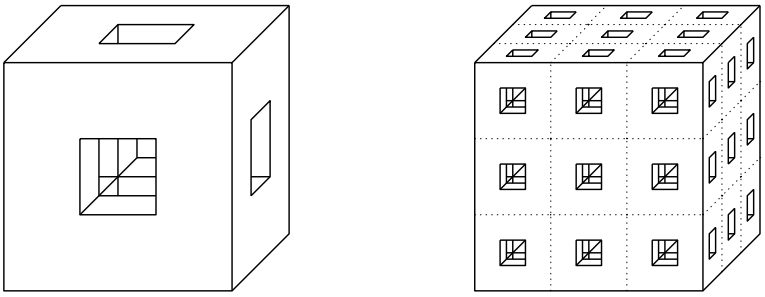


Fig. 5.6 M_1^3 and M_2^3

$$\begin{aligned} M_k^{2n+1} &= \left\{x \in \mathbf{I}^{2n+1} \;\middle|\; x(i) \in \mathbf{I} \setminus R_k \;\text{ for at least } n+1\, i\right\} \\ &= \left\{x \in \mathbf{I}^{2n+1} \;\middle|\; x(i) \in R_k \;\text{ for at most } n\, i\right\}, \end{aligned}$$

where it should be noted that

$$\begin{aligned} \mathbf{I}^{2n+1} \setminus M_k^{2n+1} &= \left\{x \in \mathbf{I}^{2n+1} \;\middle|\; x(i) \in \mathbf{I} \setminus R_k \;\text{ for at most } n\, i\right\} \\ &= \left\{x \in \mathbf{I}^{2n+1} \;\middle|\; x(i) \in R_k \;\text{ for at least } n+1\, i\right\}. \end{aligned}$$

Now, we define $\mu^n = \bigcap_{k\in\mathbb{N}} M_k^{2n+1}$. Since each M_k^{2n+1} is compact, μ^n is also compact. See Figs. 5.5, 5.6, and 5.7.

In the rest of this section, we use the metric $d \in \mathrm{Metr}(\mathbf{I}^{2n+1})$ defined by

$$d(x, y) = \max\left\{|x(i) - y(i)| \;\middle|\; i = 1, \ldots, 2n+1\right\}.$$

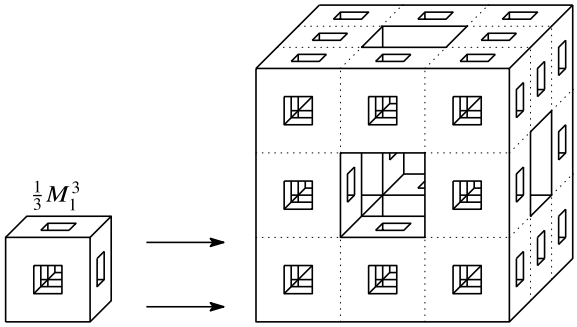


Fig. 5.7 $M_1^3 \cap M_2^3$

Proposition 5.10.6 *For each* $n \in \mathbb{N}$, $\dim \mu^n = n$.

Proof Since μ^n contains every n-face of $\mathbf{I}^{2n+1}$, it follows that $\dim \mu^n \geqslant n$. We will apply Proposition 5.8.11 to see $\dim \mu^n \leqslant n$. For each $\varepsilon > 0$, choose $k \in \mathbb{N}$ so large that $2/3^{k-1} < \varepsilon$. Let K be the cell complex consisting of all faces of $(2n+1)$-cubes

$$\prod_{i=1}^{2n+1} \left[\frac{m_i - 1}{3^{k-1}}, \frac{m_i}{3^{k-1}} \right] \subset \mathbf{I}^{2n+1}, \ m_i = 1, \ldots, 3^{k-1}.$$

Since $\mu^n \subset M_k^{2n+1}$, it suffices to construct an ε-map of M_k^{2n+1} to $|K^{(n)}|$, where $K^{(n)}$ is the n-skeleton of K.

For each $C \in K$ with $\dim C > n$, let $r_C : C \setminus \{\hat{C}\} \to \partial C$ be the radial retraction, where $\hat{C}$ is the barycenter of C and ∂C is the radial boundary of C. Observe that

$$M_k^{2n+1} \cap C \subset C \setminus \{\hat{C}\} \ \text{ and } \ r_C(M_k^{2n+1} \cap C) \subset M_k^{2n+1} \cap \partial C.$$

For each $m = n, n+1, \ldots, 2n$, we can define a retraction

$$r_m : M_k^{2n+1} \cap |K^{(m+1)}| \to M_k^{2n+1} \cap |K^{(m)}|$$

by $r_m | M_k^{2n+1} \cap C = r_C | M_k^{2n+1} \cap C$ for each $(m+1)$-cell $C \in K$. Since $|K^{(n)}| \subset M_k^{2n+1}$, we have a retraction

$$r = r_n \cdots r_{2n} : M_k^{2n+1} \to |K^{(n)}|.$$

By construction, $r^{-1}(x) \subset \mathrm{st}(x, K)$ for each $x \in |K^{(n)}|$. Since $\mathrm{mesh}\, K = 1/3^{k-1} < \varepsilon/2$, it follows that r is an ε-map. □

For each $k \in \mathbb{N}$, $\mu^n \approx 3^{-k}\mu^n \subset [0, 3^{-k}]^{2n+1}$. Let

$$V_k^n = \left\{ v \in 3^{-k}\mathbb{Z}^{2n+1} \;\middle|\; [0, 3^{-k}]^{2n+1} + v \subset \textstyle\bigcap_{j=1}^{k} M_j^{2n+1} \right\}.$$

Then, $\bigcap_{j=1}^{k} M_j^{2n+1} = \bigcup_{v \in V_k^n} ([0, 3^{-k}]^{2n+1} + v)$ and $\mu^n = 3^{-k}\mu^n + V_k^n$. Thus, the Menger compactum μ^n has the following property:

Proposition 5.10.7 *Every neighborhood of each point of μ^n contains a small copy of μ^n.* □

Remark 5.10.2 The Menger compactum μ^n has a property called the *self-similarity*.[17] In fact, for each $v \in V_1^n$, let $\varphi_v : \mu^n \to \mu^n$ be a 3^{-1}-Lipschitz embedding defined by $\varphi_v(x) = 3^{-1}x + v$, where $\varphi_v(\mu^n) = 3^{-1}\mu^n + v$. Then, $\mu^n = \bigcup_{v \in V_1^n} \varphi_v(\mu^n)$.

We will derive the universality of μ^n from that of $\nu^n \approx \nu^n \cap (0, 1)^{2n+1}$. To this end, we need some more notations. For each $k \in \mathbb{N}$, let

$$Q_k = \left\{ m/3^{k-1} + 1/3^{k-1}2 \;\middle|\; m = 0, \ldots, 3^{k-1} - 1 \right\},$$

which are the mid-points of components of R_k. We define

$$\begin{aligned} L_k &= \left\{ x \in \mathbf{I}^{2n+1} \;\middle|\; x(i) \in Q_k \;\text{ for at least } n+1\, i \right\} \\ &= \left\{ x \in \mathbf{I}^{2n+1} \;\middle|\; x(i) \in \mathbf{I} \setminus Q_k \;\text{ for at most } n\, i \right\}. \end{aligned}$$

For each $0 < s \leqslant 1/3^k 2$, we also define $R_k(s) = \bigcup_{q \in Q_k} (q - s, q + s) \subset \mathbf{I}$. Then,

$$\begin{aligned} \mathrm{N}_d(L_k, s) &= \left\{ x \in \mathbf{I}^{2n+1} \;\middle|\; x(i) \in R_k(s) \;\text{ for at least } n+1\, i \right\} \\ &= \left\{ x \in \mathbf{I}^{2n+1} \;\middle|\; x(i) \in \mathbf{I} \setminus R_k(s) \;\text{ for at most } n\, i \right\}; \\ \mathbf{I}^{2n+1} \setminus \mathrm{N}_d(L_k, s) &= \left\{ x \in \mathbf{I}^{2n+1} \;\middle|\; x(i) \in \mathbf{I} \setminus R_k(s) \;\text{ for at least } n+1\, i \right\} \\ &= \left\{ x \in \mathbf{I}^{2n+1} \;\middle|\; x(i) \in R_k(s) \;\text{ for at most } n\, i \right\}. \end{aligned}$$

Since $R_k(1/3^k 2) = R_k$, it follows that

$$\mathbf{I}^{2n+1} \setminus M_k^{2n+1} = \mathrm{N}_d(L_k, 1/3^k 2), \;\text{ i.e., }\; M_k^{2n+1} = \mathbf{I}^{2n+1} \setminus \mathrm{N}_d(L_k, 1/3^k 2).$$

Moreover, it should be noted that $L_k \subset \mathbf{I}^{2n+1} \setminus \nu^n$, i.e., $\mathbf{I}^{2n+1} \cap \nu^n \subset \mathbf{I}^{2n+1} \setminus L_k$.

Now, we will prove the following:

Theorem 5.10.8 *The n-dimensional Menger compactum μ^n is a universal space for separable metrizable spaces with* $\dim \leqslant n$.

Proof Note that $\nu^n \cap \mathbf{I}^{2n+1}$ contains $\nu^n \cap (0, 1)^{2n+1}$, which is a universal space for separable metrizable spaces with $\dim \leqslant n$ by Proposition 5.10.2 and Theorem 5.10.3. The Dimension-Preserving Compactification Theorem 5.7.2 reduces the proof to showing that every compact set X in $\nu^n \cap \mathbf{I}^{2n+1}$ can be embedded in μ^n.

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[17] This is a property studied in Fractal Geometry. A metric space $X = (X, d)$ is said to be **self-similar** if there exist λ-Lipschitz embeddings $f_s : X \to X$, $s \in S$, such that $X = \bigcup_{s \in S} f_s(X)$ and $\lambda < 1$, where S is finite.

Given a homeomorphism $g : \mathbf{I} \to \mathbf{I}$, let $\hat{g} : \mathbf{I}^{2n+1} \to \mathbf{I}^{2n+1}$ be a homeomorphism defined by $\hat{g}(x) = (g(x(1)), \dots, g(x(2n+1)))$. For $k \in \mathbb{N}$, we inductively choose $s_k \in (0, 1/3^k 2] \cap \mathbb{Q}$ and define a PL homeomorphism $h_k : \mathbf{I} \to \mathbf{I}$ so as to satisfy the following conditions:

(1) $\hat{h}_{k-1}(X) \cap \mathrm{N}_d(L_k, s_k) = \emptyset$, i.e., $\hat{h}_{k-1}(X) \subset \mathbf{I}^{2n+1} \setminus \mathrm{N}_d(L_k, s_k)$,
(2) $\hat{h}_k(\mathbf{I}^{2n+1} \cap \nu^n) = \mathbf{I}^{2n+1} \cap \nu^n$ and $\hat{h}_k(X) \subset \bigcap_{i=1}^k M_i^{2n+1} \cap \nu^n$ $(\subset \mathbf{I}^{2n+1} \setminus L_{k+1})$,
(3) $\sup_{x \in \mathbf{I}} |h_k(x) - h_{k-1}(x)| = 1/3^k 2 - s_k$,
(4) $h_k | \partial \mathbf{I} \cup h_{k-1}^{-1}(\bigcup_{i=1}^{k-1} \mathrm{bd}\, R_i) = h_{k-1} | \partial \mathbf{I} \cup h_{k-1}^{-1}(\bigcup_{i=1}^{k-1} \mathrm{bd}\, R_i)$,
(5) h_k is affine on the closure of each component of $\mathbf{I} \setminus h_k^{-1}(\bigcup_{i=1}^{k} \mathrm{bd}\, R_i)$,
(6) the length (diameter) of each component of $\mathbf{I} \setminus h_k^{-1}(\bigcup_{i=1}^{k} \mathrm{bd}\, R_i)$ is at most $1/2^k$,

where $h_0 = \mathrm{id}_{\mathbf{I}}$ in (1) for $k = 1$.

Suppose $h_{k-1} : \mathbf{I} \to \mathbf{I}$ has been defined. Since $\hat{h}_{k-1}(X) \subset \mathbf{I}^{2n+1} \setminus L_k$ and $\hat{h}_{k-1}(X)$ is compact, we can choose $s_k \in (0, 1/3^k 2] \cap \mathbb{Q}$ so as to satisfy (1) and define a PL homeomorphism $g_k : \mathbf{I} \to \mathbf{I}$ by the following conditions:

(7) $g_k(m/3^{k-1}) = m/3^{k-1}$ for every $m = 0, 1, \dots, 3^{k-1}$, and $g_k(q \pm s_k) = q \pm 1/3^k 2$ for each $q \in Q_k$, i.e., $g_k | \partial \mathbf{I} \cup \bigcup_{i=1}^{k-1} \mathrm{bd}\, R_i = \mathrm{id}$ and $g_k(\mathrm{bd}\, R_k(s_k)) = \mathrm{bd}\, R_k$,
(8) g_k is affine on the closure of each component of $\mathbf{I} \setminus (\bigcup_{i=1}^{k-1} \mathrm{bd}\, R_i \cup \mathrm{bd}\, R_k(s_k))$.

We will show that the PL homeomorphism $h_k = g_k h_{k-1}$ satisfies (2)–(6), where $\hat{h}_k = \hat{g}_k \hat{h}_{k-1}$. Because $s_k \in \mathbb{Q}$, we have $g_k(\mathbf{I} \cap \mathbb{Q}) = \mathbf{I} \cap \mathbb{Q}$ by the definition of g_k, so

$$\hat{h}_k(\mathbf{I}^{2n+1} \cap \nu^n) = \hat{g}_k \hat{h}_{k-1}(\mathbf{I}^{2n+1} \cap \nu^n) = \hat{g}_k(\mathbf{I}^{2n+1} \cap \nu^n) = \mathbf{I}^{2n+1} \cap \nu^n.$$

Since $g_k(R_k(s_k)) = R_k$ by the definition of g_k, it follows from (1) that

$$\hat{h}_k(X) = \hat{g}_k(\hat{h}_{k-1}(X)) \subset \hat{g}_k(\mathbf{I}^{2n+1} \setminus \mathrm{N}_d(L_k, s_k)) = M_k^{2n+1}.$$

Moreover, for each $i = 1, \dots, k-1$, we have $g_k(R_i) = R_i$ by the definition of g_k, which implies

$$\hat{h}_k(X) = \hat{g}_k(\hat{h}_{k-1}(X)) \subset \hat{h}_k(M_i^{2n+1}) \subset M_i^{2n+1}.$$

Thus, $\hat{h}_k$ satisfies (2). By the definitions of g_k and h_k, we have (3), that is,

$$\sup_{x \in \mathbf{I}} |h_k(x) - h_{k-1}(x)| = \sup_{x \in \mathbf{I}} |g_k(x) - x| = 1/3^k 2 - s_k.$$

Furthermore, (4) comes from (7) as follows:

$$\begin{aligned} h_k | \partial \mathbf{I} \cup h_{k-1}^{-1}(\textstyle\bigcup_{i=1}^{k-1} \mathrm{bd}\, R_i) &= \big(g_k | \partial \mathbf{I} \cup \textstyle\bigcup_{i=1}^{k-1} \mathrm{bd}\, R_i\big)\big(h_{k-1} | \partial \mathbf{I} \cup h_{k-1}^{-1}(\textstyle\bigcup_{i=1}^{k-1} \mathrm{bd}\, R_i)\big) \\ &= h_{k-1} | \partial \mathbf{I} \cup h_{k-1}^{-1}(\textstyle\bigcup_{i=1}^{k-1} \mathrm{bd}\, R_i). \end{aligned}$$

Since $g_k^{-1}(\operatorname{bd} R_k) = \operatorname{bd} R_k(s_k)$ and $g_k^{-1}(\bigcup_{i=1}^{k-1} \operatorname{bd} R_i) = \bigcup_{i=1}^{k-1} \operatorname{bd} R_i$, (8) means that g_k is affine on the closure of each component of $\mathbf{I} \setminus g_k^{-1}(\bigcup_{i=1}^{k} \operatorname{bd} R_i)$. Since h_{k-1} is affine on the closure of each component of $\mathbf{I} \setminus h_{k-1}^{-1}(\bigcup_{i=1}^{k-1} \operatorname{bd} R_i)$, it follows that h_{k-1} is affine on the closure of each component of

$$\begin{aligned}\mathbf{I} \setminus h_{k-1}^{-1}\Big(\bigcup_{i=1}^{k-1} \operatorname{bd} R_i \cup \operatorname{bd} R_k(s_k)\Big) &= \mathbf{I} \setminus h_{k-1}^{-1} g_k^{-1}\Big(\bigcup_{i=1}^{k} \operatorname{bd} R_i\Big) \\ &= \mathbf{I} \setminus h_k^{-1}\Big(\bigcup_{i=1}^{k} \operatorname{bd} R_i\Big).\end{aligned}$$

Hence, (5) holds, that is, $h_k = g_k h_{k-1}$ is affine on the closure of each component of $\mathbf{I} \setminus h_k^{-1}(\bigcup_{i=1}^{k} \operatorname{bd} R_i)$. Recall that $R_k(s_k) = \bigcup_{q \in Q_k} (q - s_k, q + s_k)$ and Q_k is the mid-points of components of $\mathbf{I} \setminus \bigcup_{i=1}^{k-1} \operatorname{bd} R_i$. Then,

$$h_k^{-1}(\operatorname{bd} R_k) = h_{k-1}^{-1} g_k^{-1}(\operatorname{bd} R_k) = h_{k-1}^{-1}(\operatorname{bd} R_k(s_k)) = \big\{ h_{k-1}^{-1}(q \pm s_k) \;\big|\; q \in Q_k \big\},$$

and $h_{k-1}^{-1}(Q_k)$ is the mid-points of components of $\mathbf{I} \setminus h_k^{-1}(\bigcup_{i=1}^{k-1} \operatorname{bd} R_i)$, where each $h_{k-1}^{-1}(q \pm s_k)$ is contained on both sides of each component of $\mathbf{I} \setminus h_k^{-1}(\bigcup_{i=1}^{k-1} \operatorname{bd} R_i)$ divided into two parts at the mid-point. Since the length of each component of $\mathbf{I} \setminus h_{k-1}^{-1}(\bigcup_{i=1}^{k-1} \operatorname{bd} R_i)$ is at most $1/2^{k-1}$, it follows that the length of each component of $\mathbf{I} \setminus h_k^{-1}(\bigcup_{i=1}^{k} \operatorname{bd} R_i)$ is at most $1/2^k$, that is, (6) holds.

The sequence $(h_k)_{k \in \mathbb{N}}$ uniformly converges to a map $h : \mathbf{I} \to \mathbf{I}$ because it is uniformly Cauchy by (3). Due to (4),

$$h|\partial\mathbf{I} \cup h_k^{-1}\Big(\bigcup_{i=1}^{k} \operatorname{bd} R_i\Big) = h_k|\partial\mathbf{I} \cup h_k^{-1}\Big(\bigcup_{i=1}^{k} \operatorname{bd} R_i\Big) \;\text{ for each } k \in \mathbb{N}. \qquad (*)$$

In particular, $h(\partial\mathbf{I}) = \partial\mathbf{I}$, which implies that $h(\mathbf{I}) = \mathbf{I}$, that is, h is surjective.

To see that h is injective, let $x < y \in \mathbf{I}$. By (6), we can take $k \in \mathbb{N}$ and $x', y' \in h_k^{-1}(\bigcup_{i=1}^{k} \operatorname{bd} R_i)$ so that $x < x' < y' < y$. Since h_k is increasing (order-preserving), we have $h(x') = h_k(x') < h_k(y') = h(y')$ by (*). Consequently, $h(x) \leqslant h(x') < h(y') \leqslant h(y)$. Hence, h is injective. Thus, we conclude that $h : \mathbf{I} \to \mathbf{I}$ is a homeomorphism, where $\hat{h} : \mathbf{I}^{2n+1} \to \mathbf{I}^{2n+1}$ is a homeomorphism.

Finally, we will show that $\hat{h}(X) \subset \mu^n = \bigcap_{k \in \mathbb{N}} M_k^{2n+1}$. For each $x \in X$ and $k \in \mathbb{N}$, since $\hat{h}_j(x) \in \bigcap_{i=1}^{j} M_i^{2n+1} \subset M_k^{2n+1}$ if $j \geqslant k$, it follows that $\hat{h}(x) = \lim_{j \to \infty} \hat{h}_j(x) \in M_k^{2n+1}$. Hence, $\hat{h}(x) \in \bigcap_{k \in \mathbb{N}} M_k^{2n+1} = \mu^n$. □

Remark 5.10.3 In Theorem 5.10.8, suppose that $X \subset \nu^n \cap \bigcap_{j=1}^{k-1} M_j^{2n+1}$. Then, in the above proof, we can take $s_1 = 1/3{\cdot}2$, …, $s_{k-1} = 1/3^{k-1}2$, which implies $g_1 = \cdots = g_{k-1} = \mathrm{id}$, and hence $h_0 = h_1 = \cdots = h_{k-1} = \mathrm{id}$. If $j \geqslant k$, for each $s \in \mathbf{I}$, we have

$$
\begin{aligned}
|h_j(s) - s| = |h_j(s) - h_k(s)| &\leqslant \sum_{i=k}^{j} |h_i(s) - h_{i-1}(s)| \\
&= \sum_{i=k}^{j} |g_i h_{i-1}(s) - h_{i-1}(s)| = \sum_{i=k}^{j} (1/3^i 2 - s_i) \\
&< \sum_{i=k}^{j} 1/3^i 2 \leqslant \sum_{i=k}^{\infty} 1/3^i 2 = 1/3^{k-1} 4,
\end{aligned}
$$

which implies $|h(s) - s| \leqslant 1/3^{k-1}4$. Thus, the homeomorphism $\hat{h} : \mathbf{I}^{2n+1} \to \mathbf{I}^{2n+1}$ induced by h satisfies

$$
d_\infty(h, \mathrm{id}) = \sup_{x \in \mathbf{I}^{2n+1}} d(h(x), x) < 1/3^{k-1}4.
$$

Using this remark, we can also prove the following theorem:

Theorem 5.10.9 *Let X be a compactum with* $\dim X \leqslant n$. *Then, every map $f : X \to \mu^n$ can be approximated by embeddings into μ^n.*

Proof For each $\varepsilon > 0$, choose $k \in \mathbb{N}$ so that $1/3^{k-1} < \varepsilon$. Applying Proposition 5.10.4 (with Proposition 5.10.2), we can obtain an embedding $f' : X \to (0, 1)^{2n+1} \cap \nu^n$ such that $d_\infty(f, f') < 1/3^k 2$. Then, $f'(X) \subset \mathrm{N}_d(\mu^n, 1/3^k 2) \cap \nu^n$, where

$$
\overline{\mathrm{N}}_d(\mu^n, 1/3^k 2) \subset \bigcap_{j=1}^{k-1} \overline{\mathrm{N}}_d\big(M_j^{2n+1}, 1/3^k 2\big).
$$

We will construct a homeomorphism $\varphi : \mathbf{I}^{2n+1} \to \mathbf{I}^{2n+1}$ such that

$$
\begin{gathered}
d_\infty(\varphi, \mathrm{id}) \leqslant 1/3^k 2,\ \ \varphi((0, 1)^{2n+1} \cap \nu^n) \subset (0, 1)^{2n+1} \cap \nu^n,\ \text{ and} \\
\varphi\big(\mathrm{N}_d\big(M_j^{2n+1}, 1/3^k 2\big)\big) \subset M_j^{2n+1} \ \text{ for } j = 1, \ldots, k-1.
\end{gathered}
$$

Then, $\varphi f'(X) \subset \nu^n \cap \bigcap_{j=1}^{k-1} M_j^{2n+1}$. By Theorem 5.10.8 with Remark 5.10.3, we have a homeomorphism $h : \mathbf{I}^{2n+1} \to \mathbf{I}^{2n+1}$ such that $d(h, \mathrm{id}) < 1/3^{k-1}4$ and $h(\varphi f'(X)) \subset \mu^n$. Thus, we can obtain the embedding $h\varphi f' : X \to \mu^n$ with $d_\infty(f, h\varphi f') < 1/3^{k-1} < \varepsilon$.

Now, we will construct the homeomorphism φ. For each $j = 1, \ldots, k-1$, let

$$
R_j^* = \bigcup_{m=0}^{3^{j-1}-1} \left(\frac{m}{3^{j-1}} + \frac{1}{3^j} + \frac{1}{3^k 2}, \frac{m}{3^{j-1}} + \frac{2}{3^j} - \frac{1}{3^k 2} \right) \subset R_j.
$$

Since $\overline{\mathrm{N}}(\mathbf{I} \setminus R_j, 1/3^k 2) = \mathbf{I} \setminus R_j^*$, we have

$$\overline{\mathrm{N}}_d(M_j^{2n+1}, 1/3^k 2) = \left\{x \in \mathbf{I}^{2n+1} \;\middle|\; x(i) \in \mathbf{I} \setminus R_j^* \text{ for at least } n+1\, i\right\}.$$

Observe that

$$\begin{aligned}\operatorname{bd} R_j &= \left\{m/3^{j-1} + 1/3^j, m/3^{j-1} + 2/3^j \;\middle|\; m = 0, \ldots, 3^{j-1} - 1\right\} \\ &= \left\{m/3^j \;\middle|\; m = 1, \ldots, 3^j - 1\right\} \setminus \left\{m/3^{j-1} \;\middle|\; m = 1, \ldots, 3^{j-1} - 1\right\}.\end{aligned}$$

Then, it follows that

$$\begin{aligned}\bigcup_{j=1}^{k-1} \operatorname{bd} R_j &= \left\{\frac{m}{3^{k-1}} \;\middle|\; m = 1, \ldots, 3^{k-1} - 1\right\} \\ &\subset \bigcup_{m=1}^{3^{k-1}-1} \left[\frac{m}{3^{k-1}} - \frac{1}{3^k}, \frac{m}{3^{k-1}} + \frac{1}{3^k}\right] \\ &= \left(\frac{1}{3^k}, 1 - \frac{1}{3^k}\right) \setminus \bigcup_{m=0}^{3^{k-1}-1} \left(\frac{m}{3^{k-1}} + \frac{1}{3^k}, \frac{m}{3^{k-1}} + \frac{2}{3^k}\right) \\ &= \mathbf{I} \setminus (R_k \cup [0, 1/3^k] \cup [1 - 1/3^k, 1]).\end{aligned}$$

We define a PL homeomorphism $\psi : \mathbf{I} \to \mathbf{I}$ as follows:

$$\psi | R_k \cup [0, 1/3^k] \cup [1 - 1/3^k, 1] = \mathrm{id},$$

and for each $j = 1, \ldots, k-1$ and $m = 0, \ldots, 3^{j-1} - 1$,

$$\begin{aligned}\psi(m/3^{j-1} + 1/3^j + 1/3^k 2) &= m/3^{j-1} + 1/3^j, \\ \psi(m/3^{j-1} + 2/3^j - 1/3^k 2) &= m/3^{j-1} + 2/3^j,\end{aligned}$$

and ψ is affine on each of the following segments:

$$\left[\frac{m}{3^{j-1}} + \frac{1}{3^j} - \frac{1}{3^k}, \frac{m}{3^{j-1}} + \frac{1}{3^j} + \frac{1}{3^k 2}\right], \left[\frac{m}{3^{j-1}} + \frac{1}{3^j} + \frac{1}{3^k 2}, \frac{m}{3^{j-1}} + \frac{1}{3^j} + \frac{1}{3^k}\right],$$
$$\left[\frac{m}{3^{j-1}} + \frac{2}{3^j} - \frac{1}{3^k}, \frac{m}{3^{j-1}} + \frac{2}{3^j} - \frac{1}{3^k 2}\right], \left[\frac{m}{3^{j-1}} + \frac{2}{3^j} - \frac{1}{3^k 2}, \frac{m}{3^{j-1}} + \frac{2}{3^j} + \frac{1}{3^k}\right].$$

In the above, note that

$$\frac{m}{3^{j-1}} + \frac{1}{3^j} = \frac{3^{k-1-j}(3m+1)}{3^{k-1}} \text{ and } \frac{m}{3^{j-1}} + \frac{2}{3^j} = \frac{3^{k-1-j}(3m+2)}{3^{k-1}}.$$

Then, $\sup_{x \in \mathbf{I}} |\psi(x) - x| = 1/3^k 2$, $\psi(\mathbf{I} \cap \mathbb{Q}) = \mathbf{I} \cap \mathbb{Q}$, and $\psi(\mathbf{I} \setminus R_j^*) = \mathbf{I} \setminus R_j$ for each $j = 1, \cdots, k-1$. The desired homeomorphism $\varphi : \mathbf{I}^{2n+1} \to \mathbf{I}^{2n+1}$ can be

defined as follows:

$$\varphi(x) = (\psi(x(1)), \ldots, \psi(x(2n+1))) \text{ for each } x \in \mathbf{I}^{2n+1}. \qquad \square$$

5.11 Total Disconnectedness and the Cantor Set

A space X is said to be **totally disconnected** provided that, for any two distinct points $x \neq y \in X$, there is a clopen set H in X such that $x \in H$ but $y \notin H$ (i.e., the empty set is a partition between any two distinct points). Equivalently, for each $x \in X$ the intersection of all clopen sets containing x is the singleton $\{x\}$. It is easy to see that the zero-dimensionality implies the total disconnectedness.

Indeed, when $\dim X = 0$, for any two points $x \neq y \in X$, X has a finite open cover $\mathcal{U}$ such that $\mathcal{U} \prec \{\{X \setminus \{x\}, \{X \setminus \{y\}\}$ and $\operatorname{ord} \mathcal{U} = 1$. Since every $U \in \mathcal{U}[x]$ is clopen in X and $U \subset X \setminus \{y\}$, we have a clopen set $\operatorname{st}(x, \mathcal{U})$ contained in $X \setminus \{y\}$, that is, $y \notin \operatorname{st}(x, \mathcal{U})$.

We say that X is **hereditarily disconnected** if every non-degenerate subset of X is disconnected (i.e., every component of X is a singleton). Clearly, the total disconnectedness implies the hereditary disconnectedness. Therefore, we have the following fact:

Fact *Every zero-dimensional space is totally disconnected, and every totally disconnected space is hereditarily disconnected.*

The converse assertions are true for compact spaces. To see this, we prove the following lemma:

Lemma 5.11.1 *Let X be compact, $x \in X$, and C be the intersection of all clopen sets in X containing x.*

(1) *For each open neighborhood U of C in X, there is a clopen set H in X such that $C \subset H \subset U$.*
(2) *C is the component of X containing x.*

Proof (1): Let $\mathcal{H}$ be the collection of all clopen sets in X containing x. Since $X \setminus U$ is compact and $\{X \setminus H \mid H \in \mathcal{H}\}$ is its open cover in X, there are $H_1, \ldots, H_n \in \mathcal{H}$ such that

$$X \setminus U \subset \bigcup_{i=1}^{n} (X \setminus H_i) = X \setminus \bigcap_{i=1}^{n} H_i.$$

Thus, we have $H = \bigcap_{i=1}^{n} H_i \in \mathcal{H}$ and $C \subset H \subset U$.

(2): Since C clearly contains the component of X containing x, it suffices to show that C is connected. Now assume that $C = A \cup B$, where A and B are disjoint closed sets in C and $x \in A$. From the normality, it follows that there are disjoint open sets U and V in X such that $A \subset U$ and $B \subset V$. By (1), we have a clopen set

H in X such that $C \subset H \subset U \cup V$. Since $H \cap U$ is open in X and $H \setminus V$ is closed in X, $H \cap U = H \setminus V$ is clopen in X. Then, $C \subset H \cap U \subset U$, which implies that $B \subset C \cap V = \emptyset$. Thus, C is connected. □

Theorem 5.11.2 *For every non-empty compact space X, the following are equivalent:*

(a) $\dim X = 0$;
(b) *X is totally disconnected;*
(c) *X is hereditarily disconnected.*

Proof The implications (a) ⇒ (b) ⇒ (c) follow from the above Fact. Here, we will prove the converse implications.

(c) ⇒ (b): For each $x \in X$, the intersection of all clopen sets in X containing x is a component of X by Lemma 5.11.1(2). It is, in fact, the singleton $\{x\}$, which means that X is totally disconnected.

(b) ⇒ (a): Let $\mathcal{U}$ be a finite open cover of X. Each $x \in X$ belongs to some $U \in \mathcal{U}$. Because of the total disconnectedness of X, the singleton $\{x\}$ is the intersection of all clopen sets in X containing x. By Lemma 5.11.1(1), we have a clopen set H_x in X such that $x \in H_x \subset U$. From the compactness, it follows that $X = \bigcup_{i=1}^{n} H_{x_i}$ for some $x_1, \ldots, x_n \in X$. Let

$$V_1 = H_{x_1}, V_2 = H_{x_2} \setminus H_{x_1}, \ldots, V_n = H_{x_n} \setminus (H_{x_1} \cup \cdots \cup H_{x_{n-1}}).$$

Then, $\mathcal{V} = \{V_1, \ldots, V_n\}$ is an open refinement of $\mathcal{U}$ and $\operatorname{ord} \mathcal{V} = 1$. Consequently, $\dim X = 0$. □

The implications (c) ⇒ (b) ⇒ (a) in Theorem 5.11.2 do not hold in general. In the next section, we will show the existence of nonzero-dimensional totally disconnected spaces, i.e., counter-examples for (b) ⇒ (a). Here, we give a counter-example for (c) ⇒ (b) via the following theorem:

Theorem 5.11.3 *There exists a separable metrizable space that is hereditarily disconnected but not totally disconnected.*

Proof (*Example*) Let D be a countable dense subset of the Cantor set μ^0. We define

$$X = D \times \mathbb{Q} \cup (\mu^0 \setminus D) \times (\mathbb{R} \setminus \mathbb{Q}) \subset \mu^0 \times \mathbb{R}.$$

Let $p : X \to \mu^0$ be the restriction of the projection of $\mu^0 \times \mathbb{R}$ onto μ^0.

First, we show that X is hereditarily disconnected. Let $A \subset X$ be a non-degenerate subset. When $\operatorname{card} p(A) > 1$, since μ^0 is hereditarily disconnected, $p(A)$ is disconnected, which implies that A is disconnected. When $\operatorname{card} p(A) = 1$, $A \subset p(A) \times \mathbb{Q} \approx \mathbb{Q}$ or $A \subset p(A) \times (\mathbb{R} \setminus \mathbb{Q}) \approx \mathbb{R} \setminus \mathbb{Q}$. Since both $\mathbb{Q}$ and $\mathbb{R} \setminus \mathbb{Q}$ are hereditarily disconnected, A is disconnected.

Next, we prove that X is not totally disconnected. Assume that X is totally disconnected and let $x_0 \in D \subset \mu^0$. Because $(x_0, 0), (x_0, 1) \in X$, we have closed sets F_0

and F_1 in $\mu^0 \times \mathbb{R}$ such that $X \subset F_0 \cup F_1$, $F_0 \cap F_1 \cap X = \emptyset$, and $(x_0, i) \in F_i$. Then, $F_0 \cap X$ and $F_1 \cap X$ are clopen in X. Choose an open neighborhood U_0 of x_0 in μ^0 so that

$$(U_0 \cap D) \times \{i\} = U_0 \times \{i\} \cap X \subset F_i \quad \text{for } i = 0, 1.$$

Since F_i is closed in $\mu^0 \times \mathbb{R}$ and D is dense in μ^0, it follows that $U_0 \times \{i\} \subset F_i$. For each $r \in \mathbb{Q}$, let

$$C_r = \left\{ x \in U_0 \;\middle|\; (x, r) \in F_0 \cap F_1 \right\}.$$

Then, each C_r is closed and nowhere dense in U_0. Indeed, for each $x \in U_0 \setminus C_r$, because $(x, r) \notin F_0 \cap F_1$ and $F_0 \cap F_1$ is closed in $\mu^0 \times \mathbb{R}$, x has a neighborhood U in U_0 such that $U \times \{r\} \cap F_0 \cap F_1 = \emptyset$. Then, $(y, r) \notin F_0 \cap F_1$ for all $y \in U$, i.e., $U \cap C_r = \emptyset$, so C_r is closed in U_0. Since $F_0 \cap F_1 \cap X = \emptyset$ and $(x, r) \in X$ for $x \in D$, we have $C_r \subset U_0 \setminus D$, which implies that C_r is nowhere dense in U_0.

We will show that $U_0 \setminus D = \bigcup_{r \in \mathbb{Q}} C_r$. Then,

$$U_0 = \bigcup_{r \in \mathbb{Q}} C_r \cup \bigcup_{x \in D \cap U_0} \{x\},$$

which is contrary to the Baire Category Theorem 2.5.1. Thus, it would follow that X is not totally disconnected. For each $x \in U_0 \setminus D$,

$$\{x\} \times \mathbb{R} = \mathrm{cl}_{\mu^0 \times \mathbb{R}} \{x\} \times (\mathbb{R} \setminus \mathbb{Q}) \subset \mathrm{cl}_{\mu^0 \times \mathbb{R}} X \subset F_0 \cup F_1.$$

If $x \notin \bigcup_{r \in \mathbb{Q}} C_r$, then $F_0 \cap F_1 \cap \{x\} \times \mathbb{Q} = \emptyset$ because $x \notin C_r$ for all $r \in \mathbb{Q}$. Therefore,

$$F_0 \cap F_1 \cap \{x\} \times \mathbb{R} = F_0 \cap F_1 \cap \{x\} \times (\mathbb{R} \setminus \mathbb{Q}) \subset F_0 \cap F_1 \cap X = \emptyset.$$

Because $(x, i) \in F_i \cap \{x\} \times \mathbb{R}$, this contradicts the connectedness of $\mathbb{R}$. Therefore, $x \in \bigcup_{r \in \mathbb{Q}} C_r$ and the proof is complete. □

In the remainder of this section, we give a characterization of the Cantor set μ^0 and show that every compactum is a continuous image of μ^0. Recall that $\mu^0 \approx \mathbf{2}^{\mathbb{N}}$, where $\mathbf{2} = \{0, 1\}$ is the discrete space of two points. In the following, μ^0 can be replaced by $\mathbf{2}^{\mathbb{N}}$ (cf. Sect. 1.1).

Theorem 5.11.4 (Characterization of the Cantor Set) *A space X is homeomorphic to the Cantor set μ^0 if and only if X is a totally disconnected non-empty compactum with no isolated points.*

Proof It suffices to show the "if" part. Since $\mu^0 \approx \mathbf{2}^{\mathbb{N}}$, we will construct a homeomorphism $h : \mathbf{2}^{\mathbb{N}} \to X$. Let $d \in \mathrm{Metr}(X)$ with $\mathrm{diam}\, X < 1$. First, note that:

$(*)$ Each non-empty open set in X can be written as the disjoint union of an arbitrary finite number of non-empty open sets.

In fact, because X has no isolated points, each non-empty open set U in X is non-degenerate and $\dim U = 0$ by Theorem 5.11.2 and the Subset Theorem 5.4.4. We apply Theorem 5.2.4 iteratively to obtain the fact $(*)$.

Using the fact $(*)$, we will inductively construct an increasing sequence $1 = n_0 < n_1 < \cdots$ in $\mathbb{N}$ and

$$\mathcal{E}_n = \left\{ E(x) \;\middle|\; x \in \mathbf{2}^n \right\} \in \operatorname{cov}(X), \;\; n \in \mathbb{N},$$

so as to satisfy the following conditions:

(1) Each $E(x) \in \mathcal{E}_n$ is non-empty, so non-degenerate;
(2) $\operatorname{mesh} \mathcal{E}_{n_i} < 2^{-i}$;
(3) $E(x) \cap E(y) = \emptyset$ if $x \neq y \in \mathbf{2}^n$; and
(4) $E(x) = E(x(1), \ldots, x(n), 0) \cup E(x(1), \ldots, x(n), 1)$ for all $x \in \mathbf{2}^n$.

By $(*)$, we have $\mathcal{E}_1 = \{E(0), E(1)\} \in \operatorname{cov}(X)$ such that $E(0)$ and $E(1)$ are non-empty, $E(0) \cap E(1) = \emptyset$, and $\operatorname{mesh} \mathcal{E}_1 \leqslant \operatorname{diam} X < 1 = 2^0$. Assume that $1 = n_0 < \cdots < n_{i-1}$ and $\mathcal{E}_1, \ldots, \mathcal{E}_{n_{i-1}}$ have been defined. For each $x \in \mathbf{2}^{n_{i-1}}$, $E(x) \in \mathcal{E}_{n_{i-1}}$ is a compactum as a clopen set in X. Since $\dim E(x) = 0$, $E(x)$ has a finite open cover $\mathcal{U}_x$ with $\operatorname{ord} \mathcal{U}_x = 1$ and $\operatorname{mesh} \mathcal{U}_x < 2^{-i}$ (Theorem 5.4.1). Choose $m \in \mathbb{N}$ so that $\operatorname{card} \mathcal{U}_x \leqslant 2^m$ for each $x \in \mathbf{2}^{n_{i-1}}$. Using the fact $(*)$, as a refinement of $\mathcal{U}_x$, we can obtain

$$\mathcal{E}_x = \left\{ E(x, y) \;\middle|\; y \in \mathbf{2}^m \right\} \in \operatorname{cov}(E(x))$$

such that $E(x, y) \neq \emptyset$ for every $y \in \mathbf{2}^m$ and $E(x, y) \cap E(x, y') = \emptyset$ for $y \neq y' \in \mathbf{2}^m$. Then, $\operatorname{mesh} \mathcal{E}_x < 2^{-i}$. We define $n_i = m + n_{i-1} > n_{i-1}$ and

$$\mathcal{E}_{n_i} = \bigcup_{x \in \mathbf{2}^{n_{i-1}}} \mathcal{E}_x = \left\{ E(x, y) \;\middle|\; (x, y) \in \mathbf{2}^{n_{i-1}} \times \mathbf{2}^m = \mathbf{2}^{n_i} \right\}.$$

Thus, we have $\mathcal{E}_{n_i} \in \operatorname{cov}(X)$ with $\operatorname{mesh} \mathcal{E}_{n_i} < 2^{-i}$. By the downward induction using formula (4), we can define $\mathcal{E}_{n_i - 1}, \ldots, \mathcal{E}_{n_{i-1}+1} \in \operatorname{cov}(X)$. Therefore, we obtain $\mathcal{E}_1, \ldots, \mathcal{E}_{n_i} \in \operatorname{cov}(X)$.

For each $x \in \mathbf{2}^{\mathbb{N}}$, $\bigcap_{n \in \mathbb{N}} E(x(1), \ldots, x(n)) \neq \emptyset$ because of the compactness of X. Since

$$\lim_{n \to \infty} \operatorname{diam} E(x(1), \ldots, x(n)) = 0,$$

we can define $h : \mathbf{2}^{\mathbb{N}} \to X$ by

$$\{h(x)\} = \bigcap_{n \in \mathbb{N}} E(x(1), \ldots, x(n)).$$

To show that h is a homeomorphism, it suffices to prove that h is a continuous bijection because $\mathbf{2}^{\mathbb{N}}$ is compact. For each $\varepsilon > 0$, choose $i \in \mathbb{N}$ so that $2^{-i} < \varepsilon$. Then, $\operatorname{mesh} \mathcal{E}_{n_i} < \varepsilon$ by (2). For each $x, y \in \mathbf{2}^{\mathbb{N}}$, $x(1) = y(1), \ldots, x(n_i) = y(n_i)$ imply

$h(x), h(y) \in E(x(1), \ldots, x(n_i)) \in \mathcal{E}_{n_i}$, so $d(h(x), h(y)) < \varepsilon$. Hence, h is continuous. It easily follows from (3) that h is injective. By (4), for each $y \in X$, we can inductively choose $x(n) \in \mathbf{2}$, $n \in \mathbb{N}$, so that $y \in E(x(1), \ldots, x(n))$. Then, we have $x \in \mathbf{2}^{\mathbb{N}}$ such that $y \in \bigcap_{n\in\mathbb{N}} E(x(1), \ldots, x(n))$, i.e., $y = h(x)$. Hence, h is surjective. This completes the proof. □

The Cantor set is very important because of the following theorem:

Theorem 5.11.5 *Every non-empty compactum X is a continuous image of the Cantor set, that is, there exists a surjective map $f : \mu^0 \to X$.*

This is a combination of the following two propositions.

Proposition 5.11.6 *Every separable metrizable space is a continuous image of a subspace of the Cantor set and every compactum is a continuous image of a closed subspace of the Cantor set.*

Proof We have a natural surjective map $\varphi : \mu^0 \to \mathbf{I}$ defined by $\varphi(\sum_{i\in\mathbb{N}} 2x_i/3^i) = \sum_{i\in\mathbb{N}} x_i/2^i$, where $x_i \in \mathbf{2} = \{0, 1\}$. Since $(\mu^0)^{\mathbb{N}} \approx \mu^0$, the Hilbert cube $\mathbf{I}^{\mathbb{N}}$ is a continuous image of the Cantor set. The result follows from the fact that every separable metrizable space can be embedded in $\mathbf{I}^{\mathbb{N}}$ (Corollary 2.3.9). As a particular case, every compactum can be embedded in the Cantor set as a closed set. □

Proposition 5.11.7 *Any non-empty closed set A in μ^0 is a retract of μ^0, that is, there is a map $r : \mu^0 \to A$ with $r|A = \mathrm{id}$.*

Proof Since $\mu^0 \approx \mathbf{2}^{\mathbb{N}}$, we may replace μ^0 by $\mathbf{2}^{\mathbb{N}}$. For each $x \in \mathbf{2}^{\mathbb{N}}$ and $n \in \mathbb{N}$, we inductively define $x^A(n) \in \mathbf{2}$ as follows:

$$x^A(n) = \begin{cases} x(n) & \text{if } (x^A(1), \ldots, x^A(n-1), x(n)) \in p_n(A), \\ 1 - x(n) & \text{otherwise,} \end{cases}$$

where $p_n : \mathbf{2}^{\mathbb{N}} \to \mathbf{2}^n$ is the projection onto the first n factors. Since $A \neq \emptyset$, $(x^A(1), \ldots, x^A(n)) \in p_n(A)$ for each $n \in \mathbb{N}$. Since A is closed in $\mathbf{2}^{\mathbb{N}}$, it follows that $x^A = (x^A(n))_{n\in\mathbb{N}} \in A$. It is obvious that $x^A = x$ for $x \in A$. We can define a retraction $r : \mathbf{2}^{\mathbb{N}} \to A$ by $r(x) = x^A$. For each $x, y \in \mathbf{2}^{\mathbb{N}}$,

$$p_n(x) = p_n(y) \Rightarrow p_n(r(x)) = p_n(x^A) = p_n(y^A) = p_n(r(y)),$$

hence r is continuous. □

5.12 Totally Disconnected Examples and Infinite-Dimensional Examples

In this section, we will construct totally disconnected separable completely metrizable spaces X with $\dim X \neq 0$, not only finite-dimensional but also strongly infinite-dimensional. Using such an infinite-dimensional space, we will construct two

infinite-dimensional compacta, one is weakly infinite-dimensional but not countable-dimensional and the other is hereditarily infinite-dimensional.

The first example called the **Erdős space** is constructed in the proof of the following theorem. This space is also an example of spaces X such that $\dim X^2 \neq 2 \dim X$.

Theorem 5.12.1 *There exists a one-dimensional totally disconnected separable metrizable space X that is homeomorphic to $X^2 = X \times X$.*

Proof (*Example*) The desired space X is a subspace of the Hilbert space ℓ_2 defined as follows:

$$X = \left\{ x \in \ell_2 \;\middle|\; x(i) \in \mathbb{Q} \text{ for all } i \in \mathbb{N} \right\}.$$

The space $\ell_2 \times \ell_2$ has the norm $\|(x, y)\| = (\|x\|^2 + \|y\|^2)^{1/2}$, where the norm $\|\cdot\|$ of ℓ_2 defined by the inner product. Then, the map $h : \ell_2 \times \ell_2 \to \ell_2$ defined by $h(x, y)(2i - 1) = x(i)$ and $h(x, y)(2i) = y(i)$ is an isometry, hence it is a homeomorphism. Since $h(X \times X) = X$, we have $X \times X \approx X$.

To prove the total disconnectedness of X, let $x \neq y \in X$. Then, $x(i_0) \neq y(i_0)$ for some $i_0 \in \mathbb{N}$. Without loss of generality, we may assume that $x(i_0) < y(i_0)$. Choose $t \in \mathbb{R} \setminus \mathbb{Q}$ so that $x(i_0) < t < y(i_0)$. Then, $H = \{z \in X \mid z(i_0) < t\}$ is clopen in X and $x \in H$ but $y \notin H$. Hence, X is totally disconnected.

Note that $\dim X = \operatorname{ind} X$ by the Coincidence Theorem 5.5.4. Next, we show that $\operatorname{ind} X > 0$ and $\operatorname{ind} X \leqslant 1$. If so, we would have $\dim X = \operatorname{ind} X = 1$.

To show that $\operatorname{ind} X > 0$, it suffices to prove that $\operatorname{bd} U \neq \emptyset$ for every open neighborhood U of $\mathbf{0}$ contained in $\mathrm{B}(\mathbf{0}, 1) = \{x \in X \mid \|x\| < 1\}$. We can inductively choose $a_1, a_2, \cdots \in \mathbb{Q}$ so that

$$x_n = (a_1, \ldots, a_n, 0, 0, \ldots) \in U \;\text{ and }\; d(x_n, X \setminus U) < 1/n.$$

In fact, when $a_1, \ldots, a_n$ have been chosen, let

$$k_0 = \min \left\{ k \in \mathbb{N} \;\middle|\; (a_1, \ldots, a_n, k/(n+2), 0, 0, \ldots) \notin U \right\}.$$

Then, $(k_0 - 1)/(n + 2) \in \mathbb{Q}$ is the desired a_{n+1}. Since $\sum_{i=1}^{n} a_i^2 < 1$ for each n, it follows that $\sum_{i=1}^{\infty} a_i^2 \leqslant 1 < \infty$, hence $x_0 = (a_i)_{i \in \mathbb{N}} \in X$. Since $x_n \to x_0$ $(n \to \infty)$, it follows that $x_0 \in \operatorname{cl} U$. On the other hand, since $d(x_n, X \setminus U) < 1/n$, we have $x_0 \in \operatorname{cl}(X \setminus U)$. Therefore, $x_0 \in \operatorname{bd} U$.

To show that $\operatorname{ind} X \leqslant 1$, it suffices to prove that each $F_n = \{x \in X \mid \|x\| = 1/n\}$ is zero-dimensional. Note that $F_n \subset \mathbb{Q}^{\mathbb{N}}$ as sets. Furthermore, the topology on F_n coincides with the product inherited from the product space $\mathbb{Q}^{\mathbb{N}}$ (Proposition 1.2.6). Since $\dim \mathbb{Q}^{\mathbb{N}} = 0$, we have $\dim F_n = 0$ by the Subset Theorem 5.4.4. The proof is complete. □

To construct totally disconnected metrizable spaces X of arbitrarily large dimensions, we need the following lemmas:

Lemma 5.12.2 *Let $(A_\gamma, B_\gamma)_{\gamma\in\Gamma}$ be an essential family of pairs of disjoint closed sets in a compact space X and $\gamma_0 \in \Gamma$. For each $\gamma \in \Gamma \setminus \{\gamma_0\}$, let L_γ be a partition between A_γ and B_γ in X and $L = \bigcap_{\gamma\in\Gamma\setminus\{\gamma_0\}} L_\gamma$. Then, L has a component that meets both A_{γ_0} and B_{γ_0}.*

Proof Assume that L has no components that meet both A_{γ_0} and B_{γ_0}. Let D be the union of all components of L that meet A_{γ_0}, where we allow the case $D = \emptyset$ or $D = L$. For each $x \in L \setminus D$, the component C_x of L containing x misses A_{γ_0}. By Lemma 5.11.1(1), we have a clopen set E_x in L such that $C_x \subset E_x \subset L \setminus A_{\gamma_0}$. For each $y \in E_x$, the component C_y of L with $y \in C_y$ is contained in E_x, hence $C_y \cap A_{\gamma_0} = \emptyset$. Then, it follows that $E_x \subset L \setminus D$. Therefore, $L \setminus D$ is open in L, that is, D is closed in L, so it is compact.

For each $x \in D$, the component of L containing x misses B_{γ_0} by the assumption. As above, we have a clopen set E_x in L such that $x \in E_x \subset L \setminus B_{\gamma_0}$. Since D is compact, $D \subset \bigcup_{i=1}^n E_{x_i}$ for some $x_1, \ldots, x_n \in D$. Then, $E = \bigcup_{i=1}^n E_{x_i}$ is clopen in L and $A_{\gamma_0} \cap L \subset D \subset E \subset L \setminus B_{\gamma_0}$.

By the normality of X, we have disjoint open sets U and V in X such that $A_{\gamma_0} \cup E \subset U$ and $B_{\gamma_0} \cup (L \setminus E) \subset V$. Then, $L_{\gamma_0} = X \setminus (U \cup V)$ is a partition between A_{γ_0} and B_{γ_0} in X and $\bigcap_{\gamma\in\Gamma} L_\gamma = L \cap L_{\gamma_0} = \emptyset$. This is contrary to the essentiality of $(A_\gamma, B_\gamma)_{\gamma\in\Gamma}$. Therefore, L has a component that meets both A_{γ_0} and B_{γ_0}. □

Lemma 5.12.3 *Let X be a compactum and $f : X \to Y$ be a continuous surjection. Then, X has a G_δ-subset S that meets each fiber of f at precisely one point, that is,*

$$\mathrm{card}(f^{-1}(y) \cap S) = 1 \quad \textit{for each} \quad y \in Y.$$

Proof We may assume that $X \subset \mathbf{I}^{\mathbb{N}}$. For each $y \in Y$, since $f^{-1}(y)$ is non-empty and compact, we can define $g(y) \in X$ as follows:

$$g(y)(1) = \min \mathrm{pr}_1(f^{-1}(y)) \quad \text{and}$$
$$g(y)(n) = \min \mathrm{pr}_n\big(f^{-1}(y) \cap \textstyle\bigcap_{i=1}^{n-1} \mathrm{pr}_i^{-1}(g(y)(i))\big) \quad \text{for } n > 1.$$

Then, $\emptyset \neq f^{-1}(y) \cap \bigcap_{i=1}^n \mathrm{pr}_i^{-1}(g(y)(i)) \subset \mathrm{pr}_n^{-1}(g(y)(n))$. By the compactness of $f^{-1}(y)$, we have

$$\emptyset \neq f^{-1}(y) \cap \bigcap_{i\in\mathbb{N}} \mathrm{pr}_i^{-1}(g(y)(i)) \subset \bigcap_{n\in\mathbb{N}} \mathrm{pr}_n^{-1}(g(y)(n)) = \{g(y)\},$$

which means $g(y) \in f^{-1}(y)$. Thus, the set $S = \{g(y) \mid y \in Y\}$ meets each fiber of f at precisely one point.

For each $n, m \in \mathbb{N}$, let

$$F_{n,m} = \Big\{x \in X \;\Big|\; \exists z \in X \text{ such that } z(i) = x(i) \text{ for } i < n,\ z(n) \leqslant x(n) - \tfrac{1}{m} \text{ and } f(z) = f(x)\Big\}.$$

Since X is a compactum, it is easy to see that $F_{n,m}$ is closed in X, hence $U_{n,m} = X \setminus F_{n,m}$ is open in X. We show that $S = \bigcap_{n,m\in\mathbb{N}} U_{n,m}$, which is a G_δ-set in X. For each $y \in Y$, if $z \in X$, $z(i) = g(y)(i)$ for all $i < n$ and $z(n) \leqslant g(y)(n) - \frac{1}{m}$, then $f(z) \neq y = f(g(y))$; otherwise $g(y)(n) \leqslant z(n)$ $(< g(y)(n))$ by the definition of $g(y)$. Thus, $g(y) \in U_{n,m}$ for all $n, m \in \mathbb{N}$, i.e., $S \subset \bigcap_{n,m\in\mathbb{N}} U_{n,m}$. Conversely, for each $x \in \bigcap_{n,m\in\mathbb{N}} U_{n,m}$, let $y = f(x)$ (i.e., $x \in f^{-1}(y)$). Then, $x = g(y) \in S$. Otherwise, let $n = \min\{i \in \mathbb{N} \mid x(i) \neq g(y)(i)\}$. Since $g(y)(n) < x(n)$ by the definition of $g(y)$, it follows that $g(y)(n) \leqslant x(n) - \frac{1}{m}$ for some $m \in \mathbb{N}$, i.e., $x \in F_{n,m} = X \setminus U_{n,m}$, which is a contradiction. □

Recall that $\mathrm{Comp}(X)$ is the space of all non-empty compact sets in X with the Vietoris topology defined in Sect. 2.12. When X is metrizable, the Hausdorff metric d_H defined by $d \in \mathrm{Metr}(X)$ is admissible for $\mathrm{Comp}(X)$ (Proposition 2.12.7).

Theorem 5.12.4 *For each $n \in \mathbb{N}$, there exists an n-dimensional totally disconnected separable completely metrizable space. In addition, there exists a strongly infinite-dimensional totally disconnected separable completely metrizable space.*

Proof (*Example*) To construct the examples simultaneously, let $X = \mathbf{I} \times \mathbf{I}^\Gamma$ and $d \in \mathrm{Metr}(X)$ where $\Gamma = \{1, \ldots, n\}$ in the n-dimensional case and $\Gamma = \mathbb{N}$ in the infinite-dimensional case. Let $p_0 : X \to \mathbf{I}$ be the projection onto the first factor. Put $A_0 = p_0^{-1}(0)$ and $B_0 = p_0^{-1}(1)$ and define

$$\mathcal{E} = \left\{ E \in \mathrm{Comp}(X) \;\middle|\; E \text{ is connected},\ E \cap A_0 \neq \emptyset,\ E \cap B_0 \neq \emptyset \right\}.$$

Then, $\mathcal{E}$ is closed in $\mathrm{Comp}(X)$. Indeed, let $D \in \mathrm{Comp}(X) \setminus \mathcal{E}$. When D is not connected, it can be written as the disjoint union of two non-empty closed subsets D_1 and D_2. Let $\varepsilon = \frac{1}{2}\,\mathrm{dist}_d(D_1, D_2) > 0$. Then, every $E \in \mathrm{B}_{d_H}(D, \varepsilon)$ is not connected because E is contained in $\mathrm{N}_d(D_1, \varepsilon) \cup \mathrm{N}_d(D_2, \varepsilon)$ and meets both $\mathrm{N}_d(D_1, \varepsilon)$ and $\mathrm{N}_d(D_2, \varepsilon)$. Hence, $\mathrm{B}_{d_H}(D, \varepsilon) \cap \mathcal{E} = \emptyset$. If $D \cap A_0 = \emptyset$, then $\mathrm{N}_d(D, \delta) \cap A_0 = \emptyset$, where $\delta = \mathrm{dist}_d(A_0, D) > 0$. Every $E \in \mathrm{B}_{d_H}(D, \delta)$ also misses A_0, which implies $\mathrm{B}_{d_H}(D, \delta) \cap \mathcal{E} = \emptyset$. The case $D \cap B_0 = \emptyset$ is identical.

Since $\mathrm{Comp}(X)$ is compact by Theorem 2.12.8 (or 2.12.9), $\mathcal{E}$ is also compact. Then, we have a map $\alpha : \mu^0 \to \mathcal{E}$ of the Cantor set μ^0 onto $\mathcal{E}$ by Theorem 5.11.5. We define

$$Y = \left\{ y \in p_0^{-1}(\mu^0) \;\middle|\; y \in \alpha p_0(y) \right\} \subset X.$$

Obviously, $p_0(Y) \subset \mu^0$. For each $t \in \mu^0$, since $\alpha(t)$ is a continuum that meets both A_0 and B_0, it follows that $p_0\alpha(t) = \mathbf{I}$, so $t = p_0(y)$ for some $y \in \alpha(t)$, where $y \in Y$ (cf. Fig. 5.8). Thus, we have $p_0(Y) = \mu^0$. Moreover, Y is closed in X, so is compact. Indeed, let $(y_i)_{i\in\mathbb{N}}$ be a sequence in Y converging to $y \in X$. Since $p_0(y_i) \in \mu^0$ for every $i \in \mathbb{N}$ and $(p_0(y_i))_{i\in\mathbb{N}}$ converges to $p_0(y)$, we have $p_0(y) \in \mu^0$. Since $y_i \in \alpha p_0(y_i)$ for every $i \in \mathbb{N}$ and $(\alpha p_0(y_i))_{i\in\mathbb{N}}$ converges to $\alpha p_0(y)$ in $\mathcal{E}$, it easily follows that $y \in \alpha p_0(y)$, hence $y \in Y$.

By Lemma 5.12.3, Y has a G_δ-subset S such that

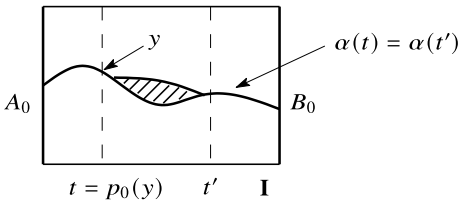


Fig. 5.8 $\alpha(t), t \in \mu^0$

$$\mathrm{card}(p_0^{-1}(t) \cap S) = 1 \quad \text{for each } t \in \mu^0.$$

Since Y is compact, S is completely metrizable. Since $p_0|S : S \to \mu^0$ is a continuous bijection and μ^0 is totally disconnected, it follows that S is also totally disconnected. Moreover, $S \cap E \neq \emptyset$ for every $E \in \mathcal{E}$. Indeed, because $\mathcal{E} = \alpha p_0(S)$, we can find $y \in S \subset Y$ such that $E = \alpha p_0(y)$, where $y \in \alpha p_0(y) = E$.

Now, for each $i \in \Gamma$, let $p_i : X \to \mathbf{I}$ be the projection onto the i-th coordinates of the second factor $\mathbf{I}^\Gamma$. Since $p_i^{-1}([0, 1/3]), p_i^{-1}([1/3, 1]) \in \mathcal{E}$, it follows that $A_i = S \cap p_i^{-1}([0, 1/3]) \neq \emptyset$ and $B_i = S \cap p_i^{-1}([1/3, 1]) \neq \emptyset$. Then, $(A_i, B_i)_{i\in\Gamma}$ is essential in S. In fact, by the Partition Extension Lemma 5.4.8, for each partition L_i between A_i and B_i in S, we have a partition $\tilde{L}_i$ between $p_i^{-1}(0)$ and $p_i^{-1}(1)$ in X such that $\tilde{L}_i \cap S \subset L_i$. According to Lemma 5.12.2, an intersection of the partitions $\tilde{L}_i$ has a component $E \in \mathcal{E}$. Then, $\bigcap_{i\in\Gamma} L_i \supset E \cap S \neq \emptyset$. Therefore, S is s.i.d. when $\Gamma = \mathbb{N}$. In the case that $\Gamma = \{1, \ldots, n\}$, $\dim S \geqslant n$ by Theorem 5.3.9. Since $S \subset \mu^0 \times \mathbf{I}^\Gamma$, $\dim S \leqslant n$ by the Subset Theorem 5.4.4 and the Product Theorem 5.4.15, hence $\dim S = n$. □

Using the infinite-dimensional example of Theorem 5.12.4, we will construct two more infinite-dimensional compacta. One is weakly infinite-dimensional but not countable-dimensional. The other is hereditarily infinite-dimensional. First, we present the following theorem:

Theorem 5.12.5 *There exists a weakly infinite-dimensional compact metrizable space that contains a strongly infinite-dimensional subspace, and hence it is not countable-dimensional.*

Proof (*Example*) Let S be a s.i.d. totally disconnected separable completely metrizable space (Theorem 5.12.4) and let $X = \gamma S$ be a metrizable compactification of S with the c.d. remainder (Theorem 5.7.4). We show that X is the required example.

First, X contains the s.i.d. subset S, so X is not c.d. (Theorem 5.6.3). To see that X is w.i.d., let $(A_i, B_i)_{i\in\omega}$ be a family of pairs of disjoint closed sets in X. Since $X \setminus S$ is c.d., $X \setminus S = \bigcup_{i\in\mathbb{N}} X_i$, where $\dim X_i = 0$ for each $i \in \mathbb{N}$. For each $i \in \mathbb{N}$, by Theorem 5.4.9 and the Partition Extension Lemma 5.4.8, X has a partition L_i between A_i and B_i such that $L_i \cap X_i = \emptyset$. Then,

$$L = \bigcap_{i\in\mathbb{N}} L_i \subset \bigcap_{i\in\mathbb{N}} X \setminus X_i = X \setminus \bigcup_{i\in\mathbb{N}} X_i = S.$$

If $L \neq \emptyset$, then L is compact and totally disconnected, which implies $\dim L = 0$ by Theorem 5.11.2. Again by Theorem 5.4.9 and the Partition Extension Lemma 5.4.8, X has a partition L_0 between A_0 and B_0 such that $L_0 \cap L = \emptyset$, which means $\bigcap_{i\in\omega} L_i = \emptyset$. □

For a compact space X and a metric space $Y = (Y, d)$, let $\mathrm{C}(X, Y)$ be the space of all maps from X to Y admitting the topology induced by the sup-metric $d_\infty(f, g) = \sup_{x\in X} d(f(x), g(x))$, which is identical to the compact-open topology because X is compact (cf. Proposition 2.10.8 (6)). Then, from Proposition 2.10.8 (5), we have the following lemma:

Lemma 5.12.6 *Let X be a compactum and $Y = (Y, d)$ be a separable metric space. The space $\mathrm{C}(X, Y)$ is separable.*

The direct proof of this lemma is rather simple.

Sketch of Proof Let $\{U_i \mid i \in \mathbb{N}\}$ and $\{V_j \mid j \in \mathbb{N}\}$ be open bases for X and Y, respectively. For each $i, j \in \mathbb{N}$, we have the following open set in $\mathrm{C}(X, Y)$:

$$W_{i,j} = \left\{ f \in \mathrm{C}(X, Y) \;\middle|\; f(\mathrm{cl}\, U_i) \subset V_j \right\} = \langle \mathrm{cl}\, U_i; V_j \rangle.$$

To see that $\{W_{i,j} \mid i, j \in \mathbb{N}\}$ is an open subbase for $\mathrm{C}(X, Y)$, let $f \in \mathrm{C}(X, Y)$ and $\varepsilon > 0$. For each $x \in X$, choose $i(x), j(x) \in \mathbb{N}$ so that $f(x) \in V_{j(x)}$, $\mathrm{diam}\, V_{j(x)} < \varepsilon$, $x \in U_{i(x)}$, and $f(\mathrm{cl}\, U_{i(x)}) \subset V_{j(x)}$, i.e., $f \in W_{i(x),j(x)}$. Because of the compactness of X, we have $x_1, \ldots, x_n \in X$ such that $X = U_{i(x_1)} \cup \cdots U_{i(x_n)}$. Then, observe that $f \in \bigcap_{k=1}^{n} W_{i(x_k),j(x_k)} \subset \mathrm{B}_{d_\infty}(f, \varepsilon)$.

To construct a hereditarily infinite-dimensional space, we need the following key lemma:

Lemma 5.12.7 *Let $C \subset \mathbf{I}$ be homeomorphic to the Cantor set and $n \in \mathbb{N}$. For any infinite set $\Gamma \subset \mathbb{N} \setminus \{n\}$, there exists a collection $\{S_i \mid i \in \Gamma\}$ of partitions S_i between $A_i = \mathrm{pr}_i^{-1}(0)$ and $B_i = \mathrm{pr}_i^{-1}(1)$ in $\mathbf{I}^{\mathbb{N}}$ such that every subset $X \subset \bigcap_{i\in\Gamma} S_i$ is strongly infinite-dimensional if $C \subset \mathrm{pr}_n(X)$, where $\mathrm{pr}_i : \mathbf{I}^{\mathbb{N}} \to \mathbf{I}$ is the projection of $\mathbf{I}^{\mathbb{N}}$ onto the i-th factor.*

Proof Without loss of generality, we may assume that $n = 1$ and $\Gamma = \{2i \mid i \in \mathbb{N}\}$. For each $i \in \mathbb{N}$, let $C_i = \mathrm{pr}_i^{-1}([0, 1/4])$ and $D_i = \mathrm{pr}_i^{-1}([3/4, 1])$. We define

$$\Omega = \left\{ f \in \mathrm{C}(\mathbf{I}^{\mathbb{N}}, \mathbf{I}^{\mathbb{N}}) \;\middle|\; \forall i \in \mathbb{N},\ f^{-1}(A_i) = C_{2i},\ f^{-1}(B_i) = D_{2i} \right\}.$$

Since Ω is separable by Lemma 5.12.6, there exist $T \subset C$ and a continuous surjection $\psi : T \to \Omega$ by Proposition 5.11.6. Let $E = \mathrm{pr}_1^{-1}(T) \subset \mathbf{I}^{\mathbb{N}}$ and define a map $\varphi : E \to \mathbf{I}^{\mathbb{N}}$ by $\varphi(x) = (\psi \mathrm{pr}_1(x))(x)$. For each $i \in \mathbb{N}$,

$$\begin{aligned} \varphi^{-1}(A_i) &= \left\{ x \in E \;\middle|\; \varphi(x) = (\psi \mathrm{pr}_1(x))(x) \in A_i \right\} \\ &= \left\{ x \in E \;\middle|\; x \in (\psi \mathrm{pr}_1(x))^{-1}(A_i) = C_{2i} \right\} = E \cap C_{2i} \end{aligned}$$

and similarly $\varphi^{-1}(B_i) = E \cap D_{2i}$. Since $\mathrm{pr}_i^{-1}(1/2)$ is a partition between A_i and B_i in $\mathbf{I}^{\mathbb{N}}$, $\varphi^{-1}(\mathrm{pr}_i^{-1}(1/2))$ is a partition between $C_{2i} \cap E$ and $D_{2i} \cap E$ in E. By the

Partition Extension Lemma 5.4.8, we have a partition S_{2i} between A_{2i} and B_{2i} in $\mathbf{I}^{\mathbb{N}}$ such that $S_{2i} \cap E \subset \varphi^{-1}(\mathrm{pr}_i^{-1}(1/2))$. It should be noted that $(A_{2i} \cap \mathrm{pr}_1^{-1}(x), B_{2i} \cap \mathrm{pr}_1^{-1}(x))_{i\in\mathbb{N}}$ is essential in $\mathrm{pr}_1^{-1}(x)$ for every $x \in C$. Then, $\mathrm{pr}_1^{-1}(x) \cap \bigcap_{i\in\mathbb{N}} S_{2i} \neq \emptyset$ for every $x \in C$, hence $C \subset \mathrm{pr}_1\big(\bigcap_{i\in\mathbb{N}} S_{2i}\big)$.

Take $X \subset \bigcap_{i\in\mathbb{N}} S_{2i}$ such that $C \subset \mathrm{pr}_1(X)$. We will show that X is s.i.d., that is, X has an infinite essential family of pairs of disjoint closed sets. For each $i \in \mathbb{N}$, let $C_i' = \mathrm{pr}_i^{-1}([0, 1/3]) \cap X$ and $D_i' = \mathrm{pr}_i^{-1}([2/3, 1]) \cap X$. To see that $(C_{2i}', D_{2i}')_{i\in\mathbb{N}}$ is essential, let L_i be a partition between C_{2i}' and D_{2i}' in X. By the Partition Extension Lemma 5.4.8, we have a partition H_i between C_{2i} and D_{2i} in $\mathbf{I}^{\mathbb{N}}$ such that $H_i \cap X \subset L_i$. There is a map $f_i : \mathbf{I}^{\mathbb{N}} \to \mathbf{I}$ such that $f_i^{-1}(0) = C_{2i}$, $f_i^{-1}(1) = D_{2i}$, and $f_i^{-1}(\frac{1}{2}) = H_i$.[18] Indeed, let U_i and V_i be disjoint open sets in $\mathbf{I}^{\mathbb{N}}$ such that $C_{2i} \subset U_i$, $D_{2i} \subset V_i$, and $X \setminus H_i = U_i \cup V_i$. We can take maps $g_i : X \setminus V_i \to \mathbf{I}$ and $h_i : X \setminus U_i \to \mathbf{I}$ such that $g_i^{-1}(0) = C_{2i}$, $g_i^{-1}(1) = H_i$, $h_i^{-1}(0) = H_i$, and $h_i^{-1}(1) = D_{2i}$ (cf. Theorem 2.2.7). The desired f_i can be defined by

$$f_i(x) = \begin{cases} g_i(x)/2 & \text{if } x \in X \setminus V_i, \\ (h_i(x) + 1)/2 & \text{if } x \in X \setminus U_i. \end{cases}$$

Now, we define a map $f : \mathbf{I}^{\mathbb{N}} \to \mathbf{I}^{\mathbb{N}}$ by $f(x) = (f_i(x))_{i\in\mathbb{N}}$. For each $i \in \mathbb{N}$, $f^{-1}(A_i) = f^{-1}(\mathrm{pr}_i^{-1}(0)) = f_i^{-1}(0) = C_{2i}$ and similarly $f^{-1}(B_i) = D_{2i}$, which implies that $f \in \Omega = \psi(T)$, hence $f = \psi(t)$ for some $t \in T$. Since $T \subset C \subset \mathrm{pr}_1(X)$, we have $x \in X$ such that $t = \mathrm{pr}_1(x)$. Then, $\varphi(x) = (\psi\,\mathrm{pr}_1(x))(x) = f(x)$. On the other hand, since $x \in \mathrm{pr}_1^{-1}(T) = E$, we have

$$x \in X \cap E \subset \bigcap_{i\in\mathbb{N}} S_{2i} \cap E \subset \bigcap_{i\in\mathbb{N}} \varphi^{-1}(\mathrm{pr}_i^{-1}(1/2)) = \varphi^{-1}(1/2, 1/2, \dots).$$

Then, $f(x) = \varphi(x) = (1/2, 1/2, \dots)$, i.e., $f_i(x) = 1/2$ for each $i \in \mathbb{N}$, hence $x \in \bigcap_{i\in\mathbb{N}} H_i \cap X \subset \bigcap_{i\in\mathbb{N}} L_i$. Therefore, $(C_{2i}', D_{2i}')_{i\in\mathbb{N}}$ is essential. □

Theorem 5.12.8 *There exists a hereditarily infinite-dimensional compact metrizable space.*

Proof (*Example*) For each $j \in \mathbb{N}$, let $A_j = \mathrm{pr}_j^{-1}(0)$ and $B_j = \mathrm{pr}_j^{-1}(1)$, where $\mathrm{pr}_j : \mathbf{I}^{\mathbb{N}} \to \mathbf{I}$ is the projection onto the j-th factor. Let $\{C_n \mid n \in \mathbb{N}\}$ be a collection of Cantor sets in $\mathbf{I}$ such that every non-degenerate subinterval of $\mathbf{I}$ contains some C_n. Let $\Gamma_{i,n}$ $(i, n \in \mathbb{N})$ be disjoint infinite subsets of $\mathbb{N} \setminus \{1\}$ such that $i \notin \Gamma_{i,n}$. For each $i, n \in \mathbb{N}$, by Lemma 5.12.7, we have a compact set $S_{i,n} \subset \mathbf{I}^{\mathbb{N}}$ that is an intersection of partitions between A_j and B_j $(j \in \Gamma_{i,n})$ and has the property that $X \subset S_{i,n}$ is s.i.d. if $C_n \subset \mathrm{pr}_i(X)$.

We will show that $S = \bigcap_{i,n\in\mathbb{N}} S_{i,n}$ is h.i.d. Since S is an intersection of partitions between A_j and B_j, $j \in \bigcup_{i,n\in\mathbb{N}} \Gamma_{i,n}$, and $(A_j, B_j)_{j\in\mathbb{N}}$ is essential, S meets every partition between A_1 and B_1 in $\mathbf{I}^{\mathbb{N}}$. By the Partition Extension Lemma 5.4.8, if P is

[18] Refer to the last Remark of Sect. 2.2.

a partition between $A_1 \cap S$ and $B_1 \cap S$ in S, then there is a partition P' between A_1 and B_1 in $\mathbf{I}^{\mathbb{N}}$ such that $P' \cap S \subset P$. Then, it follows that $P \neq \emptyset$. Due to Theorem 5.3.9, this means that $\dim S > 1$, that is, $\dim S \neq -1, 0$. Now, let $\emptyset \neq X \subset S$. In the case that $\dim \mathrm{pr}_i(X) = 0$ for every $i \in \mathbb{N}$, since $\dim \prod_{i\in\mathbb{N}} \mathrm{pr}_i(X) = 0$ by Theorem 5.4.7 and $X \subset \prod_{i\in\mathbb{N}} \mathrm{pr}_i(X)$, we have $\dim X = 0$ by the Subset Theorem 5.4.4. When $\dim \mathrm{pr}_i(X) \neq 0$ for some $i \in \mathbb{N}$, $\mathrm{pr}_i(X)$ contains a non-degenerate subinterval of $\mathbf{I}$, hence it contains some C_n. Then, it follows that X is s.i.d. □

Remark 5.12.1 Due to Theorem 5.6.6, a hereditarily infinite-dimensional metrizable space is not countable-dimensional. It is known that there exist separable, first-countable and connected compact spaces B_n, $n \geqslant 2$, such that $\dim B_n = n$ and yet B_n contains no closed set A with $0 < \dim A < n$.—Refer to Notes for this chapter.

5.13 Addendum: The Hahn–Mazurkiewicz Theorem

The content of this section is not part of Dimension Theory but is related to the content of Sect. 5.11. According to Theorem 5.11.5, every non-empty compactum is the continuous image of the Cantor (ternary) set μ^0. In this section, we will prove the following characterization of the continuous image of the interval $\mathbf{I}$:

Theorem 5.13.1 (HAHN–MAZURKIEWICZ) *A space X is the continuous image of the interval $\mathbf{I}$ if and only if X is a locally connected non-empty continuum.*[19]

Here, X is **locally connected** if each point of X has a neighborhood basis consisting of connected neighborhoods. Every (connected) component of a space X is closed in X, but if X is locally connected, it is also open (so clopen) in X. Moreover, as is easily observed, every open subspace of a locally connected space X is locally connected. Then, it follows that each point of a locally connected space X has a neighborhood basis consisting of connected *open* neighborhoods. Because of Theorem 5.13.1, a locally connected continuum is called a **Peano continuum** in honor of the first mathematician who showed that the square $\mathbf{I}^2$ is the continuous image of the interval $\mathbf{I}$.

The continuous image of a continuum is also a continuum, where the metrizability follows from Proposition 2.4.4 (1). Since every surjective closed map is a quotient map, the "only if" part of Theorem 5.13.1 comes from the following proposition:

Proposition 5.13.2 *Let $f : X \to Y$ be a quotient map. If X is locally connected, then so is Y. Namely, the quotient space of a locally connected space is also locally connected.*

Proof Let $y \in Y$. For each open neighborhood U of y in Y, let C be the connected component of U with $y \in C$. Since X is locally connected, each $x \in f^{-1}(C)$ has a connected neighborhood $V_x \subset f^{-1}(U)$. Note that $f(V_x)$ is connected, $f(V_x) \subset U$,

[19] Recall that a continuum is a compact connected metrizable space.

and $f(V_x) \cap C \neq \emptyset$. Since C is a connected component of U, it follows that $f(V_x) \subset C$, hence $V_x \subset f^{-1}(C)$. Therefore, $f^{-1}(C)$ is open in X, which means that C is open in Y. Thus, C is a connected neighborhood of y in Y with $C \subset U$. □

To prove the "if" part of Theorem 5.13.1, we introduce a simple chain in a locally compact metric space $X = (X, d)$. A **chain** is a finite sequence $\mathsf{C} = (U_1, \ldots, U_n)$ of *connected open* sets in X such that each $\mathrm{cl}\, U_i$ is *compact* and

$$U_i \cap U_{i+1} \neq \emptyset \ \text{ for each } i = 1, \ldots, n-1,$$

where $n \geqslant 1$ is called the **length** of C.[20] For $\varepsilon > 0$, C is called an **ε-chain** if $\operatorname{diam} U_i < \varepsilon$ for each $i = 1, \ldots, n$. Moreover, C is said to be **simple** provided that

$$\mathrm{cl}\, U_i \cap \mathrm{cl}\, U_j = \emptyset \ \text{ if } |i - j| > 1.$$

This condition is stronger than usual, and is adopted to simplify our argument.[21] Let A, B be non-empty subsets X. It is said that A and B are **connected by a simple (ε-)chain** $\mathsf{C} = (U_1, \ldots, U_n)$ if

$$A \cap U_1 \neq \emptyset, \ \mathrm{cl}\, A \cap \mathrm{cl}(U_2 \cup \cdots \cup U_n) = \emptyset,$$
$$B \cap U_n \neq \emptyset, \ \text{and} \ \ \mathrm{cl}\, B \cap \mathrm{cl}(U_1 \cup \cdots \cup U_{n-1}) = \emptyset,$$

where C is called a **simple (ε-)chain from A to B**. Letting $A = \{a\}$ or $B = \{b\}$, we can define the notions of a simple (ε-)chain from a to b, where

$$a \in U_1 \setminus \mathrm{cl}(U_2 \cup \cdots \cup U_n) \ \text{ and } \ b \in U_n \setminus \mathrm{cl}(U_1 \cup \cdots \cup U_{n-1}).$$

Let $\mathsf{C} = (U_1, \ldots, U_n)$ be a simple chain from A to B in X. Suppose that $\mathsf{D} = (V_1, \ldots, V_k)$ is a simple chain in some U_i from $U_{i-1} \cap U_i$ to $U_i \cap U_{i+1}$, where $U_0 = A$ if $i = 1$ and $U_{n+1} = B$ if $i = n$. Then, the following new simple chain C' from A to B in X can be defined:

$$\mathsf{C}' = (U_1, \ldots, U_{i-1}, V_1, \ldots, V_k, U_{i+1}, \ldots, U_n).$$

In this situation, we say that C' is obtained from C by replacing U_i with D.

Lemma 5.13.3 *Let $X = (X, d)$ be a connected, locally connected, locally compact metric space. Then, the following holds:*

(1) *For any $a, b \in X$ and $\varepsilon > 0$, there exists an ε-chain $\mathsf{C} = (U_1, \ldots, U_n)$ with $a \in U_1$ and $b \in U_n$.*

---

[20] In much of the literature, each U_i is not assumed to be connected, open, or to have compact closure. But, our argument can be simplified by these assumptions.

[21] Usually, it is said that $(U_1, \ldots, U_n)$ is a simple chain if $U_i \cap U_j \neq \emptyset \Leftrightarrow |i - j| \leqslant 1$. However, in our definition, $U_i \cap U_j \neq \emptyset \Leftrightarrow \mathrm{cl}\, U_i \cap \mathrm{cl}\, U_j \neq \emptyset \Leftrightarrow |i - j| \leqslant 1$ for a chain $(U_1, \ldots, U_n)$.

(2) *Every two points $a, b \in X$ are contained in some connected open set U in X such that* $\operatorname{cl} U$ *is compact.*
(3) *For any $\varepsilon > 0$, each pair of non-empty subsets $A, B \subset X$ is connected by a simple ε-chain in X.*
(4) *For each non-empty subsets $A, B \subset X$ with $\operatorname{dist}(A, B) > 0$, for any $n \in \mathbb{N}$, A and B are connected by a simple chain in X of length n.*

Proof Clearly, (1) implies (2), and (1) is easy to show.

Indeed, let W be the set of all $x \in X$ such that there exists an ε-chain $\mathsf{C} = (U_1, \ldots, U_n)$ with $a \in U_1$ and $x \in U_n$. Show that W is clopen in X, which means $W = X$ by the connectedness of X.

(3): By (1), there is an ε-chain $\mathsf{C} = (U_1, \ldots, U_n)$ in X such that $A \cap U_1 \neq \emptyset$ and $B \cap U_n \neq \emptyset$. Choose C so that the length n is the smallest among such chains. Then, $U_{i-1} \cap U_j = \emptyset$ if $i < j \leqslant n+1$, where $U_0 = A$ and $U_{n+1} = B$.

Indeed, if $U_{i-1} \cap U_j \neq \emptyset$ for some $i < j \leqslant n+1$, we can reduce the length of C by removing $U_i, \ldots, U_{j-1}$ from C, which is a contradiction.

For each $i = 1, \ldots, n+1$, take a point $p_i \in U_{i-1} \cap U_i$. Applying (2) to $p_i, p_{i+1} \in U_i$, $i = 1, \ldots, n$, we can obtain a connected open set V_i such that $p_i, p_{i+1} \in V_i$ and $\operatorname{cl}_{U_i} V_i$ is compact, where $\operatorname{cl}_X V_i = \operatorname{cl}_{U_i} V_i \subset U_i$ because of compactness. Then, it follows that $(V_1, \ldots, V_n)$ is a simple ε-chain in X from A to B.

(4): For each $n \in \mathbb{N}$, let $\varepsilon = \operatorname{dist}(A, B)/n > 0$. By (3), we can take a simple ε-chain $(V_1, \ldots, V_m)$ in X from A to B. Then, $n < m$ because

$$\operatorname{dist}(A, B) \leqslant \operatorname{diam} V_1 + \cdots + \operatorname{diam} V_m < m\varepsilon = m \operatorname{dist}(A, B)/n.$$

Hence, we have a simple chain $(V_1, \ldots, V_{n-1}, \bigcup_{i=n}^{m} V_i)$ from A to B, whose length is n. □

Recall that X is **path-connected** if every pair of points $x, y \in X$ can be connected by a path, i.e., there is a path $f : \mathbf{I} \to X$ with $f(0) = x$ and $f(1) = y$. It is said that X is **arcwise connected** if every two distinct points $x, y \in X$ can be connected by an arc, i.e., there is an arc $f : \mathbf{I} \to X$ with $f(0) = x$ and $f(1) = y$.[22] It should be noticed that, for three distinct points x, y, z in an arbitrary space X, if x, y can be connected by an arc in X and y, z can be connected by an arc in X, then x, z can be connected by an arc in X.

Indeed, let $f, g : \mathbf{I} \to X$ be arcs from x to y and from y to z, respectively. Let $t_0 = \min\{t \in \mathbf{I} \mid f(t) \in g(\mathbf{I})\}$ and $t_1 = g^{-1}(f(t_0))$ (i.e., $f(t_0) = g(t_1)$). Then, we can define an arc $h : \mathbf{I} \to X$ from x to z as follows:

$$h(t) = \begin{cases} f(2t) & \text{if } t \leqslant t_0/2, \\ g((2(1-t_1)t + 2t_1 - t_0)/(2 - t_0)) & \text{if } t \geqslant t_0/2. \end{cases}$$

[22] An arc is an injective path, i.e., an embedding of $\mathbf{I}$.

A space X is **locally path-connected** (or **locally arcwise connected**) if each neighborhood U of each point $x \in X$ contains a neighborhood V of x such that every two (distinct) points $y, y' \in V$ can be connected by a path (or an arc) in U. According to the following lemma, the local path-connectedness and the local arcwise connectedness can be defined in the same manner as the local connectedness.

Lemma 5.13.4 *For a locally path-connected (or locally arcwise connected) space X, the following holds:*

(1) *Every component of X is open and path-connected (or arcwise connected).*
(2) *Each point of X has a neighborhood basis consisting of path-connected (or arcwise connected) open neighborhoods.*

Proof (1): For each $x \in X$, let W be a subset of X consisting of all points connected with x by a path (or an arc) in X (and x itself). Then, it is easy to see that W is a connected clopen set in X, and hence it is a component of X. Since W is path-connected (or arcwise connected), we have the result.

(2): Every open neighborhood U of each $x \in X$ is also locally path-connected (or locally arcwise connected). It follows from (1) that the component of U containing x is a path-connected (or arcwise connected) open neighborhood of x. □

Obviously, every (locally) arcwise connected space is (locally) path-connected, and every (locally) path-connected space is (locally) connected. However, according to the following theorem, for connected locally compact metrizable spaces, the local connectedness implies the local arcwise connectedness.

Theorem 5.13.5 *Every connected, locally connected, locally compact metrizable space X is arcwise connected and locally arcwise connected.*

Proof By Corollary 2.5.7, we can assume that $X = (X, d)$ is a complete metric space with $\operatorname{diam} X < 1$. Let $a, b \in X$ be two distinct points. By Lemma 5.13.3 (2), a and b are contained in some connected open set U_0^0 such that $\operatorname{cl} U_0^0$ is compact. Then, $\mathsf{C}_0 = (U_0^0)$ is a simple 2^0-chain in X from a to b. By induction on $i \in \mathbb{N}$, we will construct a simple 2^{-i}-chain $\mathsf{C}_i = (U_0^i, \dots, U_{2^{n(i)}-1}^i)$ from a to b so that

(1) $n(0) = 0 < n(1) < n(2) < \cdots$ and
(2) $U_k^i \subset U_j^{i-1}$ for $2^{n(i)-n(i-1)} j \leqslant k < 2^{n(i)-n(i-1)}(j+1)$. Cf. Fig. 5.9.

Suppose that a simple $2^{-(i-1)}$-chain $\mathsf{C}_{i-1} = (U_0^{i-1}, \dots, U_{2^{n(i-1)}-1}^{i-1})$ from a to b has been obtained. Notice that each U_j^{i-1} is connected, locally connected, and locally compact. Using Lemma 5.13.3 (3) inductively, we can obtain simple 2^{-i}-chains $\mathsf{D}_j = (V_1^j, \dots, V_{k(j)}^j)$, $j = 0, \dots, 2^{n(i-1)} - 1$, from $V_{k(j-1)}^{j-1} \cap U_j^{i-1}$ to $U_j^{i-1} \cap U_{j+1}^{i-1}$ in U_j^{i-1}, where $V_{k(-1)}^{-1} = \{a\}$ and $U_{2^{n(i-1)}}^{i-1} = \{b\}$. We may assume $k(j) \geqslant 2$ by choosing $V_1^j, \dots, V_{k(j)}^j$ with sufficiently small diameters. Let C_i' be the 2^{-i}-chain formed by connecting $\mathsf{D}_0, \mathsf{D}_1, \dots, \mathsf{D}_{2^{n(i-1)}-1}$ in this order, that is,

$$\mathsf{C}_i' = \big(V_1^0, \dots, V_{k(0)}^0, V_1^1, \dots, V_{k(1)}^1, \dots, V_1^{2^{n(i-1)}-1}, \dots, V_{k(2^{n(i-1)}-1)}^{2^{n(i-1)}-1}\big).$$

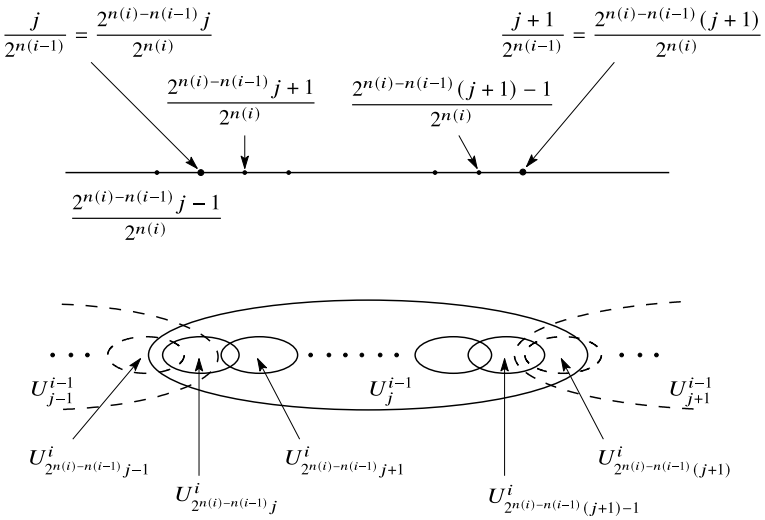


Fig. 5.9 Illustration of condition (2) for C_i

Then, C'_i is a simple chain from a to b because it can be obtained from C_{i-1} successively replacing the open set U^{i-1}_j with the simple chain D_j. Choose $n(i) \in \mathbb{N}$ so that

$$2^{n(i)-n(i-1)} > \max\left\{k(1), \ldots, k(2^{n(i-1)})\right\}.$$

For each $j = 1, \ldots, 2^{n(i-1)}$, using Lemma 5.13.3 (4), we can obtain a simple chain D'_j of length $2^{n(i)-n(i-1)} - k(j) + 1$ from $V^j_{k(j)-1} \cap V^j_{k(j)}$ to $V^j_{k(j)} \cap V^{j+1}_1$ in $V^j_{k(j)}$, where $V^{2^{n(i-1)}}_1 = \{b\}$. We can define C_i to be the simple 2^{-i}-chain from a to b obtained from C'_i by successively replacing $V^j_{k(j)}$ with D'_j. Then, the length of C_i is $2^{n(i)}$, so we can write $\mathsf{C}_i = (U^i_0, \ldots, U^i_{2^{n(i)}-1})$. By our construction, the conditions (1) and (2) are satisfied.

Now, using the above chains C_i, $i \in \mathbb{N}$, we will construct map $f : \mathbf{2}^{\mathbb{N}} \to X$. For each $x \in \mathbf{2}^{\mathbb{N}} = \{0, 1\}^{\mathbb{N}}$, observe that $0 \leqslant \sum_{j=1}^{n(i)} 2^{n(i)-j} x(j) \leqslant 2^{n(i)} - 1$ and

$$\sum_{j=1}^{n(i)} 2^{n(i)-j} x(j) = 2^{n(i)-n(i-1)} \sum_{j=1}^{n(i-1)} 2^{n(i-1)-j} x(j) \; + \sum_{j=n(i-1)+1}^{n(i)} 2^{n(i)-j} x(j) \;\text{ if } i \geqslant 2,$$

where $0 \leqslant \sum_{j=n(i-1)+1}^{n(i)} 2^{n(i)-j} x(j) < 2^{n(i)-n(i-1)}$. Then, it follows from (2) that

$$U^i_{\sum_{j=1}^{n(i)} 2^{n(i)-j} x(j)} \subset U^{i-1}_{\sum_{j=1}^{n(i-1)} 2^{n(i-1)-j} x(j)}.$$

Since $\operatorname{diam} U^i_m < 2^{-i}$ and X is complete, the following is a singleton:

$$F_x = \bigcap_{i\in\mathbb{N}} \operatorname{cl} U^i_{\sum_{j=1}^{n(i)} 2^{n(i)-j}x(j)} \neq \emptyset.$$

Then, we have a map $f : \mathbf{2}^{\mathbb{N}} \to X$ such that $\{f(x)\} = F_x$, where $f(0, 0, \dots) = a$ and $f(1, 1, \dots) = b$. For $x, y \in \mathbf{2}^{\mathbb{N}}$, if $x(j) = y(j)$ for $j \leqslant n(i)$, then

$$f(x),\, f(y) \in U^i_{\sum_{j=1}^{n(i)} 2^{n(i)-j}x(j)} = U^i_{\sum_{j=1}^{n(i)} 2^{n(i)-j}y(j)},$$

hence $d(f(x),\, f(y)) < 2^{-i}$. Therefore, f is continuous.

Let $\varphi : \mathbf{2}^{\mathbb{N}} \to \mathbf{I}$ be the quotient map defined by $\varphi(x) = \sum_{i=1}^{\infty} 2^{-i}x(i)$. For each $x, y \in \mathbf{2}^{\mathbb{N}}$, we will show that $\varphi(x) = \varphi(y)$ if and only if $f(x) = f(y)$, hence f induces the embedding $h : \mathbf{I} \to X$ with $h(0) = a$ and $h(1) = b$.

First, suppose that $\varphi(x) = \varphi(y)$, i.e., $\sum_{i=1}^{\infty} 2^{-i}x(i) = \sum_{i=1}^{\infty} 2^{-i}y(i)$. When $x \neq y$, let $k = \min\{i \in \mathbb{N} \mid x(i) \neq y(i)\}$, where we may assume that $x(k) = 1$ and $y(k) = 0$. Then,

$$\begin{aligned}\sum_{i=1}^{\infty} 2^{-i}x(i) \geqslant \sum_{i=1}^{k} 2^{-i}x(i) &= \sum_{i=1}^{k-1} 2^{-i}x(i) + 2^{-k} \\ &= \sum_{i=1}^{k} 2^{-i}y(i) + \sum_{j=k+1}^{\infty} 2^{-j} \geqslant \sum_{i=1}^{\infty} 2^{-i}y(i),\end{aligned}$$

which implies that $x(i) = 0$ and $y(i) = 1$ for every $i > k$. Thus, we have

$$\begin{aligned}\sum_{j=1}^{m} 2^{m-j}x(j) &= \sum_{j=1}^{m} 2^{m-j}y(j) \ \text{ for every } m < k, \text{ and} \\ \sum_{j=1}^{m} 2^{m-j}x(j) &= \sum_{j=1}^{m} 2^{m-j}y(j) + 1 \ \text{ for every } m \geqslant k.\end{aligned}$$

Then, it follows that

$$U^i_{\sum_{j=1}^{n(i)} 2^{n(i)-j}x(j)} \cap U^i_{\sum_{j=1}^{n(i)} 2^{n(i)-j}y(j)} \neq \emptyset \ \text{ for every } i \in \mathbb{N},$$

which implies $d(f(x),\, f(y)) = 0$ because $\operatorname{diam} U^i_m < 2^{-i}$, hence $f(x) = f(y)$.

Conversely, suppose that $f(x) = f(y)$. For every $i \in \mathbb{N}$,

$$U^i_{\sum_{j=1}^{n(i)} 2^{n(i)-j}x(j)} \cap U^i_{\sum_{j=1}^{n(i)} 2^{n(i)-j}y(j)} \neq \emptyset,$$

which means $\left| \sum_{j=1}^{n(i)} 2^{n(i)-j}x(j) - \sum_{j=1}^{n(i)} 2^{n(i)-j}y(j) \right| \leqslant 1$. Therefore,

$$
\begin{aligned}
|\varphi(x) - \varphi(y)| &= \left| \sum_{j=1}^{\infty} 2^{-j} x(j) - \sum_{j=1}^{\infty} 2^{-j} y(j) \right| \\
&= \lim_{i \to \infty} \left| \sum_{j=1}^{n(i)} 2^{-j} x(j) - \sum_{j=1}^{n(i)} 2^{-j} y(j) \right| \\
&= \lim_{i \to \infty} 2^{-n(i)} \left| \sum_{j=1}^{n(i)} 2^{n(i)-j} x(j) - \sum_{j=1}^{n(i)} 2^{n(i)-j} y(j) \right| \\
&\leqslant \lim_{i \to \infty} 2^{-n(i)} = 0,
\end{aligned}
$$

that is, $\varphi(x) = \varphi(y)$. Thus, we have proved that X is arcwise connected.

Finally, note that every neighborhood of each point $x \in X$ contains a connected open neighborhood U in X. Since U is also locally compact, it follows that U is also arcwise connected. This means that X is locally arcwise connected. □

By the "only if" part of Theorems 5.13.1 and 5.13.5, we have the following corollary:

Corollary 5.13.6 *Let X be an arbitrary space. Then, each pair of distinct points $x \neq y \in X$ is connected by a path in X if and only if the points are connected by an arc in X. In this case, the image of the arc is contained in the image of the path.*

Proof The "if" part is obvious. To see the "only if" part, let $f : \mathbf{I} \to X$ be a path with $f(0) = x$ and $f(1) = y$. Since the image $f(\mathbf{I})$ is a locally connected continuum (i.e., a Peano continuum) by the "only if" part of Theorem 5.13.1, we have an arc from x to y in $f(\mathbf{I})$ $(\subset X)$ by Theorem 5.13.5. □

Thus, we know that there is no difference between the (local) path-connectedness and the (local) arcwise connectedness of an arbitrary space. This allows us to state the following:

Corollary 5.13.7 *An arbitrary space X is path-connected if and only if X is arcwise connected. Moreover, X is locally path-connected if and only if X is locally arcwise connected.* □

A metric space $X = (X, d)$ is said to be **uniformly locally path-connected** provided that, for every $\varepsilon > 0$, there is $\delta > 0$ such that each pair of points $x, y \in X$ with $d(x, y) < \delta$ can be connected by a path with diam $< \varepsilon$, that is, a path $f : \mathbf{I} \to X$ with diam $f(\mathbf{I}) < \varepsilon$.

Proposition 5.13.8 *A compact metric space X is uniformly locally path-connected if it is locally path-connected.*

Proof For each $\varepsilon > 0$, we apply Lemma 5.13.4 (2) to obtain $\mathcal{U} \in \mathrm{cov}(X)$ consisting of path-connected open sets with mesh $\mathcal{U} < \varepsilon$. Let $\delta > 0$ be a Lebesgue number for $\mathcal{U}$. Then, each pair of points $x, y \in X$ with $d(x, y) < \delta$ can be connected by a path with diam $< \varepsilon$. □

We are now ready to prove the "if" part of the Hahn–Mazurkiewicz Theorem 5.13.1.

Proof (*The "if" Part of Theorem* 5.13.1) We may assume that $X = (X, d)$ is a non-empty, compact, connected, locally connected metric space. Let μ^0 be the Cantor (ternary) set in $\mathbf{I}$. By Theorem 5.11.5, there exists a continuous surjection $f : \mu^0 \to X$. By Theorem 5.13.5, X is path-connected and locally path-connected (arcwise connected and locally arcwise connected). According to Proposition 5.13.8, we have $\delta_1 > \delta_2 > \cdots > 0$ such that every two distinct points within δ_n can be connected by a path with diam $< 1/n$, where we may assume that $\delta_n \leqslant 1/n$.

Because of the construction of μ^0, the complement $\mathbf{I} \setminus \mu^0$ has only finitely many components $C_i = (a_i, b_i)$, $i = 1, \ldots, m$, such that $d(f(a_i), f(b_i)) \geqslant \delta_1$. Indeed, there is some $k \in \mathbb{N}$ such that

$$a, b \in \mu^0,\ |a - b| < 3^{-k} \Rightarrow d(f(a), f(b)) < \delta_1,$$
$$\big(\text{i.e., } d(f(a), f(b)) \geqslant \delta_1 \Rightarrow |a - b| \geqslant 3^{-k}\big),$$

which implies that $m \leqslant \sum_{i=1}^{k} 2^{i-1}$. For each $i = 1, \ldots, m$, let $f_i : \operatorname{cl} C_i = [a_i, b_i] \to X$ be a path with $f_i(a_i) = f(a_i)$ and $f_i(b_i) = f(b_i)$. Then, we can extend f to the map

$$f' : M = \mu^0 \cup \bigcup_{i=1}^{m} \operatorname{cl} C_i \to X$$

that is defined by $f'|\operatorname{cl} C_i = f_i$ for each $i = 1, \ldots, m$.

For each component $C = (a, b)$ of $\mathbf{I} \setminus M$ (which is a component of $\mathbf{I} \setminus \mu^0$), $f(a) = f(b)$ or $0 < d(f(a), f(b)) < \delta_1$. In the former case, let $f_C : \operatorname{cl} C = [a, b] \to X$ be the constant path with $f_C([a, b]) = \{f(a)\}$ $(= \{f(b)\})$. In the latter case, choose $n \in \mathbb{N}$ so that $\delta_{n+1} \leqslant d(f(a), f(b)) < \delta_n$ and take a path $f_C : \operatorname{cl} C = [a, b] \to X$ such that $f_C(a) = f(a)$, $f_C(b) = f(b)$, and diam $f_C([a, b]) < 1/n$. Then, f' can be extended to the map $f^* : \mathbf{I} \to X$ by $f^*|\operatorname{cl} C = f_C$ for every component C of $\mathbf{I} \setminus M$.

It remains to verify the continuity of f^*. Since each component of $\mathbf{I} \setminus \mu^0$ is an open interval, the continuity of f^* at a point of $\mathbf{I} \setminus \mu^0$ comes from the continuity of f_C or f_i. We will show the continuity of f^* at a point $x \in \mu^0$. For each $\varepsilon > 0$, choose $n \in \mathbb{N}$ so that $1/n < \varepsilon/2$. If there exists a component C_0 of $\mathbf{I} \setminus M$ such that $x \in \operatorname{cl} C_0$ (such a C_0 is unique in this case), we define $M' = M \cup \operatorname{cl} C_0$. Otherwise, we define $M' = M$. Since $f^*|M'$ is continuous, we have an open neighborhood U of x in $\mathbf{I}$ such that diam $f^*(U \cap M') < \varepsilon/2$. In the same way as we saw above, we know that $\mathbf{I} \setminus M'$ has only finitely many component $C_i' = (a_i', b_i')$, $i = 1, \ldots, m'$, such that $d(f(a_i'), f(b_i')) \geqslant \delta_n$. Then, $K = \bigcup_{i=1}^{m'} \operatorname{cl} C_i'$ is compact and $x \notin K$. Let U' be the connected component of $U \setminus K$ containing x. Then, U' is an open neighborhood of x in $\mathbf{I}$.

We show that $d(f^*(x), f^*(x')) < \varepsilon$ for every $x' \in U'$. If $x' \in U' \cap M'$, then

$$d(f^*(x), f^*(x')) \leqslant \operatorname{diam} f^*(U \cap M') < \varepsilon/2 < \varepsilon.$$

If $x' \in U' \setminus M'$ ($\subset U \setminus K$), the component $C = (a, b)$ of $\mathbf{I} \setminus M'$ containing x' satisfies $d(f(a), f(b)) < \delta_n$. Note that $x \notin C$ because $x \in \mu^0 \subset M'$. Then, it follows that $x \leqslant a < x' < b$ or $a < x' < b \leqslant x$. Since $x, x' \in U'$ and U' is an open interval, we have $a \in U'$ or $b \in U'$. In the case of $a \in U'$,

$$\begin{aligned} d(f^*(x), f^*(x')) &\leqslant d(f^*(x), f^*(a)) + d(f^*(a), f^*(x')) \\ &= d(f^*(x), f^*(a)) + d(f_C(a), f_C(x')) \\ &\leqslant \operatorname{diam} f^*(U \cap M') + \operatorname{diam} f_C([a, b]) \\ &< \varepsilon/2 + 1/n < \varepsilon. \end{aligned}$$

Similarly, we can obtain $d(f^*(x), f^*(x')) < \varepsilon$ in the case of $b \in U'$. Thus, the proof is complete. □

Notes for Chap. 5

Below, we list only four textbooks on Dimension Theory:

- M.G. Charalambous, *Dimension Theory, A Selection of Theorems and Counterexamples*, Atlantis Studies in Math. **7** (Springer Nature Switzerland AG, 2019).
- R. Engelking, *Theory of Dimensions, Finite and Infinite*, Sigma Ser. in Pure Math. **10** (Heldermann Verlag, Lembo, 1995).
- W. Hurewicz and H. Wallman, *Dimension Theory* (Princeton Univ. Press, Princeton, 1941).
- K. Nagami, *Dimension Theory* (Academic Press, Inc., New York, 1970).

For a more comprehensive study of Dimension Theory, we refer to the first two books. Engelking's book contains excellent historical notes. Charalambous' book presents relatively recent counterexamples in Dimension Theory, which was published after the publication of the first edition of this book. Nagami's book is quite readable and contains an appendix titled "Cohomological Dimension Theory" by Kodama. The classical book by Hurewicz and Wallman is still a worthwhile read. Nothing fundamental has yet changed in the framework of Dimension Theory since its publication. In their book, Hurewicz and Wallman discuss the Hausdorff dimension, which is useful in the field of Fractal Geometry. However, we do not discuss this here. In the following textbook by van Mill, Chap. 4 is devoted to Dimension Theory, and was used to prepare Sect. 5.12 of this chapter.

- J. van Mill, *Infinite-Dimensional Topology, Prerequisites and Introduction*, North-Holland Math. Library **43** (Elsevier Sci. Publ. B.V., Amsterdam, 1989).

The definition of dim, which is due to Čech [11], is based on a property of covers of $\mathbf{I}^n$ discovered by Lebesgue [34]. The Brouwer Fixed Point Theorem 5.1.1 was established in [8]. The proof using Sperner's Lemma 5.1.2 in [64] is due to Knaster, Kuratowski, and Mazurkiewicz [31]. The proof of Lemma 5.1.7 presented here was given by Yamashita in a personal communication with the author.

The equivalence between (a), (b), and (d) in Theorem 5.2.4 was established by Hemmingsen [23] and the equivalence between (a) and (d) was proved independently by Alexandroff [2] and Dowker [14]. The equivalence between (a) and (f) was first established for compact metrizable spaces by Hurewicz [26] and for normal spaces by Alexandroff [2], Hemmingsen [23], and Dowker [14], independently. The compact case of Theorem 5.2.7 was established by Freudenthal [20]. The Countable Sum Theorem 5.2.8 was established by Čech [11] and the Locally Finite Sum Theorem 5.2.9 was proved independently by Morita [47] and Katětov [27]. In Remark 5.2.2, we mentioned

the existence of a one-dimensional compact (non-metrizable) space which cannot be represented as the limit of an inverse system of polyhedra with dim $\leqslant 1$. That was shown by Mardešić [41].

In [25], a map $f : X \to \mathbf{I}^n$ is called a **universal map** if it satisfies condition (b) in Theorem 5.3.7. The equivalence between (b) and (c) in Theorem 5.3.7 is due to Holszyński [25]. The equivalence between (a) and (b) in Theorem 5.3.9 was established by Alexandroff [1]. The equivalence between (a) and (c) in Theorem 5.3.9 was first established by Eilenberg and Otto [16] in the separable metrizable case and extended to normal spaces by Hemmingsen [23].

Theorem 5.4.1 was established by Vopěnka [67] and Theorem 5.4.2 was proved by Nagami [49]. Theorem 5.4.3 is implicitly contained in Freudenthal's paper [20], whose generalization mentioned in Remark 5.4.1 was announced by Pasynkov [53] and given a proof by Katuta [29] and Yajima [69]. The Subset Theorem 5.4.4 was proved by Dowker [15]. The Addition Theorem 5.4.14 was proved by Smirnov [63]. The Decomposition and Product Theorems 5.4.11, 5.4.15 were proved independently by Katětov [27] and Morita [48].

An inductive definition of dimension was outlined by Poincaré [55]. The first precise definition of a dimension function was introduced by Brouwer [9]. His function coincides with Ind in the class of locally connected compact metrizable spaces. The definition of Ind was formulated by Čech [10]. On the other hand, the definition of ind was formulated by Urysohn [65] and Menger [45]. The first example in Theorem 5.5.5 was constructed by Roy [58, 59] but the example presented here was constructed by Kulesza [33] and the proof of $\dim > 0$ was simplified by Levin [37]. The example of Theorem 5.5.6 was described by Lokucievskiĭ [40]. The examples mentioned in Remark 5.5.4 are discussed in Chaps. 15, 31, and 32 of the above Charalambous' book. Due to notes in the book, (1) and (2) were first constructed by Vopěnka [66] and Lifanov [39], respectively. A simpler construction of (1) was obtained by Krzempek [32] and a better construction of (2) was obtained by Charalambous [12, 13]. For (2), before Lifanov [39], Nagami [50] provided a normal space X with $\operatorname{ind} X = 0$, $\dim X = 1$, and $\operatorname{Ind} X = 2$, which is seen in the above Nagami's book.

The weak infinite dimension was first introduced by Alexandroff in [3]. In Remark 5.6.2, we mentioned the weak infinite dimension in the sense of Smirnov, which was first studied in [38, 62]. Theorem 5.7.4 is due to Lelek [36] and the simple proof presented here is taken from Engelking and Pol [17]. Our proofs of Theorems 5.8.12, 5.8.13, and Corollary 5.8.15 are based on the book by Hurewicz and Wallman, Chap. IV, Sect. 4 but that of Theorem 5.8.13 was modified by Yamashita.

In [52], Nöbeling introduced the spaces ν^n and showed their universality. The spaces μ^n were introduced by Menger [46], who showed that the universality μ^1 is a universal space for compacta with $\dim \leqslant 1$. Theorem 5.10.8 is due to Bothe [6]. In [35], Lefschetz constructed a universal space for compacta with $\dim \leqslant n$. In [5], Bestvina gave the topological characterization of μ^n. Using Bestvina's characterization, we can see that Lefschetz' universal space is homeomorphic to μ^n; the result for $n = 1$ had been obtained by Anderson [4]. In [51], Nagórko established the topological characterization of ν^n.

The total disconnectedness and the hereditary disconnectedness were, respectively, introduced by Sierpiński [61] and Hausdorff [22]. The example of Theorem 5.11.3 is due to Knaster and Kuratowski [30] (their example is the one in the Remark).

The example of Theorem 5.12.1 was described by Erdős [18]. Lemma 5.12.3 is due to Bourbaki [7, Chapter 9] and the proof presented here is due to van Mill (Chap. 4 in his book listed above). The first completely metrizable nonzero-dimensional totally disconnected space was constructed by Sierpiński [61] (his example is one-dimensional). Theorem 5.12.4 was established by Mazurkiewicz [44] but the example and proof presented here is due to Rubin, Schori, and Walsh [60] with some help from [56]. The example of Theorem 5.12.5 is presented by Pol [56]. Theorem 5.12.8 is due to Walsh [68] but the example given here is due to Pol [57]. The earlier example of a compact metrizable space, whose compact subsets are all either zero-dimensional or infinite-dimensional, was constructed by Henderson [24]. The example in Remark 5.12.1 was constructed by Fedorčuk [19], which is also contained in Charalambous' book (Chap. 30).

For more details, consult the historical and bibliographical notes at the end of each section of Engelking's book. There are also some other informations contained in Charalambous' book.

As an addendum, in the last section, we discussed Peano continua. In 1890, Peano [54] showed that the square $\mathbf{I}^2$ is the continuous image of $\mathbf{I}$. The Hahn–Mazurkiewicz Theorem 5.13.1 was independently proved by Hahn [21] for planar sets and by Mazurkiewicz [42] for subspaces of Euclidean space. In [43], Mazurkiewicz gave a systematic exposition. For Peano continua, Chapter VIII of the following book can be consulted:

- S.B. Nadler, Jr., *Continuum Theory, An Introduction*, Monographs and Textbooks in Pure and Applied Math. **158** (Marcel Dekker, Inc., New York, 1992).

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Chapter 6
Retracts and Extensors



A subset A of a space X is called a **retract** of X if there is a map $r : X \to A$ such that $r|A = \mathrm{id}$, which is called a **retraction**. As is easily observed, every retract of X is *closed* in X. A **neighborhood retract** of X is a *closed* set in X that is a retract of some neighborhood in X. A *metrizable* space X is called an **absolute neighborhood retract** (**ANR**) (resp. **absolute retract** (**AR**)) if X is a neighborhood retract (resp. retract) of an arbitrary metrizable space that contains X as a closed subspace. A space Y is called an **absolute extensor** (**AE**) (resp. **absolute neighborhood extensor** (**ANE**)) **for metrizable spaces** if each map $f : A \to Y$ from any closed set A in an arbitrary metrizable space X extends over X (resp. some neighborhood U of A in X). In this book, the phrase "*for metrizable spaces*" is omitted. As is easily observed, every *metrizable* AE (resp. ANE) is an AR (resp. ANR). As will be shown, the converse is also true. Thus, a *metrizable* space is an AE (resp. ANE) if and only if it is an AR (resp. ANR). An **AE** (**ANE**) **for normal spaces** or **for compact spaces** or **for paracompact spaces**, etc., are also similarly defined, where the phrases *for normal spaces*, *for compact spaces*, etc., are not omitted. Moreover, an **absolute extensor** (resp. **absolute neighborhood extensor**) **for metrizable spaces of dim** $\leqslant \boldsymbol{n}$ are similarly defined, which is abbreviated as an **AE(*n*)** (resp. **ANE(*n*)**). It will be shown that an n-dimensional metrizable space is an AR (resp. ANR) if and only if it is an AE$(n+1)$ (resp. ANE$(n+1)$).

This chapter is devoted to lectures on ANR Theory (Theory of Retracts). Basic properties and fundamental theorems are proved and various characterizations of ANRs and ANE(n)s are given. It is shown that the n-dimensional Menger compactum μ^n and Nöbeling space ν^n are AE(n)s, and that every n-dimensional metrizable space can be embedded as a closed set into an $(n+1)$-dimensional AR (and an n-dimensional AE(n)) of the same weight. For a Peano continuum X, we also show that the hyperspaces $\mathrm{Comp}(X)$ and $\mathrm{Cont}(X)$ with the Vietoris topology consisting of all compacta and all continua in X are ARs.

The results in Chaps. 2 and 4 are used frequently. For topological linear spaces, refer to Chap. 3. For countable-dimensional or finite-dimensional spaces, we need some results in Chap. 5.

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6.1 The Dugundji Extension Theorem and ANEs

In this section, we show basic properties of ANEs and AEs. Recall that a topological linear space E is locally convex if $\mathbf{0} \in E$ has a neighborhood basis consisting of convex sets. First, we prove the DUGUNDJI EXTENSION THEOREM:

Theorem 6.1.1 (DUGUNDJI EXTENSION) *Let E be a locally convex topological linear space, and X be a metrizable space with a closed set $A \neq \emptyset$. Then, each map $f : A \to E$ can be extended to a map $\tilde{f} : X \to E$ such that the image $\tilde{f}(X)$ is contained in the convex hull $\langle f(A) \rangle$ of $f(A)$.*

Due to Theorem 6.1.1, every locally convex topological linear space and its non-empty convex set are AEs, and, if they are metrizable, then they are ARs.

Recall that every metrizable space X is paracompact, that is, any open cover of X has a locally finite open refinement (Theorem 2.3.1), and that, for each locally finite open cover $\mathcal{U} \in \mathrm{cov}(X)$, X has a partition of unity $(\lambda_U)_{U \in \mathcal{U}}$ such that $\operatorname{supp} \lambda_U \subset U$ for each $U \in \mathcal{U}$ (Theorem 2.7.7), where $\operatorname{supp} \lambda_U = \operatorname{cl} \lambda_U^{-1}((0, 1])$.

Note For a locally finite open cover $\mathcal{U}$ of a metric space $X = (X, d)$, a partition of unity can be directly defined as follows: For each $U \in \mathcal{U}$, define a map $\lambda_U : X \to \mathbf{I}$ by

$$\lambda_U(x) = \frac{d(x, X \setminus U)}{\sum_{V \in \mathcal{U}} d(x, X \setminus V)}.$$

Then, $(\lambda_U)_{U \in \mathcal{U}}$ is a partition of unity on X such that $\lambda_U^{-1}((0, 1]) = U$ (hence $\operatorname{supp} \lambda_U = \operatorname{cl} U$) for each $U \in \mathcal{U}$. By shrinking $\mathcal{U}$ (the Open Cover Shrinking Lemma 2.7.6), we can require $\operatorname{supp} \lambda_U \subset U$.

The following is a key to the proof of Theorem 6.1.1:

Lemma 6.1.2 *Let $X = (X, d)$ be a metric space and $A \neq \emptyset$ be a proper closed set in X. Then, there exists a locally finite open cover $\mathcal{U}$ of $X \setminus A$ with $a_U \in A$, $U \in \mathcal{U}$, such that $x \in U \in \mathcal{U}$ implies $d(x, a_U) \leqslant 2d(x, A)$.*

Proof Let $\mathcal{U}$ be a locally finite open cover of $X \setminus A$ such that

$$\mathcal{U} \prec \left\{ \mathrm{B}(x, d(x, A)/4) \;\middle|\; x \in X \setminus A \right\}.$$

For each $U \in \mathcal{U}$, choose $x_U \in X \setminus A$ so that $U \subset \mathrm{B}(x_U, d(x_U, A)/4)$, and then choose $a_U \in A$ so that $d(x_U, a_U) < 5d(x_U, A)/4$. If $x \in U \in \mathcal{U}$ then

$$d(x, a_U) \leqslant d(x, x_U) + d(x_U, a_U) < 3d(x_U, A)/2 < 2d(x, A)$$

because $d(x, A) \geqslant d(x_U, A) - d(x, x_U) > 3d(x_U, A)/4$. □

We call the above $(a_U)_{U \in \mathcal{U}}$ a **Dugundji system** for $A \subset X$. Now, using this system, we prove Theorem 6.1.1:

Proof (*Theorem* 6.1.1) By the above lemma, we have a Dugundji system $(a_U)_{U\in\mathcal{U}}$ for $A \subset X$. Let $(\lambda_U)_{U\in\mathcal{U}}$ be a partition of unity on $X \setminus A$ such that $\operatorname{supp}\lambda_U \subset U$ for each $U \in \mathcal{U}$. (Here, it is enough to require that $\lambda_U^{-1}((0,1]) \subset U$ for each $U \in \mathcal{U}$.) We define $\tilde{f} : X \to E$ as follows:

$$\tilde{f}(x) = \begin{cases} f(x) & \text{if } x \in A, \\ \sum_{U\in\mathcal{U}} \lambda_U(x) f(a_U) & \text{if } x \in X \setminus A. \end{cases}$$

By the local finiteness of $\mathcal{U}$, $\tilde{f}$ is continuous at $x \in X \setminus A$. To prove the continuity of $\tilde{f}$ at $a \in A$, let V be any convex neighborhood of $f(a)$ in E. From the continuity of f, we have $\delta > 0$ such that $f(\mathrm{B}(a,\delta) \cap A) \subset V$. For each $x \in \mathrm{B}(a,\delta/3) \setminus A$, if $\lambda_U(x) \neq 0$ then $x \in U$, hence

$$d(a, a_U) \leqslant d(a,x) + d(x, a_U) \leqslant d(a,x) + 2d(x,A) \leqslant 3d(x,a) < \delta,$$

which implies $f(a_U) \in V$. From the convexity of V, it follows that $\tilde{f}(x) \in V$. Thus, $\tilde{f}(x) \in V$ for all $x \in \mathrm{B}(a,\delta/3)$. □

Remark 6.1.1 It is easy to show that $\mathrm{C}(X,E)$ and $\mathrm{C}(A,E)$ are topological linear spaces with respect to the compact-open topology. If E is locally convex, so is $\mathrm{C}(X,E)$.

The continuity of addition on $\mathrm{C}(X,E)$ with respect to the compact-open topology can be shown as follows: For each $f, f' \in \mathrm{C}(X,E)$, let K be a compact set in X and U be an open set in E such that $f + f' \in \langle K; U\rangle$, i.e., $(f+f')(K) \subset U$. For each $x \in K$, $f(x)$ and $f'(x)$ have open neighborhoods V_x and V'_x in E such that $V_x + V'_x \subset U$. Then, x has an open neighborhood W_x in X such that $\operatorname{cl} W_x \subset f^{-1}(V_x) \cap f'^{-1}(V'_x)$ Since K is compact, we have finite $x_1, \ldots, x_n \in K$ such that $K \subset \bigcup_{i=1}^n W_{x_i}$. Then, each $K_i = K \cap \operatorname{cl} W_{x_i}$ is compact and $K = \bigcup_{i=1}^n K_i$. Since $f(K_i) \subset f(\operatorname{cl} W_{x_i}) \subset V_{x_i}$, we have $f \in \langle K_i, V_{x_i}\rangle$. Hence, $\mathcal{V} = \bigcap_{i=1}^n \langle K_i, V_{x_i}\rangle$ is a neighborhood of f. Similarly, $\mathcal{V}' = \bigcap_{i=1}^n \langle K_i, V'_{x_i}\rangle$ is a neighborhood of f'. It is easy to verify that $g + g' \in \langle K; U\rangle$ for every $g \in \mathcal{V}$ and $g' \in \mathcal{V}'$.

The continuity of scalar multiplication on $\mathrm{C}(X,E)$ with respect to the compact-open topology is left to the reader.

For an open neighborhood basis $\mathcal{W}$ at $\mathbf{0} \in E$, it is easy to see that the following $\mathfrak{W}$ is an open neighborhood basis at $\mathbf{0} \in \mathrm{C}(X,E)$ with respect to the compact-open topology:

$$\mathfrak{W} = \{\langle K; V\rangle \mid K \subset X \text{ is compact}, \ V \in \mathcal{W}\}.$$

If V is convex, then so is $\langle K; V\rangle$. Thus, if E is locally convex, so is $\mathrm{C}(X,E)$.

Let $R : \mathrm{C}(X,E) \to \mathrm{C}(A,E)$ be the restriction operator, i.e., $R(f) = f|A$ for each $f \in \mathrm{C}(X,E)$. Then, R is linear and continuous. Fixing a Dugundji system $(a_U)_{U\in\mathcal{U}}$ for $A \subset X$ and a partition of unity $(\lambda_U)_{U\in\mathcal{U}}$, the extension operator $L : \mathrm{C}(A,E) \to \mathrm{C}(X,E)$ is defined by $L(f) = \tilde{f}$ as above. Then, $R \circ L = \mathrm{id}$ and L is linear by definition. Moreover, L is continuous.

To prove the continuity of L at $\mathbf{0} \in \mathrm{C}(A,E)$, let $\mathfrak{V}$ be a neighborhood of $\mathbf{0} \in \mathrm{C}(X,E)$. Then, we have a compact set K in X and an open convex neighborhood W of $\mathbf{0} \in E$ such that $\langle K; W\rangle \subset \mathfrak{V}$. We will show that the following K_A is compact:

$$K_A = \left\{ a_U \;\middle|\; U \in \mathcal{U}[K] \right\} \cup (A \cap K).$$

Then, it will follow that for $f \in \mathrm{C}(A, E)$, $f(K_A) \subset W$ implies $\tilde{f}(K) \subset W$ by the convexity of W and the definition of $\tilde{f}$. Hence, L is continuous at $\mathbf{0}$.

To prove the compactness of K_A, it suffices to show that every sequence $(a_k)_{k\in\mathbb{N}}$ in K_A has a subsequence that converges to a point of K_A. If $a_k \in A \cap K$ for infinitely many $k \in \mathbb{N}$, then we can find a convergent subsequence $(a_k)_{k\in\mathbb{N}}$ by the compactness of $A \cap K$. Otherwise, we may assume that $a_k \in A \setminus K$ for all $k \in \mathbb{N}$, i.e., $a_k = a_{U_k}$ for some $U_k \in \mathcal{U}[K]$, where we may also assume that $U_k \neq U_{k'}$ if $k \neq k'$. For each $k \in \mathbb{N}$, we have $x_k \in U_k \cap K$. By the compactness of K, $(x_k)_{k\in\mathbb{N}}$ has a subsequence $(x_{k_j})_{j\in\mathbb{N}}$ converging to some $x_0 \in K$. Since $\{U_{k_j} \mid j \in \mathbb{N}\}$ is locally finite in $X \setminus A$, it follows that $x_0 \in A$. Hence,

$$d(x_{k_j}, a_{k_j}) = d(x_{k_j}, a_{U_{k_j}}) \leqslant 2d(x_{k_j}, A) \leqslant 2d(x_{k_j}, x_0) \to 0,$$

which implies that $(a_{k_j})_{j\in\mathbb{N}}$ also converges to $x_0 \in K \cap A \subset K_A$. Therefore, K_A is compact.

Remark 6.1.2 When $E = (E, \|\cdot\|)$ is a normed linear space, let $\mathrm{C}^B(X, E)$ be the normed linear space of all bounded maps with the sup-norm $\|f\| = \sup_{x\in X} \|f(x)\|$. For a closed set A in X, given a Dugundji system $(a_U)_{U\in\mathcal{U}}$ and a partition of unity $(\lambda_U)_{U\in\mathcal{U}}$, Theorem 6.1.1 gives the extension operator $L : \mathrm{C}^B(A, E) \to \mathrm{C}^B(X, E)$, which is an isometry (i.e., $\|L(f)\| = \|f\|$ for every $f \in \mathrm{C}^B(A, E)$). In this case, the restriction operator $R : \mathrm{C}^B(X, E) \to \mathrm{C}^B(A, E)$ satisfies $\|R(f)\| \leqslant \|f\|$ for every $f \in \mathrm{C}^B(X, E)$.

Remark 6.1.3 In Theorem 6.1.1, the local convexity is essential. Actually, there exists a metric linear space that is not an AE (i.e., an AR). Such an example will be constructed in Sect. 7.12 (Theorem 7.12.1).

Remark 6.1.4 Theorem 6.1.1 is valid even if E is a Fréchet space and X is a paracompact space, where it cannot be guaranteed that $\tilde{f}(X) \subset \langle f(A)\rangle$ but $\tilde{f}(X) \subset \mathrm{cl}\langle f(A)\rangle$ holds. In other words, every Fréchet space is an AE for paracompact spaces. Indeed, applying the Michael Selection Theorem 3.8.8 to the set-valued function $\varphi : X \ni x \mapsto \mathrm{cl}\langle f(A)\rangle \in \mathrm{Conv}(E)$, we can extend f to a map $\tilde{f} : X \to \mathrm{cl}\langle f(A)\rangle \subset E$. Here, a complete linear metric on E is not fully used. In fact, it suffices to assume that the restriction to $\mathrm{cl}\langle f(A)\rangle$ of a linear metric on E is complete. As a result, every non-empty complete convex set in a locally convex metric linear space is an AE for paracompact spaces.[1]

Using the Tietze–Urysohn Extension Theorem 2.2.2, the following can be easily proved in the same way as Theorem 5.1.6 (the detail is left to the reader):

Theorem 6.1.3 *The Hilbert cube* $\mathbf{I}^{\mathbb{N}}$ *and the space* $\mathbb{R}^{\mathbb{N}}$ *are AEs for normal spaces.* □

[1] This will be extended to Theorem 6.2.7.

A polyhedron (with the Whitehead topology) is another example of an ANE.

Theorem 6.1.4 *The polyhedron $|K|$ of any simplicial complex K is an ANE.*

Proof Let X be a metrizable space and $f : A \to |K|$ be a map from a closed set A in X to $|K|$. Let $(\beta_v(x))_{v\in K^{(0)}}$ be the barycentric coordinate of $x \in |K|$. Since $\beta_v : |K| \to \mathbf{I}$ is continuous, each $U_v = f^{-1}(O_K(v))$ is open in A, where $O_K(v) = \beta_v^{-1}((0, 1])$ is the open star at v with respect to K. Then, $\{U_v \mid v \in K^{(0)}\}$ is locally finite in A. Indeed, due to Proposition 4.2.11(5), each $a \in A$ has an open neighborhood V in A such that $f(V) \subset |L|$ for some finite subcomplex L of K, where $L^{(0)}$ is finite and for $v \in K^{(0)} \setminus L^{(0)}$, $f(V) \cap O_K(v) = \emptyset$, that is, $V \cap U_v = \emptyset$.

Now, for each $x \in A$, choose an open neighborhood V_x of x in A so that $V_x \cap U_v = \emptyset$ except for finitely many $v \in K^{(0)}$. Let V be an open neighborhood of A in X such that $\mathrm{cl}\, V \subset \bigcup_{x\in A} E(V_x)$, where $E(U)$ is defined in the Canonical Open Set Extension Lemma 5.4.10. For each $v \in K^{(0)}$, we define $\widetilde{U}_v = E(U_v) \cap V$. Then, $\{\widetilde{U}_v \mid v \in K^{(0)}\}$ is locally finite in X. We can apply the Tietze–Urysohn Extension Theorem 2.2.2 to construct a map $g_v : X \to \mathbf{I}$ such that $g_v|A = \beta_v \circ f$ and $g_v(X \setminus \widetilde{U}_v) = 0$. Then, $g = \sum_{v\in K^{(0)}} g_v : X \to \mathbb{R}_+$ is well-defined and continuous. Since $g(x) = \sum_{v\in K^{(0)}} \beta_v f(x) = 1$ for each $x \in A$, we have an open neighborhood $W = g^{-1}((0, \infty))$ of A in X. We can extend f to $\tilde{f} : W \to |K|$ by

$$\tilde{f}(x) = \sum_{v\in K^{(0)}} \frac{g_v(x)}{g(x)} v \quad \left(\text{i.e., } \beta_v(\tilde{f}(x)) = \frac{g_v(x)}{g(x)}\right).$$

Then, $\tilde{f}$ is well-defined and continuous. Indeed, for each $x \in W$, only finitely many $g_v(x)$ are nonzero, say

$$\left\{v \in K^{(0)} \;\middle|\; g_v(x) \neq 0\right\} = \{v_1, \cdots, v_n\}.$$

Then, it follows that

$$E\left(\textstyle\bigcap_{i=1}^n U_{v_i}\right) = E(U_{v_1}) \cap \cdots \cap E(U_{v_n}) \supset \widetilde{U}_{v_1} \cap \cdots \cap \widetilde{U}_{v_n} \neq \emptyset,$$

which implies $\bigcap_{i=1}^n U_{v_i} \neq \emptyset$, and hence $\bigcap_{i=1}^n O_K(v_i) \neq \emptyset$. Therefore, $\tilde{f}(x) \in \langle v_1, \cdots, v_n\rangle \in K$. Thus, we have $\tilde{f}(x) \in |K|$. Each $x \in W$ has a neighborhood W_x in W that meets only finitely many $g_v^{-1}((0, \infty))$, $v \in K^{(0)}$, i.e., $\tilde{f}(W_x) \subset |L|$ for a finite subcomplex L of K. Since $\tilde{f}|W_x : W_x \to |L|_{\mathrm{m}} = |L|$ is continuous, so is $\tilde{f} : W \to |K|$. □

Remark 6.1.5 Since every metrizable ANE is an ANR, it follows that $|K|\,(= |K|_{\mathrm{m}})$ is an ANR for every locally finite simplicial complex K. If K is not locally finite, then there exists a map from a compactum to $|K|_{\mathrm{m}}$ that is not continuous with respect to the Whitehead topology. Indeed, K contains a subcomplex L such that $|L|_{\mathrm{m}}$ is homeomorphic to the hedgehog

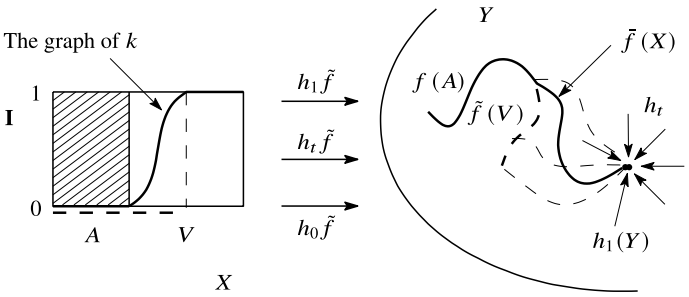


Fig. 6.1 Extending a map using a contraction

$$J(\mathbb{N}) = \left\{ x \in \ell_1 \cap \mathbf{I}^{\mathbb{N}} \;\middle|\; x(i) \neq 0 \text{ for at most one } i \in \mathbb{N} \right\} \subset \ell_1.$$

It is easy to construct a map $f : \mathbf{I} \to |L|_{\mathrm{m}} = J(\mathbb{N})$ such that $f(0) = f(1/(2i-1)) = \mathbf{0}$ and $f(1/2i) = 2^{-i}\mathbf{e}_i$ for each $i \in \mathbb{N}$. Then, $f : \mathbf{I} \to |L|$ ($\subset |K|$) is not continuous because $f(\mathbf{I})$ is not contained in the polyhedron of any finite subcomplex of L. In the next section, we will show that $|K|_{\mathrm{m}}$ is an ANR for an arbitrary simplicial complex K (Theorem 6.2.17).

Let X be a space and $\emptyset \neq A \subset X$. It is said that A is **contractible** in X if the inclusion map $A \subset X$ is null-homotopic, i.e., there is a homotopy $h : A \times \mathbf{I} \to X$ such that $h_0 = \mathrm{id}$ and h_1 is constant. Such a homotopy h is called a **contraction** of A in X. If $A = X$, we simply say that X is **contractible** and h is called a **contraction** of X. A space X is contractible if and only if X has the homotopy type of a singleton.

Theorem 6.1.5 *A contractible ANE is an AE.*

This theorem is a direct consequence of the following proposition:

Proposition 6.1.6 *Let $f : A \to Y$ be a map from a closed set A in a normal space X to a contractible space Y. If f extends over a neighborhood U of A in X, then f extends over X.*

Proof Let $h : Y \times \mathbf{I} \to Y$ be a contraction of Y and $\tilde{f} : U \to Y$ a map with $\tilde{f}|A = f$. Choose an open neighborhood V of A in X so that $\mathrm{cl}\, V \subset U$, and let $k : X \to \mathbf{I}$ be a Urysohn map with $k(A) = 0$ and $k(X \setminus V) = 1$. Then, we can extend f to a map $\bar{f} : X \to Y$ by $\bar{f}(X \setminus V) = h_1(Y)$ and

$$\bar{f}(x) = h(\tilde{f}(x), k(x)) \quad \text{for each } x \in \mathrm{cl}\, V.$$

See Fig. 6.1. □

By Theorem 5.1.6 (2) and the No Retraction Theorem 5.1.5, the unit sphere $\mathbf{S}^n$ is an ANE but not an AE. Then, we have the following corollary:

Corollary 6.1.7 *For every $n \in \omega$, the unit sphere $\mathbf{S}^n$ is not contractible.* □

For any full complex K, $|K|$ is contractible. Indeed, fixing $v_0 \in K^{(0)}$, we can define a contraction $h : |K| \times \mathbf{I} \to |K|$ by

$$\beta_v(h(x,t)) = (1-t)\beta_v(x) + t\beta_v(v_0),\ v \in K^{(0)}.$$

Thus, we have the following:

Corollary 6.1.8 *The polyhedron $|K|$ of a full complex K is an AE.* □

Recall that a tower $X_1 \subset X_2 \subset \cdots$ of spaces is said to be **closed** if each X_n is closed in X_{n+1}.

Theorem 6.1.9 *For a closed tower $Y_1 \subset Y_2 \subset \cdots$ of ANEs, the direct limit $Y = \varinjlim Y_n$ is also an ANE.*

Proof Let $f : A \to Y$ be a map from a closed set A in a metric space $X = (X, d)$. For each $n \in \mathbb{N}$, let $A_n = f^{-1}(Y_n)$ and define

$$X_n = \left\{ x \in X \;\middle|\; d(x, A_n) \leqslant d(x, A \setminus A_n) \right\}.$$

Then, $A = \bigcup_{n\in\mathbb{N}} A_n$, each X_n is closed in X, and $X_n \cap A = A_n$. By induction, we can obtain $N_1 \subset N_2 \subset \cdots \subset X$ and maps $f_n : A \cup N_n \to Y$ such that N_n is a closed neighborhood of A_n in X_n, $f_n(N_n) \subset Y_n$, and $f_n | A \cup N_{n-1} = f_{n-1}$, where $N_0 = \emptyset$, $f_0 = f$. Indeed, assume that f_n has been obtained. Since Y_{n+1} is an ANE and $A_{n+1} \cup N_n$ is a closed subset of X_{n+1}, $A_{n+1} \cup N_n$ has a closed neighborhood N_{n+1} in X_{n+1} such that $f_n | A_{n+1} \cup N_n : A_{n+1} \cup N_n \to Y_{n+1}$ extends to a map $f' : N_{n+1} \to Y_{n+1}$. The map f_{n+1} is defined by $f_{n+1}|A = f$ and $f_{n+1}|N_{n+1} = f'$. Then, we define $\tilde{f} : \bigcup_{n\in\mathbb{N}} \operatorname{int}_X N_n \to Y$ by $\tilde{f} | \operatorname{int}_X N_n = f_n | \operatorname{int}_X N_n$. If $A \subset \bigcup_{n\in\mathbb{N}} \operatorname{int}_X N_n$, then $\bigcup_{n\in\mathbb{N}} \operatorname{int}_X N_n$ is a neighborhood of A in X and the continuity of $\tilde{f}$ follows from that of each f_n.

Since each N_n is a neighborhood of A_n in X_n, it suffices to show that $A \subset \bigcup_{n\in\mathbb{N}} \operatorname{int}_X X_n$. On the contrary, assume that there is an $a \in A \setminus \bigcup_{n\in\mathbb{N}} \operatorname{int}_X X_n$. Then, a is contained in some A_k. For each $i \in \mathbb{N}$, we can choose $x_i \in X \setminus X_{k+i}$ so that $\lim_{i\to\infty} x_i = a$. By the definition of X_{k+i}, $d(x_i, a_i) < d(x_i, A_{k+i})$ for some $a_i \in A \setminus A_{k+i}$. Since $d(x_i, a_i) < d(x_i, a)$, it follows that $\lim_{i\to\infty} a_i = a$, hence $f(a) = \lim_{i\to\infty} f(a_i)$. On the other hand, $\{f(a_i) \mid i \in \mathbb{N}\}$ is closed in Y because each Y_{k+i} contains only finitely many points $f(a_j)$. Therefore, $f(a) = f(a_i)$ for some $i \in \mathbb{N}$, hence $f(a) \notin Y_{k+i}$, which is a contradiction. Thus, the proof is complete. □

Remark 6.1.6 In the above proof, it should be noted that $X \neq \bigcup_{n\in\mathbb{N}} X_n$ in general. Indeed, let $X = \mathbf{I}^2 \setminus \{\mathbf{0}\}$ be a metric space with the metric d induced from the Euclidean norm of $\mathbb{R}^2$, $A = (0, 1] \times \{0\}$ a closed set in X. We define a map $f : A \to \mathbb{R}$ by $f(x, 0) = x^{-1}$ for each $x \in (0, 1]$, where $\mathbb{R}$ is the direct limit of

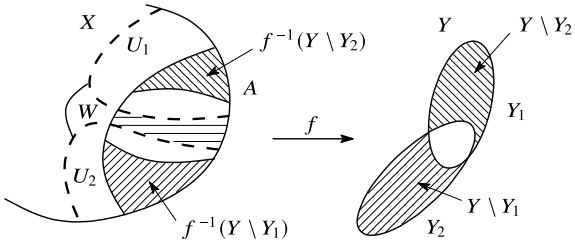


Fig. 6.2 The union of two open ANEs

a tower $[-2, 2] \subset [-2^2, 2^2] \subset [-2^3, 2^3] \subset \cdots$ of closed intervals. Then, $A_n = f^{-1}([-2^n, 2^n]) = [2^{-n}, 1] \times \{0\}$ and

$$X_n = \left\{x \in X \;\middle|\; d(x, A_n) \leqslant d(x, A \setminus A_n)\right\} = [2^{-n}, 1] \times \mathbf{I}.$$

Hence, $\bigcup_{n\in\mathbb{N}} X_n = (0, 1] \times \mathbf{I} \neq X$.

Now, we list the basic properties of ANEs, which can be easily proved.

Proposition 6.1.10 (Basic Properties of ANEs))

(1) *An arbitrary product of AEs is an AE and a finite product of ANEs is an ANE.*

Sketch of Proof Extend coordinate-wise.

(2) *A retract of an AE is an AE and a neighborhood retract of an ANE is an ANE.*

Sketch of Proof Compose an extension with a retraction.

(3) *Any open set in an ANE is also an ANE.*

Sketch of Proof Restrict an extension.

(4) *A topological sum of ANEs is an ANE.*

Sketch of Proof Let $f : A \to \bigoplus_{\lambda\in\Lambda} Y_\lambda$ be a map from a closed set A in a metrizable space X, where each Y_λ is an ANE. Extend each $f|f^{-1}(Y_\lambda)$ over a neighborhood in $E(f^{-1}(Y_\lambda))$, where E is the extension operator in the Canonical Open Set Extension Lemma 5.4.10.

(5) *Let $Y = Y_1 \cup Y_2$, where Y_i is open in Y. If Y_1 and Y_2 are ANEs, then so is Y.*

Sketch of Proof Let $f : A \to Y$ be a map of a closed set $A \subset X$. Separate $f^{-1}(Y_1 \setminus Y_2)$ and $f^{-1}(Y_2 \setminus Y_1)$ by open sets U_1 and U_2 in X. By extending $f| : A \setminus (U_1 \cup U_2) \to Y_1 \cap Y_2$ over a closed neighborhood W in $X \setminus (U_1 \cup U_2)$, we first extend f to $f' : A \cup W \to Y$. Then, extending $f'| : (A \cap U_i) \cup W \to Y_i$ over a neighborhood in $U_i \cup W$ $(i = 1, 2)$, we can obtain an extension of f. Refer to Fig. 6.2.

(6) (HANNER'S THEOREM) *A paracompact space is an ANE if it is locally an ANE, i.e., each point has an ANE neighborhood.*

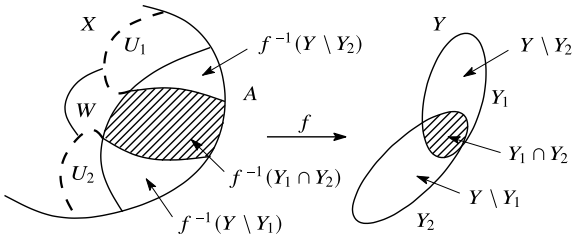


Fig. 6.3 The union of two closed ANEs

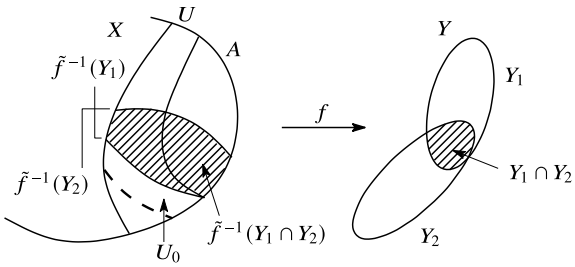


Fig. 6.4 Two closed sets whose union and intersection are ANEs

Sketch of Proof Applying Michael's Theorem 2.6.7 on local properties, the proof follows from (3), (4), and (5).

Note In the above, (4) and (5) are special cases of (6). In (5), it can be similarly proved that if Y_1, Y_2, and $Y_1 \cap Y_2$ are AEs, then so is Y. But, instead of assuming that $Y_1 \cap Y_2$ is an AE, it suffices to assume that $Y_1 \cap Y_2$ is contractible (see the above (3) and Theorem 6.1.5). Moreover, combining (6) with Theorem 6.1.1, every n-manifold (with boundary) is an ANE.[2] This will be pointed out again in Proposition 6.2.9 (4).

(7) *Let $Y = Y_1 \cup Y_2$, where Y_i is closed in Y. If Y_1, Y_2, and $Y_1 \cap Y_2$ are ANEs (AEs) then so is Y. If Y and $Y_1 \cap Y_2$ are ANEs (AEs), then so are Y_1 and Y_2.*

Sketch of Proof The first assertion is similar to (5). Now, U_1 and U_2 are disjoint open sets such that $U_1 \cap A = f^{-1}(Y_1 \setminus Y_2)$ and $U_2 \cap A = f^{-1}(Y_2 \setminus Y_1)$. See Fig. 6.3.

For the second assertion, let $f : A \to Y_1$ be a map of a closed set $A \subset X$. First, extend f to a map $\tilde{f} : U \to Y$ of a neighborhood U of A in X. Then, extending $\tilde{f}| : \tilde{f}^{-1}(Y_1 \cap Y_2) \to Y_1 \cap Y_2$ over a neighborhood U_0 of $\tilde{f}^{-1}(Y_1 \cap Y_2)$ in $\tilde{f}^{-1}(Y_2)$, we can extend $\tilde{f}|\tilde{f}^{-1}(Y_1)$ over $U_1 = \tilde{f}^{-1}(Y_1) \cup U_0$. See Fig. 6.4.

(8) *For a compactum X and an ANE (resp. AE) Y, the space $\mathrm{C}(X, Y)$ of all maps from X to Y with the compact-open topology is an ANE (resp. AE).*

Sketch of Proof Let Z be a metrizable space with C a closed set. A map $f : C \to \mathrm{C}(X, Y)$ induces a map $g : C \times X \to Y$ by Proposition 2.10.8 (4). Then, g extends over a neighborhood W of $C \times X$ in $Z \times X$. By the compactness of X, $U \times X \subset W$ for some neighborhood U of C in Z. The extension of g induces a map from U to $\mathrm{C}(X, Y)$ by Proposition 2.10.3.

[2] For the definition of an n-manifold, refer to Sect. 1.1.

(9) *For a pair (X, A) of compacta and a pair (Y, B) of ANEs (resp. AEs), the space $\mathrm{C}((X, A), (Y, B))$ of all maps from (X, A) to (Y, B) with the compact-open topology is an ANE (resp. AE), where B is not necessarily closed in Y.*

Sketch of Proof Let Z be a metrizable space with C a closed set. A map $f : C \to \mathrm{C}((X, A), (Y, B))$ induces a map $g : (C \times X, C \times A) \to (Y, B)$ (cf. Proposition 2.10.8(4)). As in (8), extend $g|C \times A$ to a map $g' : V \times A \to B$, where V is a closed neighborhood of C in Z. Define $g'' : (C \times X) \cup (V \times A) \to Y$ by $g''|C \times X = g$ and $g''|V \times A = g'$. Next, extend g'' over a neighborhood W of $(C \times X) \cup (V \times A)$ in $Z \times X$ and find a neighborhood U of C in Z so that $U \subset V$ and $U \times X \subset W$. The restriction of this extension of g'' to $U \times X$ induces an extension $\tilde{f} : U \to \mathrm{C}((X, A), (Y, B))$ of f (cf. Proposition 2.10.3).

Theorem 5.1.6 can be generalized as follows:

Theorem 6.1.11 *For every finite simplicial complex K, $|K|$ is an ANE for normal spaces. When $|K|$ is contractible, $|K|$ is an AE for normal spaces.*

Sketch of Proof Let $f : A \to |K|$ be a map from a closed set A in a normal space X. We can regard K as a subcomplex of a full complex F with $\mathrm{card}\, F^{(0)} = \mathrm{card}\, K^{(0)}$. Then, $|F|$ is an n-simplex, where $n + 1 = \mathrm{card}\, F^{(0)}$. Due to Proposition 4.3.12, K is a full subcomplex of $\mathrm{Sd}_K F$. We apply Proposition 4.12.1 to obtain a retraction $r : N \to |K|$ of a neighborhood N of $|K|$ in $|F|$. By Theorem 5.1.6 (1), f extends to a map $\tilde{f} : X \to |F|$. Then, $\tilde{f}^{-1}(N)$ is a neighborhood of A in X and $r\tilde{f}|\tilde{f}^{-1}(N)$ is an extension of f. For the additional statement, apply Proposition 6.1.6.

Combining Proposition 4.2.3 with Theorem 6.1.11 above, we can easily obtain the following:

Corollary 6.1.12 *For every simplicial complex K, $|K|$ is an ANE for compact spaces. When $|K|$ is contractible, $|K|$ is an AE for compact spaces.* □

Remark 6.1.7 In the above corollary, it can be asserted that every map $f : A \to |K|$ from a compact set A in a normal space X can be extended over a neighborhood of A in X. When $|K|$ is contractible, f can be extended over X.

6.2 Embeddings of Metric Spaces and ANRs

In this section, we first prove the following embedding theorem:

Theorem 6.2.1 (Arens–Eells Embedding Theorem) *Every (complete) metric space X can be isometrically embedded in a (complete) normed linear space E with* $\mathrm{dens}\, E = \aleph_0 \mathrm{dens}\, X$ *as a linearly independent closed set.*

Proof Let $X^* = X \cup \{y_0\}$, where $y_0 \notin X$. Fix a point $x_0 \in X$ and extend the metric d of X to a metric on X^* by defining

$$d(x, y_0) = d(y_0, x) = d(x, x_0) + 1 \quad \text{for } x \in X.$$

By $\mathrm{Fin}(X^*)$, we denote the set of all non-empty finite subsets of X^*, and by $\ell_\infty(\mathrm{Fin}(X^*))$ the Banach space of all bounded real functions on $\mathrm{Fin}(X^*)$ with the sup-norm $\|\cdot\|$. For each $x \in X^*$, we define $\varphi(x) \in \ell_\infty(\mathrm{Fin}(X^*))$ by

$$\varphi(x)(F) = d(x, F) - d(y_0, F) \quad \text{for} \quad F \in \mathrm{Fin}(X^*),$$

where $|\varphi(x)(F)| \leqslant d(x, y_0) = d(x, x_0) + 1$. Then, $\|\varphi(x) - \varphi(y)\| = d(x, y)$ for each $x, y \in X^*$. Indeed, for each $F \in \mathrm{Fin}(X^*)$,

$$\begin{aligned} |\varphi(x)(F) - \varphi(y)(F)| &= |d(x, F) - d(y, F)| \\ &\leqslant d(x, y) = |\varphi(x)(\{y\}) - \varphi(y)(\{y\})|. \end{aligned}$$

Thus, we have an isometry $\varphi : X^* \to \ell_\infty(\mathrm{Fin}(X^*))$, where it should be noted that $\varphi(y_0) = \mathbf{0}$. Let E be the linear subspace of $\ell_\infty(\mathrm{Fin}(X^*))$ spanned by $\varphi(X)$, i.e.,

$$E = \Big\{ \textstyle\sum_{i=1}^n \lambda_i \varphi(x_i) \;\Big|\; n \in \mathbb{N},\ \lambda_i \in \mathbb{R} \ \text{ and } \ x_i \in X \Big\}.$$

We show that $\mathrm{dens}\, E = \aleph_0 \,\mathrm{dens}\, X$. Take a dense set D in X with $\mathrm{card}\, D = \mathrm{dens}\, X$, and define

$$E_D = \Big\{ \textstyle\sum_{i=1}^n \lambda_i \varphi(x_i) \;\Big|\; n \in \mathbb{N},\ \lambda_i \in \mathbb{Q} \ \text{ and } \ x_i \in D \Big\}.$$

Then, E_D is dense in E and $\mathrm{card}\, E_D \leqslant \aleph_0 \,\mathrm{card}\, D = \aleph_0 \,\mathrm{dens}\, X$. Hence, we have $\mathrm{dens}\, E \leqslant \aleph_0 \,\mathrm{dens}\, X$. On the other hand, since X is embedded into E, it follows that $\aleph_0 \,\mathrm{dens}\, X \leqslant \mathrm{dens}\, E$.

We show that $\varphi(X)$ is closed in E. Let $f = \sum_{i=1}^m \lambda_i \varphi(y_i) \in E$ and $x_n \in X$, $n \in \mathbb{N}$, such that $\varphi(x_n)$ converges to f. Let $F = \{y_0, y_1, \cdots, y_m\} \in \mathrm{Fin}(X^*)$. Then, $d(x_n, F) = \varphi(x_n)(F)$ converges to

$$f(F) = \sum_{i=1}^m \lambda_i \varphi(y_i)(F) = \sum_{i=1}^m \lambda_i d(y_i, F) = 0.$$

Since F is finite and $d(y_0, X) = 1$, $(x_n)_{n\in\mathbb{N}}$ has a subsequence that converges to some $y \in F \setminus \{y_0\} \subset X$, so $f = \varphi(y) \in \varphi(X)$. Therefore, $\varphi(X)$ is closed in E. If X is complete, $\varphi(X)$ is closed in $\ell_\infty(\mathrm{Fin}(X^*))$, so is closed in the closure $\mathrm{cl}\, E$ of E in $\ell_\infty(\mathrm{Fin}(X^*))$.

Finally, we show that $\varphi(X)$ is linearly independent. Let $\sum_{i=1}^m \lambda_i \varphi(x_i) = 0$, where $\lambda_i \in \mathbb{R}$, $x_i \in X$, and $x_i \neq x_j$ if $i \neq j$. Then,

$$\sum_{i=1}^m \lambda_i (d(x_i, F) - d(y_0, F)) = 0 \quad \text{for each } F \in \mathrm{Fin}(X^*).$$

For each $i = 1, \ldots, m$, applying this to

$$F_i = \{y_0, x_1, \cdots, x_m\} \setminus \{x_i\} \in \mathrm{Fin}(X^*),$$

we have $\lambda_i d(x_i, F_i) = 0$, hence $\lambda_i = 0$ because $d(x_i, F_i) \neq 0$. □

Remark 6.2.1 According to Sketch of Proof of Theorem 2.3.10, every metric space X can be embedded into the Banach space $\mathrm{C}^B(X)$ by the isometry $\varphi : X \to \mathrm{C}^B(X)$ defined as $\varphi(x)(z) = d(x, z) - d(x_0, z)$, where $x_0 \in X$ is fixed. In this case, $\varphi(X)$ is closed in the convex hull $\langle\varphi(X)\rangle$.[3]

Indeed, for each $f \in \langle\varphi(X)\rangle \setminus \varphi(X)$, write $f = \sum_{i=1}^n t_i\varphi(x_i)$, where $n > 1$, $x_i \in X$, $t_i > 0$, and $\sum_{i=1}^n t_i = 1$. Choose $\delta > 0$ so that $2\delta < \|f - \varphi(x_i)\|$ for any $i = 1, \ldots, n$. Then, $\mathrm{B}(f, \delta) \cap \varphi(X) = \emptyset$. Otherwise, $\|f - \varphi(y)\| \leqslant \delta$ for some $y \in X$. Since

$$d(x_i, y) = \|\varphi(x_i) - \varphi(y)\| \geqslant \|f - \varphi(x_i)\| - \|f - \varphi(y)\| > 2\delta - \delta = \delta,$$

we have

$$\|f - \varphi(y)\| \geqslant |f(y) - \varphi(y)(y)| = \sum_{i=1}^n t_i d(x_i, y) > \delta,$$

which is a contradiction.

However, in general, $\varphi(X)$ is not closed in the flat hull (affine hull) $\mathrm{fl}\,\varphi(X)$, where $\mathrm{fl}\,\varphi(X)$ coincides with the linear span of $\varphi(X)$ because $\mathbf{0} = \varphi(x_0) \in \varphi(X)$. For example, let $X = [0, 1) \times \{0\} \cup \{p, q\} \subset \mathbb{R}^2$, where $p = (1/2, 1/2), q = (1/2, -1/2) \in \mathbb{R}^2$. Give X the metric d induced from the norm $\|x\|_\infty = \max\{|x(1)|, |x(2)|\}$ on $\mathbb{R}^2$ and $x_0 = \mathbf{0} \in X$. Then, $\varphi(s, 0)$ converges to $\varphi(p) + \varphi(q)$ as $s \to 1$ and $\varphi(p) + \varphi(q) \notin \varphi(X)$, which means that $\varphi(X)$ is not closed in the linear span of $\varphi(X)$.

From the definition, the following are easily verified:

$$\varphi(p)(x) = \begin{cases} 1/2 - t & \text{if } x = (t, 0), \\ -1/2 & \text{if } x = p, \\ 1/2 & \text{if } x = q, \end{cases} \qquad \varphi(q)(x) = \begin{cases} 1/2 - t & \text{if } x = (t, 0), \\ 1/2 & \text{if } x = p, \\ -1/2 & \text{if } x = q, \end{cases}$$

$$(\varphi(p) + \varphi(q))(x) = \begin{cases} 1 - 2t & \text{if } x = (t, 0), \\ 0 & \text{if } x = p, q, \end{cases}$$

$$\varphi(s, 0)(x) = \begin{cases} s - 2t & \text{if } x = (t, 0), t \leqslant s, \\ -s & \text{if } x = (t, 0), t > s, \\ 0 & \text{if } x = p, q, \end{cases}$$

The following is a very useful procedure to extend homeomorphisms.

Theorem 6.2.2 (KLEE'S TRICK) *Let E and F be metrizable topological linear spaces that are AEs and let A and B be closed sets in E and F, respectively. Then, each homeomorphism $f : A \times \{\mathbf{0}\} \to \{\mathbf{0}\} \times B$ extends to a homeomorphism $\tilde{f} : E \times F \to E \times F$.*

[3] See Footnote 9 of Sect. 2.3 and Notes for Chap. 2 and this chapter.

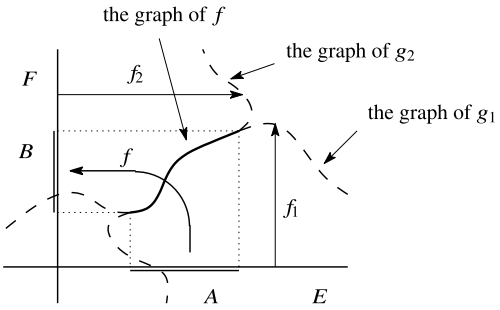


Fig. 6.5 Klee's trick

Proof Let $i_E : E \to E \times \{\mathbf{0}\} \subset E \times F$ and $i_F : F \to \{\mathbf{0}\} \times F \subset E \times F$ be the natural injections. Then, $\mathrm{pr}_F \circ f \circ i_E | A : A \to F$ and $\mathrm{pr}_E \circ f^{-1} \circ i_F | B : B \to E$ extend to maps $g_1 : E \to F$ and $g_2 : F \to E$, respectively. We define homeomorphisms $f_1, f_2 : E \times F \to E \times F$ as follows:

$$f_1(x, y) = (x, y + g_1(x)) \quad \text{and} \quad f_2(x, y) = (x + g_2(y), y).$$

Then, the homeomorphism $\tilde{f} = f_2^{-1} \circ f_1 : E \times F \to E \times F$ is an extension of f (cf. Fig. 6.5). Indeed, for each $x \in A$, $f(x, \mathbf{0}) = (\mathbf{0}, \mathrm{pr}_F f(x, \mathbf{0}))$ and then

$$\begin{aligned}
\tilde{f}(x, \mathbf{0}) &= f_2^{-1}(x, g_1(x)) = (x - g_2(g_1(x)), g_1(x)) \\
&= (x - \mathrm{pr}_E \circ f^{-1}(\mathbf{0}, \mathrm{pr}_F \circ f(x, \mathbf{0})), \mathrm{pr}_F \circ f(x, \mathbf{0})) \\
&= (x - \mathrm{pr}_E \circ f^{-1}(f(x, \mathbf{0})), \mathrm{pr}_F \circ f(x, \mathbf{0})) \\
&= (\mathbf{0}, \mathrm{pr}_F \circ f(x, \mathbf{0})) = f(x, \mathbf{0}).
\end{aligned}$$

□

Using Klee's Trick, we can easily prove HAUSDORFF'S METRIC EXTENSION THEOREM:

Theorem 6.2.3 (HAUSDORFF'S METRIC EXTENSION) *Let A be a closed set in a (completely) metrizable space X. Every admissible (complete) metric on A extends to an admissible (complete) metric on X.*

Proof By Theorem 6.2.1, we have a closed embedding $g : X \to E$ of X into a (complete) normed linear space $E = (E, \|\cdot\|_E)$. Let d be an admissible (complete) metric on A. Again, by Theorem 6.2.1, we have a closed isometry $h : A \to F$ of $A = (A, d)$ into a (complete) normed linear space $F = (F, \|\cdot\|_F)$. Since E and F are AEs by the Dugundji Extension Theorem 6.1.1, we can apply Theorem 6.2.2 to obtain a homeomorphism $f : E \times F \to E \times F$ such that

$$f(g(x), \mathbf{0}) = (\mathbf{0}, h(x)) \quad \text{for all } x \in A.$$

Let $i : E \to E \times \{\mathbf{0}\} \subset E \times F$ be the natural injection. Then, $f \circ i \circ g : X \to E \times F$ is a closed embedding of X into the product normed linear space $E \times F$ with the norm

$$\|(x, y)\| = \|x\|_E + \|y\|_F.$$

Since $f \circ i \circ g|A$ is an isometry with respect to d and $\|\cdot\|$, we can extend d to a (complete) metric $\tilde{d}$ on X as follows:

$$\tilde{d}(x, y) = \|f \circ i \circ g(x) - f \circ i \circ g(y)\|. \qquad \square$$

In Theorem 6.2.1, if an embedding is not required to be an isometry, we have the following theorem on embedding into a (pre-)Hilbert space:

Theorem 6.2.4 *Every completely metrizable space can be embedded in a Hilbert space with the same density as a closed set. Moreover, every metrizable space can be embedded in a pre-Hilbert space (that is, a linear subspace of a Hilbert space) with the same density as a closed set.*

Proof Let $X = (X, d)$ be a metric space. We may assume that X is infinite. For each $n \in \mathbb{N}$, X has a locally finite partition of unity $(f_\gamma)_{\gamma \in \Gamma(n)}$ such that $\operatorname{diam} \operatorname{supp} f_\gamma < 2^{-n}$ for each $\gamma \in \Gamma(n)$, which implies that $f_\gamma(x) f_\gamma(y) = 0$ if $d(x, y) \geqslant 2^{-n}$. Let $\Gamma = \bigcup_{n \in \mathbb{N}} \Gamma(n)$, where it can be assumed that $\Gamma(n) \cap \Gamma(m) = \emptyset$ if $n \neq m$. For each $\gamma \in \Gamma$, let $g_\gamma = 2^{-n} f_\gamma$, where $\gamma \in \Gamma(n)$. We define $h : X \to \ell_2(\Gamma)$ by $h(x) = (g_\gamma(x)^{\frac{1}{2}})_{\gamma \in \Gamma}$. Observe that $h(X)$ is contained in the unit sphere of $\ell_2(\Gamma)$. Then, the continuity of h follows from the continuity of g_γ, $\gamma \in \Gamma$ (Proposition 1.2.6). It is easy to show that

$$2^{-n} \leqslant d(x, y) \ (< 2^{-n+1}) \ \Rightarrow \ \|h(x) - h(y)\|^2 \geqslant 2^{-n+1} \ (> d(x, y)).$$

Hence, the inverse h^{-1} of h is uniformly continuous. Thus, h is an embedding. Since $(f_\gamma)_{\gamma \in \Gamma(n)}$ is locally finite, we have $\operatorname{card} \Gamma(n) \leqslant w(X) = \operatorname{dens} X$, hence $\operatorname{card} \Gamma \leqslant \operatorname{dens} X$. Observe that

$$\operatorname{dens} X \leqslant \operatorname{dens} \ell_2(\Gamma) = \operatorname{card} \Gamma \leqslant \operatorname{dens} X,$$

so we have $\operatorname{dens} \ell_2(\Gamma) = \operatorname{dens} X$. When X is complete, $h(X)$ is also complete, so closed in $\ell_2(\Gamma)$.

When X is not complete, let E be the linear subspace of $\ell_2(\Gamma)$ generated by $h(X)$. We now show that $h(X)$ is closed in E. Let $y \in \operatorname{cl}_E h(X)$. Then, $y = \sum_{i=1}^m t_i h(x_i)$ for some $x_i \in X$ and $t_i \in \mathbb{R}$. We may assume that $x_i \neq x_j$ if $i \neq j$ and $t_i \neq 0$ for all i. Suppose that $m > 1$. Choose $n \in \mathbb{N}$ so that $d(x_i, x_j) \geqslant 2^{-n+1}$ if $i \neq j$. For each $i = 1, \ldots, m$, we have $\gamma_i \in \Gamma(n)$ such that $f_{\gamma_i}(x_i) > 0$, i.e., $h(x_i)(\gamma_i) > 0$. If $i \neq j$ then $f_{\gamma_i}(x_j) = 0$, i.e., $h(x_j)(\gamma_i) = 0$. Hence, $y(\gamma_i) > 0$. Since $y \in \operatorname{cl} h(X)$, we have $x \in X$ such that $h(x)(\gamma_i) > 0$, i.e., $f_{\gamma_i}(x) > 0$ for all i. This implies that $d(x, x_i) < 2^{-n}$ for each i, hence $d(x_1, x_2) < 2^{-n+1}$, which is a contradiction. Therefore, $y = t_1 h(x_1)$.

Observe that $y(\gamma) \geqslant 0$ for every $\gamma \in \Gamma$ and $\|y\| = 1$. Thus, we have $y = h(x_1)$. Hence, $h(X)$ is closed in E. □

Note By Corollary 3.8.12 and the Arens–Eells Embedding Theorem 6.2.1, every completely metrizable space X can be embedded in $\ell_1(\Gamma)$ as a closed set, where $\operatorname{card}\Gamma = \aleph_0$ dens X. Recall $\ell_1(\Gamma) \approx \ell_2(\Gamma)$ (Mazur's Theorem 1.2.7). Thus, we have an alternative proof for Theorem 6.2.4 for completely metrizable spaces.

As is easily observed, a metrizable AE (resp. ANE) is an AR (resp. ANR). The converse is also true by the Arens–Eells Embedding Theorem 6.2.1 (or Theorem 6.2.4), the Dugundji Extension Theorem 6.1.1, and Proposition 6.1.10 (2). Thus, we have the following theorem:

Theorem 6.2.5 *For a metrizable space X, the following hold:*

(1) *X is an AR if and only if X is an AE.*
(2) *X is an ANR if and only if X is an ANE.* □

Since every separable completely metrizable space can be embedded into the space $\mathbb{R}^{\mathbb{N}}$ as a closed set (Theorem 2.5.4), the following is obtained from Theorem 6.1.3 (the detail is left to the reader):

Theorem 6.2.6 *Every separable completely metrizable ANR (resp. AR) is an ANE (resp. AE) for normal spaces.* □

The following is a non-separable version of the above theorem, which generalizes the last statement of Remark 6.1.4:

Theorem 6.2.7 *Every completely metrizable ANR (resp. AR) is an ANE (resp. AE) for paracompact spaces.*

To prove Theorem 6.2.7, we need the following lemma:

Lemma 6.2.8 *Let X be a paracompact space and A a closed set in X. For each continuous pseudo-metric ρ on A, there is a continuous pseudo-metric ϱ on X such that for each $x, y \in A$ and $n \in \mathbb{N}$, $\varrho(x, y) < 2^{-n}$ implies $\rho(x, y) < 2^{-n}$.*

Proof Since ρ is continuous, $\mathrm{B}_\rho(x, r)$ is an open set in A for each $x \in A$ and $r > 0$. Let $\mathcal{U}_0 = \{X \setminus A\} \cup \{U_x \mid x \in A\}$, where each U_x is an open set in X with $U_x \cap A = \mathrm{B}_\rho(x, 2^{-1})$. Since X is paracompact, using Theorem 2.6.3 (c), we can easily construct open covers $\mathcal{U}_n \in \operatorname{cov}(X), n \in \mathbb{N}$, so that $\mathcal{U}_n|A \prec \{\mathrm{B}_\rho(x, 2^{-n-1}) \mid x \in A\}$ and $\operatorname{st}\mathcal{U}_n \prec \mathcal{U}_{n-1}$, Then, as mentioned immediately before Proposition 2.4.5, there is a continuous pseudo-metric ϱ on X such that

$$\operatorname{st}(x, \mathcal{U}_n) \subset \overline{\mathrm{B}}_\varrho(x, 2^{-n}) \subset \operatorname{st}(x, \mathcal{U}_{n-1}) \quad \text{for each } x \in X \text{ and } n \in \mathbb{N},$$

Hence, for each $x \in A$,

$$\overline{\mathrm{B}}_\varrho(x, 2^{-n}) \cap A \subset \operatorname{st}(x, \mathcal{U}_{n-1}|A) \subset \bigcup_{\rho(y,x)<2^{-n-1}} \mathrm{B}_\rho(y, 2^{-n-1}) \subset \mathrm{B}_\rho(x, 2^{-n}),$$

which implies that for each $x, y \in A$, $\varrho(x, y) < 2^{-n}$ implies $\rho(x, y) < 2^{-n}$. □

Proof (*Theorem* 6.2.7) Let Y be an ANR with an admissible complete metric d and $f : A \to Y$ a map from a closed set A in a paracompact space X. Then, A has a continuous pseudo-metric ρ defined as follows:

$$\rho(x, y) = d(f(x), f(y)) \text{ for each } x, y \in A.$$

By Lemma 6.2.8, X has a continuous pseudo-metric ϱ such that

$$\forall x, y \in A,\ \forall n \in \mathbb{N},\ \varrho(x, y) < 2^{-n} \ \Rightarrow\ \rho(x, y) < 2^{-n}.$$

Consider the quotient $X^* = X/\sim$, where $\sim$ is the equivalence relation defined as follows:

$$x \sim y \underset{\text{def}}{\Leftrightarrow} \varrho(x, y) = 0.$$

Let $q : X \to X^*$ be the natural surjection, that is, $q(x) = \{y \in X \mid \varrho(x, y) = 0\}$. Then, X^* has the metric ϱ^* induced by ϱ, that is, $\varrho^*(q(x), q(y)) = \varrho(x, y)$ for each $x, y \in X$. For $x, y \in A$, $\varrho(x, y) = 0$ implies $d(f(x), f(y)) = \rho(x, y) = 0$. Hence, we can define $f' : q(A) \to Y$ by $f'(q(x)) = f(x)$ for each $x \in A$. Then, f' is continuous with respect to ϱ^*. Indeed, for each $x, y \in A$,

$$\begin{aligned}\varrho^*(q(x), q(y)) = \varrho(x, y) < 2^{-n}\\ \Rightarrow\ d(f'(q(x)), f'(q(y))) = d(f(x), f(y)) = \rho(x, y) < 2^{-n}.\end{aligned}$$

Note that $q : X \to X^* = (X^*, \varrho^*)$ is continuous but $q(A)$ is not closed in X^* in general. We will extend f' to a map $f'' : A^* = \mathrm{cl}_{X^*} q(A) \to Y$ as follows: For each $x \in X$ with $q(x) \in A^* = \mathrm{cl}_{X^*} q(A)$, take $x_n \in A$, $n \in \mathbb{N}$, so that $q(x) = \lim_{n\to\infty} q(x_n)$. Since $(q(x_n))_{n\in\mathbb{N}}$ is ϱ^*-Cauchy, for each $n \in \mathbb{N}$, we have $m \in \mathbb{N}$ such that $i, j \geqslant m$ implies $\varrho(x_i, x_j) = \varrho^*(q(x_i), q(x_j)) < 2^{-n}$, hence $\rho(x_i, x_j) = d(f(x_i), f(x_j)) < 2^{-n}$. Hence, $(f'(q(x_n))_{n\in\mathbb{N}} = (f(x_n))_{n\in\mathbb{N}}$ is d-Cauchy, so it is convergent in Y because d is complete. Then, $f''(x) = \lim_{n\to\infty} f(x_n)$ is determined independently from choice of $x_n \in A$, $n \in \mathbb{N}$.

> Indeed, take other $x'_n \in A$, $n \in \mathbb{N}$, so that $q(x) = \lim_{n\to\infty} q(x'_n)$. For each $n \in \mathbb{N}$, choose $m \in \mathbb{N}$ so that $i \geqslant m$ implies $\varrho^*(q(x), q(x_i)) < 2^{-n-1}$ and $\varrho^*(q(x), q(x'_i)) < 2^{-n-1}$, hence $\varrho^*(q(x_i), q(x'_i)) < 2^{-n}$, which implies $d(f(x_i), f(x'_i)) = \rho(x_i, x'_i) < 2^{-n}$. Hence, we have $\lim_{n\to\infty} f(x_n) = \lim_{n\to\infty} f(x'_n)$.

Thus, we have a map $f'' : A^* \to Y$ from a closed set A^* in X^*, where $f''q|A = f'q|A = f$. Since Y is an ANE (for metrizable spaces), f'' extends to a map $f^* : U \to Y$ from an open neighborhood U of A^* in X^*. Then, $q^{-1}(U)$ is an open neighborhood of A in X and $f^*q|q^{-1}(U) : q^{-1}(U) \to Y$ is an extension of f. Therefore, X is an ANE for paracompact spaces.

In the above, if Y is an AR, we can take $U = X^*$. Then, f extend over $q^{-1}(U) = X$. Consequently, X is an AE for paracompact spaces. □

Note Theorems 6.2.6 and 6.2.7 can be strengthened as follows:

- *An ANR (resp. AR) X is an ANE (resp. AE) for normal spaces if and only if X is separable completely metrizable;*
- *An ANR (resp. AR) X is an ANE (resp. AE) for paracompact spaces if and only if X is completely metrizable.*

For proofs of the above "only if" parts and other analogous results, refer to Chap. II, Sect. 16 of Hu's book *Theory of Retracts* (cf. Notes for this chapter).

By applying Theorem 6.2.5, Proposition 6.1.10 can be translated as follows:

Proposition 6.2.9 (Basic Properties of ANRs)

(1) *A countable product of ARs is an AR and a finite product of ANRs is an ANR.*

The metrizability requires a countable product.

(2) *A retract of an AR is an AR and a neighborhood retract of an ANR is an ANR.*
(3) *Any open set in an ANR is also an ANR.*
(4) (HANNER'S THEOREM) *A paracompact space is an ANR if it is locally an ANR, that is, each point has an ANR neighborhood.*

In the above, the metrizability of X follows from Proposition 2.6.9 (5).

Note As a direct consequence of (4) above, every n-manifold is an ANR (cf. Note after Proposition 6.1.10 (6)).[4] Moreover, as in Note after Proposition 6.1.10 (6), we can assert that if $X = X_1 \cup X_2$, X_1 and X_2 are open sets in X being ARs, and $X_1 \cap X_2$ is contractible, then X is an AR.

(5) *Let $X = X_1 \cup X_2$, where X_i is closed in X, $i = 1, 2$. If X_1, X_2, and $X_1 \cap X_2$ are ANRs (ARs) then so is X. If X and $X_1 \cap X_2$ are ANRs (ARs), then so are X_1 and X_2.*

Sketch of Proof In the first assertion, the metrizability of X follows from Proposition 2.4.4 (2). The second assertion can also be proved by showing that X_1 and X_2 are neighborhood retracts of X (cf. (2)).

(6) *The space $\mathrm{C}(X, Y)$ of all maps from a compactum X to an ANR (resp. AR) Y with the compact-open topology is an ANR (resp. AR).*

For the metrizability of $\mathrm{C}(X, Y)$, refer to Proposition 2.10.8 (6).

(7) *For a pair (X, A) of compacta and a pair (Y, B) of ANRs (resp. ARs), the space $\mathrm{C}((X, A), (Y, B))$ of all maps from (X, A) to (Y, B) with the compact-open topology is an ANR (resp. AR), where B is not necessarily closed in Y.*

[4] The definition of an n-manifold is in Sect. 1.1.

Let X be a subspace of Y. A homotopy $h : X \times \mathbf{I} \to Y$ with $h_0 = \mathrm{id}$ is called a **deformation** of X in Y. When $X = Y$, h is called a **deformation** of X. A subset A of X is said to be a **deformation retract** of X if there is a deformation $h : X \times \mathbf{I} \to X$ such that h_1 is a retraction of X onto A, where h_1 is called a **deformation retraction** of X onto A. When $h_t|A = \mathrm{id}$ for all $t \in \mathbf{I}$, we call A a **strong deformation retract** of X and h_1 a **strong deformation retraction** of X onto A. A deformation retraction (resp. a strong deformation retraction) $r : X \to A\ (\subset X)$ is a retraction with $r \simeq \mathrm{id}_X$ (resp. $r \simeq \mathrm{id}_X$ rel. A). A *closed* set A in X is called a **neighborhood deformation retract** of X if A has a neighborhood U in X with a deformation $h : U \times \mathbf{I} \to X$ such that h_1 is a retraction of U onto A, where h_1 is called a **deformation retraction** of U onto A in X. When $h_t|A = \mathrm{id}$ for all $t \in \mathbf{I}$, we call A a **strong neighborhood deformation retract** of X and h_1 a **strong deformation retraction** of U onto A in X. Moreover, it is said that X is **deformable into** $A\ (\subset X)$ if there is a deformation $h : X \times \mathbf{I} \to X$ with $h_1(X) \subset A$. The following is easily proved:

Proposition 6.2.10 *A subset A of a space X is a deformation retract of X if and only if X is deformable into A and A is a retract of X.*

Sketch of Proof To see the "if" part, let $h : X \times \mathbf{I} \to X$ be a deformation with $h_1(X) \subset A$ and let $r : X \to A$ be a retraction. Using the fact that $rh_1 = h_1$, we can define a homotopy from id_X to r.

Proposition 6.2.11 *Any retract of an AR is a strong deformation retract and any neighborhood retract of an ANR is a strong neighborhood deformation retract. Consequently, a closed set of an AR (ANR) is a strong (neighborhood) deformation retract if it is an AR (ANR).*

Proof Let X be an ANR and A be a closed set in X with $r : U \to A$ a retraction of an open neighborhood U of A in X. Since X is an ANE by Theorem 6.2.5, we have a neighborhood W of $U \times \{0, 1\} \cup A \times \mathbf{I}$ in $U \times \mathbf{I}$ and a map $h : W \to X$ such that $h(x, 0) = x$, $h(x, 1) = r(x)$ for $x \in U$, and $h(x, t) = x$ for $x \in A$. Choose a neighborhood V of A in U so that $V \times \mathbf{I} \subset W$. Then, $h|V \times \mathbf{I}$ is the desired homotopy. When X is an AR, letting $U = X$, we can take $W = X \times \mathbf{I}$ in the above. □

In particular, every singleton in an AR X is a strong deformation retract of X and every singleton in an ANR X is a strong neighborhood deformation retract of X.

Remark 6.2.2 By Proposition 6.2.11, every (deformation) retract of an ANR is a strong *neighborhood* deformation retract. In fact, every deformation retract of an ANR is a strong deformation retract, which will be proved in Sect. 6.4 (Theorem 6.4.4).

A space X is said to be **locally contractible** if each neighborhood U of any point $x \in X$ contains a neighborhood V of x that is contractible in U. Then, the following is easy:

Proposition 6.2.12 *A space X is locally contractible if every singleton in X is a strong neighborhood deformation retract of X. Moreover, X is contractible if some singleton in X is a strong deformation retract of X.*

Sketch of Proof For the first assertion, assume that every $x \in X$ has a neighborhood W with a homotopy $h : W \times \mathbf{I} \to X$ such that $h_0 = \mathrm{id}$, $h_1(W) = \{x\}$, and $h(\{x\} \times \mathbf{I}) = \{x\}$ (i.e., h is a contraction of W in X with $h_1(W) = \{x\} = h(\{x\} \times \mathbf{I})$). As is easily observed, each neighborhood U of x contains a neighborhood V of x such that $V \subset W$ and $h(V \times \mathbf{I}) \subset U$, hence X is locally contractible.

Combining Propositions 6.2.11 and 6.2.12, we have the following:

Corollary 6.2.13 *Every ANR is locally contractible. Additionally, every AR is contractible.* □

By Theorems 6.1.5 and 6.2.5, a contractible ANR is an AR. Thus, we have also the following characterization of ARs:

Corollary 6.2.14 *A space is an AR if and only if it is a contractible ANR.* □

When X is not compact, Proposition 6.2.9 (6) does not hold in general, that is:

Proposition 6.2.15 *There exists a locally compact separable metrizable space X such that the space $\mathrm{C}(X, \mathbf{S}^1)$ with the compact-open topology is not locally path-connected, so it is not an ANR.*

Proof (*Example*) Let $X = \bigcup_{i \in \mathbb{N}} (\mathbf{S}^1 + 2i\mathbf{e}_1)$ be a subspace of $\mathbb{R}^2$ and define a map $f : X \to \mathbf{S}^1$ by

$$f(x + 2i, y) = ((-1)^i x, y) \quad \text{for every } (x, y) \in \mathbf{S}^1.$$

For each $n \in \mathbb{N}$, let $X_n = \bigcup_{i=1}^{n} (\mathbf{S}^1 + 2i\mathbf{e}_1)$ and $r_n : X \to X_n$ be the retraction defined by $r_n(X \setminus X_n) = \{(2n + 1)\mathbf{e}_1\}$. Each neighborhood $\mathcal{U}$ of f in $\mathrm{C}(X, \mathbf{S}^1)$ contains some fr_n. But $fr_n \not\simeq f$ in $\mathbf{S}^1$, which means that fr_n and f cannot be connected by any path in $\mathrm{C}(X, \mathbf{S}^1)$. □

Remark 6.2.3 For a separable polyhedron X and an ANR Y, it is known that $\mathrm{C}(X, Y)$ is an ANR if and only if it has the homotopy type of an ANR (equiv. the homotopy type of a simplicial complex—Corollary 6.6.5).

When Y is an AR, Proposition 6.2.9 (6) holds even if X is not compact, that is:

Proposition 6.2.16 *For a locally compact separable metrizable space X and an AR Y, the space $\mathrm{C}(X, Y)$ of all maps from X to Y with the compact-open topology is an AR.*

Sketch of Proof Regard Y as a retract of a normed linear space E by the Arens–Eells Embedding Theorem 6.2.1. Then, $\mathrm{C}(X, Y)$ can be regarded as a retract of $\mathrm{C}(X, E)$ by Proposition 2.10.8 (1), where $\mathrm{C}(X, E)$ is a locally convex topological linear space by Remark 6.1.1, and hence is an AE by the Dugundji Extension Theorem 6.1.1. Since $\mathrm{C}(X, E)$ is metrizable by Proposition 2.10.8 (6), it is an AR, so $\mathrm{C}(X, Y)$ is an AR.

Note Proposition 6.2.9 (6) can be directly proved in the same way as the above.

The following theorem corresponds to Theorem 6.1.4:

Theorem 6.2.17 *For any simplicial complex K, $|K|_{\mathrm{m}}$ is an ANR.*

Proof We can regard K as a subcomplex of the full simplicial complex F with $F^{(0)} = K^{(0)}$. Note that the canonical representation $\beta^F : |F| \to \ell_1(F^{(0)})$ is an isometry with respect to the metric ρ_F (cf. Sect. 4.6) and $\beta^F(|F|)$ is a convex set in $\ell_1(F^{(0)})$. Then, $|F|_{\mathrm{m}}$ is an AE by Theorem 6.1.1. Combining Proposition 6.1.10 (2) with Theorem 6.2.5 (2), it suffices to show that $|K|_{\mathrm{m}}$ is a neighborhood retract of $|F|_{\mathrm{m}}$ (refer to Proposition 6.2.9 (2)).

By Theorem 4.7.1, $|K|_{\mathrm{m}} = |\operatorname{Sd}^2 K|_{\mathrm{m}}$ and $|F|_{\mathrm{m}} = |\operatorname{Sd}^2 F|_{\mathrm{m}}$ as spaces. Let

$$N = \bigcup_{v \in (\operatorname{Sd}^2 K)^{(0)}} \operatorname{St}(v, \operatorname{Sd}^2 F).$$

Then, N is a subcomplex of $\operatorname{Sd}^2 F$ and $|N|$ is a neighborhood of $|K|$ in $|F|_{\mathrm{m}}$. Each vertex $v \in N^{(0)}$ is the barycenter of $\sigma \in \operatorname{Sd} F$ with $\sigma \cap |K| \neq \emptyset$, where it should be noted that $\sigma \cap |K| = \langle \sigma^{(0)} \cap |K| \rangle \in \operatorname{Sd} K$. Let $r_0(v)$ be the barycenter of $\sigma \cap |K|$. Then, for $v_1, \ldots, v_n \in N^{(0)}$,

$$\langle v_1, \ldots, v_n \rangle \in N \Rightarrow \langle r_0(v_1), \ldots, r_0(v_n) \rangle \in \operatorname{Sd}^2 K.$$

Hence, the function $r_0 : N^{(0)} \to (\operatorname{Sd}^2 K)^{(0)}$ defines the simplicial map $r : N \to \operatorname{Sd}^2 K$. By definition, $r|\operatorname{Sd}^2 K = \mathrm{id}$, which means that r is a retraction. Thus, $|K|_{\mathrm{m}}$ is a neighborhood retract of $|F|_{\mathrm{m}}$. This completes the proof. □

In the above, if K is locally finite, then $|K| = |K|_{\mathrm{m}}$ is locally compact, and if K contains no infinite full complexes as subcomplexes, then $|K|_{\mathrm{m}}$ is completely metrizable (Theorem 4.6.5). Hence, combining Theorem 6.2.17 above with Theorems 6.2.6 and 6.2.7, we have the following:

Corollary 6.2.18 *Let K be a simplicial complex.*

(1) *If K is countable and locally finite, then $|K| = |K|_{\mathrm{m}}$ is an ANE for normal spaces.*
(2) *If K contains no infinite full complexes as subcomplexes, then $|K|_{\mathrm{m}}$ is an ANE for paracompact spaces.* □

For a compact polyhedron X and a polyhedron Y, we now consider the space $\mathrm{PL}(X, Y)$ of PL maps from X to Y with the compact-open topology. Let L be a triangulation of Y. Since $\mathrm{id} : |L| \to |L|_{\mathrm{m}}$ is continuous, it follows that $\mathrm{C}(X, |L|) \subset \mathrm{C}(X, |L|_{\mathrm{m}})$ as sets and the inclusion $\mathrm{C}(X, |L|) \subset \mathrm{C}(X, |L|_{\mathrm{m}})$ is continuous. The space $\mathrm{PL}(X, Y)$ is a subspace of $\mathrm{C}(X, Y) = \mathrm{C}(X, |L|)$. Let $\mathrm{PL}(X, |L|_{\mathrm{m}})$ denote the space $\mathrm{PL}(X, Y)$ with the subspace topology inherited from $\mathrm{C}(X, |L|_{\mathrm{m}})$.[5] Then,

[5] Note that the set $\mathrm{PL}(X, Y) = \mathrm{PL}(X, |L|_{\mathrm{m}})$ depends on the topology of Y but the topology for $\mathrm{PL}(X, |L|_{\mathrm{m}})$ is defined by using the topology of $|L|_{\mathrm{m}}$. Refer to Remark 4.11.1.

$\mathrm{PL}(X, Y) \neq \mathrm{PL}(X, |L|_{\mathrm{m}})$ as spaces if Y is not locally compact (equivalently, L is not locally finite). Indeed, for each $y \in Y$, let $c_y \in \mathrm{PL}(X, Y)$ be the constant map with $c_y(X) = \{y\}$. As is easily observed, Y can be embedded in $\mathrm{PL}(X, Y)$ by the correspondence $Y \ni y \mapsto c_y \in \mathrm{PL}(X, Y)$, which means that $\mathrm{PL}(X, Y)$ is non-metrizable. But, $\mathrm{PL}(X, |L|_{\mathrm{m}})$ is metrizable because so is $\mathrm{C}(X, |L|_{\mathrm{m}})$ (Proposition 2.10.2 (2)). When Y is locally compact (equivalently, L is locally finite), since $Y = |L| = |L|_{\mathrm{m}}$ as spaces, $\mathrm{C}(X, |L|) = \mathrm{C}(X, |L|_{\mathrm{m}})$ as spaces, so $\mathrm{PL}(X, |L|) = \mathrm{PL}(X, |L|_{\mathrm{m}})$.

Note If $\dim X > 0$ and Y is not locally compact (i.e., L is not locally finite), then $\mathrm{C}(X, |L|) \subsetneqq \mathrm{C}(X, |L|_{\mathrm{m}})$. Indeed, X contains infinite distinct points $x_n \in X, n \in \omega$, such that $\lim_{n\to\infty} x_n = x_0$, and L has infinite distinct vertices $v_n \in L^{(0)}, n \in \omega$, such that $\langle v_0, v_n\rangle \in L$ for every $n \in \mathbb{N}$. Then, $L_0 = \bigcup_{n\in\mathbb{N}} F(\langle v_0, v_n\rangle)$ is a subcomplex of L. For each $n \in \mathbb{N}$, let

$$y_n = (1 - n^{-1})v_0 + n^{-1}v_n \quad \text{for each } n \in \mathbb{N}.$$

Since X is metrizable and $|L_0|_{\mathrm{m}}$ is an AR (6.1.4 and 6.1.5), we have a map $f : X \to |L_0|_{\mathrm{m}} \subset |L|_{\mathrm{m}}$ such that $f(x_0) = v_0$ and $f(x_n) = y_n$ for each $n \in \mathbb{N}$. However, $f : X \to |L|$ is not continuous because $(f(x_n))_{n\in\mathbb{N}}$ does not converge to $v_0 = f(x_0)$ in $|L_0|$.

For the space $\mathrm{PL}(X, Y)$ with the compact-open topology, we have the following:

Proposition 6.2.19 *Let X be a compact polyhedron and Y a locally compact polyhedron. Then, $\mathrm{PL}(X, Y)$ is an ANR.*

Proof Let K and L be triangulations of X and Y, respectively. We have the canonical representation $\beta^L : |L| \to \ell_1(L^{(0)})$ of L (cf. Sect. 4.6). Identifying L with the simplicial complex $\beta^L(L)$ in $\ell_1(L^{(0)})$, we can regard L as a subcomplex of the full complex $F = \mathrm{F}_{L^{(0)}}$. Then, $\mathrm{PL}(X, |F|_{\mathrm{m}})$ is a subspace of the Banach space $\mathrm{C}(X, \ell_1(L^{(0)}))$ with the sup-norm because the compact-open topology on $\mathrm{C}(X, |F|_{\mathrm{m}})$ coincides with the uniform convergence topology (Proposition 2.10.2 (2)). Since F is a full complex, a function $f : X \to |F|$ is PL if and only if f satisfies the following condition:

- $\exists K' \lhd K$ such that $\forall \sigma \in K'$, $f|\sigma$ is affine,

where $f(\sigma)$ is contained in some $\tau \in F$. Then, it is easy to see that $\mathrm{PL}(X, |F|_{\mathrm{m}})$ is a convex set in $\mathrm{C}(X, \ell_1(L^{(0)}))$. Hence, $\mathrm{PL}(X, |F|_{\mathrm{m}})$ is an AR. As shown in the proof of Theorem 6.2.17, we have a subcomplex N of $\mathrm{Sd}^2 F$ and a simplicial map $r : N \to \mathrm{Sd}^2 L$ such that $|N|$ is a neighborhood of $|L|$ in $|\mathrm{Sd}^2 F|_{\mathrm{m}} = |F|_{\mathrm{m}}$ and $r|\mathrm{Sd}^2 L = \mathrm{id}$, i.e., r is a retraction. Then, r induces a retraction

$$r_* : \mathrm{PL}(X, |N|_{\mathrm{m}}) \to \mathrm{PL}(X, |\mathrm{Sd}^2 L|_{\mathrm{m}}) = \mathrm{PL}(X, Y),$$

which is defined by $r_*(f) = rf$ for each $f \in \mathrm{PL}(X, |N|_{\mathrm{m}})$. Thus, $\mathrm{PL}(X, Y)$ is a neighborhood retract of $\mathrm{PL}(X, |F|_{\mathrm{m}})$, which is an ANR. □

6.3 Small Homotopies and LEC Spaces

For a space X, let Δ_X denote the diagonal of X^2, that is,

$$\Delta_X = \left\{(x, x) \;\middle|\; x \in X\right\} \subset X^2.$$

For each $A \subset X^2$ and $x \in X$, we define $A(x) = \{y \in X \mid (x, y) \in A\}$. Each neighborhood U of Δ_X in X^2 gives every $x \in X$ its neighborhood $U(x)$ simultaneously. Given an open cover $\mathcal{U}$ of X, we have an open neighborhood $W = \bigcup_{U \in \mathcal{U}} U^2$ of Δ_X in X^2, where $W(x) = \mathrm{st}(x, \mathcal{U})$ for each $x \in X$. Such open neighborhoods of Δ_X in X^2 form a neighborhood basis of Δ_X. Indeed, let U be an open neighborhood of Δ_X in X^2. Each $x \in X$ has an open neighborhood V_x in X such that $V_x^2 \subset U$. Thus, we have an open cover $\mathcal{V} = \{V_x \mid x \in X\}$ such that $\bigcup_{x \in X} V_x^2 \subset U$.

A space X said to be **locally equi-connected** (**LEC**) if the diagonal Δ_X of X^2 has a neighborhood U and there is a map $\lambda : U \times \mathbf{I} \to X$ such that

$$\lambda(x, y, 0) = x \;\text{ and }\; \lambda(x, y, 1) = y \;\text{ for each } (x, y) \in U, \text{ and}$$
$$\lambda(x, x, t) = x \;\text{ for each } x \in X \text{ and } t \in \mathbf{I},$$

where $\mathbf{I} \ni t \mapsto \lambda(x, y, t) \in X$ is a path from x to y in X. Such a map λ is called an **equi-connecting map** for X. If $U = X^2$, X is said to be **equi-connected** (**EC**). For example, every convex set X in a topological linear space is EC, where a natural EC map λ is defined by

$$\lambda(x, y, t) = (1 - t) \cdot x + t \cdot y.$$

In particular, every topological linear space is EC. More generally, a contractible topological group X is also EC and a semi-locally contractible topological group X is LEC, where X is said to be **semi-locally contractible** if each point of X has a neighborhood that is contractible in X. In fact, let $\varphi : V \times \mathbf{I} \to X$ be a contraction of a neighborhood of the unit $1 \in X$ ($V = X$ when X is contractible). Then, $U = \{(x, y) \mid x \cdot y^{-1} \in V\}$ is a neighborhood of Δ_X in X^2 ($U = X^2$ if $V = X$). We can define an equi-connecting map $\lambda : U \times \mathbf{I} \to X$ by

$$\lambda(x, y, t) = \varphi(1, t)^{-1} \cdot \varphi(x \cdot y^{-1}, t) \cdot y.$$

The following proposition is easily proved:

Proposition 6.3.1 *An AR is EC and an ANR is LEC.*

Sketch of Proof An equi-connecting map λ for an AR (or an ANR) X can be obtained as an extension of the map of $(X^2 \times \{0, 1\}) \cup (\Delta_X \times \mathbf{I})$ to X defined by the conditions of an equi-connecting map.

The converse of Proposition 6.3.1 does not hold (cf. Remark 6.1.3 on Theorem 6.1.1). Such an example will be given in Sect. 7.12 (Theorem 7.12.1). The following is also easy:

Proposition 6.3.2 *For an arbitrary space X, if X is EC, then X is contractible; if X is LEC, then every singleton in X is a strong neighborhood deformation retract of X, hence X is locally contractible.*

Sketch of Proof For the second assertion, let $\lambda : U \times \mathbf{I} \to X$ be an equi-connecting map, where U is a neighborhood of Δ_X in X^2. For each $x \in X$, $U(x) = \{y \in X \mid (x, y) \in U\}$ is a neighborhood of x in X. Then, we have a contraction $h : U(x) \times \mathbf{I} \to X$ defined by $h(y, t) = \lambda(x, y, t)$ for each $(y, t) \in U(x) \times \mathbf{I}$, where $h(\{x\} \times \mathbf{I}) = \{x\}$. Thus, $\{x\}$ is a strong neighborhood deformation retract of X, which implies that X is locally contractible by Proposition 6.2.12.

Now, we will characterize LEC spaces via the following theorem.

Theorem 6.3.3 *For an arbitrary space X,*

(1) *X is EC if and only if Δ_X is a strong deformation retract of X^2, and*
(2) *X is LEC if and only if Δ_X is a strong neighborhood deformation retract of X^2.*

Proof To prove the "only if" part of both (1) and (2), let $\lambda : U \times \mathbf{I} \to X$ be an equi-connecting map for X, where $U = X^2$ for (1) or U is a neighborhood of Δ_X in X^2 for (2). Let $h : U \times \mathbf{I} \to X^2$ be the homotopy defined by $h(x, y, t) = (x, \lambda(x, y, 1 - t))$. Then, $h_0 = \mathrm{id}$, $h_t|\Delta_X = \mathrm{id}$ for each $t \in \mathbf{I}$, and h_1 is a retraction of U onto Δ_X.

To show the "if" part of (1) and (2), let $h : U \times \mathbf{I} \to X^2$ be a homotopy such that $h_0 = \mathrm{id}$, $h_t|\Delta_X = \mathrm{id}$ for each $t \in \mathbf{I}$ and h_1 is a retraction of U onto Δ_X, where $U = X^2$ or U is a neighborhood of Δ_X in response to (1) or (2). An equi-connecting map $\lambda : U \times \mathbf{I} \to X$ for X can be defined as follows:

$$\lambda(x, y, t) = \begin{cases} \mathrm{pr}_1 h(x, y, 2t) & \text{for } 0 \leqslant t \leqslant 1/2, \\ \mathrm{pr}_2 h(x, y, 2 - 2t) & \text{for } 1/2 \leqslant t \leqslant 1, \end{cases}$$

where $\mathrm{pr}_1, \mathrm{pr}_2 : X^2 \to X$ are the projections onto the first and the second coordinates, respectively. □

By the proof of Theorem 6.3.3, for every LEC space X, the diagonal Δ_X has a neighborhood U in X^2 with a pr_1-preserving homotopy $h : U \times \mathbf{I} \to X^2$ such that $h_0 = \mathrm{id}$, $h_1 : U \to \Delta_X$ is a retraction and $h_t|\Delta_X = \mathrm{id}$ for all $t \in \mathbf{I}$, where h is **pr_1-preserving** if $\mathrm{pr}_1 h_t = \mathrm{pr}_1|U$ for each $t \in \mathbf{I}$. Thus, every LEC space X has the following property:

(∗) Each neighborhood U of Δ_X in X^2 contains a neighborhood V of Δ_X with a pr_1-preserving homotopy $h : V \times \mathbf{I} \to U$ such that $h_0 = \mathrm{id}$ and $h_1(V) = \Delta_X$.

In the above (∗), for each $x \in X$, $U(x) = \{y \in X \mid (x, y) \in U\}$ is a neighborhood of x in X. We say that X is **unified locally contractible** (**ULC**) when X has the above property (∗).[6] It is easy to see that a ULC space is locally contractible. However, as will be seen, the converse does not hold. As seen in the above, an LEC space is ULC but the converse is unknown, that is:

[6] This concept was introduced by Ancel [2]. He adopted the term "uniformly locally contractible" but here we replace "uniformly" by "unified" because we say that a metric space X is **uniformly locally contractible** if, for each $\varepsilon > 0$, there is some $\delta > 0$ such that $\mathrm{B}(x, \delta)$ is contractible in $\mathrm{B}(x, \varepsilon)$ for each $x \in X$.

Open Question on LEC-ness
If X is ULC, is X LEC?

Theorem 6.3.4 *Let $X \neq \emptyset$ be a space such that X^2 is normal.*[7] *Then, X is EC if and only if X is contractible and LEC.*

Proof An equi-connecting map $\lambda : X^2 \times \mathbf{I} \to X$ induces the contraction $h : X \times \mathbf{I} \to X$ defined by $h(x, t) = \lambda(x, x_0, t)$, where $x_0 \in X$ is fixed. Thus, we have the "only if" part.

To show the "if" part, let X be contractible and LEC. By Theorem 6.3.3, Δ_X has an open neighborhood U in X^2 and there is a homotopy $h : U \times \mathbf{I} \to X^2$ such that $h_0 = \mathrm{id}$, $h_t|\Delta_X = \mathrm{id}$ for each $t \in \mathbf{I}$, and $h_1(U) = \Delta_X$. On the other hand, X has a contraction $\varphi : X \times \mathbf{I} \to X$. Note that $(\varphi(x, t), \varphi(x, t)) \in \Delta_X$ for each $x \in X$ and $t \in \mathbf{I}$. Hence, there exists an open neighborhood V of Δ_X in X^2 such that $(\varphi(x, t), \varphi(y, t)) \in U$ for $(x, y) \in V$ and $t \in \mathbf{I}$. Let $k : X^2 \to \mathbf{I}$ be a Urysohn map with $k(\Delta_X) = 0$ and $k(X^2 \setminus V) = 1$. Observe that

$$(\varphi(x, k(x, y)), \varphi(y, k(x, y))) \in U \text{ for every } (x, y) \in X^2.$$

Then, we can define a homotopy $\tilde{h} : X^2 \times \mathbf{I} \to X^2$ as follows:

$$\tilde{h}(x, y, t) = \begin{cases} (\varphi(x, 2k(x, y)t), \varphi(y, 2k(x, y)t)) & \text{for } 0 \leqslant t \leqslant 1/2, \\ h(\varphi(x, k(x, y)), \varphi(y, k(x, y)), 2t - 1) & \text{for } 1/2 \leqslant t \leqslant 1. \end{cases}$$

Observe that $\tilde{h}_0 = \mathrm{id}$, $\tilde{h}_t|\Delta_X = \mathrm{id}$ for each $t \in \mathbf{I}$, and $\tilde{h}_1(X^2) = \Delta_X$. Therefore, Δ_X is a strong deformation retract of X^2, which implies that X is EC by Theorem 6.3.3. □

Let $\mathcal{V}$ be an open refinement of an open cover $\mathcal{U}$ of a space X. We call $\mathcal{V}$ an ***h*-refinement** (resp. ***ħ*-refinement**) of $\mathcal{U}$ if any two $\mathcal{V}$-close maps $f, g : Y \to X$ defined on an arbitrary space Y are $\mathcal{U}$-homotopic (resp. $\mathcal{U}$-homotopic rel. $\{y \in Y \mid f(y) = g(y)\}$), where we write

$$\mathcal{V} \underset{h}{\prec} \mathcal{U} \quad \text{or} \quad \mathcal{U} \underset{h}{\succ} \mathcal{V} \quad \Big(\text{resp. } \mathcal{V} \underset{\hbar}{\prec} \mathcal{U} \quad \text{or} \quad \mathcal{U} \underset{\hbar}{\succ} \mathcal{V}\Big).$$

Using $\hbar$-refinements, we can characterize LEC spaces as follows:

Theorem 6.3.5 *A space X is LEC if and only if each open cover of X has an $\hbar$-refinement.*

[7] As we saw in Proposition 2.9.10, the Sorgenfrey line S is (perfectly) normal but S^2 is not normal.

Proof (*The "only if" Part*) Let $\lambda : U \times \mathbf{I} \to X$ be an equi-connecting map for X and $\mathcal{U} \in \mathrm{cov}(X)$. For each $x \in X$, choose $U_x \in \mathcal{U}$ so that $x \in U_x$. From the continuity of λ, each $x \in X$ has an open neighborhood V_x such that $V_x^2 \subset U$ and $\lambda(V_x^2 \times \mathbf{I}) \subset U_x$. Then, $\mathcal{V} = \{V_x \mid x \in X\}$ is the desired refinement of $\mathcal{U}$. Indeed, any two $\mathcal{V}$-close maps $f, g : Y \to X$ of an arbitrary space Y are $\mathcal{U}$-homotopic by a homotopy $h : Y \times \mathbf{I} \to X$ defined by $h(y, t) = \lambda(f(y), g(y), t)$.

(*The "if" Part*) By the condition, the open cover $\mathcal{U} = \{X\}$ has an $\hbar$-refinement $\mathcal{V}$. Then, $U = \bigcup_{V \in \mathcal{V}} V^2$ is a neighborhood of Δ_X in X^2. Since $\mathrm{pr}_1|U$ and $\mathrm{pr}_2|U$ are $\mathcal{V}$-close and $\mathrm{pr}_1|\Delta_X = \mathrm{pr}_2|\Delta_X$, we have a homotopy $\lambda : U \times \mathbf{I} \to X$ such that $\lambda_0 = \mathrm{pr}_1|U$, $\lambda_1 = \mathrm{pr}_2|U$ and $\lambda_t|\Delta_X = \mathrm{pr}_1|\Delta_X = \mathrm{pr}_2|\Delta_X$ for each $t \in \mathbf{I}$. This homotopy λ is an equi-connecting map for X. □

Combining the above theorem with Proposition 6.3.1, we have the following:

Corollary 6.3.6 *Every open cover of an ANR has an $\hbar$-refinement (hence, it has an h-refinement).* □

For ULC spaces, we have the following characterization using h-refinements:

Theorem 6.3.7 *A paracompact space X is ULC if and only if each open cover of X has an h-refinement.*

Proof (*The "only if" Part*) For each $\mathcal{U} \in \mathrm{cov}(X)$, let $\mathcal{U}'$ be a star-refinement of $\mathcal{U}$. Then, $W = \bigcup_{U \in \mathcal{U}'} U^2$ contains a neighborhood W_0 of Δ_X in X^2 with a pr_1-preserving homotopy $h : W_0 \times \mathbf{I} \to W$ such that $h_0 = \mathrm{id}$ and $h_1(W_0) = \Delta_X$. Choose $\mathcal{V} \in \mathrm{cov}(X)$ so that $\bigcup_{V \in \mathcal{V}} V^2 \subset W_0$. Observe that $V \subset \mathrm{st}(x, \mathcal{U}')$ for each $V \in \mathcal{V}$ and $x \in V$. Then, it follows that $\mathcal{V} \prec \mathcal{U}$. Moreover, $\mathcal{V}$ is an h-refinement of $\mathcal{U}$. Indeed, let $f, g : Y \to X$ be $\mathcal{V}$-close maps. Since $(f(y), g(y)) \in W_0$ for each $y \in Y$, we can define a homotopy $h^* : Y \times \mathbf{I} \to X$ by $h^*(y, t) = \mathrm{pr}_2 h(f(y), g(y), t)$. Then, $h_0^* = g$ and $h_1^* = f$. For each $y \in Y$ and $t \in \mathbf{I}$, since $h(f(y), g(y), t) \in W$, we have $U \in \mathcal{U}'$ such that $(f(y), h^*(y, t)) = h(f(y), g(y), t) \in U^2$, resulting in $U \in \mathcal{U}'[f(y)]$ and $h^*(y, t) \in U$. Therefore, $h^*(\{y\} \times \mathbf{I}) \subset \mathrm{st}(f(y), \mathcal{U}')$. Thus, h^* is a $\mathcal{U}$-homotopy.

(*The "if" Part*) Let W be a neighborhood of Δ_X in X^2. Choose $\mathcal{U} \in \mathrm{cov}(X)$ so that $\bigcup_{U \in \mathcal{U}} U^2 \subset W$. Taking an h-refinement $\mathcal{V}$ of $\mathcal{U}$, define $W_0 = \bigcup_{V \in \mathcal{V}} V^2$. Then, W_0 is a neighborhood of Δ_X in W. Since $\mathrm{pr}_1|W_0$ and $\mathrm{pr}_2|W_0$ are $\mathcal{V}$-close, there is a $\mathcal{U}$-homotopy $h : W_0 \times \mathbf{I} \to X$ such that $h_0 = \mathrm{pr}_1|W_0$ and $h_1 = \mathrm{pr}_2|W_0$. We define a pr_1-preserving homotopy $\tilde{h} : W_0 \times \mathbf{I} \to X^2$ by $\tilde{h}(x, y, t) = (x, h(x, y, 1 - t))$. Then, $\tilde{h}_0 = \mathrm{id}$ and $\tilde{h}_1(W_0) = \Delta_X$. For each $(x, y) \in W_0$, $h(\{(x, y)\} \times \mathbf{I})$ is contained in some $U \in \mathcal{U}$, which implies $\tilde{h}(\{(x, y)\} \times \mathbf{I}) \subset U^2$. Therefore, $\tilde{h}(W_0 \times \mathbf{I}) \subset W$. □

Next, consider the following related theorem:

Theorem 6.3.8 *A space X has an open cover $\mathcal{U}$ such that any two $\mathcal{U}$-close maps $f, g : Y \to X$ defined on an arbitrary space Y are homotopic (i.e., $\mathcal{U}$ is an h-refinement of $\{X\}$) if and only if Δ_X is a neighborhood deformation retract of X^2.*

Proof (*The "only if" Part*) Given an open cover $\mathcal{U}$ in the condition, $W = \bigcup_{U\in\mathcal{U}} U^2$ is a neighborhood of Δ_X in X^2. Since $\mathrm{pr}_1|W$ and $\mathrm{pr}_2|W$ are $\mathcal{U}$-close, there is a homotopy $h : W \times I \to X$ such that $h_0 = \mathrm{pr}_1|W$ and $h_1 = \mathrm{pr}_2|W$. Then, we can define $\tilde{h} : W \times I \to X^2$ by $\tilde{h}(x, y, t) = (x, h(x, y, 1 - t))$. Observe that $\tilde{h}_0 = \mathrm{id}$ and $\tilde{h}_1 : W \to \Delta_X$ is a retraction. Thus, Δ_X is a neighborhood deformation retract of X^2.

(*The "if" Part*) Let W be a neighborhood of Δ_X in X^2 and $\varphi : W \times \mathbf{I} \to X^2$ a homotopy such that $\varphi_0 = \mathrm{id}$ and $\varphi_1 : W \to \Delta_X$ is a retraction. Then, we have $\mathcal{U} \in \mathrm{cov}(X)$ such that $U^2 \subset W$ for each $U \in \mathcal{U}$. Any two $\mathcal{U}$-close maps $f, g : Y \to X$ of an arbitrary space Y are homotopic by a homotopy $h : Y \times \mathbf{I} \to X$ defined as follows:

$$h(y,t) = \begin{cases} \mathrm{pr}_1\varphi(f(y), g(y), 2t) & \text{for } 0 \leqslant t \leqslant 1/2, \\ \mathrm{pr}_2\varphi(f(y), g(y), 2-2t) & \text{for } 1/2 \leqslant t \leqslant 1. \end{cases}$$

The proof is completed. □

In general, a locally contractible space is not ULC, as shown in the following theorem:

Theorem 6.3.9 (BORSUK) *There exists a continuum X such that each point has a neighborhood basis consisting of contractible neighborhoods but the cover $\{X\}$ has no h-refinements (hence, X is not ULC).*

Proof (*Example*) Let $X_0 = \{x \in \mathbf{I}^{\mathbb{N}} \mid x(1) = 0\}$ and, for each $n \in \mathbb{N}$, let $X_n = \partial C_n$ be the boundary n-sphere of the $(n + 1)$-cube

$$C_n = \left[(n+1)^{-1}, n^{-1}\right] \times \mathbf{I}^n \times \{0\} \times \{0\} \times \cdots \subset \mathbf{I}^{\mathbb{N}}.$$

Then, we prove that $X = \bigcup_{n\in\omega} X_n$ ($\subset \mathbf{I}^{\mathbb{N}}$) is the desired continuum.

First, we show that each point of X has a neighborhood basis consisting of contractible neighborhoods. Since $\bigcup_{n\in\mathbb{N}} X_n = X \setminus X_0$ is a polyhedron that is open in X, each $x \in \bigcup_{n\in\mathbb{N}} X_n$ has such a neighborhood basis. When $x \in X_0$, for each neighborhood U of x in X, we can find $m \geqslant 2$ and a convex neighborhood W of $(x(2), \ldots, x(m))$ in $\mathbf{I}^{m-1}$ such that $p_m^{-1}([0, m^{-1}] \times W) \subset U$, where $p_m : X \to \mathbf{I}^m$ is the restriction of the projection onto the first m coordinates. When $x(m + 1) \leqslant 1/2$ (resp. $\geqslant 1/2$), we define a neighborhood V of x in X as follows:

$$V = p_{m+1}^{-1}([0, m^{-1}] \times W \times [0, 1)) \quad \big(\text{resp. } p_{m+1}^{-1}([0, m^{-1}] \times W \times (0, 1])\big).$$

Then, $V \subset U$ and V is contractible. Indeed, let

$$V_0 = p_{m+1}^{-1}(\{0\} \times W \times \{0\}) \subset X_0 \quad \big(\text{resp. } p_{m+1}^{-1}(\{0\} \times W \times \{1\})\big) \text{ and}$$
$$V_1 = p_{m+1}^{-1}([0, m^{-1}] \times W \times \{0\}) \quad \big(\text{resp. } p_{m+1}^{-1}([0, m^{-1}] \times W \times \{1\})\big).$$

Then, V_1 is a strong deformation retract of V by a deformation $h : V \times \mathbf{I} \to V$ sliding along the $(m + 1)$-th coordinate, and V_0 is a strong deformation retract of V_1 by a

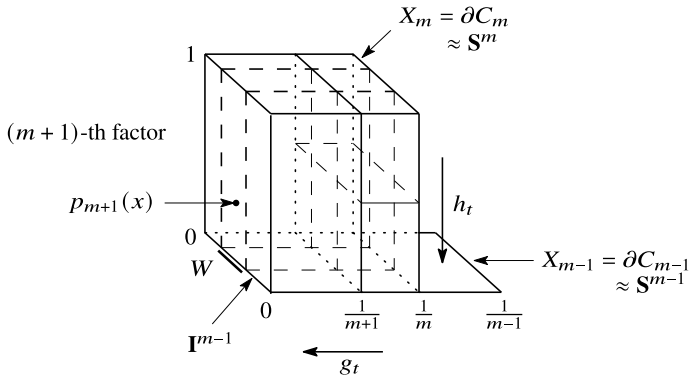


Fig. 6.6 Borsuk's example

deformation $g : V_1 \times \mathbf{I} \to V_1$ sliding along the first coordinate (Fig. 6.6). Since V_0 is contractible, V is also contractible.

Next, we show that $\{X\}$ has no h-refinements. On the contrary, assume that $\{X\}$ has an h-refinement $\mathcal{U} \in \mathrm{cov}(X)$. By the compactness of X, we can take $m \in \mathbb{N}$ such that $\{p_m^{-1}(x) \mid x \in \mathbf{I}^m\} \prec \mathcal{U}$. Then, the map $f : X \to X$ defined by $f(x) = (x(1), \ldots, x(m), 0, 0, \ldots)$ is $\mathcal{U}$-close to id, hence $f \simeq \mathrm{id}$. Let $r : X \to X_m$ be a retraction defined as follows:

$$r(x) = \begin{cases} ((m+1)^{-1}, x(2), \ldots, x(m+1), 0, 0, \ldots) & \text{if } x(1) \leqslant (m+1)^{-1}, \\ (m^{-1}, x(2), x(3), \ldots) & \text{if } x(1) \geqslant m^{-1}, \\ x & \text{otherwise.} \end{cases}$$

Then, $\mathrm{id} = r|X_m \simeq rf|X_m : X_m \to X_m$. Moreover,

$$rf(X_m) = f(X_m) = [(m+1)^{-1}, m^{-1}] \times \mathbf{I}^{m-1} \times \{0\} \times \cdots,$$

hence $rf|X_m \simeq 0$. Thus, $\mathbf{S}^m \approx X_m$ is contractible, which contradicts Corollary 6.1.7. □

Remark 6.3.1 A space X is said to be **strongly locally contractible** if each point has a neighborhood basis consisting of contractible neighborhoods (as an continuum obtained in Theorem 6.3.9). Obviously, every strongly locally contractible space is locally contractible. However, the converse does not hold. In fact, there exists an ANR that is not strongly locally contractible. Such an example is described in Chap. VI, Sect. 4 in Borsuk's book, *Theory of Retracts*.[8] Due to Proposition 6.3.2,[9] any LEC space X has the following property:

[8] For this book, refer to Notes for this chapter.

[9] For an ANR X, see Proposition 6.2.11.

- *Every singleton in X is a strong neighborhood deformation retract of X.*

This property implies local contractibility (Proposition 6.2.12), but the converse does not hold. Observe that Borsuk's example in the proof of Theorem 6.3.9 does not have this property. Thus, even strong local contractibility does not imply this property. Moreover, this property does not imply strong local contractibility because there is an ANR that is not strongly locally contractible as mentioned above.

Proposition 6.3.10 (Basic Properties of LEC Spaces)

(1) *An arbitrary product of EC spaces is EC and a finite product of LEC spaces is LEC.*

Sketch of Proof Define an equi-connecting map coordinate-wise.

(2) *A retract of an EC space is EC and a neighborhood retract of an LEC space is LEC.*

Sketch of Proof Compose a retraction with the restriction of an equi-connecting map.

(3) *Any open set in an LEC space is LEC.*

Sketch of Proof Restrict an equi-connecting map.

(4) *Let $X = X_1 \cup X_2$, where X_i is open in X. Suppose that X^2 is normal. If X_1 and X_2 are LEC, then so is X.*

Sketch of Proof Apply Theorem 6.3.3. Note that $X_0 = X_1 \cap X_2$ is also LEC by (3). For $i = 0, 1, 2$, we have a neighborhood U_i of Δ_{X_i} in X_i^2 and a homotopy $h^i : U_i \times \mathbf{I} \to X_i^2$ such that $h_0^i = \mathrm{id}$, h_1^i is a retraction onto Δ_{X_i} and $h_t^i | \Delta_{X_i} = \mathrm{id}$ for all $t \in \mathbf{I}$. Choose open sets V_0, V_0', V_1, V_2 in X^2 so that

$$\mathrm{cl}\, V_1 \cap \mathrm{cl}\, V_2 = \emptyset,\ \Delta_X \setminus X_2^2 \subset V_1 \subset U_1,\ \Delta_X \setminus X_1^2 \subset V_2 \subset U_2,$$
$$\Delta_X \setminus (V_1 \cup V_2) \subset V_0 \subset \mathrm{cl}\, V_0 \subset V_0' \subset \mathrm{cl}\, V_0' \subset U_0$$

and $h^0(\mathrm{cl}\, V_0' \times \mathbf{I}) \subset U_1 \cap U_2$. Then, $V = V_0 \cup V_1 \cup V_2$ is a neighborhood of Δ_X in X^2. Using h^i $(i = 0, 1, 2)$ and a Urysohn map $k : X^2 \to \mathbf{I}$ with $k(X^2 \setminus V_0') = 0$ and $k(\mathrm{cl}\, V_0) = 1$, define a homotopy $h : V \times \mathbf{I} \to X^2$ such that $h_0 = \mathrm{id}$, h_1 is a retraction onto Δ_X, and $h_t | \Delta_X = \mathrm{id}$ for all $t \in \mathbf{I}$. See Fig. 6.7.

(5) *A metrizable space X is LEC if each point of X has an LEC neighborhood in X.*

Sketch of Proof Apply Michael's Theorem 2.6.7 on local properties.

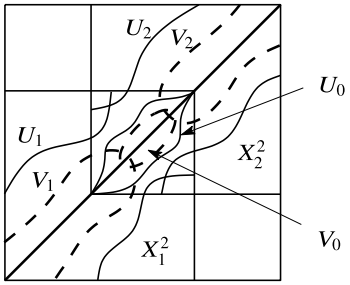


Fig. 6.7 The union of two open LEC spaces

6.4 The Homotopy Extension Property

It is said that a space Y has the **homotopy extension property** (**HEP**) **for metrizable spaces** provided, for every homotopy $h : A \times \mathbf{I} \to Y$ of a closed set A in a metrizable space X, if h_0 extends to a map $f : X \to Y$, then h extends to a homotopy $\tilde{h} : X \times \mathbf{I} \to Y$ such that $\tilde{h}_0 = f$. We prove the following HOMOTOPY EXTENSION THEOREM:

Theorem 6.4.1 (HOMOTOPY EXTENSION) *Let Y be an ANE, $\mathcal{U}$ be an open cover of Y, and $h : A \times \mathbf{I} \to Y$ be a $\mathcal{U}$-homotopy of a closed set A in a metrizable space X. If h_0 extends to a map $f : X \to Y$, then h extends to a $\mathcal{U}$-homotopy $\tilde{h} : X \times \mathbf{I} \to Y$ with $\tilde{h}_0 = f$. Consequently, every ANE has the HEP for metrizable spaces.*

Proof Since Y is an ANE, h extends to a map $h' : W \to Y$ from a neighborhood W of $X \times \{0\} \cup A \times \mathbf{I}$ in $X \times \mathbf{I}$ such that $h'(x, 0) = f(x)$ for each $x \in X$. For each $a \in A$, choose $U_a \in \mathcal{U}$ so that $h'(\{a\} \times \mathbf{I}) = h(\{a\} \times \mathbf{I}) \subset U_a$. Then, each $a \in A$ has an open neighborhood V_a in X such that $V_a \times \mathbf{I} \subset W$ and $h'(V_a \times \mathbf{I}) \subset U_a$. Let $V = \bigcup_{a \in A} V_a$ and let $k : X \to \mathbf{I}$ be a Urysohn map with $k(X \setminus V) = 0$ and $k(A) = 1$. Then, the desired homotopy $\tilde{h} : X \times \mathbf{I} \to Y$ can be defined by $\tilde{h}(x, t) = h'(x, tk(x))$ (see Fig. 6.8). To verify $\tilde{h}$ is an $\mathcal{U}$-homotopy, let $x \in X$. If $x \in X \setminus V$, then $\tilde{h}(\{x\} \times \mathbf{I}) = \{f(x)\}$. If $x \in V$, then $x \in V_a$ for some $a \in A$, which implies

$$\tilde{h}(\{x\} \times \mathbf{I}) \subset h'(\{x\} \times \mathbf{I}) \subset h'(V_a \times \mathbf{I}) \subset U_a. \qquad \square$$

Remark 6.4.1 The **HEP for compact spaces** or **for paracompact spaces**, etc., are similarly defined. Here, we do not consider the HEP "*for normal spaces*" because $X \times \mathbf{I}$ is not normal for some normal space X (Remark 2.9.2). By the same proof as above, we can show that if Y is an ANE for compact (resp. paracompact) spaces, then Y has the HEP for compact (resp. paracompact) spaces. Thus, by Corollary 6.1.12, for every simplicial complex K, $|K|$ has the HEP for compact spaces. Moreover, by Corollary 6.2.18, if K is countable locally finite, then $|K| = |K|_{\mathrm{m}}$ has the HEP for paracompact spaces; if K contains no infinite full complexes as subcomplexes, then $|K|_{\mathrm{m}}$ has the HEP for paracompact spaces.

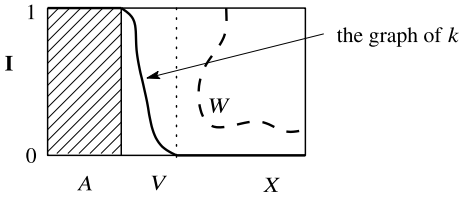


Fig. 6.8 An extension of a homotopy

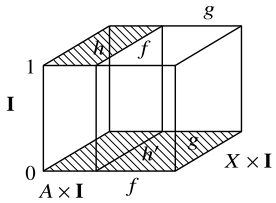


Fig. 6.9 An extension of a homotopy

Using Corollary 6.3.6, we can prove another Homotopy Extension Theorem:

Theorem 6.4.2 *Any open cover $\mathcal{U}$ of an ANR Y has an open refinement $\mathcal{V}$ satisfying the following condition:*

(∗) *For any $\mathcal{V}$-homotopy $h : A \times \mathbf{I} \to Y$ of a closed set A in a metrizable space X, if h_0 and h_1 extend to $\mathcal{V}$-close maps $f, g : X \to Y$, respectively, then h extends to a $\mathcal{U}$-homotopy $\tilde{h} : X \times \mathbf{I} \to Y$ such that $h_0 = f$ and $h_1 = g$.*

Proof By Corollary 6.3.6, we can take open refinements of $\mathcal{U}$ as follows:

$$\mathcal{V} \underset{h}{\prec} \mathcal{W}' \overset{*}{\prec} \mathcal{W} \underset{h}{\prec} \mathcal{U}' \overset{*}{\prec} \mathcal{U}.$$

We will show that $\mathcal{V}$ is the desired refinement. Let $h : A \times \mathbf{I} \to Y$ be a $\mathcal{V}$-homotopy of a closed set A in a metrizable space X such that h_0 and h_1 extend to $\mathcal{V}$-close maps $f, g : X \to Y$, respectively. Then, we have a $\mathcal{W}'$-homotopy $h' : X \times \mathbf{I} \to Y$ such that $h'_0 = f$ and $h'_1 = g$. On the other hand, h extends to a map $h'' : A \times \mathbf{I} \cup X \times \{0, 1\} \to Y$ defined by $h''(x, 0) = f(x)$ and $h''(x, 1) = g(x)$ for all $x \in X$. Since $h'|A \times \mathbf{I} \cup X \times \{0, 1\}$ and h'' are $\mathcal{W}$-close, there is a $\mathcal{U}'$-homotopy

$$\varphi : (A \times \mathbf{I} \cup X \times \{0, 1\}) \times \mathbf{I} \to Y$$

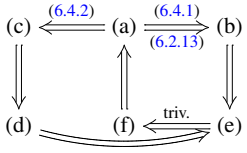
such that $\varphi_0 = h'|A \times \mathbf{I} \cup X \times \{0, 1\}$ and $\varphi_1 = h''$ (cf. Fig. 6.9). By Theorem 6.4.1, φ extends to a $\mathcal{U}'$-homotopy $\tilde{\varphi} : (X \times \mathbf{I}) \times \mathbf{I} \to Y$ with $\tilde{\varphi}_0 = h'$. Then, $\tilde{h} = \tilde{\varphi}_1 : X \times \mathbf{I} \to Y$ is a $\mathcal{U}$-homotopy such that $\tilde{h}_0 = f$, $\tilde{h}_1 = g$, and $\tilde{h}|A \times \mathbf{I} = h$. □

Using the HEP, we can characterize ANRs as follows:

Theorem 6.4.3 *For a metrizable space X, the following are equivalent:*

(a) *X is an ANR;*
(b) *X is semi-locally contractible and has the HEP for metrizable spaces;*
(c) *X has an open cover $\mathcal{V}$ such that for any $\mathcal{V}$-homotopy $h : A \times \mathbf{I} \to X$ of a closed set A in a metrizable space Y, if h_0 and h_1 respectively extend to $\mathcal{V}$-close maps $f, g : Y \to X$, then h extends to a homotopy $\tilde{h} : Y \times \mathbf{I} \to X$ such that $\tilde{h}_0 = f$ and $\tilde{h}_1 = g$;*
(d) *X is locally contractible and has an open cover $\mathcal{V}$ such that for any $\mathcal{V}$-homotopy $h : A \times \mathbf{I} \to X$ of a closed set A in a metrizable space Y, if h_0 extends to a map $f : Y \to X$ then h extends to a homotopy $\tilde{h} : Y \times \mathbf{I} \to X$ such that $\tilde{h}_0 = f$;*
(e) *Each $x \in X$ has a neighborhood V in X such that any map $f : A \to V$ of a closed set A in a metrizable space Y extends to a map $\tilde{f} : Y \to X$;*
(f) *Each $x \in X$ has a neighborhood V in X such that any map $f : A \to V$ of a closed set A in a metrizable space Y extends to a map $\tilde{f} : W \to X$ of a neighborhood W of A in Y to X.*

Sketch of Proof Observe the following implications:



(b) (or (d)) $\Rightarrow$ (e): Choose a neighborhood V of $x \in X$ so as to be contractible in X (or in some member of $\mathcal{V}$ in (d)).

(c) $\Rightarrow$ (d): Take $\mathcal{V}$ such as in (c). Each $x \in X$ is contained in some $V \in \mathcal{V}$. Using (c), construct a contraction $g : V \times \mathbf{I} \to X$ so that $g_t(x) = x$ for each $t \in \mathbf{I}$. Restricting g to a small neighborhood of x, we can show the local contractibility. For f and h of (d), applying (c) to the $\mathcal{V}$-homotopy $h^* : A \times \mathbf{I} \to X$ defined by $h^*_t = h_{2t}$ for $t \leqslant 1/2$ and $h^*_t = h_{2-2t}$ for $t \geqslant 1/2$, we can obtain an extension of h.

(f) $\Rightarrow$ (a): In condition (f), int V is an ANE, hence an ANR. Then, Hanner's Theorem 6.2.9 (4) can be applied.

Using the Homotopy Extension Theorem 6.4.1, we can prove the following theorem, which was announced in Remark 6.2.2 (cf. Proposition 6.2.11):

Theorem 6.4.4 *A deformation retract A of an ANR X is a strong deformation retract, where a deformation retraction $r : X \to A$ is a strong deformation retraction.*

Proof There is a homotopy $h : X \times \mathbf{I} \to X$ such that $h_0 = \mathrm{id}$ and $h_1 = r$. Since $h_1 h_1 = h_1$ and $h_1 | A = \mathrm{id}$, we can define a homotopy

$$\varphi : (A \times \mathbf{I} \cup X \times \{0, 1\}) \times \mathbf{I} \to X$$

as follows:

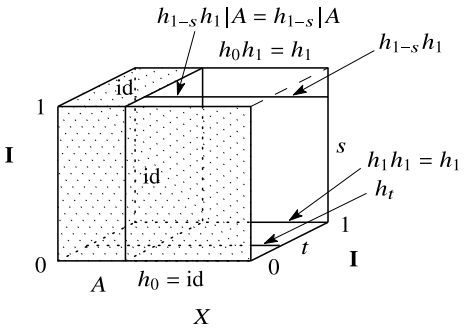


Fig. 6.10 The homotopy φ

$$\varphi(x,t,s) = \begin{cases} h_{t(1-s)}(x) & \text{if } x \in A, \\ h_{1-s}h_1(x) & \text{if } t = 1, \\ x & \text{if } t = 0, \\ h_t(x) & \text{if } s = 0. \end{cases}$$

Refer to Fig. 6.10. Since φ_0 extends to h, φ extends a homotopy $\tilde{\varphi} : (X \times \mathbf{I}) \times \mathbf{I} \to X$ by Homotopy Extension Theorem 6.4.1. Then, $h' = \tilde{\varphi}_1 : X \times \mathbf{I} \to X$ is a homotopy such that $h'_0 = \mathrm{id}$, $h'_1 = h_0h_1 = h_1 = r$, and $h'_t|A = h_0|A = \mathrm{id}$ for every $t \in \mathbf{I}$. Thus, h'_1 is a strong deformation retraction of X onto A, so A is a strong deformation retract of X. □

Remark 6.4.2 In the above proof, if h is a $\mathcal{U}$-homotopy for some $\mathcal{U} \in \mathrm{cov}(X)$, then h' can be taken as an st $\mathcal{U}$-homotopy. Indeed, by the definition, φ is also a $\mathcal{U}$-homotopy. Then, we have a $\mathcal{U}$-homotopy $\tilde{\varphi} : (X \times \mathbf{I}) \times \mathbf{I} \to X$ with $\tilde{\varphi}_0 = h$ by the Homotopy Extension Theorem 6.4.1. In particular, $h' = \varphi_1 : X \times \mathbf{I} \to X$ is $\mathcal{U}$-close to h. Since h is a $\mathcal{U}$-homotopy, it follows that h' is an st $\mathcal{U}$-homotopy. Thus, we have

$$r \simeq_{r^{-1}(\mathcal{U})} \mathrm{id}_X \ \text{ in } X \ \Rightarrow \ r \simeq_{r^{-1}(\mathrm{st}\,\mathcal{U})} \mathrm{id}_X \ \text{ rel. } A \text{ in } X.$$

Combining Theorem 6.4.4 with Proposition 6.2.10, we can easily prove the following:

Theorem 6.4.5 *Let A be an ANR that is a closed set in an ANR X. Then, A is a strong deformation retract of X if and only if the inclusion $A \subset X$ is a homotopy equivalence.*

Proof The "only if"part is trivial. In fact, a strong deformation retraction of X onto A is a homotopy inverse of the inclusion $A \subset X$. To see the "if part," let $f : X \to A$ be a homotopy inverse of the inclusion $A \subset X$. Since $f|A \simeq \mathrm{id}_A$ in A, we can apply the Homotopy Extension Theorem 6.4.1 to obtain a retraction $r : X \to A$ such that $r \simeq f$ in A. Moreover, X is deformable into A because $f \simeq \mathrm{id}_X$ in X. It follows from Proposition 6.2.10 that A is a deformation retract of X, which means that A is a strong deformation retract of X by Theorem 6.4.4. □

In the remainder of this section, we discuss another type of the homotopy extension property. Let (X, A) be a pair of a space X and its closed set A. It is said that (X, A) has the **homotopy extension property** (**HEP**) provided, for every homotopy $h : A \times \mathbf{I} \to Y$ to an arbitrary space Y, if h_0 extends to a map $f : X \to Y$, then h extends to a homotopy $\tilde{h} : X \times \mathbf{I} \to Y$ such that $\tilde{h}_0 = f$. According to Theorem 4.5.3, for a cell complex K and its subcomplex L, the pair $(|K|, |L|)$ has the HEP. The following is easy:

Proposition 6.4.6 *For a space X and its closed set A, the following are equivalent:*

(a) (X, A) *has the HEP;*
(b) *Every map from* $(X \times \{0\}) \cup (A \times \mathbf{I})$ *to an arbitrary space Y extends over* $X \times \mathbf{I}$*;*
(c) $(X \times \{0\}) \cup (A \times \mathbf{I})$ *is a retract of* $X \times \mathbf{I}$.

Sketch of Proof The equivalence (a) $\Leftrightarrow$ (b) is trivial. To obtain (b) $\Rightarrow$ (c), just apply (b) to the identity map of $(X \times \{0\}) \cup (A \times \mathbf{I})$ onto itself. To see (c) $\Rightarrow$ (b), let $r : X \times \mathbf{I} \to (X \times \{0\}) \cup (A \times \mathbf{I})$ be a retraction. Then, every map $f : (X \times \{0\}) \cup (A \times \mathbf{I}) \to Y$ has an extension $fr : X \times \mathbf{I} \to Y$.

Proposition 6.4.7 *For a space X and its closed set A, if (X, A) has the HEP, then* (i) *A is a neighborhood retract of X, and* (ii) *$(X \times Y, A \times Y)$ has the HEP for an arbitrary space Y.*

Sketch of Proof As an immediate consequence of Proposition 6.4.6, we have (ii). To see (i), let $r : X \times \mathbf{I} \to (X \times \{0\}) \cup (A \times \mathbf{I})$ be a retraction and take an open neighborhood U of A in X so that $U \times \{0\} \subset r^{-1}(A \times (0, 1])$. Then, $\mathrm{pr}_A r i_1 : U \to A$ is a retraction, where $i_1 : U \to U \times \{1\} \subset X \times \mathbf{I}$ is the natural injection defined by $i_1(x) = (x, 1)$.

By the following theorem, for a simplicial complex K and its subcomplex L, the pair $(|K|_{\mathrm{m}}, |L|_{\mathrm{m}})$ has also the HEP (cf. Theorem 4.5.3).

Theorem 6.4.8 *Let X and A be ANRs such that A is closed in X. Then, (X, A) has the HEP.*

Proof By Proposition 6.2.9 (1) and (5), $(X \times \{0\}) \cup (A \times \mathbf{I})$ is an ANR, which is a retract of a neighborhood W in $X \times \mathbf{I}$. Let $r : W \to (X \times \{0\}) \cup (A \times \mathbf{I})$ be a retraction. Since $\mathbf{I}$ is compact, we can take an open neighborhood U of A in X such that $U \times \mathbf{I} \subset W$. Let $f : X \to \mathbf{I}$ be a Urysohn map such that $f(X \setminus U) \subset \{0\}$ and $f(A) \subset \{1\}$. Then, we can define a retraction $\tilde{r} : X \times \mathbf{I} \to (X \times \{0\}) \cup (A \times \mathbf{I})$ as follows:

$$\tilde{r}(x, t) = \begin{cases} r(x, f(x)) & \text{if } t \geqslant f(x), \\ r(x, t) & \text{if } t \leqslant f(x). \end{cases}$$

Thus, $(X \times \{0\}) \cup (A \times \mathbf{I})$ is a retract of $X \times \mathbf{I}$, which means that (X, A) has the HEP by Proposition 6.4.6. □

Concerning pairs with the HEP, the following holds:

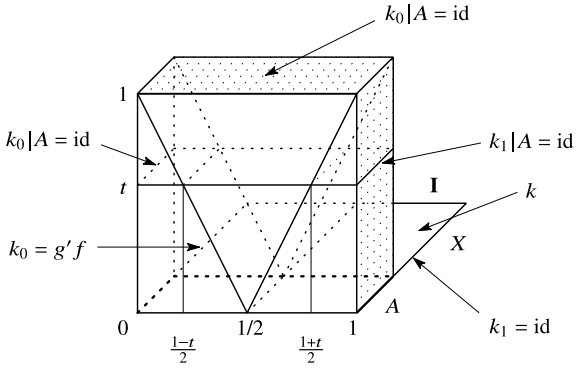


Fig. 6.11 The homotopy φ

Proposition 6.4.9 *Let X and Y be spaces with a common closed subspace A. If (X, A) and (Y, A) have the HEP and $f : X \to Y$ is a homotopy equivalence with $f|A = \mathrm{id}$, then f is a homotopy equivalence relative to A, that is, there is a map $g : Y \to X$ with $g|A = \mathrm{id}$, $gf \simeq \mathrm{id}_X$ rel. A, and $fg \simeq \mathrm{id}_Y$ rel. A.*

Proof Let $g : Y \to X$ be a homotopy inverse of f and $h : X \times \mathbf{I} \to X$ a homotopy with $h_0 = gf$ and $h_1 = \mathrm{id}_X$. Because $f|A = \mathrm{id}$, the restriction $h|A \times \mathbf{I}$ is a homotopy from $g|A$ to id_A in X. Since (Y, A) has the HEP, $h|A \times \mathbf{I}$ extends to a homotopy $h' : Y \times \mathbf{I} \to X$ with $h'_0 = g$. Then, we have a map $g' = h'_1 : Y \to X$ such that $g' \simeq g$ and $g'|A = \mathrm{id}$.

Now, we define a homotopy $k : X \times \mathbf{I} \to X$ as follows:

$$k(x, t) = \begin{cases} h'(f(x), 1 - 2t) & \text{if } t \leqslant 1/2, \\ h(x, 2t - 1) & \text{if } t \geqslant 1/2, \end{cases}$$

where $k_0 = g'f$, $k_1 = \mathrm{id}_X$, $k_0|A = k_1|A = \mathrm{id}$, and $k_t|A = k_{1-t}|A$ for each $t \in \mathbf{I}$.

In the above, note that if the last equality holds for $t \leqslant 1/2$, then it holds for every $t \in \mathbf{I}$. For $t \leqslant 1/2$, we have $1 - t \geqslant 1/2$ and $1 - 2t = 2(1 - t) - 1$, which implies

$$k_t|A = h'_{1-2t}|A = h_{1-2t}|A = h_{2(1-t)-1}|A = k_{1-t}|A.$$

Since $k_{(1-t)/2}|A = k_{(1+t)/2}|A$ for each $t \in \mathbf{I}$, we can define a homotopy $\varphi : (A \times \mathbf{I}) \times \mathbf{I} \to X$ as follows:

$$\varphi(x, s, t) = \begin{cases} k(x, s) & \text{if } s \leqslant (1 - t)/2, \\ k(x, (1 - t)/2) = k(x, (1 + t)/2) & \text{if } (1 - t)/2 \leqslant s \leqslant (1 + t)/2. \\ k(x, s) & \text{if } s \geqslant (1 + t)/2. \end{cases}$$

Refer to Fig. 6.11. Then, for each $(x, s) \in A \times \mathbf{I}$, $\varphi_0(x, s) = k(x, s)$ and $\varphi_1(x, s) = k_0(x) = k_1(x) = x$. Since $(X \times \mathbf{I}, A \times \mathbf{I})$ has the HEP (Proposition 6.4.7), φ extends

to a homotopy $\tilde{\varphi} : (X \times \mathbf{I}) \times \mathbf{I} \to X$ such that $\tilde{\varphi}_0 = k$. Let $k' : X \times \mathbf{I} \to X$ be a homotopy defined as follows:

$$k'(x,t) = \begin{cases} \tilde{\varphi}(x,0,3t) & \text{if } t \leqslant 1/3, \\ \tilde{\varphi}(x,3t-1,1) & \text{if } 1/3 \leqslant t \leqslant 2/3, \\ \tilde{\varphi}(x,1,3-3t) & \text{if } t \geqslant 2/3. \end{cases}$$

Then, $k'_0 = k_0 = g'f$ and $k'_1 = k_1 = \mathrm{id}_X$. Moreover, $k'_t|A = k_0|A = \mathrm{id}$ if $t \leqslant 1/2$, and $k'_t|A = k_1|A = \mathrm{id}$ if $t \geqslant 1/2$. Hence, this homotopy realizes $g'f \simeq \mathrm{id}$ rel. A.

Since $g' \simeq g$, it follows that $fg' \simeq fg \simeq \mathrm{id}$, where $g'|A = \mathrm{id}$. Replacing the pair f, g with g', f, we can repeat the above argument to obtain a map $f' : X \to Y$ such that $f' \simeq f$, $f'|A = \mathrm{id}$, and $f'g' \simeq \mathrm{id}$ rel. A. Then, it follows that $f' \simeq f'(g'f) = (f'g')f \simeq f$ rel. A, which implies that $fg' \simeq f'g' \simeq \mathrm{id}$ rel. A. □

A pointed space (X, x_0) can be identified with a pair $(X, \{x_0\})$. Then, we have the following corollary:

Corollary 6.4.10 *Let (X, x_0) and (Y, y_0) be pointed spaces that have the HEP and $f : X \to Y$ a homotopy equivalence with $f(x_0) = y_0$. Then, $f : (X, x_0) \to (Y, y_0)$ is a homotopy equivalence between pointed spaces.* □

6.5 Complementary Pairs of ANRs

First, we prove the following lemma, which is often used in extending maps:

Lemma 6.5.1 (C.H. DOWKER) *Let X and Y be metrizable spaces and $f : A \to Y$ be a map of a closed set A in X. Suppose there exists a homotopy $h : A \times \mathbf{I} \to Y$ such that $h_0 = f$ and $h|A \times (0,1]$ extends over a neighborhood W of $A \times (0,1]$ in $X \times (0,1]$. Then, f extends over a neighborhood of A in X. If $X \times \{1\} \subset W$, then f extends over X.*

Proof For simplicity, let d stand for admissible metrics of X and Y as well as the metric on $X \times \mathbf{I}$ defined as follows:

$$d((x,t),(x',t')) = \max\{d(x,x'),\ |t-t'|\}.$$

We may assume that h is defined on the set $W \cup A \times \{0\}$ and that $h|A \times \mathbf{I}$ and $h|W$ are continuous. However, it is not assumed that h is continuous. First, we find a neighborhood W^* of $A \times (0,1]$ in $X \times (0,1]$ such that $W^* \subset W$ and $h|W^* \cup A \times \{0\}$ is continuous. To this end, for each $(a,t) \in A \times (0,1]$, choose a neighborhood $W(a,t)$ of (a,t) in W so that $\operatorname{diam} W(a,t) < t/2$ and $\operatorname{diam} h(W(a,t)) < t/2$. Then, $W^* = \bigcup_{(a,t) \in A \times (0,1]} W(a,t)$ is a neighborhood of $A \times (0,1]$ in $X \times (0,1]$. To verify the continuity of $h|W^* \cup A \times \{0\}$ at $(a,0) \in A \times \{0\}$, let $\varepsilon > 0$. By the

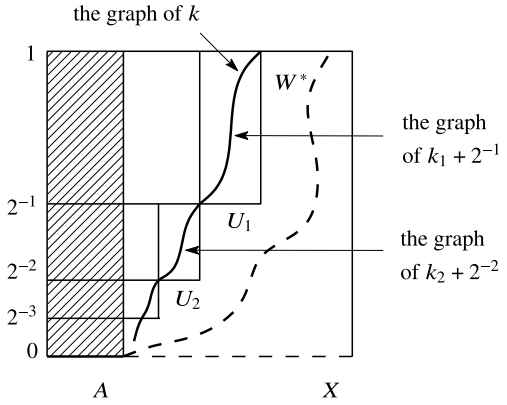


Fig. 6.12 Extending a map

continuity of $h|A \times \mathbf{I}$ at $(a, 0)$, we can choose $\delta > 0$ so that, if $(a', t') \in A \times \mathbf{I}$ and $d((a, 0), (a', t')) < \delta$, then $d(h(a, 0), h(a', t')) < \varepsilon/2$. Let $(x, t) \in W^*$ with $d((a, 0), (x, t)) < \min\{\delta/2, \varepsilon/2\}$. Then, we have $(a', t') \in A \times (0, 1]$ such that $(x, t) \in W(a', t')$, resulting in $d((x, t), (a', t')) < t'/2$ and $d(h(x, t), h(a', t')) < t'/2$. Since $|t - t'| < t'/2$, it follows that $t'/2 < t < \min\{\delta/2, \varepsilon/2\}$. Thus, $d((x, t), (a', t')) < \delta/2$ and $d(h(x, t), h(a', t')) < \varepsilon/2$. Since

$$d((a, 0), (a', t')) \leqslant d((a, 0), (x, t)) + d((x, t), (a', t')) < \delta,$$

it follows that $d(h(a, 0), h(a', t')) < \varepsilon/2$. Then,

$$d(h(a, 0), h(x, t)) \leqslant d(h(a, 0), h(a', t')) + d(h(a', t'), h(x, t)) < \varepsilon.$$

Therefore, $h|W^* \cup A \times \{0\}$ is continuous at $(a, 0)$.

Choose open neighborhoods U_n of A in X, $n \in \mathbb{N}$, so that $U_n \times [2^{-n}, 1] \subset W^*$, $\operatorname{cl} U_{n+1} \subset U_n$, and $\bigcap_{n\in\mathbb{N}} U_n = A$. Take maps $k_n : X \to [0, 2^{-n}]$, $n \in \mathbb{N}$, so that $k_n(\operatorname{cl} U_{n+1}) = 0$ and $k_n(X \setminus U_n) = 2^{-n}$, and define a map $k : X \to \mathbf{I}$ by $k(x) = \sum_{n\in\mathbb{N}} k_n(x)$. Observe that $k^{-1}(0) = A$ and $k(x) \in [2^{-n}, 2^{-n+1}]$ for each $x \in U_n \setminus U_{n+1}$, which implies that $(x, k(x)) \in A \times \{0\} \cup W^*$ for all $x \in U_1$. Then, f extends to a map $\tilde{f} : U_1 \to Y$ defined by $\tilde{f}(x) = h(x, k(x))$. See Fig. 6.12.

If $X \times \{1\} \subset W$, we define $W^* = \bigcup_{(a,t)\in A\times(0,1]} W(a, t) \cup X \times \{1\}$. Then, the map $\tilde{f}$ above can be defined over X. □

Remark 6.5.1 In the above proof, it is not enough to take a map k with $k^{-1}(0) = A$. Indeed, we define a function

$$h : \mathbf{I} \times (0, 1] \cup \{0\} \times \{0\} \to \mathbb{R}$$

by $h(x,t) = x/t$ if $t \neq 0$ and $h(0,0) = 0$, where $(X, A) = (\mathbf{I}, \{0\})$, $Y = \mathbb{R}$ and $W = \mathbf{I} \times (0, 1]$. Note that f is non-continuous but the restrictions $f|\{0\} \times \mathbf{I}$ and $h|\mathbf{I} \times (0, 1]$ are continuous. Moreover, let $k = \mathrm{id} : \mathbf{I} \to \mathbf{I}$. Then, $k^{-1}(0) = \{0\}$. Using this k, we define $\tilde{f} : \mathbf{I} \to \mathbb{R}$ as in the above proof, i.e., $\tilde{f}(x) = h(x, k(x))$ for each $x \in \mathbf{I}$. Then, $\tilde{f}$ is not continuous at $0 \in \mathbf{I}$ because $\tilde{f}(0) = h(0,0) = 0$, but $\tilde{f}(x) = h(x, k(x)) = x/k(x) = 1$ if $x \neq 0$.

Applying Lemma 6.5.1, we prove the following theorem:

Theorem 6.5.2 (KRUSE–LIEBNITZ) *Let X be metrizable and A be a strong neighborhood deformation retract of X. If A and $X \setminus A$ are ANRs, then so is X.*

Proof From the assumption, we have an open neighborhood U of A in X and a homotopy $h : \mathrm{cl}\, U \times \mathbf{I} \to X$ such that $h_0 = \mathrm{id}$, h_1 is a retraction of $\mathrm{cl}\, U$ onto A, and $h_t|A = \mathrm{id}$ for every $t \in \mathbf{I}$. Given an admissible metric d for X, we may assume that $\mathrm{diam}\, h(\{x\} \times \mathbf{I}) < 1$ for every $x \in \mathrm{cl}\, U$. Let $f : B \to X$ be a map from a closed set B in an arbitrary metrizable space Y. We apply Lemma 6.5.1 to extend f over a neighborhood of B in Y.

Note that $f^{-1}(U)$ and $f^{-1}(X \setminus A)$ are closed in $Y \setminus f^{-1}(X \setminus U)$ and $Y \setminus f^{-1}(A)$, respectively. Since A and $X \setminus A$ are ANRs, maps $h_1 f|f^{-1}(U)$ and $f|f^{-1}(X \setminus A)$ extend to maps $f' : V' \to A$ and $f'' : V'' \to X \setminus A$, respectively, where V' and V'' are open sets in $Y \setminus f^{-1}(X \setminus U)$ and $Y \setminus f^{-1}(A)$, respectively, such that $V' \cap B = f^{-1}(U)$ and $V'' \cap B = f^{-1}(X \setminus A)$. Observe that

$$V' \cap V'' \cap B = f^{-1}(U \setminus A) \quad \text{and} \quad f'|V' \cap V'' \cap B = h_1 f''|V' \cap V'' \cap B.$$

Note that every open cover of an ANR has an $\hbar$-refinement by Corollary 6.3.6. Then, we can find an open set V_0 in $V' \cap V''$ such that $V_0 \cap B = V' \cap V'' \cap B$ and $f'|V_0 \simeq h_1 f''|V_0$ rel. $V_0 \cap B$. Let $\varphi : V_0 \times \mathbf{I} \to A$ be a homotopy such that

$$\varphi_0 = f'|V_0, \ \varphi_1 = h_1 f''|V_0 \quad \text{and} \quad \varphi_t|V_0 \cap B = f'|V_0 \cap B \quad \text{for all } t \in \mathbf{I}.$$

Choose disjoint open sets V_1 and V_2 in Y so that $f^{-1}(A) \subset V_1 \subset V'$ and $B \setminus f^{-1}(U) \subset V_2 \subset V''$. Then, $V = V_0 \cup V_1 \cup V_2$ is an open neighborhood of B in Y. Cf. Fig. 6.13.

The function $\mathrm{cl}\, U \ni x \mapsto \mathrm{diam}\, h(\{x\} \times \mathbf{I}) \in \mathbf{I}$ is continuous. Indeed, for each $x \in \mathrm{cl}\, U$ and $\varepsilon > 0$, using the compactness of $\mathbf{I}$, we can find $\delta > 0$ such that $d(x, x') < \delta$ implies $d(h(x,t), h(x',t)) < \varepsilon/2$ for every $t \in \mathbf{I}$. Let $x' \in \mathrm{cl}\, U$ with $d(x, x') < \delta$. For each $t_0, t_1 \in \mathbf{I}$,

$$\begin{aligned} d(h(x', t_0), h(x', t_1)) &\leqslant d(h(x, t_0), h(x, t_1)) \\ &\quad + d(h(x, t_0), h(x', t_0)) + d(h(x, t_1), h(x', t_1)) \\ &< \mathrm{diam}\, h(\{x\} \times \mathbf{I}) + \varepsilon, \end{aligned}$$

which implies that

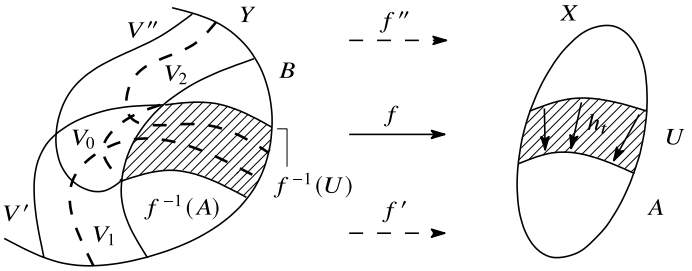


Fig. 6.13 Extensions of $h_1 f|f^{-1}(U)$ and $f|f^{-1}(X \setminus A)$

$$\operatorname{diam} h(\{x'\} \times \mathbf{I}) \leqslant \operatorname{diam} h(\{x\} \times \mathbf{I}) + \varepsilon.$$

Replacing x' with x in the above, we have

$$\operatorname{diam} h(\{x\} \times \mathbf{I}) \leqslant \operatorname{diam} h(\{x'\} \times \mathbf{I}) + \varepsilon.$$

Let $\alpha : (V \setminus V_0) \cup B \to \mathbf{I}$ be a Urysohn map with $\alpha(V_1 \setminus V_0) = 0$ and $\alpha(V_2 \setminus V_0) = 1$. We define the map $\beta : (V \setminus V_0) \cup B \to \mathbf{I}$ by $\beta(V_1 \setminus V_0) = 0$, $\beta(V_2 \setminus V_0) = 1$, and

$$\beta(x) = \min\left\{1,\ \operatorname{diam} h(\{f(x)\} \times \mathbf{I}) + \alpha(x)\right\} \text{ for } x \in f^{-1}(\operatorname{cl} U).$$

Note that $\operatorname{diam} h(\{f(x)\} \times \mathbf{I}) \leqslant \beta(x)$ for every $x \in f^{-1}(\operatorname{cl} U)$ and $\beta^{-1}(0) \cap B = f^{-1}(A)$. By the Tietze–Urysohn Extension Theorem 2.2.2, β extends to a map $\gamma : V \to \mathbf{I}$. Observe that $\gamma^{-1}((0, 1)) \subset V_0$, $\gamma^{-1}([0, 1)) \subset V'$, and $\gamma^{-1}((0, 1]) \subset V''$. Then, we can define a map $g : V \times (0, 1] \to X$ as follows:

$$g(x,t) = \begin{cases} f''(x) & \text{if } t \leqslant \gamma(x), \\ \tilde{h}(f''(x), \gamma(x)^{-1}t - 1) & \text{if } \gamma(x) \leqslant t \leqslant 2\gamma(x), \\ \varphi(x, 3 - \gamma(x)^{-1}t) & \text{if } 2\gamma(x) \leqslant t \leqslant 3\gamma(x), \\ f'(x) & \text{if } t \geqslant 3\gamma(x), \end{cases}$$

where $\tilde{h} : \operatorname{cl} U \times \mathbf{I} \cup X \times \{0\} \to X$ is the map defined by $\tilde{h}|\operatorname{cl} U \times \mathbf{I} = h$ and $\tilde{h}(x, 0) = x$ for each $x \in X$ (cf. Fig. 6.14).

When $\gamma(x) < 1$, since $\gamma^{-1}([0, 1)) \subset V'$, it follows that $x \in V' \cap B = f^{-1}(U)$. Hence, $f''(x) = f(x) \in U \subset \operatorname{cl} U$. However, when $\gamma(x) = 1$, since $f''(x) \in \operatorname{cl} U$ is not guaranteed, it is necessary to extend h to $\tilde{h}$.

For each $(x, t) \in B \times (0, 1]$, we have $d(g(x, t), f(x)) \leqslant t$. Indeed, recall $\beta|B = \gamma|B$ and $\varphi_t|V_0 \cap B = h_1 f|V_0 \cap B$ for every $t \in \mathbf{I}$. Observe the following:

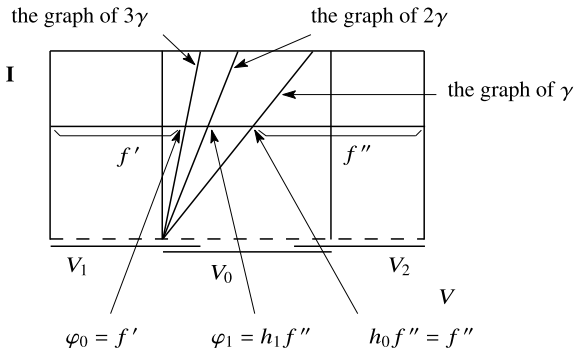


Fig. 6.14 The map g

$$t \leqslant \gamma(x) \Rightarrow g(x,t) = f(x);$$
$$\gamma(x) < t \leqslant 2\gamma(x) \Rightarrow d(g(x,t), f(x)) \leqslant \operatorname{diam} h(\{f(x)\} \times \mathbf{I}) \leqslant \gamma(x) \leqslant t;$$
$$t \geqslant 2\gamma(x) \Rightarrow d(g(x,t), f(x)) = d(h_1 f(x), f(x)) \leqslant \gamma(x) \leqslant t.$$

Therefore, $g|B \times (0,1]$ can be extended to the homotopy $\tilde{g} : B \times \mathbf{I} \to X$ with $\tilde{g}_0 = f$. By Lemma 6.5.1, f extends over a neighborhood of B in Y. □

Recall that a perfect map $f : X \to Y$ is a closed map such that $f^{-1}(y)$ is compact for every $y \in Y$. The perfect image of a metrizable space is also metrizable (Proposition 2.4.4 (1)). As a corollary of Theorem 6.5.2, we have the following theorem on the adjunction spaces:

Theorem 6.5.3 (BORSUK–WHITEHEAD–HANNER) *Let X and Y be ANRs and $f : A \to Y$ be a map from a closed set A in X. Then, the adjunction space $Y \cup_f X$ is an ANR if f is perfect and A is an ANR.*

Proof Since A is a strong neighborhood deformation retract of X by Proposition 6.2.11, it follows that Y is a strong neighborhood deformation retract of $Y \cup_f X$. The result follows from Theorem 6.5.2. □

As special cases of Theorem 6.5.3, we have the following:

Corollary 6.5.4 *For any perfect map $f : X \to Y$ between ANRs, the mapping cylinder M_f is an ANR.* □

When Y is a singleton in the above, the mapping cylinder M_f is regarded as the cone $(X \times \mathbf{I})/(X \times \{0\})$. Since a contractible ANR is an AR (Corollary 6.2.14), we have the following corollary:

Corollary 6.5.5 *The cone $(X \times \mathbf{I})/(X \times \{0\})$ over a compact ANR X is an AR.* □

Mapping cylinders are very useful, as seen in Sect. 4.19. However, they are not, in general, metrizable. We introduce the mapping cylinder that is different from the quotient space. For a map $f : X \to Y$, let

$$M(f) = Y \cup \big(X \times (0, 1]\big) \quad \text{(the disjoint union)},$$

whose topology is generated by open sets in $X \times (0, 1]$ and sets $U \cup (f^{-1}(U) \times (0, \varepsilon))$, where U is open in Y and $0 < \varepsilon < 1$. Then, we have the natural continuous bijection $\phi_f : M_f \to M(f)$, where $\phi_f | Y = \mathrm{id}$ and $\phi_f | X \times (0, 1] = \mathrm{id}$.

When X and Y are bounded closed sets in normed linear spaces E and F, respectively, we can define a closed embedding $\varphi : M(f) \to E \times F \times \mathbb{R}$ as follows: $\varphi(y) = (\mathbf{0}, y, 0)$ for each $y \in Y$ and

$$\varphi(x, t) = (tx, (1 - t)f(x), t) \ \text{ for each } (x, t) \in X \times (0, 1].$$

To verify the continuity of φ at $y \in \mathrm{cl}\, f(X)$, let $\varepsilon > 0$. Choose $0 < \delta_1 < \varepsilon$ so that $(1 - t)\mathrm{B}(y, \delta_1) \subset \mathrm{B}(y, \varepsilon)$ if $0 < t < \delta_1$. Since X is bounded in E, we have $0 < \delta < \delta_1$ $(< \varepsilon)$ such that $\delta X \subset \mathrm{B}(\mathbf{0}, \varepsilon)$. Let $U = Y \cap \mathrm{B}(y, \delta)$. Then, $U \cup f^{-1}(U) \times (0, \delta)$ is an open neighborhood of y in $M(f)$ and

$$\varphi\big(U \cup (f^{-1}(U) \times (0, \delta))\big) \subset \mathrm{B}(\mathbf{0}, \varepsilon) \times \mathrm{B}(y, \varepsilon) \times (-\varepsilon, \varepsilon).$$

To verify the closedness of φ, let $(z_i)_{i \in \mathbb{N}}$ be a sequence in $M(f)$ such that $(\varphi(z_i))_{i \in \mathbb{N}}$ converges to $(x, y, t) \in E \times F \times \mathbb{R}$. We will show that $(z_i)_{i \in \mathbb{N}}$ has a convergent subsequence. Then, we may assume that $z_i \in Y$ for every $i \in \mathbb{N}$ or $z_i \in X \times (0, 1]$ for every $i \in \mathbb{N}$. In the first case, since $\varphi(z_i) = (\mathbf{0}, z_i, 0)$, $(z_i)_{i \in \mathbb{N}}$ converges to y. In the second case, let $z_i = (x_i, t_i)$ for each $i \in \mathbb{N}$. Since $(t_i x_i, (1 - t_i)f(x_i), t_i) = \varphi(z_i) \to (x, y, t)$, we have $t = \lim_{i \to \infty} t_i \in \mathbf{I}$. If $t > 0$, we may assume that $t_i > 0$ for every $i \in \mathbb{N}$. Since $x_i = t_i^{-1}(t_i x_i) \to t^{-1}x_i$ and X is closed in E, we have $t^{-1}x \in X$. Hence, $(z_i)_{i \in \mathbb{N}}$ converges to $(t^{-1}x, t)$ in $M(f)$. If $t = 0$, we may assume that $t_i < 1$ for every $i \in \mathbb{N}$. Since $f(x_i) = (1 - t_i)^{-1}((1 - t_i)f(x_i)) \to y$ and Y is closed in F, we have $y \in Y$. To see that $(z_i)_{i \in \mathbb{N}}$ converges to y in $M(f)$, let U be an open neighborhood of y in Y and $0 < \varepsilon < 1$. Then, we have $i_0 \in \mathbb{N}$ such that $i \geqslant i_0$ implies $t_i < \varepsilon$ and $f(x_i) \in U$, i.e., $z_i = (x_i, t_i) \in U \cup f^{-1}(U) \times (0, \varepsilon)$, which means that $(z_i)_{i \in \mathbb{N}}$ converges to y in $M(f)$.

Due to the Arens–Eells Embedding Theorem 6.2.1, every metrizable (resp. completely metrizable) space can be embedded into a normed linear space (resp. a Banach space) as a closed set. Then, $M(f)$ is (completely) metrizable if X and Y are. The metrizability of $M(f)$ can also be shown by applying the Bing Metrization Theorem 2.3.4.

For any map $f : X \to Y$, we can identify $M(f)$ with M_f as sets (by the natural map ϕ_f), but the topology of M_f is finer than the topology of $M(f)$. Let $q_f : Y \oplus (X \times \mathbf{I}) \to M_f$ be the quotient map. If $f : X \to Y$ is *perfect*, then $M(f) = M_f$ as spaces, where the natural map $\phi_f : M_f \to M(f)$ is a homeomorphism.

Indeed, let U be a neighborhood of $y \in f(X)$ in M_f. Then, $f^{-1}(y) \times \{0\} \subset q_f^{-1}(U) \cap (X \times \mathbf{I})$. Because $f^{-1}(y)$ is compact, we have a neighborhood W of $f^{-1}(y)$ in X with $\delta \in (0, 1)$

such that $W \times [0, \delta) \subset q_f^{-1}(U) \cap (X \times \mathbf{I})$. Since f is closed, there is a neighborhood V of y in Y such that $f^{-1}(V) \subset W$ and $V \subset q_f^{-1}(U) \cap Y$. Then, $V \cup (f^{-1}(V) \times (0, \delta)) \subset \phi_f(U)$. Thus, $\phi_f(U)$ is a neighborhood of y in $M(f)$.

We also call $M(f)$ the **mapping cylinder** of f (or the **metrizable mapping cylinder** of f when we need to distinguish it from M_f). As with M_f, we define the **collapsing** $c_f : M(f) \to Y$, which is continuous. Then, $c_f \simeq \mathrm{id}$ rel. Y in $M(f)$ by the homotopy $h^f : M(f) \times \mathbf{I} \to M(f)$ defined in the same manner as $h^f : M_f \times \mathbf{I} \to M_f$. Hence, Y is a strong deformation retract of $M(f)$. The natural map $\phi_f q_f : Y \oplus (X \times \mathbf{I}) \to M(f)$ is abbreviated by q_f, the same notation as the natural quotient map.

From Theorem 6.5.2, we deduce the following:

Corollary 6.5.6 *For any map $f : X \to Y$ between ANRs, the mapping cylinder $M(f)$ is an ANR.* □

The mapping cylinder $M(\mathrm{id}_X) = M_{\mathrm{id}_X}$ of the identity map id_X of X is regarded as the product space $X \times \mathbf{I}$. When X is a subspace of Y, the mapping cylinder $M(i)$ of the inclusion map $i : X \subset Y$ can be regarded as a subspace $(Y \times \{0\}) \cup (X \times \mathbf{I})$ of the product space $Y \times \mathbf{I}$, but M_i cannot be regarded thus unless X is *closed* in Y (cf. Sect. 4.18). When $Y = \{0\}$, there is only one defined map $f : X \to \{0\}$, so we write $C(X) = M(f)$, i.e.,

$$C(X) = \{0\} \cup (X \times (0, 1]),$$

which has the topology generated by open sets in the product space $X \times (0, 1]$ and sets $\{0\} \cup (X \times (0, \varepsilon))$, where $0 < \varepsilon \leqslant 1$. We call $C(X)$ the (**metrizable**) **cone** over X. The following subspace of $C(X)$ is called the (**metrizable**) **open cone** over X:

$$C^o(X) = \{0\} \cup (X \times (0, 1)).$$

Then, $C(X)$ and $C^o(X)$ are contractible.

Corollary 6.5.7 *The cone $C(X)$ over any ANR X is an AR. Hence, so is the open cone $C^o(X)$.* □

For a map to a paracompact space, we have the following lemma:

Lemma 6.5.8 *Let Y be a paracompact space and $f : X \to Y$ be a map. For each open cover $\mathcal{U}$ of $M(f)$, Y has a locally finite open cover $\mathcal{V}$ with a map $\alpha : Y \to (0, 1)$ such that, for each $V \in \mathcal{V}$, $V \cup (f^{-1}(V) \times (0, \sup \alpha(V)])$ is contained in some member of $\mathcal{U}$.*

Proof For each $y \in Y$, let

$$\gamma(y) = \sup \Big\{ s \in (0, 1) \;\Big|\; \exists U \in \mathcal{U},\ \exists V : \text{open in } Y \text{ such that } y \in V \cup (f^{-1}(V) \times (0, s)) \subset U \Big\} > 0$$

Then, $\gamma : Y \to (0, 1)$ is lower semi-continuous (l.s.c.). By Theorem 2.7.8, we have a map $\alpha : Y \to (0, 1)$ such that $\alpha(y) < \gamma(y)$ for every $y \in Y$. For each $y \in Y$, we have $s_y > \alpha(y)$ and an open neighborhood V_y of y in Y such that $V_y \cup (f^{-1}(V_y) \times (0, s_y])$ is contained in some $U \in \mathcal{U}$ and $\alpha(y') < s_y$ for every $y' \in V_y$. Let $\mathcal{V} \in \mathrm{cov}(Y)$ be a locally finite open refinement of the open cover $\{V_y \mid y \in Y\}$. Then, α and $\mathcal{V}$ are as required. □

As seen above, the topology of $M(f)$ is different from that of M_f, but we have the following theorem:

Theorem 6.5.9 *For each map $f : X \to Y$, the natural bijection $\phi_f : M_f \to M(f)$ is a homotopy equivalence with a homotopy inverse $\psi : M(f) \to M_f$ such that*

$$\psi\phi_f \simeq \mathrm{id} \text{ rel. } Y \cup (X \times \{1\}) \text{ and } \phi_f\psi \simeq \mathrm{id} \text{ rel. } Y \cup (X \times \{1\}),$$

hence $M(f) \simeq M_f$ rel. $Y \cup X \times \{1\}$. If Y is paracompact, for each open cover $\mathcal{U}$ of $M(f)$, the homotopy inverse ψ can be chosen such that

$$\psi\phi_f \simeq_{\phi_f^{-1}(\mathcal{U})} \mathrm{id} \text{ rel. } Y \cup (X \times \{1\}) \text{ and } \phi_f\psi \simeq_{\mathcal{U}} \mathrm{id} \text{ rel. } Y \cup (X \times \{1\}).$$

Proof We first prove the case where Y is paracompact. By Lemma 6.5.8, for each open cover $\mathcal{U}$ of $M(f)$, we have a map $\alpha : Y \to (0, 1)$ such that

$$\left\{\{y\} \cup (f^{-1}(y) \times (0, \alpha(y)]) \;\middle|\; y \in Y\right\} \prec \mathcal{U}.$$

Let $q_f : Y \oplus (X \times \mathbf{I}) \to M_f$ be the quotient map. Then, we can define a map $\psi : M(f) \to M_f$ by $\psi|Y = q_f|Y$ and

$$\psi(x, s) = \begin{cases} q_f(x, s) = (x, s) & \text{if } x \in X,\ s \geqslant \alpha(f(x)), \\ q_f(x, 2s - \alpha(f(x))) & \text{if } x \in X,\ \alpha(f(x))/2 \leqslant s \leqslant \alpha(f(x)), \\ q_f(x, 0) = f(x) & \text{if } x \in X,\ 0 < s \leqslant \alpha(f(x))/2. \end{cases}$$

The continuity of ψ follows from that of $\psi|X \times (0, 1]$ and $\psi|W$, where

$$W = Y \cup \left\{(x, s) \in X \times (0, 1] \;\middle|\; s < \alpha(f(x))/2\right\}.$$

Now, we define a homotopy $h : (Y \oplus (X \times \mathbf{I})) \times \mathbf{I} \to Y \oplus (X \times \mathbf{I})$ as follows: $h_0 = \mathrm{id}$, $h_t|Y = \mathrm{id}$ for each $t \in \mathbf{I}$, and

$$h_t(x, s) = \begin{cases} (x, s) & \text{if } x \in X,\ s \geqslant t\alpha(f(x)), \\ (x,\ 2s - t\alpha(f(x))) & \text{if } x \in X,\ t\alpha(f(x))/2 \leqslant s \leqslant t\alpha(f(x)), \\ (x, 0) & \text{if } x \in X,\ s \leqslant t\alpha(f(x))/2. \end{cases}$$

Then, as is easily shown, h induces a $\mathcal{U}$-homotopy $h' : M(f) \times \mathbf{I} \to M(f)$ and the $\phi_f^{-1}(\mathcal{U})$-homotopy $h'' : M_f \times \mathbf{I} \to M_f$ such that

$$h'_0 = \mathrm{id},\ h'_1 = \phi_f\psi,\ \text{and}\ h'_t|Y \cup (X \times \{1\}) = \mathrm{id}\ \text{ for each } t \in \mathbf{I};$$
$$h''_0 = \mathrm{id},\ h''_1 = \psi\phi_f,\ \text{and}\ h''_t|Y \cup (X \times \{1\}) = \mathrm{id}\ \text{ for each } t \in \mathbf{I}.$$

The continuity of h' can be verified as follows: Since $\phi_f : M_f \to M(f)$ is continuous, the continuity of $h'|M(f) \times (0, 1]$ can be shown in a manner similar to that of ψ. Observe that $X \times (0, 1] \times \{0\}$ has the following open neighborhood in $M(f) \times \mathbf{I}$:

$$W' = \big\{(x, s, t) \in X \times (0, 1] \times \mathbf{I} \,\big|\, s > t\alpha(f(x))\big\}.$$

The continuity of $h'|W'$ is obvious. For each neighborhood V of y in Y and $\varepsilon \in (0, 1)$, it is easy to show that

$$h'((V \cup (f^{-1}(V) \times (0, \varepsilon))) \times \mathbf{I}) \subset V \cup (f^{-1}(V) \times (0, \varepsilon)).$$

Thus, h' is also continuous at $(y, 0)$.

In the general case, since there are no covering estimations, we can take a constant map as α (e.g., $\alpha(y) = 1/2$) in the above proof. □

For maps $f_i : X_i \to X_{i-1}, i = 1, \ldots, n$, let $M(f_1, \ldots, f_n) = \bigcup_{i=1}^{n} M(f_i)$ be the adjunction space, where each $X_i \times \{1\} \subset M(f_i)$ is identified with $X_i \subset M(f_{i+1})$. We call $M(f_1, \ldots, f_n)$ the **mapping telescope** of $f_1, \ldots, f_n$. By Theorem 6.5.9, Theorems 4.18.1 and 4.18.3 are also valid for the metrizable mapping cylinders.

Corollary 6.5.10 *For maps $f, g : X \to Y$, the following conditions are equivalent:*

(a) $f \simeq g$;
(b) $M(f) \simeq M(g)$ *rel.* $Y \cup X \times \{1\}$;
(c) *There is a map* $\varphi : M(f) \to M(g)$ *with* $\varphi|Y \cup X \times \{1\} = \mathrm{id}$. □

Corollary 6.5.11 *For maps $f : X \to Y$ and $g : Y \to Z$,*

$$M(gf) \simeq M(g, f)\ \ \textit{rel.}\ Z \cup X \times \{1\}.$$ □

The above corollaries can also be proved directly, which is a good exercise for the reader.

6.6 Realizations of Simplicial Complexes

Let X be a space and $\mathcal{U} \in \mathrm{cov}(X)$. Let K be a simplicial complex and L be a subcomplex of K with $K^{(0)} \subset L$. Then, a map $f : |L| \to X$ is called a **partial $\mathcal{U}$-realization** of K in X if $\{f(\sigma \cap |L|) \mid \sigma \in K\} \prec \mathcal{U}$. A **full $\mathcal{U}$-realization** of K in X is a map $f : |K| \to X$ such that $\{f(\sigma) \mid \sigma \in K\} \prec \mathcal{U}$. We call $\mathcal{V} \in \mathrm{cov}(X)$ a **Lefschetz refinement** of $\mathcal{U}$ and write

$$\mathcal{V} \underset{L}{\prec} \mathcal{U} \quad \text{or} \quad \mathcal{U} \underset{L}{\succ} \mathcal{V}$$

if $\mathcal{V} \prec \mathcal{U}$ and any partial $\mathcal{V}$-realization of an arbitrary simplicial complex in X extends to a full $\mathcal{U}$-realization in X. The following is called LEFSCHETZ'S CHARACTERIZATION of ANRs.

Theorem 6.6.1 (LEFSCHETZ) *A metrizable space X is an ANR if and only if any open cover of X admits a Lefschetz refinement.*

Proof (*The "only if" Part*) By the Arens–Eells Embedding Theorem 6.2.1, we may assume that X is a closed set in a normed linear space E. Then, we have an open neighborhood W of X in E and a retraction $r : W \to X$. For each open cover $\mathcal{U}$ of X, $r^{-1}(\mathcal{U})$ is an open cover of W, which has a refinement $\mathcal{V}$ consisting of open convex sets in E. We show that $\mathcal{V}|X$ is a Lefschetz refinement of $\mathcal{U}$.

Let K be a simplicial complex, L be a subcomplex of K with $K^{(0)} \subset L$, and let $f_0 : |L| \to X$ be a partial $(\mathcal{V}|X)$-realization of K. By induction, we can obtain maps $f_n : |L \cup K^{(n)}| \to W$, $n \in \mathbb{N}$, so that

$$f_n\big||L \cup K^{(n-1)}| = f_{n-1} \quad \text{and}$$
$$f_n(\sigma \cap |L \cup K^{(n)}|) \subset \langle f_0(\sigma \cap |L|)\rangle \quad \text{for each } \sigma \in K.$$

Indeed, given f_{n-1}, then for each $\sigma \in K^{(n)} \setminus (L \cup K^{(n-1)})$, $f_{n-1}|\partial\sigma$ extends to a map $f_\sigma : \sigma \to \langle f_0(\sigma \cap |L|)\rangle$ by the Dugundji Extension Theorem 6.1.1. Thus, f_n can be defined by $f_n|\sigma = f_\sigma$ for $\sigma \in K^{(n)} \setminus (L \cup K^{(n-1)})$. For each $\sigma \in K$, we write $\sigma \cap |L \cup K^{(n)}| = \bigcup_{i=1}^m \sigma_i$, where $\sigma_i \in L \cup K^{(n)}$ for each $i = 1, \cdots, m$. Then,

$$f_n(\sigma \cap |L \cup K^{(n)}|) = \bigcup_{i=1}^m f_{\sigma_i}(\sigma_i) \subset \bigcup_{i=1}^m \langle f_0(\sigma_i \cap |L|)\rangle \subset \langle f_0(\sigma \cap |L|)\rangle.$$

Let $f : |K| \to W$ be the map defined by $f||L \cup K^{(n)}| = f_n$. For each $\sigma \in K$, $f_0(\sigma \cap |L|)$ is contained in some $V \in \mathcal{V}$, which implies that

$$f(\sigma) \subset \langle f_0(\sigma \cap |L|)\rangle \subset V.$$

Thus, f is a full $\mathcal{V}$-realization of K in W. Since $\mathcal{V} \prec r^{-1}(\mathcal{U})$, $r \circ f : |K| \to X$ is a full $\mathcal{U}$-realization of K in X, which is an extension of f_0.

(*The "if" Part*) It suffices to show that X is an ANE. Let $f : A \to X$ be a map from a closed set A in a metrizable space Y. Given $d \in \mathrm{Metr}(X)$, take open covers of X as follows:

$$\mathcal{U}_1 \underset{L}{\succ} \mathcal{V}_1 \overset{*}{\succ} \mathcal{U}_2 \underset{L}{\succ} \mathcal{V}_2 \overset{*}{\succ} \mathcal{U}_3 \underset{L}{\succ} \mathcal{V}_3 \overset{*}{\succ} \cdots,$$

where $\mathrm{mesh}\,\mathcal{U}_n < 2^{-n}$ for each $n \in \mathbb{N}$. For each $n \in \mathbb{N}$, let

$$\mathcal{W}_n = \big\{ f^{-1}(U) \times J_n \;\big|\; U \in \mathcal{U}_{n+2} \big\},$$

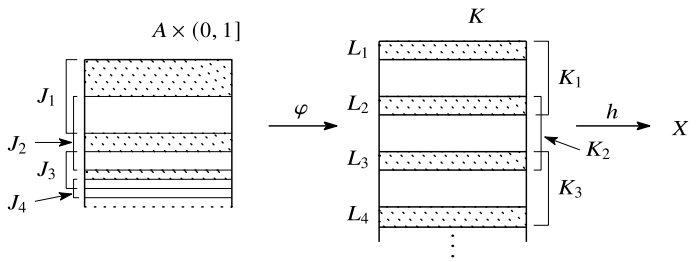


Fig. 6.15 The nerves K_n and L_n

where $J_1 = (2^{-1}, 1]$ and $J_n = (2^{-n}, 2^{-n}3)$ for $n > 1$. Then, $\mathcal{W} = \bigcup_{n\in\mathbb{N}} \mathcal{W}_n$ is an open cover of $A \times (0, 1]$. Let K be the nerve of $\mathcal{W}$ with $\varphi : A \times (0, 1] \to |K|$ a canonical map. For each $n \in \mathbb{N}$, let K_n and L_n be the nerves of $\mathcal{W}_n \cup \mathcal{W}_{n+1}$ and $\mathcal{W}_n$, respectively, which are naturally regarded as subcomplexes of K. Then, we have

$$K = \bigcup_{n\in\mathbb{N}} K_n, \ \ K_n \cap K_{n+1} = L_{n+1} \ \text{ and } \ L_n \cap L_{n+1} = \emptyset.$$

For each $n \in \mathbb{N}$, let $g'_n : |L_n^{(0)}| \to X$ be a map such that $g'_n(W) \in f\mathrm{pr}_A(W)$ for each $W \in \mathcal{W}_n$. Observe that g'_n is a partial $\mathcal{V}_{n+1}$-realization of L_n in X. Then, g'_n extends to a full $\mathcal{U}_{n+1}$-realization $g_n : |L_n| \to X$. We define $h'_n : |L_n \cup L_{n+1}| \to X$ by $h'_n||L_n| = g_n$ and $h'_n||L_{n+1}| = g_{n+1}$. Observe that h'_n is a partial $\mathcal{V}_n$-realization of K_n in X. Then, h'_n extends to a full $\mathcal{U}_n$-realization $h_n : |K_n| \to X$. We can define a map $h : |K| \to X$ by $h||K_n| = h_n$. Thus, we have the map $h\varphi : A \times (0, 1] \to X$. Refer to Fig. 6.15.

For each $(a, t) \in A \times (0, 1]$, let $\sigma \in K$ be the carrier of $\varphi(a, t)$ and $W \in \sigma^{(0)}$. Since φ is a canonical map, we have $(a, t) \in W$ (Proposition 4.15.2). When $\sigma \in K_n$, $h(\sigma) = h_n(\sigma)$ is contained in some $U \in \mathcal{U}_n$, hence $h\varphi(a, t) \in U$. Moreover, since $(a, t) \in W \in \mathcal{W}_n \cup \mathcal{W}_{n+1}$, it follows that $t > 2^{-n-1}$ and $f\mathrm{pr}_A(W)$ is contained in some $U' \in \mathcal{U}_{n+2} \cup \mathcal{U}_{n+3}$. Note that

$$h(W) = g'_i(W) \in f\mathrm{pr}_A(W) \cap U \subset U' \cap U \ \text{ and } \ f(a) \in U',$$

where $i = n$ or $i = n + 1$. Then, it follows that

$$\begin{aligned} d(h\varphi(a, t), f(a)) &\leqslant d(h\varphi(a, t), h(W)) + d(h(W), f(a)) \\ &\leqslant \operatorname{diam} U + \operatorname{diam} U' \\ &< 2^{-n} + 2^{-n-2} = 2^{-n-1}3 < 3t. \end{aligned}$$

Therefore, $h\varphi$ can be extended to a homotopy $\tilde{h} : A \times \mathbf{I} \to X$ by $\tilde{h}_0 = f$.

On the other hand, since $|K|$ is an ANE, φ extends over a neighborhood of $A \times (0, 1]$ in $Y \times (0, 1]$, hence so does $h\varphi$. Then, we can apply Lemma 6.5.1 to extend f over a neighborhood of A in Y. □

Recall that a space X is **homotopy dominated** by a space Y (or Y **homotopy dominates** X) if there are maps $f : X \to Y$ and $g : Y \to X$ such that $gf \simeq \mathrm{id}_X$. If $gf \simeq_{\mathcal{U}} \mathrm{id}_X$ for an open cover $\mathcal{U}$ of X (resp. $gf \simeq_{\varepsilon} \mathrm{id}_X$ for $\varepsilon > 0$), we say that X is **$\mathcal{U}$-homotopy dominated** (resp. **ε-homotopy dominated**) by Y. The following theorem is HANNER'S CHARACTERIZATION of ANRs:

Theorem 6.6.2 (HANNER) *For a metric space $X = (X, d)$, the following are equivalent:*

(a) *X is an ANR;*
(b) *For each open cover $\mathcal{U}$ of X, there is a simplicial complex K such that X is $\mathcal{U}$-homotopy dominated by $|K|$;*
(c) *For any $\varepsilon > 0$, X is ε-homotopy dominated by an ANE.*

Proof The implication (b) ⇒ (c) follows from the fact that $|K|$ is an ANE for any simplicial complex K. We will prove the implications (a) ⇒ (b) and (c) ⇒ (a).

(a) ⇒ (b): For each open cover $\mathcal{U}$ of X, applying Theorem 6.6.1 and Theorem 6.3.7 with Proposition 6.3.1, we can take open refinements as follows:

$$\mathcal{U} \underset{h}{\succ} \mathcal{V} \overset{*}{\succ} \mathcal{V}' \underset{L}{\succ} \mathcal{W} \overset{*}{\succ} \mathcal{W}'.$$

Let K be the nerve of $\mathcal{W}'$ with a canonical map $f : X \to |K|$. For each $W \in \mathcal{W}' = K^{(0)}$, choosing $g'(W) \in W$, we can obtain a partial $\mathcal{W}$-realization $g' : K^{(0)} \to X$ of K, which extends to a full $\mathcal{V}'$-realization $g : |K| \to X$ of K. Observe that gf is $\mathcal{V}$-close to id_X, which means $gf \simeq_{\mathcal{U}} \mathrm{id}_X$. Thus, X is $\mathcal{U}$-homotopy dominated by K.

(c) ⇒ (a): We show that X is an ANE. Let $f : A \to X$ be a map of a closed set A in a metrizable space Y. For each $n \in \mathbb{N}$, we have an ANE X_n, maps $j_n : X \to X_n$ and $k_n : X_n \to X$, and a 3^{-n}-homotopy $h^n : X \times \mathbf{I} \to X$ with $h^n_0 = \mathrm{id}_X$ and $h^n_1 = k_n j_n$. For each $n \in \mathbb{N}$, let $I'_n = [3^{-n}2, 3^{-(n-1)}]$ and $I''_n = [3^{-n}, 3^{-n}2]$. Since each X_n is an ANE, $j_n f$ extends to a map $f_n : B_n \to X_n$ from a closed neighborhood B_n of A in Y. We may assume that $B_{n+1} \subset B_n$ for each $n \in \mathbb{N}$. For each $n \in \mathbb{N}$, we define a map $g'_n : B_n \times I'_n \to X$ by

$$g'_n(y, t) = h^{n+1}(k_n f_n(y), 3 - 3^n t).$$

Then, for each $y \in B_n$,

$$g'_n(y, 3^{-(n-1)}) = k_n f_n(y) \quad \text{and} \quad g'_n(y, 3^{-n}2) = k_{n+1} j_{n+1} k_n f_n(y).$$

Also, for each $(a, t) \in A \times I'_n$,

$$d(g'_n(a,t), f(a)) = d(h^{n+1}(k_n j_n f(a), 3-3^n t), f(a)) \\ \leqslant 3^{-(n+1)} + 3^{-n} < 3^{-n}2 \leqslant t.$$

For each $n \in \mathbb{N}$, let

$$A_n = A \times I''_n \cup B_{n+1} \times \partial I''_n \subset B_{n+1} \times I''_n.$$

We define a map $h'_n : A_n \to X_{n+1}$ as follows:

$$h'_n(y,t) = \begin{cases} j_{n+1}k_n f_n(y) & \text{if } (y,t) \in B_{n+1} \times \{3^{-n}2\}, \\ f_{n+1}(y) & \text{if } (y,t) \in B_{n+1} \times \{3^{-n}\}, \\ j_{n+1}h^n(f(y), 3^n t - 1) & \text{if } (y,t) \in A \times I''_n. \end{cases}$$

Since X_{n+1} is an ANE, h'_n extends to a map $h''_n : C_n \to X_{n+1}$ of a closed neighborhood C_n of A_n in $B_{n+1} \times I''_n$. Thus, we obtain the map $g''_n = k_{n+1}h''_n : C_n \to X$. Observe that, for each $y \in B_{n+1}$,

$$g''_n(y, 3^{-n}2) = k_{n+1}j_{n+1}k_n f_n(y) \quad \text{and} \quad g''_n(y, 3^{-n}) = k_{n+1}f_{n+1}(y)$$

and for each $(a,t) \in A \times I''_n$,

$$d(g''_n(a,t), f(a)) = d(k_{n+1}j_{n+1}h^n(f(a), 3^n t - 1), f(a)) \\ < 3^{-(n+1)} + 3^{-n} < 3^{-n}2 \leqslant 2t.$$

Then, $W = \bigcup_{n\in\mathbb{N}}((B_n \times I'_n) \cup C_n)$ is a neighborhood of $A \times (0,1]$ in $Y \times (0,1]$. We define a map $g : W \to X$ by

$$g|B_n \times I'_n = g'_n \quad \text{and} \quad g|C_n = g''_n.$$

See Fig. 6.16. Since $d(g(a,t), f(a)) < 2t$ for each $(a,t) \in A \times (0,1]$, we can apply Lemma 6.5.1 to extend f over a neighborhood of A in X. □

Remark 6.6.1 In condition (b) of Theorem 6.6.2, we can take K as a locally finite-dimensional simplicial complex with $\operatorname{card} K^{(0)} \leqslant w(X)$; as a countable and locally finite simplicial complex if X is separable; as a finite simplicial complex if X is compact. Indeed, each open cover of X has a locally finite σ-discrete open refinement with the locally finite-dimensional nerve (Theorem 4.15.11); a countable star-finite open refinement if X is separable (Theorem 4.15.13); a finite open refinement if X is compact. Then, in the proof of (a) $\Rightarrow$ (b), we can take $\mathcal{W}'$ such that $\mathcal{W}'$ is σ-discrete, the nerve K of $\mathcal{W}'$ is locally finite-dimensional, and

$$\operatorname{card} K^{(0)} = \operatorname{card} \mathcal{W}' \leqslant \aleph_0 w(X) = w(X).$$

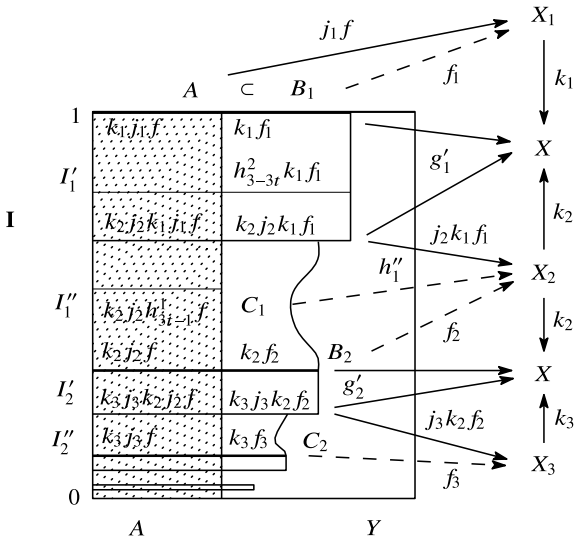


Fig. 6.16 Extension using Lemma 6.5.1

When X is separable, we can take a countable star-finite $\mathcal{W}'$, whose nerve K is countable and locally finite. When X is compact, we can take a finite $\mathcal{W}'$, whose nerve K is finite.

Corollary 6.6.3 *Let $f : X \to Y$ be a map from a paracompact space X to an ANR Y and $\mathcal{U}$ an open cover of Y. Then, each open cover $\mathcal{V}$ of X has an open refinement $\mathcal{W}$ with a map $\psi : |N(\mathcal{W})| \to Y$ such that $\psi\varphi \simeq_{\mathcal{U}} f$ for any canonical map $\varphi : X \to |N(\mathcal{W})|$.*

Proof Let $\mathcal{U}' \in \mathrm{cov}(Y)$ be such that st $\mathcal{U}'$-close maps are $\mathcal{U}$-homotopic. By Theorem 6.6.2, we have a simplicial complex K with maps $\varphi' : Y \to |K|$ and $\psi' : |K| \to Y$ such that $\psi'\varphi' \simeq_{\mathcal{U}'} \mathrm{id}_Y$, hence $f \simeq_{\mathcal{U}'} \psi'\varphi' f$. Replacing K with a small subdivision, we may assume that $O_K \prec \psi'^{-1}(\mathcal{U}')$ (Theorem 4.8.12). Let $\mathcal{W} \in \mathrm{cov}(X)$ be a common refinement of $(\varphi' f)^{-1}(O_K)$ and $\mathcal{V}$. Assigning to each $W \in \mathcal{W} = N(\mathcal{W})^{(0)}$ a $g(W) \in K^{(0)}$ such that $W \subset (\varphi' f)^{-1}(O_K(g(W)))$, we can obtain a simplicial map $g : N(\mathcal{W}) \to K$. Thus, we have a map $\psi = \psi' g : |N(\mathcal{W})| \to Y$. Let $\varphi : X \to |N(\mathcal{W})|$ be a canonical map, that is, $\varphi^{-1}(O_{N(\mathcal{W})}(W)) \subset W$ for each $W \in \mathcal{W}$. For each $x \in X$, choose $W \in \mathcal{W}$ so that $x \in \varphi^{-1}(O_{N(\mathcal{W})}(W))$. Then, $g\varphi(x) \in g(O_{N(\mathcal{W})}(W)) \subset O_K(g(W))$ and $\varphi' f(x) \in \varphi' f(W) \subset O_K(g(W))$. Since $\psi'(O_K) \prec \mathcal{U}'$, it follows that $\psi\varphi = \psi' g\varphi$ is $\mathcal{U}'$-close to $\psi'\varphi' f$, hence $\psi\varphi$ is st $\mathcal{U}'$-close to f, which implies that $\psi\varphi \simeq_{\mathcal{U}} f$. □

Remark 6.6.2 In Corollary 6.6.3, $\mathcal{W}$ can be taken to be locally finite and σ-discrete in X and to have a locally finite-dimensional nerve $N(\mathcal{W})$ (cf. Theorem 4.15.11). If X is regular Lindelöf, we can take a star-finite countable open refinement $\mathcal{W}$, which has the countable locally finite nerve $N(\mathcal{W})$ (cf. Theorem 4.15.13). When X is compact, we can take a finite open refinement $\mathcal{W}$.

By Theorem 6.6.2 and Corollary 4.19.4, we have the following corollary:

Corollary 6.6.4 *Every ANR X has the homotopy type of a locally finite-dimensional simplicial complex K with* $\operatorname{card} K^{(0)} \leqslant w(X)$. *In particular, every separable ANR has the homotopy type of a countable locally finite simplicial complex.* □

For every simplicial complex K, we have $|K| \simeq |K|_{\mathrm{m}}$ by Theorem 4.15.7 and $|K|_{\mathrm{m}}$ is an ANR by Theorem 6.2.17. Then, we have the following corollary:

Corollary 6.6.5 *A space X has the homotopy type of a simplicial complex if and only if it has the homotopy type of an ANR.* □

Moreover, by Corollary 4.20.10, we can obtain:

Corollary 6.6.6 *Let X and Y be ANRs. Then, every weak homotopy equivalence $f : X \to Y$ is a homotopy equivalence.* □

Recall that a space X is **proper homotopy dominated** by a space Y (or Y **proper homotopy dominates** X) if there are proper maps $f : X \to Y$ and $g : Y \to X$ such that $gf \simeq^p \mathrm{id}_X$. If $gf \simeq^p_{\mathcal{U}} \mathrm{id}_X$ for an open cover $\mathcal{U}$ of X (resp. $gf \simeq^p_{\varepsilon} \mathrm{id}_X$ for $\varepsilon > 0$), we say that X is **proper $\mathcal{U}$-homotopy dominated** (resp. **proper ε-homotopy dominated**) by Y.

For locally compact ANRs, the following can be proved in the same way as the implication (a) $\Rightarrow$ (b) of Theorem 6.6.2:

Theorem 6.6.7 *Let X be a locally compact ANR. For each open cover $\mathcal{U}$ of X, there is a locally finite simplicial complex K such that X is proper $\mathcal{U}$-homotopy dominated by $|K|$.*

Sketch of Proof Take open refinements as follows:

$$\mathcal{U} \succ \mathcal{U}' \underset{h}{\succ} \mathcal{V} \overset{*}{\succ} \mathcal{V}' \underset{L}{\succ} \mathcal{W} \overset{*}{\succ} \mathcal{W}',$$

where $\mathcal{U}'$ is locally finite, every $U \in \mathcal{U}'$ has the compact closure, and $\mathcal{W}'$ is star-finite (cf. Theorem 4.15.12). Then, the nerve K of $\mathcal{W}'$ is locally finite. Let $f : X \to |K|$ be a canonical map. Choosing $g'(W) \in W \in \mathcal{W}'$, we have a partial $\mathcal{W}$-realization $g' : K^{(0)} \to X$, which can be extended to a full $\mathcal{V}'$-realization $g : |K| \to X$. Then, observe that f and g are proper, and $gf \simeq^p_{\mathcal{U}'} \mathrm{id}$ (cf. Propositions 4.15.1 and 2.1.19).

Combining Theorems 4.19.5 and 6.6.7, we have the following:

Theorem 6.6.8 *For every locally compact ANR X, the product space $X \times \mathbb{R}_+$ has the proper homotopy type of a locally finite simplicial complex.* □

A subset $A \subset X$ is said to be **homotopy dense** in X if there exists a homotopy $h : X \times \mathbf{I} \to X$ such that $h_0 = \mathrm{id}$ and $h(X \times (0, 1]) \subset A$. When X is paracompact, for each open cover $\mathcal{U} \in \mathrm{cov}(X)$, X is $\mathcal{U}$-homotopy dominated by A and A is $\mathcal{U}|A$-homotopy dominated by X. In fact, we have a lower semi-continuous function $\gamma : X \to (0, 1]$ defined by

$$\gamma(x) = \sup\left\{t \in \mathbf{I} \;\middle|\; h(\{x\} \times [0,t]) \subset U \text{ for some } U \in \mathcal{U}\right\}.$$

By Theorem 2.7.8, there is a map $\alpha : X \to (0,1]$ such that $\alpha(x) < \gamma(x)$ for each $x \in X$. We can define a $\mathcal{U}$-homotopy $h^\alpha : X \times \mathbf{I} \to X$ by $h^\alpha(x,t) = h(x, t\alpha(x))$. Then, $h_1^\alpha : X \to A$, $h_1^\alpha \simeq_{\mathcal{U}} \mathrm{id}_X$ in X, and $h_1^\alpha | A \simeq_{\mathcal{U}} \mathrm{id}_A$ in A. Thus, the following can be obtained as a corollary of Theorem 6.6.2:

Theorem 6.6.9 *Let X be a metrizable space and A be a homotopy dense subset of X. Then, X is an ANR if and only if A is an ANR.* □

Evidently, if $A \subset B \subset X$ and A is homotopy dense in X, then B is also homotopy dense in X. For any $A \subset \partial\mathbf{I}^n$, $\mathbf{I}^n \setminus A$ is homotopy dense in $\mathbf{I}^n$. As is easily observed, the radial interior rint $\boldsymbol{Q} = \bigcup_{n\in\mathbb{N}}[-1+2^{-n}, 1-2^{-n}]^{\mathbb{N}}$ and the pseudo-interior $(-1,1)^{\mathbb{N}}$ of the Hilbert cube $\boldsymbol{Q} = [-1,1]^{\mathbb{N}}$ are homotopy dense in $\boldsymbol{Q}$. By the following proposition, $\boldsymbol{Q}_f = [-1,1]_f^{\mathbb{N}}$ and $(-1,1)_f^{\mathbb{N}}$ are also homotopy dense in $\boldsymbol{Q}$.

Proposition 6.6.10 *Let $x_0 \in A \subset X$. If A is homotopy dense in X and X is contractible, then the following set $A_f^{\mathbb{N}}$ is homotopy dense in $X^{\mathbb{N}}$:*

$$A_f^{\mathbb{N}} = \left\{x \in A^{\mathbb{N}} \;\middle|\; x(n) = x_0 \text{ except for finitely many } n \in \mathbb{N}\right\}.$$

Proof We have a homotopy $h : X \times \mathbf{I} \to X$ such that $h_0 = \mathrm{id}$, $h_t(X) \subset A$ for every $t > 0$, and $h_1(X) = \{x_0\}$. Indeed, let $f : X \times \mathbf{I} \to X$ be a contraction and $g : X \times \mathbf{I} \to X$ be a homotopy such that $g_0 = \mathrm{id}$ and $g_t(X) \subset A$ for every $t > 0$. Then, h can be defined by

$$h_t(x) = \begin{cases} g_{2t} f_{2t}(x) & \text{for } 0 \leqslant t \leqslant 1/2, \\ g_{2-2t} f_{2-2t}(x_0) & \text{for } 1/2 \leqslant t \leqslant 1. \end{cases}$$

Now, we define the homotopy $\varphi : X^{\mathbb{N}} \times \mathbf{I} \to X^{\mathbb{N}}$ as follows: $\varphi_0 = \mathrm{id}$,

$$\begin{aligned}
\varphi_1(x) &= (x_0, x_0, x_0, \dots) \\
\varphi_{2^{-1}}(x) &= (h_{2^{-1}}(x(1)), x_0, x_0, \dots) \\
\varphi_{2^{-2}}(x) &= (h_{2^{-2}}(x(1)), h_{2^{-1}}(x(2)), x_0, x_0, \dots) \\
&\vdots \\
\varphi_{2^{-n}}(x) &= (h_{2^{-n}}(x(1)), h_{2^{-n+1}}(x(2)), \dots, h_{2^{-1}}(x(n)), x_0, x_0, \dots) \\
&\vdots
\end{aligned}$$

and for $2^{-n} < t < 2^{-n+1}$,

$$\varphi_t(x) = (h_t(x(1)), \dots, h_{2^{n-2}t}x(n-1), h_{2^{n-1}t}(x(n)), x_0, x_0, \dots).$$

Then, $\varphi_t(X^{\mathbb{N}}) \subset A_f^{\mathbb{N}}$ for every $t > 0$. □

A space $X \neq \emptyset$ is said to be **homotopically trivial** if, for each $n \in \mathbb{N}$, every map $f : \mathbf{S}^{n-1} \to X$ is null-homotopic, that is, f extends over $\mathbf{B}^n$. As is easily observed, X is homotopically trivial if and only if the map from X to the singleton $\{0\}$ is a weak homotopy equivalence.

Proposition 6.6.11 *Let K be a simplicial (or cell) complex and L be a subcomplex of K. If X is homotopically trivial then any map $f : |L| \to X$ extends to a map $\tilde{f} : |K| \to X$.*

Proof For each $n \in \omega$, let $K_n = K^{(n)} \cup L$. Then, $K = \bigcup_{n\in\omega} K_n$. Since X is homotopically trivial, we can inductively construct maps $f_n : |K_n| \to X$, $n \in \omega$, so that $f_n||K_{n-1}| = f_{n-1}$, where $K_{-1} = L$ and $f_{-1} = f$. The desired extension $\tilde{f}$ is defined by $\tilde{f}||K_n| = f_n$ for each $n \in \omega$. □

Every contractible space is homotopically trivial. For an ANR X, the converse is also true. In fact, if X is homotopically trivial, it follows from Corollary 6.6.6 that X is homotopy equivalent to the singleton $\{0\}$, which means that X is contractible. As a consequence, the following characterization follows from Corollary 6.2.14:

Theorem 6.6.12 *A space is an AR if and only if it is a homotopically trivial ANR.* □

For a metrizable LEC space X, the condition of Lefschetz's Characterization of ANRs (Theorem 6.6.1) can be weakened as follows:

Theorem 6.6.13 (DUGUNDJI) *A metrizable LEC space X is an ANR if and only if each open cover $\mathcal{V}$ has an open refinement $\mathcal{W}$ such that for an arbitrarily simplicial complex K, every partial $\mathcal{W}$-realization $f : K^{(0)} \to X$ of K extends to a full $\mathcal{V}$-realization $\tilde{f} : |K| \to X$ of K.*

Proof It is enough to show the "if" part. By Hanner's Characterization of ANRs (Theorem 6.6.2), it suffices to show that for each $\mathcal{V} \in \mathrm{cov}(X)$, there is a simplicial complex K such that X is $\mathcal{V}$-homotopy dominated by $|K|$. Take $\mathcal{V}', \mathcal{V}'' \in \mathrm{cov}(X)$ so that $\mathrm{st}\,\mathcal{V}' \prec \mathcal{V}''$ and $\mathcal{V}''$ is an $\hbar$-refinement of $\mathcal{V}$. By the condition, we have $\mathcal{W}' \in \mathrm{cov}(X)$ such that for an arbitrary simplicial complex K, every partial $\mathcal{W}'$-realization $g : K^{(0)} \to X$ extends to a full $\mathcal{V}'$-realization $\tilde{g} : |K| \to X$. Let $\mathcal{W} \in \mathrm{cov}(X)$ be a star-refinement of $\mathcal{W}'$.

Now, let K be the nerve of $\mathcal{W}$ with $f : X \to |K|$ a canonical map. We will show that X is $\mathcal{V}$-homotopy dominated by $|K|$. Choosing a point $g_0(W) \in W$ for each $W \in \mathcal{W} = K^{(0)}$, we define a map $g_0 : K^{(0)} \to X$. For each simplex $\sigma = \langle W_0, \ldots, W_n\rangle \in K$, we can take $x_0 \in \bigcap_{i=0}^n W_i$ $(\neq \emptyset)$. Then, $g_0(\sigma^{(0)}) \subset \bigcup_{i=0}^n W_i \subset \mathrm{st}(x_0, \mathcal{W})$. Since $\mathcal{W}$ is a star-refinement of $\mathcal{W}'$, $g_0(\sigma^{(0)})$ is contained in some member of $\mathcal{W}'$. Thus, g_0 is a partial $\mathcal{W}'$-realization of K. Hence, g_0 extends to a full $\mathcal{V}'$-realization $g : |K| \to X$ of K. To see that $gf \simeq_{\mathcal{V}} \mathrm{id}$, it suffices to show that gf is $\mathcal{V}''$-close to id. Due to Proposition 4.15.2, for each $x \in X$, $c_K(f(x))^{(0)} = \{W_0, \ldots, W_n\} \subset \mathcal{W}[x]$. As we saw above, $g(c_K(f(x))^{(0)}) \subset \mathrm{st}(x, \mathcal{W})$. Since g is a full $\mathcal{V}'$-realization of K, $g(c_K(f(x)))$ is contained in some $V \in \mathcal{V}'$. Hence, $x, gf(x) \in \mathrm{st}(V, \mathcal{W}) \subset \mathrm{st}(V, \mathcal{V}')$. Since $\mathrm{st}\,\mathcal{V}' \prec \mathcal{V}''$, it follows that gf is $\mathcal{V}''$-close to id. The proof is completed. □

Remark 6.6.3 In the above proof, it suffices to take $\mathcal{V}''$ as an h-refinement of $\mathcal{V}$ instead of $\hbar$-refinement. Therefore, Theorem 6.6.13 is valid even if "LEC" can be replaced by "ULC" (cf. Theorem 6.3.7).

Let X be an LEC space with an equi-connecting map $\lambda : U \times \mathbf{I} \to X$. A subset $A \subset X$ is said to be **λ-convex** if $A^2 \subset U$ and $\lambda(A^2 \times \mathbf{I}) \subset A$, that is, $\lambda(x, y, t) \in A$ for every $x, y \in A$ and $t \in \mathbf{I}$. For $A \subset X$, if A is contained in some $B \subset X$ such that $A \times B \subset U$ and $\lambda(A \times B \times \mathbf{I}) \subset B$, then the **$\lambda$-extension** A^λ of A is defined as the intersection of all such subsets of X, which is the smallest subset of X such that $A \times A^\lambda \subset U$ and $\lambda(A \times A^\lambda \times \mathbf{I}) \subset A^\lambda$, where $\lambda(A \times A^\lambda \times \mathbf{I}) = A^\lambda$ because $A^\lambda \subset \lambda(A \times A^\lambda \times \mathbf{I}) \subset A^\lambda$.

We can also construct A^λ as follows: First, let $A_0 = A$ and inductively define $A_n = \lambda(A \times A_{n-1} \times \mathbf{I})$ if $A \times A_{n-1} \subset U$, where $A \subset A_n$ can be seen by induction. When $A \times A_{n-1} \not\subset U$, A_n is not defined. If all A_n, $n \in \omega$, are defined, then we can define $A^\lambda = \bigcup_{n\in\omega} A_n$. Indeed,

$$\lambda(A \times A^\lambda \times \mathbf{I}) = \bigcup_{n\in\omega} \lambda(A \times A_n \times \mathbf{I}) \subset \bigcup_{n\in\mathbb{N}} A_n \subset A^\lambda.$$

If $\lambda(A \times B \times \mathbf{I}) \subset B$, then $A_0 = A \subset B$ and it inductively follows that

$$A_n = \lambda(A \times A_{n-1} \times \mathbf{I}) \subset \lambda(A \times B \times \mathbf{I}) \subset B.$$

Hence, $A^\lambda = \bigcup_{n\in\omega} A_n \subset B$.

For $A \subset B \subset X$, it is said that A is **λ-stable** in B if A^λ is defined and $A^\lambda \subset B$. Evidently, each λ-convex set A is λ-stable in itself, hence in any B containing A. We define a **λ-stable refinement** of an open cover $\mathcal{V}$ of X as an open refinement $\mathcal{W}$ such that each $W \in \mathcal{W}$ is λ-stable in some $V \in \mathcal{V}$.

Theorem 6.6.14 (DUGUNDJI) *Let X be a metrizable LEC space with an equi-connecting map $\lambda : U \times \mathbf{I} \to X$. If every open cover $\mathcal{V} \in \mathrm{cov}(X)$ has a λ-stable refinement $\mathcal{W} \in \mathrm{cov}(X)$, then X is an ANR.*

Proof By virtue of Theorem 6.6.13, it suffices to show that if $\mathcal{W}$ is a λ-stable refinement of $\mathcal{V} \in \mathrm{cov}(X)$ and K is a simplicial complex, then every partial $\mathcal{W}$-realization $f : |K^{(0)}| \to X$ extends to a full $\mathcal{V}$-realization $\tilde{f} : |K| \to X$.

We may assume that K is an ordered simplicial complex. Each $\sigma \in K$ with $\dim \sigma > 0$ can be written $\sigma = v_\sigma \sigma_*$, where $v_\sigma = \min \sigma^{(0)}$ and $\sigma_* < \sigma$ is the opposite face of σ to v_σ. For each $\sigma \in K$, we can define a map $f_\sigma : \sigma \to X$ by induction on $\dim \sigma$, so as to satisfy the following conditions:

(1) $\tau \leqslant \sigma \ \Rightarrow \ f_\sigma|\tau = f_\tau$,
(2) $f_\sigma(\sigma) \subset f(\sigma^{(0)})^\lambda$,
(3) $f_\sigma((1-t)v_\sigma + tx) = \lambda(f(v_\sigma), f_{\sigma_*}(x), t)$ for each $x \in \sigma_*$ and $t \in \mathbf{I}$,

where $f(\sigma^{(0)})^\lambda$ in (2) is defined because f is a partial $\mathcal{W}$-realization of K and $\mathcal{W}$ is a λ-stable refinement of $\mathcal{V}$. In (3), $(f(v_\sigma), f_{\sigma_*}(x)) \in U$ because

$$f(\sigma^{(0)}) \times f_{\sigma_*}(\sigma_*) \subset f(\sigma^{(0)}) \times f(\sigma_*^{(0)})^\lambda \subset f(\sigma^{(0)}) \times f(\sigma^{(0)})^\lambda \subset U,$$

where the first inclusion comes from (2) for f_{σ_*}, which is assumed by induction. Hence, (3) gives the definition of f_σ. To verify (1), let $\tau < \sigma$. We distinguish three cases: (i) $v_\sigma \notin \tau^{(0)}$, (ii) $\tau = v_\sigma$, and (iii) $\tau \neq v_\sigma \in \tau^{(0)}$. In case (i), since $\tau \leqslant \sigma_*$ and $f_\sigma|\sigma_* = f_{\sigma_*}$ by (3), we have $f_\sigma|\tau = f_{\sigma_*}|\tau = f_\tau$ by (2) for f_{σ_*}. In case (ii), we have $f_\sigma|v_\sigma = f_{v_\sigma} = f|v_\sigma$ by (3). In case (iii), $v_\tau = v_\sigma$, $\tau = v_\tau \tau^* = v_\sigma \tau_*$, and $\tau_* < \sigma_*$. Since $f_{\sigma_*}|\tau_* = f_{\tau_*}$, we have $f_\sigma|\tau = f_\tau$ by (3) for τ.

Now, by (1), we can define a map $\tilde{f} : |K| \to X$ by $\tilde{f}|\sigma = f_\sigma$ for each $\sigma \in K$. Then, $\tilde{f}$ is an extension of f. For each n-simplex $\sigma \in K$, since f is a partial $\mathcal{W}$-realization of K and $\mathcal{W}$ is a λ-stable refinement of $\mathcal{V}$, there exist $W \in \mathcal{W}$ and $V \in \mathcal{V}$ such that $f(\sigma^{(0)}) \subset W$ and $W^\lambda \subset V$. By (2), we have

$$\tilde{f}(\sigma) = f_\sigma(\sigma) \subset f(\sigma^{(0)})^\lambda \subset W^\lambda \subset V.$$

Thus, $\tilde{f}$ is a full $\mathcal{V}$-realization of K. □

Let X be a metrizable LEC space with an equi-connecting map $\lambda : U \times \mathbf{I} \to X$. Suppose that each point $x \in X$ has arbitrarily small λ-convex neighborhoods, that is, for each open neighborhood V of x, there is a λ-convex neighborhood W of x such that $W \subset V$. Then, $(\operatorname{int} W)^\lambda \subset W^\lambda = W$. Hence, every open cover $\mathcal{V}$ has a λ-stable open refinement $\mathcal{W}$. Thus, we have the following:

Corollary 6.6.15 *A metrizable LEC (resp. EC) space $X \neq \emptyset$ is an ANR (resp. AR) if X has an equi-connecting map λ such that each point of X has arbitrarily small λ-convex neighborhoods.* □

6.7 Fine Homotopy Equivalences

For an open cover $\mathcal{U}$ of a space Y, a map $f : X \to Y$ is called a **$\mathcal{U}$-homotopy equivalence** if there is a map $g : Y \to X$ such that $gf \simeq_{f^{-1}(\mathcal{U})} \mathrm{id}_X$ and $fg \simeq_{\mathcal{U}} \mathrm{id}_Y$, where g is called a **$\mathcal{U}$-homotopy inverse** of f. For every simplicial complex K, $\mathrm{id} : |K| \to |K|_{\mathrm{m}}$ is an $\mathcal{O}_K$-homotopy equivalence (cf. Theorem 4.15.7). We call $f : X \to Y$ a **fine homotopy equivalence** if f is a $\mathcal{U}$-homotopy equivalence for each open cover $\mathcal{U}$ of Y. For example, if $X = Y \times Z$ and Z is contractible, then the projection $\mathrm{pr}_Y : X \to Y$ is a fine homotopy equivalence. Note that the image of a fine homotopy equivalence is dense in its range (or codomain).

Proposition 6.7.1 *A subset X of a metrizable space Y is homotopy dense in Y if and only if the inclusion map $X \subset Y$ is a fine homotopy equivalence.*

Proof The "only if" part was shown in the arguments before Theorem 6.6.9.

To prove the "if" part, assume that the inclusion map $X \subset Y$ is a fine homotopy equivalence and let $d \in \mathrm{Metr}(Y)$. For each $n \in \mathbb{N}$, we have a map $f_n : Y \to X$ with 3^{-n}-homotopies $\varphi^{(n)} : Y \times \mathbf{I} \to Y$ and $\psi^{(n)} : X \times \mathbf{I} \to X$ such that $\varphi_0^{(n)} = \mathrm{id}_Y$,

$\varphi_1^{(n)} = f_n$, $\psi_0^{(n)} = \mathrm{id}_X$, and $\psi_1^{(n)} = f_n|X$. We can define a homotopy $h : Y \times \mathbf{I} \to Y$ as follows: $h_0 = \mathrm{id}$ and for $t > 0$,

$$h_t = \begin{cases} f_i \varphi_{3-3^i t}^{(i+1)} & \text{if } 3^{-i}2 \leqslant t \leqslant 3^{-i+1}, \\ \psi_{3^i t-1}^{(i)} f_{i+1} & \text{if } 3^{-i} \leqslant t \leqslant 3^{-i}2. \end{cases}$$

Then, $d(h_t, \mathrm{id}) < 2t$ because

$$d(h_t, \mathrm{id}) < 3^{-i} + 3^{-i-1} < 3^{-i}2 \text{ if } 3^{-i} \leqslant t \leqslant 3^{-i+1}.$$

Hence, h is continuous at points of $Y \times \{0\}$. Since $h(Y \times (0, 1]) \subset X$, it follows that X is homotopy dense in Y. □

Proposition 6.7.2 *For each map $f : X \to Y$, both of the collapsings $c_f : M(f) \to Y$ and $c_f : M_f \to Y$ are fine homotopy equivalences.*

Proof Let $j : Y \to M(f)$ be the inclusion map. Then, $c_f j = \mathrm{id}_Y$. The homotopy $h^f : M(f) \times \mathbf{I} \to M(f)$ is a $c_f^{-1}(\mathcal{U})$-homotopy from $c_f = jc_f$ to $\mathrm{id}_{M(f)}$ for every $\mathcal{U} \in \mathrm{cov}(Y)$. Thus, $c_f : M(f) \to Y$ is a fine homotopy equivalence. The same proof is valid for $c_f : M_f \to Y$. □

The following has been shown in Theorem 6.5.9 as an additional statement:

Proposition 6.7.3 *Let Y be a paracompact space. For each map $f : X \to Y$, the natural bijection $\phi_f : M_f \to M(f)$ is a fine homotopy equivalence.* □

From Theorem 6.6.2, it follows that the range of a fine homotopy equivalence of an ANR is an ANR if it is metrizable. This extends as follows:

Theorem 6.7.4 (G. KOZLOWSKI) *Let $f : X \to Y$ be a map from an ANR X to a metrizable space Y such that $f(X)$ is dense in Y and, for each open cover $\mathcal{U}$ of Y, there is a map $g : Y \to X$ with $gf \simeq_{f^{-1}(\mathcal{U})} \mathrm{id}_X$. Then, f is a fine homotopy equivalence and Y is an ANR.*

Proof First, we prove the following two claims:

Claim (1) *For a map $g : Y \to X$, if gf is $f^{-1}(\mathcal{U})$-close to id_X, then fg is $\mathrm{st}\,\mathcal{U}$-close to id_Y.*

For each $y \in Y$, choose $U_1, U_2 \in \mathcal{U}$ so that $y \in U_1$ and $fg(y) \in U_2$, i.e., $y \in U_1 \cap (fg)^{-1}(U_2)$. Since $f(X)$ is dense in Y, we have an $x \in X$ such that $f(x) \in U_1 \cap (fg)^{-1}(U_2)$. Since gf is $f^{-1}(\mathcal{U})$-close to id_X, we have a $U_3 \in \mathcal{U}$ such that x, $gf(x) \in f^{-1}(U_3)$, i.e., $f(x)$, $fgf(x) \in U_3$. Then, $U_1 \cap U_3 \neq \emptyset$ and $U_2 \cap U_3 \neq \emptyset$. Hence, it follows that y, $fg(y) \in \mathrm{st}(U_3, \mathcal{U})$.

Claim (2) *Each open cover $\mathcal{U}$ of Y has an open refinement $\mathcal{V}$ such that $f^{-1}(\mathcal{V})$ is an h-refinement of $f^{-1}(\mathcal{U})$.*

For an open star-refinement $\mathcal{W}$ of $\mathcal{U}$, we have a map $g : Y \to X$ such that $gf \simeq_{f^{-1}(\mathcal{W})} \mathrm{id}_X$. By Corollary 6.3.6, $f^{-1}(\mathcal{W})$ has an h-refinement $\mathcal{W}'$. Let $\mathcal{V} \in \mathrm{cov}(Y)$ be a common refinement of $g^{-1}(\mathcal{W}')$ and $\mathcal{U}$. Then, $\mathcal{V}$ is the desired refinement. Indeed, let $k, k' : Z \to X$ be $f^{-1}(\mathcal{V})$-close maps. Then, gfk and gfk' are $\mathcal{W}'$-close, so $gfk \simeq_{f^{-1}(\mathcal{W})} gfk'$. Since $gf \simeq_{f^{-1}(\mathcal{W})} \mathrm{id}_X$, it follows that $k \simeq_{\mathrm{st}\, f^{-1}(\mathcal{W})} k'$, so $k \simeq_{f^{-1}(\mathcal{U})} k'$.

Now, we will prove the theorem. Because of Theorem 6.6.2, it suffices to show that f is a fine homotopy equivalence. For each open cover $\mathcal{U}$ of Y, Y has an admissible metric ρ such that $\{\overline{\mathrm{B}}_\rho(y, 1) \mid y \in Y\} \prec \mathcal{U}$ (Proposition 2.7.9 (1)). Using Claim (2) inductively, we can take $\mathcal{U}_i \in \mathrm{cov}(Y)$, $i \in \omega$, such that

$$\mathcal{U}_0 \overset{*}{\succ} \mathcal{U}_1 \overset{*}{\succ} \cdots, \ \mathrm{mesh}\, \mathcal{U}_n < 2^{-n-1} \ \text{ and } \ f^{-1}(\mathcal{U}_n) \underset{h}{\prec} f^{-1}(\mathcal{U}_{n-1}).$$

By the condition on f, we have maps $g_n : Y \to X$, $n \in \mathbb{N}$, such that $g_n f \simeq_{f^{-1}(\mathcal{U}_{n+2})} \mathrm{id}_X$. Then, fg_n is $\mathcal{U}_{n+1}$-close to id_Y by Claim (1). Since fg_n and fg_{n+1} are $\mathcal{U}_n$-close, that is, g_n and g_{n+1} are $f^{-1}(\mathcal{U}_n)$-close, we have an $f^{-1}(\mathcal{U}_{n-1})$-homotopy $h_n : Y \times \mathbf{I} \to X$ such that $h_{n,0} = g_n$ and $h_{n,1} = g_{n+1}$. Let $h : Y \times \mathbf{I} \to Y$ be the homotopy defined as follows:

$$h(y, t) = \begin{cases} fh_n(y, 2 - 2^n t) & \text{if } \ 2^{-n} \leqslant t \leqslant 2^{-n+1}, \ n \in \mathbb{N} \\ y & \text{if } \ t = 0. \end{cases}$$

Then, h is a $\mathcal{U}$-homotopy with $h_0 = \mathrm{id}_Y$ and $h_1 = fg_1$. Indeed, for each $y \in Y$,

$$\begin{aligned} \mathrm{diam}\, h(\{y\} \times \mathbf{I}) &= \mathrm{diam}\, h(\{y\} \times (0, 1]) = \mathrm{diam} \bigcup_{n \in \mathbb{N}} fh_n(\{y\} \times \mathbf{I}) \\ &\leqslant \sum_{n \in \mathbb{N}} \mathrm{mesh}\, \mathcal{U}_{n-1} < \sum_{n \in \mathbb{N}} 2^{-n} = 1. \end{aligned}$$

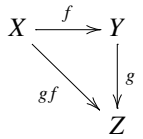
Thus, we have $fg_1 \simeq_{\mathcal{U}} \mathrm{id}_Y$. Recall $g_1 f \simeq_{f^{-1}(\mathcal{U}_3)} \mathrm{id}_X$. Then, it follows that g_1 is a $\mathcal{U}$-homotopy inverse of f. Consequently, f is a fine homotopy equivalence. □

In view of Proposition 6.7.1, Theorem 6.7.4 yields the following:

Corollary 6.7.5 *Let X be an ANR that is a dense set in a metrizable space Y. Suppose that, for each open cover $\mathcal{U}$ of Y, there is a map $g : Y \to X$ with $g|X \simeq_{\mathcal{U}} \mathrm{id}_X$ in X. Then, X is homotopy dense in Y and Y is an ANR.* □

Here, we note the following:

Proposition 6.7.6 *Let $f : X \to Y$ be a fine homotopy equivalence and Z be a paracompact space. Then, a map $g : Y \to Z$ is a fine homotopy equivalence if and only if gf is a fine homotopy equivalence.*



Proof (*The "only if" Part*) Assume that g is a fine homotopy equivalence. For each $\mathcal{U} \in \mathrm{cov}(Z)$, we have $\mathcal{V} \in \mathrm{cov}(Z)$ such that $\mathrm{st}^2 \mathcal{V} \prec \mathcal{U}$. Then, g has a $\mathcal{V}$-homotopy inverse $g' : Z \to Y$ and f has a $g^{-1}(\mathcal{V})$-homotopy inverse $f' : Y \to X$. We show that $f'g'$ is a $\mathcal{U}$-homotopy inverse of gf. Since $g(g^{-1}(\mathcal{V})) \prec \mathcal{V}$, we have $gff'g' \simeq_{\mathcal{V}} gg' \simeq_{\mathcal{V}} \mathrm{id}_Z$, hence $gff'g' \simeq_{\mathcal{U}} \mathrm{id}_Z$. On the other hand,

$$f'g'gf \simeq_{f'(g^{-1}(\mathcal{V}))} f'f \simeq_{f^{-1}(g^{-1}(\mathcal{V}))} \mathrm{id}_X,$$

where $f'(g^{-1}(\mathcal{V})) \prec f^{-1}(g^{-1}(\mathrm{st}\,\mathcal{V}))$. Hence, $f'g'gf \simeq_{f^{-1}(g^{-1}(\mathcal{U}))} \mathrm{id}_X$. Thus, gf is a fine homotopy equivalence.

(*The "if" Part*) Assume that gf is a fine homotopy equivalence. For each $\mathcal{U} \in \mathrm{cov}(Z)$, we have $\mathcal{V} \in \mathrm{cov}(Z)$ such that $\mathrm{st}^3 \mathcal{V} \prec \mathcal{U}$. Then, f has a $g^{-1}(\mathcal{V})$-homotopy inverse $f' : Y \to X$ and gf has a $\mathcal{V}$-homotopy inverse $h : Z \to X$. Since $fhg(g^{-1}(\mathcal{V})) \prec fh(\mathcal{V})$, we have

$$fhg \simeq_{fh(\mathcal{V})} fhgff' \simeq_{g^{-1}(\mathcal{V})} ff' \simeq_{g^{-1}(\mathcal{V})} \mathrm{id}_Y,$$

where $fh(\mathcal{V}) \prec g^{-1}(\mathrm{st}\,\mathcal{V})$ because $gfh(\mathcal{V}) \prec \mathrm{st}\,\mathcal{V}$. Hence, $fhg \simeq_{g^{-1}(\mathcal{U})} \mathrm{id}_Y$. On the other hand, $gfh \simeq_{\mathcal{V}} \mathrm{id}_Z$. Thus, g is a fine homotopy equivalence. □

Remark 6.7.1 Let $f : X \to Y$ and $g : Y \to Z$ be maps such that $f(X)$ is dense in Y, $g(Y)$ is dense in Z, and Z is paracompact. Consider the following conditions:

(a) For each $\mathcal{U} \in \mathrm{cov}(Y)$, there is a map $f' : Y \to X$ such that $f'f \simeq_{f^{-1}(\mathcal{U})} \mathrm{id}_X$;
(a′) For each $\mathcal{U} \in \mathrm{cov}(Y)$, there is a map $f' : Y \to X$ such that $ff' \simeq_{\mathcal{U}} \mathrm{id}_Y$;
(b) For each $\mathcal{V} \in \mathrm{cov}(Z)$, there is a map $g' : Z \to Y$ such that $g'g \simeq_{g^{-1}(\mathcal{V})} \mathrm{id}_Y$;
(c) For each $\mathcal{V} \in \mathrm{cov}(Z)$, there is a map $h : Z \to X$ such that $hgf \simeq_{(gf)^{-1}(\mathcal{V})} \mathrm{id}_X$.

Then, in the same way as the proof of Proposition 6.7.6 but using Claim (1) in the proof of Theorem 6.7.4, we can show that (a) + (b) ⇒ (c) and (a′) + (c) ⇒ (b). Thus, if f is a fine homotopy equivalence, then (b) and (c) are equivalent, i.e., (b) ⇔ (c).

Sketch of Proof (a) + (b) ⇒ (c): By (a) and (b), for each $\mathcal{V} \in \mathrm{cov}(Z)$, we can take maps $f' : Y \to X$ and $g' : Z \to Y$ so that $f'f \simeq_{f^{-1}(g^{-1}(\mathcal{V}))} \mathrm{id}_X$ and $g'g \simeq_{g^{-1}(\mathcal{V})} \mathrm{id}_Y$. Since ff' is $g^{-1}(\mathrm{st}\,\mathcal{V})$-close to id_Y by Claim (1) in the proof of Theorem 6.7.4, we have $f'(g^{-1}(\mathcal{V})) \prec f^{-1}(g^{-1}(\mathrm{st}^3 \mathcal{V}))$. Then, it follows that

$$f'g'gf \simeq_{f^{-1}(g^{-1}(\mathrm{st}^3 \mathcal{V}))} f'f \simeq_{f^{-1}(g^{-1}(\mathcal{V}))} \mathrm{id}_X.$$

Hence, $f'g'gf \simeq_{(gf)^{-1}(\mathrm{st}^4 \mathcal{V})} \mathrm{id}_X$.

(a′) + (c) ⇒ (b): By (a′) and (c), for each $\mathcal{V} \in \mathrm{cov}(Z)$, we can take maps $f' : Y \to X$ and $h : Z \to X$ so that $ff' \simeq_{g^{-1}(\mathcal{V})} \mathrm{id}_Y$ and $hgf \simeq_{(gf)^{-1}(\mathcal{V})} \mathrm{id}_X$. Since gfh is $\mathrm{st}\,\mathcal{V}$-close to

id_Z by Claim (1) in the proof of Theorem 6.7.4, we have $fh(\mathcal{V}) \prec g^{-1}(\mathrm{st}^3\mathcal{V})$. Then, it follows that

$$fhg \simeq_{g^{-1}(\mathrm{st}^3\mathcal{V})} fhgff' \simeq_{g^{-1}(\mathcal{V})} ff' \simeq_{g^{-1}(\mathcal{V})} \mathrm{id}_Y,$$

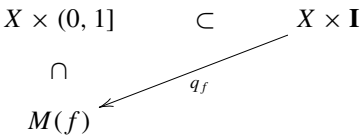
Hence, $fhg \simeq_{g^{-1}(\mathrm{st}^5\mathcal{V})} \mathrm{id}_Y$.

Using the mapping cylinders, we can characterize fine homotopy equivalences between ANRs as follows:

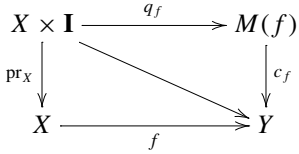
Theorem 6.7.7 *For a map $f : X \to Y$ between ANRs, the following are equivalent:*

(a) *f is a fine homotopy equivalence;*
(b) *$X \times (0, 1]$ is homotopy dense in $M(f)$;*
(c) *The natural map $q_f : X \times \mathbf{I} \to M(f)$ is a fine homotopy equivalence.*

Proof Because of Proposition 6.7.1, (b) means that the inclusion $X \times (0, 1] \subset M(f)$ is a fine homotopy equivalence. The equivalence (b) $\Leftrightarrow$ (c) follows from Proposition 6.7.6.



(c) $\Rightarrow$ (a): Since the projection $\mathrm{pr}_X : X \times \mathbf{I} \to X$ and the collapsing $c_f : M(f) \to Y$ are fine homotopy equivalences (Proposition 6.7.2), we can apply Proposition 6.7.6 to see that if the natural map $q_f : X \times \mathbf{I} \to M(f)$ is a fine homotopy equivalence, then $f : X \to Y$ is a fine homotopy equivalence.



(a) $\Rightarrow$ (b): Due to Corollary 6.7.5, for each open cover $\mathcal{U}$ of $M(f)$, it suffices to construct a map $k : M(f) \to X \times (0, 1]$ such that $k|X \times (0, 1]$ is $\mathcal{U}$-homotopic to id in $X \times (0, 1]$.[10] By Lemma 6.5.8, we have an open cover $\mathcal{V}$ of Y with a map $\alpha : Y \to (0, 1)$ such that, for each $V \in \mathcal{V}$,

$$V \cup (f^{-1}(V) \times (0, \sup \alpha(V)])$$

is contained in some member of $\mathcal{U}$. Then, there exist a map $g : Y \to X$ and an $f^{-1}(\mathcal{V})$-homotopy $h : X \times \mathbf{I} \to X$ such that $h_0 = \mathrm{id}_X$ and $h_1 = gf$. We define a map $k : M(f) \to X \times (0, 1]$ as follows:

[10] It is not required that $k \simeq_{\mathcal{U}} \mathrm{id}$ in $M(f)$.

$$k(x,s)=\begin{cases}(x,s) & \text{if } x\in X \text{ and } s\geqslant \alpha f(x),\\ (h(x,(2\alpha f(x)-2s)/\alpha f(x)),\ s) & \\ & \text{if } x\in X \text{ and } \alpha f(x)/2\leqslant s\leqslant \alpha f(x),\\ (gf(x),\alpha f(x)/2) & \text{if } x\in X \text{ and } s\leqslant \alpha f(x)/2,\end{cases}$$

$$k(y)=(g(y),\alpha(y)/2) \text{ for each } y\in Y.$$

Then, $k|X\times(0,1]$ is $\mathcal{U}$-homotopic to id in $X\times(0,1]$ by the homotopy $\varphi: X\times(0,1]\times \mathbf{I}\to X\times(0,1]$ defined as follows:

$$\varphi_t(x,s)=\begin{cases}(x,s) & \text{if } x\in X \text{ and } s\geqslant \alpha f(x),\\ (h(x,t(2\alpha f(x)-2s)/\alpha f(x)),\ s) & \\ & \text{if } x\in X \text{ and } \alpha f(x)/2\leqslant s\leqslant \alpha f(x),\\ (h_t(x),(1-t)s+t\alpha f(x)/2) & \\ & \text{if } x\in X \text{ and } s\leqslant \alpha f(x)/2.\end{cases}$$

This completes the proof. □

Remark 6.7.2 In Theorem 6.7.7, it is assumed that Y is an ANR. This assumption is not used in the proof, but if f is a fine homotopy equivalence (i.e., (a) holds), then Y is an ANR because X is an ANR (Theorem 6.7.4). In fact, as can be seen from the above proof, (b) $\Leftrightarrow$ (c) $\Rightarrow$ (a) hold for any metrizable spaces X and Y.[11] Moreover, using Remark 6.7.1 instead of Proposition 6.7.6, we can prove the following:

Proposition 6.7.8 *Let $f: X\to Y$ be a map between metrizable spaces such that $f(X)$ is dense in Y. Then, the following are equivalent:*

(a′) *For each open cover $\mathcal{U}$ of Y, there is a map $g: Y\to X$ such that $gf\simeq_{f^{-1}(\mathcal{U})}\mathrm{id}_X$;*

(b′) *For each open cover $\mathcal{U}$ of $M(f)$, there is a map $g: M(f)\to X\times(0,1]$ such that $g|X\times(0,1]\simeq_{\mathcal{U}}\mathrm{id}$ in $X\times(0,1]$;*

(c′) *For each open cover $\mathcal{U}$ of $M(f)$, there is a map $g: M(f)\to X\times\mathbf{I}$ such that $gq_f\simeq_{q_f^{-1}(\mathcal{U})}\mathrm{id}_{X\times\mathbf{I}}$.* □

It should be remarked that if X is an ANR and $f(X)$ is dense in Y, then the above condition (a′) implies that f is a fine homotopy equivalence and that Y is also an ANR by Theorem 6.7.4.

Regarding the inverse limit of an inverse sequence of ANRs, we have:

Theorem 6.7.9 *Let $X=\varprojlim(X_i,\varphi_i)$ be the inverse limit of an inverse sequence $(X_i,\varphi_i)_{i\in\mathbb{N}}$ of completely metrizable ANRs such that each bonding map $\varphi_i: X_{i+1}\to X_i$ is a fine homotopy equivalence. Then, X is an ANR. Moreover, if X_1 is an AR (so every X_i is an AR), then X is an AR.*

[11] In the proof of (a) $\Rightarrow$ (b) of Theorem 6.7.7, Corollary 6.7.5 can be applied because $X\times(0,1]$ is an ANR.

Proof Let $f : A \to X$ be a map from a closed set A in a metrizable space Y. For each $i \in \mathbb{N}$, let $p_i : X \to X_i$ be the inverse limit projection. Take an admissible complete metric d_i for X_i and choose $\mathcal{U}_i \in \mathrm{cov}(X_i)$ so that $\mathrm{mesh}_{d_j} \varphi_{j,i}(\mathcal{U}_i) < 2^{-i}/3$ for each $j \leqslant i$, where $\varphi_{i,i} = \mathrm{id}$ and $\varphi_{j,i} = \varphi_j \varphi_{j+1} \cdots \varphi_{i-1} : X_i \to X_j$ for $j < i$. Since φ_i is a fine homotopy inverse, it has a $\mathcal{U}_i$-homotopy inverse $\psi_i : X_i \to X_{i+1}$.

Now, since X_1 is an ANR, the map $p_1 f$ extends to a map $f_1 : U \to X_1$ of a neighborhood U of A in Y. By induction, we can obtain maps $f_i : U \to X_i$, $i \in \mathbb{N}$, such that $f_i|A = p_i f$ and $d_j(\varphi_{j,i} f_i, \varphi_{j,i+1} f_{i+1}) < 2^{-i}$ for each $j \leqslant i$. Indeed, suppose that f_i has been obtained. Then,

$$p_{i+1} f \simeq_{\varphi_i^{-1}(\mathcal{U}_i)} \psi_i \varphi_i p_{i+1} f = \psi_i p_i f = \psi_i f_i | A.$$

By the Homotopy Extension Theorem 6.4.1, $p_{i+1} f$ extends to a map $f_{i+1} : U \to X_{i+1}$ that is $\varphi_i^{-1}(\mathcal{U}_i)$-homotopic to $\psi_i f_i$, hence $\varphi_i f_{i+1}$ is $\mathrm{st}\,\mathcal{U}_i$-close to f_i, which implies that $d_j(\varphi_{j,i+1} f_{i+1}, \varphi_{j,i} f_i) < 2^{-i}$.

For each $j \in \mathbb{N}$, since d_j is complete, we have a map $\tilde{f}_j = \lim_{i\to\infty} \varphi_{j,i} f_i : U \to X_j$. Then, it follows that

$$\varphi_j \tilde{f}_{j+1} = \varphi_j \big(\lim_{i\to\infty} \varphi_{j+1,i} f_i \big) = \lim_{i\to\infty} \varphi_{j,i} f_i = \tilde{f}_j.$$

Therefore, we have a map $\tilde{f} : U \to X$ such that $p_j \tilde{f} = \tilde{f}_j$ for every $j \in \mathbb{N}$. Since $p_j \tilde{f}|A = \tilde{f}_j|A = p_j f$ for every $j \in \mathbb{N}$, it follows that $\tilde{f}|A = f$.

If X_1 is an AR, we can take $U = X_1$ in the above. Then, f extends over Y. □

Remark 6.7.3 In Theorem 6.7.9, it suffices to require of each bonding map φ_i the condition that for any $\mathcal{U}_i \in \mathrm{cov}(X_i)$, there is a map $\psi_i : X_i \to X_{i+1}$ such that $\psi_i \varphi_i \simeq_{\varphi_i^{-1}(\mathcal{U}_i)} \mathrm{id}_{X_{i+1}}$ and $\varphi_i \psi_i$ is $\mathcal{U}_i$-close to id_{X_i}. In this case, since $\varphi_i(X_{i+1})$ is dense in X_i, the map $\varphi_i : X_{i+1} \to X_i$ is a fine homotopy equivalence by Theorem 6.7.4.

6.8 Completions of ANRs and Uniform ANRs

For a metric space $X = (X, d)$, a sequence $\mathcal{U} = (\mathcal{U}_n)_{n\in\mathbb{N}}$ of open covers of X is called a **zero-sequence** if $\lim_{n\to\infty} \mathrm{mesh}\,\mathcal{U}_n = 0$. For a zero-sequence $\mathcal{U}$, we define the **telescope** $\mathrm{Tel}(\mathcal{U})$ as the following simplicial complex:

$$\mathrm{Tel}(\mathcal{U}) = \bigcup_{n\in\mathbb{N}} N(\mathcal{U}_n \cup \mathcal{U}_{n+1}),$$

where $N(\mathcal{U}_n \cup \mathcal{U}_{n+1})$ is the nerve of $\mathcal{U}_n \cup \mathcal{U}_{n+1}$ and we regard $\mathcal{U}_n \cap \mathcal{U}_m = \emptyset$ for $n \neq m$ as the sets of vertices of the nerves. The following characterization of ANRs is due to Nguyen To Nhu:

Theorem 6.8.1 (NGUYEN TO NHU) *A metric space $X = (X, d)$ is an ANR if and only if X has a zero-sequence $\mathcal{U} = (\mathcal{U}_n)_{n\in\mathbb{N}}$ of open covers with a map $f : |\operatorname{Tel}(\mathcal{U})| \to X$ satisfying the following conditions:*

(i) $f(U) \in U$ *for each* $U \in \bigcup_{n\in\mathbb{N}} \mathcal{U}_n = \operatorname{Tel}(\mathcal{U})^{(0)}$;
(ii) $\lim_{n\to\infty} \operatorname{mesh}\{f(\sigma) \mid \sigma \in N(\mathcal{U}_n \cup \mathcal{U}_{n+1})\} = 0$.

Proof (*The "only if" Part*) By the Arens–Eells Embedding Theorem 6.2.1, we may assume that X is a neighborhood retract of a normed linear space E. Let $r : W \to X$ be a retraction of an open neighborhood W of X in E onto X. For each $n \in \mathbb{N}$, let $\mathcal{V}_n$ be a collection of open convex sets in E such that $X \subset \bigcup \mathcal{V}_n \subset W$ and $\operatorname{mesh} r(\mathcal{V}_n) < 2^{-n}$. Then, we have a zero-sequence $\mathcal{U} = (\mathcal{U}_n)_{n\in\mathbb{N}}$ of open covers of X such that $\operatorname{st}\mathcal{U}_n \prec \mathcal{V}_n$ and $\mathcal{U}_{n+1} \prec \mathcal{U}_n$. Choosing $f_0(U) \in U$ for each $U \in \bigcup_{n\in\mathbb{N}} \mathcal{U}_n = \operatorname{Tel}(\mathcal{U})^{(0)}$, we have a map $f_0 : \operatorname{Tel}(\mathcal{U})^{(0)} \to X \subset E$, which extends to a map $f : |\operatorname{Tel}(\mathcal{U})| \to E$ such that $f|\sigma$ is affine on each simplex $\sigma \in \operatorname{Tel}(\mathcal{U})$. For each $\sigma \in \operatorname{Tel}(\mathcal{U})$, we have $n(\sigma) \in \mathbb{N}$ and $U_\sigma \in \sigma^{(0)}$ such that $\sigma \in N(\mathcal{U}_{n(\sigma)} \cup \mathcal{U}_{n(\sigma)+1})$ and $U_\sigma \in \mathcal{U}_{n(\sigma)}$. Since $\mathcal{U}_{n(\sigma)+1} \prec \mathcal{U}_{n(\sigma)}$, it follows that $f_0(\sigma^{(0)}) \subset \operatorname{st}(U_\sigma, \mathcal{U}_{n(\sigma)})$, which is contained in some $V_\sigma \in \mathcal{V}_{n(\sigma)}$. By the convexity of V_σ, $f(\sigma) \subset V_\sigma \in \mathcal{V}_{n(\sigma)}$. Hence, $f(|\operatorname{Tel}(\mathcal{U})|) \subset W$. Thus, we have the map $rf : |\operatorname{Tel}(\mathcal{U})| \to X$, which satisfies condition (i) because rf is an extension of f_0. For each $\sigma \in \operatorname{Tel}(\mathcal{U})$, $\operatorname{diam} rf(\sigma) \leqslant \operatorname{mesh} r(\mathcal{V}_{n(\sigma)}) < 2^{-n(\sigma)}$, which means that rf satisfies condition (ii).

(*The "if" Part*) Let $\mathcal{U} = (\mathcal{U}_n)_{n\in\mathbb{N}}$ be a zero-sequence of open covers of X and $f : |\operatorname{Tel}(\mathcal{U})| \to X$ be a map satisfying conditions (i) and (ii). For each $n \in \mathbb{N}$, let $\varphi_n : X \to |N(\mathcal{U}_n)|$ be a canonical map. For each $x \in X$, let $\sigma_{n,x}$ be the simplex of $N(\mathcal{U}_n \cup \mathcal{U}_{n+1})$ spanned by vertices of the carrier of $\varphi_n(x)$ in $N(\mathcal{U}_n)$ and the carrier of $\varphi_{n+1}(x)$ in $N(\mathcal{U}_{n+1})$. Then, $\varphi_n(x), \varphi_{n+1}(x) \in \sigma_{n,x}$. Thus, we have a homotopy $g^n : X \times \mathbf{I} \to |N(\mathcal{U}_n \cup \mathcal{U}_{n+1})|$ such that

$$g^n_0 = \varphi_n,\ g^n_1 = \varphi_{n+1} \text{ and } g^n(\{x\} \times \mathbf{I}) \subset \sigma_{n,x} \text{ for each } x \in X.$$

Then, each fg^n_t is α_n-close to id, where

$$\alpha_n = \sup_{i\geqslant n} \Big(\operatorname{mesh} \mathcal{U}_i + \operatorname{mesh} \big\{ f(\sigma) \;\big|\; \sigma \in N(\mathcal{U}_i \cup \mathcal{U}_{i+1}) \big\} \Big).$$

Indeed, each $x \in X$ is contained in some $U \in \mathcal{U}_n \cap \sigma^{(0)}_{n,x}$ and $f(U) \in U$, hence

$$\begin{aligned} d(x, fg^n(x,t)) &\leqslant d(x, f(U)) + d(f(U), fg^n(x,t)) \\ &\leqslant \operatorname{diam} U + \operatorname{diam} f(\sigma_{n,x}) \\ &\leqslant \operatorname{mesh} \mathcal{U}_n + \operatorname{diam} f(\sigma_{n,x}) \leqslant \alpha_n. \end{aligned}$$

Thus, we can define a homotopy $h : X \times \mathbf{I} \to X$ as follows:

$$h(x,t) = \begin{cases} fg^n(x, 2-2^n t) & \text{if } 2^{-n} \leqslant t \leqslant 2^{-n+1},\ n \in \mathbb{N}, \\ x & \text{if } t = 0. \end{cases}$$

Observe that $h_0 = \mathrm{id}$, $h_{2^{-n+1}} = f\varphi_n$, and $\operatorname{diam} h(\{x\} \times [0, 2^{-n+1}]) \leqslant 2\alpha_n$ for each $x \in X$. Hence, $f\varphi_n$ is $2\alpha_n$-homotopic to id, which implies that X is $2\alpha_n$-homotopy dominated by $|N(\mathcal{U}_n)|$. Since $\lim_{n\to\infty} \alpha_n = 0$, X is an ANR by Theorem 6.6.2. □

Remark 6.8.1 In Theorem 6.8.1, if $f(|\operatorname{Tel}(\mathcal{U})|)$ is contained in some $A \subset X$, then the homotopy h in the proof above satisfies $h(X \times (0,1]) \subset A$, hence A is homotopy dense in X.

Corollary 6.8.2 *Let X be a subset of a metric space $M = (M, d)$. If X is an ANR, then X is homotopy dense in some G_δ-set Y in M, hence Y is also an ANR.*

Proof By Theorem 6.8.1, X has a zero-sequence $\mathcal{U} = (\mathcal{U}_n)_{n\in\mathbb{N}}$ of open covers with a map $f : |\operatorname{Tel}(\mathcal{U})| \to X$ satisfying conditions (i) and (ii) in Theorem 6.8.1. Using the operator E in the Canonical Open Set Extension Lemma 5.4.10 to extend open sets in X to open sets in M, we define

$$Y = \operatorname{cl}_M X \cap \bigcap_{n\in\mathbb{N}} \bigcup_{U\in\mathcal{U}_n} E(U).$$

Then, Y is a G_δ-set in M and $X \subset Y \subset \operatorname{cl}_M X$. For each $n \in \mathbb{N}$, let

$$\widetilde{\mathcal{U}}_n = \{Y \cap E(U) \mid U \in \mathcal{U}_n\} \in \operatorname{cov}(Y).$$

Then, $\widetilde{\mathcal{U}} = (\widetilde{\mathcal{U}}_n)_{n\in\mathbb{N}}$ is a zero-sequence because $\operatorname{mesh} \widetilde{\mathcal{U}}_n = \operatorname{mesh} \mathcal{U}_n \to 0$ (as $n \to \infty$). Observe that $\operatorname{Tel}(\widetilde{\mathcal{U}}) \equiv \operatorname{Tel}(\mathcal{U})$ by the correspondence $Y \cap E(U) \leftrightarrow U$. Then, the map $f : |\operatorname{Tel}(\widetilde{\mathcal{U}})| \to X \subset Y$ satisfies conditions (i) and (ii) in Theorem 6.8.1. By the above remark, we have the result. □

Since every metrizable space X can be embedded into a completely metrizable space M (e.g., the metric completion of X for some admissible metric), Corollary 6.8.2 yields:

Corollary 6.8.3 *Every ANR is contained in a completely metrizable ANR as a homotopy dense subset.* □

Let $X = (X, d)$ and $Y = (Y, \rho)$ be metric spaces and A be a closed set in X. A map $f : X \to Y$ is said to be **uniformly continuous at** A if, for each $\varepsilon > 0$, there is a $\delta > 0$ such that for each $x \in X$ and $a \in A$, $d(x, a) < \delta$ implies $\rho(f(x), f(a)) < \varepsilon$.[12] We call A a **uniform retract** of X if there is a retraction $r : X \to A$ that is uniformly continuous at A. A **uniform neighborhood** of A in X is a subset U of X with $\operatorname{dist}(A, X \setminus U) > 0$, that is, there is a $\delta > 0$ such that $\mathrm{B}(a, \delta) \subset U$ for all $a \in A$. We

[12] We do not use the preposition "on" but "at" here. When we say that $f : X \to Y$ is uniformly continuous on A, it means that $f|A$ is uniformly continuous.

call A a **uniform neighborhood retract** of X if A is a uniform retract of a uniform neighborhood of A in X. It is easy to see that every compact (neighborhood) retract of X is a uniform (neighborhood) retract. Observe that the following closed set in the Euclidean plane $\mathbb{R}^2$ is homeomorphic to the real line $\mathbb{R}$ but is not a uniform (neighborhood) retract of $\mathbb{R}^2$:

$$\left\{(x, y) \in \mathbb{R}^2 \;\middle|\; y = x^{-1},\ x \geqslant 1\right\} \cup \left(\{1\} \times \mathbf{I}\right) \cup \left([1, \infty) \times \{0\}\right).$$

A metric space X is called a **uniform AR** (resp. a **uniform ANR**) if X is a uniform retract (resp. a uniform neighborhood retract) of an arbitrary metric space that contains X as a closed metric subspace (i.e., the inclusion of X is an isometrically closed embedding). Using the Arens–Eells Embedding Theorem 6.2.1, we can see that a uniform AR (resp. a uniform ANR) is an AR (resp. an ANR). It is easy to see that every compact AR (resp. ANR) is a uniform AR (resp. a uniform ANR) with respect to any admissible metric.

A metric space Y is called a **uniform AE** (a **uniform ANE**) if for any closed set A in an arbitrary metric space X, any uniformly continuous map $f : A \to Y$ extends to a map $\tilde{f} : X \to Y$ (a map $\tilde{f} : U \to Y$ of a uniform neighborhood of A in X) that is uniformly continuous at A. It should be noted that a metric space uniformly homeomorphic to a uniform AE (or a uniform ANE) is also a uniform AE (or a uniform ANE). Like ANRs and ANEs, we have the following theorem:

Theorem 6.8.4 *A metric space X is a uniform AR (resp. a uniform ANR) if and only if X is a uniform AE (resp. a uniform ANE).*

Proof The "if" part is trivial. We prove the "only if" part for the case where X is a uniform ANR. By the Arens–Eells Embedding Theorem 6.2.1, we may assume that X is a closed set in a normed linear space $E = (E, \|\cdot\|)$. Then, there is a uniform retraction $r : G \to X$ of a uniform neighborhood G of X in E. It is easy to prove the following:

(∗) For each $\varepsilon > 0$, there exists a $\gamma(\varepsilon) > 0$ such that if $S \subset X$ and diam $S \leqslant \gamma(\varepsilon)$ then $\langle S\rangle \subset G$ and diam $r(\langle S\rangle) < \varepsilon$.

Let $Y = (Y, d)$ be a metric space and $f : A \to X$ be a uniformly continuous map from a closed set A in Y. For each $\varepsilon > 0$, we have $\beta(\varepsilon) > 0$ such that for any $a, a' \in A$, $d(a, a') < \beta(\varepsilon)$ implies $\|f(a) - f(a')\| < \varepsilon$. For each $a \in A$, let

$$V_a = \left\{y \in Y \setminus A \;\middle|\; d(y, a) < 2d(y, A)\right\}.$$

Then, $Y \setminus A$ has a locally finite partition of unity $(\lambda_a)_{a\in A}$ subordinated to the open cover $\{V_a \mid a \in A\}$. For each $\varepsilon > 0$, we define $\delta(\varepsilon) = \beta(\gamma(\varepsilon)/2)/3 > 0$ and

$$U = \left\{y \in Y \;\middle|\; d(y, A) < \delta(1)\right\}.$$

Then, U is a uniform neighborhood of A in Y. We extend f to $\tilde{f} : U \to X$ by

$$\tilde{f}(y) = \begin{cases} f(y) & \text{if } y \in A, \\ r\big(\sum_{a\in A} \lambda_a(y) f(a)\big) & \text{if } y \in U \setminus A. \end{cases}$$

Then, $\tilde{f}$ is well-defined and uniformly continuous at A. Indeed, let $y \in U \setminus A, a \in A$, and $d(y,a) < \delta(\varepsilon)$. If $\lambda_{a'}(y) \neq 0$ then $d(y,a') < 2d(y,A) < 2\delta(\varepsilon)$, so $d(a,a') < \beta(\gamma(\varepsilon)/2)$, which implies $\|f(a) - f(a')\| < \gamma(\varepsilon)/2$. Thus,

$$\operatorname{diam}\big(\{f(a') \mid \lambda_{a'}(y) \neq 0\} \cup \{f(a)\}\big) \leqslant \gamma(\varepsilon).$$

When $\varepsilon = 1$, $\sum_{a'\in A} \lambda_{a'}(y) f(a') \in G$ by $(*)$, which shows the well-definedness of $\tilde{f}$. Since $\tilde{f}(y) = r\big(\sum_{a'\in A} \lambda_{a'}(y) f(a')\big)$, we have $\|\tilde{f}(y) - f(a)\| < \varepsilon$ by $(*)$, hence $\tilde{f}$ is uniformly continuous at A.

If X is a uniform AR, we can take $G = E$ in the above. Then, $\tilde{f}$ can be defined over Y, which means that X is a uniform AE. □

Remark 6.8.2 The above proof is also valid even if X is a closed set in a convex set C in E and G is a neighborhood of X in C. Moreover, in the case where E is not a normed linear space, if condition $(*)$ is satisfied, then the proof can be carried out.

If X itself is a convex set in a locally convex metric linear space E, let $G = C = X$ and $r = \mathrm{id}$ in the above remark. Condition $(*)$ is satisfied with respect to the linear metric for E.[13] Indeed, by the local convexity of E, each $\varepsilon/2$-neighborhood of $\mathbf{0} \in E$ contains a convex neighborhood of $\mathbf{0}$, which contains some $\gamma(\varepsilon)$-neighborhood of $\mathbf{0}$. If $S \subset X$ with $\operatorname{diam} S < \gamma(\varepsilon)$, then $\langle S\rangle \subset C = X$ and $\operatorname{diam}\langle S\rangle \leqslant \varepsilon$. Thus, we have the following corollary:

Corollary 6.8.5 *Every non-empty convex set in a locally convex metric linear space is a uniform AE, and is hence a uniform AR.* □

According to Corollary 6.3.6, every open cover $\mathcal{U}$ of an ANR X has an h-refinement $\mathcal{V}$, that is, any two $\mathcal{V}$-close maps into X are $\mathcal{U}$-homotopic. The following is a uniform version of this:

Theorem 6.8.6 *Let X be a uniform ANR. Then, for each $\varepsilon > 0$, there is $\delta > 0$ such that any two δ-close maps of an arbitrary space to X are ε-homotopic.*

Proof By the Arens–Eells Embedding Theorem 6.2.1, we can consider X as a closed set in a normed linear space $E = (E, \|\cdot\|)$. Since X is a uniform ANR, X has a uniform neighborhood U with a retraction $r : U \to X$ that is uniformly continuous at X. For each $\varepsilon > 0$, we have $\delta > 0$ such that $\delta < \operatorname{dist}(X, E \setminus U)$ and, for each $x \in X$ and $y \in U$, $\|x - y\| < \delta$ implies $\|x - r(y)\| < \varepsilon/2$. Then, any two δ-close maps $f, g : Y \to X$ are ε-homotopic by the homotopy $h : Y \times \mathbf{I} \to X$ defined as follows:

$$h(y,t) = r((1-t)f(y) + tg(y)).$$ □

[13] The linear metric can be replaced by any admissible invariant metric.

The following is a variant of Theorem 6.6.12.

Theorem 6.8.7 *A uniform ANR is a uniform AR if and only if it is homotopically trivial.*

Proof If a uniform ANR X is homotopically trivial, then X is an AR by Theorem 6.6.12. Let $Y = (Y, d)$ be a metric space and $f : A \to X$ be a uniformly continuous map from a closed set A in Y. Since X is a uniform ANE by Theorem 6.8.4, f extends to a map $\tilde{f} : U \to X$ of a uniform open neighborhood U in Y that is uniformly continuous at A. Choose a uniform neighborhood V of A in Y so that $\operatorname{cl} V \subset U$. Since X is an AE, $\tilde{f}|\operatorname{cl} V$ extends to a map $\bar{f} : Y \to X$. Obviously, $\bar{f}$ is uniformly continuous at A. Therefore, X is a uniform AE, hence it is a uniform AR by Theorem 6.8.4. □

When several metric spaces with common points appear, we denote by $\mathrm{B}_X(x, r)$ the open ball in the space X (centered at x with radius r).

Lemma 6.8.8 *A uniform ANR X is homotopy dense in every metric space Z that contains X isometrically as a dense subset.*

Proof In the proof of the "only if" part of Theorem 6.8.1, since the retraction $r : W \to X$ can be assumed to be a retraction of a uniform open neighborhood W of X in a normed linear space E that is uniformly continuous at X, we can take $\delta_1 > \delta_2 > \cdots > 0$ so that $\lim_{n\to\infty} \delta_n = 0$, $\mathrm{B}_E(x, \delta_n) \subset W$, and $\operatorname{diam} r(\mathrm{B}_E(x, \delta_n)) < 2^{-n}$ for every $x \in X$ and $n \in \mathbb{N}$. Let $\mathcal{V}_n = \{\mathrm{B}_E(x, \delta_n) \mid x \in X\}$ and $\mathcal{U}_n = \{\mathrm{B}_X(x, \delta_n/3) \mid x \in X\}$. Then, $\mathcal{U} = (\mathcal{U}_n)_{n\in\mathbb{N}}$ is a zero-sequence of open covers of X. Observe that $\operatorname{st} \mathcal{U}_n \prec \mathcal{V}_n$ and $\mathcal{U}_{n+1} \prec \mathcal{U}_n$. Take any map $f_0 : \mathrm{Tel}(\mathcal{U})^{(0)} \to X$ such that $f_0(U) \in U$. Similar to the proof of Theorem 6.8.1, we can extend f_0 to a map f satisfying conditions (i) and (ii) in Theorem 6.8.1.

For each $n \in \mathbb{N}$, let $\widetilde{\mathcal{U}}_n = \{\mathrm{B}_Z(x, \delta_n/3) \mid x \in X\}$. Then, $\widetilde{\mathcal{U}} = (\widetilde{\mathcal{U}}_n)_{n\in\mathbb{N}}$ is a zero-sequence of open covers of Z. By the same argument as in Corollary 6.8.2, we can show that X is homotopy dense in Z. □

Theorem 6.8.9 *For a metric space X, the following are equivalent:*

(a) *X is a uniform ANR;*
(b) *For every metric space Z containing X isometrically as a dense subset, Z is a uniform ANR and X is homotopy dense in Z;*
(c) *X is isometrically embedded in some uniform ANR Z as a homotopy dense subset.*

Proof The implications (a) ⇒ (c) and (b) ⇒ (a) are obvious by considering $Z = X$.

(a) ⇒ (b): Let Z be a metric space containing X isometrically as a dense subset. Then, X is homotopy dense in Z by Lemma 6.8.8. To prove that Z is a uniform ANR, assume that Z is a closed set in a metric space Y. Let $Y' = Y \setminus (Z \setminus X)$. Since X is a closed set in Y', there is a retraction $r : U \to X$ of a uniform neighborhood U of X in Y' that is uniformly continuous at X. Then, $V = U \cup Z$ is a uniform neighborhood

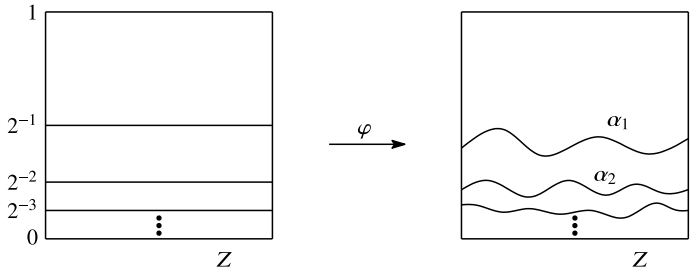


Fig. 6.17 The homeomorphism φ

of Z in Y. We extend r to $\tilde{r} : V \to Z$ by $\tilde{r}|Z \setminus X = \mathrm{id}$. It is easy to see that $\tilde{r}$ is uniformly continuous at Z. Hence, Z is a uniform ANR.

(c) $\Rightarrow$ (a): Let $Y = (Y, d)$ be a metric space and $f : A \to X$ be a uniformly continuous map from a closed set A in Y. Since Z is a uniform ANE by Theorem 6.8.4, f extends to a map $\tilde{f} : U \to Z$ of a uniform open neighborhood U of A in Y that is uniformly continuous at A, where we may assume that $d(y, A) \leqslant 1$ for all $y \in U$. On the other hand, we have a homotopy $h : Z \times \mathbf{I} \to Z$ such that $h_0 = \mathrm{id}$ and $h_t(Z) \subset X$ for all $t > 0$. Pushing the image of $\tilde{f}$ into X by this homotopy, we can define a map $U \ni y \mapsto h(\tilde{f}(y), d(y, A)) \in X$ extending f. Unfortunately, in general, this is not uniformly continuous at A, so we need to adjust the homotopy h.

By applying Theorem 2.7.8, it is easy to construct maps $\alpha_n : Z \to (0, 1)$, $n \in \mathbb{N}$, such that $\alpha_{n+1}(z) < \alpha_n(z) \leqslant 2^{-n}$ and $\operatorname{diam} h(\{z\} \times [0, \alpha_n(z)]) < 2^{-n}$.

Indeed, the following $\gamma_n : Z \to (0, \infty)$ is l.s.c.:

$$Z \ni z \mapsto \gamma_n(z) = \sup\left\{r > 0 \;\middle|\; \operatorname{diam} h(\{z\} \times [0, r]) < 2^{-n}\right\} \in (0, \infty).$$

Take a homeomorphism $\varphi : Z \times \mathbf{I} \to Z \times \mathbf{I}$ so that $\mathrm{pr}_Z \varphi = \mathrm{pr}_Z, \varphi|Z \times \{0, 1\} = \mathrm{id}$, and

$$\varphi(z, 2^{-n}) = (z, \alpha_n(z)) \quad \text{for each } z \in Z \text{ and } n \in \mathbb{N}.$$

See Fig. 6.17. Then, it follows that $d(z, h\varphi(z, t)) < 2^{-n}$ if $t < 2^{-n}$. We define an extension $f' : U \to X$ of f by $f'(y) = h\varphi(\tilde{f}(y), d(y, A))$ for each $y \in U$. For each $\varepsilon > 0$, choose $n \in \mathbb{N}$ so that $2^{-n+1} < \varepsilon$. Since $\tilde{f}$ is uniformly continuous at A, we have $0 < \delta < 2^{-n}$ such that if $y \in U$, $a \in A$, and $d(y, a) < \delta$ $(< 2^{-n})$, then $d(\tilde{f}(y), f(a)) < 2^{-n}$, resulting in

$$\begin{aligned} d(f'(y), f(a)) &\leqslant d(h\varphi(\tilde{f}(y), d(y, A)), \tilde{f}(y)) + d(\tilde{f}(y), f(a)) \\ &< 2^{-n} + 2^{-n} = 2^{-n+1} < \varepsilon. \end{aligned}$$

Therefore, f' is also uniformly continuous at A. Thus, X is a uniform ANE, hence a uniform ANR by Theorem 6.8.4. □

By Theorem 6.8.9 (combined with Theorem 6.8.7), we have the following corollary, which shows an advantage of the concepts of uniform ARs and uniform ANRs.

Corollary 6.8.10 *A metric space X is a uniform ANR (resp. a uniform AR) if and only if the metric completion $\widetilde{X}$ of X is a uniform ANR (resp. a uniform AR) and X is homotopy dense in $\widetilde{X}$.* □

Regarding admissible metrics on ANRs, we prove the following theorem:

Theorem 6.8.11 *For any admissible metric d on an AR (resp. ANR) X, X has an admissible metric $\rho \geqslant d$ such that (X, ρ) is a uniform AR (resp. uniform ANR), where if d is bounded then so is ρ.*

Proof Since an AR is a contractible ANR (Corollary 6.2.14), by Theorem 6.8.7, it is enough to prove the case where X is an ANR. By the Arens–Eells Embedding Theorem 6.2.1, we may assume that X is a neighborhood retract of a normed linear space E, i.e., there is a retraction $r : G \to X$ of a neighborhood G of X in E. Then, as noted in Remark 6.8.2 on the proof of Theorem 6.8.4, it suffices to construct an admissible metric $\rho \geqslant d$ on X that satisfies condition $(*)$ in the proof of Theorem 6.8.4.

By induction, we will construct admissible metrics ρ_i, $i \in \mathbb{N}$, for X such that, for each $n \in \mathbb{N}$ and $A \subset X$,

$$\operatorname{diam}_{\rho_i} A \leqslant 2^{-n} \Rightarrow \langle A \rangle \subset G,\ \operatorname{diam}_{\rho_{i-1}} r(\langle A \rangle) \leqslant n^{-1}2^{-n},$$

where $\rho_0 = d$. Assume ρ_{i-1} has been defined. By the continuity of r, X has open covers $\mathcal{U}_n^i$, $n \in \mathbb{N}$, such that

$$\langle U \rangle \subset G \ \text{ and } \ \operatorname{diam}_{\rho_{i-1}} r(\langle U \rangle) \leqslant n^{-1}2^{-n} \ \text{ for each } U \in \mathcal{U}_n^i.$$

Then, X has admissible metrics d_n^i, $n \in \mathbb{N}$, such that $\{\overline{\mathrm{B}}_{d_n^i}(x, 1) \mid x \in X\} \prec \mathcal{U}_n^i$ (cf. Proposition 2.7.9 (1)). We now define ρ_i as follows:

$$\rho_i(x, y) = \sum_{n=1}^{\infty} \min\left\{d_n^i(x, y),\ 2^{-n}\right\}.$$

It is easy to prove that ρ_i is an admissible metric for X. Since $\rho_i(x, y) \leqslant 2^{-n}$ implies $d_n^i(x, y) \leqslant 2^{-n}$, it follows that $A \subset X$ with $\operatorname{diam}_{\rho_i} A \leqslant 2^{-n}$ is contained in some $U \in \mathcal{U}_n^i$. Then, ρ_i satisfies the condition.

Now, we define a metric on X as follows:

$$\rho(x, y) = \rho_0(x, y) + \sum_{i=1}^{\infty} \min\left\{\rho_i(x, y), 2^{-i+1}\right\}.$$

It is easy to see that ρ is an admissible metric for X and, if $d = \rho_0$ is bounded, then so is ρ. Let $A \subset X$ such that $0 < \operatorname{diam}_{\rho} A \leqslant 1/2$. Since $\operatorname{diam}_{\rho_1} A \leqslant 1/2$, $\langle A \rangle \subset G$.

Choose $n \in \mathbb{N}$ so that $2^{-n-1} < \operatorname{diam}_\rho A \leqslant 2^{-n}$. Since $\operatorname{diam}_{\rho_i} A \leqslant 2^{-n}$ for each $i \leqslant n$, $\operatorname{diam}_{\rho_i} r(\langle A \rangle) \leqslant n^{-1}2^{-n}$ for each $i < n$. Hence,

$$\operatorname{diam}_\rho r(\langle A \rangle) \leqslant 2^{-n} + \sum_{i \geqslant n} 2^{-i+1} = 10 \cdot 2^{-n-1} < 10 \operatorname{diam}_\rho A.$$

Then, ρ satisfies condition $(*)$. □

6.9 Homotopy Types of Open Sets in ANRs

In this section, we will prove the following characterization of ANRs:

Theorem 6.9.1 (CAUTY) *A metrizable space X is an ANR if and only if every open set in X has the homotopy type of an ANR, i.e., the homotopy type of a simplicial complex (cf. Corollary 6.6.5).*

The "only if" part of this theorem is just Proposition 6.2.9 (3). To prove the "if" part, we use the mapping cylinders and the mapping telescopes discussed in Sect. 6.5. For maps $f_i : X_i \to X_{i-1}$, $i = 1, \ldots, n$, we regard

$$M(f_1, \ldots, f_n) = X_0 \cup X_1 \times (0, 1] \cup \cdots \cup X_n \times (n-1, n],$$

where $M(f_i)$ is identified with $X_{i-1} \times \{i-1\} \cup X_i \times (i-1, i]$ by reparameterizing. The following fact states in detail Corollary 6.5.11.

Fact 1 *For maps $f : X \to Y$ and $g : Y \to Z$, let $\varphi : M(g, f) \to M(gf)$ and $\psi : M(gf) \to M(g, f)$ be the maps defined as follows: $\varphi|Z = \psi|Z = \mathrm{id}_Z$,*

$$\varphi(x, s) = \begin{cases} (x, s-1) & \text{for } (x, s) \in X \times (1, 2], \\ g(x) & \text{for } (x, s) \in Y \times (0, 1]; \end{cases}$$

$$\psi(x, s) = \begin{cases} (x, 2s) & \text{for } (x, s) \in X \times (1/2, 1], \\ (f(x), 2s) & \text{for } (x, s) \in X \times (0, 1/2]. \end{cases}$$

Then, $\psi\varphi \simeq \mathrm{id}$ rel. $Z \oplus X \times \{2\}$ and $\varphi\psi \simeq \mathrm{id}$ rel. $Z \oplus X \times \{1\}$ by the homotopies $\xi : M(g, f) \times \mathbf{I} \to M(g, f)$ and $\zeta : M(gf) \times \mathbf{I} \to M(gf)$ defined as follows: $\xi_t|Z = \zeta_t|Z = \mathrm{id}$ and

$$\xi_t(x,s)=\begin{cases}\left(x,\dfrac{2(s-t)}{2-t}\right) & \text{for } (x,s)\in X\times(1+t/2,2],\\ (f(x),2s-1-t) & \text{for } (x,s)\in X\times(1,1+t/2],\\ (x,s-t) & \text{for } (x,s)\in Y\times(t,1],\\ g(x) & \text{for } (x,s)\in Y\times(0,t];\end{cases}$$

$$\zeta_t(x,s)=\begin{cases}\left(x,\dfrac{2s-t}{2-t}\right) & \text{for } (x,s)\in X\times(t/2,1],\\ gf(x) & \text{for } (x,s)\in X\times(0,t/2].\end{cases}$$

Then, we can state

(1) $\xi_t(\{x\}\times(1,2])\subset\{f(x)\}\times(0,1]\cup\{x\}\times(1,2]$ *for* $x\in X$ *and* $t\in\mathbf{I}$;
(2) $\xi_t(\{y\}\times(0,1])\subset\{g(y)\}\cup\{y\}\times(0,1]$ *for* $y\in Y$ *and* $t\in\mathbf{I}$;
(3) $\zeta_t(\{x\}\times(0,1])\subset\{gf(x)\}\cup\{x\}\times(0,1]$ *for* $x\in X$ *and* $t\in\mathbf{I}$.

We can now also state the following fact:

Fact 2 *Let* $f:X\to Y$, $g:Y\to X$ *be maps and* $h:X\times\mathbf{I}\to X$ *be a homotopy such that* $h_0=\mathrm{id}$ *and* $h_1=gf$. *Define maps* $\tilde{f}:X\times\mathbf{I}\to M(f)$, $\tilde{g}:M(f)\to X\times\mathbf{I}$, $\bar{f}:X\times\mathbf{I}\to M(g)$, *and* $\bar{g}:M(g)\to X\times\mathbf{I}$ *as follows:*

$$\tilde{f}(x,t)=\begin{cases}(x,t) & \text{for } t>0,\\ f(x) & \text{for } t=0,\end{cases}\quad\big(\text{i.e., } \tilde{f}=q_f|X\times\mathbf{I}\big);$$

$$\begin{cases}\tilde{g}(x,s)=(h_{2-2s}(x),s) & \text{for } (x,s)\in X\times(1/2,1],\\ \tilde{g}(x,s)=(gf(x),s) & \text{for } (x,s)\in X\times(0,1/2],\\ \tilde{g}(y)=(g(y),0) & \text{for } y\in Y;\end{cases}$$

$$\bar{f}(x,s)=\begin{cases}(f(x),2s-1) & \text{for } (x,s)\in X\times(1/2,1],\\ h_{2s}(x) & \text{for } (x,s)\in X\times[0,1/2];\end{cases}$$

$$\begin{cases}\bar{g}(y,s)=(g(y),s) & \text{for } (y,s)\in Y\times(0,1],\\ \bar{g}(x)=(x,0) & \text{for } x\in X.\end{cases}$$

Furthermore, define the homotopies $\tilde{h},\bar{h}:X\times\mathbf{I}\times\mathbf{I}\to X\times\mathbf{I}$ *as follows:*

$$\tilde{h}_t(x,s)=\begin{cases}(h_{(2-2s)t}(x),s) & \text{for } (x,s,t)\in X\times[1/2,1]\times\mathbf{I},\\ (h_t(x),s) & \text{for } (x,s,t)\in X\times[0,1/2]\times\mathbf{I};\end{cases}$$

$$\bar{h}_t(x,s)=\begin{cases}(h_t(x),(1+t)s-t) & \text{for } (x,s,t)\in X\times[1/2,1]\times\mathbf{I},\\ (h_{2st}(x),(1-t)s) & \text{for } (x,s,t)\in X\times[0,1/2]\times\mathbf{I}.\end{cases}$$

Then,

(1) $\tilde{h}_0 = \mathrm{id}$ *and* $\tilde{h}_1 = \tilde{g}\tilde{f}$*;*
(2) $\tilde{f}|X \times \{1\} = \tilde{g}|X \times \{1\} = \mathrm{id}$ *and* $\tilde{h}_t|X \times \{1\} = \mathrm{id}$ *for every* $t \in \mathbf{I}$*;*
(3) $\tilde{h}_t(x, 0) = (h_t(x), 0)$ *for each* $x \in X$ *and* $t \in \mathbf{I}$*;*
(4) $\tilde{h}(\{(x, s)\} \times \mathbf{I}) \subset h(\{x\} \times \mathbf{I}) \times \{s\}$ *for each* $(x, s) \in X \times \mathbf{I}$*;*
(5) $\tilde{g}(\{f(x)\} \cup \{x\} \times (0, 1]) \subset h(\{x\} \times \mathbf{I}) \times \mathbf{I}$ *for each* $x \in X$*;*
(6) $\bar{h}_0 = \mathrm{id}$ *and* $\bar{h}_1 = \bar{g}\bar{f}$*;*
(7) $\bar{h}_t|X \times \{0\} = \mathrm{id}$ *for every* $t \in \mathbf{I}$*;*
(8) $\bar{h}_t(x, 1) = (h_t(x), 1)$ *for each* $x \in X$ *and* $t \in \mathbf{I}$*;*
(9) $\bar{h}(\{x\} \times \mathbf{I} \times \mathbf{I}) \subset h(\{x\} \times \mathbf{I}) \times \mathbf{I}$ *for each* $x \in X$*;*
(10) $\bar{g}(\{g(y)\} \cup \{y\} \times (0, 1]) = \{g(y)\} \times \mathbf{I}$ *for each* $y \in Y$.

Remark 6.9.1 We can define $\tilde{g} : M(f) \to X \times \mathbf{I}$ as follows:

$$\begin{cases} \tilde{g}(x, s) = (h_{1-s}(x), s) & \text{for } (x, s) \in X \times (0, 1], \\ \tilde{g}(y) = (g(y), 0) & \text{for } y \in Y. \end{cases}$$

This definition is natural, but the continuity of $\tilde{g}$ is not guaranteed because of the topology of $M(f)$.

The following is the key lemma for Theorem 6.9.1.

Lemma 6.9.2 *Suppose that X is a metrizable space such that each open set in X has the homotopy type of some ANR. Let Y_0 be an ANR, X_0 be a closed set in X, W_0 be an open set in X with $X_0 \subset W_0$, $\mathcal{V}_0$ be an open cover of W_0, and let $f_0 : W_0 \to Y_0$, $g_0 : Y_0 \to W_0$ be maps and $h^0 : W_0 \times \mathbf{I} \to W_0$ be a $\mathcal{V}_0$-homotopy such that $h^0_0 = \mathrm{id}$ and $h^0_1 = g_0 f_0$. Given a discrete open collection $\mathcal{W}$ in X, let $W_1 = \bigcup \mathcal{W}$ and*

$$\mathcal{V} = \mathcal{V}_0 \cup \mathcal{W} \cup \left\{ V \cup W \;\middle|\; V \in \mathcal{V}_0,\ W \in \mathcal{W}[V] \right\}.$$

Then, there exists an ANR Y such that Y contains Y_0 as a closed set, $f_0|X_0$ and g_0 extend to maps $\tilde{f} : W_0 \cup W_1 \to Y$ and $\tilde{g} : Y \to W_0 \cup W_1$, respectively, and there is a $\mathcal{V}$-homotopy $\tilde{h} : (W_0 \cup W_1) \times \mathbf{I} \to W_0 \cup W_1$ such that $\tilde{h}_0 = \mathrm{id}$, $\tilde{h}_1 = \tilde{g}\tilde{f}$, and $\tilde{h}|X_0 \times \mathbf{I} = h^0|X_0 \times \mathbf{I}$.

Proof Choose a closed neighborhood N of X_0 in X so that $N \subset W_0$, and let $\alpha : W_0 \cup W_1 \to [0, 1]$ be a Urysohn map with $\alpha(N \cup (W_0 \setminus W_1)) = 0$ and $\alpha(W_1 \setminus W_0) = 1$. We define

$$S = W_0 \times [0, 1/6] \cup (W_0 \cap W_1) \times [1/6, 5/6] \cup W_1 \times [5/6, 1] \subset X \times \mathbf{I}$$

and an embedding $j : W_0 \cup W_1 \to S$ by $j(x) = (x, \alpha(x))$. Let $r = \mathrm{pr}_X|S : S \to W_0 \cup W_1$. Then, $rj = \mathrm{id}$ and $j(N \cup (W_0 \setminus W_1)) \subset W_0 \times \{0\}$.

By the hypothesis, there are ANRs Y_1, Y_* and homotopy equivalences $f_1 : W_1 \to Y_1$, $f_* : W_0 \cap W_1 \to Y_*$ with homotopy inverses g_1 and g_*, respectively. Let

$$Y = M(f_0 g_*) \cup_{Y_* \times \{1\}} M(f_1 g_*).$$

Then, Y is an ANR by Corollary 6.5.6 with Proposition 6.2.9 (5). By identifying $Y_0 = Y_0 \times \{0\} \subset M(f_0 g_*)$, Y contains Y_0 as a closed set. We will construct maps $f : S \to Y$, $g : Y \to S$ and an $r^{-1}(\mathcal{V})$-homotopy $h : S \times \mathbf{I} \to S$ such that $h_0 = \mathrm{id}$, $h_1 = gf$, $f(x, 0) = f_0(x)$ for each $x \in W_0$, $g(y) = (g_0(y), 0)$ for each $y \in Y_0$, and

$$h_t(x, 0) = \begin{cases} (h_1^0(x), 0) & \text{for } x \in W_0 \text{ and } 1/2 \leqslant t \leqslant 1, \\ (h_{2t}^0(x), 0) & \text{for } x \in W_0 \text{ and } 0 \leqslant t \leqslant 1/2. \end{cases}$$

Then, the maps $\tilde{f} = fj$ and $\tilde{g} = rg$ are the desired ones and the $\mathcal{V}$-homotopy $\tilde{h}$ is defined as follows:

$$\tilde{h}_t(x) = rh_{(1+\beta(x))^{-1}t} j(x) = rh_{(1+\beta(x))^{-1}t}(x, \alpha(x)),$$

where $\beta : W_0 \cup W_1 \to \mathbf{I}$ is a Urysohn map with $\beta((W_0 \cup W_1) \setminus N) = 0$ and $\beta(X_0) = 1$. Indeed, $\tilde{f}(x) = f(x, 0) = f_0(x)$ for $x \in X_0$, $\tilde{g}(y) = rg(y) = g_0(y)$ for $y \in Y_0$ and $\tilde{h}_t(x) = rh_{2^{-1}t}(x, 0) = h_t^0(x)$ for $(x, t) \in X_0 \times \mathbf{I}$. For every $x \in W_0 \cup W_1$, $\tilde{h}_0(x) = rh_0(x, \alpha(x)) = x$. If $x \notin N$, then $\tilde{h}_1(x) = rh_1 j(x) = rgfj(x) = \tilde{g}\tilde{f}(x)$. When $x \in N$, since $(1 + \beta(x))^{-1} \geqslant 1/2$, it follows that

$$\tilde{h}_1(x) = rh_{(1+\beta(x))^{-1}}(x, 0) = h_1^0(x) = g_0 f_0(x) = \tilde{g}\tilde{f}(x).$$

For $i = 0, 1$, let $j_i : W_0 \cap W_1 \to W_i$ be the inclusion. We define

$$T = M(f_0, j_0, g_*) \cup_{Y_* \times \{3\}} M(f_1, j_1, g_*).$$

Using Fact 2, we can obtain maps $f' : S \to T$, $g' : T \to S$ and a homotopy $h' : S \times \mathbf{I} \to S$ such that $h'_0 = \mathrm{id}$, $h'_1 = g'f'$, $h'_t(x, 0) = (h_t^0(x), 0)$ for each $x \in W_0$ and $t \in \mathbf{I}$ (cf. Fig. 6.18). The following are consequences of the construction:

(1) $h'_t | (W_0 \cap W_1) \times \big([1/6, 1/3] \cup [2/3, 5/6]\big) = \mathrm{id}$ for each $t \in \mathbf{I}$;
(2) Each h'_t preserves components of $(W_0 \cap W_1) \times [1/3, 1/2]$, $(W_0 \cap W_1) \times [1/2, 2/3]$, and $W_1 \times [5/6, 1]$;
(3) $h'(\{x\} \times [0, 1/6] \times \mathbf{I}) \subset h^0(\{x\} \times \mathbf{I}) \times [0, 1/6]$ for each $x \in W_0$;
(4) $g'(\{g_*(y)\} \times (0, 2] \cup \{y\} \times (2, 3]) \subset h^0(\{g_*(y)\} \times \mathbf{I} \times (0, 1/6] \cup \{g_*(y)\} \times (1/6, 1/2])$ for each $y \in Y_*$;
(5) $g'(\{f_0(x)\} \times \{0\} \cup \{x\} \times (0, 2]) \subset h^0(\{x\} \times \mathbf{I}) \times [0, 1/6] \cup \{x\} \times (1/6, 1/3]$ for each $x \in W_0 \cap W_1$;
(6) $g'(\{f_0(x)\} \times \{0\} \cup \{x\} \times (0, 1]) \subset h^0(\{x\} \times \mathbf{I}) \times [0, 1/6]$ for each $x \in W_0$.

Now, we can use Fact 1 to obtain maps $f'' : T \to Y$, $g'' : Y \to T$ and a homotopy $h'' : T \times \mathbf{I} \to T$ such that $h''_0 = \mathrm{id}$ and $h''_1 = g''f''$. Then, the following statements hold:

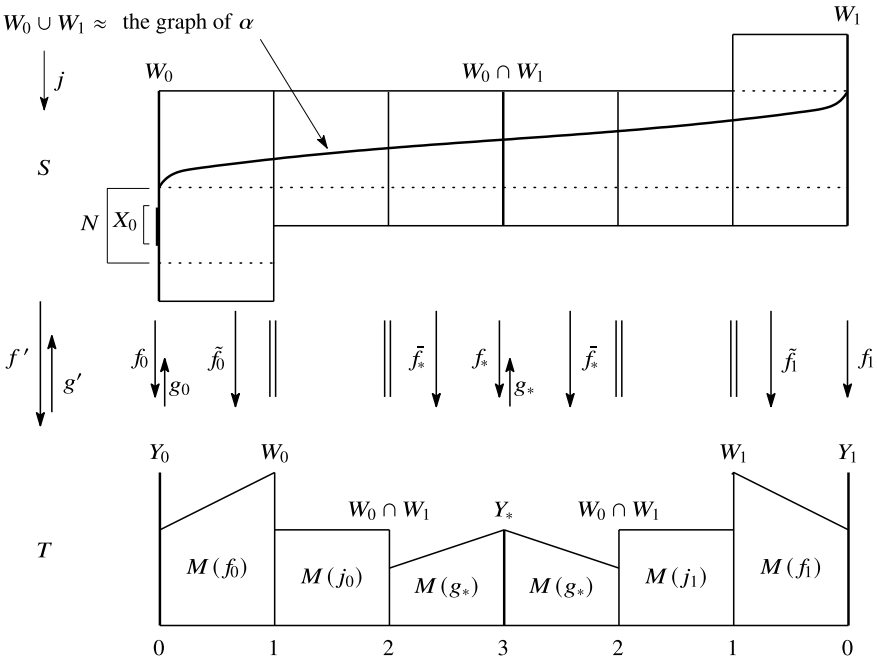


Fig. 6.18 The maps f' and g'

(7) $h''_t|Y_0 \times \{0\} \oplus Y_* \times \{3\} \oplus Y_1 \times \{0\} = \mathrm{id}$ for each $t \in \mathbf{I}$;
(8) Each h''_t preserves components of $M(f_1, j_1, g_*)$;
(9) $h''(\{y\} \times (2, 3] \times \mathbf{I}) \subset \{g_*(y)\} \times (0, 2] \cup \{y\} \times (2, 3]$ for $y \in Y_*$;
(10) $h''(\{x\} \times (0, 2] \times \mathbf{I}) \subset \{f_0(x)\} \times \{0\} \cup \{x\} \times (0, 2]$ for $x \in W_0 \cap W_1$;
(11) $h''(\{x\} \times (0, 1] \times \mathbf{I}) \subset \{f_0(x)\} \times \{0\} \cup \{x\} \times (0, 1]$ for $x \in W_0$.

Thus, we have the maps $f = f''f' : S \to Y$, $g = g'g'' : Y \to S$ and the homotopy $h : S \times \mathbf{I} \to S$ defined by

$$h_t = \begin{cases} g'h''_{2t-1}f' & \text{for } t \geqslant 1/2, \\ h'_{2t} & \text{for } t \leqslant 1/2. \end{cases}$$

Then, $h_0 = \mathrm{id}$ and $h_1 = gf$. Since h^0 is a $\mathcal{V}_0$-homotopy, it is easy to prove that h is a $r^{-1}(\mathcal{V})$-homotopy. Thus, the proof is complete. □

Now, we can complete the proof of Theorem 6.9.1 by proving the "if" part:

Proof (*The "if" Part of Theorem* 6.9.1) For any open cover $\mathcal{U}$ of X, we will show that X is $\mathcal{U}$-homotopy dominated by an ANE. Then, it will follow from Theorem 6.6.2 that X is an ANR.

Let K be the nerve of $\mathcal{U}$ with $\varphi : X \to |K|$ a canonical map. For each simplex $\sigma \in K$, let $X_\sigma = \varphi^{-1}(|\operatorname{St}(\hat{\sigma}, \operatorname{Sd}^2 K)|)$, where $\hat{\sigma}$ is the barycenter of σ. Then, $X = \bigcup_{\sigma \in K} X_\sigma$. For each $n \in \omega$, choose an open set W_n in X so that

$$\bigcup_{\sigma \in K^{(n)} \setminus K^{(n-1)}} X_\sigma \subset W_n \subset \operatorname{cl} W_n \subset \bigcup_{\sigma \in K^{(n)} \setminus K^{(n-1)}} \varphi^{-1}(O_{\operatorname{Sd} K}(\hat{\sigma})),$$

where $\{\varphi^{-1}(O_{\operatorname{Sd} K}(\hat{\sigma})) \mid \sigma \in K^{(n)} \setminus K^{(n-1)}\}$ is pairwise disjoint. For each $\sigma \in K^{(n)} \setminus K^{(n-1)}$, let $W_\sigma = W_n \cap \varphi^{-1}(O_{\operatorname{Sd} K}(\hat{\sigma}))$. Then,

$$X_\sigma \subset W_\sigma \subset \operatorname{cl} W_\sigma \subset \varphi^{-1}(O_{\operatorname{Sd} K}(\hat{\sigma}))$$

and $\mathcal{W}_n = \{W_\sigma \mid \sigma \in K^{(n)} \setminus K^{(n-1)}\}$ is a discrete open collection in X with $\bigcup \mathcal{W}_n = W_n$. For each $n \in \omega$, we define

$$V_n = \bigcup_{i=0}^{n} W_i = \bigcup_{\sigma \in K^{(n)}} W_\sigma \text{ and } X_n = \bigcup_{\sigma \in K^{(n)}} X_\sigma \subset V_n.$$

Note that $V_n = V_{n-1} \cup W_n$. For each $\sigma \in K$,

$$\begin{aligned} \sigma &\subset \bigcup_{v \in \sigma \cap (\operatorname{Sd}^2 K)^{(0)}} O(v, \operatorname{Sd}^2 K) \\ &\subset \bigcup_{v \in \sigma \cap (\operatorname{Sd}^2 K)^{(0)}} |\operatorname{St}(v, \operatorname{Sd}^2 K)| = \bigcup_{\tau \leqslant \sigma} |\operatorname{St}(\hat{\tau}, \operatorname{Sd}^2 K)|. \end{aligned}$$

Hence, we have $X = \bigcup_{n \in \omega} \operatorname{int} X_n$. We inductively define

$$\mathcal{V}_n = \mathcal{V}_{n-1} \cup \mathcal{W}_n \cup \left\{ V \cup W \;\middle|\; V \in \mathcal{V}_{n-1},\ W \in \mathcal{W}_n[V] \right\},$$

where $\mathcal{V}_0 = \mathcal{W}_0$. Then, each $\mathcal{V}_n$ is an open cover of V_n. By induction, it can be seen that for each $V \in \mathcal{V}_n$, there are some $\sigma_0, \sigma_1, \ldots, \sigma_k \in K^{(n)}$ such that $\dim \sigma_0 < \cdots < \dim \sigma_k$ (hence $k \leqslant n$), $\sigma_0 \leqslant \sigma_i$ for each $i \leqslant k$, and $V = \bigcup_{i=0}^{k} W_{\sigma_i}$. Then, for $v \in \sigma_0^{(0)}$,

$$V \subset \varphi^{-1}\left(\bigcup_{i=0}^{k} O_{\operatorname{Sd} K}(\hat{\sigma}_i)\right) \subset \varphi^{-1}(O_K(v)),$$

which means that $\mathcal{V}_n \prec \mathcal{U}$.

For each $v \in K^{(0)}$, W_v has the homotopy type of an ANR Y_v. Let $Y_0 = \bigoplus_{v \in K^{(0)}} Y_v$. Since $\mathcal{V}_0 = \mathcal{W}_0$ is discrete in X and $V_0 = W_0$, there exist two maps $f_0 : V_0 \to Y_0$, $g_0 : Y_0 \to V_0$ and a $\mathcal{V}_0$-homotopy $h^0 : V_0 \times \mathbf{I} \to V_0$ with $h_0^0 = g_0 f_0$ and $h_1^0 = \mathrm{id}$. By Lemma 6.9.2, we have an ANR Y_1, maps $f_1 : V_1 \to Y_1$, $g_1 : Y_1 \to V_1$ and a $\mathcal{V}_1$-homotopy $h^1 : V_1 \times \mathbf{I} \to V_1$ such that $h_0^1 = g_1 f_1$, $h_1^1 = \mathrm{id}$, Y_1 contains Y_0 as a closed set, $f_1 | X_0 = f_0 | X_0$, $g_1 | Y_0 = g_0$, and $h^1 | X_0 \times \mathbf{I} = h^0 | X_0 \times \mathbf{I}$. Again, using Lemma 6.9.2, we obtain an ANR Y_2, maps $f_2 : V_2 \to Y_2$ and $g_2 : Y_2 \to V_2$, and a

$\mathcal{V}_2$-homotopy $h^2 : V_2 \times \mathbf{I} \to V_2$ such that $h^2_0 = g_2 f_2$, $h^2_1 = \mathrm{id}$, Y_2 contains Y_1 as a closed set, $f_2|X_1 = f_1|X_1$, $g_2|Y_1 = g_1$, and $h^2|X_1 \times \mathbf{I} = h^1|X_1 \times \mathbf{I}$. Thus, we apply Lemma 6.9.2 inductively to obtain a closed tower $Y_0 \subset Y_1 \subset Y_2 \subset \cdots$ of ANRs with maps $f_n : V_n \to Y_n$, $g_n : Y_n \to V_n$ and $\mathcal{V}_n$-homotopies $h^n : V_n \times \mathbf{I} \to V_n$ such that $h^n_0 = g_n f_n$, $h^n_1 = \mathrm{id}$, $f_n|X_{n-1} = f_{n-1}|X_{n-1}$, $g_n|Y_{n-1} = g_{n-1}$, and $h^n|X_{n-1} \times \mathbf{I} = h^{n-1}|X_{n-1} \times \mathbf{I}$. Let $Y = \varinjlim Y_n$. We can define maps $f : X \to Y$, $g : Y \to X$, and a $\mathcal{U}$-homotopy $h : X \times \mathbf{I} \to X$ as follows: $f|X_n = f_n|X_n$, $g|Y_n = g_n$, and $h|X_n \times \mathbf{I} = h^n|X_n \times \mathbf{I}$ for $n \geqslant 0$. Then, $h_0 = gf$ and $h_1 = \mathrm{id}$, hence X is $\mathcal{U}$-homotopy dominated by Y. Moreover, Y is an ANE by Theorem 6.1.9. This completes the proof. □

6.10 Countable-Dimensional ANRs

Recall that a metrizable space X is **countable-dimensional** (**c.d.**) if it is a countable union of finite-dimensional subspaces.[14] Obviously, every finite-dimensional metrizable space is countable-dimensional. Due to Proposition 5.6.2, a metrizable space X is countable-dimensional if and only if $X = \bigcup_{i\in\mathbb{N}} A_i$ for some countably many subsets $A_i \subset X$ with $\dim A_i \leqslant 0$. In this section, we prove the following theorem:

Theorem 6.10.1 *Every countable-dimensional locally contractible metrizable space is an ANR.*

First, we introduce a covering property related to countable dimensionality. A space X has **property** $\boldsymbol{C}$ provided for any open covers $\mathcal{U}_n$ of $X, n \in \mathbb{N}$, there exists an open cover $\mathcal{V} = \bigcup_{n\in\mathbb{N}} \mathcal{V}_n$ of X such that each $\mathcal{V}_n$ is pairwise disjoint and $\mathcal{V}_n \prec \mathcal{U}_n$.

Lemma 6.10.2 *Let X be a metrizable space. If $X = \bigcup_{i\in\mathbb{N}} A_i$ and each A_i has property C, then X also has property C.*

Proof Let $\mathcal{U}_n \in \mathrm{cov}(X)$, $n \in \mathbb{N}$. Take a bijection $k : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$. Having property C, each A_i has an open cover $\mathcal{V}'_i = \bigcup_{j\in\mathbb{N}} \mathcal{V}'_{(i,j)}$ such that each $\mathcal{V}'_{(i,j)}$ is pairwise disjoint and $\mathcal{V}'_{(i,j)} \prec \mathcal{U}_{k(i,j)}$. For each $V \in \mathcal{V}'_{(i,j)}$, choose $U(V) \in \mathcal{U}_{k(i,j)}$ so that $V \subset U(V)$, and define

$$\widetilde{V} = \big\{x \in U(V) \;\big|\; d(x, V) < d(x, A_i \setminus V)\big\} = U(V) \cap E(V),$$

where $E(V)$ is the open set in X defined in the proof of the Canonical Open Set Extension Lemma 5.4.10. Then, $\widetilde{V}$ is open in X and $\widetilde{V} \cap A_i = V$. For each $n \in \mathbb{N}$, let $\mathcal{V}_n = \{\widetilde{V} \mid V \in \mathcal{V}'_{k^{-1}(n)}\}$. Then, each $\mathcal{V}_n$ is pairwise disjoint and $\mathcal{V}_n \prec \mathcal{U}_n$. Since $\bigcup_{j\in\mathbb{N}} \mathcal{V}_{k(i,j)}$ is an open cover of A_i in X, $\mathcal{V} = \bigcup_{n\in\mathbb{N}} \mathcal{V}_n$ is an open cover of $X = \bigcup_{i\in\mathbb{N}} A_i$. □

[14] In Sect. 5.6, we defined a *normal* space X to be **countable-dimensional** if it is a countable union of finite-dimensional normal subspaces.

Lemma 6.10.3 *Let X be a metrizable space that has property C. Then, every F_σ set in X also has property C.*

Proof It is easy to see that any closed set in X has property C. Then, we can apply Lemma 6.10.2 to complete the proof. □

Theorem 6.10.4 *Every countable-dimensional metrizable space has property C.*

Proof Since any open cover of a 0-dimensional metrizable space has a discrete open refinement by Theorem 5.2.5, any 0-dimensional metrizable space has property C. Then, applying Lemma 6.10.2, we complete the proof. □

Lemma 6.10.5 *If X has property C, then $X \times \mathbf{I}$ also has property C.*

Proof Let $\mathcal{U}_n \in \mathrm{cov}(X \times \mathbf{I})$, $n \in \mathbb{N}$. Using the compactness of $\mathbf{I}$, we can easily find $\mathcal{W}_n \in \mathrm{cov}(X)$, $n \in \mathbb{N}$, and partitions $0 = t_0^W < t_1^W < \cdots < t_{k(W)}^W = 1$, $W \in \mathcal{W}_n$, so that

$$\mathcal{W}_n^* = \left\{ W \times J_i^W \;\middle|\; W \in \mathcal{W}_n,\ i = 0, 1, \cdots, k(W) \right\} \prec \mathcal{U}_{2n} \wedge \mathcal{U}_{2n-1},$$

where $J_0^W = [0, t_1^W)$, $J_{k(W)}^W = (t_{k(W)-1}^W, 1]$ and $J_i^W = (t_{i-1}^W, t_{i+1}^W)$ for $0 < i < k(W)$. Since X has property C, X has an open cover $\mathcal{V} = \bigcup_{n\in\mathbb{N}} \mathcal{V}_n$ such that each $\mathcal{V}_n$ is pairwise disjoint and $\mathcal{V}_n \prec \mathcal{W}_n$. For each $V \in \mathcal{V}_n$, choose $W(V) \in \mathcal{W}_n$ so that $V \subset W(V)$. For each $n \in \mathbb{N}$, let

$$\mathcal{V}_{2n}^* = \left\{ V \times J_{2i}^{W(V)} \;\middle|\; V \in \mathcal{V}_n,\ 2i \leqslant k(W(V)) \right\} \text{ and}$$
$$\mathcal{V}_{2n-1}^* = \left\{ V \times J_{2i+1}^{W(V)} \;\middle|\; V \in \mathcal{V}_n,\ 2i+1 \leqslant k(W(V)) \right\}.$$

We have $\mathcal{V}^* = \bigcup_{n\in\mathbb{N}} \mathcal{V}_n^* \in \mathrm{cov}(X \times \mathbf{I})$, where each $\mathcal{V}_n^*$ is pairwise disjoint. Since

$$\mathcal{V}_{2n-1}^*, \mathcal{V}_{2n}^* \prec \mathcal{W}_n^* \prec \mathcal{U}_{2n-1} \wedge \mathcal{U}_{2n},$$

it follows that $\mathcal{V}_n^* \prec \mathcal{U}_n$ for each $n \in \mathbb{N}$. Hence, $X \times \mathbf{I}$ has property C. □

By Lemmas 6.10.3 and 6.10.5, we have the following proposition:

Proposition 6.10.6 *If X is metrizable and has property C, then $X \times (0, 1]$ also has property C.* □

By Theorem 6.10.4, we can obtain Theorem 6.10.1 as a corollary of the following Extension Theorem:

Theorem 6.10.7 *Let A be a closed set in a metrizable space X such that A has property C, and let Y be a locally contractible metrizable space. Then, any map $f : A \to Y$ extends over a neighborhood of A in X.*

Proof Inductively take open covers of Y:

$$\mathcal{U}_1 \succ \mathcal{V}_1 \overset{*}{\succ} \mathcal{U}_2 \succ \mathcal{V}_2 \overset{*}{\succ} \mathcal{U}_3 \succ \cdots,$$

so that $\operatorname{mesh} \mathcal{U}_n < 2^{-n}$ and each member of $\mathcal{V}_n$ is contractible in some member of $\mathcal{U}_n$. For each $n \in \mathbb{N}$, let

$$\mathcal{U}_n^* = \left\{ f^{-1}(U) \times J_m \;\middle|\; m \in \mathbb{N},\ U \in \mathcal{U}_{2n+m} \right\} \in \operatorname{cov}(A \times (0, 1]),$$

where $J_1 = (1/2, 1]$ and $J_m = (2^{-m}, 2^{-m}3)$ for $m > 1$. Since $A \times (0, 1]$ has property C, it has an open cover $\mathcal{W} = \bigcup_{n\in\mathbb{N}} \mathcal{W}_n$ such that each $\mathcal{W}_n$ is pairwise disjoint and $\mathcal{W}_n \prec \mathcal{U}_n^*$.

For each $W \in \mathcal{W}$, choose $n(W)$, $m(W)$, $k(W) \in \mathbb{N}$, $U(W) \in \mathcal{U}_{k(W)}$, $V(W) \in \mathcal{V}_{k(W)-1}$, and $\widetilde{U}(W) \in \mathcal{U}_{k(W)-1}$ as follows:

(1) $W \in \mathcal{W}_{n(W)}$,
(2) $W \subset f^{-1}(U(W)) \times J_{m(W)} \in \mathcal{U}^*_{n(W)}$,
(3) $U(W) \in \mathcal{U}_{k(W)}$, $k(W) = 2n(W) + m(W)$,
(4) $\operatorname{st}(U(W), \mathcal{U}_{k(W)}) \subset V(W) \in \mathcal{V}_{k(W)-1}$,
(5) $V(W)$ is contractible in $\widetilde{U}(W) \in \mathcal{U}_{k(W)-1}$,

and let $\psi^W : V(W) \times \mathbf{I} \to \widetilde{U}(W)$ be a contraction, that is, $\psi_0^W = \mathrm{id}$ and $\psi_1^W(V(W)) = \{x_W\}$, where $x_W \in \widetilde{U}(W)$.

Let K be the nerve of $\mathcal{W}$ with $\varphi : A \times (0, 1] \to |K|$ a canonical map. Since $|K|$ is an ANE, φ extends to a map $\tilde{\varphi} : N \to |K|$ over a neighborhood N of $A \times (0, 1]$ in $X \times (0, 1]$. We will construct a map $g : |K| \to Y$ so that $d(g\varphi(a, t), f(a)) < t$ for each $(a, t) \in A \times (0, 1]$. Then, $g\varphi : A \times (0, 1] \to Y$ would be extended over $A \times \mathbf{I}$ by f. Hence, we could apply Lemma 6.5.1 to extend f over a neighborhood of A in X.

For each $\sigma \in K$, take a vertex $W(\sigma) \in \sigma^{(0)}$ $(\subset \mathcal{W})$ such that $n(W(\sigma)) = \min\{n(W) \mid W \in \sigma^{(0)}\}$. Then, such a vertex $W(\sigma) \in \sigma^{(0)}$ is unique. Indeed, $n(W) < n(W')$ or $n(W') < n(W)$ for $W \neq W' \in \sigma^{(0)}$ because each $\mathcal{W}_n$ is pairwise disjoint. Next, when $\dim \sigma > 0$, let σ_0 be the opposite face of σ to $W(\sigma)$, i.e., $\sigma = \sigma_0 W(\sigma)$. Then, each point of σ can be represented as

$$(1 - t)z + tW(\sigma),\ t \in \mathbf{I},\ z \in \sigma_0.$$

Here, it should be noticed that, if $\tau < \sigma$, $\dim \tau > 0$, and $W(\sigma) \in \tau^{(0)}$, then $W(\tau) = W(\sigma)$ and $\tau_0 < \sigma_0$.

We will show the following:

(6) $\widetilde{U}(W) \subset V(W(\sigma))$ for every $W \in \sigma^{(0)} \setminus \{W(\sigma)\}$.

Since $J_{m(W(\sigma))} \cap J_{m(W)} \neq \emptyset$, we have $|m(W(\sigma)) - m(W)| \leqslant 1$, which implies $k(W(\sigma)) < k(W)$ by (3). Since $\mathcal{U}_{k(W)-1} \prec \mathcal{U}_{k(W(\sigma))}$, $\widetilde{U}(W)$ is contained in some member of $\mathcal{U}_{k(W(\sigma))}$. On the other hand, $U(W) \cap U(W(\sigma)) \neq \emptyset$ by (2). Since $U(W) \subset \widetilde{U}(W)$, we have $\widetilde{U}(W) \cap U(W(\sigma)) \neq \emptyset$. Thus, it follows from (4) that

$$\widetilde{U}(W) \subset \mathrm{st}(U(W(\sigma)), \mathcal{U}_{k(W(\sigma))}) \subset V(W(\sigma)).$$

Now, let $g_0 : |K^{(0)}| \to Y$ be a map such that $g_0(W) = x_W \in \widetilde{U}(W)$ for each $W \in K^{(0)}$. Assume that we have maps $g_i : |K^{(i)}| \to Y, i < n$, such that $g_i||K^{(i-1)}| = g_{i-1}$ and, if $\sigma \in K^{(i)} \setminus K^{(0)}, t \in \mathbf{I}$, and $z \in \sigma_0$, then $g_{i-1}(z) \in \widetilde{U}(W(\sigma_0)) \subset V(W(\sigma))$ and

$$g_i((1-t)z + tW(\sigma)) = \psi^{W(\sigma)}(g_{i-1}(z), t) \in \widetilde{U}(W(\sigma)).$$

For each n-simplex $\sigma \in K$, we have $g_{n-1}(\sigma_0) \subset \widetilde{U}(W(\sigma_0)) \subset V(W(\sigma))$ by the above assumption and (6). Then, we can define a map $g_\sigma : \sigma \to \widetilde{U}(W(\sigma)) \subset Y$ as follows:

$$g_\sigma((1-t)z + tW(\sigma)) = \psi^{W(\sigma)}(g_{n-1}(z), t).$$

It is easy to prove that $g_\sigma|\partial\sigma = g_{n-1}|\partial\sigma$. Hence, g_{n-1} extends to the map $g_n : |K^{(n)}| \to Y$ defined by $g_n|\sigma = g_\sigma$ for each n-simplex $\sigma \in K$, where if $\sigma \in K^{(n)} \setminus K^{(0)}, t \in \mathbf{I}$, and $z \in \sigma_0$, then

$$g_n((1-t)z + tW(\sigma)) = \psi^{W(\sigma)}(g_n(z), t) \in \widetilde{U}(W(\sigma)).$$

By induction, we have maps $g_n, n \in \mathbb{N}$, satisfying the above condition. Let $g : |K| \to Y$ be the map defined by $g||K^{(n)}| = g_n$. Then, $g(\sigma) \subset \widetilde{U}(W(\sigma))$ for each $\sigma \in K$.

It remains to show that $d(g\varphi(a,t), f(a)) < t$ for each $(a,t) \in A \times (0,1]$. Let $\sigma \in K$ be the carrier of $\varphi(a,t)$. Then, $(a,t) \in W(\sigma)$, which implies that $t \in J_{m(W(\sigma))}$, $f(a) \in U(W(\sigma)) \subset \widetilde{U}(W(\sigma))$ and $g\varphi(a,t) \in \widetilde{U}(W(\sigma))$. Hence, $t > 2^{-m(W(\sigma))} \geqslant 2^{-k(W(\sigma))+2}$ and

$$d(g\varphi(a,t), f(a)) \leqslant \mathrm{diam}\, \widetilde{U}(W(\sigma)) \leqslant \mathrm{mesh}\, \mathcal{U}_{k(W(\sigma))-1} < 2^{-k(W(\sigma))+1} < t.$$

This completes the proof. □

6.11 The Local n-Connectedness

In this section, we need some results in Dimension Theory discussed in Chap. 5: Theorem 5.2.4, the No Retraction Theorem 5.1.5, and the universal spaces ν^n and μ^n discussed in Sect. 5.10. In addition, we use the General Position Lemma 4.14.1 proved in Chap. 4.

A space X is said to be **n-connected** ($\boldsymbol{C^n}$) if each map $f : \mathbf{S}^i \to X, i \leqslant n$, extends over $\mathbf{B}^{i+1}$.[15] *Being* 0*-connected means being path-connected.* We also say that X is **simply connected** instead of 1-connected. Note that $\mathbf{S}^n$ is $(n-1)$-connected but not n-connected by Theorem 5.2.4 and the No Retraction Theorem 5.1.5. We also say that X is **locally n-connected** ($\boldsymbol{LC^n}$) if for each $x \in X$, each neighborhood U of

[15] In terms of homotopy groups, this means that $\pi_i(X, x_0) = \{0\}$ for every $x_0 \in X$ and $i \leqslant n$.

x contains a neighborhood V of x such that any map $f : \mathbf{S}^i \to V$, $i \leqslant n$, extends to a map $\tilde{f} : \mathbf{B}^{i+1} \to U$. *To be LC^0 means to be locally path-connected.* A space X is said to be **locally simply connected** instead of locally 1-connected. We say that X is $\boldsymbol{C^\infty}$ (or $\boldsymbol{LC^\infty}$) if X is C^n (or LC^n) for all $n \in \omega$. *Being C^∞ means being homotopically trivial.* Every locally contractible space is LC^∞ but, in general, the converse does not hold, as the following example shows. The example also shows that LC^∞-ness does not imply that every point has an arbitrarily small homotopically trivial (C^∞) neighborhood.

Theorem 6.11.1 (BORSUK) *There exists an LC^∞ continuum X that has a point whose any neighborhood is not contractible in X nor homotopically trivial.*

Proof (*Example*) For each $i \in \mathbb{N}$, let $u_i = 2^{-i}\mathbf{e}_1$, $v_i = 2^{-i-1}3\mathbf{e}_1 \in \ell_2$ and let S_i be the i-dimensional sphere in ℓ_2 centered at v_i with radius 2^{-i-1}, i.e., $S_i = v_i + 2^{-i-1}\mathbf{S}^i$, where we identify $\mathbf{S}^i = \mathbf{S}^i \times \{\mathbf{0}\} \subset \ell_2$. Note that $S_i \cap S_{i+1} = \{u_i\}$ and $S_i \cap S_j = \emptyset$ if $|i - j| > 1$. For each $n \in \mathbb{N}$, let $X_n = \{\mathbf{0}\} \cup \bigcup_{i=n}^{\infty} S_i \subset \ell_2$. Then, $X = X_1$ is the desired continuum. Each X_n is the closed 2^{-n+1}-neighborhood of the point $\mathbf{0}$ in X. As is easily observed, every S_n is a retract of X.

Any neighborhood V of $\mathbf{0}$ in X contains some S_n. Then, any homeomorphism $f : \mathbf{S}^n \to S_n$ ($\subset V$) cannot extend to a map $\tilde{f} : \mathbf{B}^{n+1} \to X$. Indeed, if f extends to a map $\tilde{f} : \mathbf{B}^{n+1} \to X$, then we have a retraction $f^{-1}r\tilde{f}$ of $\mathbf{B}^{n+1}$ onto $\mathbf{S}^n$, where $r : X \to S_n$ is a retraction. This contradicts the No Retraction Theorem (Corollary 5.1.5). Hence, f cannot extend to a map $\tilde{f} : \mathbf{B}^{n+1} \to X$. Then, it follows that V is not contractible in X nor homotopically trivial. On the other hand, X is locally contractible at any point of $X \setminus \{\mathbf{0}\}$. It remains to show that X is LC^∞ at the point $\mathbf{0}$.

To prove that X_1 is LC^∞ at $\mathbf{0}$, it suffices to show that X_m is n-connected for each $m > n$, that is, each map $f : \mathbf{S}^i \to X_m$, $i \leqslant n$, extends over $\mathbf{B}^{i+1}$. First, we take open sets U_j in $\mathbf{B}^{i+1}$, $j \geqslant m$, such that

$$f^{-1}(X_j \setminus \{u_{j-1}\}) \subset U_j \subset \operatorname{cl} U_j \subset U_{j-1} \setminus f^{-1}(S_{j-1} \setminus \{u_{j-1}\}),$$

where the last set is $\mathbf{B}^{i+1}$ if $j = m$. In fact, let $U_m = \mathbf{B}^{i+1} \setminus f^{-1}(u_{m-1})$ and suppose that U_{j-1} has been obtained. Since $\operatorname{cl} f^{-1}(X_j \setminus \{u_{j-1}\}) \subset f^{-1}(X_j)$ and

$$\operatorname{cl}\left((\mathbf{B}^{i+1} \setminus U_{j-1}) \cup f^{-1}(S_{j-1} \setminus \{u_{j-1}\})\right) \subset (\mathbf{B}^{i+1} \setminus U_{j-1}) \cup f^{-1}(S_{j-1}),$$

$f^{-1}(X_j \setminus \{u_{j-1}\})$ and $(\mathbf{B}^{i+1} \setminus U_{j-1}) \cup f^{-1}(S_{j-1} \setminus \{u_{j-1}\})$ are separated in $\mathbf{B}^{i+1}$. By complete normality,[16] $\mathbf{B}^{i+1}$ has the desired open set U_j. See Fig. 6.19.

Next, for each $j \geqslant m$, observe that $f(\mathbf{S}^i \cap \operatorname{bd} U_j) \subset \{u_{j-1}\}$ and

$$f(\mathbf{S}^i \cap (\operatorname{cl} U_j \setminus U_{j+1})) \subset X_j \setminus (X_{j+1} \setminus \{u_j\}) = S_j.$$

Since $\dim(\operatorname{cl} U_j \setminus U_{j+1}) \leqslant i + 1 \leqslant j$, we can apply Theorem 5.2.4 to obtain a map $f_j : \operatorname{cl} U_j \setminus U_{j+1} \to S_j$ ($\approx \mathbf{S}^j$) such that

[16] Due to Theorem 2.2.6, X is completely normal if and only if X is hereditarily normal.

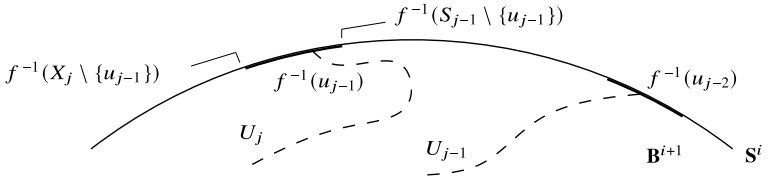


Fig. 6.19 Open sets U_j

$$f_j(\operatorname{bd} U_j) \subset \{u_{j-1}\},\ f_j(\operatorname{bd} U_{j+1}) \subset \{u_j\},\ \text{and}$$
$$f_j|\mathbf{S}^i \cap (\operatorname{cl} U_j \setminus U_{j+1}) = f|\mathbf{S}^i \cap (\operatorname{cl} U_j \setminus U_{j+1}).$$

Now, we can define a map $\tilde{f} : \mathbf{B}^{i+1} \to X_m$ as follows:

$$\tilde{f}|\operatorname{cl} U_j \setminus U_{j+1} = f_j \ \text{ for } \ j \geqslant m,$$
$$\tilde{f}(\mathbf{B}^{i+1} \setminus U_m) \subset \{u_{m-1}\},\ \text{and}\ \ \tilde{f}\big(\textstyle\bigcap_{j\geqslant m} U_j\big) \subset \{\mathbf{0}\}.$$

Then, $\tilde{f}$ is an extension of f. Hence, X_m is n-connected. □

Remark 6.11.1 In the above example, for a fixed $n \in \mathbb{N}$, let S_i be the n-dimensional sphere in ℓ_2 centered at v_i with radius 2^{-i-1}, i.e., $S_i = v_i + 2^{-i-1}\mathbf{S}^n$. Then, by a similar proof, we can show that $\{\mathbf{0}\} \cup \bigcup_{i\in\mathbb{N}} S_i$ is LC^{n-1} but not LC^n at the point $\mathbf{0}$, which is n-dimensional.

The n-dimensional universal spaces ν^n and μ^n are also LC^{n-1} but not LC^n at any point. Before showing these facts, we will recall the definition of ν^n and μ^n in Sect. 5.10. The n-dimensional universal Nöbeling space ν^n is defined as follows:

$$\begin{aligned}\nu^n &= \big\{x \in \mathbb{R}^{2n+1} \;\big|\; x(i) \in \mathbb{R} \setminus \mathbb{Q} \ \text{ for at least } n+1\ i\big\}\\ &= \big\{x \in \mathbb{R}^{2n+1} \;\big|\; x(i) \in \mathbb{Q} \ \text{ for at most } n\ i\big\}.\end{aligned}$$

For each $k \in \mathbb{N}$, let

$$R_k = \bigcup_{m=0}^{3^{k-1}-1} (m/3^{k-1} + 1/3^k, m/3^{k-1} + 2/3^k) \subset \mathbf{I}.$$

The intersection $\bigcap_{k\in\mathbb{N}}(\mathbf{I} \setminus R_k)$ $(= \mathbf{I} \setminus \bigcup_{k\in\mathbb{N}} R_k)$ is the Cantor (ternary) set μ^0. The n-dimensional universal Menger compactum μ^n is defined as an intersection $\mu^n = \bigcap_{k\in\mathbb{N}} M_k^{2n+1}$, where

$$\begin{aligned}M_k^{2n+1} &= \big\{x \in \mathbf{I}^{2n+1} \;\big|\; x(i) \in \mathbf{I} \setminus R_k \ \text{ for at least } n+1\ i\big\}\\ &= \big\{x \in \mathbf{I}^{2n+1} \;\big|\; x(i) \in R_k \ \text{ for at most } n\ i\big\}.\end{aligned}$$

Theorem 6.11.2 *The n-dimensional universal Nöbeling space ν^n and the n-dimensional universal Menger compactum μ^n are C^{n-1} and LC^{n-1}, but they are neither C^n nor LC^n at any point.*

Proof To prove that ν^n is not C^n nor LC^n at any point, it suffices to show that for any neighborhood V of $x \in \nu^n$, there is a map $f : \mathbf{S}^n \to V$ that is not null-homotopic in ν^n. Choose $a_i < b_i \in \mathbb{R} \setminus \mathbb{Q}$, $i = 1, \ldots, 2n+1$, so that $\nu^n \cap \prod_{i=1}^{2n+1}[a_i, b_i] \subset V$, and define $C = \prod_{i=1}^{n+1}[a_i, b_i] \subset \mathbb{R}^{n+1}$. Take a point $v \in \mathbb{Q}^{n+1} \cap \operatorname{int} C \subset \mathbb{R}^{n+1}$. Then, we have a retraction $r : \mathbb{R}^{n+1} \setminus \{v\} \to \partial C$. Since $\{v\} \times \mathbb{R}^n \subset \mathbb{R}^{2n+1} \setminus \nu^n$, we have the map $rp : \nu^n \to \partial C$, where $p : \nu^n \to \mathbb{R}^{n+1} \setminus \{v\}$ is the restriction of the projection of $\mathbb{R}^{2n+1}$ onto the first $n+1$ factors. Now, let $u = (a_{n+2}, \ldots, a_{2n+1}) \in (\mathbb{R} \setminus \mathbb{Q})^n$. Then, $\partial C \times \{u\} \subset \nu^n$ by the definition of ν^n. Let $f : \mathbf{S}^n \to \partial C \times \{u\}$ be a homeomorphism. Assume that f is null-homotopic in ν^n, that is, f extends to a map $\tilde{f} : \mathbf{B}^{n+1} \to \nu^n$. The map $f^{-1}rp\tilde{f} : \mathbf{B}^{n+1} \to \mathbf{S}^n$ is a retraction, which contradicts the No Retraction Theorem 5.1.5. Hence, f is not null-homotopic in ν^n.

By analogy with ν^n, we can see that μ^n is not C^n nor LC^n at any point, that is, for any neighborhood V of $x \in \mu^n$, we can similarly construct a map $f : \mathbf{S}^n \to V$ that is not null-homotopic in μ^n. In the above, replacing ν^n and $[a_i, b_i]$ with μ^n and $[(m_i - 1)/3^k, m_i/3^k]$, respectively, where $m_i = 1, \ldots, 3^k$ and

$$\prod_{i=1}^{2n+1}\left[\frac{m_i - 1}{3^k}, \frac{m_i}{3^k}\right] \subset \bigcap_{j=1}^{k} M_j^{2n+1},$$

and defining the points v, u by

$$v = ((m_1 - \tfrac{1}{2})/3^k, \ldots, (m_{n+1} - \tfrac{1}{2})/3^k) \text{ and}$$
$$u = (m_{n+2}/3^k, \ldots, m_{2n+1}/3^k),$$

we can perform the same arguments as above.

We will show that ν^n and μ^n are C^{n-1} and LC^{n-1}. Here, suppose that ν^n and μ^n inherit the metric d of $\mathbb{R}^{2n+1}$ induced by the norm

$$\|x\|_\infty = \max\left\{|x(1)|, \ldots, |x(2n+1)|\right\}.$$

To show that ν^n and μ^n are C^{n-1} and LC^{n-1}, it suffices to show that every map $f : \partial\Delta^m \to \nu^n$ (resp. $f : \partial\Delta^m \to \mu^n$), $m \leqslant n$, extends to a map $\tilde{f} : \Delta^m \to \nu^n$ (resp. $\bar{f} : \Delta^m \to \mu^n$) with $\operatorname{diam} \tilde{f}(\Delta^m) \leqslant 2\delta$ (resp. $\operatorname{diam} \bar{f}(\Delta^m) \leqslant 13\delta$), where we may assume that f is non-constant, that is,

$$\delta = \operatorname{diam} f(\partial\Delta^m) > 0.$$

Let $D = \langle f(\partial\Delta^m)\rangle$. Then, $\operatorname{diam} D = \operatorname{diam} f(\partial\Delta^m) = \delta$ by Proposition 3.2.2 (ii). Since D is an AE, f extends to a map $f_0 : \Delta^m \to D$, where it should be noted that $\operatorname{diam} f_0(\Delta^m) = \delta$. We will modify f_0 to a map $\tilde{f} : \Delta^m \to \nu^n$ (resp. $\bar{f} : \Delta^m \to \mu^n$)

so that $\tilde{f}|\partial\Delta^m = f$ (resp. $\bar{f}|\partial\Delta^m = f$) and $d(\tilde{f}, f_0) \leqslant \delta/2$ (resp. $d(\bar{f}, f_0) \leqslant 6\delta$). Then, we will have

$$\begin{gathered}\operatorname{diam} \tilde{f}(\Delta^m) \leqslant \operatorname{diam} f_0(\Delta^m) + 2d(\tilde{f}, f_0) \leqslant 2\delta \\ \big(\text{resp. } \operatorname{diam} \bar{f}(\Delta^m) \leqslant \operatorname{diam} f_0(\Delta^m) + 2d(\bar{f}, f_0) \leqslant 13\delta\big).\end{gathered}$$

The Case of ν^n: We can write $\nu^n = \mathbb{R}^{2n+1} \setminus \bigcup_{i\in\mathbb{N}} H_i$, where each H_i is an n-dimensional flat in $\mathbb{R}^{2n+1}$ (cf. Proof of Theorem 5.10.4). By induction, we will construct maps $f_k : \Delta^m \to \mathbb{R}^{2n+1} \setminus \bigcup_{i=1}^{k} H_i$, $k \in \mathbb{N}$, so that $f_k|\partial\Delta^m = f$ and $d(f_k, f_{k-1}) \leqslant \varepsilon_k/2$, where $\varepsilon_k > 0$ is defined as follows:

$$\begin{gathered}\varepsilon_1 = \min\big\{\delta/2,\ \operatorname{dist}(f(\partial\Delta^m), H_1)\big\}, \text{ and} \\ \varepsilon_k = \min\big\{\varepsilon_{k-1}/2,\ \operatorname{dist}(f(\partial\Delta^m), H_k),\ 2^{-1}\operatorname{dist}\big(f_{k-1}(\Delta^m), H_{k-1}\big)\big\} \text{ if } k > 1.\end{gathered}$$

Then, $(f_i)_{i\in\mathbb{N}}$ is Cauchy because $\sum_{i=k}^{\infty} \varepsilon_i/2 \leqslant \varepsilon_k \leqslant 2^{-k}\delta$, so it converges to a map $\tilde{f} : \Delta^m \to \mathbb{R}^{2n+1}$ by the completeness of $\mathrm{C}(\Delta^m, \mathbb{R}^{2n+1})$, where $\tilde{f}|\partial\Delta^m = f$ and $d(\tilde{f}, f_0) \leqslant \varepsilon_1 \leqslant \delta/2$. Since $d(\tilde{f}, f_i) \leqslant \varepsilon_{i+1}$ for each $i \in \mathbb{N}$, it follows that

$$\operatorname{dist}(\tilde{f}(\Delta^m), H_i) \geqslant \operatorname{dist}(f_i(\Delta^m), H_i) - \varepsilon_{i+1} \geqslant \varepsilon_{i+1} > 0,$$

hence $\tilde{f}(\Delta^m) \subset \mathbb{R}^{2n+1} \setminus \bigcup_{i\in\mathbb{N}} H_i = \nu^n$.

Now, assume that $f_0, \ldots, f_{k-1}$ have been obtained. Then, $\varepsilon_k > 0$ is defined as above. Take $K \lhd F(\Delta^m)$ and $L \lhd \partial F(\Delta^m)$ so that $\operatorname{diam} f_{k-1}(\sigma) < \varepsilon_k/4$ for every $\sigma \in K$, and L is a full subcomplex of K. For each $\sigma \in K[\partial\Delta^m]$, we have

$$\operatorname{dist}(f_{k-1}(\sigma), H_k) \geqslant \operatorname{dist}(\mathrm{N}_d(f(\partial\Delta^m), \varepsilon_k/4), H_k) \geqslant 3\varepsilon_k/4 > \varepsilon_k/2.$$

Since H_k is an n-dimensional flat in $\mathbb{R}^{2n+1}$ and $\dim K = m \leqslant n$, we can apply the General Position Lemma 4.14.1 to find distinct points $p_v \in \mathbb{R}^{2n+1}$, $v \in K^{(0)} \setminus L^{(0)}$, such that $\|p_v - f_{k-1}(v)\|_\infty < \varepsilon_k/4$ and

$$\operatorname{fl}\{p_{v_1}, \ldots, p_{v_{m+1}}\} \cap H_k = \emptyset \text{ for every } v_1, \ldots, v_{m+1} \in K^{(0)} \setminus L^{(0)}.$$

Using the barycentric coordinates, we can define a map $f_k : \Delta^m = |K| \to \mathbb{R}^{2n+1} \setminus H_k$ as follows:

$$f_k(x) = \begin{cases} \displaystyle\sum_{v\in K^{(0)}\setminus L^{(0)}} \beta_v^K(x)\, p_v & \text{if } \alpha(x) = 0, \\ \alpha(x)\, f\big(\sum_{v\in L^{(0)}} \alpha(x)^{-1}\beta_v^K(x)v\big) + \displaystyle\sum_{v\in K^{(0)}\setminus L^{(0)}} \beta_v^K(x)\, p_v & \text{if } \alpha(x) > 0, \end{cases}$$

where $\alpha(x) = \sum_{v\in L^{(0)}} \beta_v^K(x)$. Then, $f_k|\partial\Delta^m = f_k||L| = f$. In the above definition, replacing each p_v with $f_{k-1}(v)$, we define the following auxiliary point $y_x \in \mathbb{R}^{2n+1}$:

$$y_x = \begin{cases} \displaystyle\sum_{v \in K^{(0)} \setminus L^{(0)}} \beta_v^K(x) f_{k-1}(v) & \text{if } \alpha(x) = 0, \\ \alpha(x) f\Big(\sum_{v \in L^{(0)}} \alpha(x)^{-1} \beta_v^K(x) v \Big) + \displaystyle\sum_{v \in K^{(0)} \setminus L^{(0)}} \beta_v^K(x) f_{k-1}(v) & \text{if } \alpha(x) > 0. \end{cases}$$

Since $\|p_v - f_{k-1}(v)\|_\infty < \varepsilon_k/4$ for each $v \in K^{(0)} \setminus L^{(0)}$, it follows that

$$\|f_k(x) - y_x\|_\infty \leqslant \sum_{v \in K^{(0)} \setminus L^{(0)}} \beta_v^K(x) \|p_v - f_{k-1}(v)\|_\infty < \varepsilon_k/4.$$

Moreover, we can also show that $\|y_x - f_{k-1}(x)\|_\infty < \varepsilon_k/4$. When $\alpha(x) > 0$, since $\sum_{v \in K^{(0)} \setminus L^{(0)}} \beta_v^K(x) = 1 - \alpha(x)$, it follows that

$$\begin{aligned} \|y_x - f_{k-1}(x)\|_\infty &\leqslant \alpha(x) \Big\| f\Big(\textstyle\sum_{v \in L^{(0)}} \alpha(x)^{-1} \beta_v^K(x) v \Big) - f_{k-1}(x) \Big\|_\infty \\ &\quad + \sum_{v \in \sigma^{(0)} \setminus \sigma_L^{(0)}} \beta_v^K(x) \|f_{k-1}(v) - f_{k-1}(x)\|_\infty \\ &< \alpha(x)\varepsilon_k/4 + (1 - \alpha(x))\varepsilon_k/4 = \varepsilon_k/4. \end{aligned}$$

When $\alpha(x) = 0$, remove the first terms of the right-hand sides of $\leqslant$ and $<$ from the above inequality. Consequently, $\|f_k(x) - f_{k-1}(x)\|_\infty < \varepsilon_k/2$ for each $x \in |K|$. Thus, we have $d(f_k, f_{k-1}) \leqslant \varepsilon_k/2$. This completes the induction.

The Case of μ^n: We use the following notations introduced before Theorem 5.10.8. For each $k \in \mathbb{N}$, let

$$Q_k = \big\{ (m/3^{k-1} + 1/3^{k-1}2 \;\big|\; m = 0, \ldots, 3^{k-1} - 1 \big\},$$

which are the mid-points of components of R_k. We define

$$\begin{aligned} L_k &= \big\{ x \in \mathbf{I}^{2n+1} \;\big|\; x(i) \in Q_k \;\text{ for at least } n+1 \; i \big\} \\ &= \big\{ x \in \mathbf{I}^{2n+1} \;\big|\; x(i) \in \mathbf{I} \setminus Q_k \;\text{ for at most } n \; i \big\}. \end{aligned}$$

As is easily observed, L_k is contained in a union of finitely many n-dimensional flats in $\mathbb{R}^{2n+1}$. Let K_k be the cell complex consisting of all faces of cells

$$\prod_{i=1}^{2n+1} \left[\frac{m_i - 1}{3^k}, \frac{m_i}{3^k} \right], \; m_i = 1, \ldots, 3^k.$$

Then, $|K_k| = \mathbf{I}^{2n+1}$, mesh $K_k = 1/3^k$, and $M_k^{2n+1} = |C(L_k, K_k)|$.

For each cell $C \in K_k$, choose $v_C \in \operatorname{rint} C$ so that $v_C(i)$ is the midpoint of $\operatorname{pr}_i(C)$ ($\{v_C(i)\} = \operatorname{pr}_i(C)$ if $\operatorname{pr}_i(C)$ is a singleton). Using these points v_C, $C \in K_k$, we apply Proposition 4.3.11 to obtain a derived subdivision $K_k' \lhd K_k$. Then, mesh $K_k' = 1/3^k 2$ and L_k, M_k^{2n+1}, and $\bigcap_{i \leqslant k} M_i^{2n+1}$ are triangulated by full subcomplexes of K_k'. In

particular, $L_k = |C(M_k^{2n+1}, K_k')|$. Moreover,

$$C(\textstyle\bigcap_{i\leqslant k} M_i^{2n+1}, K_k') = C(\bigcap_{i<k} M_i^{2n+1}, K_k') \cup C(M_k^{2n+1}, K_k'),$$
$$\text{equiv.,}\ \ K_k'[\textstyle\bigcap_{i<k} M_i^{2n+1}] \cap K_k'[M_k^{2n+1}] = K_k'[\bigcap_{i\leqslant k} M_i^{2n+1}].$$

Indeed, $\supset$ is obvious. To see $\subset$, let

$$\sigma = \langle v_{C_1}, \dots, v_{C_\ell}\rangle \in K_k'[\textstyle\bigcap_{i<k} M_i^{2n+1}] \cap K_k'[M_k^{2n+1}],$$

where $C_1 < \cdots < C_\ell \in K_k$. Since $\sigma \cap \bigcap_{i<k} M_i^{2n+1} \neq \emptyset$, some v_{C_j} is contained in $\bigcap_{i<k} M_i^{2n+1}$, which means $C_j \subset \bigcap_{i<k} M_i^{2n+1}$. Hence, $C_1 \subset \bigcap_{i<k} M_i^{2n+1}$, which means $v_{C_1} \in \bigcap_{i<k} M_i^{2n+1}$. Similarly, it follows that $v_{C_1} \in M_k^{2n+1}$. Hence, $v_{C_1} \in \bigcap_{i\leqslant k} M_i^{2n+1}$, which implies $\sigma \cap M_k^{2n+1} \neq \emptyset$, i.e., $\sigma \in K_k'[\bigcap_{i\leqslant k} M_i^{2n+1}]$.

By Proposition 4.7.6, we have a retraction

$$r_k : |K_k'| \setminus |C(\textstyle\bigcap_{i\leqslant k} M_i^{2n+1}, K_k')| \to \bigcap_{i\leqslant k} M_i^{2n+1}$$

such that $r_k(x) \in c_{K_k'}(x)$ for each $x \in |K_k'| \setminus |C(\bigcap_{i\leqslant k} M_i^{2n+1}, K_k')|$, which implies

$$d(r_k, \mathrm{id}) \leqslant \operatorname{mesh} K_k' = 1/3^k 2.$$

When $\delta \geqslant 1/6$, let $k_0 = 0$ and $M_0^{2n+1} = \mathbf{I}^{2n+1}$. Then, $f_{k_0} = f_0 : \Delta^m \to M_0^{2n+1}$. In this case, it is not necessary to care about estimation of $d(f, f_0)$ because $6\delta > 1 = \operatorname{diam} \mathbf{I}^{2n+1}$. When $\delta < 1/6$, we can take $k_0 \in \mathbb{N}$ so that $1/3^{k_0+1}2 \leqslant \delta < 1/3^{k_0}2$. Note that $\bigcap_{i\leqslant k_0} M_i^{2n+1}$ is the polyhedron of a subcomplex of the cell complex K_{k_0}, which is the union of cubes $\prod_{i=1}^{2n+1}[(m_i - 1)/3^{k_0}, m_i/3^{k_0}]$. For each vertex $v \in K_{k_0}^{(0)}$,

$$|\operatorname{St}(v, K_{k_0}')| = \left(v + \prod_{i=1}^{2n+1}\left[\frac{-1}{3^{k_0}2}, \frac{1}{3^{k_0}2}\right]\right) \cap \mathbf{I}^{2n+1} = \overline{\mathrm{B}}(v, 1/3^{k_0}2) \cap \mathbf{I}^{2n+1}.$$

Since $K_{k_0}' = \bigcup_{v\in K_{k_0}^{(0)}} \operatorname{St}(v, K_{k_0}')$, it follows that

$$\left|N\left(\textstyle\bigcap_{i\leqslant k_0} M_i^{2n+1}, K_{k_0}'\right)\right| = \overline{\mathrm{N}}\left(\bigcap_{i\leqslant k_0} M_i^{2n+1}, 1/3^{k_0}2\right) \text{ and}$$
$$\left|C\left(\textstyle\bigcap_{i\leqslant k_0} M_i^{2n+1}, K_{k_0}'\right)\right| = \mathbf{I}^{2n+1} \setminus \mathrm{N}\left(\bigcap_{i\leqslant k_0} M_i^{2n+1}, 1/3^{k_0}2\right),$$

which implies that

$$f_0(\Delta^m) \subset \left|K_{k_0}'\right| \setminus \left|C\left(\textstyle\bigcap_{i\leqslant k_0} M_i^{2n+1}, K_{k_0}'\right)\right|.$$

Then, we have a map $f_{k_0} = r_{k_0} f_0 : \Delta^m \to \bigcap_{i\leqslant k_0} M_i^{2n+1}$, where

$$f_{k_0}|\partial\Delta^m = f_0|\partial\Delta^m = f \text{ and } d(f_{k_0}, f_0) \leqslant d(r_{k_0}, \mathrm{id}) \leqslant 1/3^{k_0}2 \leqslant 3\delta.$$

By induction, we will construct maps $f_k : \Delta^m \to \bigcap_{i\leqslant k} M_i^{2n+1}$, $k > k_0$, so that $f_k|\partial\Delta^m = f$ and $d(f_k, f_{k-1}) < 1/3^k$. Then, $(f_k)_{k>k_0}$ uniformly converges to a map $\bar{f} : \Delta^m \to \bigcap_{i\leqslant k_0} M_i^{2n+1}$, where $\bar{f}(\Delta^m) \subset \bigcap_{i\leqslant k} M_i^{2n+1}$ for each $k > k_0$ because $\bar{f}$ is a uniform limit of $(f_i)_{i\geqslant k}$. Hence, we can obtain a map $\bar{f} : \Delta^m \to \mu^n = \bigcap_{k\in\mathbb{N}} M_k^{2n+1}$ such that $\bar{f}|\partial\Delta^m = f$ and $d(\bar{f}, f_{k_0}) \leqslant \sum_{k>k_0} 1/3^k = 1/3^{k_0}2 \leqslant 3\delta$. Then, it follows that

$$d(\bar{f}, f_0) \leqslant d(\bar{f}, f_{k_0}) + d(f_{k_0}, f_0) \leqslant 6\delta.$$

Now, assume that $f_{k-1} : \Delta^m \to \bigcap_{i<k} M_i^{2n+1}$ has been defined, where $k > k_0$. Since L_k is contained in a finite union of n-dimensional flats in $\mathbb{R}^{2n+1}$, in the same way as the case of ν^n, we can apply the General Position Lemma 4.14.1 to obtain a map $f'_k : \Delta^m \to \mathbf{I}^{2n+1} \setminus L_k$ such that $f'_k|\partial\Delta^m = f$ and $d(f'_k, f_{k-1}) < 1/3^k 2$. As observed in the case of K'_{k_0}, we have

$$\big|C\big(\textstyle\bigcap_{i<k} M_i^{2n+1}, K'_k\big)\big| = \mathbf{I}^{2n+1} \setminus \mathrm{N}\big(\textstyle\bigcap_{i<k} M_i^{2n+1}, 1/3^k 2\big).$$

Hence, it follows that

$$\begin{aligned} f'_k(\Delta^m) &\subset |K'_k| \setminus \big(|C(\textstyle\bigcap_{i<k} M_i^{2n+1}, K'_k)| \cup L_k\big) \\ &= |K'_k| \setminus \big(|C(\textstyle\bigcap_{i<k} M_i^{2n+1}, K'_k)| \cup |C(M_k^{2n+1}, K'_k)|\big) \\ &= |K'_k| \setminus \big(|C(\textstyle\bigcap_{i\leqslant k} M_i^{2n+1}, K'_k)|. \end{aligned}$$

Thus, we have a map $f_k = r_k f'_k : \Delta^m \to \bigcap_{i\leqslant k} M_i^{2n+1}$, where $f_k|\partial\Delta^m = f'_k|\partial\Delta^m = f$ and

$$d(f_k, f_{k-1}) \leqslant d(r_k, \mathrm{id}) + d(f'_k, f_{k-1}) < 1/3^k 2 + 1/3^k 2 = 1/3^k.$$

This completes the induction. □

6.12 Finite-Dimensional ANRs

A space Y is called an **absolute neighborhood extensor for metrizable spaces of dimension $\leqslant n$** (or an **ANE(n)**) if each map $f : A \to Y$ from any closed set A in a metrizable space X of $\dim X \leqslant n$ extends over some neighborhood U of A in X. When f can always be extended over X (i.e., $U = X$ in the above), we call Y an **absolute extensor for metrizable spaces of dimension $\leqslant n$** (or an **AE(n)**). In this section, we show that the local n-connectedness characterizes ANE$(n+1)$s and n-dimensional ANRs.

For a metrizable space X, $\dim X \leqslant n$ if and only if each open cover $\mathcal{U}$ of X has an open refinement $\mathcal{V}$ with $\operatorname{ord}\mathcal{V} \leqslant n+1$.[17] We use the following facts of Dimension Theory:

(i) If $\dim X \leqslant n$, then $\dim X \times \mathbf{I} \leqslant n+1$ — the Product Theorem (5.4.15).
(ii) For any subset A of a metrizable space X, $\dim A \leqslant \dim X$—the Subset Theorem (5.4.4).

In Lemma 6.1.2, given a proper closed set $A \neq \emptyset$ in a metric space $X = (X, d)$, we obtained a Dugundji system $(a_U)_{U\in\mathcal{U}}$, i.e., $\mathcal{U}$ is a locally finite open cover of $X \setminus A$ with $a_U \in A$, $U \in \mathcal{U}$, such that $x \in U \in \mathcal{U}$ implies $d(x, a_U) \leqslant 2d(x, A)$. If $\dim X \setminus A \leqslant n+1$, we can take $\mathcal{U}$ with the additional condition that $\operatorname{ord}\mathcal{U} \leqslant n+2$, i.e., $\dim N(\mathcal{U}) \leqslant n+1$, where $N(\mathcal{U})$ is the nerve of $\mathcal{U}$. Thus, we have the following variant of Lemma 6.1.2:

Lemma 6.12.1 *Let $X = (X, d)$ be a metric space and $A \neq \emptyset$ be a proper closed set in X such that $\dim X \setminus A \leqslant n+1$. Then, there exists a Dugundji system $(a_U)_{U\in\mathcal{U}}$ for $A \subset X$ such that $\dim N(\mathcal{U}) \leqslant n+1$.* □

An open refinement $\mathcal{V}$ of $\mathcal{U} \in \operatorname{cov}(X)$ is called a **C^n-refinement** of $\mathcal{U}$ if each $V \in \mathcal{V}$ is contained in some $U \in \mathcal{U}$ such that any map $f : \mathbf{S}^i \to V$, $i \leqslant n$, extends to a map $\tilde{f} : \mathbf{B}^{i+1} \to U$. We call $\mathcal{V}$ an **n-Lefschetz refinement** of $\mathcal{U}$ if any partial $\mathcal{V}$-realization of an arbitrary simplicial complex with $\dim \leqslant n$ in X extends to full $\mathcal{U}$-realization in X. We write

$$\mathcal{V} \underset{C^n}{\prec} \mathcal{U} \text{ or } \mathcal{U} \underset{C^n}{\succ} \mathcal{V} \quad \left(\text{resp. } \mathcal{V} \underset{L^n}{\prec} \mathcal{U} \text{ or } \mathcal{U} \underset{L^n}{\succ} \mathcal{V}\right)$$

when $\mathcal{V}$ is a C^n-refinement (resp. an n-Lefschetz refinement) of $\mathcal{U}$.

Lemma 6.12.2 *For an LC^n paracompact space Y, each open cover $\mathcal{U}$ of Y has an $(n+1)$-Lefschetz refinement.*

Proof Since Y is LC^n, $\mathcal{U}$ has the following open refinements:

$$\mathcal{U} = \mathcal{V}_{n+1} \overset{*}{\succ} \mathcal{U}_n \underset{C^n}{\succ} \mathcal{V}_n \overset{*}{\succ} \mathcal{U}_{n-1} \underset{C^{n-1}}{\succ} \mathcal{V}_{n-1} \overset{*}{\succ} \cdots \overset{*}{\succ} \mathcal{U}_0 \underset{C^0}{\succ} \mathcal{V}_0.$$

Let K be a simplicial complex with $\dim K \leqslant n+1$, L be a subcomplex of K with $K^{(0)} \subset L$, and $f : |L| \to Y$ be a partial $\mathcal{V}_0$-realization of K. By induction, we can define partial $\mathcal{V}_i$-realizations $f_i : |L \cup K^{(i)}| \to Y$ of K such that $f_i||L \cup K^{(i-1)}| = f_{i-1}$, where $f_0 = f$. Indeed, assume that f_{i-1} has been defined. For each $\sigma \in K^{(i)} \setminus (L \cup K^{(i-1)})$, choose $V_\sigma \in \mathcal{V}_{i-1}$ so that $f_{i-1}(\partial\sigma) \subset V_\sigma$. Since $\mathcal{V}_{i-1}$ is a C^{i-1}-refinement of $\mathcal{U}_{i-1}$, $f_{i-1}|\partial\sigma$ extends to a map $f_\sigma : \sigma \to U_\sigma$ for some $U_\sigma \in \mathcal{U}_{i-1}$. Thus, f_i can be defined by $f_i|\sigma = f_\sigma$ for $\sigma \in K^{(i)} \setminus (L \cup K^{(i-1)})$. For

[17] In the definition of dimension, $\mathcal{U}$ and $\mathcal{V}$ are required to be *finite*. However, by Theorem 5.2.5, this requirement is not necessary for paracompact spaces (so metrizable spaces).

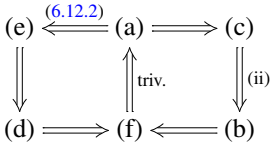
each $\sigma \in K$, we can write $\sigma \cap |L \cup K^{(i)}| = \bigcup_{j=1}^{k} \sigma_j$, where $\sigma_j \in L \cup K^{(i)}$. Since each $f_i(\sigma_j)$ is contained in some $U_j \in \mathcal{U}_{i-1}$ and $f(\sigma^{(0)})$ is contained in some $U_0 \in \mathcal{U}_{i-1}$, it follows that $f_i(\sigma \cap |L \cup K^{(i)}|) \subset \mathrm{st}(U_0, \mathcal{U}_{i-1})$. Since $\mathcal{U}_i$ is a star-refinement of $\mathcal{V}_i$, f_i is a partial $\mathcal{V}_i$-realization of K. Note that f_{n+1} is a full $\mathcal{U}$-realization of K. Therefore, $\mathcal{V}_0$ is an $(n+1)$-Lefschetz refinement of $\mathcal{U}$. □

Now, we prove the following characterization of ANE(n)s:

Theorem 6.12.3 *Let $n \in \omega$. For a metrizable space Y, the following are equivalent:*

(a) *Y is LC^n;*
(b) *Y is an ANE$(n+1)$;*
(c) *If A is a closed set in a metrizable space X with $\dim X \setminus A \leqslant n+1$, then every map $f : A \to Y$ extends over a neighborhood of A in X;*
(d) *If A is a closed set in a metrizable space X with $\dim A \leqslant n$, then every map $f : A \to Y$ extends over a neighborhood of A in X;*
(e) *Each open cover $\mathcal{U}$ of Y has an $(n+1)$-Lefschetz refinement;*
(f) *Each neighborhood U of any $y \in Y$ contains a neighborhood V of y in Y such that every map $f : X \to V$ of a metrizable space X with $\dim X \leqslant n$ is null-homotopic in U.*

Proof The implication (a) ⇒ (e) is Lemma 6.12.2. The implication (f) ⇒ (a) is trivial and (c) ⇒ (b) follows from (ii). We show the implications (a) ⇒ (c), (b) ⇒ (f), and (e) ⇒ (d) ⇒ (f).



(a) ⇒ (c): We may assume that X and Y are metric spaces. For simplicity, let d stand for metrics of both X and Y. By (a), we can take open covers of Y as follows:

$$\mathcal{V}_0 = \{Y\} \overset{*}{\succ} \mathcal{U}_1 \underset{C^n}{\succ} \mathcal{V}_1 \overset{*}{\succ} \mathcal{U}_2 \underset{C^n}{\succ} \mathcal{V}_2 \overset{*}{\succ} \cdots$$

and mesh $\mathcal{V}_i < 2^{-i}$ for each $i \in \mathbb{N}$. By Lemma 6.12.1, we have a Dugundji system $(a_W)_{W \in \mathcal{W}}$ for $A \subset X$ such that $\dim N(\mathcal{W}) \leqslant n+1$. Let $\varphi : X \setminus A \to |N(\mathcal{W})|$ be a canonical map, i.e., $\varphi(x) \in \langle \mathcal{W}[x] \rangle$ for each $x \in X \setminus A$ (cf. Proposition 4.15.2). For each $\sigma \in N(\mathcal{W})$, let $A(\sigma) = \{a_W \mid W \in \sigma^{(0)}\} \subset A$. Let K_0 be the subcomplex of $N(\mathcal{W})$ consisting of all simplexes $\sigma \in N(\mathcal{W})$ such that $f(A(\sigma))$ is a singleton. Obviously, $K_0^{(0)} = N(\mathcal{W})^{(0)} = \mathcal{W}$. For each $\sigma \in N(\mathcal{W}) \setminus K_0$, let

$$k(\sigma) = \max \left\{ i \in \omega \;\middle|\; f(A(\sigma)) \subset V \text{ for some } V \in \mathcal{V}_i \right\},$$

and choose $V_0(\sigma) \in \mathcal{V}_{k(\sigma)}$ so that $f(A(\sigma)) \subset V_0(\sigma)$. Observe that if $\sigma \in N(\mathcal{W}) \setminus K_0$ and $\sigma \prec \tau$, then $\tau \in N(\mathcal{W}) \setminus K_0$ and $k(\sigma) \geqslant k(\tau)$. We define a subcomplex $K \subset N(\mathcal{W})$ as follows:

$$K = K_0 \cup \left\{\sigma \in N(\mathcal{W}) \;\middle|\; k(\sigma) \geqslant n+1\right\}.$$

For each $i \in \mathbb{N}$, let $K_i = K_0 \cup K^{(i)}$. Note that $K = K_{n+1}$. Let $g_0 : |K_0| \to Y$ be the map defined by $g_0(\sigma) = f(A(\sigma))$ for each $\sigma \in K_0$. Then, observe that

$$g_0(\sigma \cap |K_0|) = f(A(\sigma)) \subset V_0(\sigma) \in \mathcal{V}_{k(\sigma)} \text{ for each } \sigma \in K \setminus K_0.$$

For each 1-simplex $\sigma \in K_1 \setminus K_0$, we have $U_0(\sigma) \in \mathcal{U}_{k(\sigma)}$ such that $g_0|\partial\sigma = g_0|\sigma^{(0)}$ extends to a map $g_\sigma : \sigma \to U_0(\sigma)$. Then, we can extend g_0 to a map $g_1 : |K_1| \to Y$ defined by $g_1|\sigma = g_\sigma$ for $\sigma \in K_1 \setminus K_0$. For each $\sigma \in K \setminus K_0$,

$$g_1(\sigma \cap |K_1|) \subset \operatorname{st}(V_0(\sigma), \mathcal{U}_{k(\sigma)}) \in \operatorname{st}(\mathcal{V}_{k(\sigma)}, \mathcal{U}_{k(\sigma)}) \prec \operatorname{st}\mathcal{U}_{k(\sigma)} \prec \mathcal{V}_{k(\sigma)-1},$$

hence $g_1(\sigma \cap |K_1|)$ is contained in some $V_1(\sigma) \in \mathcal{V}_{k(\sigma)-1}$. Next, for each 2-simplex $\sigma \in K_2 \setminus K_1$, we have $U_1(\sigma) \in \mathcal{U}_{k(\sigma)-1}$ such that $g_1|\partial\sigma$ extends to a map $g_\sigma : \sigma \to U_1(\sigma)$. Then, as above, we can extend g_1 to a map $g_2 : |K_2| \to Y$ such that, for each $\sigma \in K \setminus K_0$, $g_2(\sigma \cap |K_2|)$ is contained in some $V_2(\sigma) \in \mathcal{V}_{k(\sigma)-2}$. We continue this process $n+1$ times to obtain a map $g = g_{n+1} : |K| = |K_{n+1}| \to Y$ such that, for each $\sigma \in K \setminus K_0 = K_{n+1} \setminus K_0$, $g(\sigma)$ is contained in some $V_{n+1}(\sigma) \in \mathcal{V}_{k(\sigma)-n-1}$.

Now, we define $\tilde{f} : A \cup \varphi^{-1}(|K|) \to Y$ by $\tilde{f}|A = f$ and $\tilde{f}|\varphi^{-1}(|K|) = g\varphi|\varphi^{-1}(|K|)$. It remains to be shown that $A \cup \varphi^{-1}(|K|)$ is a neighborhood of A in X and $\tilde{f}$ is continuous. To this end, it suffices to prove that $A \cup \varphi^{-1}(|K|)$ is a neighborhood of each $a \in \operatorname{bd}_X A$ in X and $\tilde{f}$ is continuous at each $a \in \operatorname{bd}_X A$. We will prove these claims at the same time. For each $a \in \operatorname{bd}_X A$ and $\varepsilon > 0$, choose $i \geqslant n+1$ and $V \in \mathcal{V}_i$ so that $2^{-i}(1+2^{n+1}) < \varepsilon$ and $f(a) \in V$. Then, we have $\delta > 0$ such that $f(A \cap \mathrm{B}(a,\delta)) \subset V$. For $x \in X \setminus A$ with $d(x,a) < \delta/3$, let $\sigma_x \in N(\mathcal{W})$ be the carrier of $\varphi(x)$. For every $W \in \sigma_x^{(0)}$, since $x \in W \in \mathcal{W}$, we have $d(x, a_W) \leqslant 2d(x, A) \leqslant 2d(x, a)$, hence

$$d(a, a_W) \leqslant d(x,a) + d(x, a_W) \leqslant 3d(x,a) < \delta.$$

Then, $f(A(\sigma_x)) \subset V \in \mathcal{V}_i$, which implies that $k(\sigma_x) \geqslant i \geqslant n+1$ if $\sigma_x \notin K_0$, hence $\sigma_x \in K$. Therefore, $\varphi(x) \in |K|$. Thus, $A \cup \varphi^{-1}(|K|)$ is a neighborhood of a in X. On the other hand, when $\sigma_x \notin K_0$, since

$$f(a) \in V,\ g(\sigma_x^{(0)}) = f(A(\sigma_x)) \subset V, \text{ and } g(\sigma_x) \subset V_{n+1}(\sigma_x) \in \mathcal{V}_{k(\sigma_x)-n-1},$$

it follows that

$$d(f(a), g\varphi(x)) \leqslant \operatorname{diam} V + \operatorname{diam} V_{n+1}(\sigma_x) < 2^{-i} + 2^{-k(\sigma_x)+n+1} \leqslant 2^{-i}(1 + 2^{n+1}) < \varepsilon.$$

When $\sigma_x \in K_0$, since $g(\sigma_x) = g_0(\sigma_x) = f(A(\sigma_x))$ is a singleton, it follows that

$$d(f(a), g\varphi(x)) \leqslant \operatorname{diam} V < 2^{-i} < \varepsilon.$$

Therefore, $\tilde{f}$ is continuous at $a \in \mathrm{bd}_X A$. The continuity of $\tilde{f}$ on $\mathrm{int}_X A \cup \varphi^{-1}(|K|)$ is obvious. Thus, f is extended over a neighborhood of A in X.

(e) $\Rightarrow$ (d): Given $d \in \mathrm{Metr}(Y)$, we take open covers of Y as follows:

$$\mathcal{U}_1 \underset{L^{n+1}}{\succ} \mathcal{V}_1 \overset{*}{\succ} \mathcal{U}_2 \underset{L^{n+1}}{\succ} \mathcal{V}_2 \overset{*}{\succ} \mathcal{U}_3 \underset{L^{n+1}}{\succ} \mathcal{V}_3 \overset{*}{\succ} \cdots$$

and $\operatorname{mesh} \mathcal{U}_i < 2^{-i-1}$ for each $i \in \mathbb{N}$. Since $\dim A \leqslant n$, A has open covers $\mathcal{W}_i$, $i \in \mathbb{N}$, such that $\operatorname{ord} \mathcal{W}_i \leqslant n+1$ and $\mathcal{W}_i \prec f^{-1}(\mathcal{U}_{i+2}) \wedge \mathcal{W}_{i-1}$, where $\mathcal{W}_0 = \{A\}$. For each $i \in \mathbb{N}$, let K_i be the nerve of $\mathcal{W}_i$ with $\varphi_i : A \to |K_i|$ a canonical map. Then, $\dim K_i \leqslant n$. Extending a partial $\mathcal{V}_{i+1}$-realization of K_i defined on $K_i^{(0)}$, we can obtain a full $\mathcal{U}_{i+1}$-realization $p_i : |K_i| \to Y$ such that

$$p_i(W) \in f(W) \text{ for each } W \in K_i^{(0)} = \mathcal{W}_i.$$

Since $\mathcal{W}_{i+1} \prec \mathcal{W}_i$, there is a simplicial map $\psi_i : K_{i+1} \to K_i$ such that

$$W \subset \psi_i(W) \in K_i^{(0)} = \mathcal{W}_i \text{ for each } W \in K_{i+1}^{(0)} = \mathcal{W}_{i+1}.$$

Choosing an order on $K_{i+1}^{(0)}$ so that K_{i+1} is an ordered simplicial complex, we have the simplicial mapping cylinder Z_{ψ_i} of ψ_i. Then, $\dim Z_{\psi_i} \leqslant n+1$ and $Z_{\psi_i}^{(0)} \subset K_i \oplus K_{i+1}$. Observe that $p_i \oplus p_{i+1} : |K_i| \oplus |K_{i+1}| \to Y$ is a partial $\mathcal{V}_i$-realization of Z_{ψ_i}. Then, we have a full $\mathcal{U}_i$-realization $g_i : |Z_{\psi_i}| \to Y$ such that $g_i||K_i| = p_i$ and $g_i||K_{i+1}| = p_{i+1}$. On the other hand, we also have a map $h_i : A \times \mathbf{I} \to |Z_{\psi_i}|$ such that $h_i(a, 0) = \varphi_i(a)$ and $h_i(a, 1) = \varphi_{i+1}(a)$. Indeed, φ_i and $\psi_i\varphi_{i+1}$ are canonical maps for the cover $\mathcal{W}_i$, hence they are contiguous (Corollary 4.15.3) and h_i can be defined as follows:

$$h_i(a,t) = \begin{cases} \bar{q}_{\psi_i}(\varphi_{i+1}(a), 2t-1) & \text{if } t \geqslant 1/2, \\ (1-2t)\varphi_i(a) + 2t\psi_i\varphi_{i+1}(a) & \text{if } t \leqslant 1/2, \end{cases}$$

where $\bar{q}_{\psi_i} : |K_i| \oplus |K_{i+1} \times_s I| \to |Z_{\psi_i}|$ is the quotient map (Fig. 6.20). For each $(a, t) \in A \times [0, 1)$, the carrier of $h_i(a, t)$ in Z_{ψ_i} has a vertex $W \in \mathcal{W}_i[a]$. Then,

$$g_i(W) = p_i(W) \in f(W) \subset \mathrm{st}(f(a), \mathcal{U}_{i+2}).$$

Since g_i is a full $\mathcal{U}_i$-realization of Z_{ψ_i}, it follows that $g_i h_i(a, t) \in \mathrm{st}(f(a), \mathcal{V}_{i-1})$.

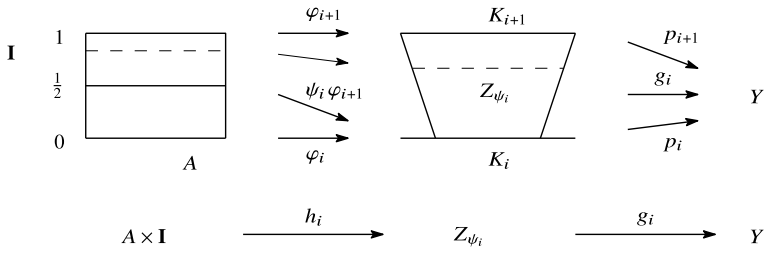


Fig. 6.20 The maps h_i and g_i

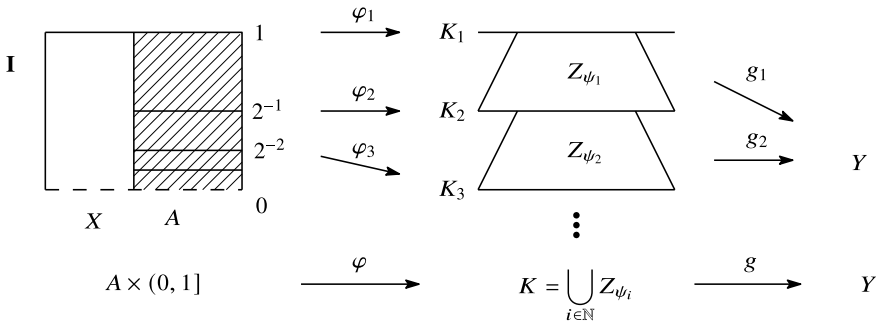


Fig. 6.21 The maps φ and g

Now, we have a simplicial complex $K = \bigcup_{i\in\mathbb{N}} Z_{\psi_i}$ and a map $g : |K| \to Y$ defined by $g||Z_{\psi_i}| = g_i$ for each $i \in \mathbb{N}$. We also define a map $\varphi : A \times (0, 1] \to |K|$ as follows:

$$\varphi(a, t) = h_i(a, 2 - 2^i t) \ \text{ for } 2^{-i} \leqslant t \leqslant 2^{-i+1}.$$

See Fig. 6.21. Since $|K|$ is an ANE, the map $g\varphi : A \times (0, 1] \to Y$ extends over a neighborhood of $A \times (0, 1]$ in $X \times (0, 1]$. On the other hand, for each $a \in A$ and $2^{-i} < t \leqslant 2^{-i+1}$,

$$g\varphi(a, t) = g_i h_i(a, 2 - 2^i t) \in \mathrm{st}(f(a), \mathcal{U}_{i-1}),$$

which implies that $d(g\varphi(a, t), f(a)) < 2^{-i} < t$. Hence, $g\varphi$ can be extended over $A \times \mathbf{I}$ by f. Thus, we can apply Lemma 6.5.1 to extend f over a neighborhood of A in X.

(b) (or (d)) $\Rightarrow$ (f): Let $d \in \mathrm{Metr}(Y)$ and assume that condition (f) does not hold. Then, we have a point $y_0 \in Y$, an open neighborhood U of y_0 in Y and maps $f_i : X_i \to U$, $i \in \mathbb{N}$, such that each X_i is a metrizable space, $\dim X_i \leqslant n$, $f_i(X_i) \subset \mathrm{B}(y_0, 2^{-i})$, and each f_i is not null-homotopic. Take $v_0 \notin \bigoplus_{i\in\mathbb{N}} X_i \times \mathbf{I}$ and let

$$X = \{v_0\} \cup \bigoplus_{i\in\mathbb{N}} X_i \times \mathbf{I} \ \text{ and } \ A = \{v_0\} \cup \bigoplus_{i\in\mathbb{N}} X_i \times \{0, 1\},$$

where X admits the topology generated by the open sets in $X_i \times \mathbf{I}$ and $\{v_0\} \cup \bigoplus_{j \geqslant i} X_j \times \mathbf{I}$, $i \in \mathbb{N}$. Then, X is metrizable by the Bing Metrization Theorem 2.3.4 and A is closed in X. By facts (i) and (ii) mentioned at the beginning of this section, $\dim X_i \times \mathbf{I} \leqslant n+1$ and $\dim X_i \times \{0,1\} \leqslant n$ for every $i \in \mathbb{N}$. By definition, it is easy to show that $\dim X \leqslant n+1$ and $\dim A \leqslant n$. We define a map $f : A \to U$ by $f(v_0) = y_0$, $f(x,0) = f_i(x)$, and $f(x,1) = y_0$ for $x \in X_i$. Then, f extends over a neighborhood W of A in X by (b) (or (d)). Note that W contains some $X_i \times \mathbf{I}$ that maps into U by the extension of f. This means that f_i is null-homotopic in U, which is a contradiction. □

Remark 6.12.1 *The following conditions can also be added to the list of Theorem* 6.12.3, as variants of (a) (or (f)) corresponding to (b), (c), and (d).

(b′) *Each neighborhood U of any point y in Y contains a neighborhood V of y in Y such that every map $f : A \to V$ defined in a closed set A in a metrizable space X with* $\dim X \leqslant n+1$ *extends to a map $\tilde{f} : X \to U$;*

(c′) *Each neighborhood U of any point y in Y contains a neighborhood V of y in Y such that every map $f : A \to V$ defined in a closed set A in a metrizable space X with* $\dim X \setminus A \leqslant n+1$ *extends to a map $\tilde{f} : X \to U$;*

(d′) *Each neighborhood U of any point y in Y contains a neighborhood V of y in Y such that every map $f : A \to V$ defined in a closed set A in a metrizable space X with* $\dim A \leqslant n$ *extends to a map $\tilde{f} : X \to U$.*

Sketch of Proof For $y \in V \subset U$ in each of these conditions, given a map $f : X \to V$ of a metrizable space X with $\dim X \leqslant n$, let $h : X \times \{0,1\} \to V$ be the map defined by $h(x,0) = f(x)$ and $h(x,1) = y$ for $x \in X$. Then, h can be extended to a homotopy giving $f \simeq 0$ in U. Thus, each of (b′), (c′), and (d′) implies condition (f) in Theorem 6.12.3.

The proofs of the implications (b) ⇒ (b′), (c) ⇒ (c′), and (d) ⇒ (d′) are similar to the proof of (b) (or (d)) ⇒ (f) in Theorem 6.12.3.

As in Proposition 6.6.11, we can prove the following lemma:

Lemma 6.12.4 *Let K be a simplicial complex with* $\dim K \leqslant n+1$ *and a subcomplex L of K. If X is n-connected then any map $f : |L| \to X$ extends over $|K|$.* □

Using this lemma and Theorem 6.12.3, we can obtain the following characterization of AE(n)s:

Theorem 6.12.5 *Let $n \in \omega$. For a metrizable space Y, the following are equivalent:*

(a) *Y is C^n and LC^n;*

(b) *Y is an $AE(n+1)$;*

(c) *If A is a closed set in a metrizable space X with* $\dim X \setminus A \leqslant n+1$, *then every map $f : A \to Y$ extends over X;*

(d) *If A is a closed set in a metrizable space X with* $\dim A \leqslant n$, *then every map $f : A \to Y$ extends over X.*

Proof The implication (c) $\Rightarrow$ (b) follows from (ii). Since each of conditions (b) and (d) implies that Y is n-connected, the implications (b) $\Rightarrow$ (a) and (d) $\Rightarrow$ (a) can be obtained by Theorem 6.12.3.

In the proof of Theorem 6.12.3 (a) $\Rightarrow$ (c), if Y is n-connected, the map $g : |K| \to Y$ can be extended over $|N(\mathcal{W})|$ by Lemma 6.12.4, hence $\tilde{f}$ can be defined over X. Thus, the implication (a) $\Rightarrow$ (c) of this theorem also holds.

In the proof of Theorem 6.12.3 (e) $\Rightarrow$ (d), we may assume that $\operatorname{diam} Y < 2^{-4}$. If Y is n-connected, we can take $\mathcal{U}_i = \mathcal{V}_i = \{Y\}$ for $i = 1, 2, 3$, so we can also take $\mathcal{W}_1 = \{A\}$. Now, since $\varphi|A \times \{1\}$ is a constant map and $|K|$ is an ANE, $g\varphi$ extends over a neighborhood of $A \times (0, 1] \cup X \times \{1\}$ in $X \times (0, 1]$. Therefore, f extends over X by Lemma 6.5.1. Since condition (a) of this theorem implies condition (e) of Theorem 6.12.3, we have the implication (a) $\Rightarrow$ (d). □

By Theorems 6.12.3 and 6.12.5, we have the following characterization of ARs and ANRs of dimension $\leqslant n$:

Corollary 6.12.6 *Let X be a metrizable space with $\dim X \leqslant n$.*

(1) *X is an AR if and only if X is C^n and LC^n (i.e., X is an $AE(n+1)$).*
(2) *X is an ANR if and only if X is LC^n (i.e., X is an $ANE(n+1)$).* □

Theorem 6.12.7 *Let $f : X \to Y$ be a map from a paracompact space X with $\dim X \leqslant n$ to an LC^{n-1} paracompact (resp. an LC^n metrizable) space Y and let $\mathcal{U}$ be an open cover of Y. Then, each open cover $\mathcal{V}$ of X has an open refinement $\mathcal{W}$ with $\operatorname{ord}\mathcal{W} \leqslant n+1$ (i.e., $\dim N(\mathcal{W}) \leqslant n$) and a full $\mathcal{U}$-realization $\psi : |N(\mathcal{W})| \to Y$ such that $\psi\varphi$ is $\mathcal{U}$-close (resp. $\mathcal{U}$-homotopic) to f for any canonical map $\varphi : X \to |N(\mathcal{W})|$.*

Proof By Lemma 6.12.2 and the paracompactness of Y, we can take the following open refinements of $\mathcal{U} \in \operatorname{cov}(Y)$:

$$\mathcal{U} \overset{*}{\succ} \mathcal{U}_1 \underset{L^n}{\succ} \mathcal{U}_0 \overset{*}{\succ} \mathcal{U}'.$$

By Theorem 5.2.5, $f^{-1}(\mathcal{U}')$ and $\mathcal{V}$ have a common open refinement $\mathcal{W} \in \operatorname{cov}(X)$ with $\operatorname{ord}\mathcal{W} \leqslant n+1$ (i.e., $\dim N(\mathcal{W}) \leqslant n$). For simplicity, we denote $K = N(\mathcal{W})$. For each $W \in \mathcal{W} = K^{(0)}$, choosing $U_W \in \mathcal{U}'$ so that $f(W) \subset U_W$ and taking a point $\psi_0(W) \in U_W$, we have a partial $\mathcal{U}_0$-realization $\psi_0 : K^{(0)} \to Y$ of K, which extends to a full $\mathcal{U}_1$-realization $\psi : |K| \to Y$ of K. It is easy to prove that $\psi\varphi$ is $\mathcal{U}$-close to f for any canonical map $\varphi : X \to |N(\mathcal{W})|$.

When Y is a metrizable $\mathrm{ANE}(n+1)$, we can take $d \in \mathrm{Metr}(Y)$ so that $\{\overline{\mathrm{B}}_d(y, 1) \mid y \in Y\} \prec \mathcal{U}$ by Proposition 2.7.9 (1). For each $i \in \mathbb{N}$, take $\mathcal{U}_i, \mathcal{V}_i \in \operatorname{cov}(Y)$ such that

$$\mathcal{V}_{i+1} \overset{*}{\prec} \mathcal{U}_i \overset{*}{\prec} \mathcal{V}_i \underset{L^{n+1}}{\prec} \mathcal{B}_i = \{\mathrm{B}_d(y, 2^{-i-3}) \mid y \in Y\}.$$

By the above argument, we can obtain $\mathcal{W}_i \in \operatorname{cov}(X)$ and a full $\mathcal{U}_i$-realization $\psi_i : |K_i| \to Y$ of the nerve K_i of $\mathcal{W}_i$ such that $\mathcal{W}_1 \prec \mathcal{V}, \mathcal{W}_{i+1} \prec \mathcal{W}_i, \operatorname{ord}\mathcal{W}_i \leqslant n+1$,

and $\psi_i\varphi_i$ is $\mathcal{V}_i$-close to f, where $\varphi_i : X \to |K_i|$ is any canonical map for $\mathcal{W}_i$. Since $\mathcal{W}_{i+1} \prec \mathcal{W}_i$, there is a simplicial map $k_i : K_{i+1} \to K_i$ such that $k_i\varphi_{i+1}$ is also a canonical map for $\mathcal{W}_i$ (cf. Sect. 4.15). Since φ_i and $k_i\varphi_{i+1}$ are contiguous, we can define a map $\varphi_i^* : X \times \mathbf{I} \to |Z_{k_i}|$ by

$$\varphi_i^*(x,t) = \begin{cases} \bar{q}_{k_i}(\varphi_{i+1}(x), 2t-1) & \text{if } \frac{1}{2} \leqslant t \leqslant 1, \\ (1-2t)\varphi_i(x) + 2tk_i\varphi_{i+1}(x) & \text{if } 0 \leqslant t \leqslant \frac{1}{2}, \end{cases}$$

where $\bar{q}_{k_i} : |K_i| \oplus |K_{i+1} \times_s I| \to Z_{k_i}$ is the natural simplicial map from the product simplicial complex $K_{i+1} \times_s I$ of K_{i+1} and $I = \{\mathbf{I}, 0, 1\}$ to the simplicial mapping cylinder Z_{k_i} of k_i (cf. Sect. 4.19). Since $\psi_i k_i$ is a full $\mathcal{U}_i$-realization of K_{i+1} and ψ_{i+1} is a full $\mathcal{U}_{i+1}$-realization of K_{i+1}, it follows that ψ_i and ψ_{i+1} induce a partial $\mathcal{V}_i$-realization of Z_{k_i}. Since $\dim Z_{k_i} \leqslant n+1$, we have a full $\mathcal{B}_i$-realization $\psi_i^* : |Z_{k_i}| \to Y$ such that $\psi_i^*||K_i| = \psi_i$ and $\psi_i^*||K_{i+1}| = \psi_{i+1}$. Observe that

$$\psi_i^*\varphi_i^*(\{x\} \times \mathbf{I}) \subset \mathrm{st}^2(\psi_i\varphi_i(x), \mathcal{B}_i),$$

hence $\operatorname{diam} \psi_i^*\varphi_i^*(\{x\} \times \mathbf{I}) < 2^{-i}$ for each $x \in X$. Then, we can define a homotopy $h : X \times \mathbf{I} \to Y$ as follows:

$$h_0 = f \quad \text{and} \quad h(x,t) = \psi_i^*\varphi_i^*(x, 2 - 2^i t) \quad \text{for } 2^{-i} \leqslant t \leqslant 2^{-i+1}.$$

Since $\operatorname{diam} h(\{x\} \times \mathbf{I}) < 1$ for each $x \in X$, we have $f \simeq_{\mathcal{U}} h_1 = \psi_1\varphi_1$. □

Remark 6.12.2 In Theorem 6.12.7, $\mathcal{W}$ can be taken as in Corollary 6.6.3, so as to be locally finite and σ-discrete in X (cf. Theorem 5.2.5). When X is separable and metrizable, a star-finite countable open refinement $\mathcal{W}$ can be taken, hence $|N(\mathcal{W})|$ is separable and locally compact (cf. Corollary 5.2.6). If X is compact, then $\mathcal{W}$ is finite, hence $|N(\mathcal{W})|$ is compact.

The following is easily seen by the same argument as in the above proof:

Proposition 6.12.8 *Let $\mathcal{U}, \mathcal{U}', \mathcal{V}, \mathcal{W} \in \mathrm{cov}(Y)$ such that*

$$\mathcal{W} \overset{*}{\prec} \mathcal{V} \underset{L^{n+1}}{\prec} \mathcal{U}' \overset{*}{\prec} \mathcal{U}.$$

Let P be a polyhedron with $\dim P \leqslant n$ and Q be a subpolyhedron of P. If two maps $f, g : P \to Y$ are $\mathcal{W}$-close and $f|Q = g|Q$, then $f \simeq_{\mathcal{U}} g$ rel. Q.

Proof Given maps $f, g : P \to Y$ that are $\mathcal{W}$-close and $f|Q = g|Q$, we define a map

$$h' : (Q \times \mathbf{I}) \cup (P \times \{0,1\}) \to Y$$

by $h'(x,0) = f(x)$ and $h'(x,1) = g(x)$ for each $x \in P$ and $h'|Q \times \mathbf{I} = f\mathrm{pr}_Q$. Let K be a triangulation of P such that Q is triangulated by a subcomplex of K and

$K \prec f^{-1}(\mathcal{W}) \wedge g^{-1}(\mathcal{W})$. Give an order on $K^{(0)}$ so that K is an ordered simplicial complex. Then, the product simplicial complex $K \times_s I$ is a triangulation of $P \times \mathbf{I}$, where $I = \{\mathbf{I}, 0, 1\}$ is the ordered simplicial complex with $|I| = \mathbf{I}$. Moreover,

$$(K \times_s I)^{(0)} \subset (Q \times \mathbf{I}) \cup (P \times \{0, 1\}).$$

Since h' is a partial $\mathcal{V}$-realization of $K \times_s I$, it extends to a full $\mathcal{U}'$-realization $h : P \times \mathbf{I} \to Y$ of $K \times_s I$. Then, h is a $\mathcal{U}$-homotopy realizing $f \simeq_{\mathcal{U}} g$ rel. Q. Thus, the proof is complete. □

An open refinement $\mathcal{V}$ of $\mathcal{U} \in \mathrm{cov}(Y)$ is called an ***h*^{n}-refinement** of $\mathcal{U}$ if any two $\mathcal{V}$-close maps $f, g : X \to Y$ defined on an arbitrary *metrizable* space X with $\dim X \leqslant n$ are $\mathcal{U}$-homotopic, where we denote

$$\mathcal{V} \underset{h^n}{\prec} \mathcal{U} \quad \text{or} \quad \mathcal{U} \underset{h^n}{\succ} \mathcal{V}.$$

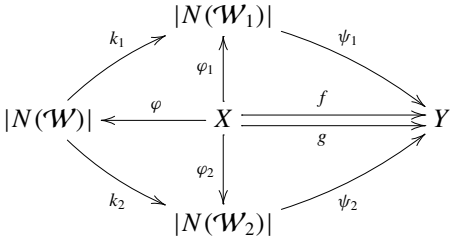
Theorem 6.12.9 *A metrizable space Y is LC^n (i.e., an ANE$(n+1)$) if and only if every open cover of Y has an h^n-refinement.*

Proof (*The "if" Part*) For each $y \in Y$ and each open neighborhood U of y in Y, the open cover $\mathcal{U} = \{U, Y \setminus \{y\}\}$ of Y has an h^n-refinement $\mathcal{V}$. Let $V \in \mathcal{V}$ such that $y \in V$. Then, $V \subset U$. For each $i \leqslant n$, every map $f : \mathbf{S}^i \to V$ is $\mathcal{V}$-close to the constant map c_y with $c_y(\mathbf{S}^i) = \{y\}$, hence $f \simeq_{\mathcal{U}} c_y$, which means $f \simeq c_y$ in U. Therefore, Y is LC^n.

(*The "only if" Part*) Each $\mathcal{U} \in \mathrm{cov}(Y)$ has the following open refinements:

$$\mathcal{U} \overset{*}{\succ} \mathcal{U}_1 \overset{*}{\succ} \mathcal{U}_0 \underset{L^{n+1}}{\succ} \mathcal{V}_0 \succ \mathrm{st}^2 \mathcal{V} \succ \mathcal{V}.$$

Let X be a metrizable space with $\dim X \leqslant n$ and $f, g : X \to Y$ be maps that are $\mathcal{V}$-close. By Theorem 6.12.7, X has open covers $\mathcal{W}_i$ and full $\mathcal{V}$-realizations $\psi_i : |N(\mathcal{W}_i)| \to Y, i = 1, 2$, such that $\dim N(\mathcal{W}_i) \leqslant n, \psi_1 \varphi_1 \simeq_{\mathcal{V}} f$, and $\psi_2 \varphi_2 \simeq_{\mathcal{V}} g$ for any canonical maps $\varphi_i : X \to |N(\mathcal{W}_i)|$. Take a common refinement $\mathcal{W} \in \mathrm{cov}(X)$ of $\mathcal{W}_1$ and $\mathcal{W}_2$ with $\dim N(\mathcal{W}) \leqslant n$ and let $\varphi : X \to |N(\mathcal{W})|$ be a canonical map. We have refining simplicial maps $k_i : N(\mathcal{W}) \to N(\mathcal{W}_i)$, $i = 1, 2$, i.e., $W \subset k_i(W)$ for each $W \in \mathcal{W} = N(\mathcal{W})^{(0)}$. Then, $k_i \varphi$ is also a canonical map for $N(\mathcal{W}_i)$ (Proposition 4.15.4 (2)).



For each principal simplex $\sigma \in N(\mathcal{W})$, take $x \in \bigcap_{W \in \sigma^{(0)}} W$. Since φ is a canonical map, it follows that $\varphi(x) \in \sigma$. On the other hand, we have $V \in \mathcal{V}$ such that $f(x), g(x) \in V$. Then, $\psi_1 k_1 \varphi(x), \psi_2 k_2 \varphi(x) \in \mathrm{st}(V, \mathcal{V})$. Since ψ_i is a full $\mathcal{V}$-realization of $N(\mathcal{W}_i)$, $\psi_i k_i$ is a full $\mathcal{V}$-realization of $N(\mathcal{W})$, so it follows that

$$\psi_1 k_1(\sigma) \cup \psi_2 k_2(\sigma) \subset \mathrm{st}^2(V, \mathcal{V}) \in \mathrm{st}^2 \mathcal{V} \prec \mathcal{V}_0.$$

Thus, $\psi_1 k_1$ and $\psi_2 k_2$ induce a partial $\mathcal{V}_0$-realization of the product simplicial complex $N(\mathcal{W}) \times_s I$, where we regard $N(\mathcal{W})$ as an ordered complex by giving an order on $\mathcal{W} = N(\mathcal{W})^{(0)}$. Then, we have a full $\mathcal{U}_0$-realization

$$h : |N(\mathcal{W}) \times_s I| = |N(\mathcal{W})| \times \mathbf{I} \to Y,$$

which is a $\mathcal{U}_1$-homotopy with $h_0 = \psi_1 k_1$ and $h_1 = \psi_2 k_2$. Thus, we have

$$f \simeq_{\mathcal{V}} \psi_1 k_1 \varphi \simeq_{\mathcal{U}_1} \psi_2 k_2 \varphi \simeq_{\mathcal{V}} g.$$

Therefore, $f \simeq_{\mathcal{U}} g$. □

Remark 6.12.3 The above proof is valid even if X is paracompact, but we need the metrizability of Y (cf. Theorem 6.12.7).

A subset Z in a space Y is said to be ***n*-homotopy dense** provided that, for each map $f : X \to Y$ from an arbitrary *metrizable* space X with $\dim X \leqslant n$, there exists a homotopy $h : X \times \mathbf{I} \to Y$ such that $h_0 = f$ and $h(X \times (0, 1]) \subset Z$. The following two theorems are the ANE(n) versions of Theorem 6.6.9:

Theorem 6.12.10 *Every n-homotopy dense set Z in an ANE(n) Y is also an ANE(n). If Y is an AE(n) then so is Z.*

Proof Let A be a closed set in a metrizable space X with $\dim X \leqslant n$. Each map $f : A \to Z$ extends to a map $g : U \to Y$, where U is a neighborhood U of A in X. Since $\dim U \leqslant n$, there exists a homotopy $h : U \times \mathbf{I} \to Y$ such that $h_0 = g$ and $h(A \times (0, 1]) \subset Z$. Take $d \in \mathrm{Metr}(X)$ such that $\mathrm{diam}_d X \leqslant 1$ and define a map $\tilde{f} : U \to Z$ by $\tilde{f}(x) = h(x, d(x, A))$. Then, $\tilde{f}$ is an extension of f. Thus, Z is an ANE(n). When Y is an AE(n), we can take $U = X$ in the above, hence Z is an AE(n). □

Theorem 6.12.11 *If a metrizable space Y contains an $(n-1)$-homotopy dense set Z that is an ANE($n+1$), then Y is an ANE(n). If Z is an $(n-1)$-connected ANE($n+1$), then Y is an AE(n).*

Proof We will verify condition (d) of Theorem 6.12.3. Let A be a closed set in a metrizable space X with $\dim A \leqslant n-1$. For each map $f : A \to Y$, there is a homotopy $h : A \times \mathbf{I} \to Y$ such that $h_0 = f$ and $h(A \times (0, 1]) \subset Z$. Since $\dim A \times (0, 1] \leqslant n$ and $A \times (0, 1]$ is closed in $X \times (0, 1]$, the restriction $h|A \times (0, 1]$ extends

over a neighborhood W of $A \times (0, 1]$ in $X \times (0, 1]$. Due to Lemma 6.5.1, this implies that f extends over a neighborhood of A in X.

Since Z is $(n-1)$-homotopy dense in Y, every map $g : \mathbf{S}^i \to Y$, $i \leqslant n-1$, is homotopic to a map $g' : \mathbf{S}^i \to Z$. When Z is $(n-1)$-connected, since g' is null-homotopic in Z, g is null-homotopic in Y, hence Y is also $(n-1)$-connected. Since an $(n-1)$-connected ANE(n) is AE(n) by Theorem 6.12.5, we have the additional statement. □

6.13 Embeddings into Finite-Dimensional ARs

Since a normed linear space is an AR by Theorems 6.1.1 and 6.2.5, the Arens–Eells Embedding Theorem 6.2.1 means that every metrizable space can be embedded in an AR as a closed set. In this section, we consider the finite-dimensional version of this fact. Here, we need more theorems from Dimension Theory, e.g., Theorems 5.2.7, 5.4.5, 5.4.2, etc.

The n-dimensional Menger compactum μ^n and the n-dimensional Nöbeling space ν^n are AE(n)s by Theorems 6.11.2 and 6.12.5. Then, Theorems 5.10.3 and 5.10.8 imply that every compact metrizable space with $\dim \leqslant n$ can be embedded in an n-dimensional separable metrizable AE(n) as a closed set.[18] These results will be extended to non-separable cases.

We use the simplicial mapping cylinder defined in Sect. 4.18. Let $f : K \to L$ be a simplicial map between simplicial complexes. Giving an order on $K^{(0)}$ so that K is an ordered simplicial complex, we defined the simplicial mapping cylinder Z_f with the collapsing $\bar{c}_f : Z_f \to L$. Then, $\bar{c}_f \simeq \mathrm{id}_{|Z_f|}$ rel. $|L|$ in $|Z_f|$ by the naturally defined homotopy $\bar{h}^f : |Z_f| \times \mathbf{I} \to |L|$. Since $\bar{c}_f$ is a simplicial map, it follows that $\bar{c}_f : |Z_f|_{\mathrm{m}} \to |L|_{\mathrm{m}}$ is continuous. However, we do not know whether $\bar{h}^f : |Z_f|_{\mathrm{m}} \times \mathbf{I} \to |L|_{\mathrm{m}}$ is continuous or not. Even so, we have the following:

Proposition 6.13.1 *For a simplicial map $f : K \to L$, the collapsing $\bar{c}_f : Z_f \to L$ is a strong deformation retraction of $|Z_f|_{\mathrm{m}}$ onto $|L|_{\mathrm{m}}$, and hence $|L|_{\mathrm{m}}$ is a strong deformation retract of $|Z_f|_{\mathrm{m}}$. Moreover, for each open cover $\mathcal{U} \in \mathrm{cov}(|L|_{\mathrm{m}})$,*

$$\bar{c}_f \simeq_{\bar{c}_f^{-1}(\mathcal{U})} \mathrm{id}_{|Z_f|_{\mathrm{m}}} \text{ rel. } |L|_{\mathrm{m}} \text{ in } |Z_f|_{\mathrm{m}},$$

which means that $\bar{c}_f$ is a fine homotopy equivalence.

Proof Due to Theorems 4.9.5 and 4.15.7, the identity $\varphi = \mathrm{id} : |Z_f| \to |Z_f|_{\mathrm{m}}$ is a fine homotopy equivalence.[19] For each open cover $\mathcal{U} \in \mathrm{cov}(|L|_{\mathrm{m}})$, take $\mathcal{V} \in$

[18] In fact, every separable completely metrizable space with $\dim \leqslant n$ can be embedded in ν^n as a closed set (cf. Remark 5.10.1 on Theorem 5.10.3).

[19] See Notes for Chap. 4.

$\mathrm{cov}(|L|_\mathrm{m})$ so that $\mathrm{st}^2\,\mathcal{V} \prec \mathcal{U}$. Then, φ has a $\bar{c}_f^{-1}(\mathcal{V})$-homotopy inverse $\psi : |Z_f|_\mathrm{m} \to |Z_f|$, that is, $\psi\varphi \simeq_{\bar{c}_f^{-1}(\mathcal{V})} \mathrm{id}$ and $\varphi\psi \simeq_{\bar{c}_f^{-1}(\mathcal{V})} \mathrm{id}$. See the following diagram:

$$\begin{array}{ccccc} |Z_f|_\mathrm{m} & \xrightarrow{\bar{c}_f} & |L|_\mathrm{m} & \subset & |Z_f|_\mathrm{m} \\ {\scriptstyle\psi}\downarrow\uparrow{\scriptstyle\varphi} & & \uparrow{\scriptstyle\varphi|} & & {\scriptstyle\varphi}\uparrow\downarrow{\scriptstyle\psi} \\ |Z_f| & \xrightarrow[\bar{c}_f]{} & |L| & \subset & |Z_f| \end{array}$$

Since $\varphi\psi \simeq_{\bar{c}_f^{-1}(\mathcal{V})} \mathrm{id}$ and $\bar{c}_f \simeq_{\bar{c}_f^{-1}(\mathcal{V})} \mathrm{id}$ in $|Z_f|$, it follows that

$$\bar{c}_f \simeq_{\mathcal{V}} \bar{c}_f\varphi\psi = \varphi\bar{c}_f\psi \simeq_{\bar{c}_f^{-1}(\mathcal{V})} \varphi\psi \simeq_{\bar{c}_f^{-1}(\mathcal{V})} \mathrm{id} \ \text{ in } |Z_f|_\mathrm{m},$$

which implies $\bar{c}_f \simeq_{\mathrm{st}\,\bar{c}_f^{-1}(\mathcal{V})} \mathrm{id}$ in $|Z_f|_\mathrm{m}$. Observe that

$$\mathrm{st}^2(\bar{c}_f^{-1}(\mathcal{V})) = \bar{c}_f^{-1}(\mathrm{st}^2\,\mathcal{V}) \prec \bar{c}_f^{-1}(\mathcal{U}).$$

Then, applying Theorem 6.4.4 with Remark 6.4.2, we have $\bar{c}_f \simeq_{\bar{c}_f^{-1}(\mathcal{U})} \mathrm{id}$ rel. $|L|_\mathrm{m}$ in $|Z_f|_\mathrm{m}$, which means that $\bar{c}_f : |Z_f|_\mathrm{m} \to |L|_\mathrm{m}$ is a strong deformation retraction and a fine homotopy equivalence. □

Using the above proposition, we prove the following theorem:

Theorem 6.13.2 *Every n-dimensional metrizable space X can be embedded as a closed set in an $(n+1)$-dimensional AR $T(X)$ of the same weight. If X is completely metrizable, compact, separable, or separable locally compact, then so is $T(X)$, respectively.*

Proof When X is finite, X can be embedded in the interval $\mathbf{I}$. Then, we may assume that X is infinite. By Theorem 5.4.5, we may also assume that X is a dense subspace of a complete metric space $\widetilde{X}$ with $\dim \widetilde{X} = n$, where $\widetilde{X} = X$ if X is completely metrizable. By Theorem 5.2.7, $\widetilde{X}$ is homeomorphic to the inverse limit $\varprojlim \mathbf{K}$ of an inverse sequence $\mathbf{K} = (|K_i|_\mathrm{m}, f_i)_{i\in\mathbb{N}}$ of metric polyhedra and PL maps such that $\mathrm{card}\, K_i \leqslant w(X)$, $\dim K_i \leqslant n$, and each $f_i : K_{i+1} \to \mathrm{Sd}\, K_i$ is simplicial. Note that $\dim K_i = n$ for sufficiently large $i \in \mathbb{N}$. Otherwise, $\dim \widetilde{X} = \dim \varprojlim \mathbf{K} < n$ by Theorem 5.4.2. We identify $\widetilde{X} = \varprojlim \mathbf{K}$. As is shown in Remark 4.17.1,

$$\mathrm{mesh}_{\rho_i}\, f_{i,i+k}(K_{i+k}) \leqslant \mathrm{mesh}_{\rho_i}\, f_{i,i+k-1}(\mathrm{Sd}\, K_{i+k-1}) \leqslant 2 \cdot \left(\frac{n}{n+1}\right)^k,$$

where $\rho_i = \rho_{K_i}$ and $f_{i,i+k} = f_i \cdots f_{i+k-1} : |K_{i+k}| \to |K_i|$.[20] Therefore,

[20] In general, $f_{i,i+2} = f_i f_{i+1}$ is not simplicial with respect to K_{i+2} and $\mathrm{Sd}^2\, K_i$ because $f_i(\hat{\tau})$ is not necessarily the barycenter of the simplex $f_i(\tau) \in \mathrm{Sd}\, K_i$ for $\tau \in K_{i+1}$.

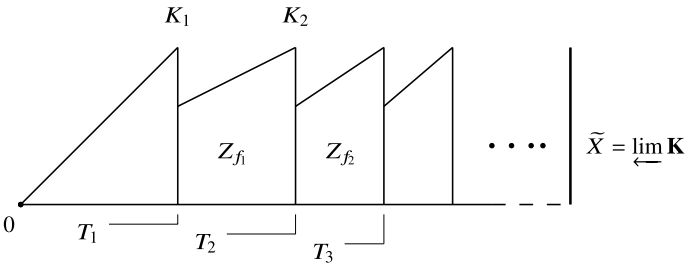


Fig. 6.22 The space $T(\widetilde{X})$

$$\lim_{k\to\infty} \operatorname{mesh}_{\rho_i} f_{i,i+k}(K_{i+k}) = 0 \quad \text{and} \quad \lim_{k\to\infty} \operatorname{mesh}_{\rho_i} f_{i,i+k}(O_{K_{i+k}}) = 0.$$

To consider the simplicial mapping cylinder Z_{f_i} of each simplicial map $f_i : K_{i+1} \to \operatorname{Sd} K_i$, we regard K_{i+1} as an ordered simplicial complex. Let $T_1 = 0 * \operatorname{Sd} K_1$ be the simplicial cone over $\operatorname{Sd} K_1$. Moreover, we may assume that $|Z_{f_i}| \cap |Z_{f_j}| = \emptyset$ for $|i - j| > 1$ and $|T_1| \cap |Z_{f_i}| = \emptyset$ for $i > 1$. We define

$$T_k = T_1 \cup \bigcup_{i=1}^{k-1} \operatorname{Sd}_{\operatorname{Sd} K_i} Z_{f_i} \quad \text{for } k > 1 \text{ and}$$
$$T = \bigcup_{i\in\mathbb{N}} T_i = T_1 \cup \bigcup_{i\in\mathbb{N}} \operatorname{Sd}_{\operatorname{Sd} K_i} Z_{f_i},$$

where $\operatorname{Sd}_{\operatorname{Sd} K_i} Z_{f_i}$ is the barycentric subdivision of Z_{f_i} relative to the subcomplex $\operatorname{Sd} K_i \subset Z_{f_i}$. Observe that T is a simplicial complex with $\operatorname{card} T^{(0)} \leqslant \aleph_0 w(X)$ and $T_1 \subset T_2 \subset \cdots$ are subcomplexes of T, where $\dim T = n + 1$ because $\dim T_i = n + 1$ for sufficiently large $i \in \mathbb{N}$. It should be noted that $\bigcup_{i\in\mathbb{N}} Z_{f_i}$, in general, is not a simplicial complex. See Fig. 6.22.

Considering $\widetilde{X} \cap |T| = \emptyset$, we define $\gamma_i : |T| \cup \widetilde{X} \to |T_i|$, $i \in \mathbb{N}$, as follows:

$$\gamma_i(x) = \begin{cases} x & \text{if } x \in |T_i|, \\ \bar{c}_{f_i} \cdots \bar{c}_{f_j}(x) & \text{if } x \in |Z_{f_j}|,\ j \geqslant i, \\ p_i(x) & \text{if } x \in \widetilde{X}, \end{cases}$$

where $\bar{c}_{f_i} : Z_{f_i} \to \operatorname{Sd} K_i$ is the simplicial collapsing map and $p_i : \widetilde{X} = \varprojlim \mathbf{K} \to |K_i|_{\mathrm{m}}$ is the projection. Note that $\gamma_i \gamma_j = \gamma_i$ for $i \leqslant j$. Let $T(\widetilde{X})$ be the space defined as $|T| \cup \widetilde{X}$ with the coarsest topology such that every $\gamma_i : |T| \cup \widetilde{X} \to |T_i|_{\mathrm{m}}$ is continuous, that is, the topology generated by sets $\gamma_i^{-1}(U)$, where $i \in \mathbb{N}$ and U is open in $|T_i|_{\mathrm{m}}$. It should be noted that if $U \subset |T_i| \setminus |K_i|$ then $\gamma_i^{-1}(U) = U$, which implies that $|T_i|_{\mathrm{m}}$ is a subspace of $T(\widetilde{X})$. Hence, $|T|_{\mathrm{m}}$ is an open subspace of $T(\widetilde{X})$ and $\widetilde{X}$ is a closed subspace of $T(\widetilde{X})$. Evidently, $|T|_{\mathrm{m}}$ is dense in $T(\widetilde{X})$.

Due to Proposition 6.13.1, the collapsing $\bar{c}_{f_i}$ is a strong deformation retraction of $|Z_{f_i}|_{\mathrm{m}}$ onto $|\operatorname{Sd} K_i|_{\mathrm{m}}$, which extends to a strong deformation retraction $c_i : |T_{i+1}|_{\mathrm{m}} = |T_i| \cup |Z_{f_i}| \to |T_i|_{\mathrm{m}}$ by $c_i||T_i| = \mathrm{id}$. Then, we have the following commutative diagram:

$$\begin{array}{ccccccc}
|T_1|_{\mathrm{m}} & \xleftarrow[\subset]{c_1} & |T_2|_{\mathrm{m}} & \xleftarrow[\subset]{c_2} & |T_3|_{\mathrm{m}} & \xleftarrow[\subset]{c_3} & \cdots \\
\cup & & \cup & & \cup & & \cdots \\
|K_1|_{\mathrm{m}} & \xleftarrow{f_1} & |K_2|_{\mathrm{m}} & \xleftarrow{f_2} & |K_3|_{\mathrm{m}} & \xleftarrow{f_3} & \cdots
\end{array}$$

We regard $\widetilde{X}$ as the inverse limit of the lower sequence. The space $T(\widetilde{X})$ can be regarded as the inverse limit of the upper sequence with $\gamma_i : T(\widetilde{X}) \to |T_i|_{\mathrm{m}}$ the projection (cf. Theorem 4.16.7).

According to Theorem 4.6.5, each $|T_i|_{\mathrm{m}}$ is completely metrizable because T_i is finite-dimensional. Therefore, $T(\widetilde{X})$ is also completely metrizable by Proposition 4.17.1. As already noted, $|T|_{\mathrm{m}}$ is an open dense subspace of $T(\widetilde{X})$, $\widetilde{X}$ is a closed subspace of $T(\widetilde{X})$. Hence, $w(T(\widetilde{X})) = w(|T|_{\mathrm{m}}) = w(X)$. Since $\dim |T_i|_{\mathrm{m}} = \dim T_i$ for each $i \in \mathbb{N}$ (Corollary 5.2.14), we have $\dim T(\widetilde{X}) = n + 1$ by Theorem 5.4.2 (or the Countable Sum Theorem 5.2.8).

We will prove that $T(\widetilde{X})$ is an AR. Note that $T(\widetilde{X})$ is contractible because so is $h_1(T(\widetilde{X})) = |0 * \operatorname{Sd} K_1|$. Due to Corollary 6.2.14, it suffices to show that $T(\widetilde{X})$ is an ANR. Since $|T|_{\mathrm{m}}$ is an ANR, applying Corollary 6.6.9, we may show that $|T|_{\mathrm{m}}$ is homotopy dense in $T(\widetilde{X})$. By Proposition 6.13.1, we have

$$\gamma_i = c_i \gamma_{i+1} \simeq_{c_i^{-1}(\mathcal{O}_{K_i})} \gamma_{i+1} \ \text{ rel. } |T_i| \text{ in } |T_{i+1}|,$$

which is realized by a $c_i^{-1}(\mathcal{O}_{K_i})$-homotopy $h^i : T(\widetilde{X}) \times \mathbf{I} \to |T_{i+1}|$ with $h^i_0 = \gamma_i$ and $h^i_1 = \gamma_{i+1}$. Connecting homotopies h^i, $i \in \mathbb{N}$, we can easily construct a homotopy $h : T(\widetilde{X}) \times \mathbf{I} \to T(\widetilde{X})$ so that $h_0 = \mathrm{id}$, $h_{2^{-i+1}} = \gamma_i$ for each $i \in \mathbb{N}$, and $h(T(\widetilde{X}) \times (0, 1]) \subset |T|$, which shows that $|T|_{\mathrm{m}}$ is homotopy dense in $T(\widetilde{X})$. We have to verify the continuity of h at $(x, 0) \in \widetilde{X} \times \{0\}$ $(\subset T(\widetilde{X}) \times \mathbf{I})$. For each neighborhood U of $x = h(x, 0)$ in $T(\widetilde{X})$, we have $k \in \mathbb{N}$ and $\varepsilon > 0$ such that $\gamma_k^{-1}(\mathrm{B}_{\rho_k}(p_k(x), \varepsilon)) \subset U$. Take $j > k$ and $\delta > 0$ so that

$$\operatorname{mesh}_{\rho_k} f_{k,j}(K_j) < \varepsilon \ \text{ and } \ \mathrm{B}_{\rho_j}(p_j(x), \delta) \subset O_{K_j}(p_j(x)).$$

Moreover, choose $i_0 > j$ so that $\operatorname{mesh}_{\rho_j} f_{j,i}(\mathcal{O}_{K_i}) < \delta/2$ for any $i \geqslant i_0$. Then, $\gamma_j^{-1}(\mathrm{B}_{\rho_j}(p_j(x), \delta/2)) \times [0, 2^{-i_0+1}]$ is a neighborhood of $(x, 0)$ in $T(\widetilde{X}) \times \mathbf{I}$. We will show that

$$h(\gamma_j^{-1}(\mathrm{B}_{\rho_j}(p_j(x), \delta/2)) \times [0, 2^{-i_0+1}]) \subset \gamma_k^{-1}(\mathrm{B}_{\rho_k}(p_k(x), \varepsilon)) \subset U.$$

Let $y \in \gamma_i^{-1}(\mathrm{B}_{\rho_j}(p_j(x), \delta/2))$ and $i \geqslant i_0$. If $y \in |T_i|$, then $h^i(\{y\} \times \mathbf{I})) = \{y\} \subset \gamma_i^{-1}(\mathrm{B}_{\rho_j}(p_j(x), \delta/2))$, that is, $\gamma_j(h^i(\{y\} \times \mathbf{I})) \subset \mathrm{B}_{\rho_j}(p_j(x), \delta/2)$. Otherwise, since h^i is a $c_i^{-1}(\mathcal{O}_{K_i})$-homotopy, $h^i(\{y\} \times \mathbf{I}) \subset c_i^{-1}(O_{K_i}(v))$ for some $v \in K_i^{(0)}$. Then,

$$\begin{aligned}\gamma_j(h^i(\{y\} \times \mathbf{I})) &\subset \gamma_j(c_i^{-1}(O_{K_i}(v))) = \gamma_j\gamma_i(c_i^{-1}(O_{K_i}(v))) \\ &\subset \gamma_j(O_{K_i}(v)) = f_{j,i}(O_{K_i}(v)) \in f_{j,i}(\mathcal{O}_{K_i}),\end{aligned}$$

which implies that $\operatorname{diam}_{\rho_j} \gamma_j(h^i(\{y\} \times \mathbf{I})) < \delta/2$. Since $\gamma_j h_0^i(y) = \gamma_j\gamma_i(y) = \gamma_j(y) \in \mathrm{B}_{\rho_j}(p_j(x), \delta/2)$, it follows that $\gamma_j(h^i(\{y\} \times \mathbf{I})) \subset \mathrm{B}_{\rho_j}(p_j(x), \delta)$. In either case, we have

$$\gamma_j(h^i(\{y\} \times \mathbf{I})) \subset \mathrm{B}_{\rho_j}(p_j(x), \delta) \subset O_{K_j}(p_j(x)) \subset \mathrm{st}(p_j(x), K_j).$$

Then, it follows that

$$\begin{aligned}\gamma_k(h^i(\{y\} \times \mathbf{I})) &= \gamma_k\gamma_j(h^i(\{y\} \times \mathbf{I})) \subset \gamma_k(\mathrm{st}(p_j(x), K_j)) \\ &\subset \mathrm{st}(p_k(x), f_{k,j}(K_j)) \subset \mathrm{B}_{\rho_k}(p_k(x), \varepsilon).\end{aligned}$$

Consequently, we have

$$h^i\big(\gamma_i^{-1}(\mathrm{B}_{\rho_j}(p_j(x), \delta/2)) \times \mathbf{I}\big) \subset \gamma_k^{-1}(\mathrm{B}_{\rho_k}(p_k(x), \varepsilon)) \quad \text{for each } i \geqslant i_0.$$

Thus, we obtain

$$\begin{aligned}&h(\gamma_j^{-1}(\mathrm{B}_{\rho_j}(p_j(x), \delta/2)) \times [0, 2^{-i_0+1}]) \\ &\quad = \bigcup_{i \geqslant i_0} h^i\big(\gamma_j^{-1}(\mathrm{B}_{\rho_j}(p_j(x), \delta/2)) \times \mathbf{I}\big) \cup \gamma_j^{-1}(\mathrm{B}_{\rho_j}(p_j(x), \delta/2)) \\ &\quad \subset \gamma_k^{-1}(\mathrm{B}_{\rho_k}(p_k(x), \varepsilon)).\end{aligned}$$

Except for the case where X is separable and locally compact, the desired space $T(X)$ is defined as the subspace $T(X) = |T| \cup X$ of $T(\widetilde{X})$. Indeed, X is closed in $T(X)$, $w(T(X)) = w(X)$, and $\dim T(X) = n + 1$. In the above, $h(T(X) \times \mathbf{I}) \subset T(X)$, so $T(X)$ is an AR for the same reason as $T(\widetilde{X})$.

When X is completely metrizable, since $\widetilde{X} = X$, $T(X) = T(\widetilde{X})$ is completely metrizable. When X is compact, since each K_i is finite due to Theorem 5.2.7, each $|T_i|_{\mathrm{m}}$ is compact, hence so is $T(X) = T(\widetilde{X})$ by Proposition 4.16.2. If X is separable then so is $\widetilde{X}$, hence each K_i is countable due to Theorem 5.2.7, which implies that T is also countable, so $T(X)$ is separable. If X is separable and locally compact, the above $T(X)$ is not locally compact at $0 \in |0 * \mathrm{Sd}\, K_1| = |T_1|$ unless K_1 is finite. Thus, some modification is necessary for the additional statement to be valid.

(*The case that X is separable and locally compact*) In the above construction, each K_i is countable and locally finite and each f_i is proper by Theorem 5.2.7. If K_1 is disconnected, let $L_1, L_2, \ldots$be the components of K_1 and take vertices $v_1 \in L_1^{(0)}$,

$v_2 \in L_2^{(0)}, \ldots$. Adding new 1-simplexes $\langle v_1, v_2\rangle, \langle v_2, v_3\rangle, \ldots$to K_1, we can assume that K_1 is connected. Moreover, instead of $T_1 = 0 * \mathrm{Sd}\, K_1$, let $T_1 = Z_{f_0}$ be the simplicial mapping cylinder of a proper simplicial map

$$f_0 : \mathrm{Sd}\, K_1 \to K_0 = \omega \cup \{[i-1, i] \mid i \in \mathbb{N}\},$$

where K_0 is the natural triangulation of the half line $\mathbb{R}_+ = [0, \infty)$. Then, T would be countable and locally finite, hence $T(X)$ would be separable and locally compact.

To construct a proper simplicial map $f_0 : \mathrm{Sd}\, K_1 \to K_0$, for each $i \in \mathbb{N}$, let

$$V_i = \left\{v \in (\mathrm{Sd}\, K_1)^{(0)} \;\middle|\; \langle v, u\rangle \in \mathrm{Sd}\, K_1 \text{ for some } u \in V_{i-1}\right\},$$

where $V_0 = \{v_0\}, v_0 \in (\mathrm{Sd}\, K_1)^{(0)}$. Then, $(\mathrm{Sd}\, K_1)^{(0)} = \bigcup_{i\in\omega} V_i$ because $\mathrm{Sd}\, K_1$ is connected. We can define a simplicial map $f_0 : \mathrm{Sd}\, K_1 \to K_0$ by

$$f_0(v) = \min\{i \in \omega \mid v \in V_i\} \in \omega = K_0^{(0)} \text{ for each } v \in (\mathrm{Sd}\, K_1)^{(0)}.$$

Since $\mathrm{Sd}\, K_1$ is locally finite, each V_i is finite, hence the map $f_0 : |K_1|_{\mathrm{m}} = |\mathrm{Sd}\, K_1|_{\mathrm{m}} \to |K_0|_{\mathrm{m}} = \mathbb{R}_+$ is proper.

Now, each $|T_i|_{\mathrm{m}}$ is locally compact and each c_i is also proper. Therefore, $T(X) = T(\widetilde{X})$ is locally compact by Proposition 4.16.6. Since $|K_0|_{\mathrm{m}} = \mathbb{R}_+$ is a deformation retract of $T(X)$, it follows that $T(X)$ is contractible, hence $T(X)$ is an AR. □

In Theorem 6.13.2 above, we cannot take $\dim T(X) = \dim X = n$, that is, an n-dimensional metrizable space cannot, in general, be embedded in an n-dimensional AR (nor an n-dimensional ANR) as a closed set. Neither the n-dimensional universal Nöbeling space ν^n nor the n-dimensional universal Menger compactum μ^n can be embedded in any n-dimensional ANR as a closed set. Otherwise, they would be retracts of an n-dimensional ANR because they are n-dimensional AE(n)s by Theorems 6.11.2 and 6.12.5. But this is impossible because they are not ANRs by Theorem 6.11.2 (cf. Proposition 6.2.9 (2)).

It should be noted that, due to Theorems 5.10.3 and 5.10.8, every separable n-dimensional metrizable space can be embedded into ν^n and μ^n, which are n-dimensional AE(n)s as noted above. Without separability, we can obtain the same embedding theorem:

Theorem 6.13.3 *Every n-dimensional metrizable space X can be embedded in an n-dimensional metrizable AE(n) $S(X)$ with $w(S(X)) = w(X)$ as a closed set. If X is completely metrizable, compact, separable, or separable locally compact, then so is $S(X)$, respectively.*

To prove this theorem, we employ the following lemmas:

Lemma 6.13.4 *Let $f : K \to L$ be a simplicial map between simplicial complexes and X be a metrizable space with $\dim X \leqslant n-1$. Then, for each map $g : X \to |Z_f^{(n)}|_{\mathrm{m}}$, $g \simeq_{\bar{c}_f^{-1}(\mathcal{W})} \bar{c}_f g$ rel. $g^{-1}(|L|)$ in $|Z_f^{(n)}|_{\mathrm{m}}$, where $\bar{c}_f : Z_f \to L$ is a simplicial collapsing and $\mathcal{W} = \{\mathrm{st}(\sigma, O_L) \mid \sigma \in L\}$.*

Proof Let $f' = f|K^{(n-1)} : K^{(n-1)} \to L^{(n)}$. Then, observe that

$$Z_f^{(n-1)} \subset Z_{f'} \subset Z_f^{(n)} \text{ and } |Z_{f'}| \setminus |K| = |Z_f^{(n)}| \setminus |K|.$$

By Theorem 5.2.13, we have a map $g' : X \to |Z_f^{(n-1)}|_{\mathrm{m}}$ such that $g'(x), g(x) \in \sigma_x$ for each $x \in X$, where $\sigma_x = c_{Z_f}(g(x)) \in Z_f^{(n)}$ is the carrier of $g(x)$. Taking a Urysohn map $k : X \to \mathbf{I}$ with $k(g^{-1}(|L|)) = 0$ and $k(g^{-1}(|K|)) = 1$, we can define a homotopy $h : X \times \mathbf{I} \to |Z_f^{(n)}|_{\mathrm{m}}$ as follows:

$$h(x,t) = (1 - k(x)t)g(x) + k(x)tg'(x) \text{ for each } (x,t) \in X \times \mathbf{I}.$$

Then, $h_0 = g$, $h_t|g^{-1}(|L|) = g|g^{-1}(|L|)$ for each $t \in \mathbf{I}$, and $h_1(X) \subset |Z_{f'}|$. Using Proposition 6.13.1 and the above homotopy, we have

$$g \simeq_{Z_f} h_1 \simeq_{\bar{c}_f^{-1}(O_L)} \bar{c}_{f'}h_1 = \bar{c}_f h_1 \simeq_L \bar{c}_f g \text{ rel. } g^{-1}(|L|) \text{ in } |Z_f^{(n)}|_{\mathrm{m}}.$$

For each $x \in X \setminus g^{-1}(|L|)$, $h(\{x\} \times \mathbf{I}) \subset \sigma_x$ and $\bar{c}_f h(\{x\} \times \mathbf{I}) \subset \bar{c}_f(\sigma_x)$. This completes the proof. □

Lemma 6.13.5 *Let $0 * K$ be the simplicial cone over a simplicial complex K. Then, the n-skeleton $|(0 * K)^{(n)}|_{\mathrm{m}}$ is $(n-1)$-connected.*

Proof Regarding $0 * K$ as the simplicial mapping cylinder of the constant map of K, we can apply Lemma 6.13.4 to show that every map $g : \mathbf{S}^i \to |(0 * K)^{(n)}|_{\mathrm{m}}$ $(i \leqslant n-1)$ is homotopic to the map of $\mathbf{S}^i$ to $\{0\}$, hence $g \simeq 0$. □

Proof (*Theorem* 6.13.3) Modifying the proof of Theorem 6.13.2, we will construct $S(X)$ as a closed subspace of $T(X)$. Let $S_1 = T_1^{(n)} = (0 * \mathrm{Sd}\, K_1)^{(n)}$,

$$S_k = S_1 \cup \bigcup_{i=1}^{k-1} \mathrm{Sd}_{\mathrm{Sd}\, K_i} Z_{f_i}^{(n)} \subset T_k^{(n)} \text{ for } k > 1 \text{ and}$$

$$S = \bigcup_{i \in \mathbb{N}} S_i = S_1 \cup \bigcup_{i \in \mathbb{N}} \mathrm{Sd}_{\mathrm{Sd}\, K_i} Z_{f_i}^{(n)} \subset T^{(n)}.$$

Since $Z_{f_i}^{(n)} = Z_{f_i|K_{i+1}^{(n-1)}} \cup K_{i+1}$, it follows that

$$\mathrm{Sd}_{\mathrm{Sd}\, K_i} Z_{f_i}^{(n)} = \mathrm{Sd}_{\mathrm{Sd}\, K_i} Z_{f_i|K_{i+1}^{(n-1)}} \cup \mathrm{Sd}\, K_{i+1},$$

which is not equal to $(\mathrm{Sd}_{\mathrm{Sd}\, K_i} Z_{f_i})^{(n)}$ if $\dim K_{i+1} = n$. Restricting the strong deformation retraction $c_k : |T_{k+1}|_{\mathrm{m}} \to |T_k|_{\mathrm{m}}$, we have the retraction $c_k' : |S_{k+1}|_{\mathrm{m}} \to |S_k|_{\mathrm{m}}$, which is no longer a strong deformation retraction.

Let $S(\widetilde{X}) = |S| \cup \widetilde{X}$ be the n-dimensional closed subspace of $T(\widetilde{X})$, which is regarded as the inverse limit of the middle sequence in the following commutative diagram:

$$\begin{array}{ccccccc}
|T_1|_{\mathrm{m}} & \xleftarrow[\subset]{c_1} & |T_2|_{\mathrm{m}} & \xleftarrow[\subset]{c_2} & |T_3|_{\mathrm{m}} & \xleftarrow[\subset]{c_3} & \cdots \\
\cup & & \cup & & \cup & & \cdots \\
|S_1|_{\mathrm{m}} & \xleftarrow[\subset]{c_1'} & |S_2|_{\mathrm{m}} & \xleftarrow[\subset]{c_2'} & |S_3|_{\mathrm{m}} & \xleftarrow[\subset]{c_3'} & \cdots \\
\cup & & \cup & & \cup & & \cdots \\
|K_1|_{\mathrm{m}} & \xleftarrow[f_1]{} & |K_2|_{\mathrm{m}} & \xleftarrow[f_2]{} & |K_3|_{\mathrm{m}} & \xleftarrow[f_3]{} & \cdots
\end{array}$$

Then, the projection $\gamma_i' : S(\widetilde{X}) \to |S_i|_{\mathrm{m}}$ is the restriction of the projection $\gamma_i : T(\widetilde{X}) \to |T_i|_{\mathrm{m}}$, where we no longer have that $\gamma_i' = c_i' \gamma_i' \simeq \gamma_{i+1}'$ rel. $|S_i|$.

The space $S(X)$ is the closed subspace $|S| \cup X = S(\widetilde{X}) \cap T(X)$ of $T(X)$, hence, if X is completely metrizable, compact or separable, then so is $S(X)$, respectively. When X is separable and locally compact, in the same way as $T(X)$, we can modify $S(X)$ to be separable and locally compact.

We will now apply Theorem 6.12.11 to prove that $S(X)$ is an AE(n). Since $S(X) \setminus X = |S|_{\mathrm{m}}$ is an ANE as a polyhedron with the metric topology, it suffices to show that $|S|_{\mathrm{m}}$ is $(n-1)$-homotopy dense in $S(X)$ and $(n-1)$-connected.

(*The* $(n-1)$*-Homotopy Denseness*) Let $g : Z \to S(X)$ be a map of a metrizable space Z with $\dim Z \leqslant n-1$. For each $i \in \mathbb{N}$, let $\mathcal{W}_i = \mathrm{st}(\mathrm{Sd}\, K_i, \mathcal{O}_{\mathrm{Sd}\, K_i}) \in \mathrm{cov}(|K_i|_{\mathrm{m}})$. Due to Lemma 6.13.4, we have

$$\gamma_i' g = c_i' \gamma_{i+1}' g \simeq_{c_i'^{-1}(\mathcal{W}_i)} \gamma_{i+1}' g \ \text{ rel. } g^{-1}(|S_i|)$$

by a $c_i'^{-1}(\mathcal{W}_i)$-homotopy $h^i : Z \times \mathbf{I} \to |S_{i+1}|_{\mathrm{m}} \subset S(X)$ with $h_0^i = \gamma_i' g$ and $h_1^i = \gamma_{i+1}' g$. Connecting these homotopies, we can define a homotopy $h : Z \times \mathbf{I} \to S(X)$ such that $h_0 = g$, $h_{2^{-i+1}} = \gamma_i' g$ for each $i \in \mathbb{N}$, and $h(Z \times (0,1]) \subset |S|$, which shows that $|S|_{\mathrm{m}}$ is $(n-1)$-homotopy dense in $S(X)$. The continuity of h at a point $(z,0) \in g^{-1}(X) \times \{0\} \subset Z \times \mathbf{I}$ can be verified by an analogous argument as in the proof of Theorem 6.13.2 using the fact that

$$\lim_{j\to\infty} \mathrm{mesh}_{\rho_i}\, f_{i,i+j}(K_{i+j}) = 0 \ \text{ and } \ \lim_{j\to\infty} \mathrm{mesh}_{\rho_i}\, f_{i,i+j}(\mathcal{W}_{i+j}) = 0,$$

which is obtained by Remark 4.17.1.

(*The* $(n-1)$*-Connectedness*) In the above, let $Z = \mathbf{S}^i$, $i \leqslant n-1$. Then, every map $g : \mathbf{S}^i \to |S|_{\mathrm{m}}$ is homotopic to a map $g' : \mathbf{S}^i \to |S_1|_{\mathrm{m}}$. Since $S_1 = T_1^{(n)} = (0 * \mathrm{Sd}\, K_1)^{(n)}$ is $(n-1)$-connected by Lemma 6.13.5, we have $g \simeq g' \simeq 0$. This completes the proof. □

6.14 Hyperspaces and Whitney Maps

In Sect. 6.2, we considered the case when function spaces (spaces of maps) are ANRs (Propositions 6.2.9 (6), (7), 6.2.16, and 6.2.19). In this section, we consider hyperspaces (spaces of closed sets), and show that they are ANRs under some suitable conditions. For topology of hyperspaces, refer to Sect. 2.12.

For a space X, as seen in Theorem 2.12.9, the hyperspace $\mathrm{Cld}(X)$ of non-empty closed sets in X with the Vietoris topology is metrizable if and only if X is a compactum (= a compact metrizable space). For a compactum X, $\mathrm{Comp}(X) = \mathrm{Cld}(X)$ and $\mathrm{Cont}(X) = \mathrm{Conn}(X)$. Due to Proposition 2.12.7, given an admissible metric d for X, the Hausdorff metric d_H is admissible for the space $\mathrm{Comp}(X)$ with the Vietoris topology. Here, we prove the WOJDYSAWSKI THEOREM 6.14.14, that is, for a compactum X, $\mathrm{Comp}(X)$ (or $\mathrm{Cont}(X)$) is an AR if and only if X is a Peano continuum (= a locally connected continuum).

As a preparation, we will introduce Whitney maps and show some their properties. For a subspace $\mathcal{H}$ of $\mathrm{Cld}(X)$, a map $w : \mathcal{H} \to \mathbb{R}_+$ is called a **Whitney map** if it satisfies the following conditions:

(W1) For any $A, B \in \mathcal{H}$, $A \subsetneqq B$ implies $w(A) < w(B)$;
(W2) For any $A \in \mathcal{H}$, $w(A) = 0$ if and only if $A \in F_1(X)$.

For example, a Whitney map $w : \mathrm{Cont}(\mathbf{I}) \to \mathbf{I}$ can be defined by $w([a, b]) = b - a$ $(= \mathrm{diam}[a, b])$ for each $[a, b] \in \mathrm{Cont}(\mathbf{I})$.

By the following theorem, every metrizable space X has a Whiteney map for $\mathrm{Comp}(X)$, that is:

Theorem 6.14.1 *For every metrizable space X, there exists a Whitney map $w : \mathrm{Comp}(X) \to \mathbb{R}_+$.*

Proof Let $d \in \mathrm{Metr}(X)$. Then, the Hausdorff metric d_H defined by d is admissible for the Vietoris topology of $\mathrm{Comp}(X)$ by Proposition 2.12.7. For each integer $n \geqslant 2$, we define $w_n : \mathrm{Comp}(X) \to \mathbb{R}_+$ as follows:[21]

$$w_n(A) \underset{\text{def}}{=} \sup_{a \in A^n} \min_{i \neq j} d(a(i), a(j)).$$

Then, $w_2(A) = 0$ if and only if $A \in F_1(X)$ (note that $w_3(A) = 0$ if $\mathrm{card}\, A = 2$); $w_n(A) \leqslant w_n(B)$ for $A \subset B \in \mathrm{Comp}(X)$; and w_n satisfies the following Lipschitz condition, so it is continuous:

$$|w_n(A) - w_n(B)| \leqslant 2d_H(A, B) \text{ for each } A, B \in \mathrm{Comp}(X).$$

[21] Because of the compactness of A, $\sup_{a \in A^n} \min_{i \neq j} d(a(i), a(j)) = \max_{a \in A^n} \min_{i \neq j} d(a(i), a(j))$.

Indeed, for any $\varepsilon > 0$, there are $a_1, \ldots, a_n \in A$ such that $d(a_i, a_j) \geqslant w_n(A) - \varepsilon$ if $i \neq j$. Then, we can take $b_1, \ldots, b_n \in B$ so that $d(a_i, b_i) < d_H(A, B) + \varepsilon$. For $i \neq j$,

$$d(b_i, b_j) \geqslant d(a_i, a_j) - 2d_H(A, B) - 2\varepsilon \geqslant w_n(A) - 2d_H(A, B) - 3\varepsilon.$$

It follows that $w_n(B) \geqslant w_n(A) - 2d_H(A, B)$. Similarly, $w_n(A) \geqslant w_n(B) - 2d_H(A, B)$. Therefore, $|w_n(A) - w_n(B)| \leqslant 2d_H(A, B)$.

As is easily observed, $w_n(A) \leqslant w_2(A) = \operatorname{diam} A$ for each $A \in \operatorname{Comp}(X)$ and $n > 2$. Then, a Whitney map $w : \operatorname{Comp}(X) \to \mathbb{R}_+$ can be defined as follows:

$$w(A) \underset{\text{def}}{=} \sum_{n=2}^{\infty} 2^{-n+1} w_n(A) \quad \text{for each } A \in \operatorname{Comp}(X),$$

where $w(A) \leqslant \operatorname{diam} A$ and the following Lipschitz condition holds:

$$|w(A) - w(B)| \leqslant 2d_H(A, B) \quad \text{for each } A, B \in \operatorname{Comp}(X).$$

Obviously, w satisfies (W2). Before verifying (W1), we show that $\lim_{n\to\infty} w_n(A) = 0$ for each $A \in \operatorname{Comp}(X)$. Otherwise, $\inf_{n>1} w_n(A) > r$ for some $r > 0$. Because A is totally bounded, there are $x_1, \ldots, x_n \in A$ such that $A \subset \bigcup_{i=1}^{n} \mathrm{B}(x_i, r/2)$. On the other hand, since $w_{n+1}(A) > r$, we have $a \in A^{n+1}$ such that $\min_{i\neq j} d(a(i), a(j)) > r$. Each $a(i)$ is contained in some $\mathrm{B}(x_{k(i)}, r/2)$. Then, $k(i) = k(j)$ for some $i \neq j$, that is, $a(i), a(j) \in B(x_{k(i)}, r/2)$, which contradicts $d(a(i), a(j)) > r$.

To verify (W1), let $A, B \in \operatorname{Comp}(X)$ with $A \subsetneqq B$. Then, note $w(B) > 0$. When $\operatorname{card} B = m < \infty$, since $w_m(A) = 0 < w_m(B)$, it follows that $w(A) < w(B)$. Now, consider the case when B is infinite. Evidently, $w_n(A) \leqslant w_n(B)$ for each $n \geqslant 2$, which implies $w(A) \leqslant w(B)$. Assume that $w(A) = w(B)$. Since B has infinitely many points, we have $w_n(A) = w_n(B) > 0$ for every $n \geqslant 2$. Take $x \in B \setminus A$, where $d(x, A) > 0$. Because $\lim_{n\to\infty} w_n(B) = 0$ as just seen above, we can find $n \geqslant 2$ such that $d(x, A) > w_n(B) > w_{n+1}(B)$. Since $w_n(A) = w_n(B)$, we can take $a \in A^n$ so that

$$w_{n+1}(B) < \min\left\{d(a(i), a(j)) \;\middle|\; i \neq j\right\} \leqslant w_n(A).$$

Let $b = (a, x) \in B^n \times B = B^{n+1}$. Since $d(a(i), x) \geqslant d(b, A) > w_n(A)$ for every $i = 1, \ldots, n$, it follows that

$$\min\left\{d(a(i), a(j)) \;\middle|\; i \neq j\right\} = \min\left\{d(b(i), b(j)) \;\middle|\; i \neq j\right\} \leqslant w_{n+1}(B),$$

which is a contradiction. Consequently, $w(A) < w(B)$. □

Remark 6.14.1 In the above proof,

$$w_n(A) = \sup\left\{r \in \mathbb{R}_+ \;\middle|\; \exists a_1, \ldots, a_n \in A \text{ such that } i \neq j \Rightarrow d(a_i, a_j) \geqslant r\right\}.$$

To see this equality, refer to Footnote 21.

Remark 6.14.2 One of the simplest constructions of a Whitney map $w : \mathrm{Cld}(X) \to \mathbf{I}$ for a separable metrizable space X is as follows: Let $\mathcal{B}$ be a countable open base for X. For each pair $U, V \in \mathcal{B}$ with $\mathrm{cl}\, U \subset V$, take a map $f_{U,V} : X \to \mathbf{I}$ so that $f_{U,V}(\mathrm{cl}\, U) = \{0\}$ and $f_{U,V}(X \setminus V) = \{1\}$. Enumerate these functions as $(f_n)_{n\in\mathbb{N}}$, that is,

$$\left\{f_n \;\middle|\; n \in \mathbb{N}\right\} = \left\{f_{U,V} \;\middle|\; U, V \in \mathcal{B},\ \mathrm{cl}\, U \subset V\right\}.$$

Then, a Whitney map $w : \mathrm{Cld}(X) \to \mathbf{I} \subset \mathbb{R}_+$ can be defined as follows:

$$w(A) = \sum_{n=1}^{\infty} 2^{-n} \operatorname{diam} f_n(A) \quad \text{for each } A \in \mathrm{Cld}(X).$$

Concerning Whitney maps for compact hyperspaces, we have the following:

Lemma 6.14.2 *Let $X = (X, d)$ be a metric space and $w : \mathcal{H} \to \mathbb{R}_+$ be a Whitney map for a compact subspace $\mathcal{H}$ of* $\mathrm{Comp}(X)$. *Then, for each $\varepsilon > 0$, there is $\delta > 0$ such that for $A, B \in \mathcal{H}$,*

$$A \subset B,\ w(B) - w(A) < \delta \Rightarrow d_H(A, B) < \varepsilon.$$

Proof Assume that the lemma does not hold. Then, there exists $\varepsilon_0 > 0$, $A_n, B_n \in \mathcal{H}$, $n \in \mathbb{N}$, such that $A_n \subset B_n$, $w(B_n) - w(A_n) \to 0$ as $n \to \infty$, and $d_H(A_n, B_n) \geqslant \varepsilon_0$. Since $\mathcal{H}$ is compact, we may assume that A_n and B_n converge to A_0 and B_0 in $\mathcal{H}$, respectively. Then, it follows that $A_0 \subset B_0$ and $w(A_0) = w(B_0)$, which implies $A_0 = B_0$ by (W1). Hence, $d_H(A_n, B_n)$ converges to $d_H(A_0, B_0) = 0$, which is a contradiction. Thus, the lemma holds. □

Lemma 6.14.3 *Let $\mathcal{A}$ be a collection of closed sets in a space X such that at least one $A_0 \in \mathcal{A}$ is compact and $\bigcap \mathcal{A} \neq \emptyset$. Then, for each open neighborhood U of $\bigcap \mathcal{A}$ in X, there are finitely many $A_1, \ldots, A_n \in \mathcal{A}$ such that $\bigcap_{i=1}^n A_i \subset U$.*

Proof When $A_0 \subset U$, there is nothing to do. When $A_0 \not\subset U$, $A_0 \setminus U$ is compact and

$$\bigcup_{A\in\mathcal{A}} (A_0 \setminus A) = A_0 \cap \bigcup_{A\in\mathcal{A}} (X \setminus A) = A_0 \cap \left(X \setminus \bigcap \mathcal{A}\right) \supset A_0 \setminus U.$$

Hence, there are finitely many $A_1, \ldots, A_n \in \mathcal{A}$ such that $\bigcup_{i=1}^n (A_0 \setminus A_i) \supset A_0 \setminus U$, which means $\bigcap_{i=0}^n A_i \subset U$. □

The intersection of all clopen sets in X containing a point x is called the **quasi-component** of x in X. Evidently, quasi-components are closed in X.

Proposition 6.14.4 *The component of a space X containing $x \in X$ is contained in the quasi-component of x in X. When X is compact, the quasi-component of x in X is a component of X.*

Proof Let C be a component of X with $x \in C$. Since C is connected, C is contained in every clopen set A in X with $x \in A$, that is, C is contained in the quasi-component of x in X.

To see the additional statement in the case when X is compact, let Q be the quasi-component of x in X. It suffices to show that Q is connected. Let P and R be disjoint closed sets in Q with $Q = P \cup R$, where $x \in P$. Since P and R are closed in X and X is normal, we have disjoint open sets U and V in X such that $P \subset U$ and $R \subset V$. Then, $Q \subset U \cup V$, $x \in U$, and $U \cap V = \emptyset$. Since X is compact, we can apply Lemma 6.14.3 to obtain finitely many clopen sets $A_1, \ldots, A_n$ in X such that $x \in \bigcap_{i=1}^{n} A_i \subset U \cup V$. Then,

$$\bigcap_{i=1}^{n} A_i \cap \operatorname{cl} U = \bigcap_{i=1}^{n} A_i \cap (U \cup V) \cap \operatorname{cl} U = \bigcap_{i=1}^{n} A_i \cap U,$$

which is clopen in X and contains x. Hence, $Q \subset \bigcap_{i=1}^{n} A_i \cap U \subset U$, which means $V \cap Q = \emptyset$. Thus, it follows that Q is connected. □

By Lemma 6.14.3 and Proposition 6.14.4, we have the following:

Corollary 6.14.5 *Let C be a component of a compact space X. For each open neighborhood U of C in X, there is a clopen set H in X such that $C \subset H \subset U$.* □

Remark 6.14.3 When X is non-compact, the quasi-component of x in X is not a component of X in general. Consider the following subspace X of Euclidean plane $\mathbb{R}^2$:

$$X = (\{0\} \times \partial \mathbf{I}) \cup \bigcup_{n \in \mathbb{N}} \{n^{-1}\} \times \mathbf{I} \subset \mathbb{R}^2.$$

The quasi-component of $\mathbf{0}$ in X is $\{0\} \times \partial \mathbf{I}$, which is disconnected.

For a subspace $\mathcal{H}$ of $\operatorname{Comp}(X)$, let $w : \mathcal{H} \to \mathbb{R}_+$ be a Whitney map. Given $A, B \in \mathcal{H}$ with $A \subset B$, a path $\alpha : \mathbf{I} \to \mathcal{H}$ is called a ***w*-segment** from A to B if α has the following properties:

(s1) $\alpha(0) = A$ and $\alpha(1) = B$;
(s2) $w(\alpha(t)) = (1 - t)w(A) + tw(B)$ for each $t \in \mathbf{I}$;
(s3) $t_1 \leqslant t_2 \Rightarrow \alpha(t_1) \subset \alpha(t_2)$.

The following should be remarked:

- When $A \subsetneqq B$, α is non-constant and injective, i.e., an arc.

For a compactum X with a Whitney map $w : \operatorname{Comp}(X) \to \mathbb{R}_+$, the following gives a condition on A and B for the existence of a w-segment from A to B in $\operatorname{Comp}(X)$:

Theorem 6.14.6 *Let X be a compactum and $w : \mathrm{Comp}(X) \to \mathbb{R}_+$ a Whitney map. Given $A, B \in \mathrm{Comp}(X)$ with $A \subset B$, there exists a w-segment $\alpha : \mathbf{I} \to \mathrm{Comp}(X)$ from A to B if and only if every component of B meets A.*

Proof (*The "only if" Part*) Let $\alpha : \mathbf{I} \to \mathrm{Comp}(X)$ be a w-segment from A to B. Then, $A = \alpha(0) \subset \alpha(t) \subset \alpha(1) = B$ for each $t \in \mathbf{I}$. Suppose that B has a component B_0 with $B_0 \cap A = \emptyset$. Applying Corollary 6.14.5, we can obtain a clopen set C in B such that $B_0 \subset C \subset B \setminus A$. Then, there are disjoint open sets U and V in X so that $B \setminus C \subset U$ and $C \subset V$, which implies that $A \subset U$, $B_0 \subset V$, and $B \subset U \cup V$. Observe that U^+ is an open neighborhood of $A = \alpha(0)$ in $\mathrm{Comp}(X)$, V^- is an open neighborhood of $B = \alpha(1)$ in $\mathrm{Comp}(X)$, and $U^+ \cap V^- = \emptyset$. Since $B \subset U \cup V$, it follows that $\alpha(\mathbf{I}) \subset \mathrm{Comp}(B) \subset U^+ \cup V^-$, which contradicts the connectedness of $\alpha(\mathbf{I})$.

(*The "if" Part*) Let $\mathcal{A}$ be a maximal totally ordered subset of $\mathrm{Comp}(X) = (\mathrm{Comp}(X), \subset)$ with the following property:

$(*)$ $A = \min \mathcal{A}$, $B = \max \mathcal{A}$, and every component of each $C \in \mathcal{A}$ meets A.

The existence of $\mathcal{A}$ is guaranteed by Zorn's Lemma.

> Indeed, let $\mathfrak{W} = (\mathfrak{W}, \subset)$ be a totally ordered collection consisting of all totally ordered subsets of $\mathrm{Comp}(X)$ with the property $(*)$. Evidently, $\{A, B\} \in \mathfrak{W}$, so $\mathfrak{W} \neq \emptyset$. Observe that $\bigcup \mathfrak{W}$ is also a totally ordered subset of $\mathrm{Comp}(X)$ satisfying $(*)$.

We will prove that $\mathcal{A}$ is closed in $\mathrm{Comp}(X)$, i.e., $\mathrm{cl}\,\mathcal{A} = \mathcal{A}$. By the maximality of $\mathcal{A}$, it suffices to show that $\mathrm{cl}\,\mathcal{A}$ is totally ordered and has the above property $(*)$.

First, we show that $\mathrm{cl}\,\mathcal{A}$ is totally ordered, that is, $C_1 \subset C_2$ or $C_2 \subset C_1$ for each $C_1, C_2 \in \mathrm{cl}\,\mathcal{A}$. If $C_1 \not\subset C_2$ and $C_2 \not\subset C_1$, then we have $x_1 \in C_1 \setminus C_2$ and $x_2 \in C_2 \setminus C_1$. Then, we have open sets U_i, V_i, $i = 1, 2$, in X such that

$$x_i \in U_i, \ C_i \subset V_i, \ \text{ and } \ U_i \cap V_{3-i} = \emptyset.$$

Since $U_i^- \cap V_i^+$ is an open neighborhood of C_i in $\mathrm{Comp}(X)$, we can take $D_i \in U_i^- \cap V_i^+ \cap \mathcal{A}$. Then, $D_i \setminus V_{3-i} \supset D_i \cap U_i \neq \emptyset$ and $D_i \subset V_i$. Hence, $D_i \setminus D_{3-i} \neq \emptyset$, that is, $D_1 \not\subset D_2$ and $D_2 \not\subset D_1$, which contradicts the fact that $\mathcal{A}$ is totally ordered.

To show $A = \min \mathrm{cl}\,\mathcal{A}$ and $B = \max \mathrm{cl}\,\mathcal{A}$, it suffices to show that $A \subset C \subset B$ for each $C \in \mathrm{cl}\,\mathcal{A}$. If $A \not\subset C$, then we have $a \in A \setminus C$. Since $(X \setminus \{a\})^+$ is an open neighborhood of C in $\mathrm{Comp}(X)$, we have $C' \in (X \setminus \{a\})^+ \cap \mathcal{A}$. Then, $C' \subset X \setminus \{a\}$, i.e., $a \notin C'$, which contradicts $A = \min \mathcal{A}$. If $C \not\subset B$, then $C \setminus B \neq \emptyset$. Since $(X \setminus B)^-$ is an open neighborhood of C in $\mathrm{Comp}(X)$, we have $C'' \in (X \setminus B)^- \cap \mathcal{A}$. Then, $C'' \not\subset B$, which contradicts $B = \max \mathcal{A}$.

Assume that $\mathrm{cl}\,\mathcal{A}$ does not have the property $(*)$, that is, there is a component C_0 of some $C \in \mathrm{cl}\,\mathcal{A}$ which does not meet A. Then, C has a clopen set H such that $C_0 \subset H \subset C \setminus A$. Take disjoint open sets U_1, U_2 in X so that $C \setminus H \subset U_1$ and $H \subset U_2$. Since $\mathrm{V}(U_1, U_2)$ $(= (U_1 \cup U_2)^+ \cap U_1^- \cap U_2^-)$ is a neighborhood of C in $\mathrm{Comp}(X)$, we have $D \in \mathrm{V}(U_1, U_2) \cap \mathcal{A}$. Then, D has a component D_0 such that $D_0 \subset U_2$ because $D \subset U_1 \cup U_2$ and $D \cap U_i \neq \emptyset$, $i = 1, 2$. Since $A \subset C \setminus H \subset U_1$,

it follows that $D_0 \cap A = \emptyset$, which is a contradiction. Thus, every component of $C \in \operatorname{cl}\mathcal{A}$ meets A.

Now, we will prove that $w(\mathcal{A}) = [w(A), w(B)]$. Since $A = \min\mathcal{A}$ and $B = \max\mathcal{A}$, it follows that $w(\mathcal{A}) \subset [w(A), w(B)]$. Suppose that there exists some $w(A) < t_0 < w(B)$ such that $t_0 \notin w(\mathcal{A})$. As observed in the above, $\mathcal{A}$ is closed in $\operatorname{Comp}(X)$, where $\operatorname{Comp}(X)$ is a compactum because so is X (Theorem 2.12.8). Thus, $\mathcal{A}$ is compact, hence so is $w(\mathcal{A})$. Then, we have $A_1, A_2 \in \mathcal{A}$ such that

$$w(A_1) = \max\big(w(\mathcal{A}) \cap [w(A), t_0)\big) \text{ and } w(A_2) = \min\big(w(\mathcal{A}) \cap (t_0, w(B)]\big).$$

Since $\mathcal{A}$ is totally ordered, it follows that $A \subset A_1 \subsetneqq A_2 \subset B$. Take a point $x \in A_2 \setminus A_1$. We have an open neighborhood U of A_1 in X such that $x \notin \operatorname{cl} U$. The component C_x of A_2 containing x meets A. Then, $C_x \cap \operatorname{cl} U$ has a component C_0 which meets both A and $\operatorname{bd} U$. Otherwise, using Corollary 6.14.5 and the compactness of $A \cap C_x$, we can easily obtain finitely many clopen sets $H_1, \ldots, H_n$ in $C_x \cap \operatorname{cl} U$ such that

$$A \cap C_x \subset H_1 \cup \cdots \cup H_n \subset U.$$

Then, $H_1 \cup \cdots \cup H_n$ is a clopen proper subset of C_x, which contradicts the connectedness of C_x. Let $A_3 = A_1 \cup C_0 \in \operatorname{Comp}(X)$. Then, $A_1 \subsetneqq A_3 \subsetneqq A_2$ and every component of A_3 meets A. Hence, $\mathcal{A}' = \mathcal{A} \cup \{A_3\}$ is totally ordered and $A = \min\mathcal{A}'$ and $B = \max\mathcal{A}'$, which contradicts the maximality of $\mathcal{A}$.

Since $\mathcal{A}$ is totally ordered, the restriction $w|\mathcal{A}$ is injective by (W1). Since $\mathcal{A}$ is compact, $w|\mathcal{A} : \mathcal{A} \to [w(A), w(B)]$ is homeomorphism. Thus, we can define a w-segment $\alpha : \mathbf{I} \to \mathcal{A} \subset \operatorname{Comp}(X)$ from A to B as follows:

$$\alpha(t) = (w|\mathcal{A})^{-1}((1-t)w(A) + tw(B)) \text{ for each } t \in \mathbf{I}.$$

This completes the proof. □

In Theorem 6.14.6, when $A \in \operatorname{Cont}(X)$, if every component of B meets A, then $B \in \operatorname{Cont}(X)$ and the set $\mathcal{A}$ in the above proof is contained in $\operatorname{Cont}(X)$. Thus, we have the following:

Corollary 6.14.7 *Let X be a compactum and $w : \operatorname{Cont}(X) \to \mathbb{R}_+$ a Whitney map. Given $A, B \in \operatorname{Cont}(X)$ with $A \subset B$, there exists a w-segment $\alpha : \mathbf{I} \to \operatorname{Cont}(X)$ from A to B.* □

Recall that each map $f : X \to Y$ induces the map $\hat{f} : \mathfrak{P}_0(X) \to \mathfrak{P}_0(Y)$ defined by $\hat{f}(A) = f(A) = \{f(a) \mid a \in A\}$, where $\mathfrak{P}_0(X) = \mathfrak{P}(X) \setminus \{\emptyset\}$ (cf. the last part of Sect. 2.12). The following maps are restrictions of $\hat{f}$ but written by using the same notations:

$$\hat{f} : \operatorname{Comp}(X) \to \operatorname{Comp}(Y), \quad \hat{f} : \operatorname{Cont}(X) \to \operatorname{Cont}(Y).$$

Concerning homotopies, we have the following:

Lemma 6.14.8 *A homotopy $h : X \times \mathbf{I} \to Y$ induces homotopies:*

$$\hat{h} : \mathrm{Comp}(X) \times \mathbf{I} \to \mathrm{Comp}(Y),\ \bar{h} : X \times \mathbf{I} \to \mathrm{Cont}(Y),$$

which are defined as follows:[22]

$$\hat{h}(A,t) \underset{def}{=} h_t(A) = h(A \times \{t\}) = \big\{h(a,t) \bigm| a \in A\big\},$$
$$\bar{h}(x,t) \underset{def}{=} h(\{x\} \times [0,t]) = \big\{h(x,s) \bigm| s \leqslant t\big\}.$$

Proof Obviously, $\hat{h}$ and $\bar{h}$ are well-defined. It is necessary to verify their continuity.[23] Let U be an open set in Y. It suffices to show that $\hat{h}^{-1}(U^+_{\mathrm{Comp}})$ and $\hat{h}^{-1}(U^-_{\mathrm{Comp}})$ are open in $\mathrm{Comp}(X) \times \mathbf{I}$ and that $\bar{h}^{-1}(U^+_{\mathrm{Cont}})$ and $\bar{h}^{-1}(U^-_{\mathrm{Cont}})$ are open in $X \times \mathbf{I}$.

To see that $\hat{h}^{-1}(U^+_{\mathrm{Comp}})$ is open in $\mathrm{Comp}(X) \times \mathbf{I}$, let $(A,t) \in \hat{h}^{-1}(U^+_{\mathrm{Comp}})$. Then, $h^{-1}(U)$ is open in $X \times \mathbf{I}$ and $A \times \{t\} \subset h^{-1}(U)$. Since A is compact, we can take an open neighborhood V of A in X and a neighborhood W of t in $\mathbf{I}$ so that $V \times W \subset h^{-1}(U)$. Then, $V^+_{\mathrm{Comp}} \times W$ is a neighborhood of (A,t) in $\mathrm{Comp}(X) \times \mathbf{I}$. As is easily observed, $V^+_{\mathrm{Comp}} \times W \subset \hat{h}^{-1}(U^+_{\mathrm{Comp}})$. Therefore, $\hat{h}^{-1}(U^+_{\mathrm{Comp}})$ is open in $\mathrm{Comp}(X) \times \mathbf{I}$.

To see that $\hat{h}^{-1}(U^-_{\mathrm{Comp}})$ is open in $\mathrm{Comp}(X) \times \mathbf{I}$, let $(A,t) \in \hat{h}^{-1}(U^-_{\mathrm{Comp}})$, i.e., $h(a,t) \in U$ for some $a \in A$. We have an open neighborhood V of a in X and a neighborhood W of t in $\mathbf{I}$ such that $h(V \times W) \subset U$. Then, $V^-_{\mathrm{Comp}} \times W$ is a neighborhood of (A,t) in $\mathrm{Comp}(X) \times \mathbf{I}$. As is easily observed, $V^-_{\mathrm{Comp}} \times W \subset \hat{h}^{-1}(U^-_{\mathrm{Comp}})$. Hence, $\hat{h}^{-1}(U^-_{\mathrm{Comp}})$ is open in $\mathrm{Comp}(X) \times \mathbf{I}$.

To see that $\bar{h}^{-1}(U^+_{\mathrm{Cont}})$ is open in $X \times \mathbf{I}$, let $(x,t) \in \bar{h}^{-1}(U^+_{\mathrm{Cont}})$. Then, $h^{-1}(U)$ is open in $X \times \mathbf{I}$ and $\{x\} \times [0,t] \subset h^{-1}(U)$. Since $[0,t]$ is compact, we can find an open neighborhood V of x in X and $t \leqslant t_0 \leqslant 1$ such that $[0,t_0]$ is a neighborhood of t in $\mathbf{I}$ and $V \times [0,t_0] \subset h^{-1}(U)$. Then, $V \times [0,t_0]$ is a neighborhood of (x,t) in $X \times \mathbf{I}$. As is easily observed, $V \times [0,t_0] \subset \bar{h}^{-1}(U^+_{\mathrm{Cont}})$. Hence, $\bar{h}^{-1}(U^+_{\mathrm{Cont}})$ is open in $X \times \mathbf{I}$.

To see that $\bar{h}^{-1}(U^-_{\mathrm{Cont}})$ is open in $X \times \mathbf{I}$, let $(x,t) \in \bar{h}^{-1}(U^-_{\mathrm{Cont}})$, i.e., $h(x,s) \in U$ for some $0 \leqslant s \leqslant t$. We have an open neighborhood V of x in X and $0 \leqslant s_0 < s_1 \leqslant 1$ such that $[s_0,s_1]$ is a neighborhood of s in $\mathbf{I}$ and $h(V \times [s_0,s_1]) \subset U$. Then, $V \times [s_0,1]$ is a neighborhood of (x,t) in $X \times \mathbf{I}$. Observe that

$$h(x',s_0) \in h(\{x'\} \times [0,t']) \cap U \quad \text{for each } (x',t') \in V \times [s_0,1],$$

which means that $\bar{h}(x',t') \in U^-_{\mathrm{Cont}}$ for each $(x',t') \in V \times [s_0,1]$, i.e., $V \times [s_0,1] \subset \bar{h}^{-1}(U^-_{\mathrm{Cont}})$. Thus, $\bar{h}^{-1}(U^-_{\mathrm{Cont}})$ is open in $X \times \mathbf{I}$. □

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[22] Each $(\hat{h})_t$ is equal to the map induced by h_t, i.e., $(\hat{h})_t = (h_t)\hat{}$, which is denoted by $\hat{h}_t$.

[23] It should be noted that the continuity of $\hat{h}_t : \mathrm{Comp}(X) \to \mathrm{Comp}(Y)$, $t \in \mathbf{I}$, does not imply the continuity of $\hat{h}$.

For a compactum X, the following continuous union operator was obtained in Proposition 2.12.15:

$$u_{\mathrm{Comp}} : \mathrm{Comp}(\mathrm{Comp}(X)) \to \mathrm{Comp}(X).$$

We can restate Lemma 2.12.14 as follows:

- Let X be a compactum. If $\mathcal{A} \in \mathrm{Cont}(\mathrm{Comp}(X))$ and $\mathcal{A} \cap \mathrm{Cont}(X) \neq \emptyset$, then $u_{\mathrm{Comp}}(\mathcal{A}) \in \mathrm{Cont}(X)$.

Using the union operator u_{Comp}, we can prove the following theorem concerning contractibility of hyperspaces:

Theorem 6.14.9 *For a compactum X, the following are equivalent:*

(a) $F_1(X)$ *is contractible in* $\mathrm{Comp}(X)$*;*
(b) $\mathrm{Comp}(X)$ *is contractible;*
(c) $\mathrm{Cont}(X)$ *is contractible.*

Proof Since (c) $\Rightarrow$ (a) is trivial, we prove the implications (a) $\Rightarrow$ (b) $\Rightarrow$ (c).

(a) $\Rightarrow$ (b): By (a), we have a homotopy $h : X \times \mathbf{I} \to \mathrm{Comp}(X)$ such that $h_0(x) = \{x\}$ for each $x \in X$ and h_1 is a constant map, that is, $h_1(x) = A_0 \in \mathrm{Comp}(X)$ for every $x \in X$. Due to Lemma 6.14.8, h induces a homotopy

$$\hat{h} : \mathrm{Comp}(X) \times \mathbf{I} \to \mathrm{Comp}(\mathrm{Comp}(X)).$$

Composing this $\hat{h}$ with u_{Comp}, we can obtain a homotopy

$$\tilde{h} = u_{\mathrm{Comp}}\hat{h} : \mathrm{Comp}(X) \times \mathbf{I} \to \mathrm{Comp}(X),$$

that is, $\tilde{h}(A, t) = \bigcup_{a \in A} h_t(a)$ for $A \in \mathrm{Comp}(X)$ and $t \in \mathbf{I}$. For each $A \in \mathrm{Comp}(X)$,

$$\tilde{h}_0(A) = \bigcup_{a \in A} \{a\} = A \quad \text{and} \quad \tilde{h}_1(A) = \bigcup_{a \in A} h_1(a) = A_0.$$

Hence, $\mathrm{Comp}(X)$ is contractible.

(b) $\Rightarrow$ (c): Because $\mathrm{Cont}(X) \subset \mathrm{Comp}(X)$, it follows from (b) that $\mathrm{Cont}(X)$ is contractible in $\mathrm{Comp}(X)$, that is, there is a homotopy $h : \mathrm{Cont}(X) \times \mathbf{I} \to \mathrm{Comp}(X)$ such that $h_0 = \mathrm{id}$ and h_1 is a constant map. Moreover, since $\mathrm{Comp}(X)$ is path-connected by (b), we may assume that $h_1(A) = X$ for each $A \in \mathrm{Comp}(X)$. Due to Lemma 6.14.8, h induces a homotopy

$$\bar{h} : \mathrm{Cont}(X) \times \mathbf{I} \to \mathrm{Cont}(\mathrm{Comp}(X)) \subset \mathrm{Comp}(\mathrm{Comp}(X)).$$

Since $\bar{h}_t(A) \cap \mathrm{Cont}(X) \ni h_0(A) = A$ for $A \in \mathrm{Cont}(X)$ and $t \in \mathbf{I}$, we have a homotopy

$$\tilde{h} = u_{\mathrm{Comp}}\bar{h} : \mathrm{Cont}(X) \times \mathbf{I} \to \mathrm{Cont}(X),$$

where $\tilde{h}_0(A) = h_0(A) = A$ and $\tilde{h}_1(A) = \bigcup_{t\in\mathbf{I}} h_t(A) = X$ because $h_1(X) = X$. Therefore, Cont(X) is contractible. □

Theorem 6.14.10 *Let $X = (X, d)$ be a metric continuum. Suppose the following condition:*

(KP) *For each $\varepsilon > 0$, there exists $\delta > 0$ such that*

$$a, b \in X,\ d(a, b) < \delta,\ a \in A \in \mathrm{Cont}(X) \Rightarrow\ \exists B \in \mathrm{Cont}(X)\ \text{such that}\ b \in B,\ d_H(A, B) < \varepsilon.$$

Then, Comp(X) *and* Cont(X) *are contractible.*

Proof Due to Theorem 6.14.9. it suffices to show that $F_1(X)$ is contractible in Comp(X). By Theorem 6.14.1, we have a Whiteney map $w : \mathrm{Cont}(X) \to \mathbb{R}_+$. Multiplying w by $w(X)^{-1}$, we may assume that $w(X) = 1$. For each $t \in \mathbf{I}$ and $x \in X$, since $w^{-1}(t)$ is closed in Cont(X), it follows that

$$w^{-1}(t)[x] = w^{-1}(t) \cap \left(\mathrm{Comp}(\mathrm{Cont}(X)) \setminus (X \setminus \{x\})^+\right) \in \mathrm{Comp}(\mathrm{Cont}(X)).$$

Hence, we have a homotopy $h : X \times \mathbf{I} \to \mathrm{Comp}(\mathrm{Cont}(X))$ defined by

$$h(x, t) = w^{-1}(t)[x] = \left\{A \in \mathrm{Cont}(X) \;\middle|\; x \in A,\ w(A) = t\right\}.$$

Observe that $h_0(x) = \{\{x\}\}$ and $h_1(x) = \{X\}$ for each $x \in X$.

To verify the continuity of h, let $\varepsilon > 0$. By Lemma 6.14.2 and the uniform continuity of w, we have $0 < \eta' < \eta < \varepsilon/2$ such that the following hold for $A, B \in$ Comp(X):

$$A \subset B,\ w(B) - w(A) < \eta \ \Rightarrow\ d_H(A, B) < \varepsilon/2,\ \text{and}$$
$$d_H(A, B) < \eta' \ \Rightarrow\ |w(A) - w(B)| < \eta/2.$$

Take $\delta > 0$ satisfying the condition (KP) for $\eta' > 0$. Assume that $d(x, y) < \delta$ and $|t - s| < \eta/2$. For each $A \in h(x, t)$, since $x \in A \in \mathrm{Cont}(X)$, we have $B' \in \mathrm{Cont}(X)$ such that $y \in B'$ and $d_H(A, B') < \eta'$ $(< \varepsilon/2)$. Then,

$$|t - w(B')| = |w(A) - w(B')| < \eta/2.$$

When $w(B') = s$, let $B = B' \in h(y, s)$. When $w(B') < s$ (resp. $w(B') > s$), we apply Corollary 6.14.7 to obtain a w-segment $\alpha_y : \mathbf{I} \to \mathrm{Cont}(X)$ from B' to X (resp. from $\{y\}$ to B'). Then, we can find $B \in \mathrm{Cont}(X)$ with $B' \subset B$ (resp. $y \in B \subset B'$) and $w(B) = s$, which implies $B \in h(y, s)$. Moreover, $d_H(B, B') < \varepsilon/2$ because

$$|w(B) - w(B')| = |s - t| + |t - w(B')| < \eta.$$

Thus, we have $d_H(A, B) < \varepsilon$. Hence, $d_H(A, h(y, s)) < \varepsilon$ for each $A \in h(x, t)$. Similarly, it follows that $d_H(B, h(x, t)) < \varepsilon$ for each $B \in h(y, s)$. Consequently, we have $(d_H)_H(h(x, t), h(y, s)) < \varepsilon$.

Now, note that $\bigcup\{A \in \mathrm{Cont}(X) \mid x \in A, w(A) = t\}$ is connected for each $x \in X$ and $t \in \mathbf{I}$. Thus, we have a homotopy $h^* = u_{\mathrm{Comp}}h : X \times \mathbf{I} \to \mathrm{Cont}(X)$, where $h_0^*(x) = \{x\}$ and $h_1^*(x) = X$ for each $x \in X$. Therefore, $F_1(X)$ is contractible in $\mathrm{Comp}(X)$. □

Corollary 6.14.11 *For every Peano continuum X,* $\mathrm{Comp}(X)$ *and* $\mathrm{Cont}(X)$ *are contractible.*

Proof Given $d \in \mathrm{Metr}(X)$, it suffices to show that $X = (X, d)$ satisfies (KP) in Theorem 6.14.10. Let $\varepsilon > 0$. Since X is locally connected, each $x \in X$ has a connected closed neighborhood N_x in X with $\operatorname{diam} N_x < \varepsilon$. Because of the compactness of X, we have a Lebesgue number $\delta > 0$ for the cover $\{\operatorname{int} N_x \mid x \in X\}$. Let $a, b \in X$, $d(a, b) < \delta$, and $a \in A \in \mathrm{Cont}(X)$. Then, $a, b \in N_x$ for some $x \in X$. Let $B = A \cup N_x$. Then, $b \in B \in \mathrm{Cont}(X)$. Since $\operatorname{diam} N_x < \varepsilon$, it follows that $d_H(A, B) < \varepsilon$. □

Lemma 6.14.12 *Let $X = (X, d)$ be a metric compactum. For each compactum $\mathcal{A} \subset \mathrm{Comp}(X)$, $u_{\mathrm{Comp}}(\mathrm{Comp}(\mathcal{A})) = \{u_{\mathrm{Comp}}(C) \mid C \in \mathrm{Comp}(\mathcal{A})\}$ has the same diameter as $\mathcal{A}$, that is,*

$$\operatorname{diam}_{d_H} \mathcal{A} = \operatorname{diam}_{d_H} u_{\mathrm{Comp}}(\mathrm{Comp}(\mathcal{A})).$$

Proof Since $u_{\mathrm{Comp}}(\{A\}) = A$ for each $A \in \mathcal{A}$, we have $\mathcal{A} \subset u_{\mathrm{Comp}}(\mathrm{Comp}(\mathcal{A}))$, which implies

$$\operatorname{diam}_{d_H} \mathcal{A} \leqslant \operatorname{diam}_{d_H} u_{\mathrm{Comp}}(\mathrm{Comp}(\mathcal{A})).$$

To see the converse, let $\mathcal{C}, \mathcal{D} \in \mathrm{Comp}(\mathcal{A})$, where $\mathcal{C}, \mathcal{D} \subset \mathcal{A}$. Take $D \in \mathcal{D}$. Then,

$$\sup_{x \in u_{\mathrm{Comp}}(\mathcal{C})} d(x, u_{\mathrm{Comp}}(\mathcal{D})) \leqslant \sup_{C \in \mathcal{C}} \sup_{x \in C} d(x, D) \leqslant \sup_{C \in \mathcal{C}} d_H(C, D) \leqslant \operatorname{diam}_{d_H} \mathcal{A}.$$

Similarly, $\sup_{y \in u_{\mathrm{Comp}}(\mathcal{D})} d(y, u_{\mathrm{Comp}}(\mathcal{C})) \leqslant \operatorname{diam}_{d_H} \mathcal{A}$. Hence, we have

$$d_H(u_{\mathrm{Comp}}(\mathcal{C}), u_{\mathrm{Comp}}(\mathcal{D})) \leqslant \operatorname{diam}_{d_H} \mathcal{A}.$$

Thus, $\operatorname{diam}_{d_H}(u_{\mathrm{Comp}}(\mathrm{Comp}(\mathcal{A}))) \leqslant \operatorname{diam}_{d_H} \mathcal{A}$. □

Remark 6.14.4 In the above, since $\mathrm{Cont}(\mathcal{A}) \subset \mathrm{Comp}(\mathcal{A})$, we have

$$\operatorname{diam}_{d_H} \mathcal{A} = \operatorname{diam}_{d_H}(u_{\mathrm{Comp}}(\mathrm{Cont}(\mathcal{A}))) = \operatorname{diam}_{d_H}(u_{\mathrm{Comp}}(\mathrm{Comp}(\mathcal{A}))).$$

When $\mathcal{A}$ is a Peano continuum, $\mathrm{Comp}(\mathcal{A})$ and $\mathrm{Cont}(\mathcal{A})$ are contractible by Corollary 6.14.11, which implies that $\mathcal{A}$ is contractible in $\mathrm{Cont}(\mathcal{A})$ if $\mathcal{A}$ is identified with $F_1(\mathcal{A}) = \{\{A\} \mid A \in \mathcal{A}\}$.

Lemma 6.14.13 *For every Peano continuum X,* $\mathrm{Comp}(X)$ *and* $\mathrm{Cont}(X)$ *are locally path-connected (i.e., LC^0).*

Proof Let $d \in \mathrm{Metr}(X)$, $A \in \mathrm{Comp}(X)$ (or $\mathrm{Cont}(X)$), and $\varepsilon > 0$. Each $x \in X$ has a connected closed neighborhood N_x in X with $N_x \subset \mathrm{B}_d(x, \varepsilon/3)$. Since A is compact, we have $x_1, \ldots, x_n \in A$ such that $A \subset \bigcup_{i=1}^n \operatorname{int} N_{x_i}$. Then, $A^* = \bigcup_{i=1}^n N_{x_i}$ is a neighborhood of A in X, $A^* \in \mathrm{Comp}(X)$ (or $\mathrm{Cont}(X)$), and $d_H(A, A^*) < \varepsilon/3$. Since each component of A^* contains some N_{x_i}, each component of A^* meets A. Choose $\delta > 0$ so that $\mathrm{N}_d(A, \delta) \subset A^*$ and $\mathrm{B}_d(x_i, \delta) \subset N_{x_i}$ for each $i = 1, \ldots, n$. Note that $\delta \leqslant \varepsilon/3$. For each $B \in \mathrm{Comp}(X)$ (or $\mathrm{Cont}(X)$) with $d_H(A, B) < \delta$, since $B \subset \mathrm{N}_d(A, \delta)$, it follows that $B \subset A^*$. Moreover,

$$d_H(B, A^*) \leqslant d_H(B, A) + d_H(A, A^*) < 2\varepsilon/3.$$

For each $i = 1, \ldots, n$, since $d(x_i, B) \leqslant d_H(A, B) < \delta$, it follows that N_{x_i} meets B. Hence, each component of A^* meets B. By Theorem 6.14.6 (or Corollary 6.14.7), we have paths $\alpha, \beta : \mathbf{I} \to \mathrm{Comp}(X)$ (or $\mathrm{Cont}(X)$) such that $\alpha(0) = A$, $\beta(0) = B$, $\alpha(1) = \beta(1) = A^*$, $A \subset \alpha(t) \subset A^*$, and $B \subset \beta(t) \subset A^*$ for each $t \in \mathbf{I}$. Then,

$$\operatorname{diam}_{d_H} \alpha(\mathbf{I}) \leqslant d_H(A, A^*) < \varepsilon/3 \quad \text{and} \quad \operatorname{diam}_{d_H} \beta(\mathbf{I}) \leqslant d_H(B, A^*) < 2\varepsilon/3.$$

Joining α and β, we can obtain a path from A to B in $\mathrm{Comp}(X)$ (or $\mathrm{Cont}(X)$) with $\operatorname{diam}_{d_H} < \varepsilon$. □

Now, we prove the following WOJDYSAWSKI THEOREM:

Theorem 6.14.14 (WOJDYSAWSKI) *For a compactum X, the following are equivalent:*

(a) *X is a Peano continuum;*
(b) $\mathrm{Comp}(X)$ *is a Peano continuum;*
(b′) $\mathrm{Cont}(X)$ *is a Peano continuum;*
(c) $\mathrm{Comp}(X)$ *is an AR;*
(c′) $\mathrm{Cont}(X)$ *is an AR.*

Proof First, note that $\mathrm{Comp}(X)$ is a compactum by Theorem 2.12.8. Since $\mathrm{Cont}(X) = \mathrm{Conn}(X) \cap \mathrm{Comp}(X)$ is closed in $\mathrm{Comp}(X)$ by Proposition 2.12.1(7), it follows that $\mathrm{Cont}(X)$ is also a compactum. Then, (c) ⇒ (b) and (c′) ⇒ (b′) are trivial. By Lemma 6.14.13, we can obtain (a) ⇒ (b), (b′). We will show the implications (b) or (b′) ⇒ (a); (b) ⇒ (c); (b′) ⇒ (c′).

(b) or (b′) ⇒ (a): From Lemmas 2.12.12 and 2.12.13, it follows that $X = \bigcup \mathrm{Comp}(X)$ (or $= \bigcup \mathrm{Cont}(X)$) is a continuum. To see the local connectedness of X, let $x \in X$ and U an open neighborhood of x in X. It should be noted that $\{x\}$ has an open neighborhood basis in $\mathrm{Cld}(X)$ consisting of W^+, where W is an open neighborhood of x in X.

Indeed, for open sets $U_1, \dots, U_n \neq \emptyset$ in X, if $\{x\} \in \mathrm{V}(U_1, \dots, U_n)$, then

$$\{x\} \in (U_1 \cap \cdots \cap U_n)^+ \subset \mathrm{V}(U_1, \dots, U_n).$$

Since $\mathrm{Comp}(X)$ (or $\mathrm{Cont}(X)$) is locally path-connected by Theorem 5.13.5, we have an open neighborhood V of x in X such that $V \subset U$ and for each $y \in V$, there is a path $f : \mathbf{I} \to U^+_{\mathrm{Comp}}$ (or $f : \mathbf{I} \to U^+_{\mathrm{Cont}}$) from $\{x\}$ to $\{y\}$. Then, $\bigcup f(\mathbf{I}) = \bigcup_{t\in\mathbf{I}} f(t)$ is a continuum by Lemma 2.12.14 and $x, y \in \bigcup f(\mathbf{I}) \subset U$. Hence, X is locally connected.

(b) $\Rightarrow$ (c) (resp. (b′) $\Rightarrow$ (c′)): Note that X is a Peano continuum because (b) or (b′) $\Rightarrow$ (a). Since $\mathrm{Comp}(X)$ (resp. $\mathrm{Cont}(X)$) is contractible by Corollary 6.14.11, it suffices to show that $\mathrm{Comp}(X)$ (resp. $\mathrm{Cont}(X)$) is an ANR (Corollary 6.2.14). Due to Lefschetz's Characterization of ANRs (Theorem 6.6.1), it suffices to show that every open cover $\mathcal{U}$ of $\mathrm{Comp}(X)$ (resp. $\mathrm{Cont}(X)$) has a Lefschetz refinement. Since $\mathrm{Comp}(X)$ (resp. $\mathrm{Cont}(X)$) is compact, we have a Lebesgue number $\delta > 0$ for $\mathcal{U}$ with respect to the Hausdorff metric d_H defined by $d \in \mathrm{Metr}(X)$. Take an open cover $\mathcal{V}$ of $\mathrm{Comp}(X)$ (resp. $\mathrm{Cont}(X)$) with mesh $\mathcal{V} < \delta/3$. Since $\mathrm{Comp}(X)$ (resp. $\mathrm{Cont}(X)$) is LC^0 by Theorem 5.13.5, $\mathcal{V}$ has a 1-Lefschetz refinement $\mathcal{W}$ by Theorem 6.12.3. We show that an open cover $\mathcal{W}$ is a Lefschetz refinement of $\mathcal{U}$.

Let $f : |L| \to \mathrm{Comp}(X)$ (resp. $\mathrm{Cont}(X)$) be a partial $\mathcal{W}$-realization of a simplicial complex K, where L is a subcomplex of K with $K^{(0)} \subset L$. Since $f||L^{(1)}|$ is a partial $\mathcal{W}$-realization of $K^{(1)}$, $f||L^{(1)}|$ extends to a full $\mathcal{V}$-realization $f' : |K^{(1)}| \to \mathrm{Comp}(X)$ (resp. $\mathrm{Cont}(X)$). For simplicity, we write $L \cup K^{(n)} = K_n$ for each $n \in \mathbb{N}$. Then, a partial st$\mathcal{V}$-realization $\bar{f} : |K_1| \to \mathrm{Comp}(X)$ (resp. $\mathrm{Cont}(X)$) of K can be defined by $\bar{f}||L| = f$ and $\bar{f}||K^{(1)}| = f'$.

We define a map $f_1 : |K_1| \to \mathrm{Cont}(\mathrm{Comp}(X))$ (resp. $\mathrm{Cont}(\mathrm{Cont}(X))$) by $f_1(x) = \{\bar{f}(x)\}$ for each $x \in |K_1|$. We will inductively define maps $f_n : |K_n| \to \mathrm{Cont}(\mathrm{Comp}(X))$ (resp. $\mathrm{Cont}(\mathrm{Cont}(X))$), $n > 1$, so that $f_n||K_{n-1}| = f_{n-1}$ and

$(*)$ $f_n(\sigma \cap |K_n|) \subset \mathrm{Cont}(\bar{f}(\sigma \cap |K_1|))$ for each $\sigma \in K$,

where $\bar{f}(\sigma \cap |K_1|) \subset \mathrm{Comp}(X)$ is a Peano continuum by Proposition 5.13.2. Assume that f_{n-1} has been defined. Observe that each $\sigma \in K_n \setminus K_{n-1}$ is an n-simplex with $\partial\sigma = \sigma \cap |K_{n-1}|$ and $f_{n-1}(\partial\sigma) \subset \mathrm{Cont}(\bar{f}(\sigma \cap |K_1|))$. Since $\mathrm{Cont}(\bar{f}(\sigma \cap |K_1|))$ is contractible by Corollary 6.14.11, $f_{n-1}|\partial\sigma : \partial\sigma \to \mathrm{Cont}(\bar{f}(\sigma \cap |K_1|))$ can be extended to a map $f_\sigma : \sigma \to \mathrm{Cont}(\bar{f}(\sigma \cap |K_1|))$. Then, f_{n-1} extends to a map $f_n : |K_n| \to \mathrm{Cont}(\mathrm{Comp}(X))$ (resp. $\mathrm{Cont}(\mathrm{Cont}(X))$) defined by $f_n|\sigma = f_\sigma$ for each $\sigma \in K_n \setminus K_{n-1}$. We can show that f_n satisfies $(*)$ as follows: For $\sigma \in K \setminus K_n$,

$$\begin{aligned} f_n(\sigma \cap |K_n|) &= f_{n-1}(\sigma \cap |K_{n-1}|) \cup \bigcup_{\tau\in(K_n\setminus K_{n-1})\cap F(\sigma)} f_n(\tau) \\ &\subset \mathrm{Cont}(\bar{f}(\sigma \cap |K_1|)) \cup \bigcup_{\tau\in(K_n\setminus K_{n-1})\cap F(\sigma)} \mathrm{Cont}(\bar{f}(\tau \cap |K_1|) \\ &= \mathrm{Cont}(\bar{f}(\sigma \cap |K_1|)). \end{aligned}$$

Using the above f_n, $n \in \mathbb{N}$, we define a map $f_* : |K| \to \mathrm{Cont}(\mathrm{Comp}(X))$ (resp. $\mathrm{Cont}(\mathrm{Cont}(X))$) by $f_*||K_n| = f_n$ for each $n \in \mathbb{N}$. The map $\tilde{f} = uf_* : |K| \to \mathrm{Comp}(X)$ (resp. $\mathrm{Cont}(X)$) is an extension of f because $uf_*||K_1| = \bar{f}$ where $u = u_{\mathrm{Comp}}$ (resp. u_{Cont}). For each $\sigma \in K$ with $\dim \sigma = n$,

$$\tilde{f}(\sigma) = uf_n(\sigma \cap |K_n|) \subset u(\mathrm{Cont}(\bar{f}(\sigma \cap |K_1|)).$$

Then, applying Lemma 6.14.12, we have

$$\mathrm{diam}_{d_H}\, \tilde{f}(\sigma) \leqslant \mathrm{diam}_{d_H}\, u(\mathrm{Cont}(\bar{f}(\sigma \cap |K_1|)) = \mathrm{diam}_{d_H}\, \bar{f}(\sigma \cap |K_1|).$$

Since $\bar{f}$ is a partial $\mathrm{st}\,\mathcal{V}$-realization of K and $\mathrm{mesh}\,\mathrm{st}\,\mathcal{V} < \delta$, it follows that $\mathrm{diam}_{d_H}\, \tilde{f}(\sigma) < \delta$, which implies that $\tilde{f}(\sigma)$ is contained some member of $\mathcal{U}$. Hence, $\tilde{f}$ is a full $\mathcal{U}$-realization of K. Consequently, $\mathcal{W}$ is a Lefschetz refinement of $\mathcal{U}$. □

Notes for Chap. 6

For supplementary results and examples, refer to the following classical books by Borsuk and Hu. One can find interesting examples in Borsuk's book, Chapter VI. In Hu's book, ANEs are discussed not only for the class of metrizable spaces but also for more general classes.

- K. Borsuk, *Theory of Retracts*, MM **44** (Polish Sci. Publ., Warsaw, 1967)
- S.-T. Hu, *Theory of Retracts* (Wayne State Univ. Press, Detroit, 1965)

The theory has developed considerably since the publication of these books. Many important results have been gained, and many problems have been solved. Some of them have been treated in this chapter, while others have not. Many interesting examples can be seen in numerous existing articles. Especially, in 1994, Cauty constructed a metric linear space that is not an AR. This very important result will be proved in Sect. 7.12. For the history, we can refer to Mardešić's article [40].

Theorem 6.1.1 was established in [15]. Remarks 6.1.1 and 6.1.2 are taken from Michael [41]. Proposition 6.1.10 (6) is due to Hanner [22].

Theorem 6.2.1 was established in [4], but the short proof presented here is due to Toruńczyk [50]. For another short proof, see [43]. Concerning Remark 6.2.1, Wojdysławski proved in [56, Proposition 7] that the image of the isometry φ defined in Footnote 10 of Chap. 2 is closed in its convex hull (cf. Notes for Chap. 2). Theorem 6.2.2 was the trick used in [31]. Theorem 6.2.3 was established in [23] and the proof presented here is due to Toruńczyk [50]. The first assertion of Theorem 6.2.4 was proved in [13]. Theorem 6.2.7 and Lemma 6.2.8 were shown in [3]. Proposition 6.2.15 was remarked in [49] and the result in Remark 6.2.3 was also proved there.

The concept of LEC was introduced by Fox [19] and called ULC (an abbreviation for "Uniformly Locally Contractible") by Serre [48]. Theorem 6.3.3 was established in [17] and Theorem 6.3.5 was proved independently in [17, 26]. For conditions that LEC spaces are ANRs, refer to [12]. The notion of ULC defined in Sect. 6.3 was introduced by Ancel [2], where he used the term "uniformly locally contractible" but the word "uniformly" is here changed to "unified" because it is different from Serre's notion and evokes another concept.[24] The example for Theorem 6.3.9 was constructed in [8] as a locally contractible compactum that is not an ANR, and it was shown in [17] that this is not LEC. The example mentioned in Remark 6.3.1 was constructed in [6].

The first version of Theorem 6.4.1 is due to Borsuk [7]. Lemma 6.5.1 is proved by Dowker [14] in a more general setting (X is countably paracompact normal and A is a closed G_δ). Theorem 6.5.2 was

[24] See Footnote 6.

established in [36] and Theorem 6.5.3 was essentially proved in successive stages by [5, 22, 52]. Theorems 6.6.1 and 6.6.2 are due to Lefschetz [39] and Hanner [22], respectively. The present proof of the implication (c) $\Rightarrow$ (a) in Theorem 6.6.2 follows Dowker's idea [14]. Theorems 6.6.13 and 6.6.14 were proved by Dugundji [17]. In [42] (cf. [26]), the concept of convex structure was introduced, which was not discussed in this book but gives a characterization of ANRs.

Theorem 6.7.4 was established in [34] (cf. [18]). The compact case of Theorem 6.7.9 can be obtained as a corollary of the result in [11].

Theorem 6.8.1 was established by Nguyen To Nhu [45] and the proof presented here is due to Sakai [46]. The concept of a uniform retract (or a uniform retraction) was introduced by Michael [44]; it is called a *regular retract* (or *retraction*) in [51]. Theorem 6.8.11 was proved by Michael [44] and Toruńczyk [51], independently. For characterizations of (finite-dimensional) uniform ARs and ANRs, refer to [44].

Theorem 6.9.1 was conjectured by Geoghegan [20] and proved by Cauty [10]. The proof presented here is due to Sakai [47].

Theorem 6.10.1 was first proved in [24] for σ-compact spaces and then generalized in [21]. The property C was named in [25] and the definition is due to [1].

The examples of Theorem 6.11.1 and Remark 6.11.1 were given in Borsuk's book *Theory of Retracts*, Chap. 1, Sect. 17.

Theorem 6.12.3 and 6.12.5 (Corollary 6.12.6) were first proved for separavle metrizable spaces by Kuratowski [37] and then extended to non-separable metrizable spaces by Kodama [32] and Dugundji [16] (cf. [38] and [39]).

Theorem 6.13.2 was proved in [9]. The proof presented here is due to Kodama [33]. Theorem 6.13.3 was implicitly proved in [29].

A Whitney map was introduced by Whitney [53]. The construction in Remark 6.14.2 is due to Krasinkiewicz [35]. Other constructions can be seen in [27, 54], etc. Thus, there are various ways to construct Whitney maps. For Whitney maps, refer to the book by Illanes and Nadler [28]. Given a Whitney map w, Kelley [30] introduced a segment, which we call a w-segment because it is a concept with respect to w. He proved Theorem 6.14.6. Wojdysławski [55] proved the case $\mathrm{Comp}(X)$ of Corollary 6.14.11 and also established Theorem 6.14.14 in [56]. Our proofs of these results are based on Kelley's proofs given in [30].

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Chapter 7
Cell-Like Maps and Related Topics



A compact set $A \neq \emptyset$ in X is said to be **cell-like** in X if A is contractible in every neighborhood of A in X. A compactum X is **cell-like** if X is cell-like in some *metrizable* space that contains X as a subspace. It will be seen that X is cell-like in every ANR that contains X as a subspace (Theorem 7.1.2). A **cell-like** (**CE**) **map** is a perfect (surjective) map $f : X \to Y$ such that each fiber $f^{-1}(y)$ is cell-like. The concept of cell-like maps is very important in Geometric Topology. It has been mainly developed in Shape Theory and Decomposition Theory. For infinite-dimensional manifolds (in particular Hilbert cube manifolds), this concept is one of the main tools.

In the first six sections of this chapter, we will discuss several fundamental properties of cell-like maps and related concepts. The remainder of the chapter will be devoted to some associated topics. In Sect. 7.7, we construct a compactum X that is not cell-like but has a cell-like map onto the Hilbert cube $\boldsymbol{Q}$ and a cell-like map of $\boldsymbol{Q}$ onto a compactum that is not an ANR. Simplicial Eilenberg–MacLane complexes are constructed in Sect. 7.8 and used in Sect. 7.9 to define the cohomological dimension. Using the cohomological dimension, it can be proved that $\dim X \times P = \dim X + \dim P$ for every metrizable space X and every locally compact polyhedron P. In Sect. 7.10, we construct an infinite-dimensional compactum with finite cohomological dimension, which is an affirmative answer to the long-standing Alexandorff's Problem in Dimension Theory. It is also proved that this result positively solves the CE Problem, that is, there exists a cell-like map of a finite-dimensional compactum onto an infinite-dimensional compactum. In Sect. 7.12, using this example, we construct a linear metric space that is not an AR, which answers a long-standing problem in ANR Theory.

This chapter is based on results in Chaps. 2–6. Sect. 7.7 needs the Toruńczyk's characterization of the Hilbert cube. In Sects. 7.7 and 7.10, we will use some results from Algebraic Topology without proofs. Thus, these sections are not self-contained. Moreover, the construction of Sect. 7.12 requires an open cell-like map of a finite-dimensional compactum onto an infinite-dimensional compactum. By using Walsh's result on open maps, the cell-like map in Sect. 7.10 can be remade to be open. However, we will not give the proof of Walsh's result because it is beyond the scope of this book.

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7.1 Trivial Shape and Related Properties

In this section, we introduce some properties related to cell-like compacta. First, note that every contractible compactum is cell-like but that the converse is not true. The $\sin(1/x)$-curve

$$\{0\} \times [-1, 1] \cup \left\{ (x, y) \in \mathbb{R}^2 \;\middle|\; y = \sin \frac{1}{x},\ 0 < x \leqslant 1 \right\}$$

is a typical example of a non-contractible cell-like compactum. The cell-likeness of this compactum comes from the following:

Proposition 7.1.1 *Let $X_1 \supset X_2 \supset \cdots$ be a nested sequence of contractible non-empty compacta. Then, $X = \bigcap_{i \in \mathbb{N}} X_i$ is cell-like.*

Proof Every neighborhood U of X in X_1 contains some X_i. Since X_i is contractible, X is contractible in U. □

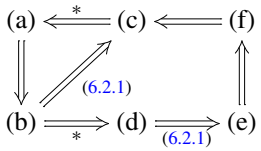
It is said that a non-empty closed set A in X has **property UV^*** in X (simply A is **UV^*** in X) if each neighborhood U of A in X contains a neighborhood V of A that is contractible in U. A non-empty metrizable space X is said to be **UV^*** if X is UV^* in some ANR that contains X as a closed subspace.[1] It is said that X has **trivial shape** if any map from X to an arbitrary ANR is null-homotopic; equivalently, any map from X to an arbitrary polyhedron is null-homotopic (Corollary 6.6.5).

Theorem 7.1.2 *For a metrizable space X ($\neq \emptyset$), the following are equivalent:*

(a) *X has trivial shape;*
(b) *X is contractible in every ANR that contains X as a closed set;*
(c) *X is contractible in every neighborhood in some metrizable space that contains X as a closed set;*
(d) *X is UV^* in any ANR that contains X as a closed set;*
(e) *X is UV^*;*
(f) *X is UV^* in some metrizable space that contains X as a closed set.*

In particular, a compactum is cell-like if and only if it has trivial shape.

Proof Consider the following diagram of implications. Every implication is trivial except for the two marked with asterisks.



[1] In other literature, **property UV^*** is called **property UV^∞**.

(c) $\Rightarrow$ (a): Assume that X is closed in a metrizable space M and X is contractible in every neighborhood in M. For any ANR Y, every map $f : X \to Y$ extends over an open neighborhood U of X in M. Let $\tilde{f} : U \to Y$ be an extension of f. There exists a contraction $h : X \times \mathbf{I} \to U$. Then, $\tilde{f}h : X \times \mathbf{I} \to Y$ is a homotopy such that $\tilde{f}h_0 = f$ and $\tilde{f}h_1$ is constant.

(b) $\Rightarrow$ (d): Let Y be an ANR containing X as a closed subspace. For each open neighborhood U of X in Y, we have a contraction $h : X \times \mathbf{I} \to U$ by (b). Since U is an ANR, we have a neighborhood V of X in U and a homotopy $\tilde{h} : V \times \mathbf{I} \to U$ such that $\tilde{h}_0 = \mathrm{id}$ and $\tilde{h}_1$ is a constant map. Thus, X is UV^* in Y. □

Due to Theorem 7.1.2, a UV^* compactum is equal to a cell-like compactum. The term "cell-like" is used only for compacta but "UV^*" is used without compactness.

Note When X is a compactum, the term "metrizable" in condition (c) can be replaced by the term "normal." Indeed, assume that X is contractible in every neighborhood in some normal space Y that contains X. We can regard X as a closed subspace of the Hilbert cube $\boldsymbol{Q} = [-1, 1]^{\mathbb{N}}$. Then, it suffices to show that X is contractible in every neighborhood U in $\boldsymbol{Q}$. By the coordinate-wise application of the Tietze Extension Theorem 2.2.2, we have a map $f : Y \to \boldsymbol{Q}$ with $f|X = \mathrm{id}$. Because $f^{-1}(U)$ is a neighborhood of X in Y, we have a contraction $h : X \times \mathbf{I} \to f^{-1}(U)$. Hence, $fh : X \times \mathbf{I} \to U$ is a contraction, that is, X is contractible in U.

Let $n \in \omega$. A non-empty closed set $A \subset X$ has **property $\boldsymbol{UV^n}$** in X (or simply A is $\boldsymbol{UV^n}$ in X) if each neighborhood U of A in X contains a neighborhood V of A such that, for each $0 \leqslant i \leqslant n$, every map $f : \mathbf{S}^i \to V$ is null-homotopic in U. We say that A has **property $\boldsymbol{UV^\infty}$** in X (or A is $\boldsymbol{UV^\infty}$ in X) if it is UV^n in X for every $n \in \omega$. A non-empty metrizable space X is said to be $\boldsymbol{UV^n}$ (resp. $\boldsymbol{UV^\infty}$) if X is UV^n (resp. UV^∞) in some ANR that contains X as a closed subspace.[2]

Proposition 7.1.3 *A metrizable space X is UV^n (resp. UV^∞) if and only if X is UV^n (resp. UV^∞) in every ANR that contains X as a closed subspace.*

Proof Since the UV^∞ case follows from the UV^n case and the "if" part is trivial, it suffices to show the "only if" part of the UV^n case.

Let Y_1 and Y_2 be ANRs containing X as a closed subspace and assume that X is UV^n in Y_1. Then, we show that X is UV^n in Y_2. We have maps $\varphi : W_1 \to Y_2$ and $\psi : W_2 \to Y_1$ of neighborhoods W_i of X in Y_i such that $\varphi|X = \psi|X = \mathrm{id}$. For each open neighborhood U of X in Y_2, $\varphi^{-1}(U)$ is a neighborhood of X in Y_1, which contains a neighborhood V' of X such that for each $0 \leqslant i \leqslant n$, every map $g : \mathbf{S}^i \to V'$ is null-homotopic in $\varphi^{-1}(U)$. Because U is an ANR, there is a $\mathcal{U} \in \mathrm{cov}(U)$ that is an $\hbar$-refinement of $\{U\}$ (Corollary 6.3.6). We can easily find a neighborhood V of X in U such that $\varphi\psi|V$ is $\mathcal{U}$-close to id and $V \subset \psi^{-1}(V')$. Then, for each map $f : \mathbf{S}^i \to V$, the map $\psi f : \mathbf{S}^i \to V'$ is null-homotopic in $\varphi^{-1}(U)$, hence

$$f \simeq (\varphi\psi|V)f = \varphi\psi f \simeq 0 \ \text{ in } U.$$

Therefore, X is UV^n in Y_2. Thus, we have the "only if" part. □

[2] Our **property $\boldsymbol{UV^\infty}$** is weaker than in most other literature (cf. the previous footnote).

Note that X is UV^n in X itself if and only if X is n-connected. Hence, X is UV^∞ in X itself if and only if X is homotopically trivial. Then, we have the following:

Corollary 7.1.4 *An ANR is UV^n if and only if it is n-connected. Consequently, a UV^∞ ANR is the same as an AR.*[3] □

We show that property UV^0 characterizes connectedness.

Proposition 7.1.5 *A metrizable space X is UV^0 if and only if X is connected.*

Proof Regard X as a closed set in a normed linear space Y (Theorem 6.2.1). It is easy to show that if X is not connected then X is not UV^0 in Y, that is, the "only if" part holds. Each open neighborhood U of X in Y has a cover $\mathcal{B}$ consisting of open balls in Y. If X is connected, then, for every two $B, B' \in \mathcal{B}[X]$, there are $B_0, B_1, \ldots, B_n \in \mathcal{B}$ such that each B_i meets B_{i-1}, $B_0 = B$, and $B_n = B'$, which implies that $\mathrm{st}(X, \mathcal{B})$ is path-connected. Thus, we have the "if" part. □

According to Theorem 7.1.2, property UV^* characterizes the trivial shape. For a finite-dimensional space, we have the following:

Theorem 7.1.6 *An n-dimensional metrizable space X has trivial shape if and only if X is UV^n. Consequently, an n-dimensional compactum is cell-like if and only if it is UV^n.*

Proof Since the "only if" part is trivial, it suffices to show the "if" part. Assuming that X is UV^n in an ANR Y, we will show that X is contractible in each open neighborhood U of X in Y. Choose open neighborhoods $U = V_{n+1} \supset \cdots \supset V_1 \supset V_0$ of X in Y so that every map $f : \mathbf{S}^i \to V_i$ is null-homotopic in V_{i+1}. Since V_0 is an ANR and $\dim X = n$, we can apply Theorem 6.12.7 and Corollary 6.3.6 to obtain a simplicial complex L with $\dim L \leqslant n$ and maps $f : X \to |L|$, $g : |L| \to V_0$ such that $gf \simeq \mathrm{id}_X$ in V_0. Let $K = v_0 * L$ be the simplicial cone over L with v_0 the cone vertex. Then, $\dim K \leqslant n + 1$.

For each $0 \leqslant i \leqslant n + 1$, we denote $K_i = K^{(i)} \cup L$. Note that $K_0 = \{v_0\} \cup L$. Taking any point $h_0(v_0) \in V_0$, we can extend g to a map $h_0 : |K_0| \to V_0$. Suppose that g extends to a map $h_i : |K_i| \to V_i$. For each $\tau \in K_{i+1} \setminus K_i$, τ is an $(i + 1)$-simplex with $\partial\tau \in |K_i|$, and $h_i|\partial\tau : \partial\tau \to V_i$ extends to a map $h_\tau : \tau \to V_{i+1}$. We define a map $h_{i+1} : |K_{i+1}| \to V_{i+1}$ by $h_{i+1}||K_i| = h_i$ and $h_{i+1}|\tau = h_\tau$ for each $\tau \in K_{i+1} \setminus K_i$, which is an extension of g. By induction, we have a map $h = h_{n+1} : |K| = |K_{n+1}| \to V_{n+1} = U$ such that $h||L| = g$. Since $|K|$ is contractible, we have $g \simeq 0$ in U, hence $\mathrm{id}_X \simeq gf \simeq 0$ in U, that is, X is contractible in U. By virtue of Theorem 7.1.2, X has trivial shape. □

Due to Corollary 4.17.8, every compactum X is the inverse limit of an inverse sequence (X_i, f_i) of compact polyhedra and PL maps. The following lemma is useful when treating the inverse limits:

[3] Cf. Theorem 6.6.12.

Lemma 7.1.7 *Let X be the inverse limit of an inverse sequence $(X_i, f_i)_{i\in\mathbb{N}}$ of compacta with projections $p_i : X \to X_i$, $i \in \mathbb{N}$, and let P be a space with the homotopy type of a simplicial complex.*

(1) *For every map $g : X \to P$, there exist some $i_0 \in \mathbb{N}$ and maps $g_i : X_i \to P$, $i \geqslant i_0$, such that $g_i p_i \simeq g$.*
(2) *Let $g, g' : X_{i_0} \to P$ be maps. If $gp_{i_0} \simeq g' p_{i_0}$, then there is some $j_0 \geqslant i_0$ such that $gf_{i_0,j} \simeq g' f_{i_0,j}$ for every $j \geqslant j_0$, where $f_{i,j} = f_i \cdots f_{j-1}$ for $i < j$ and $f_{i,i} = \mathrm{id}_{X_i}$.*

Proof Since P has the homotopy type of an ANR (Corollary 6.6.5), we may assume that P is an ANR. Recall $X = \varprojlim(X_i, f_i) \subset \prod_{i\in\mathbb{N}} X_i$ and $p_i = \mathrm{pr}_i|X$, $i \in \mathbb{N}$ (cf. Sect. 4.17). Then, $f_i p_{i+1} = p_i$. For each $i \in \mathbb{N}$, let $z_i \in X_i$ be fixed and define an embedding $\varphi_i : X_i \to \prod_{i\in\mathbb{N}} X_i$ by

$$\varphi_i(x) = (f_{1,i}(x), \ldots, f_{i-1,i}(x), x, z_{i+1}, z_{i+2}, \ldots).$$

Regard $\prod_{i\in\mathbb{N}} X_i$ as a metric space given an admissible metric d. Then, $\mathrm{ulim}_{i\to\infty}\varphi_i p_i = \mathrm{id}$.[4] For each $n \in \mathbb{N}$, let

$$X_n^* = \left\{x \in \prod_{i\in\mathbb{N}} X_i \;\middle|\; x(i) = f_{i,n}(x(n)) \text{ for each } i \leqslant n\right\}.$$

Then, $\bigcap_{i\in\mathbb{N}} X_i^* = X$. Using this representation, we will show (1) and (2).

(1): Because P is an ANR, g extends to a map $\tilde{g} : U \to P$ over an open neighborhood U of X in $\prod_{i\in\mathbb{N}} X_i$. Then, $X_n^* \subset U$ for some $n \in \mathbb{N}$. For each $i \geqslant n$, since $\varphi_i(X_i) \subset X_i^* \subset X_n^*$, we have a map $g_i = \tilde{g}\varphi_i : X_i \to P$. Since $\mathrm{ulim}_{i\to\infty}\varphi_i p_i = \mathrm{id}$, we have $\mathrm{ulim}_{n\leqslant i\to\infty} g_i p_i = g$. Hence, $g_i p_i \simeq g$ for sufficiently large $i \geqslant n$.

(2): Extending a homotopy from gp_{i_0} to $g' p_{i_0}$ over the following set

$$\Xi = (X \times \mathbf{I}) \cup (\textstyle\prod_{i\in\mathbb{N}} X_i \times \{0, 1\}),$$

we have a map $h : \Xi \to P$ such that $h(x, 0) = g(x(i_0)) = g\mathrm{pr}_{i_0}(x)$ and $h(x, 1) = g'(x(i_0)) = g'\mathrm{pr}_{i_0}(x)$ for each $x \in \prod_{i\in\mathbb{N}} X_i$. Since P is an ANR, h extends to a map $\tilde{h} : U \times \mathbf{I} \to P$, where U is an open neighborhood of X in $\prod_{i\in\mathbb{N}} X_i$. Then, $X_{j_0}^* \subset U$ for some $j_0 \geqslant i_0$. For every $j \geqslant j_0$, $\varphi_j(X_j) \times \mathbf{I} \subset X_j^* \times \mathbf{I} \subset U$. Consequently, we have a homotopy $h' = \tilde{h}(\varphi_j \times \mathrm{id}_{\mathbf{I}}) : X_j \times \mathbf{I} \to P$. For each $x \in X_j$,

$$h'_0(x) = \tilde{h}(\varphi_j(x), 0) = h(\varphi_j(x), 0) = g\mathrm{pr}_{i_0}\varphi_j(x) = gf_{i_0,j}(x),$$

i.e., $h'_0 = gf_{i_0,j}$. Similarly, we have $h'_1 = g' f_{i_0,j}$. Therefore, $gf_{i_0,j} \simeq g' f_{i_0,j}$. □

Remark 7.1.1 As we saw in the proof above, in Lemma 7.1.7 (1), when P is an ANR, there exist some $n \in \mathbb{N}$ and maps $g_i : X_i \to P, i \geqslant n$, such that $\mathrm{ulim}_{n\leqslant i\to\infty} g_i p_i = g$.

[4] See Sect. 1.1.

The cell-likeness of the inverse limits can be characterized as follows:

Theorem 7.1.8 *Let X be the inverse limit of an inverse sequence $(X_i, f_i)_{i\in\mathbb{N}}$ of compact ANRs and $p_i : X \to X_i$, $i \in \mathbb{N}$, be the projections. Then, the following are equivalent:*

(a) *X is cell-like;*
(b) *$p_i \simeq 0$ for every $i \in \mathbb{N}$;*
(c) *For each $i \in \mathbb{N}$, there is a $j \geqslant i$ such that $f_{i,j} \simeq 0$.*

Proof The implication (b) $\Rightarrow$ (c) follows from Lemma 7.1.7 (2). Because $p_i = f_{i,j}p_j$, the implication (c) $\Rightarrow$ (b) is trivial.

(a) $\Rightarrow$ (b): By Theorem 7.1.2, X has trivial shape. Since every X_i is an ANR, the implication (a) $\Rightarrow$ (b) is trivial.

(b) $\Rightarrow$ (a): Let $g : X \to P$ be a map from X to an ANR P. By Lemma 7.1.7 (1), there exist some $i \in \mathbb{N}$ and a map $g' : X_i \to P$ such that $g'p_i \simeq g$. Since $p_i \simeq 0$, it follows that $g \simeq 0$. Thus, X has trivial shape, hence it is cell-like by Theorem 7.1.2. □

Remark 7.1.2 The implications (c) $\Rightarrow$ (b) $\Rightarrow$ (a) hold without the assumption that each X_i is an ANR. This is a generalization of Proposition 7.1.1.

Concerning the UV^n properties, we have the following characterization:

Theorem 7.1.9 *Let X be the inverse limit of an inverse sequence $(X_i, f_i)_{i\in\mathbb{N}}$ of compact ANRs and $p_i : X \to X_i$, $i \in \mathbb{N}$, be the projections. Then, the following are equivalent:*

(a) *X is UV^n;*
(b) *For each $i \in \mathbb{N}$, there is a $j \geqslant i$ such that $f_{i,j}\alpha \simeq 0$ for every map $\alpha : \mathbf{S}^k \to X_j$, $0 \leqslant k \leqslant n$.*

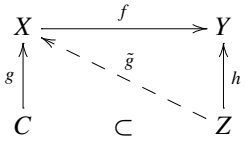
Proof We can regard X as a closed set in the Hilbert cube $\boldsymbol{Q}$.

(a) $\Rightarrow$ (b): For each $i \in \mathbb{N}$, since X_i is an ANR, $p_i : X \to X_i$ extends to a map $\tilde{p}_i : U \to X_i$ over a neighborhood U of X in $\boldsymbol{Q}$. Then, U contains an open neighborhood V of X such that, for each $0 \leqslant k \leqslant n$, every map from $\mathbf{S}^k$ to V is null-homotopic in U. Because V is an ANR, applying Lemma 7.1.7 (1) to the inclusion $X \subset V$, we can obtain some $j' \geqslant i$ and a map $g : X_{j'} \to V$ such that $gp_{j'} \simeq \mathrm{id}_X$ in V. Since $\tilde{p}_i gp_{j'} \simeq \tilde{p}_i|X = p_i = f_{i,j'}p_{j'}$, it follows from Lemma 7.1.7 (2) that $\tilde{p}_i gf_{j',j} \simeq f_{i,j'}f_{j',j} = f_{i,j}$ for some $j \geqslant j'$. For $0 \leqslant k \leqslant n$ and each map $\alpha : \mathbf{S}^k \to X_j$, we have $gf_{j',j}\alpha \simeq 0$ in U, hence $f_{i,j}\alpha \simeq \tilde{p}_i gf_{j',j}\alpha \simeq 0$ in X_i.

(b) $\Rightarrow$ (a): For each open neighborhood U of X in $\boldsymbol{Q}$, applying Lemma 7.1.7(1) to the inclusion $X \subset U$, we can obtain some $i \in \mathbb{N}$ and a map $g : X_i \to U$ such that $gp_i \simeq \mathrm{id}_X$ in U. Then, there is some $j \geqslant i$ such that $f_{i,j}\alpha \simeq 0$ for any map $\alpha : \mathbf{S}^k \to X_j$, $0 \leqslant k \leqslant n$. Since X_j is an ANR, $p_j : X \to X_j$ extends to a map $\tilde{p}_j : V' \to X_j$ over a neighborhood V' of X in $\boldsymbol{Q}$. Because $gf_{i,j}\tilde{p}_j|X = gf_{i,j}p_j = gp_i \simeq \mathrm{id}_X$ in the ANR U, X has a neighborhood V in $\boldsymbol{Q}$ such that $V \subset U \cap V'$ and $gf_{i,j}\tilde{p}_j|V \simeq \mathrm{id}_V$ in U. For $0 \leqslant k \leqslant n$ and each map $\alpha : \mathbf{S}^k \to V$, we have $\alpha \simeq gf_{i,j}\tilde{p}_j\alpha \simeq 0$ in U. □

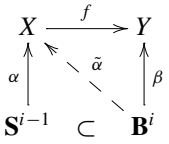
7.2 Soft Maps and the 0-Dimensional Selection Theorem

A map $f : X \to Y$ is said to be **soft** (***n*-soft**) provided that, for any metrizable space Z (with $\dim Z \leqslant n$) and any map $g : C \to X$ of a closed set C of Z, if fg extends to a map $h : Z \to Y$ then g extends to a map $\tilde{g} : Z \to X$ with $f\tilde{g} = h$.



In the above, if Y is an AE or an ANE (an AE(n) or an ANE(n)) then so is X. Replacing the pair (Z, C) with a polyhedral pair[5] (with $\dim \leqslant n$), we can introduce a **polyhedrally soft map** (a **polyhedrally *n*-soft map**). By the following proposition, every soft (resp. n-soft) map is polyhedrally soft (n-soft).

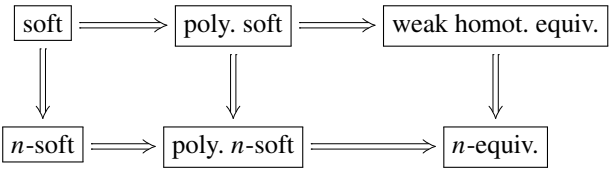
Proposition 7.2.1 *For a map $f : X \to Y$ to be polyhedrally soft (resp. polyhedrally n-soft), it is necessary and sufficient that, for any map $\alpha : \mathbf{S}^{i-1} \to X$, $i \in \omega$ (resp. $i \leqslant n$), if $f\alpha$ extends to a map $\beta : \mathbf{B}^i \to Y$ then α extends to a map $\tilde{\alpha} : \mathbf{B}^i \to X$ with $f\tilde{\alpha} = \beta$, where $\mathbf{B}^0 = \{0\}$ and $\mathbf{S}^{-1} = \emptyset$.*



Consequently, a map $f : X \to Y$ is polyhedrally soft if and only if f is polyhedrally n-soft for every $n \in \omega$.

To show the sufficiency, let K be a simplicial complex with L a subcomplex of K, and let $g : |L| \to X$ and $h : |K| \to Y$ be maps with $fg = h||L|$. We inductively construct maps $g_i : |K^{(i)} \cup L| \to X, i \in \omega$ (resp. $i \leqslant n$), so that each g_i is an extension of g_{i-1} and $fg_i = h||K^{(i)} \cup L|$, where $g_{-1} = g$. This can be done by applying the condition to each simplex in $K^{(i)} \setminus (K^{(i-1)} \cup L)$. Then, the desired extension $\tilde{g} : |K| \to X$ of g can be defined by $\tilde{g}||K^{(i)}| = g_i||K^{(i)}|$ for each $i \in \omega$ (resp. $\tilde{g} = g_n$).

Evidently, every polyhedrally soft map is a weak homotopy equivalence and every polyhedrally n-soft map is an n-equivalence (cf. Sect. 4.20).



[5] That is, $Z = |K|$ and $C = |L|$ for some simplicial complex K and a subcomplex $L \subset K$.

We have the following characterization of polyhedrally 0-soft maps:

Proposition 7.2.2 *A map $f : X \to Y$ is polyhedrally 0-soft if and only if f is surjective.*

Proof Because 0-dimensional polyhedra are discrete, it is trivial that a surjective map is polyhedrally 0-soft. To prove the converse, for each $y \in Y$, consider the pair $(\{y\}, \emptyset)$ in the definition of a polyhedrally 0-soft map. Then, we can find $x \in X$ such that $f(x) = y$. □

Proposition 7.2.3 *When Y is metrizable, every 0-soft map $f : X \to Y$ is a surjective open map.*

Proof According to Proposition 7.2.2, f is surjective. Suppose that f is not open. Then, X has an open set U with $x \in U$ such that $f(x) \notin \operatorname{int} f(U)$. We have $y_n \in Y \setminus f(U)$, $n \in \mathbb{N}$, such that $f(x) = \lim_{n\to\infty} y_n$. Since $Z = \{f(x)\} \cup \{y_n \mid n \in \mathbb{N}\}$ is 0-dimensional, we have a map $g : Z \to X$ such that $g(f(x)) = x$ and $fg = \mathrm{id}_Z$, which causes a contradiction. □

If X is completely metrizable, the converse of Proposition 7.2.3 above is also true. This will be shown as a corollary of the following 0-Dimensional Selection Theorem, which can be proved by the same strategy as the Michael Selection Theorem 3.8.8.

Theorem 7.2.4 (0- DIMENSIONAL SELECTION) *Let X be a paracompact space with $\dim X = 0$ and $Y = (Y, d)$ be a metric space. Then, every lower semi-continuous (l.s.c.) closed-valued function $\varphi : X \to \mathrm{Cld}(Y)$ admits a selection if each $\varphi(x)$ is d-complete. In addition, if A is a closed set in X then each selection $f : A \to Y$ for $\varphi|A$ can extend to a selection $\tilde{f} : X \to Y$ for φ.*

Proof We may assume that $\operatorname{diam} Y < 1$. By induction, we will construct maps $f_i : X \to Y$, $i \in \mathbb{N}$, such that

$$d(f_i(x), \varphi(x)) < 2^{-i} \text{ and } d(f_{i+1}(x), f_i(x)) < 2^{-i+2} \text{ for every } x \in X.$$

Assume that f_{n-1} has been obtained, where f_0 is any map. We define a closed-valued function $\psi : X \to \mathrm{Cld}(Y)$ as follows:

$$\psi(x) = \mathrm{cl}_Y\big(\varphi(x) \cap \mathrm{B}(f_{n-1}(x), 2^{-n+1})\big) \text{ for each } x \in X.$$

Let $W = \bigcup_{y\in Y} \mathrm{B}(y, 2^{-n+1}) \times \{y\}$. Since $\mathrm{B}(y, 2^{-n})^2 \subset W$ for every $y \in Y$, W is a neighborhood of the diagonal Δ_Y in Y^2. Therefore, ψ is l.s.c. by Lemmas 3.8.5 and 3.8.3.

For each $y \in Y$, let

$$V_y = \{x \in X \mid \psi(x) \cap \mathrm{B}(y, 2^{-n}) \neq \emptyset\}.$$

Since ψ is l.s.c., each V_y is open in X. By the 0-dimensionality, X has an open cover $\mathcal{U} \prec \{V_y \mid y \in Y\}$ with $\operatorname{ord} \mathcal{U} = 1$ (cf. Theorem 5.2.5). For each $U \in \mathcal{U}$, choose

$y(U) \in Y$ so that $U \subset V_{y(U)}$. Then, we have a map $f_n : X \to Y$ such that $f_n(x) = y(U)$ for every $x \in U \in \mathcal{U}$. Observe that $d(f_n(x), \psi(x)) < 2^{-n}$ for every $x \in X$, which means that $d(f_n(x), \varphi(x)) < 2^{-n}$ and $d(f_n(x), f_{n-1}(x)) < 2^{-n+3}$.

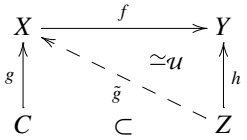
The rest of the proof is similar to that of the Michael Selection Theorem 3.8.8. □

Now, as a corollary of Theorem 7.2.4, we can obtain the following characterization of 0-soft maps:

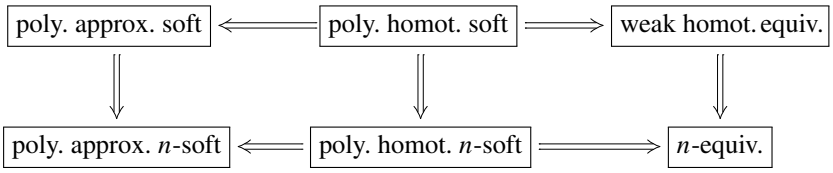
Proposition 7.2.5 *Let X is be a completely metrizable and Y be a metrizable space. Then, a map $f : X \to Y$ is 0-soft if and only if f is surjective and open.*

Proof The "only if" part is Proposition 7.2.3. To see the "if" part, let Z be a 0-dimensional metrizable space and C a closed set in Z. Given maps $g : C \to X$ and $h : Z \to Y$ such that $fg = h|C$, note that $f^{-1}h : Z \to \mathrm{Cld}(X)$ is an l.s.c. closed-valued function (cf. Proposition 3.8.1) and g is a selection for $f^{-1}h|C$. We can apply Theorem 7.2.4 to extend g to a selection $\tilde{g} : Z \to X$. Then, $f\tilde{g} = h$. □

A map $f : X \to Y$ is said to be **homotopically soft** (**homotopically n-soft**) (resp. **approximately soft** (**approximately n-soft**)) provided that, for any metrizable space Z (with $\dim Z \leqslant n$), any map $g : C \to X$ of a closed set C in Z and each open cover $\mathcal{U}$ of Y, if fg extends to a map $h : Z \to Y$ then g extends $\tilde{g} : Z \to X$ such that $f\tilde{g} \simeq_{\mathcal{U}} h$ rel. C (resp. $f\tilde{g}$ is $\mathcal{U}$-close to h).



Replacing the pair (Z, C) with a pair of polyhedra (with $\dim \leqslant n$), we can introduce a **polyhedrally homotopically soft map** (a **polyhedrally homotopically n-soft map**) or a **polyhedrally approximately soft map** (a **polyhedrally approximately n-soft map**).



A polyhedrally approximately 0-soft map and a polyhedrally homotopically 0-soft map are characterized as follows:

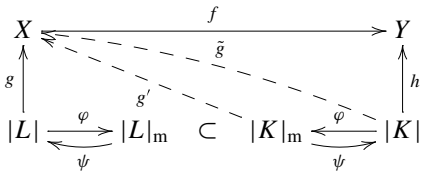
Proposition 7.2.6 (1) *A map $f : X \to Y$ is polyhedrally approximately 0-soft if and only if $f(X)$ is dense in Y (i.e., f has the dense image).*

(2) *A map $f : X \to Y$ is polyhedrally homotopically 0-soft if and only if for each $y \in Y$ and each neighborhood U of y in Y, there is some $x \in f^{-1}(U)$ such that $f(x)$ is connected to y in U.*

Sketch of Proof. For the "if" part of (1) and (2), note that 0-dimensional polyhedra are discrete. To see the "only if" part (1) (or (2)), for each $y \in Y$ and each open neighborhood U of y in Y, consider $(Z, C) = (\{y\}, \emptyset)$, $h = \mathrm{id}$, and $\mathcal{U} = \{Y \setminus \{y\}, U\}$ in the definition of a polyhedrally approximately 0-soft map (or polyhedrally homotopically 0-soft map).

Proposition 7.2.7 *When Y is paracompact, every approximately (n-)soft map $f : X \to Y$ is polyhedrally approximately (n-)soft.*

Proof Let K be a simplicial complex (with $\dim K \leqslant n$) and L be a subcomplex of K with a map $g : |L| \to X$ such that fg extends to a map $h : |K| \to Y$. For each open cover $\mathcal{U}$ of Y, we will show that g extends to a map $\tilde{g} : |K| \to X$ such that $f\tilde{g}$ is $\mathcal{U}$-close to h. Because Y is paracompact, $\mathcal{U}$ has an open star-refinement $\mathcal{V}$. Due to Whitehead's Theorem 4.8.12 on small subdivisions, replacing K with a subdivision, we may assume that $K \prec h^{-1}(\mathcal{V})$. We denote $\varphi = \mathrm{id}_{|K|} : |K| \to |K|_{\mathrm{m}}$. By Theorem 4.15.7, we have a map $\psi : |K|_{\mathrm{m}} \to |K|$ such that $\psi\varphi \simeq_K \mathrm{id}$ and $\varphi\psi \simeq_K \mathrm{id}$ by the straight-line homotopy. Due to Remark 4.15.3, it follows that $\psi\varphi||L| \simeq_L \mathrm{id}$ and $\varphi\psi||L|_{\mathrm{m}} \simeq_L \mathrm{id}$ by the straight-line homotopy. Because f is approximately (n-)soft, the map $g\psi||L|_{\mathrm{m}}$ extends a map $g' : |K|_{\mathrm{m}} \to X$ such that fg' is $\mathcal{V}$-close to $h\psi$.



Since $K \prec h^{-1}(\mathcal{V})$, it follows that $L \prec g^{-1}f^{-1}(\mathcal{V})$, consequently

$$g'\varphi||L| = g\psi\varphi||L| \simeq_{f^{-1}(\mathcal{V})} g.$$

By the Homotopy Extension Theorem 4.5.3, g extends to a map $\tilde{g} : |K| \to X$ such that $\tilde{g} \simeq_{f^{-1}(\mathcal{V})} g'\varphi$, hence $f\tilde{g} \simeq_{\mathcal{V}} fg'\varphi$. Since $fg'\varphi$ is $\mathcal{V}$-close to $h\psi\varphi$ and $h\psi\varphi \simeq_{\mathcal{V}} h$, it follows that $f\tilde{g}$ is $\mathcal{U}$-close to h. □

Remark 7.2.1 In the above proof, if $f : X \to Y$ is homotopically (n-)soft, then $g\psi||L|_{\mathrm{m}}$ can be extended to a map $g' : |K|_{\mathrm{m}} \to X$ such that $fg' \simeq_{\mathcal{V}} h\psi$ rel. $|L|_{\mathrm{m}}$, so $fg'\varphi \simeq_{\mathcal{V}} h\psi\varphi$ rel. $|L|$. But $f\tilde{g}||L| = fg \neq fg\psi\varphi||L| = fg'\varphi||L|$ and $h\psi\varphi||L| \neq h||L|$ in general. Then, we cannot conclude that $f\tilde{g} \simeq_{\mathcal{U}} h$ rel. $|L|$. However, every homotopically (n-)soft map $f : X \to Y$ is polyhedrally homotopically (n-)soft when Y is paracompact. This will be shown in the next section.

The following lemma is useful for constructing approximations of maps to paracompact spaces:

Lemma 7.2.8 *Given a sequence of open covers of Y as follows:*

$$\mathcal{U} \overset{*}{\succ} \mathcal{U}_1 \overset{*}{\succ} \mathcal{U}_2 \overset{*}{\succ} \cdots,$$

let $\mathcal{W}_{i+1} = \mathrm{st}(\mathcal{W}_i, \mathcal{U}_{i+1})$, where $\mathcal{W}_1 = \mathcal{U}_1$. Then, $\mathrm{st}(\mathcal{W}_i, \mathcal{U}_i) \prec \mathcal{U}$ for every $i \in \mathbb{N}$.

Sketch of Proof To show by induction, observe that

$$\mathrm{st}(\mathcal{W}_i, \mathcal{U}_i) = \mathrm{st}(\mathrm{st}(\mathcal{W}_{i-1}, \mathcal{U}_i), \mathcal{U}_i) \prec \mathrm{st}(\mathcal{W}_{i-1}, \mathrm{st}\,\mathcal{U}_i) \prec \mathrm{st}(\mathcal{W}_{i-1}, \mathcal{U}_{i-1}).$$

Theorem 7.2.9 *Let Y be a paracompact space. A map $f : X \to Y$ is polyhedrally homotopically soft if and only if f is polyhedrally homotopically n-soft for every $n \in \omega$.*

Proof The "only if" part is obvious. To prove the "if" part, let K be a simplicial complex with L a subcomplex of K, and let $g : |L| \to X$ and $h : |K| \to Y$ be maps with $fg = h||L|$. Since Y is paracompact, each open cover $\mathcal{U} \in \mathrm{cov}(Y)$ has open refinements as follows:

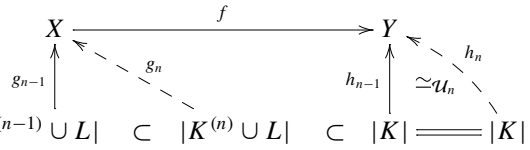
$$\mathcal{U} \overset{*}{\succ} \mathcal{U}_0 \overset{*}{\succ} \mathcal{U}_1 \overset{*}{\succ} \mathcal{U}_2 \overset{*}{\succ} \cdots.$$

For each $i \in \mathbb{N}$, let $\mathcal{W}_i = \mathrm{st}(\mathcal{W}_{i-1}, \mathcal{U}_i)$, where $\mathcal{W}_0 = \mathcal{U}_0$. By Lemma 7.2.8, $\mathrm{st}(\mathcal{W}_i, \mathcal{U}_i) \prec \mathcal{U}$ for every $i \in \mathbb{N}$.

We will inductively construct maps $g_i : |K^{(i)} \cup L| \to X$, $h_i : |K| \to Y$ and $\mathcal{U}_i$-homotopies $\varphi^{(i)} : |K| \times \mathbf{I} \to Y$, $i \in \omega$, such that

$$g_i||K^{(i-1)} \cup L| = g_{i-1},\ fg_i = h_i||K^{(i)} \cup L|,\ \varphi_0^{(i)} = h_{i-1},\ \varphi_1^{(i)} = h_i,$$
$$\varphi_t^{(i)}||K^{(i-1)} \cup L| = h_{i-1}||K^{(i-1)} \cup L| = fg_{i-1} \ \text{ for every } t \in \mathbf{I},$$

where $g_{-1} = g$ and $h_{-1} = h$. Suppose that g_{n-1}, h_{n-1}, and $\varphi^{(n-1)}$ have been defined.



By the polyhedral homotopy n-softness of f, the map g_{n-1} extends to a map $g_n : |K^{(n)} \cup L| \to X$ such that

$$fg_n \simeq_{\mathcal{U}_n} h_{n-1}||K^{(n)} \cup L| \ \text{ rel. } |K^{(n-1)} \cup L|.$$

Using the Homotopy Extension Theorem 4.5.3, we have a $\mathcal{U}_n$-homotopy $\varphi^{(n)} : |K| \times \mathbf{I} \to Y$ such that

$$\varphi_0^{(n)} = h_{n-1},\ \varphi_1^{(n)}||K^{(n)} \cup L| = fg_n \text{ and}$$
$$\varphi_t^{(n)}||K^{(n-1)} \cup L| = h_{n-1}||K^{(n-1)} \cup L| \text{ for each } t \in \mathbf{I}.$$

The map h_n is defined by $h_n = \varphi_1^{(n)} : |K| \to Y$.

Now, we can extend g to the map $\tilde{g} : |K| \to X$ defined by $\tilde{g}||K^{(i)} \cup L| = g_i$ for each $i \in \omega$. Let $\varphi : |K| \times \mathbf{I} \to Y$ be a homotopy defined as follows: $\varphi_0 = f\tilde{g}$ and

$$\varphi(x,t) = \varphi^{(i)}(x, 2 - 2^{i+1}t) \text{ if } 2^{-i-1} \leqslant t \leqslant 2^{-i},\ i \in \omega.$$

Then, $\varphi_1 = \varphi_0^{(0)} = h_{-1} = h$ and $\varphi_t||L| = h||L| = fg = f\tilde{g}||L|$ for every $t \in \mathbf{I}$. The continuity of φ at each point in $|K| \times \{0\}$ follows from the continuity of $\varphi|\sigma \times \mathbf{I}$ for each simplex $\sigma \in K$, where if $\dim \sigma = k$ then

$$\varphi_t|\sigma = h_k|\sigma = fg_k|\sigma = f\tilde{g}|\sigma = \varphi_0|\sigma \text{ for every } t \leqslant 2^{-k-1}.$$

For each $x \in |K|$ and $n \in \mathbb{N}$,

$$\begin{aligned}\varphi(\{x\} \times [2^{-n-1}, 1]) &= \varphi(\{x\} \times [2^{-n}, 1]) \cup \varphi^{(n)}(\{x\} \times \mathbf{I}) \\ &\subset \mathrm{st}(\varphi(\{x\} \times [2^{-n}, 1]), \mathcal{U}_n).\end{aligned}$$

Since $\varphi(\{x\} \times [2^{-1}, 1]) = \varphi^{(0)}(\{x\} \times \mathbf{I})$ is contained in some member of $\mathcal{U}_0 = \mathcal{W}_0$, it follows that $\varphi(\{x\} \times [2^{-n-1}, 1])$ is contained in some member of $\mathcal{W}_n$. In addition, note that $\varphi(\{x\} \times [0, 2^{-k-1}]) = \{f\tilde{g}(x)\}$ for sufficiently large $k \in \mathbb{N}$. Then, $\varphi(\{x\} \times \mathbf{I})$ is contained in some member of $\mathcal{U}$. Hence, φ is a $\mathcal{U}$-homotopy. Thus, we have $f\tilde{g} \simeq_{\mathcal{U}} h$ rel. $|L|$. □

Remark 7.2.2 If f is polyhedrally approximately n-soft for every $n \in \omega$, the arguments above are not valid to prove that f is polyhedrally approximately soft because we cannot apply the Homotopy Extension Theorem 4.5.3.

Every open cover $\mathcal{U}$ of an ANR (resp. a metrizable ANE$(n+1)$) Y has an $\hbar$-refinement (resp. h^n-refinement) $\mathcal{V} \in \mathrm{cov}(Y)$ by Corollary 6.3.6 (resp. Theorem 6.12.9 with Remark 6.12.3). Using this fact and Theorem 7.2.9, we have the following:

Proposition 7.2.10 *Let Y be a metrizable space and $f : X \to Y$ be a map.*

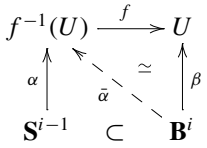
(1) *When Y is LC^n, i.e., Y is an ANE$(n+1)$, f is (polyhedrally) homotopically n-soft if and only if f is (polyhedrally) approximately n-soft.*
(2) *When Y is LC^∞, f is polyhedrally homotopically soft if and only if f is polyhedrally approximately soft.*
(3) *When Y is an ANR, f is homotopically soft if and only if f is approximately soft.* □

7.3 Hereditary n-Equivalence and Local Connections

A map $f : X \to Y$ is called a **hereditary n-equivalence** if $f|f^{-1}(U) : f^{-1}(U) \to U$ is an n-equivalence for every open set U in Y, that is, it satisfies the following condition $(\pi)_i$ for each $i = 0, \ldots, n$:

$(\pi)_i$ For each map $\alpha : \mathbf{S}^{i-1} \to f^{-1}(U)$, if $f\alpha$ extends to a map $\beta : \mathbf{B}^i \to U$, then α extends to a map $\bar{\alpha} : \mathbf{B}^i \to f^{-1}(U)$ such that $f\bar{\alpha} \simeq \beta$ rel. $\mathbf{S}^{i-1}$ in U,

where $\mathbf{B}^0 = \{0\}$ and $\mathbf{S}^{-1} = \emptyset$.



A map $f : X \to Y$ is a hereditary 0-equivalence if and only if every $y \in Y$ can be connected with an arbitrarily close point of $f(X)$ by a small path. A **hereditary weak homotopy equivalence** is a map $f : X \to Y$ such that $f|f^{-1}(U) : f^{-1}(U) \to U$ is a weak homotopy equivalence for every open set U in Y; equivalently, it is a hereditary n-equivalence for every $n \in \omega$. The following statements are easily proved:

- *Every polyhedrally homotopically n-soft map is a hereditary n-equivalence.*
- *Every polyhedrally homotopically soft map is a hereditary weak homotopy equivalence.*

Sketch of Proof. The second statement follows from the first statement. Let $f : X \to Y$ be a polyhedrally homotopically n-soft map and U be an open set in Y. For maps $\alpha : \mathbf{S}^{i-1} \to f^{-1}(U)$ and $\beta : \mathbf{B}^i \to U$ with $\beta|\mathbf{S}^{i-1} = f\alpha$, consider $\mathcal{U} = \{U, Y \setminus \beta(\mathbf{B}^i)\} \in \mathrm{cov}(Y)$.

If Y is paracompact, the converse statements are also true.

Theorem 7.3.1 *Let Y be a paracompact space. Then, a map $f : X \to Y$ is a hereditary n-equivalence if and only if f is polyhedrally homotopically n-soft.*

Proof As mentioned in the above, the "if" part is easily proved. To prove the "only if" part, let (P, Q) be a pair of polyhedra with $\dim P \leqslant n$, and let $g : Q \to X$ and $h : P \to Y$ be maps with $fg = h|Q$. For each $\mathcal{U} \in \mathrm{cov}(Y)$, take open refinements as follows:

$$\mathcal{U} = \mathcal{U}_n \overset{*}{\succ} \mathcal{U}_{n-1} \overset{*}{\succ} \cdots \overset{*}{\succ} \mathcal{U}_0 \overset{*}{\succ} \mathcal{V}.$$

Then, P has a triangulation K such that Q is triangulated by a subcomplex of K and $K \prec h^{-1}(\mathcal{V})$ (Theorem 4.8.12). We will inductively define maps $g_i : Q \cup |K^{(i)}| \to X$ with homotopies $\varphi^{(i)} : (Q \cup |K^{(i)}|) \times \mathbf{I} \to Y$, $i = 0, \ldots, n$, such that

$$g_i|Q \cup |K^{(i-1)}| = g_{i-1},\ \varphi^{(i)}|(Q \cup |K^{(i-1)}|) \times \mathbf{I} = \varphi^{(i-1)},$$
$$\{\varphi^{(i)}(\sigma \times \mathbf{I}) \mid \sigma \in K^{(i)}\} \prec \mathcal{U}_i,\ \varphi_0^{(i)} = h|Q \cup |K^{(i)}| \text{ and } \varphi_1^{(i)} = fg_i,$$

where $g_{-1} = g$ and $\varphi^{(-1)} = h\mathrm{pr}_Q : Q \times \mathbf{I} \to Y$. Then, g can be extended to a map $g_n : P \to X$ such that $fg_n \simeq_{\mathcal{U}} h$ rel. Q.

Suppose that g_{i-1} and $\varphi^{(i-1)}$ have been defined. Then, $\varphi^{(i-1)}$ can be extended to the map

$$\bar{\varphi}^{(i-1)} : (P \times \{0\}) \cup ((Q \cup |K^{(i-1)}|) \times \mathbf{I}) \to Y$$

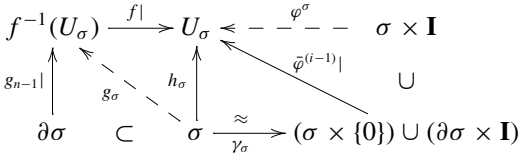
defined by $\bar{\varphi}^{(i-1)}(x, 0) = h(x)$ for each $x \in P$. For each i-simplex $\sigma \in K$ with $\sigma \notin Q$, since $K \prec h^{-1}(\mathcal{V})$ and $\mathrm{st}(\mathcal{V}, \mathcal{U}_{i-1}) \prec \mathcal{U}_i$, we can choose $U_\sigma \in \mathcal{U}_i$ so that

$$\bar{\varphi}^{(i-1)}((\sigma \times \{0\}) \cup (\partial\sigma \times \mathbf{I})) \subset \mathrm{st}(h(\sigma), \mathcal{U}_{i-1}) \subset U_\sigma.$$

Then, $fg_{i-1}(\partial\sigma) \subset U_\sigma$, i.e., $g_{i-1}(\partial\sigma) \subset f^{-1}(U_\sigma)$. Take a homeomorphism

$$\gamma_\sigma : \sigma \to (\sigma \times \{0\}) \cup (\partial\sigma \times \mathbf{I})$$

such that $\gamma_\sigma(x) = (x, 1)$ for each $x \in \partial\sigma$. Consider the map $h_\sigma = \bar{\varphi}^{(i-1)}\gamma_\sigma : \sigma \to U$. Then, $h_\sigma|\partial\sigma = fg_{i-1}|\partial\sigma$. Because $f|f^{-1}(U_\sigma) : f^{-1}(U_\sigma) \to U_\sigma$ is an n-equivalence, $g_{i-1}|\partial\sigma$ extends to a map $g_\sigma : \sigma \to f^{-1}(U_\sigma)$ such that $fg_\sigma \simeq h_\sigma$ rel. $\partial\sigma$.



Using a homotopy from h_σ to fg_σ rel. $\partial\sigma$, we can easily construct a homotopy $\varphi^\sigma : \sigma \times \mathbf{I} \to U_\sigma$ such that $\varphi^\sigma_1 = fg_\sigma$ and

$$\varphi^\sigma|(\sigma \times \{0\}) \cup (\partial\sigma \times \mathbf{I}) = h_\sigma\gamma_\sigma^{-1}.$$

The last condition means that $\varphi^\sigma_0 = h|\sigma$ and $\varphi^\sigma|\partial\sigma \times \mathbf{I} = \varphi^{(i-1)}|\partial\sigma \times \mathbf{I}$. Summing up these g_σ and φ^σ, we can obtain the desired extensions g_i and $\varphi^{(i)}$ of g_{i-1} and $\varphi^{(i-1)}$, respectively. □

Combining Theorems 7.3.1 and 7.2.9, we have the following:

Corollary 7.3.2 *When Y is paracompact, a map $f : X \to Y$ is a hereditary weak homotopy equivalence if and only if f is a polyhedrally homotopically soft map.* □

As in the case of polyhedrally homotopically (n-)soft maps, it is easy to see that every homotopically n-soft map is a hereditary n-equivalence and every homotopically soft map is a hereditary weak homotopy equivalence. Then, by Theorem 7.3.1 and Corollary 7.3.2, we have the following:

Corollary 7.3.3 *Let Y be a paracompact space. Then, every homotopically (n-)soft map $f : X \to Y$ is a polyhedrally homotopically (n-)soft map.* □

A map $f : X \to Y$ is called a **local n-connection** (resp. a **strong local n-connection**) if $f(X)$ is dense in Y and every neighborhood U of each $y \in Y$ contains a neighborhood V of y such that, for each $0 \leqslant i < n$, every map $g : \mathbf{S}^i \to f^{-1}(V)$ is null-homotopic in $f^{-1}(U)$ and, for each map $g : \mathbf{S}^n \to f^{-1}(V)$, $fg \simeq 0$ in U (resp. $g \simeq 0$ in $f^{-1}(U)$, i.e., for each $0 \leqslant i \leqslant n$, every map $g : \mathbf{S}^i \to f^{-1}(V)$ is null-homotopic in $f^{-1}(U)$), where $g \simeq 0$ in $f^{-1}(U)$ if and only if g extends to a map $\tilde{g} : \mathbf{B}^{i+1} \to f^{-1}(U)$ ($fg \simeq 0$ in U if and only if fg extends to a map $h : \mathbf{B}^{i+1} \to U$).

$$\begin{array}{ccccc} f^{-1}(V) & \subset & f^{-1}(U) & \xrightarrow{f} & U \\ \uparrow{\scriptstyle g} & & \uparrow{\scriptstyle \tilde{g}} & \nearrow{\scriptstyle h} & \\ \mathbf{S}^i & \subset & \mathbf{B}^{i+1} & & \end{array}$$

In the above, f can be called a local strong (-1)-connection if $f(X)$ is dense in Y. In fact, for $(\mathbf{B}^0, \mathbf{S}^{-1}) = (\{0\}, \emptyset)$, the existence of $\tilde{g}$ means $f^{-1}(U) \neq \emptyset$, i.e., $f(X) \cap U \neq \emptyset$. The following are direct consequences of the definitions:

- *Every strong local n-connection is a local n-connection.*
- *Every local n-connection is a strong local $(n-1)$-connection.*
- *A local n-connection is a strong local n-connection if it is a hereditary $(n+1)$-equivalence.*

We call f a **local ∞-connection** if f is a (strong) local n-connection for every $n \in \omega$. A map $f : X \to Y$ is called a **local $*$-connection** if $f(X)$ is dense in Y and every neighborhood U of each $y \in Y$ contains a neighborhood V of y such that $f^{-1}(V)$ is contractible in $f^{-1}(U)$. If f is a closed map, each non-empty fiber $f^{-1}(y)$ has trivial shape. A perfect local $*$-connection is simply a cell-like map.

A strong local n-connection can be characterized as follows:

Proposition 7.3.4 *Let Y be a regular space. Then, a map $f : X \to Y$ is a strong local n-connection if and only if $f(X)$ is dense in Y and each open cover $\mathcal{U}$ of Y has an open refinement $\mathcal{V}$ such that $f^{-1}(\mathcal{V})$ is a C^n-refinement of $f^{-1}(\mathcal{U})$.*

Proof The "only if" part is trivial. We will show the "if" part. Each open neighborhood U of $y \in Y$ contains the closure of some open neighborhood U_0 of y. Let $\mathcal{V} \in \mathrm{cov}(Y)$ be a refinement of $\mathcal{U} = \{U_0, U \setminus \{y\}, Y \setminus \mathrm{cl}\, U_0\} \in \mathrm{cov}(Y)$ such that $f^{-1}(\mathcal{V})$ is a C^n-refinement of $f^{-1}(\mathcal{U})$. Take $V \in \mathcal{V}$ with $y \in V$, and choose $U' \in \mathcal{U}$ so that $f^{-1}(V) \subset f^{-1}(U')$ and, for each $0 \leqslant i \leqslant n$, every map $g : \mathbf{S}^i \to f^{-1}(V)$ is null-homotopic in $f^{-1}(U')$, where $f^{-1}(V) \neq \emptyset$ because $f(X)$ is dense in Y. Observe that

$$U_0 \supset V \supset f(f^{-1}(V)) \subset f(f^{-1}(U')) \subset U',$$

which implies that $U_0 \cap U' \neq \emptyset$. Since $U' \neq Y \setminus \mathrm{cl}\, U_0$, it follows that $U' \subset U$. $\square$

Proposition 7.3.5 *Let Y be a paracompact space. Then, every strong local n-connection $f : X \to Y$ is polyhedrally approximately $(n+1)$-soft.*

Proof Since f is a strong local n-connection, each $\mathcal{U} \in \mathrm{cov}(Y)$ has the following open refinements:

$$\mathcal{U} \overset{*}{\succ} \mathcal{V}_{n+1} \overset{*}{\succ} \mathcal{U}_{n+1} \succ \mathcal{V}_n \overset{*}{\succ} \cdots \overset{*}{\succ} \mathcal{U}_1 \succ \mathcal{V}_0 \overset{*}{\succ} \mathcal{U}_0$$

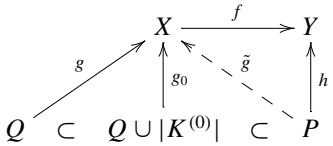
such that $f^{-1}(\mathcal{U}_{i+1}) \underset{C^i}{\succ} f^{-1}(\mathcal{V}_i)$. Then, it follows that

$$f^{-1}(\mathcal{V}_{n+1}) \overset{*}{\succ} f^{-1}(\mathcal{U}_{n+1}) \underset{C^n}{\succ} f^{-1}(\mathcal{V}_n) \overset{*}{\succ} \cdots \overset{*}{\succ} f^{-1}(\mathcal{U}_1) \underset{C^0}{\succ} f^{-1}(\mathcal{V}_0) \overset{*}{\succ} f^{-1}(\mathcal{U}_0).$$

By the same arguments as in the proof of Lemma 6.12.2, we can prove that

$$f^{-1}(\mathcal{V}_0) \underset{L^{n+1}}{\prec} f^{-1}(\mathcal{V}_{n+1}).$$

Let (P, Q) be a pair of polyhedra with $\dim P \leqslant n+1$, and let $g : Q \to X$ and $h : P \to Y$ be maps such that $fg = h|Q$. Then, P has a triangulation K such that Q is triangulated by a subcomplex of K and $K \prec h^{-1}(\mathcal{U}_0)$ (Theorem 4.8.12). Since $f(X)$ is dense in Y, we can extend g to a map $g_0 : Q \cup |K^{(0)}| \to X$ such that fg_0 is $\mathcal{U}_0$-close to $h|Q \cup |K^{(0)}|$. Then, g_0 is a partial $f^{-1}(\mathcal{V}_0)$-realization of K, which extends to a full $f^{-1}(\mathcal{V}_{n+1})$-realization $\tilde{g} : P = |K| \to X$.



Each $x \in P$ is contained in some $\tau \in K$. Then, we have $V \in \mathcal{V}_{n+1}$ such that $\tilde{g}(\tau) \subset f^{-1}(V)$, i.e., $f\tilde{g}(\tau) \subset V$. On the other hand, $h(\tau)$ is contained in some $V_0 \in \mathcal{V}_0$. Take $v \in \tau^{(0)}$. Then, $f\tilde{g}(v) = fg_0(v)$ and $h(v)$ are contained in the same $U_0 \in \mathcal{U}_0$, hence $f\tilde{g}(x), h(x) \in \mathrm{st}(U_0, \mathcal{V}_{n+1})$. Consequently, $f\tilde{g}$ is $\mathcal{U}$-close to h. □

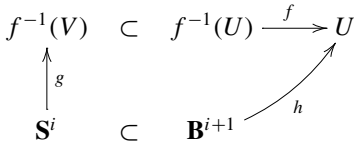
If Y is LC^n, we have the following proposition:

Proposition 7.3.6 *Let Y be LC^n and $f : X \to Y$ be a map.*

(1) *If f is polyhedrally approximately n-soft, then f is a local n-connection.*
(2) *If f is polyhedrally approximately $(n+1)$-soft, then f is a strong local n-connection.*

Proof First, note that a polyhedrally approximately 0-soft map has the dense image by Proposition 7.2.6 (1). Since Y is LC^0, every $y \in Y$ can be connected with an arbitrarily close point of $f(X)$ by a small path, which means that f is a local 0-connection.

Since Y is LC^n, every open neighborhood U of each $y \in Y$ contains a neighborhood V of y such that, for $0 \leqslant i \leqslant n$, every map $g : \mathbf{S}^i \to V$ extends to a map $\tilde{g} : \mathbf{B}^{i+1} \to U$. Now, let $0 \leqslant i \leqslant n$ and $g : \mathbf{S}^i \to f^{-1}(V)$ be a map. Then, fg extends to a map $h : \mathbf{B}^{i+1} \to U$.



When f is polyhedrally approximately n-soft, if $0 \leqslant i \leqslant n-1$ then g extends to a map $\tilde{g} : \mathbf{B}^{i+1} \to X$ such that $f\tilde{g}$ is $\mathcal{U}$-close to h, where

$$\mathcal{U} = \{Y \setminus h(\mathbf{B}^{i+1}),\ U\} \in \mathrm{cov}(Y).$$

Then, $f\tilde{g}(\mathbf{B}^{i+1}) \subset U$, i.e., $\tilde{g}(\mathbf{B}^{i+1}) \subset f^{-1}(U)$. This means that f is a local n-connection. If f is polyhedrally approximately $(n+1)$-soft, this argument is valid for all $0 \leqslant i \leqslant n$, which shows that f is a strong local n-connection. □

Combining Propositions 7.3.5 and 7.3.6 (2), Theorems 7.2.10 (1) and 7.3.1, we have the following characterization:

Theorem 7.3.7 *Let $n \in \omega$ and Y be an LC^n metrizable space. Then, for a map $f : X \to Y$, the following statements are equivalent:*

(a) *f is a strong local n-connection;*
(b) *f is polyhedrally approximately $(n+1)$-soft;*
(c) *f is polyhedrally homotopically $(n+1)$-soft;*
(d) *f is a hereditary $(n+1)$-equivalence.* □

When Y is metrizable, every local n-connection $f : X \to Y$ is polyhedrally homotopically n-soft, that is:

Theorem 7.3.8 *Let $n \in \omega$, Y be a metrizable space, and $f : X \to Y$ be a local n-connection. Then, f is polyhedrally homotopically n-soft and Y is LC^n.*

Proof We may assume that $Y = (Y, d)$ is a metric space.

(*The Case $n = 0$*) It suffices to show that for each $\varepsilon > 0$, each $y \in Y$ has an open neighborhood V in Y such that for every $x \in f^{-1}(V)$, $f(x)$ can be connected with y by an ε-path in Y (such a path is contained in $\mathrm{B}(y, \varepsilon)$) (cf. Proposition 7.2.6 (2)).

Indeed, each $y' \in V$ has an open neighborhood V' in Y so that for each $x' \in f^{-1}(V')$, $f(x')$ can be connected with y' by an ε-path in Y. Since $V \cap V'$ is a neighborhood of y, we can take $x \in f^{-1}(V' \cap V)$. Then, $f(x)$ can be connected to y and y' by ε-paths in Y.

We can choose open neighborhoods $V_1 \supset V_2 \supset \cdots$ of y in Y such that, for every pair of points $x, x' \in f^{-1}(V_i)$, $f(x)$ and $f(x')$ are connected by a path in $\mathrm{B}(y, 2^{-i+1}\varepsilon)$.

Since $f(X)$ is dense in Y, we have points $x_i \in f^{-1}(V_i)$. Taking paths $h_i : \mathbf{I} \to \mathrm{B}(y, 2^{-i+1}\varepsilon)$ with $h_i(0) = f(x_i)$ and $h_i(1) = f(x_{i+1})$, we can define a path $h : \mathbf{I} \to \mathrm{B}(y, \varepsilon)$ as follows:

$$h(0) = y \ \text{ and } \ h(t) = h_i(2 - 2^i t) \ \text{ for } 2^{-i} \leqslant t \leqslant 2^{-i+1}.$$

(*The Case* $n \geqslant 1$) Assuming the theorem is valid for $n-1$, we prove the theorem for n. Let (P, Q) be a pair of polyhedra with $\dim P = n$, and let $g : Q \to X$ and $h : P \to Y$ be maps such that $fg = h|Q$. Each $\mathcal{U} \in \mathrm{cov}(Y)$ has refinements

$$\mathcal{U} \overset{*}{\succ} \mathcal{U}_1 \succ \mathcal{V}_1 \overset{*}{\succ} \mathcal{U}_2 \succ \mathcal{V}_2 \overset{*}{\succ} \cdots,$$

such that $\mathrm{mesh}\,\mathcal{U}_i < 2^{-i}$ and the following condition holds:

- Each $V \in \mathcal{V}_i$ is contained in some $U \in \mathcal{U}_i$ such that, for each map $\alpha : \mathbf{S}^n \to f^{-1}(V)$, $f\alpha \simeq 0$ in U, that is, $f\alpha$ extends to a map $\beta : \mathbf{B}^{n+1} \to U$.

For convenience, we denote $\mathcal{W}_i = \mathcal{U}_{i+2}$. Then,

$$\mathrm{st}^3 \mathcal{W}_i = \mathrm{st}(\mathrm{st}^2 \mathcal{W}_i, \mathcal{W}_i) = \mathrm{st}(\mathcal{W}_i, \mathrm{st}\,\mathcal{W}_i) \prec \mathcal{V}_i \ \text{ and } \ \mathcal{W}_{i+1} \prec \mathcal{W}_i.$$

For each $i \in \mathbb{N}$, let K_i be a triangulation of P such that Q is triangulated by some subcomplex of K_i, $K_i \prec h^{-1}(\mathcal{W}_i)$, and $K_{i+1} \lhd K_i$. Then, note that $|K_i^{(n-1)}| \subset |K_{i+1}^{(n-1)}|$.

By the inductive assumption, we have a map $g_i' : Q \cup |K_i^{(n-1)}| \to X$ such that $g_i'|Q = g$ and $fg_i' \simeq_{\mathcal{W}_i} h|Q \cup |K_i^{(n-1)}|$ rel. Q. Let

$$\psi^{(i)} : (Q \cup |K_i^{(n-1)}|) \times \mathbf{I} \to Y$$

be a $\mathcal{W}_i$-homotopy such that $\psi_0^{(i)} = h|Q \cup |K_i^{(n-1)}|$, $\psi_1^{(i)} = fg_i'$, and $\psi_t^{(i)}|Q = h|Q = fg$ for each $t \in \mathbf{I}$. Applying the Homotopy Extension Theorem 4.5.3, we can extend fg_i' to a map $h_i : P \to Y$ such that $h_i \simeq_{\mathcal{W}_i} h$ rel. Q. Since f is polyhedrally approximately n-soft by Theorem 7.3.5, we can extend g_i' to a map $g_i : P \to X$ such that fg_i is $\mathcal{W}_i$-close to h_i, hence fg_i is $\mathrm{st}\,\mathcal{W}_i$-close to h. Let

$$\varphi_i : (P \times \{0, 1\}) \cup (Q \times \mathbf{I}) \to X$$

be the map defined by $\varphi_i(x, t) = g(x)$ for each $(x, t) \in Q \times \mathbf{I}$, $\varphi_i(x, 0) = g_i(x)$, and $\varphi_i(x, 1) = g_{i+1}(x)$ for each $x \in P$. Then, $f\varphi_i$ extends to the map

$$\psi_i : (P \times \{0, 1\}) \cup ((Q \cup |K_i^{(n-1)}|) \times \mathbf{I}) \to Y$$

defined as follows: $\psi_i|P \times \{0, 1\} = f\varphi_i|P \times \{0, 1\}$ and

$$\psi_i(x,t) = \begin{cases} \psi^{(i)}(x, 1-2t) & \text{if } t \leqslant \frac{1}{2}, \\ \psi^{(i+1)}(x, 2t-1) & \text{if } t \geqslant \frac{1}{2}. \end{cases}$$

Since f is polyhedrally approximately n-soft, we can extend φ_i to a map

$$\tilde{\varphi}_i : (P \times \{0,1\}) \cup ((Q \cup |K_i^{(n-1)}|) \times \mathbf{I}) \to X$$

such that $f\tilde{\varphi}_i$ is $\mathcal{W}_i$-close to ψ_i.

For each n-simplex $\sigma \in K_i$ with $\sigma \not\subset Q$, $h(\sigma)$ is contained in some $W \in \mathcal{W}_i$. Since fg_i is $\operatorname{st}\mathcal{W}_i$-close to h, we have

$$fg_i(\sigma) \cup fg_{i+1}(\sigma) \subset \operatorname{st}(W, \operatorname{st}\mathcal{W}_i) \in \operatorname{st}^3\mathcal{W}_i \prec \mathcal{V}_i.$$

On the other hand, $\psi_i(\partial\sigma \times \mathbf{I}) \subset \operatorname{st}(W, \mathcal{W}_i)$ by the definition of ψ_i. Since $f\tilde{\varphi}_i$ is $\mathcal{W}_i$-close to ψ_i, it follows that

$$f\tilde{\varphi}_i(\partial\sigma \times \mathbf{I}) \subset \operatorname{st}(\operatorname{st}(W, \mathcal{W}_i), \mathcal{W}_i) \in \operatorname{st}^2\mathcal{W}_i \prec \mathcal{V}_i.$$

Consequently, there is some $V \in \mathcal{V}_i$ such that

$$f\tilde{\varphi}_i((\sigma \times \{0,1\}) \cup (\partial\sigma \times \mathbf{I})) = fg_i(\sigma) \cup fg_{i+1}(\sigma) \cup f\tilde{\varphi}_i(\partial\sigma \times \mathbf{I}) \subset V.$$

Then, we have $U \in \mathcal{U}_i$ and a map $\psi_\sigma : \sigma \times \mathbf{I} \to U$ such that

$$\psi_\sigma|(\sigma \times \{0,1\}) \cup (\partial\sigma \times \mathbf{I}) = f\tilde{\varphi}_i|(\sigma \times \{0,1\}) \cup (\partial\sigma \times \mathbf{I}).$$

Pasting these ψ_σ, we can obtain a $\mathcal{U}_i$-homotopy $\tilde{\psi}_i : P \times \mathbf{I} \to Y$, that is, $\tilde{\psi}_i$ is an extension of $f\tilde{\varphi}_i$ such that $\tilde{\psi}_i|\sigma \times \mathbf{I} = \psi_\sigma|\sigma \times \mathbf{I}$ for each n-simplex $\sigma \in K_i$ with $\sigma \not\subset Q$. Observe that

$$\tilde{\psi}_i(x,t) = f\tilde{\varphi}_i(x,t) = f\varphi_i(x,t) = fg(x) \text{ for each } (x,t) \in Q \times \mathbf{I}.$$

Now, we define $\psi^* : P \times \mathbf{I} \to Y$ by $\psi_0^* = h$ and

$$\psi^*(x,t) = \tilde{\psi}_i(x, 2 - 2^i t) \text{ if } 2^{-i} \leqslant t \leqslant 2^{-i+1},\ i \in \mathbb{N}.$$

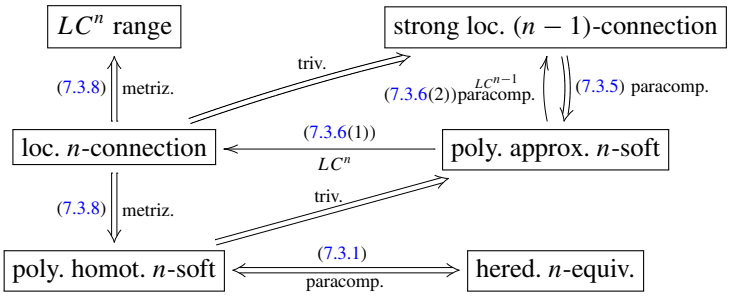
Clearly ψ^* is continuous at each point of $P \times (0,1]$. To verify the continuity of ψ^* at each point of $P \times \{0\}$, let $x \in P$ and $\varepsilon > 0$. By the continuity of h, we have a neighborhood U of x in P such that $\operatorname{diam} h(U) < \varepsilon/2$. Choose $k \in \mathbb{N}$ so that $2^{-k} < \varepsilon/2$. For each $x' \in U$ and $t \in (0, 2^{-k})$,

$$\begin{aligned} d(\psi^*(x',t),\psi^*(x,0)) &\leqslant d(\psi^*(x',t),\psi^*(x',0)) + d(h(x'),h(x)) \\ &\leqslant \operatorname{diam}\psi^*(\{x'\}\times[0,2^{-k}]) + \operatorname{diam}h(U) \\ &\leqslant \sum_{i=k+1}^{\infty}\operatorname{diam}\tilde{\psi}_i(\{x'\}\times\mathbf{I}) + \varepsilon/2 \\ &< \sum_{i=k+1}^{\infty}\operatorname{mesh}\mathcal{U}_i + \varepsilon/2 < 2^{-k} + \varepsilon/2 < \varepsilon. \end{aligned}$$

Thus, ψ^* is continuous. Observe that ψ^* is a $\mathcal{U}$-homotopy and $\psi_t^*|Q = fg$ for every $t \in \mathbf{I}$. Then, $fg_1 = \psi_1^* \simeq_{\mathcal{U}} h$ rel. Q. Hence, f is polyhedrally homotopically n-soft.

To see that Y is LC^n, let $y \in Y$. Each open neighborhood U of y in Y contains an open neighborhood V of y such that $fg \simeq 0$ in U for each map $g : \mathbf{S}^i \to f^{-1}(V)$, $0 \leqslant i \leqslant n$. For $0 \leqslant i \leqslant n$ and for each map $h : \mathbf{S}^i \to V$, let $\mathcal{U} = \{V, Y \setminus h(\mathbf{S}^i)\} \in \operatorname{cov}(Y)$. Since f is polyhedrally homotopically n-soft by Theorem 7.3.8, we have a map $g : \mathbf{S}^i \to X$ such that $fg \simeq_{\mathcal{U}} h$, which means that $fg \simeq h$ in V and $g(\mathbf{S}^i) \subset f^{-1}(V)$. Since $fg \simeq 0$ in U, it follows that $h \simeq 0$ in U. □

All the results can be summarized as follows:

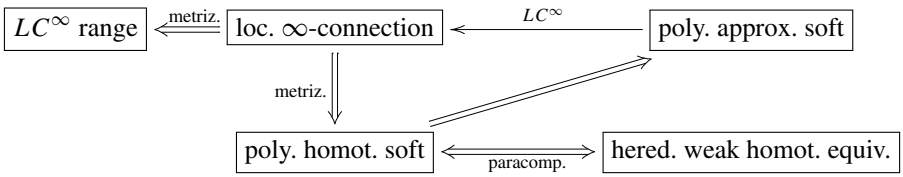


Corollary 7.3.9 *When Y be an LC^n metrizable space, the following five conditions for a map $f : X \to Y$ are equivalent:*

(a) *f is a local n-connection;*
(b) *f is a strong local $(n-1)$-connection ($f(X)$ is dense in Y when $n = 0$);*
(c) *f is polyhedrally approximately n-soft;*
(d) *f is polyhedrally homotopically n-soft;*
(e) *f is a hereditary n-equivalence.*

If Y is an LC^n paracompact space, conditions (a), (b), *and* (c) *are equivalent. If Y is an LC^{n-1} paracompact space, conditions* (b) *and* (c) *are equivalent. If Y is paracompact, conditions* (d) *and* (e) *are equivalent.* □

According to Theorem 7.2.9, when Y is metrizable, a map $f : X \to Y$ is polyhedrally homotopically soft if and only if f is polyhedrally homotopically n-soft for every $n \in \omega$. Then, the above implies the following:



Corollary 7.3.10 *When Y is an LC^∞ metrizable space, the following four conditions for a map $f : X \to Y$ are equivalent:*

(a) *f is a local ∞-connection;*
(b) *f is polyhedrally approximately soft;*
(c) *f is polyhedrally homotopically soft;*
(d) *f is a hereditary weak homotopy equivalence.*

When Y is paracompact, conditions (c) *and* (d) *are equivalent.* □

7.4 Fine Homotopy Equivalences Between ANRs

Recall that a map $f : X \to Y$ is a fine homotopy equivalence if f has a $\mathcal{U}$-homotopy inverse $g : Y \to X$ for each open cover $\mathcal{U}$ of Y, i.e., $gf \simeq_{f^{-1}(\mathcal{U})} \mathrm{id}_X$ and $fg \simeq_{\mathcal{U}} \mathrm{id}_Y$.

Proposition 7.4.1 *Let $f : X \to Y$ be a fine homotopy equivalence and Y be regular.*

(1) *If Y is locally contractible, then f is a local $*$-connection.*
(2) *If Y is LC^n, then f is a strong local n-connection.*
(3) *If Y is LC^∞, then f is a local ∞-connection.*

Proof Assertion (3) is a direct consequence of (2). The following is a proof of (1) (resp. (2)).

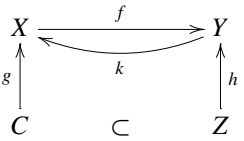
For each $y \in Y$ and each open neighborhood U of y in Y, choose open neighborhoods $V \subset W$ of y so that $\mathrm{cl}\, W \subset U$ and V is contractible in W (resp. $\alpha \simeq 0$ in W for every map $\alpha : \mathbf{S}^i \to V, 0 \leqslant i \leqslant n$) and let $\mathcal{U} = \{U,\ Y \setminus \mathrm{cl}\, W\} \in \mathrm{cov}(Y)$. Then, f has a $\mathcal{U}$-homotopy inverse $g : Y \to X$. Observe that $gf|f^{-1}(V) \simeq \mathrm{id}$ in $f^{-1}(U)$. On the other hand, $gf|f^{-1}(V) \simeq 0$ in $g(W)$ (resp. $gf\alpha \simeq 0$ in $g(W)$ for every map $\alpha : \mathbf{S}^i \to f^{-1}(V), 0 \leqslant i \leqslant n$). Since $fg(W) \subset \mathrm{st}(W, \mathcal{U}) = U$, it follows that $g(W) \subset f^{-1}(U)$. Therefore, $f^{-1}(V)$ is contractible in $f^{-1}(U)$ (resp. $\alpha \simeq 0$ in $f^{-1}(U)$ for every map $\alpha : \mathbf{S}^i \to f^{-1}(V), 0 \leqslant i \leqslant n$). □

Proposition 7.4.2 *Let $f : X \to Y$ be a fine homotopy equivalence and Y be paracompact.*

(1) *If X is an ANE, then f is approximately soft.*
(2) *If X is an ANE and Y is an ANR, then f is homotopically soft.*

Proof Because (2) is a combination of (1) and Proposition 7.2.10 (3), it suffices to show (1).

Let Z be a metrizable space and $g : C \to X$ be a map of a closed set C in Z such that fg extends to a map $h : Z \to Y$. For each $\mathcal{U} \in \mathrm{cov}(Y)$, let $\mathcal{V}$ be an open star-refinement of $\mathcal{U}$. We have a map $k : Y \to X$ such that $fk \simeq_{\mathcal{V}} \mathrm{id}_Y$ and $kf \simeq_{f^{-1}(\mathcal{V})} \mathrm{id}_X$. Then, $g \simeq_{f^{-1}(\mathcal{V})} kfg = kh|C$.



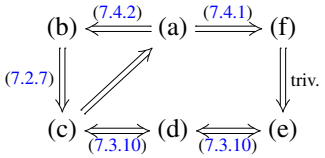
By the Homotopy Extension Theorem 6.4.1, g can be extended to a map $\tilde{g} : Z \to X$ that is $f^{-1}(\mathcal{V})$-homotopic to kh. Then, $f\tilde{g} \simeq_{\mathcal{V}} fkh \simeq_{\mathcal{V}} h$, so $f\tilde{g} \simeq_{\mathcal{U}} h$.[6] □

A fine homotopy equivalence between ANRs is characterized as follows:

Theorem 7.4.3 *For a map $f : X \to Y$ between ANRs, the following are equivalent:*

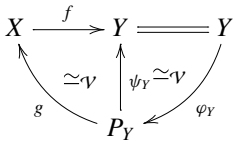
(a) *f is a fine homotopy equivalence;*
(b) *f is approximately (= homotopically) soft;*
(c) *f is polyhedrally approximately (= homotopically) soft;*
(d) *f is a hereditary weak homotopy equivalence;*
(e) *f is a local ∞-connection;*
(f) *f is a local $*$-connection.*

Proof In the following diagram of implications, the equivalence among (c), (d), and (e) has been observed in Corollary 7.3.10. The implication (f) $\Rightarrow$ (e) is trivial and (b) $\Rightarrow$ (c) has been shown in Proposition 7.2.7. Moreover, (a) $\Rightarrow$ (b) and (a) $\Rightarrow$ (f) have been obtained in Propositions 7.4.2 and 7.4.1, respectively. Thus, it remains to show the implication (c) $\Rightarrow$ (a).



(c) $\Rightarrow$ (a): For each $\mathcal{U} \in \mathrm{cov}(Y)$, let $\mathcal{V} \in \mathrm{cov}(Y)$ such that $\mathrm{st}^3 \mathcal{V} \prec \mathcal{U}$. Since Y is an ANR, we have a polyhedron P_Y with maps $\varphi_Y : Y \to P_Y$ and $\psi_Y : P_Y \to Y$ such that $\psi_Y \varphi_Y \simeq_{\mathcal{V}} \mathrm{id}_Y$, and apply (c) to obtain a map $g : P_Y \to X$ such that $fg \simeq_{\mathcal{V}} \psi_Y$.

[6] It is not shown that $f\tilde{g} \simeq_{\mathcal{U}} h$ rel. C.



Since $fg\varphi_Y \simeq_{\mathcal{V}} \psi_Y\varphi_Y \simeq_{\mathcal{V}} \mathrm{id}_Y$, we have a st$\mathcal{V}$-homotopy $h : Y \times \mathbf{I} \to Y$ such that $h_0 = \mathrm{id}_Y$ and $h_1 = fg\varphi_Y$.

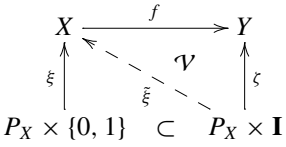
Choose $\mathcal{W} \in \mathrm{cov}(X)$ so that

$$\mathcal{W} \prec f^{-1}(\mathcal{V}) \text{ and } \mathcal{W} \prec (g\varphi_Y f)^{-1}(f^{-1}(\mathcal{V})).$$

Because X is an ANR, we have a polyhedron P_X with maps $\varphi_X : X \to P_X$ and $\psi_X : P_X \to X$ such that $\psi_X\varphi_X \simeq_{\mathcal{W}} \mathrm{id}_X$. We define a st$\mathcal{V}$-homotopy $\zeta : P_X \times \mathbf{I} \to Y$ and a map $\xi : P_X \times \{0, 1\} \to X$ as follows:

$$\zeta(x, t) = h(f\psi_X(x), t);\ \xi(x, 0) = \psi_X(x),\ \xi(x, 1) = g\varphi_Y f\psi_X(x).$$

Then, $\zeta | P_X \times \{0, 1\} = f\xi$. Indeed, $\zeta(x, 0) = h_0 f\psi_X(x) = f\psi_X(x)$ and $\zeta(x, 1) = h_1\psi_X(x) = fg\varphi_Y f\psi_X(x)$.



By (c), we can obtain an $f^{-1}(\mathrm{st}^2\mathcal{V})$-homotopy $\tilde{\xi} : P_X \times \mathbf{I} \to X$ that extends ξ. Let $\tilde{h} : X \times \mathbf{I} \to X$ be the $f^{-1}(\mathrm{st}^2\mathcal{V})$-homotopy defined by $\tilde{h}(x, t) = \tilde{\xi}(\varphi_X(x), t)$. Then,

$$\tilde{h}_0 = \tilde{\xi}_0\varphi_X = \psi_X\varphi_X \simeq_{\mathcal{W}} \mathrm{id}_X \text{ and}$$
$$\tilde{h}_1 = \tilde{\xi}_1\varphi_X = g\varphi_Y f\psi_X\varphi_X \simeq_{g\varphi_Y f(\mathcal{W})} g\varphi_Y f.$$

Since $\mathcal{W}$, $g\varphi_Y f(\mathcal{W}) \prec f^{-1}(\mathcal{V})$ and $\mathrm{st}^3\mathcal{V} \prec \mathcal{U}$, it follows that $g\varphi_Y f \simeq_{f^{-1}(\mathcal{U})} \mathrm{id}_X$. □

Remark 7.4.1 In Theorem 7.4.3 above, instead of assuming that X is an ANR, it suffices to assume that X is an ANE with the following property:

(∗) For each $\mathcal{U} \in \mathrm{cov}(X)$, X is $\mathcal{U}$-homotopy dominated by a polyhedron.

Indeed, the implications of the diagram in the above proof hold except (c) ⇒ (a). By Hanner's Characterization 6.6.2, every ANR has property (∗), which is the property of X required in the proof of (c) ⇒ (a).

Thus, Theorem 7.4.3 is valid for a (non-metrizable) polyhedron X and an ANR Y.

Since every open set in an ANR is an ANR, Theorem 7.4.3 yields the following corollary:

Corollary 7.4.4 *Let $f : X \to Y$ be a fine homotopy equivalence between ANRs. Then, for every open set U in Y, $f|f^{-1}(U) : f^{-1}(U) \to U$ is also a fine homotopy equivalence.* □

Corollary 7.4.5 *Let $f : X \to Y$ be a fine homotopy equivalence between ANRs. If $A \subset Y$ is contractible in an open neighborhood U in Y, then $f^{-1}(A)$ is contractible in $f^{-1}(U)$.*

Proof By Corollary 7.4.4, $f|f^{-1}(U) : f^{-1}(U) \to U$ has a homotopy inverse $g : U \to f^{-1}(U)$. Let $h : f^{-1}(U) \times \mathbf{I} \to f^{-1}(U)$ be a homotopy with $h_0 = \mathrm{id}$ and $h_1 = gf|f^{-1}(U)$. On the other hand, we have a contraction $k : A \times \mathbf{I} \to U$, that is, $k_0 = \mathrm{id}$ and k_1 is constant. Then, we can define a contraction $\varphi : f^{-1}(A) \times \mathbf{I} \to f^{-1}(U)$ as follows:

$$\varphi(x,t) = \begin{cases} h(x,2t) & \text{if } t \leqslant 1/2, \\ gk(f(x),2t-1) & \text{if } t \geqslant 1/2. \end{cases}$$ □

If $f_i : X_i \to Y_i$, $i = 1, 2$, are local $*$-connections, then $f_1 \times f_2 : X_1 \times X_2 \to Y_1 \times Y_2$ is clearly a local $*$-connection. Then, we have the following:

Corollary 7.4.6 *Let $f_i : X_i \to Y_i$, $i = 1, 2$, be fine homotopy equivalences between ANRs. Then, $f_1 \times f_2 : X_1 \times X_2 \to Y_1 \times Y_2$ is a fine homotopy equivalence.* □

Remark 7.4.2 The above corollary is valid even if X_1 and X_2 are ANEs with the property $(*)$ in Remark 7.4.1.

Due to Proposition 6.7.1, a subset X of a metrizable space Y is homotopy dense in Y if and only if the inclusion $X \subset Y$ is a fine homotopy equivalence. Applying Theorem 7.4.3 to the inclusion $X \subset Y$, we have the following:

Corollary 7.4.7 *Let X and Y be ANRs such that X is a dense subset of Y. Then, the following are equivalent:*

(a) *X is homotopy dense in Y;*
(b) *For each open set U in Y, the inclusion $U \cap X \subset U$ is a weak homotopy equivalence, i.e., for each $n \in \mathbb{N}$, if a map $\alpha : \mathbf{S}^{n-1} \to U \cap X$ extends to a map $\beta : \mathbf{B}^n \to U$, then α extends to a map $\tilde{\alpha} : \mathbf{B}^n \to U \cap X$ such that $\tilde{\alpha} \simeq \beta$ in U rel. $\mathbf{S}^{n-1}$;*
(c) *Every neighborhood U of each y in Y contains a neighborhood V of y in Y such that every map $\alpha : \mathbf{S}^{n-1} \to V \cap X$ is null-homotopic in $U \cap X$ for each $n \in \mathbb{N}$.* □

From the equivalence of (a) and (b) in the above corollary, the following is easily derived:

Corollary 7.4.8 *Let X be a homotopy dense set in an ANR Y. Then, for every open set U in Y, $X \cap U$ is homotopy dense in U.* □

Applying Corollary 7.4.7, we can show that every locally homotopy dense subset of an ANR is homotopy dense, that is:

Proposition 7.4.9 *Let X be a subset of an ANR Y. If there is an open cover $\mathcal{U}$ of X in Y such that for each $U \in \mathcal{U}$, $X \cap U$ is homotopy dense in U, then X is homotopy dense in Y.*

Proof First of all, note that each $U \in \mathcal{U}$ is an ANR as an open set in an ANR. It follows from Theorem 6.6.9 that $X \cap U$ is an ANR. By Proposition 6.2.9 (4) (Hanner's Theorem), X is an ANR.

Let V be an open neighborhood of $y \in Y$. Take $U \in \mathcal{U}$ such that $y \in U$. As we saw above, U and $X \cap U$ are ANRs. By Corollary 7.4.7, $V \cap U$ contains a neighborhood W of y in U such that every map $\alpha : \mathbf{S}^{n-1} \to W \cap (X \cap U)$ $(= W \cap X)$ is null-homotopic in $(V \cap U) \cap (X \cap U)$ $(= V \cap U \cap X)$ for each $n \in \mathbb{N}$. Then, every map $\alpha : \mathbf{S}^{n-1} \to W \cap X$ is null-homotopic in $V \cap X$ for each $n \in \mathbb{N}$. Due to Corollary 7.4.7, this means that X is homotopy dense in Y. □

To Corollary 7.4.7, we can add one more condition, that is:

Proposition 7.4.10 *Let X and Y be ANRs with $X \subset Y$ and $d \in \mathrm{Metr}(Y)$. Then, X is homotopy dense in Y if and only if, for each compact set A in Y and $\varepsilon > 0$, there is a map $f : A \to X$ that is ε-close to* id.

Proof The "only if" part is trivial. To see the "if" part, take U, α, and β as in condition (b) of Corollary 7.4.7. Since U is an ANR, we have an $\hbar$-refinement $\mathcal{V}$ of $\{U\}$ by Corollary 6.3.6. Since $\beta(\mathbf{B}^n)$ is compact, we can find $\delta > 0$ such that $\{\mathrm{B}_d(y, \delta) \mid y \in \beta(\mathbf{B}^n)\} \prec \mathcal{V}$, which implies that if a map $\beta' : \mathbf{B}^n \to U$ is δ-close to β and $\beta'|\mathbf{S}^{n-1} = \beta|\mathbf{S}^{n-1}$, then $\beta' \simeq \beta$ rel. $\mathbf{S}^{n-1}$ in U. Moreover, since $U \cap X$ is also an ANR, we can find $\varepsilon > 0$ such that $\varepsilon < \mathrm{dist}_d(\beta(\mathbf{B}^n), Y \setminus U)$ and if a map $\alpha' : \mathbf{S}^{n-1} \to U \cap X$ is ε-close to α, then $\alpha' \simeq_{\delta/2} \alpha$ in $U \cap X$ (so $\varepsilon < \delta/2$).

Applying the condition to $\beta(\mathbf{B}^n)$ and $\varepsilon > 0$, we can find a map $f : \beta(\mathbf{B}^n) \to X$ that is ε-close to id. Thus, we can obtain a map $f\beta : \mathbf{B}^n \to X \cap U$ such that $f\beta|\mathbf{S}^{n-1}$ is ε-close to $\beta|\mathbf{S}^{n-1} = \alpha$, which implies that $f\beta|\mathbf{S}^{n-1} \simeq_{\delta/2} \alpha$ in $U \cap X$. We can apply the Homotopy Extension Theorem 6.4.1 to extend α to a map $\tilde{\alpha} : \mathbf{B}^n \to U \cap X$ such that $f\beta \simeq_{\delta/2} \tilde{\alpha}$ in $U \cap X$. Observe that $\tilde{\alpha}$ is δ-close to β and $\tilde{\alpha}|\mathbf{S}^{n-1} = \alpha = \beta|\mathbf{S}^{n-1}$, which implies $\tilde{\alpha} \simeq \beta$ rel. $\mathbf{S}^{n-1}$ in U. Therefore, X is homotopy dense in Y by Corollary 7.4.7. □

Now, we prove the following theorem:

Theorem 7.4.11 *Let X and Y be ANRs and $f = \lim_{i \to \infty} f_i : X \to Y$ be the uniform limit of fine homotopy equivalences with respect to some $d \in \mathrm{Metr}(Y)$. Then, f is also a fine homotopy equivalence.*

Proof According to Theorem 7.4.3, it suffices to show that f is a local ∞-connection. Since each $f_i(X)$ is dense in Y, it follows that $f(X)$ is dense. Every open neighborhood U of each y in Y contains an open neighborhood V of y such that V is contractible in U. For each $n \in \mathbb{N}$, let $\alpha : \mathbf{S}^n \to f^{-1}(V)$ be a map. Then, $f\alpha$ extends to a map $\beta : \mathbf{B}^{n+1} \to U$. Let

$$\delta = \operatorname{dist}(\beta(\mathbf{B}^{n+1}), Y \setminus U) > 0.$$

Since Y is an ANR, the open cover $\{\mathrm{B}(y, \delta/6) \mid y \in Y\}$ of Y has an h-refinement $\mathcal{V} \in \operatorname{cov}(Y)$ (Corollary 6.3.6). Because $f\alpha(\mathbf{S}^n)$ is compact, we have $\varepsilon > 0$ such that $\{\mathrm{B}(f\alpha(x), \varepsilon) \mid x \in \mathbf{S}^n\} \prec \mathcal{V}$. Indeed, find $x_i \in \mathbf{S}^n$ and $\varepsilon_i > 0$, $i = 1, \ldots, k$, such that

$$\{\mathrm{B}(f\alpha(x_i), 2\varepsilon_i) \mid i = 1, \ldots, k\} \prec \mathcal{V} \text{ and } f\alpha(\mathbf{S}^n) \subset \bigcup_{i=1}^{k} \mathrm{B}(f\alpha(x_i), \varepsilon_i).$$

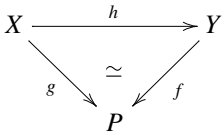
Then, $\min\{\varepsilon_i \mid i = 1, \ldots, k\} > 0$ is the desired $\varepsilon > 0$. Choose $i \in \mathbb{N}$ so that f_i is ε-close to f, which implies that $f_i\alpha \simeq_{\delta/3} f\alpha = \beta|\mathbf{S}^n$. By the Homotopy Extension Theorem 6.4.1, $f_i\alpha$ extends to a map $\beta' : \mathbf{B}^{n+1} \to Y$ such that $\beta' \simeq_{\delta/3} \beta$. Since f_i is polyhedrally approximately soft by Theorem 7.4.3, α extends to a map $\tilde{\alpha} : \mathbf{B}^{n+1} \to X$ such that $d(f_i\tilde{\alpha}, \beta') < \delta/3$. Observe that

$$d(f\tilde{\alpha}, \beta) \leqslant d(f\tilde{\alpha}, f_i\tilde{\alpha}) + d(f_i\tilde{\alpha}, \beta') + d(\beta', \beta) < \delta.$$

Then, $f\tilde{\alpha}(\mathbf{B}^{n+1}) \subset U$, i.e., $\tilde{\alpha}(\mathbf{B}^{n+1}) \subset f^{-1}(U)$. □

7.5 Hereditary Shape Equivalences and UV^n Maps

Recall that $[X, Y]$ is the set of the homotopy classes of maps from X to Y. For any space P, a map $h : X \to Y$ induces a function $h^* : [Y, P] \to [X, P]$ defined by $h^*([f]) = [fh]$ for each map $f : Y \to P$.



Fact *For every space X, Y, Z, and P, the following statements hold:*

(1) *The identity* $\mathrm{id}_X : X \to X$ *induces the identity* $\mathrm{id}_{[X,P]} : [X, P] \to [X, P]$, *i.e.,* $(\mathrm{id}_X)^* = \mathrm{id}_{[X,P]}$.

(2) *The composition $h_2h_1 : X \to Z$ of two maps $h_1 : X \to Y$ and $h_2 : Y \to Z$ induces the composition $h_1^*h_2^* : [Z, P] \to [X, P]$ of $h_2^* : [Z, P] \to [Y, P]$ and $h_1^* : [Y, P] \to [X, P]$, i.e., $(h_2h_1)^* = h_1^*h_2^*$.*

A map $h : X \to Y$ is said to be a **shape equivalence** if the function $h^* : [Y, P] \to [X, P]$ is a bijection for every ANR P, equivalently for every polyhedron P (Corollary 6.6.5). Observe the following:

(i) *Every homotopy equivalence is a shape equivalence.*
(ii) *The composition of shape equivalences is also a shape equivalence.*

Remark 7.5.1 We do not give the definition that X and Y have the same shape type, but it is not defined by the existence of a shape equivalence $h : X \to Y$. It should be noted that if there is a shape equivalence $h : X \to Y$ then X and Y have the same shape, but the converse does not hold.

The following is easy to prove:

Proposition 7.5.1 *The following are equivalent for a space $X \neq \emptyset$:*

(a) *X has trivial shape;*
(b) *The map of X to the singleton $\{0\}$ is a shape equivalence;*
(c) *For every $x \in X$, the inclusion $\{x\} \hookrightarrow X$ is a shape equivalence:*
(d) *There is some $x \in X$ such that the inclusion $\{x\} \hookrightarrow X$ is a shape equivalence.* □

Corollary 7.5.2 *Let $h : X \to Y$ be a shape equivalence. Then, X has trivial shape if and only if Y has trivial shape. When X and Y are compacta, X is cell-like if and only if Y is cell-like.* □

A map $h : X \to Y$ is said to be a **hereditary shape equivalence** if $h|h^{-1}(A) : h^{-1}(A) \to A$ is a shape equivalence for any closed set A in Y. In this case, $h(X) = Y$ and each fiber $h^{-1}(y)$ has trivial shape. Then, every hereditary shape equivalence is cell-like if it is perfect.

Theorem 7.5.3 *Every closed fine homotopy equivalence $h : X \to Y$ between metrizable spaces is a hereditary shape equivalence.*

Proof Let A be a closed set in Y and P be an ANR. Each map $f : h^{-1}(A) \to P$ extends to a map $\tilde{f} : U \to P$ from an open neighborhood U of $h^{-1}(A)$ in X. Because h is a closed map, there is an open neighborhood V of A in Y such that $h^{-1}(V) \subset U$. Let $\mathcal{V} = \{V, Y \setminus A\} \in \mathrm{cov}(Y)$. Since h is a fine homotopy equivalence, it has a $\mathcal{V}$-homotopy inverse $g : Y \to X$, i.e., $hg \simeq_{\mathcal{V}} \mathrm{id}$ and $gh \simeq_{h^{-1}(\mathcal{V})} \mathrm{id}$. Observe that $gh|h^{-1}(A) \simeq \mathrm{id}_{h^{-1}(A)}$ in $h^{-1}(V) \subset U$. Then, we have a map $\tilde{f}g|A : A \to P$ such that $\tilde{f}gh|h^{-1}(A) \simeq \tilde{f}|h^{-1}(A) = f$.

Now, let $f_0, f_1 : A \to P$ be maps such that $f_0 h|h^{-1}(A) \simeq f_1 h|h^{-1}(A)$. Since P is an ANR, A has an open neighborhood W in Y with maps $\tilde{f}_0, \tilde{f}_1 : W \to P$ that are extensions of f_0 and f_1, respectively. Then, $\tilde{f}_0 h|h^{-1}(A) \simeq \tilde{f}_1 h|h^{-1}(A)$. We can choose an open neighborhood U of $h^{-1}(A)$ in X so that $U \subset h^{-1}(W)$ and $\tilde{f}_0 h|U \simeq \tilde{f}_1 h|U$. As before, there is an open neighborhood V of A in Y such that $V \subset W$, $h^{-1}(V) \subset U$, and h has a $\mathcal{V}$-homotopy inverse $g : Y \to X$, where $\mathcal{V} = \{V, Y \setminus A\} \in \mathrm{cov}(Y)$. Since $hg|A \simeq \mathrm{id}_A$ in V and $g(A) \subset h^{-1}(V) \subset U$, it follows that

$$\begin{aligned} f_0 = \tilde{f}_0|A &\simeq \tilde{f}_0 hg|A = (\tilde{f}_0 h|U)(g|A) \\ &\simeq (\tilde{f}_1 h|U)(g|A) = \tilde{f}_1 hg|A \simeq \tilde{f}_1|A = f_1. \end{aligned}$$ □

Let X be a metrizable space. A surjective map $f : X \to Y$ is called a ***UV* map*** if each fiber $f^{-1}(y)$ is UV^*; equivalently, each fiber $f^{-1}(y)$ has trivial shape by Theorem 7.1.2. Thus, we can assert as follows:

- A perfect UV^* map is the same as a cell-like map.

For $n \in \omega \cup \{\infty\}$, a surjective map $f : X \to Y$ is a ***UV^n* map** if each fiber $f^{-1}(y)$ is UV^n. For a **closed** surjective map $f : X \to Y$, if X is an ANR, then the following equivalences hold:

$$\begin{aligned} f \text{ is } UV^n &\Leftrightarrow f \text{ is a strong local } n\text{-connection;} \\ f \text{ is } UV^\infty &\Leftrightarrow f \text{ is a local } \infty\text{-connection;} \\ f \text{ is } UV^* &\Leftrightarrow f \text{ is a local } *\text{-connection.} \end{aligned}$$

For closed surjective maps, we have the following version of Theorem 7.4.3:

Theorem 7.5.4 *For a closed surjective map $f : X \to Y$ between ANRs X and Y, the following are equivalent:*

(a) *f is a fine homotopy equivalence;*
(b) *f is a UV^∞ map (= a local ∞-connection);*
(c) *f is a UV^* map (= a local $*$-connection);*
(d) *f is a hereditary shape equivalence.*

When f is a perfect map, the following is also equivalent to the above:

(e) *f is a cell-like map.*

Proof The equivalence among (a) through (c) has been obtained by Theorem 7.4.3. The implication (a) $\Rightarrow$ (d) has been shown in Theorem 7.5.3. Condition (d) implies that each fiber $f^{-1}(y)$ has trivial shape. Because X is an ANR, f is a closed UV^* map. Thus, we have (d) $\Rightarrow$ (c). □

Remark 7.5.2 Let X be the $\sin(1/x)$-curve given at the beginning of Sect. 7.1 and $f = \mathrm{pr}_1|X : X \to Y = \mathbf{I}$, where $\mathrm{pr}_1 : \mathbb{R}^2 \to \mathbb{R}$ is the projection onto the first factor. Then, f is cell-like (UV^*) but not a (fine) homotopy equivalence nor polyhedrally

approximately 1-soft. Hence, in Theorem 7.5.4, it is essential that X is an ANR. Moreover, in general, a cell-like map is not a shape equivalence, hence it is not a hereditary shape equivalence. Such examples will be given in Theorems 7.7.5 and 7.7.8.

Corollary 7.5.5 *Let X, Y, and Z be ANRs. For every two cell-like maps $f : X \to Y$ and $g : Y \to Z$, the composition $gf : X \to Y$ is also a cell-like map.*

Proof It is easy to show that the composition of hereditary shape equivalences is also a hereditary shape equivalence. Then, the result follows from Theorem 7.5.4. □

Remark 7.5.3 In general, the composition of cell-like maps is not cell-like. Such an example will be seen in Remark 7.7.4 in Sect. 7.7.

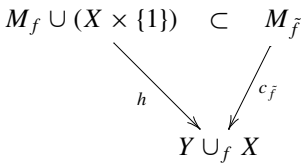
The following proposition will be used in Sect. 7.7.

Proposition 7.5.6 *For each map $f : A \to Y$ of a closed set A in a metrizable space X, let*

$$h : M_f \cup (X \times \{1\}) \to Y \cup_f X$$

be the map defined by $h|M_f = c_f$ and $h(x, 1) = x$ for each $x \in X \setminus A$, where c_f is the collapsing of the mapping cylinder. Then, h is a shape equivalence.

Proof First, we extend f to the map $\tilde{f} : X \to Y \cup_f X$ by $\tilde{f}|X \setminus A = \mathrm{id}$, that is, $\tilde{f}$ is the restriction of the natural quotient map from $Y \oplus X$ onto $Y \cup_f X$. Then, $M_f \cup (X \times \{1\}) \subset M_{\tilde{f}}$ and $h = c_{\tilde{f}}|M_f \cup (X \times \{1\})$, where the collapsing $c_{\tilde{f}}$ is a homotopy equivalence, and is therefore a shape equivalence.



Since the composition of shape equivalences is a shape equivalence, it suffices to show that the inclusion $i : M_f \cup (X \times \{1\}) \hookrightarrow M_{\tilde{f}}$ is a shape equivalence, that is,

$$i^* : [M_{\tilde{f}}, P] \to [M_f \cup (X \times \{1\}),\ P]$$

is bijective for each ANR P. Identifying $X \setminus A = (X \setminus A) \times \{0\}$, we can regard the mapping cylinder $M_{\tilde{f}}$ as follows:

$$M_{\tilde{f}} = Y \cup_{f \circ \mathrm{pr}_A | A \times \{0\}} (X \times \mathbf{I}).$$

Let $q : Y \oplus (X \times \mathbf{I}) \to M_{\tilde{f}}$ be the natural quotient map. See Fig. 7.1.

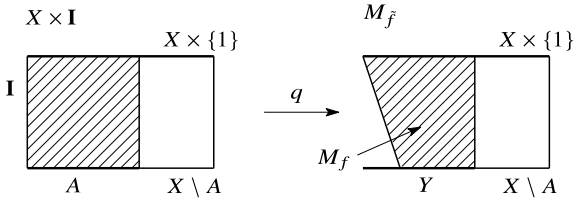


Fig. 7.1 $M_f \cup (X \times \{1\}) \subset M_{\tilde{f}}$

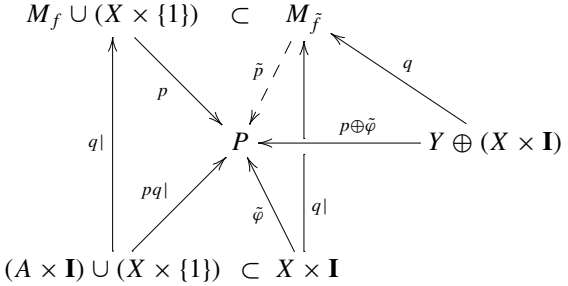
To show that i^* is surjective, let $p : M_f \cup (X \times \{1\}) \to P$ be a map. Regarding $\varphi = pq|A \times \mathbf{I}$ as a homotopy such that φ_1 extends over X, we apply the Homotopy Extension Theorem 6.4.1 to obtain a map $\tilde{\varphi} : X \times \mathbf{I} \to P$ such that

$$\tilde{\varphi}|(A \times \mathbf{I}) \cup (X \times \{1\}) = pq|(A \times \mathbf{I}) \cup (X \times \{1\}).$$

Then, it follows that

$$\tilde{\varphi}|A \times \{0\} = pq|A \times \{0\} = (p|Y)f\mathrm{pr}_A|A \times \{0\}.$$

Thus, we have the map $\tilde{p} : M_{\tilde{f}} \to P$ such that $\tilde{p}|Y = p|Y$ and $\tilde{p}q|X \times \mathbf{I} = \tilde{\varphi}$, hence $\tilde{p}i = \tilde{p}|M_f \cup X \times \{1\} = p$.



To show that i^* is injective, let $p, p' : M_{\tilde{f}} \to P$ such that $pi \simeq p'i$. Pasting p, p' and a homotopy from pi to $p'i$, we define a map $\psi : D \to P$ on the set

$$D = \big(\big(M_f \cup (X \times \{1\})\big) \times \mathbf{I}\big) \cup \big(M_{\tilde{f}} \times \{0, 1\}\big),$$

which is regarded as the adjunction space $(Y \times \mathbf{I}) \cup_{\check{f}} \Pi$, where

$$\Pi = (A \times \mathbf{I} \times \mathbf{I}) \cup (X \times \{1\} \times \mathbf{I}) \cup (X \times \mathbf{I} \times \{0, 1\}) \subset X \times \mathbf{I} \times \mathbf{I} \text{ and}$$
$$\check{f} = (f \circ \mathrm{pr}_A|A \times \{0\}) \times \mathrm{id}_{\mathbf{I}} : A \times \{0\} \times \mathbf{I} \to Y \times \mathbf{I}.$$

We make the following identification:

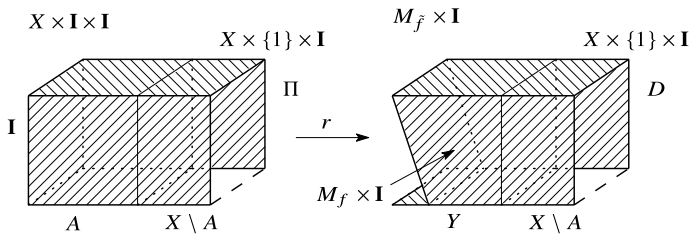


Fig. 7.2 $\big((M_f \cup (X \times \{1\})) \times \mathbf{I}\big) \cup \big(M_{\check{f}} \times \{0, 1\}\big) \subset M_{\tilde{f}} \times \mathbf{I}$

$$M_{\tilde{f}} \times \mathbf{I} = (Y \times \mathbf{I}) \cup_{\check{f}} (X \times \mathbf{I} \times \mathbf{I}),$$

where the natural quotient map is denoted by

$$r = q \times \mathrm{id}_{\mathbf{I}} : (Y \times \mathbf{I}) \oplus (X \times \mathbf{I} \times \mathbf{I}) = \big(Y \oplus (X \times \mathbf{I})\big) \times \mathbf{I} \to M_{\tilde{f}} \times \mathbf{I}.$$

Then, $r((Y \times \mathbf{I}) \oplus \Pi)$ is the domain of ψ. See Fig. 7.2.

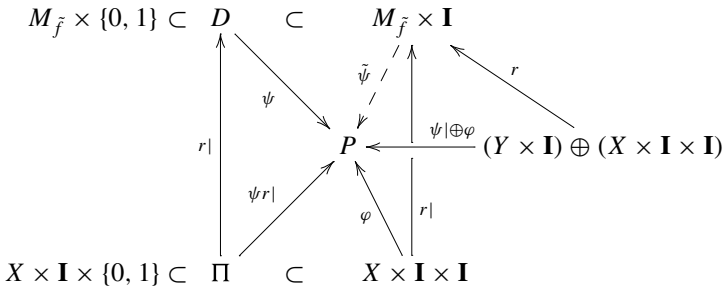
Now, we regard the map $\psi r | (A \times \mathbf{I} \times \mathbf{I}) \cup (X \times \mathbf{I} \times \{0, 1\})$ as a homotopy that realizes

$$\begin{aligned} &\psi r | (A \times \{0\} \times \mathbf{I}) \cup (X \times \{0\} \times \{0, 1\}) \\ &\qquad \simeq \psi r | (A \times \{1\} \times \mathbf{I}) \cup (X \times \{1\} \times \{0, 1\}). \end{aligned}$$

Since $\psi r | (A \times \{1\} \times \mathbf{I}) \cup (X \times \{1\} \times \{0, 1\})$ extends over $X \times \{1\} \times \mathbf{I}$, we apply the Homotopy Extension Theorem 6.4.1 to obtain a map $\varphi : X \times \mathbf{I} \times \mathbf{I} \to P$ such that $\varphi | \Pi = \psi r | \Pi$. Then, it follows that

$$\varphi | A \times \{0\} \times \mathbf{I} = \psi r | A \times \{0\} \times \mathbf{I} = (\psi | Y \times \mathbf{I}) \check{f}.$$

Therefore, we have the homotopy $\tilde{\psi} : M_{\tilde{f}} \times \mathbf{I} \to P$ such that $\tilde{\psi} r | X \times \mathbf{I} \times \mathbf{I} = \varphi$ and $\tilde{\psi} | Y \times \mathbf{I} = \psi | Y \times \mathbf{I}$.



Since $\tilde{\psi} | M_{\tilde{f}} \times \{0, 1\} = \psi | M_{\tilde{f}} \times \{0, 1\}$, $\tilde{\psi}$ is a homotopy realizing $p \simeq p'$. □

7.6 The Near-Selection Theorem

Recall that X has property C if, for any open covers $\mathcal{U}_n$ of X, $n \in \mathbb{N}$, X has an open cover $\mathcal{V} = \bigcup_{n\in\mathbb{N}} \mathcal{V}_n$ such that each $\mathcal{V}_n$ is pairwise disjoint and $\mathcal{V}_n \prec \mathcal{U}_n$ (see Sect. 6.10). Every countable-dimensional metrizable space has property C (Proposition 6.10.4). In this section, we prove the following Haver's Near-Selection Theorem, where $\mathrm{Cont}(Y)$ is the hyperspace of continua in Y with the Vietoris topology (cf. Sect. 2.12):

Theorem 7.6.1 (HAVER'S NEAR-SELECTION) *Let X be a σ-compact metrizable space with property C, $Y = (Y, d)$ be a metric ANR, and $\varphi : X \to \mathrm{Cont}(Y)$ be a continuous set-valued function such that each $\varphi(x)$ is cell-like. For each map $\alpha : X \to (0, \infty)$, there exists a map $h : X \to Y$ such that $d(h(x), \varphi(x)) < \alpha(x)$ for every $x \in X$.*

Proof We may assume that X is a metric space. Let $X = \bigcup_{n\in\mathbb{N}} X_n$, where $X_1 \subset X_2 \subset \cdots$ are compact. We can inductively choose $\delta_1 > \delta_2 > \cdots > 0$ so that $\delta_n \leqslant 2^{-n-3}$ and $\mathrm{N}(\varphi(x), 2\delta_n)$ is contractible in $\mathrm{N}(\varphi(x), \delta_{n-1})$ for each $x \in X_n$, where $\delta_0 = 2^{-3}$. Indeed, assume that $\delta_{n-1} > 0$ is chosen. Because $\varphi(x)$ is compact and $U V^*$, we have $\delta_x > 0$ such that $\mathrm{N}(\varphi(x), \delta_x)$ is contractible in $\mathrm{N}(\varphi(x), \delta_{n-1}/2)$, where $\delta_x \leqslant \delta_{n-1}/2$. Since φ is continuous, each $x \in X_n$ has an open neighborhood V_x such that $d_H(\varphi(x), \varphi(x')) < \delta_x/2$ for any $x' \in V_x$. Observe that $x' \in V_x$ implies

$$\mathrm{N}(\varphi(x'), \delta_x/2) \subset \mathrm{N}(\varphi(x), \delta_x) \text{ and } \mathrm{N}(\varphi(x), \delta_{n-1}/2) \subset \mathrm{N}(\varphi(x'), \delta_{n-1}).$$

By the compactness of X_n, we have $x_1, \ldots, x_k \in X_n$ such that $X_n \subset \bigcup_{i=1}^{k} V_{x_i}$. Then, the desired $\delta_n > 0$ can be defined as follows:

$$\delta_n = \frac{1}{4} \min \left\{ \delta_{x_i} \;\middle|\; i = 1, \ldots, k \right\} > 0.$$

By the continuity of φ and the compactness of X_n, we can take $\varepsilon_1 > \varepsilon_2 > \cdots > 0$ such that if $x \in X_n$ and $x' \in X$ with $d(x, x') < 2\varepsilon_n$ then $d_H(\varphi(x), \varphi(x')) < \delta_n$, that is, $\varphi(x) \subset \mathrm{N}(\varphi(x'), \delta_n)$ and $\varphi(x') \subset \mathrm{N}(\varphi(x), \delta_n)$.

We will construct collections $\mathcal{V}_1, \mathcal{V}_2, \ldots$ of open sets in X satisfying the following conditions:

(1) $\mathcal{V} = \bigcup_{i\in\mathbb{N}} \mathcal{V}_i \in \mathrm{cov}(X)$;
(2) Each $\mathcal{V}_i$ is pairwise disjoint;
(3) $\mathrm{mesh}\, \mathcal{V}_i < \varepsilon_i$;
(4) Each open set in $\mathcal{V}_i$ meets X_i;
(5) $\alpha(x) > 2^{-i-1}$ for $x \in \bigcup \mathcal{V}_i$;
(6) $\mathcal{V}_i \cap \mathcal{V}_j = \emptyset$ if $i \neq j$.

Let $k : \mathbb{N}^2 \to \mathbb{N}$ be a bijection such that $k(i, n) \geqslant i$ for every $(i, n) \in \mathbb{N}^2$ (e.g., $k(i, n) = 2^{n-1}(2i - 1) \geqslant i$). Without loss of generality, we can assume that $\alpha(x) \leqslant 1$ for all $x \in X$. For each $i, j \in \mathbb{N}$, let

$$X_{i,j} = X_i \cap \alpha^{-1}([2^{-j}, 2^{-j+1}]).$$

Since each $X_{i,j}$ has property C by Lemma 6.10.3, we have collections $\mathcal{V}_{j,k(i,n)}$, $n \in \mathbb{N}$, of open sets in

$$\alpha^{-1}\big((2^{-j} - 2^{-j-2}, 2^{-j+1} + 2^{-j-2})\big)$$

such that $\bigcup_{n\in\mathbb{N}} \mathcal{V}_{j,k(i,n)}$ covers $X_{i,j}$, each $\mathcal{V}_{j,k(i,n)}$ is pairwise disjoint, and

$$\operatorname{mesh} \mathcal{V}_{j,k(i,n)} < \varepsilon_{j+2k(i,n)-2}.$$

It can be assumed that the open sets of $\mathcal{V}_{j,k(i,n)}$ meet $X_{i,j}$, so they meet $X_{j+2k(i,n)-2}$ because $X_{i,j} \subset X_i \subset X_{j+2k(i,n)-2}$. Then, the following conditions hold:

(7) $\bigcup_{m\in\mathbb{N}} \mathcal{V}_{j,m}$ covers $\alpha^{-1}([2^{-j}, 2^{-j+1}])$;
(8) Each $\mathcal{V}_{j,m}$ is pairwise disjoint;
(9) $\operatorname{mesh} \mathcal{V}_{j,m} < \varepsilon_{j+2m-2}$;
(10) Every open set of $\mathcal{V}_{j,m}$ meets X_{j+2m-2};
(11) Every open set of $\mathcal{V}_{j,m}$ is contained in

$$\alpha^{-1}\big((2^{-j} - 2^{-j-2}, 2^{-j+1} + 2^{-j-2})\big).$$

We define $\mathcal{V}_i$ as follows:

$$\mathcal{V}_{2i-1} = \bigcup_{m=1}^{i} \mathcal{V}_{2i-2m+1,m} \quad \text{and} \quad \mathcal{V}_{2i} = \bigcup_{m=1}^{i} \mathcal{V}_{2i-2m+2,m}.$$

For $j, m \in \mathbb{N}$, let $n = j + 2m - 2$. Then,

$$n = 2i - 1 \;\Rightarrow\; j = n - 2m + 2 = 2i - 2m + 1;$$
$$n = 2i \;\Rightarrow\; j = n - 2m + 2 = 2i - 2m + 2.$$

Consequently, $\mathcal{V}_{j,m} \subset \mathcal{V}_n$. Therefore, (1) follows from (7); (2) follows from (8) and (11); (3) follows from (9); (4) follows from (10); (5) follows from (11). By replacing $\mathcal{V}_n$ with $\mathcal{V}_n \setminus \bigcup_{i<n} \mathcal{V}_i$, condition (6) is satisfied without failing the other conditions.

Let K be the nerve of $\mathcal{V} = \bigcup_{i\in\mathbb{N}} \mathcal{V}_i \in \operatorname{cov}(X)$ with $p : X \to |K|$ a canonical map. For each $V \in \mathcal{V}_i$, take $x_V \in V \cap X_i$ (cf. (4)) and $y_V \in \varphi(x_V)$. Then, we have a contraction

$$r^V : \mathrm{N}(\varphi(x_V), 2\delta_i) \times \mathbf{I} \to \mathrm{N}(\varphi(x_V), \delta_{i-1})$$

such that $r_1^V(\mathrm{N}(\varphi(x_V), 2\delta_i)) = \{y_V\}$ ($r_0^V = \mathrm{id}$). For each n-simplex $\sigma \in K$, let $V(\sigma) \in \mathcal{V}_{i(\sigma)}$ be the vertex of σ such that $i(\sigma) = \min\{i \in \mathbb{N} \mid \sigma^{(0)} \cap \mathcal{V}_i \neq \emptyset\}$ (cf. (2)), and let σ_0 be the $(n-1)$-face of σ with $V(\sigma) \notin \sigma_0$. For each $n \in \mathbb{N}$, let

$$K_n = \Big\{\sigma \in K \;\Big|\; \sigma^{(0)} \subset \textstyle\bigcup_{i=1}^{n} \mathcal{V}_i\Big\}.$$

Then, each K_n is a subcomplex of K with $K_n \subset K_{n+1}$ and $K = \bigcup_{n\in\mathbb{N}} K_n$.

We will inductively construct maps $g_n : |K_n| \to Y$, $n \in \mathbb{N}$, such that $g_n||K_{n-1}| = g_{n-1}$ and, for $\sigma \in K_n$, $t \in \mathbf{I}$, and $z \in \sigma_0$,

$(*)$ $g_n((1-t)z + tV(\sigma)) = r^{V(\sigma)}(g_n(z), t) \in \mathrm{N}(\varphi(x_{V(\sigma)}), \delta_{i(\sigma)-1})$.

Since $K_1 = K_1^{(0)} = \mathcal{V}_1$ by (2), we can first define $g_1 : |K_1| \to Y$ by $g_1(V) = y_V$ for each $V \in \mathcal{V}_1$. Suppose that $g_1, \ldots, g_{n-1}$ have been defined. For $j \leqslant n$, let

$$K_{n,j} = \left\{\sigma \in K \;\middle|\; \sigma^{(0)} \subset \textstyle\bigcup_{i=j}^{n} \mathcal{V}_i\right\} \subset K_n.$$

Then, $K_{n,j}$ is a subcomplex of K_n. Observe that $K_{n,n} = K_{n,n}^{(0)} = \mathcal{V}_n$ by (2), $K_{n,j} \subset K_{n,j-1}$, and $K_n = K_{n,1}$. We define $g_{n,n} : |K_{n,n}| \to Y$ by $g_{n,n}(V) = y_V$ for each $V \in \mathcal{V}_n$. Now, suppose that $g_{n,i} : |K_{n,i}| \to Y$, $i > j$, have been defined such that

(12) $g_{n,i}||K_{n,i+1}| = g_{n,i+1}$;
(13) $g_{n,i}||K_{n-1,i}| = g_{n-1}||K_{n-1,i}|$;
(14) $g_{n,i}(\sigma) \subset \mathrm{N}(\varphi(x_{V(\sigma)}), \delta_{i(\sigma)-1})$ for $\sigma \in K_{n,i}$;
(15) $g_{n,i}$ satisfies $(*)$ for $\sigma \in K_{n,i}$, $t \in \mathbf{I}$, and $z \in \sigma_0$.

For each $\sigma \in K_{n,j} \setminus K_{n,j+1}$, $V(\sigma) \in \mathcal{V}_j$ and $\sigma_0 \in K_{n,j+1}$. By the inductive hypothesis on $g_{n,j+1}$,

$$g_{n,j+1}(\sigma_0) \subset \mathrm{N}(\varphi(x_{V(\sigma_0)}), \delta_{i(\sigma_0)-1}) \subset \mathrm{N}(\varphi(x_{V(\sigma_0)}), \delta_j).$$

On the other hand, since $V(\sigma_0) \cap V(\sigma) \neq \emptyset$, it follows that

$$d(x_{V(\sigma_0)}, x_{V(\sigma)}) \leqslant \operatorname{diam} V(\sigma_0) + \operatorname{diam} V(\sigma) < \varepsilon_{i(\sigma_0)} + \varepsilon_j < 2\varepsilon_j.$$

Since $x_{V(\sigma)} \in X_j$, we have $\varphi(x_{V(\sigma_0)}) \subset \mathrm{N}(\varphi(x_{V(\sigma)}), \delta_j)$, hence $g_{n,j+1}(\sigma_0) \subset \mathrm{N}(\varphi(x_{V(\sigma)}), 2\delta_j)$. Then, $g_{n,j+1}$ can be extended to the map $g_{n,j} : |K_{n,j}| \to Y$ by

$$g_{n,j}((1-t)z + tV(\sigma)) = r^{V(\sigma)}(g_{n,j+1}(z), t) \in \mathrm{N}(\varphi(x_{V(\sigma)}), \delta_{j-1})$$
$$\text{for } \sigma \in K_{n,j} \setminus K_{n,j+1},\ t \in \mathbf{I} \text{ and } z \in \sigma_0,$$

where $j = i(\sigma)$. Then, $g_{n,j}$ satisfies (12), (14), and (15) by definition. Since g_{n-1} satisfies $(*)$, it follows that $g_{n,j}||K_{n-1,j}| = g_{n-1}||K_{n-1,j}|$, i.e., (13) is also satisfied. By downward induction on j, we can obtain the map $g_n = g_{n,1} : |K_n| = |K_{n,1}| \to Y$ that extends g_{n-1} and satisfies $(*)$.

Now, let $g : |K| \to Y$ be the map defined by $g||K_n| = g_n$. For each $x \in X$, let $\sigma \in K$ be the carrier of $p(x)$. Since p is a canonical map for the nerve K of $\mathcal{V}$, we have $x \in V(\sigma) \in \mathcal{V}_{i(\sigma)}$, hence $d(x, x_{V(\sigma)}) \leqslant \operatorname{mesh} \mathcal{V}_{i(\sigma)} < \varepsilon_{i(\sigma)}$ by (3) and $\alpha(x) > 2^{-i(\sigma)-1}$ by (5). Since $x_{V(\sigma)} \in X_{i(\sigma)}$, it follows that

$$\begin{aligned} gp(x) \in g(\sigma) &\subset \mathrm{N}(\varphi(x_{V(\sigma)}), \delta_{i(\sigma)-1}) \\ &\subset \mathrm{N}(\mathrm{N}(\varphi(x), \delta_{i(\sigma)}), \delta_{i(\sigma)-1}) \subset \mathrm{N}(\varphi(x), 2\delta_{i(\sigma)-1}). \end{aligned}$$

Therefore, we have

$$d(gp(x), \varphi(x)) < 2\delta_{i(\sigma)-1} \leqslant 2^{-i(\sigma)-1} < \alpha(x).$$

Thus, $gp : X \to Y$ is the desired map. □

7.7 The Suspensions and the Taylor Example

In this section, using a K-theory result of J.F. Adams, we will construct the Taylor example, that is, a compactum that is not cell-like but has the Hilbert cube $\boldsymbol{Q} = [-1, 1]^{\mathbb{N}}$ as its cell-like image. Moreover, using the Taylor example, we construct a cell-like map of $\boldsymbol{Q}$ onto a compactum that is not an ANR. Here, we use the TORUŃCZYK CHARACTERIZATION OF THE HILBERT CUBE[7]:

Theorem 7.7.1 (TORUŃCZYK CHARACTERIZATION) *A space X is homeomorphic to the Hilbert cube $\boldsymbol{Q}$ if and only if X is a compact AR that has the following property called the* **disjoint cells property***:*

- *for each $n \in \mathbb{N}$, each pair of maps $f, g : \mathbf{I}^n \to X$ can be approximated by maps $f', g' : \mathbf{I}^n \to X$ with disjoint images, i.e., $f'(\mathbf{I}^n) \cap g'(\mathbf{I}^n) = \emptyset$.* □

The **suspension** ΣX of a space X is the following quotient space of $[-1, 1] \times X$:

$$\Sigma X = [-1, 1] \times X / \{\{-1\} \times X, \{1\} \times X\},$$

where $\{-1\} \times X$ and $\{1\} \times X$ are regarded as two distinct points. Let $q_X : [-1, 1] \times X \to \Sigma X$ be the quotient map. For each map $f : X \to Y$, there exists a unique map $\Sigma f : \Sigma X \to \Sigma Y$ such that $q_Y \circ (\mathrm{id} \times f) = (\Sigma f) \circ q_X$, that is, the following diagram is commutative:

$$\begin{array}{ccc} [-1, 1] \times X & \xrightarrow{\mathrm{id} \times f} & [-1, 1] \times Y \\ {\scriptstyle q_X}\downarrow & & \downarrow{\scriptstyle q_Y} \\ \Sigma X & \xrightarrow[\Sigma f]{} & \Sigma Y. \end{array}$$

It should be noted that $\Sigma \mathrm{id}_X = \mathrm{id}_{\Sigma X}$ and $\Sigma g \Sigma f = \Sigma(gf)$ for each pair of maps $f : X \to Y$ and $g : Y \to Z$. Moreover, each homotopy $h : X \times \mathbf{I} \to Y$ induces a homotopy $\Sigma h : \Sigma X \times \mathbf{I} \to \Sigma Y$ defined by $(\Sigma h)_t = \Sigma h_t$ for every $t \in \mathbf{I}$, that is, the following diagram commutes:

[7] For this proof, refer to the author's other book: *Topology of Infinite-Dimensional Manifolds*, or van Mill's book: *Infinite-Dimensional Topology*, mentioned in the Preface. See Notes of this chapter.

$$\begin{array}{ccc} [-1,1]\times X\times \mathbf{I} & \xrightarrow{\mathrm{id}\times h} & [-1,1]\times Y \\ \downarrow{\scriptstyle q_X\times \mathrm{id}_{\mathbf{I}}} & & \downarrow{\scriptstyle q_Y} \\ \Sigma X\times \mathbf{I} & \xrightarrow[\Sigma h]{} & \Sigma Y. \end{array}$$

In the above, $q_X \times \mathrm{id}_{\mathbf{I}}$ is a quotient map by Proposition 2.1.9, from which the continuity of Σh follows. Observe that the suspension ΣX is a union of two cones over X,

$$\mathbf{I}\times X/\{1\}\times X \quad \text{and} \quad [-1,0]\times X/\{-1\}\times X.$$

Since the cone over a compact ANR is an AR by Corollary 6.5.5, we apply Proposition 6.2.9 (5) to obtain the next proposition:

Proposition 7.7.2 *For every compact ANR X, the suspension ΣX is an ANR.* □

We will show the following:

Proposition 7.7.3 *For each cell-like compactum X, the suspension ΣX is also cell-like.*

Proof Embed X into the Hilbert cube $\boldsymbol{Q}$. Then, it suffices to prove that ΣX is cell-like in $\Sigma \boldsymbol{Q}$. For each neighborhood U of ΣX in $\Sigma \boldsymbol{Q}$, we can choose a neighborhood V of X in $\boldsymbol{Q}$ so that $q_{\boldsymbol{Q}}([-1,1]\times V) \subset U$, where $q_{\boldsymbol{Q}} : [-1,1]\times \boldsymbol{Q} \to \Sigma \boldsymbol{Q}$ is the quotient map. Since X is cell-like in $\boldsymbol{Q}$, there is a contraction $h : X\times \mathbf{I} \to V$. Then, as observed in the above, h induces the homotopy

$$\Sigma h : \Sigma X\times \mathbf{I} \to q_{\boldsymbol{Q}}([-1,1]\times V) \subset U \subset \Sigma \boldsymbol{Q},$$

where $(\Sigma h)_0 = \Sigma h_0 = \mathrm{id}$ and

$$(\Sigma h)_1(\Sigma X) = \Sigma h_1(\Sigma X) = [-1,1]\times h_1(X) \approx [-1,1],$$

Hence, ΣX is contractible in U. Consequently, ΣX is cell-like in $\Sigma \boldsymbol{Q}$. □

Next, we will show that the suspension of the inverse limit is the inverse limit of suspensions, that is:

Proposition 7.7.4 *Let $X = \varprojlim(X_i, f_i)$ be the inverse limit of an inverse sequence $(X_i, f_i)_{i\in\mathbb{N}}$ of compacta with $p_i : X \to X_i$, $i \in \mathbb{N}$, the projections of the inverse limit. Then, there exists the unique homeomorphism $h : \Sigma X \to \varprojlim(\Sigma X_i, \Sigma f_i)$ such that $q_i h = \Sigma p_i$ for each $i \in \mathbb{N}$, where each $q_i : \varprojlim(\Sigma X_i, \Sigma f_i) \to \Sigma X_i$ is the inverse limit projection.*

Proof Let $q_X : [-1,1]\times X \to \Sigma X$ and $q_{X_i} : [-1,1]\times X_i \to \Sigma X_i$, $i \in \mathbb{N}$, be the quotient maps. For $(t,x) \in [-1,1]\times X$ and $i \in \mathbb{N}$,

$$(\Sigma f_i) q_{X_{i+1}}(t, x(i+1)) = q_{X_i}(t, f_i(x(i+1))) = q_{X_i}(t, x(i)).$$

Then, we can define a map $h : \Sigma X \to \varprojlim(\Sigma X_i, \Sigma f_i)$ as follows:

$$hq_X(t, x) = \big(q_{X_i}(t, x(i))\big)_{i\in\mathbb{N}} = \big((\Sigma p_i)q_X(t, x)\big)_{i\in\mathbb{N}},$$

which means that $q_i h = \Sigma p_i$ for each $i \in \mathbb{N}$. The uniqueness of the map h follows from the last condition.[8] We will show that h is bijective, which implies that h is a homeomorphism because ΣX is compact.

$$\begin{array}{ccccc} [-1,1]\times X & \xrightarrow{q_X} & \Sigma X & \xrightarrow{h} & \varprojlim(\Sigma X_i, \Sigma f_i) \quad \subset \quad \prod_{i\in\mathbb{N}} \Sigma X_i \\ {\scriptstyle \mathrm{id}\times p_i}\downarrow & & {\scriptstyle \Sigma p_i}\downarrow & \swarrow_{q_i} & \\ [-1,1]\times X_i & \xrightarrow[q_{X_i}]{} & \Sigma X_i & & \end{array}$$

To prove that h is surjective, let $(q_{X_i}(t_i, x_i))_{i\in\mathbb{N}} \in \varprojlim(\Sigma X_i, \Sigma f_i)$. For each $i \in \mathbb{N}$,

$$q_{X_i}(t_i, x_i) = (\Sigma f_i)q_{X_{i+1}}(t_{i+1}, x_{i+1}) = q_{X_i}(t_{i+1}, f_i(x_{i+1})),$$

which implies that $t_1 = t_2 = \cdots = t$. If $t \neq \pm 1$ then $f_i(x_{i+1}) = x_i$ for every $i \in \mathbb{N}$. When $t = \pm 1$, we may assume that $f_i(x_{i+1}) = x_i$. Then, $x = (x_i)_{i\in\mathbb{N}} \in X$. It follows that $hq_X(t, x) = (q_{X_i}(t_i, x_i))_{i\in\mathbb{N}}$.

To show that h is injective, let $(t, x), (t', x') \in [-1, 1] \times X$ and assume that $hq_X(t, x) = hq_X(t', x')$. For each $i \in \mathbb{N}$,

$$q_{X_i}(t, p_i(x)) = q_i hq_X(t, x) = q_i hq_X(t', x') = q_{X_i}(t', p_i(x')).$$

Then, we have $t = t'$. If $t = t' = \pm 1$ then $q_X(t, x) = q_X(t', x')$. When $t = t' \neq \pm 1$, it follows that $p_i(x) = p_i(x')$ for every $i \in \mathbb{N}$, which means $x = x'$, hence $q_X(t, x) = q_X(t', x')$. □

For each $n \in \mathbb{N}$, the ***n*-fold suspension** $\Sigma^n X$ is inductively defined by $\Sigma^n X = \Sigma(\Sigma^{n-1}X)$, where $\Sigma^0 X = X$. For a map $f : X \to Y$, let

$$\Sigma^n f = \Sigma(\Sigma^{n-1} f) : \Sigma^n X = \Sigma(\Sigma^{n-1}X) \to \Sigma^n Y = \Sigma(\Sigma^{n-1}Y),$$

where $\Sigma^0 f = f$. The 2-fold suspension $\Sigma^2 X$ is also called the **double suspension**. Observe that

$$\begin{aligned} \Sigma^2 X &= [-1,1]^2 \times X / \big\{\{\pm 1\} \times [-1,1] \times X,\ \{(t, \pm 1)\} \times X \ \big|\ -1 < t < 1\big\} \\ &= \Sigma[-1,1] \times X / \big\{\{z\} \times X \ \big|\ z \in \partial\Sigma[-1,1]\big\}, \end{aligned}$$

where $\Sigma[-1, 1] \approx \mathbf{B}^2$ and

[8] We can apply Theorem 4.16.7 to show the existence and the uniqueness of h.

$$\partial\Sigma[-1,1] = \big(\{\pm 1\}\times[-1,1]\big)\cup\big((-1,1)\times\{\pm 1\}\big) \approx \mathbf{S}^1.$$

Furthermore, regarding $[-1,1] = \Sigma\{0\}$, we can write $\Sigma[-1,1] = \Sigma^2\{0\}$. Then, we can also write

$$\Sigma^2 X = \Sigma^2\{0\}\times X/\big\{\{z\}\times X \;\big|\; z\in\partial\Sigma^2\{0\}\big\}.$$

By induction, we can show that

$$\Sigma^n X = \Sigma^n\{0\}\times X/\big\{\{z\}\times X \;\big|\; z\in\partial\Sigma^n\{0\}\big\},$$

where $\partial\Sigma^n\{0\}$ is the boundary $(n-1)$-sphere of $\Sigma^n\{0\}\approx\mathbf{B}^n$. Indeed, assume the above equality holds. Then, it follows that

$$\begin{aligned}\Sigma^{n+1}X &= \Sigma\big(\Sigma^n X\big) = \Sigma\big(\Sigma^n\{0\}\times X/\big\{\{z\}\times X \;\big|\; z\in\partial\Sigma^n\{0\}\big\}\big)\\ &= [-1,1]\times\Sigma^n\{0\}\times X/\big\{\{\pm 1\}\times\Sigma^n\{0\}\times X,\ \{t\}\times\{z\}\times X\\ &\qquad\qquad \big|\ -1<t<1,\ z\in\partial\Sigma^n\{0\}\big\}\\ &= \Sigma^{n+1}\{0\}\times X/\big\{\{z\}\times X \;\big|\; z\in\partial\Sigma^{n+1}\{0\}\big\},\end{aligned}$$

where the last equality comes from the following facts:

$$\begin{aligned}\Sigma^{n+1}\{0\} &= [-1,1]\times\Sigma^n\{0\}/\big\{\{\pm 1\}\times\Sigma^n\{0\}\big\} \text{ and}\\ \partial\Sigma^{n+1}\{0\} &= \big\{\{\pm 1\}\times\Sigma^n\{0\}\big\}\cup\big((-1,1)\times\partial\Sigma^n\{0\}\big).\end{aligned}$$

As we have just seen, the n-fold suspension $\Sigma^n X$ is the quotient space of $\Sigma^n\{0\}\times X$, where the quotient map is denoted by $q_X^n : \Sigma^n\{0\}\times X\to\Sigma^n X$. For each map $f : X\to Y$, the map $\Sigma^n f : \Sigma^n X\to\Sigma^n Y$ is induced by the map $\mathrm{id}\times f : \Sigma^n\{0\}\times X\to\Sigma^n\{0\}\times Y$, that is, $\Sigma^n f$ is defined by the following commutative diagram:

$$\begin{array}{ccc}\Sigma^n\{0\}\times X & \xrightarrow{\mathrm{id}\times f} & \Sigma^n\{0\}\times Y\\ \Big\downarrow{\scriptstyle q_X^n} & & \Big\downarrow{\scriptstyle q_Y^n}\\ \Sigma^n X & \xrightarrow[\Sigma^n f]{} & \Sigma^n Y\end{array}$$

Remark 7.7.1 Since $\Sigma^n\{0\}\approx\mathbf{I}^n$, we can regard

$$\Sigma^n X = \mathbf{I}^n\times X/\big\{\{z\}\times X \;\big|\; z\in\partial\mathbf{I}^n\big\}.$$

In this case, $\Sigma^m\Sigma^n X\approx\Sigma^{m+n}X$ but $\Sigma^m\Sigma^n X\neq\Sigma^{m+n}X$ because

$$\Sigma^{m+n}X = \mathbf{I}^{m+n} \times X \Big/ \Big\{\{z\} \times X \;\Big|\; z \in \partial\mathbf{I}^{m+n}\Big\} \text{ but}$$
$$\Sigma^m\Sigma^n X = \mathbf{I}^{m+n} \times X \Big/ \Big\{\{y\} \times \mathbf{I}^n \times X,\ \{y'\} \times \{z\} \times X \;\Big|\; y \in \partial\mathbf{I}^m,\ y' \in \mathbf{I}^m \setminus \partial\mathbf{I}^m,\ z \in \partial\mathbf{I}^n\Big\}.$$

In his paper [1], J.F. Adams constructed a compact polyhedron A with a map $\alpha : \Sigma^r A \to A$ from the r-fold suspension of A onto A such that every composition

$$\alpha \circ \Sigma^r\alpha \circ \cdots \circ \Sigma^{(i-1)r}\alpha : \Sigma^{ir}A \to A$$

is essential, i.e., it is not null-homotopic. Using this map, we can construct a cell-like map that is not a shape equivalence, which is the subject of the following theorem:

Theorem 7.7.5 (TAYLOR) *There exists a cell-like map $f : X \to \boldsymbol{Q}$ of a compactum X onto the Hilbert cube $\boldsymbol{Q}$ such that X is not cell-like, that is, f is not a shape equivalence (cf. Corollary 7.5.2), where X is homeomorphic to the r-fold suspension $\Sigma^r X$.*

Proof Let X be the inverse limit of the inverse sequence:

$$A \xleftarrow{\alpha} \Sigma^r A \xleftarrow{\Sigma^r\alpha} \Sigma^{2r}A \xleftarrow{\Sigma^{2r}\alpha} \cdots$$

where A is the compact polyhedron with the map $\alpha : \Sigma^r A \to A$ constructed by Adams. Let $\rho_i : X \to \Sigma^{(i-1)r}A$, $i \in \mathbb{N}$, be the inverse limit projections, where $\Sigma^0 A = A$. Note that every $\Sigma^{ir}A$ is a compact ANR by Proposition 7.7.2. If X is cell-like, by Theorem 7.1.8, we can find some $j \in \mathbb{N}$ such that

$$\alpha \circ \Sigma^r\alpha \circ \cdots \circ \Sigma^{(j-1)r}\alpha \simeq 0,$$

which is a contradiction. Thus, X is not cell-like. Moreover, by Proposition 7.7.4, $\Sigma^r X$ is homeomorphic to the inverse limit of

$$\Sigma^r A \xleftarrow{\Sigma^r\alpha} \Sigma^{2r}A \xleftarrow{\Sigma^{2r}\alpha} \Sigma^{3r}A \xleftarrow{\Sigma^{3r}\alpha} \cdots,$$

hence $\Sigma^r X \approx X$ (Proposition 4.17.3).

We regard $\Sigma^{ir}A$ as the quotient space of $\Sigma^{ir}\{0\} \times A$ with the quotient map $q_A^{ir} : \Sigma^{ir}\{0\} \times A \to \Sigma^{ir}A$ such that $q_A^{ir}(\{z\} \times A)$ is a singleton for each $z \in \partial\Sigma^{ir}\{0\}$. Let $f_i : \Sigma^{ir}A \to \Sigma^{ir}\{0\}$ be the map induced by the projection of $\Sigma^{ir}\{0\} \times A$ onto $\Sigma^{ir}\{0\}$. Because $\Sigma^{(i+1)r}\{0\} = \Sigma^{ir}\Sigma^r\{0\}$ is the quotient space of $\Sigma^{ir}\{0\} \times \Sigma^r\{0\}$, we have the map

$$\varphi_i : \Sigma^{(i+1)r}\{0\} = \Sigma^{ir}\Sigma^r\{0\} \to \Sigma^{ir}\{0\},$$

which is induced by the projection of $\Sigma^{ir}\{0\} \times \Sigma^r\{0\}$ onto $\Sigma^{ir}\{0\}$, that is, the map φ_i is defined by the following commutative diagram:

$$\begin{array}{ccc} & & \Sigma^{ir}\{0\} \times \Sigma^{r}\{0\} \\ & \swarrow^{\mathrm{pr}_{\Sigma^{ir}\{0\}}} & \downarrow{\scriptstyle q^{ir}_{\Sigma^{r}\{0\}}} \\ \Sigma^{ir}\{0\} & \xleftarrow[\varphi_i]{} & \Sigma^{ir}\Sigma^{r}\{0\} = \Sigma^{(i+1)r}\{0\} \end{array}$$

In the above, $\mathrm{pr}_{\Sigma^{ir}\{0\}}$ is a fine homotopy equivalence. Since $q^{ir}_{\Sigma^{r}\{0\}}$ is cell-like, it is also a fine homotopy equivalence. By Proposition 6.7.6, φ_i is a fine homotopy equivalence. On the other hand, we have the embedding $\psi_i : \Sigma^{ir}\{0\} \to \Sigma^{(i+1)r}\{0\}$ defined by $\psi_i(x) = q^{ir}_{\Sigma^{r}\{0\}}(x, 0)$. Therefore, $\varphi_i \psi_i = \mathrm{id}$.

By diagram chasing, we can see $f_i \Sigma^{ir}\alpha = \varphi_i f_{i+1}$. In the diagram below, the circumference and the upper right square commute, where

$$\begin{aligned} \Sigma^{(i+1)r}A &= q_A^{(i+1)r}(q^{ir}_{\Sigma^{r}\{0\}} \times \mathrm{id})(\Sigma^{ir}\{0\} \times \Sigma^{r}\{0\} \times A) \\ &= q^{ir}_{\Sigma^{r}A}(\mathrm{id} \times q_A^{r})(\Sigma^{ir}\{0\} \times \Sigma^{r}\{0\} \times A). \end{aligned}$$

Hence, the bottom rectangle also commutes.

$$\begin{array}{ccccc} \Sigma^{ir}\{0\} \times \Sigma^{r}\{0\} \times A & & \xrightarrow{q^{ir}_{\Sigma^{r}\{0\}} \times \mathrm{id}} & & \Sigma^{ir}\Sigma^{r}\{0\} \times A \\ \downarrow{\scriptstyle \mathrm{id} \times q_A^{r}} & & & & \| \\ \Sigma^{ir}\{0\} \times A & \xleftarrow[\mathrm{id} \times \alpha]{} & \Sigma^{ir}\{0\} \times \Sigma^{r}A & & \Sigma^{(i+1)r}\{0\} \times A \\ \downarrow{\scriptstyle q_A^{ir}} & & \downarrow{\scriptstyle q^{ir}_{\Sigma^{r}A}} & & \downarrow{\scriptstyle q_A^{(i+1)r}} \\ \Sigma^{ir}A & \xleftarrow{\Sigma^{ir}\alpha} & \Sigma^{ir}\Sigma^{r}A & = & \Sigma^{(i+1)r}A \\ \downarrow{\scriptstyle f_i} & & & & \downarrow{\scriptstyle f_{i+1}} \\ \Sigma^{ir}\{0\} & & \xleftarrow[\varphi_i]{} & & \Sigma^{(i+1)r}\{0\} \end{array}$$

(with $\mathrm{pr}_{\Sigma^{ir}\{0\}}$ and $\mathrm{pr}_{\Sigma^{(i+1)r}\{0\}}$ as the outer arrows)

Thus, we have the following commutative diagram:

$$\begin{array}{ccccccccc} A & \xleftarrow{\alpha} & \Sigma^{r}A & \xleftarrow{\Sigma^{r}\alpha} & \Sigma^{2r}A & \xleftarrow{\Sigma^{2r}\alpha} & \Sigma^{3r}A & \xleftarrow{\Sigma^{3r}\alpha} & \cdots \\ & & \downarrow{\scriptstyle f_1} & & \downarrow{\scriptstyle f_2} & & \downarrow{\scriptstyle f_3} & & \\ & & \Sigma^{r}\{0\} & \xleftarrow[\varphi_1]{} & \Sigma^{2r}\{0\} & \xleftarrow[\varphi_2]{} & \Sigma^{3r}\{0\} & \xleftarrow[\varphi_3]{} & \cdots \end{array}$$

Let $Y = \varprojlim \Sigma^{ir}\{0\}$ be the inverse limit of the bottom sequence and $p_i : Y \to \Sigma^{ir}\{0\}$, $i \in \mathbb{N}$, the inverse limit projections. Then, we have the map $f = \varprojlim f_i : X \to Y$ defined by the maps f_i. It remains to show that $Y \approx \boldsymbol{Q}$ and $f : X \to Y$ is cell-like.

To show $Y \approx \boldsymbol{Q}$, we use the Toruńczyk characterization of $\boldsymbol{Q}$ (Theorem 7.7.1). As observed in the above, each bonding map φ_i is a fine homotopy equivalence,

hence Y is a compact AR by Theorem 6.7.9. Recall that there are embeddings $\psi_i : \Sigma^{ir}\{0\} \to \Sigma^{(i+1)r}\{0\}$, $i \in \mathbb{N}$, such that $\varphi_i \psi_i = \mathrm{id}$. We can define an embedding $\bar{\psi}_i : \Sigma^{ir}\{0\} \to Y$, $i \in \mathbb{N}$, as follows:

$$\bar{\psi}_i(x) = (\varphi_{1,i}(x), \dots, \varphi_{i-1,i}(x), x, \psi_{i,i+1}(x), \psi_{i,i+2}(x), \dots),$$

where $\varphi_{j,i} = \varphi_j \cdots \varphi_{i-1}$ for $j < i$ and $\psi_{i,j} = \psi_{j-1} \cdots \psi_i$ for $j > i$. Then, $p_i \bar{\psi}_i = \mathrm{id}$ for each $i \in \mathbb{N}$ and $\lim_{i\to\infty} \bar{\psi}_i p_i = \mathrm{id}_Y$. Since $\Sigma^{ir}\{0\} \approx \mathbf{I}^{ir}$ for each $i \in \mathbb{N}$, we can apply Theorem 5.8.4 to see that Y has the disjoint cells property, that is, for each $n \in \mathbb{N}$ and $\mathcal{U} \in \mathrm{cov}(Y)$, every maps $f, g : \mathbf{I}^n \to Y$ are $\mathcal{U}$-close to maps $f', g' : \mathbf{I}^n \to Y$ with $f'(\mathbf{I}^n) \cap g'(\mathbf{I}^n) = \emptyset$. By the Toruńczyk characterization of the Hilbert cube, we have $Y \approx \boldsymbol{Q}$.

To prove that f is cell-like, let $y = (y_i)_{i\in\mathbb{N}} \in Y = \varprojlim \Sigma^{ir}\{0\}$. Consider the following commutative diagram:

$$\begin{array}{ccccccc}
\Sigma^r\{0\} \times A & \xleftarrow{\varphi_1\times\mathrm{id}} & \Sigma^{2r}\{0\} \times A & \xleftarrow{\varphi_2\times\mathrm{id}} & \Sigma^{3r}\{0\} \times A & \xleftarrow{\varphi_3\times\mathrm{id}} & \cdots \\
\cup & & \cup & & \cup & & \\
\{y_1\} \times A & \longleftarrow & \{y_2\} \times A & \longleftarrow & \{y_3\} \times A & \longleftarrow & \cdots \\
\downarrow{\scriptstyle q_A^r} & & \downarrow{\scriptstyle q_A^{2r}} & & \downarrow{\scriptstyle q_A^{3r}} & & \\
f_1^{-1}(y_1) & \longleftarrow & f_2^{-1}(y_2) & \longleftarrow & f_3^{-1}(y_3) & \longleftarrow & \cdots \\
\cap & & \cap & & \cap & & \\
\Sigma^r A & \xleftarrow[\Sigma^r\alpha]{} & \Sigma^{2r} A & \xleftarrow[\Sigma^{2r}\alpha]{} & \Sigma^{3r} A & \xleftarrow[\Sigma^{3r}\alpha]{} & \cdots
\end{array}$$

The inverse limit of the third sequence is $f^{-1}(y)$. Recall that $X = \varprojlim \Sigma^{ir} A$ is a subspace of the product space $\prod_{i\in\mathbb{N}} \Sigma^{ir} A$. For each neighborhood U of $f^{-1}(y)$ in $\prod_{i\in\mathbb{N}} \Sigma^{ir} A$, we will show that $f^{-1}(y)$ is contractible in U, which implies that $f^{-1}(y)$ is cell-like by Theorem 7.1.2.

For each $j \in \mathbb{N}$, let

$$Y_j = \Big\{ x \in \textstyle\prod_{i\leqslant j} \Sigma^{ir} A \;\Big|\; x(j) \in f_j^{-1}(y_j), \\ \forall i < j,\ x(i) = \big(\Sigma^{ir}\alpha \circ \cdots \circ \Sigma^{(j-1)r}\alpha\big)(x(j)) \Big\}.$$

Then, $Y_j \times \prod_{i>j} \Sigma^{ir} A$ is compact and

$$f^{-1}(y) = \bigcap_{j\in\mathbb{N}} \Big(Y_j \times \prod_{i>j} f_i^{-1}(y_i) \Big) = \bigcap_{j\in\mathbb{N}} \Big(Y_j \times \prod_{i>j} \Sigma^{ir} A \Big) \subset U \subset \prod_{i\in\mathbb{N}} \Sigma^{ir} A.$$

Hence, we can find $j \in \mathbb{N}$ such that $Y_j \times \prod_{i>j} \Sigma^{ir} A \subset U$. For each $i > j$, $f_i^{-1}(y_i)$ is contractible in $\Sigma^{ir} A$. Indeed, recall

$$\Sigma^{ir} A = (\Sigma^{ir}\{0\} \times A) \big/ \big\{\{z\} \times A \;\big|\; z \in \partial \Sigma^{ir}\{0\}\big\}.$$

Then, $f_i^{-1}(y_i)$ is a singleton if $y_i \in \partial \Sigma^{ir}\{0\}$. When $y_i \in \Sigma^{ir}\{0\} \setminus \partial \Sigma^{ir}\{0\}$, take a path $\gamma : \mathbf{I} \to \Sigma^{ir}\{0\}$ from $\gamma(0) = y_i$ to $\gamma(1) \in \partial \Sigma^{ir}\{0\}$ and define a homotopy $h : \{y_i\} \times A \times \mathbf{I} \to \Sigma^{ir}\{0\} \times A$ by $h(y_i, x, t) = (\gamma(t), x)$. Then, h induces a contraction $\tilde{h}$ of $f_i^{-1}(y_i)$ in $\Sigma^{ir} A$:

$$\begin{array}{ccc} \{y_i\} \times A \times \mathbf{I} & \xrightarrow{h} & \Sigma^{ir}\{0\} \times A \\ \Big\downarrow{\scriptstyle q_A^{ir} \times \mathrm{id}} & & \Big\downarrow{\scriptstyle q_A^{ir}} \\ f_i^{-1}(y_i) \times \mathbf{I} & \xrightarrow[\tilde{h}]{} & \Sigma^{ir} A \end{array}$$

It follows that $Y_{j+1} \times \prod_{i>j+1} f_i^{-1}(y_i)$ is contractible in $Y_j \times \prod_{i>j} \Sigma^{ir} A$, which implies that $f^{-1}(y)$ is contractible in U. Therefore, $f^{-1}(y)$ is cell-like. □

Remark 7.7.2 In the above proof, as mentioned in Remark 7.7.1, we can regard

$$\Sigma^n A = \mathbf{I}^n \times A \big/ \big\{\{z\} \times A \;\big|\; z \in \partial \mathbf{I}^n\big\}.$$

In this case, recall that $\Sigma^{ir} \Sigma^r A \neq \Sigma^{(i+1)r} A$. For each $i \in \mathbb{N}$, the projection of $\mathbf{I}^{ir} \times A$ onto $\mathbf{I}^{ir}$ induces the map $f_i : \Sigma^{ir} A \to \mathbf{I}^{ir}$. In the literature explaining Taylor's example (as in Taylor's original paper [31]), the map $f : X \to \boldsymbol{Q}$ is constructed using the following diagram:

$$\begin{array}{ccccccccccc} A & \xleftarrow{\alpha} & \Sigma^r A & \xleftarrow{\Sigma^r \alpha} & \Sigma^{2r} A & \xleftarrow{\Sigma^{2r} \alpha} & \Sigma^{3r} A & \xleftarrow{\Sigma^{3r} \alpha} & \cdots \\ & & \Big\downarrow{\scriptstyle f_1} & & \Big\downarrow{\scriptstyle f_2} & & \Big\downarrow{\scriptstyle f_3} & & \\ & & \mathbf{I}^r & \xleftarrow[\mathrm{pr}]{} & \mathbf{I}^{2r} & \xleftarrow[\mathrm{pr}]{} & \mathbf{I}^{3r} & \xleftarrow[\mathrm{pr}]{} & \cdots \end{array}$$

However, the domain of the map $\Sigma^{ir} \alpha$ is $\Sigma^{ir} \Sigma^r A$, which is not $\Sigma^{(i+1)r} A$. If $\Sigma^{ir} \Sigma^r A$ is identified with $\Sigma^{(i+1)r} A$ by a homeomorphism then the above diagram is not commutative. Although $\Sigma^{ir} \Sigma^r A$ can be regarded as the quotient space of $\mathbf{I}^{(i+1)r} \times A$, the projection of $\mathbf{I}^{(i+1)r} \times A$ onto $\mathbf{I}^{(i+1)r}$ does not induce a map of $\Sigma^{ir} \Sigma^r A$ onto $\mathbf{I}^{(i+1)r}$.

Remark 7.7.3 We can also regard

$$\Sigma^n A = \mathbf{B}^n \times A \big/ \big\{\{z\} \times A \;\big|\; z \in \mathbf{S}^{n-1}\big\}.$$

As above, $\Sigma^{(i+1)r} A \neq \Sigma^{ir} \Sigma^r A$. For each $i \in \mathbb{N}$, the projection of $\mathbf{B}^{ir} \times A$ onto $\mathbf{B}^{ir}$ induces the map $f_i : \Sigma^{ir} A \to \mathbf{B}^{ir}$. Let $p_i : \mathbf{B}^{(i+1)r} \to \mathbf{B}^{ir}$ be the restriction of the

projection $\mathrm{pr}_{\mathbf{B}^{ir}} : \mathbf{B}^{ir} \times \mathbf{B}^r \to \mathbf{B}^{ir}$ to $\mathbf{B}^{(i+1)r}$ ($\subset \mathbf{B}^{ir} \times \mathbf{B}^r$). Then, we can construct a homeomorphism $\bar{\theta}_i : \Sigma^{ir}\Sigma^r A \to \Sigma^{ir} A$ so that the following diagram commutes:

$$\begin{array}{ccccc}
\Sigma^{ir} A & \xleftarrow{\Sigma^{ir}\alpha} & \Sigma^{ir}\Sigma^r A & \xrightarrow[\approx]{\bar{\theta}_i} & \Sigma^{(i+1)r} A \\
\Big\downarrow{\scriptstyle f_i} & & & & \Big\downarrow{\scriptstyle f_{i+1}} \\
\mathbf{B}^{ir} & & \xleftarrow[p_i]{} & & \mathbf{B}^{(i+1)r}
\end{array}$$

In fact, we have a surjective map $\theta_i : \mathbf{B}^{ir} \times \mathbf{B}^r \to \mathbf{B}^{(i+1)r}$ defined by

$$\theta_i(y, z) = \left(y, \sqrt{1 - \|y\|^2}\, z\right) \text{ for each } (y, z) \in \mathbf{B}^{ir} \times \mathbf{B}^r.$$

Then, $\theta_i | (\mathbf{B}^{ir} \setminus \mathbf{S}^{ir-1}) \times \mathbf{B}^r$ is injective,

$$\theta_i^{-1}(\mathbf{S}^{(i+1)r-1}) = (\mathbf{S}^{ir-1} \times \mathbf{B}^r) \cup (\mathbf{B}^{ir} \times \mathbf{S}^{r-1}),$$
$$\theta_i^{-1}(\theta_i(y, z)) = \theta_i^{-1}(y, 0) = \{y\} \times \mathbf{B}^r \text{ for each } (y, z) \in \mathbf{S}^{ir-1} \times \mathbf{B}^r$$

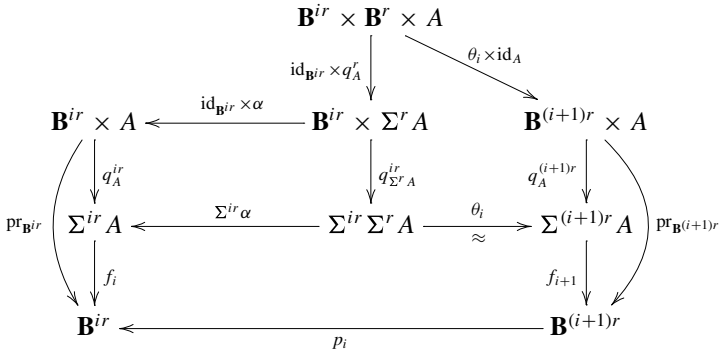
and θ_i is $\mathbf{B}^{ir}$-preserving, that is, the following diagram commutes:

$$\begin{array}{ccc}
\mathbf{B}^{ir} \times \mathbf{B}^r & \xrightarrow{\theta_i} & \mathbf{B}^{(i+1)r} \\
& {\scriptstyle \mathrm{pr}_{\mathbf{B}^{ir}}} \searrow \quad \swarrow {\scriptstyle p_i} & \\
& \mathbf{B}^{ir} &
\end{array}$$

Observe that

$$\begin{aligned}
\Sigma^{ir}\Sigma^r A &= \mathbf{B}^{ir} \times \mathbf{B}^r \times A \Big/ \Big\{ \{y\} \times \mathbf{B}^r \times A,\ \{y'\} \times \{z\} \times A \ \Big| \\
&\qquad\qquad y \in \mathbf{S}^{ir-1},\ y' \in \mathbf{B}^{ir} \setminus \mathbf{S}^{ir-1},\ z \in \mathbf{S}^{r-1} \Big\} \\
&= \mathbf{B}^{ir} \times \mathbf{B}^r \times A \Big/ \Big\{ \theta_i^{-1}(z) \times A \ \Big|\ z \in \mathbf{S}^{(i+1)r-1} \Big\}.
\end{aligned}$$

Then, $\theta_i \times \mathrm{id}_A$ induces the desired homeomorphism $\bar{\theta}_i : \Sigma^{ir}\Sigma^r A \to \Sigma^{(i+1)r} A$. Indeed, let $q_X^n : \mathbf{B}^n \times X \to \Sigma^n X$ be the quotient map. Then, the following diagram commutes:



In this case, Y is defined as the inverse limit of the inverse sequence

$$\mathbf{B}^r \xleftarrow{p_1} \mathbf{B}^{2r} \xleftarrow{p_2} \mathbf{B}^{3r} \xleftarrow{p_3} \cdots$$

However, it is not trivial that Y is homeomorphic to $\boldsymbol{Q}$.

Remark 7.7.4 Using the Taylor example, we can prove that the composition of cell-like maps is not a cell-like map in general. In fact, the composition $c \circ f : X \to \{0\}$ of the constant map $c : \boldsymbol{Q} \to \{0\}$ and Taylor's cell-like map $f : X \to \boldsymbol{Q}$ is not a cell-like map.

Recall that Taylor's cell-like map is not a shape equivalence by Corollary 7.5.2. Thus, in general, a cell-like map need not be a shape equivalence but we do have the following theorem:

Theorem 7.7.6 *Let $f : X \to Y$ be a cell-like map between metrizable spaces. If $f^{-1}(y)$ is a singleton except for finitely many $y \in Y$, then f is a hereditary shape equivalence.*

In this theorem, $Y \approx X/\{f^{-1}(y_1), \ldots, f^{-1}(y_k)\}$ for some finite $y_1, \ldots, y_k \in Y$. Since the composition of shape equivalences is a shape equivalence, this theorem can be easily reduced to the following special case:

Theorem 7.7.7 *For each cell-like compactum A in a metrizable space X, the quotient map $p : X \to X/A$ is a shape equivalence.*

Proof In Proposition 7.5.6, when Y is a singleton, we can identify $Y \cup_f X = X/A$ and $M_f = A \times \mathbf{I}/A \times \{0\}$, where $M_f \cup (X \times \{1\})$ can be regarded as the following space:

$$Z = (A \times \mathbf{I}/A \times \{0\}) \cup (X \times \{1\}).$$

Then, hi is the quotient map of X onto X/A, where $i(x) = (x, 1)$ for each $x \in X$. Thus, it suffices to show that $i : X \to Z$ is a shape equivalence, that is, $i^* : [Z, P] \to [X, P]$ is bijective for each ANR P.

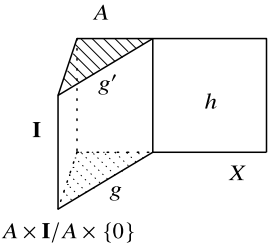


Fig. 7.3 Extending a homotopy $h : gi \simeq g'i$

Let $f : X \to P$ be a map. According to Theorem 7.1.2, A has trivial shape, hence $f|A \simeq 0$. Then, we have a map $\tilde{f} : A \times \mathbf{I}/A \times \{0\} \to P$ such that $\tilde{f}i|A = f$, which extends to a map $g : Z \to P$ such that $gi = f$. Therefore, i^* is surjective.

To show that i^* is injective, let $g, g' : Z \to P$ be maps with a homotopy $h : X \times \mathbf{I} \to P$ such that $h_0 = gi$ and $h_1 = g'i$. Consider the space

$$W = (A \times \{1\} \times \mathbf{I}) \cup ((A \times \mathbf{I}/A \times \{0\}) \times \{0, 1\})$$

and define a map $h' : W \to P$ by $h'(x, 1, t) = h(x, t)$ for each $x \in A$ and $t \in \mathbf{I}$; $h'(z, 0) = g(z)$ and $h'(z, 1) = g'(z)$ for each $z \in A \times \mathbf{I}/A \times \{0\}$. Notice $W \approx \Sigma A$, which is cell-like by Proposition 7.7.3, so W has trivial shape by Theorem 7.1.2. Thus, $h' : W \to P$ is null-homotopic. Since $(A \times \mathbf{I}/A \times \{0\}) \times \mathbf{I}$ can be regarded as the cone over W, the map h' extends to a map $h'' : (A \times \mathbf{I}/A \times \{0\}) \times \mathbf{I} \to P$. Then, h'' extends to a homotopy $\tilde{h} : Z \times \mathbf{I} \to P$ such that $\tilde{h}i = h$, where $\tilde{h}_0 = g$ and $\tilde{h}_1 = g'$. See Fig. 7.3. □

Using the Taylor example, we can also obtain the following theorem:

Theorem 7.7.8 *There exists a cell-like map $g : \boldsymbol{Q} \to Y$ of the Hilbert cube $\boldsymbol{Q}$ onto a compactum Y that is not cell-like nor an ANR, which means that g is not a shape equivalence.*

Proof Let Q_0 and Q_1 be copies of $\boldsymbol{Q}$, $f : X \to Q_0$ be Taylor's cell-like map obtained in Theorem 7.7.5, and embed X into Q_1. We define $Y = Q_0 \cup_f Q_1$. The quotient map $g : Q_1 \to Y$ is cell-like. Indeed, for each $y \in Y$, if $y \in Q_1 \setminus X$ then $g^{-1}(y) = \{y\}$, and if $y \in Q_0$ then $g^{-1}(y) = f^{-1}(y)$ is cell-like.

Assume that Y is cell-like. Since we have a shape equivalence $h : M_f \cup (Q_1 \times \{1\}) \to Y$ by Proposition 7.5.6, it follows from Corollary 7.5.2 that $M_f \cup (Q_1 \times \{1\})$ is cell-like. On the other hand, the natural map

$$q : M_f \cup (Q_1 \times \{1\}) \to (M_f \cup (Q_1 \times \{1\}))/\{Q_0, Q_1 \times \{1\}\} \approx \Sigma X$$

is a shape equivalence by Theorem 7.7.6. Hence, ΣX is also cell-like. Using Proposition 7.7.3 inductively, it follows that $\Sigma^r X$ is cell-like. But $\Sigma^r X \approx X$ is not cell-like, which is a contradiction. Thus, Y is not cell-like. Moreover, if Y is an ANR, then g

is a hereditary shape equivalence by Theorem 7.5.4, which is a contradiction. Thus, Y is not an ANR. □

7.8 Eilenberg–MacLane Complexes

For each $n \in \mathbb{N}$, $\pi_n(\mathbf{S}^n) \neq \{0\}$ by the No Retraction Theorem 5.1.5. In fact, it is known that $\pi_n(\mathbf{S}^n)$ is an infinite cyclic group generated by $[\mathrm{id}_{\mathbf{S}^n}]$, that is, $\pi_n(\mathbf{S}^n) \cong \mathbb{Z}$. It follows from Theorem 5.2.4 that $\pi_m(\mathbf{S}^n) = \{0\}$ for $m < n$, and it is known that $\pi_m(\mathbf{S}^1) = \{0\}$ for any $m > 1$. However, for each $n > 1$, there is some $m > n$ such that $\pi_m(\mathbf{S}^n) \neq \{0\}$. For instance, $\pi_3(\mathbf{S}^2) \cong \mathbb{Z}$.[9]

There exists a path-connected space $K(\mathbb{Z}, n)$ homotopy dominated by a simplicial complex such that

$$\pi_n(K(\mathbb{Z}, n)) \cong \mathbb{Z} \text{ and } \pi_m(K(\mathbb{Z}, n)) = \{0\} \text{ if } m \neq n.$$

Such a space $K(\mathbb{Z}, n)$ is called an **Eilenberg–MacLane space (of type** $(\mathbb{Z}, n)$**)**, which is unique up to homotopy type. The unit circle $\mathbf{S}^1$ is $K(\mathbb{Z}, 1)$. It is easy to see the homotopical uniqueness of $K(\mathbb{Z}, 1)$, that is, if X is a path-connected space homotopy dominated by a simplicial complex such that $\pi_1(X) \cong \mathbb{Z}$ and $\pi_m(X) = \{0\}$ for any $m > 1$, then X has the homotopy type of $\mathbf{S}^1$. Indeed, let $f : \mathbf{S}^1 \to X$ be a map such that $[f] \in \pi_1(X)$ is a generator of $\pi_1(X) \cong \mathbb{Z}$. Then, f induces an isomorphism $f_\sharp : \pi_1(\mathbf{S}^1) \to \pi_1(X)$ because $f_\sharp([\mathrm{id}_{\mathbf{S}^1}]) = [f]$. For each $m > 1$, $f_\sharp : \pi_m(\mathbf{S}^1) \to \pi_m(X)$ is an isomorphism because $\pi_m(\mathbf{S}^1) = \pi_m(X) = \{0\}$. Therefore, f is a weak homotopy equivalence by Theorem 8.2.8, hence it is a homotopy equivalence by Corollary 4.20.10. In general, given an Abelian group G, there exists a path-connected space $K(G, n)$ homotopy dominated by a simplicial complex such that $\pi_n(K(G, n)) \cong G$ and $\pi_m(K(G, n)) = \{0\}$ if $m \neq n$. The space $K(G, n)$ is also called an **Eilenberg–MacLane space (of type** (G, n)**)**, which is unique up to homotopy type.[10]

In this section, for each $n \in \mathbb{N}$, we will construct a *countable simplicial complex* $K(\mathbb{Z}, n)$ called a **(simplicial) Eilenberg–MacLane complex (of type** $(\mathbb{Z}, n)$**)** whose polyhedron $|K(\mathbb{Z}, n)|$ is an Eilenberg–MacLane space of type $(\mathbb{Z}, n)$.[11] Because $\mathbf{S}^1 \approx \partial\Delta^2 = |\partial F(\Delta^2)|$, we can define

$$K(\mathbb{Z}, 1) = \partial F(\Delta^2).$$

[9] For these facts, refer to any textbook on Homotopy Theory or Algebraic Topology.

[10] Usually, Eilenberg–MacLane space $K(G, n)$ is constructed as a CW complex, which has the homotopy type of a simplicial complex.

[11] The notation $K(\mathbb{Z}, n)$, which denotes the Eilenberg–MacLane space in the previous paragraph, is abused here. But, hereafter, the Eilenberg–MacLane space is the polyhedron $|K(\mathbb{Z}, n)|$ of Eilenberg–MacLane complex $K(\mathbb{Z}, n)$.

For $n \geqslant 2$, $K(\mathbb{Z}, n)$ will be constructed so that $\partial F(\Delta^{n+1})$ is its subcomplex and $\partial\Delta^{n+1} = |\partial F(\Delta^{n+1})|$ is a mapping $(n+1)$-deformation retract of $|K(\mathbb{Z}, n)|$ for (the class of) metrizable spaces, where a closed set A in a space X is a **mapping $(n+1)$-deformation retract** of X for a class C of spaces if the following condition is satisfied:

(D_{n+1}) For any space $Z \in C$ with $\dim Z \leqslant n+1$ and each map $f : Z \to X$, there is a map $g : Z \to A$ such that $g|f^{-1}(A) = f|f^{-1}(A)$ and $f \simeq g$ rel. $f^{-1}(A)$, that is, there exists a homotopy $h : Z \times \mathbf{I} \to X$ such that $h_0 = f$, $h_1 = g$, and $h_t|f^{-1}(A) = f|f^{-1}(A)$ for every $t \in \mathbf{I}$.

$$\begin{array}{ccccc} A & & \subset & & X \\ {\scriptstyle f|}\uparrow & \nearrow & \simeq & & \uparrow{\scriptstyle f} \\ & {\scriptstyle \exists g} & & & \\ f^{-1}(A) & & \subset & & Z \in C \quad \dim Z \leqslant n+1 \end{array}$$

Observe that the above (D_{n+1}) implies $(\pi)_i$, $i = 0, 1, \ldots, n+1$, introduced in Sect. 4.20, that is, we have the following fact:

- *If A is a mapping $(n+1)$-deformation retract of X for compact polyhedra, then the inclusion $i : A \subset X$ is an $(n+1)$-equivalence.*

In the above, when X is path-connected, the inclusion map $i : A \subset X$ induces the isomorphisms $i_\sharp : \pi_m(A) \to \pi_m(X)$, $m \leqslant n$, and the epimorphism $i_\sharp : \pi_{n+1}(A) \to \pi_{n+1}(X)$ (Theorem 8.2.8). Using this fact, we will prove the homotopical uniqueness of $K(\mathbb{Z}, n)$ (Theorem 7.8.6).

For polyhedra, we have the following:

Proposition 7.8.1 *Let K be a simplicial complex with L a subcomplex of K. If $|L|$ is a mapping $(n+1)$-deformation retract of $|K|$ for metrizable spaces, then $|L^{(n+1)}|$ is a retract of $|K^{(n+1)}|$.*

Proof By Theorem 4.15.7 with Remark 4.15.3, $\varphi = \mathrm{id} : |K| \to |K|_{\mathrm{m}}$ is a homotopy equivalence that has a homotopy inverse $\psi : |K|_{\mathrm{m}} \to |K|$ such that $\psi||L^{(n+1)}|_{\mathrm{m}}$ is a homotopy inverse of $\varphi||L^{(n+1)}| = \mathrm{id} : |L^{(n+1)}| \to |L^{(n+1)}|_{\mathrm{m}}$. Since $|L|$ is a mapping $(n+1)$-deformation retract of $|K|$ for metrizable spaces, there is a map $f : |K^{(n+1)}|_{\mathrm{m}} \to |L|$ such that $f||L^{(n+1)}|_{\mathrm{m}} = \psi||L^{(n+1)}|_{\mathrm{m}}$ and $f \simeq \psi||K^{(n+1)}|_{\mathrm{m}}$ rel. $|L^{(n+1)}|_{\mathrm{m}}$ in $|K|$. Thus, we have a map $f\varphi : |K^{(n+1)}| \to |L|$ such that

$$f\varphi||L^{(n+1)}| = (\psi||L^{(n+1)}|_{\mathrm{m}})(\varphi||L^{(n+1)}|) \simeq \mathrm{id} \ \text{ in } |L^{(n+1)}|.$$

Applying the Homotopy Extension Theorem 4.5.3, we can obtain a retraction $r : |K^{(n+1)}| \to |L^{(n+1)}|$ such that $r \simeq f\varphi||K^{(n+1)}|$ in $|L|$. □

To construct Eilenberg–MacLane complexes, we need the following:

Lemma 7.8.2 *Suppose that a space X is written as $X = Y \cup \bigcup_{\lambda\in\Lambda} B_\lambda$, where Y is closed in X, $(B_\lambda)_{\lambda\in\Lambda}$ is discrete in X, and each pair $(B_\lambda, Y \cap B_\lambda)$ is homeomorphic to $(\mathbf{B}^{n+2}, \mathbf{S}^{n+1})$. Then, Y is a mapping $(n+1)$-deformation retract of X for normal spaces.*

Proof Let $f : Z \to X$ be a map from a normal space Z with $\dim Z \leqslant n+1$. Since $\dim f^{-1}(B_\lambda) \leqslant n+1$ and $Y \cap B_\lambda \approx \mathbf{S}^{n+1}$, we can apply Theorem 5.2.4 to obtain a map $f_\lambda : f^{-1}(B_\lambda) \to Y \cap B_\lambda$ such that $f_\lambda | f^{-1}(Y \cap B_\lambda) = f | f^{-1}(Y \cap B_\lambda)$. By identifying B_λ with $\mathbf{B}^{n+1}$ and using the straight-line homotopy, we have $f|f^{-1}(B_\lambda) \simeq f_\lambda$ rel. $f^{-1}(Y \cap B_\lambda)$ in B_λ. Because $(f^{-1}(B_\lambda))_{\lambda\in\Lambda}$ is discrete in Z, we can define a map $f' : Z \to Y$ by $f'|f^{-1}(Y) = f|f^{-1}(Y)$ and $f'|f^{-1}(B_\lambda) = f_\lambda$ for each $\lambda \in \Lambda$. Then, $f' \simeq f$ rel. $f^{-1}(Y)$. □

Proposition 7.8.3 *For every connected simplicial complex K and $n \in \mathbb{N}$, there exists a connected simplicial complex $\widetilde{K}$ containing K as a subcomplex such that $\pi_{n+1}(|\widetilde{K}|) = \{0\}$ and $|K|$ is a mapping $(n+1)$-deformation retract of $|\widetilde{K}|$ for normal spaces, hence the inclusion map $i : |K| \subset |\widetilde{K}|$ induces an isomorphism $i_\sharp : \pi_m(|K|) \to \pi_m(|\widetilde{K}|)$ for $m \leqslant n$. Moreover, if K is countable, then so is $\widetilde{K}$.*

Proof Here, we identify $\mathbf{S}^{n+1} = \partial\Delta^{n+2}$. By Remark 4.10.2 for the Simplicial Approximation Theorem 4.10.11, we can write $\pi_{n+1}(|K|) = \{[h_\lambda] \mid \lambda \in \Lambda\}$,[12] where $h_\lambda : L_\lambda \to K$ is a simplicial map of $L_\lambda = \mathrm{Sd}^{n_\lambda} \partial F(\Delta^{n+2})$ for some $n_\lambda \in \mathbb{N}$. If K is countable, then Λ is also countable. For each $\lambda \in \Lambda$, we have $K_\lambda \lhd F(\Delta^{n+2})$ with $L_\lambda \subset K_\lambda$. For example, take a point $v \in \mathrm{rint}\,\Delta^{n+2}$ and let $K_\lambda = L_\lambda \cup \{v\} \cup \{\langle \sigma \cup \{v\}\rangle \mid \sigma \in L_\lambda\}$ (cf. Proof of Theorem 4.3.11). Then, $(|K_\lambda|, |L_\lambda|) \approx (\mathbf{B}^{n+2}, \mathbf{S}^{n+1})$. Regarding $K_\lambda, \lambda \in \Lambda$, as a pairwise disjoint collection, we have a simplicial complex $\bigoplus_{\lambda\in\Lambda} K_\lambda$ and its subcomplex $\bigoplus_{\lambda\in\Lambda} L_\lambda$. Let $h : \bigoplus_{\lambda\in\Lambda} L_\lambda \to K$ be the simplicial map defined by $h|L_\lambda = h_\lambda$ for each $\lambda \in \Lambda$. The desired simplicial complex $\widetilde{K}$ can be defined as $\widetilde{K} = Z_h \cup \bigoplus_{\lambda\in\Lambda} K_\lambda$, where Z_h is the simplicial mapping cylinder of h. If K is countable, then so is $\widetilde{K}$ because Λ is countable and each K_λ is finite. By Lemma 7.8.2, $|Z_h|$ is a mapping $(n+1)$-deformation retract of $|\widetilde{K}|$ for normal spaces. Since $|K|$ is a strong deformation retract of $|Z_h|$, it follows that $|K|$ is also a mapping $(n+1)$-deformation retract of $|\widetilde{K}|$ for normal spaces. Hence, the inclusion $i : |K| \subset |\widetilde{K}|$ is an $(n+1)$-equivalence. In particular, the inclusion i induces the epimorphism $i_\sharp : \pi_{n+1}(|K|) \to \pi_{n+1}(|\widetilde{K}|)$. Since every $h_\lambda \simeq 0$ in $|\widetilde{K}|$, it follows that $\pi_{n+1}(|\widetilde{K}|) = i_\sharp(\pi_{n+1}(|K|)) = \{0\}$. □

Now, we will construct the **Eilenberg–MacLane complex** $K(\mathbb{Z}, n)$ for each $n \geqslant 2$. We apply Proposition 7.8.3 inductively to obtain a tower

$$F(\partial\Delta^{n+1}) = K_0 \subset K_1 \subset K_2 \subset \cdots$$

of countable simplicial complexes such that each $|K_{k-1}|$ is a mapping $(n+k)$-deformation retract of $|K_k|$ for normal spaces, $\pi_m(|K_k|) = \{0\}$ for $n \neq m \leqslant n+k$,

[12] It suffices to take a set $\{[h_\lambda] \mid \lambda \in \Lambda\}$ generating the group $\pi_{n+1}(|K|)$. Then, if $\pi_{n+1}(|K|)$ is finitely generated, we can take a finite set as Λ.

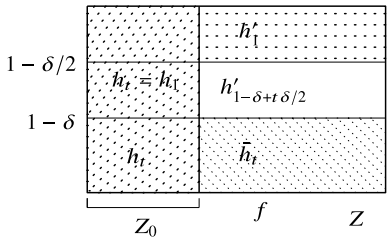


Fig. 7.4 Extending a homotopy h

and the inclusion $i_k : \partial\Delta^{n+1} \subset |K_k|$ induces the isomorphism $(i_k)_\sharp : \pi_n(\partial\Delta^{n+1}) \to \pi_n(|K_k|)$. Then, the countable simplicial complex $K(\mathbb{Z}, n) = \bigcup_{k\in\omega} K_k$ is the desired complex. Indeed, due to Proposition 4.2.3, each map $f : \mathbf{S}^m \to |K(\mathbb{Z}, n)|$ has the image $f(\mathbf{S}^m)$ contained in some $|K_k|$ and if $f \simeq g$ in $|K(\mathbb{Z}, n)|$ for maps $f, g : \mathbf{S}^m \to |K(\mathbb{Z}, n)|$ then $f \simeq g$ in some $|K_k|$. So, it follows that

$$\pi_n(|K(\mathbb{Z}, n)|) \cong \pi_n(\partial\Delta^{n+1}) \cong \mathbb{Z} \text{ and } \pi_m(|K(\mathbb{Z}, n)|) = \{0\} \text{ if } m \neq n.$$

By the same argument, it is easy to prove that $\partial\Delta^{n+1}$ is a mapping $(n+1)$-deformation retract of $|K(\mathbb{Z}, n)|$ for *compact spaces*.

To show that $\partial\Delta^{n+1}$ is a mapping $(n+1)$-deformation retract of $|K(\mathbb{Z}, n)|$ for *metrizable spaces*, we use the following lemma:

Lemma 7.8.4 *Suppose that A is a mapping $(n+1)$-deformation retract of an ANE X for metrizable spaces. Let $f : Z \to X$ be a map of a metrizable space Z with $\dim Z \leqslant n+1$ and a homotopy $h : Z_0 \times \mathbf{I} \to X$ of a closed set Z_0 in Z with $\delta > 0$ such that $h_0 = f|Z_0$, $h_1(Z_0) \subset A$, $h_t = h_1$ for every $t \in [1-\delta, 1]$, and $h_t|Z_0 \cap f^{-1}(A) = f|Z_0 \cap f^{-1}(A)$ for every $t \in \mathbf{I}$. Then, h extends to a homotopy $\tilde{h} : Z \times \mathbf{I} \to X$ such that $\tilde{h}_0 = f$, $\tilde{h}_1(Z) \subset A$, $\tilde{h}_t = \tilde{h}_1$ for every $t \in [1-\delta/2, 1]$, and $\tilde{h}_t|f^{-1}(A) = f|f^{-1}(A)$ for every $t \in \mathbf{I}$.*

Proof Because X is an ANE, h can be extended to a homotopy $\bar{h} : Z \times \mathbf{I} \to X$ such that $\bar{h}_0 = f$ and $\bar{h}_t|f^{-1}(A) = f|f^{-1}(A)$ for every $t \in \mathbf{I}$. Since A is a mapping $(n+1)$-deformation retract of X for metrizable spaces, we have a homotopy $h' : Z \times \mathbf{I} \to X$ such that

$$h'_0 = \bar{h}_{1-\delta},\ h'_1(Z) \subset A \text{ and } h'_t|\bar{h}^{-1}_{1-\delta}(A) = \bar{h}_{1-\delta}|\bar{h}^{-1}_{1-\delta}(A) \text{ for every } t \in \mathbf{I}.$$

The desired homotopy $\tilde{h} : Z \times \mathbf{I} \to X$ can be defined as follows:

$$\tilde{h}_t = \begin{cases} \bar{h}_t & \text{if } t \leqslant 1-\delta, \\ h'_{2(t-1+\delta)/\delta} & \text{if } 1-\delta \leqslant t \leqslant 1-\delta/2, \\ h'_1 & \text{if } t \geqslant 1-\delta/2. \end{cases}$$

Refer to Fig. 7.4. Indeed, $\tilde{h}_0 = \bar{h}_0 = f$ and $\tilde{h}_t = h'_1$ for every $t \in [1-\delta/2, 1]$. Since $Z_0 \cup f^{-1}(A) \subset \bar{h}^{-1}_{1-\delta}(A)$, it follows that $h'_t|Z_0 = \bar{h}_{1-\delta}|Z_0 = h_1$ and $h'_t|f^{-1}(A) =$

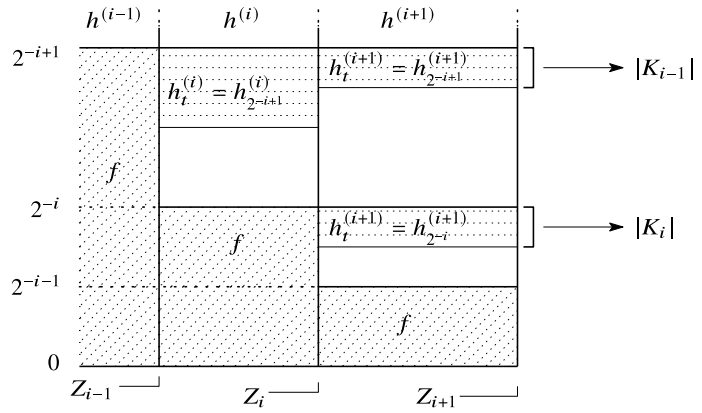


Fig. 7.5 Homotopies $h^{(i)}$

$\bar{h}_1|f^{-1}(A) = f|f^{-1}(A)$ for every $t \in \mathbf{I}$. Then, $\tilde{h}|Z_0 \times \mathbf{I} = h$ and $\tilde{h}_t|f^{-1}(A) = f|f^{-1}(A)$ for every $t \in \mathbf{I}$. □

Proposition 7.8.5 *For each $n \geqslant 2$, $\partial\Delta^{n+1}$ is a mapping $(n+1)$-deformation retract of $|K(\mathbb{Z}, n)|$ for metrizable spaces.*

Proof Observe that $|K(\mathbb{Z}, n)| = \varinjlim |K_k|$, where $(K_i)_{i\in\omega}$ is the tower defining $K(\mathbb{Z}, n)$. Then, each $|K_i|$ is a mapping $(n+1)$-deformation retract of $|K_{i+1}|$ for normal spaces, hence for metrizable spaces. Let $f : Z \to |K(\mathbb{Z}, n)|$ be a map from a metrizable space Z with $\dim Z \leqslant n+1$. For each $i \in \omega$, let $Z_i = f^{-1}(|K_i|)$. Then, $Z_0 \subset Z_1 \subset \cdots$ and $Z = \bigcup_{i\in\omega} Z_i$. By induction, we will define homotopies $h^{(i)} : Z_i \times \mathbf{I} \to |K_i|$, $i \in \omega$, such that

$$h^{(i)}|Z_{i-1} \times \mathbf{I} = h^{(i-1)} \text{ for } i > 0,\ h_t^{(i)} = f|Z_i \text{ for } 0 \leqslant t \leqslant 2^{-i},$$
$$h_1^{(i)}(Z_i) \subset \partial\Delta^{n+1},\ h^{(i)}(Z_i \times [2^{-j}, 2^{-j+1}]) \subset |K_j| \text{ for } 0 < j \leqslant i, \text{ and}$$
$$h_t^{(i)} = h^{(i)}_{2^{-j}} \text{ for } 2^{-j} - 2^{-i-1} \leqslant t \leqslant 2^{-j},\ j < i.$$

Refer to Fig. 7.5. The homotopy $h^{(0)} : Z_0 \times \mathbf{I} \to |K_0|$ is defined by $h_t^{(0)} = f|Z_0$ for $t \in \mathbf{I}$. Assume that $h^{(i)}$ has been defined. For each $j = 0, 1, \ldots, i+1$, let

$$L_j = \big(Z_i \times \mathbf{I}\big) \cup \big(Z_{i+1} \times [0, 2^{-j}]\big).$$

The homotopy $h^{(i)}$ can be extended over L_{i+1} by $(f|Z_{i+1}) \times \mathrm{id}$. Then, using Lemma 7.8.4 iteratively, we extend this over $L_i \subset \cdots \subset L_1 \subset L_0 = Z_{i+1} \times \mathbf{I}$ step-by-step. Thus, we can obtain the homotopy $h^{(i+1)} : Z_{i+1} \times \mathbf{I} \to |K_{i+1}|$ satisfying the conditions.

According to Theorem 2.8.8 (2), each $z \in Z$ has a neighborhood V in Z such that $f(V) \subset |K_i|$ for some $i \in \omega$, which means that $V \subset Z_i$, so $z \in \operatorname{int} Z_i$. Therefore,

$Z = \bigcup_{i\in\omega} \operatorname{int} Z_i$. Thus, we can define a homotopy $h : Z \times \mathbf{I} \to |K(\mathbb{Z}, n)|$ by $h|Z_i \times \mathbf{I} = h^{(i)}$ for each $i \in \omega$. Then, $h_0 = f$, $h_1(Z) \subset \partial\Delta^{n+1}$ and $h_t|f^{-1}(\partial\Delta^{n+1}) = f|f^{-1}(\partial\Delta^{n+1})$ for every $t \in \mathbf{I}$. □

The homotopical uniqueness of Eilenberg–MacLane spaces can be derived from the following theorem:

Theorem 7.8.6 *Let $n \in \mathbb{N}$ and suppose that X is a path-connected space homotopy dominated by a simplicial complex such that $\pi_n(X) \cong \mathbb{Z}$ and $\pi_m(X) = \{0\}$ for every $m \neq n$. Then, X has the homotopy type of Eilenberg–MacLane complex $K(\mathbb{Z}, n)$ constructed above.*

Proof The case $n = 1$ has been seen already. For $n \geqslant 2$, let $(K_i)_{i\in\omega}$ be the tower in the definition of $K(\mathbb{Z}, n)$. Take a homeomorphism $\varphi : |K_0| \to \mathbf{S}^n$ and a map $\alpha : \mathbf{S}^n \to X$ such that $[\alpha]$ is a generator of $\pi_n(X) \cong \mathbb{Z}$. Let $f_0 = \alpha\varphi : |K_0| \to X$. To extend f_0 to a map $f : |K(\mathbb{Z}, n)| \to X$, we inductively construct maps $f_i : |K_i| \to X$, $i \in \mathbb{N}$, such that $f_i||K_{i-1}| = f_{i-1}$, and then f can be defined by $f||K_i| = f_i$. In fact, the construction of K_i from K_{i-1} is as follows: Taking a simplicial map $h : \bigoplus_{\lambda\in\Lambda} L_\lambda \to K_{i-1}$ such that each L_λ is a triangulation of $\mathbf{S}^{n+i}$ and $\{[h|L_\lambda] \mid \lambda \in \Lambda\} = \pi_{n+i}(|K_{i-1}|)$ (or $\{[h|L_\lambda] \mid \lambda \in \Lambda\}$ generates $\pi_{n+i}(|K_{i-1}|)$), we define $K_i = Z_h \cup \bigoplus_{\lambda\in\Lambda} K_\lambda$, where K_λ is a triangulation of $\mathbf{B}^{n+i+1}$ with $L_\lambda \subset K_\lambda$. For each $\lambda \in \Lambda$, the map $f_{i-1}h|L_\lambda : \mathbf{S}^{n+i} \to X$ is null-homotopic because of $\pi_{n+i}(X) = \{0\}$. Hence, f_{i-1} extends over $Z_{h|L_\lambda} \cup K_\lambda$. Thus, f_{i-1} extends to a map $f_i : |K_i| \to X$.

It remains to show that f is a homotopy equivalence. By virtue of Corollary 4.20.10, it suffices to verify that f is a weak homotopy equivalence. For $m \neq n$, $f_\sharp : \pi_m(|K(\mathbb{Z}, n)|) \to \pi_m(X)$ is an isomorphism because $\pi_m(|K(\mathbb{Z}, n)|) = \pi_m(X) = \{0\}$. Recall that the inclusion $|K_0| = \partial\Delta^{n+1} \subset |K(\mathbb{Z}, n)|$ is an $(n + 1)$-equivalence, hence it induces the isomorphism from $\pi_n(|K_0|)$ onto $\pi_n(|K(\mathbb{Z}, n)|)$ (Theorem 8.2.8). On the other hand, $(f_0)_\sharp : \pi_n(|K_0|) \to \pi_n(X)$ is an isomorphism because it sends a generator of $\pi_n(|K_0|) \cong \mathbb{Z}$ to that of $\pi_n(X) \cong \mathbb{Z}$. Then, it follows that $f_\sharp : \pi_n(|K(\mathbb{Z}, n)|) \to \pi_n(X)$ is also an isomorphism. Consequently, f is a weak homotopy equivalence by Theorem 8.2.8. □

Remark 7.8.1 Recall that $K(\mathbb{Z}, 0) = \mathbb{Z}$ and $K(\mathbb{Z}, 1) = \partial F(\Delta^2)$. For $n \geqslant 2$, depending on how it is made, $K(\mathbb{Z}, n)$ is not locally finite, but it has the homotopy type of a countable locally finite simplicial complex by Theorem 4.19.3. Because of the homotopy uniqueness of $K(\mathbb{Z}, n)$, we can replace $K(\mathbb{Z}, n)$ with a countable locally finite simplicial complex of the same homotopy type. Then, by Corollary 6.2.18 (1), $|K(\mathbb{Z}, n)| = |K(\mathbb{Z}, n)|_{\mathrm{m}}$ is an ANE for normal spaces, and hence it has the HEP for paracompact spaces (Remark 6.4.1). However, in this case, it is not necessarily guaranteed that $\partial\Delta^{n+1}$ is a mapping $(n + 1)$-deformation retract of $|K(\mathbb{Z}, n)|$ for metrizable spaces. Refer to Sect. 8.4.

For a pair (X, A) of spaces, a map $f : (Y, B) \to (Z, C)$ induces the map

$$f_* : \mathrm{C}((X, A), (Y, B)) \to \mathrm{C}((X, A), (Z, C)),$$

which is defined by $f_*(k) = f \circ k$ (cf. Proposition 2.10.8 (1)).

Lemma 7.8.7 *Let $f, f' : (Y, B) \to (Z, C)$ with $f \simeq f'$ (as maps of pairs). For each pair (X, A) of spaces,*

$$f_* \simeq f'_* : \mathrm{C}((X, A), (Y, B)) \to \mathrm{C}((X, A), (Z, C)).$$

Proof Let $h : (Y \times \mathbf{I}, B \times \mathbf{I}) \to (Z, C)$ be a homotopy from f to f'. We define

$$\tilde{h} : \mathrm{C}((X, A), (Y, B)) \times \mathbf{I} \to \mathrm{C}((X, A), (Z, C))$$

by $\tilde{h}(g, t) = h_t \circ g$. To see the continuity of $\tilde{h}$, for each compact set $K \subset X$ and each open set $U \subset Z$, let $(g, t) \in \tilde{h}^{-1}(\langle K; U\rangle)$. Since $h(g(K) \times \{t\}) = h_t(g(K)) \subset U$, we can choose open sets $V \subset Y$ and $W \subset \mathbf{I}$ so that $g(K) \times \{t\} \subset V \times W$ and $h(V \times W) \subset U$. Then, $g \in \langle K; V\rangle$ and $t \in W$. Moreover,

$$\forall g' \in \langle K; V\rangle \cap \mathrm{C}((X, A), (Y, B)), \forall s \in W,$$
$$h_s(g'(K)) = h(g'(K) \times \{s\}) \subset h(V \times W) \subset U, \text{ i.e., } \tilde{h}(g', s) \in \langle K; U\rangle.$$

Thus, we have $\langle K; V\rangle \times W \subset \tilde{h}^{-1}(\langle K; U\rangle)$. Consequently, $\tilde{h}^{-1}(\langle K; U\rangle)$ is open in $\mathrm{C}((X, A), (Y, B)) \times \mathbf{I}$. □

Proposition 7.8.8 *For each pair (X, A) of spaces, $(Y, B) \simeq (Y', B')$ implies*

$$\mathrm{C}((X, A), (Y, B)) \simeq \mathrm{C}((X, A), (Y', B')).$$

Proof Let $f : (Y, B) \to (Y', B')$ be a homotopy equivalence with g a homotopy inverse. Then, f and g induce the maps

$$f_* : \mathrm{C}((X, A), (Y, B)) \simeq \mathrm{C}((X, A), (Y', B')) \text{ and}$$
$$g_* : \mathrm{C}((X, A), (Y', B')) \simeq \mathrm{C}((X, A), (Y, B))$$

defined by $f_*(k) = f \circ k$ and $g_*(k) = g \circ k$, respectively. By virtue of Lemma 7.8.7, we have $g_* f_* = (gf)_* \simeq \mathrm{id}$ and $f_* g_* = (fg)_* \simeq \mathrm{id}$. □

A path $\alpha : \mathbf{I} \to X$ is called a **loop** at x_0 if $\alpha(0) = \alpha(1) = x_0$, i.e., $\alpha \in \mathrm{C}((\mathbf{I}, \partial\mathbf{I}), (X, x_0))$. For each pointed space (X, x_0), the **loop space** $\Omega(X, x_0)$ is defined as the space $\mathrm{C}((\mathbf{I}, \partial\mathbf{I}), (X, x_0))$ admitting the compact-open topology,[13] where the constant loop $c_{x_0} : \mathbf{I} \to \{x_0\} \subset X$ is the base point of $\Omega(X, x_0)$. Refer to Sect. 8.1.

Corollary 7.8.9 *For each simplicial complex K with $v_0 \in K^{(0)}$, the loop space $\Omega(|K|, v_0)$ has the homotopy type of a simplicial complex and $\Omega(|K|, v_0) \simeq \Omega(|K|_{\mathrm{m}}, v_0)$, where $\Omega(|K|, v_0)$ is an ANE and $\Omega(|K|_{\mathrm{m}}, v_0)$ is an ANR.*

[13] For the compact-open topology, refer to Sect. 2.10.

Proof By virtue of Proposition 7.8.8,

$$\Omega(|K|, v_0) = \mathrm{C}((\mathbf{I}, \partial\mathbf{I}), (|K|, v_0)) \simeq \mathrm{C}((\mathbf{I}, \partial\mathbf{I}), (|K|_{\mathrm{m}}, v_0)) = \Omega(|K|_{\mathrm{m}}, v_0),$$

where the first space is an ANE and the last is an ANR by Propositions 6.1.10 (9) and 6.2.9 (7), respectively. The first assertion follows from Corollary 6.6.5. □

For any pointed space (X, x_0) and $n \in \mathbb{N}$, we have

$$\pi_n(\Omega(X, x_0), c_{x_0}) \cong \pi_{n+1}(X, x_0)$$

(Theorem 8.1.4). Then, combining Theorem 7.8.6 with Corollary 7.8.9, we have the following corollary:

Corollary 7.8.10 *For each $n \in \mathbb{N}$, the loop space $\Omega(|K(\mathbb{Z}, n+1)|, v)$ has the homotopy type of $|K(\mathbb{Z}, n)|$, where $v \in |K(\mathbb{Z}, n+1)|$ is any point.* □

Normally, $K(\mathbb{Z}, 0)$ is not defined, but it is convenient to define it as follows:

$$K(\mathbb{Z}, 0) = \mathbb{Z} \ \text{(a countable zero-dimensional complex)}.$$

In fact, Corollary 7.8.10 above holds even for $n = 0$, i.e., $\Omega(|K(\mathbb{Z}, 1)|, v) \approx \Omega(\mathbf{S}^1, \mathbf{e}_1) \simeq \mathbb{Z}$.

The homotopy equivalence $\Omega(\mathbf{S}^1, \mathbf{e}_1) \simeq \mathbb{Z}$ can be shown as follows: Due to Theorem 8.1.4,

$$\pi_n(\Omega(\mathbf{S}^1, \mathbf{e}_1), c_{\mathbf{e}_1}) \cong \pi_{n+1}(\mathbf{S}^1, \mathbf{e}_1) = \{0\} \ \text{ for each } n \in \mathbb{N},$$

where $c_{\mathbf{e}_1} \in \Omega(\mathbf{S}^1, \mathbf{e}_1)$ is the constant loop. Let Ω_0 be the path-component of $\Omega(\mathbf{S}^1, \mathbf{e}_1)$ containing $c_{\mathbf{e}_1}$. Then, Ω_0 is weak homotopy equivalent to a singleton because $\pi_n(\Omega_0, c_{\mathbf{e}_1}) = \{0\}$ for every $n \in \mathbb{N}$. Since $\Omega(\mathbf{S}^1, \mathbf{e}_1) = \mathrm{C}((\mathbf{I}, \partial\mathbf{I}), (\mathbf{S}^1, \mathbf{e}_1))$ is an ANE (so an ANR) by Proposition 6.2.9 (7), it follows that Ω_0 is open in $\Omega(\mathbf{S}^1, \mathbf{e}_1)$, so it is also an ANR. Hence, Ω_0 has the homotopy type of a singleton by Corollary 6.6.6, that is, Ω_0 is contractible. By the way, we can regard $\mathbf{S}^1 = \{z \in \mathbb{C} \mid |z| = 1\}$ $(\mathbf{e}_1 = 1 \in \mathbb{C})$ which is a subgroup of $\mathbb{C} \setminus \{0\}$ with multiplication. Then, $\Omega(\mathbf{S}^1, 1)$ is a topological group with pointwise multiplication. For each $n \in \mathbb{N}$, we define $\alpha_n \in \Omega(\mathbf{S}^1, 1)$ by $\alpha_n(t) = e^{2n\pi it}$, $t \in \mathbf{I}$. As is well known, we have the following isomorphism:

$$\mathbb{Z} \ni n \mapsto [\alpha_n] \in \pi_1(\mathbf{S}^1, 1) = \Omega(\mathbf{S}^1, 1)/\simeq .$$

This means that each path-component contains some α_n. Since $\Omega(\mathbf{S}^1, 1)$ is a topological group, the path components of $\Omega(\mathbf{S}^1, 1)$ are $\alpha_n\Omega_0$, $n \in \mathbb{Z}$, from which it follows that

$$\Omega(\mathbf{S}^1, 1) = \bigoplus_{n\in\mathbb{N}} \alpha_n\Omega_0 \approx \mathbb{Z} \times \Omega_0 \simeq \mathbb{Z}.$$

Remark 7.8.2 It should be noted that $K(\mathbb{Z}, n)$ is connected for each $n \geqslant 1$, but $K(\mathbb{Z}, 0)$ is not. Moreover, $\partial\Delta^{n+1}$ is a mapping $(n+1)$-deformation retract of $|K(\mathbb{Z}, n)|$, but $\partial\Delta^1$ *is not a mapping* 1*-deformation retract of* $|K(\mathbb{Z}, 0)|$.

The **wedge sum** (or the **one-point union**) $X_1 \vee \cdots \vee X_k$ of pointed spaces $X_1 = (X_1, x_1)$, ..., $X_k = (X_k, x_k)$ is defined as the quotient space

$$X_1 \vee \cdots \vee X_k = (X_1 \oplus \cdots \oplus X_k)/\{x_1, \ldots, x_n\},$$

which is also defined as the following subspace of the product space $\prod_{i=1}^n X_i$:

$$X_1 \vee \cdots \vee X_k = \left\{ x \in \textstyle\prod_{i=1}^k X_i \;\middle|\; x(i) = x_i \text{ except for one } i \right\}.$$

When $(X_i, x_i) = (X, x_0)$ for every $i = 1, \ldots, k$, we write $\bigvee^k X$ instead of $X \vee \cdots \vee X$.

We use the following theorem but leave the proof to any textbook on Homotopy Theory or Algebraic Topology.

Theorem 7.8.11 *For every $n \geqslant 2$ and $k \in \mathbb{N}$, $\pi_i(\bigvee^k \mathbf{S}^n) = \{0\}$ for $i < n$ and*

$$\pi_n(\textstyle\bigvee^k \mathbf{S}^n) \cong \mathbb{Z}^k \cong \underbrace{\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}}_{k}.$$

The case $n = 1$ is THEOREM OF SEIFERT AND VAN KAMPEN:

$$\pi_1(\textstyle\bigvee^k \mathbf{S}^1) \cong \underbrace{\mathbb{Z} * \cdots * \mathbb{Z}}_{k} \quad \text{(the free product)}.$$

Proposition 7.8.12 *For each $m > n$, $|F(\Delta^m)^{(n)}| \simeq \bigvee^k \mathbf{S}^n$, where $k = \binom{m}{n+1}$. Consequently, $\pi_i(|F(\Delta^m)^{(n)}|) = \{0\}$ for $i < n$ and $\pi_n(|F(\Delta^m)^{(n)}|) \cong \mathbb{Z}^k$.*

Proof Since $P = |\operatorname{St}(\mathbf{e}_{m+1}, F(\Delta^m)^{(n)})|$ is a compact AR, the quotient space $|F(\Delta^m)^{(n)}|/P$ is a compact ANR by Theorem 6.5.3, where the quotient map $q : |F(\Delta^m)^{(n)}| \to |F(\Delta^m)^{(n)}|/P$ is a (fine) homotopy equivalence by Theorem 7.5.4. Thus, $|F(\Delta^m)^{(n)}| \simeq |F(\Delta^m)^{(n)}|/P$. The latter is homeomorphic to $\bigvee^k \mathbf{S}^n$. □

In the construction of Eilenberg–MacLane complex $K(\mathbb{Z}, n)$, replacing the starting complex $F(\partial\Delta^{n+1})$ with a connected simplicial complex L, we can obtain a connected simplicial complex $K(L, n)$ such that $\pi_i(|K(L, n)|) = \{0\}$ for $i > n$ and $|L|$ is a mapping $(n+1)$-deformation retract of $|K(L, n)|$ for metrizable spaces. Then, the inclusion $|L| \subset |K(L, n)|$ is an $(n+1)$-equivalence, hence $\pi_i(|K(L, n)|) \cong \pi_i(|L|)$ for $i \leqslant n$ (Theorem 8.2.8).

Now, for each $n \geqslant 2$ and $k \in \mathbb{N}$, let L be a triangulation of $\bigvee^k \mathbf{S}^n$. We can obtain an **Eilenberg–MacLane complex** $K(\mathbb{Z}^k, n)$ such that $|L|$ is a mapping $(n+1)$-deformation retract of $|K(\mathbb{Z}^k, n)|$ for metrizable spaces. It should be noted that $K(\mathbb{Z}^k, n)$ is also countable. In the same way as Theorem 7.8.6, we can prove the homotopical uniqueness of $K(\mathbb{Z}^k, n)$. Indeed, let X be a path-connected space homotopy dominated by a simplicial complex such that $\pi_n(X) \cong \mathbb{Z}^k$ and $\pi_i(X) = \{0\}$ for every $i \neq n$. Using generators of $\pi_n(X)$, we define a map $\varphi : |L| \approx \bigvee^k \mathbf{S}^n \to X$. By the construction of $K(\mathbb{Z}^k, n)$, we can inductively extend φ to a map $f : |K(\mathbb{Z}^k, n)| \to X$. Then, f is a weak homotopy equivalence, so f is a homotopy equivalence by Corollary 4.20.10.

Since $\pi_n(|K(\mathbb{Z},n)|^k) \cong \mathbb{Z}^k$ and $\pi_i(|K(\mathbb{Z},n)|^k) = \{0\}$ for $i \neq n$ by Proposition 8.1.1, we have the following:

Proposition 7.8.13 *For each $n \geqslant 2$ and $k \in \mathbb{N}$, the product space $|K(\mathbb{Z},n)|^k$ has the homotopy type of $|K(\mathbb{Z}^k,n)|$.* □

In the rest of this section, we will construct the so-called an Edwards–Walsh complex, which will be used in Sect. 7.10.

Proposition 7.8.14 *Let K be a simplicial complex and $n \geqslant 2$. There exist countable simplicial complexes K_σ, $\sigma \in K$, satisfying the following conditions:*

(i) $|K_\sigma| = \sigma$ *if* $\dim \sigma \leqslant n$;
(ii) $K_{\sigma'} \subset K_\sigma$ *if* $\sigma' < \sigma$;
(iii) $|K_\sigma| \cap |K_{\sigma'}| = |K_{\sigma\cap\sigma'}|$ *for each* $\sigma, \sigma' \in K$, *where* $K_\emptyset = \emptyset$;
(iv) $(|K_\sigma|, \partial\sigma) \approx (|K(\mathbb{Z},n)|, \partial\Delta^{n+1})$ *for each* $(n+1)$*-simplex* $\sigma \in K$;
(v) $|K_\sigma| \simeq |K(\mathbb{Z},n)|^{k(m)}$ *if* $\dim \sigma = m > n+1$, *where* $k(m) = \binom{m}{n+1}$;
(vi) *If* $\dim \sigma \geqslant n+1$, *then* $|K_{\partial\sigma}|$ *is a mapping* $(n+1)$*-deformation retract of* $|K_\sigma|$ *for metrizable spaces, where* $K_{\partial\sigma} = \bigcup_{\sigma'<\sigma} K_{\sigma'}$.

Proof For every $\sigma \in K^{(n)}$, let $K_\sigma = F(\sigma)$. Suppose that K_σ have been defined for all $\sigma \in K^{(m)}$. For each $(m+1)$-simplex $\sigma \in K$, let

$$L_{\sigma,i} = \bigcup \{K_{\sigma'} \mid \sigma' < \sigma,\ \dim \sigma' = i\},\ n \leqslant i \leqslant m.$$

Then, $F(\sigma)^{(n)} = L_{\sigma,n} \subset L_{\sigma,n+1} \subset \cdots \subset L_{\sigma,m} = K_{\partial\sigma}$ are simplicial complexes by (ii) and (iii). Using (vi), we can show that $|L_{\sigma,i}|$ is a mapping $(n+1)$-deformation retract of $|L_{\sigma,i+1}|$ for metrizable spaces. As observed in the above, we can construct a simplicial complex $K_\sigma = K(K_{\partial\sigma}, n)$ such that $\pi_i(|K_\sigma|) = \{0\}$ for $i > n$ and $|K_{\partial\sigma}|$ is a mapping $(n+1)$-deformation retract of $|K_\sigma|$ for metrizable spaces. Then, it follows that $\sigma^{(n)} = |L_{\sigma,n}|$ is a mapping $(n+1)$-deformation retract of $|K_\sigma|$ for metrizable spaces, which implies that the inclusion $\sigma^{(n)} \subset |K_\sigma|$ is an $(n+1)$-equivalence. Since $\pi_i(|K_\sigma|) \cong \pi_i(\sigma^{(n)})$ for $i \leqslant n$, we have $\pi_i(|K_\sigma|) = \{0\}$ for $i < n$ and $\pi_n(|K_\sigma|) \cong \mathbb{Z}^{k(m+1)}$ by Proposition 7.8.12. From the homotopy uniqueness and Proposition 7.8.13, it follows that

$$|K_\sigma| \simeq |K(\mathbb{Z}^{k(m+1)}, n)| \simeq |K(\mathbb{Z},n)|^{k(m+1)}.$$

Thus, we have a simplicial complex K_σ satisfying (ii), (v), and (vi). Moreover, we can construct K_σ for every $(m+1)$-simplex $\sigma \in K$ so that $|K_\sigma| \cap |K_{\sigma'}| = |K_{\sigma\cap\sigma'}|$, that is, (iii) is satisfied. By induction, we obtain the desired result. □

In Proposition 7.8.14, for a simplicial complex K and $n \geqslant 2$, we have the following simplicial complex:

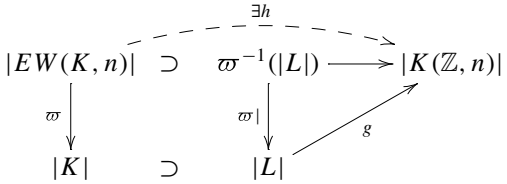
$$EW(K,n) = \bigcup_{\sigma\in K} K_\sigma = K^{(n)} \cup \bigcup_{\sigma\in K\setminus K^{(n)}} K_\sigma,$$

which is called a (**simplicial**) **Edwards–Walsh complex**. Since each K_σ is countable, if K is countable, then so is $EW(K,n)$. There exists a map $\varpi : |EW(K,n)| \to |K|$ such that $\varpi||K^{(n)}| = \mathrm{id}$ and

$$\varpi^{-1}(\sigma) = |K_\sigma| \simeq |K(\mathbb{Z},n)|^{k(\dim\sigma)} \quad \text{for each } \sigma \in K \setminus K^{(n)},$$

which is called an **associate map** of $EW(K,n)$. Indeed, we inductively define maps $\varpi_\sigma : |K_\sigma| \to \sigma, \sigma \in K$, such that $\varpi_\sigma^{-1}(\partial\sigma) = |K_{\partial\sigma}|$ and $\varpi_\sigma||K_{\sigma'}| = \varpi_{\sigma'}$ for each $\sigma' < \sigma$. Let $\varpi_\sigma = \mathrm{id}$ for every $\sigma \in K^{(n)}$ and assume that $\varpi_\sigma : |K_\sigma| \to \sigma$ have been defined for every $\sigma \in K^{(m)}$. For each $(m+1)$-simplex $\sigma \in K$, let $\varpi_{\partial\sigma} : |K_{\partial\sigma}| \to \partial\sigma$ be the map defined by $\varpi_{\partial\sigma}||K_{\sigma'}| = \varpi_{\sigma'}$ for each $\sigma' < \sigma$. Since $\varpi_{\partial\sigma} \simeq 0$ in σ, $\varpi_{\partial\sigma}$ can be extended to a map $\varpi_\sigma : |K_\sigma| \to \sigma$ by the Homotopy Extension Theorem 4.5.3, where we can modify ϖ_σ to satisfy the condition $\varpi_\sigma^{-1}(\partial\sigma) = |K_{\partial\sigma}|$ because $\mathrm{rint}\,\sigma$ is homotopy dense in σ and $|K_{\partial\sigma}|$ is a zero set in $|K_\sigma|$, i.e., $f^{-1}(0) = |K_{\partial\sigma}|$ for some map $f : |K_\sigma| \to \mathbf{I}$ (cf. Proposition 4.2.2).[14] Then, the map ϖ can be defined by $\varpi||K_\sigma| = \varpi_\sigma$ for each $\sigma \in K$. We prove the following proposition regarding the map $\varpi : |EW(K,n)| \to |K|$:

Proposition 7.8.15 *For each simplicial complex K and $n \geqslant 2$, let $\varpi : |EW(K,n)| \to |K|$ be an associate map of Edwards–Walsh complex $EW(K,n)$. Then, for each subcomplex L of K and a map $g : |L| \to |K(\mathbb{Z},n)|$, the composition $g\varpi|\varpi^{-1}(|L|)$ extends over $|EW(K,n)|$.*



Proof Because $|K(\mathbb{Z},n)|$ is $(n-1)$-connected, we can extend g over $|L \cup K^{(n)}|$ by skeleton-wise induction. Replacing L with $L \cup K^{(n)}$, we can assume that $K^{(n)} \subset L$. For simplicity, we write $\widetilde{K}_m = \bigcup_{\sigma \in L \cup K^{(m)}} K_\sigma$ for $m \geqslant n$. Then, we have $|\widetilde{K}_m| = \varpi^{-1}(|L \cup K^{(m)}|)$. Inductively, we will construct maps $h_m : |\widetilde{K}_m| \to |K(\mathbb{Z},n)|, m \geqslant n$, so that $h_{m+1}||\widetilde{K}_m| = h_m$. Assume that h_m has been obtained. For each $(m+1)$-simplex $\sigma \in K \setminus L$, since $|K_{\partial\sigma}|$ is a mapping $(n+1)$-deformation retract of $|K_\sigma|$ for metrizable spaces (Proposition 7.8.14 (vi)), $|K_{\partial\sigma}^{(n+1)}|$ is a retract of $|K_\sigma^{(n+1)}|$ by Proposition 7.8.1. Therefore, $h_m||K_{\partial\sigma}|$ extends to a map $h'_\sigma : |K_{\partial\sigma} \cup K_\sigma^{(n+1)}| \to |K(\mathbb{Z},n)|$. Since $\pi_i(|K(\mathbb{Z},n)|) = \{0\}$ for every $i > n$, we can further extend h'_σ to a map $h''_\sigma : |K_\sigma| \to |K(\mathbb{Z},n)|$ by skeleton-wise induction. Then, h_m extends to the map $h_{m+1} : |\widetilde{K}_{m+1}| \to |K(\mathbb{Z},n)|$ defined by $h_{m+1}||K_\sigma| = h''_\sigma$ for each $(m+1)$-simplex $\sigma \in K \setminus L$. The desired extension $h : |EW(K,n)| \to |K(\mathbb{Z},n)|$ can be defined by $h||\widetilde{K}_m| = h_m$ for each $m \geqslant n$. □

[14] Since $\mathrm{Sd}\,L$ is a full subcomplex of $\mathrm{Sd}\,K_\sigma$, such a simplicial map f can be defined by $f((\mathrm{Sd}\,L)^{(0)}) = \{0\}$ and $f((\mathrm{Sd}\,K_\sigma)^{(0)} \setminus (\mathrm{Sd}\,L)^{(0)}) = \{1\}$.

7.9 Cohomological Dimension

In this section, using Eilenberg–MacLane complexes defined in the previous section, we define the cohomological dimension and discuss its relationship with the (covering) dimension. We apply the cohomological dimension to prove that $\dim X \times P = \dim X + \dim P$ for every metrizable space X and every locally compact (or metric) polyhedron P (Theorem 7.9.8). This result was announced in Sect. 5.3.

In Theorem 5.2.4, the (covering) dimension is characterized as follows:

- For a normal space X, $\dim X \leqslant n$ if and only if, for each $k \geqslant n$ and each closed set A in X, every map $f : A \to \mathbf{S}^k$ extends over X.

Replacing $\mathbf{S}^k$ by $|K(\mathbb{Z}, k)|$, we define $\mathrm{cd}_{\mathbb{Z}} X \leqslant n$ if X satisfies the following condition $(\mathrm{e})_k$ for every $k \geqslant n$:[15]

$(\mathrm{e})_k$ Every map $f : A \to |K(\mathbb{Z}, k)|$ of each closed set A in X extends over X.

It is clear from this definition that $\mathrm{cd}_{\mathbb{Z}} X \leqslant n$ implies $\mathrm{cd}_{\mathbb{Z}} X \leqslant m$ for any $m > n$. Then, it can be defined that $\mathrm{cd}_{\mathbb{Z}} X = n$ if $\mathrm{cd}_{\mathbb{Z}} X \leqslant n$ and $\mathrm{cd}_{\mathbb{Z}} X \not\leqslant n$, where $\mathrm{cd}_Z X = -1$ if $X = \emptyset$. Moreover, $\mathrm{cd}_{\mathbb{Z}} X = \infty$ means that $\mathrm{cd}_{\mathbb{Z}} X \not\leqslant n$ for any $n \in \mathbb{N}$. We call $\mathrm{cd}_{\mathbb{Z}} X$ the **cohomological dimension** of X.[16] Recall we defined $K(\mathbb{Z}, 0) = \mathbb{Z}$. Thus, we have:

$(\mathrm{e})_0$ Every map $f : A \to \mathbb{Z}$ of each closed set A in X extends over X.

Remark 7.9.1 In the above definition of $\mathrm{cd}_{\mathbb{Z}} X$, $|K(\mathbb{Z}, k)|$ is not topologically unique. Then, it is necessary to verify that the condition $(\mathrm{e})_k$ is equivalent to the condition $(\mathrm{e})'_k$ obtained by replacing $|K(\mathbb{Z}, k)|$ with another simplicial complex K or more generally an ANE Ω (resp. an ANE Ω for compact spaces) of the same homotopy type as $|K(\mathbb{Z}, k)|$.[17] For a metrizable (resp. compact) space X, the equivalence $(\mathrm{e})_k \Leftrightarrow (\mathrm{e})'_k$ can be shown as follows: Let $\varphi : \Omega \to |K(\mathbb{Z}, k)|$ be a homotopy equivalence with a homotopy inverse ψ. Assume $(\mathrm{e})_k$ and let $f : A \to \Omega$ be a map of a closed set A in X. Then, $\varphi f : A \to |K(\mathbb{Z}, k)|$ extends to a map $h : X \to |K(\mathbb{Z}, k)|$. Since Ω is an ANE (resp. ANE for compact spaces), $f \simeq \psi\varphi f$, and $\psi\varphi f : A \to \Omega$ extends to $\psi h : X \to \Omega$, we can apply the Homotopy Extension Theorem 6.4.1 to extend f over X. Thus, the implication $(\mathrm{e})_k \Rightarrow (\mathrm{e})'_k$ is true. The converse implication $(\mathrm{e})'_k \Rightarrow (\mathrm{e})_k$ also true by the same argument. Thus, the above definition of $\mathrm{cd}_{\mathbb{Z}} X$ is valid for a metrizable (resp. compact) space X.

- *In the definition of* $\mathrm{cd}_{\mathbb{Z}} X$, *it is necessary to assume that* X *is metrizable (or compact).*

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[15] In Chigogidze's book, *Inverse Spectra*, the notation $\dim_{\mathbb{Z}} X$ is used.

[16] The cohomological dimension $\mathrm{cd}_{\mathbb{Z}} X$ was originally defined as the greatest number $n \in \omega$ such that $\check{H}^n(X, A) \neq 0$ for some closed set A in X, where $\check{H}^n(X, A)$ is the n-th relative Čech cohomology group. This is why $\mathrm{cd}_{\mathbb{Z}}$ is called the cohomological dimension. Refer to Sect. 8.4.

[17] For each simplicial complex K, $|K|$ is an ANE (for metrizable spaces) and an ANE for compact spaces by Theorem 6.1.4 (resp. Corollary 6.1.12).

Remark 7.9.2 For an Abelian group G, using Eilenberg–MacLane complex $K(G, n)$ instead of $K(\mathbb{Z}, n)$, we can also define the cohomological dimension $\mathrm{cd}_G X$ with respect to G.[18]

Due to Theorem 5.2.4, to assert $\dim X \leqslant n$, it suffices to examine the extensibility for maps from closed sets in X to the n-dimensional sphere $\mathbf{S}^n$, and it is not necessary to examine for $\mathbf{S}^k$ for $k > n$. For the cohomological dimension, the same situation holds, which is stated as follows:

Theorem 7.9.1 *For a metrizable space X, $\mathrm{cd}_{\mathbb{Z}} X \leqslant n$ if and only if X satisfies the condition $(\mathrm{e})_n$, i.e., every map $f : A \to |K(\mathbb{Z}, n)|$ of each closed set A in X extends over X.*

Proof The "only if" part of Theorem 7.9.1 is obvious. The "if" part can be obtained by induction if the implication $(\mathrm{e})_k \Rightarrow (\mathrm{e})_{k+1}$ holds. Since every constant map of $A \subset X$ extends over X, if every map $f : A \to |K(\mathbb{Z}, k+1)|$ of each closed set A in X is null-homotopic (i.e., homotopic to a constant map), then f extends over X by the Homotopy Extension Theorem 6.4.1. Thus, the "if" part can be deduced from the next lemma. □

Lemma 7.9.2 *For a metrizable space X, $(\mathrm{e})_k$ implies that every map $f : A \to |K(\mathbb{Z}, k+1)|$ of each closed set A in X is null-homotopic.*

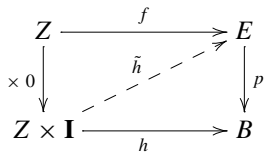
To prove this lemma, we use a path space. The path space $P(X, x_0)$ on a pointed space (X, x_0) is defined as follows:

$$P(X, x_0) = \mathrm{C}((\mathbf{I}, 0), (X, x_0)),$$

and it admits the compact-open topology. Let $e_1 : P(X, x_0) \to X$ be the map defined by $e_1(\alpha) = \alpha(1)$ (cf. Proposition 2.10.8 (3)). Then, $e_1^{-1}(x_0)$ is simply the loop space $\Omega(X, x_0)$ (cf. Sect. 8.1).

A map $p : E \to B$ is called a **Hurewicz fibration** if it has the following **homotopy lifting property** for an arbitrary space Z:

(HL) Given a homotopy $h : Z \times \mathbf{I} \to B$ and a map $f : Z \to E$ with $pf = h_0$, there exists a homotopy $\tilde{h} : Z \times \mathbf{I} \to E$ such that $p\tilde{h} = h$ and $\tilde{h}_0 = f$.



[18] Although $K(G, n)$ might be uncountable, it is countable if G is countable.

Proposition 7.9.3 *For every pointed space (X, x_0), the following statements hold:*

(1) *The path space $P(X, x_0)$ is contractible;*
(2) *The map $e_1 : P(X, x_0) \to X$ is a Hurewicz fibration whose image is the path-component of X containing x_0;*
(3) *If $x \in X$ belongs to the same path-component as x_0 then $e_1^{-1}(x) \simeq e_1^{-1}(x_0) = \Omega(X, x_0)$;*
(4) *If A is a subset of the path-component of X containing x_0 and there is a contraction $h : A \times \mathbf{I} \to A$ such that $h_1(A) = \{a\}$ and $h_t(a) = a$ for every $t \in \mathbf{I}$, then $e_1^{-1}(A) \simeq \Omega(X, x_0)$.*

Proof (1): We have the contraction $h : P(X, x_0) \times \mathbf{I} \to P(X, x_0)$ defined by $h(\alpha, t)(s) = \alpha((1-t)s)$ for each $(\alpha, t) \in P(X, x_0) \times \mathbf{I}$ and $s \in \mathbf{I}$ (cf. Propositions 2.10.3 and 2.10.8 (4)).

(2): It is trivial that the image of e_1 is the path-component of X containing x_0. To prove that $e_1 : P(X, x_0) \to X$ is a Hurewicz fibration, let $h : Z \times \mathbf{I} \to X$ be a homotopy and $f : Z \to P(X, x_0)$ a map with $e_1 f = h_0$, that is, $h_0(z) = e_1 f(z) = f(z)(1)$ for every $z \in Z$. We can define a map $h' : Z \times \mathbf{I} \times \mathbf{I} \to X$ as follows:

$$h'(z, t, s) = \begin{cases} f(z)((t+1)s) & \text{if } 0 \leqslant s \leqslant (t+1)^{-1}, \\ h(z, (t+1)s - 1) & \text{if } (t+1)^{-1} \leqslant s \leqslant 1. \end{cases}$$

Then, h' induces the map $\tilde{h} : Z \times \mathbf{I} \to \mathrm{C}(\mathbf{I}, X)$ defined by $\tilde{h}(z, t)(s) = h'(z, t, s)$. Observe that

$$\tilde{h}(z, t)(0) = f(z)(0) = x_0,\ \tilde{h}(z, 0)(s) = f(z)(s) \ \text{ and}$$
$$e_1\tilde{h}(z, t) = \tilde{h}(z, t)(1) = h(z, t).$$

Thus, we have a homotopy $\tilde{h} : Z \times \mathbf{I} \to P(X, x_0)$ such that $\tilde{h}_0 = f$ and $e_1\tilde{h} = h$.

(3): Take any $\omega \in e_1^{-1}(x)$ and define maps $f : e_1^{-1}(x) \to e_1^{-1}(x_0)$ and $g : e_1^{-1}(x_0) \to e_1^{-1}(x)$ by $f(\alpha) = \alpha * \omega^{\leftarrow}$ and $g(\beta) = \beta * \omega$. It is easy to see that $gf \simeq \mathrm{id}$ and $fg \simeq \mathrm{id}$. Refer to Sect. 8.1.

(4): Due to (3), it suffices to show that $e_1^{-1}(A) \simeq e_1^{-1}(a)$. By virtue of (2), we can obtain a homotopy $\tilde{h} : e_1^{-1}(A) \times \mathbf{I} \to e_1^{-1}(A)$ such that $\tilde{h}_0 = \mathrm{id}$ and $e_1\tilde{h} = h(e_1 \times \mathrm{id})$. Therefore,

$$\tilde{h}|e_1^{-1}(a) \times \mathbf{I} : e_1^{-1}(a) \times \mathbf{I} \to e_1^{-1}(a)$$

is a homotopy from id to $\tilde{h}_1|e_1^{-1}(a)$. Hence, the inclusion $e_1^{-1}(a) \subset e_1^{-1}(A)$ is a homotopy equivalence. □

The following can be proved by analogy to Proposition 6.1.10 (9):

Lemma 7.9.4 *Let X_1, X_2 be disjoint closed sets in a compactum X, and let Y be an ANE with $Y_1, Y_2 \subset Y$ that are also ANEs. Then, the space $\mathrm{C}((X, X_1, X_2), (Y, Y_1, Y_2))$ with the compact-open topology is an ANE.* □

Remark 7.9.3 When $X_1 \cap X_2 \neq \emptyset$, the above lemma holds if $Y_1 \cap Y_2$ is an ANE. The proof is left to the reader as a good exercise, where it is not necessary to assume that Y_1 and Y_2 are closed in Y (cf. Proposition 6.1.10 (7)).

It should be noted that if $X = X_1 = X_2$, then $\mathrm{C}((X, X_1, X_2), (Y, Y_1, Y_2)) \approx \mathrm{C}(X, Y_1 \cap Y_2)$.

Now, we will show Lemma 7.9.2.

Proof (*Lemma* 7.9.2) Let $f : A \to |K(\mathbb{Z}, k+1)|$ be a map from a closed set A in a metrizable space X. To show that f is null-homotopic, consider the fibration

$$e_1 : E = P(|K(\mathbb{Z}, k+1)|, v_0) \to |K(\mathbb{Z}, k+1)|,$$

where $v_0 \in K(\mathbb{Z}, k+1)^{(0)}$. We will construct a map $\tilde{f} : A \to E$ such that $e_1 \tilde{f} \simeq f$. Then, $\tilde{f} \simeq 0$ because E is contractible by Propositions 7.9.3 (1). Thus, we have $f \simeq 0$.

Before construction of $\tilde{f}$, it should be noted that for each subcomplex L of $K(\mathbb{Z}, k+1)$, $e_1^{-1}(|L|)$ is an ANE by Lemma 7.9.4 because

$$e_1^{-1}(|L|) = \mathrm{C}((\mathbf{I}, \{0\}, \{1\}), (|K(\mathbb{Z}, k+1)|, \{v_0\}, |L|)).$$

In particular, $e_1^{-1}(\sigma) = e_1^{-1}(|F(\sigma)|)$ is an ANE for every $\sigma \in K(\mathbb{Z}, k+1)$. For each $i \in \omega$, we define

$$E_i = e_1^{-1}(|K(\mathbb{Z}, k+1)^{(i)}|) \subset E \quad \text{and} \quad A_i = f^{-1}(|K(\mathbb{Z}, k+1)^{(i)}|) \subset A.$$

Then, $E = \bigcup_{i \in \omega} E_i$ and $A = \bigcup_{i \in \omega} A_i$.

$$\begin{array}{ccccccc} & & e_1^{-1}(\sigma) & \subset & E_i & \subset & E = P(|K(\mathbb{Z}, k+1)|, v_0) \\ & {\scriptstyle f_\sigma} \nearrow & & {\scriptstyle f_i} \nearrow & & & \downarrow {\scriptstyle e_1} \\ f^{-1}(\sigma) & \subset & A_i & \subset & A & \xrightarrow[f]{} & |K(\mathbb{Z}, k+1)| \end{array}$$

We will inductively construct maps $f_i : A_i \to E_i$, $i \in \omega$, such that $f_i | A_{i-1} = f_{i-1}$ and $f_i(f^{-1}(\sigma)) \subset e_1^{-1}(\sigma)$ for every $\sigma \in K(\mathbb{Z}, k+1)^{(i)}$. Let $f_0 : A_0 \to E_0$ be a map such that $e_1 f_0 = f | A_0$. For instance, it can be defined so that $f_0(f^{-1}(v))$ is a singleton for each vertex $v \in K(\mathbb{Z}, k+1)^{(0)}$. Assume that f_{i-1} has been constructed. For each i-simplex $\sigma \in K(\mathbb{Z}, k+1)$, $f_{i-1}(f^{-1}(\partial\sigma)) \subset e_1^{-1}(\partial\sigma)$. Due to Proposition 7.9.3 (4) and Corollary 7.8.10,

$$e_1^{-1}(\sigma) \simeq \Omega(|K(\mathbb{Z}, k+1)|, v_0) \simeq |K(\mathbb{Z}, k)|,$$

where $e_1^{-1}(\sigma)$ is an ANE as noted in the above. By the same argument of Remark 7.9.1, applying the condition $(\mathrm{e})_k$ and the Homotopy Extension Theorem 6.4.1, we can

extend $f_{i-1}|f^{-1}(\partial\sigma)$ over X as a map into $e_1^{-1}(\sigma)$. By restricting this extension, we have a map $f_\sigma : f^{-1}(\sigma) \to e_1^{-1}(\sigma)$ such that $f_\sigma|f^{-1}(\partial\sigma) = f_{i-1}|f^{-1}(\partial\sigma)$. Then, the desired map $f_i : A_i \to E_i$ can be defined by $f_i|f^{-1}(\sigma) = f_\sigma$ for each i-simplex $\sigma \in K(\mathbb{Z}, k+1)$. Since A_i is metrizable, each $x \in A_i$ has a neighborhood U_x in A_i such that $f(U_x) \subset |L_x|$ for some finite subcomplexes $L_x \subset K(\mathbb{Z}, k+1)$ (cf. Proposition 4.2.11 (5)). Then, we have finitely many i-simplexes $\sigma_1, \ldots, \sigma_m \in L_x$ such that

$$U_x \setminus A_{i-1} \subset f^{-1}(\sigma_1) \cup \cdots \cup f^{-1}(\sigma_m).$$

Since each $f_i|U_x \cap f^{-1}(\sigma_j) = f_{\sigma_j}|U_x \cap f^{-1}(\sigma_j)$ is continuous, it follows that $f_i|U_x$ is continuous. Hence, f_i is continuous at each $x \in A_i$.

Now, we define $\tilde{f} : A \to E$ by $\tilde{f}|A_i = f_i$ for every $i \in \omega$. Because of the metrizability of A, each $x \in A$ has a neighborhood U_x in A such that $f(U_x) \subset |K(\mathbb{Z}, k+1)^{(j)}|$ for some $j \in \omega$ (cf. Proposition 4.2.11 (5)). In other words, $U_x \subset A_j$, which means $\tilde{f}|U_x = f_j|U_x$. Therefore, $\tilde{f}$ is continuous at each $x \in A$. Moreover, for each $x \in A$, $f(x)$ is contained in an i-simplex $\sigma \in K(\mathbb{Z}, k+1)$. Then,

$$\tilde{f}(x) = f_i(x) \in f_i(f^{-1}(\sigma)) \subset e_1^{-1}(\sigma),$$

i.e., $e_1\tilde{f}(x) \in \sigma$. Thus, $e_1\tilde{f}$ is contiguous to f, which implies $e_1\tilde{f} \simeq f$. The proof is completed. □

For a finite-dimensional metrizable space, we will show the cohomological dimension coincides with the (covering) dimension. Before proving this, we show the following:

Theorem 7.9.5 *For every metrizable space X, the following equivalences hold:*

$$\dim X \leqslant 1 \;\Leftrightarrow\; \mathrm{cd}_{\mathbb{Z}} X \leqslant 1\,;\; \dim X \leqslant 0 \;\Leftrightarrow\; \mathrm{cd}_{\mathbb{Z}} X \leqslant 0.$$

In conclusion, when $n \leqslant 1$, the equivalence $\dim X = n \Leftrightarrow \mathrm{cd}_{\mathbb{Z}} X = n$ holds.

Proof By virtue of Theorem 7.9.1, the above first equivalence is obvious because $|K(\mathbb{Z}, 1)| = \partial\Delta^1 \approx \mathbf{S}^1$. Because $\mathbf{S}^0 = \{-1, 1\}$ is a retract of $\mathbb{Z}$, the implication $\mathrm{cd}_{\mathbb{Z}} X \leqslant 0 \;\Rightarrow\; \dim X \leqslant 0$ is easy to see. All that remains is to show the implication $\dim X \leqslant 0 \;\Rightarrow\; \mathrm{cd}_{\mathbb{Z}} X \leqslant 0$.

Let $f : A \to \mathbb{Z} = |K(\mathbb{Z}, 0)|$ be a map of a closed set A of X. Then, $\{f^{-1}(n) \mid n \in \mathbb{Z}\}$ is a discrete clopen cover of A, which is a discrete collection of closed sets in X. Since X is collectionwise normal (Theorem 2.6.1), we have a discrete collection $\{U_n \mid n \in \mathbb{Z}\}$ of open sets in X such that $f^{-1}(n) \subset U_n$ for each $n \in \mathbb{Z}$.. Because $\dim X \leqslant 0$, the open cover $\{U_n \mid n \in \mathbb{Z}\} \cup \{X \setminus A\}$ of X has an open refinement $\mathcal{V}$ with $\mathrm{ord}\,\mathcal{V} \leqslant 1$ by Theorem 5.2.5, where $\mathcal{V}$ is a pairwise disjoint clopen cover of X. For each $n \in \mathbb{Z}$, let $V_n = \mathrm{st}(f^{-1}(n), \mathcal{V})$ $(\subset U_n)$. Moreover, let $V_* = \bigcup\{V \in \mathcal{V} \mid V \cap A = \emptyset\}$. Then, $\{V_n \mid n \in \mathbb{Z}\} \cup \{V_*\}$ is a pairwise disjoint clopen cover of X and $V_n \cap A = f^{-1}(n)$ for every $n \in \mathbb{Z}$. We can extend f to a map $\tilde{f} : X \to \mathbb{Z}$ defined by $\tilde{f}(x) = n$ if $x \in V_n$, $n \in \mathbb{Z}$, and $\tilde{f}(x) = 0$ if $x \in V_*$. □

Now, we show the following:

Theorem 7.9.6 *For every metrizable space X, $\mathrm{cd}_{\mathbb{Z}} X \leqslant \dim X$. If X is finite-dimensional, then $\mathrm{cd}_{\mathbb{Z}} X = \dim X$.*

Proof The case $\dim X = \infty$ is trivial and the case $\dim X \leqslant 1$ has been proved in Theorem 7.9.5. Then, it suffices to show the implication $\dim X = n \Rightarrow \mathrm{cd}_{\mathbb{Z}} X = n$ when $2 \leqslant n < \infty$.

Let $f : A \to |K(\mathbb{Z}, n)|$ be a map from a closed set A in X. Since $\partial\Delta^{n+1}$ is a mapping $(n+1)$-deformation retract of $|K(\mathbb{Z}, n)|$ for metrizable spaces,[19] f is homotopic to a map $g : A \to \partial\Delta^{n+1}$. Because of $\dim X = n$, g extends to a map $\tilde{g} : X \to \partial\Delta^{n+1} \approx \mathbf{S}^n$. Since $|K(\mathbb{Z}, n)|$ is an ANE, f also extends over X by the Homotopy Extension Theorem 6.4.1. Hence, we have $\mathrm{cd}_{\mathbb{Z}} X \leqslant n$.

Now, assume that $\mathrm{cd}_{\mathbb{Z}} X \leqslant n-1$. Let A be a closed set in X. Then, every map $f : A \to \partial\Delta^n \approx \mathbf{S}^{n-1}$ extends to a map $\tilde{f} : X \to |K(\mathbb{Z}, n-1)|$. Since $\partial\Delta^n$ is a mapping n-deformation retract of $|K(\mathbb{Z}, n-1)|$ for metrizable spaces, we have a map $g : X \to \partial\Delta^n$ such that $g|A = \tilde{f}|A = f$, that is, g is an extension of f. This means that $\dim X \leqslant n-1$, which is a contradiction. Consequently, $\mathrm{cd}_{\mathbb{Z}} X = n$. □

Now, using the cohomological dimension, we can prove the following lemma:

Lemma 7.9.7 *For every non-empty metrizable space X, $\dim X \times \mathbf{I} = \dim X + 1$, where we mean $\infty + 1 = \infty$.*

Proof We may assume that $\dim X = n < \infty$. Since $\dim X \times \mathbf{I} \leqslant n+1$ by the Product Theorem 5.4.15, it suffices to show that $\dim X \times \mathbf{I} \geqslant n+1$. Since $X \times \mathbf{I}$ contains a copy of $\mathbf{I}$, we have the case $n = 0$. Consequently, we may assume that $n > 0$. Since $\dim X = \mathrm{cd}_{\mathbb{Z}} X = n$ by Theorem 7.9.6, there is a map $f : A \to |K(\mathbb{Z}, n-1)|$ of a closed set A in X that cannot extend over X. Take a point $v \in |K(\mathbb{Z}, n)|$. By Corollary 7.8.10, we have a homotopy equivalence $h : |K(\mathbb{Z}, n-1)| \to \Omega(|K(\mathbb{Z}, n)|, v)$ with its homotopy inverse $g : \Omega(|K(\mathbb{Z}, n)|, v) \to |K(\mathbb{Z}, n-1)|$. Then, $hf : A \to \Omega(|K(\mathbb{Z}, n)|, v)$ cannot extend over X. Indeed, if a map $hf : A \to \Omega(|K(\mathbb{Z}, n)|, v)$ extends to a map $\tilde{f} : X \to \Omega(|K(\mathbb{Z}, n)|, v)$, then $f \simeq ghf = gh\tilde{f}|A$. Since $|K(\mathbb{Z}, n-1)|$ in an ANE, we can apply the Homotopy Extension Theorem 6.4.1 to extend f over X, which is a contradiction.

Take $x_0 \in A$ so that $hf(x_0)$ is contained in the path-component of $\Omega(|K(\mathbb{Z}, n)|, v)$ containing c_v, where $\Omega(|K(\mathbb{Z}, n)|, v)$ is path-connected for $n > 1$ but it is not if $n = 1$. Since $\Omega(|K(\mathbb{Z}, n)|, v)$ is an ANE (Corollary 7.8.9), using the Homotopy Extension Theorem 6.4.1, we can easily obtain a map $f' : A \to \Omega(|K(\mathbb{Z}, n)|, v)$ such that $f'(x_0) = c_v$ and $hf \simeq f'$. Then, by the Homotopy Extension Theorem 6.4.1, f' also cannot be extended over X. The following map induced by the inclusion $i : A \subset X$ is not surjective:

$$i^* : \mathrm{C}((X, x_0), (\Omega(|K(\mathbb{Z}, n)|, v), c_v)) \to \mathrm{C}((A, x_0), (\Omega(|K(\mathbb{Z}, n)|, v), c_v)).$$

[19] As mentioned in Remark 7.8.2, $\partial\Delta^1$ is not a mapping 1-deformation retract of $|K(\mathbb{Z}, 0)|$.

According to Proposition 8.1.2, we have the following commutative diagram:

$$\begin{array}{ccc} \mathrm{C}((X\times\mathbf{I},H_X),(|K(\mathbb{Z},n)|,v)) & \xrightarrow{\varphi} & \mathrm{C}((X,x_0),(\Omega(|K(\mathbb{Z},n)|,v),c_v)) \\ \downarrow{\scriptstyle (i\times \mathrm{id}_{\mathbf{I}})^*} & & \downarrow{\scriptstyle i^*} \\ \mathrm{C}((A\times\mathbf{I},H_A),(|K(\mathbb{Z},n)|,v)) & \xrightarrow[\varphi]{} & \mathrm{C}((A,x_0),(\Omega(|K(\mathbb{Z},n)|,v),c_v)), \end{array}$$

where $H_X = X\times\partial\mathbf{I}\cup\{x_0\}\times\mathbf{I}$, $H_A = A\times\partial\mathbf{I}\cup\{x_0\}\times\mathbf{I}$. In the above diagram, since both φ are bijective and the right i^* is not surjective, the left $(i\times\mathrm{id}_{\mathbf{I}})^*$ is not surjective, that is, there exists a map $g : A\times\mathbf{I}\to|K(\mathbb{Z},n)|$ such that $g(H_A) = \{w\}$ and g cannot be extended to a map $\tilde{g} : X\times\mathbf{I}\to|K(\mathbb{Z},n)|$ with $g(H_X) = \{w\}$. Consider the closed set $H_{X,A} = X\times\partial\mathbf{I}\cup A\times\mathbf{I}$ in $X\times\mathbf{I}$ and the map $\bar{g} : H_{X,A}\to|K(\mathbb{Z},n)|$ defined by $\bar{g}|A\times\mathbf{I} = g$ and $\bar{g}(X\times\partial\mathbf{I}) = \{w\}$. Evidently, $\bar{g}$ cannot extend over $X\times\mathbf{I}$, which means that $\mathrm{cd}_{\mathbb{Z}}\, X\times\mathbf{I} > n$. Because $X\times\mathbf{I}$ is finite-dimensional, $\dim X\times\mathbf{I} = \mathrm{cd}_{\mathbb{Z}}\, X\times\mathbf{I}\geqslant n+1$ by Theorem 7.9.6. □

Theorem 7.9.8 *Let P be a locally compact polyhedron or $P = |K|_{\mathrm{m}}$ be the polyhedron of an arbitrary simplicial complex K with the metric topology. For every non-empty metrizable space X,*

$$\dim X\times P = \dim X + \dim P.$$

Proof Since $\dim X\times P\leqslant\dim X+\dim P$ by the Product Theorem 5.4.15, it suffices to show that $\dim X\times P\geqslant\dim X+\dim P$. If $\dim P=\infty$ then $\dim X\times P=\infty$ because P can be embedded into $X\times P$ as a closed set. Then, we may assume that $\dim P = n<\infty$. In this case, P contains an n-simplex σ, which is homeomorphic to $\mathbf{I}^n$. Inductively applying Lemma 7.9.7, we have

$$\dim X\times\sigma = \dim X\times\mathbf{I}^n = \dim X+n.$$

Since $X\times\sigma$ is a closed set in $X\times P$, it follows that $\dim X\times P\geqslant\dim X+n$. □

In the remainder of this section, we will prove the following theorem, which means that any cell-like map does not raise the cohomological dimension:

Theorem 7.9.9 *Let $f : X\to Y$ be a cell-like map between compacta. Then,* $\mathrm{cd}_{\mathbb{Z}}\, Y\leqslant\mathrm{cd}_{\mathbb{Z}}\, X$.

This is the direct consequence of the following theorem:

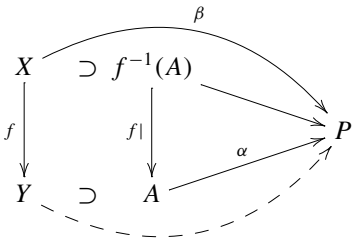
Theorem 7.9.10 *Let $f : X\to Y$ be a UV^n map between compacta. Then,* $\mathrm{cd}_{\mathbb{Z}}\, X\leqslant n$ *implies* $\mathrm{cd}_{\mathbb{Z}}\, Y\leqslant n$.

To prove this theorem, we need the following Vietoris–Begle-type mapping theorem:

Theorem 7.9.11 *Let $f : X \to Y$ be a UV^{n-1} map between compacta, $n \in \mathbb{N}$, and P a connected ANR (or a connected polyhedron) with $\pi_i(P) = \{0\}$ for $i \geqslant n$. Then, $f^* : [Y, P] \to [X, P]$ is a bijection.*

Note that it suffices to prove only the ANR case. To prove Theorem 7.9.11, we will prove the following lemma:

Lemma 7.9.12 *Let $f : X \to Y$ be a UV^{n-1} map between compacta, A a closed set in Y, and P a connected ANR with $\pi_i(P) = \{0\}$ for $i \geqslant n$. Then, a map $\alpha : A \to P$ extends over Y if $\alpha f | f^{-1}(A)$ extends to a map $\beta : X \to P$.*



Proof Let $\mathrm{pr}_i : \boldsymbol{Q} = \mathbf{I}^{\mathbb{N}} \to \mathbf{I}$, $i \in \mathbb{N}$, be the projections of the Hilbert cube $\boldsymbol{Q}$ onto the i-th factor, i.e., $\mathrm{pr}_i(x) = x(i)$ for each $x \in \boldsymbol{Q}$. Embed X and Y as closed sets in $\boldsymbol{Q}$ so that $Y \subset \mathrm{pr}_1^{-1}(1)$, and extend f to a map $\bar{f} : \boldsymbol{Q} \to \boldsymbol{Q}$. Take a map $k : \boldsymbol{Q} \to \mathbf{I}$ with $k^{-1}(1) = X$ and define a map $\tilde{f} : \boldsymbol{Q} \to \boldsymbol{Q}$ as follows:

$$\tilde{f}(x) = (k(x)\mathrm{pr}_1 \bar{f}(x), \mathrm{pr}_2 \bar{f}(x), \mathrm{pr}_3 \bar{f}(x), \dots) \text{ for each } x \in \boldsymbol{Q}.$$

Then, $\tilde{f}|X = f$ and $\tilde{f}^{-1}(Y) = X$. The maps α and β extend to maps $\tilde{\alpha} : R \to P$ and $\tilde{\beta} : M \to P$, where M and R are open neighborhoods of X and A in $\boldsymbol{Q}$, respectively. Because $\tilde{f}$ is closed and $\tilde{f}^{-1}(Y) = X \subset M$, we have an open neighborhood N of Y in $\boldsymbol{Q}$ such that $\tilde{f}^{-1}(N) \subset M$.

Take open covers of P as follows:

$$\{P\} \underset{h}{\succ} \mathcal{W}_0 \overset{*}{\succ} \mathcal{W}_1 \underset{L}{\succ} \mathcal{W}_2 \overset{*}{\succ} \mathcal{W}_3.$$

Replacing R by a smaller one, we can assume that $\tilde{f}^{-1}(R) \subset M$ and that $\tilde{\beta}|\tilde{f}^{-1}(R)$ and $\tilde{\alpha}\tilde{f}|\tilde{f}^{-1}(R)$ are $\mathcal{W}_3$-close because $\tilde{f}$ is closed and

$$\tilde{\beta}|\tilde{f}^{-1}(A) = \beta|f^{-1}(A) = \alpha f|f^{-1}(A) = \tilde{\alpha}\tilde{f}|\tilde{f}^{-1}(A).$$

Choose an open cover $\mathcal{V}_0$ of Y in $\boldsymbol{Q}$ so that

$$\mathrm{st}(A, \mathcal{V}_0) \subset R \text{ and } \mathcal{V}_0[A] \prec \tilde{\alpha}^{-1}(\mathcal{W}_3).$$

Let $\mathcal{U}_0$ be an open cover of Y in N with $\mathrm{st}\,\mathcal{U}_0 \prec \mathcal{V}_0$. We can inductively choose open covers $\mathcal{U}_i, \mathcal{V}_i$ of Y in N, $i = 1, \dots, n$, as follows:

$$\tilde{f}^{-1}(\mathcal{V}_i) \underset{C^{n-1}}{\prec} \tilde{f}^{-1}(\mathcal{U}_{i-1}) \text{ and } \operatorname{st}\mathcal{U}_i \prec \mathcal{V}_i \prec \mathcal{U}_{i-1}.$$

Indeed, assume that $\mathcal{V}_{i-1}$ has been obtained and take an open star-refinement $\mathcal{U}_{i-1}$ of $\mathcal{V}_{i-1}$. Each $y \in Y$ is contained in some $U_y \in \mathcal{U}_{i-1}$. Then, $f^{-1}(y) = \tilde{f}^{-1}(y) \subset \tilde{f}^{-1}(U_y)$. Since $f^{-1}(y)$ is UV^{n-1} in $\boldsymbol{Q}$ (Proposition 7.1.3), $f^{-1}(y)$ has an open neighborhood V'_y in M such that any map from $\mathbf{S}^j$ to V'_y extends to a map from $\mathbf{B}^{j+1}$ to $\tilde{f}^{-1}(U_y)$ for each $j \leqslant n-1$. Because $\tilde{f}$ is closed, y has an open neighborhood V_y in N such that $V_y \subset U_y$ and $\tilde{f}^{-1}(V_y) \subset V'_y$. Then, $\mathcal{V}_i = \{V_y \mid y \in Y\}$ is an open cover of Y in N that has the desired property.

Now, consider the nerve $K = N(\mathcal{U})$ of the following open cover of Y:

$$\mathcal{U} = \mathcal{U}_n|Y = \left\{U \cap Y \;\middle|\; U \in \mathcal{U}_n[Y]\right\}$$

and let $\varphi : Y \to |K|$ be a canonical map. For each vertex $U \in K^{(0)} = \mathcal{U}$, choose a point $\psi_0(U) \in f^{-1}(U)$, where $\psi_0(U) \in f^{-1}(U \cap A)$ for $U \in \mathcal{U}[A]$. Thus, we have a map $\psi_0 : |K^{(0)}| \to X \subset M$. Since $\operatorname{st}\mathcal{U} \prec \mathcal{V}_n$, it follows that ψ_0 is a partial $\tilde{f}^{-1}(\mathcal{V}_n)$-realization of K. Assume that ψ_0 extends to a partial $\tilde{f}^{-1}(\mathcal{V}_{n-i+1})$-realization $\psi_{i-1} : |K^{(i-1)}| \to M$ of K. For each i-simplex $\sigma \in K$, there is some $V_\sigma \in \mathcal{V}_{n-i+1}$ such that $\psi_{i-1}(\partial\sigma) \subset \tilde{f}^{-1}(V_\sigma)$. Then, we have some $U_\sigma \in \mathcal{U}_{n-i}$ such that $\psi_{i-1}|\partial\sigma$ extends to a map $\psi_\sigma : \sigma \to \tilde{f}^{-1}(U_\sigma)$. Now, we can extend ψ_{i-1} to a map $\psi_i : |K^{(i)}| \to M$ by $\psi_i|\sigma = \psi_\sigma$ for each i-simplex $\sigma \in K$. In the following, for simplicity, we denote $\sigma^{(j)} = |F(\sigma)^{(j)}|$ $(= \sigma \cap |K^{(j)}|)$ for each $\sigma \in K$ and $j \in \omega$. For every $\sigma \in K$, there is $V_\sigma \in \mathcal{V}_{n-i+1}$ such that $\psi_{i-1}(\sigma^{(i-1)}) \subset \tilde{f}^{-1}(V_\sigma)$ because ψ_{i-1} is a partial $\tilde{f}^{-1}(\mathcal{V}_{n-i+1})$-realization of K. Then, it follows that

$$\begin{aligned}\psi_i(\sigma^{(i)}) &\subset \operatorname{st}(\tilde{f}^{-1}(V_\sigma), \tilde{f}^{-1}(\mathcal{U}_{n-i}))\\ &\in \operatorname{st}(\tilde{f}^{-1}(\mathcal{V}_{n-i+1}), \tilde{f}^{-1}(\mathcal{U}_{n-i}))\\ &\prec \operatorname{st}\tilde{f}^{-1}(\mathcal{U}_{n-i}) \prec \tilde{f}^{-1}(\mathcal{V}_{n-i}).\end{aligned}$$

Thus, ψ_i is a partial $\tilde{f}^{-1}(\mathcal{V}_{n-i})$-realization of K. By induction, we can obtain a partial $\tilde{f}^{-1}(\mathcal{V}_0)$-realization $\psi_n : |K^{(n)}| \to M$.

The nerve $L = N(\mathcal{U}[A])$ of $\mathcal{U}[A]$ is a subcomplex of K. For each $\sigma \in L$, we have $V \in \mathcal{V}_0$ such that $\psi_n(\sigma^{(n)}) \subset \tilde{f}^{-1}(V)$, where $V \cap A \neq \emptyset$ because $\psi_n(\sigma^{(0)}) \subset \tilde{f}^{-1}(V \cap A)$. Then, $\tilde{\alpha}(V)$ is contained in some $W \in \mathcal{W}_3$, hence

$$\tilde{\alpha}\tilde{f}\psi_n(\sigma^{(n)}) \subset \tilde{\alpha}(V) \subset W.$$

Since $\tilde{\beta}|\tilde{f}^{-1}(R)$ and $\tilde{\alpha}\tilde{f}|\tilde{f}^{-1}(R)$ are $\mathcal{W}_3$-close and $V \subset R$, it follows that

$$\tilde{\beta}\psi_n(\sigma^{(n)}) \subset \operatorname{st}(W, \mathcal{W}_3) \in \operatorname{st}\mathcal{W}_3 \prec \mathcal{W}_2.$$

Therefore, $\tilde{\beta}\psi_n||L^{(n)}|$ is a partial $\mathcal{W}_2$-realization of L, which extends to a full $\mathcal{W}_1$-realization $\gamma : |L| \to P$.

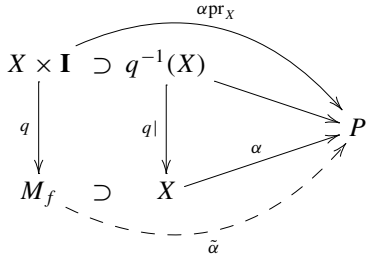
For each $a \in A$, let $\sigma \in L$ be the carrier of $\varphi(a)$ (i.e., $\sigma = c_L(\varphi(a))$), where $\gamma(\sigma)$ is contained in some $W \in \mathcal{W}_1$. Note that $\mathcal{U}[A] \prec \tilde{\alpha}^{-1}(\mathcal{W}_3)$ because $\mathcal{V}_0[A] \prec \tilde{\alpha}^{-1}(\mathcal{W}_3)$ and $\mathcal{U} \prec \mathcal{U}_n \prec \mathcal{V}_0$. Since $\sigma^{(0)} \subset \mathcal{U}[a]$, it follows that

$$\begin{aligned}\gamma(\sigma^{(0)}) &= \tilde{\beta}\psi_n(\sigma^{(0)}) = \beta\psi_0(\sigma^{(0)}) \\ &\subset \beta f^{-1}(\mathrm{st}(a, \mathcal{U}) \cap A) \subset \alpha(\mathrm{st}(a, \mathcal{U}) \cap A) \\ &\subset \mathrm{st}(\alpha(a), \tilde{\alpha}(\mathcal{U}[A])) \subset \mathrm{st}(\alpha(a), \mathcal{W}_3),\end{aligned}$$

hence $W \cap \mathrm{st}(\alpha(a), \mathcal{W}_3) \neq \emptyset$. Then, α and $\gamma\varphi|A$ are $\mathcal{W}_0$-close, which implies $\alpha \simeq \gamma\varphi|A$.

On the other hand, we can define a map $\gamma_n : |K^{(n)}| \cup |L| \to P$ by $\gamma_n||K^{(n)}| = \tilde{\beta}\psi_n$ and $\gamma_n||L| = \gamma$. Since $\pi_i(P) = \{0\}$ for every $i \geqslant n$, by skeleton-wise induction, we can construct maps $\gamma_i : |K^{(i)}| \cup |L| \to P$, $i \geqslant n$, such that $\gamma_{i+1}||K^{(i)}| \cup |L| = \gamma_i$. Then, we have a map $\tilde{\gamma} : |K| \to P$ such that $\tilde{\gamma}||K^{(i)}| \cup |L| = \gamma_i$ for each $i \geqslant n$. Notice the map $\tilde{\gamma}\varphi : Y \to P$ satisfies $\tilde{\gamma}\varphi|A = \gamma\varphi|A \simeq \alpha$. By the Homotopy Extension Theorem 6.4.1, α can be extended over Y. □

Proof (*Theorem* 7.9.11) First, we show that f^* is surjective. Consider the mapping cylinder M_f, where we identify $X = X \times \{1\} \subset M_f$. Let $q : X \times \mathbf{I} \to M_f$ be the restriction of the quotient map. Then, q is a UV^{n-1} map and $q^{-1}(X) = X \times \{1\} \subset X \times \mathbf{I}$. For every map $\alpha : X \to P$, the map $\alpha\mathrm{pr}_X : X \times \mathbf{I} \to P$ is an extension of $\alpha q|q^{-1}(X)$.



By Lemma 7.9.12, α extends to a map $\tilde{\alpha} : M_f \to P$. Since the collapsing $c_f : M_f \to Y$ is a strong deformation retraction, it follows that

$$(\tilde{\alpha}|Y)f = (\tilde{\alpha}|Y)c_f|X \simeq \tilde{\alpha}|X = \alpha.$$

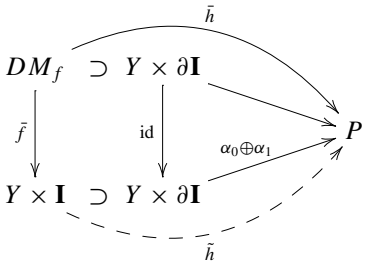
Thus, we have $f^*[\tilde{\alpha}|Y] = [\alpha]$.

Next, we show that f^* is injective. The double mapping cylinder DM_f is defined as the adjunction space

$$DM_f = (Y \times \partial\mathbf{I}) \cup_{f \times \mathrm{id}_{\partial\mathbf{I}}} (X \times \mathbf{I}).$$

The map $f \times \mathrm{id}_{\mathbf{I}} : X \times \mathbf{I} \to Y \times \mathbf{I}$ induces the map $\bar{f} : DM_f \to Y \times \mathbf{I}$, which is a UV^{n-1} map and $\bar{f}|Y \times \partial\mathbf{I} = \mathrm{id}$. Let $\alpha_0, \alpha_1 : Y \to P$ be maps with $f^*[\alpha_0] = f^*[\alpha_1]$,

i.e., $\alpha_0 f \simeq \alpha_1 f$. Then, a homotopy $h : X \times \mathbf{I} \to P$ from $\alpha_0 f$ to $\alpha_1 f$ induces the map $\bar{h} : DM_f \to P$.



Applying Lemma 7.9.12, we have a map $\tilde{h} : Y \times \mathbf{I} \to P$ such that $\tilde{h}|Y \times \partial\mathbf{I} = \bar{h}|Y \times \partial\mathbf{I}$, which means that $\tilde{h}$ is a homotopy from α_0 to α_1. Consequently, it follows that $[\alpha_0] = [\alpha_1]$. □

Now, we prove Theorem 7.9.10.

Proof (*Theorem* 7.9.10) When $n = 0$, since $f : X \to Y$ is a UV^0 map and $\mathrm{cd}_{\mathbb{Z}} X \leqslant 0$, for each $y \in Y$, $f^{-1}(y)$ is connected and $\dim f^{-1}(y) \leqslant \dim X \leqslant 0$, which means that $f^{-1}(y)$ is a singleton. Hence, f is a homeomorphism, which implies $\mathrm{cd}_{\mathbb{Z}} Y \leqslant 0$.

When $n > 0$, let $f : X \to Y$ be a UV^n map and $\mathrm{cd}_{\mathbb{Z}} X \leqslant n$. For each closed set A in Y, we define $f_A = f|f^{-1}(A) : f^{-1}(A) \to A$. Then, f_A is also a UV^n map. Let $i : A \subset Y$ and $j : f^{-1}(A) \subset X$ be the inclusions. We have the following commutative diagram:

$$\begin{array}{ccc} [Y, |K(\mathbb{Z}, n)|] & \xrightarrow{i^*} & [A, |K(\mathbb{Z}, n)|] \\ \downarrow{\scriptstyle f^*} & & \downarrow{\scriptstyle f_A^*} \\ [X, |K(\mathbb{Z}, n)|] & \xrightarrow[j^*]{} & [f^{-1}(A), |K(\mathbb{Z}, n)|]. \end{array}$$

According to Theorem 7.9.11, both vertical f^* and f_A^* are bijective. Because $\mathrm{cd}_{\mathbb{Z}} X \leqslant n$, j^* is surjective, consequently so is i^*. This means that $\mathrm{cd}_{\mathbb{Z}} Y \leqslant n$. □

As a corollary of Theorem 7.9.11, we have the following:

Corollary 7.9.13 *Let $f : X \to Y$ be a cell-like map between compacta and P a connected ANR (or a connected polyhedron) with $\pi_i(P) = \{0\}$ except for finitely many $i \in \mathbb{N}$. Then, $f^* : [Y, P] \to [X, P]$ is a bijection.* □

Remark 7.9.4 As we saw in Sect. 7.7, a cell-like map $f : X \to Y$ is not a shape equivalence even if X or Y is a compact AR. Therefore, the homotopy condition on P is essential in Corollary 7.9.13.

7.10 Alexandroff's Problem and the CE Problem

In the previous section (Sect. 7.9), we have shown that $\mathrm{cd}_{\mathbb{Z}} X \leqslant \dim X$ for every metrizable space X and $\mathrm{cd}_{\mathbb{Z}} X = \dim X$ if X is finite-dimensional (Theorem 7.9.6). Then, it is natural to ask the following question:

- Does there exist an infinite-dimensional compactum with finite cohomological dimension $\mathrm{cd}_{\mathbb{Z}}$?

This problem is called **Alexandroff's Problem**. Recall that it has been shown in Sect. 7.9 that any cell-like map between compacta does not raise the cohomological dimension (Theorem 7.9.9). So, it is also natural to ask the following question:

- Does a cell-like map of compacta raise the dimension?

This is called the **CE Problem**. These two problems are equivalent. In fact, the following theorem holds:

Theorem 7.10.1 *For every compactum X, the following are equivalent:*

(a) $\mathrm{cd}_{\mathbb{Z}} X \leqslant n$;
(b) *There exists a cell-like map $f : Y \to X$ for some compactum Y with* $\dim Y \leqslant n$.

This will be proved in the first half of this section. In the second half, we will give an affirmative answer to Alexandroff's Problem, so the CE Problem is also positively answered.

To prove Theorem 7.10.1, we first establish a criterion to estimate the cohomological dimension of the inverse limit via the bonding maps. For each map $f : X \to Y$ from a compact space X to a metric space $Y = (Y, d)$ and $n, m \in \mathbb{N}$, we define $\alpha_n(f)$ and $\alpha_n^m(f)$ as follows:

$$\alpha_n(f) = \inf\left\{d(f, g) \,\middle|\, g \in \mathrm{C}(X, Y),\ \dim g(X) \leqslant n\right\};$$
$$\alpha_n^m(f) = \sup\left\{\alpha_n(f|A) \,\middle|\, A \in \mathrm{Cld}(X),\ \dim A \leqslant m\right\}.$$

Theorem 7.10.2 *Let $X = \varprojlim(X_i, f_i)$ be the inverse limit of an inverse sequence of compact metric polyhedra and $n \geqslant 2$. Then, the following conditions are equivalent:*

(a) $\mathrm{cd}_{\mathbb{Z}} X \leqslant n$;
(b) $\lim_{j\to\infty} \alpha_n^{n+1}(f_{i,j}) = 0$ *for each* $i \in \mathbb{N}$,

where $f_{i,j} = f_i \cdots f_{j-1} : X_j \to X_i$ for $i < j$.

Proof (b) $\Rightarrow$ (a): Let A be a closed set in X and $g : A \to |K(\mathbb{Z}, n)|$ be a map. For each $i \in \mathbb{N}$, let $A_i = p_i(A)$, where $p_i : X \to X_i$ is the inverse limit projection. Then, $A = \varprojlim(A_i, f_i|A_{i+1})$ and each $p_i|A$ is the inverse limit projection by Proposition 4.16.5 (2). By Lemma 7.1.7, we have some $i \in \mathbb{N}$ and a map $g_i : A_i \to |K(\mathbb{Z}, n)|$ such that $g_i p_i|A \simeq g$. Since $|K(\mathbb{Z}, n)|$ is an ANE, g_i extends to a map $\tilde{g}_i : U \to |K(\mathbb{Z}, n)|$ over a neighborhood U of A_i in X_i. Let $\varepsilon = \frac{1}{4}\operatorname{dist}(A_i, X_i \setminus U) > 0$. Because X_i is

a compact ANR, we can find $0 < \delta < \varepsilon$ such that every pair of δ-close maps to X_i is ε-homotopic in X_i. By (b), we can choose $j > i$ so that $\alpha_n^{n+1}(f_{i,j}) < \delta$. Let K be a triangulation of X_j with mesh $f_{i,j}(K) < \varepsilon$ (Corollary 4.8.7) and let L be the subcomplex of K with $|L| = \mathrm{st}(A_j, K)$. Then, we have a map $h' : |K^{(n+1)}| \to X_i$ such that $\dim h'(|K^{(n+1)}|) \leqslant n$ and $d(h', f_{i,j}||K^{(n+1)}|) < \delta$, hence $h' \simeq_\varepsilon f_{i,j}||K^{(n+1)}|$. By the Homotopy Extension Theorem 6.4.1, h' extends to a map $h : |K| = X_j \to X_i$ such that $h \simeq_\varepsilon f_{i,j}$. Then, mesh $h(K) < 3\varepsilon$, hence $h(|L|) \subset U$ and $h||L| \simeq f_{i,j}||L|$ in U. Thus, we have the map $\tilde{g}_i h||L| : |L| \to |K(\mathbb{Z}, n)|$ and $\tilde{g}_i h||L| \simeq \tilde{g}_i f_{i,j}||L|$.

Recall that $\partial\Delta^{n+1}$ is a mapping $(n+1)$-deformation retract of $|K(\mathbb{Z}, n)|$ for metrizable spaces (see Sect. 7.8). Since $\dim h(|L^{(n+1)}|) \leqslant n$ and $h(|L^{(n+1)}|) \subset U$, we have a map $g' : h(|L^{(n+1)}|) \to \partial\Delta^{n+1} \approx \mathbf{S}^n$ such that $g'|D = \tilde{g}_i|D$ and $g' \simeq \tilde{g}_i|h(|L^{(n+1)}|)$ rel. D, where

$$D = \tilde{g}_i^{-1}(\partial\Delta^{n+1}) \cap h(|L^{(n+1)}|).$$

Because $\dim h(|K^{(n+1)}|) \leqslant n$, the map g' extends to a map $\tilde{g}' : h(|K^{(n+1)}|) \to \partial\Delta^{n+1} \approx \mathbf{S}^n$. Then, $\tilde{g}'h||L^{(n+1)}| = g'h||L^{(n+1)}| \simeq \tilde{g}_i h||L^{(n+1)}|$. By the Homotopy Extension Theorem 6.4.1, we can obtain a map $g'' : |L| \to |K(\mathbb{Z}, n)|$ such that $g''||L^{(n+1)}| = \tilde{g}'h||L^{(n+1)}|$ and $g'' \simeq \tilde{g}_i h||L|$. Thus, we have a map $g''' : |K^{(n+1)}| \cup |L| \to |K(\mathbb{Z}, n)|$ defined by $g'''||K^{(n+1)}| = \tilde{g}'h||K^{(n+1)}|$ and $g'''||L| = g''$. On the other hand, for every $m > n$, every map from $\mathbf{S}^m$ to $|K(\mathbb{Z}, n)|$ extends over $\mathbf{B}^{m+1}$ because $\pi_m(|K(\mathbb{Z}, n)|) = 0$. By skeleton-wise induction, we can extend g''' to a map $g^* : |K| = X_j \to |K(\mathbb{Z}, n)|$. Then, it follows that

$$g^* p_j|A = g'' p_j|A \simeq \tilde{g}_i h p_j|A \simeq \tilde{g}_i f_{i,j} p_j|A = \tilde{g}_i p_i|A \simeq g,$$

hence g can be extended over X by the Homotopy Extension Theorem 6.4.1.

(a) $\Rightarrow$ (b): For each $j \in \mathbb{N}$ and $\varepsilon > 0$, we have to find $k_0 \geqslant j$ so that $\alpha_n^{n+1}(f_{j,k}) < \varepsilon$ for every $k \geqslant k_0$. We assume that $\dim X_j > n$ because the other case is obvious. Let K be a triangulation of X_j with mesh $K < \varepsilon/2$ (Corollary 4.8.7). Let $EW(K, n)$ be Edwards–Walsh complex for K (see Sect. 7.8). Recall that $K^{(n)} \subset EW(K, n) = \bigcup_{\sigma \in K} K_\sigma$, where K_σ, $\sigma \in K$, are simplicial complexes obtained by Proposition 7.8.14. Suppose that the map $p_j|p_j^{-1}(|K^{(n)}|) : p_j^{-1}(|K^{(n)}|) \to |EW(K, n)|$ extends to a map $g_m : p_j^{-1}(|K^{(m)}|) \to |EW(K, n)|$ for $m \geqslant n$ such that $g_m(p_j^{-1}(\sigma)) \subset |K_\sigma|$ for each $\sigma \in K^{(m)}$. For each $(m+1)$-simplex $\sigma \in K$, observe that

$$g_m(p_j^{-1}(\partial\sigma)) \subset \bigcup_{\sigma' < \sigma} |K_{\sigma'}| \subset |K_\sigma| \simeq |K(\mathbb{Z}, n)|^k \text{ for some } k \in \mathbb{N}.$$

Since $\mathrm{cd}_{\mathbb{Z}}\, p_j^{-1}(\sigma) \leqslant \mathrm{cd}_{\mathbb{Z}}\, X \leqslant n$, $g_m|p_j^{-1}(\partial\sigma)$ extends to a map $g_\sigma : p_j^{-1}(\sigma) \to |K_\sigma|$. Then, g_m extends to a map g_{m+1} defined by $g_{m+1}|p_j^{-1}(\sigma) = g_\sigma$ for every $(m+1)$-simplex $\sigma \in K$.

Now, we have obtained the map $g_{\dim K} : X \to |EW(K, n)|$, which extends to a map $g : U \to |EW(K, n)|$ over a neighborhood U of X in $\prod_{i \in \mathbb{N}} X_i$. Observe that

$$g(\mathrm{pr}_j^{-1}(\sigma) \cap X) = g(p_j^{-1}(\sigma)) \subset |K_\sigma| \text{ for each } \sigma \in K.$$

Then, X has a neighborhood $V \subset U$ in $\prod_{i\in\mathbb{N}} X_i$ such that

$$g(\mathrm{pr}_j^{-1}(\sigma) \cap V) \subset \left|EW(K,n)\right| \setminus \left|C(K_\sigma, EW(K,n))\right| \text{ for each } \sigma \in K.$$

We use the same notation X_i^* as in the proof of Lemma 7.1.7. Because $\prod_{i\in\mathbb{N}} X_i$ is compact, we can choose $k_0 > j$ so that

$$X_{k_0}^* = \left\{x \in \textstyle\prod_{i\in\mathbb{N}} X_i \;\middle|\; x(i) = f_{i,k}(x(k)) \text{ for } i < k\right\} \subset V.$$

Let $z \in \prod_{i\in\mathbb{N}} X_i$ be fixed and define an embedding $\varphi_i : X_i \to X_i^*$ by

$$\varphi_i(x) = (f_{1,i}(x), \dots, f_{i-1,i}(x), x, z(i+1), z(i+2), \dots).$$

Then, for every $k \geqslant k_0$, the following statement holds:

(*) If $x \in X_k$, $f_{j,k}(x) \in \sigma \in K$, and $g\varphi_k(x) \in |K_{\sigma'}|$, $\sigma' \in K$, then $\sigma \cap \sigma' \neq \emptyset$.

Indeed, since $f_{j,k} = \mathrm{pr}_j \varphi_k$, we have $\varphi_k(x) \in \mathrm{pr}_j^{-1}(\sigma) \cap V$, which implies

$$g(\varphi_k(x)) \in \left|EW(K,n)\right| \setminus \left|C(K_\sigma, EW(K,n))\right|.$$

It follows that $|K_\sigma| \cap |K_{\sigma'}| \neq \emptyset$, so $\sigma \cap \sigma' \neq \emptyset$ by Proposition 7.8.14 (iii).

Let A be a closed set in X_k with $\dim A \leqslant n+1$ and consider the restriction $f_{j,k}|A : A \to X_j = |K|$. Using Proposition 7.8.14 (vi), we can obtain a map $\psi : A \to |K^{(n)}|$ such that each $\psi(x)$ is contained in $\sigma \cap |K^{(n)}|$ for some $\sigma \in K$ with $g\varphi_k(x) \in |K_\sigma|$ and

$$\psi|(g\varphi_k)^{-1}(|K^{(n)}|) \cap A = g\varphi_k|(g\varphi_k)^{-1}(|K^{(n)}|) \cap A,$$

where the carrier of $f_{j,k}(x)$ in K meets σ by (*). Then, $d(f_{j,k}(x), \psi(x)) \leqslant 2\,\mathrm{mesh}\, K < \varepsilon$. This means that $\alpha_n^{n+1}(f_{j,k}) < \varepsilon$. □

Let $X = (X, d_X)$ and $Y = (Y, d_Y)$ be metric spaces. A map $f : X \to Y$ is said to be **non-expansive** if $d_Y(f(x), f(x')) \leqslant d_X(x, x')$ for every $x, x' \in X$. An inverse sequence $(X_i, f_i)_{i\in\mathbb{N}}$ of metric spaces $X_i = (X_i, d_i)$ is said to be **non-expansive** if every bonding map f_i is non-expansive. For every inverse sequence $(X_i, f_i)_{i\in\mathbb{N}}$ of metrizable spaces, we have $d_i \in \mathrm{Metr}(X_i)$, $i \in \mathbb{N}$, such that $(X_i, f_i)_{i\in\mathbb{N}}$ is non-expansive. Indeed, for each $i \in \mathbb{N}$, let $d_i' \in \mathrm{Metr}(X_i)$. The desired metric d_i for X_i can be defined as follows:

$$d_i(x, y) = \sum_{j\leqslant i} d_j'(f_{j,i}(x), f_{j,i}(y)),$$

where $f_{j,i} = f_j \cdots f_{i-1} : X_i \to X_j$ and $f_{i,i} = \mathrm{id}$.

Lemma 7.10.3 *Let X and Y be the inverse limits of inverse sequences $(X_i, f_i)_{i\in\mathbb{N}}$ and $(Y_i, g_i)_{i\in\mathbb{N}}$ with $p_i : X \to X_i$ and $q_i : Y \to Y_i$ the projections, respectively. Suppose that each Y_i has an admissible complete metric d_i such that $(Y_i, g_i)_{i\in\mathbb{N}}$ is non-expansive. Then, given maps $h_i : X_i \to Y_i$, $i \in \mathbb{N}$, such that $\sum_{i\in\mathbb{N}} d_i(h_i f_i, g_i h_{i+1}) < \infty$, there exists a map $h : X \to Y$ such that $d_i(h_i p_i, q_i h) \leqslant \sum_{j\geqslant i} d_j(h_j f_j, g_j h_{j+1})$.*

Proof Since $g_{i,j}$ is non-expansive for $i < j$, it follows that

$$\begin{aligned} d_i(g_{i,j}h_j p_j, g_{i,j+1}h_{j+1}p_{j+1}) &\leqslant d_j(h_j p_j, g_j h_{j+1} p_{j+1}) \\ &= d_j(h_j f_j p_{j+1}, g_j h_{j+1} p_{j+1}) \leqslant d_j(h_j f_j, g_j h_{j+1}). \end{aligned}$$

Then, for each $i < j < k$,

$$\begin{aligned} d_i(g_{i,j}h_j p_j, g_{i,k}h_k p_k) &\leqslant d_i(g_{i,j}h_j p_j, g_{i,j+1}h_{j+1}p_{j+1}) \\ &\quad + \cdots + d_i(g_{i,k-1}h_{k-1}p_{k-1}, g_{i,k}h_k p_k) \\ &\leqslant d_j(h_j f_j, g_j h_{j+1}) + \cdots + d_{k-1}(h_{k-1}f_{k-1}, g_{k-1}h_k). \end{aligned}$$

Now, for each $\varepsilon > 0$, choose $k \in \mathbb{N}$ so that $\sum_{j\geqslant k} d_j(h_j f_j, g_j h_{j+1}) < \varepsilon$. Then, for each $j' > j \geqslant k$,

$$\begin{aligned} d_i(g_{i,j}h_j p_j, g_{i,j'}h_{j'}p_{j'}) &\leqslant d_j(h_j f_j, g_j h_{j+1}) \\ &\quad + \cdots + d_{j'-1}(h_{j'-1}f_{j'-1}, g_{j'-1}h_{j'}) < \varepsilon. \end{aligned}$$

Therefore, $(g_{i,j}h_j p_j)_{j\geqslant i}$ is a Cauchy sequence in $\mathrm{C}(X, Y_i)$ with the sup-metric. Since $Y_i = (Y_i, d_i)$ is complete, so is $\mathrm{C}(X, Y_i)$, hence $(g_{i,j}h_j p_j)_{j\geqslant i}$ converges to a map $h'_i : X \to Y_i$. It should be noted that $g_{i-1}h'_i = h'_{i-1}$ for each $i \geqslant 2$. We can define a map $h : X \to Y$ by $h(x) = (h'_i(x))_{i\in\mathbb{N}}$. For each $i \in \mathbb{N}$ and $\varepsilon' > 0$, choose $j > i$ so that $d_i(g_{i,j}h_j p_j, h'_i) < \varepsilon'$. Then, it follows that

$$\begin{aligned} d_i(h_i p_i, q_i h) &= d_i(h_i p_i, h'_i) \\ &\leqslant d_i(h_i p_i, g_i h_{i+1} p_{i+1}) + d_i(g_{i,i+1}h_{i+1}p_{i+1}, g_{i,i+2}h_{i+2}p_{i+2}) \\ &\quad + \cdots + d_i(g_{i,j-1}h_{j-1}p_{j-1}, g_{i,j}h_j p_j) + d_i(g_{i,j}h_j p_j, h'_i) \\ &\leqslant d_i(h_i f_i, g_i h_{i+1}) + d_{i+1}(h_{i+1}f_{i+1}, g_{i+1}h_{i+2}) \\ &\quad + \cdots + d_{j-1}(h_{j-1}f_{j-1}, g_{j-1}h_j) + \varepsilon'. \end{aligned}$$

Therefore, $d_i(h_i p_i, q_i h) \leqslant \sum_{j\geqslant i} d_j(h_j f_j, g_j h_{j+1})$ for each $i \in \mathbb{N}$. This completes the proof. □

Now, we can prove Theorem 7.10.1.

Proof (*Theorem* 7.10.1) The implication (b) ⇒ (a) is obtained by combining Theorems 7.9.6 and 7.9.9.

(a) ⇒ (b): When $n \leqslant 1$, since $\dim X = \mathrm{cd}_{\mathbb{Z}} X$, we can take $Y = X$ and $f = \mathrm{id} : Y \to X$. For the case $n \geqslant 2$, by Corollary 4.17.8, we may assume that $X =$

$\varprojlim(X_i, f_i)$, where each X_i is a compact polyhedron. By $p_i : X \to X_i$, we denote the inverse limit projection. As observed in the above, each X_i has an admissible metric d_i such that every $f_i : X_{i+1} \to X_i$ is non-expansive.

By the uniformly local contractibility of compact metric polyhedra and Theorem 7.10.2, we can inductively take two sequences $1 = \varepsilon_0 > \varepsilon_1 > \varepsilon_2 > \cdots > 0$ and $1 = k(1) < k(2) < \cdots \in \mathbb{N}$ so that:

(1) $2\varepsilon_i < \varepsilon_{i-1}$ and the $2\varepsilon_i$-neighborhood $\mathrm{B}(x, 2\varepsilon_i)$ of each point $x \in X_{k(i)}$ is contractible in the ε_{i-1}-neighborhood $\mathrm{B}(x, \varepsilon_{i-1})$ $(\subset X_{k(i)})$;
(2) $\alpha_n^{n+1}(f_{k(i),k(i+1)}) < \varepsilon_i/3$.

Then, (1) implies that $\varepsilon_i < 2^{-(i-j)}\varepsilon_j$ for every $i > j$. Hence, $\sum_{i>j} \varepsilon_i < \varepsilon_j$.

Triangulate each $X_{k(i)}$ by a simplicial complex K_i with mesh $K_i < \varepsilon_i/3$. Since $\alpha_n(f_{k(i),k(i+1)}||K_{i+1}^{(n+1)}|) < \varepsilon_i/3$ by (2), we have a map $g_i' : |K_{i+1}^{(n+1)}| \to |K_i|$ such that

$$d_{k(i)}\big(g_i', f_{k(i),k(i+1)}\big|\big|K_{i+1}^{(n+1)}\big|\big) < \varepsilon_i/3 \quad \text{and} \quad \dim g_i'\big(\big|K_{i+1}^{(n+1)}\big|\big) \leqslant n.$$

Due to Theorem 5.2.13, g_i' is contiguous to a map $g_i'' : |K_{i+1}^{(n+1)}| \to |K_i^{(n)}|$. Then, $d_{k(i)}(g_i'', g_i') \leqslant \operatorname{mesh} K_i < \varepsilon_i/3$. For each $i \in \mathbb{N}$, let $Y_i = |K_i^{(n)}|$ and $g_i = g_i''|Y_{i+1} : Y_{i+1} \to Y_i$. Thus, we have an inverse sequence $(Y_i, g_i)_{i\in\mathbb{N}}$ with $Y = \varprojlim(Y_i, g_i) \neq \emptyset$ (cf. Proposition 4.16.2), where the inverse limit projection is denoted by $q_i : Y \to Y_i$. According to Theorem 5.4.2, we have $\dim Y \leqslant n$. By Proposition 4.17.3, we can regard $X = \varprojlim(X_{k(i)}, f_{k(i),k(i+1)})$. Because $d_{k(i)}(g_i, f_{k(i),k(i+1)}|Y_{i+1}) < 2\varepsilon_i/3$, we can apply Lemma 7.10.3 to the inclusions $h_i : Y_i \subset X_{k(i)}$, $i \in \mathbb{N}$, to obtain a map $h : Y \to X$ such that

$$d_{k(i)}(p_{k(i)}h, q_i) \leqslant \sum_{j\geqslant i} 2\varepsilon_j/3 < 2\varepsilon_{i-1}/3 \quad \text{for each } i \in \mathbb{N}.$$

We will now show that h is cell-like. For each point $x \in X$, we write $x_i = p_{k(i)}(x) \in X_{k(i)}$ for each $i \in \mathbb{N}$. Recall that $f_{k(i),k(i+1)}$ is non-expansive. For each $y \in Y_{i+1}$, we have

$$\begin{aligned} d_{k(i)}(x_i, g_i(y)) &\leqslant d_{k(i)}(f_{k(i),k(i+1)}(x_{i+1}), f_{k(i),k(i+1)}(y)) \\ &\quad + d_{k(i)}(f_{k(i),k(i+1)}(y), g_i(y)) \\ &< d_{k(i+1)}(x_{i+1}, y) + 2\varepsilon_i/3. \end{aligned}$$

Then, it follows that

$$g_i(\overline{\mathrm{B}}(x_{i+1}, \varepsilon_i) \cap Y_{i+1}) \subset \mathrm{B}(x_i, \varepsilon_i + 2\varepsilon_i/3) \cap Y_i \subset \overline{\mathrm{B}}(x_i, \varepsilon_{i-1}) \cap Y_i.$$

Since $Y_i = |K_i^{(n)}|$ and $\operatorname{mesh} K_i < \varepsilon_i/3$, we have $Q_i = \overline{\mathrm{B}}(x_i, \varepsilon_{i-1}) \cap Y_i \neq \emptyset$. Thus, we have an inverse sequence $(Q_i, g_i|Q_{i+1})_{i\in\mathbb{N}}$. The inverse limit $Q = \varprojlim(Q_i, g_i|Q_{i+1}) \neq \emptyset$ is a subspace of Y and $q_i|Q$, $i \in \mathbb{N}$, are the projections (cf. Proposition 4.16.2).

We will show that $h^{-1}(x) = Q$. For each $y \in Q$,

$$\begin{aligned} d_{k(i)}(p_{k(i)}h(y), x_i) &\leqslant d_{k(i)}(p_{k(i)}h(y), q_i(y)) + d_{k(i)}(q_i(y), x_i) \\ &< 2\varepsilon_{i-1}/3 + \varepsilon_{i-1} < 2\varepsilon_{i-1}. \end{aligned}$$

Then, $\lim_{i\to\infty} d_{k(i)}(p_{k(i)}h(y), x_i) = 0$. Since $f_{k(i),k(i+1)}$ is non-expansive, we have $h(y) = x$, that is, $y \in h^{-1}(x)$. Conversely, if $y \in Y \setminus Q$ then $q_i(y) \notin Q_i$ for some $i \in \mathbb{N}$, which means that $d_{k(i)}(q_i(y), x_i) > \varepsilon_{i-1}$. Since $d_{k(i)}(p_{k(i)}h(y), q_i(y)) < 2\varepsilon_{i-1}/3$, it follows that $p_{k(i)}h(y) \neq x_i = p_{k(i)}(x)$, hence $h(y) \neq x$, i.e., $y \in Y \setminus h^{-1}(x)$.

It remains to prove that Q is cell-like. By Remark 7.1.2 of Theorem 7.1.8, for each $i \in \mathbb{N}$, it suffices to find $j > i$ so that $g_{i,j}|Q_j \simeq 0$ in Q_i. Observe that

$$\begin{aligned} g_{i+1,i+3}(Q_{i+3}) &= g_{i+1}g_{i+2}(\overline{\mathrm{B}}(x_{i+3}, \varepsilon_{i+2}) \cap Y_{i+3}) \\ &\subset g_{i+1}(\overline{\mathrm{B}}(x_{i+2}, \varepsilon_{i+1}) \cap Y_{i+2}) \subset \mathrm{B}(x_{i+1}, 2\varepsilon_{i+1}) \cap Y_{i+1}, \end{aligned}$$

which implies that $g_{i+1,i+3}(Q_{i+3})$ is contractible in $\overline{\mathrm{B}}(x_{i+1}, \varepsilon_i)$ by (1). Since $\dim g_{i+1,i+3}(Q_{i+3}) \times \mathbf{I} \leqslant n+1$, pushing this contraction into $|K_{i+1}^{(n+1)}|$ by Theorem 5.2.13, we have a homotopy $h : g_{i+1,i+3}(Q_{i+3}) \times \mathbf{I} \to |K_{i+1}^{(n+1)}|$ such that h_0 is contiguous to id in $K_{i+1}^{(n)}$, $h_1(g_{i+1,i+3}(Q_{i+3}))$ is contained in a simplex $\sigma \in K_{i+1}^{(n+1)}$, and

$$h(g_{i+1,i+3}(Q_{i+3}) \times \mathbf{I}) \subset \mathrm{st}\left(\overline{\mathrm{B}}(x_{i+1}, \varepsilon_i), K_{i+1}^{(n+1)}\right).$$

Since $h_0 \simeq_{\varepsilon_{i+1}/3} \mathrm{id}$ and $h_1 \simeq 0$ in σ, we can conclude that $g_{i+1,i+3}(Q_{i+3})$ is contractible in $\overline{\mathrm{B}}(x_{i+1}, \varepsilon_i + 2\varepsilon_{i+1}/3) \cap |K_{i+1}^{(n+1)}|$. On the other hand, similar to g_i,

$$d_{k(i)}(x_i, g_i''(y)) < d_{k(i+1)}(x_{i+1}, y) + 2\varepsilon_i/3 \ \text{ for each } y \in |K_{i+1}^{(n+1)}|.$$

Then, it follows that

$$\begin{aligned} g_i''\left(\overline{\mathrm{B}}(x_{i+1}, \varepsilon_i + 2\varepsilon_{i+1}/3) \cap |K_{i+1}^{(n+1)}|\right) &\subset \overline{\mathrm{B}}(x_i, \varepsilon_i + 2\varepsilon_{i+1}/3 + 2\varepsilon_i/3) \cap Y_i \\ &\subset \overline{\mathrm{B}}(x_i, \varepsilon_{i-1}) \cap Y_i = Q_i. \end{aligned}$$

Thus, $g_{i,i+3}(Q_{i+3})$ is contractible in Q_i. This completes the proof. □

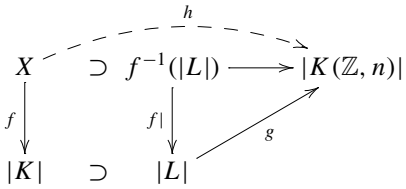
In the rest of this section, we will give an affirmative answer to Alexandroff's Problem, that is, we will prove the DRANISHNIKOV THEOREM:

Theorem 7.10.4 (DRANISHNIKOV) *There exists an infinite-dimensional compactum X with finite cohomological dimension.*

Remark 7.10.1 In Theorem 7.10.4, $\mathrm{cd}_{\mathbb{Z}} X = 3$ for the first example constructed by A.N. Dranishnikov, but J. Dydak and J.J. Walsh constructed an example of $\mathrm{cd}_{\mathbb{Z}} X = 2$. One should recall that $\mathrm{cd}_{\mathbb{Z}} X \leqslant 1$ if and only if $\dim X \leqslant 1$.

Given a metrizable space X and a simplicial complex K, we introduce the following extension property for a map $f : X \to |K|$:

$(\mathrm{EW})_n$ For each subcomplex L of K and each map $g : |L| \to |K(\mathbb{Z}, n)|$, the composition $gf|f^{-1}(|L|)$ extends to a map $h : X \to |K(\mathbb{Z}, n)|$.



Then, Proposition 7.8.15 can be reformulated as follows:

Proposition 7.10.5 (7.8.15) *For each simplicial complex K and $n \geqslant 2$, an associate map $\varpi : |EW(K, n)| \to |K|$ of Edwards–Walsh complex $EW(K, n)$ has property* $(\mathrm{EW})_n$.

Note For a map $f : X \to |K|$, we define $\mathrm{cd}_{\mathbb{Z}}(f, K) \leqslant n$ if f has the following property:

$(\mathrm{EW})'_n$ For each subcomplex L' of $K' \lhd K$ and each map $g : |L'| \to |K(\mathbb{Z}, n)|$, the composition $gf|f^{-1}(|L'|)$ extends to a map $h : X \to |K(\mathbb{Z}, n)|$.

The implication $(\mathrm{EW})'_n \Rightarrow (\mathrm{EW})_n$ is obvious. Moreover, $(\mathrm{EW})'_n \Rightarrow (\mathrm{EW})'_{n+1}$ can be proved. Thus, because $\mathrm{cd}_{\mathbb{Z}}(f, K) \leqslant n \Rightarrow \mathrm{cd}_{\mathbb{Z}}(f, K) \leqslant n + 1$ holds, it is possible to define $\mathrm{cd}_{\mathbb{Z}}(f, K) = n$ if $\mathrm{cd}_{\mathbb{Z}}(f, K) \leqslant n$ and $\mathrm{cd}_{\mathbb{Z}}(f, K) \not\leqslant n - 1$.

Sketch of Proof of $(\mathrm{EW})'_n \Rightarrow (\mathrm{EW})'_{n+1}$: Assume that $f : X \to |K|$ has $(\mathrm{EW})'_n$. To show that f has $(\mathrm{EW})'_{n+1}$, let L' be a subcomplex of $K' \lhd K$ and $g : |L'| \to |K(\mathbb{Z}, n + 1)|$ a map. In the same way as the proof of Theorem 7.9.1, it suffices to show that $gf|f^{-1}(|L'|) \simeq 0$. To this end, as the proof of Lemma 7.9.2, we may construct a map $\tilde{h} : f^{-1}(|L'|) \to P(|K(\mathbb{Z}, n + 1)|)$ so that $e_1\tilde{h} \simeq gf|f^{-1}(|L'|)$. By the Simplicial Approximation Theorem 4.10.11 (with Proposition 4.3.11) and the Homotopy Extension Theorem 6.4.1, we may assume that g is simplicial with respect to L' and $K(\mathbb{Z}, n + 1)$. For each simplex $\sigma \in K(\mathbb{Z}, n + 1)$, $g^{-1}(\sigma)$ is triangulated by some subcomplex of L', which is a subcomplex of K'. Then, we can perform the same construction in the proof of Lemma 7.9.2. The details are left to the reader.

The following is trivial by definition:

Lemma 7.10.6 *Let X be a metrizable space and K a simplicial complex. If a map $f : X \to |K|$ has property* $(\mathrm{EW})_n$, *then the restriction $f|Y$ has also property* $(\mathrm{EW})_n$ *for any $Y \subset X$.* □

Now, we show the following lemma:

Lemma 7.10.7 *Let X be the inverse limit of an inverse sequence $(|K_i|, f_i)_{i \in \mathbb{N}}$ of compact metric polyhedra. If $\lim_{j \to \infty} \mathrm{mesh}\, f_{i,j}(K_j) = 0$ and each bonding map $f_i : |K_{i+1}| \to |K_i|$ has property* $(\mathrm{EW})_n$, *then* $\mathrm{cd}_{\mathbb{Z}}\, X \leqslant n$.

Proof Let $g : A \to |K(\mathbb{Z}, n)|$ be a map from a closed set A in X. For each $i \in \mathbb{N}$, let $A_i = p_i(A)$, where $p_i : X \to |K_i|$ is the inverse limit projection. Then, $A = \varprojlim(A_i, f_i|A_{i+1})$ and each $p_i|A$ is the inverse limit projection. By Lemma 7.1.7, we have some $i \in \mathbb{N}$ and a map $g_i : A_i \to |K(\mathbb{Z}, n)|$ such that $g_i p_i|A \simeq g$. Then, g_i extends to a map $\tilde{g}_i : U \to |K(\mathbb{Z}, n)|$ over a neighborhood U of A_i in $|K_i|$. Since $\lim_{j\to\infty} \operatorname{mesh} f_{i,j}(K_j) = 0$, we can find $j \geqslant i$ so that $f_{i,j}(\operatorname{st}(A_j, K_j)) \subset \operatorname{st}(A_i, f_{i,j}(K_j)) \subset U$, where $\operatorname{st}(A_j, K_j) = |L|$ for some subcomplex L of K_j. Observe that

$$A_{j+1} = p_{j+1}(A) \subset f_j^{-1}(A_j) \subset f_j^{-1}(\operatorname{st}(A_j, K_j)) = f_j^{-1}(|L|) \subset |K_{j+1}|.$$

Because $\operatorname{cd}_{\mathbb{Z}}(f_j, K_j) \leqslant n$, there is a map $g_{j+1} : |K_{j+1}| \to |K(\mathbb{Z}, n)|$ such that

$$g_{j+1}|f_j^{-1}(\operatorname{st}(A_j, K_j)) = \tilde{g}_i f_{i,j+1}|f_j^{-1}(\operatorname{st}(A_j, K_j)).$$

Thus, we have a map $g_{j+1}p_{j+1} : X \to |K(\mathbb{Z}, n)|$. Then, it follows that

$$g_{j+1}p_{j+1}|A = \tilde{g}_i f_{i,j+1}p_{j+1}|A = g_i p_i|A \simeq g.$$

Therefore, g extends over X by the Homotopy Extension Theorem 6.4.1. □

For a pointed space (Y, y_0), let $\Omega(Y, y_0)$ be the loop space, that is,

$$\Omega(Y, y_0) = \mathbf{C}((\mathbf{I}, \partial\mathbf{I}), (Y, y_0)) = \mathbf{C}((\mathbf{S}^1, \mathbf{e}_1), (Y, y_0)),$$

where the constant loop c_{y_0} is the base point of $\Omega(Y, y_0)$. For each $i > 1$, $\Omega^i(Y, y_0) = \Omega(\Omega^{i-1}(Y, y_0), c_*)$, where c_* is the constant loop at the base point $* \in \Omega^{i-1}(Y, y_0)$ and $\Omega^0(Y, y_0)$ means $Y = (Y, y_0)$ itself. Here, omitting the base point, we simply write $\Omega^i Y$ instead of $\Omega^i(Y, y_0)$. To prove Theorem 7.10.4, we need three more lemmas whose proofs are not given because more algebraic preliminaries would be necessary. For their proofs, refer to the paper by Dydak and Walsh [13].

Lemma 7.10.8 [20] *Let (X, x_0), (Y, y_0) be pointed polyhedra, K be a countable simplicial complex, and $p : X \to |K|$ be a map with the following properties:*

(i) *$p^{-1}(|L|)$ is a subpolyhedron of X for every subcomplex L of K;*
(ii) *There is some $i_0 \geqslant 2$ such that $[p^{-1}(\sigma), \Omega^i Y] = \{0\}$ for every $\sigma \in K$ and $i \geqslant i_0$.*

Then, p induces the isomorphisms

$$p^* : [(|K|, p(x_0)), (\Omega^i Y, *)] \to [(X, x_0), (\Omega^i Y, *)], \ i \geqslant i_0.$$

[20] This lemma is formulated in a more general setting in [13] and called the COMBINATORIAL VIETORIS–BEGLE THEOREM, where X is a k-space (i.e., compactly generated) and property (1) is that the inclusion $p^{-1}(|L|) \subset X$ is a cofibration for every subcomplex L of K. Due to the Homotopy Extension Theorem 4.5.3, for an arbitrary simplicial complex K and any subcomplex L of K, the inclusion $|L| \subset |K|$ is a cofibration.

Let $EW(K,n)$ be Edwards–Walsh complex for a simplicial complex K and $n \geqslant 2$. It should be noted that if K is countable then so is $EW(K,n)$. Then, an associate map $\varpi : |EW(K,n)| \to |K|$ has property (i) of Lemma 7.10.8 because $\varpi^{-1}(|L|)$ is a subpolyhedron of $|EW(K,n)|$ for every subcomplex L of K. Recall that $\varpi^{-1}(\sigma) = \sigma$ for each $\sigma \in K^{(n)}$ and $\varpi^{-1}(\sigma) \simeq |K(\mathbb{Z},n)|^{k(\dim\sigma)}$ for each $\sigma \in K \setminus K^{(n)}$.

Lemma 7.10.9 *For each $n \geqslant 2$ and $k \geqslant 1$, $[|K(\mathbb{Z}^k,n)|, \Omega^i\mathbf{S}^m] = \{0\}$ if m is odd and $i \geqslant m$ or if m is even and $i \geqslant 2m-1$.*

In Lemma 7.10.9, when $n=2$ and $m=3$, $[|K(\mathbb{Z}^k,2)|, \Omega^i\mathbf{S}^3] = \{0\}$ for each $k \geqslant 1$ and $i \geqslant 3$. Then, an associate map $\varpi : |EW(K,2)| \to |K|$ satisfies condition (ii) of Lemma 7.10.8, where $Y = \mathbf{S}^3$, $n=2$, and $i_0 = 3$. Consequently, ϖ induces isomorphisms

$$\varpi^* : [|K|, \Omega^3\mathbf{S}^3] \to [|EW(K,2)|, \Omega^3\mathbf{S}^3].$$

Therefore, for a map $f : |K| \to \Omega^3\mathbf{S}^3$,

$$f \not\simeq 0 \;\Leftrightarrow\; f\varpi \not\simeq 0.$$

Lemma 7.10.10 *Let $K = \bigcup_{i\in\mathbb{N}} K_i$ be a countable simplicial complex, where $K_1 \subset K_2 \subset \cdots$ are finite subcomplexes of K. Let (Y, y_0) be a pointed polyhedron and $i_0 \geqslant 3$ such that $\pi_i(Y,y_0)$ is finite for each $i \geqslant i_0$. For a map $f : |K| \to \Omega^i(Y,y_0)$, $i \geqslant i_0 - 1$, if $f||K_j| \simeq 0$ for every $j \in \mathbb{N}$, then $f \simeq 0$.*

It is known that $\pi_i(\mathbf{S}^{2n-1})$ is finite for every $i > 2n-1$.[21] In particular, $\pi_i(\Omega^3\mathbf{S}^3) \cong \pi_{i+3}(\mathbf{S}^3)$ is finite for every $i \in \mathbb{N}$. Moreover, $\pi_m(\Omega^3\mathbf{S}^3) \cong \pi_{m+3}(\mathbf{S}^3) \neq \{0\}$ for some $m \geqslant 3$.[22] Then, we have a map $f : \mathbf{S}^m \to \Omega^3\mathbf{S}^3$ with $f \not\simeq 0$ for some $m \geqslant 3$.

Theorem 7.10.11 (DYDAK–WALSH) *For a map $f : \mathbf{S}^m \to \Omega^3\mathbf{S}^3$ with $f \not\simeq 0$ ($m \geqslant 3$), there exists a map $g : X \to \mathbf{S}^m$ of a compactum X with $\mathrm{cd}_{\mathbb{Z}} X \leqslant 2$ such that $fg \nsim 0$, which implies $g \not\simeq 0$.*

Proof Let $K_1 = F(\partial\Delta^{m+1})$ and identify $|K_1| = \mathbf{S}^m$. Thus, we can regard $f : |K_1| \to \Omega^3\mathbf{S}^3$. We will inductively construct finite simplicial complexes K_i and maps $g_i : |K_{i+1}| \to |K_i|$, $i \in \mathbb{N}$, so that $(|K_i|, g_i)_{i\in\mathbb{N}}$ satisfies the conditions of Lemma 7.10.7 for $n=2$, and the compositions $fg_{1,i} = fg_1 \cdots g_{i-1} : |K_i| \to \Omega^3\mathbf{S}^3$ are not null-homotopic (i.e., $fg_1 \cdots g_i \not\simeq 0$). Then, $X = \varprojlim(|K_i|, g_i)$ and the inverse limit projection $p_1 : X \to |K_1| = \mathbf{S}^m$ is the desired map. Indeed, $\mathrm{cd}_{\mathbb{Z}} X \leqslant 2$ by Lemma 7.10.7. Moreover, $fp_1 \not\simeq 0$ by Lemma 7.1.7 (2).

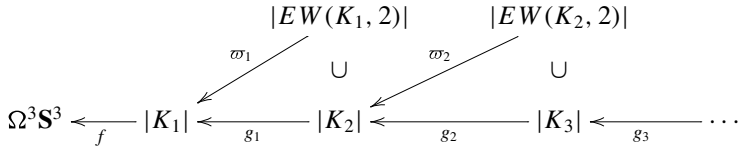
Let $EW(K_1,2)$ be Edwards–Walsh complex for K_1. Then, as observed above, an associate map $\varpi_1 : |EW(K_1,2)| \to |K_1|$ satisfies conditions (i) and (ii) of Lemma

[21] For example, see Theorem 7.1 in Chapter XI of Hu's book *Homotopy Theory*. More generally, $\pi_i(\mathbf{S}^n)$ is finite for every $i > n$ except for $\pi_{4k-1}(\mathbf{S}^{2k})$ (cf. Hatcher's book *Algebraic Topology*, p. 339).

[22] In fact, $\pi_6(\mathbf{S}^3) \cong \mathbb{Z}_{12}$ by Theorem 16.1 in Chapter XI of Hu's book *Homotopy Theory*.

7.10.8, hence it follows that $f\varpi_1 \not\simeq 0$. By Lemma 7.10.10, $f\varpi_1||K_2| \not\simeq 0$ for some finite subcomplex K_2 of $EW(K_1, 2)$. Let $g_1 = \varpi_1||K_2|$ and replace K_2 by a subdivision such that mesh $g_1(K_2) < 2^{-1}$. Then, since $\varpi_1 : |EW(K_1, 2)| \to |K_1|$ has property $(\mathrm{EW})_2$ by Proposition 7.10.5, the map $g_1 : |K_2| \to |K_1|$ has property $(\mathrm{EW})_2$ by Lemma 7.10.6.

Next, let $\varpi_2 : |EW(K_2, 2)| \to |K_2|$ be an associate map of Edwards–Walsh complex $EW(K_2, 2)$ for K_2. By the analogy of g_1, we can apply Lemma 7.10.8 to have $(fg_1)\varpi_2 \not\simeq 0$. By Lemma 7.10.10, $(fg_1)\varpi_2||K_3| \not\simeq 0$ for some finite subcomplex K_3 of $EW(K_2, 2)$. Let $g_2 = \varpi_2||K_3|$ and replace K_3 by a subdivision such that mesh $g_1g_2(K_3) < 2^{-2}$ and mesh $g_2(K_3) < 2^{-2}$. Then, using Proposition 7.10.5 and Lemma 7.10.6 in the same way as the above g_1, we have $g_2 : |K_3| \to |K_2|$ has property $(\mathrm{EW})_2$.



Inductively, let $\varpi_i : |EW(K_i, 2)| \to |K_i|$ be an associate map of Edwards–Walsh complex $EW(K_i, 2)$ for K_i. Applying Lemma 7.10.8 as above, we have $(fg_1 \cdots g_{i-1})\varpi_i \not\simeq 0$. By Lemma 7.10.10, $(fg_1 \cdots g_{i-1})\varpi_i||K_{i+1}| \not\simeq 0$ for some finite subcomplex K_{i+1} of $EW(K_i, 2)$. Let $g_i = \varpi_i||K_{i+1}|$ and replace K_{i+1} by a subdivision such that mesh $g_j \cdots g_i(K_{i+1}) < 2^{-i}$ for every $j \leqslant i$. Then, in the same way as g_1 and g_2, each $g_i : |K_{i+1}| \to |K_i|$ has property $(\mathrm{EW})_2$. This completes the proof. □

Now, we can prove Theorem 7.10.4.

Proof (*Theorem* 7.10.4) An example X in Theorem 7.10.11 is an infinite-dimensional compactum with $\mathrm{cd}_{\mathbb{Z}} X \leqslant 2$. Since X has a map $g : X \to \mathbf{S}^m$ with $g \not\simeq 0$, we have $\dim X \geqslant m \geqslant 3$ by Proposition 5.3.1, which implies that X is infinite-dimensional by Theorem 7.9.6. Recall that $\mathrm{cd}_{\mathbb{Z}} X \leqslant 1$ if and only if $\dim X \leqslant 1$. Then, it follows that $\mathrm{cd}_{\mathbb{Z}} X = 2$. □

Combining Theorem 7.10.4 with Theorem 7.10.1, we have the following positive answer to the CE Problem:

Corollary 7.10.12 *There exists a cell-like map $f : X \to Y$ from a 2-dimensional compactum X onto an infinite-dimensional compactum Y.* □

A map $f : X \to Y$ is said to be **monotone** if $f^{-1}(y)$ is connected for every $y \in Y$. The following profound theorem on approximations by open maps was established by J. J. Walsh. It is beyond the scope of this book to present the proof; refer instead to his paper [34].

Theorem 7.10.13 (J. J. WALSH) *Let M be a compact connected (topological) n-manifold (possibly with boundary), where $n \geqslant 3$. Every monotone map $f : M \to Y$*

of M onto a metrizable space Y can be approximated by open maps $g : M \to Y$ such that $g^{-1}(y)$ and $f^{-1}(y)$ have the same shape type for each $y \in Y$.[23] *In particular, every cell-like map $f : M \to Y$ can be approximated by a cell-like open map.*

Let $f : X \to Y$ be a cell-like map from a 2-dimensional compactum X onto an infinite-dimensional compactum Y as in Corollary 7.10.12. Embed the compactum X into $\mathbf{S}^5$ (or $\mathbf{I}^5$), and consider the adjunction space $Y \cup_f \mathbf{S}^5$ (or $Y \cup_f \mathbf{I}^5$). Restricting the quotient map, we can obtain a cell-like map $g : \mathbf{S}^5 \to Y \cup_f \mathbf{S}^5$ (or $g : \mathbf{I}^5 \to Y \cup_f \mathbf{I}^5$). Because $\mathbf{S}^5$ (or $\mathbf{I}^5$) is a compact connected 5-manifold (with boundary), we can apply the above Walsh's Theorem 7.10.13 to obtain the following corollary that will be used in Sect. 7.12:

Corollary 7.10.14 *There exists a cell-like open map from $\mathbf{S}^5$ (or $\mathbf{I}^5$) onto an infinite-dimensional compactum.* □

7.11 Continuous Metrics on Free Topological Linear Spaces

For a space X, the free topological linear space $L(X)$ was introduced in Sect. 3.9. The **free topological linear space** $L(X)$ over X is defined as a topological linear space that contains X as a subspace and has the following extension property:

(LE) For an arbitrary topological linear space F, every map $f : X \to F$ of X uniquely extends to a linear map[24] $\tilde{f} : L(X) \to F$.

A topological linear space $L(X)$ with property (LE) is uniquely determined up to linear homeomorphism if it exists. For every Tychonoff space X, such a space $L(X)$ exists (Theorem 3.9.3). In the next section, using the free topological linear space $L(X)$ over a compactum X, we will construct a metric linear space that is not an AR,

In this section, as preparation, we will study topological and geometrical structures of $L(X)$ and continuous metrics d on $L(X)$ such that $(L(X), d)$ is a metric linear space, where it should be noted that $L(X)$ is non-metrizable even if X is a compactum. Throughout this section, we assume

- *X is a compactum (= a compact metrizable space).*

First, we will review the definitions and notations. As discussed in Sect. 3.9, $L(X)$ can be obtained by topologizing the following linear space:

$$\mathbb{R}_f^X = \left\{ z \in \mathbb{R}^X \;\middle|\; z(x) = 0 \text{ except for finitely many } x \in X \right\},$$

[23] More generally, this theorem is valid for a quasi-open (or quasi-monotone) map, where f is said to be **quasi-open** or **quasi-monotone** if $f(U) = V$ for each open connected set V in Y and each component U of $f^{-1}(V)$.

[24] I.e., a continuous linear function.

where the algebraic operations are coordinate-wisely defined and each $x \in X$ is identified with $\delta_x \in \mathbb{R}_f^X$ defined by $\delta_x(x) = 1$ and $\delta_x(y) = 0$ for $y \in X \setminus \{x\}$ (see arguments after Proposition 3.9.8). Namely, we regard $L(X) = \mathbb{R}_f^X$ as linear spaces. For each $z \in \mathbb{R}_f^X$, we define

$$\operatorname{supp} z = z^{-1}(\mathbb{R} \setminus \{0\}) \quad \text{and} \quad \nu(z) = \operatorname{card} z^{-1}(\mathbb{R} \setminus \{0\}).$$

When $\operatorname{supp} z = \{x_1, \dots, x_{\nu(z)}\}$, the condition that $x_i \neq x_j$ if $i \neq j$ is implicit in this description. Then, $z \in \mathbb{R}_f^X$ can be represented as follows:

$$z = \sum_{x \in X} z(x)x = \sum_{x \in \operatorname{supp} z} z(x)x = \sum_{i=1}^{\nu(z)} z(x_i)x_i,$$

To topologize $\mathbb{R}_f^X$, take $* \in X$ as a base point and make the following identification:

$$X^n \times \mathbb{R}^n = (X^n \times \{(*, *, \dots)\}) \times (\mathbb{R}^n \times \{(0, 0, \dots)\}) \subset X_f^{\mathbb{N}} \times \mathbb{R}_f^{\mathbb{N}}.$$

Then, we have the direct limit $\varinjlim(X^n \times \mathbb{R}^n)$ of the following closed tower:

$$X \times \mathbb{R} \subset X^2 \times \mathbb{R}^2 \subset X^3 \times \mathbb{R}^3 \subset \cdots.$$

The free topological linear space $L(X)$ can be defined as the space $\mathbb{R}_f^X$ with the quotient topology induced by the surjection $\gamma_\infty : \varinjlim(X^n \times \mathbb{R}^n) \to \mathbb{R}_f^X$ defined as follows:

$$\gamma_\infty(x, \alpha) = \sum_{i=1}^{n} \alpha(i)x(i) \quad \text{for } (x, \alpha) \in X^n \times \mathbb{R}^n.$$

Then, $\gamma_\infty | X$ is the identification $x \mapsto \delta_x$, that is, $\gamma_\infty | X = \mathrm{id}_X$, where we regard as follows:

$$X = X \times \{1\} \subset X \times \mathbb{R} \subset \varinjlim(X^n \times \mathbb{R}^n),$$

For each $n \in \mathbb{N}$,

$$\begin{aligned} \gamma_n &= \gamma_\infty | X^n \times \mathbb{R}^n : X^n \times \mathbb{R}^n \to L_n(X), \\ L_n(X) &= \gamma_\infty(X^n \times \mathbb{R}^n) = \left\{ z \in L(X) \;\middle|\; \nu(z) \leqslant n \right\}, \quad \text{and} \\ S_n(X) &= L_n(X) \setminus L_{n-1}(X) = \left\{ z \in L(X) \;\middle|\; \nu(z) = n \right\}, \end{aligned}$$

where $L_0(X) = \{\mathbf{0}\}$. According to Proposition 3.9.6, each $\gamma_n | \gamma_n^{-1}(S_n(X))$ is a covering projection over $S_n(X)$ such that $\operatorname{card} \gamma_n^{-1}(z) = n!$ for each $z \in S_n(X)$, and that each point of $S_n(X)$ has an open neighborhood in $S_n(X)$ that is homeomorphic to an open set in $X^n \times \mathbb{R}^n$. Then, $L_n(X)$ and $S_n(X)$ have the following properties:

Proposition 7.11.1 (1) *Each $L_n(X)$ is closed in $L(X)$ and it is nowhere dense in $L(X)$ if* $\operatorname{card} X \geqslant n+1$.

This holds for an arbitrary Tychonoff space X (Propositions 3.9.4 and 3.9.8).

(2) *Each $S_n(X)$ is locally compact, separable metrizable, so σ-compact.*

This is a particular case of Corollary 3.9.7.

(3) *$L(X) = \varinjlim \gamma_n(X^n \times [-n,n]^n)$ is the direct limit of a tower of compacta which is a perfectly normal, paracompact k-space (so hereditarily normal = completely normal).*

This is a particular case of Corollary 3.9.14 (cf. Theorem 3.9.13).

(4) *If $\dim X < \infty$, then $\dim S_n(X) < \infty$. In this case, $L(X)$ is a countable union of finite-dimensional compacta, hence it is strongly countable-dimensional.*

Sketch of Proof Combine Proposition 3.9.6 with Product Theorem 5.4.15 and Corollary 5.2.11. The additional statement follows from (2) above.

(5) *If X is strongly countable-dimensional, then so is $L(X)$.*

Sketch of Proof The same as (4) above.

(6) *If X is an ANR, then each $S_n(X)$ is an ANR.*

Sketch of Proof Combine Proposition 3.9.6 with the Hanner Theorem 6.2.9 (4).

To look into the geometrical structure of $L(X)$, the following ℓ_1-norm of $x \in \mathbb{R}^n$ is useful:

$$\|x\|_1 \underset{\text{def}}{=} \sum_{i=1}^{n} |x(i)| = |x(1)| + \cdots + |x(n)|.$$

The unit closed ball with respect to this norm is denoted by $\Diamond^n$, that is,

$$\Diamond^n \underset{\text{def}}{=} \left\{ x \in \mathbb{R}^n \;\middle|\; \|x\|_1 \leqslant 1 \right\}.$$

For each $z \in L(X)$,

$$\begin{aligned} \gamma_\infty^{-1}(z) &= \bigcup_{m \geqslant \nu(z)} \left\{ (x,\alpha) \in X^m \times \mathbb{R}^m \;\middle|\; \forall y \in X, z(y) = \textstyle\sum_{x(i)=y} \alpha(i) \right\} \\ &\subset \varinjlim \left(X^m \times \mathbb{R}^m \right). \end{aligned}$$

For each $n \in \mathbb{N}$ and $r > 0$, we define

$$L_n(X,r) \underset{\text{def}}{=} \gamma_n(X^n \times r\Diamond^n) = \left\{ z \in L_n(X) \;\middle|\; \textstyle\sum_{x \in \operatorname{supp} z} |z(x)| \leqslant r \right\}.$$

The second equality in the above can be verified as follows: The last set is obviously contained in the middle. To see the converse inclusion, let $z = \gamma_n(y, \alpha)$, where $(y, \alpha) \in X^n \times r\Diamond^n$. Then, $\sum_{x \in \operatorname{supp} z} z(x)x = \sum_{i=1}^n \alpha(i)y(i)$ and $\sum_{i=1}^n |\alpha(i)| \leqslant r$. Since $z(x) = \sum_{y(i)=x} \alpha(i)$ for each $x \in \operatorname{supp} z$, it follows that

$$\sum_{x \in \operatorname{supp} z} |z(x)| = \sum_{x \in \operatorname{supp} z} \left| \sum_{y(i)=x} \alpha(i) \right| \leqslant \sum_{x \in \operatorname{supp} z} \sum_{y(i)=x} |\alpha(i)| = \sum_{i=1}^n |\alpha(i)| \leqslant r.$$

Each $L_n(X, r)$ is a compactum because $\gamma_n | X^n \times r\Diamond^n : X^n \times r\Diamond^n \to L_n(X, r)$ is a perfect map (see Proposition 2.4.4 (1)).

The following is trivial:

Lemma 7.11.2 *For each $n < m$ and $r > 0$, $L_n(X) \cap L_m(X, r) = L_n(X, r)$.* □

Remark 7.11.1 Lemma 7.11.2 is one of the reasons for adopting $\Diamond^n$ instead of $[-1, 1]^n$ (cf. Sect. 3.9). In fact, $L_m(X, r) = \gamma_\infty(X^m \times r\Diamond^m)$ cannot be replaced by $\gamma_\infty(X^m \times [-r, r]^m)$. Precisely,

$$L_n(X) \cap \gamma_\infty(X^m \times [-r, r]^m) \supsetneqq \gamma_\infty(X^n \times [-r, r]^n).$$

Indeed, take $(x, \alpha) \in X^m \times [-r, r]^m$ so that $x(i) \neq x(j)$ for $1 \leqslant i < j \leqslant n$, $x(i) = x(n)$ for $n \leqslant i \leqslant m$, and $\alpha(i) = r$ for $1 \leqslant i \leqslant m$. Then,

$$\begin{aligned} \gamma_\infty(x, \alpha) &= \sum_{i=1}^{n-1} rx(i) + (m - n + 1)rx(n) \\ &\in \left(L_n(X) \cap \gamma_\infty(X^m \times [-r, r]^m\right) \setminus \gamma_\infty(X^n \times [-r, r]^n). \end{aligned}$$

Because $\Diamond^n \subset [-1, 1]^n \subset n\Diamond^n$, it follows that

$$L_n(X, n) \subset \gamma_\infty(X^n \times [-n, n]^n) \subset L_n(X, n^2) \subset L_{n^2}(X, n^2).$$

Since $L(X) = \varinjlim \gamma_n(X^n \times [-n, n]^n)$ (Proposition 7.11.1 (3)), we have:

Proposition 7.11.3 $L(X) = \varinjlim L_n(X, n)$. □

Observe that $L_n(X) \cap L_m(X, m) = L_n(X, m)$ for every $m \geqslant n$ by Lemma 7.11.2. Since $L_n(X)$ is closed in $L(X)$ (Lemma 3.9.4), we can apply Proposition 2.8.2 to obtain the following:

Proposition 7.11.4 *For each $n \in \mathbb{N}$, $L_n(X) = \varinjlim L_n(X, m)$.* □

Since $L_n(X, n) \subset L_n(X)$, the following is obtained from Proposition 7.11.3:

Proposition 7.11.5 $L(X) = \varinjlim L_n(X)$. □

The following lemma will be used in the next section:

Lemma 7.11.6 *For each $n \in \mathbb{N}$ and $0 < r < s$, the following hold:*

(i) $\sum_{i=1}^n t_i x_i \in S_n(X) \cap L_n(X, r)$ *implies* $\sum_{i=1}^n |t_i| \leqslant r$;
(ii) $S_n(X) \cap L_n(X, s)$ *is a neighborhood of* $S_n(X) \cap L_n(X, r)$ *in* $S_n(X)$.

Proof Since (i) is trivial, we prove only (ii). For each $z = \sum_{i=1}^n t_i x_i \in S_n(X) \cap L_n(X, r)$, we have disjoint open sets $U_1, \ldots, U_n$ in X such that $x_i \in U_i$. Because $0 < \sum_{i=1}^n |t_i| \leqslant r < s$ by (i), we can choose $0 < a_i < t_i < b_i$ so that $0 < \sum_{i=1}^n |t_i'| < s$ if $a_i < t_i' < b_i$. Then, the following is an open neighborhood of z in $S_n(X)$:

$$U_z = \gamma_n\big(\textstyle\prod_{i=1}^n U_i \times \prod_{i=1}^n (a_i, b_i)\big) \subset S_n(X) \cap L_n(X, s).$$

Thus, we have the desired result. □

Let $\mathfrak{M}_X$ be the set of all continuous metrics d on $L(X)$ such that $(L(X), d)$ is a metric linear space.[25] Conventionally, open sets, neighborhoods, closures, etc., in $(L(X), d)$ are called d-open sets, d-neighborhoods, d-closures, etc. In addition, continuous maps with respect to d are said to be d-continuous. We will show some auxiliary results on $\mathfrak{M}_X$ for the next section.

Lemma 7.11.7 *Each neighborhood U of $\mathbf{0} \in L(X)$ is a d-neighborhood of $\mathbf{0}$ for some $d \in \mathfrak{M}_X$. In particular, $\mathfrak{M}_X \neq \emptyset$.*

Proof Since $L(X)$ is a perfectly normal topological linear space (Proposition 7.11.1 (3)), $L(X)$ has open sets $U_1 \supset U_2 \supset \cdots$ such that

$$U_1 \subset U,\ U_{i+1} + U_{i+1} \subset U_i,\ [-1, 1]U_{i+1} \subset U_i \ \text{ and } \ \{\mathbf{0}\} = \bigcap_{i \in \mathbb{N}} U_i.$$

For each $z \in L(X)$, because each U_i is a neighborhood of $\mathbf{0} = 0z \in L(X)$, we can find $s > 0$ such that $sz \in U_i$, i.e., $z \in s^{-1}U_i$. By Proposition 3.4.1, $\{U_i \mid i \in \mathbb{N}\}$ is a neighborhood basis at $\mathbf{0}$ in some topology of $L(X)$, which makes $L(X)$ a topological linear space. By Theorem 3.6.1, this topology is metrizable and coarser than the original topology. Moreover, U is a neighborhood of $\mathbf{0}$ in this topology because $U_1 \subset U$. Thus, we have the desired result. □

Lemma 7.11.8 *For a compactum X, the following statements hold:*

(1) *Given $d_i \in \mathfrak{M}_X$, $i \in \mathbb{N}$, there is some $d \in \mathfrak{M}_X$ such that $\mathrm{id}_{L(X)} : (L(X), d) \to (L(X), d_i)$ is continuous for every $i \in \mathbb{N}$;*
(2) *Each open set U in $L(X)$ is d-open for some $d \in \mathfrak{M}_X$, equivalently each closed set A in $L(X)$ is d-closed for some $d \in \mathfrak{M}_X$;*
(3) *For an arbitrary metric space Y and $A \subset L(X)$, each map $f : A \to Y$ is d-continuous for some $d \in \mathfrak{M}_X$;*

[25] Here, a continuous metric d on $L(X)$ means a metric on the set $L(X)$ such that $d : L(X) \times L(X) \to \mathbb{R}_+$ is continuous, or equivalently, the topology of the metric space $(L(X), d)$ is coarser than or equal to the original topology of $L(X)$.

(4) *For each metrizable subset* $A \subset L(X)$ *and* $d_0 \in \mathfrak{M}_X$, *there is some* $d \in \mathfrak{M}_X$ *such that* $d \geqslant d_0$ *and* $d|A^2 \in \mathrm{Metr}(A)$.

Proof (1): Note that the product space $F = \prod_{i\in\mathbb{N}}(L(X), d_i)$ is a metrizable topological linear space. Let $\mathfrak{d} : L(X) \to F$ be the diagonal injection (i.e., $\mathfrak{d}(x)(i) = x$ for each $i \in \mathbb{N}$). Then, $\mathfrak{d}$ is continuous and linear. The desired metric can be defined by $\mathfrak{d}$ and an admissible linear metric for F. (For example, $d(z, z') = \sup_{i\in\mathbb{N}} \min\{i^{-1},\ d_i(z, z')\}$ or $d(z, z') = \sum_{i\in\mathbb{N}} \min\{2^{-i},\ d_i(z, z')\}$.)

(2): For each $x \in U$, $U - x$ is a neighborhood of $\mathbf{0}$ in $L(X)$. By Lemma 7.11.7, $U - x$ is a d_x-neighborhood of $\mathbf{0}$ for some $d_x \in \mathfrak{M}_X$. Choose $r_x > 0$ so that $\mathrm{B}_{d_x}(\mathbf{0}, r_x) \subset U - x$ and let $V_x = \mathrm{B}_{d_x}(\mathbf{0}, r_x) + x \subset U$. Note that V_x is also open in $L(X)$. On the other hand, we have a tower $D_1 \subset D_2 \subset \cdots$ of compacta with $L(X) = \varinjlim D_i$. Then, $U = \bigcup_{i\in\mathbb{N}}(U \cap D_i)$ and each $U \cap D_i$ is separable metrizable. It is easy to prove that U is covered by a countable subcollection of $\{V_x \mid x \in U\}$, say $U = \bigcup_{i\in\mathbb{N}} V_{x_i}$. By (1), we have $d \in \mathfrak{M}_X$ such that $\mathrm{id}_{L(X)} : (L(X), d) \to (L(X), d_{x_i})$ is continuous for every $i \in \mathbb{N}$. Since each V_{x_i} is d-open, $U = \bigcup_{i\in\mathbb{N}} V_{x_i}$ is also d-open.

(3): For each $x \in A$ and $i \in \mathbb{N}$, choose an open neighborhood $V_i(x)$ of x in $L(X)$ so that $\mathrm{diam}\, f(V_i(x) \cap A) < 1/i$. Then, $V_i(x)$ is d_i^x-open for some $d_i^x \in \mathfrak{M}_X$ by (2). Here, we write $L(X) = \varinjlim D_i$ as in the proof of (2). Because $A = \bigcup_{n\in\mathbb{N}}(A \cap D_n)$ and each $A \cap D_n$ is separable metrizable, we can obtain $\{x_j \mid j \in \mathbb{N}\} \subset A$ such that $A \subset \bigcup_{j\in\mathbb{N}} V_i(x_j)$ for every $i \in \mathbb{N}$. By (1), we have $d \in \mathfrak{M}_X$ such that $\mathrm{id}_{L(X)} : (L(X), d) \to (L(X), d_i^{x_j})$ is continuous, hence every $V_i(x_j)$ is d-open. For each $x \in A$ and $i \in \mathbb{N}$, choose $j \in \mathbb{N}$ so that $x \in V_i(x_j)$. Since $V_i(x_j)$ is a d-open neighborhood of x and $\mathrm{diam}\, f(V_i(x_j) \cap A) < 1/i$, it follows that f is d-continuous.

(4): Take $d' \in \mathrm{Metr}(A)$ and apply (3) to obtain $d'' \in \mathfrak{M}_X$ such that $\mathrm{id}_A : A \to (A, d')$ is d''-continuous. Note that $\mathrm{id}_A : A = (A, d') \to (A, d''|A^2)$ is also continuous, hence it is a homeomorphism. Therefore, $d''|A^2 \in \mathrm{Metr}(A)$. We define a metric on $L(X)$ as follows:

$$d(z, z') = \max\left\{d''(z, z'),\ d_0(z, z')\right\}.$$

Then, it is easy to see that $d \in \mathfrak{M}_X$ is the desired metric. □

7.12 A Non-AR Metric Linear Space

By the Dugundji Extension Theorem 6.1.1, every locally convex topological linear space is an AE. In this section, we show that the local convexity is essential. This involves the following theorem:

Theorem 7.12.1 (CAUTY) *There exists a* σ*-compact metric linear space that is not an AE, hence it is not an AR.*

As mentioned in the previous section, we will construct such a space by using the free topological linear space $L(X)$ over a compactum X. We need the following lemma:

Lemma 7.12.2 *Let E and F be metrizable topological linear spaces and $h : E \to F$ a continuous linear surjection. If E and F are ARs, then for each open set U in F, $h|h^{-1}(U) : h^{-1}(U) \to U$ is a fine homotopy equivalence.*

Proof Due to Corollary 7.4.4, if h is a fine homotopy equivalence, then for each open set U in F, $h|h^{-1}(U) : h^{-1}(U) \to U$ is also a fine homotopy equivalence. By Theorem 7.4.3, to show that h is a fine homotopy equivalence, it suffices to prove that h is a local $*$-connection.

For every neighborhood V of each $y \in F$, choose an open neighborhood W of $\mathbf{0} \in F$ so that $tW \subset W$ for $t \in \mathbf{I}$ and $W + y \subset V$. Then,

$$h^{-1}(W + y) = h^{-1}(W) + x \quad \text{for each } x \in h^{-1}(y).$$

Since $th^{-1}(W) \subset h^{-1}(W)$ for $t \in \mathbf{I}$, it follows that $h^{-1}(W)$ is contractible. Thus, $h^{-1}(W + y)$ is a contractible open neighborhood of $h^{-1}(y)$ in $h^{-1}(V)$. Hence, h is a local $*$-connection. □

In Corollary 7.10.14, we showed that there exists a cell-like open map from a finite-dimensional compact ANR onto an infinite-dimensional compactum. Hence, Theorem 7.12.1 can be derived from the following theorem:

Theorem 7.12.3 (CAUTY) *Let X be an infinite-dimensional compactum with a cell-like open map $f : Y \to X$ of a finite-dimensional compact ANR Y. Then, the free topological linear space $L(X)$ over X has a continuous metric d such that $(L(X), d)$ is a metric linear space but is not an AR.*

Proof Let $\dim Y \leqslant m$. Because X is infinite-dimensional, there is a closed set A in X with a map $g : A \to \mathbf{S}^m$ that cannot extend over X (Theorem 5.2.4). Since $\mathbf{S}^m$ is an ANE for normal spaces (Theorem 5.1.6 (2)), A has an open neighborhood W in $L(X)$ such that $\mathbf{0} \notin \operatorname{cl} W$ and g extends to a map $\bar{g} : \operatorname{cl} W \to \mathbf{S}^m$. Observe that

$$\bar{g} f_\sharp | Y \cap f_\sharp^{-1}(\operatorname{cl} W) = \bar{g} f | Y \cap f_\sharp^{-1}(\operatorname{cl} W).$$

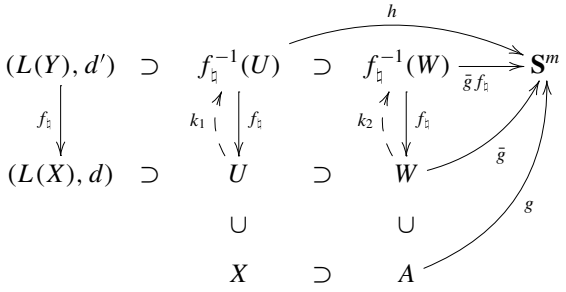
Later, using Haver's Near-Selection Theorem 7.6.1 and the fact that $\dim Y \leqslant m$ and f is a cell-like open map, we will prove the following claim:

Claim *There exists an open neighborhood U of $X \cup \operatorname{cl} W$ in $L(X)$ and the map $\bar{g} f_\sharp | f_\sharp^{-1}(\operatorname{cl} W)$ extends to a map $h : f_\sharp^{-1}(U) \to \mathbf{S}^m$.*

Combining Lemma 7.11.8 (2) and (3), we have $d \in \mathfrak{M}_X$ such that U and W are d-open and $\bar{g}$ is d-continuous. Then, by Lemma 7.11.8 (1), we have $d' \in \mathfrak{M}_Y$ such that the following are continuous:

$$f_\natural : (L(Y), d') \to (L(X), d) \text{ and } h : (f_\natural^{-1}(U), d') \to \mathbf{S}^m.$$

Thus, we have the following commutative diagram of metric spaces and maps:



Due to Proposition 7.11.1 (4), $L(Y)$ is a countable union of finite-dimensional compacta. Since each compact subspace in $L(Y)$ is also a subspace of $(L(Y), d')$, it follows that $(L(Y), d')$ is also a countable union of finite-dimensional compacta, hence it is (strongly) countable-dimensional. Therefore, $(L(Y), d')$ is an AR by Corollaries 6.10.1 and 6.2.14.

Suppose that $(L(X), d)$ is an AR. Then, $f_\natural | f_\natural^{-1}(U)$ and $f_\natural | f_\natural^{-1}(W)$ are homotopy equivalences by Lemma 7.12.2. Let $k_1 : U \to f_\natural^{-1}(U)$ and $k_2 : W \to f_\natural^{-1}(W)$ be their homotopy inverses. Observe that

$$hk_1 | W \simeq hk_1 f_\natural k_2 \simeq hk_2 = \bar{g} f_\natural k_2 \simeq \bar{g} | W.$$

Thus, we have $hk_1 | A \simeq g$. By the Homotopy Extension Theorem 6.4.1, g extends over U. This contradicts the fact that g cannot extend over X. Consequently, $(L(X), d)$ is not an AR. □

To prove the claim, we need the following lemma, which is easily obtained by combining Lemma 7.11.8 (1), (2), and (3):

Lemma 7.12.4 *There is some $d_Y \in \mathfrak{M}_Y$ such that $f_\natural^{-1}(\mathrm{cl}\, W)$ and $f_\natural^{-1}(L_n(X))$, $n \in \mathbb{N}$, are d_Y-closed and $\bar{g} f_\natural | f_\natural^{-1}(\mathrm{cl}\, W) : f_\natural^{-1}(\mathrm{cl}\, W) \to \mathbf{S}^m$ is d_Y-continuous.* □

Recall that $\mathrm{Comp}(Z)$ is the space of all non-empty compact sets in a space Z with the Vietoris topology. When $Z = (Z, d)$ is a metric space, the Vietoris topology of $\mathrm{Comp}(Z)$ is induced by the Hausdorff metric d_H (Proposition 2.12.7). Then, by Theorem 2.12.8, if Z is compact metrizable, then so is $\mathrm{Comp}(Z)$. One should note that $S_n(X) = L_n(X) \setminus L_{n-1}(X)$ is separable metrizable by Proposition 7.11.1 (2).

Lemma 7.12.5 *For each $z \in S_n(X)$, $\varphi_n(z) = f_\natural^{-1}(z) \cap S_n(Y)$ is a cell-like compactum, and the set-valued function $\varphi_n : S_n(X) \to \mathrm{Comp}(S_n(Y))$ is continuous.*

Proof We write $z = \sum_{i=1}^n t_i x_i$ by the irreducible representation. Since $f^{-1}(x_i) \cap f^{-1}(x_j) = \emptyset$ for $i \neq j$, it follows that

$$\varphi_n(z) = \sum_{i=1}^{n} t_i f^{-1}(x_i) = \left\{ \sum_{i=1}^{n} t_i y_i \;\middle|\; y_i \in f^{-1}(x_i) \right\} \approx \prod_{i=1}^{n} f^{-1}(x_i),$$

where each $f^{-1}(x_i)$ is a cell-like compactum. Then, $\varphi_n(z)$ is a cell-like compactum.

To show that φ_n is upper semi-continuous (u.s.c.), let U be an open set in $S_n(Y)$ such that $\varphi_n(z) = \sum_{i=1}^{n} t_i f^{-1}(x_i) \subset U$. Choose disjoint open sets $V_1, \ldots, V_n$ in Y and $\delta > 0$ so that

$$f^{-1}(x_i) \subset V_i,\ \delta < \min\left\{ |t_i| \;\middle|\; i = 1, \ldots, n \right\} \text{ and } \sum_{i=1}^{n} (t_i - \delta, t_i + \delta) V_i \subset U.$$

Because f is a closed map, each x_i has an open neighborhood W_i in X such that $f^{-1}(W_i) \subset V_i$. Let

$$W = \prod_{i=1}^{n} W_i \times \prod_{i=1}^{n} (t_i - \delta, t_i + \delta) \subset X^n \times \mathbb{R}^n.$$

Then, $\gamma_n(W)$ is an open neighborhood of z in $S_n(X)$. For each $(x, \lambda) \in W$,

$$\varphi_n(\gamma_n(x, \lambda)) = \sum_{i=1}^{n} \lambda(i) f^{-1}(x(i)) \subset \sum_{i=1}^{n} (t_i - \delta, t_i + \delta) V_i \subset U.$$

Therefore, φ_n is u.s.c.

Next, to prove that φ_n is lower semi-continuous (l.s.c.), let U be an open set in $S_n(Y)$ that meets $\varphi_n(z) = \sum_{i=1}^{n} t_i f^{-1}(x_i)$. Choose $y_i \in f^{-1}(x_i)$, $i = 1, \ldots, n$, so that $\sum_{i=1}^{n} t_i y_i \in U$. Since $f(y_i) = x_i \neq x_j = f(y_j)$ for $i \neq j$, we can take open neighborhoods V_i of y_i in Y, $i = 1, \ldots, n$ and $\delta > 0$ so that $f(V_i) \cap f(V_j) = \emptyset$ for $i \neq j$,

$$\delta < \min\left\{ |t_i| \;\middle|\; i = 1, \ldots, n \right\} \text{ and } \sum_{i=1}^{n} (t_i - \delta, t_i + \delta) V_i \subset U.$$

Because f is an open map, each $f(V_i)$ is an open neighborhood of $x_i = f(y_i)$ in X. Now, let

$$W = \prod_{i=1}^{n} f(V_i) \times \prod_{i=1}^{n} (t_i - \delta, t_i + \delta) \subset X^n \times \mathbb{R}^n.$$

Then, $\gamma_n(W)$ is an open neighborhood of z in $S_n(X)$. For each $(x, \lambda) \in W$, $\varphi_n(\gamma_n(x, \lambda)) \cap U \neq \emptyset$. Indeed, choose $y_i' \in V_i$, $i = 1, \ldots, n$, so that $f(y_i') = x(i)$. Then, it follows that

$$\sum_{i=1}^{n} \lambda(i) y_i' \in \sum_{i=1}^{n} \lambda(i) f^{-1}(x(i)) = \varphi_n(\gamma_n(x, \lambda)) \quad \text{and}$$

$$\sum_{i=1}^{n} \lambda(i) y_i' \in \sum_{i=1}^{n} (t_i - \delta, t_i + \delta) V_i \subset U.$$

Therefore, φ_n is l.s.c. □

Now, we will prove the claim.

Proof (*Claim*) In the following, we will inductively construct closed sets $V_n \subset V_n' \subset L_n(X)$ and maps $h_n : f_\natural^{-1}(V_n' \cup \operatorname{cl} W) \to \mathbf{S}^m$, $n \in \mathbb{N}$, so as to satisfy the conditions below:

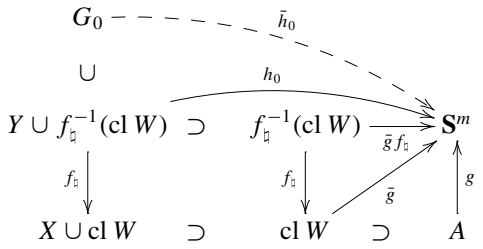
(1) $L_n(X) \cap (X \cup \operatorname{cl} W) \subset \operatorname{int}_n V_n \subset V_n \subset \operatorname{int}_n V_n'$;
(2) $L_n(X) \cap V_{n+1} = V_n$ and $L_n(X) \cap \operatorname{int}_{n+1} V_{n+1} = \operatorname{int}_n V_n$;
(3) $h_n | f_\natural^{-1}(\operatorname{cl} W) = \bar{g} f_\natural | f_\natural^{-1}(\operatorname{cl} W)$;
(4) $h_{n+1} | f_\natural^{-1}(V_n \cup \operatorname{cl} W) = h_n | f_\natural^{-1}(V_n \cup \operatorname{cl} W)$,

where we describe $\operatorname{int}_n C = \operatorname{int}_{L_n(X)} C$ for $C \subset L_n(X)$. Recall $L(X) = \varinjlim L_n(X)$ (Proposition 7.11.5). Then, $V = \bigcup_{n \in \mathbb{N}} V_n$ is a closed neighborhood of $X \cup \operatorname{cl} W$ in $L(X)$ by (1) and (2). Since $L(Y) = \varinjlim L_n(Y)$ and $f_\natural^{-1}(L_n(X)) \supset L_n(Y)$, it follows from Proposition 2.8.2 and (2) that

$$f_\natural^{-1}(V) = \varinjlim \left(f_\natural^{-1}(V) \cap L_n(Y) \right) = \varinjlim \left(f_\natural^{-1}(V_n) \cap L_n(Y) \right),$$

which implies $f_\natural^{-1}(V) = \varinjlim f_\natural^{-1}(V_n)$. By (4), we have the map $h_\infty : f_\natural^{-1}(V) \to \mathbf{S}^m$ such that $h_\infty | f_\natural^{-1}(V_n) = h_n | f_\natural^{-1}(V_n)$, hence $h_\infty | f_\natural^{-1}(\operatorname{cl} W) = \bar{g} f_\natural | f_\natural^{-1}(\operatorname{cl} W)$ by (3). Thus, we have $U = \operatorname{int} V$ and $h = h_\infty | f_\natural^{-1}(U)$, which are needed in the claim.

(*The First Step*) Because $\dim Y \leqslant m$, the map $\bar{g} f | Y \cap f_\natural^{-1}(\operatorname{cl} W)$ extends over Y (Theorem 5.2.4), hence the map $\bar{g} f_\natural | f_\natural^{-1}(\operatorname{cl} W)$ extends to a map $h_0 : Y \cup f_\natural^{-1}(\operatorname{cl} W) \to \mathbf{S}^m$. Since $\mathbf{S}^m$ is an ANE for normal spaces (Theorem 5.1.6 (2)), h_0 extends to a map $\bar{h}_0 : G_0 \to \mathbf{S}^m$ from an open neighborhood G_0 of $Y \cup f_\natural^{-1}(\operatorname{cl} W)$ in $L(Y)$. Consider the following diagram:



Recall that $\varphi_1 : S_1(X) \to \operatorname{Comp}(S_1(Y))$ is the continuous set-valued function defined by $\varphi_1(z) = f_\natural^{-1}(z) \cap S_1(Y)$ (Lemma 7.12.5). Then, we have the following

open set in $S_1(X)$:

$$M_1 = \big\{ z \in S_1(X) \bigm| \varphi_1(z) \subset G_0 \big\},$$

which is open in $L_1(X)$ because $S_1(X)$ is open in $L_1(X)$. Observe that

$$\varphi_1(x) = f^{-1}(x) \subset Y \subset G_0 \quad \text{for } x \in X \text{ and}$$
$$\varphi_1(z) \subset f_\natural^{-1}(z) \subset f_\natural^{-1}(\mathrm{cl}\, W) \subset G_0 \quad \text{for } z \in L_1(X) \cap \mathrm{cl}\, W,$$

hence $L_1(X) \cap (X \cup \mathrm{cl}\, W) \subset M_1$. Therefore, we can find closed sets $V_1 \subset V_1' \subset L_1(X)$ such that

$$L_1(X) \cap (X \cup \mathrm{cl}\, W) \subset \mathrm{int}_1 V_1 \subset V_1 \subset \mathrm{int}_1 V_1' \subset V_1' \subset M_1.$$

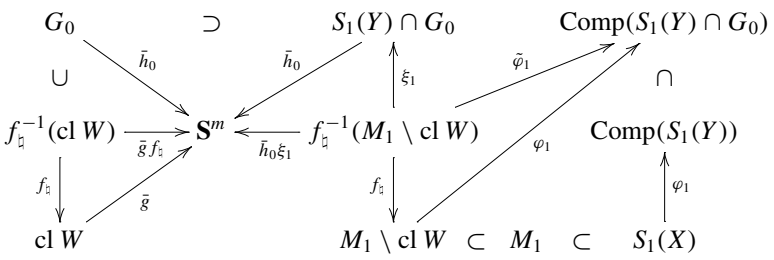
Since $f_\natural$ is continuous, we have the continuous set-valued function

$$\tilde{\varphi}_1 = \varphi_1 f_\natural | f_\natural^{-1}(M_1) : f_\natural^{-1}(M_1) \to \mathrm{Comp}(S_1(Y) \cap G_0),$$

where $S_1(Y) \cap G_0$ is an ANR because so is $S_1(Y)$ (Proposition 7.11.1 (6)). For each $y \in f_\natural^{-1}(M_1)$, $\tilde{\varphi}_1(y) = \varphi_1(f_\natural(y))$ is a cell-like compactum by Lemma 7.12.5. Since $S_1(Y)$ is metrizable, we can apply Lemma 7.11.8 (3), (4) to obtain $d_1 \in \mathfrak{M}_Y$ such that $d_1 \geqslant d_Y$, $d_1 | S_1(Y)^2 \in \mathrm{Metr}(S_1(Y))$, and $\tilde{\varphi}_1$ is d_1-continuous, where d_Y is the metric obtained by Lemma 7.12.4. Note that $f_\natural^{-1}(\mathrm{cl}\, W)$ is d_1-closed and $\bar{g} f_\natural | f_\natural^{-1}(\mathrm{cl}\, W)$ is d_1-continuous. Since $L(X)$ is perfectly normal (Proposition 7.11.1 (3)), $f_\natural^{-1}(M_1 \setminus \mathrm{cl}\, W) = f_\natural^{-1}(M_1) \setminus f_\natural^{-1}(\mathrm{cl}\, W)$ is F_σ in $L(Y)$. Since $L(Y)$ is a countable union of finite-dimensional compacta (Proposition 7.11.1 (4)), so is $f_\natural^{-1}(M_1 \setminus \mathrm{cl}\, W)$. Note that every compact subspace of $L(Y)$ is a subspace of $(L(Y), d_1)$. Then, it follows that $(f_\natural^{-1}(M_1 \setminus \mathrm{cl}\, W), d_1)$ is also a countable union of finite-dimensional compacta. We can apply Haver's Near-Selection Theorem 7.6.1 to obtain a map

$$\xi_1 : \big(f_\natural^{-1}(M_1 \setminus \mathrm{cl}\, W), d_1\big) \to S_1(Y) \cap G_0$$

such that $d_1(\xi_1(y), \tilde{\varphi}_1(y)) < d_1(y, f_\natural^{-1}(\mathrm{cl}\, W))$ for every $y \in f_\natural^{-1}(M_1 \setminus \mathrm{cl}\, W)$, where it should be noticed that $\xi_1 : f_\natural^{-1}(M_1 \cup \mathrm{cl}\, W) \to S_1(Y) \cap G_0$ is also continuous.



Then, we can define $h_1 : f_\natural^{-1}(V_1' \cup \operatorname{cl} W) \to \mathbf{S}^m$ as follows:

$$h_1(y) = \begin{cases} \bar{g} f_\natural(y) & \text{if } \ y \in f_\natural^{-1}(\operatorname{cl} W), \\ \bar{h}_0 \xi_1(y) & \text{if } \ y \in f_\natural^{-1}(V_1' \setminus \operatorname{cl} W). \end{cases}$$

We will verify the continuity of h_1. Since $f_\natural^{-1}(V_1' \setminus \operatorname{cl} W)$ is an open set in $f_\natural^{-1}(V_1' \cup \operatorname{cl} W)$, we have to show the continuity of h_1 at each $y \in f_\natural^{-1}(\operatorname{cl} W)$. To this end, we will show that h_1 is d_1-continuous at y because $\mathrm{id} : L(Y) \to (L(Y), d_1)$ is continuous. Since $h_1 | f_\natural^{-1}(\operatorname{cl} W) = \bar{g} f_\natural | f_\natural^{-1}(\operatorname{cl} W)$ is d_1-continuous, it suffices to prove that

$$\lim_{i\to\infty} d_1(y_i, y) = 0, \ \ y_i \in f_\natural^{-1}(V_1' \setminus \operatorname{cl} W) \ \Rightarrow \ \lim_{i\to\infty} \bar{h}_0 \xi_1(y_i) = \bar{g} f_\natural(y).$$

Since $\tilde{\varphi}_1 : f_\natural^{-1}(M_1) \to \operatorname{Comp}(S_1(Y) \cap G_0)$ is u.s.c. and

$$d_1(\xi_1(y_i), \tilde{\varphi}_1(y_i)) < d_1(y_i, f_\natural^{-1}(\operatorname{cl} W)) \leqslant d_1(y_i, y),$$

it follows that $\{\xi_1(y_i) \mid i \in \mathbb{N}\} \cup \tilde{\varphi}_1(y)$ is compact, hence $(\xi_1(y_i))_{i\in\mathbb{N}}$ has a convergent subsequence $(\xi_1(y_{i_j}))_{j\in\mathbb{N}}$. Since $d_1(\xi_1(y_{i_j}), \tilde{\varphi}_1(y)) \to 0$, $(\xi_1(y_{i_j}))_{j\in\mathbb{N}}$ converges to some

$$z \in \tilde{\varphi}_1(y) = \varphi_1(f_\natural(y)) = f_\natural^{-1}(f_\natural(y)) \cap S_1(Y) \subset f_\natural^{-1}(\operatorname{cl} W).$$

Then, $(\bar{h}_0 \xi_1(y_{i_j}))_{j\in\mathbb{N}}$ converges to $\bar{h}_0(z) = \bar{g} f_\natural(z) = \bar{g} f_\natural(y)$. By the same argument, any subsequence of $(\bar{h}_0 \xi_1(y_i))_{i\in\mathbb{N}}$ has a subsequence converging to $\bar{g} f_\natural(y)$. This means that $\lim_{i\to\infty} \bar{h}_0 \xi_1(y_i) = \bar{g} f_\natural(y)$.

(*The Inductive Step*) Assume that V_{n-1}, V_{n-1}', and h_{n-1} have been constructed. Because $\mathbf{S}^m$ is an ANE for normal spaces, h_{n-1} extends to a map $\bar{h}_{n-1} : G_{n-1} \to \mathbf{S}^m$ from an open neighborhood G_{n-1} of $f_\natural^{-1}(V_{n-1}' \cup \operatorname{cl} W)$ in $L(Y)$. Recall that $\varphi_n : S_n(X) \to \operatorname{Comp}(S_n(Y))$ is the continuous set-valued function defined by $\varphi_n(z) = f_\natural^{-1}(z) \cap S_n(Y)$ (Lemma 7.12.5). Let

$$M_n = \left\{ z \in S_n(X) \;\middle|\; \varphi_n(z) \subset G_{n-1} \right\} \ \text{ and } \ N_n = M_n \cup \operatorname{int}_{n-1} V_{n-1}'.$$

Then, M_n is open in $S_n(X)$, hence in $L_n(X)$.

We will show that N_n is open in $L_n(X)$. Suppose that N_n is not open in $L_n(X)$. Then, $N_n \cap L_n(X, m)$ is not open in $L_n(X, m)$ for some $m \in \mathbb{N}$ because $L_n(X) = \varinjlim L_n(X, m)$ (Proposition 7.11.4). Since X is compact metrizable, $L_n(X, m) = \overrightarrow{\gamma_n}(X^n \times m\Diamond^n)$ is also compact metrizable. Then, there is some $z \in N_n \cap L_n(X, m)$ that is the limit of $z_i \in L_n(X, m) \setminus N_n$. Since M_n is open in $L_n(X)$, it follows that $z \in \operatorname{int}_{n-1} V_{n-1}' \subset L_{n-1}(X)$. Then, $L_{n-1}(X)$ contains only finitely many z_i. Otherwise, infinitely many z_i would be contained in $\operatorname{int}_{n-1} V_{n-1}' \subset N_n$. Consequently, we may assume that $z_i \in S_n(X) = L_n(X) \setminus L_{n-1}(X)$ for every $i \in \mathbb{N}$. Since $z_i \notin M_n$,

we have $y_i \in \varphi_n(z_i) \setminus G_{n-1}$. Recall that $\varphi_n(z_i) = f_\natural^{-1}(z_i) \cap S_n(Y)$. Then,

$$f_\natural(y_i) = z_i \in L_n(X, m) \cap S_n(X) = \gamma_n(X^n \times m\Diamond^n) \cap S_n(X).$$

Since $y_i \in S_n(Y)$, it follows that $y_i \in L_n(Y, m) = \gamma_n(Y^n \times m\Diamond^n)$. Because $L_n(Y, m)$ is compact metrizable, we may assume that $\lim_{i\to\infty} y_i = y \in L_n(Y, m)$. Then,

$$f_\natural(y) = \lim_{i\to\infty} f_\natural(y_i) = \lim_{i\to\infty} z_i = z,$$

hence $y \in f_\natural^{-1}(z) \subset f_\natural^{-1}(V'_{n-1}) \subset G_{n-1}$. Therefore, $y_i \in G_{n-1}$ for sufficiently large $i \in \mathbb{N}$, which is a contradiction. Thus, N_n is open in $L_n(X)$.

For each $z \in S_n(X) \cap \operatorname{cl} W$, we have

$$\varphi_n(z) \subset f_\natural^{-1}(z) \subset f_\natural^{-1}(\operatorname{cl} W) \subset G_{n-1}.$$

Hence, $S_n(X) \cap \operatorname{cl} W \subset M_n$. Then, it follows that

$$\begin{aligned} L_n(X) \cap (X \cup \operatorname{cl} W) &= (S_n(X) \cap \operatorname{cl} W) \cup \big(L_{n-1}(X) \cap (X \cup \operatorname{cl} W)\big) \\ &\subset M_n \cup \operatorname{int}_{n-1} V'_{n-1} = N_n. \end{aligned}$$

Since $L(X)$ is hereditarily normal (= completely normal) (Proposition 7.11.1 (3)), so is $L_n(X)$, hence, we can find an open set $\widetilde{V}$ in $L_n(X)$ such that:

(5) $\big(L_n(X) \cap (X \cup \operatorname{cl} W)\big) \cup \operatorname{int}_{n-1} V_{n-1} \subset \widetilde{V}$ and
(6) $\operatorname{cl} \widetilde{V} \cap \big((L_{n-1}(X) \setminus V_{n-1}) \cup (L_n(X) \setminus N_n)\big) = \emptyset$.

Let $V_n = V_{n-1} \cup \operatorname{cl} \widetilde{V} \subset N_n$. Then, it follows from (6) that

$$L_{n-1}(X) \cap V_n \subset V_{n-1} \subset L_{n-1}(X) \cap V_n,$$

hence $L_{n-1}(X) \cap V_n = V_{n-1}$. Moreover, by (5),

$$\begin{aligned} \operatorname{int}_{n-1} V_{n-1} &\subset L_{n-1}(X) \cap \widetilde{V} \subset L_{n-1}(X) \cap \operatorname{int}_n V_n \\ &\subset \operatorname{int}_{n-1}(L_{n-1}(X) \cap V_n) = \operatorname{int}_{n-1} V_{n-1}. \end{aligned}$$

Thus, we have $L_{n-1}(X) \cap \operatorname{int}_n V_n = \operatorname{int}_{n-1} V_{n-1}$. Choose a closed set $V'_n \subset L_n(X)$ so that $V_n \subset \operatorname{int}_n V'_n \subset V'_n \subset N_n$. Thus, we have obtained V_n and V'_n satisfying conditions (1) and (2) (Fig. 7.6).

Similar to $\tilde{\varphi}_1$, the following is a continuous set-valued function:

$$\tilde{\varphi}_n = \varphi_n f_\natural | f_\natural^{-1}(M_n) : f_\natural^{-1}(M_n) \to \operatorname{Comp}(S_n(Y) \cap G_{n-1}),$$

where $S_n(Y) \cap G_{n-1}$ is an ANR because so is $S_n(Y)$ (Proposition 7.11.1 (6)). For each $y \in f_\natural^{-1}(M_1)$, $\tilde{\varphi}_1(y) = \varphi_1(f_\natural(y))$ is a cell-like compactum by Lemma 7.12.5. In the same way as d_1, we apply Lemma 7.11.8 (3), (4) to obtain $d_n \in \mathfrak{M}_Y$ such that $d_n \geqslant d_Y$,

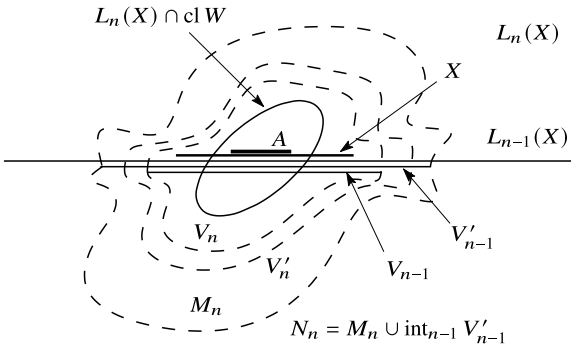


Fig. 7.6 M_n, N_n, and V_n

$d_n|S_n(Y)^2 \in \operatorname{Metr}(S_n(Y))$, and $\bar{\varphi}_n$ is d_n-continuous. Then, $f_\natural^{-1}(L_{n-1}(X) \cup \operatorname{cl} W)$ is d_n-closed and $\bar{g} f_\natural | f_\natural^{-1}(\operatorname{cl} W)$ is d_n-continuous. Since $L(X)$ is perfectly normal (Proposition 7.11.1 (3)), $f_\natural^{-1}(M_n \setminus \operatorname{cl} W)$ is F_σ in $L(Y)$. Then, $f_\natural^{-1}(M_n \setminus \operatorname{cl} W)$ is a countable union of finite-dimensional compacta because so is $L(Y)$ (Proposition 7.11.1 (4)). Since every compact subspace of $L(Y)$ is a subspace of $(L(Y), d_1)$, it follows that $(f_\natural^{-1}(M_n \setminus \operatorname{cl} W), d_1)$ is a countable union of finite-dimensional compacta.

Due to Lemma 7.11.6 (ii), $S_n(Y) \cap L_n(Y, m+1)$ is a neighborhood of $S_n(Y) \cap L_n(Y, m)$ in $S_n(Y)$ for each $m \in \mathbb{N}$. Then, we have an l.s.c. function $\zeta_n : S_n(Y) \to (0, \infty)$ defined by

$$\zeta_n(y) = \tfrac{1}{2} d_n(y, S_n(Y) \setminus L_n(Y, m+1)) \quad \text{if } y \in S_n(Y) \cap L_n(Y, m) \setminus L_n(Y, m-1),\ m \in \mathbb{N}.$$

Indeed, assume that $y \in S_n(Y) \cap L_n(Y, m) \setminus L_n(Y, m-1)$ and $\zeta_n(y) > t$, i.e., $\frac{1}{2} d_n(y, S_n(Y) \setminus L_n(Y, m+1)) > t$. Then, the following V_y is a neighborhood of y in $S_n(Y)$:

$$V_y = \left\{ y' \in S_n(Y) \;\middle|\; \tfrac{1}{2} d_n(y', S_n(Y) \setminus L_n(Y, m+1)) > t \right\} \cap L_n(Y, m+1) \setminus L_n(Y, m-1).$$

For each $y' \in V_y$, if $y' \in L_n(Y, m)$ then

$$\zeta_n(y') = \tfrac{1}{2} d_n\big(y', S_n(Y) \setminus L_n(Y, m+1)\big) > t,$$

and if $y' \notin L_n(Y, m)$ then

$$\begin{aligned} \zeta_n(y') &= \tfrac{1}{2} d_n(y', S_n(Y) \setminus L_n(Y, m+2)) \\ &\geqslant \tfrac{1}{2} d_n(y', S_n(Y) \setminus L_n(Y, m+1)) > t. \end{aligned}$$

Applying Theorem 2.7.8, we have a map $\eta_n : S_n(Y) \to (0, 1)$ such that $0 < \eta_n(y) < \zeta_n(y)$. Then, it follows that

$$\eta_n(y) < \tfrac{1}{2} d_n(y, S_n(Y) \setminus L_n(Y, m+1)) \quad \text{for each } y \in S_n(Y) \cap L_n(Y, m) \text{ and } m \in \mathbb{N}.$$

Because each $\tilde{\varphi}_n(y)$ is compact, we can define $\bar{\eta}_n : f_\natural^{-1}(M_n \setminus \operatorname{cl} W) \to (0, 1)$ by $\bar{\eta}_n(y) = \min \eta_n \tilde{\varphi}_n(y)$. To prove that $\bar{\eta}_n$ is d_n-continuous, observe that the following set-valued function is d_n-continuous:

$$f_\natural^{-1}(M_n \setminus \operatorname{cl} W) \ni y \mapsto \eta_n \tilde{\varphi}_n(y) \in \operatorname{Comp}((0, 1)).$$

For each $y \in f_\natural^{-1}(M_n \setminus \operatorname{cl} W)$ and $\varepsilon > 0$, we have $\delta > 0$ such that

$$d_n(y, y') < \delta \Rightarrow d_H(\eta_n \tilde{\varphi}_n(y), \eta_n \tilde{\varphi}_n(y')) < \varepsilon,$$

where d_H is the Hausdorff metric on $\operatorname{Comp}((0, 1))$ induced by the metric $d(t, t') = |t - t'|$. When $d_H(\eta_n \tilde{\varphi}_n(y), \eta_n \tilde{\varphi}_n(y')) < \varepsilon$, we can find $t \in \eta_n \tilde{\varphi}_n(y)$ and $t' \in \eta_n \tilde{\varphi}_n(y')$ such that $|t - \bar{\eta}_n(y')| < \varepsilon$ and $|t' - \bar{\eta}_n(y)| < \varepsilon$. Then,

$$\bar{\eta}_n(y) \leqslant t < \bar{\eta}_n(y') + \varepsilon \quad \text{and} \quad \bar{\eta}_n(y') \leqslant t' < \bar{\eta}_n(y) + \varepsilon,$$

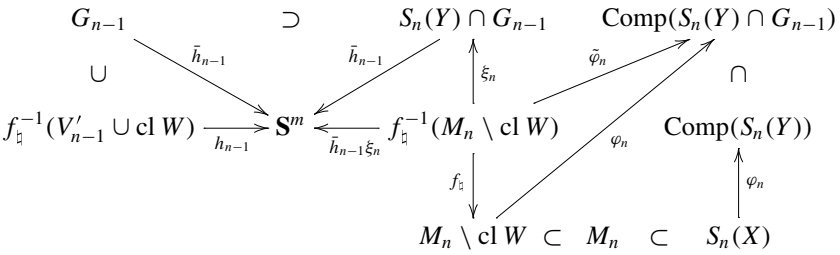
which means that $|\bar{\eta}_n(y) - \bar{\eta}_n(y')| < \varepsilon$.

As in the first step, we apply Haver's Near-Selection Theorem 7.6.1 to obtain a map

$$\xi_n : \left(f_\natural^{-1}(M_n \setminus \operatorname{cl} W), d_n\right) \to S_n(Y) \cap G_{n-1}$$

satisfying

$$d_n(\xi_n(y), \tilde{\varphi}_n(y)) < \min\left\{d_n(y, f_\natural^{-1}(L_{n-1}(X) \cup \operatorname{cl} W)),\ \bar{\eta}_n(y)\right\}.$$



Then, $\xi_n : f_\natural^{-1}(M_n \setminus \operatorname{cl} W) \to S_n(Y) \cap G_{n-1}$ is also continuous. Here, it should be remarked that the distance between $\bar{h}_{n-1}\xi_n(y)$ and $\bar{h}_{n-1}(y)$ converges to 0 as y tends to $f_\natural^{-1}(\operatorname{cl} W)$ but it does not always decrease as y tends to $f_\natural^{-1}(V'_{n-1} \setminus \operatorname{cl} W)$.

We will show that the line segment from $\xi_n(y)$ to y is contained in G_{n-1} if y is close to $f_\natural^{-1}(V'_{n-1} \setminus \operatorname{cl} W)$. To see this, let

$$R_n = \left\{ y \in f_\natural^{-1}(M_n \setminus \operatorname{cl} W) \;\middle|\; \langle y, \xi_n(y) \rangle \subset G_{n-1} \right\} \quad \text{and}$$
$$T_n = R_n \cup f_\natural^{-1}(\operatorname{int}_{n-1} V'_{n-1} \cup \operatorname{cl} W).$$

Note that $f_\natural^{-1}(M_n \setminus \operatorname{cl} W)$ is open in $f_\natural^{-1}(L_n(X) \cup \operatorname{cl} W)$. Since G_{n-1} is open in $L(Y)$ and ξ_n is continuous, it follows that R_n is open in $f_\natural^{-1}(M_n \setminus \operatorname{cl} W)$, hence so in $f_\natural^{-1}(L_n(X) \cup \operatorname{cl} W)$.

We now show that T_n is open in $f_\natural^{-1}(L_n(X) \cup \operatorname{cl} W)$. Assume that T_n is not open in $f_\natural^{-1}(L_n(X) \cup \operatorname{cl} W)$. Because $f_\natural^{-1}(L_n(X) \cup \operatorname{cl} W)$ is closed in $L(Y) = \varinjlim L_m(Y, m)$ (Proposition 7.11.3), we have

$$f_\natural^{-1}(L_n(X) \cup \operatorname{cl} W) = \varinjlim \left(f_\natural^{-1}(L_n(X) \cup \operatorname{cl} W) \cap L_m(Y, m) \right).$$

Then, $T_n \cap L_m(Y, m)$ is not open in $f_\natural^{-1}(L_n(X) \cup \operatorname{cl} W) \cap L_m(Y, m)$ for some $m \in \mathbb{N}$, hence we have $z \in T_n \cap L_m(Y, m)$ which is the limit of

$$z_i \in f_\natural^{-1}(L_n(X) \cup \operatorname{cl} W) \cap L_m(Y, m) \setminus T_n, \;\; i \in \mathbb{N}.$$

Since R_n is open in $f_\natural^{-1}(L_n(X) \cup \operatorname{cl} W)$ and $z_i \notin R_n$, $i \in \mathbb{N}$, it follows that $z \in f_\natural^{-1}(\operatorname{int}_{n-1} V'_{n-1} \cup \operatorname{cl} W)$. Note that N_n is open in $L_n(X)$, $L_n(X) \cap \operatorname{cl} W \subset N_n$, and $\operatorname{int}_{n-1} V'_{n-1} \subset N_n$, hence $N_n \cup \operatorname{cl} W$ is a neighborhood of $f_\natural(z)$ in $L_n(X) \cup \operatorname{cl} W$. Therefore, we may assume that $z_i \in f_\natural^{-1}(N_n \cup \operatorname{cl} W)$ for every $i \in \mathbb{N}$. Then, $z_i \in f_\natural^{-1}(M_n \setminus \operatorname{cl} W)$ because $z_i \notin f_\natural^{-1}(\operatorname{int}_{n-1} V'_{n-1} \cup \operatorname{cl} W)$. Since $z_i \notin R_n$, we have $\langle z_i, \xi_n(z_i) \rangle \not\subset G_{n-1}$.

On the other hand, since $z_i \in L_m(Y, m)$, it follows that $f_\natural(z_i) \in L_m(X, m)$. Note that $f_\natural(z_i) \in M_n \subset S_n(X)$. Then, $f_\natural(z_i) \in S_n(X) \cap L_m(X, m)$. For $z' = \sum_{j=1}^n t_j y_j \in S_n(Y)$, by Lemmas 7.11.2 and 7.11.6 (i),

$$f_\natural(z') = \sum_{j=1}^n t_j f(y_j) \in S_n(X) \cap L_m(X, m) = S_n(X) \cap L_n(X, m)$$
$$\Rightarrow z' \in L_n(Y, m).$$

Thus, it follows that

$$\tilde{\varphi}_n(z_i) = \varphi_n(f_\natural(z_i)) = f_\natural^{-1}(f_\natural(z_i)) \cap S_n(Y) \subset S_n(Y) \cap L_n(Y, m),$$

hence $\xi_n(z_i) \in L_n(Y, m+1)$ because

$$\begin{aligned} d_n(\xi_n(z_i), \tilde{\varphi}_n(z_i)) &< \bar{\eta}_n(z_i) = \min \eta_n \tilde{\varphi}_n(z_i) \\ &< \tfrac{1}{2} \min \left\{ d_n(y, S_n(Y) \setminus L_n(Y, m+1)) \;\middle|\; y \in \tilde{\varphi}_n(z_i) \right\}. \end{aligned}$$

Since $L_n(Y, m+1)$ is compact metrizable, so is $\mathrm{Comp}(L_n(Y, m+1))$ (Theorem 2.12.8). Then, we may assume that $\xi_n(z_i)$ converges to some $a \in L_n(Y, m+1)$ and also $\tilde{\varphi}_n(z_i)$ converges to some $K \in \mathrm{Comp}(L_n(Y, m+1))$. Since $f_\natural(\tilde{\varphi}_n(z_i)) = f_\natural(z_i)$ converges to $f_\natural(z)$, it follows that $K \subset f_\natural^{-1}(f_\natural(z))$. Observe that

$$\begin{aligned} \lim_{i\to\infty} d_n(\xi_n(z_i), \tilde{\varphi}_n(z_i)) &\leqslant \lim_{i\to\infty} d_n\big(z_i, f_\natural^{-1}(L_{n-1}(X) \cup \mathrm{cl}\, W)\big) \\ &\leqslant \lim_{i\to\infty} d_n(z_i, z) = 0. \end{aligned}$$

Hence, $\lim_{i\to\infty} d_n(a, \tilde{\varphi}_n(z_i)) = 0$. Since $L_n(Y, m+1)$ is compact, d_n is admissible on $L_n(Y, m+1)$, hence $a \in K \subset f_\natural^{-1}(f_\natural(z))$. By the linearity of $f_\natural$, $f_\natural^{-1}(f_\natural(z))$ is a flat, which is convex. Then, it follows that

$$\langle z, a \rangle \subset f_\natural^{-1}(f_\natural(z)) \subset f_\natural^{-1}(V'_{n-1} \cup \mathrm{cl}\, W) \subset G_{n-1}.$$

Since $\langle z_i, \xi_n(z_i) \rangle$ converges to $\langle z, a \rangle$ in $\mathrm{Comp}(L_n(Y, m+1))$, $\langle z_i, \xi_n(z_i) \rangle \subset G_{n-1}$ for sufficiently large $i \in \mathbb{N}$, which is a contradiction. Consequently, T_n is open in $f_\natural^{-1}(L_n(X) \cup \mathrm{cl}\, W)$.

Now, since $V'_n \subset N_n$ and $N_n \cap L_{n-1}(X) = \mathrm{int}_{n-1} V'_{n-1}$, it follows that

$$\begin{aligned} f_\natural^{-1}((V'_n \cap L_{n-1}(X)) \cup \mathrm{cl}\, W) &\subset f_\natural^{-1}((N_n \cap L_{n-1}(X)) \cup \mathrm{cl}\, W) \\ &= f_\natural^{-1}(\mathrm{int}_{n-1} V'_{n-1} \cup \mathrm{cl}\, W) \subset T_n. \end{aligned}$$

Then, choose an open set B in $f_\natural^{-1}(L_n(X) \cup \mathrm{cl}\, W)$ so that

$$f_\natural^{-1}((V'_n \cap L_{n-1}(X)) \cup \mathrm{cl}\, W) \subset B \subset \mathrm{cl}\, B \subset T_n$$

and let $\beta : f_\natural^{-1}(L_n(X) \cup \mathrm{cl}\, W) \to \mathbf{I}$ be a Urysohn map with

$$\beta(\mathrm{cl}\, B) = 0 \;\text{ and }\; \beta(f_\natural^{-1}(L_n(X) \cup \mathrm{cl}\, W) \setminus T_n) = 1.$$

From the definition of T_n, it follows that

$$(1 - \beta(y))y + \beta(y)\xi_n(y) \in G_{n-1} \;\text{ for } y \in f_\natural^{-1}(M_n \setminus \mathrm{cl}\, W).$$

On the other hand,

$$\begin{aligned} f_\natural^{-1}(V_n' \cup \operatorname{cl} W) \setminus B &\subset f_\natural^{-1}(V_n' \cup \operatorname{cl} W) \setminus f_\natural^{-1}((V_n' \cap L_{n-1}(X)) \cup \operatorname{cl} W) \\ &\subset f_\natural^{-1}((N_n \setminus L_{n-1}(X)) \setminus \operatorname{cl} W) \setminus f_\natural^{-1}(V_{n-1}' \cup \operatorname{cl} W) \\ &= f_\natural^{-1}(M_n \setminus \operatorname{cl} W). \end{aligned}$$

Then, we can define a map $h_n : f_\natural^{-1}(V_n' \cup \operatorname{cl} W) \to \mathbf{S}^m$ by

$$h_n(y) = \begin{cases} \bar{h}_{n-1}(y) & \text{if } y \in \operatorname{cl} B, \\ \bar{h}_{n-1}((1-\beta(y))y + \beta(y)\xi_n(y)) & \text{if } y \notin B. \end{cases}$$

Because $\bar{h}_{n-1}$ is an extension of h_{n-1}, the map h_n satisfies conditions (3) and (4). This completes the proof. □

Notes for Chap. 7

As mentioned at the beginning of this chapter, the concept of cell-like maps is profoundly related to Shape Theory and Decomposition Theory. For insight into these theories, refer to the following textbooks:

- S. Mardešić and J. Segal, *Shape Theory*, North-Holland Math. Library **26** (Elsevier Sci. B.V., Amsterdam, 1982)
- A. Chigogidze, *Inverse Spectra*, North-Holland Math. Library **53** (Elsevier Sci. B.V., Amsterdam, 1996)
- R.J. Daverman, *Decomposition of Manifolds*, Pure and Appl. Math. **124** (Academic Press, Inc., Orlando, 1986)

Shape Theory was founded by K. Borsuk in 1968. As a textbook on Shape Theory, his own book gives a good introduction, as is the above book by Mardešić and Segal:

- K. Borsuk, *Theory of Shape*, Monog. Mat. **59** (Polish Sci. Publ., Warsaw, 1975)

To study cell-like maps, a background in Algebraic Topology is required. There exist many textbooks on Algebraic Topology. Among them, we recommend the following:

- E. Spanier, *Algebraic Topology* (McGraw-Hill, New York, 1966)
- S.-T. Hu, *Homotopy Theory* (Academic Press, Inc., New York, 1959)
- A. Hatcher, *Algebraic Topology* (Cambridge Univ. Press, Cambridge, 2002)

We used in Sect. 7.7 the Toruńczyk characterization of Hilbert cube (Theorem 7.7.1). For the proof, one can refer to the following books:

- J. van Mill, *Infinite-Dimensional Topology, Prerequisites and Introduction*, North-Holland Math. Library **43** (Elsevier Sci. Publ. B.V., Amsterdam, 1989)
- K. Sakai, *Topology of Infinite-Dimensional Manifolds*, Springer Monog. Math. (Springer Nature, Singapore, 2020)

The paper [23] gives a good survey of cell-like maps up to the mid-70 s. A compactum A in an n-manifold M is called *cellular* if A has an arbitrarily small neighborhood in M that is homeomorphic to $\mathbf{B}^n$. In other words, A can be written as $A = \bigcap_{i\in\mathbb{N}} B_i$, where $B_i \approx \mathbf{B}^n$ and $B_{i+1} \subset \operatorname{int} B_i$. Every cellular compactum is cell-like but the converse does not hold. The Whitehead continuum is an example of non-cellular cell-like compacta (see Daverman's book, pp. 68–69).

The concept of the (n-)soft map was introduced by Shchepin in [29] and generalized to the (polyhedrally) approximately (n-)soft map in [30]. These are discussed in Chigogidze's book above. The 0-Dimensional Selection Theorem 7.2.4 was established by Michael [24, 25]. In [19], Kozlowski introduced the concept of the local n-connection and proved Theorem 7.3.8.

As noted in Remark 7.4.1, Theorem 7.4.3 is valid for a (non-metrizable) polyhedron X and an ANR Y. More generally, it is valid if X and Y are ANRs for stratifiable spaces [4, Proposition 2], where a stratifiable space was first introduced by Ceder as an M_3-space in [7] and later renamed by Borges in [5]. Every metrizable space and every CW complex are stratifiable spaces.

The Toruńczyk characterization of the Hilbert cube $\boldsymbol{Q}$ (Theorem 7.7.1) was established in [33]. The result of Adams used in Sect. 7.7 appeared in [1]. It also follows from the work of Toda [32]. The Taylor example in Theorem 7.7.5 was constructed in [31] (cf. [28]). The example of Theorem 7.7.8 is due to Keesling [17]. Using the Taylor example, many counter-examples are constructed in ANR Theory. For example, in [26], J. van Mill constructed a map $f : \boldsymbol{Q} \to Y$ of the Hilbert cube $\boldsymbol{Q}$ onto a non-AR compactum Y such that $f^{-1}(y) \approx \boldsymbol{Q}$ for every $y \in Y$, and in [27] he also constructed a separable metrizable space X such that, for each compact set A in an arbitrary metrizable space Y, every map $f : A \to X$ extends over Y but X is not an ANR.

Commonly, Eilenberg–MacLane complexes are constructed as CW complexes, but we constructed them as simplicial complexes in Sect. 7.8. Eilenberg–MacLane complexes was first introduced in 1943 [15] ([16] in detail). Edwards–Walsh complex was first constructed in [35] (cf. notes before Lemma 8 in [13]).

In Sect. 7.9, we defined the cohomological dimension by using Eilenberg–MacLane complexes. As mentioned in Footnote 16, this is different from the usual definition. There are good surveys on Cohomological Dimension Theory [11, 12, 18, 22]. Theorem 7.9.1 was originally proved by Cohen [8] but the proof presented here is based on an idea of S. Ferry, which appeared in Appendix A in [35]. Theorem 7.9.6 was established by Alexandroff [3]. The proof of Theorem 7.9.13 (7.9.11) appeared in [35], and is based on Kozlowski's technique in [20].

Alexandroff's Problem, mentioned in Sect. 7.10, was posed in [2]. Theorems 7.10.1 and 7.10.2 are due to Edwards [14] (see [35]). The existence of dimension-raising cell-like maps was first shown by A. Dranishnikov [9, 10]. Lemma 7.10.8 and Theorem 7.10.11 were established by Dydak and Walsh [13]. In [21], Kozlowski and Walsh proved that the image of a cell-like map of a compact 3-dimensional manifold is always finite-dimensional. It remains unsolved whether there exists a cell-like map of $\mathbf{I}^4$ onto an infinite-dimensional compactum. Walsh's Theorem 7.10.13 on approximations by open maps was proved in [34].

The example of Theorem 7.12.1 (i.e., Theorem 7.12.3) was constructed by R. Cauty [6]. Sections 7.11 and 7.12 were written based on his paper [6].

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Chapter 8
Appendix: Homotopy Groups and Čech Cohomology Groups



As mentioned in the Preface, knowledge of homotopy groups is not required to read Sect. 4.20 but the second half of Chap. 7 requires basic results for homotopy groups. We review here homotopy groups and characterize n-equivalences and weak homotopy equivalences in terms of homotopy groups. Moreover, we also prove some theorems which are required in the second half of Chap. 7. In Sect. 7.9, avoiding the use of cohomology groups, we defined the cohomological dimension $\mathrm{cd}_{\mathbb{Z}} X$ by using the Eilenberg–MacLane complex $K(\mathbb{Z}, n)$. However, as noted in Footnote 16 of the previous chapter (Sect. 7.9), $\mathrm{cd}_{\mathbb{Z}} X$ is originally defined in terms of the cohomology groups. Here, introducing a group structure of $[X, |K(\mathbb{Z}, n)|]$, we define a cohomology theory satisfying the Eilenberg–Steenrod Axioms, which is isomorphic to Čech cohomology.

8.1 Definition and Basic Properties of Homotopy Groups

In this section, we review several definitions related to the homotopy groups together with their basic results, and then we show that $\pi_n(\Omega(X, x_0), c_{x_0})$ is isomorphic to $\pi_{n+1}(X, x_0)$. Some of them are stated without proof. For details, refer to any textbook on Homotopy Theory or Algebraic Topology.

For a pair of paths $\alpha, \beta : \mathbf{I} \to X$ with $\alpha(1) = \beta(0)$, we can define the **join** $\alpha * \beta : \mathbf{I} \to X$ as follows:

$$\alpha * \beta(t) = \begin{cases} \alpha(2t) & \text{if } 0 \leqslant t \leqslant 1/2, \\ \beta(2t-1) & \text{if } 1/2 \leqslant t \leqslant 1. \end{cases}$$

The **inverse** $\alpha^{\leftarrow} : \mathbf{I} \to X$ of a path α is defined by $\alpha^{\leftarrow}(t) = \alpha(1-t)$ for each $t \in \mathbf{I}$. For another pair of paths α', β', the following holds:

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K. Sakai, *Geometric Aspects of General Topology*, Springer Asia Pacific Mathematics Series 2, https://doi.org/10.1007/978-981-96-8570-7_8

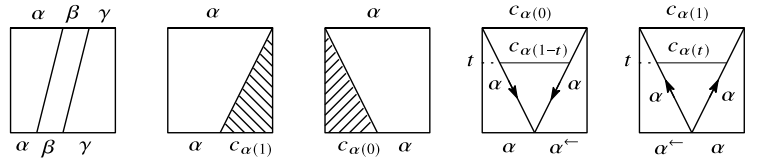


Fig. 8.1 Construction of five homotopies

$$\alpha' \simeq \alpha,\ \beta' \simeq \beta \ \text{ rel. } \partial\mathbf{I} \ \Rightarrow\ \alpha * \beta \simeq \alpha' * \beta',\ \alpha^{\leftarrow} \simeq \alpha'^{\leftarrow} \ \text{ rel. } \partial\mathbf{I}.$$

Let $c_x : \mathbf{I} \to X$ be the constant path with $c_x(\mathbf{I}) = \{x\}$.[1] For three paths $\alpha, \beta, \gamma : \mathbf{I} \to X$ with $\alpha(1) = \beta(0)$ and $\beta(1) = \gamma(0)$,

$$\begin{gathered}(\alpha * \beta) * \gamma \simeq \alpha * (\beta * \gamma) \ \text{ rel. } \partial\mathbf{I};\\ \alpha * c_{\alpha(1)} \simeq \alpha \ \text{ rel. } \partial\mathbf{I};\ c_{\alpha(0)} * \alpha \simeq \alpha \ \text{ rel. } \partial\mathbf{I};\\ \alpha * \alpha^{\leftarrow} \simeq c_{\alpha(0)} \ \text{ rel. } \partial\mathbf{I};\ \alpha^{\leftarrow} * \alpha \simeq c_{\alpha(1)} \ \text{ rel. } \partial\mathbf{I}.\end{gathered}$$

Referring to Fig. 8.1, we can easily construct these five homotopies.

A path $\alpha : \mathbf{I} \to X$ is called a **loop** (at x_0) if $\alpha(0) = \alpha(1)\ (= x_0)$. For each pointed space (X, x_0), $\mathrm{C}((\mathbf{I}, \partial\mathbf{I}), (X, x_0))$ is the set of all loops in X at x_0. The **loop space** $\Omega(X, x_0)$ is defined as the space $\mathrm{C}((\mathbf{I}, \partial\mathbf{I}), (X, x_0))$ admitting the compact-open topology, where the constant loop $c_{x_0} : \mathbf{I} \to \{x_0\} \subset X$ is the base point of $\Omega(X, x_0)$.[2] Each pointed map $f : (X, x_0) \to (Y, y_0)$ induces the pointed map $f_* : \Omega(X, x_0) \to \Omega(Y, y_0)$ defined by $f_*(\alpha) = f\alpha$ (cf. Proposition 2.10.8 (1)). Now, we can define the **fundamental group** $\pi_1(X, x_0)$ or the **first homotopy group** of X at x_0 (or (X, x_0)) as follows:

$$\pi_1(X, x_0) = [(\mathbf{I}, \partial\mathbf{I}), (X, x_0)] = \Omega(X, x_0)/\simeq$$

with the operation $[\alpha][\beta] = [\alpha * \beta]$ and the inverse $[\alpha]^{-1} = [\alpha^{\leftarrow}]$, where $[\alpha]$ is the equivalence class of $\alpha \in \Omega(X, x_0)$ with respect to $\simeq$.[3] In general, $\pi_1(X, x_0)$ is not commutative (i.e., non-Abelian). By the way, we have $\pi_1(X, x_0) = \pi_0(\Omega(X, x_0))$ by Proposition 2.10.5.

For each $n > 1$ and each $\alpha, \beta \in \mathrm{C}((\mathbf{I}^n, \partial\mathbf{I}^n), (X, x_0))$, we define $\alpha * \beta$ and $\alpha^{\leftarrow} \in \mathrm{C}((\mathbf{I}^n, \partial\mathbf{I}^n), (X, x_0))$ as follows:

[1] We use the notation c_x for any constant map with the value x even if it is not a path.

[2] The loop space $\Omega(X, x_0)$ is defined just before Corollary 7.8.9.

[3] For loops $\alpha, \beta \in \Omega(X, x_0)$, $\alpha \simeq \beta$ means $\alpha \simeq \beta$ rel. $\partial\mathbf{I}$.

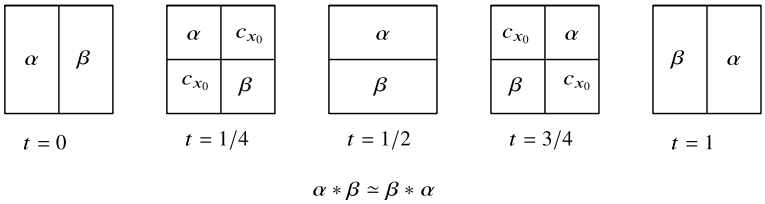


Fig. 8.2 A homotopy $\alpha * \beta \simeq \beta * \alpha$ in the case when $n = 2$

$$\alpha * \beta(t_1, \ldots, t_n) = \begin{cases} \alpha(2t_1, t_2, \ldots, t_n) & \text{if } 0 \leqslant t_1 \leqslant 1/2, \\ \beta(2t_1 - 1, t_2, \ldots, t_n) & \text{if } 1/2 \leqslant t_1 \leqslant 1; \end{cases}$$
$$\alpha^{\leftarrow}(t_1, \ldots, t_n) = \alpha(1 - t_1, t_2, \ldots, t_n).$$

Then, for each $\alpha, \alpha', \beta, \beta', \gamma \in \mathrm{C}((\mathbf{I}^n, \partial\mathbf{I}^n), (X, x_0))$,

$$\alpha \simeq \alpha',\ \beta \simeq \beta' \Rightarrow \alpha * \beta \simeq \alpha' * \beta',\ \alpha^{\leftarrow} \simeq \alpha'^{\leftarrow};$$
$$(\alpha * \beta) * \gamma \simeq \alpha * (\beta * \gamma)\ ;\ c_{x_0} * \alpha \simeq \alpha * c_{x_0} \simeq \alpha\ ;\ \alpha * \alpha^{\leftarrow} \simeq \alpha^{\leftarrow} * \alpha \simeq c_{x_0}.$$

Moreover,[4] we can also show that $\alpha * \beta \simeq \beta * \alpha$.

This homotopy can be easily constructed with reference to Fig. 8.2.

Thus, for each $n > 1$, we have the additive group

$$\pi_n(X, x_0) = [(\mathbf{I}^n, \partial\mathbf{I}^n), (X, x_0)] = \mathrm{C}((\mathbf{I}^n, \partial\mathbf{I}^n), (X, x_0))/\simeq,$$

with the operation of addition $[\alpha] + [\beta] = [\alpha * \beta]$ and the inverse $-[\alpha] = [\alpha^{\leftarrow}]$. The group $\pi_n(X, x_0)$ is called the ***n*-th homotopy group** of X at x_0 (or (X, x_0)).

Every pointed map $f : (X, x_0) \to (Y, y_0)$ induces the homomorphisms $f_\sharp : \pi_n(X, x_0) \to \pi_n(Y, y_0)$, $n \in \mathbb{N}$, defined by $f_\sharp[\alpha] = [f\alpha]$. For another pointed map $f' : (X, x_0) \to (Y, y_0)$, $f'_\sharp = f_\sharp$ if $f' \simeq f$.[5] For each pair of pointed maps $f : (X, x_0) \to (Y, y_0)$ and $g : (Y, y_0) \to (Z, z_0)$, we have $(gf)_\sharp = g_\sharp f_\sharp$ and $(\mathrm{id}_X)_\sharp = \mathrm{id}_{\pi_n(X, x_0)}$. If $f : (X, x_0) \to (Y, y_0)$ is a pointed homotopy equivalence (i.e., there is a pointed map $g : (Y, y_0) \to (X, x_0)$ such that $gf \simeq \mathrm{id}_X$ rel. x_0 and $fg \simeq \mathrm{id}_Y$ rel. y_0), then $f_\sharp$ is an isomorphism (with $(f_\sharp)^{-1} = g_\sharp$).

[4] For $\alpha, \beta \in \mathrm{C}((\mathbf{I}^n, \partial\mathbf{I}^n), (X, x_0))$, $\alpha \simeq \beta$ if and only if $\alpha \simeq \beta$ rel. $\partial\mathbf{I}^n$ (cf. Sect. 1.1).

[5] For pointed maps $f, f' \in \mathrm{C}((X, x_0), (Y, y_0))$, $f' \simeq f$ means $f' \simeq f$ rel. x_0, i.e., there exists a homotopy $h : X \times \mathbf{I} \to Y$ from f to f' such that $h_t \in \mathrm{C}((X, x_0), (Y, y_0))$ for each $t \in \mathbf{I}$, equiv., $h(\{x_0\} \times \mathbf{I}) \subset \{y_0\}$.

Recall $\pi_0(X)$ denotes the set of all path-components of a space X, which was defined immediately before Proposition 4.20.2. For a pointed space (X, x_0), $\pi_0(X, x_0)$ is defined as the pointed set $\pi_0(X)$ whose base point is the path-component of X containing x_0. Then, a pointed map $f : (X, x_0) \to (Y, y_0)$ induces the pointed function $f_\sharp : \pi_0(X, x_0) \to \pi_0(Y, y_0)$.[6]

In the following, we use the norm $\|z\|_\infty = \max\{|z(i)| \mid i = 1, \ldots, n\}$ for $z \in \mathbb{R}^n$. Suppose that $x_0, x_1 \in X$ are contained in the same path-component of X. Let $\omega : \mathbf{I} \to X$ be a path from x_0 to x_1 in X, $n \in \mathbb{N}$, and $\boldsymbol{v} = (1/2, \ldots, 1/2) \in \mathbb{R}^n$. For each $n \geqslant 1$ and each $\alpha \in \mathrm{C}((\mathbf{I}^n, \partial\mathbf{I}^n), (X, x_1))$, we define $\alpha^\omega \in \mathrm{C}((\mathbf{I}^n, \partial\mathbf{I}^n), (X, x_0))$ as follows:

$$\alpha^\omega(z) = \begin{cases} \alpha(2z - \boldsymbol{v}) & \text{if } \|z - \boldsymbol{v}\|_\infty \leqslant 1/4, \\ \omega(2 - 4\|z - \boldsymbol{v}\|_\infty) & \text{if } \|z - \boldsymbol{v}\|_\infty \geqslant 1/4. \end{cases}$$

Then, ω induces the isomorphism $\omega_* : \pi_n(X, x_1) \to \pi_n(X, x_0)$ defined by $\omega_*[\alpha] = [\alpha^\omega]$, where $(\omega_*)^{-1} = \omega_*^{\leftarrow}$. For a path $\omega' : \mathbf{I} \to X$ with $\omega'(0) = x_0$ and $\omega'(1) = x_1$, if $\omega \simeq \omega'$ rel. $\partial\mathbf{I}$ then $\omega_* = \omega'_*$. If $\omega'' : \mathbf{I} \to X$ is another path with $\omega''(0) = x_1$ and $\omega''(1) = x_2$, then $(\omega * \omega'')_* = \omega_*\omega''_* : \pi_n(X, x_2) \to \pi_n(X, x_0)$. When X is path-connected, $\pi_n(X, x_0) \cong \pi_n(X, x_1)$ for any pair of points $x_0, x_1 \in X$. In this case, it is acceptable to write $\pi_n(X)$ without the base point x_0.

For each pair of maps $f, f' : X \to Y$ with $f \simeq f'$, let $h : X \times \mathbf{I} \to Y$ be a homotopy from f to f' (i.e., $h_0 = f$ and $h_1 = f'$). For each $x_0 \in X$, the homotopy h gives a path $\omega : \mathbf{I} \to Y$ from $f(x_0)$ to $f'(x_0)$ defined by $\omega(t) = h(x_0, t)$. For each $\alpha \in \mathrm{C}((\mathbf{I}^n, \partial\mathbf{I}^n), (X, x_0))$, $h(\alpha \times \mathrm{id})$ is a homotopy from $f\alpha$ to $f'\alpha$ and $h(\alpha(\partial\mathbf{I}^n) \times \{t\}) = \{\omega(t)\}$. We define a homotopy $h^\alpha : \mathbf{I}^n \times \mathbf{I} \to Y$ as follows:

$$h^\alpha(x, t) = \begin{cases} h_t\alpha((2 - t)^{-1}2(z - \boldsymbol{v}) + \boldsymbol{v}) & \text{if } \|z - \boldsymbol{v}\|_\infty \leqslant (2 - t)/4, \\ \omega(2 - 4\|z - \boldsymbol{v}\|_\infty), & \text{if } \|z - \boldsymbol{v}\|_\infty \geqslant (2 - t)/4, \end{cases}$$

which is well-defined. Indeed, when $\|z - \boldsymbol{v}\|_\infty = (2 - t)/4$,

$$\|((2 - t)^{-1}2(z - \boldsymbol{v}) + \boldsymbol{v}) - \boldsymbol{v}\|_\infty = (2 - t)^{-1}2\|z - \boldsymbol{v}\|_\infty = 1/2,$$

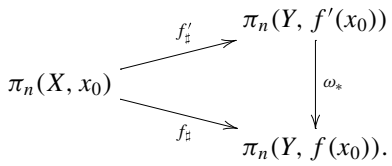
which means $(2 - t)^{-1}2(z - \boldsymbol{v}) + \boldsymbol{v} \in \partial\mathbf{I}^n$. Then, it follows that

$$h_t\alpha((2 - t)^{-1}2(z - \boldsymbol{v}) + \boldsymbol{v}) = h_t(x_0) = \omega(t) = \omega(2 - 4\|z - \boldsymbol{v}\|_\infty).$$

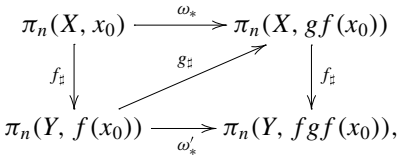
Observe that $h_0^\alpha = f\alpha$, $h_1^\alpha = (f'\alpha)^\omega$, and $h_t^\alpha(\partial\mathbf{I}^n) = \omega(0) = f(x_0)$ for each $t \in \mathbf{I}$. Hence, $f\alpha \simeq (f'\alpha)^\omega$ as maps in $\mathrm{C}((\mathbf{I}^n, \partial\mathbf{I}^n), (Y, f(x_0)))$. Thus, we have $f_\sharp = \omega_* f'_\sharp$, that is, the following diagram commutes:

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[6] As for a pointed space, a **pointed set** (S, s_0) is defined as a set S given a particular point $s_0 \in S$ called a **base point**, and a **pointed function** $f : (S, s_0) \to (T, t_0)$ is defined as a function $f : S \to T$ such that $f(s_0) = t_0$.



Using this fact, we can show that every homotopy equivalence $f : X \to Y$ induces the isomorphisms $f_\sharp : \pi_n(X, x_0) \to \pi_n(Y, f(x_0))$, $n \in \mathbb{N}$, $x_0 \in X$. In fact, let $g : Y \to X$ be a homotopy inverse of f, that is, $gf \simeq \mathrm{id}_X$ and $fg \simeq \mathrm{id}_Y$. A homotopy from gf to id gives a path $\omega : \mathbf{I} \to X$ from $gf(x_0)$ to x_0 and a homotopy from fg to id_Y gives a path $\omega' : \mathbf{I} \to Y$ from $fgf(x_0)$ to $f(x_0)$. For each $x_0 \in X$ and $n \in \mathbb{N}$, we have the following commutative diagram:



where it should be noted that the left and the right $f_\sharp$ in the diagram are different from each other. Since $g_\sharp f_\sharp = \omega_*$ and $f_\sharp g_\sharp = \omega'_*$ are isomorphisms, $g_\sharp$ is also an isomorphism. Then, it follows that $f_\sharp : \pi_n(X, x_0) \to \pi_n(Y, f(x_0))$ is an isomorphism.

For each $n \geqslant 2$, we define

$$\mathbf{J}^{n-1} \underset{\text{def}}{=} (\partial \mathbf{I}^{n-1} \times \mathbf{I}) \cup (\mathbf{I}^{n-1} \times \{1\}).$$

Identifying $\mathbf{I}^{n-1} = \mathbf{I}^{n-1} \times \{0\} \subset \mathbf{I}^n$, we have

$$\mathbf{I}^{n-1} \cup \mathbf{J}^{n-1} = \partial \mathbf{I}^n \quad \text{and} \quad \mathbf{I}^{n-1} \cap \mathbf{J}^{n-1} = \partial \mathbf{I}^{n-1}.$$

For a space X with $x_0 \in A \subset X$, the **relative n-th homotopy group** of (X, A) at x_0 is defined as the group

$$\pi_n(X, A, x_0) \underset{\text{def}}{=} [(\mathbf{I}^n, \partial \mathbf{I}^n, \mathbf{J}^{n-1}), (X, A, x_0)]$$

with the operations defined by the analogy of $\pi_n(X, x_0)$. In general, the group $\pi_2(X, A, x_0)$ is non-commutative, so we describe it as a multiplicative group like the fundamental group. On the other hand, for $n \geqslant 3$, $\pi_n(X, A, x_0)$ is represented as the additive group because it is commutative. It should be noted that $\pi_n(X, \{x_0\}, x_0) = \pi_n(X, x_0)$. We also define

$$\pi_1(X, A, x_0) \underset{\text{def}}{=} [(\mathbf{I}, \partial \mathbf{I}, 0), (X, A, x_0)],$$

which is regarded as the pointed set whose base point is the homotopy class $[c_{x_0}]$. As in the case with pointed spaces, every map $f : (X, A, x_0) \to (Y, B, y_0)$ induces the homomorphisms $f_\sharp : \pi_n(X, A, x_0) \to \pi_n(Y, B, y_0), n \geqslant 2$, and the pointed function $f_\sharp : \pi_1(X, A, x_0) \to \pi_1(Y, B, y_0)$. For another map $f' : (X, A, x_0) \to (Y, B, y_0)$, $f'_\sharp = f_\sharp$ if $f' \simeq f$.[7] For each pair of maps $f : (X, A, x_0) \to (Y, B, y_0)$ and $g : (Y, B, y_0) \to (Z, C, z_0)$, we have $(gf)_\sharp = g_\sharp f_\sharp$ and $(\mathrm{id}_X)_\sharp = \mathrm{id}_{\pi_n(X,A,x_0)}$. If $f : (X, A, x_0) \to (Y, B, y_0)$ is a homotopy equivalence (i.e., there is a map $g : (Y, B, y_0) \to (X, A, x_0)$ such that $gf \simeq \mathrm{id}_X$ and $fg \simeq \mathrm{id}_Y$), then $f_\sharp$ is an isomorphism (with $(f_\sharp)^{-1} = g_\sharp$).

The details are left to the reader as an exercise.

For each $n \geqslant 2$, let $\partial : \pi_n(X, A, x_0) \to \pi_{n-1}(A, x_0)$ be the homomorphism induced by the restriction operator, i.e., $\partial[\alpha] = [\alpha|\mathbf{I}^{n-1}]$, which is called the **boundary operator**. In addition, we define the pointed function $\partial : \pi_1(X, A, x_0) \to \pi_0(A, x_0)$ as follows: $\partial[\alpha]$ is the path-component of $\alpha(1)$ for each $\alpha \in \mathrm{C}((\mathbf{I}, \partial\mathbf{I}, 0), (X, A, x_0))$. Then, the following diagram commutes for each map $f : (X, A, x_0) \to (Y, B, y_0)$:

$$\begin{array}{ccccccccc}
\pi_n(A, x_0) & \xrightarrow{i_\sharp} & \pi_n(X, x_0) & \xrightarrow{j_\sharp} & \pi_n(X, A, x_0) & \xrightarrow{\partial} & \pi_{n-1}(A, x_0) \\
\downarrow{\scriptstyle (f|A)_\sharp} & & \downarrow{\scriptstyle f_\sharp} & & \downarrow{\scriptstyle f_\sharp} & & \downarrow{\scriptstyle (f|A)_\sharp} \\
\pi_n(B, y_0) & \xrightarrow[i_\sharp]{} & \pi_n(Y, y_0) & \xrightarrow[j_\sharp]{} & \pi_n(Y, B, y_0) & \xrightarrow[\partial]{} & \pi_{n-1}(B, y_0),
\end{array}$$

where the $i_\sharp$ are the homomorphisms induced by the inclusions $i : (A, x_0) \subset (X, x_0)$ and $i : (B, y_0) \subset (Y, y_0)$, and the $j_\sharp$ are the ones induced by the inclusions $j : (X, \{x_0\}, x_0) \subset (X, A, x_0)$ and $j : (Y, \{y_0\}, y_0) \subset (Y, B, y_0)$. Moreover, $\mathrm{im}\, i_\sharp = \ker j_\sharp$, $\mathrm{im}\, j_\sharp = \ker \partial$ and $\mathrm{im}\, \partial = \ker i_\sharp$.[8] Namely, the following sequence is exact:

$$\begin{aligned}
\cdots \xrightarrow{\partial} \pi_n(A, x_0) \xrightarrow{i_\sharp} \pi_n(X, x_0) \xrightarrow{j_\sharp} \pi_n(X, A, x_0) \xrightarrow{\partial} \\
\pi_{n-1}(A, x_0) \xrightarrow{i_\sharp} \cdots \xrightarrow{\partial} \pi_1(A, x_0) \xrightarrow{i_\sharp} \pi_1(X, x_0) \\
\xrightarrow{j_\sharp} \pi_1(X, A, x_0) \xrightarrow{\partial} \pi_0(A, x_0) \xrightarrow{i_\sharp} \pi_0(X, x_0).
\end{aligned}$$

This sequence is called the **homotopy exact sequence** of (X, A, x_0).

The details are left to the reader as an exercise.

Let $\omega : \mathbf{I} \to A$ be a path from x_0 to x_1. For each $n \geqslant 2$ and

[7] For homotopy $\simeq$ on $\mathrm{C}((X, A, x_0), (Y, B, y_0))$, refer to Sect. 1.1.

[8] For a pointed function $f : (A, a_0) \to (B, b_0)$, we mean $\ker f = f^{-1}(b_0)$. Of course, $\mathrm{im}\, f = f(A)$ goes without saying.

$$\alpha \in \mathrm{C}((\mathbf{I}^n, \partial\mathbf{I}^n, \mathbf{J}^{n-1}), (X, A, x_1)),$$

we define α^ω somewhat differently from the case of pointed spaces, that is,

$$\alpha^\omega(z) = \begin{cases} \alpha(2z' - \boldsymbol{v}', 2z(n)) & \text{if } \|(z' - \boldsymbol{v}', z(n)/2)\|_\infty \leqslant 1/4, \\ \omega(2 - 4\|(z' - \boldsymbol{v}', z(n)/2)\|_\infty) & \text{if } \|(z' - \boldsymbol{v}', z(n)/2)\|_\infty \geqslant 1/4, \end{cases}$$
$$\text{where } z' = (z(1), \ldots, z(n-1)),\ \boldsymbol{v}' = (1/2, \ldots, 1/2) \in \mathbf{I}^{n-1}.$$

As in the case of pointed spaces, the isomorphism $\omega_* : \pi_n(X, A, x_1) \to \pi_n(X, A, x_0)$ can be defined by $\omega_*[\alpha] = [\alpha^\omega]$. For a path $\omega' : \mathbf{I} \to A$ with $\omega'(0) = x_0$ and $\omega'(1) = x_1$, if $\omega \simeq \omega'$ rel. $\partial\mathbf{I}$, then $\omega_* = \omega'_*$. Moreover, the following diagram commutes:

$$\begin{array}{ccccccc} \pi_n(A, x_1) & \xrightarrow{i_\sharp} & \pi_n(X, x_1) & \xrightarrow{j_\sharp} & \pi_n(X, A, x_1) & \xrightarrow{\partial} & \pi_{n-1}(A, x_1) \\ \downarrow{\scriptstyle \omega_*} & & \downarrow{\scriptstyle \omega_*} & & \downarrow{\scriptstyle \omega_*} & & \downarrow{\scriptstyle \omega_*} \\ \pi_n(A, x_0) & \xrightarrow[i_\sharp]{} & \pi_n(X, x_0) & \xrightarrow[j_\sharp]{} & \pi_n(X, A, x_0) & \xrightarrow[\partial]{} & \pi_{n-1}(A, y_0). \end{array}$$

Thus, if A is path-connected, then $\pi_n(X, A, x_0) \cong \pi_n(X, A, x_1)$ for any pair of points $x_0, x_1 \in A$. In this case, it is acceptable to write $\pi_n(X, A)$ without the base point $x_0 \in A$.

The details of the above are left to the reader as an exercise.

Let $f, f' : (X, A) \to (Y, B)$ be maps such that $f \simeq f'$ by a homotopy $h : (X \times \mathbf{I}, A \times \mathbf{I}) \to (Y, B)$ (i.e., $h_0 = f$ and $h_1 = f'$). For each $x_0 \in A$, the homotopy h gives a path $\omega : \mathbf{I} \to B$ defined by $\omega(t) = h(x_0, t)$. Let

$$\alpha \in \mathrm{C}((\mathbf{I}^n, \partial\mathbf{I}^n, \mathbf{J}^{n-1}), (X, A, x_0)).$$

Then, $h(\alpha \times \mathrm{id})$ is a homotopy from $f\alpha$ to $f'\alpha$,

$$h(\alpha(\partial\mathbf{I}^n) \times \{t\}) \subset B, \text{ and } h(\alpha(\mathbf{J}^{n-1}) \times \{t\}) = \{\omega(t)\}.$$

By the analogy of the case of $f, f' \in \mathrm{C}(X, Y)$, we can show that $f\alpha \simeq (f'\alpha)^\omega$ as maps in $\mathrm{C}((\mathbf{I}^n, \partial\mathbf{I}^n, \mathbf{J}^{n-1}), (Y, B, f(x_0)))$. Thus, it follows that $f_\sharp = \omega_* f'_\sharp$, that is, the following diagram commutes:

$$\begin{array}{ccc} & \overset{f'_\sharp}{\nearrow} & \pi_n(Y, B, f'(x_0)) \\ \pi_n(X, A, x_0) & & \downarrow{\scriptstyle \omega_*} \\ & \underset{f_\sharp}{\searrow} & \pi_n(Y, B, f(x_0)). \end{array}$$

Using this fact, we can show that every homotopy equivalence $f : (X, A) \to (Y, B)$ induces the isomorphisms $f_\sharp : \pi_n(X, A, x_0) \to \pi_n(Y, B, f(x_0))$, $n \in \mathbb{N}$, $x_0 \in X$.

The above details are left to the reader as an exercise.

The proof of the following proposition is quite elementary:

Proposition 8.1.1 *For pointed spaces $(X_1, x_1), \ldots, (X_k, x_k)$ and $n \in \mathbb{N}$,*

$$\pi_n(X_1 \times \cdots \times X_k, (x_1, \ldots, x_k)) \cong \pi_n(X_1, x_1) \times \cdots \times \pi_n(X_k, x_k).$$

Proof We have the following isomorphism:

$$\varphi : \pi_n(X_1 \times \cdots \times X_k, (x_1, \ldots, x_k)) \to \pi_n(X_1, x_1) \times \cdots \times \pi_n(X_k, x_k);$$
$$\varphi[\alpha] = ((\mathrm{pr}_1)_\sharp[\alpha], \ldots, (\mathrm{pr}_k)_\sharp[\alpha]) = ([\mathrm{pr}_1\alpha], \ldots, [\mathrm{pr}_k\alpha]).$$

Indeed, for each set of maps $\alpha_i : (\mathbf{I}^n, \partial\mathbf{I}^n) \to (X_i, x_i)$, $i = 1, \ldots, k$, we have the map

$$\alpha : (\mathbf{I}^n, \partial\mathbf{I}^n) \to (X_1 \times \cdots \times X_k, (x_1, \ldots, x_k))$$

defined by $\alpha(x) = (\alpha_1(x), \ldots, \alpha_k(x))$. Then, $\varphi([\alpha]) = ([\alpha_1], \ldots, [\alpha_k])$. Hence, φ is an epimorphism. If every α_1, ..., α_k are null-homotopic, then the map α is null-homotopic, which means that φ is a monomorphism. □

For additive groups $G_1, \ldots, G_k$, the direct product $G_1 \times \cdots \times G_k$ is regarded as the direct sum $G_1 \oplus \cdots \oplus G_k$. Thus, when $n \geqslant 2$ in Proposition 8.1.1,

$$\pi_n(X_1 \times \cdots \times X_k, (x_1, \ldots, x_k)) \cong \pi_n(X_1, x_1) \oplus \cdots \oplus \pi_n(X_k, x_k).$$

To consider homotopy groups of loop spaces, the following proposition is a key:

Proposition 8.1.2 *For a pair (X, A) of spaces and a pointed space (Y, y_0), there exists a natural homeomorphism*

$$\varphi : \mathrm{C}((X \times \mathbf{I}, H_{X,A}), (Y, y_0)) \to \mathrm{C}((X, A), (\Omega(Y, y_0), c_{y_0}))$$

with respect to the compact-open topology, where $H_{X,A} = (X \times \partial\mathbf{I}) \cup (A \times \mathbf{I})$, and φ is natural in the following sense: given maps $f : (X', A') \to (X, A)$ and $g : (Y, y_0) \to (Y', y_0')$, the diagram below is commutative:

$$\begin{array}{ccc}
\mathrm{C}((X' \times \mathbf{I}, H_{X',A'}), (Y, y_0)) & \xrightarrow{\varphi} & \mathrm{C}((X', A'), (\Omega(Y, y_0), c_{y_0})) \\
\uparrow{\scriptstyle (f\times\mathrm{id})^*} & & \uparrow{\scriptstyle f^*} \\
\mathrm{C}((X \times \mathbf{I}, H_{X,A}), (Y, y_0)) & \xrightarrow{\varphi} & \mathrm{C}((X, A), (\Omega(Y, y_0), c_{y_0})) \\
\downarrow{\scriptstyle g_*} & & \downarrow{\scriptstyle (g_*)_*} \\
\mathrm{C}((X \times \mathbf{I}, H_{X,A}), (Y', y_0')) & \xrightarrow[\varphi]{} & \mathrm{C}((X, A), (\Omega(Y', y_0'), c_{y_0'})),
\end{array}$$

where $(f \times \mathrm{id})^*$, f^*, g_*, *and* $(g_*)_*$ *are maps naturally induced by* $f \times \mathrm{id}$, f, g, *and* g_*, *respectively.*[9]

Proof Restricting φ of Proposition 2.10.3 (which is a homeomorphism according to Corollary 2.10.7), we have a homeomorphism

$$\varphi : \mathrm{C}((X \times \mathbf{I}, H_{X,A}), (Y, y_0)) \ni k \mapsto \varphi(k) \in \mathrm{C}((X, A), (\Omega(Y, y_0), c_{y_0})),$$

which is defined by $\varphi(k)(x)(s) = k(x, s)$. To show the commutativity of the diagram, let $k \in \mathrm{C}((X \times \mathbf{I}, H_{X,A}), (Y, y_0))$. For each $x' \in X'$ and $s \in \mathbf{I}$,

$$\begin{aligned}\varphi((f \times \mathrm{id})^*(k))(x')(s) &= \varphi(k(f \times \mathrm{id}))(x')(s) = k(f \times \mathrm{id})(x', s) \\ &= k(f(x'), s) = \varphi(k)(f(x'))(s) = f^*(\varphi(k))(x')(s).\end{aligned}$$

Hence, $\varphi((f \times \mathrm{id})^*(k)) = f^*(\varphi(k))$. On the other hand, for each $x \in X$ and $s \in \mathbf{I}$,

$$\begin{aligned}\varphi(g_*(k))(x)(s) &= \varphi(gk)(x)(s) = gk(x, s) = g(\varphi(k)(x)(s)) \\ &= g_*(\varphi(k)(x))(s) = (g_*)_*(\varphi(k))(x)(s).\end{aligned}$$

Hence, $\varphi(g_*(k)) = (g_*)_*(\varphi(k))$. □

In the above situation, given a homotopy $h : (X \times \mathbf{I}) \times \mathbf{I} \to Y$ such that $h_t(H_{X,A}) = \{y_0\}$ for every $t \in \mathbf{I}$, we have the homotopy $\varphi(h) : X \times \mathbf{I} \to \Omega(Y, y_0)$ defined by $\varphi(h)_t = \varphi(h_t)$, that is,

$$\varphi(h)(x, t)(s) = \varphi(h_t)(x)(s) = h_t(x, s) = h(x, s, t).$$

Then, φ induces the function

$$\Phi : [(X \times \mathbf{I}, H_{X,A}), (Y, y_0)] \to [(X, A), (\Omega(Y, y_0), c_{y_0})].$$

The inverse of φ in Proposition 8.1.2 is a restriction of ψ in Proposition 2.10.6 (Corollary 2.10.7), that is,

$$\psi : \mathrm{C}((X, A), (\Omega(Y, y_0), c_{y_0})) \ni k \mapsto \psi(k) \in \mathrm{C}((X \times \mathbf{I}, H_{X,A}), (Y, y_0)),$$

where $\psi(k)(x, s) = k(x)(s)$. Given a homotopy $h : X \times \mathbf{I} \to \Omega(Y, y_0)$ such that $h_t(A) = \{c_{y_0}\}$ for every $t \in \mathbf{I}$, we have the homotopy $\psi(h) : X \times \mathbf{I} \times \mathbf{I} \to Y$ defined by $\psi(h)_t = \psi(h_t)$, that is,

$$\psi(h)(x, s, t) = \psi(h_t)(x, s) = h_t(x)(s) = h(x, t)(s).$$

Then, ψ induces the function

[9] Cf. Proposition 2.10.8 (1).

$$\Psi : [(X, A), (\Omega(Y, y_0), c_{y_0})] \to [(X \times \mathbf{I}, H_{X,A}), (Y, y_0)].$$

Furthermore, given a homotopy $h : X \times \mathbf{I} \times \mathbf{I} \to Y$ such that $h_t(H_{X,A}) = \{y_0\}$ for every $t \in \mathbf{I}$, $\psi(\varphi(h))_t = \psi(\varphi(h_t)) = h_t$. Similarly, given a homotopy $h : X \times \mathbf{I} \to \Omega(Y, y_0)$ such that $h_t(A) = \{c_{y_0}\}$ for every $t \in \mathbf{I}$, $\varphi(\psi(h))_t = \varphi(\psi(h_t)) = h_t$. Then, it follows that Φ is a bijection with $\Phi^{-1} = \Psi$. Thus, we have the following:

Proposition 8.1.3 *For a pair (X, A) of spaces and a pointed space (Y, y_0), there exists a natural bijection*

$$\Phi : [(X \times \mathbf{I}, H_{X,A}), (Y, y_0)] \to [(X, A), (\Omega(Y, y_0), c_{y_0})],$$

where $H_{X,A} = (X \times \partial\mathbf{I}) \cup (A \times \mathbf{I})$ and Φ is natural in the similar sense to Proposition 8.1.2. □

In the above, replace (X, A) and (Y, y_0) with $(\mathbf{I}^n, \partial\mathbf{I}^n)$ and (X, x_0), respectively. Then, observe that

$$H_{\mathbf{I}^n, \partial\mathbf{I}^n} = (\mathbf{I}^n \times \partial\mathbf{I}) \cup (\partial\mathbf{I}^n \times \mathbf{I}) = \partial\mathbf{I}^{n+1}.$$

Namely, $(X \times \mathbf{I}, H_{X,A})$ corresponds to $(\mathbf{I}^{n+1}, \partial\mathbf{I}^{n+1})$. Hence, we have a bijection

$$\Phi : [(\mathbf{I}^{n+1}, \partial\mathbf{I}^{n+1}), (X, x_0)] \to [(\mathbf{I}^n, \partial\mathbf{I}^n), (\Omega(X, x_0), c_{x_0})],$$

that is, $\Phi : \pi_{n+1}(X, x_0) \to \pi_n(\Omega(X, x_0), c_{x_0})$ is a bijection. As can be easily seen, $\varphi(\alpha * \alpha') = \varphi(\alpha) * \varphi(\alpha')$ and $\varphi(\alpha^{\leftarrow}) = \varphi(\alpha)^{\leftarrow}$ for each $\alpha, \alpha' \in \mathrm{C}((\mathbf{I}^{n+1}, \partial\mathbf{I}^{n+1}), (X, x_0))$. Therefore, $\Phi : \pi_{n+1}(X, x_0) \to \pi_n(\Omega(X, x_0), c_{x_0})$ is an isomorphism. Thus, we have the following theorem:

Theorem 8.1.4 *For every pointed space (X, x_0) and $n \in \mathbb{N}$,*

$$\pi_n(\Omega(X, x_0), c_{x_0}) \cong \pi_{n+1}(X, x_0).$$ □

Note Recall that the loop space $\Omega(X, x_0)$ has the operations $\alpha * \beta$ and $\alpha^{\leftarrow}$, which induce the operations of the fundamental group $\pi_1(X, x_0)$. Using these operations of $\Omega(X, x_0)$, we can define the operations $\alpha \hat{*} \beta$ and $\hat{\alpha^{\leftarrow}}$ on $\mathrm{C}((\mathbf{I}^n, \partial\mathbf{I}^n), (\Omega(X, x_0), c_{x_0}))$ as follows:

$$(\alpha \hat{*} \beta)(z) = \alpha(z) * \beta(z) \quad \text{and} \quad \hat{\alpha^{\leftarrow}}(z) = \alpha(z)^{\leftarrow} \quad \text{for each } z \in \mathbf{I}^n.$$

The operations of the homotopy group $\pi_n(\Omega(X, x_0), c_{x_0})$ are also induced by the above operations. In fact, we can define operations $\alpha \underline{*} \beta$ and $\alpha_{\leftarrow}$ on $\mathrm{C}((\mathbf{I}^{n+1}, \partial\mathbf{I}^{n+1}), (X, x_0))$ as follows:

$$\alpha \underline{*} \beta(t_1, \ldots, t_{n+1}) = \begin{cases} \alpha(t_1, \ldots, t_n, 2t_{n+1}) & \text{if } 0 \leqslant t_{n+1} \leqslant 1/2, \\ \beta(t_1, \ldots, t_n, 2t_{n+1} - 1) & \text{if } 1/2 \leqslant t_{n+1} \leqslant 1, \end{cases}$$

$$\alpha_{\leftarrow}(t_1, \ldots, t_{n+1}) = \alpha(t_1, \ldots, t_n, 1 - t_{n+1}),$$

which is different from $\alpha * \beta$ and $\alpha^{\leftarrow}$ inducing the homotopy group operations of $\pi_{n+1}(X, x_0)$ but $\alpha \underline{*} \beta \simeq \alpha * \beta$ and $\alpha_{\leftarrow} \simeq \alpha^{\leftarrow}$. Let

$$\psi : \mathrm{C}((\mathbf{I}^n, \partial\mathbf{I}^n), (\Omega(X, x_0), c_{x_0})) \to \mathrm{C}((\mathbf{I}^{n+1}, \partial\mathbf{I}^{n+1}), (X, x_0))$$

be the inverse of the natural homeomorphism φ of Proposition 8.1.2, i.e., $\psi(\alpha)(x, t) = \alpha(x)(t)$. For each $\alpha, \beta \in \mathrm{C}((\mathbf{I}^n, \partial\mathbf{I}^n), (\Omega(X, x_0), c_{x_0}))$,

$$\psi(\alpha \mathbin{\hat{*}} \beta) = \psi(\alpha) \mathbin{\underline{*}} \psi(\beta) \simeq \psi(\alpha) * \psi(\beta) = \psi(\alpha * \beta) \text{ and}$$
$$\psi(\alpha^{\widehat{}}) = \psi(\alpha)_{\leftarrow} \simeq \psi(\alpha)^{\leftarrow} = \psi(\alpha^{\leftarrow}),$$

where $\alpha * \beta$ and $\alpha^{\leftarrow}$ are the operations inducing the homotopy group operations of $\pi_n(\Omega(X, x_0))$. Hence, $\alpha \mathbin{\hat{*}} \beta \simeq \alpha * \beta$ and $\alpha_{\leftarrow} \simeq \alpha^{\leftarrow}$.

Inductively, we can define $\Omega^n(X, x_0) = \Omega(\Omega^{n-1}(X, x_0), c_{x_0})$, where $\Omega^0(X, x_0) = X$. Then, we have

$$\begin{aligned}\pi_n(X, x_0) &= \pi_{n-1}(\Omega(X, x_0), c_{x_0}) = \cdots \\ &= \pi_1(\Omega^{n-1}(X, x_0), c_{x_0}) = \pi_0(\Omega^n(X, x_0), c_{x_0}).\end{aligned}$$

8.2 Characterizing n-Equivalences and Weak Homotopy Equivalences

In this section, we will characterizing n-equivalences and weak homotopy equivalences in terms of the homotopy groups.

Let $\mathbf{e}_1 = (1, 0, \ldots, 0) \in \mathbb{R}^{n+1}$. For each $n \in \mathbb{N}$,

$$(\mathbf{I}^n/\partial\mathbf{I}^n, \partial\mathbf{I}^n/\partial\mathbf{I}^n) \approx (\mathbf{S}^n, \mathbf{e}_1) \text{ and}$$
$$(\mathbf{I}^{n+1}/\mathbf{J}^n, \partial\mathbf{I}^{n+1}/\mathbf{J}^n, \mathbf{J}^n/\mathbf{J}^n) \approx (\mathbf{B}^{n+1}, \mathbf{S}^n, \mathbf{e}_1).$$

By these homeomorphisms, the following identifications are possible:

$$\mathrm{C}((\mathbf{I}^n, \partial\mathbf{I}^n), (X, x_0)) = \mathrm{C}((\mathbf{S}^n, \mathbf{e}_1), (X, x_0)) \text{ and}$$
$$\mathrm{C}((\mathbf{I}^n, \partial\mathbf{I}^n, \mathbf{J}^{n-1}), (X, A, x_0)) = \mathrm{C}((\mathbf{B}^n, \mathbf{S}^{n-1}, \mathbf{e}_1), (X, A, x_0)).$$

In particular, $\mathrm{C}((\mathbf{S}^1, \mathbf{e}_1), (X, x_0))$ can be regarded as the loop space $\Omega(X, x_0)$ $(= \mathrm{C}((\mathbf{I}, \partial\mathbf{I}), (X, x_0)))$. Thus, we can make the following identifications:

$$\pi_n(X, x_0) = [(\mathbf{S}^n, \mathbf{e}_1), (X, x_0)] \text{ and}$$
$$\pi_{n+1}(X, A, x_0) = [(\mathbf{B}^{n+1}, \mathbf{S}^n, \mathbf{e}_1), (X, A, x_0)].$$

In this case, the boundary operator $\partial : \pi_{n+1}(X, A, x_0) \to \pi_n(A, x_0)$ is defined by $\partial[\alpha] = [\alpha|\mathbf{S}^n]$.

When X (resp. $A \subset X$) is path-connected, as observed before, it is acceptable to denote $\pi_n(X)$ (resp. $\pi_n(X, A)$) without a base point. However, this does not mean that $\pi_n(X)$ (resp. $\pi_n(X, A)$) can be identified with $[\mathbf{S}^n, X]$ (resp. $[(\mathbf{B}^n, \mathbf{S}^{n-1}), (X, A)]$).

It is said that X is **simply connected** if it is path-connected and $\pi_1(X, x_0) = \{0\}$ for any/some $x_0 \in X$. The latter condition is equivalent to the condition that every map $\alpha : \mathbf{S}^1 \to X$ is null-homotopic or α extends over $\mathbf{B}^2$, which will be shown in Proposition 8.2.2. When X (resp. A) is simply connected, $\pi_n(X)$ (resp. $\pi_n(X, A)$) can be identified with $[\mathbf{S}^n, X]$ (resp. $[(\mathbf{B}^n, \mathbf{S}^{n-1}), (X, A)]$). In fact, we have the following proposition:

Proposition 8.2.1 (1) *If X is simply connected, for any $x_0 \in X$ and $n \geqslant 2$, the inclusion*

$$i : \mathrm{C}((\mathbf{S}^n, \mathbf{e}_1), (X, x_0)) \subset \mathrm{C}(\mathbf{S}^n, X)$$

induces a bijection from $\pi_n(X, x_0)$ onto $[\mathbf{S}^n, X]$.

(2) *If $A \subset X$ is simply connected, for any $x_0 \in A$ and $n \geqslant 2$, the inclusion*

$$i : \mathrm{C}((\mathbf{B}^n, \mathbf{S}^{n-1}, \mathbf{e}_1), (X, A, x_0)) \subset \mathrm{C}((\mathbf{B}^n, \mathbf{S}^{n-1}), (X, A))$$

induces a bijection from $\pi_n(X, A, x_0)$ onto $[(\mathbf{B}^n, \mathbf{S}^{n-1}), (X, A)]$.

Proof (1): For each $\alpha \in \mathrm{C}(\mathbf{S}^n, X)$, take a path $\omega : \mathbf{I} \to X$ from $\alpha(\mathbf{e}_1)$ to x_0. Applying the Homotopy Extension Theorem 4.5.3, we can obtain a homotopy $h : \mathbf{S}^n \times \mathbf{I} \to X$ such that $h_0 = \alpha$ and $h(\mathbf{e}_1, t) = \omega(t)$ for each $t \in \mathbf{I}$. Thus, α is homotopic to the map $h_1 \in \mathrm{C}((\mathbf{S}^n, \mathbf{e}_1), (X, x_0))$. This means that i induces a surjection.

Let $\alpha, \beta \in \mathrm{C}((\mathbf{S}^n, \mathbf{e}_1), (X, x_0))$ and assume that there is a homotopy $h : \mathbf{S}^n \times \mathbf{I} \to X$ from α to β. Since X is simply connected, we have a map $k : \mathbf{I}^2 \to X$ such that $k(t, 0) = h(\mathbf{e}_1, t)$ and $k(0, t) = k(1, t) = k(t, 1) = x_0$ for each $t \in \mathbf{I}$. Applying the Homotopy Extension Theorem 4.5.3, we have a homotopy $\tilde{\varphi} : \mathbf{S}^n \times \mathbf{I} \times \mathbf{I} \to X$ that is an extension of the homotopy

$$\varphi : ((\mathbf{S}^n \times \{0, 1\}) \cup (\{\mathbf{e}_1\} \times \mathbf{I})) \times \mathbf{I} \to X$$

defined by $\varphi_t(x, 0) = h(x, 0) = \alpha(x)$ and $\varphi_t(x, 1) = h(x, 1) = \beta(x)$ for each $x \in \mathbf{S}^n$, and $\varphi_t(\mathbf{e}_1, s) = k(s, t)$ for each $s \in \mathbf{I}$. Then, $\tilde{\varphi}_0 = h$ and $\tilde{\varphi}_1 : \mathbf{S}^n \times \mathbf{I} \to X$ is a homotopy from α to β with $\tilde{\varphi}_1(\mathbf{e}_1, t) = x_0$ for every $t \in \mathbf{I}$. This means that i induces an injection.

(2): For each $\alpha \in \mathrm{C}((\mathbf{B}^n, \mathbf{S}^{n-1}), (X, A))$, take a path $\omega : \mathbf{I} \to A$ from $\alpha(\mathbf{e}_1)$ to x_0. Applying the Homotopy Extension Theorem 4.5.3 twice, we can obtain a homotopy $h : \mathbf{B}^n \times \mathbf{I} \to X$ such that $h_0 = \alpha, h(\mathbf{S}^{n-1} \times \mathbf{I}) \subset A$ and $h(\mathbf{e}_1, t) = \omega(t)$ for each $t \in \mathbf{I}$. Thus, α is homotopic to the map $h_1 \in \mathrm{C}((\mathbf{B}^n, \mathbf{S}^{n-1}, \mathbf{e}_1), (X, A, x_0))$. This means that i induces a surjection.

Let $\alpha, \beta \in \mathrm{C}((\mathbf{B}^n, \mathbf{S}^{n-1}, \mathbf{e}_1), (X, A, x_0))$ and assume that there is a homotopy $h : \mathbf{B}^n \times \mathbf{I} \to X$ from α to β such that $h(\mathbf{S}^{n-1} \times \mathbf{I}) \subset A$. Since A is simply connected, we have a map $k : \mathbf{I}^2 \to A$ such that $k(t, 0) = h(\mathbf{e}_1, t)$ and $k(0, t) = k(1, t) = k(t, 1) = x_0$ for each $t \in \mathbf{I}$. Applying the Homotopy Extension Theorem 4.5.3 twice, we have a homotopy $\tilde{\varphi} : \mathbf{B}^n \times \mathbf{I} \times \mathbf{I} \to X$ such that $\tilde{\varphi}(\mathbf{S}^{n-1} \times \mathbf{I} \times \mathbf{I}) \subset A$ and $\tilde{\varphi}$ is an extension of the homotopy

$$\varphi : ((\mathbf{B}^n \times \{0, 1\}) \cup (\{\mathbf{e}_1\} \times \mathbf{I})) \times \mathbf{I} \to X$$

defined by $\varphi_t(x, 0) = h(x, 0) = \alpha(x)$ and $\varphi_t(x, 1) = h(x, 1) = \beta(x)$ for each $x \in \mathbf{B}^n$, and $\varphi_t(\mathbf{e}_1, s) = k(s, t)$ for each $s \in \mathbf{I}$. Then, $\tilde{\varphi}_0 = h$ and $\tilde{\varphi}_1 : \mathbf{B}^n \times \mathbf{I} \to X$ is a homotopy from α to β with $\tilde{\varphi}_1(\mathbf{e}_1, t) = x_0$ for every $t \in \mathbf{I}$. This means that i induces an injection. □

If X is not simply connected, for two maps $\alpha, \alpha' : \mathbf{S}^n \to X$ with $\alpha(\mathbf{e}_1) = \alpha'(\mathbf{e}_1)$, $\alpha \simeq \alpha'$ does not imply $\alpha \simeq \alpha'$ rel. $\mathbf{e}_1$. However, we have the following proposition:

Proposition 8.2.2 *For a map $\alpha : \mathbf{S}^n \to X$, the following are equivalent:*

(a) *α extends over $\mathbf{B}^{n+1}$;*
(b) *α is null-homotopic, i.e., $\alpha \simeq 0$;*
(c) *$\alpha \simeq c_{\alpha(\mathbf{e}_1)}$ rel. $\mathbf{e}_1$, i.e., $[\alpha] = 0 \in \pi_n(X, \alpha(\mathbf{e}_1))$.*

Proof The implication (c) ⇒ (b) is obvious. For (b) ⇒ (a), let $h : \mathbf{S}^n \times \mathbf{I} \to X$ be a homotopy from α to a constant map. Then, α extends to the map $\beta : \mathbf{B}^{n+1} \to X$ defined by

$$\beta(x) = \begin{cases} h(\|x\|^{-1}x, 1 - \|x\|) & \text{if } x \neq 0, \\ h_1(\mathbf{e}_1) & \text{if } x = 0. \end{cases}$$

For (a) ⇒ (c), using an extension $\beta : \mathbf{B}^{n+1} \to X$ of α, we can define the homotopy $h : \mathbf{S}^n \times \mathbf{I} \to X$ by $h(x, t) = \beta((1 - t)x + t\mathbf{e}_1)$, which realizes $\alpha \simeq 0$ rel. $\mathbf{e}_1$. □

Even if $A \subset X$ is not simply connected, we have the following:

Proposition 8.2.3 *For a map $\alpha : (\mathbf{B}^{n+1}, \mathbf{S}^n) \to (X, A)$, the following are equivalent:*

(a) *There is a map $\beta : \mathbf{B}^{n+1} \to A$ such that $\beta|\mathbf{S}^n = \alpha|\mathbf{S}^n$ and $\alpha \simeq \beta$ rel. $\mathbf{S}^n$.*
(b) *$\alpha \simeq 0$ by a homotopy $h : \mathbf{B}^{n+1} \times \mathbf{I} \to X$ such that $h(\mathbf{S}^n \times \mathbf{I}) \subset A$.*
(c) *$\alpha \simeq 0$ by a homotopy $h : \mathbf{B}^{n+1} \times \mathbf{I} \to X$ such that $h(\mathbf{S}^n \times \mathbf{I}) \subset A$ and $h(\{\mathbf{e}_1\} \times \mathbf{I}) = \{\alpha(\mathbf{e}_1)\}$, i.e., $[\alpha] = 0 \in \pi_{n+1}(X, A, \alpha(\mathbf{e}_1))$.*

Sketch of Proof The implication (c) ⇒ (b) is obvious. For (b) ⇒ (a), observe that

$$(\mathbf{B}^{n+1}, \mathbf{S}^n) \approx \big((\mathbf{S}^n \times \mathbf{I}) \cup (\mathbf{B}^{n+1} \times \{1\}), \mathbf{S}^n \times \{0\}\big).$$

The restriction $h|(\mathbf{S}^n \times \mathbf{I}) \cup (\mathbf{B}^{n+1} \times \{1\})$ of h in (a) induces the map β in (b). Given the homotopy in condition (a), we can connect it with a similar homotopy to that in the proof of (a) ⇒ (c) in Proposition 8.2.2 to obtain the homotopy in (c).

Concerning the homomorphism induced by a map, we can state as follows:

Proposition 8.2.4 *For a map $f : X \to Y$, the following are equivalent:*

(a) *$f_\sharp : \pi_n(X, x_0) \to \pi_n(Y, f(x_0))$ is a monomorphism for every $x_0 \in X$.*
(b) *For each map $\alpha : \mathbf{S}^n \to X$, $f\alpha \simeq 0$ rel. $\mathbf{e}_1$ implies $\alpha \simeq 0$ rel. $\mathbf{e}_1$.*

(c) *Any map* $\alpha : \mathbf{S}^n \to X$ *is null-homotopic if* $f\alpha \simeq 0$.
(d) *Any map* $\alpha : \mathbf{S}^n \to X$ *extends over* $\mathbf{B}^{n+1}$ *if* $f\alpha$ *extends over* $\mathbf{B}^{n+1}$.

Sketch of Proof Identifying $(\mathbf{I}^n/\partial\mathbf{I}^n, \partial\mathbf{I}^n/\partial\mathbf{I}^n) = (\mathbf{S}^n, \mathbf{e}_1)$, we can easily obtain the equivalence (a) $\Leftrightarrow$ (b). Equivalence between (b), (c), and (d) can be obtained by Proposition 8.2.2.

Proposition 8.2.5 *For a map* $f : X \to Y$, *the following are equivalent:*

(a) $f_\sharp : \pi_n(X, x_0) \to \pi_n(Y, f(x_0))$ *is an epimorphism for every* $x_0 \in X$.
(b) *For each map* $\beta : \mathbf{S}^n \to Y$ *with* $\beta(\mathbf{e}_1) \in f(X)$, *there is a map* $\alpha : \mathbf{S}^n \to X$ *such that* $f\alpha(\mathbf{e}_1) = \beta(\mathbf{e}_1)$ *and* $f\alpha \simeq \beta$ *rel.* $\mathbf{e}_1$.

Sketch of Proof This can be easily proved by using the identification mentioned in Sketch of Proof of Proposition 8.2.4.

The following proposition can be obtained by using the homotopy exact sequence:

Proposition 8.2.6 *For a pair* (X, A) *of spaces, let* $i : A \subset X$ *be the inclusion map. For each* $n \in \mathbb{N}$ *and* $x_0 \in A$, *the following hold:*

(1) $i_\sharp : \pi_n(A, x_0) \to \pi_n(X, x_0)$ *is a monomorphism if* $\pi_{n+1}(X, A, x_0) = \{0\}$;
(2) $i_\sharp : \pi_n(A, x_0) \to \pi_n(X, x_0)$ *is an epimorphism if* $\pi_n(X, A, x_0) = \{0\}$;
(3) $i_\sharp : \pi_n(A, x_0) \to \pi_n(X, x_0)$ *is an isomorphism if* $\pi_{n+1}(X, A, x_0) = \{0\}$ *and* $\pi_n(X, A, x_0) = \{0\}$;
(4) $\pi_{n+1}(X, A, x_0) = \{0\}$ *if* $i_\sharp : \pi_n(A, x_0) \to \pi_n(X, x_0)$ *is a monomorphism and* $i_\sharp : \pi_{n+1}(A, x_0) \to \pi_{n+1}(X, x_0)$ *is an epimorphism.*

It is easy to prove this proposition directly.

Proof First of all, note that (3) is a combination of (1) and (2).

(1): Let $\alpha : (\mathbf{S}^n, \mathbf{e}_1) \to (A, x_0)$ be a map such that $\alpha \simeq 0$ rel. $\mathbf{e}_1$ in X. According to Proposition 8.2.2, α extends to a map $\beta : (\mathbf{B}^{n+1}, \mathbf{S}^n, \mathbf{e}_1) \to (X, A, x_0)$. Then, $[\beta] \in \pi_{n+1}(X, A, x_0) = \{0\}$, so there is a homotopy $h : \mathbf{B}^{n+1} \times \mathbf{I} \to X$ such that $h(\mathbf{S}^n \times \mathbf{I}) \subset A$, $h_0 = \beta$, $h_1(\mathbf{B}^{n+1}) = \{x_0\}$, and $h_t(\mathbf{e}_1) = x_0$ for each $t \in \mathbf{I}$. Hence, $\alpha \simeq 0$ rel. $\mathbf{e}_1$ in A by the restriction $h|\mathbf{S}^n \times \mathbf{I}$. Thus, $i_\sharp$ is a monomorphism.

(2): Each map $\alpha : (\mathbf{I}^n, \partial\mathbf{I}^n) \to (X, x_0)$ can be regarded as

$$\alpha \in \mathrm{C}((\mathbf{I}^n, \partial\mathbf{I}^n, \mathbf{J}^{n-1}), (X, A, x_0)),$$

and then $[\alpha] \in \pi_n(X, A, x_0) = \{0\}$. Hence, there is a homotopy $h : \mathbf{I}^n \times \mathbf{I} \to X$ such that $h(\partial\mathbf{I}^n \times \mathbf{I}) \subset A$, $h(\mathbf{J}^{n-1} \times \mathbf{I}) = \{x_0\}$, $h_0 = \alpha$, and $h_1(\mathbf{I}^n) = \{x_0\}$. We define $\alpha' : \mathbf{I}^n \to A$ as follows:

$$\alpha'(z) = h((z(1), \ldots, z(n-1), 0), z(n)) \quad \text{for each } z \in \mathbf{I}^n.$$

Then, $\alpha'(\partial\mathbf{I}^n) = \{x_0\}$, that is, $\alpha' : (\mathbf{I}^n, \partial\mathbf{I}^n) \to (A, x_0)$. It is easy to see that $\alpha' \simeq \alpha$ rel. $\partial\mathbf{I}^n$ in X. Therefore, $i_\sharp$ is an epimorphism.

(4): Let $\alpha \in \mathrm{C}((\mathbf{I}^{n+1}, \partial\mathbf{I}^{n+1}, \mathbf{J}^n), (X, A, x_0))$. Note that α itself is a homotopy realizing $\alpha|\mathbf{I}^n \simeq 0$ rel. $\partial\mathbf{I}^n$ in X. Since $i_\sharp : \pi_n(A, x_0) \to \pi_n(X, x_0)$ is a monomorphism, it follows that $\alpha|\mathbf{I}^n \simeq 0$ rel. $\partial\mathbf{I}^n$ in A. By the Homotopy Extension Theorem 4.5.3, we have a homotopy

$h : \mathbf{I}^{n+1} \times \mathbf{I} \to X$ such that $h_0 = \alpha$, $h(\mathbf{I}^n \times \mathbf{I}) \subset A$, $h_1(\mathbf{I}^n) = \{x_0\}$, and $h(\mathbf{J}^n \times \mathbf{I}) = \{x_0\}$. Then, $h_1 \in \mathrm{C}((\mathbf{I}^{n+1}, \partial\mathbf{I}^{n+1}), (X, x_0))$. Since $i_\sharp : \pi_{n+1}(A, x_0) \to \pi_{n+1}(X, x_0)$ is an epimorphism, we have $\alpha' \in \mathrm{C}((\mathbf{I}^{n+1}, \partial\mathbf{I}^{n+1}), (A, x_0))$ such that $\alpha' \simeq h_1$ rel. $\partial\mathbf{I}^{n+1}$ in X. Now, we define a homotopy $k : \mathbf{I}^{n+1} \times \mathbf{I} \to X$ as follows:

$$k(z, t) = \begin{cases} \alpha'(z(1), \ldots, z(n), z(n+1) + t) & \text{if } z(n+1) + t \leqslant 1, \\ x_0 & \text{otherwise.} \end{cases}$$

Then, $k_0 = \alpha'$, $k_1 = c_{x_0}$, $k(\mathbf{I}^n \times \mathbf{I}) = \alpha'(\mathbf{I}^{n+1}) \subset A$, and $k(\mathbf{J}^n \times \mathbf{I}) = \{x_0\}$. Thus, we have $[\alpha] = [h_1] = [\alpha'] = 0 \in \pi_{n+1}(X, A, x_0)$. □

Now, in the framework of the homotopy groups, we will give a characterization of n-equivalences. The condition $(\pi)_i$ defined in Sect. 4.20 can be divided into two conditions as follows:

Lemma 8.2.7 *Let $i \in \mathbb{N}$. For a map $f : X \to Y$, condition $(\pi)_i$ is equivalent to the combination of the following two conditions:*

$(\pi_{i-1}^{\mathrm{mono}})$ $\quad f_\sharp : \pi_{i-1}(X, x_0) \to \pi_{i-1}(Y, f(x_0))$ *is a monomorphism for every $x_0 \in X$;*
(π_i^{epi}) $\quad f_\sharp : \pi_i(X, x_0) \to \pi_i(Y, f(x_0))$ *is an epimorphism for every $x_0 \in X$.*

Proof $(\pi)_i \Rightarrow (\pi_{i-1}^{\mathrm{mono}}) + (\pi_i^{\mathrm{epi}})$: The case $i = 1$ directly follows from the definitions. When $i > 1$, since $(\pi)_i$ implies (d) of Proposition 8.2.4, we have $(\pi_{i-1}^{\mathrm{mono}})$.

To see (π_i^{epi}), let $x_0 \in X$ and $\beta : \mathbf{S}^i \to Y$ be a map with $\beta(\mathbf{e}_1) = f(x_0)$. Identifying $(\mathbf{B}^i/\partial\mathbf{B}^i, \partial\mathbf{B}^i/\partial\mathbf{B}^i) = (\mathbf{S}^i, \mathbf{e}_1)$, let $q : (\mathbf{B}^i, \mathbf{S}^{i-1}) \to (\mathbf{S}^i, \mathbf{e}_1)$ be the quotient map. We can apply $(\pi)_i$ to obtain a map $\bar{\alpha} : \mathbf{B}^i \to X$ such that $\bar{\alpha}(\mathbf{S}^{i-1}) = \{x_0\}$ and $f\bar{\alpha} \simeq \beta q$ rel. $\mathbf{S}^{i-1}$ in Y. Since $\bar{\alpha}(\mathbf{S}^{i-1}) = \{x_0\}$ is a singleton, the map $\bar{\alpha}$ induces a map $\tilde{\beta} : \mathbf{S}^i \to X$ such that $\bar{\alpha} = \tilde{\beta}q$. Then, $\tilde{\beta}(\mathbf{e}_1) = x_0$ and $f\tilde{\beta} \simeq \beta$ rel. $\mathbf{e}_1$ in Y because $f\tilde{\beta}q = f\bar{\alpha} \simeq \beta q$ rel. $\mathbf{S}^{i-1}$.

$(\pi_{i-1}^{\mathrm{mono}}) + (\pi_i^{\mathrm{epi}}) \Rightarrow (\pi)_i$: Let $\alpha : \mathbf{S}^{i-1} \to X$ and $\beta : \mathbf{B}^i \to Y$ be maps such that $f\alpha = \beta|\mathbf{S}^{i-1}$. Due to Proposition 8.2.4, $(\pi_{i-1}^{\mathrm{mono}})$ implies that α extends to a map $\alpha' : \mathbf{B}^i \to X$. Let $\gamma : \mathbf{S}^i \to Y$ be the map defined by $\gamma|\mathbf{S}^i_+ = f\alpha'\mathrm{pr}_{\mathbb{R}^i}|\mathbf{S}^i_+$ and $\gamma|\mathbf{S}^i_- = \beta\mathrm{pr}_{\mathbb{R}^i}|\mathbf{S}^i_-$, where

$$\mathbf{S}^i_+ = \mathbf{S}^i \cap (\mathbb{R}^i \times [0, \infty)) \quad \text{and} \quad \mathbf{S}^i_- = \mathbf{S}^i \cap (\mathbb{R}^i \times (-\infty, 0]).$$

Replacing $\mathbf{e}_1$ with $\mathbf{e}_{i+1} = (0, \ldots, 0, 1) \in \mathbb{R}^{i+1}$, we can apply (π_i^{epi}) to obtain a map $\gamma' : \mathbf{S}^i \to X$ such that $\gamma'(\mathbf{e}_{i+1}) = \alpha'(0)$ and $f\gamma' \simeq \gamma$ rel. $\mathbf{e}_{i+1}$. On the other hand, let $q : \mathbf{B}^i \to \mathbf{S}^i$ be the quotient map with $q(\mathbf{S}^{i-1}) = \{\mathbf{e}_{i+1}\}$, where we can assume that $\mathrm{pr}_{\mathbb{R}^i} q(x) = 2x$ if $\|x\| \leqslant 1/2$ and $\mathrm{pr}_{\mathbb{R}^i} q(x) = (2 - 2\|x\|)\|x\|^{-1}x$ if $\|x\| \geqslant 1/2$. Then, we can define a map $\bar{\alpha} : \mathbf{B}^i \to X$ as follows:

$$\bar{\alpha}(x) = \begin{cases} \gamma' q(2x) & \text{if } \|x\| \leqslant 1/2, \\ \alpha'((2\|x\| - 1)\|x\|^{-1}x) & \text{if } \|x\| \geqslant 1/2. \end{cases}$$

Since $f\gamma' \simeq \gamma$ rel. $\mathbf{e}_{i+1}$, it follows that $f\bar{\alpha} \simeq \beta'$ rel. $\mathbf{S}^{i-1}$, where $\beta' : \mathbf{B}^i \to Y$ is defined as follows:

$$\beta'(x) = \begin{cases} \gamma q(2x) & \text{if } \|x\| \leqslant 1/2, \\ f\alpha'((2\|x\| - 1)\|x\|^{-1}x) & \text{if } \|x\| \geqslant 1/2. \end{cases}$$

Observe that

$$\gamma q(2x) = \beta \mathrm{pr}_{\mathbb{R}^i} q(2x) = \beta(4x) \text{ if } \|x\| \leqslant 1/4, \text{ and}$$
$$\gamma q(2x) = f\alpha' \mathrm{pr}_{\mathbb{R}^i} q(2x) = f\alpha'((2 - 4\|x\|)\|x\|^{-1}x) \text{ if } 1/4 \leqslant \|x\| \leqslant 1/2.$$

Now, it is easy to see that $\beta' \simeq \beta''$ rel. $\mathbf{S}^{i-1}$, where $\beta'' : \mathbf{B}^i \to Y$ is defined by

$$\beta''(x) = \begin{cases} \beta(4x) & \text{if } \|x\| \leqslant 1/4, \\ f\alpha'(\|x\|^{-1}x) = f\alpha(\|x\|^{-1}x) & \text{if } \|x\| \geqslant 1/4. \end{cases}$$

Then, it follows that $\beta'' \simeq \beta$ rel. $\mathbf{S}^{i-1}$. Thus, we have $f\bar{\alpha} \simeq \beta$ rel. $\mathbf{S}^{i-1}$. □

Lemma 8.2.7 yields the following characterization of n-equivalences:

Theorem 8.2.8 *Let $n \in \mathbb{N} \cup \{\infty\}$. A map $f : X \to Y$ is an n-equivalence if and only if f induces the bijection $f_\sharp : \pi_0(X) \to \pi_0(Y)$ and, for every $x_0 \in X$, $f_\sharp : \pi_i(X, x_0) \to \pi_i(Y, f(x_0))$ is an isomorphism for every $i < n$ and $f_\sharp : \pi_n(X, x_0) \to \pi_n(Y, f(x_0))$ is an epimorphism (if $n < \infty$).*[10] □

Since every homotopy equivalence $f : X \to Y$ induces the isomorphisms $f_\sharp : \pi_n(X, x_0) \to \pi_n(Y, f(x_0))$, $n \in \mathbb{N}$, $x_0 \in X$, it is a weak homotopy equivalence by Theorem 8.2.8. Thus, Proposition 4.20.1 is a corollary of Theorem 8.2.8.

In the case that f is the inclusion, by combining Theorem 8.2.8 and Proposition 8.2.6, we have the following characterization:

Corollary 8.2.9 *For each $n \in \mathbb{N}$, the inclusion $X \subset Y$ is an n-equivalence if and only if each path-component of Y contains exactly one path-component of X, $\pi_i(Y, X, x_0) = \{0\}$ for each $i \leqslant n$ and $x_0 \in X$.* □

For a 1-equivalence $f : X \to Y$, conditions (π_i^{mono}) and (π_i^{epi}) can be modified to the conditions without base points in the next two propositions.

Proposition 8.2.10 *For a 1-equivalence $f : X \to Y$, condition (π_i^{mono}) is equivalent to the following condition:*

$(\pi_i^{\mathrm{mono}})'$ *f induces the injection from $[\mathbf{S}^i, X]$ to $[\mathbf{S}^i, Y]$, that is, any two maps $\alpha, \alpha' : \mathbf{S}^i \to X$ are homotopic if $f\alpha \simeq f\alpha'$ in Y.*

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[10] This is the definition of an n-equivalence in Homotopy Theory. However, the literature is not consistent on the use of the term "n-equivalence" (some texts require that $f_\sharp : \pi_n(X, x_0) \to \pi_n(Y, f(x_0))$ is an isomorphism).

Proof When α' is a constant map, $(\pi_i^{\mathrm{mono}})'$ is equal to condition (c) of Proposition 8.2.4, so we have the implication $(\pi_i^{\mathrm{mono}})' \Rightarrow (\pi_i^{\mathrm{mono}})$.

$(\pi_i^{\mathrm{mono}}) \Rightarrow (\pi_i^{\mathrm{mono}})'$: For maps $\alpha, \alpha' : \mathbf{S}^i \to X$, assume that $f\alpha \simeq f\alpha'$ in Y. Let $h : \mathbf{S}^i \times \mathbf{I} \to Y$ be a homotopy from $f\alpha'$ to $f\alpha$. Since f is a 1-equivalence, there is a path $\gamma : \mathbf{I} \to X$ from $\alpha'(\mathbf{e}_1)$ to $\alpha(\mathbf{e}_1)$ such that $f\gamma \simeq hj$ rel. $\partial\mathbf{I}$, where $j : \mathbf{I} \to \{\mathbf{e}_1\} \times \mathbf{I} \subset \mathbf{S}^i \times \mathbf{I}$ is the natural injection. We have a map $r : \mathbf{S}^i \to \mathbf{S}^i$ such that $r \simeq \mathrm{id}$ rel. $\mathbf{e}_1$ and $r(\mathbf{S}^i \cap \mathrm{pr}_1^{-1}(\mathbb{R}_+)) = \{\mathbf{e}_1\}$, where $\mathrm{pr}_1 : \mathbb{R}^{i+1} \to \mathbb{R}$ is the projection onto the first factor. We define maps $\alpha'' : \mathbf{S}^i \to X$ and $\beta : \mathbf{S}^i \to Y$ as follows:

$$\alpha''(x) = \begin{cases} \alpha' r(x) & \text{if } x(1) \leqslant 0, \\ \gamma(x(1)) & \text{if } x(1) \geqslant 0, \end{cases} \quad \text{and} \quad \beta(x) = \begin{cases} f\alpha' r(x) & \text{if } x(1) \leqslant 0, \\ hj(x(1)) & \text{if } x(1) \geqslant 0. \end{cases}$$

Then, $\alpha'' \simeq \alpha' r \simeq \alpha'$ and $\alpha''(\mathbf{e}_1) = \alpha(\mathbf{e}_1)$. Since $f\gamma \simeq hj$ rel. $\partial\mathbf{I}$, it follows that $f\alpha'' \simeq \beta$ rel. $\mathbf{e}_1$. On the other hand, we have a map $\varphi : \mathbf{S}^i \times \mathbf{I} \to \mathbf{S}^i \times \mathbf{I}$ such that $\varphi \simeq \mathrm{id}$ rel. $\mathbf{S}^i \times \{1\}$, $\varphi(\{\mathbf{e}_1\} \times \mathbf{I}) = \{(\mathbf{e}_1, 1)\}$, and

$$\varphi(x, 0) = \begin{cases} (r(x), 0) & \text{if } x(1) \leqslant 0, \\ j(x(1)) & \text{if } x(1) \geqslant 0. \end{cases}$$

Then, $h\varphi$ is a homotopy from β to $f\alpha$ with $h\varphi(\{\mathbf{e}_1\} \times \mathbf{I}) = \{\beta(\mathbf{e}_1)\}$, which means that $\beta \simeq f\alpha$ rel. $\mathbf{e}_1$. Hence, $f\alpha'' \simeq f\alpha$ rel. $\mathbf{e}_1$. It follows from (π_i^{mono}) that $\alpha'' \simeq \alpha$ rel. $\mathbf{e}_1$. Since $\alpha'' \simeq \alpha'$, we have $\alpha \simeq \alpha'$ in X. □

Proposition 8.2.11 *For a* 1*-equivalence* $f : X \to Y$*, condition* (π_i^{epi}) *is equivalent to the following:*

$(\pi_i^{\mathrm{epi}})'$ *f induces the surjection from* $[\mathbf{S}^i, X]$ *to* $[\mathbf{S}^i, Y]$*, that is, for each map* $\beta : \mathbf{S}^i \to Y$*, there is a map* $\alpha : \mathbf{S}^i \to X$ *such that* $f\alpha \simeq \beta$ *in* Y*.*

Proof $(\pi_i^{\mathrm{epi}}) \Rightarrow (\pi_i^{\mathrm{epi}})'$: Let $\beta : \mathbf{S}^i \to Y$ be a map. According to Proposition 4.20.2(1), there exists a path $\gamma : \mathbf{I} \to Y$ with $x_0 \in X$ such that $\gamma(0) = \beta(\mathbf{e}_1)$ and $\gamma(1) = f(x_0)$. Using the map $r : \mathbf{S}^i \to \mathbf{S}^i$ from the proof of $(\pi_i^{\mathrm{mono}}) \Rightarrow (\pi_i^{\mathrm{mono}})'$ in Proposition 8.2.10, we define a map $\beta' : \mathbf{S}^i \to Y$ by

$$\beta'(x) = \begin{cases} \beta r(x) & \text{if } x(1) \leqslant 0, \\ \gamma(x(1)) & \text{if } x(1) \geqslant 0. \end{cases}$$

Then, $\beta' \simeq \beta r \simeq \beta$ and $\beta'(\mathbf{e}_1) = f(x_0)$. By (π_i^{epi}), we have a map $\alpha : \mathbf{S}^i \to X$ such that $f\alpha \simeq \beta'$ rel. $\mathbf{e}_1$, hence $f\alpha \simeq \beta$.

$(\pi_i^{\mathrm{epi}})' \Rightarrow (\pi_i^{\mathrm{epi}})$: Let $x_0 \in X$ and $\beta : \mathbf{S}^i \to Y$ be a map with $\beta(\mathbf{e}_1) = f(x_0)$. By $(\pi_i^{\mathrm{epi}})'$, we have a map $\alpha' : \mathbf{S}^i \to X$ such that $f\alpha' \simeq \beta$ in Y. Let $h : \mathbf{S}^i \times \mathbf{I} \to Y$ be a homotopy from $f\alpha'$ to β. Since f is a 1-equivalence, we have a path $\gamma : \mathbf{I} \to X$ with $\gamma(0) = \alpha'(\mathbf{e}_1)$, $\gamma(1) = x_0$, and $f\gamma \simeq hj$ rel. $\partial\mathbf{I}$, where $j : \mathbf{I} \to \{\mathbf{e}_1\} \times \mathbf{I} \subset \mathbf{S}^i \times \mathbf{I}$

is the natural injection. Using the above map $r : \mathbf{S}^i \to \mathbf{S}^i$, we now define maps $\alpha : \mathbf{S}^i \to X$ and $\beta' : \mathbf{S}^i \to Y$ as follows:

$$\alpha(x) = \begin{cases} \alpha' r(x) & \text{if } x(1) \leqslant 0, \\ \gamma(x(1)) & \text{if } x(1) \geqslant 0, \end{cases} \quad \text{and} \quad \beta'(x) = \begin{cases} f\alpha' r(x) & \text{if } x(1) \leqslant 0, \\ hj(x(1)) & \text{if } x(1) \geqslant 0. \end{cases}$$

Since $f\gamma \simeq hj$ rel. $\partial\mathbf{I}$, it follows that $f\alpha \simeq \beta'$ rel. $\mathbf{e}_1$. Moreover, consider the same map $\varphi : \mathbf{S}^i \times \mathbf{I} \to \mathbf{S}^i \times \mathbf{I}$ used in the proof of $(\pi_i^{\text{mono}}) \Rightarrow (\pi_i^{\text{mono}})'$ of Proposition 8.2.10. Then, $h\varphi$ is a homotopy from β' to β with $h\varphi(\{\mathbf{e}_1\} \times \mathbf{I}) = \{\beta(\mathbf{e}_1)\}$, which means that $\beta' \simeq \beta$ rel. $\mathbf{e}_1$. Thus, we have $f\alpha \simeq \beta$ rel. $\mathbf{e}_1$. □

Theorem 8.2.8 can be reformulated using Propositions 8.2.10 and 8.2.11. In particular, we have the following corollary:

Corollary 8.2.12 *A map $f : X \to Y$ is a weak homotopy equivalence if and only if f is a 1-equivalence and f induces the bijection between $[\mathbf{S}^n, X]$ and $[\mathbf{S}^n, Y]$ for every $n \geqslant 1$.* □

8.3 Axioms of Čech Cohomology Theory

For every paracompact space X, as observed in Notes at the end of Sect. 4.16, we have an inverse system $(|N(\mathcal{U})|, [\varphi_{\mathcal{U},\mathcal{V}}])_{\mathcal{U} \in \mathrm{cov}(X)}$ in the homotopy category HTop, i.e., the category of topological spaces and homotopy classes of maps, where each $\varphi_{\mathcal{U},\mathcal{V}} : N(\mathcal{V}) \to N(\mathcal{U})$ is a refining simplicial map, which is defined just before Proposition 4.15.4. For each $n \in \omega$, the ***n*-th Čech cohomology** $\check{H}^n(X)$ of X can be defined as follows:

$$\check{H}^n(X) \underset{\text{def}}{=} \varinjlim (H^n(N(\mathcal{U})), \varphi^*_{\mathcal{U},\mathcal{V}})_{\mathcal{U} \in \mathrm{cov}(X)}.$$

Moreover, let A be a closed set in X. For each $\mathcal{U} \in \mathrm{cov}(X)$, we have $\mathcal{U}|A \in \mathrm{cov}(A)$, where $N(\mathcal{U}|A)$ can be regarded as a subcomplex of $N(\mathcal{U})$. Then, the ***n*-th relative Čech cohomology** of (X, A) is defined as follows:

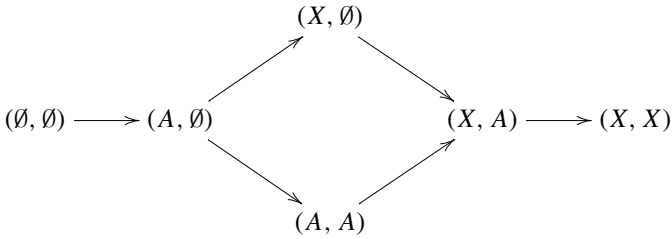
$$\check{H}^n(X, A) \underset{\text{def}}{=} \varinjlim (H^n(N(\mathcal{U}), N(\mathcal{U}|A)), \varphi^*_{\mathcal{U},\mathcal{V}})_{\mathcal{U} \in \mathrm{cov}(X)}.$$

Thus, the (relative) Čech cohomology is defined by using the (relative) cohomologies of simplicial complexes (with subcomplexes). However, the definition of the cohomology of a simplicial complex is not simple.

In their book *Foundations of Algebraic Topology* [FAT],[11] Eilenberg and Steenrod gave an axiomatic characterization of the Čech cohomology theory for compact

[11] Cf. Notes for this chapter.

spaces. Here, we survey their result and its extension. The category of Abelian groups and group homomorphisms is denoted by Ab and the category of pairs of spaces and maps of pairs is denoted by Top_2. For a pair $(X, A) \in \mathsf{Top}_2$, the following diagram of pairs and inclusion maps is called the **lattice** of (X, A):



A subcategory $\mathfrak{A}$ of Top_2 is said to be **admissible** for (co)homology theory if it satisfies the following conditions:

(i) The lattice of every $(X, A) \in \mathfrak{A}$ is a diagram in $\mathfrak{A}$ (i.e., all pairs and inclusions consisting the lattices are in $\mathfrak{A}$).
(ii) Each map $f : (X, A) \to (Y, B)$ in $\mathfrak{A}$ induces a map of any member of the lattice of (X, A) to the corresponding member of (Y, B). Then, all such maps in $\mathfrak{A}$.
(iii) For every $(X, A) \in \mathfrak{A}$, $(X, A) \times \mathbf{I} = (X \times \mathbf{I}, A \times \mathbf{I}) \in \mathfrak{A}$ and the maps $j_0, j_1 : (X, A) \to (X, A) \times \mathbf{I}$ defined by $j_i(x) = (x, i)$ are in $\mathfrak{A}$.
(iv) $\{0\} \in \mathfrak{A}$. If $X, P \in \mathfrak{A}$ and P is a singleton, then any map $f : P \to X$ is in $\mathfrak{A}$.

Here, we regard $(X, \emptyset) = X$. As is easily observed, the following categories are admissible for (co)homology theory:

- $\mathsf{Poly}_2^{\mathsf{c}}$—pairs of compact polyhedra and PL maps of pairs;
- Poly_2—pairs of polyhedra and PL maps of pairs;
- $\mathfrak{W}_2$—pairs having the homotopy type of a pair of polyhedra and maps of pairs;[12]
- Comp_2—pairs of compact spaces and closed sets and maps of pairs;
- Metr_2—pairs of metrizable spaces and closed sets and maps of pairs;
- Para_2—pairs of paracompact spaces and closed sets and maps of pairs.

Remark 8.3.1 The category Normal_2 of pairs of normal spaces and closed sets and maps of pairs is not admissible because it does not satisfy (iii).

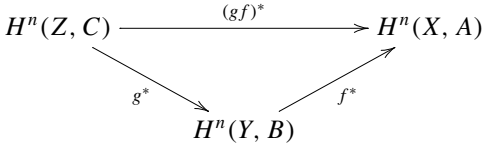
Let $\mathfrak{A}$ be an admissible category. A **cohomology theory** $H^* = (H^*, \delta)$ on $\mathfrak{A}$ is a triple of functions satisfying seven axioms below, the first is a function associating each pair $(X, A) \in \mathfrak{A}$ to a sequence of Abelian groups $H^n(X, A)$, $n \in \omega$, the second is a function associating each map $f : (X, A) \to (Y, B)$ in $\mathfrak{A}$ to a sequence of homomorphisms $H^n(f) = f^* : H^n(Y, B) \to H^n(X, A)$, $n \in \omega$, called **induced homomorphisms**, and the third is a function associating each pair $(X, A) \in \mathfrak{A}$ to

[12] Equiv., having the homotopy type of a pair of a CW complex and its subcomplex.

a sequence of homomorphisms $\delta : H^{n-1}(A) \to H^n(X, A)$, $n \in \mathbb{N}$, called **coboundary operators**.

Axiom 1 If $f = \mathrm{id} : (X, A) \to (X, A)$, then $f^* = \mathrm{id} : H^n(X, A) \to H^n(X, A)$.

Axiom 2 For any two maps $f : (X, A) \to (Y, B)$ and $g : (Y, B) \to (Z, C)$, $(gf)^* = f^*g^* : H^n(Z, C) \to H^n(X, A)$, i.e., the following diagram commutes:



Such a combination $H^n : \mathfrak{A} \to \mathsf{Ab}$ of functions in Axioms 1 and 2 is generally called a **contravariant functor**.

Axiom 3 For any map $f : (X, A) \to (Y, B)$ in $\mathfrak{A}$, $\delta(f|A)^* = f^*\delta$, i.e., the following diagram commutes:

$$\begin{array}{ccc} H^{n-1}(B) & \xrightarrow{\delta} & H^n(Y, B) \\ {\scriptstyle (f|A)^*}\downarrow & & \downarrow{\scriptstyle f^*} \\ H^{n-1}(A) & \xrightarrow[\delta]{} & H^n(X, A) \end{array}$$

The following sequence is called the **cohomology sequence** of $(X, A) \in \mathfrak{A}$:

$$\{0\} \longrightarrow H^0(X, A) \xrightarrow{j^*} H^0(X) \xrightarrow{i^*} H^0(A) \xrightarrow{\delta} H^1(X, A) \xrightarrow{j^*} \cdots$$
$$\longrightarrow H^n(X, A) \xrightarrow{j^*} H^n(X) \xrightarrow{i^*} H^n(A) \xrightarrow{\delta} H^{n+1}(X, A) \longrightarrow \cdots,$$

where $i : A \subset X$ and $j : X = (X, \emptyset) \subset (X, A)$ are the inclusions.

Axiom 4 (EXACTNESS) The cohomology sequence for each $(X, A) \in \mathfrak{A}$ is exact, i.e., $\operatorname{im} j^* = \ker i^*$, $\operatorname{im} i^* = \ker \delta$, and $\operatorname{im} \delta = \ker j^*$.

Axiom 5 (HOMOTOPY) If a map $f : (X, A) \to (Y, B)$ in $\mathfrak{A}$ is homotopic to a map $g : (X, A) \to (Y, B)$ in $\mathfrak{A}$, then $f^* = g^* : H^n(Y, B) \to H^n(X, A)$.

Axiom 6 (EXCISION) If U is an open set in X with $\operatorname{cl} U \subset \operatorname{int} A$ and the inclusion $i : (X \setminus U, A \setminus U) \subset (X, A)$ is in $\mathfrak{A}$, then $i^* : H^n(X, A) \to H^n(X \setminus U, A \setminus U)$ is an isomorphism for any $n \in \omega$.

Axiom 7 (DIMENSION) If P is a singleton, then $H^n(P) = \{0\}$ for any $n \in \mathbb{N}$.

The group $G = H^0(\{0\})$ is called the **coefficient group** of the cohomology theory H^*. Then, $H^0(P) \cong G$ for every singleton P by Axiom 5. Due to Theorem 10.1c in [FAT, Chap. 3], the EILENBERG–STEENROD UNIQUENESS THEOREM for cohomology theories on $\mathsf{Poly}_2^{\mathsf{c}}$ holds:[13]

Theorem 8.3.1 (EILENBERG–STEENROD) *A cohomology theory on $\mathsf{Poly}_2^{\mathsf{c}}$ is essentially unique. More precisely, if two cohomology theories $H^* = (H^*, \delta)$ and $\bar{H}^* = (\bar{H}^*, \bar{\delta})$ on $\mathsf{Poly}_2^{\mathsf{c}}$ have isomorphic coefficient groups, then H^* and $\bar{H}^*$ are isomorphic, that is, there exists isomorphisms $\varphi : H^n(X, A) \to \bar{H}^n(X, A)$, $n \in \omega$, for each $(X, A) \in \mathsf{Poly}_2^{\mathsf{c}}$ such that the following diagram commutes:*

$$\begin{array}{ccc} H^n(A) & \xrightarrow{\delta} & H^{n+1}(X, A) \\ \downarrow{\scriptstyle \varphi} & & \downarrow{\scriptstyle \varphi} \\ \bar{H}^n(A) & \xrightarrow[\bar{\delta}]{} & \bar{H}^{n+1}(X, A) \end{array}$$

and for each map $f : (X, A) \to (Y, B)$ in $\mathsf{Poly}_2^{\mathsf{c}}$, the following diagram commutes:

$$\begin{array}{ccc} H^n(Y, B) & \xrightarrow{f^*} & H^n(X, A) \\ \downarrow{\scriptstyle \varphi} & & \downarrow{\scriptstyle \varphi} \\ \bar{H}^n(Y, B) & \xrightarrow[f^*]{} & \bar{H}^n(X, A) \end{array}$$

where the top f^* is $H^n(f)$ and the bottom f^* is $\bar{H}^n(f)$ in the above diagram. In other words, any cohomology theory on $\mathsf{Poly}_2^{\mathsf{c}}$ is isomorphic to a simplicial cohomology theory. □

To extend the above uniqueness to the category Poly_2 (or $\mathfrak{W}_2$), one more axiom is necessary. A cohomology theory H^* on $\mathfrak{A}$ is said to be **additive** if it satisfies the following axiom:

Axiom 8 (ADDITIVITY) If $X = \bigoplus_{\lambda\in\Lambda} X_\lambda \in \mathfrak{A}$ and the inclusion $i_\lambda : X_\lambda \subset X$ is in $\mathfrak{A}$ for each $\lambda \in \Lambda$, then there is an isomorphism $\varphi : H^n(X) \to \prod_{\lambda\in\Lambda} H^n(X_\lambda)$ for each $n \in \omega$ such that $\mathrm{pr}_\lambda \varphi = i_\lambda^*$ for each $\lambda \in \Lambda$.

The Uniqueness Theorem 8.3.1 for cohomology theories on $\mathsf{Poly}_2^{\mathsf{c}}$ can be extended to Poly_2 (or $\mathfrak{W}_2$) as follows:

Theorem 8.3.2 (MILNOR) *An additive cohomology theory on Poly_2 (or $\mathfrak{W}_2$) is essentially unique. Thus, any additive cohomology theory on Poly_2 is isomorphic to a simplicial cohomology theory.*

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[13] In the book [FAT], a finite simplicial complex is simply called a simplicial complex (see Definition 3.1 of Chap. II in [FAT]) and a triangulated space (or a triangulated pair) means a topological compact polyhedron (or a pair of such polyhedra).

A cohomology theory H^* on $\mathfrak{A}$ is said to be **continuous** if it satisfies the following axiom:

Axiom 9 (CONTINUITY) For an inverse system $(X_\lambda, A_\lambda)_{\lambda\in\Lambda}$ in $\mathfrak{A}$ with $(X, A) = \varprojlim(X_\lambda, A_\lambda)$ $\left(= \left(\varprojlim X_\lambda, \varprojlim A_\lambda\right)\right) \in \mathfrak{A}$, each $H^n(X, A)$ is naturally isomorphic to the direct limit $\varinjlim H^n(X_\lambda, A_\lambda)$ by an isomorphism $\varphi : H^n(X, A) \to \varinjlim H^n(X_\lambda, A_\lambda)$ such that $\varphi p_\lambda^* : H^n(X_\lambda, A_\lambda) \to \varinjlim H^n(X_\lambda, A_\lambda)$ is the inclusion for each $\lambda \in \Lambda$, where $p_\lambda : X \to X_\lambda$ is the inverse limit projection.

Due to [FAT, Chap. X, Theorem 12.2], the Čech cohomology on Comp_2 is characterized as follows:

Theorem 8.3.3 (EILENBERG–STEENROD) *A continuous cohomology theory on* Comp_2 *is essentially unique. In other words, any continuous cohomology theory on* Comp_2 *is isomorphic to the Čech cohomology theory.*

To extend the above characterization to the category Para_2, a new axiom was introduced by Bacon [1] instead of the Continuity Axiom 9. For each $(X, A) \in \mathfrak{A}$ and $h \in H^n(X)$, we define $h|A = i^*(h)$, where $i^* : H^n(X) \to H^n(A)$ is the induced homomorphism by the inclusion $i : A \subset X$. A cohomology theory on $\mathfrak{A}$ is **point reductive** if it satisfies the following:

Axiom 10 (POINT REDUCTIVITY) For $X \in \mathfrak{A}$ with a singleton $P \subset X$ and $h \in H^n(X)$, if $h|P = 0$, then there is a neighborhood N of P in X such that the inclusion $N \subset X$ is in $\mathfrak{A}$ and $h|N = 0$.

Both Theorems 8.3.2 and 8.3.3 can be generalized as follows:

Theorem 8.3.4 (BACON) *A point reductive additive cohomology theory on* Para_2 *is essentially unique. In other words, any point reductive additive cohomology theory on* Para_2 *is isomorphic to the Čech cohomology theory.*

Note In the same way as a cohomology theory, a **homology theory** $H_* = (H_*, \partial)$ on $\mathfrak{A}$ is defined as a triple of functions satisfying seven axioms (Axioms 1–7 above adjusted properly), the first is a function associating each pair $(X, A) \in \mathfrak{A}$ to a sequence of Abelian groups $H_n(X, A), n \in \omega$, the second is a function associating each map $f : (X, A) \to (Y, B)$ in $\mathfrak{A}$ to a sequence of homomorphisms $H_n(f) = f_* : H_n(X, A) \to H_n(Y, B), n \in \omega$, called **induced homomorphisms**, and the third is a function associating each pair (X, A) to a sequence of homomorphisms $\partial : H_n(X, A) \to H_{n-1}(A), n \in \mathbb{N}$, called **boundary operators**.[14] For a homology theory, we have the following **homology sequence**:

$$\cdots \longrightarrow H_n(A) \xrightarrow{i_*} H_n(X) \xrightarrow{j_*} H_n(X, A) \xrightarrow{\partial} H_{n-1}(A) \longrightarrow$$

$$\cdots \longrightarrow H_1(X, A) \xrightarrow{\partial} H_0(A) \xrightarrow{i_*} H_0(X) \xrightarrow{j_*} H_0(X, A),$$

[14] In homology, the domains and ranges (codomains) of induced homomorphisms and (co)boundary operators are interchanged from those in cohomology.

where $i : A \subset X$ and $j : X = (X, \emptyset) \subset (X, A)$ are the inclusions. Due to Theorem 10.1 in [FAT, Chap. III], *a homology theory on* Poly_2^c *essentially unique*, that is, the EILENBERG–STEENROD UNIQUENESS THEOREM is valid for homology theories.

An **additive** homology theory is a homology theory H_* satisfying the Additivity Axiom (Axiom 8) adjusted for homology theories, that is,

Axiom 8 (ADDITIVITY) If $X = \bigoplus_{\lambda\in\Lambda} X_\lambda \in \mathfrak{A}$, for each $\lambda \in \Lambda$, $X_\lambda \in \mathfrak{A}$ and the inclusion $X_\lambda \subset X$ is in $\mathfrak{A}$, then there is an isomorphism $\varphi : \bigoplus_{\lambda\in\Lambda} H_n(X_\lambda) \to H^n(X)$ for each $n \in \mathbb{N}$ such that $\varphi | H_n(X_\lambda) = (i_\lambda)_*$ for each $\lambda \in \Lambda$.

Then, *an additive homology theory on* Poly_2 *(or* $\mathfrak{W}$*) is essentially unique*. Namely, the homology version of MILNOR UNIQUENESS THEOREM is also valid.

The direct limit preserves the exactness but the inverse limit does not. In fact, there exists a compact pair (X, A) whose homology sequence is not exact (cf. [FAT, Chap. X, Sect. 4]). Hence, we need to consider a slightly weaker axiom than Axiom 4:

Axiom 4′ (PARTIAL EXACTNESS) If $(X, A) \in \mathfrak{A}$ is a pair of polyhedra, then the homology sequence of (X, A) is exact. In general, $j_*i_* = 0, i_*\partial = 0$, and $\partial j_* = 0$ (equiv., $\operatorname{im} i_* \subset \ker j_*$, $\operatorname{im} \partial \subset \ker i_*$, and $\operatorname{im} j_* \subset \ker \partial$) in the homology sequence.

In the definition of a homology theory, replacing Axiom 4 with the above axiom, we define a **partially exact homology theory**.

The homology version of Theorem 8.3.3 [FAT, Chap. X, Theorem 12.2] states as follows:

Theorem 8.3.5 (EILENBERG–STEENROD) *Any continuous partially exact homology theory on* Comp_2 *is isomorphic to the Cěch homology theory.* □

In the above, a partially exact homology theory H_* is **continuous** if it satisfies the following:

Axiom 9 (CONTINUITY) For an inverse system $(X_\lambda, A_\lambda)_{\lambda\in\Lambda}$ in $\mathfrak{A}$ with $(X, A) = \varprojlim(X_\lambda, A_\lambda) \in \mathfrak{A}$, each $H_n(X, A)$ is naturally isomorphic to the inverse limit $\varprojlim H_n(X_\lambda, A_\lambda)$ by an isomorphism $\varphi : H_n(X, A) \to \varprojlim H_n(X_\lambda, A_\lambda)$ such that $q_\lambda\varphi = (p_\lambda)_* : H_n(X, A) \to H_n(X_\lambda, A_\lambda)$ for each $\lambda \in \Lambda$, where $p_\lambda : X \to X_\lambda$ and $q_\lambda : \varprojlim H_n(X_\lambda, A_\lambda) \to H_n(X_\lambda, A_\lambda)$ are the inverse limit projections.

8.4 Homotopical Definition of Čech Cohomology Groups

Recall that **Eilenberg–MacLane spaces** $K(\mathbb{Z}, n)$, $n \in \mathbb{N}$, are defined as path-connected spaces homotopy dominated by simplicial complexes such that

$$\pi_n(K(\mathbb{Z}, n)) \cong \mathbb{Z} \text{ and } \pi_i(K(\mathbb{Z}, n)) = \{0\} \text{ if } i \neq n.$$

It should be remarked that each $K(\mathbb{Z}, n)$ is not topologically unique but its homotopy type is unique. Eilenberg–MacLane spaces are usually defined as CW complexes, where $K(\mathbb{Z}, 1) = \mathbf{S}^1$. However, in Sect. 7.8, we constructed connected countable simplicial complexes $K(\mathbb{Z}, n), n \in \mathbb{N}$, so that $|K(\mathbb{Z}, n)|$ is Eilenberg–MacLane space,

where $K(\mathbb{Z}, 1) = \partial F(\Delta^2)$ is a finite simplicial complex. Furthermore, for the case $n = 0$, we defined $K(\mathbb{Z}, 0) = \mathbb{Z}$, a countable zero-dimensional simplicial complex.

In this section, replacing $|K(\mathbb{Z}, n)|$ with a space of the same homotopy type which has a kind of algebraic operation,[15] we introduce a group structure on $[X, |K(\mathbb{Z}, n)|]$ which can be considered as the n-th cohomology group $H^n(X)$ of X. For a pair (X, A) of a space X and its closed set A, $[(X, A), (|K(\mathbb{Z}, n)|, v_n)]$ is considered as the n-th cohomology group $H^n(X, A)$, where v_n is a base point of $|K(\mathbb{Z}, n)|$ considered as the unit element of algebraic operation. Then, we apply Bacon's Theorem 8.3.4 to conclude that this cohomology theory is isomorphic to the Čech cohomology theory. To this end, our cohomology theory must be defined on Para_2, so it is desirable that $|K(\mathbb{Z}, n)|$ has the HEP for paracompact spaces. As mentioned in Remark 7.8.1, $n \geqslant 2$, $K(\mathbb{Z}, n)$ can be replaced with a countable locally finite simplicial complex of the same homotopy type. Thus, we can require the following condition:

- *Each* $|K(\mathbb{Z}, n)| = |K(\mathbb{Z}, n)|_{\mathrm{m}}$ *is an ANR having the HEP for paracompact spaces.*

In the following, spaces are assumed to be *paracompact* unless otherwise noticed.

For each $n \in \omega$, choose a base point $v_n \in |K(\mathbb{Z}, n)|$ that is a vertex of $K(\mathbb{Z}, n)$ where $v_0 = 0 \in \mathbb{Z}$ $(= K(\mathbb{Z}, 0))$. By Corollary 7.8.10 (cf. Theorem 8.1.4), we have the following homotopy equivalence:

$$|K(\mathbb{Z}, n)| \simeq \Omega(|K(\mathbb{Z}, n+1)|, v_{n+1}).$$

In the above, $\Omega(|K(\mathbb{Z}, n+1)|, v_{n+1})$ is an ANR by Proposition 6.2.9 (7). Then, both $(|K(\mathbb{Z}, n)|, v_n)$ and $(\Omega(|K(\mathbb{Z}, n+1)|, v_{n+1}), c_{v_{n+1}})$ have the HEP by Theorems 4.5.3 and 6.4.8, respectively, where c_v is the constant loop at v.[16] Hence, applying Corollary 6.4.10, we can obtain the following pointed homotopy equivalence:

$$\big(|K(\mathbb{Z}, n)|, v_n\big) \simeq \big(\Omega(|K(\mathbb{Z}, n+1)|, v_{n+1}), c_{v_{n+1}}\big).$$

Then, we have also the following homotopy equivalence:

$$\big(\Omega(|K(\mathbb{Z}, n+1)|, v_{n+1}), c_{v_{n+1}}\big) \simeq \big(\Omega(\Omega(|K(\mathbb{Z}, n+2)|, v_{n+2}), c_{v_{n+2}}), c_{c_{v_{n+2}}}\big).$$

In the following, to avoid complication and bothersome, the base points v_n, c_{v_n}, $c_{c_{v_n}}$ are denoted by v_* or c_*.

First, we will recall the definition of the join of loops. Let

---

[15] In general, a homotopy equivalence $k : Y \to Y'$ naturally induces a bijection $k_* : [X, Y] \to [X, Y']$. Indeed, for each $f, g \in \mathrm{C}(X, Y)$, if $f \simeq g$, then $kf \simeq kg$. Hence, k induces a function $k_* : [X, Y] \to [X, Y']$ defined by $k_*([f]) = [kf]$. Similarly, a homotopy inverse $k' : Y' \to Y$ of k induces $k'_* : [X, Y'] \to [X, Y]$. Since $k'_* k_*([f]) = [k'kf] = [f]$ for each $f \in \mathrm{C}(X, Y)$ and $k_* k'_*([g]) = [kk'g] = [g]$ for each $g \in \mathrm{C}(X, Y')$, k_* is a bijection and $(k_*)^{-1} = k'_*$.

[16] Here the HEP is a different type from the HEP (for metrizable spaces), which is defined just before Proposition 6.4.6.

$$\alpha, \beta \in \Omega(|K(\mathbb{Z}, n+1)|, v_*)\ (= \mathrm{C}((\mathbf{I}, \partial\mathbf{I}), (|K(\mathbb{Z}, n+1)|, v_*)).$$

The join $\alpha * \beta \in \mathrm{C}((\mathbf{I}, \partial\mathbf{I}), (|K(\mathbb{Z}, n+1)|, v_*)$ is defined as follows:

$$(\alpha * \beta)(t) = \begin{cases} \alpha(2t) & \text{if } t \leqslant 1/2, \\ \beta(2t-1) & \text{if } t \geqslant 1/2. \end{cases}$$

This join induces the group operation on the fundamental group

$$\pi_1(|K(\mathbb{Z}, n+1)|, v_*) = [(\mathbf{I}, \partial\mathbf{I}), (|K(\mathbb{Z}, n+1)|, v_*)].$$

Note In the above, $\alpha * \beta \simeq \beta * \alpha$. Indeed, we have also

$$\begin{aligned} &\mathrm{C}((\mathbf{I}, \partial\mathbf{I}), (|K(\mathbb{Z}, n+1)|, v_*)) = \Omega(|K(\mathbb{Z}, n+1)|, v_*) \\ &\simeq \Omega(\Omega(|K(\mathbb{Z}, n+2)|, v_*), c_*) = \mathrm{C}((\mathbf{I}, \partial\mathbf{I}), (\mathrm{C}((\mathbf{I}, \partial\mathbf{I}), (|K(\mathbb{Z}, n+2)|, v_*)), c_*)) \\ &\overset{\psi}{\approx} \mathrm{C}((\mathbf{I}^2, \partial\mathbf{I}^2), (|K(\mathbb{Z}, n+2)|, v_*)), \end{aligned}$$

where the last homeomorphism ψ is naturally defined as $\psi(\gamma)(t, s) = \gamma(t)(s)$ for $(t, s) \in \mathbf{I}^2$ (Corollary 2.10.7). Compositing the above homotopy equivalences, we have the following pointed homotopy equivalence:

$$\vartheta : (\Omega(|K(\mathbb{Z}, n+1)|, v_*), c_*) \to (\mathrm{C}((\mathbf{I}^2, \partial\mathbf{I}^2), (|K(\mathbb{Z}, n+2)|, v_*)), c_*),$$

where $\gamma \simeq \gamma' \Leftrightarrow \vartheta(\gamma) \simeq \vartheta(\gamma')$ for each $\gamma, \gamma' \in \Omega(|K(\mathbb{Z}, n+1)|, v_*)$ (Proposition 2.10.6). For the above α, β, observe that $\vartheta(\alpha * \beta) = \vartheta(\alpha) * \vartheta(\beta)$, where the latter operation is the one defining the addition on $\pi_2(|K(\mathbb{Z}, n+2)|, v_*)$ (cf. Sect. 8.1). Then, we have $\vartheta(\alpha) * \vartheta(\beta) \simeq \vartheta(\beta) * \vartheta(\alpha)$, which means $\alpha * \beta \simeq \beta * \alpha$.

The ***n*-th cohomology group** of a space X is the following additive group:

$$\begin{aligned} H^n(X) &\underset{\text{def}}{=} [X, \Omega(|K(\mathbb{Z}, n+1)|, v_*)], \\ [\alpha] + [\beta] &= [\alpha * \beta],\ \alpha, \beta \in \mathrm{C}(X, \Omega(|K(\mathbb{Z}, n+1)|, v_*)). \end{aligned}$$

For a closed set A in X, the **relative *n*-th cohomology group** of (X, A) is the following additive group:

$$\begin{aligned} H^n(X, A) &\underset{\text{def}}{=} [(X, A), (\Omega(|K(\mathbb{Z}, n+1)|, v_*), c_*)], \\ [\alpha] + [\beta] &= [\alpha * \beta],\ \alpha, \beta \in \mathrm{C}((X, A), (\Omega(|K(\mathbb{Z}, n+1)|, v_*), c_*)), \end{aligned}$$

where $H^n(X, \emptyset) = H^n(X)$. In the above, $\alpha * \beta$ is defined by $(\alpha * \beta)(x) = \alpha(x) * \beta(x)$ for each $x \in X$ (the latter $*$ is the join of loops). The null element of $H^n(X, A)$ is $[c_*]$, i.e., $[c_*] = 0$. It is necessary to verify the well-definedness and commutativity of $[\alpha] + [\beta]$, which will be done a little later.[17]

[17] As noted above, $\alpha(x) * \beta(x) \simeq \beta(x) * \alpha(x)$ for each $x \in X$, but this does not mean $\alpha * \beta \simeq \beta * \alpha$.

For simplicity, we will use the following notation introduced in Proposition 8.1.2:

$$H_{X,A} = (X \times \partial \mathbf{I}) \cup (A \times \mathbf{I}) \quad \big(H_{X,\emptyset} = X \times \partial \mathbf{I}\big).$$

Moreover, let $H^2_{X,A}$ denote the following set:

$$\begin{aligned} H_{X\times\mathbf{I},H_{X,A}} &= (X \times \mathbf{I} \times \partial \mathbf{I}) \cup (H_{X,A} \times \mathbf{I}) \\ &= (X \times \mathbf{I} \times \partial \mathbf{I}) \cup (X \times \partial \mathbf{I} \times \mathbf{I}) \cup (A \times \mathbf{I}^2) \\ &= (X \times \partial \mathbf{I}^2) \cup (A \times \mathbf{I}^2). \end{aligned}$$

Then, $H^n(X, A)$ can alternatively be defined as follows:

(a) $H^n(X, A) = [(X \times \mathbf{I}, H_{X,A}), (|K(\mathbb{Z}, n+1)|, v_*)]$,
$[\alpha] + [\beta] = [\alpha * \beta]$, $\alpha, \beta \in \mathrm{C}((X \times \mathbf{I}, H_{X,A}), (|K(\mathbb{Z}, n+1)|, v_*))$,
where the join $\alpha * \beta$ is defined as follows:

$$(\alpha * \beta)(x, t) = \begin{cases} \alpha(x, 2t) & \text{if } t \leqslant 1/2, \\ \beta(x, 2t-1) & \text{if } t \geqslant 1/2. \end{cases}$$

(b) $H^n(X, A) = [(X \times \mathbf{I}^2, H^2_{X,A}), (|K(\mathbb{Z}, n+2)|, v_*)]$,
$[\alpha] + [\beta] = [\alpha * \beta]$, $\alpha, \beta \in \mathrm{C}((X \times \mathbf{I}^2, H^2_{X,A}), (|K(\mathbb{Z}, n+2)|, v_*))$, where the join $\alpha * \beta$ is defined as follows:

$$(\alpha * \beta)(x, t, s) = \begin{cases} \alpha(x, 2t, s) & \text{if } t \leqslant 1/2, \\ \beta(x, 2t-1, s) & \text{if } t \geqslant 1/2. \end{cases}$$

If we adopt Definition (a), it is very easy to see that $[\alpha] + [\beta]$ is well-defined.

Using Definition (a), in the same way as the well-definedness of group operation on the fundamental group $\pi_1(X)$, it is easy to see that

$$\alpha \simeq \alpha',\ \beta \simeq \beta' \ \Rightarrow\ \alpha * \beta \simeq \alpha' * \beta' \quad \text{for each } \alpha, \alpha', \beta, \beta' \in \mathrm{C}((X \times \mathbf{I}, H_{X,A}), (|K(\mathbb{Z}, n+1)|, v_*)),$$

which means that the addition of $H^n(X, A)$ is well-defined (cf. Sect. 8.1).

If we adopt Definition (b), the commutativity of $[\alpha] + [\beta]$ can be shown in the same way as $\pi_2(X)$.

Using Definition (b), in the same way as the commutativity of group operation on the homotopy group $\pi_2(X)$, it is easy to see that $\alpha * \beta \simeq \beta * \alpha$, which means that the addition of $H^n(X, A)$ is commutative (cf. Sect. 8.1).

We will show that the above definitions (a) and (b) are equivalent to the first definition of $H^n(X, A)$.

(a): By Proposition 8.1.2 (Corollary 2.10.7), we have a natural homeomorphism

$$\psi_1 : \mathrm{C}((X, A), (\Omega(|K(\mathbb{Z}, n+1)|, v_*), c_*)) \\ \to \mathrm{C}((X \times \mathbf{I}, H_{X,A}), (|K(\mathbb{Z}, n+1)|, v_*)),$$

which preserves the homotopy relation $\simeq$ (i.e., $\gamma \simeq \gamma' \Leftrightarrow \psi_1(\gamma) \simeq \psi_1(\gamma')$). Since ψ_1 is defined by $\psi_1(\gamma)(x, t) = \gamma(x)(t)$ for each $x \in X$ and $t \in \mathbf{I}$, it is easy to see that ψ_1 preserves the operation $*$, that is,

$$\psi_1(\alpha * \beta) = \psi_1(\alpha) * \psi_1(\beta) \quad \text{for } \alpha, \beta \in \mathrm{C}((X, A), (\Omega(|K(\mathbb{Z}, n+1)|, v_*), c_*)).$$

(b): A pointed homotopy equivalence $\theta : (|K(\mathbb{Z}, n+1)|, v_*) \to (M, v)$ induces the following homotopy equivalence:

$$\theta_* : \mathrm{C}((X \times \mathbf{I}, H_{X,A}), (|K(\mathbb{Z}, n+1)|, v_*)) \to \mathrm{C}((X \times \mathbf{I}, H_{X,A}), (M, v)),$$

which is defined by $\theta_*(\gamma)(x, t) = \theta(\gamma(x, t))$. In the same way as (a), we can define the operation $*$ on $\mathrm{C}((X \times \mathbf{I}, H_{X,A}), (M, v))$. Then, θ_* preserves the homotopy relation $\simeq$ and the operation $*$.[18] Taking $(\Omega(|K(\mathbb{Z}, n+2)|, v_*), c_*)$ as (M, v), we have the following homotopy equivalence:

$$\theta_* : \mathrm{C}((X \times \mathbf{I}, H_{X,A}), (|K(\mathbb{Z}, n+1)|, v_*)) \\ \to \mathrm{C}((X \times \mathbf{I}, H_{X,A}), (\Omega(|K(\mathbb{Z}, n+2)|, v_*), c_*)).$$

Again by Proposition 8.1.2 (Corollary 2.10.7), we have a natural homeomorphism

$$\psi' : \mathrm{C}((X \times \mathbf{I}, H_{X,A}), (\Omega(|K(\mathbb{Z}, n+2)|, v_*), c_*)) \\ \to \mathrm{C}((X \times \mathbf{I}^2, H^2_{X,A}), (|K(\mathbb{Z}, n+2)|, v_*)).$$

Composing θ_* and ψ', we have the following homotopy equivalence:

$$\psi_2 = \psi'\theta_* : \mathrm{C}((X \times \mathbf{I}, H_{X,A}), (|K(\mathbb{Z}, n+1)|, v_*)) \\ \to \mathrm{C}((X \times \mathbf{I}^2, H^2_{X,A}), (|K(\mathbb{Z}, n+2)|, v_*)),$$

which preserves the homotopy relation $\simeq$ and the operation $*$ because both θ_* and ψ' do.

---

[18] As observed in Footnote 15, θ induces a bijection

$$\theta_* : [(X \times \mathbf{I}, H_{X,A}), (|K(\mathbb{Z}, n+1)|, v_*)] \to [(X \times \mathbf{I}, H_{X,A}), (M, v)],$$

which is now a group isomorphism because

$$\theta_*([\alpha] + [\beta]) = \theta_*([\alpha * \beta]) = [\theta(\alpha * \beta)] = [\theta\alpha * \theta\beta] = [\theta\alpha] + [\theta\beta] = \theta_*([\alpha]) + \theta_*([\beta]).$$

.

Let (Y, B) be another pair of a space Y and its closed set B. For a map $f : (X, A) \to (Y, B)$, we define the **induced homomorphism** $f^* : H^n(Y, B) \to H^n(X, A)$ as follows:

$$f^*([\alpha]) \underset{\text{def}}{=} [\alpha f] \text{ for each } \alpha \in \mathrm{C}((Y, B), (\Omega(|K(\mathbb{Z}, n+1)|, v_*), c_*)).$$

Let $\alpha, \beta \in \mathrm{C}((Y, B), (\Omega(|K(\mathbb{Z}, n+1)|, v_*), c_*))$. Then, $\alpha \simeq \beta$ implies $\alpha f \simeq \beta f$, which means f^* is well-defined. Moreover, the fact that f^* is a homomorphism is shown as follows:

$$\begin{aligned} f^*([\alpha] + [\beta]) &= f^*([\alpha * \beta]) = [(\alpha * \beta) f] = [(\alpha f) * (\beta f)] \\ &= [\alpha f] + [\beta f] = f^*([\alpha]) + f^*([\beta]). \end{aligned}$$

When alternative definitions (a) and (b) are adopted, f^* is alternatively defined as follows:

(a) $f^*([\alpha]) \underset{\text{def}}{=} [\alpha(f \times \mathrm{id}_{\mathbf{I}})]$ for $\alpha \in \mathrm{C}((Y \times \mathbf{I}, H_{Y,B}), (|K(\mathbb{Z}, n+1)|, v_*))$;

(b) $f^*([\alpha]) \underset{\text{def}}{=} [\alpha(f \times \mathrm{id}_{\mathbf{I}^2})]$ for $\alpha \in \mathrm{C}((Y \times \mathbf{I}^2, H^2_{Y,B}), (|K(\mathbb{Z}, n+2)|, v_*))$.

When the group operation is not necessarily considered,

(c) $f^*([\alpha]) \underset{\text{def}}{=} [\alpha f]$ for $\alpha \in \mathrm{C}((Y, B), (|K(\mathbb{Z}, n)|, v_*))$.

From the definition, the following are obvious:

- $(\mathrm{id}_{(X,A)})^* = \mathrm{id}_{H^n(X,A)}$;
- *For maps* $f : (X, A) \to (Y, B)$ *and* $g : (Y, B) \to (Z, C)$,

$$(gf)^* = f^* g^* : H^n(Z, C) \to H^n(X, A);$$

- (HOMOTOPY AXIOM) *For maps* $f, g : (X, A) \to (Y, B)$,

$$f \simeq g \;\Rightarrow\; f^* = g^* : H^n(Y, B) \to H^n(X, A).$$

As a combination of the above three, we have the following:

- *Every homotopy equivalence* $f : (X, A) \to (Y, B)$ *induces an isomorphism* $f^* : H^n(Y, B) \to H^n(X, A)$. *Hence,*

$$(X, A) \simeq (Y, B) \;\Rightarrow\; H^n(X, A) \cong H^n(Y, B).$$

The following is easy to see:

- (EXCISION AXIOM) *Let* A *be a closed set and* U *an open set in* X *such that* $\mathrm{cl}\, U \subset \mathrm{int}\, A$. *Then,* $H^n(X \setminus U, A \setminus U) \cong H^n(X, A)$, *where the inclusion* $X \setminus U \subset X$ *induces an isomorphism.*

Indeed, let $i : (X \setminus U, A \setminus U) \subset (X, A)$ be the inclusion map. Then,

$$i^*([\alpha]) = [\alpha | X \setminus U] \text{ for each } \alpha \in \mathrm{C}((X, A), (\Omega(|K(\mathbb{Z}, n+1)|, v_*), c_*)).$$

Every $\beta \in \mathrm{C}((X \setminus U, A \setminus U), (\Omega(|K(\mathbb{Z}, n+1)|, v_*), c_*))$ uniquely extends to $\bar{\beta} \in \mathrm{C}((X, A), (\Omega(|K(\mathbb{Z}, n+1)|, v_*), c_*))$ with $\bar{\beta}(U) = \{c_*\}$ because $\beta(A \setminus U) = \{c_*\}$ and $\mathrm{cl}\, U \subset \mathrm{int}\, A$. Hence, i^* is surjective, i.e., i^* is an epimorphism.[19]

For $\alpha, \alpha' \in \mathrm{C}((X, A), (\Omega(|K(\mathbb{Z}, n+1)|, v_*), c_*))$, assume that $i^*([\alpha]) = i^*([\alpha'])$, i.e., there is a homotopy

$$h : ((X \setminus U) \times \mathbf{I}, (A \setminus U) \times \mathbf{I}) \to (\Omega(|K(\mathbb{Z}, n+1)|, v_*), c_*)$$

such that $h_0 = \alpha | X \setminus U$ and $h_1 = \alpha' | X \setminus U$. Then, h uniquely extends to a homotopy

$$\bar{h} : (X \times \mathbf{I}, A \times \mathbf{I}) \to (\Omega(|K(\mathbb{Z}, n+1)|, v_*), c_*)$$

such that $h_0 = \alpha$ and $h_1 = \alpha'$, which means that $[\alpha] = [\alpha']$. Hence, i^* is injective, i.e., i^* is a monomorphism. Consequently, i^* is an isomorphism.

When $X = \{a\}$ is a singleton, we have the following:

- (DIMENSION AXIOM) $H^n(\{a\}) = \{0\}$ *for every* $n > 0$ *and* $H^0(\{a\}) \cong \mathbb{Z}$.

Indeed, $H^n(\{a\}) = [\{a\}, |K(\mathbb{Z}, n)|]$ can be regarded as $\pi_0(|K(\mathbb{Z}, n)|)$, which is the set of path-components of $|K(\mathbb{Z}, n)|$. If $n > 0$, since $|K(\mathbb{Z}, n)|$ is path-connected, we have $H^n(\{a\}) = \{0\}$. When $n = 0$, it follows that[20]

$$H^0(\{a\}) = [\{a\}, \Omega(|K(\mathbb{Z}, 1)|, v_*)] \cong \Omega(\partial\Delta^2, v_*)/\simeq \; = \pi_1(\partial\Delta^2) \cong \mathbb{Z}.$$

Thus, the cohomology H^* satisfies Axioms 1, 2, 5, 6, and 7.

Remark 8.4.1 Given an Abelian group G, the cohomology group $H^n(X; G)$ (or $H^n(X, A; G)$) with a coefficient group G can be defined by using $K(G, n)$ instead of $K(\mathbb{Z}, n)$. Then, it should be noted that $K(G, n)$ is not necessary to be countable (cf. Remark 7.8.1). However, $K(G, n)$ can be replaced with a locally finite-dimensional simplicial complex of the same homotopy type by Theorem 4.19.3. Then, $|K(G, n)|_{\mathrm{m}}$ is an ANE for paracompact spaces by Corollary 6.2.18 (2), which has the HEP for paracompact spaces (Remark 6.4.1). It should be noted that $|K(G, n)| \simeq |K(G, n)|_{\mathrm{m}}$ but $|K(G, n)| \not\approx |K(G, n)|_{\mathrm{m}}$. In this case, the following is the Dimension Axiom:

- $H^n(\{a\}; G) = \{0\}$ *for every* $n > 0$ *and* $H^0(\{a\}; G) = G$.

For a closed set A in a space X, let $i : A \subset X$ and $j : X = (X, \emptyset) \subset (X, A)$ be the inclusion maps. Then,

$$i^*([\alpha]) = [\alpha | A] \text{ for each } \alpha \in \mathrm{C}(X, \Omega(|K(\mathbb{Z}, n+1)|, v_*)) \text{ and}$$
$$j^*([\beta]) = [\beta j] \text{ for each } \beta \in \mathrm{C}((X, A), (\Omega(|K(\mathbb{Z}, n+1)|, v_*), c_*)).$$

[19] The correspondence $\alpha \mapsto \alpha | X \setminus U$ is bijective, but this does not mean that i^* is bijective.

[20] Although $H^0(\{a\}) = [\{a\}, |K(\mathbb{Z}, 0)|] = [\{a\}, \mathbb{Z}] \equiv \mathbb{Z}$, the last $\equiv$ does not mean as groups.

The following is easily shown:

- $\ker i^* = \operatorname{im} j^*$.

  Indeed, let $\beta \in \mathrm{C}((X, A), (\Omega(|K(\mathbb{Z}, n+1)|, v_*), c_*))$. Since $\beta j i = \beta | A = c_*$, it follows that $i^* j^*([\beta]) = [\beta j i] = [c_*] = 0$. Hence, $\operatorname{im} j^* \subset \ker i^*$.

  To see $\ker i^* \subset \operatorname{im} j^*$, we adopt Definition (a). Let $\alpha \in \mathrm{C}((X \times \mathbf{I}, X \times \partial\mathbf{I}), (|K(\mathbb{Z}, n+1)|, v_*))$ such that $i^*([\alpha]) = 0$, i.e., $\alpha | A \times \mathbf{I} \simeq c_*$ rel. $A \times \partial\mathbf{I}$, which implies that

  $$\alpha | H_{X,A} = \alpha | (X \times \partial\mathbf{I}) \cup (A \times \mathbf{I}) \simeq c_* \ \text{ rel. } X \times \partial\mathbf{I}.$$

  Since $|K(\mathbb{Z}, n+1)|$ has the HEP for paracompact spaces, we can easily obtain a map $\beta : X \times \mathbf{I} \to |K(\mathbb{Z}, n+1)|$ such that $\beta(H_{X,A}) = \{v_*\}$ and $\beta \simeq \alpha$ rel. $X \times \partial\mathbf{I}$, i.e.,

  $$\beta \in \mathrm{C}((X \times \mathbf{I}, H_{X,A}), (|K(\mathbb{Z}, n+1)|, v_*), c_*)$$

  such that $\beta(j \times \mathrm{id}_{\mathbf{I}}) \simeq \alpha$ in $\mathrm{C}((X \times \mathbf{I}, X \times \partial\mathbf{I}), (|K(\mathbb{Z}, n+1)|, v_*))$, which implies that $[\alpha] = [\beta(j \times \mathrm{id}_{\mathbf{I}})] = j^*([\beta])$. Therefore, $\ker i^* \subset \operatorname{im} j^*$.

Now, for a closed set A in a space X, we will define the coboundary operator $\delta : H^n(A) \to H^{n+1}(X, A)$. To this end, adopting Definitions (b) and (a), respectively, we write

$$H^n(A) = [(A \times \mathbf{I}^2, A \times \partial\mathbf{I}^2), (|K(\mathbb{Z}, n+2)|, v_*)] \quad \text{and}$$
$$H^{n+1}(X, A) = [(X \times \mathbf{I}, H_{X,A}), (|K(\mathbb{Z}, n+2)|, v_*)].$$

Each $\alpha \in \mathrm{C}((A \times \mathbf{I}^2, A \times \partial\mathbf{I}^2), (|K(\mathbb{Z}, n+2)|, v_*))$ can be extended to a homotopy

$$\tilde{\alpha} : ((X \times \mathbf{I}) \times \mathbf{I}, (X \times \partial\mathbf{I}) \times \mathbf{I}) \to (|K(\mathbb{Z}, n+2)|, v_*)$$

with $\tilde{\alpha}_0 = c_*$ and $\tilde{\alpha}_1(H_{X,A}) = \{v_*\}$.

Indeed, extend α to a homotopy

$$\bar{\alpha} : H_{X,A} \times \mathbf{I} = ((X \times \partial\mathbf{I}) \cup (A \times \mathbf{I})) \times \mathbf{I} \to |K(\mathbb{Z}, n+2)|$$

by $\bar{\alpha}((X \times \partial\mathbf{I}) \times \mathbf{I}) = \{v_*\}$. Then, $\bar{\alpha}$ is a homotopy with $\bar{\alpha}_0 = \bar{\alpha}_1 = c_*$ and $\bar{\alpha}_0$ extends to $c_* : X \times \mathbf{I} \to |K(\mathbb{Z}, n+2)|$. Since $|K(\mathbb{Z}, n+2)|$ has the HEP for paracompact spaces, we can extend $\bar{\alpha}$ to the desired homotopy $\tilde{\alpha}$.

Then, observe that

$$\tilde{\alpha}_1 \in \mathrm{C}((X \times \mathbf{I}, H_{X,A}), (|K(\mathbb{Z}, n+2)|, v_*)) \quad \text{and}$$
$$\tilde{\alpha}_1 \simeq \tilde{\alpha}_0 = c_* \in \mathrm{C}((X \times \mathbf{I}, X \times \partial\mathbf{I}), (K(\mathbb{Z}, n+2), v_*)). \qquad (*)$$

Moreover, the following holds:

$$\alpha \simeq \alpha' \in \mathrm{C}((A \times \mathbf{I}^2, A \times \partial\mathbf{I}^2), (|K(\mathbb{Z}, n+2)|, v_*))$$
$$\Rightarrow \ \tilde{\alpha}_1 \simeq \tilde{\alpha}'_1 \in \mathrm{C}((X \times \mathbf{I}, H_{X,A}), (|K(\mathbb{Z}, n+2)|, v_*)).$$

To show this, let h be a homotopy from α to α' relative to $A \times \partial \mathbf{I}^2$, that is,

$$h : ((A \times \mathbf{I}^2) \times \mathbf{I}, (A \times \partial \mathbf{I}^2) \times \mathbf{I}) \to (|K(\mathbb{Z}, n+2)|, v_*).$$

Regarding the last coordinate of $A \times \mathbf{I}^2$ as homotopy parameter, we extended α and α' to homotopies $\tilde{\alpha}$ and $\tilde{\alpha}'$. So, we now regard the third coordinate of $X \times \mathbf{I}^3$ (not the last coordinate) as homotopy parameter, that is, $h_t(x, s, t') = h(x, s, t, t')$. Then, h_0 (or h_1) is a homotopy from α_0 to α'_0 (or from α_1 to α'_1). Arranging construction of $\tilde{\alpha}$ from α for h, we can extend h to a homotopy $\tilde{h}$ so that $\tilde{h}_1$ is a homotopy from $\tilde{\alpha}_1$ to $\tilde{\alpha}'_1$ relative to $H_{X,A}$, that is,

$$\tilde{h}_1 : ((X \times \mathbf{I}) \times \mathbf{I}, H_{X,A} \times \mathbf{I}) \to (|K(\mathbb{Z}, n+2)|, v_*).$$

Indeed, using $\tilde{\alpha}$ and $\tilde{\alpha}'$, we can extend h to a map

$$\bar{h} : (X \times \mathbf{I}^2 \times \partial \mathbf{I}) \cup (X \times \partial \mathbf{I} \times \mathbf{I}^2) \cup (A \times \mathbf{I}^3) \to |K(\mathbb{Z}, n+2)|$$

defined by $\bar{h}(x, s, t, 0) = \tilde{\alpha}(x, s, t)$, $\bar{h}(x, s, t, 1) = \tilde{\alpha}'(x, s, t)$ for $(x, s, t) \in X \times \mathbf{I}^2$, and $\bar{h}((X \times \partial \mathbf{I} \times \mathbf{I}^2)) = \{v_*\}$. Then, $\bar{h}$ can be regarded as a homotopy $(\bar{h}_t)_{t \in \mathbf{I}}$ on $H^2_{X,A} = (X \times \partial \mathbf{I}^2) \cup (A \times \mathbf{I}^2)$, where t is the third coordinate of $X \times \mathbf{I}^3$, that is, $\bar{h}_t(x, s, t') = \bar{h}(x, s, t, t')$. Observe that $\bar{h}_0(x, 0, t') = \bar{h}_0(x, 1, t') = v_*, \bar{h}_0(x, s, 0) = \tilde{\alpha}_0(x, s) = v_*$, and $\bar{h}_0(x, s, 1) = \tilde{\alpha}'_0(x, s) = v_*$. Thus, $\bar{h}_0$ can be extended over $X \times \mathbf{I}^2$ by c_*. Since $|K(\mathbb{Z}, n+2)|$ has the HEP for paracompact spaces, we can extend $\bar{h}$ to a homotopy $\tilde{h} : X \times \mathbf{I}^3 \to |K(\mathbb{Z}, n+2)|$ with $\tilde{h}_0 = c_*$, where the third coordinate of $X \times \mathbf{I}^3$ is again regarded as homotopy parameter. Then, $\tilde{h}_1 : X \times \mathbf{I}^2 \to |K(\mathbb{Z}, n+2)|$ is a homotopy from $(\tilde{h}_1)_0 = \tilde{\alpha}_1$ to $(\tilde{h}_1)_1 = \tilde{\alpha}'_1$, and $\tilde{h}_1(H_{X,A} \times \mathbf{I}) = \{v_*\}$, where $(\tilde{h}_1)_t(x, s) = \tilde{h}(x, s, 1, t)$.

Thus, we can define the **coboundary operator** $\delta : H^n(A) \to H^{n+1}(X, A)$ as follows:

$$\delta([\alpha]) = [\tilde{\alpha}_1] \text{ for each } \alpha \in \mathrm{C}((A \times \mathbf{I}^2, A \times \partial \mathbf{I}^2), (|K(\mathbb{Z}, n+2)|, v_*)).$$

Because $j^*\delta([\alpha]) = j^*([\tilde{\alpha}_1]) = [\tilde{\alpha}_1 j] = [c_*] = 0$ by $(*)$, the following holds:

- $\operatorname{im} \delta \subset \ker j^*$.

Next, we show the following:

- $\delta : H^n(A) \to H^{n+1}(X, A)$ *is a homomorphism.*

Proof For each $\alpha, \beta \in \mathrm{C}((A \times \mathbf{I}^2, A \times \partial \mathbf{I}^2), (|K(\mathbb{Z}, n+2)|, v_*))$, take a homotopy

$$\tilde{\alpha}, \tilde{\beta} : ((X \times \mathbf{I}) \times \mathbf{I}, (X \times \partial \mathbf{I}) \times \mathbf{I}) \to (|K(\mathbb{Z}, n+2)|, v_*)$$

such that $\tilde{\alpha}|A \times \mathbf{I}^2 = \alpha, \tilde{\beta}|A \times \mathbf{I}^2 = \beta, \tilde{\alpha}_0 = \tilde{\beta}_0 = c_*$, and $\tilde{\alpha}_1(H_{X,A}) = \tilde{\beta}_1(H_{X,A}) = \{v_*\}$. Then, we can define a homotopy

$$\tilde{\alpha} * \tilde{\beta} : ((X \times \mathbf{I}) \times \mathbf{I}, (X \times \partial \mathbf{I}) \times \mathbf{I}) \to (|K(\mathbb{Z}, n+2)|, v_*)$$

as follows:

$$(\tilde{\alpha} * \tilde{\beta})(x, s, t) = \begin{cases} \tilde{\alpha}(x, 2s, t) & \text{if } s \leqslant 1/2, \\ \tilde{\beta}(x, 2s-1, t) & \text{if } s \geqslant 1/2. \end{cases}$$

Then, $\tilde{\alpha} * \tilde{\beta}|A \times \mathbf{I}^2 = \alpha * \beta$ and $(\tilde{\alpha} * \tilde{\beta})_1 = \tilde{\alpha}_1 * \tilde{\beta}_1$. Since $\tilde{\alpha} * \tilde{\beta}(A \times \partial\mathbf{I}^2) = \{v_*\}$, it follows that $(\tilde{\alpha} * \tilde{\beta})_0 = c_*$, and $(\tilde{\alpha} * \tilde{\beta})_1|H_{X,A} = c_*$. By the definition of δ, we have $\delta([\alpha * \beta]) = [(\tilde{\alpha} * \tilde{\beta})_1]$. Then, it follows that

$$\begin{aligned} \delta([\alpha] + [\beta]) &= \delta([\alpha * \beta]) = [(\tilde{\alpha} * \tilde{\beta})_1] \\ &= [\tilde{\alpha}_1 * \tilde{\beta}_1] = [\tilde{\alpha}_1] + [\tilde{\beta}_1] = \delta([\alpha]) + \delta([\beta]). \end{aligned}$$

Therefore, δ is a homomorphism. □

The following can be proved straightforward:

- *For each map $f : (X, A) \to (Y, B)$, the following diagram commutes:*

$$\begin{array}{ccc} H^n(B) & \xrightarrow{\delta} & H^{n+1}(Y, B) \\ {\scriptstyle (f|A)^*}\downarrow & & \downarrow{\scriptstyle f^*} \\ H^n(A) & \xrightarrow[\delta]{} & H^{n+1}(X, A) \end{array}$$

Proof Each $\beta \in \mathrm{C}((B \times \mathbf{I}^2, B \times \partial\mathbf{I}^2), (|K(\mathbb{Z}, n+2)|, v_*))$ can be extended to a homotopy

$$\tilde{\beta} : ((Y \times \mathbf{I}) \times \mathbf{I}, (Y \times \partial\mathbf{I}) \times \mathbf{I}) \to (|K(\mathbb{Z}, n+2)|, v_*)$$

with $\tilde{\beta}_0 = c_*$ and $\tilde{\beta}_1|H_{Y,B} = c_*$. Then, we have a homotopy

$$\tilde{\beta}(f \times \mathrm{id}_{\mathbf{I}^2}) : ((X \times \mathbf{I}) \times \mathbf{I}, (X \times \partial\mathbf{I}) \times \mathbf{I}) \to (|K(\mathbb{Z}, n+2)|, v_*)$$

such that $\tilde{\beta}(f \times \mathrm{id}_{\mathbf{I}^2})|A \times \mathbf{I}^2 = \beta((f|A) \times \mathrm{id}_{\mathbf{I}^2})$, $(\tilde{\beta}(f \times \mathrm{id}_{\mathbf{I}^2}))_0 = \tilde{\beta}_0(f \times \mathrm{id}_{\mathbf{I}}) = c_*$, and

$$(\tilde{\beta}(f \times \mathrm{id}_{\mathbf{I}^2}))_1|H_{X,A} = (\tilde{\beta}_1|H_{Y,B})((f \times \mathrm{id}_{\mathbf{I}})|H_{X,A}) = c_*.$$

By the definition of δ, we have $\delta([\beta((f|A) \times \mathrm{id}_{\mathbf{I}^2})]) = [(\tilde{\beta}(f \times \mathrm{id}_{\mathbf{I}^2}))_1] = [\tilde{\beta}_1(f \times \mathrm{id}_{\mathbf{I}})]$. Then, it follows that

$$\begin{aligned} f^*\delta([\beta]) &= f^*([\tilde{\beta}_1]) = [\tilde{\beta}_1(f \times \mathrm{id}_{\mathbf{I}})] \\ &= \delta([\beta((f|A) \times \mathrm{id}_{\mathbf{I}^2})]) = \delta(f|A)^*([\beta]). \end{aligned}$$

Consequently, $f^*\delta = \delta(f|A)^*$. □

We will show the Exactness Axiom:

- (EXACTNESS) *The cohomology sequence is exact for each pair (X, A) of a space X and its closed set A.*

$$\{0\} \longrightarrow H^0(X, A) \xrightarrow{j^*} H^0(X) \xrightarrow{i^*} H^0(A) \xrightarrow{\delta} H^1(X, A) \xrightarrow{j^*} \cdots$$

$$\longrightarrow H^n(X, A) \xrightarrow{j^*} H^n(X) \xrightarrow{i^*} H^n(A) \xrightarrow{\delta} H^{n+1}(X, A) \longrightarrow \cdots$$

Already, $\ker i^* = \operatorname{im} j^*$ and $\operatorname{im} \delta \subset \ker j^*$ were shown. Moreover, the first $j^* : H^0(X, A) \to H^0(X)$ is injective.

Indeed, since $(|K(\mathbb{Z}, 0)|, v_*) = (\mathbb{Z}, 0)$ and $\mathbb{Z}$ is discrete, it follows that

$$\begin{aligned} H^0(X, A) &= [(X, A), (|K(\mathbb{Z}, 0)|, v_*)] = \mathrm{C}((X, A), (\mathbb{Z}, 0)) \\ &\subset \mathrm{C}(X, \mathbb{Z}) = [X, |K(\mathbb{Z}, 0)|] = H^0(X). \end{aligned}$$

Thus, $\ker j^* \subset \operatorname{im} \delta$ and $\ker \delta = \operatorname{im} i^*$ remain to be proved.

- $\ker j^* \subset \operatorname{im} \delta$.

Proof Let $\beta \in \mathrm{C}((X \times \mathbf{I}, H_{X,A}), (|K(\mathbb{Z}, n+2)|, v_*))$ such that $j^*([\beta]) = 0$. Then, there is a homotopy

$$h : (X \times \mathbf{I} \times \mathbf{I}, X \times \partial\mathbf{I} \times \mathbf{I}) \to (|K(\mathbb{Z}, n+2)|, v_*)$$

with $h_0 = c_*$ and $h_1 = \beta$. Since $h(A \times \mathbf{I} \times \{0\}) = c_*(A \times \mathbf{I}) = \{v_*\}$, $h(A \times \mathbf{I} \times \{1\}) = \beta(A \times \mathbf{I}) = \{v_*\}$, and $h(A \times \partial\mathbf{I} \times \mathbf{I}) = \{v_*\}$, we have

$$\alpha = h|A \times \mathbf{I}^2 \in \mathrm{C}((A \times \mathbf{I}^2, A \times \partial\mathbf{I}^2), (|K(\mathbb{Z}, n+2)|, v_*)).$$

Moreover, h can be regarded as $\tilde{\alpha}$ in the process defining $\delta([\alpha])$. Since $\tilde{\alpha}_1 = h_1 = \beta$, it follows that $\delta([\alpha]) = [\tilde{\alpha}_1] = [\beta]$. Hence, $\ker j^* \subset \operatorname{im} \delta$. □

- $\operatorname{im} i^* \subset \ker \delta$.

Proof For each $\beta \in \mathrm{C}((X \times \mathbf{I}^2, X \times \partial\mathbf{I}^2), (|K(\mathbb{Z}, n+2)|, v_*))$, let

$$\alpha = \beta|A \times \mathbf{I}^2 \in \mathrm{C}((A \times \mathbf{I}^2, A \times \partial\mathbf{I}^2), (|K(\mathbb{Z}, n+2)|, v_*)).$$

Then, β is regarded as a homotopy on $X \times \mathbf{I}$ with $\beta_0 = c_*$ and

$$\beta_1(H_{X,A}) \subset \beta_1(X \times \mathbf{I}) = \beta(X \times \mathbf{I} \times \{1\}) = \{v_*\}.$$

Since $[\beta_1] = \delta([\alpha])$ by the definition of δ, it follows that $\delta i^*[\beta] = \delta([\alpha]) = [\beta_1] = [c_*] = 0$. Thus, we have $\operatorname{im} i^* \subset \ker \delta$. □

- $\ker \delta \subset \operatorname{im} i^*$.

Proof Let $\alpha \in \mathrm{C}((A \times \mathbf{I}^2, A \times \partial\mathbf{I}^2), (|K(\mathbb{Z}, n+2)|, v_*))$ such that $\delta([\alpha]) = 0$. Then, there is a homotopy

$$\tilde{\alpha} : ((X \times \mathbf{I}) \times \mathbf{I}, (X \times \partial\mathbf{I}) \times \mathbf{I}) \to (|K(\mathbb{Z}, n+2)|, v_*)$$

with $\tilde{\alpha}_0 = c_*$ and $\tilde{\alpha}_1(H_{X,A}) = \{v_*\}$. Since $[\tilde{\alpha}_1] = \delta([\alpha]) = 0$, i.e., $\tilde{\alpha}_1 \simeq c_*$ rel. $H_{X,A}$, we have a homotopy

$$h : (X \times \mathbf{I} \times \mathbf{I}, H_{X,A} \times \mathbf{I}) \to (|K(\mathbb{Z}, n+2)|, v_*)$$

with $h_0 = \tilde{\alpha}_1$ and $h_1 = c_*$. Then, we can define $\beta : (X \times \mathbf{I}^2, X \times \partial\mathbf{I}^2) \to (|K(\mathbb{Z}, n+2)|, v_*)$ as follows:

$$\beta(x,s,t) = \begin{cases} \tilde{\alpha}(x,s,2t) & \text{if } t \leqslant 1/2, \\ h(x,s,2t-1) & \text{if } t \geqslant 1/2. \end{cases}$$

Then, it is easy to see that $\beta i = \beta | A \times \mathbf{I}^2 \simeq \alpha$ rel. $A \times \partial\mathbf{I}^2$, which means that $i^*([\beta]) = [\beta i] = [\alpha]$. Thus, we have $\ker \delta \subset \operatorname{im} i^*$. □

Thus, our cohomology H^* also satisfies Axioms 3 and 4. Consequently, H^* satisfies all of the Eilenberg–Steenrod axioms (Axioms 1–7). To apply Bacon's Theorem 8.3.4, it remains to be shown that H^* is additive and point reductive.

For the topological sum $\bigoplus_{\lambda\in\Lambda} X_\lambda$ of $(X_\lambda)_{\lambda\in\Lambda}$ and a space Y, the space $\mathrm{C}(\bigoplus_{\lambda\in\Lambda} X_\lambda, Y)$ can be naturally identified with $\prod_{\lambda\in\Lambda} \mathrm{C}(X_\lambda, Y)$ by the correspondence $f \mapsto (f|X_\lambda)_{\lambda\in\Lambda}$ (cf. Proposition 2.10.8 (7)). Since this correspondence preserve the homotopy relation $\simeq$, we can naturally regard

$$\left[\bigoplus_{\lambda\in\Lambda} X_\lambda, Y\right] = \prod_{\lambda\in\Lambda} [X_\lambda, Y].$$

Hence, the following axiom is trivial:

- (ADDITIVITY) *Let $X = \bigoplus_{\lambda\in\Lambda} X_\lambda$ be the topological sum of $(X_\lambda)_{\lambda\in\Lambda}$. For each $\lambda \in \Lambda$, let $i_\lambda : X_\lambda \subset X$ be the inclusion and $\mathrm{pr}_\lambda : \prod_{\lambda\in\Lambda} H^n(X_\lambda) \to H^n(X_\lambda)$ be the projection. Then, there exist unique isomorphisms $\varphi : H^n(X) \to \prod_{\lambda\in\Lambda} H^n(X_\lambda)$, $n \in \omega$, such that $\mathrm{pr}_\lambda \varphi = i_\lambda^*$ for each $\lambda \in \Lambda$.*

The following axiom is also easily seen:

- (POINT REDUCTIVITY) *For each $x \in X$ and $[\alpha] \in H^n(X)$, if $[\alpha|\{x\}] = 0$, then there is a neighborhood U of x in X such that $[\alpha|U] = 0$.*

Indeed, $U = \alpha^{-1}(|\operatorname{St}(c_{K(\mathbb{Z},n)}(\alpha(x)), K(\mathbb{Z}, n))|)$ is a neighborhood of x in X. The assumption $[\alpha|\{x\}] = 0$ means that $\alpha(x)$ is in the path-component containing the base points v_*.[21] Since $\alpha(U)$ is contractible in $|K(\mathbb{Z}, n)|$, we have $\alpha|U \simeq c_*$ in $|K(\mathbb{Z}, n)|$, which means that $[\alpha|U] = 0$.

Finally, we can apply Bacon's Theorem 8.3.4 to conclude as follows:

Theorem 8.4.1 *Our cohomology H^* is isomorphic to the Čech cohomology.* □

The **cohomological dimension** $\mathrm{cd}_\mathbb{Z} X$ is usually defined as follows: $\mathrm{cd}_\mathbb{Z} \emptyset = -1$ and for $X \neq \emptyset$,

$$\mathrm{cd}_\mathbb{Z} X = \sup\left\{n \;\middle|\; \exists A \in \mathrm{Cld}^*(X) \text{ such that } H^n(X, A) \neq \{0\}\right\},$$

[21] When $n > 0$, this is trivial because $|K(\mathbb{Z}, n)|$ is path-connected. Thus, this is a non-trivial statement only if $n = 0$. In fact, since we defined $K(\mathbb{Z}, 0)|$ as a countable discrete space, we have $\alpha(x) = v_*$ if $n = 0$.

where $\mathrm{Cld}^*(X) = \mathrm{Cld}(X) \cup \{\emptyset\}$.[22] Then, note that $\mathrm{cd}_Z X \geqslant 0$ because $H^0(X, \emptyset) = H^0(X) = [X, |K(\mathbb{Z}, 0)|] = [X, \mathbb{Z}]$ is not a singleton. However, in Sect. 7.9, we defined the cohomological dimension $\mathrm{cd}_{\mathbb{Z}} X$ without resorting to cohomology theory. To be more specific, for a metrizable space X, we defined $\mathrm{cd}_{\mathbb{Z}} X \leqslant n$ if X satisfies the following condition $(\mathrm{e})_k$ for every $k \geqslant n$:

$(\mathrm{e})_k$ Every map $f : A \to |K(\mathbb{Z}, k)|$ of each closed set A in X extends over X.

Using the Homotopy Extension Theorem 6.4.1, it is easy to see that $(\mathrm{e})_k$ is equivalent to the following:

$(\mathrm{h})_k$ For each closed set A in X, the inclusion $i : A \subset X$ induces an epimorphism $i^* : H^k(X) \to H^k(A)$.

By the exactness of the cohomology sequence, the following are equivalent for each closed set A in X:

(a) $i^* : H^k(X) \to H^k(A)$ is an epimorphism;
(b) $\delta(H^k(A)) = \{0\}$;
(c) $j^* : H^{k+1}(X, A) \to H^{k+1}(X)$ is a monomorphism,

where $i : A \subset X$ and $j : X = (X, \emptyset) \subset (X, A)$ are inclusions and $\delta : H^k(A) \to H^{k+1}(X, A)$ is the coboundary operator. Thus, $(\mathrm{e})_k$ is equivalent to the following:

$(\mathrm{h})'_{k+1}$ For each closed set A in X, the inclusion $j : X = (X, \emptyset) \subset (X, A)$ induces a monomorphism $j^* : H^{k+1}(X, A) \to H^{k+1}(X)$.

Here, we show that our definition of $\mathrm{cd}_{\mathbb{Z}} X$ is consistent with the usual definition.

Proof (*Consistency of Definitions of the Cohomological Dimension*) Assume that X satisfies $(\mathrm{e})_k$ for every $k \geqslant n$. Due to Lemma 7.9.2, $(\mathrm{e})_k$ implies that every map $f : A \to |K(\mathbb{Z}, k+1)|$ of each closed set A in X is null-homotopic, which means $H^{k+1}(A) = \{0\}$. In particular, $H^{k+1}(X) = \{0\}$. Since $(\mathrm{e})_k$ is equivalent to $(\mathrm{h})'_{k+1}$, $H^{k+1}(X, A) = \{0\}$. Thus, $H^k(X, A) = \{0\}$ for every $k > n$, which means $\mathrm{cd}_{\mathbb{Z}} X \leqslant n$ in the usual sense.

Conversely, assume that $\mathrm{cd}_{\mathbb{Z}} X \leqslant n$ in the usual sense and let A be a closed set in X. Since $H^k(X, A) = \{0\}$ for every $k > n$, we have $\delta(H^k(A)) = \{0\}$ for every $k \geqslant n$, which means that the inclusion $i : A \subset X$ induces an epimorphism $i^* : H^k(X) \to H^k(A)$ for every $k \geqslant n$ as observed in the above. Hence, X satisfies $(\mathrm{e})_k$ for every $k \geqslant n$. □

Notes for Chap. 8

There are many good textbooks on homotopy groups and (co)homology groups. Here, we list only five of them:

- A. Hatcher, *Algebraic Topology* (Cambridge Univ. Press, Cambridge, 2002).
- S.-T. Hu, *Homotopy Theory* (Academic Press, Inc., New York, 1959).
- E. Spanier, *Algebraic Topology* (McGraw-Hill, New York, 1966).

[22] Cf. Footnote 25 on Sect. 2.12.

- A. Dold, *Lectures on Algebraic Topology, Second Edition* (Springer-Verlag, Heidelberg, 1980).
- S. Eilenberg and N. Steenrod, *Foundations of Algebraic Topology* (Princeton Univ. Press, Princeton, 1952) — cited as [FAT] in Sect. 8.3 and below.

In their book [FAT], S. Eilenberg and N. Steenrod established Theorems 8.3.1 and 8.3.3 (and their homology versions), which was extended to Theorem 8.3.2 (and its homology version) by Milnor [3] and to Theorem 8.3.4 by Bacon [1]. The homotopical definition of cohomology introduced in Sect. 8.4 was given by Huber [2].

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Index

Symbols
$(X, X_1, \ldots, X_n) \approx (Y, Y_1, \ldots, Y_n)$, 2
$(X, X_1, \ldots, X_n) \simeq (Y, Y_1, \ldots, Y_n)$, 12
$(X, x_0) \approx (Y, y_0)$, 2
$(X, x_0) \simeq (Y, y_0)$, 12
K_L^{ε}, 304
$X \approx Y$, 2
$X \simeq Y$, 11
X^{∞}, 78
$[(X, X_1, \ldots, X_n), (Y, Y_1, \ldots, Y_n)]$, 11
$[(X, x_0), (Y, y_0)]$, 12
$[0, \omega_1]$, 82
$[X, Y]$, 11
$\cong$, 232
$\equiv$, 231
∞_X, 38
$\mathcal{A} \prec \mathcal{B}$, 8
$\mathcal{A} \wedge \mathcal{B}$, 8
$\mathcal{A}[Y]$, 8
$\mathcal{A}[x]$, 8
$\mathcal{A}^{\mathrm{cl}}$, 8
$\mathcal{A}|Y$, 8
$\sum_{\gamma \in \Gamma} x(\gamma)$, 20
$f \simeq 0$, 10
$f \simeq g$, 10
$f \simeq g$ rel. A, 12
$f \simeq^p g$, 39
$f \simeq^p g$ rel. A, 39
$f \simeq^p_{\mathcal{U}} g$, $f \simeq^p_{\varepsilon} g$, 39
$f \simeq_{\mathcal{U}} g$, 13
$f \simeq_{\varepsilon} g$, 13
2, the discrete space of two points, 4
$\mathbf{2}^{\mathbb{N}}$, 4

A
Absolute Extensor (AE), 491
— for compact spaces, 491
— for normal spaces, 395, 491
— for paracompact spaces, 491
Absolutely G_δ, 56
Absolute Neighborhood Extensor (ANE), 491
— for compact spaces, 491
— for normal spaces, 395, 491
— for paracompact spaces, 491
Absolute Neighborhood Retract (ANR), 491
Cauty's characterization, 557
— characterization by HEP, 520
Hanner's characterization, 536
Lefschetz' characterization, 534
Nguyen To Nhu's characterization, 549
— with dim $\leqslant n$, 580
Absolute Retract (AR), 491
characterization, 509, 541
— with dim $\leqslant n$, 580
Abstract complex, 338
Addition Theorem, 418
Adjunction space, 7
Admissible
— PL map, 300
— subdivision, 260
— triangulation, 270
Admissible PL Approximation Theorem, 300
Admissible Simplicial Approximation Theorem, 291
AE for paracompact spaces, 494
AE(n), 491
AE(n), absolute extensor for metrizable spaces of dim $\leqslant n$, 573
characterization, 579
Affine
— function (or map), 130, 133
— hull, 129

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— set, 127
Affinely independent, 129
A.H. Stone's Theorem, 45
Alexandroff Characterization of dimension, 410
Alexandorff compactification, αX, 38
Alexandroff's Problem, 676, 681
αf, 38
αX, Alexandorff (or one-point) compactification, 38
ANE(n), 491
ANE(n), absolute neighborhood extensor for metrizable spaces of dim $\leqslant n$, 573
characterization, 575
Arc, 10
Arcwise connected, 10, 479
Arens–Eells Embedding Theorem, 500
Ascoli–Arzelà Theorem, 102

B

$B(A, K)$, 262, 277
$B(L, K)$, 277
$\mathrm{B}_d(x, \varepsilon)$, $\mathrm{B}(x, \varepsilon)$, 3
$\overline{\mathrm{B}}_d(x, \varepsilon)$, $\overline{\mathrm{B}}(x, \varepsilon)$, 3
$\mathbf{B}^n$, the unit closed n-ball, 4
$\mathbf{B}_X$, the unit closed ball of X, 18
Baire Category Theorem, 55
Baire property, 55
Baire space, 55, 107, 108
Banach–Schauder Theorem, 178
Barycenter of a simplex, 205
Barycentric coordinate, 249
Barycentric refinement, 52
Barycentric subdivision, 230
Base point, 2
β_L^K, 265
βX, the Stone–Čech compactification of X, 31
Bonding map, 349
Borsuk–Whitehead–Hanner Theorem, 529
Boundary of a cell, 204
Boundary operator, 712, 717, 728
Brouwer Fixed Point Theorem, 392
Brouwer's Theorem on Invariance of Domain, 397

C

$\boldsymbol{c}$, 19
$\mathrm{C}((X, x_0), (Y, y_0))$, 12
$\mathrm{C}((X, X_1, \ldots, X_n), (Y, Y_1, \ldots, Y_n))$, 11
$C(A, K)$, 218, 262, 277
$\boldsymbol{c}(\Gamma)$, 18
$C(L, K)$, 277
$c(X)$, the cellularity of X, 2
$\mathrm{C}(X, Y)$, 10, 94, 103
$\mathrm{C}^B(X)$, 27
C^n, n-connected, 566
C^n-refinement, 574
$\mathrm{C}^P(X, Y)$, 39, 108, 443
$\boldsymbol{c}_0$, 19
$\boldsymbol{c}_0(\Gamma)$, 18
$\mathrm{C}_p(X, Y)$, 15, 95
$\mathrm{C}_{\mathcal{U}}(X, Y)$, 442
$\mathrm{C}_u(X, Y)$, 13, 95
Canonical map, 340, 348
Canonical Open Set Extension Lemma, 416
Canonical representation of a simplicial complex, 249
Cantor (ternary) set, 4, 468, 470
$\mathrm{card}X$, the cardinality of X, 2
Carrier, 215
Čech-complete, 59
Cell-like
— compactum, 609
— map, 609
— open map, 686
Cell-Like Mapping Problem, 676
Cell, (linear cell), 164, 204
Cellular, 703
CE map, 609
CE Problem, 676
Chain; ε-chain, 478
Characterization of LEC-ness, 513, 514
Characterization of zero-dimensionality, 413
Characterization Theorem
— for combinatorial n-manifolds, 320
Circled, 153
Cld X, 111
Clopen basis, 413
Clopen set, 2, 413
Closed combinatorial n-manifold, 326
Closed convex hull, 156
Closed Graph Theorem, 177
Closed shrinking, 62
Closed tower, 78
Coboundary operator, 726, 736, 737
Coefficient group, 727
Cohomological dimension, $\mathrm{cd}_{\mathbb{Z}}$, 665, 740
Cohomology
— group, 731
— group with a coefficient group G, 735
— sequence, 726, 738
Cohomology theory, 725

additive, 727
continuous, 728
Collapsing, 365, 531
Collectionwise normal, 61, 124
Combinatorial
— n-dimensional manifold, 317
Combinatorially equivalent, 232
Combinatorial manifold
boundary of, 326
Combinatorial n-manifold, 317
characterization, 323
Combinatorial n-manifold without boundary, 326
Comp_2, 728
Compactification preserving the (strong) countable dimensionality, 440
Compactification Theorem, Dimension-Preserving, 437
Compactification with the strongly countable-dimensional remainder, 439
Compactly generated space, 32
Compact-open topology, 95, 96, 100
Compactum (pl., compacta), 1
Completely metrizable
characterization, 56–58
Completely normal, 43
Completion Theorem, Dimension-Preserving, 413
Complex
cell, 213
countable, 214
finite, 214
finite-dimensional (f.d.), 214
infinite, 214
infinite-dimensional (i.d.), 214
locally countable, 218
locally finite, 218
locally finite-dimensional (l.f.d.), 218
ordered cell, 228
ordered simplicial, 228
simplicial, 214
$\mathrm{Comp}(X)$, 111
Cone, 34, 366
Cone, metrizable, 531
Congruent, 231, 286, 338
Connected, 223
$\mathrm{Conn}(X)$, 111
Contiguous, 246, 252, 343
Continued fraction, 5
Continuous set-valued function, 181
Continuum (pl., continua), 1
Contractible, 496
locally, 508
semi-locally, 512
strongly locally, 517
Contraction, 496
$\mathrm{Cont}(X)$, 111
$\mathrm{Conv}(E)$, 180, 182
Convergent (infinite sum), 20
Convex, 131
Convex hull, 132
Core, 144
Countable dense homogeneous, 446
Countable-dimensional (c.d.), 429, 563
Countable Sum Theorem, 402
Countably compact, 83
Countably paracompact, 91
Cover, 9
Covering dimension, (dim), 391
Covering projection, 193
$\mathrm{cov}(X)$, 9
$\mathrm{cov}^*(X)$, 357
Cozero set, 43

D

∂M, 7, 324
$d(x, A)$, 3
Decomposition Theorem, 417
Deformable into, 508
Deformation, 508
Deformation retract, 508, 521
Deformation retraction, 508, 521
Deleted Tychonoff plank, 84
δ_L^K, 265
Δ^n, the standard n-simplex, 4
Δ-refinement, 52
$\mathrm{dens}X$, the density of X, 2
Derived neighborhood, 303, 327
Derived subdivision, 230
$\mathrm{diam}_d A$, $\mathrm{diam}A$, 3
dim, 391
Dimension
characterization, 398, 402, 410, 415
— of a complex, 214
— of a convex set, 132
— of a flat, 130
— of an abstract complex, 338
— of a polyhedron, 235
— of a simplex, 204
— of a simplicial complex, 405
— of a space, 391
Dimension-Preserving
— Compactification Theorem, 437
— Completion Theorem, 413
Directed set, 20

Direct limit, 75
Discrete, 8
Disjoint cells property, 643
$\mathrm{dist}_d(A, B)$, $\mathrm{dist}(A, B)$, 3
Double suspension, 645
Dowker space, 91
Dugundji Extension Theorem, 492
Dugundji system, 492
Dydak–Walsh Theorem, 684

E
$\mathbf{e}_i$, 4
Edwards–Walsh complex, simplicial, 664
Eilenberg–MacLane complex, 654, 656, 662, 707
Eilenberg–MacLane space, 654, 729
Eilenberg–Otto Characterization of dimension, 410
Embedding Approximation Theorem, 442, 444
Embedding Theorem, 440
$\mathrm{Emb}(X, Y)$, 441, 443
ε-close, 13
ε-homotopic, 13
ε-map, 441
ε-neighborhood, 304
Equi-Connected (EC), 512
Equi-connecting map, 512
Equi-continuous, 101
Erdős space, 471
Essential family, 407
Essential map, 407
ev, 15
Evaluation, 15
Extended long line, 94
Extension, 9
Extreme point, 139, 208

F
$F = \omega^{\mathfrak{P}(\mathbf{I})}$, 87
$\mathrm{F}_n X$, 111
F_σ set, 2
Face, 138, 209
Face of a cell, 208
F-hereditary property, 67
Fine homotopy equivalence, 543, 544
$\mathrm{Fin}(\Gamma)$, 16
Finite-dimensional (f.d.) , 391
Finite intersection property (f.i.p.), 30, 57
$\mathrm{Fin}(X)$, 111
Fixed point property, 392
Flat, 127
Flat hull, 129
F-norm, 171
F-normed linear space, 172
Fréchet space, 175, 184, 494
Free topological linear space, 186, 188, 686
F-space, 175
Full complex, 221
Full realization, 533
Full simplicial complex, 221
Full subcomplex, 221, 266
Fully normal, 66, 124
Functional, 128
Fundamental group, 708

G
G_δ set, 2
General Position Lemma, 328, 339
Geometrically independent, 129
G-hereditary property, 66
Graph of a map, 9, 289

H
$H^n(X)$, 731
$H^n(X, A)$, 731
$H^n(X, A; G)$, 735
$H^n(X; G)$, 735
Hahn–Banach extension theorem, 143
Hahn–Mazurkiewicz Theorem, 477
Hanner's Theorem, 498, 507
Hauptvermutung, 302
Hausdorff's Metric Extension Theorem, 503
$\hbar$-refinement, 514
Hedgehog, 48
Henderson–Sakai Theorem, 281
Hereditarily
— disconnected, 466
— infinite-dimensional (h.i.d.), 431, 474, 476, 477
— normal, 43
— paracompact, 69
Hereditary
— n-equivalence, 621
— shape equivalence, 635
— weak homotopy equivalence, 621
— property, 81
Hilbert cube
fixed point property, 393
universality, 48
Hilbert space, 24
h^n-refinement, 582
Hölder Inequality, 22

Homeo(X), 105
Homogeneous, 153
  countable dense, 446
Homology theory, 728
Homotopic, 10
Homotopically trivial, 541
Homotopy, 10
  — class, 10
  — dense, 539
  — dominate, 375
  — dominated, 375, 536
  — dominated by a simplicial complex, 375
  — domination, 375
  — equivalence, 11
  — equivalent, 11
  — equivalent relative to, 366
  — exact sequence, 712
  — group, 709
  — inverse, 11
  — lifting property, 666
  — relative to A, 12
  — track, 10
  — type, 11
  — type of a simplicial complex, 375
Homotopy Extension Property (HEP), 519
  — for compact spaces, 519
  — for metrizable spaces, 519
  — for normal spaces, 659
  — for paracompact spaces, 519, 730
Homotopy Extension Theorem
  — for ANEs, 519
  — for cell complexes, 244
h-refinement, 514
Hurewicz fibration, 666
Hyperplane, 128, 131

I
$\mathbf{I}^{\mathbf{I}}$, 87
I — the natural triangulation of $\mathbf{I}$, 304
Ind, 419, 420
Indexed collection (or family) of spaces, 3
Indexed cover, 9
Induced function, $f_\sharp$, 381, 710
Induced homomorphism
  f^*, 725, 734
  f_*, 728
  $f_\sharp$, 711
Inductive dimension
  large, 419
  small, 420
Inessential family, 407
Infinite-dimensional (i.d.), 391
∞-equivalence, 380
Int M, 6, 324
Interior of a cell, 204
Invariance of Domain, 397
Invariant metric, 168
Inverse limit, 349
Inverse of a path, 707
Inverse sequence, 349
Inverse system, 348

J
Join, 207
  — of simplexes, 321
  — of simplicial complexes, 321
Joinable, 207, 219, 220
Join of paths, 707

K
$K(G, n)$, 654, 735
$K(\mathbb{Z}, 0)$, 661
$K(\mathbb{Z}, n)$, 654, 707, 729
K_ω, 454
$K_\omega(\Gamma)$, 454
Klee's Trick, 502
Kozlowski's Theorem on fine homotopy equivalences, 544
Kruse–Liebnitz Theorem, 527
k-space, 32, 97, 216

L
$L(X)$, 686
$\ell_2(\Gamma)$, 24
ℓ_∞, 19
$\ell_\infty(\Gamma)$, 18
$\ell_\infty^f(\Gamma)$, 18
ℓ_∞^f, 19
$\ell_p(\Gamma)$, 22
$\ell_p^f(\Gamma)$, 24
λ-convex, 542
λ-extension, 542
λ-stable, 542
λ-stable refinement, 542
Large inductive dimension, (Ind), 419
Large transfinite inductive dimension, 432
Lavrentieff G_δ-Extension Theorem, 59
Lavrentieff Homeomorphism Extension Theorem, 60
Locally Equi-Connected (LEC), 512
Lefschetz refinement, 533

Limitation topology, 103–105, 442, 443
Lindelöf, 65
Linear
— in the affine sense, 133
— manifold or variety, 127
— metric, 171
— span, 17
Linearly accessible, 135
Link in a complex, 220
Lipschitz constant, 10
Lipschitz map, 10
Local
— $*$-connection, 623
— ∞-connection, 623
— n-connection, 623, 704
Locally
— arcwise connected, 10, 480
— compact metrizable space, 59
— connected, 477
— convex, 157, 492
— n-connected, LC^n, 566
— path-connected, 10, 480, 567
— simply connected, 567
Locally contractible, 508
strongly, 517
Locally finite, 8
Locally finite-dimensional nerve, 344
Locally Finite Sum Theorem, 403
Locally n-Connected (LC^n), 566
Long line, 92
Long segment, 94
Loop, 660, 708
Loop space, $\Omega(X, x_0)$, 660, 666, 683, 708, 716
Lower semi-continuous (l.s.c.)
— real-valued function, 73
— set-valued function, 181

M
Majorant topology, 106
Manifold, 6, 47, 68, 94
boundary of, 7
closed, 7
interior of, 6
— without boundary, 7
Mapping cylinder, 365
metrizable, 531
simplicial, 370
Mapping $(n+1)$-deformation retract, 655
Mapping telescope, 533
Mazur's Theorem, 27
Menger compactum, μ^n, 458, 568, 569
$\mathrm{mesh}_d\mathcal{A}$, $\mathrm{mesh}\mathcal{A}$, 3
$\mathrm{Metr}(X)$, 2
$\mathrm{Metr}^B(X)$, 2, 104
$\mathrm{Metr}^c(X)$, 2, 104
Metric linear space, 171
Metric polyhedron, 249, 270
Metric topology of a polyhedron, 249
Metrization Theorem
Alexandroff–Urysohn, 49, 124
Bing, 47, 124
Frink, 49, 124
Nagata–Smirnov, 47, 124
Urysohn, 47, 124
Michael's Theorem on local property, 66
Minkowski functional, 145
Minkowski Inequality, 22
Minkowski norm, 146
Monotone map, 685
μ^0, the Cantor (ternary) set, 4
μ^n, 458, 568, 569

N
$\mathrm{N}_d(A, \varepsilon)$, $\mathrm{N}(A, \varepsilon)$, 3
$N(A, K)$, 218, 262, 277
$N(L, K)$, 277
$\mathbb{N}^{\mathbf{I}}$, 85
$\mathbb{N}^{\mathbb{N}}$, 4
n-connected, C^n, 566
n-dimensional, 391
n-dimensionality of $X \subset \mathbb{R}^n$, 446
Neighborhood deformation retract, 508
Neighborhood retract, 491
n-equivalence, 380, 717
Nerve, 340, 348
Nested sequence, 349
n-homotopy dense, 583
n-Lefschetz refinement, 574
n-manifold, n-dimensional manifold, 6
Nöbeling space, ν^n, 455, 568, 569
Non-degenerate, 1
Non-expansive
— inverse sequence, 678
— map, 678
No Retraction Theorem, 394
Normal cover, 54, 70, 342, 357
n-th star, 52
ν^0, the space of irrationals, 4
ν^n, 455, 568, 569
ν_n, 450
$\nu_n(\Gamma)$, 450
ν_ω, 430
$\nu_\omega(\Gamma)$, 450

Null-homotopic, 10
Numerable cover, 70, 342, 357

O
$\Omega(X, x_0)$, 660, 683, 708, 716
One-point compactification, αX, 38
One-point union, 662
Open cone, metrizable, 531
Open Cover Shrinking Lemma, 72
Open Mapping Theorem, 179
Open shrinking, 62
Open star, 218, 249, 262
Opposite face, 209
Ordered
— cell complex, 228
— simplicial complex, 228
Order of a collection, 391
Oscillation, $\operatorname{osc}_f(x)$, 60

P
$\mathfrak{P}_0(Y)$, 180
$\mathfrak{P}(Y)$, 180
Para_2, 728
Paracompact, 124
characterization, 63, 72
definition, 61
Partially exact homology theory, 729
Partial realization, 533
Partition, 406
Partition Extension Lemma, 414
Partition of unity, 69
— (weakly) subordinated to an open cover, 70
locally finite, 69
Path, 10
Path-component, 381
Path-connected, 10, 223, 479, 566
Path space, $P(X, x_0)$, 666
Peano continuum, 477
Perfectly normal, 43, 216
Perfect map, 35, 108
$\pi_0(X)$, 381, 710
$\pi_0(X, x_0)$, 710
$\pi_1(X, x_0)$, 708
$\pi_n(X)$, 710
$\pi_n(X, A, x_0)$, 383, 711
$\pi_n(X, x_0)$, 709
Piecewise Linear (PL)
— bi-collared, 314
— collar, 314
— collared, 314
— homeomorphism, 301, 302
— map, 292, 302
PL Approximation Theorem, 300
PL Embedding Approximation Theorem, 336
PL manifold
— without boundary, 324
boundary of, 324
boundary point of, 324
closed, 324
interior of, 324
interior point of, 324
n-dimensional, 317
PL n-manifold, 317
Point at infinity, 38
Pointed
— function, 710
— homotopic, 12
— homotopy, 12
— homotopy equivalence, 12
— homotopy equivalent, 12
— homotopy inverse, 12
— map, 12
— set, 710
— space, 2
Point-finite, 8, 72
Point reductive cohomology theory, 728
Pointwise convergence topology, 15, 16, 27, 95
Polyhedrally 0-soft, 616
Polyhedral neighborhood, 218
Polyhedron, 214, 234
finite-dimensional (f.d.), 235
infinite-dimensional (i.f.d.), 235
locally finite-dimensional (l.f.d.), 235
Principal cell, 215
Product
— cell complex, 241
— simplicial complex, 243
— topology, 3
Productive property, 81
Product Theorem, 418
Proper
— face, 209
— homotopic, 39
— map, 35, 108
Proper homotopy, 39
— dominated, 539
— equivalence, 39
— equivalent, 39
— equivalent relative to, 366
— inverse, 39
— type, 39
Property C, 563, 640

Property UV^*, 610
Property UV^∞, 611
Property UV^n, 611
Pseudo-arc, 364, 387
Pseudo-metric, 52

Q
$\boldsymbol{Q}$, the Hilbert cube, 4
Quasi-component, 594
Quasi-monotone map, 686
Quasi-open map, 686
Quotient
— F-normed linear space, 175
— linear space, 155
— normed linear space, 175

R
$\mathbb{R}^\Gamma_f$, 17
$\mathbb{R}^n$, Euclidean n-space, 4
$\mathbb{R}^n_+$, Euclidean half n-space, 4
Radial
— boundary, 135
— closure, 135
— interior, 135
Refine, 8
Refinement, 9
— by open balls, 53, 74
Refining simplicial map, 341
Regular neighborhood, 313, 326
Relative
— cohomology group, 731
— homotopy group, 711
Relative PL Approximation Theorem, 332
Restriction of a cover, 9
Retract, 266, 491
Retraction, 266, 491

S
$\boldsymbol{s}$, the space of sequences, 4
$\mathbf{S}^{n-1}$, the unit $(n-1)$-sphere, 4
$\boldsymbol{s}_f$, 17
$\mathbf{S}_X$, the unit sphere of X, 18
Same homotopy type, 11
Same proper homotopy type, 39
Selection, 180
Semi-locally contractible, 512
Separated, 40
Separation Theorem, 146, 156
Shape equivalence, 635
Shrinking, 62
ΣX, suspension of X, 643
Simple chain, 478
Simplex, 204
Simplicial
— approximation, 290
— complex, 214
— cone, 219, 229, 370
— Edwards–Walsh complex, 664
— embedding, 287
— homeomorphism, 233, 286
— isomorphism, 286
— map, 285
— map between abstract complexes, 339
— mapping cylinder, 370
Simplicial Approximation Theorem, 290
Simply connected, 566, 718
Skeleton
— of a cell complex, 214
— of an abstract complex, 338
Sliced neighborhood, 309
Small inductive dimension, (ind), 420
Small transfinite inductive dimension, 432
Soft (n-soft)
— map, 615
approximately, 617
homotopically, 617
polyhedrally, 615
polyhedrally approximately, 617
polyhedrally homotopy, 617
Sorgenfrey line, 90
Sorgenfrey plane, 90
Source majorant topology, 109
Sperner map, 392
Sperner's Lemma, 392
Star, 49
Star-finite, 8
— open cover, 344
Star in a complex, 217
Star-refinement, 52
Starring, 263
Stellar subdivision, 263
Stone–Čech compactification, βX, 31
Stone's Theorem, 45
Straight-line homotopy, 252
Strong
— deformation retract, 305, 508, 521
— deformation retraction, 305, 508, 521
— local n-connection, 623
— neighborhood deformation retract, 508
— open shrinking, 62
Strongly
— countable-dimensional (s.c.d.), 430

— infinite-dimensional (s.i.d.), 428
— locally contractible, 517
Strong Separation Theorem, 157
Subcomplex
— of a cell complex, 213
— of a simplicial complex, 338
Subcover, 9
Subdivision
— of a cell complex, 224
simplicial, 224
Subflat, 131
Sublinear, 142
Subpolyhedron, 235
Subsequence of an inverse sequence, 359
Subset Theorem, 413
Sup-metric, 13
Support of a map, 69
Suspension, 643
n-fold, 645
Swelling, 398
Swelling Lemma, 398

T
Telescope, 549
Tietze–Urysohn Extension Theorem, 40, 42, 124
Tight PL map, 296, 302
Topological
— group, 153
— linear space, 151
— sum, 7
Topologically bounded, 158
Totally bounded, 166, 437
Totally disconnected, 466
Transfinite inductive dimension
large, 432
small, 432
Triangulation, 234
trInd, 432
trind, 432
Trivial shape, 610
Tychonoff plank, 84
Tychonoff Theorem, 30

U
$\mathcal{U}(f)$, 103
$\mathcal{U}$-close, 12
$\mathcal{U}$-homotopic, 13
$\mathcal{U}$-homotopy, 12
$\mathcal{U}$-homotopy equivalence, 543
$U_d(f)$, 104
$\mathcal{U}$-map, 442
Unified Locally Contractible (ULC), 513
Uniform
— AE, 552
— ANE, 552
— ANR, 552
— AR, 552
— convergence topology, 13, 16
— neighborhood, 551
— neighborhood retract, 552
— retract, 551
Uniformly
— continuous at A, 551
— locally contractible, 513, 604
— locally path-connected, 483
Universal map, in the sense of Holszyński, 486
Universal space, 449
Upper semi-continuous (u.s.c.)
— real-valued function, 73
— set-valued function, 181
Urysohn map, 42
Urysohn's Lemma, 42
UV^*, 610
UV^* map, 636
UV^∞, 611
UV^∞ map, 636
UV^n, 611
UV^n map, 636

V
Vertex (pl., vertices)
— of a cell, 207
— of a simplex, 204

W
Wallace's Theorem, 30
Walsh's Theorem, 685
Weak homotopy equivalence, 380, 717, 724
Weakly infinite-dimensional (w.i.d.), 428
— in the sense of Alexandroff (A-w.i.d.), 436
— in the sense of Smirnov (S-w.i.d.), 436
Weak topology, 7
— of a polyhedron, 215
Wedge (in a linear space), 146
Wedge sum, 662
Whitehead–Milnor Theorem, 375

Whitehead Theorem on small subdivisions, 276
Whitehead topology of a polyhedron, 215
w-segment, 595
$w(X)$, the weight of X, 2

Z

Zero-sequence, 549
Zero set, 43
0-soft, 616, 617